

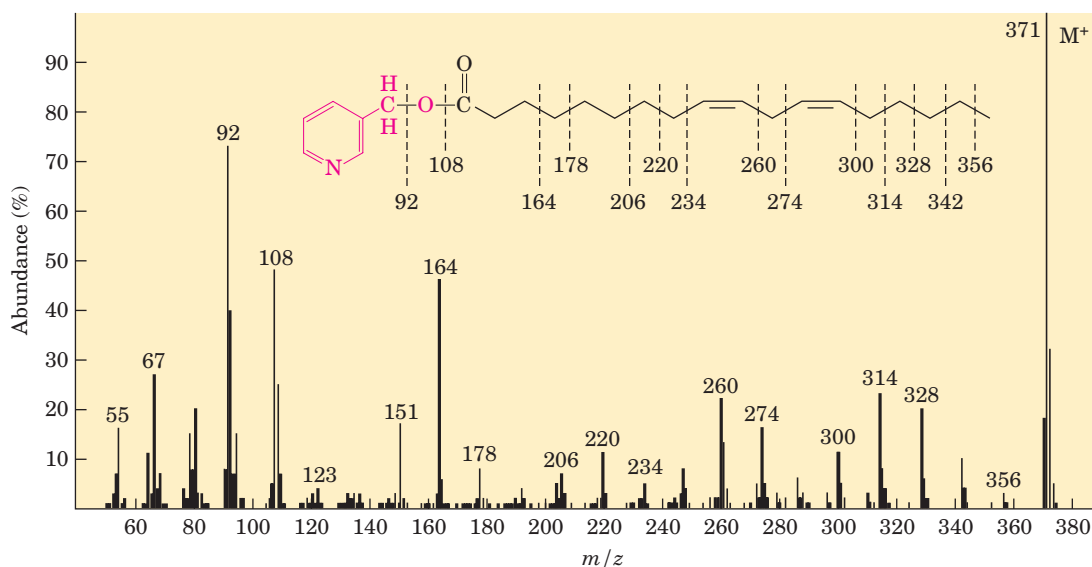


FIGURE 10-24 Determination of the structure of a fatty acid by mass spectrometry. The fatty acid is first converted to a derivative that minimizes migration of the double bonds when the molecule is fragmented by electron bombardment. The derivative shown here is a picolinyl ester of linoleic acid—18:2(Δ^{9,12}) (M_r 371)—in which the alcohol is picolinol (red). When bombarded with a stream of electrons, this molecule is volatilized and converted to a parent ion (M^+ ; M_r 371), in which the N atom bears the positive charge, and a series of smaller fragments produced by breakage of C—C bonds in the fatty acid. The mass spectrometer separates these charged fragments according to their mass/charge ratio (m/z). (To review the principles of mass spectrometry, see Box 3–2.)

The prominent ions at $m/z = 92$, 108, 151, and 164 contain the pyridine ring of the picolinol and various fragments of the carboxyl group, showing that the compound is indeed a picolinyl ester. The molecular ion ($m/z = 371$) confirms the presence of a C-18 fatty acid with two double bonds. The uniform series of ions 14 atomic mass units (amu) apart represents loss of each successive methyl and methylene group from the right end of the molecule (C-18 of the fatty acid), until the ion at $m/z = 300$ is reached. This is followed by a gap of 26 amu for the carbons of the terminal double bond, at $m/z = 274$; a further gap of 14 amu for the C-11 methylene group, at $m/z = 260$, and so forth. By this means the entire structure is determined, although these data alone do not reveal the configuration (cis or trans) of the double bonds.

SUMMARY 10.4 Working with Lipids

- In the determination of lipid composition, the lipids are first extracted from tissues with organic solvents and separated by thin-layer, gas-liquid, or high-performance liquid chromatography.

- Phospholipases specific for one of the bonds in a phospholipid can be used to generate simpler compounds for subsequent analysis.
- Individual lipids are identified by their chromatographic behavior, their susceptibility to hydrolysis by specific enzymes, or mass spectrometry.

Key Terms

Terms in bold are defined in the glossary.

fatty acid 343	plasmalogen 349
triacylglycerol 345	galactolipid 351
lipases 346	sphingolipid 352
phospholipid 348	ceramide 352
glycolipid 348	sphingomyelin 352
glycerophospholipid 349	glycosphingolipid 352
ether lipid 349	cerebroside 352
	globoside 352

neutral glycolipids 352
gangliosides 352
sterols 354
cholesterol 355
prostaglandins 359
thromboxanes 359
leukotrienes 359
vitamin 360

vitamin D ₃ 361
cholecalciferol 361
vitamin A (retinol) 361
vitamin E 362
tocopherols 362
vitamin K 363
dolichol 363

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Problems

1. Operational Definition of Lipids How is the definition of “lipid” different from the types of definitions used for other biomolecules that we have considered, such as amino acids, nucleic acids, and proteins?

2. Melting Points of Lipids The melting points of a series of 18-carbon fatty acids are: stearic acid, 69.6 °C; oleic acid, 13.4 °C; linoleic acid, –5 °C; and linolenic acid, –11 °C.

(a) What structural aspect of these 18-carbon fatty acids

can be correlated with the melting point? Provide a molecular explanation for the trend in melting points.

(b) Draw all the possible triacylglycerols that can be constructed from glycerol, palmitic acid, and oleic acid. Rank them in order of increasing melting point.

(c) Branched-chain fatty acids are found in some bacterial membrane lipids. Would their presence increase or decrease the fluidity of the membranes (that is, give them a lower or higher melting point)? Why?

3. Preparation of Béarnaise Sauce During the preparation of béarnaise sauce, egg yolks are incorporated into melted butter to stabilize the sauce and avoid separation. The stabilizing agent in the egg yolks is lecithin (phosphatidylcholine). Suggest why this works.

4. Hydrophobic and Hydrophilic Components of Membrane Lipids A common structural feature of membrane lipids is their amphipathic nature. For example, in phosphatidylcholine, the two fatty acid chains are hydrophobic and the phosphocholine head group is hydrophilic. For each of the following membrane lipids, name the components that serve as the hydrophobic and hydrophilic units: (a) phosphatidylethanolamine; (b) sphingomyelin; (c) galactosylcerebroside; (d) ganglioside; (e) cholesterol.

5. Alkali Lability of Triacylglycerols A common procedure for cleaning the grease trap in a sink is to add a product that contains sodium hydroxide. Explain why this works.



6. The Action of Phospholipases The venom of the Eastern diamondback rattler and the Indian cobra contains phospholipase A₂, which catalyzes the hydrolysis of fatty acids at the C-2 position of glycerophospholipids. The phospholipid breakdown product of this reaction is lysolecithin (lecithin is phosphatidylcholine). At high concentrations, this and other lysophospholipids act as detergents, dissolving the membranes of erythrocytes and lysing the cells. Extensive hemolysis may be life-threatening.

(a) Detergents are amphipathic. What are the hydrophilic and hydrophobic portions of lysolecithin?

(b) The pain and inflammation caused by a snake bite can be treated with certain steroids. What is the basis of this treatment?

(c) Though high levels of phospholipase A₂ can be deadly, this enzyme is necessary for a variety of normal metabolic processes. What are these processes?

7. Intracellular Messengers from Phosphatidylinositols When the hormone vasopressin stimulates cleavage of

phosphatidylinositol 4,5-bisphosphate by hormone-sensitive phospholipase C, two products are formed. What are they? Compare their properties and their solubilities in water, and predict whether either would diffuse readily through the cytosol.

8. Storage of Fat-Soluble Vitamins In contrast to water-soluble vitamins, which must be a part of our daily diet, fat-soluble vitamins can be stored in the body in amounts sufficient for many months. Suggest an explanation for this difference, based on solubilities.

9. Hydrolysis of Lipids Name the products of mild hydrolysis with dilute NaOH of (a) 1-stearoyl-2,3-dipalmitoylglycerol; (b) 1-palmitoyl-2-oleoylphosphatidylcholine.

10. Effect of Polarity on Solubility Rank the following in order of increasing solubility in water: a triacylglycerol, a diacylglycerol, and a monoacylglycerol, all containing only palmitic acid.

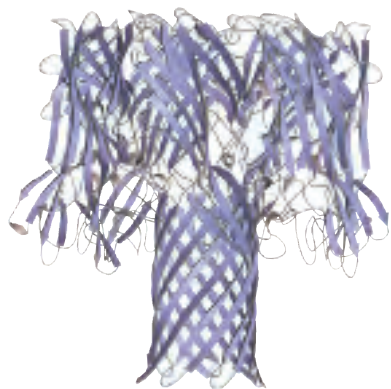
11. Chromatographic Separation of Lipids A mixture of lipids is applied to a silica gel column, and the column is then washed with increasingly polar solvents. The mixture consists of phosphatidylserine, phosphatidylethanolamine, phosphatidylcholine, cholesteryl palmitate (a sterol ester), sphingomyelin, palmitate, *n*-tetradecanol, triacylglycerol, and cholesterol. In what order do you expect the lipids to elute from the column? Explain your reasoning.

12. Identification of Unknown Lipids Johann Thudichum, who practiced medicine in London about 100 years ago, also dabbled in lipid chemistry in his spare time. He isolated a variety of lipids from neural tissue, and characterized and named many of them. His carefully sealed and labeled vials of isolated lipids were rediscovered many years later.

(a) How would you confirm, using techniques not available to Thudichum, that the vials labeled “sphingomyelin” and “cerebroside” actually contain these compounds?

(b) How would you distinguish sphingomyelin from phosphatidylcholine by chemical, physical, or enzymatic tests?

13. Ninhydrin to Detect Lipids on TLC Plates Ninhydrin reacts specifically with primary amines to form a purplish-blue product. A thin-layer chromatogram of rat liver phospholipids is sprayed with ninhydrin, and the color is allowed to develop. Which phospholipids can be detected in this way?



chapter 11

BIOLOGICAL MEMBRANES AND TRANSPORT

- 11.1 The Composition and Architecture of Membranes 370**
- 11.2 Membrane Dynamics 380**
- 11.3 Solute Transport across Membranes 389**

Good fences make good neighbors.

—Robert Frost, “Mending Wall,” in North of Boston, 1914

The first cell probably came into being when a membrane formed, enclosing a small volume of aqueous solution and separating it from the rest of the universe. Membranes define the external boundaries of cells and regulate the molecular traffic across that boundary (Fig. 11–1); in eukaryotic cells, they divide the internal space into discrete compartments to segregate processes and components. They organize complex reaction sequences and are central to both biological energy conservation and cell-to-cell communication. The biological activities of membranes flow from their remarkable physical properties. Membranes are flexible, self-sealing, and selectively permeable to polar solutes. Their flexibility permits the shape changes that accompany cell growth and movement (such as amoeboid movement). With their ability to break and reseal, two membranes can fuse, as in exocytosis, or a single membrane-enclosed compartment can undergo fission to yield two sealed compartments, as in endocytosis or cell division, without creating gross leaks through cellular surfaces. Because membranes are selectively permeable, they retain certain compounds and ions within cells and within specific cellular compartments, while excluding others.

Membranes are not merely passive barriers. They include an array of proteins specialized for promoting or

catalyzing various cellular processes. At the cell surface, transporters move specific organic solutes and inorganic ions across the membrane; receptors sense extracellular signals and trigger molecular changes in the cell; adhesion molecules hold neighboring cells together. Within the cell, membranes organize cellular processes such as the synthesis of lipids and certain proteins, and the energy transductions in mitochondria and chloroplasts. Because membranes consist of just two layers of molecules, they are very thin—essentially two-dimensional. Intermolecular collisions are far more probable in this two-dimensional space than in three-dimensional space, so the efficiency of enzyme-catalyzed processes organized within membranes is vastly increased.

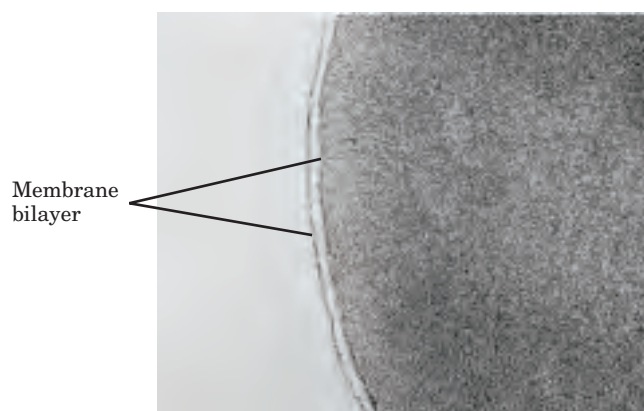


FIGURE 11–1 Biological membranes. Viewed in cross section, all cell membranes share a characteristic trilaminar appearance. When an erythrocyte is stained with osmium tetroxide and viewed with an electron microscope, the plasma membrane appears as a three-layer structure, 5 to 8 nm (50 to 80 Å) thick. The trilaminar image consists of two electron-dense layers (the osmium, bound to the inner and outer surfaces of the membrane) separated by a less dense central region.

In this chapter we first describe the composition of cellular membranes and their chemical architecture—the molecular structures that underlie their biological functions. Next, we consider the remarkable dynamic features of membranes, in which lipids and proteins move relative to each other. Cell adhesion, endocytosis, and the membrane fusion accompanying neurotransmitter secretion illustrate the dynamic role of membrane proteins. We then turn to the protein-mediated passage of solutes across membranes via transporters and ion channels. In later chapters we discuss the role of membranes in signal transduction (Chapters 12 and 23), energy transduction (Chapter 19), lipid synthesis (Chapter 21), and protein synthesis (Chapter 27).

11.1 The Composition and Architecture of Membranes

One approach to understanding membrane function is to study membrane composition—to determine, for example, which components are common to all membranes and which are unique to membranes with specific functions. So before describing membrane structure and function we consider the molecular components of membranes: proteins and polar lipids, which account for almost all the mass of biological membranes, and carbohydrates, present as part of glycoproteins and glycolipids.

Each Type of Membrane Has Characteristic Lipids and Proteins

The relative proportions of protein and lipid vary with the type of membrane (Table 11–1), reflecting the diversity of biological roles. For example, certain neurons have a myelin sheath, an extended plasma membrane that wraps around the cell many times and acts as a passive electrical insulator. The myelin sheath consists primarily of lipids, whereas the plasma membranes of bacteria and the membranes of mitochondria and

chloroplasts, the sites of many enzyme-catalyzed processes, contain more protein than lipid (in mass per total mass).

For studies of membrane composition, the first task is to isolate a selected membrane. When eukaryotic cells are subjected to mechanical shear, their plasma membranes are torn and fragmented, releasing cytoplasmic components and membrane-bounded organelles such as mitochondria, chloroplasts, lysosomes, and nuclei. Plasma membrane fragments and intact organelles can be isolated by centrifugal techniques described in Chapter 1 (see Fig. 1–8).

Chemical analyses of membranes isolated from various sources reveal certain common properties. Each kingdom, each species, each tissue or cell type, and the organelles of each cell type have a characteristic set of membrane lipids. Plasma membranes, for example, are enriched in cholesterol and contain no detectable cardiolipin (Fig. 11–2); in the inner mitochondrial membrane of the hepatocyte, this distribution is reversed: very low cholesterol and high cardiolipin. Cardiolipin is essential to the function of certain proteins of the inner mitochondrial membrane. Cells clearly have mechanisms to control the kinds and amounts of membrane lipids they synthesize and to target specific lipids to particular organelles. In many cases, we can surmise the adaptive advantages of distinct combinations of membrane lipids; in other cases, the functional significance of these combinations is as yet unknown.

The protein composition of membranes from different sources varies even more widely than their lipid composition, reflecting functional specialization. In a rod cell of the vertebrate retina, one portion of the cell is highly specialized for the reception of light; more than 90% of the plasma membrane protein in this region is the light-absorbing glycoprotein rhodopsin. The less-specialized plasma membrane of the erythrocyte has about 20 prominent types of proteins as well as scores of minor ones; many of these are transporters, each moving a specific solute across the membrane. The plasma membrane of *Escherichia coli* contains hun-

TABLE 11–1 Major Components of Plasma Membranes in Various Organisms					
	Components (% by weight)				
	Protein	Phospholipid	Sterol	Sterol type	Other lipids
Human myelin sheath	30	30	19	Cholesterol	Galactolipids, plasmalogens
Mouse liver	45	27	25	Cholesterol	—
Maize leaf	47	26	7	Sitosterol	Galactolipids
Yeast	52	7	4	Ergosterol	Triacylglycerols, steryl esters
<i>Paramecium</i> (ciliated protist)	56	40	4	Stigmasterol	—
<i>E. coli</i>	75	25	0	—	—

Note: Values do not add up to 100% in every case, because there are components other than protein, phospholipids, and sterol; plants, for example, have high levels of glycolipids.

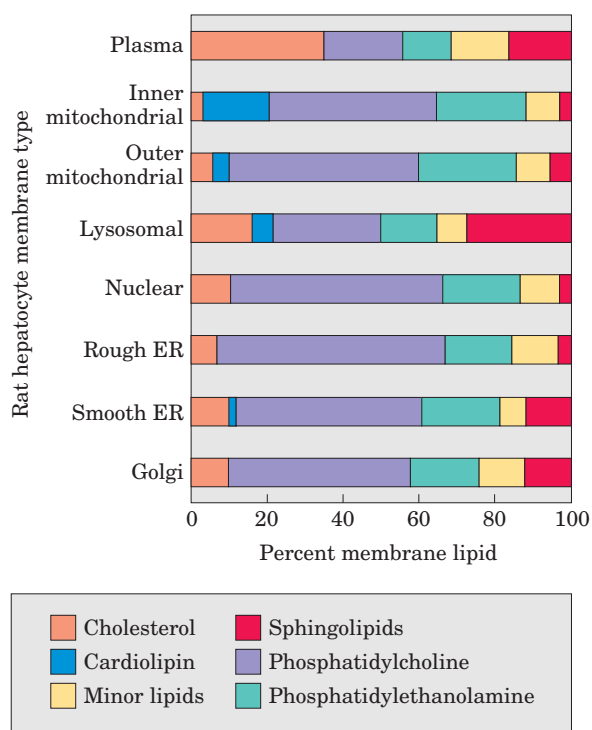


FIGURE 11-2 Lipid composition of the plasma membrane and organelle membranes of a rat hepatocyte. The functional specialization of each membrane type is reflected in its unique lipid composition. Cholesterol is prominent in plasma membranes but barely detectable in mitochondrial membranes. Cardiolipin is a major component of the inner mitochondrial membrane but not of the plasma membrane. Phosphatidylserine, phosphatidylinositol, and phosphatidylglycerol are relatively minor components (yellow) of most membranes but serve critical functions; phosphatidylinositol and its derivatives, for example, are important in signal transductions triggered by hormones. Sphingolipids, phosphatidylcholine, and phosphatidylethanolamine are present in most membranes, but in varying proportions. Glycolipids, which are major components of the chloroplast membranes of plants, are virtually absent from animal cells.

dreds of different proteins, including transporters and many enzymes involved in energy-conserving metabolism, lipid synthesis, protein export, and cell division. The outer membrane of *E. coli*, which encloses the plasma membrane, has a different function (protection) and a different set of proteins.

Some membrane proteins are covalently linked to complex arrays of carbohydrate. For example, in glycophorin, a glycoprotein of the erythrocyte plasma membrane, 60% of the mass consists of complex oligosaccharide units covalently attached to specific amino acid residues. Ser, Thr, and Asn residues are the most common points of attachment (see Fig. 7-31). At the other end of the scale is rhodopsin of the rod cell plasma membrane, which contains just one hexasaccharide. The sugar moieties of surface glycoproteins influence the folding of the proteins, as well as their sta-

bilities and intracellular destinations, and they play a significant role in the specific binding of ligands to glycoprotein surface receptors (see Fig. 7-37).

Some membrane proteins are covalently attached to one or more lipids, which serve as hydrophobic anchors that hold the proteins to the membrane, as we shall see.

All Biological Membranes Share Some Fundamental Properties

Membranes are impermeable to most polar or charged solutes, but permeable to nonpolar compounds; they are 5 to 8 nm (50 to 80 Å) thick and appear trilaminar when viewed in cross section with the electron microscope (Fig. 11-1). The combined evidence from electron microscopy and studies of chemical composition, as well as physical studies of permeability and the motion of individual protein and lipid molecules within membranes, led to the development of the **fluid mosaic model** for the structure of biological membranes (Fig. 11-3). Phospholipids form a bilayer in which the nonpolar regions of the lipid molecules in each layer face the core of the bilayer and their polar head groups face outward, interacting with the aqueous phase on either side. Proteins are embedded in this bilayer sheet, held by hydrophobic interactions between the membrane lipids and hydrophobic domains in the proteins. Some proteins protrude from only one side of the membrane; others have domains exposed on both sides. The orientation of proteins in the bilayer is asymmetric, giving the membrane “sidedness”: the protein domains exposed on one side of the bilayer are different from those exposed on the other side, reflecting functional asymmetry. The individual lipid and protein units in a membrane form a fluid mosaic with a pattern that, unlike a mosaic of ceramic tile and mortar, is free to change constantly. The membrane mosaic is fluid because most of the interactions among its components are noncovalent, leaving individual lipid and protein molecules free to move laterally in the plane of the membrane.

We now look at some of these features of the fluid mosaic model in more detail and consider the experimental evidence that supports the basic model but has necessitated its refinement in several ways.

A Lipid Bilayer Is the Basic Structural Element of Membranes

Glycerophospholipids, sphingolipids, and sterols are virtually insoluble in water. When mixed with water, they spontaneously form microscopic lipid aggregates in a phase separate from their aqueous surroundings, clustering together, with their hydrophobic moieties in contact with each other and their hydrophilic groups interacting with the surrounding water. Recall that lipid clustering reduces the amount of hydrophobic surface

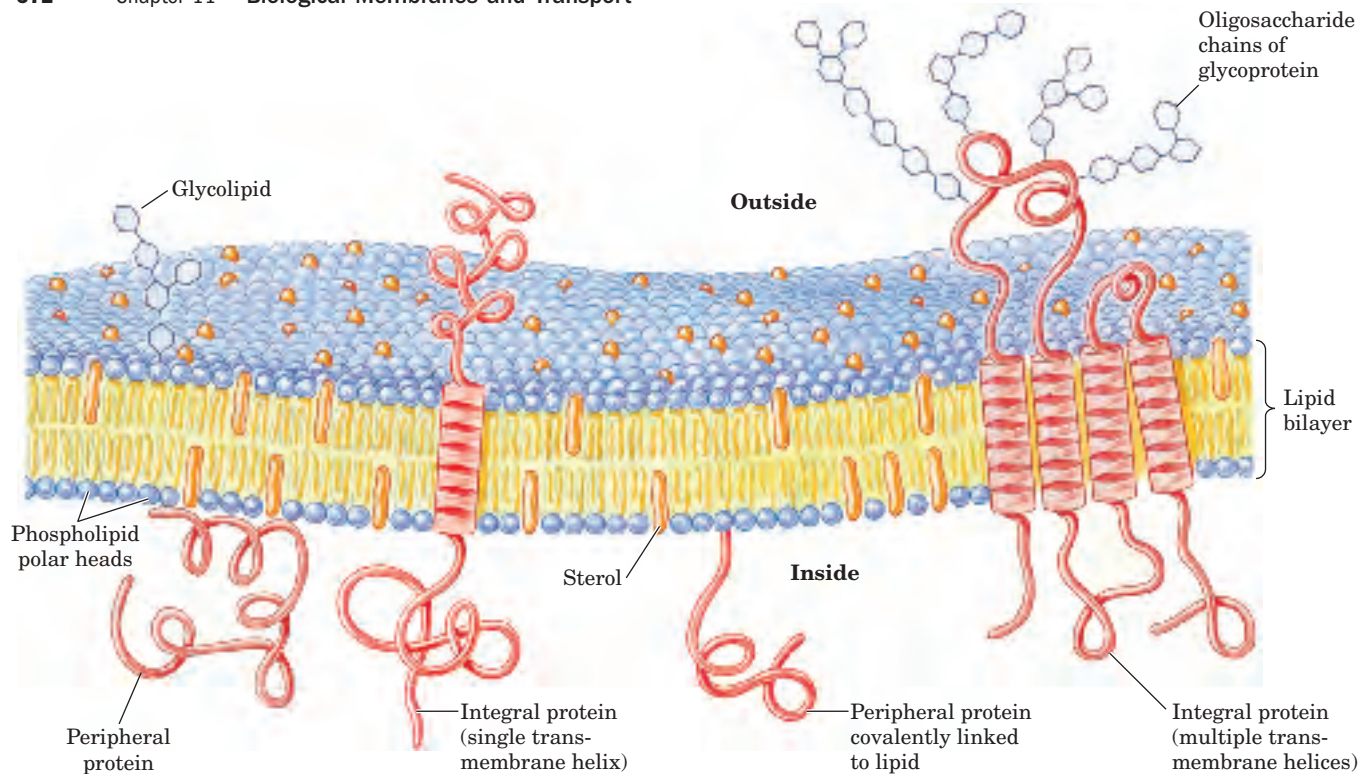


FIGURE 11-3 Fluid mosaic model for membrane structure. The fatty acyl chains in the interior of the membrane form a fluid, hydrophobic region. Integral proteins float in this sea of lipid, held by hydrophobic interactions with their nonpolar amino acid side chains. Both proteins and lipids are free to move laterally in the plane of the

bilayer, but movement of either from one face of the bilayer to the other is restricted. The carbohydrate moieties attached to some proteins and lipids of the plasma membrane are exposed on the extracellular surface of the membrane.

exposed to water and thus minimizes the number of molecules in the shell of ordered water at the lipid-water interface (see Fig. 2-7), resulting in an increase in entropy. Hydrophobic interactions among lipid molecules provide the thermodynamic driving force for the formation and maintenance of these clusters.

Depending on the precise conditions and the nature of the lipids, three types of lipid aggregates can form when amphipathic lipids are mixed with water (Fig. 11-4). **Micelles** are spherical structures that contain anywhere from a few dozen to a few thousand amphipathic molecules. These molecules are arranged with

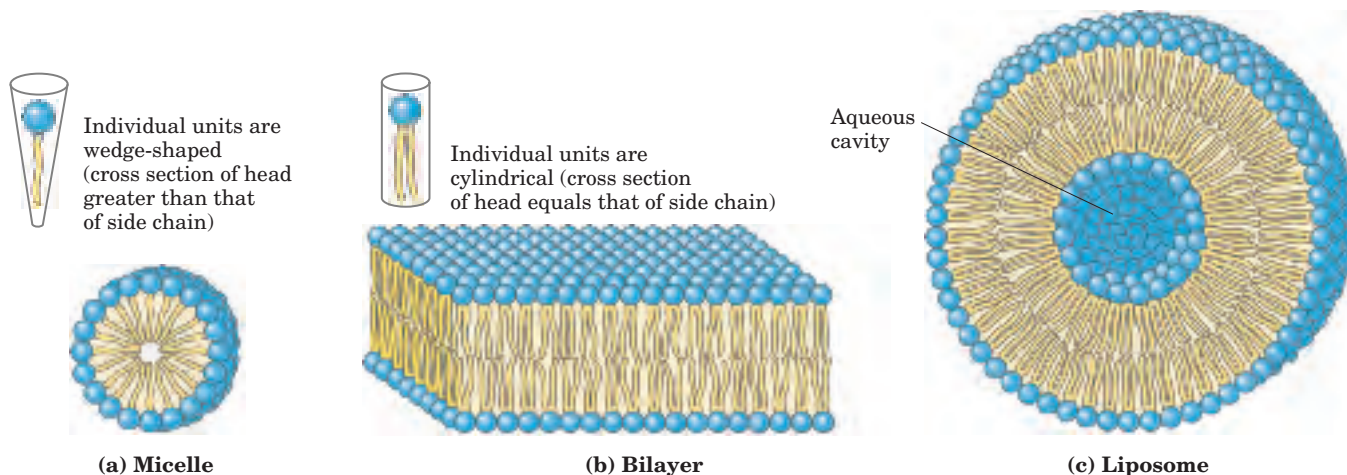


FIGURE 11-4 Amphipathic lipid aggregates that form in water. (a) In micelles, the hydrophobic chains of the fatty acids are sequestered at the core of the sphere. There is virtually no water in the hydrophobic interior. (b) In an open bilayer, all acyl side chains except those at the

edges of the sheet are protected from interaction with water. (c) When a two-dimensional bilayer folds on itself, it forms a closed bilayer, a three-dimensional hollow vesicle (liposome) enclosing an aqueous cavity.

their hydrophobic regions aggregated in the interior, where water is excluded, and their hydrophilic head groups at the surface, in contact with water. Micelle formation is favored when the cross-sectional area of the head group is greater than that of the acyl side chain(s), as in free fatty acids, lysophospholipids (phospholipids lacking one fatty acid), and detergents such as sodium dodecyl sulfate (SDS; p. 92).

A second type of lipid aggregate in water is the **bilayer**, in which two lipid monolayers (leaflets) form a two-dimensional sheet. Bilayer formation occurs most readily when the cross-sectional areas of the head group and acyl side chain(s) are similar, as in glycerophospholipids and sphingolipids. The hydrophobic portions in each monolayer, excluded from water, interact with each other. The hydrophilic head groups interact with water at each surface of the bilayer. Because the hydrophobic regions at its edges (Fig. 11-4b) are transiently in contact with water, the bilayer sheet is relatively unstable and spontaneously forms a third type of aggregate: it folds back on itself to form a hollow sphere, a vesicle or **liposome** (Fig. 11-4c). By forming vesicles, bilayers lose their hydrophobic edge regions, achieving maximal stability in their aqueous environment. These bilayer vesicles enclose water, creating a separate aqueous compartment. It is likely that the precursors to the first living cells resembled liposomes, their aqueous contents segregated from the rest of the world by a hydrophobic shell.

Biological membranes are constructed of lipid bilayers 3 nm (30 Å) thick, with proteins protruding on each side. The hydrocarbon core of the membrane, made up of the $-\text{CH}_2-$ and $-\text{CH}_3$ of the fatty acyl groups, is about as nonpolar as decane, and liposomes formed in the laboratory from pure lipids are essentially impermeable to polar solutes, as are biological membranes (although the latter, as we shall see, are permeable to solutes for which they have specific transporters).

Plasma membrane lipids are asymmetrically distributed between the two monolayers of the bilayer, although the asymmetry, unlike that of membrane proteins, is not absolute. In the plasma membrane of the erythrocyte, for example, choline-containing lipids (phosphatidylcholine and sphingomyelin) are typically found in the outer (extracellular or exoplasmic) leaflet (Fig. 11-5), whereas phosphatidylserine, phosphatidylethanolamine, and the phosphatidylinositols are much more common in the inner (cytoplasmic) leaflet. Changes in the distribution of lipids between plasma membrane leaflets have biological consequences. For example, only when the phosphatidylserine in the plasma membrane moves into the outer leaflet is a platelet able to play its role in formation of a blood clot. For many other cells types, phosphatidylserine exposure on the outer surface marks a cell for destruction by programmed cell death.

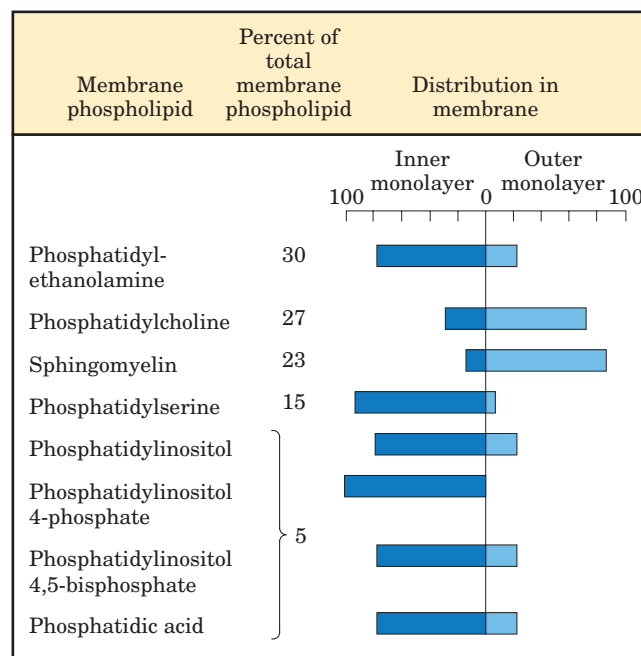


FIGURE 11-5 Asymmetric distribution of phospholipids between the inner and outer monolayers of the erythrocyte plasma membrane. The distribution of a specific phospholipid is determined by treating the intact cell with phospholipase C, which cannot reach lipids in the inner monolayer (leaflet) but removes the head groups of lipids in the outer monolayer. The proportion of each head group released provides an estimate of the fraction of each lipid in the outer monolayer.

Peripheral Membrane Proteins Are Easily Solubilized

Membrane proteins may be divided operationally into two groups (Fig. 11-6). **Integral proteins** are very firmly associated with the membrane, removable only by agents that interfere with hydrophobic interactions, such as detergents, organic solvents, or denaturants. **Peripheral proteins** associate with the membrane through electrostatic interactions and hydrogen bonding with the hydrophilic domains of integral proteins and with the polar head groups of membrane lipids. They can be released by relatively mild treatments that interfere with electrostatic interactions or break hydrogen bonds; a commonly used agent is carbonate at high pH. Peripheral proteins may serve as regulators of membrane-bound enzymes or may limit the mobility of integral proteins by tethering them to intracellular structures.

Many Membrane Proteins Span the Lipid Bilayer

Membrane protein topology (localization relative to the lipid bilayer) can be determined with reagents that react with protein side chains but cannot cross membranes—polar chemical reagents that react with primary amines of Lys residues, for example, or enzymes

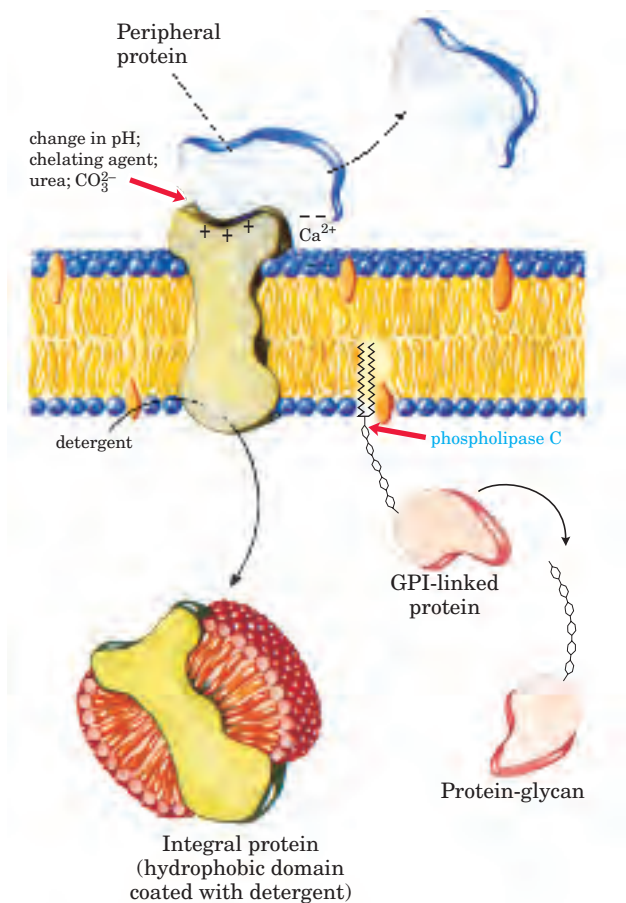


FIGURE 11-6 Peripheral and integral proteins. Membrane proteins can be operationally distinguished by the conditions required to release them from the membrane. Most peripheral proteins are released by changes in pH or ionic strength, removal of Ca^{2+} by a chelating agent, or addition of urea or carbonate. Integral proteins are extractable with detergents, which disrupt the hydrophobic interactions with the lipid bilayer and form micelle-like clusters around individual protein molecules. Integral proteins covalently attached to a membrane lipid, such as a glycosyl phosphatidylinositol (GPI; see Fig. 11-14), can be released by treatment with phospholipase C.

like trypsin that cleave proteins but cannot cross the membrane. The human erythrocyte is convenient for such studies because it has no membrane-bounded organelles; the plasma membrane is the only membrane present. If a membrane protein in an intact erythrocyte reacts with a membrane-impermeant reagent, that protein must have at least one domain exposed on the outer (extracellular) face of the membrane. Trypsin is found to cleave extracellular domains but does not affect domains buried within the bilayer or exposed on the inner surface only, unless the plasma membrane is broken to make these domains accessible to the enzyme.

Experiments with such topology-specific reagents show that the erythrocyte glycoprotein **glycophorin** spans the plasma membrane. Its amino-terminal domain (bearing the carbohydrate chains) is on the outer sur-

face and is cleaved by trypsin. The carboxyl terminus protrudes on the inside of the cell, where it cannot react with impermeant reagents. Both the amino-terminal and carboxyl-terminal domains contain many polar or charged amino acid residues and are therefore quite hydrophilic. However, a segment in the center of the protein (residues 75 to 93) contains mainly hydrophobic amino acid residues, suggesting that glycophorin has a transmembrane segment arranged as shown in Figure 11-7.

One further fact may be deduced from the results of experiments with glycophorin: its disposition in the membrane is asymmetric. Similar studies of other membrane proteins show that each has a specific orientation

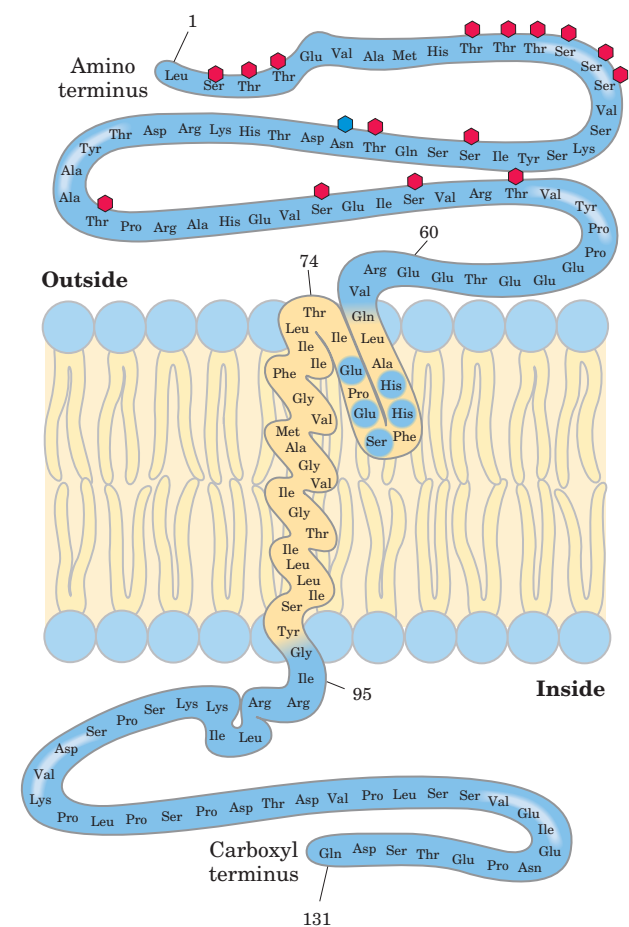


FIGURE 11-7 Transbilayer disposition of glycophorin in an erythrocyte. One hydrophilic domain, containing all the sugar residues, is on the outer surface, and another hydrophilic domain protrudes from the inner face of the membrane. Each red hexagon represents a tetrasaccharide (containing two Neu5Ac (sialic acid), Gal, and GalNAc) O-linked to a Ser or Thr residue; the blue hexagon represents an oligosaccharide chain N-linked to an Asn residue. The relative size of the oligosaccharide units is larger than shown here. A segment of 19 hydrophobic residues (residues 75 to 93) forms an α helix that traverses the membrane bilayer (see Fig. 11-11a). The segment from residues 64 to 74 has some hydrophobic residues and probably penetrates into the outer face of the lipid bilayer, as shown.

in the bilayer; one domain of a transmembrane protein always faces out, the other always faces in. Furthermore, glycoproteins of the plasma membrane are invariably situated with their sugar residues on the outer surface of the cell. As we shall see, the asymmetric arrangement of membrane proteins results in functional asymmetry. All the molecules of a given ion pump, for example, have the same orientation in the membrane and therefore pump in the same direction.

Integral Proteins Are Held in the Membrane by Hydrophobic Interactions with Lipids

The firm attachment of integral proteins to membranes is the result of hydrophobic interactions between membrane lipids and hydrophobic domains of the protein. Some proteins have a single hydrophobic sequence in the middle (as in glycophorin) or at the amino or carboxyl terminus. Others have multiple hydrophobic sequences, each of which, when in the α -helical conformation, is long enough to span the lipid bilayer (Fig. 11–8). The same techniques used to determine the three-dimensional structures of soluble proteins can, in principle, be applied to membrane proteins. In practice, however, membrane proteins have until recently proved difficult to crystallize. New techniques are overcoming this obstacle, and crystallographic structures of membrane proteins are regularly becoming available, yielding deep insights into membrane events at the molecular level.

One of the best-studied membrane-spanning proteins, bacteriorhodopsin, has seven very hydrophobic internal sequences and crosses the lipid bilayer seven times. Bacteriorhodopsin is a light-driven proton pump densely packed in regular arrays in the purple membrane of the bacterium *Halobacterium salinarum*. X-ray crystallography reveals a structure with seven α -helical segments, each traversing the lipid bilayer, connected by nonhelical loops at the inner and outer face of the membrane (Fig. 11–9). In the amino acid sequence of bacteriorhodopsin, seven segments of about 20 hydrophobic residues can be identified, each segment just long enough to form an α helix that spans the bilayer. Hydrophobic interactions between the nonpolar amino acids and the fatty acyl groups of the membrane lipids firmly anchor the protein in the membrane. The seven helices are clustered together and oriented not quite perpendicular to the bilayer plane, providing a transmembrane pathway for proton movement. As we shall see in Chapter 12, this pattern of seven hydrophobic membrane-spanning helices is a common motif in membrane proteins involved in signal reception.

The photosynthetic reaction center of a purple bacterium was the first membrane protein structure solved by crystallography. Although a more complex membrane protein than bacteriorhodopsin, it is constructed on the same principles. The reaction center has four protein

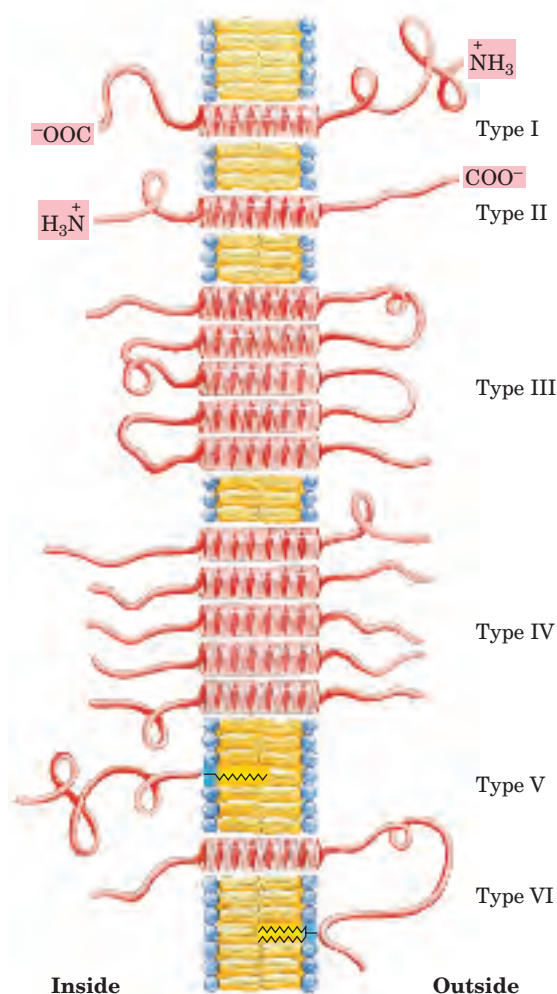


FIGURE 11–8 Integral membrane proteins. For known proteins of the plasma membrane, the spatial relationships of protein domains to the lipid bilayer fall into six categories. Types I and II have only one transmembrane helix; the amino-terminal domain is outside the cell in type I proteins and inside in type II. Type III proteins have multiple transmembrane helices in a single polypeptide. In type IV proteins, transmembrane domains of several different polypeptides assemble to form a channel through the membrane. Type V proteins are held to the bilayer primarily by covalently linked lipids (see Fig. 11–14), and type VI proteins have both transmembrane helices and lipid (GPI) anchors.

In this figure, and in figures throughout the book, we represent transmembrane protein segments in their most likely conformations: as α helices of six to seven turns. Sometimes these helices are shown simply as cylinders. As relatively few membrane protein structures have been deduced by x-ray crystallography, our representation of the extramembrane domains is arbitrary and not necessarily to scale.

subunits, three of which contain α -helical segments that span the membrane (Fig. 11–10). These segments are rich in nonpolar amino acids, their hydrophobic side chains oriented toward the outside of the molecule where they interact with membrane lipids. The architecture of the reaction center protein is therefore the inverse of that seen in most water-soluble proteins, in

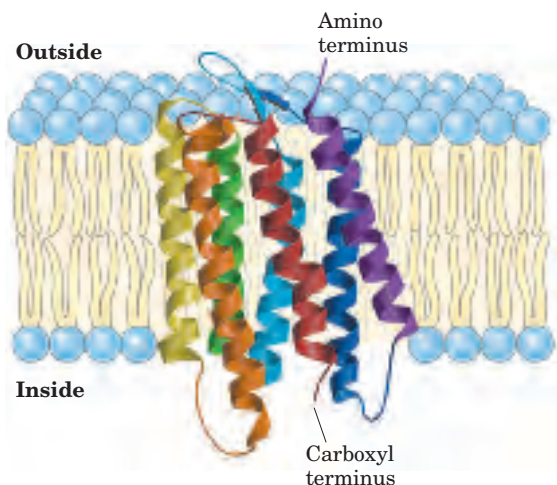


FIGURE 11-9 Bacteriorhodopsin, a membrane-spanning protein. (PDB ID 2AT9) The single polypeptide chain folds into seven hydrophobic α helices, each of which traverses the lipid bilayer roughly perpendicular to the plane of the membrane. The seven transmembrane helices are clustered, and the space around and between them is filled with the acyl chains of membrane lipids. The light-absorbing pigment retinal (see Fig. 10-21) is buried deep in the membrane in contact with several of the helical segments (not shown). The helices are colored to correspond with the hydropathy plot in Figure 11-11b.

which hydrophobic residues are buried within the protein core and hydrophilic residues are on the surface (recall the structures of myoglobin and hemoglobin, for example). In Chapter 19 we will encounter several complex membrane proteins having multiple transmembrane helical segments in which hydrophobic chains are positioned to interact with the lipid bilayer.

The Topology of an Integral Membrane Protein Can Be Predicted from Its Sequence

Determination of the three-dimensional structure of a membrane protein, or its topology, is generally much more difficult than determining its amino acid sequence, which can be accomplished by sequencing the protein or its gene. Thousands of sequences are known for membrane proteins, but relatively few three-dimensional structures have been established by crystallography or NMR spectroscopy. The presence of unbroken sequences of more than 20 hydrophobic residues in a membrane protein is commonly taken as evidence that these sequences traverse the lipid bilayer, acting as hydrophobic anchors or forming transmembrane channels. Virtually all integral proteins have at least one such sequence. Application of this logic to entire genomic sequences leads to the conclusion that in many species, 10% to 20% of all proteins are integral membrane proteins.

What can we predict about the secondary structure of the membrane-spanning portions of integral proteins? An α -helical sequence of 20 to 25 residues is just long

enough to span the thickness (30 Å) of the lipid bilayer (recall that the length of an α helix is 1.5 Å (0.15 nm) per amino acid residue). A polypeptide chain surrounded by lipids, having no water molecules with which to hydrogen-bond, will tend to form α helices or β sheets, in which intrachain hydrogen bonding is maximized. If the side chains of all amino acids in a helix are nonpolar, hydrophobic interactions with the surrounding lipids further stabilize the helix.

Several simple methods of analyzing amino acid sequences yield reasonably accurate predictions of secondary structure for transmembrane proteins. The relative polarity of each amino acid has been determined experimentally by measuring the free-energy change accompanying the movement of that amino acid side chain from a hydrophobic solvent into water. This free energy of transfer ranges from very exergonic for charged or polar residues to very endergonic for amino acids with aromatic or aliphatic hydrocarbon side chains. The overall hydrophobicity of a sequence of amino acids is estimated by summing the free energies of transfer for

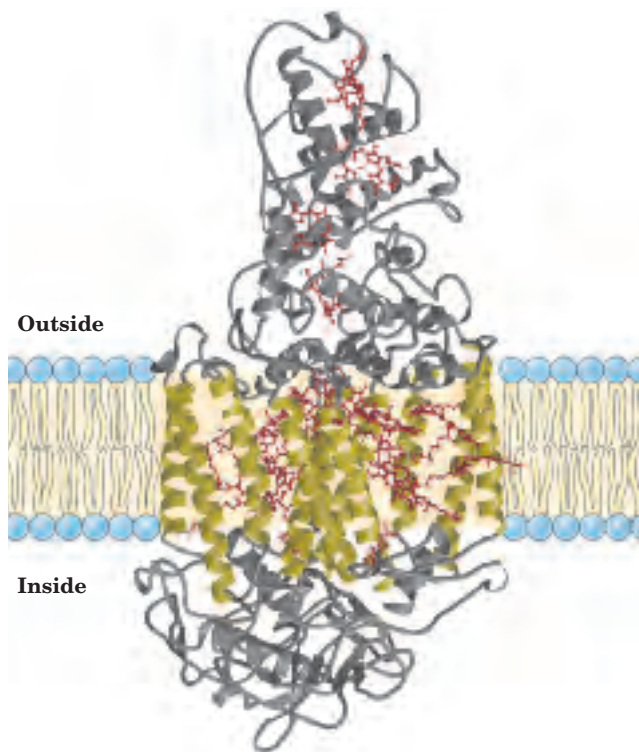


FIGURE 11-10 Three-dimensional structure of the photosynthetic reaction center of *Rhodospseudomonas viridis*, a purple bacterium. This was the first integral membrane protein to have its atomic structure determined by x-ray diffraction methods (PDB ID 1PRC). Eleven α -helical segments from three of the four subunits span the lipid bilayer, forming a cylinder 45 Å (4.5 nm) long; hydrophobic residues on the exterior of the cylinder interact with lipids of the bilayer. In this ribbon representation, residues that are part of the transmembrane helices are shown in yellow. The prosthetic groups (light-absorbing pigments and electron carriers; see Fig. 19-45) are red.

the residues in the sequence, which yields a **hydropathy index** for that region (see Table 3–1). To scan a polypeptide sequence for potential membrane-spanning segments, an investigator calculates the hydropathy index for successive segments (called windows) of a given size, from 7 to 20 residues. For a window of seven residues, for example, the indices for residues 1 to 7, 2 to 8, 3 to 9, and so on, are plotted as in Figure 11–11 (plotted for the middle residue in each window—residue 4 for residues 1 to 7, for example). A region with more than 20 residues of high hydropathy index is presumed to be a transmembrane segment. When the sequences of membrane proteins of known three-dimensional structure are scanned in this way, we find a reasonably good correspondence between predicted and known membrane-spanning segments. Hydropathy analysis predicts a single hydrophobic helix for glycophorin (Fig. 11–11a) and seven transmembrane segments for bacteriorhodopsin (Fig. 11–11b)—in agreement with experimental studies.

On the basis of their amino acid sequences and hydropathy plots, many of the transport proteins described in this chapter are believed to have multiple membrane-spanning helical regions—that is, they are type III or type IV integral proteins (Fig. 11–8). When predictions are consistent with chemical studies of protein localization (such as those described above for glycophorin and bacteriorhodopsin), the assumption that hydrophobic regions correspond to membrane-spanning domains is much better justified.

A further remarkable feature of many transmembrane proteins of known structure is the presence of Tyr and Trp residues at the interface between lipid and water (Fig. 11–12). The side chains of these residues apparently serve as membrane interface anchors, able to interact simultaneously with the central lipid phase and the aqueous phases on either side of the membrane.

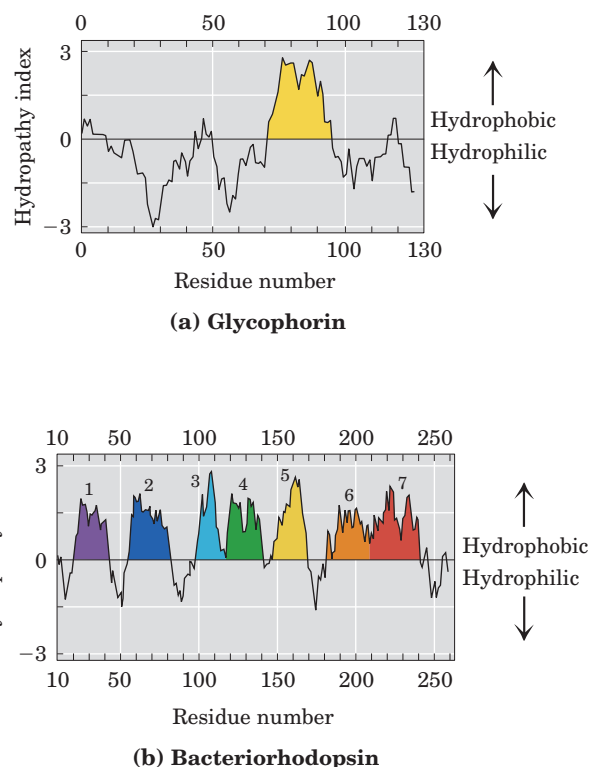


FIGURE 11–11 Hydropathy plots. Hydropathy index (see Table 3–1) is plotted against residue number for two integral membrane proteins. The hydropathy index for each amino acid residue in a sequence of defined length (called the window) is used to calculate the average hydropathy for the residues in that window. The horizontal axis shows the residue number in the middle of the window. **(a)** Glycophorin from human erythrocytes has a single hydrophobic sequence between residues 75 and 93 (yellow); compare this with Figure 11–7. **(b)** Bacteriorhodopsin, known from independent physical studies to have seven transmembrane helices (see Fig. 11–9), has seven hydrophobic regions. Note, however, that the hydropathy plot is ambiguous in the region of segments 6 and 7. Physical studies have confirmed that this region has two transmembrane segments.

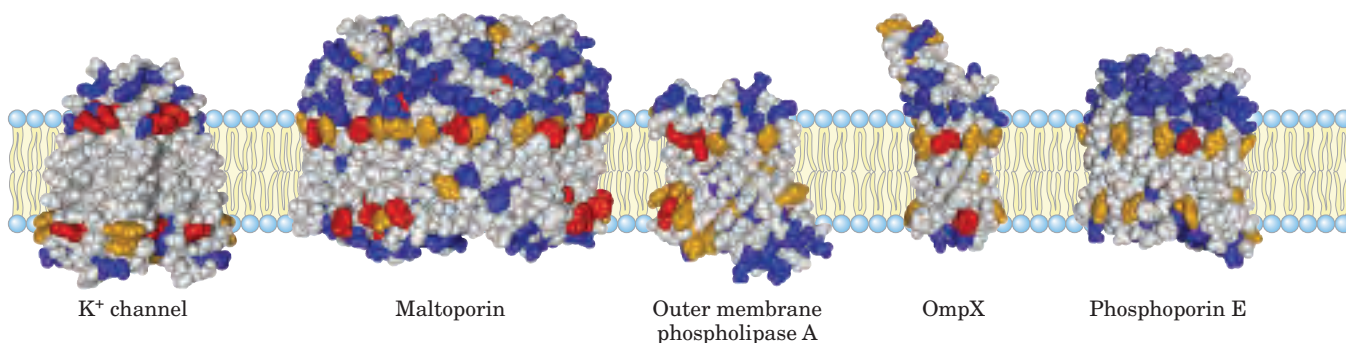


FIGURE 11–12 Tyr and Trp residues of membrane proteins clustering at the water-lipid interface. The detailed structures of these five integral membrane proteins are known from crystallographic studies. The K^+ channel (PDB ID 1BL8) is from the bacterium *Streptomyces lividans* (see Fig. 11–48); malto porin (PDB ID 1AF6), outer membrane phospholipase A (PDB ID 1QD5), OmpX (PDB ID 1QJ9), and phos-

phorin E (PDB ID 1PHO) are proteins of the outer membrane of *E. coli*. Residues of Tyr (orange) and Trp (red) are found predominantly where the nonpolar region of acyl chains meets the polar head group region. Charged residues (Lys, Arg, Glu, Asp) are shown in blue; they are found almost exclusively in the aqueous phases.

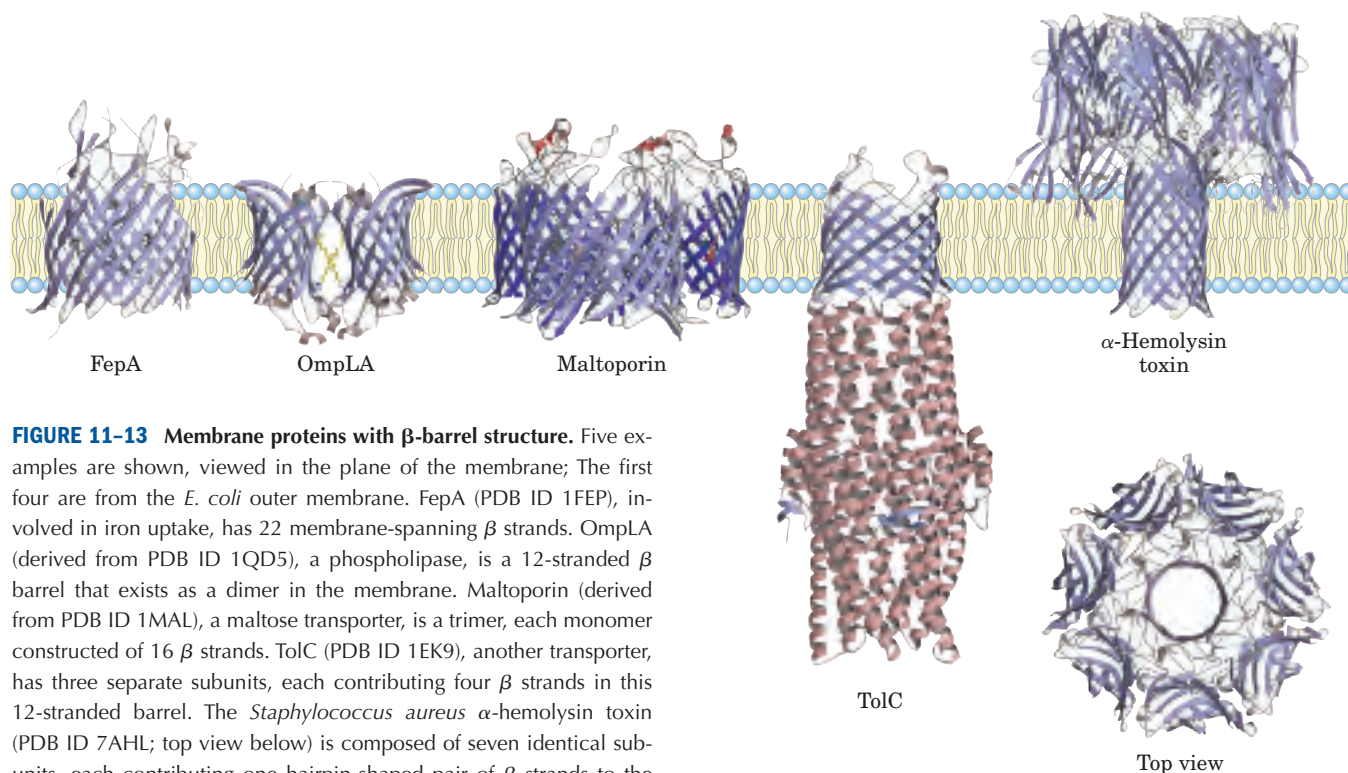


FIGURE 11-13 Membrane proteins with β -barrel structure. Five examples are shown, viewed in the plane of the membrane; The first four are from the *E. coli* outer membrane. FepA (PDB ID 1FEP), involved in iron uptake, has 22 membrane-spanning β strands. OmpLA (derived from PDB ID 1QD5), a phospholipase, is a 12-stranded β barrel that exists as a dimer in the membrane. Maltoporin (derived from PDB ID 1MAL), a maltose transporter, is a trimer, each monomer constructed of 16 β strands. TolC (PDB ID 1EK9), another transporter, has three separate subunits, each contributing four β strands in this 12-stranded barrel. The *Staphylococcus aureus* α -hemolysin toxin (PDB ID 7AHL; top view below) is composed of seven identical subunits, each contributing one hairpin-shaped pair of β strands to the 14-stranded barrel.

The hydrophobic domains of some integral membrane proteins penetrate only one leaflet of the bilayer. Cyclooxygenase, the target of aspirin action, is an example; its hydrophobic helices do not span the whole membrane but interact strongly with the acyl groups on one side of the bilayer (see Box 21-2, Fig. 1a).

Not all integral membrane proteins are composed of transmembrane α helices. Another structural motif common in membrane proteins is the **β barrel** (see Fig. 4-20d), in which 20 or more transmembrane segments form β sheets that line a cylinder (Fig. 11-13). The same factors that favor α -helix formation in the hydrophobic interior of a lipid bilayer also stabilize β barrels. When no water molecules are available to hydrogen-bond with the carbonyl oxygen and nitrogen of the peptide bond, maximal intrachain hydrogen bonding gives the most stable conformation. Planar β sheets do not maximize these interactions and are generally not found in the membrane interior; β barrels do allow all possible hydrogen bonds and are apparently common among membrane proteins. **Porins**, proteins that allow certain polar solutes to cross the outer membrane of gram-negative bacteria such as *E. coli*, have many-stranded β barrels lining the polar transmembrane passage.

A polypeptide is more extended in the β conformation than in an α helix; just seven to nine residues of β conformation are needed to span a membrane. Recall that in the β conformation, alternating side chains

project above and below the sheet (see Fig. 4-7). In β strands of membrane proteins, every second residue in the membrane-spanning segment is hydrophobic and interacts with the lipid bilayer; aromatic side chains are commonly found at the lipid-protein interface. The other residues may or may not be hydrophilic. The hydrophathy plot is not useful in predicting transmembrane segments for proteins with β barrel motifs, but as the database of known β barrel motifs increases, sequence-based predictions of transmembrane β conformations have become feasible. For example, a number of outer membrane proteins of gram-negative bacteria (Fig. 11-13) have been correctly predicted, by sequence analysis, to contain β barrels.

Covalently Attached Lipids Anchor Some Membrane Proteins

Some membrane proteins contain one or more covalently linked lipids of several types: long-chain fatty acids, isoprenoids, sterols, or glycosylated derivatives of phosphatidylinositol, GPI (Fig. 11-14). The attached lipid provides a hydrophobic anchor that inserts into the lipid bilayer and holds the protein at the membrane surface. The strength of the hydrophobic interaction between a bilayer and a single hydrocarbon chain linked to a protein is barely enough to anchor the protein securely, but many proteins have more than one attached

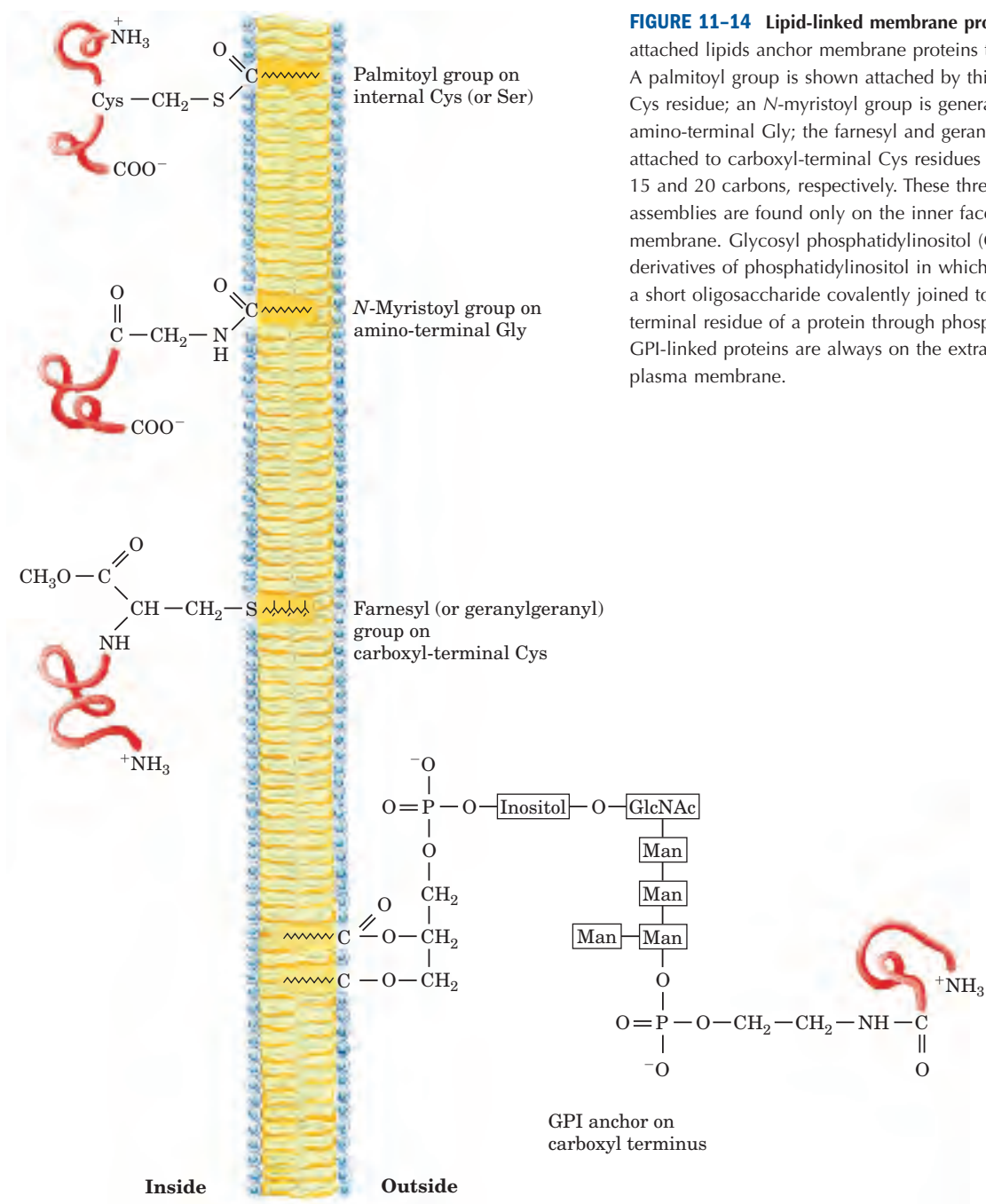


FIGURE 11-14 Lipid-linked membrane proteins. Covalently attached lipids anchor membrane proteins to the lipid bilayer. A palmitoyl group is shown attached by thioester linkage to a Cys residue; an *N*-myristoyl group is generally attached to an amino-terminal Gly; the farnesyl and geranylgeranyl groups attached to carboxyl-terminal Cys residues are isoprenoids of 15 and 20 carbons, respectively. These three lipid-protein assemblies are found only on the inner face of the plasma membrane. Glycosyl phosphatidylinositol (GPI) anchors are derivatives of phosphatidylinositol in which the inositol bears a short oligosaccharide covalently joined to the carboxyl-terminal residue of a protein through phosphoethanolamine. GPI-linked proteins are always on the extracellular face of the plasma membrane.

lipid moiety. Other interactions, such as ionic attractions between positively charged Lys residues in the protein and negatively charged lipid head groups, probably contribute to the stability of the attachment. The association of these lipid-linked proteins with the membrane is certainly weaker than that for integral membrane proteins and is, in at least some cases, reversible. But treatment with alkaline carbonate does not release GPI-linked proteins, which are therefore, by the working definition, integral proteins.

Beyond merely anchoring a protein to the membrane, the attached lipid may have a specific role. In the plasma membrane, proteins with GPI anchors are exclusively on the outer face and are confined within clusters, as we shall see below, whereas other types of lipid-linked proteins (with farnesyl or geranylgeranyl groups attached; Fig. 11-14) are exclusively on the inner face. Attachment of a specific lipid to a newly synthesized membrane protein therefore has a targeting function, directing the protein to its correct membrane location.

SUMMARY 11.1 The Composition and Architecture of Membranes

- Biological membranes define cellular boundaries, divide cells into discrete compartments, organize complex reaction sequences, and act in signal reception and energy transformations.
- Membranes are composed of lipids and proteins in varying combinations particular to each species, cell type, and organelle. The fluid mosaic model describes features common to all biological membranes. The lipid bilayer is the basic structural unit. Fatty acyl chains of phospholipids and the steroid nucleus of sterols are oriented toward the interior of the bilayer; their hydrophobic interactions stabilize the bilayer but give it flexibility.
- Peripheral proteins are loosely associated with the membrane through electrostatic interactions and hydrogen bonds or by covalently attached lipid anchors. Integral proteins associate firmly with membranes by hydrophobic interactions between the lipid bilayer and their nonpolar amino acid side chains, which are oriented toward the outside of the protein molecule.
- Some membrane proteins span the lipid bilayer several times, with hydrophobic sequences of about 20 amino acid residues forming transmembrane α helices. Detection of such hydrophobic sequences in proteins can be used to predict their secondary structure and transmembrane disposition. Multistranded β barrels are also common in integral membrane proteins. Tyr and Trp residues of transmembrane proteins are commonly found at the lipid-water interface.
- The lipids and proteins of membranes are inserted into the bilayer with specific sidedness; thus membranes are structurally and functionally asymmetric. Many membrane proteins contain covalently attached oligosaccharides. Plasma membrane glycoproteins are always oriented with the carbohydrate-bearing domain on the extracellular surface.

11.2 Membrane Dynamics

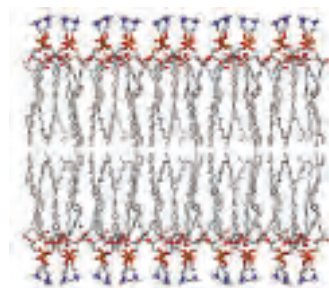
One remarkable feature of all biological membranes is their flexibility—their ability to change shape without losing their integrity and becoming leaky. The basis for this property is the noncovalent interactions among lipids in the bilayer and the motions allowed to individual lipids because they are not covalently anchored to one another. We turn now to the dynamics of membranes:

the motions that occur and the transient structures allowed by these motions.

Acyl Groups in the Bilayer Interior Are Ordered to Varying Degrees

Although the lipid bilayer structure is quite stable, its individual phospholipid and sterol molecules have some freedom of motion (Fig. 11–15). The structure and flexibility of the lipid bilayer depend on temperature and on the kinds of lipids present. At relatively low temperatures, the lipids in a bilayer form a semisolid **gel phase**, in which all types of motion of individual lipid molecules are strongly constrained; the bilayer is paracrystalline (Fig. 11–15a). At relatively high temperatures, individual hydrocarbon chains of fatty acids are in constant motion produced by rotation about the carbon–carbon bonds of the long acyl side chains. In this **liquid-disordered state**, or fluid state (Fig. 11–15b), the interior of the bilayer is more fluid than solid and the bilayer is like a sea of constantly moving lipid. At intermediate temperatures, the lipids exist in a **liquid-ordered state**; there is less thermal motion in the acyl chains of the lipid bilayer, but lateral movement in the plane of the bilayer still takes place. These differences in bilayer state are easily observed in liposomes composed of a single lipid,

(a) Paracrystalline state (gel)



Heat produces thermal motion of side chains
(gel $\rightarrow$ fluid transition)

(b) Fluid state

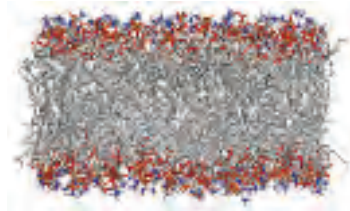


FIGURE 11–15 Two states of bilayer lipids. (a) In the paracrystalline state, or gel phase, polar head groups are uniformly arrayed at the surface, and the acyl chains are nearly motionless and packed with regular geometry; (b) in the liquid disordered state, or fluid state, acyl chains undergo much thermal motion and have no regular organization. Intermediate between these extremes is the liquid-ordered state, in which individual phospholipid molecules can diffuse laterally but the acyl groups remain extended and more or less ordered.

but biological membranes contain many lipids with a variety of fatty acyl chains and thus do not show sharp phase changes with temperature.

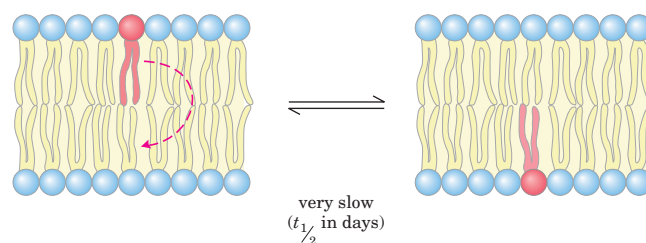
At temperatures in the physiological range (about 20 to 40 °C), long-chain saturated fatty acids (such as 16:0 and 18:0) pack well into a liquid-ordered array, but the kinks in unsaturated fatty acids (see Fig. 10–1) interfere with this packing, favoring the liquid-disordered state. Shorter-chain fatty acyl groups have the same effect. The sterol content of a membrane (which varies greatly with organism and organelle; Table 11–1) is another important determinant of lipid state. The rigid planar structure of the steroid nucleus, inserted between fatty acyl side chains, reduces the freedom of neighboring fatty acyl chains to move by rotation about their carbon–carbon bonds, forcing acyl chains into their fully extended conformation. The presence of sterols therefore reduces the fluidity in the core of the bilayer, thus favoring the liquid-ordered phase, and increases the thickness of the lipid leaflet (as described below).

Cells regulate their lipid composition to achieve a constant membrane fluidity under various growth conditions. For example, bacteria synthesize more unsaturated fatty acids and fewer saturated ones when cultured at low temperatures than when cultured at higher temperatures (Table 11–2). As a result of this adjustment in lipid composition, membranes of bacteria cultured at high or low temperatures have about the same degree of fluidity.

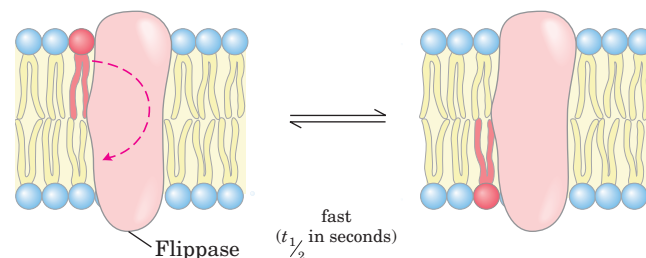
Transbilayer Movement of Lipids Requires Catalysis

At physiological temperature, transbilayer—or “flip-flop”—diffusion of a lipid molecule from one leaflet of the bilayer to the other (Fig. 11–16a) occurs very slowly unless if at all in most membranes. Transbilayer movement requires that a polar or charged head group leave its

(a) Uncatalyzed transverse (“flip-flop”) diffusion



(b) Transverse diffusion catalyzed by flippase



(c) Uncatalyzed lateral diffusion

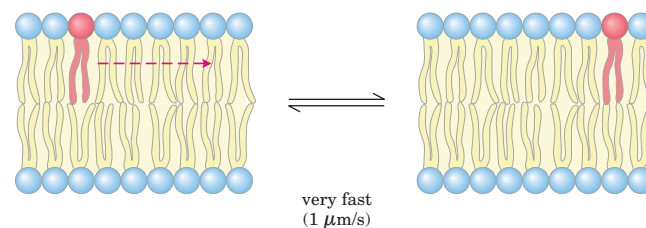


FIGURE 11–16 Motion of single phospholipids in a bilayer. (a) Movement from one leaflet to the other is very slow, unless (b) catalyzed by a flippase; in contrast, lateral diffusion within the leaflet (c) is very rapid and requires no protein catalysis.

TABLE 11–2 Fatty Acid Composition of *E. coli* Cells Cultured at Different Temperatures

	Percentage of total fatty acids*			
	10 °C	20 °C	30 °C	40 °C
Myristic acid (14:0)	4	4	4	8
Palmitic acid (16:0)	18	25	29	48
Palmitoleic acid (16:1)	26	24	23	9
Oleic acid (18:1)	38	34	30	12
Hydroxymyristic acid	13	10	10	8
Ratio of unsaturated to saturated [†]	2.9	2.0	1.6	0.38

Source: Data from Marr, A.G. & Ingraham, J.L. (1962) Effect of temperature on the composition of fatty acids in *Escherichia coli*. *J. Bacteriol.* **84**, 1260.

*The exact fatty acid composition depends not only on growth temperature but on growth stage and growth medium composition.

[†]Ratios calculated as the total percentage of 16:1 plus 18:1 divided by the total percentage of 14:0 plus 16:0. Hydroxymyristic acid was omitted from this calculation.

aqueous environment and move into the hydrophobic interior of the bilayer, a process with a large, positive free-energy change. There are, however, situations in which such movement is essential. For example, during synthesis of the bacterial plasma membrane, phospholipids are produced on the inside surface of the membrane and must undergo flip-flop diffusion to enter the outer leaflet of the bilayer. Similar transbilayer diffusion must also take place in eukaryotic cells as membrane lipids synthesized in one organelle move from the inner to the outer leaflet and into other organelles. A family of proteins, the **flippases** (Fig. 11–16b), facilitates flip-flop diffusion, providing a transmembrane path that is energetically more favorable and much faster than the uncatalyzed movement.

Lipids and Proteins Diffuse Laterally in the Bilayer

Individual lipid molecules can move laterally in the plane of the membrane by changing places with neighboring lipid molecules (Fig. 11–16c). A molecule in one monolayer, or leaflet, of the lipid bilayer—the outer leaflet of the erythrocyte plasma membrane, for example—can diffuse laterally so fast that it circumnavigates the erythrocyte in seconds. This rapid lateral diffusion within the plane of the bilayer tends to randomize the positions of individual molecules in a few seconds.

Lateral diffusion can be shown experimentally by attaching fluorescent probes to the head groups of lipids and using fluorescence microscopy to follow the probes over time (Fig. 11–17). In one technique, a small region ($5\ \mu\text{m}^2$) of a cell surface with fluorescence-tagged lipids is bleached by intense laser radiation so that the irradiated patch no longer fluoresces when viewed in the much dimmer light of the fluorescence microscope. However, within milliseconds, the region recovers its fluorescence as unbleached lipid molecules diffuse into the bleached patch and bleached lipid molecules diffuse away from it. The rate of fluorescence recovery after photobleaching, or **FRAP**, is a measure of the rate of lateral diffusion of the lipids. Using the FRAP technique,

researchers have shown that some membrane lipids diffuse laterally by up to $1\ \mu\text{m/s}$.

Another technique, single particle tracking, allows one to follow the movement of a *single* lipid molecule in the plasma membrane on a much shorter time scale. Results from these studies confirm the rapid lateral diffusion within small, discrete regions of the cell sur-

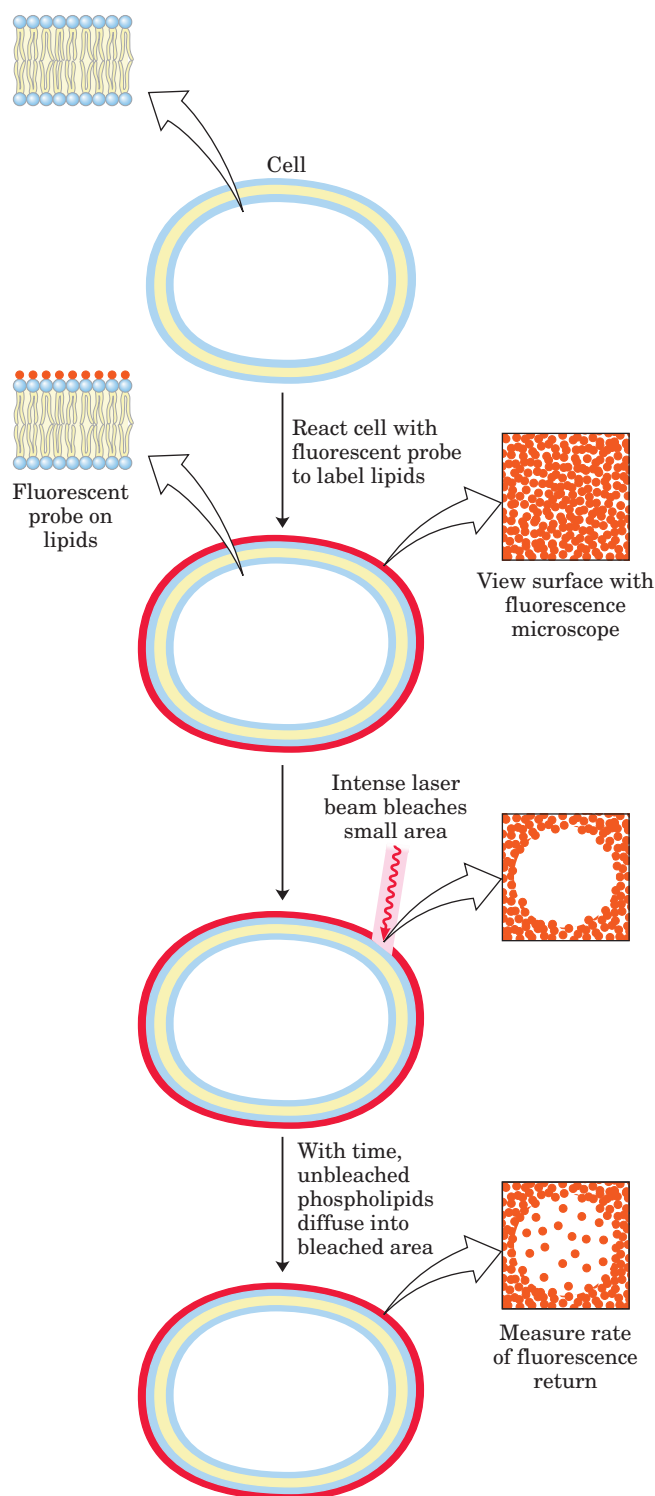


FIGURE 11–17 Measurement of lateral diffusion rates of lipids by fluorescence recovery after photobleaching (FRAP). The lipids in the outer leaflet of the plasma membrane are labeled by reaction with a membrane-impermeant fluorescent probe (red), so the surface is uniformly labeled when viewed with a fluorescence microscope. A small area is bleached by irradiation with an intense laser beam, leaving that area nonfluorescent. With the passage of time, labeled lipid molecules diffuse into the bleached region, and it again becomes fluorescent. From the time course of fluorescence return to this area, the diffusion coefficient for the labeled lipid is determined. The rates are typically high; a lipid moving at this speed could circumnavigate *E. coli* in one second. (The FRAP method can also be used to measure the lateral diffusion of membrane proteins.)

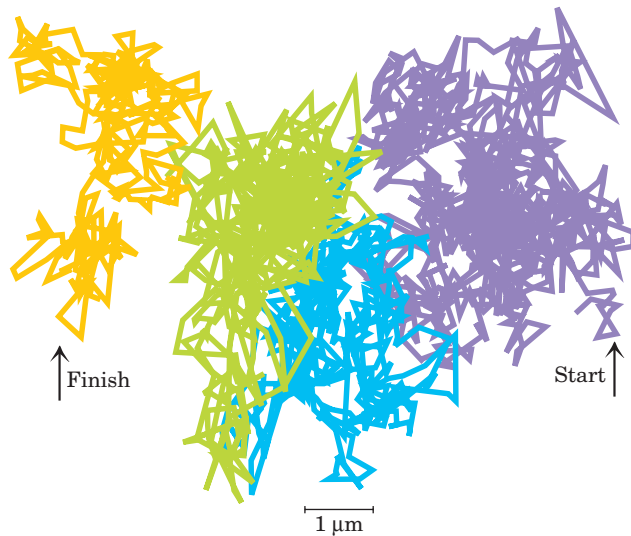


FIGURE 11-18 Hop diffusion of individual lipid molecules. The motion of a single fluorescent lipid molecule in a cell surface is recorded on video by fluorescence microscopy, with a time resolution of $25\ \mu\text{s}$ (equivalent to 40,000 frames/s). The track shown here represents a molecule followed for 56 ms (a total of 2,250 frames); the trace begins in the purple area and continues through blue, green, and orange. The pattern of movement indicates rapid diffusion within a confined region (about 250 nm in diameter, shown by a single color), with occasional hops into an adjoining region. This finding suggests that the lipids are corralled by molecular fences that they occasionally jump.

face and show that movement from one such region to a nearby region is inhibited; lipids behave as though corralled by fences that they can occasionally jump (Fig. 11-18).

Many membrane proteins seem to be afloat in a sea of lipids. Like membrane lipids, these proteins are free to diffuse laterally in the plane of the bilayer and are in

constant motion, as shown by the FRAP technique with fluorescence-tagged surface proteins. Some membrane proteins associate to form large aggregates (“patches”) on the surface of a cell or organelle in which individual protein molecules do not move relative to one another; for example, acetylcholine receptors (see Fig. 11-51) form dense patches on neuron plasma membranes at synapses. Other membrane proteins are anchored to internal structures that prevent their free diffusion. In the erythrocyte membrane, both glycophorin and the chloride-bicarbonate exchanger (p. 395) are tethered to spectrin, a filamentous cytoskeletal protein (Fig. 11-19). One possible explanation for the pattern of lateral diffusion of lipid molecules shown in Figure 11-18 is that membrane proteins immobilized by their association with spectrin are the “fences” that define the regions of relatively unrestricted lipid motion.

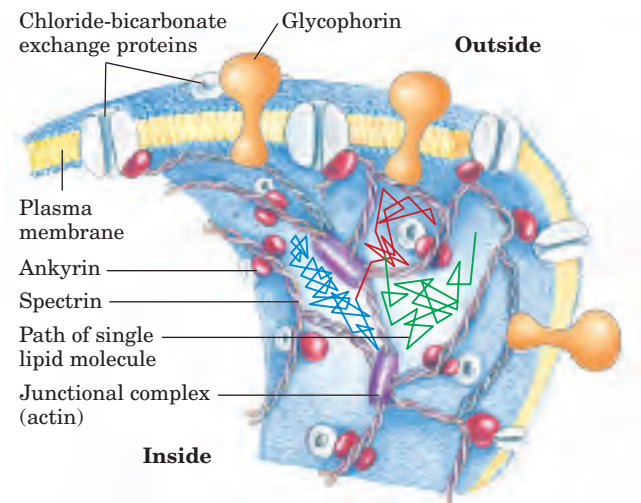


FIGURE 11-19 Restricted motion of the erythrocyte chloride-bicarbonate exchanger and glycophorin. The proteins span the membrane and are tethered to spectrin, a cytoskeletal protein, by another protein, ankyrin, limiting their lateral mobilities. Ankyrin is anchored in the membrane by a covalently bound palmitoyl side chain (see Fig. 11-14). Spectrin, a long, filamentous protein, is cross-linked at junctional complexes containing actin. A network of cross-linked spectrin molecules attached to the cytoplasmic face of the plasma membrane stabilizes the membrane against deformation. This network of anchored membrane proteins may be the “corral” suggested by the experiment shown in Figure 11-18; the lipid tracks shown here are confined to subregions defined by the tethered membrane proteins.

Sphingolipids and Cholesterol Cluster Together in Membrane Rafts

We have seen that diffusion of membrane lipids from one bilayer leaflet into the other is very slow unless catalyzed, and that the different lipid species of the plasma membrane are asymmetrically distributed in the two leaflets of the bilayer (Fig. 11-5). Even within a single leaflet, the lipid distribution is not random. Glycosphingolipids (cerebrosides and gangliosides), which typically contain long-chain saturated fatty acids, form transient clusters in the outer leaflet that largely exclude glycerophospholipids, which typically contain one unsaturated fatty acyl group and a shorter saturated fatty acyl group. The long, saturated acyl groups of sphingolipids can form more compact, more stable associations with the long ring system of cholesterol than can the shorter, often unsaturated, chains of phospholipids. The cholesterol-sphingolipid **microdomains** in the outer monolayer of the plasma membrane, visible with atomic-force microscopy (Box 11-1), are slightly thicker and more ordered (less fluid) than neighboring microdomains rich in phospholipids (Fig. 11-20) and are more difficult

BOX 11-1 WORKING IN BIOCHEMISTRY

Atomic Force Microscopy to Visualize Membrane Proteins

In atomic force microscopy (AFM), the sharp tip of a microscopic probe attached to a flexible cantilever is drawn across an uneven surface such as a membrane (Fig. 1). Electrostatic and van der Waals interactions between the tip and the sample produce a force that moves the probe up and down (in the z dimension) as it encounters hills and valleys in the sample. A laser beam reflected from the cantilever detects motions of as little as 1 Å. In one type of atomic force microscope, the force on the probe is held constant (relative to a standard force, on the order of piconewtons) by a feedback circuit that causes the platform holding the sample to rise or fall to keep the force constant. A series of scans in the x and y dimensions (the plane of the membrane) yields a three-dimensional contour map of the surface with resolution near the atomic scale—0.1 nm in the vertical dimension, 0.5 to 1.0 nm in the lateral dimensions. The membrane rafts shown in Figure 11-20b were visualized by this technique.

In favorable cases, AFM can be used to study single membrane protein molecules. Single molecules of bacteriorhodopsin in the purple membranes of the bacterium *Halobacterium salinarum* (see Fig. 11-9) are seen as highly regular structures (Fig. 2a). When a number of images of individual units are superimposed with the help of a computer, the real parts of the image reinforce each other and the noise in individual images is averaged out, yielding a high-resolution image of the protein (inset in Fig. 2a). AFM of purified *E. coli* aquaporin, reconstituted into

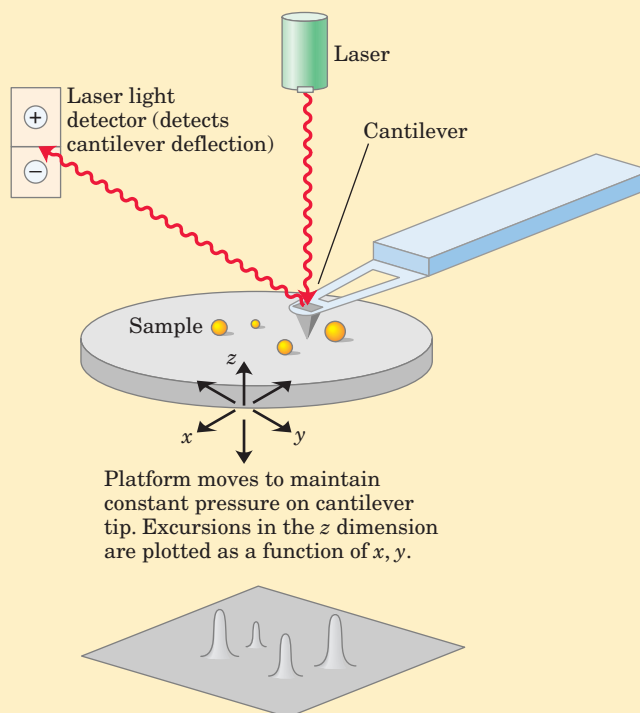
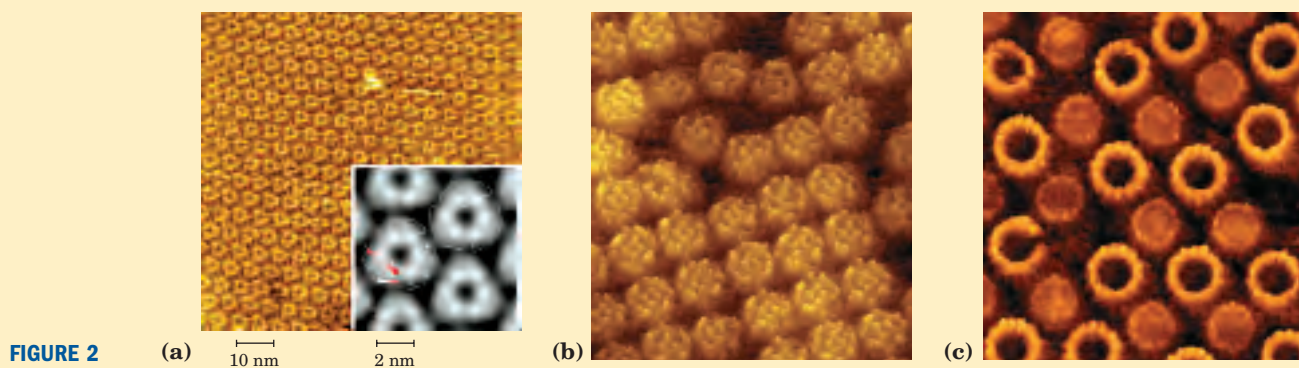


FIGURE 1

lipid bilayers and viewed as if from the outside of a cell, shows the fine details of the protein's periplasmic domains (Fig. 2b). And AFM reveals that F_0 , the proton-driven rotor of the chloroplast ATP synthase (p. 742), is composed of many subunits (14 in Fig. 2c) arranged in a circle.



to dissolve with nonionic detergents; they behave like liquid-ordered sphingolipid **rafts** adrift in a sea of liquid-disordered phospholipids.

These lipid rafts are remarkably enriched in two classes of integral membrane proteins: those anchored to the membrane by two covalently attached long-chain

saturated fatty acids (two palmitoyl groups or a palmitoyl and a myristoyl group) and GPI-anchored proteins (Fig. 11-14). Presumably these lipid anchors, like the acyl chains of sphingolipids, form more stable associations with the cholesterol and long acyl groups in rafts than with the surrounding phospholipids. (It is notable

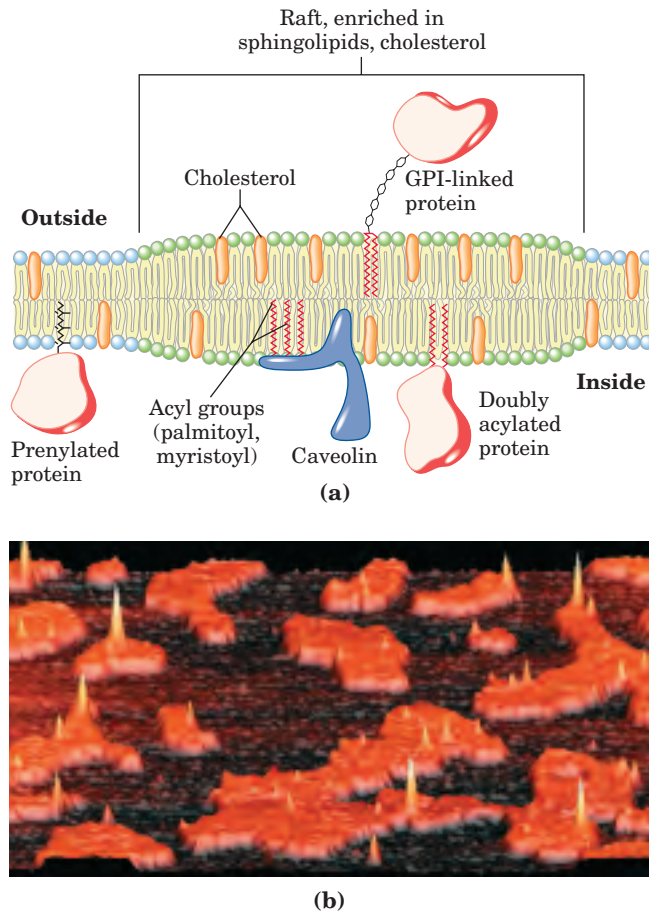


FIGURE 11-20 Microdomains (rafts) in the plasma membrane. (a) Stable associations of sphingolipids and cholesterol in the outer leaflet produce a microdomain, slightly thicker than other membrane regions, that is enriched with specific types of membrane proteins. GPI-linked proteins are commonly found in the outer leaflet of such rafts, and proteins with one or several covalently attached long-chain acyl groups are common in the inner leaflet. Caveolin is especially common in inwardly curved rafts called caveolae (see Fig. 11-21). Proteins with attached prenyl groups (such as Ras; see Fig. 12-6) tend to be excluded from rafts. (b) The greater thickness of raft regions can be visualized by atomic force microscopy (see Box 11-1). In this view of a membrane region, we can see the rafts protruding from a lipid bilayer ocean; in the rafts, sharp peaks represent GPI-linked proteins. Note that these peaks are found almost exclusively in rafts.

that other lipid-linked proteins, those with covalently attached isoprenyl groups such as farnesyl, are *not* preferentially associated with the outer leaflet of sphingolipid/cholesterol rafts (Fig. 11-20a.) The “raft” and “sea” domains of the plasma membrane are not rigidly separated; membrane proteins can move into and out of lipid rafts on a time scale of seconds. But in the shorter time scale (microseconds) more relevant to many membrane-mediated biochemical processes, many of these proteins reside primarily in a raft.

We can estimate the fraction of the cell surface occupied by rafts from the fraction of the plasma mem-

brane that resists detergent solubilization, which can be as high as 50% in some cases: the rafts cover half of the ocean (Fig. 11-20b). Indirect measurements in cultured fibroblasts suggest a diameter of roughly 50 nm for an individual raft, which corresponds to a patch containing a few thousand sphingolipids and perhaps 10 to 50 membrane proteins. Because most cells express more than 50 different kinds of plasma membrane proteins, it is likely that a single raft contains only a subset of membrane proteins and that this segregation of membrane proteins is functionally significant. For a process that involves interaction of two membrane proteins, their presence in a single raft would hugely increase the likelihood of their collision. Certain membrane receptors and signaling proteins, for example, appear to be segregated together in membrane rafts. Experiments show that signaling through these proteins can be disrupted by manipulations that deplete the plasma membrane of cholesterol and destroy lipid rafts.

Caveolins Define a Special Class of Membrane Rafts

Caveolin is an integral membrane protein with two globular domains connected by a hairpin-shaped hydrophobic domain, which binds the protein to the cytoplasmic leaflet of the plasma membrane. Three palmitoyl groups attached to the carboxyl-terminal globular domain further anchor it to the membrane. Caveolin (actually, a family of related caveolins) binds cholesterol in the membrane, and the presence of caveolin forces the associated lipid bilayer to curve inward, forming **caveolae** (“little caves”) in the surface of the cell (Fig. 11-21). Caveolae are unusual rafts: they involve *both* leaflets of the bilayer—the cytoplasmic leaflet, from which the caveolin globular domains project, and the exoplasmic leaflet, a typical sphingolipid/cholesterol raft with associated GPI-anchored proteins. Caveolae are implicated in a variety of cellular functions, including membrane trafficking within cells and the transduction of external signals into cellular responses. The receptors for insulin and other growth factors, as well as certain GTP-binding proteins and protein kinases associated with transmembrane signaling, appear to be localized in rafts and perhaps in caveolae. We discuss some possible roles of rafts in signaling in Chapter 12.

Certain Integral Proteins Mediate Cell-Cell Interactions and Adhesion

Several families of integral proteins in the plasma membrane provide specific points of attachment between cells, or between a cell and extracellular matrix proteins.

Integrins are heterodimeric proteins (two unlike subunits, α and β) anchored to the plasma membrane by a single hydrophobic transmembrane helix in each subunit (Fig. 11-22; see also Fig. 7-30). The large extracellular domains of the α and β subunits combine to form a specific binding site for extracellular proteins

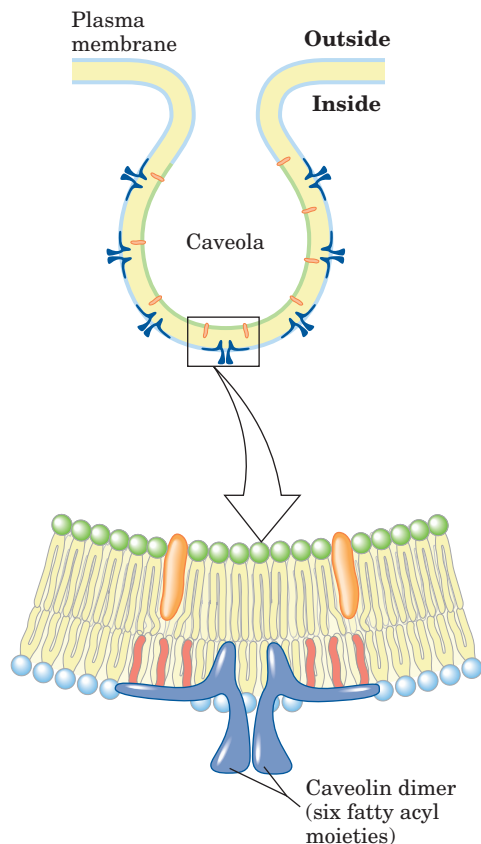


FIGURE 11-21 Caveolin forces inward curvature in membranes. The protein caveolin has a central hydrophobic domain and three long-chain acyl groups on each monomeric unit, which hold the molecule to the inside of the plasma membrane. When a number of caveolin dimers are concentrated in a small region (a raft), they force a curvature in the lipid bilayer, forming a caveola.

such as collagen and fibronectin. As there are 18 different α subunits and at least 8 different β subunits, a wide variety of specificities may be generated from various combinations of α and β . One common determinant of integrin binding in several extracellular partners of integrins is the sequence Arg–Gly–Asp (RGD).

 Integrins are not merely adhesives; they serve as receptors and signal transducers, conveying information across the plasma membrane in both directions. Integrins regulate many processes, including platelet aggregation at the site of a wound, tissue repair, the activity of immune cells, and the invasion of tissue by a tumor. Mutation in an integrin gene encoding the β subunit known as CD18 is the cause of leukocyte adhesion deficiency in humans, a rare genetic disease in which leukocytes fail to pass out of blood vessels to reach sites of infection (see Fig. 7–33). Infants with a severe defect in CD18 commonly die of infections before the age of two. ■

At least three other families of plasma membrane proteins are also involved in surface adhesion (Fig. 11–22). **Cadherins** undergo homophilic (“with same kind”) interactions with identical cadherins in an adjacent cell. **Immunoglobulin-like proteins** can undergo either homophilic interactions with their identical counterparts on another cell or heterophilic interactions with an integrin on a neighboring cell. **Selectins** have extracellular domains that, in the presence of Ca^{2+} , bind specific polysaccharides on the surface of an adjacent cell. Selectins are present primarily in the various types of blood cells and in the endothelial cells that line blood vessels (see Fig. 7–33). They are an essential part of the blood-clotting process.

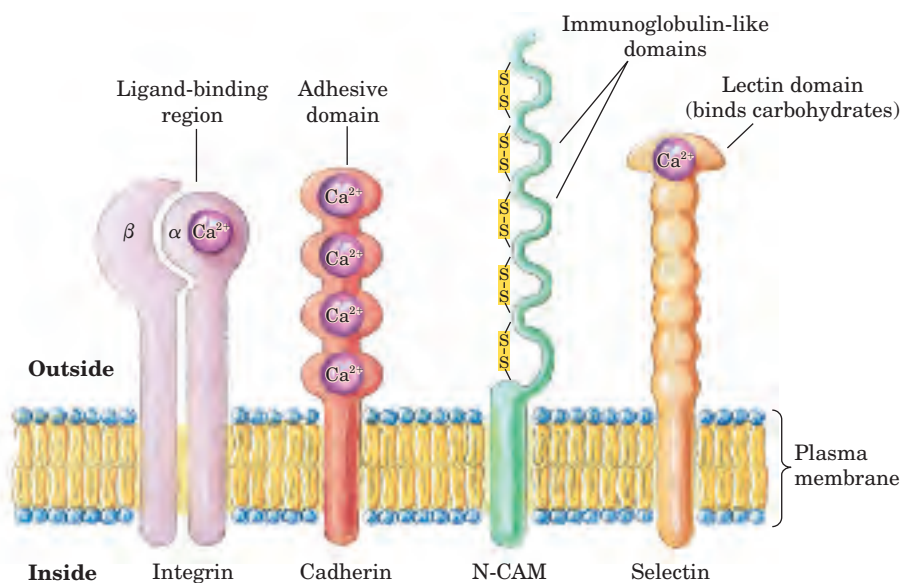


FIGURE 11-22 Four examples of integral protein types that function in cell-cell interactions.

Integrins consist of α and β transmembrane polypeptides; their extracellular domains combine to form binding sites for divalent metal ions and proteins of the extracellular matrix (such as collagen and fibronectin) or for specific surface proteins of other cells. Cadherin has four extracellular Ca^{2+} -binding domains, the most distal of which contains the site that binds to cadherin on another cell surface. N-CAM (neuronal cell adhesion molecule) is one of a family of immunoglobulin-like proteins that mediate Ca^{2+} -independent interactions with surface proteins of nearby cells. Selectins bind tightly to carbohydrate moieties in neighboring cells; this binding is Ca^{2+} -dependent.

Integral proteins play a role in many other cellular processes. They serve as transporters and ion channels (discussed in Section 11.3) and as receptors for hormones, neurotransmitters, and growth factors (Chapter 12). They are central to oxidative phosphorylation and photosynthesis (Chapter 19) and to cell-cell and antigen-cell recognition in the immune system (Chapter 5). Integral proteins are also important players in the membrane fusion that accompanies exocytosis, endocytosis, and the entry of many types of viruses into host cells.

Membrane Fusion Is Central to Many Biological Processes

A remarkable feature of the biological membrane is its ability to undergo fusion with another membrane without losing its continuity. Although membranes are stable, they are by no means static. Within the eukaryotic endomembrane system (which includes the nuclear membrane, endoplasmic reticulum, Golgi, and various small vesicles), the membranous compartments constantly reorganize. Vesicles bud from the endoplasmic reticulum to carry newly synthesized lipids and proteins to other organelles and to the plasma membrane. Exocytosis, endocytosis, cell division, fusion of egg and sperm cells, and entry of a membrane-enveloped virus into its host cell all involve membrane reorganization in which the fundamental operation is fusion of two membrane segments without loss of continuity (Fig. 11–23).

Specific fusion of two membranes requires that (1) they recognize each other; (2) their surfaces become closely apposed, which requires the removal of water molecules normally associated with the polar head groups of lipids; (3) their bilayer structures become locally disrupted, resulting in fusion of the outer leaflet of each membrane (hemifusion); and (4) their bilayers fuse to form a single continuous bilayer. Receptor mediated endocytosis, or regulated secretion, also requires that (5) the fusion process is triggered at the appropriate time or in response to a specific signal. Integral proteins called **fusion proteins** mediate these events, bringing about specific recognition and a transient local distortion of the bilayer structure that favors membrane fusion. (Note that these fusion proteins are unrelated to the products of two fused genes, also called fusion proteins, discussed in Chapter 9.)

Two cases of membrane fusion are especially well studied: the entry into a host cell of an enveloped virus such as influenza virus, and the release of neurotransmitters by exocytosis. Both processes involve complexes of fusion proteins that undergo dramatic conformational changes.

The influenza virus is surrounded by a membrane containing, among other proteins, many molecules of the hemagglutination (HA) protein (named for its abil-

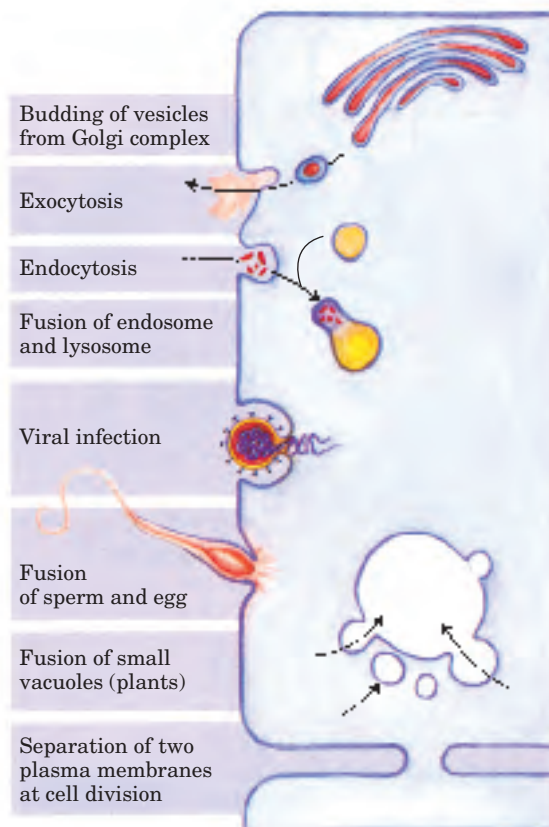


FIGURE 11–23 Membrane fusion. The fusion of two membranes is central to a variety of cellular processes involving both organelles and the plasma membrane.

ity to cause erythrocytes to clump together). The virus enters a host cell by inducing endocytosis, which encloses the virus in an endosome, a small membrane vesicle with a pH of about 5 (Fig. 11–24). At this pH, a conformational change in the HA protein occurs, exposing a sequence within the HA protein called the **fusion peptide** and enabling the protein to penetrate the endosomal membrane. The endosomal membrane and the viral membrane are now connected through the HA protein. Next, the HA protein bends at its middle to form a hairpin shape, bringing its two ends together. This pulls the two membranes into close apposition and causes fusion of the viral membrane and the endosomal membrane. The HA protein functions as a trimer (Fig. 11–24). In its low-pH form, three HA domains at the closed end of the hairpin twist about each other to form a stable, coiled structure. The fusion process involves an intermediate stage (hemifusion) in which the outer leaflet of the viral membrane is fused with the inner leaflet of the endosomal membrane, while the other two leaflets maintain their continuity. At the point of hemifusion, the lipid bilayer must be temporarily disorganized, presumably caused by the HA fusion peptide

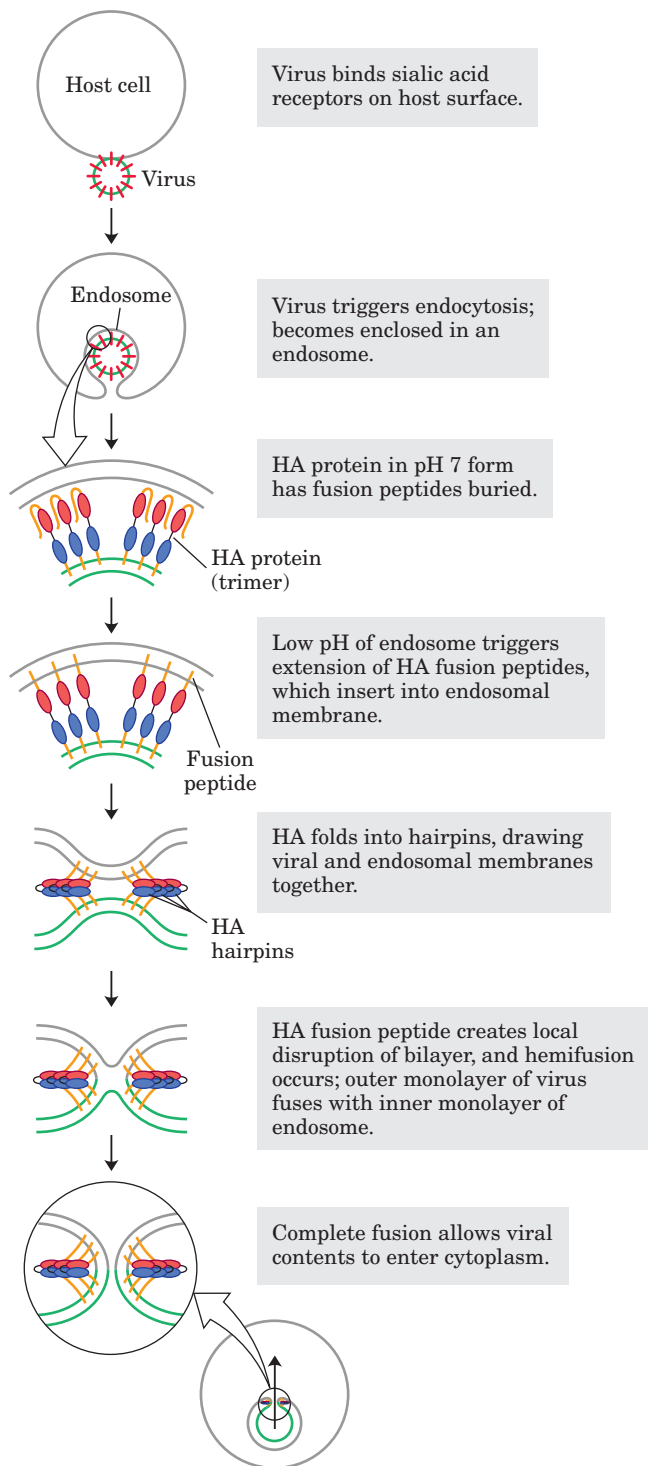


FIGURE 11-24 Fusion induced by the hemagglutinin (HA) protein during viral infection. HA protein is exposed on the membrane surface of the influenza virus. When the virus moves from the neutral pH of the interstitial fluid to the low-pH compartment (endosome) in the host cell, HA undergoes dramatic shape changes that mediate fusion of the viral and endosomal membranes, releasing the viral contents into the cytoplasm.

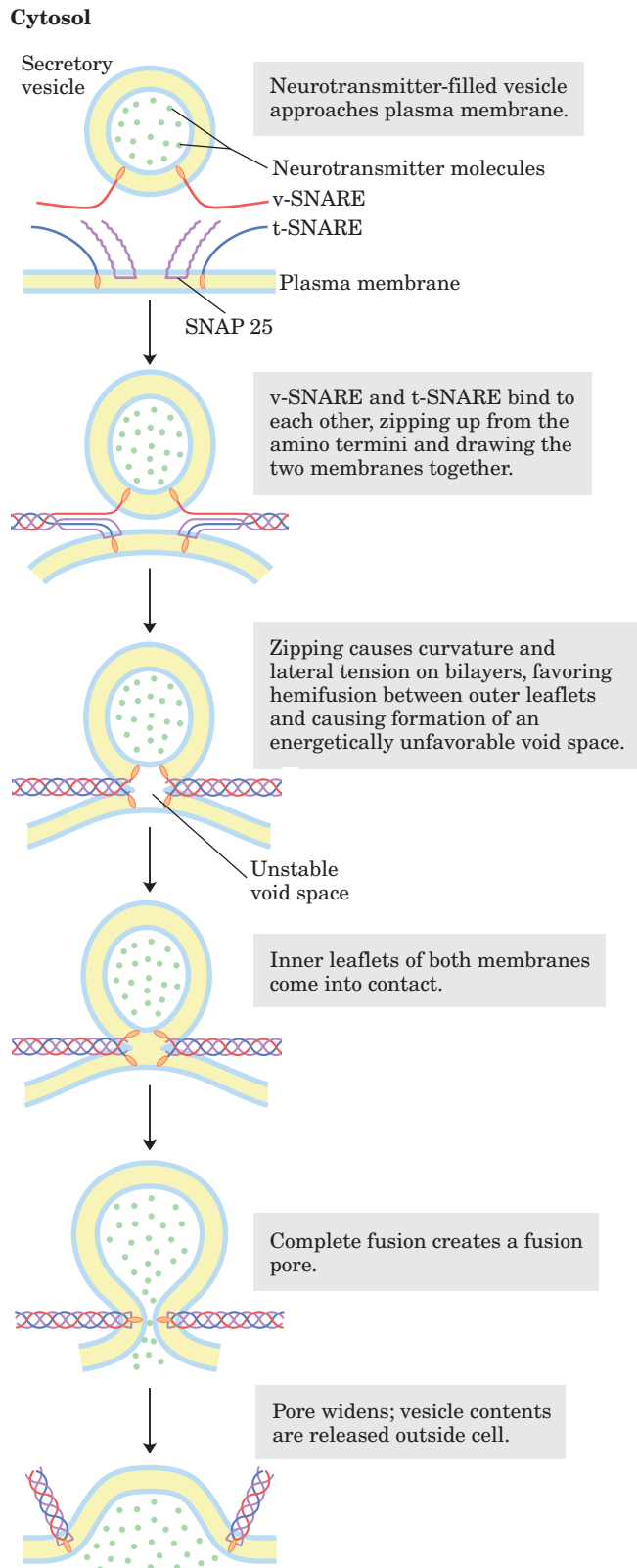


FIGURE 11-25 Fusion during neurotransmitter release at a synapse. The membrane of the secretory vesicle contains the v-SNARE synaptobrevin (red). The target (plasma) membrane contains the t-SNAREs syntaxin (blue) and SNAP25 (violet). When a local increase in $[Ca^{2+}]$ signals release of neurotransmitter, the v-SNARE, SNAP25, and t-SNARE interact, forming a coiled bundle of four α helices, pulling the two membranes together and disrupting the bilayer locally, which leads to membrane fusion and neurotransmitter release.

domains. Complete fusion results in release of the viral contents into the host cell cytoplasm.

Neurotransmitters are released at synapses when intracellular vesicles loaded with neurotransmitter fuse with the plasma membrane. This process involves a family of proteins called SNAREs (Fig. 11–25). SNAREs in the cytoplasmic face of the intracellular vesicles are called **v-SNAREs**; those in the target membranes with which the vesicles fuse (the plasma membrane during exocytosis) are **t-SNAREs**. Two other proteins, SNAP25 and NSF, are also involved. During fusion, v- and t-SNAREs bind to each other and undergo a structural change that produces a bundle of long thin rods made up of helices from both v- and t-SNAREs and two helices from SNAP25 (Fig. 11–25). The two SNAREs initially interact at their ends, then zip up into the bundle of helices. This structural change pulls the two membranes into contact and initiates the fusion of their lipid bilayers.

The complex of SNAREs and SNAP25 is the target of the powerful *Clostridium botulinum* toxin, a protease that cleaves specific bonds in these proteins, preventing neurotransmission and causing the death of the organism. Because of its very high specificity for these proteins, purified botulinum toxin has served as a powerful tool for dissecting the mechanism of neurotransmitter release in vivo and in vitro.

SUMMARY 11.2 Membrane Dynamics

- Lipids in a biological membrane can exist in liquid-ordered or liquid-disordered states; in the latter state, thermal motion of acyl chains makes the interior of the bilayer fluid. Fluidity is affected by temperature, fatty acid composition, and sterol content.
- Flip-flop diffusion of lipids between the inner and outer leaflets of a membrane is very slow except when specifically catalyzed by flippases.
- Lipids and proteins can diffuse laterally within the plane of the membrane, but this mobility is limited by interactions of membrane proteins with internal cytoskeletal structures and interactions of lipids with lipid rafts. One class of lipid rafts consists of sphingolipids and cholesterol with a subset of membrane proteins that are GPI-linked or attached to several long-chain fatty acyl moieties.
- Caveolin is an integral membrane protein that associates with the inner leaflet of the plasma membrane, forcing it to curve inward to form caveolae, probably involved in membrane transport and signaling.
- Integrins are transmembrane proteins of the plasma membrane that act both to attach cells

to each other and to carry messages between the extracellular matrix and the cytoplasm.

- Specific proteins mediate the fusion of two membranes, which accompanies processes such as viral invasion and endocytosis and exocytosis.

11.3 Solute Transport across Membranes

Every living cell must acquire from its surroundings the raw materials for biosynthesis and for energy production, and must release to its environment the byproducts of metabolism. A few nonpolar compounds can dissolve in the lipid bilayer and cross the membrane unassisted, but for polar or charged compounds or ions, a membrane protein is essential for transmembrane movement. In some cases a membrane protein simply facilitates the diffusion of a solute down its concentration gradient, but transport often occurs against a gradient of concentration, electrical charge, or both, in which case solutes must be “pumped” in a process that requires energy (Fig. 11–26). The energy may come directly from ATP hydrolysis or may be supplied in the form of movement of another solute down its electrochemical gradient with enough energy to carry another solute up its gradient. Ions may also move across membranes via ion channels formed by proteins, or they may be carried across by ionophores, small molecules that mask the charge of the ions and allow them to diffuse through the lipid bilayer. With very few exceptions, the traffic of small molecules across the plasma membrane is mediated by proteins such as transmembrane channels, carriers, or pumps. Within the eukaryotic cell, different compartments have different concentrations of metabolic intermediates and products and of ions, and these, too, must move across intracellular membranes in tightly regulated, protein-mediated processes.

Passive Transport Is Facilitated by Membrane Proteins

When two aqueous compartments containing unequal concentrations of a soluble compound or ion are separated by a permeable divider (membrane), the solute moves by **simple diffusion** from the region of higher concentration, through the membrane, to the region of lower concentration, until the two compartments have equal solute concentrations (Fig. 11–27a). When ions of opposite charge are separated by a permeable membrane, there is a transmembrane electrical gradient, a **membrane potential**, V_m (expressed in volts or millivolts). This membrane potential produces a force opposing ion movements that increase V_m and driving ion movements that reduce V_m (Fig. 11–27b). Thus the direction in which a charged solute tends to move spontaneously across a membrane depends on both the

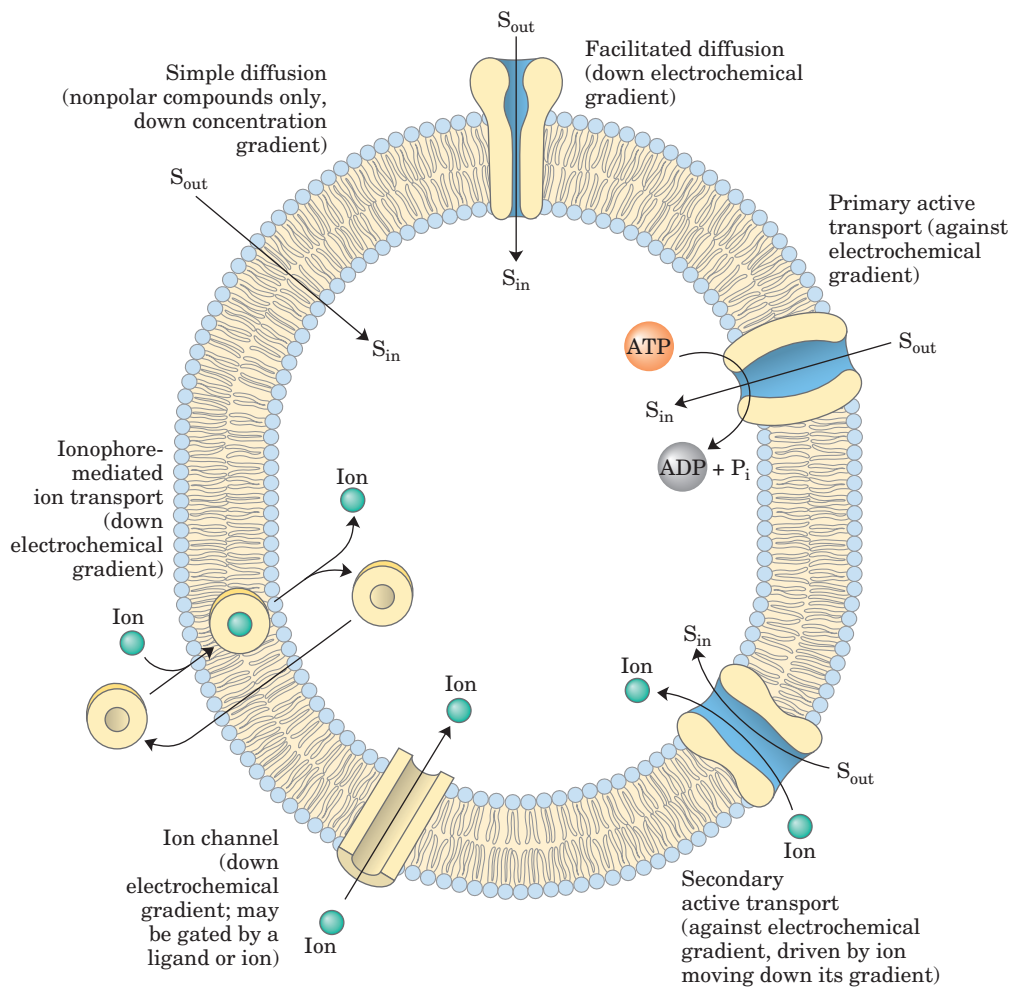


FIGURE 11-26 Summary of transport types.

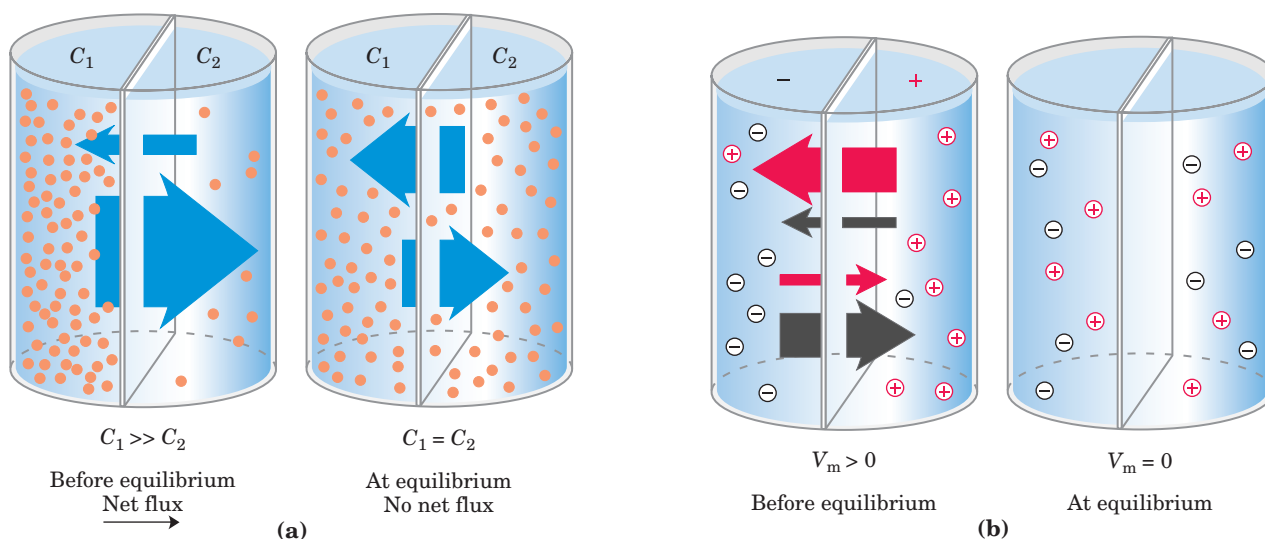


FIGURE 11-27 Movement of solutes across a permeable membrane. (a) Net movement of electrically neutral solutes is toward the side of lower solute concentration until equilibrium is achieved. The solute concentrations on the left and right sides of the membrane are designated C_1 and C_2 . The rate of transmembrane movement (indicated by

the large arrows) is proportional to the concentration gradient, C_1/C_2 . (b) Net movement of electrically charged solutes is dictated by a combination of the electrical potential (V_m) and the chemical concentration difference across the membrane; net ion movement continues until this electrochemical potential reaches zero.

chemical gradient (the difference in solute concentration) and the electrical gradient (V_m) across the membrane. Together, these two factors are referred to as the **electrochemical gradient** or **electrochemical potential**. This behavior of solutes is in accord with the second law of thermodynamics: molecules tend to spontaneously assume the distribution of greatest randomness and lowest energy.

To pass through a lipid bilayer, a polar or charged solute must first give up its interactions with the water molecules in its hydration shell, then diffuse about 3 nm (30 Å) through a solvent (lipid) in which it is poorly soluble (Fig. 11–28). The energy used to strip away the hydration shell and to move the polar compound from water into and through lipid is regained as the compound leaves the membrane on the other side and is rehydrated. However, the intermediate stage of transmembrane passage is a high-energy state comparable to the transition state in an enzyme-catalyzed chemical reaction. In both cases, an activation barrier must be overcome to reach the intermediate stage (Fig. 11–28; compare with Fig. 6–3). The energy of activation ($\Delta G^\ddagger$) for

translocation of a polar solute across the bilayer is so large that pure lipid bilayers are virtually impermeable to polar and charged species over periods of time relevant to cell growth and division.

Membrane proteins lower the activation energy for transport of polar compounds and ions by providing an alternative path through the bilayer for specific solutes. Proteins that bring about this **facilitated diffusion**, or **passive transport**, are not enzymes in the usual sense; their “substrates” are moved from one compartment to another, but are not chemically altered. Membrane proteins that speed the movement of a solute across a membrane by facilitating diffusion are called **transporters** or **permeases**.

Like enzymes, transporters bind their substrates with stereochemical specificity through multiple weak, noncovalent interactions. The negative free-energy change associated with these weak interactions, $\Delta G_{\text{binding}}$, counterbalances the positive free-energy change that accompanies loss of the water of hydration from the substrate, $\Delta G_{\text{dehydration}}$, thereby lowering $\Delta G^\ddagger$ for transmembrane passage (Fig. 11–28). Transporters span the lipid bilayer several times, forming a transmembrane channel lined with hydrophilic amino acid side chains. The channel provides an alternative path for a specific substrate to move across the lipid bilayer without its having to dissolve in the bilayer, further lowering $\Delta G^\ddagger$ for transmembrane diffusion. The result is an increase of several orders of magnitude in the rate of transmembrane passage of the substrate.

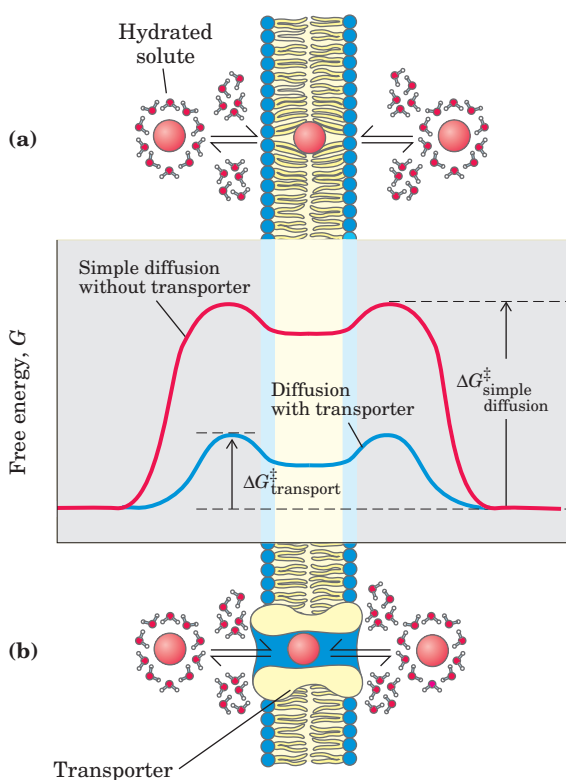


FIGURE 11–28 Energy changes accompanying passage of a hydrophilic solute through the lipid bilayer of a biological membrane. (a) In simple diffusion, removal of the hydration shell is highly endergonic, and the energy of activation ($\Delta G^\ddagger$) for diffusion through the bilayer is very high. (b) A transporter protein reduces the $\Delta G^\ddagger$ for transmembrane diffusion of the solute. It does this by forming noncovalent interactions with the dehydrated solute to replace the hydrogen bonding with water and by providing a hydrophilic transmembrane passageway.

Transporters Can Be Grouped into Superfamilies Based on Their Structures

We know from genomic studies that transporters constitute a significant fraction of all proteins encoded in the genomes of both simple and complex organisms. There are probably a thousand or more different transporters in the human genome. A few hundred transporters from various species have been studied with biochemical, genetic, and electrophysiological tools, but investigators have determined the three-dimensional structures for only a handful of these. Examination of the many transporter genes reveals obvious sequence similarities among subsets of transporters. And as experience has shown, similar amino acid sequences in proteins generally reflect similar three-dimensional structures and, often, similar mechanisms of action. It is reasonable to hope that by determining the structure and mechanism of action of at least one member of each transporter family, we can learn much about the other members of the family—about their structures, substrate specificities, transport rates, and mechanisms of energy coupling. A phylogenetic tree in which proteins are grouped together based on sequence homologies has the potential to tell us much about the transport properties of individual proteins on that tree. When this

TABLE 11–3 The Transporter Classification (TC) System

1.A. α Helix type channels
1.A.1. Voltage-gated ion channel VIC superfamily
Voltage-gated K^+ channel
1.A.3. Ryanodine/ IP_3 receptor Ca^{2+} channel
1.A.8. Major intrinsic protein family
Aquaporins
1.A.9. Ligand-gated ion channel (LIC) of neurotransmitter receptors
Acetylcholine receptor/channel
1.B. β Barrel porins
1.B.1. General bacterial porin (GBP) family
1.C. Pore-forming toxins
1.C.7. Diphtheria toxin family
1.C.18. Mellitin family (bee venoms)
2.A. Porters: uniporters, symporters, and antiporters
2.A.1. Major facilitator superfamily (MFS)
Lactose transporter/permease of <i>E. coli</i>
2.A.1.1. Sugar porter family
GLUT1 glucose transporter of erythrocyte
2.A.1.9. P_i - H^+ symporter
2.A.12. ATP-ADP antiporter (AAA) family
2.A.13. C_4 -dicarboxylate uptake (Dcu) family
2.A.21. Solute- Na^+ symporter (SSS) family
Na^+ -glucose symporter in epithelial cells
2.A.73. HCO_3^- transporters
HCO_3^- - Cl^- antiporter
2.B. Nonribosomally synthesized porters
2.B.1. Valinomycin carrier family
Valinomycin
3.A. Diphosphate bond hydrolysis-driven transporters (use PP_i , not ATP)
3.A.1. ATP-binding cassette (ABC) superfamily
CFTR Cl^- channel; multidrug transporter MDR1
3.A.2. H^+ - or Na^+ -translocating F-type, V-type, A-type ATPase superfamily
F_0F_1 ATPase proton pump; V_0V_1 ATPase; A_0A_1 ATPase
3.A.3. P-type ATPase superfamily
Na^+K^+ ATPase antiporter; SERCA Ca^{2+} pump

Note: The three broad groups correspond to groups 1, 2, and 3 in Figure 11-29. The individual transporters listed here (screened in yellow) are discussed in this chapter.

phylogeny is combined with knowledge of structure, specificity, or mechanism, we have a very useful and relatively simple representation of the huge group of transporters (Table 11–3).

Transporters can usefully be classified into super-families, whose members have considerable similarity of sequence and might therefore be expected to share structural and functional properties. There are two very broad categories of transporters: carriers and channels (Fig. 11–29). **Carriers** bind their substrates with high stereospecificity, catalyze transport at rates well below the limits of free diffusion, and are saturable in the same

sense as are enzymes: there is some substrate concentration above which further increases will not produce a greater rate of activity. **Channels** generally allow transmembrane movement at rates several orders of magnitude greater than those typical of carriers, rates approaching the limit of unhindered diffusion. Channels typically show less stereospecificity than carriers and are usually not saturable. Most channels are oligomeric complexes of several, often identical, subunits, whereas many carriers function as monomeric proteins. The classification as carrier or channel is the broadest distinction among transporters. Within each of these categories

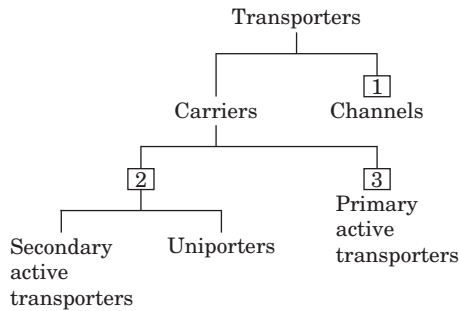


FIGURE 11-29 Classification of transporters. The numbers here correspond to the main subdivisions in Table 11-3.

are superfamilies of various types, defined not only by their primary sequences but by their secondary structures. Some channels are constructed primarily of helical transmembrane segments, others have β -barrel structures (Table 11-3). Among the carriers, some simply facilitate diffusion down a concentration gradient; they are the uniporter superfamily. Others (active transporters) can drive substrates across the membrane against a concentration gradient, some using energy provided directly by a chemical reaction (primary active transporters) and some coupling uphill transport of one substrate with the downhill transport of another (secondary active transporters). We now consider some well-studied representatives of the main transporter su-

perfamilies. You will encounter some of these transporters again in later chapters in the context of the metabolic pathways in which they participate.

The Glucose Transporter of Erythrocytes Mediates Passive Transport

Energy-yielding metabolism in erythrocytes depends on a constant supply of glucose from the blood plasma, where the glucose concentration is maintained at about 5 mM. Glucose enters the erythrocyte by facilitated diffusion via a specific glucose transporter, at a rate about 50,000 times greater than the uncatalyzed diffusion rate. The glucose transporter of erythrocytes (called GLUT1 to distinguish it from related glucose transporters in other tissues) is a type III integral protein ($M_r \sim 45,000$) with 12 hydrophobic segments, each of which is believed to form a membrane-spanning helix. The detailed structure of GLUT1 is not yet known, but one plausible model suggests that the side-by-side assembly of several helices produces a transmembrane channel lined with hydrophilic residues that can hydrogen-bond with glucose as it moves through the channel (Fig. 11-30).

The process of glucose transport can be described by analogy with an enzymatic reaction in which the “substrate” is glucose outside the cell (S_{out}), the “product” is glucose inside (S_{in}), and the “enzyme” is the transporter, T. When the rate of glucose uptake is measured

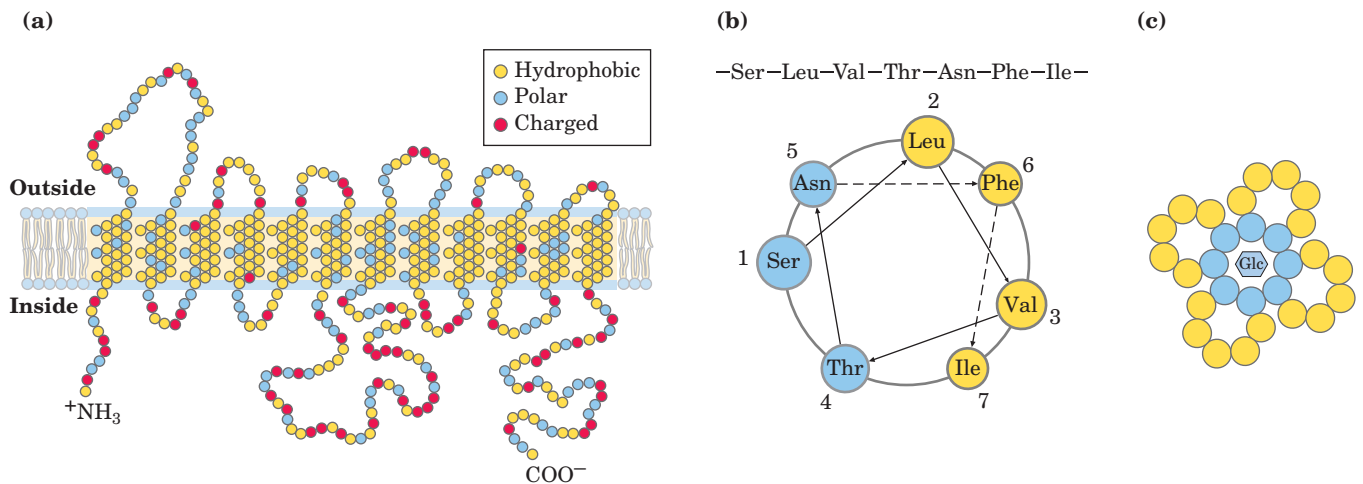


FIGURE 11-30 Proposed structure of GLUT1. (a) Transmembrane helices are represented as oblique (angled) rows of three or four amino acid residues, each row depicting one turn of the α helix. Nine of the 12 helices contain three or more polar or charged amino acid residues, often separated by several hydrophobic residues. (b) A helical wheel diagram shows the distribution of polar and nonpolar residues on the surface of a helical segment. The helix is diagrammed as though observed along its axis from the amino terminus. Adjacent residues in the linear sequence are connected with arrows, and each residue is placed around the wheel in the position it occupies in the helix; recall that 3.6 residues are required to make one complete turn of the

α helix. In this example, the polar residues (blue) are on one side of the helix and the hydrophobic residues (yellow) on the other. This is, by definition, an amphipathic helix. (c) Side-by-side association of five or six amphipathic helices, each with its polar face oriented toward the central cavity, can produce a transmembrane channel lined with polar and charged residues. This channel provides many opportunities for hydrogen bonding with glucose as it moves through the transporter. The three-dimensional structure of GLUT1 has not yet been determined by x-ray crystallography, but researchers expect that the hydrophilic transmembrane channels of this and many other transporters and ion channels will resemble this model.

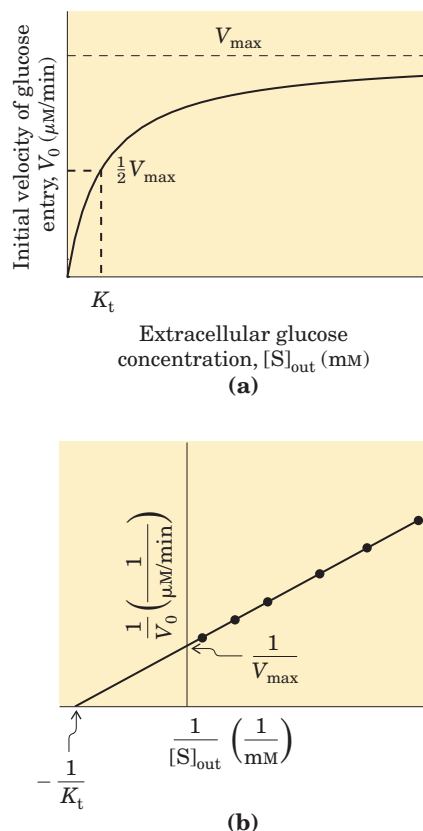
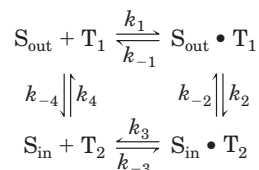


FIGURE 11-31 Kinetics of glucose transport into erythrocytes. **(a)** The initial rate of glucose entry into an erythrocyte, V_0 , depends upon the initial concentration of glucose on the outside, $[S]_{\text{out}}$. **(b)** Double-reciprocal plot of the data in **(a)**. The kinetics of facilitated diffusion is analogous to the kinetics of an enzyme-catalyzed reaction. Compare these plots with Figure 6–11, and Figure 1 in Box 6–1. Note that K_t is analogous to K_m , the Michaelis constant.

as a function of external glucose concentration (Fig. 11–31), the resulting plot is hyperbolic; at high external glucose concentrations the rate of uptake approaches V_{max} . Formally, such a transport process can be described by the equations



in which k_1 , k_{-1} , and so forth, are the forward and reverse rate constants for each step; T_2 is the transporter conformation that faces out, and T_1 the one that faces in. The steps are summarized in Figure 11–32.

The rate equations for this process can be derived exactly as for enzyme-catalyzed reactions (Chapter 6), yielding an expression analogous to the Michaelis-Menten equation:

$$V_0 = \frac{V_{\text{max}}[S]_{\text{out}}}{K_t + [S]_{\text{out}}}$$

in which V_0 is the initial velocity of accumulation of glucose inside the cell when its concentration in the surrounding medium is $[S]_{\text{out}}$, and K_t ($K_{\text{transport}}$) is a constant analogous to the Michaelis constant, a combination of rate constants that is characteristic of each transport system. This equation describes the *initial* velocity, the rate observed when $[S]_{\text{in}} = 0$. As is the case for enzyme-catalyzed reactions, the slope-intercept form of the equation describes a linear plot of $1/V_0$ against $1/[S]_{\text{out}}$, from which we can obtain values of K_t and V_{max} (Fig. 11–31b). When $[S] = K_t$, the rate of uptake is $\frac{1}{2} V_{\text{max}}$; the transport process is half-saturated. The concentration of blood glucose, 4.5 to 5 mM, is about

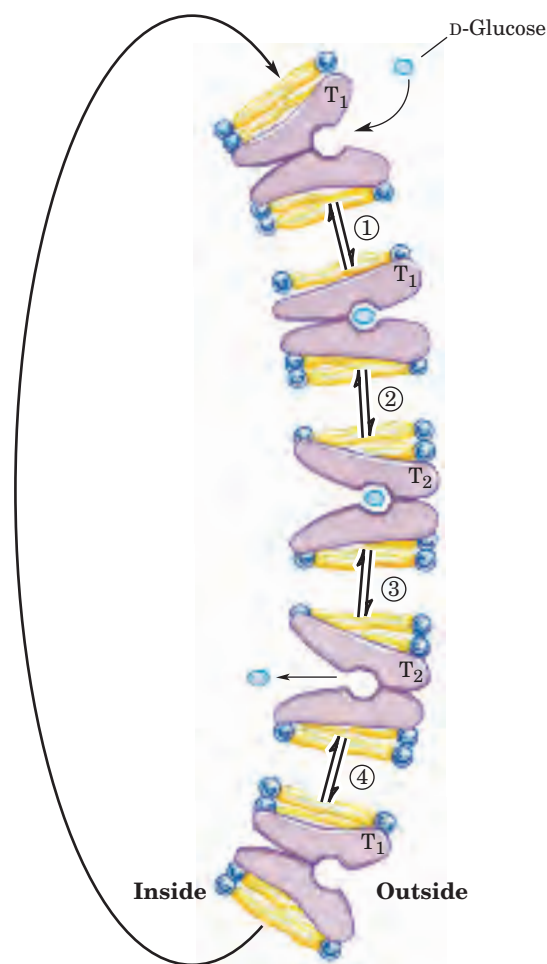


FIGURE 11-32 Model of glucose transport into erythrocytes by GLUT1. The transporter exists in two conformations: T_1 , with the glucose-binding site exposed on the outer surface of the plasma membrane, and T_2 , with the binding site exposed on the inner surface. Glucose transport occurs in four steps. ① Glucose in blood plasma binds to a stereospecific site on T_1 ; this lowers the activation energy for ② a conformational change from $S_{\text{out}} \cdot T_1$ to $S_{\text{in}} \cdot T_2$, effecting the transmembrane passage of the glucose. ③ Glucose is now released from T_2 into the cytoplasm, and ④ the transporter returns to the T_1 conformation, ready to transport another glucose molecule.

three times K_t , which ensures that GLUT1 is nearly saturated with substrate and operates near $V_{\max}$.

Because no chemical bonds are made or broken in the conversion of S_{out} to S_{in} , neither “substrate” nor “product” is intrinsically more stable, and the process of entry is therefore fully reversible. As $[S]_{\text{in}}$ approaches $[S]_{\text{out}}$, the rates of entry and exit become equal. Such a system is therefore incapable of accumulating the substrate (glucose) within a cell at concentrations above that in the surrounding medium; it simply achieves equilibration of glucose on the two sides of the membrane much faster than would occur in the absence of a specific transporter. GLUT1 is specific for D-glucose, having a measured K_t of 1.5 mM. For the close analogs D-mannose and D-galactose, which differ only in the position of one hydroxyl group, the values of K_t are 20 and 30 mM, respectively; and for L-glucose, K_t exceeds 3,000 mM. Thus GLUT1 shows the three hallmarks of passive transport: high rates of diffusion down a concentration gradient, saturability, and specificity.

Twelve glucose transporters are encoded in the human genome, each with unique kinetic properties, patterns of tissue distribution, and function (Table 11–4). In liver, GLUT2 transports glucose out of hepatocytes when liver glycogen is broken down to replenish blood glucose. GLUT2 has a K_t of about 66 mM and can therefore respond to increased levels of intracellular glucose (produced by glycogen breakdown) by increasing outward transport. Skeletal muscle and adipose tissue have yet another glucose transporter, GLUT4 ($K_t = 5$ mM), which is distinguished by its stimulation by insulin: its activity increases when release of insulin signals a high blood glucose concentration, thus increasing the rate of glucose uptake into muscle and adipose tissue (Box 11–2 describes some malfunctions of this transporter).

The Chloride-Bicarbonate Exchanger Catalyzes Electroneutral Cotransport of Anions across the Plasma Membrane

The erythrocyte contains another facilitated diffusion system, an anion exchanger that is essential in CO_2 transport to the lungs from tissues such as skeletal muscle and liver. Waste CO_2 released from respiring tissues into the blood plasma enters the erythrocyte, where it is converted to bicarbonate (HCO_3^-) by the enzyme carbonic anhydrase. (Recall that HCO_3^- is the primary buffer of blood pH; see Box 2–4). The HCO_3^- reenters the blood plasma for transport to the lungs (Fig. 11–33). Because HCO_3^- is much more soluble in blood plasma than is CO_2 , this roundabout route increases the capacity of the blood to carry carbon dioxide from the tissues to the lungs. In the lungs, HCO_3^- reenters the erythrocyte and is converted to CO_2 , which is eventually released into the lung space and exhaled. To be effective, this shuttle requires very rapid movement of HCO_3^- across the erythrocyte membrane.

The **chloride-bicarbonate exchanger**, also called the **anion exchange (AE) protein**, increases the permeability of the erythrocyte membrane to HCO_3^- more than a millionfold. Like the glucose transporter, it is an integral protein that probably spans the membrane at least 12 times. This protein mediates the simultaneous movement of two anions: for each HCO_3^- ion that moves in one direction, one Cl^- ion moves in the opposite direction (Fig. 11–33), with no net transfer of charge; the exchange is **electroneutral**. The coupling of Cl^- and HCO_3^- movements is obligatory; in the absence of chloride, bicarbonate transport stops. In this respect, the anion exchanger is typical of all systems, called **cotransport systems**, that simultaneously carry

TABLE 11–4 Glucose Transporters in the Human Genome

Transporter	Tissue(s) where expressed	Gene	Role*
GLUT1	Ubiquitous	SLC2A1	Basal glucose uptake
GLUT2	Liver, pancreatic islets, intestine	SLC2A2	In liver, removal of excess glucose from blood; in pancreas, regulation of insulin release
GLUT3	Brain (neuronal)	SLC2A3	Basal glucose uptake
GLUT4	Muscle, fat, heart	SLC2A4	Activity increased by insulin
GLUT5	Intestine, testis, kidney, sperm	SLC2A5	Primarily fructose transport
GLUT6	Spleen, leukocytes, brain	SLC2A6	Possibly no transporter function
GLUT7	Liver microsomes	SLC2A7	—
GLUT8	Testis, blastocyst, brain	SLC2A8	—
GLUT9	Liver, kidney	SLC2A9	—
GLUT10	Liver, pancreas	SLC2A10	—
GLUT11	Heart, skeletal muscle	SLC2A11	—
GLUT12	Skeletal muscle, adipose, small intestine	SLC2A12	—

*Dash indicates role uncertain.



Defective Glucose and Water Transport in Two Forms of Diabetes

When ingestion of a carbohydrate-rich meal causes blood glucose to exceed the usual concentration between meals (about 5 mM), excess glucose is taken up by the myocytes of cardiac and skeletal muscle (which store it as glycogen) and by adipocytes (which convert it to triacylglycerols). Glucose uptake into myocytes and adipocytes is mediated by the glucose transporter GLUT4. Between meals, some GLUT4 is present in the plasma membrane, but most is sequestered in the membranes of small intracellular vesicles (Fig. 1). Insulin released from the pancreas in response to high blood glucose triggers the movement of these intracellular vesicles to the plasma membrane, where they fuse, thus exposing GLUT4 molecules on the outer surface of the cell (see Fig. 12–8). With more GLUT4 molecules in action, the rate of glucose uptake increases 15-fold or more. When blood glucose levels return to normal, insulin release slows and most GLUT4 molecules are removed from the plasma membrane and stored in vesicles.

In type I (juvenile onset) diabetes mellitus, the inability to release insulin (and thus to mobilize glucose

transporters) results in low rates of glucose uptake into muscle and adipose tissue. One consequence is a prolonged period of high blood glucose after a carbohydrate-rich meal. This condition is the basis for the glucose tolerance test used to diagnose diabetes (Chapter 23).

The water permeability of epithelial cells lining the renal collecting duct in the kidney is due to the presence of an aquaporin (AQP-2) in their apical plasma membranes (facing the lumen of the duct). Antidiuretic hormone (ADH) regulates the retention of water by mobilizing AQP-2 molecules stored in vesicle membranes within the epithelial cells, much as insulin mobilizes GLUT4 in muscle and adipose tissue. When the vesicles fuse with the epithelial cell plasma membrane, water permeability greatly increases and more water is reabsorbed from the collecting duct and returned to the blood. When the ADH level drops, AQP-2 is resequenced within vesicles, reducing water retention. In the relatively rare human disease diabetes insipidus, a genetic defect in AQP-2 leads to impaired water reabsorption by the kidney. The result is excretion of copious volumes of very dilute urine.

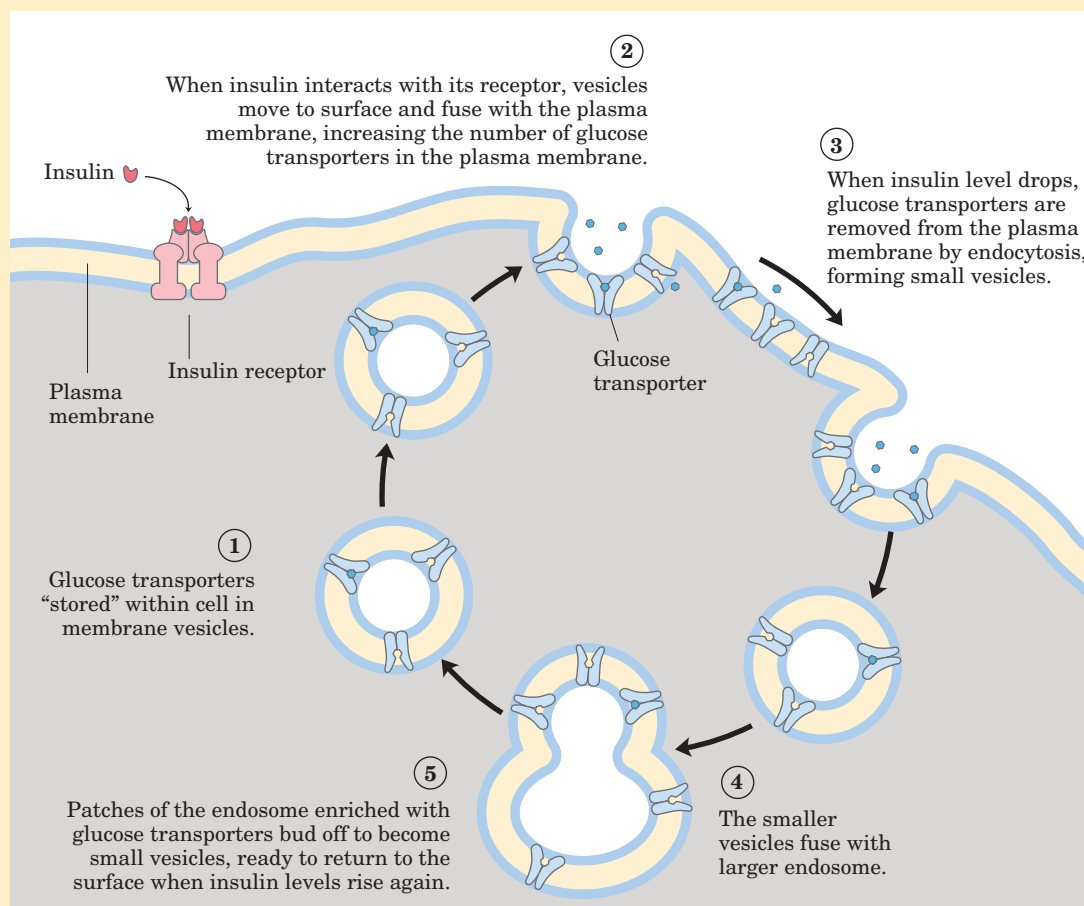


FIGURE 1 Regulation by insulin of glucose transport by GLUT4 into a myocyte.

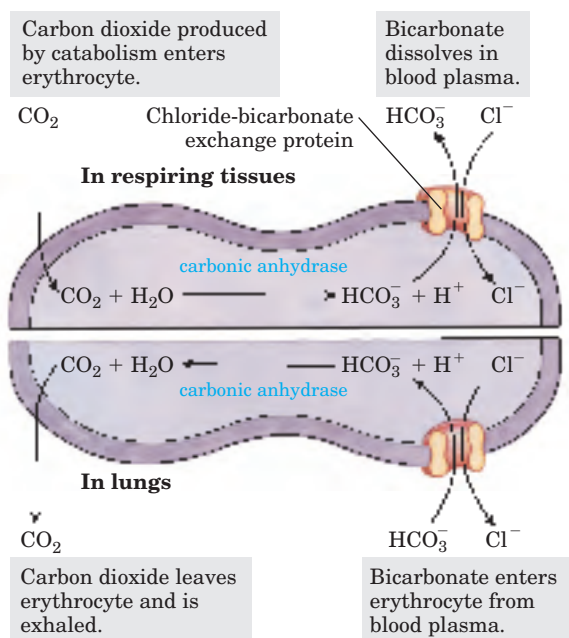


FIGURE 11-33 Chloride-bicarbonate exchanger of the erythrocyte membrane. This cotransport system allows the entry and exit of HCO_3^- without changes in the transmembrane electrical potential. Its role is to increase the CO_2 -carrying capacity of the blood.

two solutes across a membrane. When, as in this case, the two substrates move in opposite directions, the process is **antiport**. In **symport**, two substrates are moved simultaneously in the same direction. As we noted earlier, transporters that carry only one substrate, such as the erythrocyte glucose transporter, are **uniport** systems (Fig. 11-34).

The human genome has genes for three closely related chloride-bicarbonate exchangers, all with the same predicted transmembrane topology. Erythrocytes

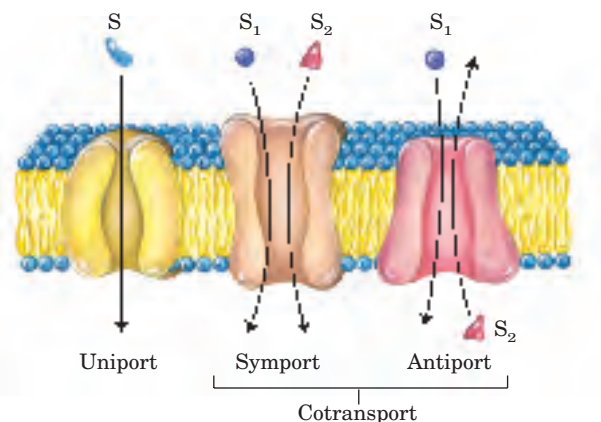


FIGURE 11-34 Three general classes of transport systems. Transporters differ in the number of solutes (substrates) transported and the direction in which each is transported. Examples of all three types of transporters are discussed in the text. Note that this classification tells us nothing about whether these are energy-requiring (active transport) or energy-independent (passive transport) processes.

contain the AE1 transporter, AE2 is prominent in liver, and AE3 is present in plasma membranes of the brain, heart, and retina. Similar anion exchangers are also found in plants and microorganisms.

Active Transport Results in Solute Movement against a Concentration or Electrochemical Gradient

In passive transport, the transported species always moves down its electrochemical gradient and is not accumulated above the equilibrium concentration. Active transport, by contrast, results in the accumulation of a solute above the equilibrium point. Active transport is thermodynamically unfavorable (endergonic) and takes place only when coupled (directly or indirectly) to an exergonic process such as the absorption of sunlight, an oxidation reaction, the breakdown of ATP, or the concomitant flow of some other chemical species down its electrochemical gradient. In **primary active transport**, solute accumulation is coupled directly to an exergonic chemical reaction, such as conversion of ATP to $\text{ADP} + \text{P}_i$ (Fig. 11-35). **Secondary active transport** occurs when endergonic (uphill) transport of one solute is coupled to the exergonic (downhill) flow of a different solute that was originally pumped uphill by primary active transport.

The amount of energy needed for the transport of a solute against a gradient can be calculated from the initial concentration gradient. The general equation for the free-energy change in the chemical process that converts S to P is

$$\Delta G = \Delta G^\circ + RT \ln [P]/[S] \quad (11-1)$$

where R is the gas constant, $8.315 \text{ J/mol} \cdot \text{K}$, and T is the absolute temperature. When the “reaction” is simply

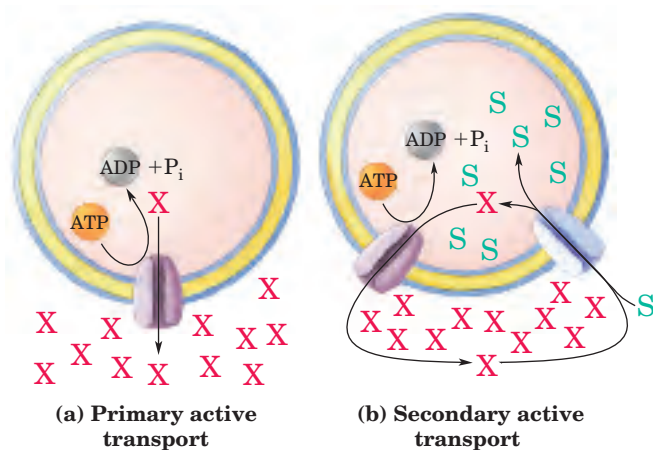


FIGURE 11-35 Two types of active transport. (a) In primary active transport, the energy released by ATP hydrolysis drives solute movement against an electrochemical gradient. (b) In secondary active transport, a gradient of ion X (often Na^+) has been established by primary active transport. Movement of X down its electrochemical gradient now provides the energy to drive cotransport of a second solute (S) against its electrochemical gradient.

transport of a solute from a region where its concentration is C_1 to a region where its concentration is C_2 , no bonds are made or broken and the standard free-energy change, $\Delta G'^\circ$, is zero. The free-energy change for transport, ΔG_t , is then

$$\Delta G_t = RT \ln \frac{C_2}{C_1} \quad (11-2)$$

If there is a tenfold difference in concentration between two compartments, the cost of moving 1 mol of an uncharged solute at 25 °C across a membrane separating the compartments is therefore

$$\begin{aligned} \Delta G_t &= (8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K})(\ln 10/1) = 5,700 \text{ J/mol} \\ &= 5.7 \text{ kJ/mol} \end{aligned}$$

Equation 11-2 holds for all uncharged solutes.

When the solute is an ion, its movement without an accompanying counterion results in the endergonic separation of positive and negative charges, producing an electrical potential; such a transport process is said to be **electrogenic**. The energetic cost of moving an ion depends on the electrochemical potential (p. 391), the sum of the chemical and electrical gradients:

$$\Delta G_t = RT \ln \left(\frac{C_2}{C_1} \right) + Z F \Delta \psi \quad (11-3)$$

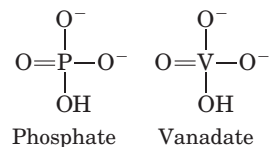
where Z is the charge on the ion, F is the Faraday constant (96,480 J/V · mol), and $\Delta \psi$ is the transmembrane electrical potential (in volts). Eukaryotic cells typically have electrical potentials across their plasma membranes of about 0.05 to 0.1 V (with the inside negative relative to the outside), so the second term of Equation 11-3 can make a significant contribution to the total free-energy change for transporting an ion. Most cells maintain more than tenfold differences in ion concentrations across their plasma or intracellular membranes, and for many cells and tissues active transport is therefore a major energy-consuming process.

The mechanism of active transport is of fundamental importance in biology. As we shall see in Chapter 19, the formation of ATP in mitochondria and chloroplasts occurs by a mechanism that is essentially ATP-driven ion transport operating in reverse. The energy made available by the spontaneous flow of protons across a membrane is calculable from Equation 11-3; remember that ΔG for flow *down* an electrochemical gradient has a negative value, and ΔG for transport of ions *against* an electrochemical gradient has a positive value.

P-Type ATPases Undergo Phosphorylation during Their Catalytic Cycles

The family of active transporters called **P-type ATPases** are ATP-driven cation transporters that are reversibly phosphorylated by ATP as part of the transport cycle; phosphorylation forces a conformational change that is central to moving the cation across the membrane. All P-type transport ATPases have similarities in amino acid

sequence, especially near the Asp residue that undergoes phosphorylation, and all are sensitive to inhibition by the phosphate analog **vanadate**.



Each P-type ATPase transporter is an integral protein with ten predicted membrane-spanning regions in a single polypeptide; some also have a second subunit. The P-type transporters are very widely distributed. In animal tissues, the Na^+K^+ ATPase (an antiporter for Na^+ and K^+) and the Ca^{2+} ATPase (a uniporter for Ca^{2+}) are ubiquitous P-type ATPases that maintain differences in the ionic composition of the cytosol and the extracellular medium. Parietal cells in the lining of the mammalian stomach have a P-type ATPase that pumps H^+ and K^+ across the plasma membrane, thereby acidifying the stomach contents. In vascular plants, a P-type ATPase pumps protons out of the cell, establishing an electrochemical difference of as much as 2 pH units and 250 mV across the plasma membrane. A similar P-type ATPase in the bread mold *Neurospora* pumps protons out of cells to establish an inside-negative membrane potential, which is used to drive the uptake of substrates and ions from the surrounding medium by secondary active transport. Bacteria use P-type ATPases to pump out toxic heavy metal ions such as Cd^{2+} and Cu^{2+} .

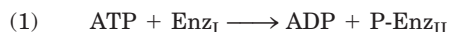
In virtually every animal cell type, the concentration of Na^+ is lower in the cell than in the surrounding medium, and the concentration of K^+ is higher (Fig. 11-36). This imbalance is maintained by a primary active transport system in the plasma membrane. The enzyme **Na^+K^+ ATPase**, discovered by Jens Skou in 1957, couples breakdown of ATP to the simultaneous movement of both Na^+ and K^+ against their electrochemical gradients. For each molecule of ATP converted to ADP and P_i , the transporter moves two K^+ ions inward and three Na^+ ions outward across the plasma membrane. The Na^+K^+ ATPase is an integral protein with two subunits ($M_r \sim 50,000$ and $\sim 110,000$), both of which span the membrane.



Jens Skou

The detailed mechanism by which ATP hydrolysis is coupled to transport awaits determination of the protein's three-dimensional structure, but a current model (Fig. 11-37) proposes that the ATPase cycles between two forms, a phosphorylated form (designated P-Enz_{II}) with high affinity for K^+ and low affinity for Na^+ , and a dephosphorylated form (Enz_I) with high affinity for Na^+

and low affinity for K^+ . The conversion of ATP to ADP and P_i takes place in two steps catalyzed by the enzyme, involving formation then hydrolysis of the phospho-enzyme:



Because three Na^+ ions move outward for every two K^+ ions that move inward, the process is electrogenic—it creates a net separation of charge across the membrane. The result is a transmembrane potential of -50 to -70 mV (inside negative relative to outside), which is characteristic of most animal cells and essential to the conduction of action potentials in neurons. The central role of the Na^+K^+ ATPase is reflected in the energy invested in this single reaction: about 25% of the total energy consumption of a human at rest!

 The steroid derivative **ouabain** (pronounced wah'-bane; from *waa bayyo*, Somali for “arrow poison”) is a potent and specific inhibitor of the Na^+K^+ ATPase. Ouabain binds preferentially to the form of the enzyme that is open to the extracellular side, locking in two Na^+ ions and preventing the changes of conformation necessary to ion transport. Another very potent toxin, palytoxin (produced by a coral on the Hawaiian shoreline), also targets the Na^+K^+ ATPase, but it binds

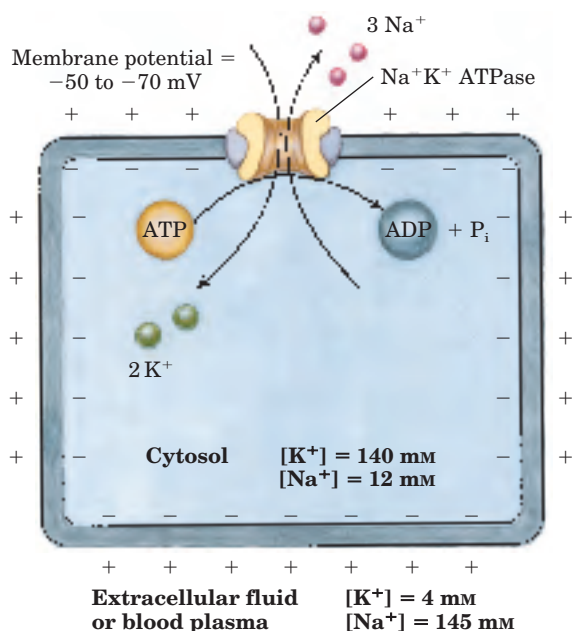


FIGURE 11-36 Na^+K^+ ATPase. In animal cells, this active transport system is primarily responsible for setting and maintaining the intracellular concentrations of Na^+ and K^+ and for generating the transmembrane electrical potential. It does this by moving three Na^+ out of the cell for every two K^+ it moves in. The electrical potential is central to electrical signaling in neurons, and the gradient of Na^+ is used to drive the uphill cotransport of solutes in many cell types.

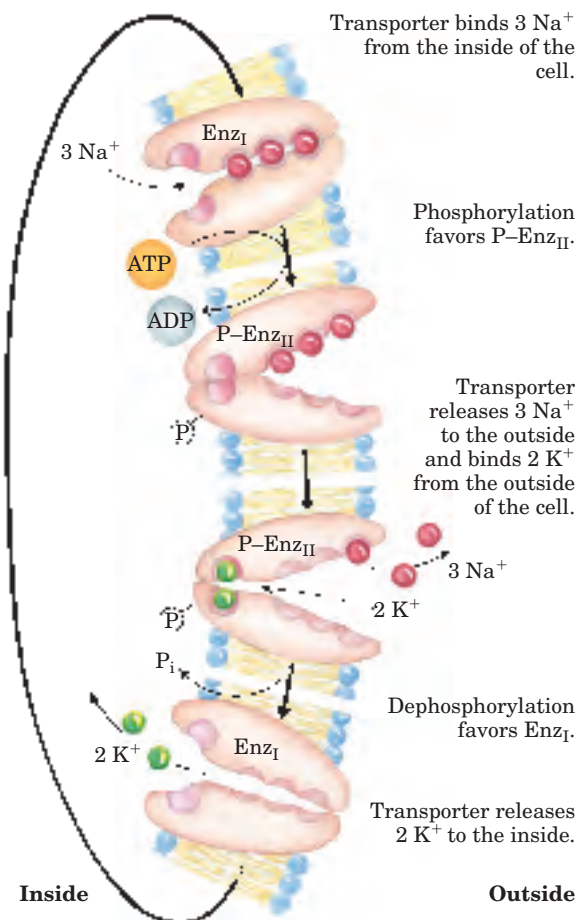
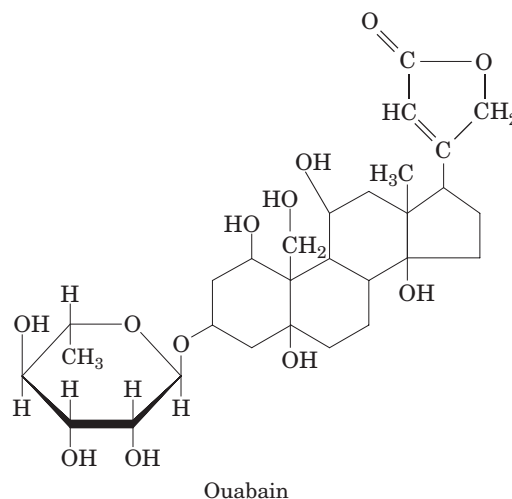


FIGURE 11-37 Postulated mechanism of Na^+ and K^+ transport by the Na^+K^+ ATPase.

to the protein so as to lock it into a position in which the ion-binding sites are permanently accessible from both sides, converting the transporter into a nonspecific ion channel. This allows exit of K^+ from cells and deflates the (essential) ion gradient across the plasma membrane, which accounts for the high toxicity of this compound.



Ouabain and another steroid derivative, **digitoxigenin**, are the active ingredients of digitalis, an extract of the leaves of the foxglove plant. (Ouabain is found in lower concentrations in a number of other plants, presumably serving to discourage herbivores.) Digitalis has been used to treat congestive heart failure since its introduction for that purpose (treatment of “dropsy”) by the British physician William Withering in 1785. It strengthens heart muscle contractions without increasing the heart rate and thus increases the efficiency of the heart. Digitalis inhibits the efflux of Na^+ , raising the intracellular $[\text{Na}^+]$ enough to activate a $\text{Na}^+/\text{Ca}^{2+}$ antiporter in cardiac muscle. The increased influx of Ca^{2+} through this antiporter produces elevated cytosolic $[\text{Ca}^{2+}]$, which strengthens the contractions of the heart. The potency of ouabain in animals led to the suggestion (50 years ago) that this plant product might act by mimicking a normal regulator of the Na^+/K^+ ATPase produced in animals, and it now appears that this may be so. Ouabain itself has been isolated from bovine adrenal glands and has been detected in the blood plasma and hypothalamus of mammals. ■

P-Type Ca^{2+} Pumps Maintain a Low Concentration of Calcium in the Cytosol

The cytosolic concentration of free Ca^{2+} is generally at or below 100 nM, far lower than that in the surrounding medium, whether pond water or blood plasma. The ubiquitous occurrence of inorganic phosphates (P_i and PP_i) at millimolar concentrations in the cytosol necessitates a low cytosolic Ca^{2+} concentration, because inorganic phosphate combines with calcium to form relatively insoluble calcium phosphates. Calcium ions are pumped out of the cytosol by a P-type ATPase, the **plasma membrane Ca^{2+} pump**. Another P-type Ca^{2+} pump in the endoplasmic reticulum moves Ca^{2+} into the ER lumen, a compartment separate from the cytosol. In myocytes, Ca^{2+} is normally sequestered in a specialized form of endoplasmic reticulum called the sarcoplasmic reticulum. **The sarcoplasmic and endoplasmic reticulum calcium (SERCA) pumps** are closely related in structure and mechanism, and both are inhibited by the tumor-promoting agent thapsigargin, which does not affect the plasma membrane Ca^{2+} pump.

The plasma membrane Ca^{2+} pump and SERCA pumps are integral proteins that cycle between phosphorylated and dephosphorylated conformations in a mechanism similar to that for Na^+/K^+ ATPase (Fig. 11–37). Phosphorylation favors a conformation with a high-affinity Ca^{2+} -binding site exposed on the cytoplasmic side, and dephosphorylation favors one with a low-affinity Ca^{2+} -binding site on the luminal side. By this mechanism, the energy released by hydrolysis of ATP during one phosphorylation-dephosphorylation cycle drives Ca^{2+} across the membrane against a large electrochemical gradient.

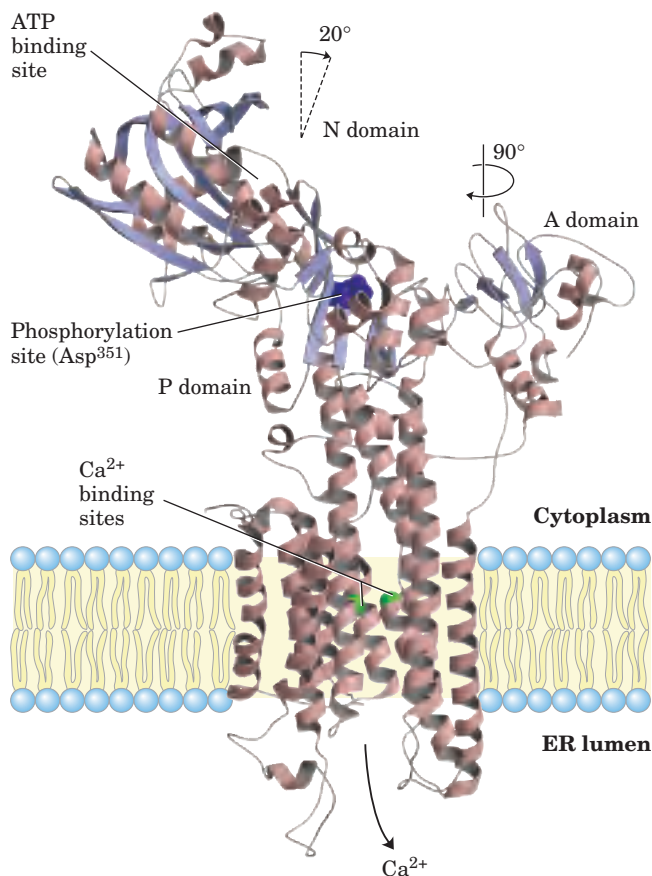


FIGURE 11–38 Structure of the Ca^{2+} pump of sarcoplasmic reticulum. (PDB ID 1EUL) Ten transmembrane helices surround the path for Ca^{2+} movement through the membrane. Two of the helices are interrupted near the middle of the bilayer, and their nonhelical regions form the binding sites for two Ca^{2+} ions (green). The carboxylate groups of an Asp residue in one helix and a Glu residue in another are central to the Ca^{2+} -binding sites. Three globular domains extend from the cytoplasmic side: the N (nucleotide-binding) domain has the binding site for ATP; the P (phosphorylation) domain contains the Asp^{351} residue (blue) that undergoes reversible phosphorylation, and the A (actuator) domain somehow mediates the structural changes that alter the Ca^{2+} affinity of the Ca^{2+} -binding site and its exposure to cytoplasm or lumen. Note the long distance between the phosphorylation site and the Ca^{2+} -binding site. There is strong evidence that during one transport cycle, the N domain tips about 20° to the right, bringing the ATP site close to Asp^{351} , and that during each catalytic cycle the A domain twists by about 90° around the normal (perpendicular) to the membrane. These conformational changes must expose the Ca^{2+} -binding site first on one side of the membrane, then on the other, changing the Ca^{2+} affinity of the site from high on the cytoplasmic side to lower on the luminal side. A complete understanding of the coupling between phosphorylation and Ca^{2+} transport awaits determination of all the conformations involved in the cycle.

The Ca^{2+} pump of the sarcoplasmic reticulum, which comprises 80% of the protein in that membrane, consists of a single polypeptide ($M_r \sim 100,000$) that spans the membrane ten times and has three cytoplasmic domains formed by loops that connect the transmembrane helices (Fig. 11–38). The two Ca^{2+} -binding sites are located near the middle of the membrane bi-

layer, 40 to 50 Å from the phosphorylated Asp residue characteristic of all P-type ATPases, so the effects of Asp phosphorylation are not direct. They must be mediated by conformational changes that alter the affinity for Ca^{2+} and open a path for Ca^{2+} release on the luminal side of the membrane.

The amino acid sequences of the SERCA pumps and the Na^+K^+ ATPase share 30% identity and 65% sequence similarity, and their topology relative to the membrane is also the same. Thus it seems likely that the Na^+K^+ ATPase structure is similar to that of the SERCA pumps and that all P-type ATPase transporters share the same basic structure.

F-Type ATPases Are Reversible, ATP-Driven Proton Pumps

The **F-type ATPase** active transporters play a central role in energy-conserving reactions in mitochondria, bacteria, and chloroplasts; we discuss that role in detail in our description of oxidative phosphorylation and photophosphorylation in Chapter 19. The F-type ATPases catalyze the uphill transmembrane passage of protons driven by ATP hydrolysis (“F-type” originated in the identification of these ATPases as energy-coupling factors). The F_o integral membrane protein complex (Fig. 11–39; subscript *o* denoting its inhibition by the drug oligomycin) provides a transmembrane pore for protons, and the peripheral protein F_1 (subscript *1* indicating that it was the first of several factors isolated from mitochondria) is a molecular machine that uses the energy

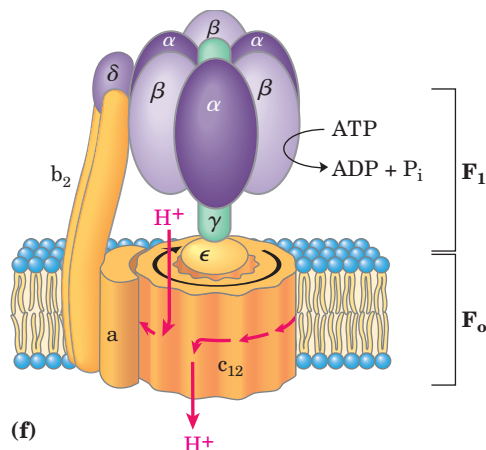


FIGURE 11–39 Structure of the F_oF_1 ATPase/ATP synthase. F-type ATPases have a peripheral domain, F_1 , consisting of three α subunits, three β subunits, one δ subunit (purple), and a central shaft (the γ subunit, green). The integral portion of F-type ATPases, F_o (yellow), has multiple copies of *c*, one *a*, and two *b* subunits. F_o provides a transmembrane channel through which about four protons are pumped (red arrows) for each ATP hydrolyzed on the β subunits of F_1 . The remarkable mechanism by which these two events are coupled is described in detail in Chapter 19. It involves rotation of F_o relative to F_1 (black arrow). The structures of V_oV_1 and A_oA_1 are essentially similar to that of F_oF_1 , and the mechanisms are probably similar, too.

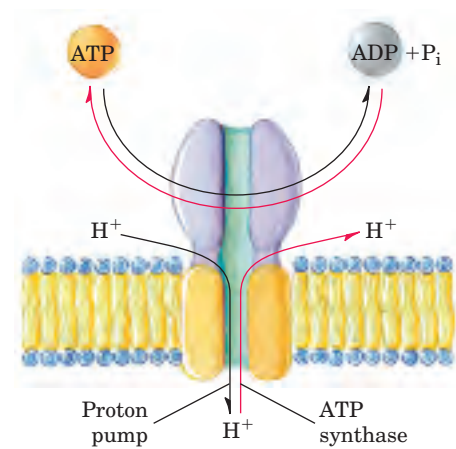


FIGURE 11–40 Reversibility of F-type ATPases. An ATP-driven proton transporter also can catalyze ATP synthesis (red arrows) as protons flow down their electrochemical gradient. This is the central reaction in the processes of oxidative phosphorylation and photophosphorylation, both described in detail in Chapter 19.

of ATP to drive protons uphill (into a region of higher H^+ concentration). The F_oF_1 organization of proton-pumping transporters must have developed very early in evolution. Eubacteria such as *E. coli* use an F_oF_1 ATPase complex in their plasma membrane to pump protons outward, and archaeobacteria have a closely homologous proton pump, the A_oA_1 ATPase.

The reaction catalyzed by F-type ATPases is reversible, so a proton gradient can supply the energy to drive the reverse reaction, ATP synthesis (Fig. 11–40). When functioning in this direction, the F-type ATPases are more appropriately named **ATP synthases**. ATP synthases are central to ATP production in mitochondria during oxidative phosphorylation and in chloroplasts during photophosphorylation, as well as in eubacteria and archaeobacteria. The proton gradient needed to drive ATP synthesis is produced by other types of proton pumps powered by substrate oxidation or sunlight. As noted above, we return to a detailed description of these processes in Chapter 19.

V-type ATPases, a class of proton-transporting ATPases structurally (and possibly mechanistically) related to the F-type ATPases, are responsible for acidifying intracellular compartments in many organisms (thus *V* for *vacuolar*). Proton pumps of this type maintain the vacuoles of fungi and higher plants at a pH between 3 and 6, well below that of the surrounding cytosol (pH 7.5). V-type ATPases are also responsible for the acidification of lysosomes, endosomes, the Golgi complex, and secretory vesicles in animal cells. All V-type ATPases have a similar complex structure, with an integral (transmembrane) domain (V_o) that serves as a proton channel and a peripheral domain (V_1) that contains the ATP-binding site and the ATPase activity. The mechanism by which V-type ATPases couple ATP hydrolysis to the uphill transport of protons is not understood in detail.

ABC Transporters Use ATP to Drive the Active Transport of a Wide Variety of Substrates

ABC transporters (Fig. 11–41) constitute a large family of ATP-dependent transporters that pump amino acids, peptides, proteins, metal ions, various lipids, bile salts, and many hydrophobic compounds, including drugs, out of cells against a concentration gradient. One ABC transporter in humans, the **multi-drug transporter (MDR1)**, is responsible for the striking resistance of certain tumors to some generally effective antitumor drugs. MDR1 has a broad substrate specificity for hydrophobic compounds, including, for example, the chemotherapeutic drugs adriamycin, doxorubicin, and vinblastine. By pumping these drugs out of the cell, the transporter prevents their accumulation within a tumor and thus blocks their therapeutic effects. MDR1 is an integral membrane protein (M_r 170,000) with 12 transmembrane segments and two ATP-binding domains (“cassettes”), which give the family its name: ATP-binding cassette transporters.

All ABC transporters have two nucleotide-binding domains (NBDs) and two transmembrane domains (Fig. 11–41). In some cases, all these domains are in a single long polypeptide; other ABC transporters have two subunits, each contributing an NBD and a domain with six (or in some cases ten) transmembrane helices. Although many of the ABC transporters are in the plasma membrane, some types are also found in the endoplasmic reticulum and in the membranes of mitochondria and lysosomes. Most ABC transporters act as pumps, but at least some members of the superfamily act as ion channels that are opened and closed by ATP hydrolysis. The CFTR transporter (Box 11–3) is a Cl^- channel operated by ATP hydrolysis.

The NBDs of all ABC proteins are similar in sequence and presumably in three-dimensional structure; they are the conserved molecular motor that can be coupled to a wide variety of pumps and channels. When coupled with

a pump, the ATP-driven motor moves solutes against a concentration gradient; when coupled with an ion channel, the motor opens and closes the channel using ATP as energy source. The stoichiometry of ABC pumps is about one ATP hydrolyzed per molecule of substrate transported, but neither the mechanism of coupling nor the site of substrate binding are known.

Some ABC transporters have very high specificity for a single substrate; others are more promiscuous. The human genome contains at least 48 genes that encode ABC transporters, many of which are involved in maintaining the lipid bilayer and in transporting sterols, sterol derivatives, and fatty acids throughout the body. The flippases that move membrane lipids from one leaflet of the bilayer to the other are ABC transporters, and the cellular machinery for exporting excess cholesterol includes an ABC transporter. Mutations in the genes that encode some of these proteins contribute to several genetic diseases, including cystic fibrosis (Box 11–3), Tangier disease (p. 827), retinal degeneration, anemia, and liver failure.

ABC transporters are also present in simpler animals and in plants and microorganisms. Yeast has 31 genes that encode ABC transporters, *Drosophila* has 56, and *E. coli* has 80, representing 2% of its entire genome. The presence of ABC transporters that confer antibiotic resistance in pathogenic microbes (*Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Candida albicans*, *Neisseria gonorrhoeae*, and *Plasmodium falciparum*) is a serious public health concern and makes these transporters attractive targets for drug design. ■

Ion Gradients Provide the Energy for Secondary Active Transport

The ion gradients formed by primary transport of Na^+ or H^+ can in turn provide the driving force for cotransport of other solutes. Many cell types contain transport

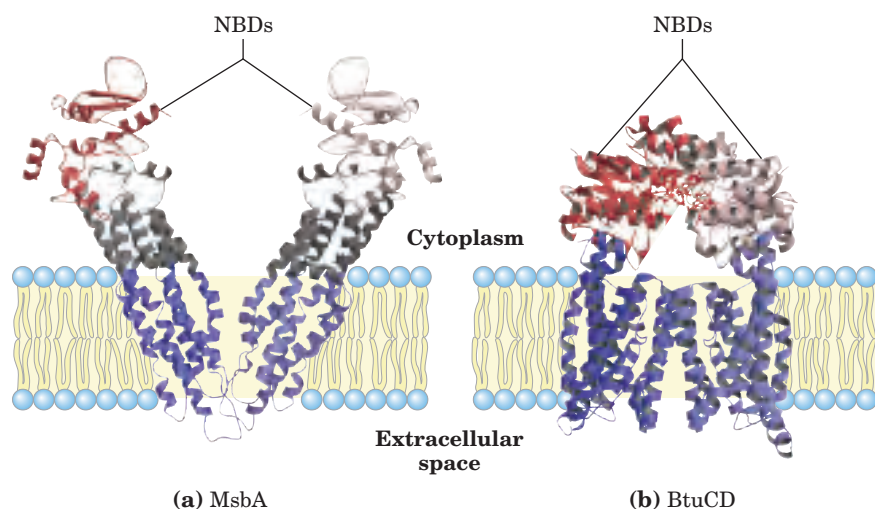


FIGURE 11–41 Structures of two ABC transporters of *E. coli*. (a) The lipid A flippase MsbA (PDB ID 1JSQ) and (b) the vitamin B₁₂ importer BtuCD (PDB ID 1L7V). Both structures are homodimers. The two nucleotide-binding domains (NBDs, in red) extend into the cytoplasm. In (b), residues involved in ATP binding and hydrolysis are shown as ball-and-stick structures. Each monomer of MsbA has six transmembrane helical segments (blue), and each monomer of BtuCD has ten.



BOX 11-3 BIOCHEMISTRY IN MEDICINE

A Defective Ion Channel in Cystic Fibrosis

Cystic fibrosis (CF) is a serious and relatively common hereditary disease of humans. About 5% of white Americans are carriers, having one defective and one normal copy of the gene. Only individuals with two defective copies show the severe symptoms of the disease: obstruction of the gastrointestinal and respiratory tracts, commonly leading to bacterial infection of the airways and death due to respiratory insufficiency before the age of 30. In CF, the thin layer of mucus that normally coats the internal surfaces of the lungs is abnormally thick, obstructing air flow and providing a haven for pathogenic bacteria, particularly *Staphylococcus aureus* and *Pseudomonas aeruginosa*.

The defective gene in CF patients was discovered in 1989. It encodes a membrane protein called cystic fibrosis transmembrane conductance regula-

tor, or CFTR. Hydropathy analysis predicted that CFTR has 12 transmembrane helices and is structurally related to the multidrug (MDR1) transporters of drug-resistant tumors (Fig. 1). The normal CFTR protein proved to be an ion channel specific for Cl^- ions. The Cl^- channel activity increases greatly when phosphoryl groups are transferred from ATP to several side chains of the protein, catalyzed by cAMP-dependent protein kinase (Chapter 12). The mutation responsible for CF in 70% of cases results in deletion of a Phe residue at position 508, with the effect that the mutant protein is not correctly folded and inserted in the plasma membrane. Other mutations yield a protein that is inserted properly but cannot be activated by phosphorylation. In each case, the fundamental problem is a nonfunctional Cl^- channel in the epithelial cells that line the airways (Fig. 2), the digestive tract, and exocrine glands (pancreas, sweat glands, bile ducts, and vas deferens).

Normally, epithelial cells that line the inner surface of the lungs secrete a substance that traps and kills bacteria, and the cilia on the epithelial cells constantly sweep away the resulting debris. When CFTR is defective or missing, this process is less efficient, and frequent infections by bacteria such as *S. aureus* and *P. aeruginosa* progressively damage the lungs and reduce respiratory efficiency.

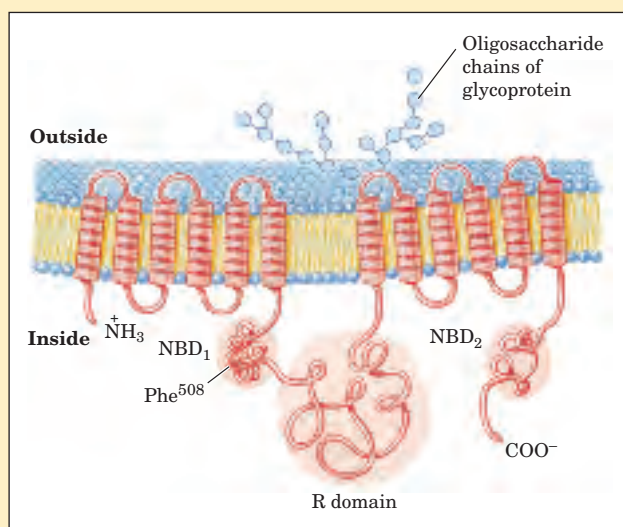


FIGURE 1 Topology of the cystic fibrosis transmembrane conductance regulator, CFTR. It has 12 transmembrane helices, and three functionally significant domains extend from the cytoplasmic surface: NBD₁ and NBD₂ are nucleotide-binding domains to which ATP binds, and a regulatory domain (R domain) is the site of phosphorylation by cAMP-dependent protein kinase. Oligosaccharide chains are attached to several residues on the outer surface of the segment between helices 7 and 8. The most commonly occurring mutation leading to CF is the deletion of Phe⁵⁰⁸, in the NBD₁ domain. The structure of CFTR is very similar to that of the multidrug transporter of tumors, described in the text.



FIGURE 2 Mucus lining the surface of the lungs traps bacteria. In healthy lungs, these bacteria are killed and swept away by the action of cilia. In CF, the bactericidal activity is impaired, resulting in recurring infections and progressive damage to the lungs.

systems that couple the spontaneous, downhill flow of these ions to the simultaneous uphill pumping of another ion, sugar, or amino acid (Table 11-5). The **lactose transporter (lactose permease)** of *E. coli* is

the well-studied prototype for proton-driven cotransporters. This protein consists of a single polypeptide chain (417 residues) that functions as a monomer to transport one proton and one lactose molecule into the

TABLE 11-5 Cotransport Systems Driven by Gradients of Na^+ or H^+

Organism/tissue/cell type	Transported solute (moving against its gradient)	Cotransported solute (moving down its gradient)	Type of transport
<i>E. coli</i>	Lactose	H^+	Symport
	Proline	H^+	Symport
	Dicarboxylic acids	H^+	Symport
Intestine, kidney (vertebrates)	Glucose	Na^+	Symport
	Amino acids	Na^+	Symport
Vertebrate cells (many types)	Ca^{2+}	Na^+	Antiport
Higher plants	K^+	H^+	Antiport
Fungi (<i>Neurospora</i>)	K^+	H^+	Antiport

cell, with the net accumulation of lactose (Fig. 11-42). *E. coli* normally produces a gradient of protons and charge across its plasma membrane by oxidizing fuels and using the energy of oxidation to pump protons outward. (This mechanism is discussed in detail in Chapter 19.) The lipid bilayer is impermeable to protons, but the lactose transporter provides a route for proton reentry, and lactose is simultaneously carried into the cell by symport. The endergonic accumulation of lactose is thereby coupled to the exergonic flow of protons into the cell, with a negative overall free-energy change.

The lactose transporter is one member of the **major facilitator superfamily (MFS)** of transporters, which comprises 28 families. Almost all proteins in this superfamily have 12 transmembrane domains (the few exceptions have 14). The proteins share rela-

tively little sequence homology, but the similarity of their secondary structures and topology suggests a common tertiary structure. The crystallographic solution of the *E. coli* lactose transporter by Ron Kaback and So Iwata in 2003 may provide a glimpse of this general structure (Fig. 11-43a). The protein has 12 transmembrane helices, and connecting loops that protrude into the cytoplasm or the periplasmic space. All six amino-terminal and six carboxyl-terminal helices form very similar domains, to produce a structure with a rough twofold symmetry. In the crystallized form of the protein, a large aqueous cavity is exposed on the cytoplasmic side of the membrane. The substrate-binding site is in this cavity, more or less in the middle of the membrane. The side of the transporter facing outward (the periplasmic face) is closed tightly, with no channel big enough for lactose to enter. The proposed mecha-

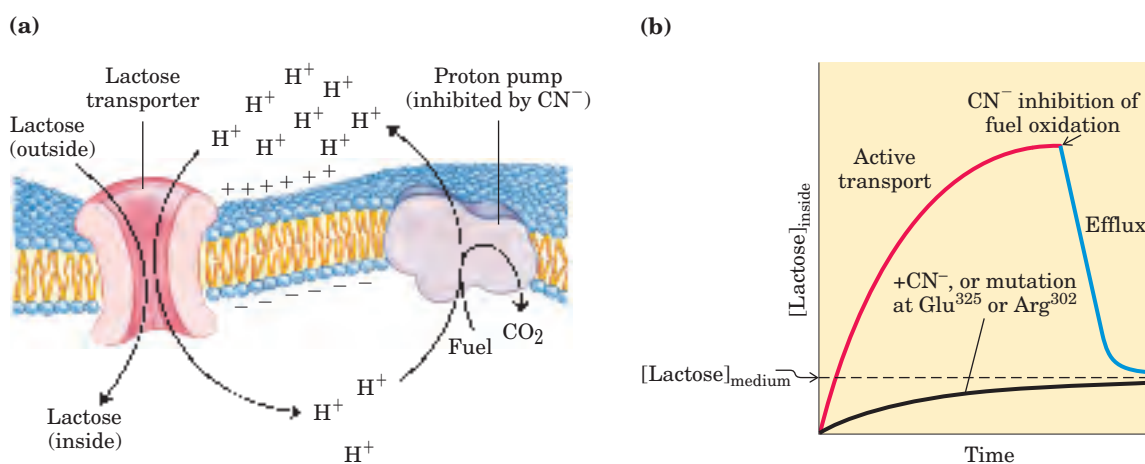


FIGURE 11-42 Lactose uptake in *E. coli*. (a) The primary transport of H^+ out of the cell, driven by the oxidation of a variety of fuels, establishes both a proton gradient and an electrical potential (inside negative) across the membrane. Secondary active transport of lactose into the cell involves symport of H^+ and lactose by the lactose transporter. The uptake of lactose against its concentration gradient is entirely dependent on this inflow of H^+ , driven by the electrochemical gradient.

(b) When the energy-yielding oxidation reactions of metabolism are blocked by cyanide (CN^-), the lactose transporter allows equilibration of lactose inside and outside the cell via passive transport. Mutations that affect Glu³²⁵ or Arg³⁰² have the same effect as cyanide. The dashed line represents the concentration of lactose in the surrounding medium.

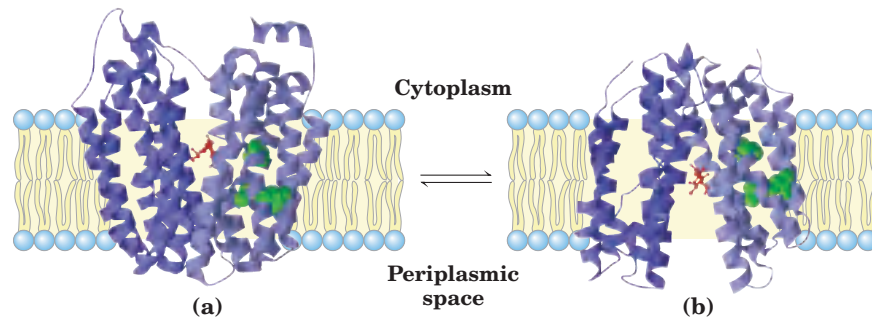


FIGURE 11-43 Structure of the lactose transporter (lactose permease) of *E. coli*. (a) Ribbon representation viewed parallel to the plane of the membrane shows the 12 transmembrane helices arranged in two nearly symmetrical domains shown in different shades of blue. In the form of the protein for which the crystal structure was determined, the substrate sugar (red) is bound near the middle of the membrane where it is exposed to the cytoplasm (derived from PDB ID 1PV7). (b) The structural changes postulated to take place during one transport

cycle. The two halves of the transporter undergo a large, reversible conformational change in which the two domains tilt relative to each other, exposing the substrate-binding site first to the periplasm (structure on the right), where lactose is picked up, then to the cytoplasm (left), where the lactose is released. The interconversion of the two forms is driven by changes in the pairing of charged (protonatable) side chains such as those of Glu³²⁵ and Arg³⁰² (green), which is affected by the transmembrane proton gradient.

nism for transmembrane passage of the substrate (Fig. 11-43b) involves a rocking motion between the two domains, driven by substrate binding and proton movement, alternately exposing the substrate-binding domain to the cytoplasm and to the periplasm. This so-called rocking banana model is similar to that shown in Figure 11-32 for GLUT1.

How is proton movement into the cell coupled with lactose uptake? Extensive genetic studies of the lactose transporter have established that of the 417 residues in the protein, only 6 are absolutely essential for cotransport of H⁺ and lactose—some for lactose binding, others for proton transport. Mutation in either of two residues (Glu³²⁵ and Arg³⁰²; Fig. 11-43) results in a protein still able to catalyze facilitated diffusion of lactose

but incapable of coupling H⁺ flow to uphill lactose transport. A similar effect is seen in wild-type (unmutated) cells when their ability to generate a proton gradient is blocked with CN⁻: the transporter carries out facilitated diffusion normally, but it cannot pump lactose against a concentration gradient (Fig. 11-42b). The balance between the two conformations of the lactose transporter is affected by changes in charge pairing between side chains.

In intestinal epithelial cells, glucose and certain amino acids are accumulated by symport with Na⁺, down the Na⁺ gradient established by the Na⁺K⁺ ATPase of the plasma membrane (Fig. 11-44). The apical surface of the intestinal epithelial cell is covered with microvilli, long thin projections of the plasma membrane

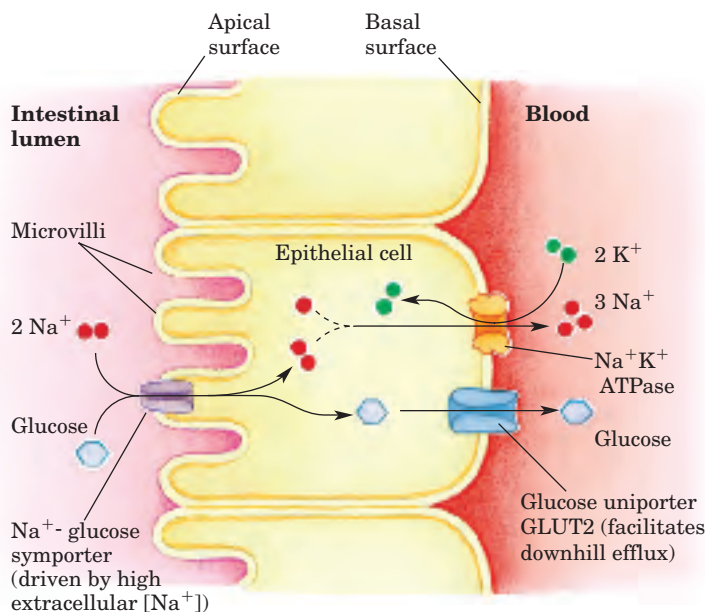


FIGURE 11-44 Glucose transport in intestinal epithelial cells. Glucose is cotransported with Na⁺ across the apical plasma membrane into the epithelial cell. It moves through the cell to the basal surface, where it passes into the blood via GLUT2, a passive glucose transporter. The Na⁺K⁺ ATPase continues to pump Na⁺ outward to maintain the Na⁺ gradient that drives glucose uptake.

that greatly increase the surface area exposed to the intestinal contents. **Na⁺-glucose symporters** in the apical plasma membrane take up glucose from the intestine in a process driven by the downhill flow of Na⁺:



The energy required for this process comes from two sources: the greater concentration of Na⁺ outside than inside (the chemical potential) and the transmembrane potential (the electrical potential), which is inside-negative and therefore draws Na⁺ inward. The electrochemical potential of Na⁺ is

$$\Delta G = RT \ln \frac{[\text{Na}^{+}]_{\text{in}}}{[\text{Na}^{+}]_{\text{out}}} + n \, \Delta E$$

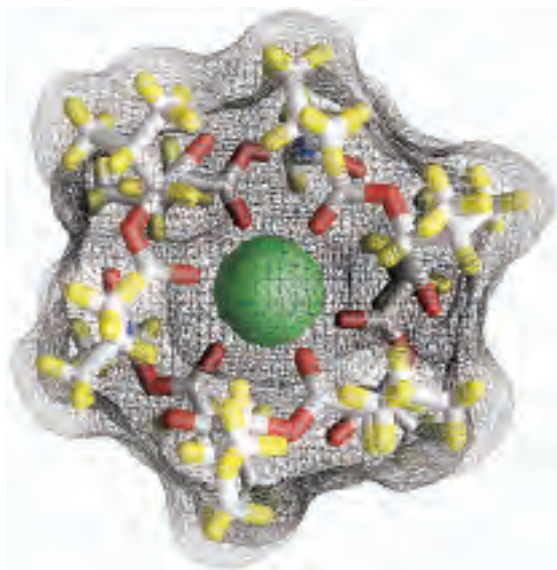
where $n = 2$, the number of Na⁺ ions cotransported with each glucose molecule. Given the typical membrane potential of -50 mV, an intracellular [Na⁺] of 12 mM, and an extracellular [Na⁺] of 145 mM, the energy, ΔG , made available as two Na⁺ ions reenter the cell is 22.5 kJ, enough to pump glucose against a large concentration gradient:

$$\Delta G_t = -22.5 \text{ kJ} = RT \ln \frac{[\text{glucose}]_{\text{in}}}{[\text{glucose}]_{\text{out}}}$$

and thus

$$\frac{[\text{Glucose}]_{\text{in}}}{[\text{Glucose}]_{\text{out}}} \approx 9,000$$

That is, the cotransporter can pump glucose inward until its concentration within the epithelial cell is about 9,000 times that in the intestine. As glucose is pumped from the intestine into the epithelial cell at the apical surface, it is simultaneously moved from the cell into the blood by passive transport through a glucose transporter (GLUT2) in the basal surface (Fig. 11-44). The crucial role of Na⁺ in symport and antiport systems such as these requires the continued outward pumping of Na⁺ to maintain the transmembrane Na⁺ gradient.



Because of the essential role of ion gradients in active transport and energy conservation, compounds that collapse ion gradients across cellular membranes are effective poisons, and those that are specific for infectious microorganisms can serve as antibiotics. One such substance is valinomycin, a small cyclic peptide that neutralizes the K⁺ charge by surrounding it with six carbonyl oxygens (Fig. 11-45). The hydrophobic peptide then acts as a shuttle, carrying K⁺ across membranes down its concentration gradient and deflating that gradient. Compounds that shuttle ions across membranes in this way are called **ionophores** (“ion bearers”). Both valinomycin and monensin (a Na⁺-carrying ionophore) are antibiotics; they kill microbial cells by disrupting secondary transport processes and energy-conserving reactions.

Aquaporins Form Hydrophilic Transmembrane Channels for the Passage of Water

A family of integral proteins discovered by Peter Agre, the **aquaporins (AQPs)**, provide channels for rapid movement of water molecules across all plasma membranes (Table 11-6 lists a few examples). Ten aquaporins are known in humans, each with its specialized role. Erythrocytes, which swell or shrink rapidly in response to abrupt changes in extracellular osmolarity as blood travels through the renal medulla, have a high density of aquaporin in their plasma membranes (2×10^5 copies of AQP-1 per cell). In the nephron (the functional unit of the kidney), the plasma membranes of proximal renal tubule cells have five different aquaporin types.



Peter Agre

FIGURE 11-45 Valinomycin, a peptide ionophore that binds K⁺. In this image, the surface contours are shown as a transparent mesh, through which a stick structure of the peptide and a K⁺ atom (green) are visible. The oxygen atoms (red) that bind K⁺ are part of a central hydrophilic cavity. Hydrophobic amino acid side chains (yellow) coat the outside of the molecule. Because the exterior of the K⁺-valinomycin complex is hydrophobic, the complex readily diffuses through membranes, carrying K⁺ down its concentration gradient. The resulting dissipation of the transmembrane ion gradient kills microbial cells, making valinomycin a potent antibiotic.

These cells reabsorb water during urine formation, a process for which water movement across membranes is essential (Box 11-3). The plant *Arabidopsis thaliana* has 38 genes that encode various types of aquaporins, reflecting the critical roles of water movement in plant physiology. Changes in turgor pressure, for example, require rapid movement of water across a membrane.

Water molecules flow through an AQP-1 channel at the rate of about 10^9 s^{-1} . For comparison, the highest known turnover number for an enzyme is that for catalase, $4 \times 10^7 \text{ s}^{-1}$, and many enzymes have turnover numbers between 1 s^{-1} and 10^4 s^{-1} (see Table 6-7). The low activation energy for passage of water through aquaporin channels ($\Delta G^\ddagger < 15 \text{ kJ/mol}$) suggests that water moves through the channels in a continuous

stream, in the direction dictated by the osmotic gradient. (For a discussion of osmosis, see p. 57.) It is essential that aquaporins not allow passage of protons (hydronium ions, H_3O^+), which would collapse membrane electrochemical potentials. And they do not. What is the basis for this extraordinary selectivity?

We find an answer in the structure of AQP-1, as determined by x-ray diffraction analysis (Fig. 11-46). AQP-1 has four monomers (each M_r 28,000) associated in a tetramer, each monomer forming a transmembrane pore with a diameter (2 to 3 Å) sufficient to allow passage of water molecules in single file. Each monomer consists of six transmembrane helical segments and two shorter helices, each of which contains the sequence Asn-Pro-Ala (NPA). The NPA-containing short helices extend toward the middle of the bilayer from opposite

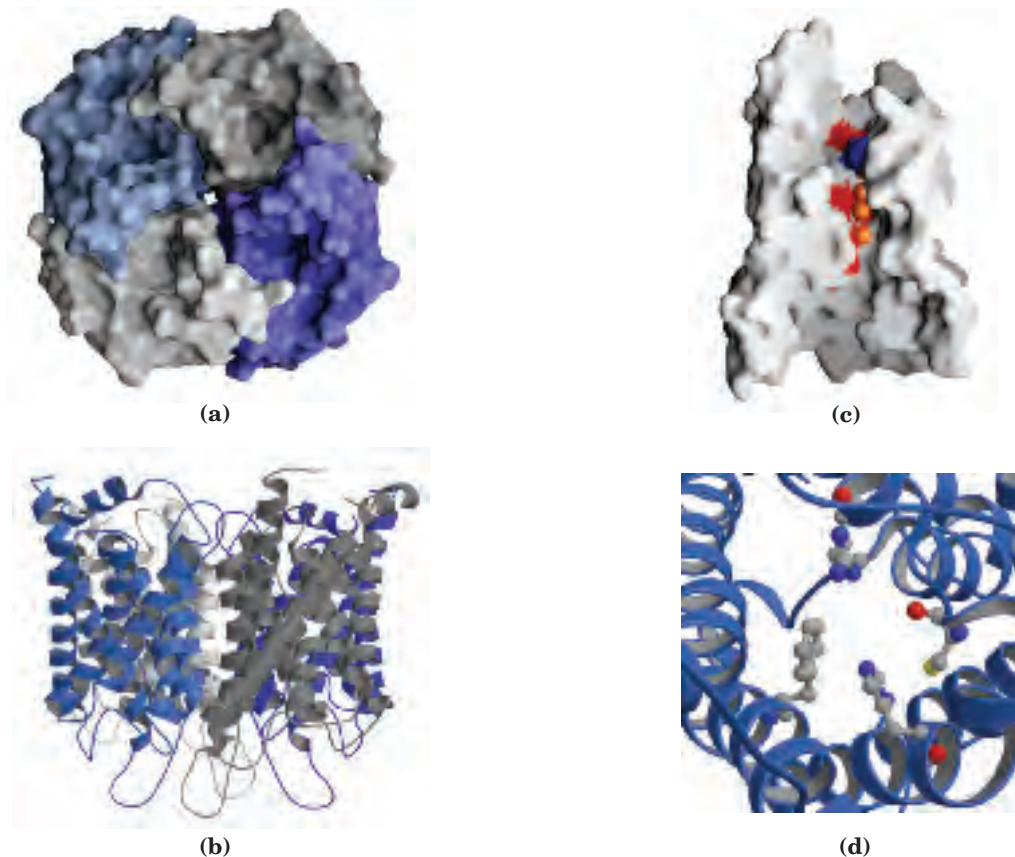


FIGURE 11-46 Structure of an aquaporin, AQP-1. The protein is a tetramer of identical monomeric units, each of which forms a transmembrane pore (derived from PDB ID 1J4N). **(a)** Surface model viewed perpendicular to the plane of the membrane. The protein contains four pores, one in each subunit. (The opening at the junction of the subunits is not a pore.) **(b)** An AQP-1 tetramer, viewed in the plane of the membrane. The helices of each subunit cluster around a central transmembrane pore. In each monomer, two short helical loops, one between helices 2 and 3 and the other between 5 and 6, contain the Asn-Pro-Ala (NPA) sequences found in all aquaporins, and form part of the water channel. **(c)** Surface representation of a single subunit, viewed in the plane of the membrane. The near side of the AQP-1

monomer has been cut away to reveal the channel running from top to bottom. The series of water molecules (orange spheres) shows the likely path of water molecules through the aquaporin channel, as predicted by molecular dynamics simulations in which investigators use the properties of water and aquaporin to calculate the lowest energy states. Hydrophilic atoms that provide selective interactions with water in the channel are colored red. A Phe residue (Phe⁵⁸) at the constriction is shown in blue. **(d)** A view down the channel, showing the constriction region of the specificity pore, which lets only a molecule as small as water pass. The side chains of Phe⁵⁸, His¹⁸², Cys¹⁹¹, and Arg¹⁹⁷ create this constriction.

TABLE 11-6 Aquaporins

<i>Aquaporin</i>	<i>Roles and/or location</i>
AQP-1	Fluid reabsorption in proximal renal tubule; secretion of aqueous humor in eye and cerebrospinal fluid in central nervous system; water homeostasis in lung
AQP-2	Water permeability in renal collecting duct (mutations produce nephrogenic diabetes insipidus)
AQP-3	Water retention in renal collecting duct
AQP-4	Cerebrospinal fluid reabsorption in central nervous system; regulation of brain edema
AQP-5	Fluid secretion in salivary glands, lachrymal glands, and alveolar epithelium of lung
AQP-6	Kidney
AQP-7	Renal proximal tubule, intestine
AQP-8	Liver, pancreas, colon, placenta
AQP-9	Liver, leukocytes
TIP	Regulation of turgor pressure in plant tonoplast
PIP	Plant plasma membrane
AQY	Yeast plasma membrane

sides, with their NPA regions overlapping in the middle of the membrane to form part of the specificity filter—the structure that allows only water to pass.

The residues that line the channel of each AQP-1 monomer are generally nonpolar, but carbonyl oxygens in the peptide backbone, projecting into the narrow part of the channel at intervals, can form hydrogen bonds with individual water molecules as they pass through; the two Asn residues (Asn⁷⁶ and Asn¹⁹²) in the NPA loops also hydrogen-bond with the water. The structure does not admit closely spaced water molecules that might form a chain to allow proton hopping (see Fig. 2-14), which would effectively move protons across the membrane. Critical Arg and His residues and electric dipoles formed by the short helices of the NPA loops provide positive charges in positions that repel any protons that might leak through the pore.

Ion-Selective Channels Allow Rapid Movement of Ions across Membranes

Ion-selective channels—first recognized in neurons and now known to be present in the plasma membranes of all cells, as well as in the intracellular membranes of eukaryotes—provide another mechanism for moving inorganic ions across membranes. Ion channels, together with ion pumps such as the Na⁺K⁺ ATPase, determine a plasma membrane's permeability to specific ions and regulate the cytosolic concentration of ions and the membrane potential. In neurons, very rapid changes in the activity of ion channels cause the changes in membrane potential (the action potentials) that carry signals from one end of a neuron to the other. In myocytes,

rapid opening of Ca²⁺ channels in the sarcoplasmic reticulum releases the Ca²⁺ that triggers muscle contraction. We discuss the signaling functions of ion channels in Chapter 12. Here we describe the structural basis for ion-channel function, using as examples a bacterial K⁺ channel, the neuronal Na⁺ channel, and the acetylcholine receptor ion channel.

Ion channels are distinguished from ion transporters in at least three ways. First, the rate of flux through channels can be several orders of magnitude greater than the turnover number for a transporter—10⁷ to 10⁸ ions/s for an ion channel, near the theoretical maximum for unrestricted diffusion. Second, ion channels are not saturable: rates do not approach a maximum at high substrate concentration. Third, they are “gated”—opened or closed in response to some cellular event. In **ligand-gated channels** (which are generally oligomeric), binding of an extracellular or intracellular small molecule forces an allosteric transition in the protein, which opens or closes the channel. In **voltage-gated ion channels**, a change in transmembrane electrical potential (V_m) causes a charged protein domain to move relative to the membrane, opening or closing the ion channel. Both types of gating can be very fast. A channel typically opens in a fraction of a millisecond and may remain open for only milliseconds, making these molecular devices effective for very fast signal transmission in the nervous system.

Ion-Channel Function Is Measured Electrically

Because a single ion channel typically remains open for only a few milliseconds, monitoring this process is be-



Erwin Neher



Bert Sakmann

yond the limit of most biochemical measurements. Ion fluxes must therefore be measured electrically, either as changes in V_m (in the millivolt range) or as electric currents I (in the microampere or picoampere range), using microelectrodes and appropriate amplifiers. In patch-clamping, a technique developed by Erwin Neher and Bert Sakmann in 1976, very small currents are measured through a tiny region of the membrane surface containing only one or a few ion-channel molecules (Fig. 11–47). The researcher can measure the size and duration of the current that flows during one opening of an ion channel and can determine how often a channel opens and how that frequency is affected by transmembrane potential, regulatory ligands, toxins, and other agents. Patch-clamp studies have revealed that as many as 10^4 ions can move through a single ion channel in 1 ms. Such an ion flux represents a huge ampli-

fication of the initial signal; for example, only two acetylcholine molecules are needed to open an acetylcholine receptor channel (as described below).

The Structure of a K^+ Channel Reveals the Basis for Its Specificity

The structure of a potassium channel from the bacterium *Streptomyces lividans*, determined crystallographically by Roderick MacKinnon in 1998, provides much insight into the way ion channels work. This bacterial ion channel is related in sequence to all other known K^+ channels and serves as the prototype for such channels, including the voltage-gated K^+ channel of neurons. Among the members of this protein family, the similarities in sequence are greatest in the “pore region,” which contains the ion selectivity filter that allows K^+ (radius 1.33 Å) to pass 10,000 times more readily than Na^+ (radius 0.95 Å)—at a rate (about 10^8 ions/s) approaching the theoretical limit for unrestricted diffusion.



Roderick MacKinnon

The K^+ channel consists of four identical subunits that span the membrane and form a cone within a cone surrounding the ion channel, with the wide end of the double cone facing the extracellular space (Fig. 11–48).

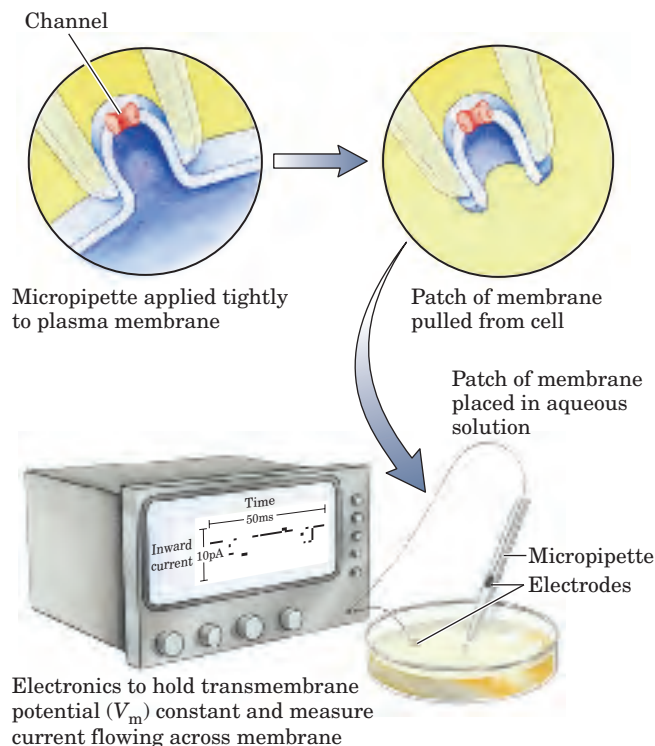
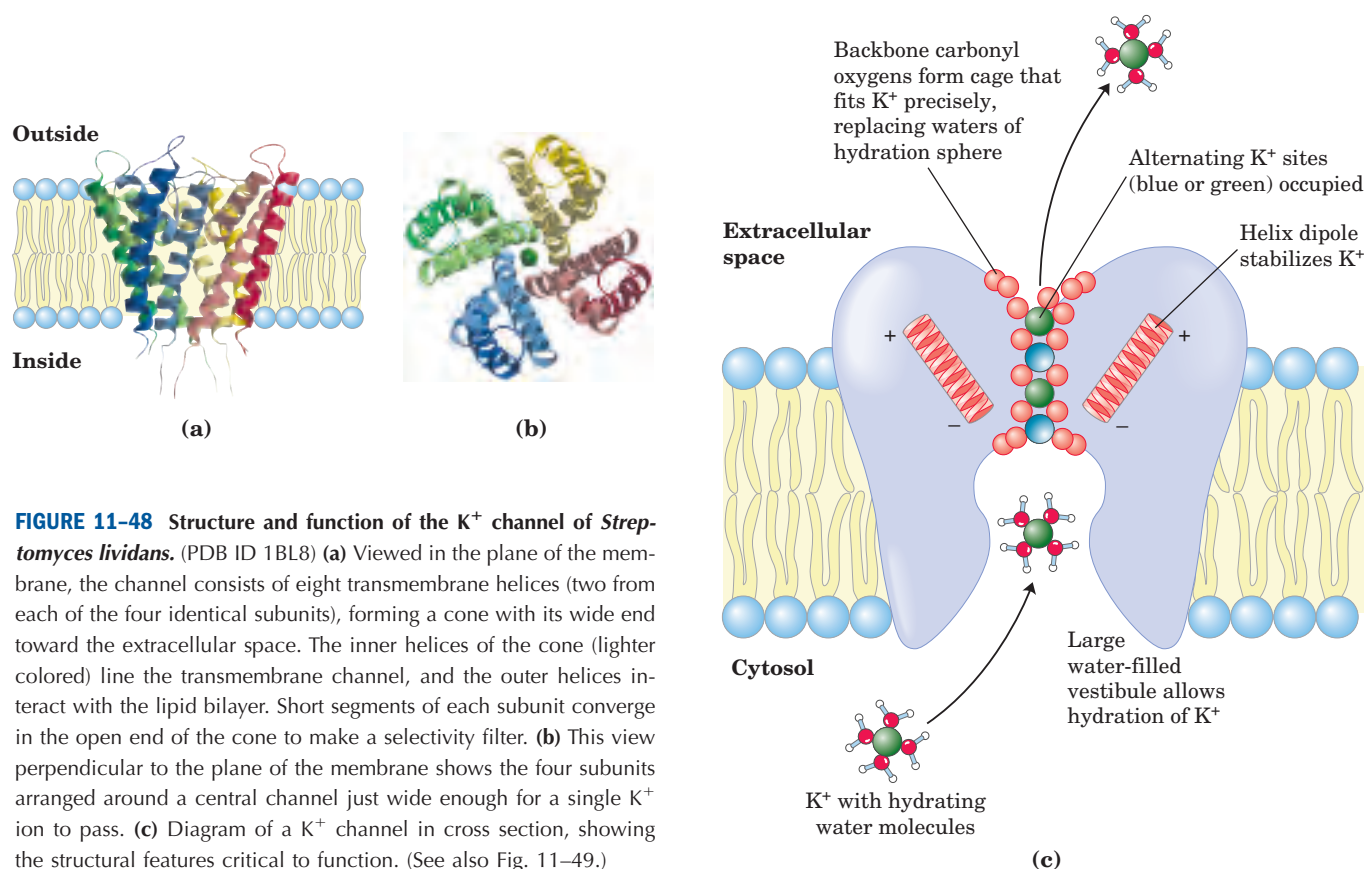


FIGURE 11–47 Electrical measurements of ion-channel function. The “activity” of an ion channel is estimated by measuring the flow of ions through it, using the patch-clamp technique. A finely drawn-out pipette (micropipette) is pressed against the cell surface, and negative pressure in the pipette forms a pressure seal between pipette and membrane. As the pipette is pulled away from the cell, it pulls off a tiny patch of membrane (which may contain one or a few ion channels). After placing the pipette and attached patch in an aqueous solution, the researcher can measure channel activity as the electric current that flows between the contents of the pipette and the aqueous solution. In practice, a circuit is set up that “clamps” the transmembrane potential at a given value and measures the current that must flow to maintain this voltage. With highly sensitive current detectors, researchers can measure the current flowing through a single ion channel, typically a few picoamperes. The trace showing the current as a function of time (in milliseconds) reveals how fast the channel opens and closes, how frequently it opens, and how long it stays open. Clamping the V_m at different values permits determination of the effect of membrane potential on these parameters of channel function.



Each subunit has two transmembrane α helices as well as a third, shorter helix that contributes to the pore region. The outer cone is formed by one of the transmembrane helices of each subunit. The inner cone, formed by the other four transmembrane helices, surrounds the ion channel and cradles the ion selectivity filter.

Both the ion specificity and the high flux through the channel are understandable from what we know of the channel's structure. At the inner and outer plasma membrane surfaces, the entryways to the channel have several negatively charged amino acid residues, which presumably increase the local concentration of cations such as K^+ and Na^+ . The ion path through the membrane begins (on the inner surface) as a wide, water-filled channel in which the ion can retain its hydration sphere. Further stabilization is provided by the short α helices in the pore region of each subunit, with the partial negative charges of their electric dipoles pointed at K^+ in the channel. About two-thirds of the way through the membrane, this channel narrows in the region of the selectivity filter, forcing the ion to give up its hydrating water molecules. Carbonyl oxygen atoms in the backbone of the selectivity filter replace the water molecules in the hydration sphere, forming a series of perfect coordination shells through which the K^+ moves. This favorable interaction with the filter is not possible for Na^+ , which is too small to make contact with all the poten-

tial oxygen ligands. The preferential stabilization of K^+ is the basis for the ion selectivity of the filter, and mutations that change residues in this part of the protein eliminate the channel's ion selectivity.

There are four potential K^+ -binding sites along the selectivity filter, each composed of an oxygen "cage" that provides ligands for the K^+ ions (Fig. 11–49). In the crystal structure, two K^+ ions are visible within the selectivity filter, about 7.5 Å apart, and two water molecules occupy the unfilled positions. K^+ ions pass through the filter in single file; their mutual electrostatic repulsion most likely just balances the interaction of each ion with the selectivity filter and keeps them moving. Movement of the two K^+ ions is concerted: first they occupy positions 1 and 3, then they hop to positions 2 and 4 (Fig. 11–48c). The energetic difference between these two configurations (1, 3 and 2, 4) is very small; energetically, the selectivity pore is not a series of hills and valleys but a flat surface, which is ideal for rapid ion movement through the channel. The structure of the channel appears to have been optimized during evolution to give maximal flow rates and high specificity.

The Neuronal Na^+ Channel Is a Voltage-Gated Ion Channel

Sodium ion channels in the plasma membranes of neurons and of myocytes of heart and skeletal muscle sense

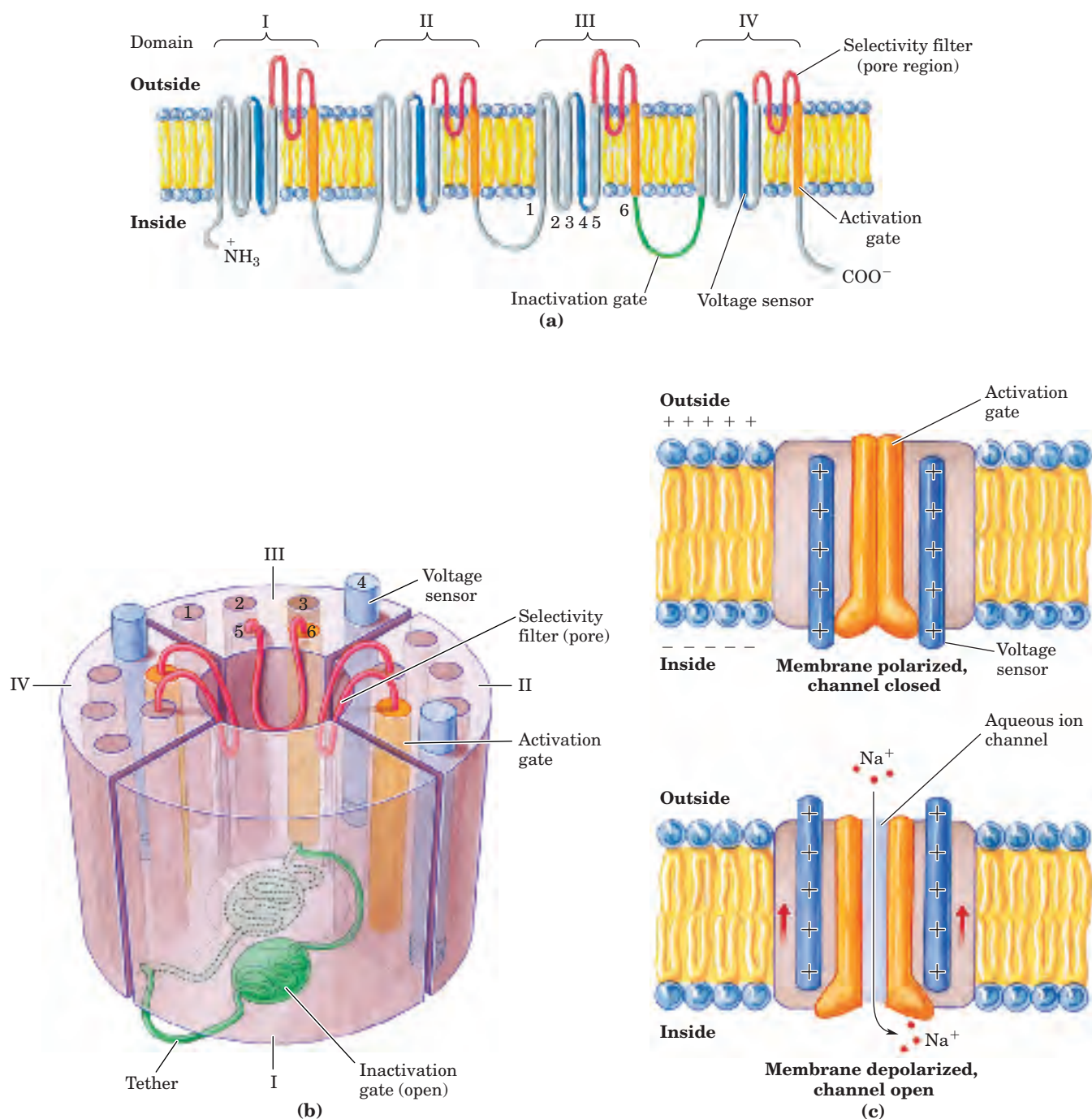
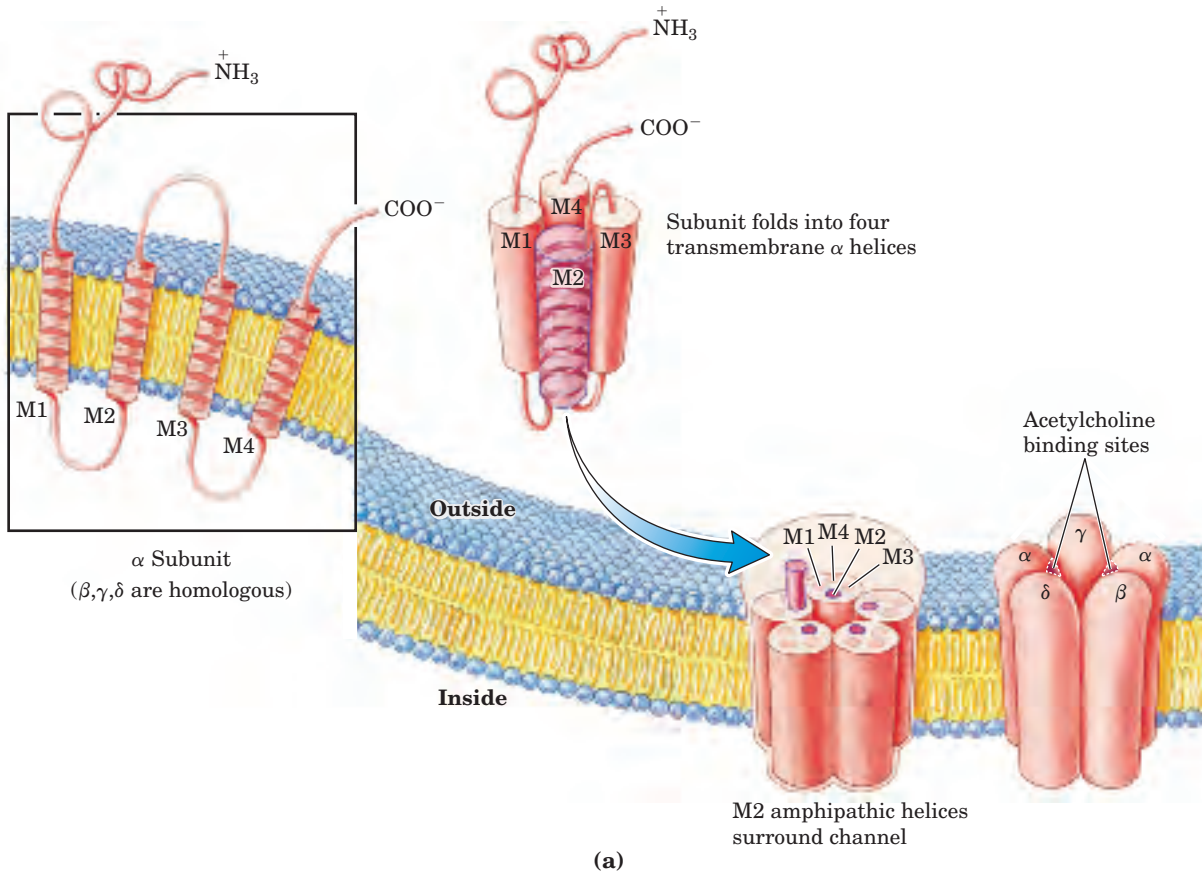


FIGURE 11-50 Voltage-gated Na^+ channels of neurons. Sodium channels of different tissues and organisms have a variety of subunits, but only the principal subunit (α) is essential. **(a)** The α subunit is a large protein with four homologous domains (I to IV), each containing six transmembrane helices (1 to 6). Helix 4 in each domain (blue) is the voltage sensor; helix 6 (orange) is thought to be the activation gate. The segments between helices 5 and 6, the pore region (red), form the selectivity filter, and the segment connecting domains III and IV (green) is the inactivation gate. **(b)** The four domains are wrapped about a central transmembrane channel lined with polar amino acid residues. The segments linking helices 5 and 6 (red) in each domain come together near the extracellular surface to form the selectivity filter, which is conserved in all Na^+ channels.

The filter gives the channel its ability to discriminate between Na^+ and other ions of similar size. The inactivation gate (green) closes (dotted lines) soon after the activation gate opens. **(c)** The voltage-sensing mechanism involves movement of helix 4 (blue) perpendicular to the plane of the membrane in response to a change in transmembrane potential. As shown at the top, the strong positive charge on helix 4 allows it to be pulled inward in response to the inside-negative membrane potential (V_m). Depolarization lessens this pull, and helix 4 relaxes by moving outward (bottom). This movement is communicated to the activation gate (orange), inducing conformational changes that open the channel in response to depolarization.

The nicotinic acetylcholine receptor has five subunits: single copies of subunits β , γ , and δ , and two identical α subunits each with an acetylcholine-binding site. All five subunits are related in sequence and tertiary structure, each having four transmembrane helical

segments (M1 to M4) (Fig. 11–51a). The five subunits surround a central pore, which is lined with their M2 helices. The pore is about 20 Å wide in the parts of the channel that protrude on the cytoplasmic and extracellular surfaces, but narrows as it passes through the



Bulky hydrophobic Leu side chains of M2 helices close the channel.

Binding of two acetylcholine molecules causes twisting of the M2 helices.

M2 helices now have smaller, polar residues lining the channel.

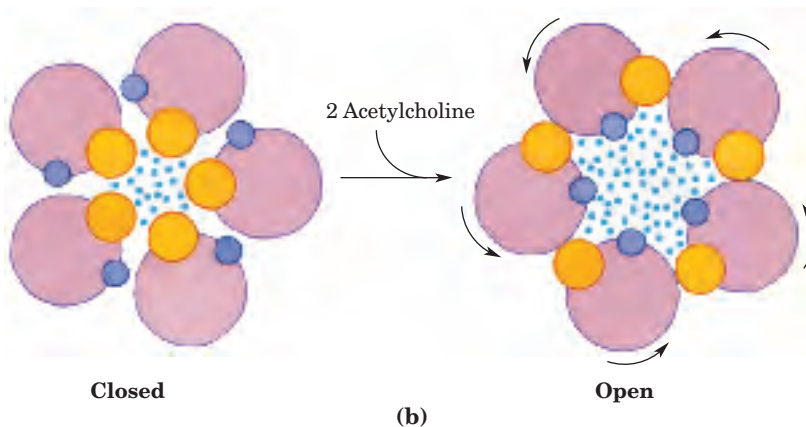


FIGURE 11-51 Structure of the acetylcholine receptor ion channel. (a) Each of the five subunits ($\alpha, \beta, \gamma, \delta$) has four transmembrane helices, M1 to M4. The M2 helices are amphipathic; the others have mainly hydrophobic residues. The five subunits are arranged around a central transmembrane channel, which is lined with the polar sides of the M2 helices. At the top and bottom of the channel are rings of negatively charged amino acid residues. (b) This top view of a cross section through the center of the M2 helices shows five Leu side chains (one from each M2 helix) protruding into the channel, constricting it to a diameter too small to allow passage of ions such as Ca^{2+} , Na^+ , and K^+ . When both acetylcholine receptor sites (one on each α subunit) are occupied, a conformational change occurs. As the M2 helices twist slightly, the five Leu residues (yellow) rotate away from the channel and are replaced by smaller, polar residues (blue). This gating mechanism opens the channel, allowing the passage of Ca^{2+} , Na^+ , or K^+ .

lipid bilayer. Near the center of the bilayer is a ring of bulky hydrophobic side chains of Leu residues in the M2 helices, positioned so close together that they prevent

ions from passing through the channel. Allosteric conformational changes induced by acetylcholine binding to the two α subunits include a slight twisting of the M2

TABLE 11–7 Transport Systems Described Elsewhere in This Text

<i>Transport system and location</i>	<i>Figure number</i>	<i>Role</i>
Adenine nucleotide antiporter of mitochondrial inner membrane	19–26	Imports substrate ADP for oxidative phosphorylation, and exports product ATP
Acyl-carnitine/carnitine transporter of mitochondrial inner membrane	17–6	Imports fatty acids into matrix for β oxidation
P_i - H^+ symporter of mitochondrial inner membrane	19–26	Supplies P_i for oxidative phosphorylation
Malate- α -ketoglutarate transporter of mitochondrial inner membrane	19–27	Shuttles reducing equivalents (as malate) from matrix to cytosol
Glutamate-aspartate transporter of mitochondrial inner membrane	19–27	Completes shuttling begun by malate- α -ketoglutarate shuttle
Citrate transporter of mitochondrial inner membrane	21–10	Provides cytosolic citrate as source of acetyl-CoA for lipid synthesis
Pyruvate transporter of mitochondrial inner membrane	21–10	Is part of mechanism for shuttling citrate from matrix to cytosol
Fatty acid transporter of myocyte plasma membrane	17–3	Imports fatty acids for fuel
Complex I, III, and IV proton transporters of mitochondrial inner membrane	19–15	Acts as energy-conserving mechanism in oxidative phosphorylation, converting electron flow into proton gradient
Thermogenin (uncoupler protein), a proton pore of mitochondrial inner membrane	19–30, 23–22	Allows dissipation of proton gradient in mitochondria as means of thermogenesis and/or disposal of excess fuel
Cytochrome <i>bf</i> complex, a proton transporter of chloroplast thylakoid	19–50, 19–54	Acts as proton pump, driven by electron flow through the Z scheme; source of proton gradient for photosynthetic ATP synthesis
Bacteriorhodopsin, a light-driven proton pump	19–59	Is light-driven source of proton gradient for ATP synthesis in halophilic bacterium
F_0F_1 ATPase/ATP synthase of mitochondrial inner membrane, chloroplast thylakoid, and bacterial plasma membrane	19–58	Interconverts energy of proton gradient and ATP during oxidative phosphorylation and photophosphorylation
P_i -triose phosphate antiporter of chloroplast inner membrane	20–15, 20–16	Exports photosynthetic product from stroma; imports P_i for ATP synthesis
Bacterial protein transporter	27–39	Exports secreted proteins through plasma membrane
Protein translocase of ER	27–33	Transports into ER proteins destined for plasma membrane, secretion, or organelles
Nuclear pore protein translocase	27–37	Shuttles proteins between nucleus and cytoplasm
LDL receptor in animal cell plasma membrane	21–42	Imports, by receptor-mediated endocytosis, lipid carrying particles
Glucose transporter of animal cell plasma membrane; regulated by insulin	12–8	Increases capacity of muscle and adipose tissue to take up excess glucose from blood
IP_3 -gated Ca^{2+} channel of endoplasmic reticulum	12–19	Allows signaling via changes of cytosolic Ca^{2+} concentration
cGMP-gated Ca^{2+} channel of retinal rod and cone cells	12–32	Allows signaling via rhodopsin linked to cAMP phosphodiesterase in vertebrate eye
Voltage-gated Na^+ channel of neuron	12–5	Creates action potentials in neuronal signal transmission

helices (Fig. 11–51b), which draws these hydrophobic side chains away from the center of the channel, opening it to the passage of ions.

Based on similarities between the amino acid sequences of other ligand-gated ion channels and the acetylcholine receptor, the receptor channels that respond to the extracellular signals γ -aminobutyric acid (GABA), glycine, and serotonin have been classified in the acetylcholine receptor superfamily, and probably share three-dimensional structure and gating mechanisms. The GABA_A and glycine receptors are anion channels specific for Cl^- or HCO_3^- , whereas the serotonin receptor, like the acetylcholine receptor, is cation-specific. The subunits of each of these channels, like those of the acetylcholine receptor, have four transmembrane helical segments and form oligomeric channels.

A second class of ligand-gated ion channels respond to *intracellular* ligands: 3',5'-cyclic guanosine mononucleotide (cGMP) in the vertebrate eye, cGMP and cAMP in olfactory neurons, and ATP and inositol 1,4,5-trisphosphate (IP₃) in many cell types. These channels are composed of multiple subunits, each with six transmembrane helical domains. We discuss the signaling functions of these ion channels in Chapter 12.

Table 11–7 shows a number of transporters not discussed in this chapter but encountered later in the book in the context of the paths in which they act.

Defective Ion Channels Can Have Adverse Physiological Consequences



The importance of ion channels to physiological processes is clear from the effects of mutations in specific ion-channel proteins (Table 11–8). Genetic defects in the voltage-gated Na^+ channel of the myocyte plasma membrane result in diseases in which muscles are periodically either paralyzed (as in hyperkalemic pe-

riodic paralysis) or stiff (as in paramyotonia congenita). As noted earlier, cystic fibrosis is the result of a mutation that changes one amino acid in the protein CFTR, a Cl^- ion channel; the defective process here is not neurotransmission but secretion by various exocrine gland cells whose activities are tied to Cl^- ion fluxes.

Many naturally occurring toxins act on ion channels, and the potency of these toxins further illustrates the importance of normal ion-channel function. Tetrodotoxin (produced by the puffer fish, *Sphaeroides rubripes*) and saxitoxin (produced by the marine dinoflagellate *Gonyaulax*, which causes “red tides”) act by binding to the voltage-gated Na^+ channels of neurons and preventing normal action potentials. Puffer fish is an ingredient of the Japanese delicacy fugu, which may be prepared only by chefs specially trained to separate

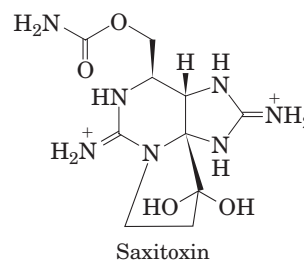
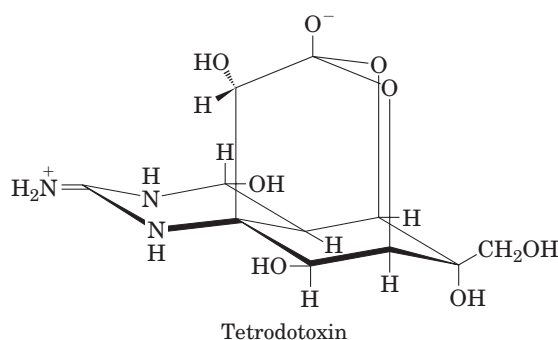
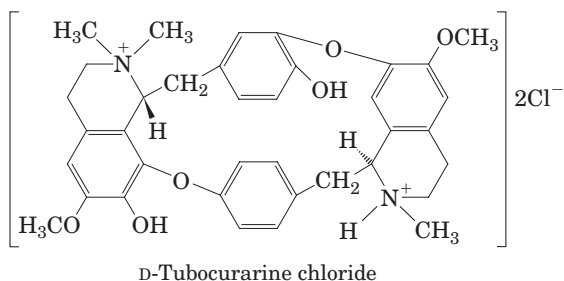


TABLE 11–8 Some Diseases Resulting from Ion Channel Defects

<i>Ion channel</i>	<i>Affected gene</i>	<i>Disease</i>
Na^+ (voltage-gated, skeletal muscle)	SCN4A	Hyperkalemic periodic paralysis (or paramyotonia congenita)
Na^+ (voltage-gated, neuronal)	SCN1A	Generalized epilepsy with febrile seizures
Na^+ (voltage-gated, cardiac muscle)	SCN5A	Long QT syndrome 3
Ca^{2+} (neuronal)	CACNA1A	Familial hemiplegic migraine
Ca^{2+} (voltage-gated, retinal)	CACNA1F	Congenital stationary night blindness
Ca^{2+} (polycystin-1)	PKD1	Polycystic kidney disease
K^+ (neuronal)	KCNQ4	Dominant deafness
K^+ (voltage-gated, neuronal)	KCNQ2	Benign familial neonatal convulsions
Nonspecific cation (cGMP-gated, retinal)	CNCG1	Retinitis pigmentosa
Acetylcholine receptor (skeletal muscle)	CHRNA1	Congenital myasthenic syndrome
Cl^-	CFTR	Cystic fibrosis



succulent morsel from deadly poison. Eating shellfish that have fed on *Gonyaulax* can also be fatal; shellfish are not sensitive to saxitoxin, but they concentrate it in their muscles, which become highly poisonous to organisms higher up the food chain. The venom of the black mamba snake contains dendrotoxin, which interferes with voltage-gated K^+ channels. Tubocurarine, the active component of curare (used as an arrow poison in the Amazon), and two other toxins from snake venoms, cobrotoxin and bungarotoxin, block the acetylcholine receptor or prevent the opening of its ion channel. By blocking signals from nerves to muscles, all these toxins cause paralysis and possibly death. On the positive side, the extremely high affinity of bungarotoxin for the acetylcholine receptor ($K_d = 10^{-15}$ M) has proved useful experimentally: the radiolabeled toxin was used to quantify the receptor during its purification. ■

SUMMARY 11.3 Solute Transport across Membranes

- Movement of polar compounds and ions across biological membranes requires protein transporters. Some transporters simply facilitate passive diffusion across the membrane from the side with higher concentration to the side with lower. Others bring about active movement of solutes against an electrochemical gradient; such transport must be coupled to a source of metabolic energy.
- Carriers, like enzymes, show saturation and stereospecificity for their substrates. Transport via these systems may be passive or active. Primary active transport is driven by ATP or electron-transfer reactions; secondary active transport, by coupled flow of two solutes, one of which (often H^+ or Na^+) flows down its electrochemical gradient as the other is pulled up its gradient.
- The GLUT transporters, such as GLUT1 of erythrocytes, carry glucose into cells by facilitated diffusion. These transporters are uniporters, carrying only one substrate. Symporters permit simultaneous passage of two

substances in the same direction; examples are the lactose transporter of *E. coli*, driven by the energy of a proton gradient (lactose- H^+ symport), and the glucose transporter of intestinal epithelial cells, driven by a Na^+ gradient (glucose- Na^+ symport). Antiporters mediate simultaneous passage of two substances in opposite directions; examples are the chloride-bicarbonate exchanger of erythrocytes and the ubiquitous Na^+K^+ ATPase.

- In animal cells, Na^+K^+ ATPase maintains the differences in cytosolic and extracellular concentrations of Na^+ and K^+ , and the resulting Na^+ gradient is used as the energy source for a variety of secondary active transport processes.
- The Na^+K^+ ATPase of the plasma membrane and the Ca^{2+} transporters of the sarcoplasmic and endoplasmic reticulum (the SERCA pumps) are examples of P-type ATPases; they undergo reversible phosphorylation during their catalytic cycle and are inhibited by the phosphate analog vanadate. F-type ATPase proton pumps (ATP synthases) are central to energy-conserving mechanisms in mitochondria and chloroplasts. V-type ATPases produce gradients of protons across some intracellular membranes, including plant vacuolar membranes.
- ABC transporters carry a variety of substrates, including many drugs, out of cells, using ATP as energy source.
- Ionophores are lipid-soluble molecules that bind specific ions and carry them passively across membranes, dissipating the energy of electrochemical ion gradients.
- Water moves across membranes through aquaporins.
- Ion channels provide hydrophilic pores through which select ions can diffuse, moving down their electrical or chemical concentration gradients; they are characteristically unsaturable and have very high flux rates. Many ion channels are highly specific for one ion, and most are gated by either voltage or a ligand. In bacterial K^+ channels, a selectivity filter provides ligands with the right geometry to replace the water of hydration of a K^+ ion as it crosses the membrane. Some K^+ channels are voltage gated. The acetylcholine receptor/channel is gated by acetylcholine, which triggers subtle conformational changes that open and close the transmembrane path.

Key Terms

Terms in bold are defined in the glossary.

fluid mosaic model	FRAP 382	facilitated diffusion	P-type ATPases 398
371	microdomains 383	391	SERCA pump 400
micelle 372	rafts 384	passive transport 391	F-type ATPases 401
bilayer 373	caveolin 385	transporters 391	ATP synthase 401
integral proteins 373	caveolae 385	carriers 392	V-type ATPases 401
peripheral proteins	fusion proteins 387	channels 392	ABC transporters 402
373	SNAREs 389	electroneutral 395	multidrug transporter 402
hydropathy index 377	simple diffusion 389	cotransport systems	ionophores 406
β barrel 378	membrane potential	395	aquaporins (AQPs) 406
gel phase 380	(V_m) 389	antiport 397	ion channel 408
liquid-disordered state	electrochemical	symport 397	
380	gradient 391	uniport 397	
liquid-ordered state 380	electrochemical	active transport 397	
flippases 382	potential 391	electrogenic 398	

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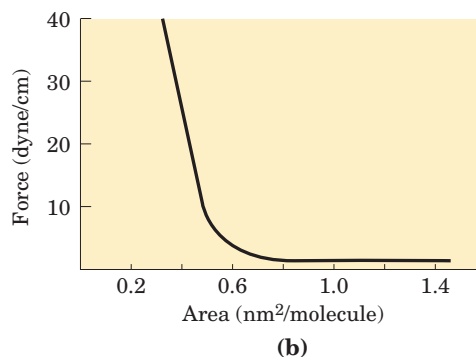
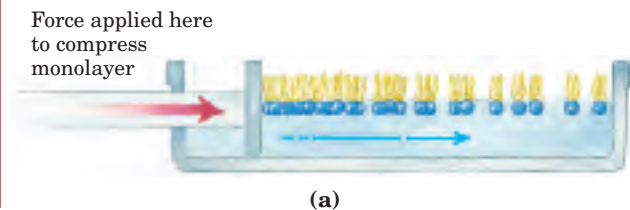
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Problems

1. Determining the Cross-Sectional Area of a Lipid Molecule When phospholipids are layered gently onto the surface of water, they orient at the air–water interface with their head groups in the water and their hydrophobic tails in the air. An experimental apparatus (a) has been devised that reduces the surface area available to a layer of lipids. By measuring the force necessary to push the lipids together, it is possible to determine when the molecules are packed tightly in a continuous monolayer; as that area is approached, the force needed to further reduce the surface area increases sharply (b). How would you use this apparatus to determine the average area occupied by a single lipid molecule in the monolayer?



2. Evidence for a Lipid Bilayer In 1925, E. Gorter and F. Grendel used an apparatus like that described in Problem 1 to determine the surface area of a lipid monolayer formed by lipids extracted from erythrocytes of several animal species. They used a microscope to measure the dimensions of individual cells, from which they calculated the average surface area of one erythrocyte. They obtained the data shown in the

table. Were these investigators justified in concluding that “chromocytes [erythrocytes] are covered by a layer of fatty substances that is two molecules thick” (i.e., a lipid bilayer)?

Animal	Volume of packed cells (mL)	Number of cells (per mm ³)	Total surface area of lipid monolayer from cells (m ²)	Total surface area of one cell (μm ²)
Dog	40	8,000,000	62	98
Sheep	10	9,900,000	6.0	29.8
Human	1	4,740,000	0.92	99.4

Source: Data from Gorter, E. & Grendel, F. (1925) On bimolecular layers of lipoids on the chromocytes of the blood. *J. Exp. Med.* **41**, 439–443.

3. Number of Detergent Molecules per Micelle When a small amount of sodium dodecyl sulfate (SDS; $\text{Na}^+\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-$) is dissolved in water, the detergent ions enter the solution as monomeric species. As more detergent is added, a concentration is reached (the critical micelle concentration) at which the monomers associate to form micelles. The critical micelle concentration of SDS is 8.2 mM. The micelles have an average particle weight (the sum of the molecular weights of the constituent monomers) of 18,000. Calculate the number of detergent molecules in the average micelle.

4. Properties of Lipids and Lipid Bilayers Lipid bilayers formed between two aqueous phases have this important property: they form two-dimensional sheets, the edges of which close upon each other and undergo self-sealing to form liposomes.

(a) What properties of lipids are responsible for this property of bilayers? Explain.

(b) What are the consequences of this property for the structure of biological membranes?

5. Length of a Fatty Acid Molecule The carbon–carbon bond distance for single-bonded carbons such as those in a saturated fatty acyl chain is about 1.5 Å. Estimate the length of a single molecule of palmitate in its fully extended form. If two molecules of palmitate were placed end to end, how would their total length compare with the thickness of the lipid bilayer in a biological membrane?

6. Temperature Dependence of Lateral Diffusion The experiment described in Figure 11–17 was performed at 37 °C. If the experiment were carried out at 10 °C, what effect would you expect on the rate of diffusion? Why?

7. Synthesis of Gastric Juice: Energetics Gastric juice (pH 1.5) is produced by pumping HCl from blood plasma (pH 7.4) into the stomach. Calculate the amount of free energy required to concentrate the H^+ in 1 L of gastric juice at 37 °C. Under cellular conditions, how many moles of ATP must be hydrolyzed to provide this amount of free energy? The free-energy change for ATP hydrolysis under cellular conditions is about –58 kJ/mol (as explained in Chapter 13). Ignore the effects of the transmembrane electrical potential.

8. Energetics of the Na^+K^+ ATPase For a typical vertebrate cell with a transmembrane potential of –0.070 V (in-

side negative), what is the free-energy change for transporting 1 mol of Na^+ out of the cell and into the blood at 37 °C? Assume the concentration of Na^+ inside the cell is 12 mM, and that in blood plasma is 145 mM.

9. Action of Ouabain on Kidney Tissue Ouabain specifically inhibits the Na^+K^+ ATPase activity of animal tissues but is not known to inhibit any other enzyme. When ouabain is added to thin slices of living kidney tissue, it inhibits oxygen consumption by 66%. Why? What does this observation tell us about the use of respiratory energy by kidney tissue?

10. Energetics of Symport Suppose that you determined experimentally that a cellular transport system for glucose, driven by symport of Na^+ , could accumulate glucose to concentrations 25 times greater than in the external medium, while the external $[\text{Na}^+]$ was only 10 times greater than the intracellular $[\text{Na}^+]$. Would this violate the laws of thermodynamics? If not, how could you explain this observation?

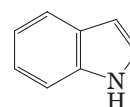
11. Location of a Membrane Protein The following observations are made on an unknown membrane protein, X. It can be extracted from disrupted erythrocyte membranes into a concentrated salt solution, and it can be cleaved into fragments by proteolytic enzymes. Treatment of erythrocytes with proteolytic enzymes followed by disruption and extraction of membrane components yields intact X. However, treatment of erythrocyte “ghosts” (which consist of just plasma membranes, produced by disrupting the cells and washing out the hemoglobin) with proteolytic enzymes followed by disruption and extraction yields extensively fragmented X. What do these observations indicate about the location of X in the plasma membrane? Do the properties of X resemble those of an integral or peripheral membrane protein?

12. Membrane Self-sealing Cellular membranes are self-sealing—if they are punctured or disrupted mechanically, they quickly and automatically reseal. What properties of membranes are responsible for this important feature?

13. Lipid Melting Temperatures Membrane lipids in tissue samples obtained from different parts of the leg of a reindeer have different fatty acid compositions. Membrane lipids from tissue near the hooves contain a larger proportion of unsaturated fatty acids than those from tissue in the upper leg. What is the significance of this observation?

14. Flip-Flop Diffusion The inner leaflet (monolayer) of the human erythrocyte membrane consists predominantly of phosphatidylethanolamine and phosphatidylserine. The outer leaflet consists predominantly of phosphatidylcholine and sphingomyelin. Although the phospholipid components of the membrane can diffuse in the fluid bilayer, this sidedness is preserved at all times. How?

15. Membrane Permeability At pH 7, tryptophan crosses a lipid bilayer at about one-thousandth the rate of the closely related substance indole:



Suggest an explanation for this observation.

16. Water Flow through an Aquaporin Each human erythrocyte has about 2×10^5 AQP-1 monomers. If water molecules flow through the plasma membrane at a rate of 5×10^8 per AQP-1 tetramer per second, and the volume of an erythrocyte is 5×10^{-11} mL, how rapidly could an erythrocyte halve its volume as it encounters the high osmolarity (1 M) in the interstitial fluid of the renal medulla? Assume that the erythrocyte consists entirely of water.

17. Labeling the Lactose Transporter A bacterial lactose transporter, which is highly specific for its substrate lactose, contains a Cys residue that is essential to its transport activity. Covalent reaction of *N*-ethylmaleimide (NEM) with this Cys residue irreversibly inactivates the transporter. A high concentration of lactose in the medium prevents inactivation by NEM, presumably by sterically protecting the Cys residue, which is in or near the lactose-binding site. You know nothing else about the transporter protein. Suggest an experiment that might allow you to determine the M_r of the Cys-containing transporter polypeptide.

18. Predicting Membrane Protein Topology from Sequence You have cloned the gene for a human erythrocyte protein, which you suspect is a membrane protein. From the nucleotide sequence of the gene, you know the amino acid sequence. From this sequence alone, how would you evaluate the possibility that the protein is an integral protein? Suppose the protein proves to be an integral protein, either type I or type II. Suggest biochemical or chemical experiments that might allow you to determine which type it is.

19. Intestinal Uptake of Leucine You are studying the uptake of L-leucine by epithelial cells of the mouse intestine. Measurements of the rate of uptake of L-leucine and several of its analogs, with and without Na^+ in the assay buffer, yield the results given in the table. What can you conclude about the properties and mechanism of the leucine transporter? Would you expect L-leucine uptake to be inhibited by ouabain?

Substrate	Uptake in presence of Na^+		Uptake in absence of Na^+	
	$V_{\max}$	K_t (mM)	$V_{\max}$	K_t (mM)
L-Leucine	420	0.24	23	0.24
D-Leucine	310	4.7	5	4.7
L-Valine	225	0.31	19	0.31

20. Effect of an Ionophore on Active Transport Consider the leucine transporter described in Problem 19. Would $V_{\max}$ and/or K_t change if you added a Na^+ ionophore to the assay solution containing Na^+ ? Explain.

21. Surface Density of a Membrane Protein *E. coli* can be induced to make about 10,000 copies of the lactose transporter (M_r 31,000) per cell. Assume that *E. coli* is a cylinder 1 μm in diameter and 2 μm long. What fraction of the plasma membrane surface is occupied by the lactose transporter molecules? Explain how you arrived at this conclusion.

Biochemistry on the Internet

22. Membrane Protein Topology The receptor for the hormone epinephrine in animal cells is an integral membrane protein (M_r 64,000) that is believed to have seven membrane-spanning regions.

(a) Show that a protein of this size is capable of spanning the membrane seven times.

(b) Given the amino acid sequence of this protein, how would you predict which regions of the protein form the membrane-spanning helices?

(c) Go to the Protein Data Bank (www.rcsb.org/pdb). Use the PDB identifier 1DEP to retrieve the data page for a portion of the β -adrenergic receptor (one type of epinephrine receptor) from a turkey. Using Chime to explore the structure, predict where this portion of the receptor is located: within the membrane or at the membrane surface. Explain.

(d) Retrieve the data for a portion of another receptor, the acetylcholine receptor of neurons and myocytes, using the PDB identifier 1A11. As in (c), predict where this portion of the receptor is located and explain your answer.

If you have not used the PDB or Chemscape Chime, you can find instructions at www.whfreeman.com/lehninger.



BIOSIGNALING

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When I first entered the study of hormone action, some 25 years ago, there was a widespread feeling among biologists that hormone action could not be studied meaningfully in the absence of organized cell structure. However, as I reflected on the history of biochemistry, it seemed to me there was a real possibility that hormones might act at the molecular level.

—Earl W. Sutherland, Nobel Address, 1971

The ability of cells to receive and act on signals from beyond the plasma membrane is fundamental to life. Bacterial cells receive constant input from membrane proteins that act as information receptors, sampling the surrounding medium for pH, osmotic strength, the availability of food, oxygen, and light, and the presence of noxious chemicals, predators, or competitors for food.

These signals elicit appropriate responses, such as motion toward food or away from toxic substances or the formation of dormant spores in a nutrient-depleted medium. In multicellular organisms, cells with different functions exchange a wide variety of signals. Plant cells respond to growth hormones and to variations in sunlight. Animal cells exchange information about the concentrations of ions and glucose in extracellular fluids, the interdependent metabolic activities taking place in different tissues, and, in an embryo, the correct placement of cells during development. In all these cases, the signal represents *information* that is detected by specific receptors and converted to a cellular response, which always involves a *chemical* process. This conversion of information into a chemical change, **signal transduction**, is a universal property of living cells.

The number of different biological signals is large (Table 12–1), as is the variety of biological responses to these signals, but organisms use just a few evolutionarily conserved mechanisms to detect extracellular signals and *transduce* them into intracellular changes. In this chapter we examine some examples of the major classes of signaling mechanisms, looking at how they are integrated in specific biological functions such as the transmission of nerve signals; responses to hormones and growth factors; the senses of sight, smell, and taste; and

TABLE 12–1
Some Signals to Which Cells Respond

Antigens	Light
Cell surface glycoproteins/ oligosaccharides	Mechanical touch
Developmental signals	Neurotransmitters
Extracellular matrix components	Nutrients
Growth factors	Odorants
Hormones	Pheromones
	Tastants

control of the cell cycle. Often, the end result of a signaling pathway is the phosphorylation of a few specific target-cell proteins, which changes their activities and thus the activities of the cell. Throughout our discussion we emphasize the conservation of fundamental mechanisms for the transduction of biological signals and the adaptation of these basic mechanisms to a wide range of signaling pathways.

12.1 Molecular Mechanisms of Signal Transduction

Signal transductions are remarkably specific and exquisitely sensitive. **Specificity** is achieved by precise molecular complementarity between the signal and receptor molecules (Fig. 12–1a), mediated by the same kinds of weak (noncovalent) forces that mediate enzyme-substrate and antigen-antibody interactions. Multicellular organisms have an additional level of specificity, because the receptors for a given signal, or the intracellular targets of a given signal pathway, are present only in certain cell types. Thyrotropin-releasing hormone, for example, triggers responses in the cells of the anterior pituitary but not in hepatocytes, which lack receptors for this hormone. Epinephrine alters glycogen metabolism in hepatocytes but not in erythrocytes; in this case, both cell types have receptors for the hormone, but whereas hepatocytes contain glycogen and the glycogen-metabolizing enzyme that is stimulated by epinephrine, erythrocytes contain neither.

Three factors account for the extraordinary sensitivity of signal transducers: the high affinity of receptors for signal molecules, cooperativity (often but not

always) in the ligand-receptor interaction, and amplification of the signal by enzyme cascades. The **affinity** between signal (ligand) and receptor can be expressed as the dissociation constant K_d , usually 10^{-10} M or less—meaning that the receptor detects picomolar concentrations of a signal molecule. Receptor-ligand interactions are quantified by Scatchard analysis, which yields a quantitative measure of affinity (K_d) and the number of ligand-binding sites in a receptor sample (Box 12–1).

Cooperativity in receptor-ligand interactions results in large changes in receptor activation with small changes in ligand concentration (recall the effect of cooperativity on oxygen binding to hemoglobin; see Fig. 5–12). **Amplification by enzyme cascades** results when an enzyme associated with a signal receptor is activated and, in turn, catalyzes the activation of many molecules of a second enzyme, each of which activates many molecules of a third enzyme, and so on (Fig. 12–1b). Such cascades can produce amplifications of several orders of magnitude within milliseconds.

The sensitivity of receptor systems is subject to modification. When a signal is present continuously, **desensitization** of the receptor system results (Fig. 12–1c); when the stimulus falls below a certain threshold, the system again becomes sensitive. Think of what happens to your visual transduction system when you walk from bright sunlight into a darkened room or from darkness into the light.

A final noteworthy feature of signal-transducing systems is **integration** (Fig. 12–1d), the ability of the system to receive multiple signals and produce a unified response appropriate to the needs of the cell or organism. Different signaling pathways converse with

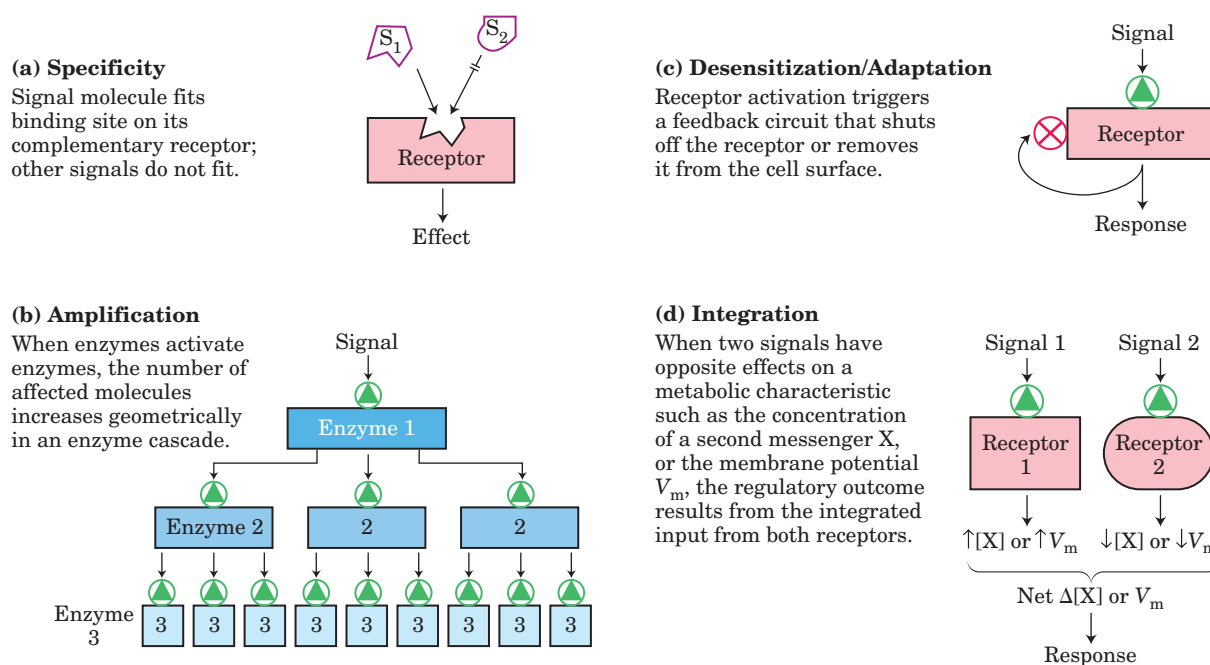
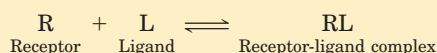


FIGURE 12–1 Four features of signal-transducing systems.

Scatchard Analysis Quantifies the Receptor-Ligand Interaction

The cellular actions of a hormone begin when the hormone (ligand, L) binds specifically and tightly to its protein receptor (R) on or in the target cell. Binding is mediated by noncovalent interactions (hydrogen-bonding, hydrophobic, and electrostatic) between the complementary surfaces of ligand and receptor. Receptor-ligand interaction brings about a conformational change that alters the biological activity of the receptor, which may be an enzyme, an enzyme regulator, an ion channel, or a regulator of gene expression.

Receptor-ligand binding is described by the equation



This binding, like that of an enzyme to its substrate, depends on the concentrations of the interacting components and can be described by an equilibrium constant:

$$\begin{array}{ccccc} \text{R} & + & \text{L} & \xrightleftharpoons[k_{-1}]{k_{+1}} & \text{RL} \\ \text{Receptor} & & \text{Ligand} & & \text{Receptor-ligand complex} \end{array}$$

$$K_a = \frac{[\text{RL}]}{[\text{R}][\text{L}]} = \frac{k_{+1}}{k_{-1}} = 1/K_d$$

where K_a is the association constant and K_d is the dissociation constant.

Like enzyme-substrate binding, receptor-ligand binding is saturable. As more ligand is added to a fixed amount of receptor, an increasing fraction of receptor molecules is occupied by ligand (Fig. 1a). A rough measure of receptor-ligand affinity is given by the concentration of ligand needed to give half-saturation of the receptor. Using **Scatchard analysis** of receptor-ligand binding, we can estimate both the dissociation constant K_d and the number of receptor-binding sites in a given preparation. When binding has reached equilibrium, the total number of possible binding sites, $B_{\max}$, equals the number of unoccupied sites, represented by [R], plus the number of occupied or ligand-bound sites, [RL]; that is, $B_{\max} = [\text{R}] + [\text{RL}]$. The number of unbound sites can be expressed in terms of total sites minus occupied sites: $[\text{R}] = B_{\max} - [\text{RL}]$. The equilibrium expression can now be written

$$K_a = \frac{[\text{RL}]}{[\text{L}](B_{\max} - [\text{RL}])}$$

Rearranging to obtain the ratio of receptor-bound ligand to free (unbound) ligand, we get

$$\begin{aligned} \frac{[\text{Bound}]}{[\text{Free}]} &= \frac{[\text{RL}]}{[\text{L}]} = K_a(B_{\max} - [\text{RL}]) \\ &= \frac{1}{K_d} (B_{\max} - [\text{RL}]) \end{aligned}$$

From this slope-intercept form of the equation, we can see that a plot of [bound ligand]/[free ligand] versus [bound ligand] should give a straight line with a slope of $-K_a$ ($-1/K_d$) and an intercept on the abscissa of $B_{\max}$, the total number of binding sites (Fig. 1b). Hormone-ligand interactions typically have K_d values of 10^{-9} to 10^{-11} M, corresponding to very tight binding.

Scatchard analysis is reliable for the simplest cases, but as with Lineweaver-Burk plots for enzymes, when the receptor is an allosteric protein, the plots deviate from linearity.

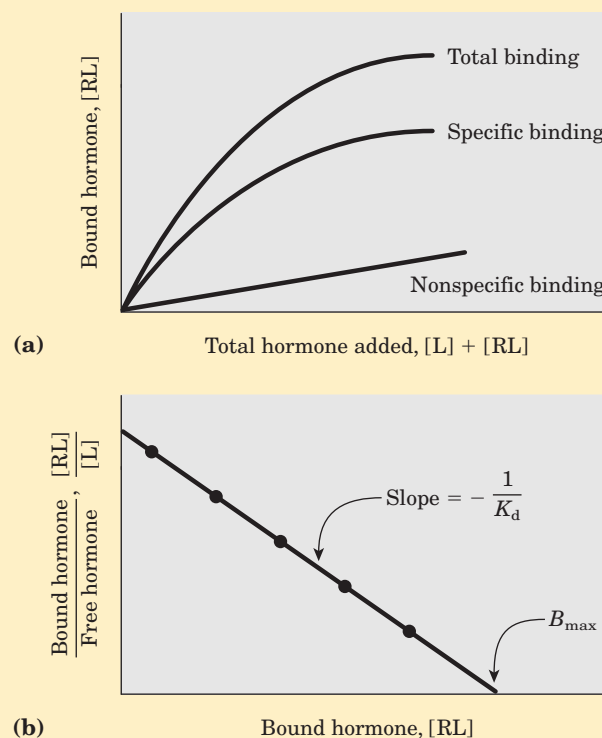


FIGURE 1 Scatchard analysis of a receptor-ligand interaction. A radiolabeled ligand (L)—a hormone, for example—is added at several concentrations to a fixed amount of receptor (R), and the fraction of the hormone bound to receptor is determined by separating the receptor-hormone complex (RL) from free hormone. (a) A plot of [RL] versus [L] + [RL] (total hormone added) is hyperbolic, rising toward a maximum for [RL] as the receptor sites become saturated. To control for nonsaturable, nonspecific binding sites (eicosanoid hormones bind nonspecifically to the lipid bilayer, for example), a separate series of binding experiments is also necessary. A large excess of unlabeled hormone is added along with the dilute solution of labeled hormone. The unlabeled molecules compete with the labeled molecules for specific binding to the saturable site on the receptor, but not for the nonspecific binding. The true value for specific binding is obtained by subtracting nonspecific binding from total binding. (b) A linear plot of [RL]/[L] versus [RL] gives K_d and $B_{\max}$ for the receptor-hormone complex. Compare these plots with those of V_0 versus [S] and $1/V_0$ versus $1/[S]$ for an enzyme-substrate complex (see Fig. 6-12, Box 6-1).

each other at several levels, generating a wealth of interactions that maintain homeostasis in the cell and the organism.

We consider here the molecular details of several representative signal-transduction systems. The trigger for each system is different, but the general features of signal transduction are common to all: a signal interacts with a receptor; the activated receptor interacts with cellular machinery, producing a second signal or a change in the activity of a cellular protein; the metabolic activity (broadly defined to include metabolism of RNA, DNA, and protein) of the target cell undergoes a change; and finally, the transduction event ends and the cell returns to its prestimulus state. To illustrate these general features of signaling systems, we provide examples of each of six basic signaling mechanisms (Fig. 12-2).

1. Gated ion channels of the plasma membrane that open and close (hence the term “gating”) in response to the binding of chemical ligands or changes in transmembrane potential. These are the simplest signal transducers. The acetylcholine receptor ion channel is an example of this mechanism (Section 12.2).
2. Receptor enzymes, plasma membrane receptors that are also enzymes. When one of these receptors is activated by its extracellular ligand, it catalyzes the production of an intracellular second messenger. An example is the insulin receptor (Section 12.3).
3. Receptor proteins (serpentine receptors) that *indirectly* activate (through GTP-binding proteins, or G proteins) enzymes that generate intracellular second messengers. This is illustrated by the β -adrenergic receptor system that detects epinephrine (adrenaline) (Section 12.4).
4. Nuclear receptors (steroid receptors) that, when bound to their specific ligand (such as the hormone estrogen), alter the rate at which specific genes are transcribed and translated into cellular proteins. Because steroid hormones function through mechanisms intimately related to the regulation of gene expression, we consider them here only briefly (Section 12.8) and defer a detailed discussion of their action until Chapter 28.
5. Receptors that lack enzymatic activity but attract and activate cytoplasmic enzymes that act on downstream proteins, either by directly converting them to gene-regulating proteins or by activating a cascade of enzymes that finally activates a gene regulator. The JAK-STAT system exemplifies the first mechanism (Section 12.3); and the TLR4 (Toll) signaling system in humans, the second (Section 12.6).

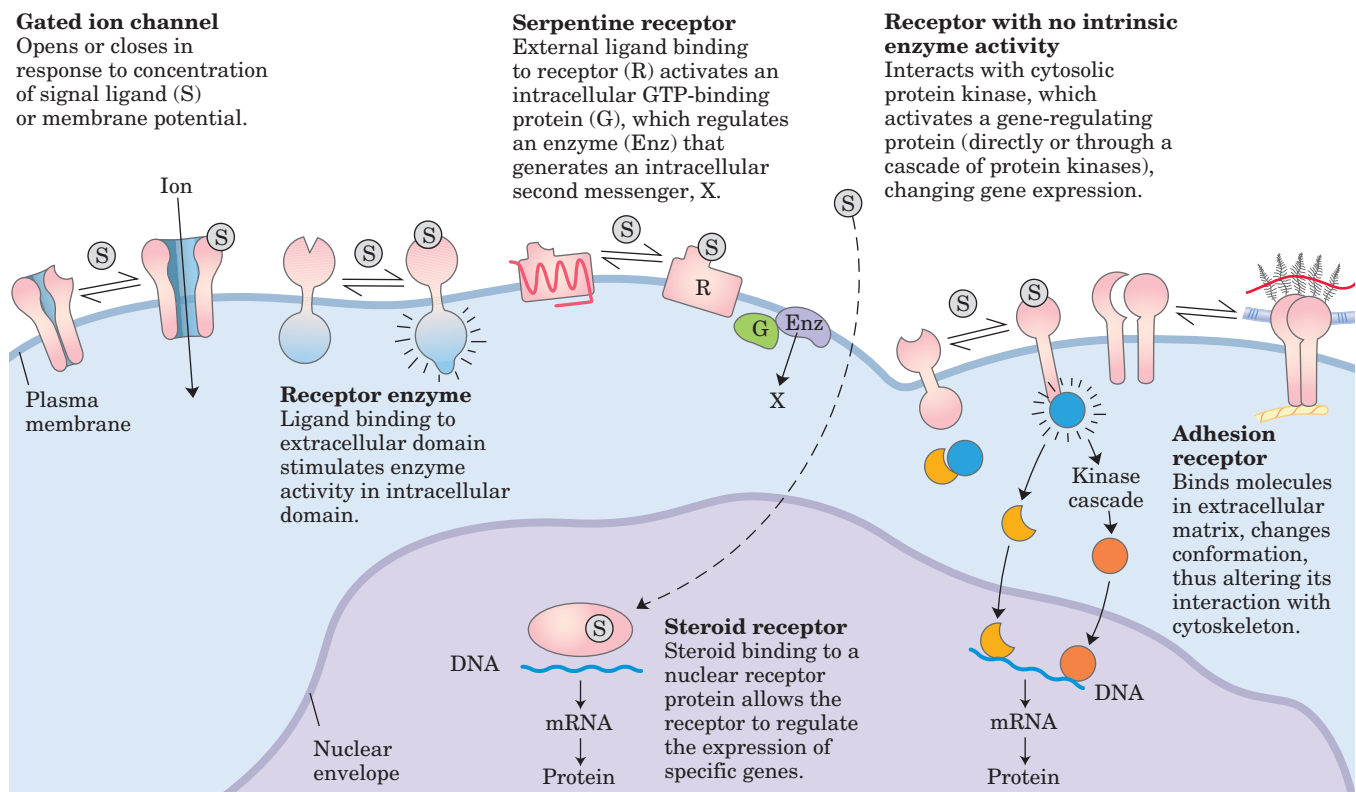


FIGURE 12-2 Six general types of signal transducers.

6. Receptors (adhesion receptors) that interact with macromolecular components of the extracellular matrix (such as collagen) and convey to the cytoskeletal system instructions on cell migration or adherence to the matrix. Integrins (discussed in Chapter 10) illustrate this general type of transduction mechanism.

As we shall see, transductions of all six types commonly require the activation of protein kinases, enzymes that transfer a phosphoryl group from ATP to a protein side chain.

SUMMARY 12.1 Molecular Mechanisms of Signal Transduction

- All cells have specific and highly sensitive signal-transducing mechanisms, which have been conserved during evolution.
- A wide variety of stimuli, including hormones, neurotransmitters, and growth factors, act through specific protein receptors in the plasma membrane.
- The receptors bind the signal molecule, amplify the signal, integrate it with input from other receptors, and transmit it into the cell. If the signal persists, receptor desensitization reduces or ends the response.
- Eukaryotic cells have six general types of signaling mechanisms: gated ion channels; receptor enzymes; membrane proteins that act through G proteins; nuclear proteins that bind steroids and act as transcription factors; membrane proteins that attract and activate soluble protein kinases; and adhesion receptors that carry information between the extracellular matrix and the cytoskeleton.

12.2 Gated Ion Channels

Ion Channels Underlie Electrical Signaling in Excitable Cells

The excitability of sensory cells, neurons, and myocytes depends on ion channels, signal transducers that provide a regulated path for the movement of inorganic ions such as Na^+ , K^+ , Ca^{2+} , and Cl^- across the plasma membrane in response to various stimuli. Recall from Chapter 11 that these ion channels are “gated”; they may be open or closed, depending on whether the associated receptor has been activated by the binding of its specific ligand (a neurotransmitter, for example) or by a change in the transmembrane electrical potential, V_m . The Na^+K^+ ATPase creates a charge imbalance across

the plasma membrane by carrying 3 Na^+ out of the cell for every 2 K^+ carried in (Fig. 12–3a), making the inside negative relative to the outside. The membrane is said to be polarized. By convention, V_m is negative when the inside of the cell is negative relative to the outside. For a typical animal cell, $V_m = -60$ to -70 mV.

Because ion channels generally allow passage of either anions or cations but not both, ion flux through a channel causes a redistribution of charge on the two sides of the membrane, changing V_m . Influx of a positively charged ion such as Na^+ , or efflux of a negatively charged ion such as Cl^- , depolarizes the membrane and brings V_m closer to zero. Conversely, efflux of K^+ hyperpolarizes the membrane and V_m becomes more negative. These ion fluxes through channels are passive, in contrast to active transport by the Na^+K^+ ATPase.

The direction of spontaneous ion flow across a polarized membrane is dictated by the electrochemical

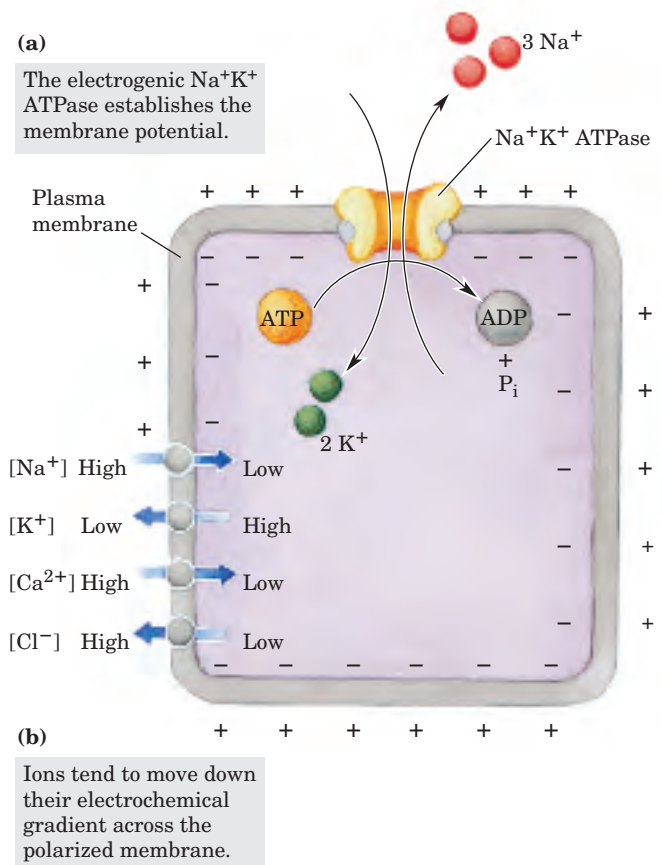


FIGURE 12–3 Transmembrane electrical potential. (a) The electrogenic Na^+K^+ ATPase produces a transmembrane electrical potential of -60 mV (inside negative). (b) Blue arrows show the direction in which ions tend to move spontaneously across the plasma membrane in an animal cell, driven by the combination of chemical and electrical gradients. The chemical gradient drives Na^+ and Ca^{2+} inward (producing depolarization) and K^+ outward (producing hyperpolarization). The electrical gradient drives Cl^- outward, against its concentration gradient (producing depolarization).

TABLE 12-2 Ion Concentrations in Cells and Extracellular Fluids (mM)

Cell type	K^+		Na^+		Ca^{2+}		Cl^-	
	In	Out	In	Out	In	Out	In	Out
Squid axon	400	20	50	440	≤ 0.4	10	40–150	560
Frog muscle	124	2.3	10.4	109	< 0.1	2.1	1.5	78

potential of that ion across the membrane. The force (ΔG) that causes a cation (say, Na^+) to pass spontaneously inward through an ion channel is a function of the ratio of its concentrations on the two sides of the membrane (C_{in}/C_{out}) and of the difference in electrical potential ($\Delta\psi$ or V_m):

$$\Delta G = RT \ln \left(\frac{C_{in}}{C_{out}} \right) + Z \mathcal{F} V_m \quad (12-1)$$

where R is the gas constant, T the absolute temperature, Z the charge on the ion, and $\mathcal{F}$ the Faraday constant. In a typical neuron or myocyte, the concentrations of Na^+ , K^+ , Ca^{2+} , and Cl^- in the cytosol are very different from those in the extracellular fluid (Table 12-2). Given these concentration differences, the resting V_m of -60 mV, and the relationship shown in Equation 12-1, opening of a Na^+ or Ca^{2+} channel will result in a spontaneous inward flow of Na^+ or Ca^{2+} (and depolarization), whereas opening of a K^+ channel will result in a spontaneous outward flux of K^+ (and hyperpolarization) (Fig. 12-3b).

A given ionic species continues to flow through a channel only as long as the combination of concentration gradient and electrical potential provides a driving force, according to Equation 12-1. For example, as Na^+ flows down its concentration gradient it depolarizes the membrane. When the membrane potential reaches $+70$ mV, the effect of this membrane potential (to resist further entry of Na^+) exactly equals the effect of the Na^+ concentration gradient (to cause more Na^+ to flow inward). At this equilibrium potential (E), the driving force (ΔG) tending to move an ion is zero. The equilibrium potential is different for each ionic species because the concentration gradients differ for each ion.

The number of ions that must flow to change the membrane potential significantly is negligible relative to the concentrations of Na^+ , K^+ , and Cl^- in cells and extracellular fluid, so the ion fluxes that occur during signaling in excitable cells have essentially no effect on the concentrations of those ions. However, because the intracellular concentration of Ca^{2+} is generally very low ($\sim 10^{-7}$ M), inward flow of Ca^{2+} can significantly alter the cytosolic $[Ca^{2+}]$.

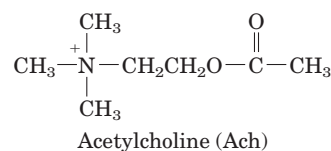
The membrane potential of a cell at a given time is the result of the types and numbers of ion channels open at that instant. In most cells at rest, more K^+ channels

than Na^+ , Cl^- , or Ca^{2+} channels are open and thus the resting potential is closer to the E for K^+ (-98 mV) than that for any other ion. When channels for Na^+ , Ca^{2+} , or Cl^- open, the membrane potential moves toward the E for that ion. The precisely timed opening and closing of ion channels and the resulting transient changes in membrane potential underlie the electrical signaling by which the nervous system stimulates the skeletal muscles to contract, the heart to beat, or secretory cells to release their contents. Moreover, many hormones exert their effects by altering the membrane potentials of their target cells. These mechanisms are not limited to complex animals; ion channels play important roles in the responses of bacteria, protists, and plants to environmental signals.

To illustrate the action of ion channels in cell-to-cell signaling, we describe the mechanisms by which a neuron passes a signal along its length and across a synapse to the next neuron (or to a myocyte) in a cellular circuit, using acetylcholine as the neurotransmitter.

The Nicotinic Acetylcholine Receptor Is a Ligand-Gated Ion Channel

One of the best-understood examples of a **ligand-gated receptor channel** is the **nicotinic acetylcholine receptor** (see Fig. 11-51). The receptor channel opens in response to the neurotransmitter acetylcholine (and to nicotine, hence the name). This receptor is found in the postsynaptic membrane of neurons at certain synapses and in muscle fibers (myocytes) at neuromuscular junctions.



Acetylcholine released by an excited neuron diffuses a few micrometers across the synaptic cleft or neuromuscular junction to the postsynaptic neuron or myocyte, where it interacts with the acetylcholine receptor and triggers electrical excitation (depolarization) of the receiving cell. The acetylcholine receptor is an allosteric protein with two high-affinity binding sites for acetylcholine, about 3.0 nm from the ion gate, on the two α

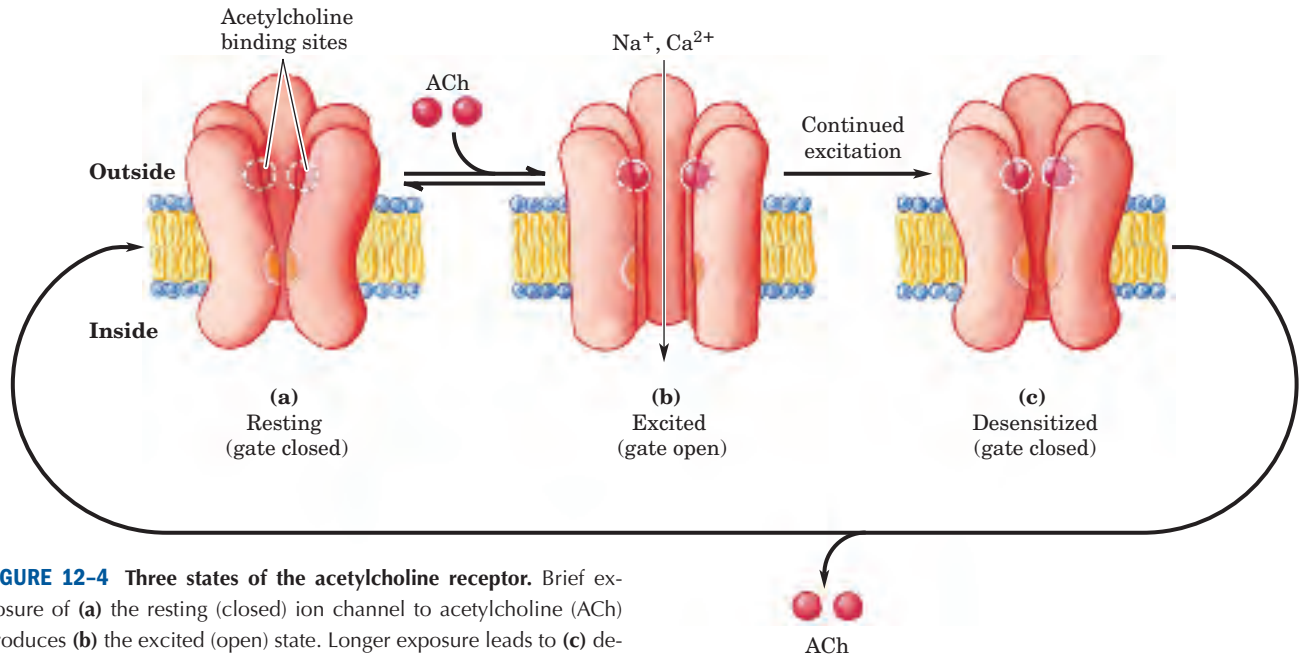


FIGURE 12-4 Three states of the acetylcholine receptor. Brief exposure of (a) the resting (closed) ion channel to acetylcholine (ACh) produces (b) the excited (open) state. Longer exposure leads to (c) desensitization and channel closure.

subunits. The binding of acetylcholine causes a change from the closed to the open conformation. The process is positively cooperative: binding of acetylcholine to the first site increases the acetylcholine-binding affinity of the second site. When the presynaptic cell releases a brief pulse of acetylcholine, both sites on the postsynaptic cell receptor are occupied briefly and the channel opens (Fig. 12-4). Either Na^+ or Ca^{2+} can now pass, and the inward flux of these ions depolarizes the plasma membrane, initiating subsequent events that vary with the type of tissue. In a postsynaptic neuron, depolarization initiates an action potential (see below); at a neuromuscular junction, depolarization of the muscle fiber triggers muscle contraction.

Normally, the acetylcholine concentration in the synaptic cleft is quickly lowered by the enzyme acetylcholinesterase, present in the cleft. When acetylcholine levels remain high for more than a few milliseconds, the receptor is desensitized (Fig. 12-1c). The receptor channel is converted to a third conformation (Fig. 12-4c) in which the channel is closed and the acetylcholine is very tightly bound. The slow release (in tens of milliseconds) of acetylcholine from its binding sites eventually allows the receptor to return to its resting state—closed and resensitized to acetylcholine levels.

Voltage-Gated Ion Channels Produce Neuronal Action Potentials

Signaling in the nervous system is accomplished by networks of neurons, specialized cells that carry an electrical impulse (action potential) from one end of the cell (the cell body) through an elongated cytoplasmic ex-

tension (the axon). The electrical signal triggers release of neurotransmitter molecules at the synapse, carrying the signal to the next cell in the circuit. Three types of **voltage-gated ion channels** are essential to this signaling mechanism. Along the entire length of the axon are **voltage-gated Na^+ channels** (Fig. 12-5; see also Fig. 11-50), which are closed when the membrane is at rest ($V_m = -60$ mV) but open briefly when the membrane is depolarized locally in response to acetylcholine (or some other neurotransmitter). The depolarization induced by the opening of Na^+ channels causes **voltage-gated K^+ channels** to open, and the resulting efflux of K^+ repolarizes the membrane locally. A brief pulse of depolarization traverses the axon as local depolarization triggers the brief opening of neighboring Na^+ channels, then K^+ channels. After each opening of a Na^+ channel, a short refractory period follows during which that channel cannot open again, and thus a unidirectional wave of depolarization sweeps from the nerve cell body toward the end of the axon. The voltage sensitivity of ion channels is due to the presence at critical positions in the channel protein of charged amino acid side chains that interact with the electric field across the membrane. Changes in transmembrane potential produce subtle conformational changes in the channel protein (see Fig. 11-50).

At the distal tip of the axon are **voltage-gated Ca^{2+} channels**. When the wave of depolarization reaches these channels, they open, and Ca^{2+} enters from the extracellular space. The rise in cytoplasmic $[\text{Ca}^{2+}]$ then triggers release of acetylcholine by exocytosis into the synaptic cleft (step ③ in Fig. 12-5). Acetylcholine diffuses to the postsynaptic cell (another

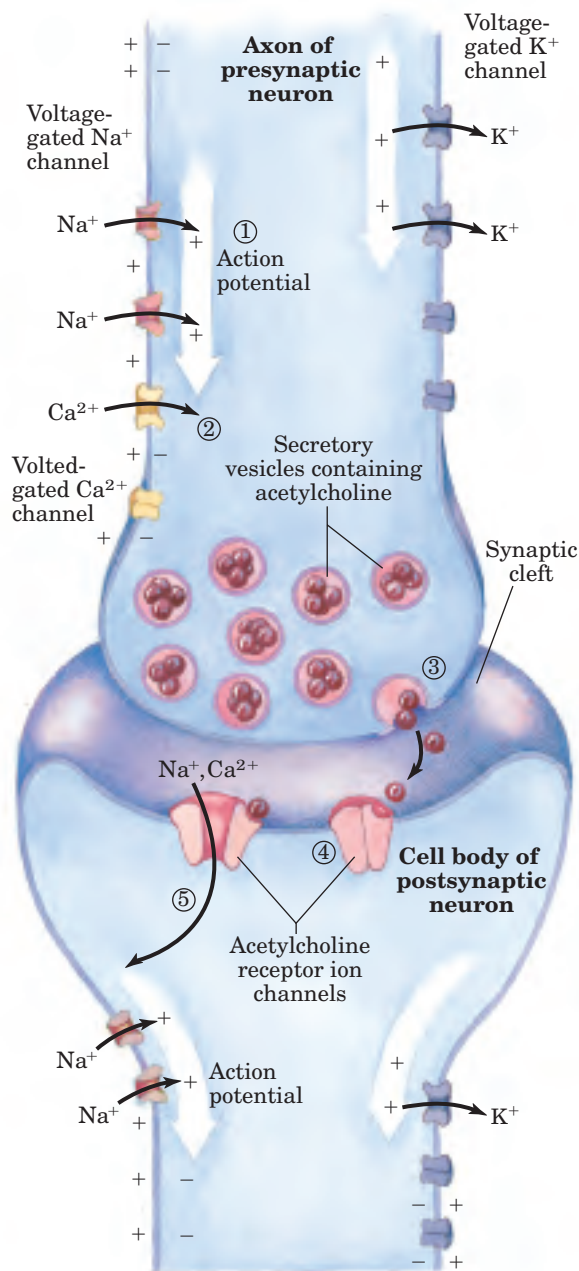


FIGURE 12-5 Role of voltage-gated and ligand-gated ion channels in neural transmission. Initially, the plasma membrane of the presynaptic neuron is polarized (inside negative) through the action of the electrogenic Na^+K^+ ATPase, which pumps 3 Na^+ out for every 2 K^+ pumped into the neuron (see Fig. 12-3). ① A stimulus to this neuron causes an action potential to move along the axon (white arrow), away from the cell body. The opening of one voltage-gated Na^+ channel allows Na^+ entry, and the resulting local depolarization causes the adjacent Na^+ channel to open, and so on. The directionality of movement of the action potential is ensured by the brief refractory period that follows the opening of each voltage-gated Na^+ channel. ② When the wave of depolarization reaches the axon tip, voltage-gated Ca^{2+} channels open, allowing Ca^{2+} entry into the presynaptic neuron. ③ The resulting increase in internal $[\text{Ca}^{2+}]$ triggers exocytic release of the neurotransmitter acetylcholine into the synaptic cleft. ④ Acetylcholine binds to a receptor on the postsynaptic neuron, causing its ligand-gated ion channel to open. ⑤ Extracellular Na^+ and Ca^{2+} enter through this channel, depolarizing the postsynaptic cell. The electrical signal has thus passed to the cell body of the postsynaptic neuron and will move along its axon to a third neuron by this same sequence of events.

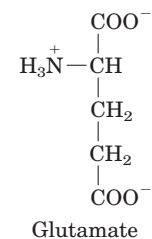
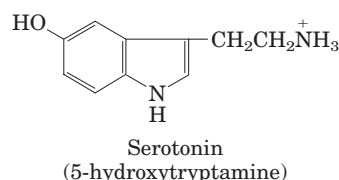
Neurons Have Receptor Channels That Respond to Different Neurotransmitters

Animal cells, especially those of the nervous system, contain a variety of ion channels gated by ligands, voltage, or both. The neurotransmitters 5-hydroxytryptamine (serotonin), glutamate, and glycine can all act through receptor channels that are structurally related to the acetylcholine receptor. Serotonin and glutamate trigger the opening of cation (K^+ , Na^+ , Ca^{2+}) channels, whereas glycine opens Cl^- -specific channels. Cation and anion channels are distinguished by subtle differences in the amino acid residues that line the hydrophilic channel. Cation channels have negatively charged Glu and Asp side chains at crucial positions. When a few of these acidic residues are experimentally replaced with basic residues, the cation channel is converted to an anion channel.

Depending on which ion passes through a channel, the ligand (neurotransmitter) for that channel either depolarizes or hyperpolarizes the target cell. A single neuron normally receives input from several (or many) other neurons, each releasing its own characteristic neurotransmitter with its characteristic depolarizing or hyperpolarizing effect. The target cell's V_m therefore reflects the *integrated* input (Fig. 12-1d) from multi-

neuron or a myocyte), where it binds to acetylcholine receptors and triggers depolarization. Thus the message is passed to the next cell in the circuit.

We see, then, that gated ion channels convey signals in either of two ways: by changing the cytosolic concentration of an ion (such as Ca^{2+}), which then serves as an intracellular **second messenger** (the hormone or neurotransmitter is the first messenger), or by changing V_m and affecting other membrane proteins that are sensitive to V_m . The passage of an electrical signal through one neuron and on to the next illustrates both types of mechanism.



ple neurons. The cell responds with an action potential only if the integrated input adds up to a net depolarization of sufficient size.

The receptor channels for acetylcholine, glycine, glutamate, and γ -aminobutyric acid (GABA) are gated by *extracellular* ligands. *Intracellular* second messengers—such as cAMP, cGMP (3',5'-cyclic GMP, a close analog of cAMP), IP₃ (inositol 1,4,5-trisphosphate), Ca²⁺, and ATP—regulate ion channels of another class, which, as we shall see in Section 12.7, participate in the sensory transductions of vision, olfaction, and gustation.

SUMMARY 12.2 Gated Ion Channels

- Ion channels gated by ligands or membrane potential are central to signaling in neurons and other cells.
- The acetylcholine receptor of neurons and myocytes is a ligand-gated ion channel.
- The voltage-gated Na⁺ and K⁺ channels of neuronal membranes carry the action potential along the axon as a wave of depolarization (Na⁺ influx) followed by repolarization (K⁺ efflux).
- The arrival of an action potential triggers neurotransmitter release from the presynaptic cell. The neurotransmitter (acetylcholine, for example) diffuses to the postsynaptic cell, binds to specific receptors in the plasma membrane, and triggers a change in V_m .

12.3 Receptor Enzymes

A fundamentally different mechanism of signal transduction is carried out by the receptor enzymes. These proteins have a ligand-binding domain on the extracellular surface of the plasma membrane and an enzyme active site on the cytosolic side, with the two domains connected by a single transmembrane segment. Commonly, the receptor enzyme is a protein kinase that phosphorylates Tyr residues in specific target proteins; the insulin receptor is the prototype for this group. In plants, the protein kinase of receptors is specific for Ser or Thr residues. Other receptor enzymes synthesize the intracellular second messenger cGMP in response to extracellular signals. The receptor for atrial natriuretic factor is typical of this type.

The Insulin Receptor Is a Tyrosine-Specific Protein Kinase

Insulin regulates both metabolism and gene expression: the insulin signal passes from the plasma membrane receptor to insulin-sensitive metabolic enzymes and to the

nucleus, where it stimulates the transcription of specific genes. The active insulin receptor consists of two identical α chains protruding from the outer face of the plasma membrane and two transmembrane β subunits with their carboxyl termini protruding into the cytosol (Fig. 12–6, step ①). The α chains contain the insulin-binding domain, and the intracellular domains of the β chains contain the protein kinase activity that transfers a phosphoryl group from ATP to the hydroxyl group of Tyr residues in specific target proteins. Signaling through the insulin receptor begins (step ①) when binding of insulin to the α chains activates the Tyr kinase activity of the β chains, and each $\alpha\beta$ monomer phosphorylates three critical Tyr residues near the carboxyl terminus of the β chain of its partner in the dimer. This **autophosphorylation** opens up the active site so that the enzyme can phosphorylate Tyr residues of other target proteins (Fig. 12–7).

One of these target proteins (Fig. 12–6, step ②) is insulin receptor substrate-1 (IRS-1). Once phosphorylated on its Tyr residues, IRS-1 becomes the point of nucleation for a complex of proteins (step ③) that carry the message from the insulin receptor to end targets in the cytosol and nucleus, through a long series of intermediate proteins. First, a $\textcircled{\text{P}}$ -Tyr residue in IRS-1 is bound by the **SH2 domain** of the protein Grb2. (SH2 is an abbreviation of *Src homology 2*; the sequences of SH2 domains are similar to a domain in another protein Tyr kinase, *Src* (pronounced sark).) A number of signaling proteins contain SH2 domains, all of which bind $\textcircled{\text{P}}$ -Tyr residues in a protein partner. Grb2 also contains a second protein-binding domain, SH3, that binds to regions rich in Pro residues. Grb2 binds to a proline-rich region of Sos, recruiting Sos to the growing receptor complex. When bound to Grb2, Sos catalyzes the replacement of bound GDP by GTP on Ras, one of a family of guanosine nucleotide-binding proteins (**G proteins**) that mediate a wide variety of signal transductions (Section 12.4). When GTP is bound, Ras can activate a protein kinase, Raf-1 (step ④), the first of three protein kinases—Raf-1, MEK, and ERK—that form a cascade in which each kinase activates the next by phosphorylation (step ⑤). The protein kinase ERK is activated by phosphorylation of both a Thr and a Tyr residue. When activated, it mediates some of the biological effects of insulin by entering the nucleus and phosphorylating proteins such as Elk1, which modulates the transcription of about 100 insulin-regulated genes (step ⑥).

The proteins Raf-1, MEK, and ERK are members of three larger families, for which several nomenclatures are employed. ERK is a member of the **MAPK** family (*mitogen-activated protein kinases*; mitogens are signals that act from outside the cell to induce mitosis and cell division). Soon after discovery of the first MAPK, that enzyme was found to be activated by another protein kinase, which came to be called MAP kinase kinase (MEK

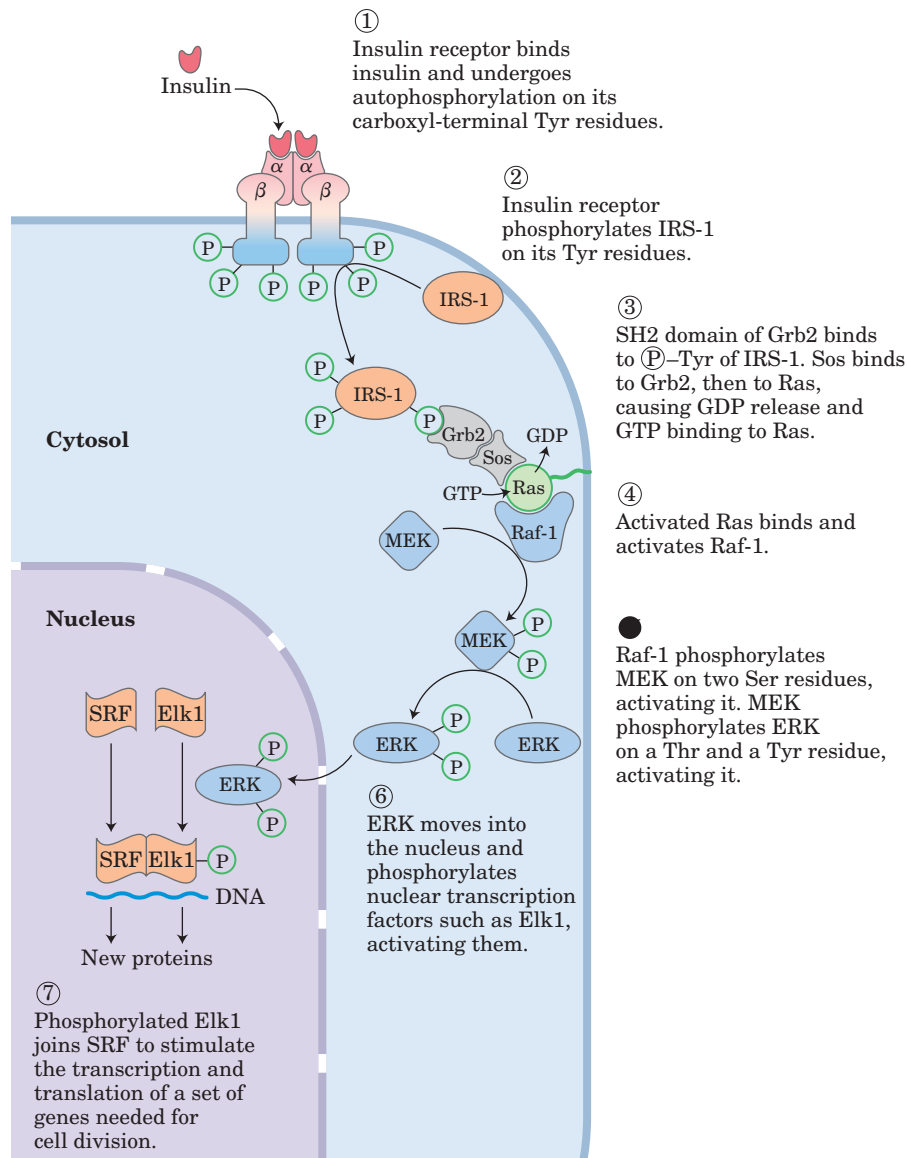


FIGURE 12-6 Regulation of gene expression by insulin. The insulin receptor consists of two α chains on the outer face of the plasma membrane and two β chains that traverse the membrane and protrude from the cytoplasmic face. Binding of insulin to the α chains triggers a conformational change that allows the autophosphorylation of Tyr residues in the carboxyl-terminal domain of the β subunits. Autophosphorylation further activates the Tyr kinase domain, which then catalyzes phosphorylation of other target proteins. The signaling pathway by which

insulin regulates the expression of specific genes consists of a cascade of protein kinases, each of which activates the next. The insulin receptor is a Tyr-specific kinase; the other kinases (all shown in blue) phosphorylate Ser or Thr residues. MEK is a dual-specificity kinase, which phosphorylates both a Thr and a Tyr residue in ERK (extracellular regulated kinase); MEK is mitogen-activated, ERK-activating kinase; SRF is serum response factor. Abbreviations for other components are explained in the text.

belongs to this family); and when a third kinase that activated MAP kinase kinase was discovered, it was given the slightly ludicrous family name MAP kinase kinase kinase (Raf-1 is a member of this family; Fig. 12-6). Slightly less cumbersome are the acronyms for these three families, MAPK, MAPKK, and MAPKKK. Kinases in the MAPK and MAPKKK families are specific for Ser or Thr residues, but MAPKKs (here, MEK) phosphorylate both a Ser and a Tyr residue in their substrate, a MAPK (here, ERK).

Biochemists now recognize the insulin pathway as but one instance of a more general theme in which hormone signals, via pathways similar to that shown in Figure 12-6, result in phosphorylation of target enzymes by protein kinases. The target of phosphorylation is often another protein kinase, which then phosphorylates a third protein kinase, and so on. The result is a cascade of reactions that amplifies the initial signal by many orders of magnitude (see Fig. 12-1b). Cascades such as that shown in Figure 12-6 are called **MAPK cascades**.

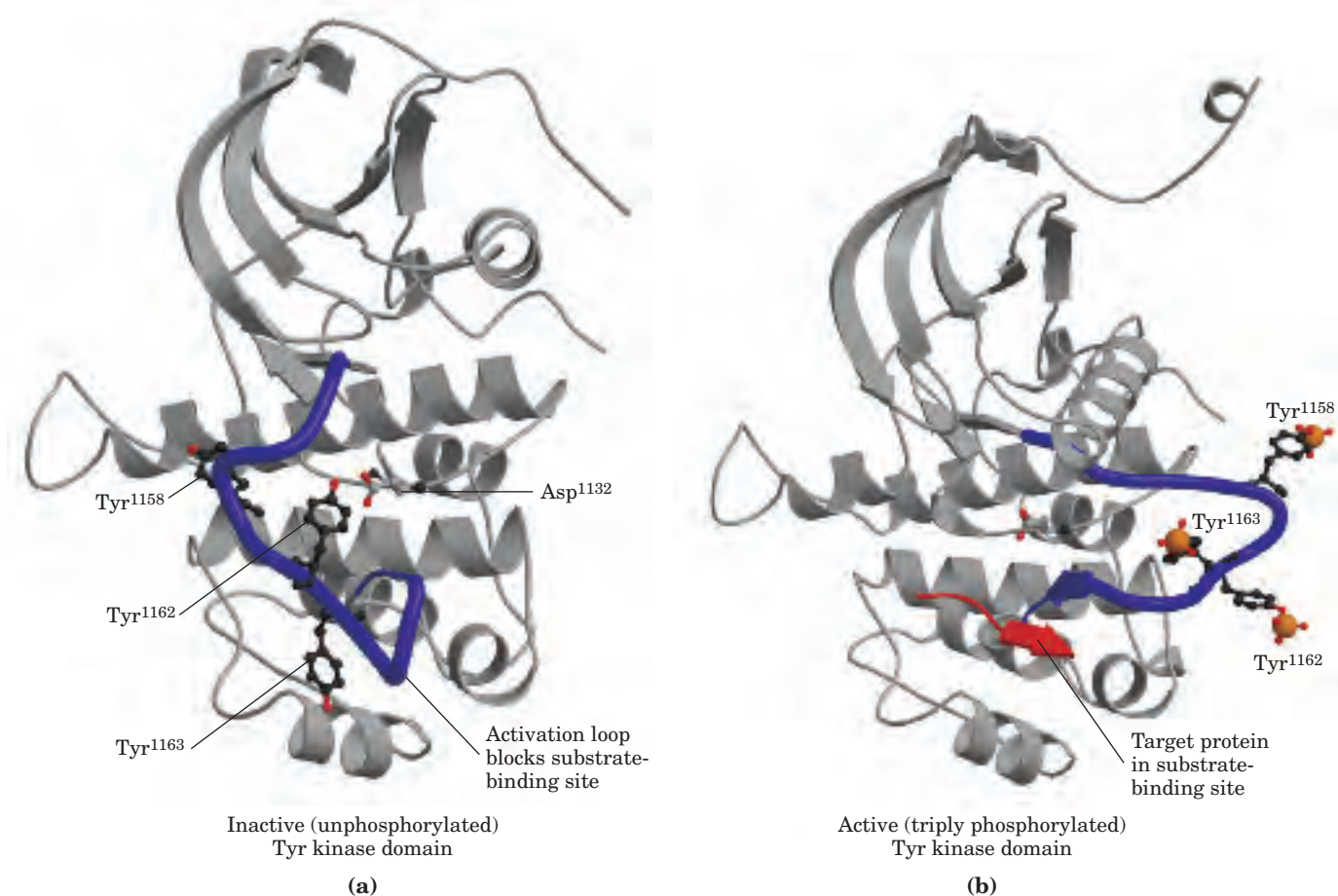
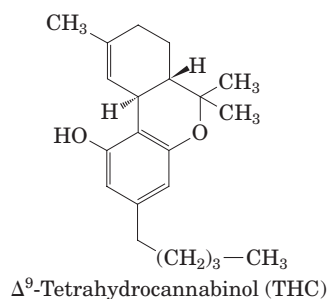


FIGURE 12-7 Activation of the insulin-receptor Tyr kinase by autophosphorylation. **(a)** In the inactive form of the Tyr kinase domain (PDB ID 1IRK), the activation loop (blue) sits in the active site, and none of the critical Tyr residues (black and red ball-and-stick structures) are phosphorylated. This conformation is stabilized by hydrogen bonding between Tyr¹¹⁶² and Asp¹¹³². **(b)** When insulin binds to the α chains of insulin receptors, the Tyr kinase of each β subunit of the dimer phosphorylates three Tyr residues (Tyr¹¹⁵⁸, Tyr¹¹⁶², and

Tyr¹¹⁶³) on the other β subunit (shown here; PDB ID 1IR3). (Phosphoryl groups are depicted here as an orange space-filling phosphorus atom and red ball-and-stick oxygen atoms.) The effect of introducing three highly charged (P)-Tyr residues is to force a 30 Å change in the position of the activation loop, away from the substrate-binding site, which becomes available to bind to and phosphorylate a target protein, shown here as a red arrow.

Grb2 is not the only protein that associates with phosphorylated IRS-1. The enzyme phosphoinositide 3-kinase (PI-3K) binds IRS-1 through the former's SH2 domain (Fig. 12-8). Thus activated, PI-3K converts the membrane lipid phosphatidylinositol 4,5-bisphosphate (see Fig. 10-15), also called PIP₂, to phosphatidylinositol 3,4,5-trisphosphate (PIP₃). When bound to PIP₃, protein kinase B (PKB) is phosphorylated and activated by yet another protein kinase, PDK1. The activated PKB then phosphorylates Ser or Thr residues on its target proteins, one of which is glycogen synthase kinase 3 (GSK3). In its active, nonphosphorylated form, GSK3 phosphorylates glycogen synthase, inactivating it and thereby contributing to the slowing of glycogen synthesis. (This mechanism is believed to be only part of the explanation for the effects of insulin on glycogen metabolism.) When phosphorylated by PKB, GSK3 is inactivated. By thus preventing inactivation of glycogen synthase in liver and muscle, the cascade of protein

phosphorylations initiated by insulin stimulates glycogen synthesis (Fig. 12-8). In muscle, PKB triggers the movement of glucose transporters (GLUT4) from internal vesicles to the plasma membrane, stimulating glucose uptake from the blood (Fig. 12-8; see also Box 11-2). PKB also functions in several other signaling pathways, including that triggered by Δ^9 -tetrahydrocannabinol (THC), the active ingredient of marijuana



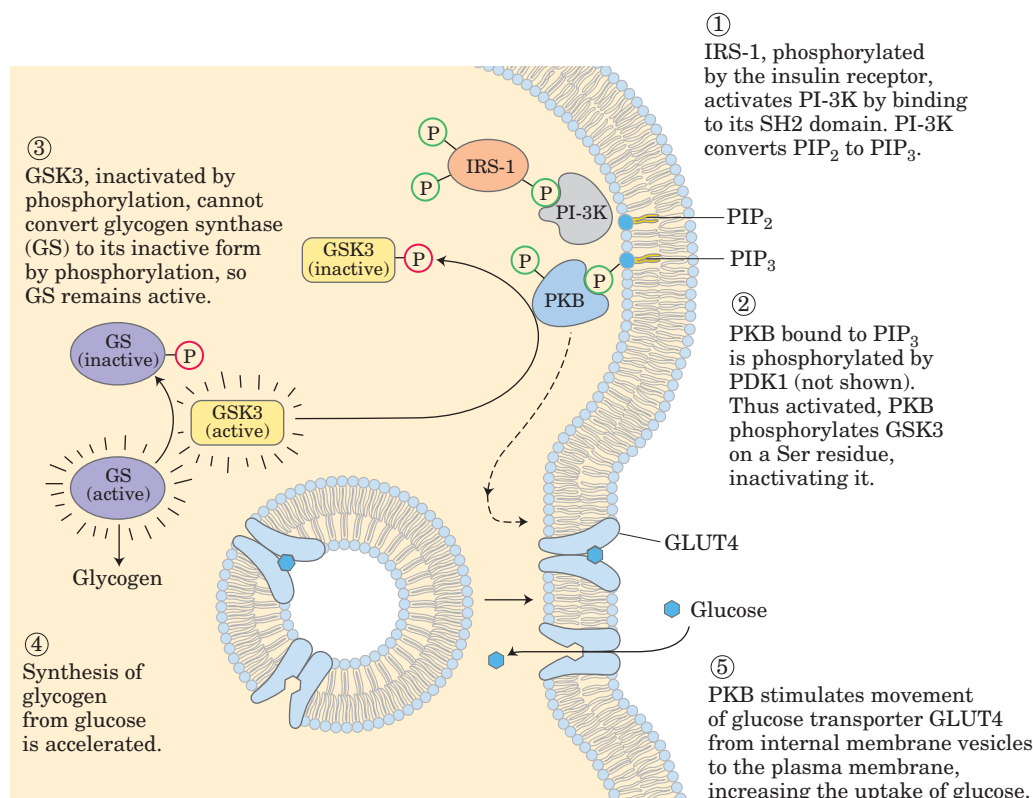
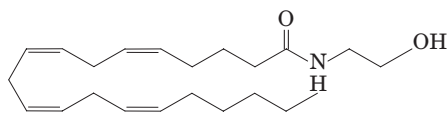


FIGURE 12-8 Activation of glycogen synthase by insulin. Transmission of the signal is mediated by PI-3 kinase (PI-3K) and protein kinase B (PKB).

and hashish. THC activates the CB_1 receptor in the plasma membrane of neurons in the brain, triggering a signaling cascade that involves MAPKs. One consequence of CB_1 activation is the stimulation of appetite, one of the well-established effects of marijuana use. The normal ligands for the CB_1 receptor are endocannabinoids such as anandamide, which serve to protect the brain from the toxicity of excessive neuronal activity—as in an epileptic seizure, for example. Hashish has for centuries been used in the treatment of epilepsy.



Anandamide (arachidonylethanolamide, an endogenous cannabinoid)



As in all signaling pathways, there is a mechanism for terminating signaling through the PI-3K–PKB pathway. A PIP_3 -specific phosphatase (PTEN in humans) removes the phosphate at the 3 position of PIP_3 to produce PIP_2 , which no longer serves as a binding site for PKB, and the signaling chain is broken. In various types of advanced cancer, tumor cells often have a defect in the PTEN gene and thus have abnormally high levels of PIP_3 and of PKB activity. The result seems to be a continuing signal for cell division and thus tumor growth. ■

What spurred the evolution of such complicated regulatory machinery? This system allows one activated receptor to activate several IRS-1 molecules, amplifying the insulin signal, and it provides for the integration of signals from several receptors, each of which can phosphorylate IRS-1. Furthermore, because IRS-1 can activate any of several proteins that contain SH2 domains, a single receptor acting through IRS-1 can trigger two or more signaling pathways; insulin affects gene expression through the Grb2–Sos–Ras–MAPK pathway and glycogen metabolism through the PI-3K–PKB pathway.

The insulin receptor is the prototype for a number of receptor enzymes with a similar structure and **receptor Tyr kinase** activity. The receptors for epidermal growth factor and platelet-derived growth factor, for example, have structural and sequence similarities to the insulin receptor, and both have a protein Tyr kinase activity that phosphorylates IRS-1. Many of these receptors dimerize after binding ligand; the insulin receptor is already a dimer before insulin binds. The binding of adaptor proteins such as Grb2 to P-Tyr residues is a common mechanism for promoting protein-protein interactions, a subject to which we return in Section 12.5.

In addition to the many receptors that act as protein Tyr kinases, a number of receptorlike plasma membrane proteins have protein Tyr phosphatase activity. Based on the structures of these proteins, we can surmise that their ligands are components of the extracellular matrix or the

surfaces of other cells. Although their signaling roles are not yet as well understood as those of the receptor Tyr kinases, they clearly have the potential to reverse the actions of signals that stimulate these kinases.

A variation on the basic theme of receptor Tyr kinases is seen in receptors that have no intrinsic protein kinase activity but, when occupied by their ligand, bind a *soluble* Tyr kinase. One example is the system that regulates the formation of erythrocytes in mammals. The **cytokine** (developmental signal) for this system is erythropoietin (EPO), a 165 amino acid protein produced in the kidneys. When EPO binds to its plasma membrane receptor (Fig. 12–9), the receptor dimerizes and can now bind the soluble protein kinase JAK (*Janus kinase*). This binding activates JAK, which phosphorylates several Tyr residues in the cytoplasmic domain of the EPO receptor. A family of transcription factors, collectively called STATs (signal transducers and activators of transcription), are also targets of the JAK kinase activity. An SH2 domain in STAT5 binds (P)–Tyr residues in the EPO receptor, positioning it for this phosphorylation by JAK. When STAT5 is phosphorylated in re-

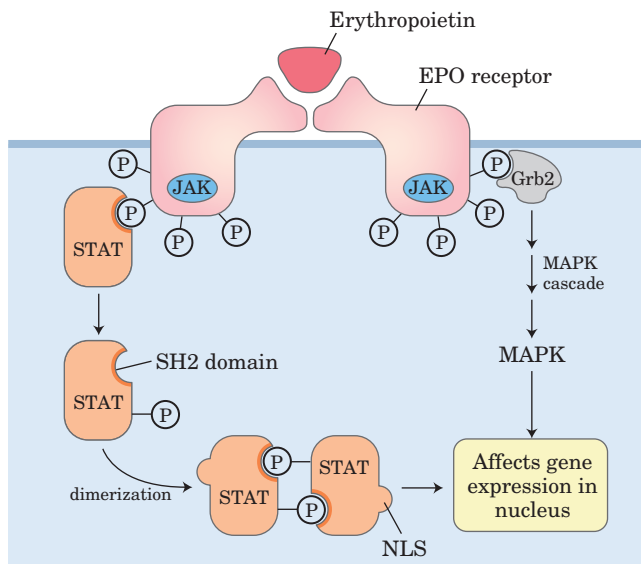


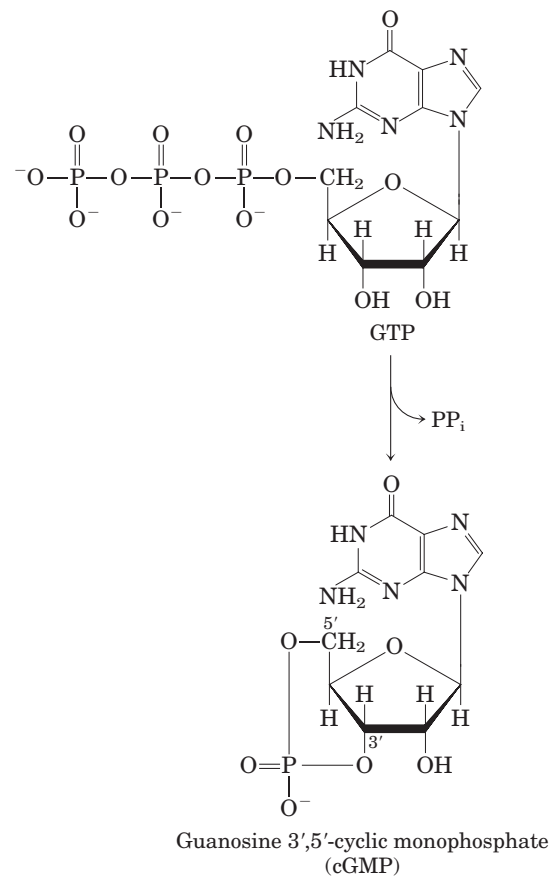
FIGURE 12–9 The JAK-STAT transduction mechanism for the erythropoietin receptor. Binding of erythropoietin (EPO) causes dimerization of the EPO receptor, which allows the soluble Tyr kinase JAK to bind to the internal domain of the receptor and phosphorylate it on several Tyr residues. The STAT protein STAT5 contains an SH2 domain and binds to the (P)–Tyr residues on the receptor, bringing the receptor into proximity with JAK. Phosphorylation of STAT5 by JAK allows two STAT molecules to dimerize, each binding the other's (P)–Tyr residue. Dimerization of STAT5 exposes a nuclear localization sequence (NLS) that targets STAT5 for transport into the nucleus. In the nucleus, STAT causes the expression of genes controlled by EPO. A second signaling pathway is also triggered by autophosphorylation of JAK that is associated with EPO binding to its receptor. The adaptor protein Grb2 binds (P)–Tyr in JAK and triggers the MAPK cascade, as in the insulin system (see Fig. 12–6).

sponse to EPO, it forms dimers, exposing a signal for its transport into the nucleus. There, STAT5 causes the expression (transcription) of specific genes essential for erythrocyte maturation. This JAK-STAT system operates in a number of other signaling pathways, including that for the hormone leptin, described in detail in Chapter 23 (see Fig. 23–34). Activated JAK can also trigger, through Grb2, the MAPK cascade (Fig. 12–6), which leads to altered expression of specific genes.

Src is another soluble protein Tyr kinase that associates with certain receptors when they bind their ligands. Src was the first protein found to have the characteristic (P)–Tyr-binding domain that was subsequently named the Src homology (SH2) domain. Yet another example of a receptor's association with a soluble protein kinase is the Toll-like receptor (TLR4) system through which mammals detect the bacterial lipopolysaccharide (LPS), a potent toxin. We return to the Toll-like receptor system in Section 12.6, in the context of the evolution of signaling proteins.

Receptor Guanylyl Cyclases Generate the Second Messenger cGMP

Guanylyl cyclases (Fig. 12–10) are another type of receptor enzyme. When activated, a guanylyl cyclase produces **guanosine 3',5'-cyclic monophosphate (cyclic GMP, cGMP)** from GTP:



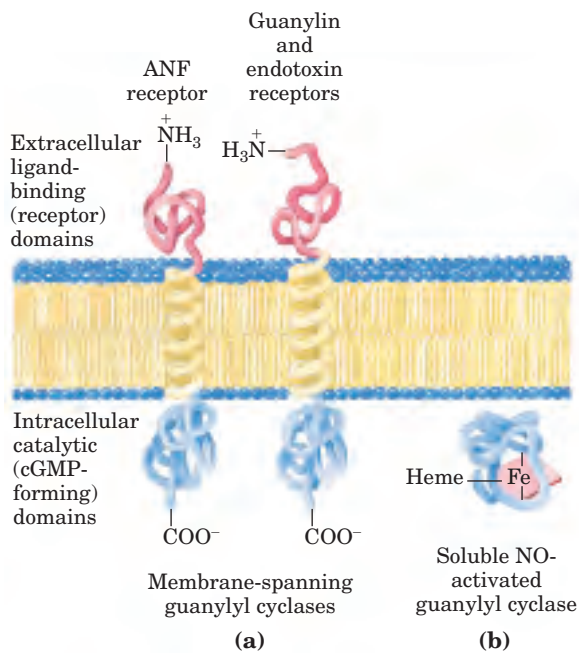


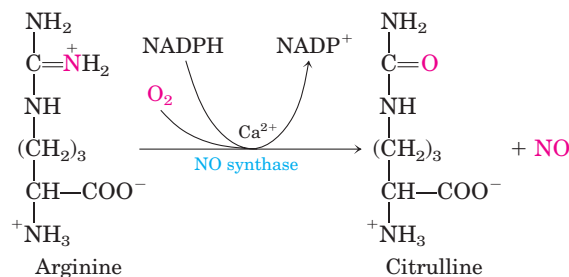
FIGURE 12-10 Two types (isozymes) of guanylyl cyclase that participate in signal transduction. (a) One isozyme exists in two similar membrane-spanning forms that are activated by their extracellular ligands: atrial natriuretic factor, ANF (receptors in cells of the renal collecting ducts and the smooth muscle of blood vessels), and guanylin (receptors in intestinal epithelial cells). The guanylin receptor is also the target of a type of bacterial endotoxin that triggers severe diarrhea. (b) The other isozyme is a soluble enzyme that is activated by intracellular nitric oxide (NO); this form is found in many tissues, including smooth muscle of the heart and blood vessels.

Cyclic GMP is a second messenger that carries different messages in different tissues. In the kidney and intestine it triggers changes in ion transport and water retention; in cardiac muscle (a type of smooth muscle) it signals relaxation; in the brain it may be involved both in development and in adult brain function. Guanylyl cyclase in the kidney is activated by the hormone **atrial natriuretic factor (ANF)**, which is released by cells in the atrium of the heart when the heart is stretched by increased blood volume. Carried in the blood to the kidney, ANF activates guanylyl cyclase in cells of the collecting ducts (Fig. 12-10a). The resulting rise in [cGMP] triggers increased renal excretion of Na^+ and, consequently, of water, driven by the change in osmotic pressure. Water loss reduces the blood volume, countering the stimulus that initially led to ANF secretion. Vascular smooth muscle also has an ANF receptor—guanylyl cyclase; on binding to this receptor, ANF causes relaxation (vasodilation) of the blood vessel, which increases blood flow while decreasing blood pressure.

A similar receptor guanylyl cyclase in the plasma membrane of intestinal epithelial cells is activated by an intestinal peptide, **guanylin**, which regulates Cl^- secretion in the intestine. This receptor is also the target of a

heat-stable peptide endotoxin produced by *Escherichia coli* and other gram-negative bacteria. The elevation in [cGMP] caused by the endotoxin increases Cl^- secretion and consequently decreases reabsorption of water by the intestinal epithelium, producing diarrhea.

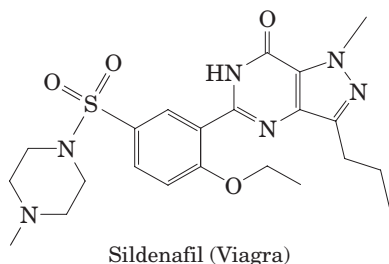
A distinctly different type of guanylyl cyclase is a cytosolic protein with a tightly associated heme group (Fig. 12-10b), an enzyme activated by nitric oxide (NO). Nitric oxide is produced from arginine by Ca^{2+} -dependent **NO synthase**, present in many mammalian tissues, and diffuses from its cell of origin into nearby cells. NO is sufficiently nonpolar to cross plasma membranes without a carrier. In the target cell, it binds to the heme group of guanylyl cyclase and activates cGMP production. In the heart, cGMP reduces the forcefulness of contractions by stimulating the ion pump(s) that expel Ca^{2+} from the cytosol.



This NO-induced relaxation of cardiac muscle is the same response brought about by nitroglycerin tablets and other nitrovasodilators taken to relieve angina, the pain caused by contraction of a heart deprived of O_2 because of blocked coronary arteries. Nitric oxide is unstable and its action is brief; within seconds of its formation, it undergoes oxidation to nitrite or nitrate. Nitrovasodilators produce long-lasting relaxation of cardiac muscle because they break down over several hours, yielding a steady stream of NO. The value of nitroglycerin as a treatment for angina was discovered serendipitously in factories producing nitroglycerin as an explosive in the 1860s. Workers with angina reported that their condition was much improved during the work week but returned on weekends. The physicians treating these workers heard this story so often that they made the connection, and a drug was born.

The effects of increased cGMP synthesis diminish after the stimulus ceases, because a specific phosphodiesterase (cGMP PDE) converts cGMP to the inactive 5'-GMP. Humans have several isoforms of cGMP PDE, with different tissue distributions. The isoform in the blood vessels of the penis is inhibited by the drug sildenafil (Viagra), which therefore causes cGMP levels to remain elevated once raised by an appropriate stimulus, accounting for the usefulness of this drug in the treatment of erectile dysfunction. ■

Most of the actions of cGMP in animals are believed to be mediated by **cGMP-dependent protein kinase**, also called **protein kinase G** or **PKG**, which, when ac-



tivated by cGMP, phosphorylates Ser and Thr residues in target proteins. The catalytic and regulatory domains of this enzyme are in a single polypeptide ($M_r \sim 80,000$). Part of the regulatory domain fits snugly in the substrate-binding site. Binding of cGMP forces this part of the regulatory domain out of the binding site, activating the catalytic domain.

Cyclic GMP has a second mode of action in the vertebrate eye: it causes ion-specific channels to open in the retinal rod and cone cells. We return to this role of cGMP in the discussion of vision in Section 12.7.

SUMMARY 12.3 Receptor Enzymes

- The insulin receptor is the prototype of receptor enzymes with Tyr kinase activity. When insulin binds to its receptor, each $\alpha\beta$ monomer of the receptor phosphorylates the β chain of its partner, activating the receptor's Tyr kinase activity. The kinase catalyzes the phosphorylation of Tyr residues on other proteins such as IRS-1.
- (P)-Tyr residues in IRS-1 serve as binding sites for proteins with SH2 domains. Some of these proteins, such as Grb2, have two or more protein-binding domains and can serve as adaptors that bring two proteins into proximity.
- Further protein-protein interactions result in GTP binding to and activation of the Ras protein, which in turn activates a protein kinase cascade that ends with the phosphorylation of target proteins in the cytosol and nucleus. The result is specific metabolic changes and altered gene expression.
- Several signals, including atrial natriuretic factor and the intestinal peptide guanylin, act through receptor enzymes with guanylyl cyclase activity. The cGMP produced acts as a second messenger, activating cGMP-dependent protein kinase (PKG). This enzyme alters metabolism by phosphorylating specific enzyme targets.
- Nitric oxide (NO) is a short-lived messenger that acts by stimulating a soluble guanylyl cyclase, raising [cGMP] and stimulating PKG.

12.4 G Protein-Coupled Receptors and Second Messengers

A third mechanism of signal transduction, distinct from gated ion channels and receptor enzymes, is defined by three essential components: a plasma membrane receptor with seven transmembrane helical segments, an enzyme in the plasma membrane that generates an intracellular second messenger, and a **guanosine nucleotide-binding protein (G protein)**. The G protein, stimulated by the activated receptor, exchanges bound GDP for GTP; the GTP-protein dissociates from the occupied receptor and binds to a nearby enzyme, altering its activity. The human genome encodes more than 1,000 members of this family of receptors, specialized for transducing messages as diverse as light, smells, tastes, and hormones. The β -adrenergic receptor, which mediates the effects of epinephrine on many tissues, is the prototype for this type of transducing system.

The β -Adrenergic Receptor System Acts through the Second Messenger cAMP

Epinephrine action begins when the hormone binds to a protein receptor in the plasma membrane of a hormone-sensitive cell. **Adrenergic receptors** ("adrenergic" reflects the alternative name for epinephrine, adrenaline) are of four general types, α_1 , α_2 , β_1 , and β_2 , defined by subtle differences in their affinities and responses to a group of agonists and antagonists. **Agonists** are structural analogs that bind to a receptor and mimic the effects of its natural ligand; **antagonists** are analogs that bind without triggering the normal effect and thereby block the effects of agonists. In some cases, the affinity of the synthetic agonist or antagonist for the receptor is greater than that of the natural agonist (Fig. 12-11). The four types of adrenergic receptors are found in different target tissues and mediate different responses to epinephrine. Here we focus on the **β -adrenergic receptors** of muscle, liver, and adipose tissue. These receptors mediate changes in fuel metabolism, as described in Chapter 23, including the increased breakdown of glycogen and fat. Adrenergic receptors of the β_1 and β_2 subtypes act through the same mechanism, so in our discussion, " **β -adrenergic**" applies to both types.

The β -adrenergic receptor is an integral protein with seven hydrophobic regions of 20 to 28 amino acid residues that "snake" back and forth across the plasma membrane seven times. This protein is a member of a very large family of receptors, all with seven transmembrane helices, that are commonly called **serpentine receptors**, **G protein-coupled receptors (GPCR)**, or **7 transmembrane segment (7tm) receptors**. The binding of epinephrine to a site on the

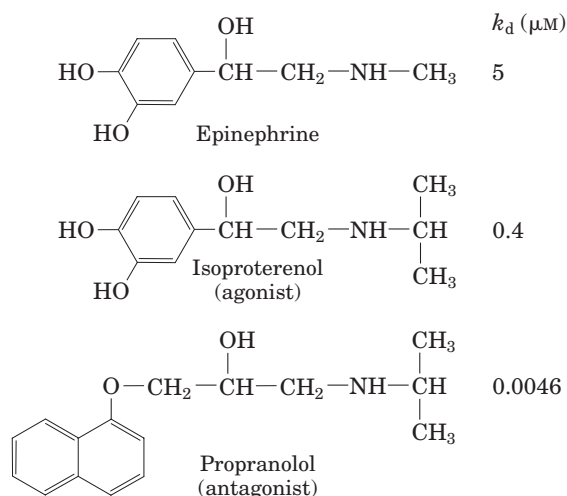


FIGURE 12-11 Epinephrine and its synthetic analogs. Epinephrine, also called adrenaline, is released from the adrenal gland and regulates energy-yielding metabolism in muscle, liver, and adipose tissue. It also serves as a neurotransmitter in adrenergic neurons. Its affinity for its receptor is expressed as a dissociation constant for the receptor-ligand complex. Isoproterenol and propranolol are synthetic analogs, one an agonist with an affinity for the receptor that is higher than that of epinephrine, and the other an antagonist with extremely high affinity.

receptor deep within the membrane (Fig. 12-12, step ①) promotes a conformational change in the receptor's intracellular domain that affects its interaction with the second protein in the signal-transduction pathway, a heterotrimeric GTP-binding **stimulatory G protein**, or **G_s**, on the cytosolic side of the plasma membrane. Alfred G. Gilman and Martin Rodbell discovered that when GTP is bound to G_s, G_s stimulates the production of cAMP by adenylyl cyclase (see below) in the plasma membrane. The function of G_s as a molecular switch resembles that of another class of G proteins typified by Ras, discussed in Section 12.3 in the context of the insulin receptor. Structurally, G_s and Ras are quite distinct; G proteins of the Ras type are monomers ($M_r \sim 20,000$), whereas the G proteins that interact with serpentine



Alfred G. Gilman



Martin Rodbell, 1925–1998

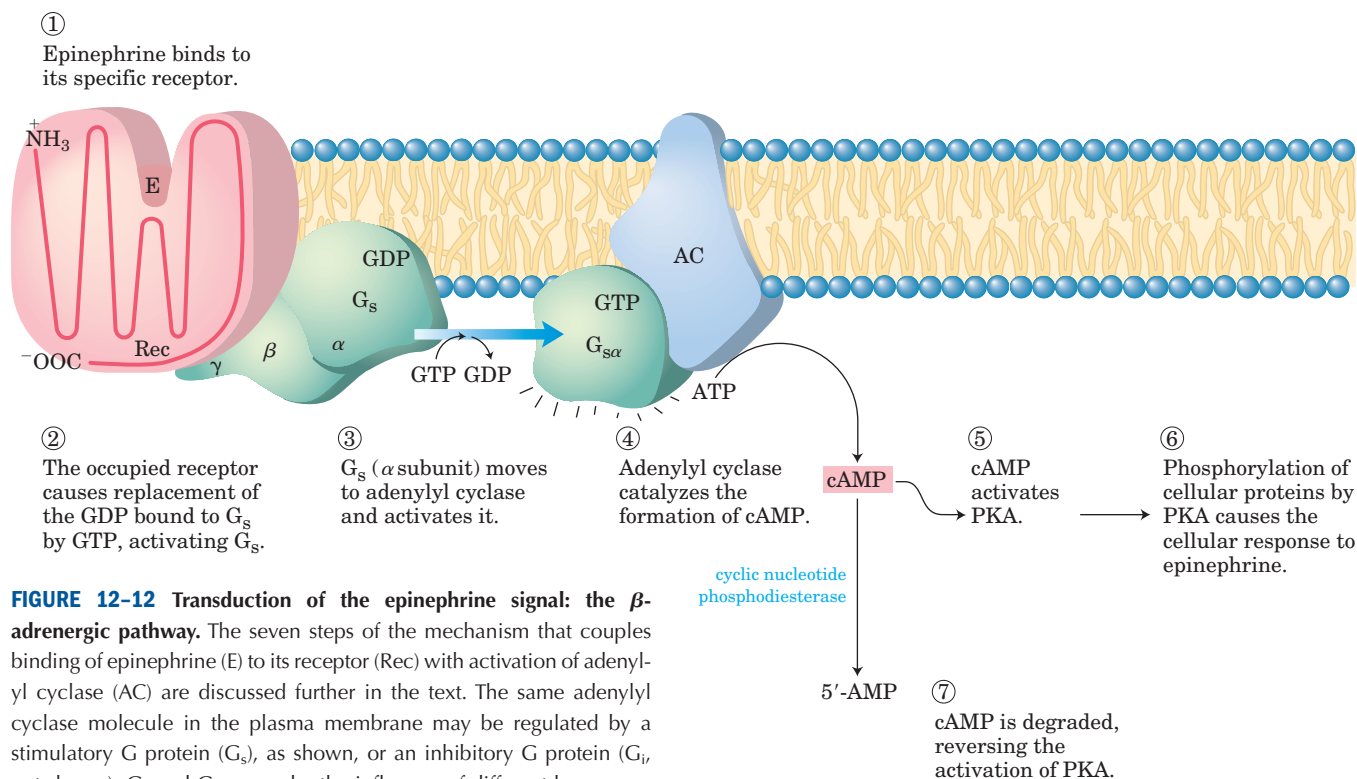
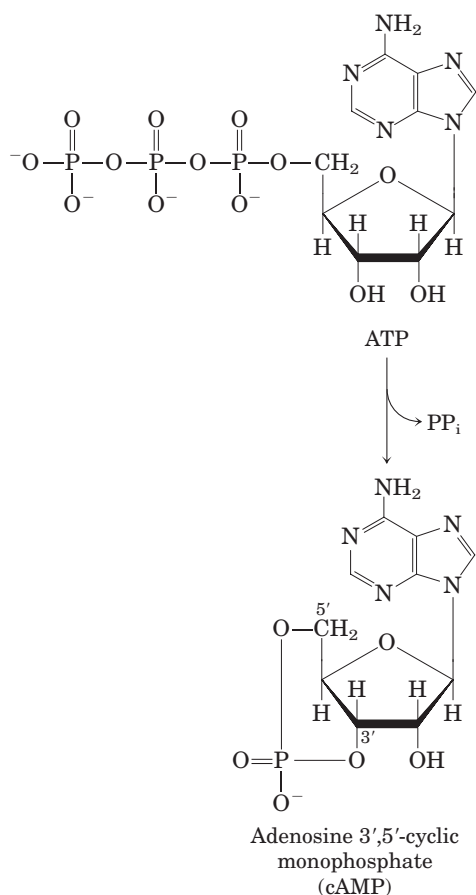


FIGURE 12-12 Transduction of the epinephrine signal: the β -adrenergic pathway. The seven steps of the mechanism that couples binding of epinephrine (E) to its receptor (Rec) with activation of adenylyl cyclase (AC) are discussed further in the text. The same adenylyl cyclase molecule in the plasma membrane may be regulated by a stimulatory G protein (G_s), as shown, or an inhibitory G protein (G_i, not shown). G_s and G_i are under the influence of different hormones. Hormones that induce GTP binding to G_i cause *inhibition* of adenylyl cyclase, resulting in lower cellular [cAMP].

receptors are trimers of three different subunits, α (M_r 43,000), β (M_r 37,000), and γ (M_r 7,500 to 10,000).

When the nucleotide-binding site of G_s (on the α subunit) is occupied by GTP, G_s is active and can activate adenylyl cyclase (AC in Fig. 12–12); with GDP bound to the site, G_s is inactive. Binding of epinephrine enables the receptor to catalyze displacement of bound GDP by GTP, converting G_s to its active form (step ②). As this occurs, the β and γ subunits of G_s dissociate from the α subunit, and $G_{s\alpha}$, with its bound GTP, moves in the plane of the membrane from the receptor to a nearby molecule of adenylyl cyclase (step ③). The $G_{s\alpha}$ is held to the membrane by a covalently attached palmitoyl group (see Fig. 11–14).

Adenylyl cyclase (Fig. 12–13) is an integral protein of the plasma membrane, with its active site on the cytosolic face. It catalyzes the synthesis of cAMP from ATP:



The association of active $G_{s\alpha}$ with adenylyl cyclase stimulates the cyclase to catalyze cAMP synthesis (Fig. 12–12, step ④), raising the cytosolic [cAMP]. This stimulation by $G_{s\alpha}$ is self-limiting; $G_{s\alpha}$ is a GTPase that turns itself off by converting its bound GTP to GDP (Fig. 12–14). The now inactive $G_{s\alpha}$ dissociates from adenylyl cyclase, rendering the cyclase inactive. After $G_{s\alpha}$ reassociates with the β and γ subunits ($G_{s\beta\gamma}$), G_s is again available to interact with a hormone-bound receptor.

 **Trimeric G Proteins: Molecular On/Off Switches**

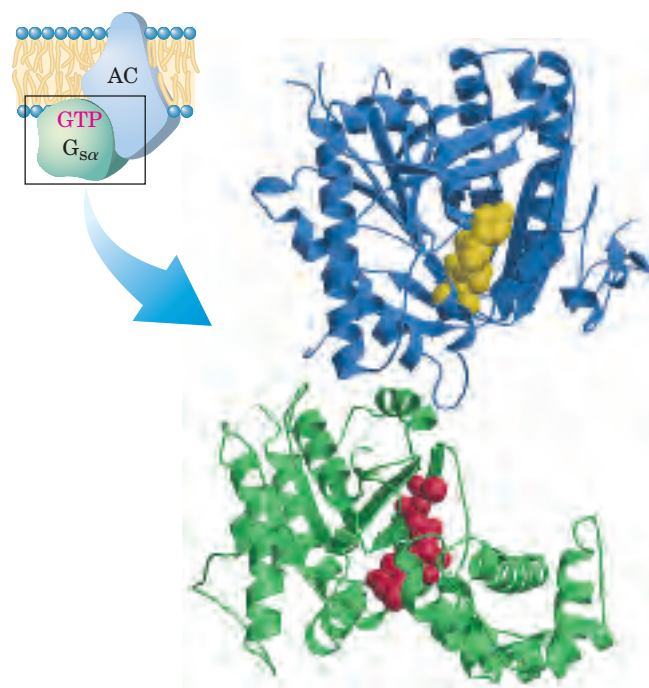


FIGURE 12-13 Interaction of $G_{s\alpha}$ with adenylyl cyclase. (PDB ID 1AZS) The soluble catalytic core of the adenylyl cyclase (AC, blue), severed from its membrane anchor, was cocrystallized with $G_{s\alpha}$ (green) to give this crystal structure. The plant terpene forskolin (yellow) is a drug that strongly stimulates the enzyme, and GTP (red) bound to $G_{s\alpha}$ triggers interaction of $G_{s\alpha}$ with adenylyl cyclase.

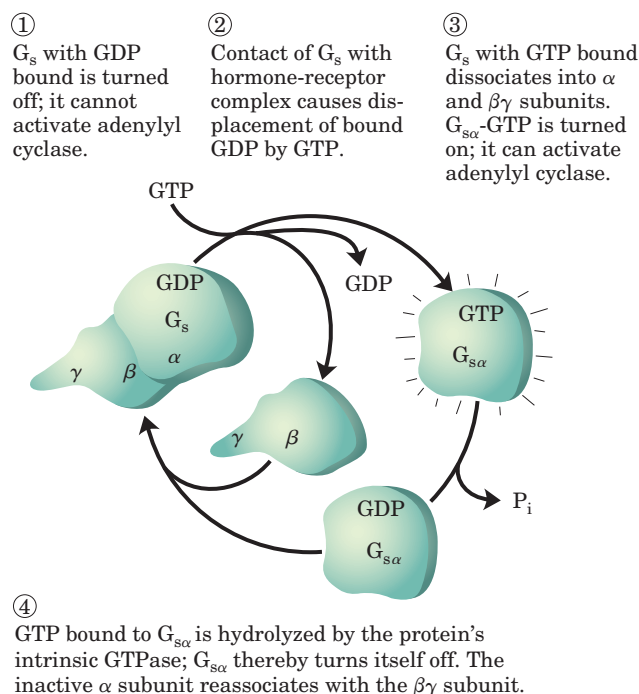


FIGURE 12-14 Self-inactivation of G_s . The steps are further described in the text. The protein's intrinsic GTPase activity, in many cases stimulated by RGS proteins (regulators of G protein signaling), determines how quickly bound GTP is hydrolyzed to GDP and thus how long the G protein remains active.

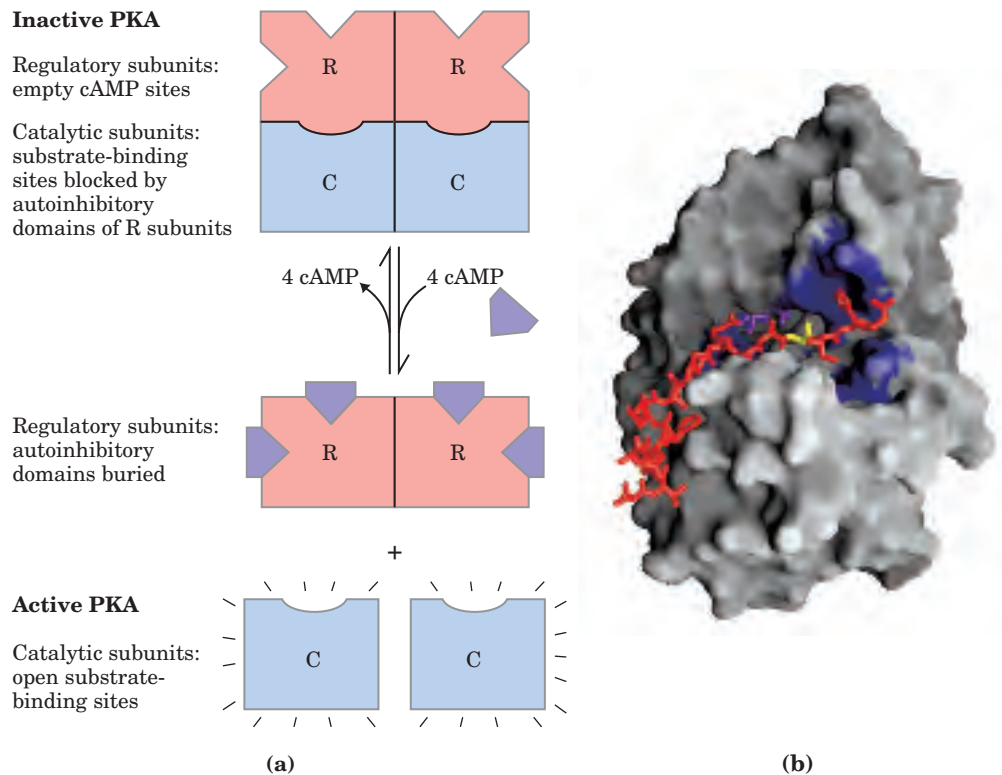


FIGURE 12-15 Activation of cAMP-dependent protein kinase, PKA.

(a) A schematic representation of the inactive R_2C_2 tetramer, in which the autoinhibitory domain of a regulatory (R) subunit occupies the substrate-binding site, inhibiting the activity of the catalytic (C) subunit. Cyclic AMP activates PKA by causing dissociation of the C subunits from the inhibitory R subunits. Activated PKA can phosphorylate a variety of protein substrates (Table 12-3) that contain the PKA consensus sequence (X-Arg-(Arg/Lys)-X-(Ser/Thr)-B, where X is any

residue and B is any hydrophobic residue), including phosphorylase *b* kinase. (b) The substrate-binding region of a catalytic subunit revealed by x-ray crystallography (derived from PDB ID 1JBP). Enzyme side chains known to be critical in substrate binding and specificity are in blue. The peptide substrate (red) lies in a groove in the enzyme surface, with its Ser residue (yellow) positioned in the catalytic site. In the inactive R_2C_2 tetramer, the autoinhibitory domain of R lies in this groove, blocking access to the substrate.

One downstream effect of epinephrine is to activate glycogen phosphorylase *b*. This conversion is promoted by the enzyme phosphorylase *b* kinase, which catalyzes the phosphorylation of two specific Ser residues in phosphorylase *b*, converting it to phosphorylase *a* (see Fig. 6-31). Cyclic AMP does not affect phosphorylase *b* kinase directly. Rather, **cAMP-dependent protein kinase**, also called **protein kinase A** or **PKA**, which is allosterically activated by cAMP (Fig. 12-12, step ⑤), catalyzes the phosphorylation of inactive phosphorylase *b* kinase to yield the active form.

The inactive form of PKA contains two catalytic subunits (C) and two regulatory subunits (R) (Fig. 12-15a), which are similar in sequence to the catalytic and regulatory domains of PKG (cGMP-dependent protein kinase). The tetrameric R_2C_2 complex is catalytically inactive, because an autoinhibitory domain of each R subunit occupies the substrate-binding site of each C subunit. When cAMP binds to two sites on each R subunit, the R subunits undergo a conformational change and the R_2C_2 complex dissociates to yield two free,

catalytically active C subunits. This same basic mechanism—displacement of an autoinhibitory domain—mediates the allosteric activation of many types of protein kinases by their second messengers (as in Figs 12-7 and 12-23, for example).

As indicated in Figure 12-12 (step ⑥), PKA regulates a number of enzymes (Table 12-3). Although the proteins regulated by cAMP-dependent phosphorylation have diverse functions, they share a region of sequence similarity around the Ser or Thr residue that undergoes phosphorylation, a sequence that marks them for regulation by PKA. The catalytic site of PKA (Fig. 12-15b) interacts with several residues near the Thr or Ser residue in the target protein, and these interactions define the substrate specificity. Comparison of the sequences of a number of protein substrates for PKA has yielded the consensus sequence—the specific neighboring residues needed to mark a Ser or Thr residue for phosphorylation (see Table 12-3).

Signal transduction by adenylyl cyclase entails several steps that amplify the original hormone signal (Fig.

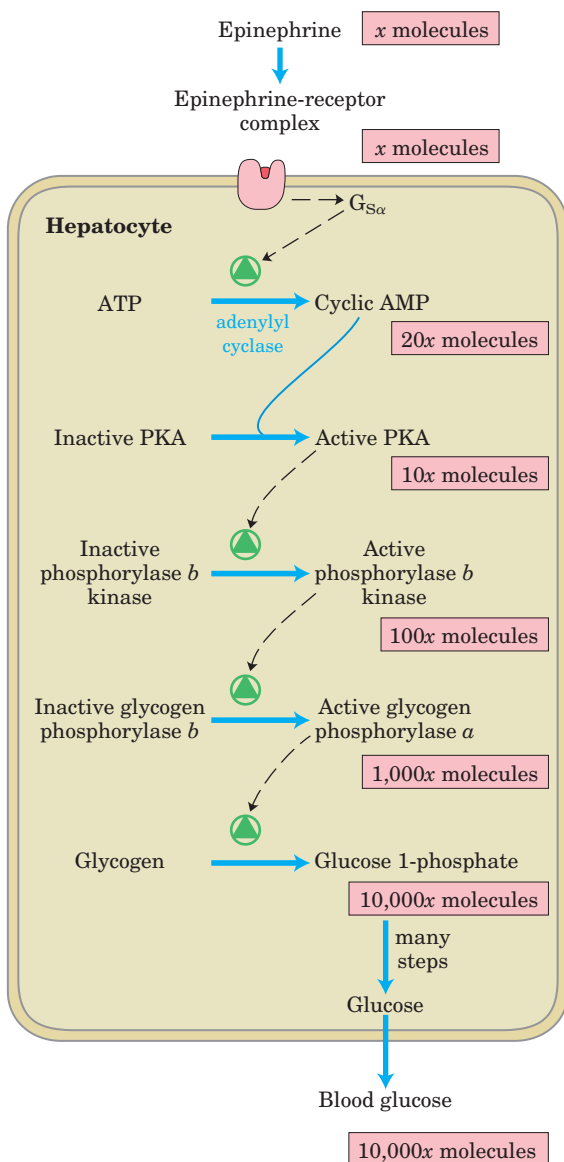
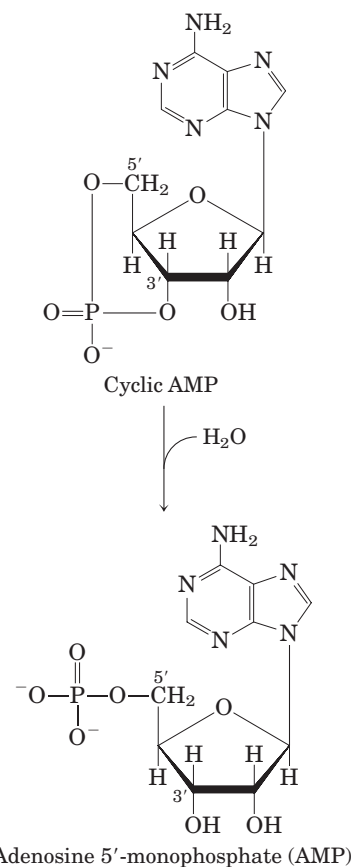


FIGURE 12-16 Epinephrine cascade. Epinephrine triggers a series of reactions in hepatocytes in which catalysts activate catalysts, resulting in great amplification of the signal. Binding of a small number of molecules of epinephrine to specific β -adrenergic receptors on the cell surface activates adenylyl cyclase. To illustrate amplification, we show 20 molecules of cAMP produced by each molecule of adenylyl cyclase, the 20 cAMP molecules activating 10 molecules of PKA, each PKA molecule activating 10 molecules of the next enzyme (a total of 100), and so forth. These amplifications are probably gross underestimates.

12-16). First, the binding of one hormone molecule to one receptor catalytically activates several G_s molecules. Next, by activating a molecule of adenylyl cyclase, each active $G_{s\alpha}$ molecule stimulates the catalytic synthesis of many molecules of cAMP. The second messenger cAMP now activates PKA, each molecule of which catalyzes the phosphorylation of many molecules of the target protein—phosphorylase b kinase in Figure 12-16. This

kinase activates glycogen phosphorylase b , which leads to the rapid mobilization of glucose from glycogen. The net effect of the cascade is amplification of the hormonal signal by several orders of magnitude, which accounts for the very low concentration of epinephrine (or any other hormone) required for hormone activity.

Cyclic AMP, the intracellular second messenger in this system, is short-lived. It is quickly degraded by **cyclic nucleotide phosphodiesterase** to 5'-AMP (Fig. 12-12, step ⑦), which is not active as a second messenger:



The intracellular signal therefore persists only as long as the hormone receptor remains occupied by epinephrine. Methyl xanthines such as caffeine and theophylline (a component of tea) inhibit the phosphodiesterase, increasing the half-life of cAMP and thereby potentiating agents that act by stimulating adenylyl cyclase.

The β -Adrenergic Receptor Is Desensitized by Phosphorylation

As noted earlier, signal-transducing systems undergo desensitization when the signal persists. Desensitization of the β -adrenergic receptor is mediated by a protein kinase that phosphorylates the receptor on the intracellular domain that normally interacts with G_s (Fig. 12-17). When the receptor is occupied by epinephrine,

TABLE 12-3 Some Enzymes and Other Proteins Regulated by cAMP-Dependent Phosphorylation (by PKA)

Enzyme/protein	Sequence phosphorylated*	Pathway/process regulated
Glycogen synthase	RASCTSSS	Glycogen synthesis
Phosphorylase <i>b</i> kinase		
α subunit	VEFRRLSI	Glycogen breakdown
β subunit	RTKRSGSV	
Pyruvate kinase (rat liver)	GVLRRASVAZL	Glycolysis
Pyruvate dehydrogenase complex (type L)	GYLRRASV	Pyruvate to acetyl-CoA
Hormone-sensitive lipase	PMRRSV	Triacylglycerol mobilization and fatty acid oxidation
Phosphofructokinase-2/fructose 2,6-bisphosphatase	LQRRRGSSIPQ	Glycolysis/gluconeogenesis
Tyrosine hydroxylase	FIGRRQSL	Synthesis of L-DOPA, dopamine, norepinephrine, and epinephrine
Histone H1	AKRKASGPPVS	DNA condensation
Histone H2B	KKAKASRKESYSVYVK	DNA condensation
Cardiac phospholamban (cardiac pump regulator)	AIRRAST	Intracellular $[Ca^{2+}]$
Protein phosphatase-1 inhibitor-1	IRRRRPTP	Protein dephosphorylation
PKA consensus sequence†	XR(R/K)X(S/T)B	Many

*The phosphorylated S or T residue is shown in red. All residues are given as their one-letter abbreviations (see Table 3-1).

†X is any amino acid; B is any hydrophobic amino acid.

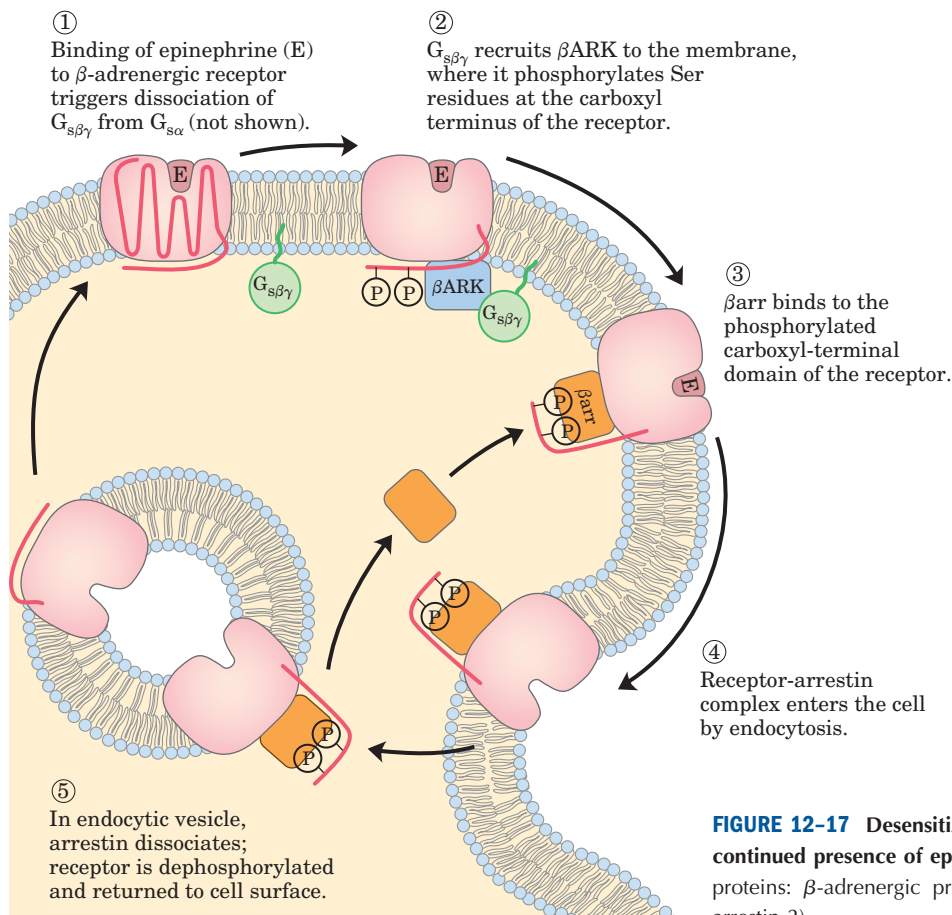


FIGURE 12-17 Desensitization of the β -adrenergic receptor in the continued presence of epinephrine. This process is mediated by two proteins: β -adrenergic protein kinase (β ARK) and β -arrestin (β arr; arrestin 2).

β -adrenergic receptor kinase (β ARK) phosphorylates Ser residues near the carboxyl terminus of the receptor. Normally located in the cytosol, β ARK is drawn to the plasma membrane by its association with the $G_{s\beta\gamma}$ subunits and is thus positioned to phosphorylate the receptor. The phosphorylation creates a binding site for the protein **β -arrestin (β arr)**, also called arrestin 2, and binding of β -arrestin effectively prevents interaction between the receptor and the G protein. The binding of β -arrestin also facilitates receptor sequestration, the removal of receptors from the plasma membrane by endocytosis into small intracellular vesicles. Receptors in the endocytic vesicles are dephosphorylated, then returned to the plasma membrane, completing the circuit and resensitizing the system to epinephrine. β -Adrenergic receptor kinase is a member of a family of **G protein-coupled receptor kinases (GRKs)**, all of which phosphorylate serpentine receptors on their carboxyl-terminal cytosolic domains and play roles similar to that of β ARK in desensitization and resensitization of their receptors. At least five different GRKs and four different arrestins are encoded in the human genome; each GRK is capable of desensitizing a subset of the serpentine receptors, and each arrestin can interact with many different types of phosphorylated receptors.

While preventing the signal from a serpentine receptor from reaching its associated G protein, arrestins can also initiate a second signaling cascade, by acting as **scaffold proteins** that bring together several protein kinases that function in a cascade. For example, the β -arrestin associated with the serpentine receptor for angiotensin, a potent regulator of blood pressure, binds the three protein kinases Raf-1, MEK1, and ERK (Fig. 12–18), serving as a scaffold that facilitates any signaling process, such as insulin signaling (Fig. 12–6), that requires these three protein kinases to interact. This is one of many known examples of cross-talk between systems triggered by different ligands (angiotensin and insulin, in this case).

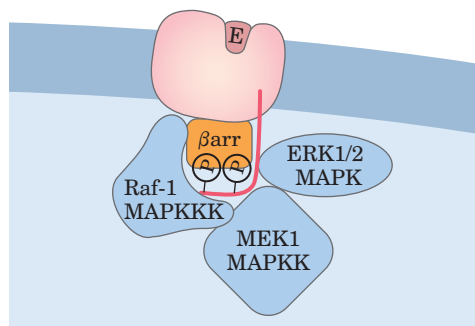


FIGURE 12–18 β -Arrestin uncouples the serpentine receptor from its G protein and brings together the three enzymes of the MAPK cascade. The effect is that one stimulus triggers two distinct response pathways: the path activated by the G protein and the MAPK cascade.

Cyclic AMP Acts as a Second Messenger for a Number of Regulatory Molecules

Epinephrine is only one of many hormones, growth factors, and other regulatory molecules that act by changing the intracellular [cAMP] and thus the activity of PKA (Table 12–4). For example, glucagon binds to its receptors in the plasma membrane of adipocytes, activating (via a G_s protein) adenylyl cyclase. PKA, stimulated by the resulting rise in [cAMP], phosphorylates and activates two proteins critical to the conversion of stored fat to fatty acids (perilipin and hormone-sensitive triacylglycerol lipase; see Fig. 17–3), leading to the mobilization of fatty acids. Similarly, the peptide hormone ACTH (adrenocorticotrophic hormone, also called corticotropin), produced by the anterior pituitary, binds to specific receptors in the adrenal cortex, activating adenylyl cyclase and raising the intracellular [cAMP]. PKA then phosphorylates and activates several of the enzymes required for the synthesis of cortisol and other steroid hormones. The catalytic subunit of PKA can also move into the nucleus, where it phosphorylates a protein that alters the expression of specific genes.

Some hormones act by *inhibiting* adenylyl cyclase, *lowering* cAMP levels, and *suppressing* protein phosphorylation. For example, the binding of somatostatin to its receptor leads to activation of an **inhibitory G protein**, or G_i , structurally homologous to G_s , that inhibits adenylyl cyclase and lowers [cAMP]. Somatostatin therefore counterbalances the effects of glucagon. In adipose tissue, prostaglandin E_1 (PGE_1 ; see Fig. 10–18b) inhibits adenylyl cyclase, thus lowering [cAMP] and slowing the

TABLE 12–4 Some Signals That Use cAMP as Second Messenger

Corticotropin (ACTH)
Corticotropin-releasing hormone (CRH)
Dopamine [D_1 , D_2]*
Epinephrine (β -adrenergic)
Follicle-stimulating hormone (FSH)
Glucagon
Histamine [H_2]*
Luteinizing hormone (LH)
Melanocyte-stimulating hormone (MSH)
Odorants (many)
Parathyroid hormone
Prostaglandins E_1 , E_2 (PGE_1 , PGE_2)
Serotonin [5-HT-1a, 5-HT-2]*
Somatostatin
Tastants (sweet, bitter)
Thyroid-stimulating hormone (TSH)

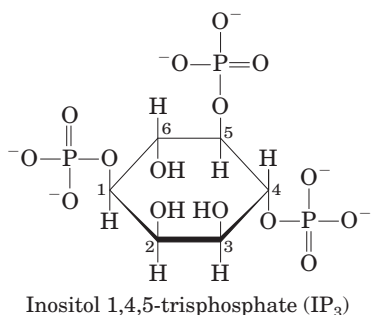
*Receptor subtypes in square brackets. Subtypes may have different transduction mechanisms. For example, serotonin is detected in some tissues by receptor subtypes 5-HT-1a and 5-HT-1b, which act through adenylyl cyclase and cAMP, and in other tissues by receptor subtype 5-HT-1c, acting through the phospholipase C- IP_3 mechanism (see Table 12–5).

mobilization of lipid reserves triggered by epinephrine and glucagon. In certain other tissues PGE₁ stimulates cAMP synthesis, because its receptors are coupled to adenylyl cyclase through a stimulatory G protein, G_s. In tissues with α_2 -adrenergic receptors, epinephrine lowers [cAMP], because the α_2 receptors are coupled to adenylyl cyclase through an inhibitory G protein, G_i. In short, an extracellular signal such as epinephrine or PGE₁ can have quite different effects on different tissues or cell types, depending on three factors: the type of receptor in each tissue, the type of G protein (G_s or G_i) with which the receptor is coupled, and the set of PKA target enzymes in the cells.

A fourth factor that explains how so many signals can be mediated by a single second messenger (cAMP) is the confinement of the signaling process to a specific region of the cell by scaffold proteins. AKAPs (*A* kinase anchoring proteins) are bivalent; one part binds to the R subunit of PKA, and another to a specific structure within the cell, confining the PKA to the vicinity of that structure. For example, specific AKAPs bind PKA to microtubules, actin filaments, Ca²⁺ channels, mitochondria, and the nucleus. Different types of cells have different AKAPs, so cAMP might stimulate phosphorylation of mitochondrial proteins in one cell and phosphorylation of actin filaments in another. In studies of the intracellular localization of biochemical changes, biochemistry meets cell biology, and techniques that cross this boundary become invaluable (Box 12-2).

Two Second Messengers Are Derived from Phosphatidylinositols

A second class of serpentine receptors are coupled through a G protein to a plasma membrane **phospholipase C (PLC)** that is specific for the plasma membrane lipid phosphatidylinositol 4,5-bisphosphate (see Fig. 10-15). This hormone-sensitive enzyme catalyzes the formation of two potent second messengers: **diacylglycerol** and **inositol 1,4,5-trisphosphate**, or **IP₃** (not to be confused with PIP₃, p. 431).



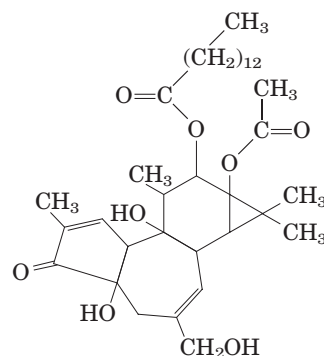
When a hormone of this class (Table 12-5) binds its specific receptor in the plasma membrane (Fig. 12-19, step ①), the receptor-hormone complex catalyzes GTP-GDP exchange on an associated G protein, **G_q**

(step ②), activating it exactly as the β -adrenergic receptor activates G_s (Fig. 12-12). The activated G_q in turn activates a specific membrane-bound PLC (step ③), which catalyzes the production of the two second messengers diacylglycerol and IP₃ by hydrolysis of phosphatidylinositol 4,5-bisphosphate in the plasma membrane (step ④).

Inositol trisphosphate, a water-soluble compound, diffuses from the plasma membrane to the endoplasmic reticulum, where it binds to specific IP₃ receptors and causes Ca²⁺ channels within the ER to open. Sequestered Ca²⁺ is thus released into the cytosol (step ⑤), and the cytosolic [Ca²⁺] rises sharply to about 10⁻⁶ M. One effect of elevated [Ca²⁺] is the activation of **protein kinase C (PKC)**. Diacylglycerol cooperates with Ca²⁺ in activating PKC, thus also acting as a second messenger (step ⑥). PKC phosphorylates Ser or Thr residues of specific target proteins, changing their catalytic activities (step ⑦). There are a number of isozymes of PKC, each with a characteristic tissue distribution, target protein specificity, and role.



The action of a group of compounds known as **tumor promoters** is attributable to their effects on PKC. The best understood of these are the phorbol esters, synthetic compounds that are potent activators of PKC. They apparently mimic cellular diacylglycerol as second messengers, but unlike naturally occurring diacylglycerols they are not rapidly metabolized. By continuously activating PKC, these synthetic tumor promoters interfere with the normal regulation of cell growth and division (discussed in Section 12.10). ■



Calcium Is a Second Messenger in Many Signal Transductions

In many cells that respond to extracellular signals, Ca²⁺ serves as a second messenger that triggers intracellular responses, such as exocytosis in neurons and endocrine cells, contraction in muscle, and cytoskeletal rearrangement during amoeboid movement. Normally, cytosolic [Ca²⁺] is kept very low (<10⁻⁷ M) by the action of Ca²⁺ pumps in the ER, mitochondria, and plasma membrane. Hormonal, neural, or other stimuli cause either an influx of Ca²⁺ into the cell through specific

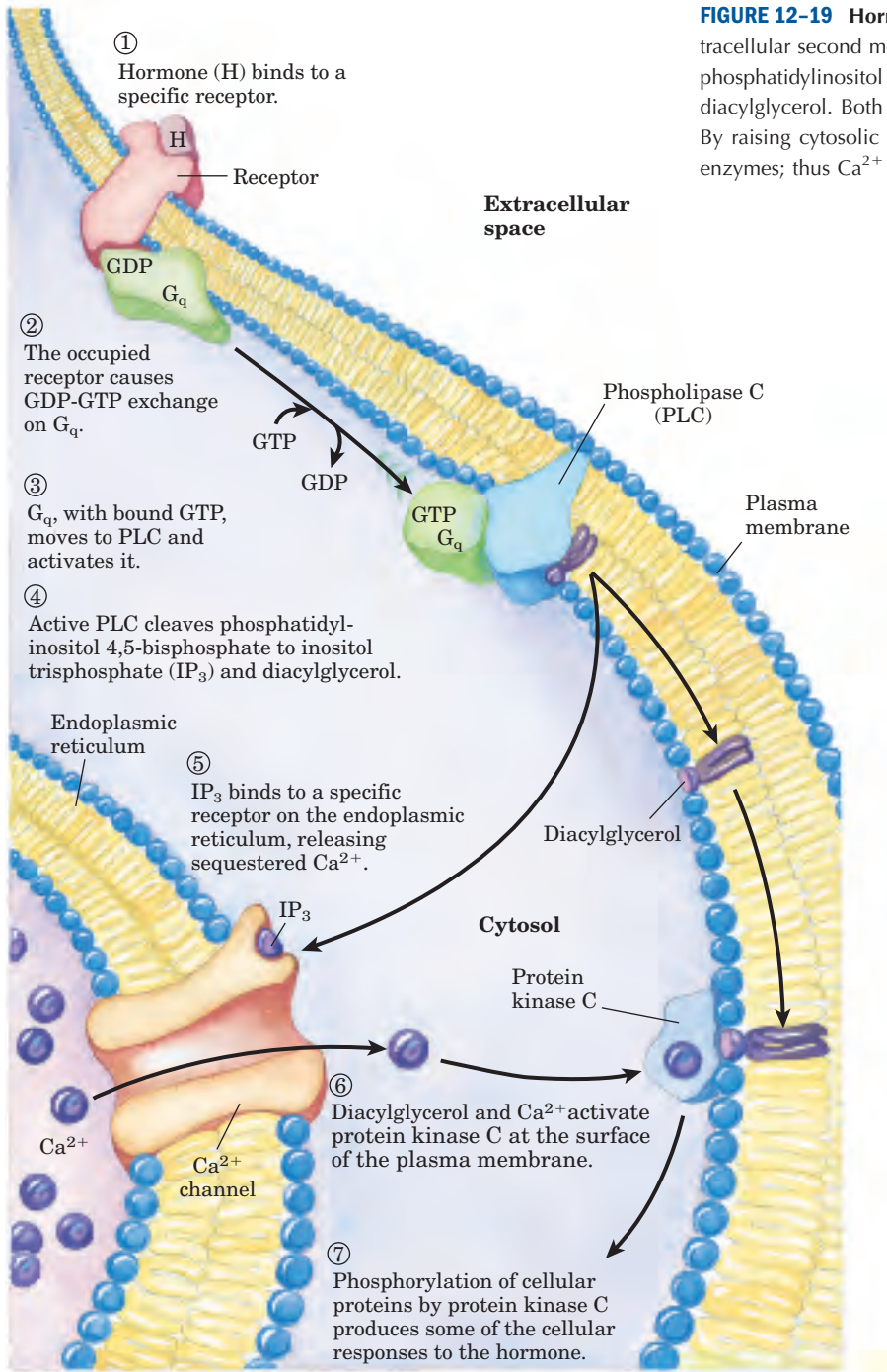


FIGURE 12-19 Hormone-activated phospholipase C and IP_3 . Two intracellular second messengers are produced in the hormone-sensitive phosphatidylinositol system: inositol 1,4,5-trisphosphate (IP_3) and diacylglycerol. Both contribute to the activation of protein kinase C. By raising cytosolic $[Ca^{2+}]$, IP_3 also activates other Ca^{2+} -dependent enzymes; thus Ca^{2+} also acts as a second messenger.

TABLE 12-5 Some Signals That Act through Phospholipase C and IP_3

Acetylcholine [muscarinic M_1]	Gastrin-releasing peptide	Platelet-derived growth factor (PDGF)
α_1 -Adrenergic agonists	Glutamate	Serotonin [5-HT-1c]*
Angiogenin	Gonadotropin-releasing hormone (GRH)	Thyrotropin-releasing hormone (TRH)
Angiotensin II	Histamine [H_1]*	Vasopressin
ATP [P_{2x} and P_{2y}]*	Light (<i>Drosophila</i>)	
Auxin	Oxytocin	

*Receptor subtypes are in square brackets; see footnote to Table 12-4.

Ca^{2+} channels in the plasma membrane or the release of sequestered Ca^{2+} from the ER or mitochondria, in either case raising the cytosolic $[\text{Ca}^{2+}]$ and triggering a cellular response.

Very commonly, $[\text{Ca}^{2+}]$ does not simply rise and then decrease, but rather oscillates with a period of a few seconds (Fig. 12–20), even when the extracellular concentration of hormone remains constant. The mechanism underlying $[\text{Ca}^{2+}]$ oscillations presumably entails feedback regulation by Ca^{2+} of either the phospholipase

that generates IP_3 or the ion channel that regulates Ca^{2+} release from the ER, or both. Whatever the mechanism, the effect is that one kind of signal (hormone concentration, for example) is converted into another (frequency and amplitude of intracellular $[\text{Ca}^{2+}]$ “spikes”).

Changes in intracellular $[\text{Ca}^{2+}]$ are detected by Ca^{2+} -binding proteins that regulate a variety of Ca^{2+} -dependent enzymes. **Calmodulin (CaM)** (M_r 17,000) is an acidic protein with four high-affinity Ca^{2+} -binding sites. When intracellular $[\text{Ca}^{2+}]$ rises to about 10^{-6} M (1 μM), the binding of Ca^{2+} to calmodulin drives a conformational change in the protein (Fig. 12–21). Calmodulin associates with a variety of proteins and, in its Ca^{2+} -bound state, modulates their activities. Calmodulin is a member of a family of Ca^{2+} -binding proteins that also includes troponin (p. 185), which triggers skeletal muscle contraction in response to increased $[\text{Ca}^{2+}]$. This family shares a characteristic Ca^{2+} -binding structure, the EF hand (Fig. 12–21c).

Calmodulin is also an integral subunit of a family of enzymes, the **Ca^{2+} /calmodulin-dependent protein kinases (CaM kinases I–IV)**. When intracellular $[\text{Ca}^{2+}]$ increases in response to some stimulus, calmodulin binds Ca^{2+} , undergoes a change in conformation, and activates the CaM kinase. The kinase then phosphorylates a number of target enzymes, regulating their activities. Calmodulin is also a regulatory subunit of phosphorylase *b* kinase of muscle, which is activated by Ca^{2+} . Thus Ca^{2+} triggers ATP-requiring muscle contractions while also activating glycogen breakdown, providing fuel for ATP synthesis. Many other enzymes are also known to be modulated by Ca^{2+} through calmodulin (Table 12–6).

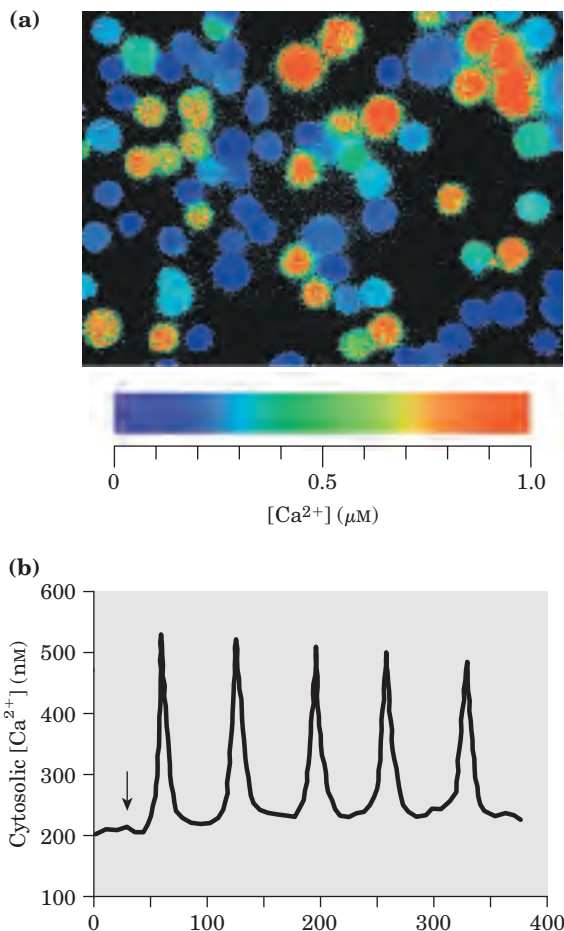


FIGURE 12–20 Triggering of oscillations in intracellular $[\text{Ca}^{2+}]$ by extracellular signals. **(a)** A dye (fura) that undergoes fluorescence changes when it binds Ca^{2+} is allowed to diffuse into cells, and its instantaneous light output is measured by fluorescence microscopy. Fluorescence intensity is represented by color; the color scale relates intensity of color to $[\text{Ca}^{2+}]$, allowing determination of the absolute $[\text{Ca}^{2+}]$. In this case, thymocytes (cells of the thymus) have been stimulated with extracellular ATP, which raises their internal $[\text{Ca}^{2+}]$. The cells are heterogeneous in their responses; some have high intracellular $[\text{Ca}^{2+}]$ (red), others much lower (blue). **(b)** When such a probe is used to measure $[\text{Ca}^{2+}]$ in a single hepatocyte, we observe that the agonist norepinephrine (added at the arrow) causes oscillations of $[\text{Ca}^{2+}]$ from 200 to 500 nM. Similar oscillations are induced in other cell types by other extracellular signals.

TABLE 12–6 Some Proteins Regulated by Ca^{2+} and Calmodulin

Adenyl cyclase (brain)
Ca^{2+} /calmodulin-dependent protein kinases (CaM kinases I to IV)
Ca^{2+} -dependent Na^+ channel (<i>Paramecium</i>)
Ca^{2+} -release channel of sarcoplasmic reticulum
Calcineurin (phosphoprotein phosphatase 2B)
cAMP phosphodiesterase
cAMP-gated olfactory channel
cGMP-gated Na^+ , Ca^{2+} channels (rod and cone cells)
Glutamate decarboxylase
Myosin light chain kinases
NAD^+ kinase
Nitric oxide synthase
Phosphoinositide 3-kinase
Plasma membrane Ca^{2+} ATPase (Ca^{2+} pump)
RNA helicase (p68)

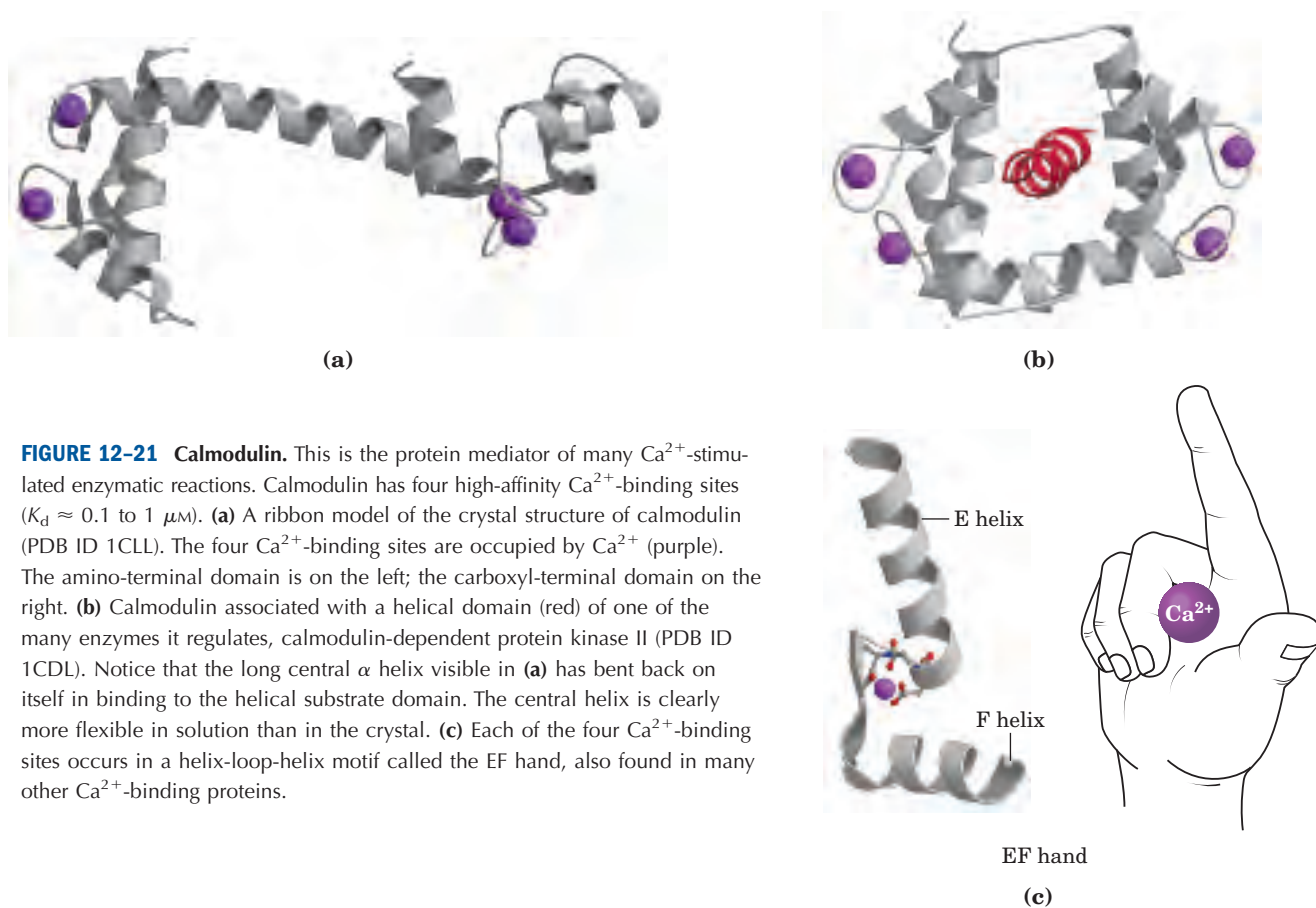


FIGURE 12-21 Calmodulin. This is the protein mediator of many Ca^{2+} -stimulated enzymatic reactions. Calmodulin has four high-affinity Ca^{2+} -binding sites ($K_d \approx 0.1$ to $1 \mu\text{M}$). **(a)** A ribbon model of the crystal structure of calmodulin (PDB ID 1CLL). The four Ca^{2+} -binding sites are occupied by Ca^{2+} (purple). The amino-terminal domain is on the left; the carboxyl-terminal domain on the right. **(b)** Calmodulin associated with a helical domain (red) of one of the many enzymes it regulates, calmodulin-dependent protein kinase II (PDB ID 1CDL). Notice that the long central α helix visible in **(a)** has bent back on itself in binding to the helical substrate domain. The central helix is clearly more flexible in solution than in the crystal. **(c)** Each of the four Ca^{2+} -binding sites occurs in a helix-loop-helix motif called the EF hand, also found in many other Ca^{2+} -binding proteins.

SUMMARY 12.4 G Protein-Coupled Receptors and Second Messengers

- A large family of plasma membrane receptors with seven transmembrane segments act through heterotrimeric G proteins. On ligand binding, these receptors catalyze the exchange of GTP for GDP bound to an associated G protein, forcing dissociation of the α subunit of the G protein. This subunit stimulates or inhibits the activity of a nearby membrane-bound enzyme, changing the level of its second messenger product.
- The β -adrenergic receptor binds epinephrine, then through a stimulatory G protein, G_s , activates adenylyl cyclase in the plasma membrane. The cAMP produced by adenylyl cyclase is an intracellular second messenger that stimulates cAMP-dependent protein kinase, which mediates the effects of epinephrine by phosphorylating key proteins, changing their enzymatic activities or structural features.
- The cascade of events in which a single molecule of hormone activates a catalyst that in turn activates another catalyst, and so on, results in large signal amplification; this is characteristic of most hormone-activated systems.
- Some receptors stimulate adenylyl cyclase through G_s ; others inhibit it through G_i . Thus cellular [cAMP] reflects the integrated input of two (or more) signals.
- Cyclic AMP is eventually eliminated by cAMP phosphodiesterase, and G_s turns itself off by hydrolysis of its bound GTP to GDP. When the epinephrine signal persists, β -adrenergic receptor-specific protein kinase and arrestin 2 temporarily desensitize the receptor and cause it to move into intracellular vesicles. In some cases, arrestin also acts as a scaffold protein, bringing together protein components of a signaling pathway such as the MAPK cascade.
- Some serpentine receptors are coupled to a plasma membrane phospholipase C that cleaves PIP_2 to diacylglycerol and IP_3 . By opening Ca^{2+} channels in the endoplasmic reticulum, IP_3 raises cytosolic $[\text{Ca}^{2+}]$. Diacylglycerol and Ca^{2+} act together to activate protein kinase C, which phosphorylates and changes the activity of specific cellular proteins. Cellular $[\text{Ca}^{2+}]$ also regulates a number of other enzymes, often through calmodulin.

BOX 12-2 WORKING IN BIOCHEMISTRY

FRET: Biochemistry Visualized in a Living Cell

Fluorescent probes are commonly used to detect rapid biochemical changes in single living cells. They can be designed to give an essentially instantaneous report (within nanoseconds) on the changes in intracellular concentration of a second messenger or in the activity of a protein kinase. Furthermore, fluorescence microscopy has sufficient resolution to reveal where in the cell such changes are occurring. In one widely used procedure, the fluorescent probes are derived from a naturally occurring fluorescent protein, the green fluorescent protein (GFP) of the jellyfish *Aequorea victoria* (Fig. 1).

When excited by absorption of a photon of light, GFP emits a photon (that is, it fluoresces) in the green region of the spectrum. GFP is an 11-stranded β barrel, and the light-absorbing/emitting center of the protein (its chromophore) comprises the tripeptide Ser⁶⁵–Tyr⁶⁶–Gly⁶⁷, located within the barrel (Fig. 2). Variants of this protein, with different fluorescence spectra, can be produced by genetic engineering of the GFP gene. For example, in the yellow fluorescent protein (YFP), Ala²⁰⁶ in GFP is replaced by a Lys residue, changing the wavelength of light absorption



FIGURE 1 *Aequorea victoria*, a jellyfish abundant in Puget Sound, Washington State.

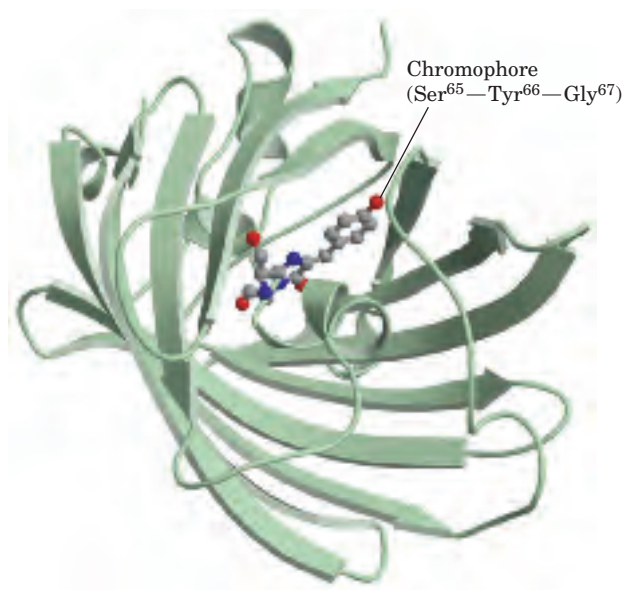


FIGURE 2 Green fluorescent protein (GFP), with the fluorescent chromophore shown in ball-and-stick form (derived from PDB ID 1GFL).

and fluorescence. Other variants of GFP fluoresce blue (BFP) or cyan (CFP) light, and a related protein (mRFP1) fluoresces red light (Fig. 3). GFP and its variants are compact structures that retain their ability to fold into their native β -barrel conformation even when fused with another protein. Investigators are using these fluorescent hybrid proteins as spectroscopic rulers to measure distances between interacting components within a cell.

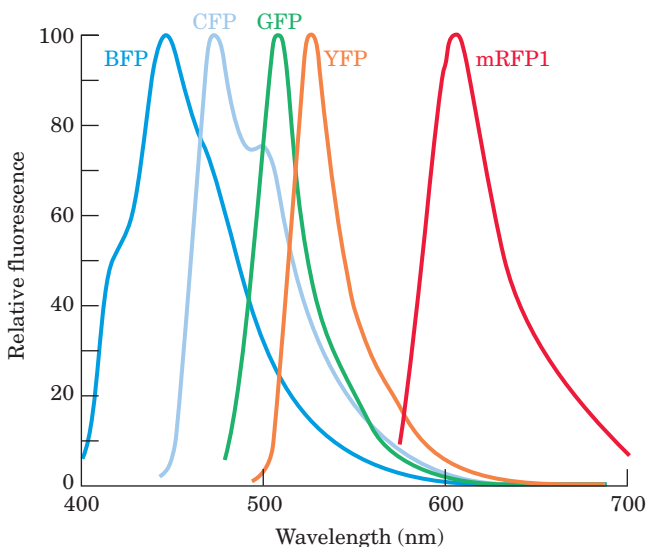


FIGURE 3 Emission spectra of GFP variants.

An excited fluorescent molecule such as GFP or YFP can dispose of the energy from the absorbed photon in either of two ways: (1) by fluorescence, emitting a photon of slightly longer wavelength (lower energy) than the exciting light, or (2) by nonradiative **fluorescence resonance energy transfer (FRET)**, in which the energy of the excited molecule (the donor) passes directly to a nearby molecule (the acceptor) *without emission of a photon*, exciting the acceptor (Fig. 4). The acceptor can now decay to its ground state by fluorescence; the emitted photon has a longer wavelength (lower energy) than both the original exciting light and the fluorescence emission of the donor. This second mode of decay (FRET) is possible only when donor and acceptor are close enough (within 1 to 50 Å); the efficiency of FRET is inversely proportional to the *sixth power* of the distance between donor and acceptor. Thus very small changes in the distance between donor and acceptor register as very large changes in FRET, measured as the fluorescence of the acceptor molecule when the donor is excited. With sufficiently sensitive light detectors, this fluorescence signal can be located to specific regions of a single, living cell.

FRET has been used to measure [cAMP] in living cells. The gene for GFP is fused with that for the regulatory subunit (R) of cAMP-dependent protein kinase, and the gene for BFP is fused with that for the

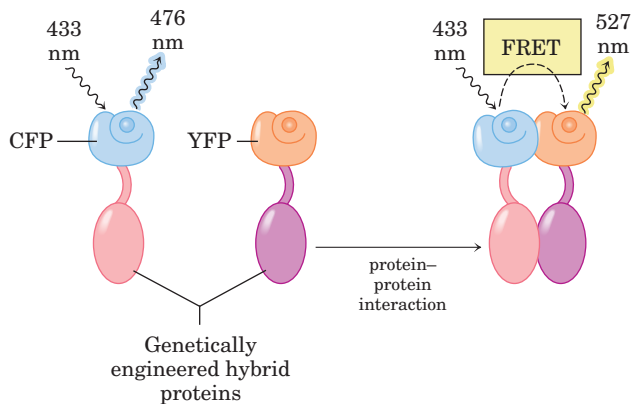


FIGURE 4 When the donor protein (CFP) is excited with monochromatic light of wavelength 433 nm, it emits fluorescent light at 476 nm (left). When the (red) protein fused with CFP interacts with the (purple) protein fused with YFP, that interaction brings CFP and YFP close enough to allow fluorescence resonance energy transfer (FRET) between them. Now, when CFP absorbs light of 433 nm, instead of fluorescing at 476 nm, it transfers energy directly to YFP, which then fluoresces at its characteristic emission wavelength, 527 nm. The ratio of light emission at 527 and 476 nm is therefore a measure of the interaction of the red and purple protein.

catalytic subunit (C) (Fig. 5). When these two hybrid proteins are expressed in a cell, BFP (donor; excitation at 380 nm, emission at 460 nm) and GFP (acceptor; excitation at 475 nm, emission at 545 nm) in the inactive PKA (R_2C_2 tetramer) are close enough to undergo FRET. Wherever in the cell [cAMP] increases, the R_2C_2 complex dissociates into R_2 and 2C and the FRET signal is lost, because donor and acceptor are now too far apart for efficient FRET. Viewed in the fluorescence microscope, the region of higher [cAMP] has a minimal GFP signal and higher BFP signal. Measuring the ratio of emission at 460 nm and 545 nm gives a sensitive measure of the change in [cAMP]. By determining this ratio for all regions of the cell, the investigator can generate a false color image of the

(continued on next page)

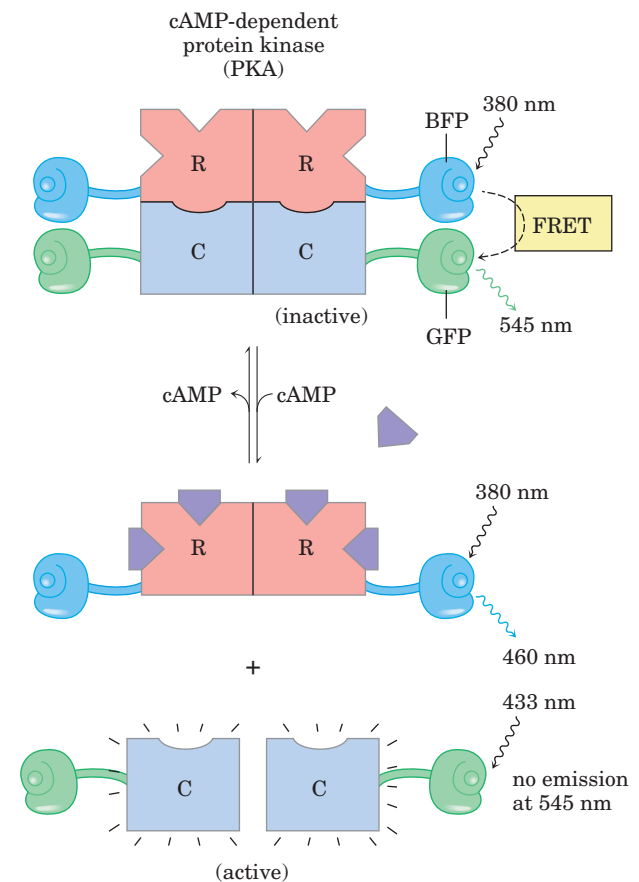


FIGURE 5 Measuring [cAMP] with FRET. Gene fusion creates hybrid proteins that exhibit FRET when the PKA regulatory and catalytic subunits are associated (low [cAMP]). When [cAMP] rises, the subunits dissociate, and FRET ceases. The ratio of emission at 460 nm (dissociated) and 545 nm (complexed) thus offers a sensitive measure of [cAMP].

BOX 12-2 WORKING IN BIOCHEMISTRY (continued from previous page)

cell in which the ratio, or relative [cAMP], is represented by the intensity of the color. Images recorded at timed intervals reveal changes in [cAMP] over time.

A variation of this technology has been used to measure the activity of PKA in a living cell (Fig. 6). Researchers create a hybrid protein containing four elements: YFP (acceptor); a short peptide with a Ser residue surrounded by the consensus sequence for PKA; a P-Ser -binding domain (called 14-3-3); and CFP (donor). When the Ser residue is not phosphorylated, 14-3-3 has no affinity for the Ser residue and the hybrid protein exists in an extended form, with the donor and acceptor too far apart to generate a FRET signal. Wherever PKA is active in the cell, it phosphorylates the Ser residue of the hybrid protein, and 14-3-3 binds to the P-Ser . In doing so, it draws YFP and CFP together and a FRET signal is detected with the fluorescence microscope, revealing the presence of active PKA.

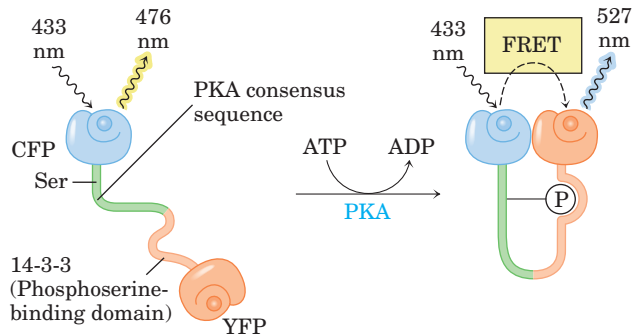


FIGURE 6 Measuring the activity of PKA with FRET. An engineered protein links YFP and CFP via a peptide that contains a Ser residue surrounded by the consensus sequence for phosphorylation by PKA, and the 14-3-3 phosphoserine binding domain. Active PKA phosphorylates the Ser residue, which docks with the 14-3-3 binding domain, bringing the fluorescence proteins close enough to allow FRET to occur, revealing the presence of active PKA.

12.5 Multivalent Scaffold Proteins and Membrane Rafts

About 10% of the 30,000 to 35,000 genes in the human genome encode signaling proteins—receptors, G proteins, enzymes that generate second messengers, protein kinases (>500), proteins involved in desensitization, and ion channels. Not every signaling protein is expressed in a given cell type, but most cells doubtless contain many such proteins. How does one protein find another in a signaling pathway, and how are their interactions regulated? As is becoming clear, the reversible phosphorylation of Tyr, Ser, and Thr residues in signaling proteins creates *docking sites* for other proteins, and many signaling proteins are *multivalent* in that they can interact with several different proteins simultaneously to form multiprotein signaling complexes. In this section we present a few examples to illustrate the general principles of protein interactions in signaling.

Protein Modules Bind Phosphorylated Tyr, Ser, or Thr Residues in Partner Proteins

We have seen that the protein Grb2 in the insulin signaling pathway (Fig. 12-6) binds through its SH2 domain to other proteins that contain exposed P-Tyr residues. The human genome encodes at least 87 SH2-containing proteins, many already known to participate in signaling. The P-Tyr residue is bound in a deep

pocket in an SH2 domain, with each of its phosphate oxygens participating in hydrogen-bonding or electrostatic interactions; the positive charges on two Arg residues figure prominently in the binding. Subtle differences in the structure of SH2 domains in different proteins account for the specificities of their interactions with various P-Tyr -containing proteins. The three to five residues on the carboxyl-terminal side of the P-Tyr residue are critical in determining the specificity of interactions with SH2 domains (Fig. 12-22).

PTB domains (phosphotyrosine-binding domains) also bind P-Tyr in partner proteins, but their critical sequences and three-dimensional structures distinguish them from SH2 domains. The human genome encodes 24 proteins that contain PTB domains, including IRS-1, which we have already met in its role as a scaffold protein in insulin-signal transduction (Fig. 12-6).

Many of the signaling protein kinases, including PKA, PKC, PKG, and members of the MAPK cascade, phosphorylate Ser or Thr residues in their target proteins, which in some cases acquire the ability to interact with partner proteins through the phosphorylated residue, triggering a downstream process. An alphabet soup of domains that bind P-Ser or P-Thr residues has been identified, and more are sure to be found. Each domain favors a certain sequence around the phosphorylated residue, so the domains represent families of highly specific recognition sites, able to bind to a specific subset of phosphorylated proteins.

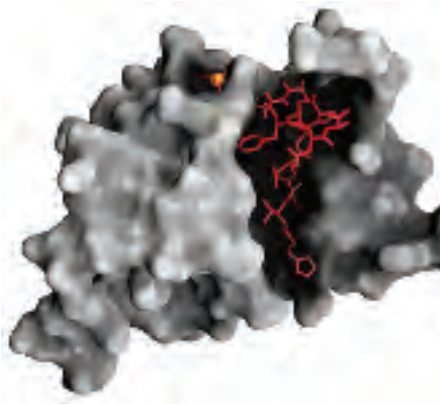
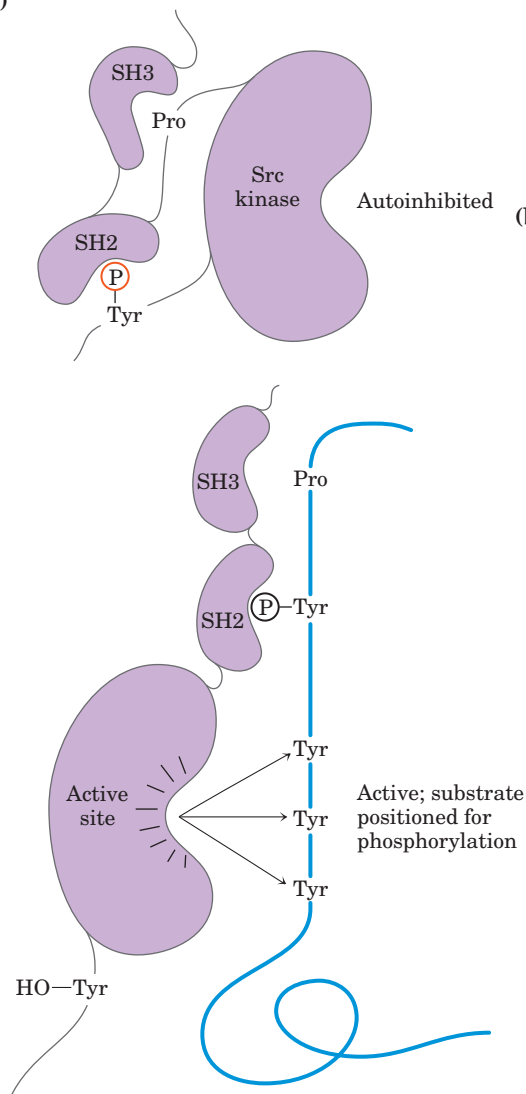


FIGURE 12-22 Structure of an SH2 domain and its interaction with a P-Tyr residue in a partner protein. (PDB ID 1SHC) The SH2 domain is shown as a gray surface contour representation. The phosphorus of the phosphate group in the interacting P-Tyr is visible as an orange sphere; most of the residue is obscured in this view. The next few residues toward the carboxyl end of the partner protein are shown in red. The SH2 domain interacts with P-Tyr (which, as the phosphorylated residue, is assigned the index position 0) and also with the next three residues toward the carboxyl terminus (designated +1, +2, +3). The residues important in the P-Tyr residue are conserved in all SH2 domains. Some SH2 domains (Src, Fyn, Hck, Nck) favor negatively charged residues in the +1 and +2 positions; others (PLC- γ 1, SHP-2) have a long hydrophobic groove that selects for aliphatic residues in positions +1 to +5. These differences define subclasses of SH2 domains that have different partner specificities.

In some cases, the domain-binding partner is internal. Phosphorylation of some protein kinases inhibits their activity by favoring the interaction of an SH2 domain with a P-Tyr in another domain of the same enzyme. For example, the soluble protein Tyr kinase Src, when phosphorylated on a critical Tyr residue, is rendered inactive as an SH2 domain needed to bind to the substrate protein instead binds to an internal P-Tyr

(Fig. 12-23). Glycogen synthase kinase 3 (GSK3) is inactive when phosphorylated on a Ser residue in its autoinhibitory domain (Fig. 12-23b). Dephosphorylation of that domain frees the enzyme to bind and phosphorylate its target proteins. Similarly, the polar head group of the phospholipid PIP_3 , protruding from the inner leaflet of the plasma membrane, provides points of attachment for proteins that contain SH3 and other domains.

(a)



(b)

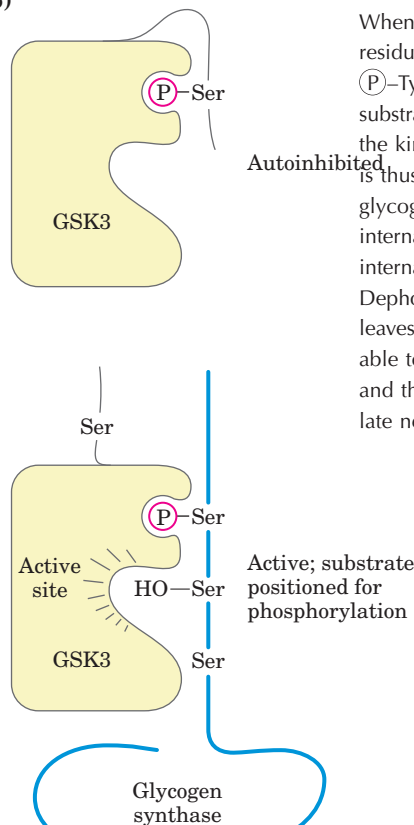


FIGURE 12-23 Mechanism of autoinhibition of Src Tyr kinase and GSK3. (a) In the active form of Src kinase, an SH2 domain binds a P-Tyr in the substrate, and an SH3 domain binds a proline-rich region of the substrate, lining up the active site of the kinase with several target Tyr residues in the substrate. When Src is phosphorylated on a specific Tyr residue, the SH2 domain binds to the internal P-Tyr instead of to the P-Tyr of the substrate, preventing productive binding of the kinase to its protein substrate; the enzyme is thus autoinhibited. (b) In the autoinhibited glycogen synthase kinase 3 (GSK3), an internal P-Ser residue is bound to an internal P-Ser -binding domain (top). Dephosphorylation of this internal Ser residue leaves the P-Ser -binding site of GSK3 available to bind P-Ser in a protein substrate, and thus to position the kinase to phosphorylate neighboring Ser residues (bottom).

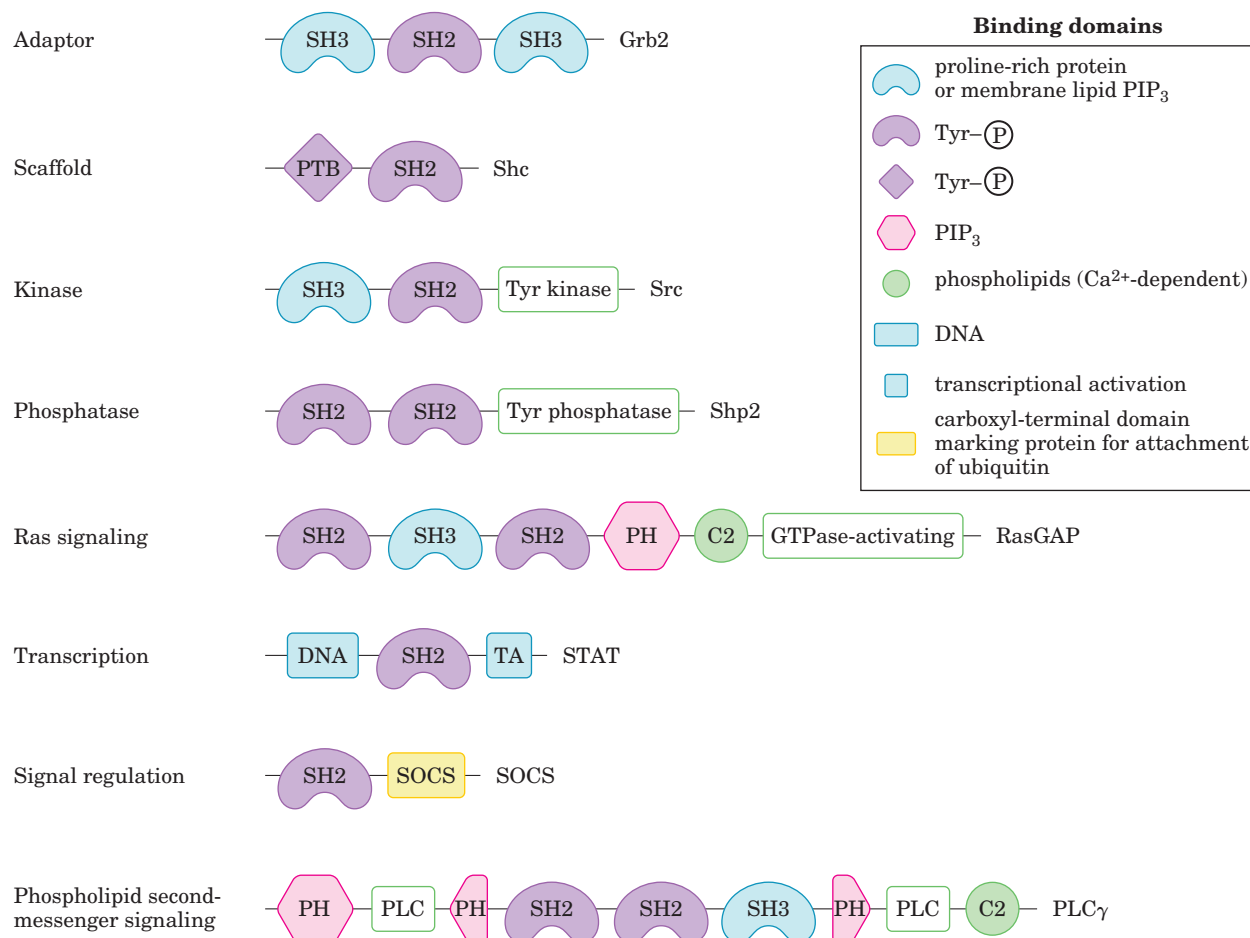


FIGURE 12-24 Some binding modules of signaling proteins. Each protein is represented by a line (with the amino terminus to the left); symbols indicate the location of conserved binding domains (with specificities as listed in the key; PH denotes plextrin homology; other abbreviations explained in the text); green boxes indicate catalytic ac-

tivities. The name of each protein is given at its carboxyl-terminal end. These signaling proteins interact with phosphorylated proteins or phospholipids in many permutations and combinations to form integrated signaling complexes.

Most of the proteins involved in signaling at the plasma membrane have one or more protein- or phospholipid-binding domains; many have three or more, and thus are multivalent in their interactions with other signaling proteins. Figure 12-24 shows a few of the many multivalent proteins known to participate in signaling.

A remarkable picture of signaling pathways has emerged from studies of many signaling proteins and the multiple binding domains they contain (Fig. 12-25). An initial signal results in phosphorylation of the receptor or a target protein, triggering the assembly of large multiprotein complexes, held together on scaffolds made from adaptor proteins with multivalent binding capacities. Some of these complexes have several protein kinases that activate each other in turn, producing a cascade of phosphorylation and a great amplification of the initial signal. Animal cells also have phosphotyrosine phosphatases (PTPases), which remove the phosphate from P-Tyr residues, reversing the effect of phosphorylation. Some of these phosphatases are receptorlike

membrane proteins, presumably controlled by extracellular factors not yet identified; other PTPases are soluble and contain SH2 domains. In addition, animal cells have protein phosphoserine and phosphothreonine phosphatases, which reverse the effects of Ser- and Thr-specific protein kinases. We can see, then, that signaling occurs in protein circuits, effectively hard-wired from signal receptor to response effector and able to be switched off instantly by the hydrolysis of a single phosphate ester bond.

The multivalency of signaling proteins allows for the assembly of many different combinations of signaling modules, each combination presumably suited to particular signals, cell types, and metabolic circumstances. The large variety of protein kinases and of phosphoprotein-binding domains, each with its own specificity (the consensus sequence required in its substrate), provides for many permutations and combinations and many different signaling circuits of extraordinary complexity. And given the variety of specific phosphatases that reverse

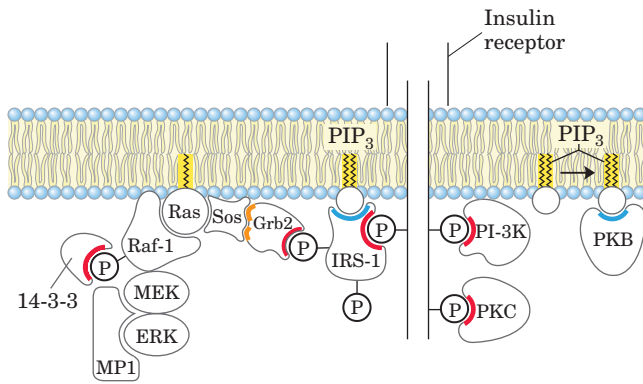


FIGURE 12-25 Insulin-induced formation of supramolecular signaling complexes. The binding of insulin to its receptor sets off a series of events that lead eventually to the formation of membrane-associated complexes involving the 12 signaling proteins shown here, as well as others. Phosphorylation of Tyr residues in the insulin receptor initiates complex formation, and dephosphorylation of any of the phosphoproteins breaks the circuit. Four general types of interaction hold the complex together: the binding of a protein to a second phosphoprotein through SH2 or PTB domains in the first (red); the binding of SH3 domains in the first with proline-rich domains in the second (orange); the binding of PH domains in one protein to the phospholipid PIP₃ in the plasma membrane (blue); or the association of a protein (RAS) with the plasma membrane through a lipid covalently bound to the protein (yellow). Two proteins shown here are not described in the text: 14-3-3, which binds a (P)-Ser in Raf and mediates its interaction with MEK; and MP1, a scaffold protein that cements the links between Raf, MEK, and ERK.

the action of protein kinases, some under specific types of external control, a cell can quickly “disconnect” the entire protein circuitry of a signaling pathway. Together, these mechanisms confer a huge capacity for cellular regulation in response to signals of many types.

Membrane Rafts and Caveolae May Segregate Signaling Proteins

Membrane rafts are regions of the membrane bilayer enriched in sphingolipids, sterols, and certain proteins, including many attached to the bilayer by GPI anchors (Chapter 11). Some receptor Tyr kinases, such as the receptors for epidermal growth factor (EGF) and platelet-derived growth factor (PDGF), appear to be localized in rafts; other signaling proteins, such as the small G protein Ras (which is prenylated) and the heterotrimeric G protein G_s (also prenylated, on the α and γ subunits), are not. Growing evidence suggests that this sequestration of signaling proteins is functionally significant. When cholesterol is removed from rafts by treatment with cyclodextrin (which binds cholesterol and removes it from membranes), the rafts are disrupted and a number of signaling pathways become defective.

How might localization in rafts influence signaling through a receptor? There are several possibilities. If a receptor Tyr kinase in a raft is phosphorylated, and the

phosphotyrosine phosphatase that reverses this phosphorylation is in another raft, then dephosphorylation of the Tyr kinase will be slowed or prevented. If two signaling proteins must interact during transduction of a signal, the probability of encounters between these proteins is greatly enhanced if both are in the same raft. Interactions between scaffold proteins might be strong enough to pull into a raft a signaling protein not normally located there, or strong enough to pull receptors out of a raft. For example, the EGF receptor in isolated fibroblasts is normally concentrated in the specialized rafts called caveolae (see Fig. 11–21), but treatment with EGF causes the receptor to leave the raft. This migration depends on the receptor’s protein kinase activity; mutant receptors lacking this activity remain in the rafts during treatment with EGF. Caveolin, an integral membrane protein localized in caveolae, is phosphorylated on Tyr residues in response to insulin, and phosphorylation may allow the now-activated EGF receptor to draw its binding partners into the raft. Finally, another example of the clustering of signaling proteins in rafts is the β -adrenergic receptor. This receptor is segregated in membrane rafts that also contain the G proteins, adenylyl cyclase, PKA, and a specific protein phosphatase, PP2, providing a highly integrated signaling unit. Spatial segregation of signaling proteins in rafts adds yet another dimension to the already complex processes initiated by extracellular signals.

SUMMARY 12.5 Multivalent Scaffold Proteins and Membrane Rafts

- Many signaling proteins have domains that bind phosphorylated Tyr, Ser, or Thr residues in other proteins; the binding specificity for each domain is determined by sequences that adjoin the phosphorylated residue.
- SH2 and PTB domains bind to proteins containing (P)-Tyr residues; other domains bind (P)-Ser and (P)-Thr residues in various contexts.
- Plextrin homology domains bind the membrane phospholipid PIP₃.
- Many signaling proteins are multivalent, with several different binding modules. By combining the substrate specificities of various protein kinases with the specificities of domains that bind phosphorylated Ser, Thr, or Tyr residues, and with phosphatases that can rapidly inactivate a pathway, cells create a large number of multiprotein signaling complexes.
- Membrane rafts and caveolae sequester groups of signaling proteins in small regions of the plasma membrane, enhancing their interactions and making signaling more efficient.

12.6 Signaling in Microorganisms and Plants

Much of what we have said here about signaling relates to mammalian tissues or cultured cells from such tissues. Bacteria, eukaryotic microorganisms, and vascular plants must also respond to a variety of external signals, such as O_2 , nutrients, light, noxious chemicals, and so on. We turn here to a brief consideration of the kinds of signaling machinery used by microorganisms and plants.

Bacterial Signaling Entails Phosphorylation in a Two-Component System

E. coli responds to a number of nutrients in its environment, including sugars and amino acids, by swimming toward them, propelled by one or a few flagella. A family of membrane proteins have binding domains on the outside of the plasma membrane to which specific **attractants** (sugars or amino acids) bind (Fig. 12–26). Ligand binding causes another domain on the inside of the plasma membrane to phosphorylate itself on a His residue. This first component of the **two-component system**, the **receptor His kinase**, then catalyzes the transfer of the phosphoryl group from the His residue to an Asp residue on a second, soluble protein, the **response regulator**; this phosphoprotein moves to the base of the flagellum, carrying the signal from the membrane receptor. The flagellum is driven by a rotary motor that can propel the cell through its medium or cause it to stall, depending on the direction of the motor's rotation. Information from the receptor allows the cell to determine whether it is moving toward or away from the source of the attractant. If its motion is toward the attractant, the response regulator signals the cell to continue in a straight line; if away from it, the cell tumbles momentarily, acquiring a new direction. Repetition of this behavior results in a random path, biased toward

movement in the direction of increasing attractant concentration.

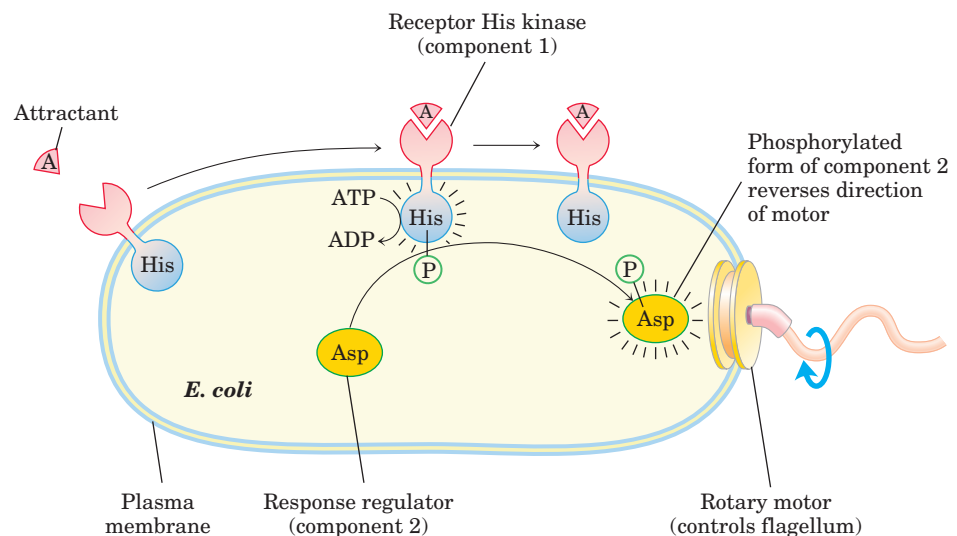
E. coli detects not only sugars and amino acids but also O_2 , extremes of temperature, and other environmental factors, using this basic two-component system. Two-component systems have been detected in many other bacteria, including gram-positive and gram-negative eubacteria and archaeobacteria, as well as in protists and fungi. Clearly this signaling mechanism developed early in the course of cellular evolution and has been conserved.

Various signaling systems used by animal cells also have analogs in the prokaryotes. As the full genomic sequences of more, and more diverse, bacteria become known, researchers have discovered genes that encode proteins similar to protein Ser or Thr kinases, Ras-like proteins regulated by GTP binding, and proteins with SH3 domains. Receptor Tyr kinases have not been detected in bacteria, but P-Tyr residues do occur in some bacterial proteins, so there must be an enzyme that phosphorylates Tyr residues.

Signaling Systems of Plants Have Some of the Same Components Used by Microbes and Mammals

Like animals, vascular plants must have a means of communication between tissues to coordinate and direct growth and development; to adapt to conditions of O_2 , nutrients, light, and temperature; and to warn of the presence of noxious chemicals and damaging pathogens (Fig. 12–27). At least a billion years of evolution have passed since the plant and animal branches of the eukaryotes diverged, which is reflected in the differences in signaling mechanisms: some plant mechanisms are conserved—that is, are similar to those in animals (protein kinases, scaffold proteins, cyclic nucleotides, electrogenic ion pumps, and gated ion channels); some are similar to bacterial two-component systems; and some are unique to plants (light-sensing mechanisms, for ex-

FIGURE 12–26 The two-component signaling mechanism in bacterial chemotaxis. When an attractant ligand (A) binds to the receptor domain of the membrane-bound receptor, a protein His kinase in the cytosolic domain (component 1) is activated and autophosphorylates on a His residue. This phosphoryl group is then transferred to an Asp residue on component 2 (in some cases a separate protein; in others, another domain of the receptor protein). After phosphorylation on Asp, component 2 moves to the base of the flagellum, where it determines the direction of rotation of the flagellar motor.



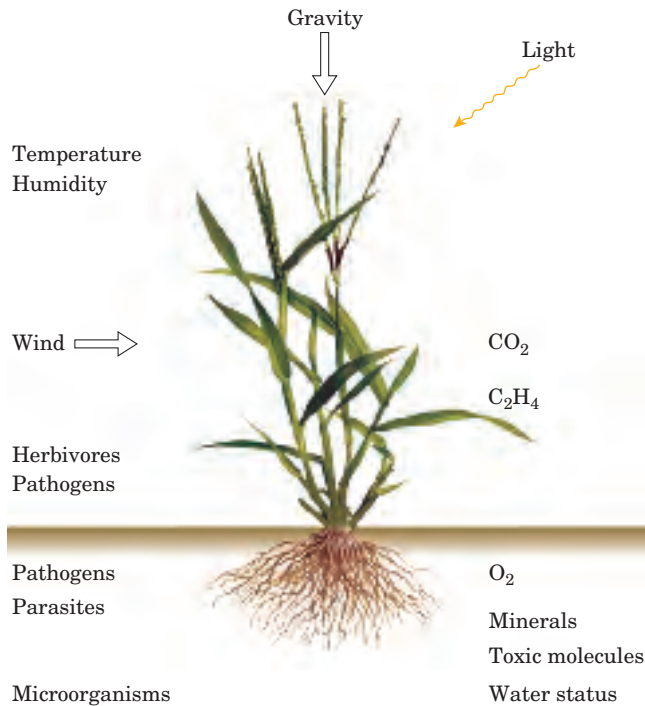


FIGURE 12-27 Some stimuli that produce responses in plants.

ample) (Table 12-7). The genome of the widely studied plant *Arabidopsis thaliana*, for example, encodes about 1,000 protein Ser/Thr kinases, including about 60 MAPKs and nearly 400 membrane-associated receptor

kinases that phosphorylate Ser or Thr residues; a variety of protein phosphatases; scaffold proteins that bring other proteins together in signaling complexes; enzymes for the synthesis and degradation of cyclic nucleotides; and 100 or more ion channels, including about 20 gated by cyclic nucleotides. Inositol phospholipids are present, as are kinases that interconvert them by phosphorylation of inositol head groups.

However, some types of signaling proteins common in animal tissues are not present in plants, or are represented by only a few genes. Cyclic nucleotide-dependent protein kinases (PKA and PKG) appear to be absent, for example. Heterotrimeric G proteins and protein Tyr kinase genes are much less prominent in the plant genome, and serpentine (G protein-coupled) receptors, the largest gene family in the human genome (>1,000 genes), are very sparsely represented in the plant genome. DNA-binding nuclear steroid receptors are certainly not prominent, and may be absent from plants. Although plants lack the most widely conserved light-sensing mechanism present in animals (rhodopsin, with retinal as pigment), they have a rich collection of other light-detecting mechanisms not found in animal tissues—phytochromes and cryptochromes, for example (Chapter 19).

The kinds of compounds that elicit signals in plants are similar to certain signaling molecules in mammals (Fig. 12-28). Instead of prostaglandins, plants have jasmonate; instead of steroid hormones, brassinosteroids.

TABLE 12-7 Signaling Components Present in Mammals, Plants, or Bacteria

Signaling protein	Mammals	Plants	Bacteria
Ion channels	+	+	+
Electrogenic ion pumps	+	+	+
Two-component His kinases	+	+	+
Adenylyl cyclase	+	+	+
Guanylyl cyclase	+	+	?
Receptor protein kinases (Ser/Thr)	+	+	?
Ca ²⁺ as second messenger	+	+	?
Ca ²⁺ channels	+	+	?
Calmodulin, CaM-binding protein	+	+	—
MAPK cascade	+	+	—
Cyclic nucleotide-gated channels	+	+	—
IP ₃ -gated Ca ²⁺ channels	+	+	—
Phosphatidylinositol kinases	+	+	—
Serpentine receptors	+	+/-	+
Trimeric G proteins	+	+/-	—
PI-specific phospholipase C	+	?	—
Tyrosine kinase receptors	+	?	—
SH2 domains	+	?	?
Nuclear steroid receptors	+	—	—
Protein kinase A	+	—	—
Protein kinase G	+	—	—

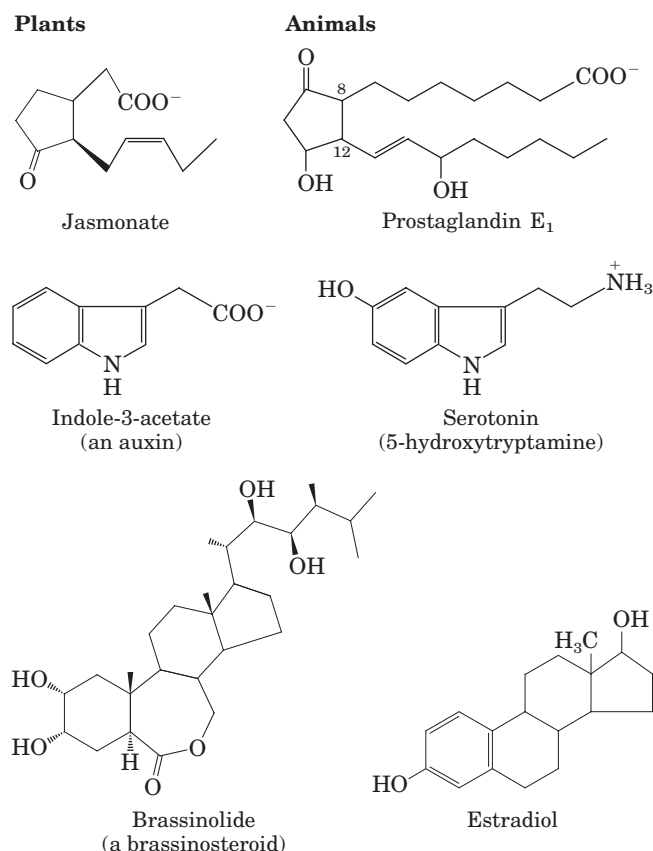


FIGURE 12-28 Structural similarities between plant and animal signals. The plant signals jasmonate, indole-3-acetate, and brassinolide resemble the mammalian signals prostaglandin E₁, serotonin, and estradiol.

About 100 different small peptides serve as plant signals, and both plants and animals use compounds derived from aromatic amino acids as signals.

Plants Detect Ethylene through a Two-Component System and a MAPK Cascade

The receptors for the plant hormone ethylene ($\text{CH}_2=\text{CH}_2$) are related in primary sequence to the receptor His kinases of the bacterial two-component systems and probably evolved from them; the cyanobacterial origin of chloroplasts (see Fig. 1–36) may have brought the bacterial signaling genes into the plant cell nucleus. In *Arabidopsis*, the two-component signaling system is contained within a single protein. The first downstream component affected by ethylene signaling is a protein Ser/Thr kinase (CTR-1; Fig. 12–29) with sequence homology to Raf, the protein kinase that begins the MAPK cascade in the mammalian response to insulin (see the comparison in Fig. 12–30). In plants, in the absence of ethylene, the CTR-1 kinase is active and *inhibits* the MAPK cascade, preventing transcription of ethylene-responsive genes. Exposure to ethylene *inac-*

tivates the CTR-1 kinase, thereby activating the MAPK cascade that leads to activation of the transcription factor EIN3. Active EIN3 stimulates the synthesis of a second transcription factor (ERF1), which in turn activates transcription of a number of ethylene-responsive genes; the gene products affect processes ranging from seedling development to fruit ripening.

Although apparently derived from the bacterial two-component signaling system, the ethylene system in *Arabidopsis* is different in that the His kinase activity that defines component 1 in bacteria is not essential to the transduction in *Arabidopsis*. The genome of the cyanobacterium *Anabaena* encodes proteins with both an ethylene-binding domain and an active His kinase domain. It seems likely that in the course of evolution, the ethylene receptor of vascular plants was derived from that of a cyanobacterial endosymbiont, and that the bacterial His kinase became a Ser/Thr kinase in the plant.

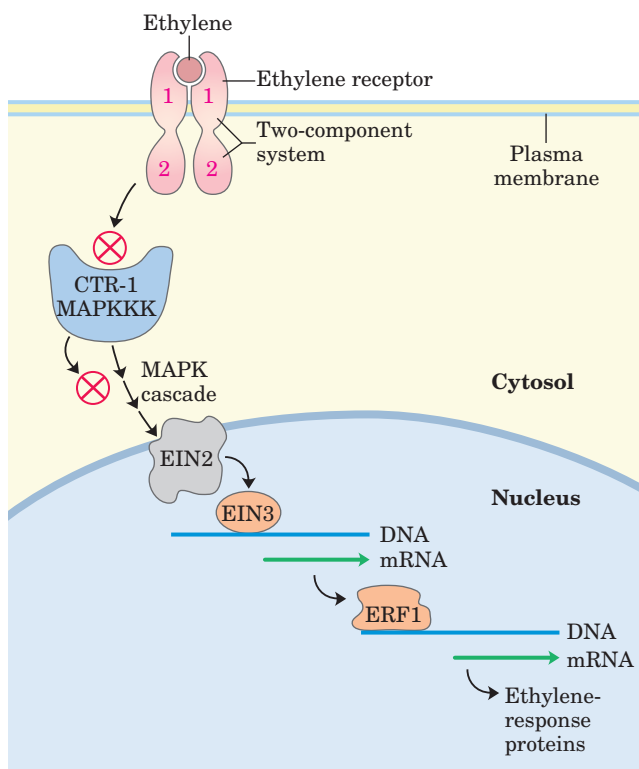


FIGURE 12-29 Transduction mechanism for detection of ethylene by plants. The ethylene receptor in the plasma membrane (red) is a two-component system contained within a single protein, which has both a receptor domain (component 1) and a response regulator domain (component 2). The receptor controls (in ways we do not yet understand) the activity of CTR1, a protein kinase similar to MAPKKKs and therefore presumed to be part of a MAPK cascade. CTR1 is a negative regulator of the ethylene response; when CTR1 is *inactive*, the ethylene signal passes through the gene product EIN2 (thought to be a nuclear envelope protein), which somehow causes increased synthesis of ERF1, a transcription factor; ERF1 in turn stimulates expression of proteins specific to the ethylene response.

Receptorlike Protein Kinases Transduce Signals from Peptides and Brassinosteroids

One common motif in plant signaling involves **receptorlike kinases (RLKs)** with a single helical segment in the plasma membrane that connects a receptor domain on the outside of the membrane with a protein Ser/Thr kinase on the cytoplasmic side. This type of receptor participates in the defense mechanism triggered by infection by a bacterial pathogen (Fig. 12–30a). The signal to turn on the genes needed for defense against infection is a peptide (flg22) released by breakdown of flagellin, the major protein of the bacterial flagellum. Binding of flg22 to the FLS2 receptor of *Arabidopsis* induces receptor dimerization and autophosphorylation on Ser and Thr residues, and the downstream effect is activation of a MAPK cascade like that described above for insulin action (Fig. 12–6). The final kinase in this

cascade activates a specific transcription factor, triggering synthesis of the proteins that defend against the bacterial infection. The steps between receptor phosphorylation and the MAPK cascade are not yet known. A phosphoprotein phosphatase (KAPP) associates with the active receptor protein and inactivates it by dephosphorylation to end the response.

The MAPK cascade in the plant's defense against bacterial pathogens is remarkably similar to the innate immune response triggered by bacterial lipopolysaccharide and mediated by the Toll-like receptors in mammals (Fig. 12–30b). Other membrane receptors use similar mechanisms to activate a MAPK cascade, ultimately activating transcription factors and turning on the genes essential to the defense response.

Most of the several hundred RLKs in plants are presumed to act in similar ways: ligand binding induces dimerization and autophosphorylation, and the

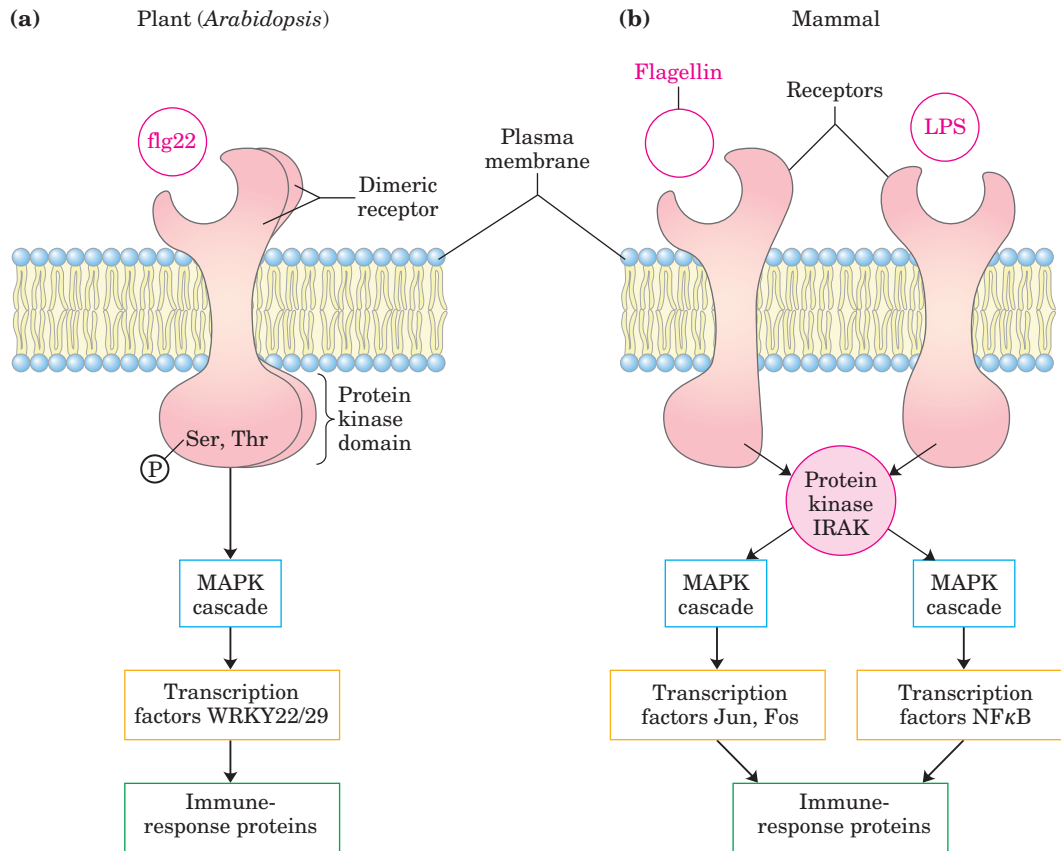


FIGURE 12–30 Similarities between the signaling pathways that trigger immune responses in plants and animals. **(a)** In the plant *Arabidopsis thaliana*, the peptide flg22, derived from the flagella of a bacterial pathogen, binds to its receptor in the plasma membrane, causing the receptors to form dimers and triggering autophosphorylation of the cytosolic protein kinase domain on a Ser or Thr residue (not a Tyr). Autophosphorylation activates the receptor protein kinase, which then phosphorylates downstream proteins. The activated receptor also activates (by means unknown) a MAPKKK. The resulting kinase cascade leads to phosphorylation of a nuclear protein that normally inhibits

the transcription factors WRKY22 and 29, triggering proteolytic degradation of the inhibitor and freeing the transcription factors to stimulate gene expression related to the immune response. **(b)** In mammals, the toxic bacterial lipopolysaccharide (LPS; see Fig. 7–32) is detected by plasma membrane receptors that associate with and activate a soluble protein kinase (IRAK). The major flagellar protein of pathogenic bacteria acts through a similar receptor to activate IRAK. Then IRAK initiates two distinct MAPK cascades that end in the nucleus, causing the synthesis of proteins needed in the immune response. Jun, Fos, and NFκB are transcription factors.

activated receptor kinase triggers downstream responses by phosphorylating key proteins at Ser or Thr residues. The ligands for these kinases have been identified in only a few cases: brassinosteroids, the peptide trigger for the self-incompatibility response that prevents self-pollination, and CLV1 peptide, a factor involved in regulating the fate of stem cells (undifferentiated cells) in plant development.

SUMMARY 12.6 Signaling in Microorganisms and Plants

- Bacteria and unicellular eukaryotes have a variety of sensory systems that allow them to sample and respond to their environment. In the two-component system, a receptor His kinase senses the signal and autophosphorylates a His residue, then phosphorylates the response regulator on an Asp residue.
- Plants respond to many environmental stimuli, and employ hormones and growth factors to coordinate the development and metabolic activities of their tissues. Plant genomes encode hundreds of signaling proteins, including some very similar to those used in signal transductions in mammalian cells.
- Two-component signaling mechanisms common in bacteria have been acquired in altered forms by plants. Cyanobacteria use typical two-component systems in the detection of chemical signals and light; plants use related proteins—which autophosphorylate on Ser/Thr, not His, residues—to detect ethylene.
- Plant receptorlike kinases (RLKs), with an extracellular ligand-binding domain, a single transmembrane segment, and a cytosolic protein kinase domain, participate in detecting a wide variety of stimuli, including peptides that originate from pathogens, brassinosteroid hormones, self-incompatible pollen, and developmental signals. RLKs autophosphorylate Ser/Thr residues, then activate downstream proteins that in some cases are MAPK cascades. The end result of many such signals is increased transcription of specific genes.

12.7 Sensory Transduction in Vision, Olfaction, and Gustation

The detection of light, smells, and tastes (vision, olfaction, and gustation, respectively) in animals is accomplished by specialized sensory neurons that use signal-transduction mechanisms fundamentally similar to those that detect hormones, neurotransmitters, and growth

factors. An initial sensory signal is amplified greatly by mechanisms that include gated ion channels and intracellular second messengers; the system adapts to continued stimulation by changing its sensitivity to the stimulus (desensitization); and sensory input from several receptors is integrated before the final signal goes to the brain.

Light Hyperpolarizes Rod and Cone Cells of the Vertebrate Eye

In the vertebrate eye, light entering through the pupil is focused on a highly organized collection of light-sensitive neurons (Fig. 12–31). The light-sensing cells are of two types: **rods** (about 10^9 per retina), which sense low levels of light but cannot discriminate colors, and **cones** (about 3×10^6 per retina), which are less sensitive to light but can discriminate colors. Both cell types are long, narrow, specialized sensory neurons with two distinct cellular compartments: the **outer segment** contains dozens of membranous disks loaded with the membrane protein rhodopsin, and the **inner segment** contains the nucleus and many mitochondria, which produce the ATP essential to phototransduction.

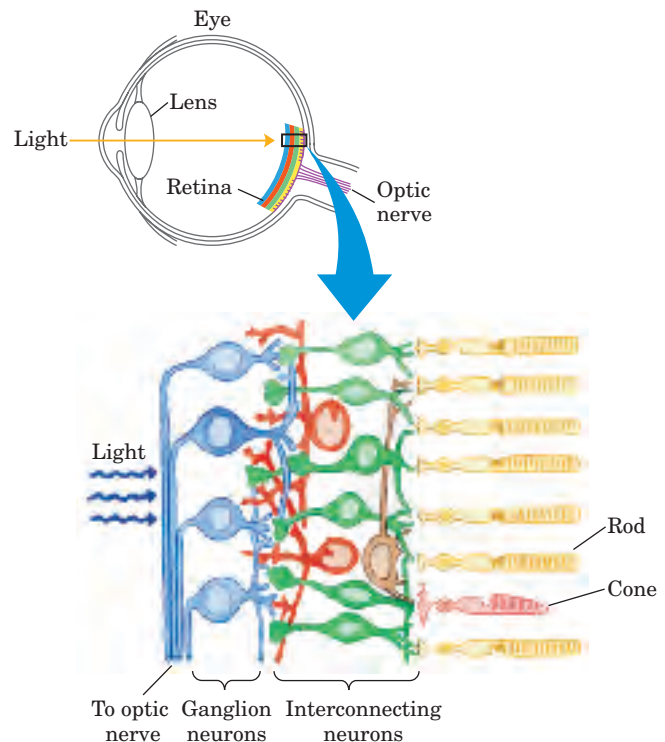


FIGURE 12–31 Light reception in the vertebrate eye. The lens of the eye focuses light on the retina, which is composed of layers of neurons. The primary photosensory neurons are rod cells (yellow), which are responsible for high-resolution and night vision, and cone cells of three subtypes (pink), which initiate color vision. The rods and cones form synapses with several ranks of interconnecting neurons that convey and integrate the electrical signals. The signals eventually pass from ganglion neurons through the optic nerve to the brain.

Like other neurons, rods and cones have a transmembrane electrical potential (V_m), produced by the electrogenic pumping of the Na^+K^+ ATPase in the plasma membrane of the inner segment (Fig. 12–32). Also contributing to the membrane potential is an ion channel in the outer segment that permits passage of either Na^+ or Ca^{2+} and is gated (opened) by cGMP. In the dark, rod cells contain enough cGMP to keep this channel open. The membrane potential is therefore determined by the net difference between the Na^+ and K^+ pumped by the inner segment (which polarizes the membrane) and the influx of Na^+ through the ion channels of the outer segment (which tends to depolarize the membrane).

The essence of signaling in the retinal rod or cone cell is a light-induced decrease in the concentration of cGMP, which causes the cGMP-gated ion channel to close. The plasma membrane then becomes hyperpolarized by the Na^+K^+ ATPase. Rod and cone cells synapse with interconnecting neurons (Fig. 12–31) that carry information about the electrical activity to the ganglion neurons near the inner surface of the retina. The ganglion neurons integrate the output from many rod or cone cells and send the resulting signal through the optic nerve to the visual cortex of the brain.

Light Triggers Conformational Changes in the Receptor Rhodopsin

Visual transduction begins when light falls on rhodopsin, many thousands of molecules of which are present in each disk of the outer segments of rod and cone cells. **Rhodopsin** (M_r 40,000) is an integral protein with seven membrane-spanning α helices (Fig. 12–33), the characteristic serpentine architecture. The amino-terminal domain projects into the disk, and the carboxyl-terminal domain faces the cytosol of the outer segment. The light-absorbing pigment (chromophore) **11-*cis*-retinal** is covalently attached to **opsin**, the protein component of rhodopsin, through a Schiff base to a Lys residue. The retinal lies near the middle of the bilayer (Fig. 12–33), oriented with its long axis approximately in the plane of the membrane. When a photon is absorbed by the retinal component of rhodopsin, the energy causes a photochemical change; 11-*cis*-retinal is converted to **all-*trans*-retinal** (see Figs 1–18b, 10–21). This change in the structure of the chromophore causes conformational changes in the rhodopsin molecule—the first stage in visual transduction.

Excited Rhodopsin Acts through the G Protein Transducin to Reduce the cGMP Concentration

In its excited conformation, rhodopsin interacts with a second protein, **transducin**, which hovers nearby on the cytoplasmic face of the disk membrane (Fig. 12–33). Transducin (T) belongs to the same family of heterotrimeric GTP-binding proteins as G_s and G_i . Although

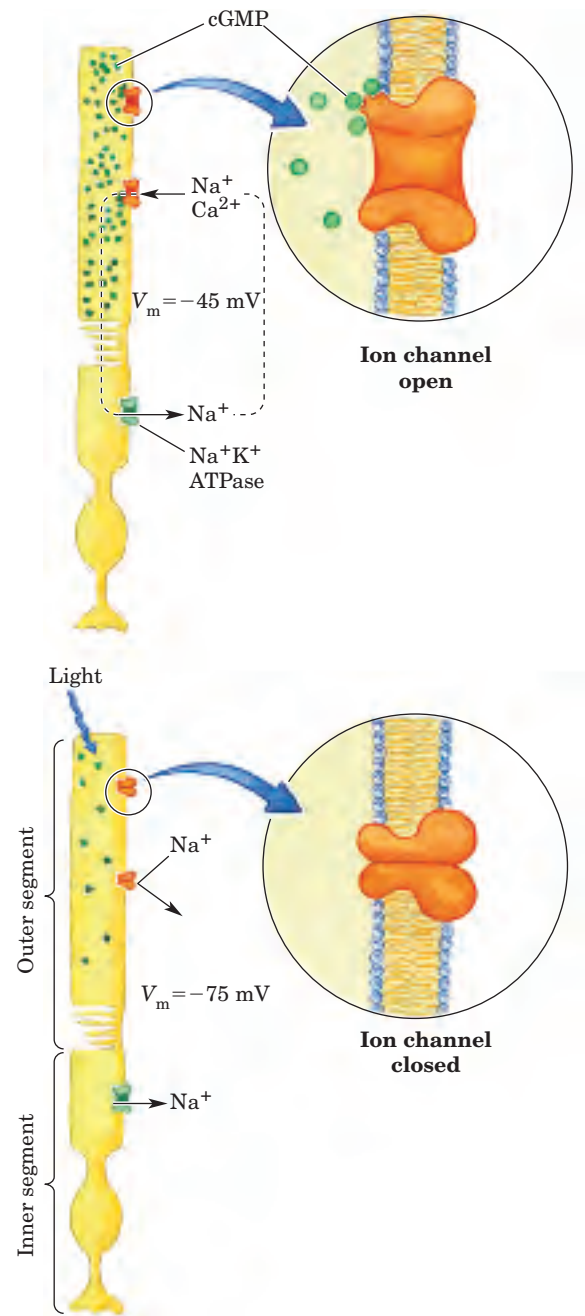


FIGURE 12–32 Light-induced hyperpolarization of rod cells. The rod cell consists of an outer segment that is filled with stacks of membranous disks (not shown) containing the photoreceptor rhodopsin and an inner segment that contains the nucleus and other organelles. Cones have a similar structure. ATP in the inner segment powers the Na^+K^+ ATPase, which creates a transmembrane electrical potential by pumping 3 Na^+ out for every 2 K^+ pumped in. The membrane potential is reduced by the flow of Na^+ and Ca^{2+} into the cell through cGMP-gated cation channels in the plasma membrane of the outer segment. When rhodopsin absorbs light, it triggers degradation of cGMP (green dots) in the outer segment, causing closure of the cation channel. Without cation influx through this channel, the cell becomes hyperpolarized. This electrical signal is passed to the brain through the ranks of neurons shown in Figure 12–31.

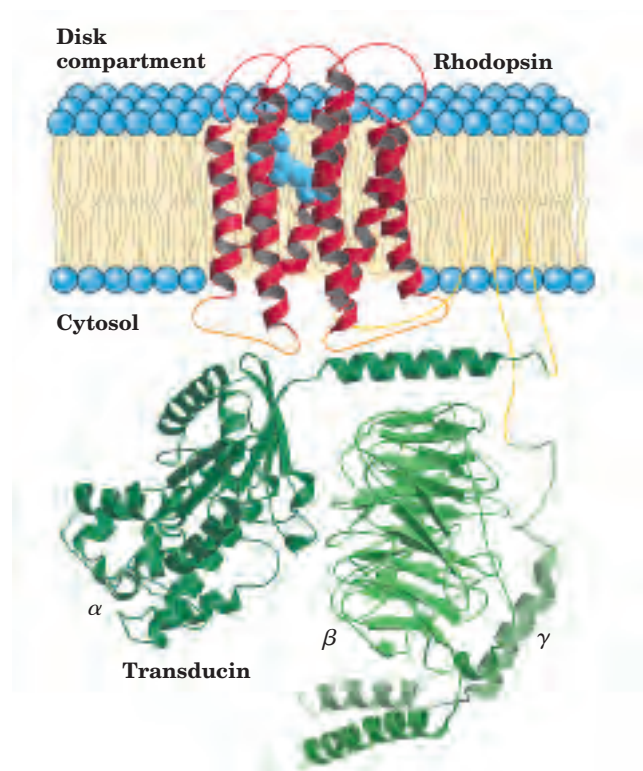


FIGURE 12-33 Likely structure of rhodopsin complexed with the G protein transducin. (PDB ID 1BAC) Rhodopsin (red) has seven transmembrane helices embedded in the disk membranes of rod outer segments and is oriented with its carboxyl terminus on the cytosolic side and its amino terminus inside the disk. The chromophore 11-*cis* retinal (blue), attached through a Schiff base linkage to Lys²⁵⁶ of the seventh helix, lies near the center of the bilayer. (This location is similar to that of the epinephrine-binding site in the β -adrenergic receptor.) Several Ser and Thr residues near the carboxyl terminus are substrates for phosphorylations that are part of the desensitization mechanism for rhodopsin. Cytosolic loops that interact with the G protein transducin are shown in orange; their exact positions are not yet known. The three subunits of transducin (green) are shown in their likely arrangement. Rhodopsin is palmitoylated at its carboxyl terminus, and both the α and γ subunits of transducin have attached lipids (yellow) that assist in anchoring them to the membrane.

specialized for visual transduction, transducin shares many functional features with G_s and G_i . It can bind either GDP or GTP. In the dark, GDP is bound, all three subunits of the protein (T_α , T_β , and T_γ) remain together, and no signal is sent. When rhodopsin is excited by light, it interacts with transducin, catalyzing the replacement of bound GDP by GTP from the cytosol (Fig. 12-34, steps ① and ②). Transducin then dissociates into T_α and $T_{\beta\gamma}$, and the T_α -GTP carries the signal from the excited receptor to the next element in the transduction pathway, cGMP phosphodiesterase (PDE); this enzyme converts cGMP to 5'-GMP (steps ③ and ④). Note that this is not the same cyclic nucleotide phosphodiesterase that hydrolyzes cAMP to terminate the β -adrenergic response. The cGMP-specific PDE is unique to the visual cells of the retina.

PDE is an integral protein with its active site on the cytoplasmic side of the disk membrane. In the dark, a tightly bound inhibitory subunit very effectively suppresses PDE activity. When T_α -GTP encounters PDE, the inhibitory subunit is released, and the enzyme's activity immediately increases by several orders of magnitude. Each molecule of active PDE degrades many molecules of cGMP to the biologically inactive 5'-GMP, lowering [cGMP] in the outer segment within a fraction of a second. At the new, lower [cGMP], the cGMP-gated ion channels close, blocking reentry of Na^+ and Ca^{2+} into the outer segment and hyperpolarizing the membrane of the rod or cone cell (step ⑤). Through this process, the initial stimulus—a photon—changes the V_m of the cell.

Amplification of the Visual Signal Occurs in the Rod and Cone Cells

Several steps in the visual-transduction process result in great amplification of the signal. Each excited rhodopsin molecule activates at least 500 molecules of transducin, each of which can activate a molecule of PDE. This phosphodiesterase has a remarkably high turnover number, each activated molecule hydrolyzing 4,200 molecules of cGMP per second. The binding of cGMP to cGMP-gated ion channels is cooperative (at least three cGMP molecules must be bound to open one channel), and a relatively small change in [cGMP] therefore registers as a large change in ion conductance. The result of these amplifications is exquisite sensitivity to light. Absorption of a single photon closes 1,000 or more ion channels and changes the cell's membrane potential by about 1 mV.

The Visual Signal Is Quickly Terminated

As your eyes move across this line, the images of the first words disappear rapidly—before you see the next series of words. In that short interval, a great deal of biochemistry has taken place. Very shortly after illumination of the rod or cone cells stops, the photosensory system shuts off. The α subunit of transducin (with bound GTP) has intrinsic GTPase activity. Within milliseconds after the decrease in light intensity, GTP is hydrolyzed and T_α reassociates with $T_{\beta\gamma}$. The inhibitory subunit of PDE, which had been bound to T_α -GTP, is released and reassociates with PDE, strongly inhibiting that enzyme. To return [cGMP] to its “dark” level, the enzyme guanylyl cyclase converts GTP to cGMP (step ⑦ in Fig. 12-34) in a reaction that is inhibited by high [Ca^{2+}] (>100 nM). Calcium levels drop during illumination, because the steady-state [Ca^{2+}] in the outer segment is the result of outward pumping of Ca^{2+} through the Na^+ - Ca^{2+} exchanger of the plasma membrane and inward movement of Ca^{2+} through open cGMP-gated channels. In the dark, this produces a [Ca^{2+}] of about 500 nM—enough to inhibit cGMP synthesis. After brief illumination, Ca^{2+} entry slows and [Ca^{2+}] declines (step

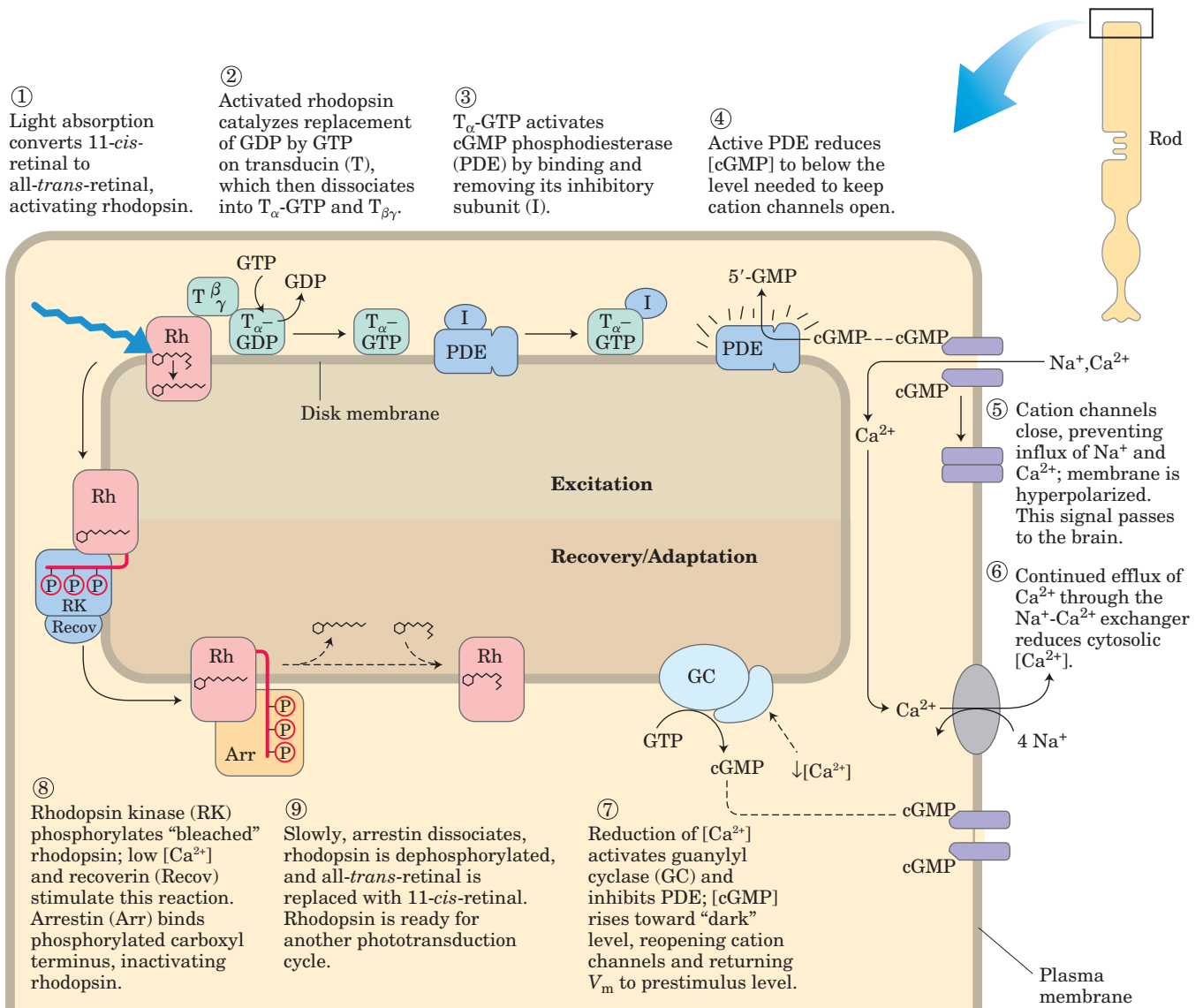


FIGURE 12-34 Molecular consequences of photon absorption by rhodopsin in the rod outer segment. The top half of the figure (steps ① to ⑤) describes excitation; the bottom (steps ⑥ to ⑨), recovery and adaptation after illumination.

⑥). The inhibition of guanylyl cyclase by Ca^{2+} is relieved, and the cyclase converts GTP to cGMP to return the system to its prestimulus state (step ⑦).

Rhodopsin Is Desensitized by Phosphorylation

Rhodopsin itself also undergoes changes in response to prolonged illumination. The conformational change induced by light absorption exposes several Thr and Ser residues in the carboxyl-terminal domain. These residues are quickly phosphorylated by **rhodopsin kinase** (step ⑧ in Fig. 12-34), which is functionally and structurally homologous to the β -adrenergic kinase (β ARK) that desensitizes the β -adrenergic receptor (Fig. 12-17). The Ca^{2+} -binding protein **recoverin** inhibits rhodopsin kinase at high [Ca^{2+}], but the inhibi-

tion is relieved when [Ca^{2+}] drops after illumination, as described above. The phosphorylated carboxyl-terminal domain of rhodopsin is bound by the protein **arrestin 1**, preventing further interaction between activated rhodopsin and transducin. Arrestin 1 is a close homolog of arrestin 2 (β arr; Fig. 12-17). On a relatively long time scale (seconds to minutes), the all-*trans*-retinal of an excited rhodopsin molecule is removed and replaced by 11-*cis*-retinal, to produce rhodopsin that is ready for another round of excitation (step ⑨ in Fig. 12-34).



Humans cannot synthesize retinal from simpler precursors and must obtain it in the diet in the form of vitamin A (see Fig. 10-21). Given the role of retinal in the process of vision, it is not surprising that dietary deficiency of vitamin A causes night blindness (poor vision at night or in dim light). ■

Cone Cells Specialize in Color Vision

Color vision in cone cells involves a path of sensory transduction essentially identical to that described above, but triggered by slightly different light receptors. Three types of cone cells are specialized to detect light from different regions of the spectrum, using three related photoreceptor proteins (opsins). Each cone cell expresses only one kind of opsin, but each type is closely related to rhodopsin in size, amino acid sequence, and presumably three-dimensional structure. The differences among the opsins, however, are great enough to place the chromophore, 11-*cis*-retinal, in three slightly different environments, with the result that the three photoreceptors have different absorption spectra (Fig. 12–35). We discriminate colors and hues by integrating the output from the three types of cone cells, each containing one of the three photoreceptors.



Color blindness, such as the inability to distinguish red from green, is a fairly common, genetically inherited trait in humans. The various types of color blindness result from different opsin mutations. One form is due to loss of the red photoreceptor; affected individuals are **red⁻ dichromats** (they see only two primary colors). Others lack the green pigment and are **green⁻ dichromats**. In some cases, the red and green photoreceptors are present but have a changed amino acid sequence that causes a change in their absorption spectra, resulting in abnormal color vision. Depending on which pigment is altered, such individuals are **red-anomalous trichromats** or **green-**

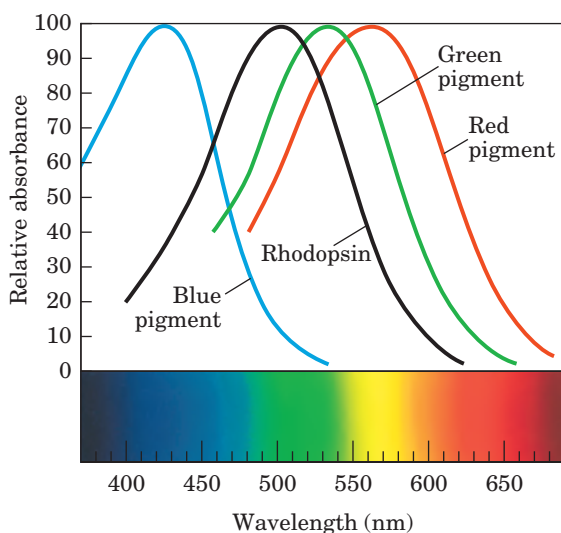


FIGURE 12–35 Absorption spectra of purified rhodopsin and the red, green, and blue receptors of cone cells. The spectra, obtained from individual cone cells isolated from cadavers, peak at about 420, 530, and 560 nm, and the maximum absorption for rhodopsin is at about 500 nm. For reference, the visible spectrum for humans is about 380 to 750 nm.

anomalous trichromats. Examination of the genes for the visual receptors has allowed the diagnosis of color blindness in a famous “patient” more than a century after his death (Box 12–3)! ■

Vertebrate Olfaction and Gustation Use Mechanisms Similar to the Visual System

The sensory cells used to detect odors and tastes have much in common with the rod and cone cells that detect light. Olfactory neurons have a number of long thin cilia extending from one end of the cell into a mucous layer that overlays the cell. These cilia present a large surface area for interaction with olfactory signals. The receptors for olfactory stimuli are ciliary membrane proteins with the familiar serpentine structure of seven transmembrane α helices. The olfactory signal can be any one of the many volatile compounds for which there are specific receptor proteins. Our ability to discriminate odors stems from hundreds of different olfactory receptors in the tongue and nasal passages and from the brain’s ability to integrate input from different types of olfactory receptors to recognize a “hybrid” pattern, extending our range of discrimination far beyond the number of receptors.

The olfactory stimulus arrives at the sensory cells by diffusion through the air. In the mucous layer overlaying the olfactory neurons, the odorant molecule binds directly to an olfactory receptor or to a specific binding protein that carries the odorant to a receptor (Fig. 12–36). Interaction between odorant and receptor triggers a change in receptor conformation that results in the replacement of bound GDP by GTP on a G protein, G_{olf} , analogous to transducin and to G_s of the β -adrenergic system. The activated G_{olf} then activates adenylyl cyclase of the ciliary membrane, which synthesizes cAMP from ATP, raising the local [cAMP]. The cAMP-gated Na^+ and Ca^{2+} channels of the ciliary membrane open, and the influx of Na^+ and Ca^{2+} produces a small depolarization called the **receptor potential**. If a sufficient number of odorant molecules encounter receptors, the receptor potential is strong enough to cause the neuron to fire an action potential. This is relayed to the brain in several stages and registers as a specific smell. All these events occur within 100 to 200 ms.

Some olfactory neurons may use a second transduction mechanism. They have receptors coupled through G proteins to PLC rather than to adenylyl cyclase. Signal reception in these cells triggers production of IP_3 (Fig. 12–19), which opens IP_3 -gated Ca^{2+} channels in the ciliary membrane. Influx of Ca^{2+} then depolarizes the ciliary membrane and generates a receptor potential or regulates Ca^{2+} -dependent enzymes in the olfactory pathway.

In either type of olfactory neuron, when the stimulus is no longer present, the transducing machinery shuts



BOX 12-3 BIOCHEMISTRY IN MEDICINE

Color Blindness: John Dalton's Experiment from the Grave

The chemist John Dalton (of atomic theory fame) was color-blind. He thought it probable that the vitreous humor of his eyes (the fluid that fills the eyeball behind the lens) was tinted blue, unlike the colorless fluid of normal eyes. He proposed that after his death, his eyes should be dissected and the color of the vitreous humor determined. His wish was honored. The day after Dalton's death in July 1844, Joseph Ransome dissected his eyes and found the vitreous humor to be perfectly colorless. Ransome, like many scientists, was reluctant to throw samples away. He placed Dalton's eyes in a jar of preservative (Fig. 1), where they stayed for a century and a half.

Then, in the mid-1990s, molecular biologists in England took small samples of Dalton's retinas and extracted DNA. Using the known gene sequences for the opsins of the red and green photopigments, they am-

plified the relevant sequences (using techniques described in Chapter 9) and determined that Dalton had the opsin gene for the red photopigment but lacked the opsin gene for the green photopigment. Dalton was a green⁻ dichromat. So, 150 years after his death, the experiment Dalton started—by hypothesizing about the cause of his color blindness—was finally finished.

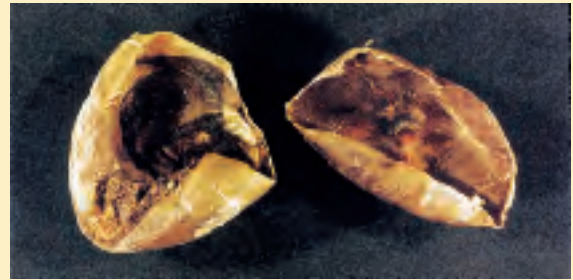


FIGURE 1 Dalton's eyes.

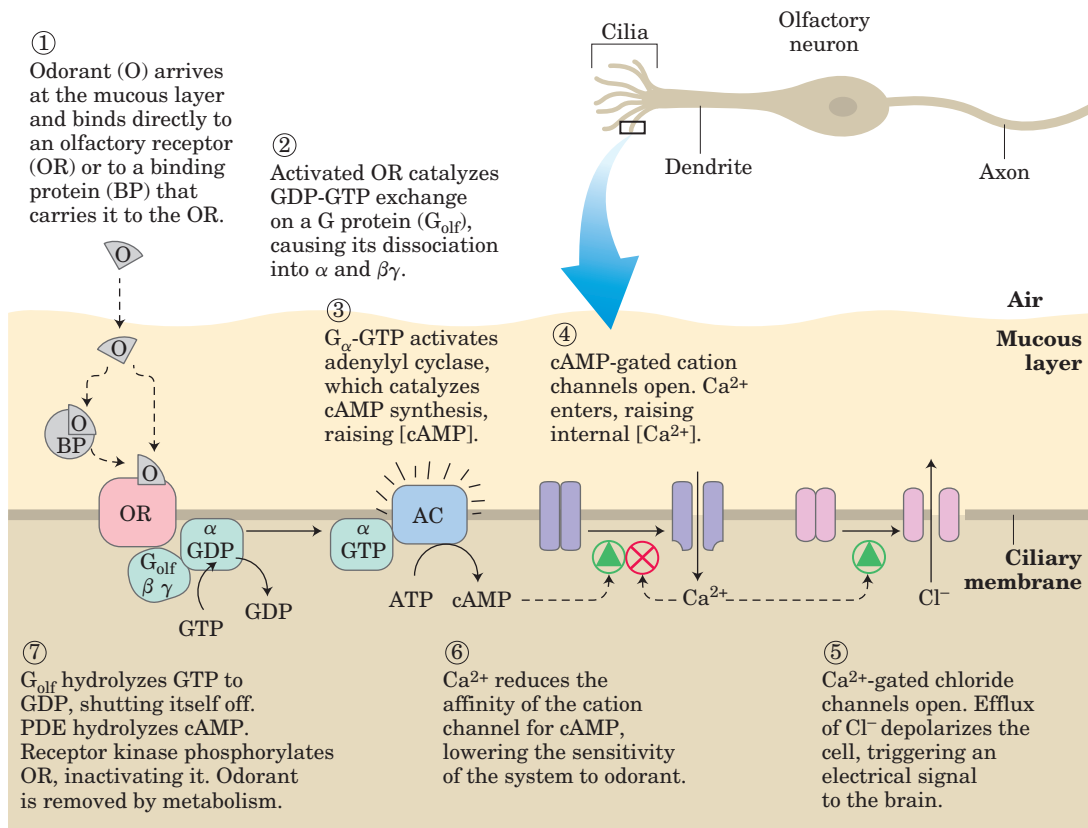


FIGURE 12-36 Molecular events of olfaction. These interactions occur in the cilia of olfactory receptor cells.

itself off in several ways. A cAMP phosphodiesterase returns [cAMP] to the prestimulus level. G_{olf} hydrolyzes its bound GTP to GDP, thereby inactivating itself. Phosphorylation of the receptor by a specific kinase prevents its interaction with G_{olf} , by a mechanism analogous to that used to desensitize the β -adrenergic receptor and rhodopsin. And lastly, some odorants are enzymatically destroyed by oxidases.

The sense of taste in vertebrates reflects the activity of gustatory neurons clustered in taste buds on the surface of the tongue. In these sensory neurons, serpentine receptors are coupled to the heterotrimeric G protein **gustducin** (very similar to the transducin of rod and cone cells). Sweet-tasting molecules are those that bind receptors in “sweet” taste buds. When the molecule (tastant) binds, gustducin is activated by replacement of bound GDP with GTP and then stimulates cAMP production by adenylyl cyclase. The resulting elevation of [cAMP] activates PKA, which phosphorylates K^+ channels in the plasma membrane, causing them to close. Reduced efflux of K^+ depolarizes the cell (Fig. 12–37). Other taste buds specialize in detecting bitter, sour, or salty tastants, using various combinations of second messengers and ion channels in the transduction mechanisms.

G Protein–Coupled Serpentine Receptor Systems Share Several Features

We have now looked at four systems (hormone signaling, vision, olfaction, and gustation) in which membrane

receptors are coupled to second messenger–generating enzymes through G proteins. It is clear that signaling mechanisms arose early in evolution; serpentine receptors, heterotrimeric G proteins, and adenylyl cyclase are found in virtually all eukaryotic organisms. Even the common brewer’s yeast *Saccharomyces* uses serpentine receptors and G proteins to detect the opposite mating type. Overall patterns have been conserved, and the introduction of variety has given modern organisms the ability to respond to a wide range of stimuli (Table 12–8). Of the 35,000 or so genes in the human genome, as many as 1,000 encode serpentine receptors, including hundreds for olfactory stimuli and a number of “orphan receptors” for which the natural ligand is not yet known.

All well-studied transducing systems that act through heterotrimeric G proteins share certain common features (Fig. 12–38). The receptors have seven transmembrane segments, a domain (generally the loop between transmembrane helices 6 and 7) that interacts with a G protein, and a carboxyl-terminal cytoplasmic domain that undergoes reversible phosphorylation on several Ser or Thr residues. The ligand-binding site (or, in the case of light reception, the light receptor) is buried deep in the membrane and includes residues from several of the transmembrane segments. Ligand binding (or light) induces a conformational change in the receptor, exposing a domain that can interact with a G protein. Heterotrimeric G proteins activate or inhibit effector enzymes (adenylyl cyclase, PDE, or PLC), which change the concentration of a second messenger (cAMP, cGMP, IP_3 , or Ca^{2+}). In the hormone-detecting

TABLE 12–8 Some Signals Transduced by G Protein–Coupled Serpentine Receptors

Acetylcholine (muscarinic)	Leukotrienes
Adenosine	Light
Angiotensin	Luteinizing hormone (LH)
ATP (extracellular)	Melatonin
Bradykinin	Odorants
Calcitonin	Opioids
Cannabinoids	Oxytocin
Catecholamines	Platelet-activating factor
Cholecystokinin	Prostaglandins
Corticotropin-releasing factor (CRF)	Secretin
Cyclic AMP (<i>Dictyostelium discoideum</i>)	Serotonin
Dopamine	Somatostatin
Follicle-stimulating hormone (FSH)	Tastants
γ -Aminobutyric acid (GABA)	Thyrotropin
Glucagon	Thyrotropin-releasing hormone (TRH)
Glutamate	Vasoactive intestinal peptide
Growth hormone-releasing hormone (GHRH)	Vasopressin
Histamine	Yeast mating factors

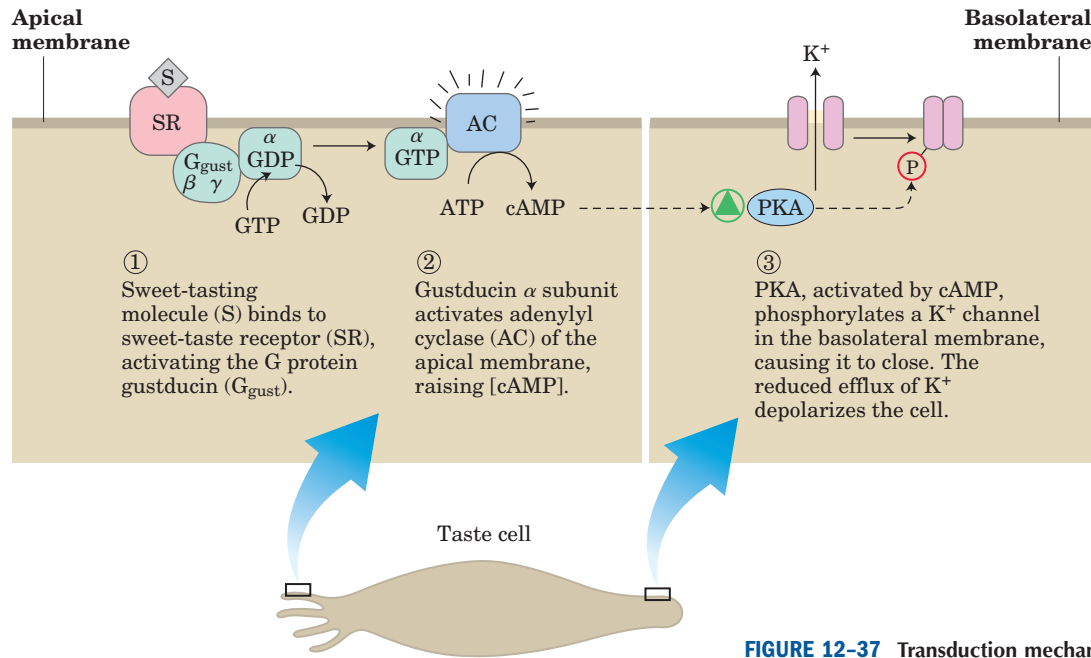


FIGURE 12-37 Transduction mechanism for sweet tastants.

systems, the final output is an activated protein kinase that regulates some cellular process by phosphorylating a protein critical to that process. In sensory neurons, the output is a change in membrane potential and a consequent electrical signal that passes to another neuron in the pathway connecting the sensory cell to the brain.

All these systems self-inactivate. Bound GTP is converted to GDP by the intrinsic GTPase activity of G proteins, often augmented by GTPase-activating proteins (GAPs) or RGS proteins (*r*egulators of *G*-protein signaling). In some cases, the effector enzymes that are the targets of G protein modulation also serve as GAPs.

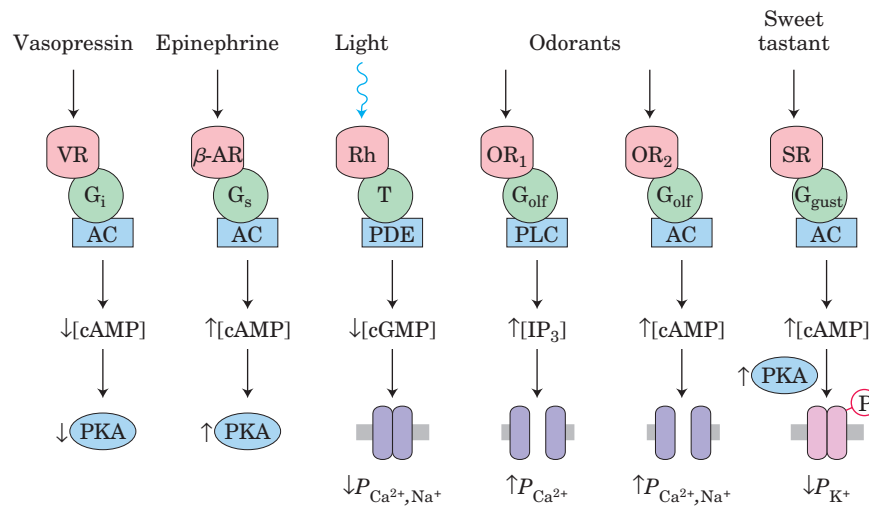


FIGURE 12-38 Common features of signaling systems that detect hormones, light, smells, and tastes. Serpentine receptors provide signal specificity, and their interaction with G proteins provides signal amplification. Heterotrimeric G proteins activate effector enzymes: adenylyl cyclase (AC), phospholipase C (PLC), and phosphodiesterases (PDE) that degrade cAMP or cGMP. Changes in concentration of the second messengers (cAMP, cGMP, IP₃) result in alterations of enzymatic activities by phosphorylation or alterations in the permeability

(P) of surface membranes to Ca^{2+} , Na^+ , and K^+ . The resulting depolarization or hyperpolarization of the sensory cell (the signal) is passed through relay neurons to sensory centers in the brain. In the best-studied cases, desensitization includes phosphorylation of the receptor and binding of a protein (arrestin) that interrupts receptor–G protein interactions. VR is the vasopressin receptor; other receptor and G protein abbreviations are as used in earlier illustrations.

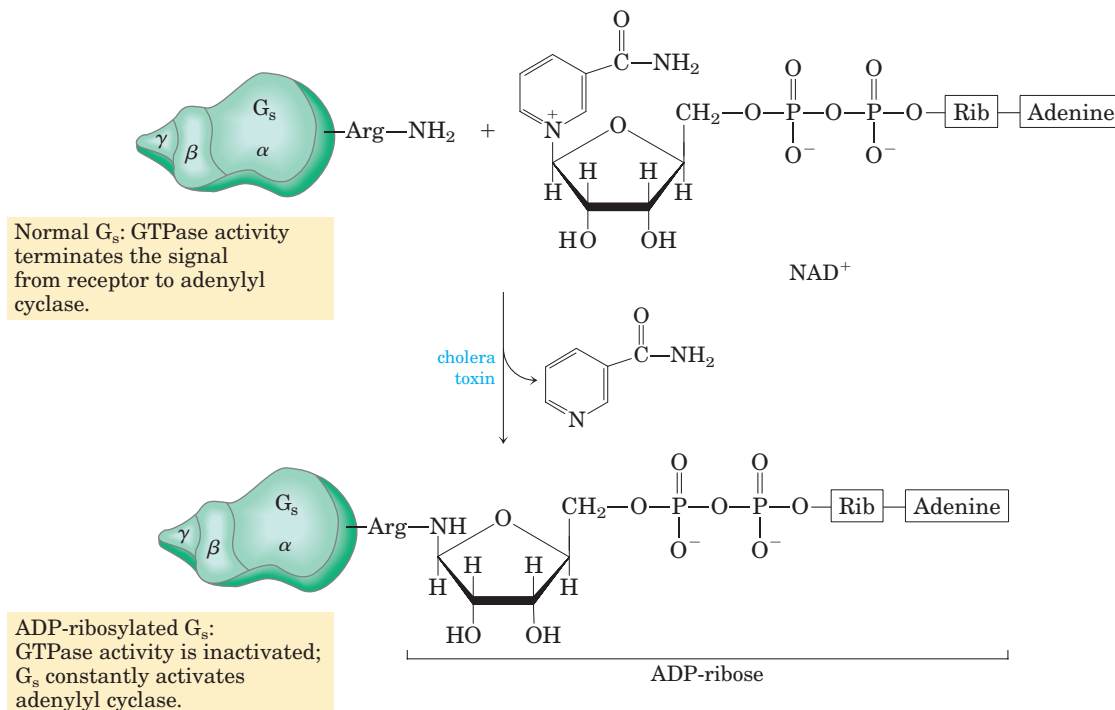


FIGURE 12-39 Toxins produced by bacteria that cause cholera and whooping cough (pertussis). These toxins are enzymes that catalyze the transfer of the ADP-ribose moiety of NAD^+ to an Arg residue (cholera toxin) or a Cys residue (pertussis toxin) of G proteins: G_s in the case

of cholera (as shown here) and G_i in whooping cough. The G proteins thus modified fail to respond to normal hormonal stimuli. The pathology of both diseases results from defective regulation of adenylyl cyclase and overproduction of cAMP.

Disruption of G-Protein Signaling Causes Disease



Biochemical studies of signal transductions have led to an improved understanding of the pathological effects of toxins produced by the bacteria that cause cholera and pertussis (whooping cough). Both toxins are enzymes that interfere with normal signal transductions in the host animal. **Cholera toxin**, secreted by *Vibrio cholerae* found in contaminated drinking water, catalyzes the transfer of ADP-ribose from NAD^+ to the α subunit of G_s (Fig. 12-39), blocking its GTPase activity and thereby rendering G_s permanently activated. This results in continuous activation of the adenylyl cyclase of intestinal epithelial cells and chronically high [cAMP], which triggers constant secretion of Cl^- , HCO_3^- , and water into the intestinal lumen. The resulting dehydration and electrolyte loss are the major pathologies in cholera. The **pertussis toxin**, produced by *Bordetella pertussis*, catalyzes ADP-ribosylation of G_i , preventing displacement of GDP by GTP and blocking inhibition of adenylyl cyclase by G_i . ■

SUMMARY 12.7 Sensory Transduction in Vision, Olfaction, and Gustation

- Vision, olfaction, and gustation in vertebrates employ serpentine receptors, which act

through heterotrimeric G proteins to change the V_m of the sensory neuron.

- In rod and cone cells of the retina, light activates rhodopsin, which stimulates replacement of GDP by GTP on the G protein transducin. The freed α subunit of transducin activates cGMP phosphodiesterase, which lowers [cGMP] and thus closes cGMP-dependent ion channels in the outer segment of the neuron. The resulting hyperpolarization of the rod or cone cell carries the signal to the next neuron in the pathway, and eventually to the brain.
- In olfactory neurons, olfactory stimuli, acting through serpentine receptors and G proteins, trigger either an increase in [cAMP] (by activating adenylyl cyclase) or an increase in $[Ca^{2+}]$ (by activating PLC). These second messengers affect ion channels and thus the V_m .
- Gustatory neurons have serpentine receptors that respond to tastants by altering [cAMP], which in turn changes V_m by gating ion channels.
- There is a high degree of conservation of signaling proteins and transduction mechanisms across species.

12.8 Regulation of Transcription by Steroid Hormones

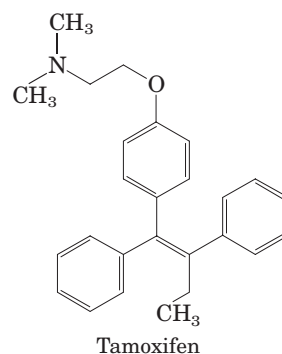
The large group of steroid, retinoic acid (retinoid), and thyroid hormones exert at least part of their effects by a mechanism fundamentally different from that of other hormones: they act in the nucleus to alter gene expression. We therefore discuss their mode of action in detail in Chapter 28, along with other mechanisms for regulating gene expression. Here we give a brief overview.

Steroid hormones (estrogen, progesterone, and cortisol, for example), too hydrophobic to dissolve readily in the blood, are carried on specific carrier proteins from their point of release to their target tissues. In target cells, these hormones pass through the plasma membranes by simple diffusion and bind to specific receptor proteins in the nucleus (Fig. 12–40). Hormone binding triggers changes in the conformation of the receptor proteins so that they become capable of interacting with specific regulatory sequences in DNA called **hormone response elements (HREs)**, thus altering gene expression (see Fig. 28–31). The bound receptor-hormone complex can either enhance or suppress the expression of specific genes adjacent to HREs. Hours or days are required for these regulators to have their full effect—the time required for the changes in RNA synthesis and

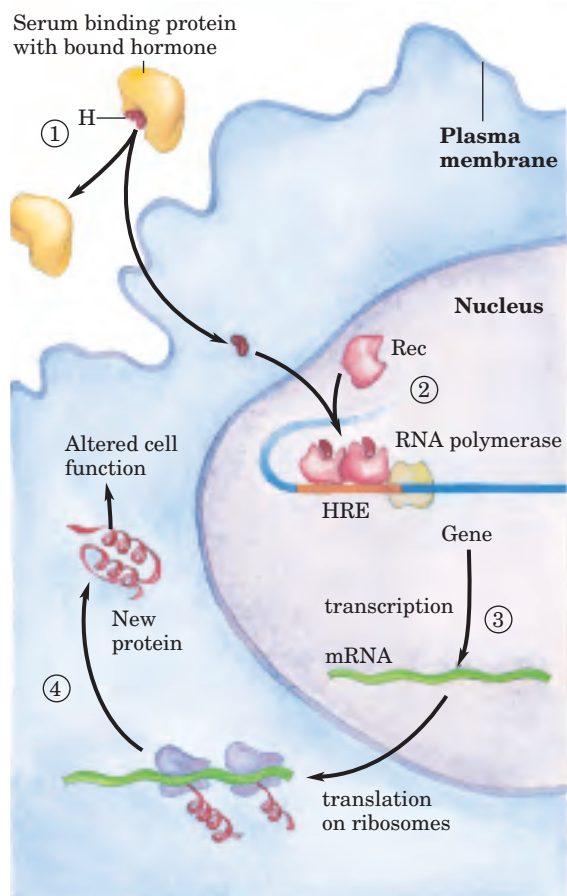
subsequent protein synthesis to become evident in altered metabolism.



The specificity of the steroid-receptor interaction is exploited in the use of the drug **tamoxifen** to treat breast cancer. In some types of breast cancer, division of the cancerous cells depends on the continued presence of the hormone estrogen. Tamoxifen competes with estrogen for binding to the estrogen receptor, but the tamoxifen-receptor complex has little or no effect on gene expression; tamoxifen is an antagonist of estrogen. Consequently, tamoxifen administered after surgery or during chemotherapy for hormone-dependent breast cancer slows or stops the growth of remaining cancerous cells.

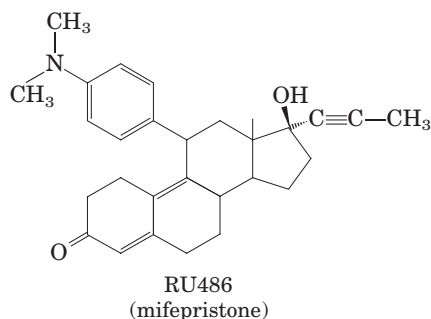


Tamoxifen



- ① Hormone (H), carried to the target tissue on serum binding proteins, diffuses across the plasma membrane and binds to its specific receptor protein (Rec) in the nucleus.
- ② Hormone binding changes the conformation of Rec; it forms homo- or heterodimers with other hormone-receptor complexes and binds to specific regulatory regions called hormone response elements (HREs) in the DNA adjacent to specific genes.
- ③ Binding regulates transcription of the adjacent gene(s), increasing or decreasing the rate of mRNA formation.
- ④ Altered levels of the hormone-regulated gene product produce the cellular response to the hormone.

FIGURE 12–40 General mechanism by which steroid and thyroid hormones, retinoids, and vitamin D regulate gene expression. The details of transcription and protein synthesis are discussed in Chapters 26 and 27. At least some steroids also act through plasma membrane receptors by a completely different mechanism.



Another steroid analog, the drug **RU486**, is used to terminate early (preimplantation) pregnancies. An antagonist of the hormone progesterone, RU486 binds to the progesterone receptor and blocks hormone actions essential to implantation of the fertilized ovum in the uterus. ■

The classic mechanism for steroid hormone action through nuclear receptors does not explain certain effects of steroids that are too fast to be the result of altered protein synthesis. For example, the estrogen-mediated dilation of blood vessels is known to be independent of gene transcription or protein synthesis, as is the steroid-induced decrease in cellular [cAMP]. Another transduction mechanism is probably responsible for some of these effects. A plasma membrane protein predicted to have seven transmembrane helical segments binds progesterone with very high affinity and mediates the inhibition of adenylyl cyclase by that hormone, accounting for the decrease in [cAMP]. A second nonclassical mechanism involves the rapid activation of the MAPK cascade by progesterone, acting through the soluble progesterone receptor. This is the same receptor that, in the nucleus, causes the much slower changes in gene expression that constitute the classic mechanism of progesterone action. How the MAPK cascade is activated is not yet clear.

SUMMARY 12.8 Regulation of Transcription by Steroid Hormones

- Steroid hormones enter cells and bind to specific receptor proteins.
- The hormone-receptor complex binds specific regions of DNA, the hormone response elements, and regulates the expression of nearby genes by interacting with transcription factors.
- Two other, faster-acting mechanisms produce some of the effects of steroids. Progesterone triggers a rapid drop in [cAMP], mediated by a plasma membrane receptor, and binding of progesterone to the classic soluble steroid receptor activates a MAPK cascade.

12.9 Regulation of the Cell Cycle by Protein Kinases

One of the most dramatic roles for protein phosphorylation is the regulation of the eukaryotic cell cycle. During embryonic growth and later development, cell division occurs in virtually every tissue. In the adult organism most tissues become quiescent. A cell's "decision" to divide or not is of crucial importance to the organism. When the regulatory mechanisms that limit cell division are defective and cells undergo unregulated division, the result is catastrophic—cancer. Proper cell division requires a precisely ordered sequence of biochemical events that assures every daughter cell a full complement of the molecules required for life. Investigations into the control of cell division in diverse eukaryotic cells have revealed universal regulatory mechanisms. Protein kinases and protein phosphorylation are central to the timing mechanism that determines entry into cell division and ensures orderly passage through these events.

The Cell Cycle Has Four Stages

Cell division in eukaryotes occurs in four well-defined stages (Fig. 12–41). In the S (synthesis) phase, the DNA is replicated to produce copies for both daughter

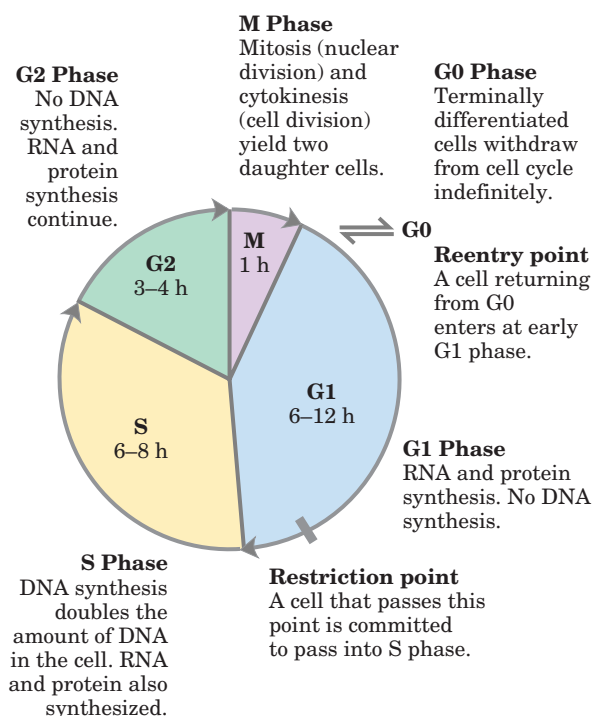


FIGURE 12–41 Eukaryotic cell cycle. The durations (in hours) of the four stages vary, but those shown are typical.

cells. In the G₂ phase (G indicates the gap between divisions), new proteins are synthesized and the cell approximately doubles in size. In the M phase (mitosis), the maternal nuclear envelope breaks down, matching chromosomes are pulled to opposite poles of the cell, each set of daughter chromosomes is surrounded by a newly formed nuclear envelope, and cytokinesis pinches the cell in half, producing two daughter cells. In embryonic or rapidly proliferating tissue, each daughter cell divides again, but only after a waiting period (G₁). In cultured animal cells the entire process takes about 24 hours.

After passing through mitosis and into G₁, a cell either continues through another division or ceases to divide, entering a quiescent phase (G₀) that may last hours, days, or the lifetime of the cell. When a cell in G₀ begins to divide again, it reenters the division cycle through the G₁ phase. Differentiated cells such as hepatocytes or adipocytes have acquired their specialized function and form; they remain in the G₀ phase.

Levels of Cyclin-Dependent Protein Kinases Oscillate

The timing of the cell cycle is controlled by a family of protein kinases with activities that change in response to cellular signals. By phosphorylating specific proteins at precisely timed intervals, these protein kinases orchestrate the metabolic activities of the cell to produce orderly cell division. The kinases are heterodimers with a regulatory subunit, **cyclin**, and a catalytic subunit, **cyclin-dependent protein kinase (CDK)**. In the absence of cyclin, the catalytic subunit is virtually inactive. When cyclin binds, the catalytic site opens up, a residue essential to catalysis becomes accessible (Fig. 12–42), and the activity of the catalytic subunit increases 10,000-fold. Animal cells have at least ten different cyclins (designated A, B, and so forth) and at least eight cyclin-dependent kinases (CDK1 through CDK8), which act in various combinations at specific points in the cell cycle. Plants also use a family of CDKs to regulate their cell division.



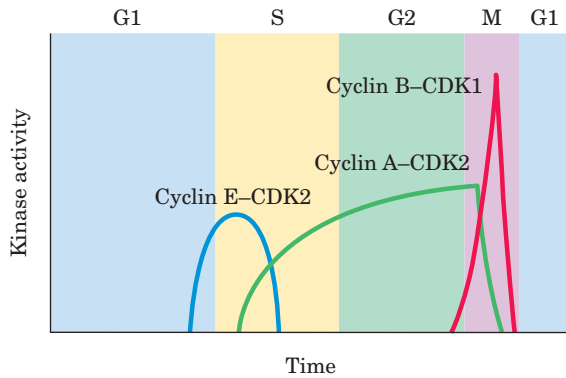


FIGURE 12-43 Variations in the activities of specific CDKs during the cell cycle in animals. Cyclin E-CDK2 activity peaks near the G1 phase–S phase boundary, when the active enzyme triggers synthesis of enzymes required for DNA synthesis (see Fig. 12-46). Cyclin A-CDK2 activity rises during the S and G2 phases, then drops sharply in the M phase, as cyclin B-CDK1 peaks.

In a population of animal cells undergoing synchronous division, some CDK activities show striking oscillations (Fig. 12-43). These oscillations are the result of four mechanisms for regulating CDK activity: phosphorylation or dephosphorylation of the CDK, controlled degradation of the cyclin subunit, periodic synthesis of CDKs and cyclins, and the action of specific CDK-inhibiting proteins.

Regulation of CDKs by Phosphorylation The activity of a CDK is strikingly affected by phosphorylation and dephosphorylation of two critical residues in the protein (Fig. 12-44a). Phosphorylation of Tyr¹⁵ near the amino terminus renders CDK2 inactive; the (P)-Tyr residue is in the ATP-binding site of the kinase, and the negatively charged phosphate group blocks the entry of ATP. A specific phosphatase dephosphorylates this (P)-Tyr residue, permitting the binding of ATP. Phosphorylation

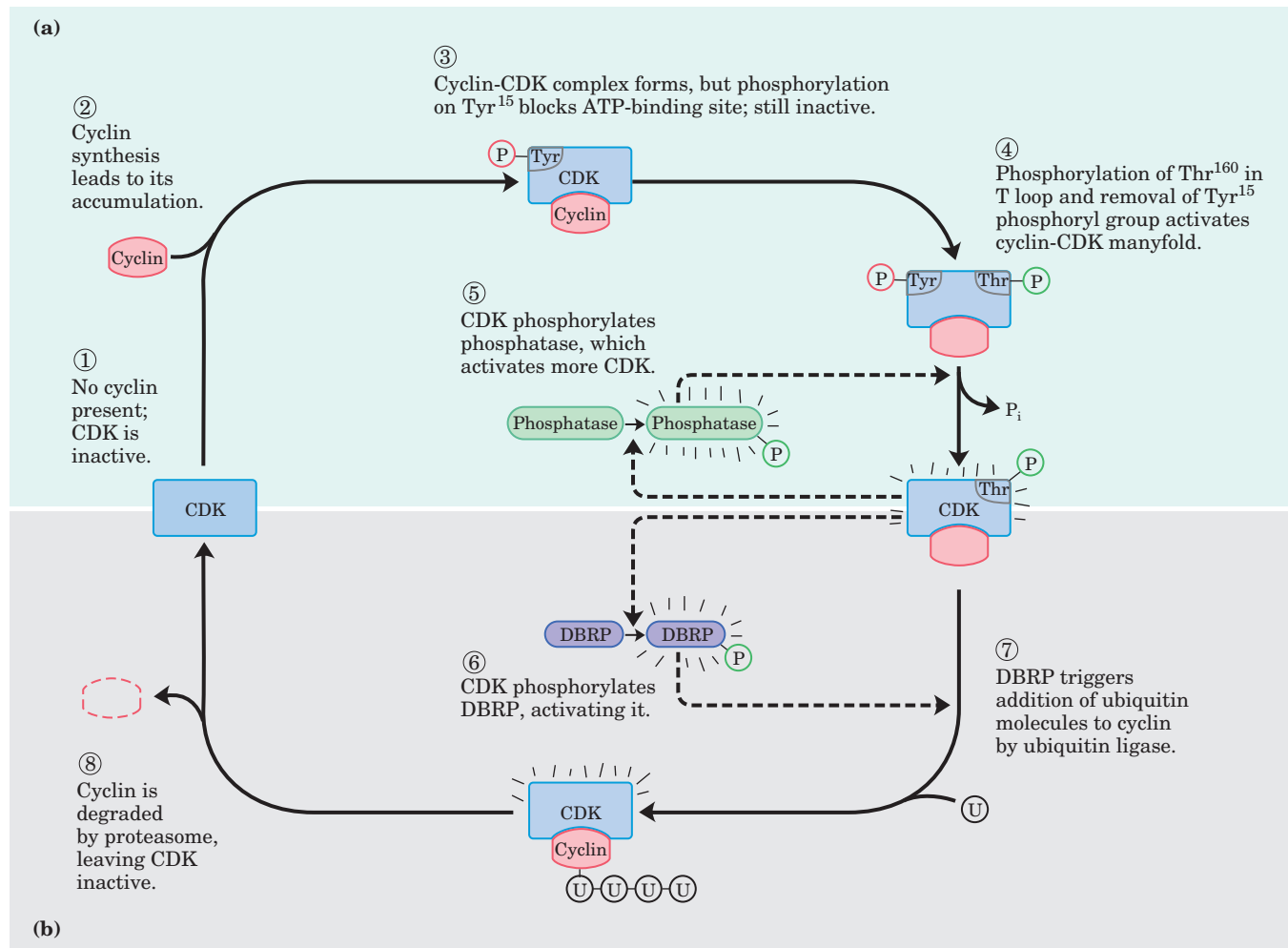


FIGURE 12-44 Regulation of CDK by phosphorylation and proteolysis. **(a)** The cyclin-dependent protein kinase activated at the time of mitosis (the M phase CDK) has a “T loop” that can fold into the substrate-binding site. When Thr¹⁶⁰ in the T loop is phosphorylated, the loop moves out of the substrate-binding site, activating the CDK

manifold. **(b)** The active cyclin-CDK complex triggers its own inactivation by phosphorylation of DBRP (destruction box recognizing protein). DBRP and ubiquitin ligase then attach several molecules of ubiquitin (U) to cyclin, targeting it for destruction by proteasomes, proteolytic enzyme complexes.

of Thr¹⁶⁰ in the “T loop” of CDK, catalyzed by the CDK-activating kinase, forces the T loop out of the substrate-binding cleft, permitting substrate binding and catalytic activity.

One circumstance that triggers this control mechanism is the presence of single-strand breaks in DNA, which leads to arrest of the cell cycle in G₂. A specific protein kinase (called Rad3 in yeast), which is activated by single-strand breaks, triggers a cascade leading to the inactivation of the phosphatase that dephosphorylates Tyr¹⁵ of CDK. The CDK remains inactive and the cell is arrested in G₂. The cell will not divide until the DNA is repaired and the effects of the cascade are reversed.

Controlled Degradation of Cyclin Highly specific and precisely timed proteolytic breakdown of mitotic cyclins regulates CDK activity throughout the cell cycle. Progress through mitosis requires first the activation then the destruction of cyclins A and B, which activate the catalytic subunit of the M-phase CDK. These cyclins contain near their amino terminus the sequence Arg–Thr–Ala–Leu–Gly–Asp–Ile–Gly–Asn, the “destruction box,” which targets them for degradation. (This usage of “box” derives from the common practice, in diagramming the sequence of a nucleic acid or protein, of enclosing within a box a short sequence of nucleotide or amino acid residues with some specific function.) The protein DBRP (destruction box recognizing protein) recognizes this sequence and initiates the process of cyclin degradation by bringing together the cyclin and another protein, **ubiquitin**. Cyclin and activated ubiquitin are covalently joined by the enzyme ubiquitin ligase (Fig. 12–44b). Several more ubiquitin molecules are then appended, providing the signal for a proteolytic enzyme complex, or **proteasome**, to degrade cyclin.

What controls the timing of cyclin breakdown? A feedback loop occurs in the overall process shown in Figure 12–44. Increased CDK activity activates cyclin proteolysis. Newly synthesized cyclin associates with and activates CDK, which phosphorylates and activates DBRP. Active DBRP then causes proteolysis of cyclin. Lowered [cyclin] causes a decline in CDK activity, and the activity of DBRP also drops through slow, constant dephosphorylation and inactivation by a DBRP phosphatase. The cyclin level is ultimately restored by synthesis of new cyclin molecules.

The role of ubiquitin and proteasomes is not limited to the regulation of cyclin; as we shall see in Chapter 27, both also take part in the turnover of cellular proteins, a process fundamental to cellular housekeeping.

Regulated Synthesis of CDKs and Cyclins The third mechanism for changing CDK activity is regulation of the rate of synthesis of cyclin or CDK or both. For example, cyclin D, cyclin E, CDK2, and CDK4 are synthesized only when a specific transcription factor, E2F, is present in

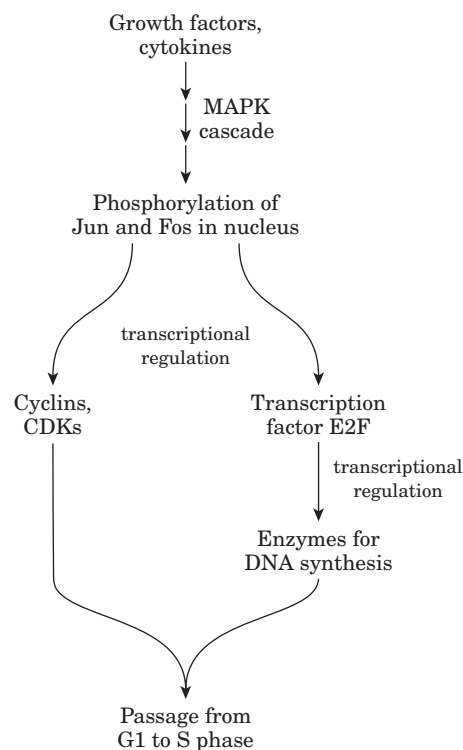


FIGURE 12–45 Regulation of cell division by growth factors. The path from growth factors to cell division leads through the enzyme cascade that activates MAPK; phosphorylation of the nuclear transcription factors Jun and Fos; and the activity of the transcription factor E2F, which promotes synthesis of several enzymes essential for DNA synthesis.

the nucleus to activate transcription of their genes. Synthesis of E2F is in turn regulated by extracellular signals such as **growth factors** and **cytokines** (inducers of cell division), compounds found to be essential for the division of mammalian cells in culture. These growth factors induce the synthesis of specific nuclear transcription factors essential to the production of the enzymes of DNA synthesis. Growth factors trigger phosphorylation of the nuclear proteins Jun and Fos, transcription factors that promote the synthesis of a variety of gene products, including cyclins, CDKs, and E2F. In turn, E2F controls production of several enzymes essential for the synthesis of deoxynucleotides and DNA, enabling cells to enter the S phase (Fig. 12–45).

Inhibition of CDKs Finally, specific protein inhibitors bind to and inactivate specific CDKs. One such protein is p21, which we discuss below.

These four control mechanisms modulate the activity of specific CDKs that, in turn, control whether a cell will divide, differentiate, become permanently quiescent, or begin a new cycle of division after a period of quiescence. The details of cell cycle regulation, such as the number of different cyclins and kinases and the

combinations in which they act, differ from species to species, but the basic mechanism has been conserved in the evolution of all eukaryotic cells.

CDKs Regulate Cell Division by Phosphorylating Critical Proteins

We have examined how cells maintain close control of CDK activity, but how does the activity of CDK control the cell cycle? The list of target proteins that CDKs are known to act upon continues to grow, and much remains to be learned. But we can see a general pattern behind CDK regulation by inspecting the effect of CDKs on the structures of laminin and myosin and on the activity of retinoblastoma protein.

The structure of the nuclear envelope is maintained in part by highly organized meshworks of intermediate filaments composed of the protein laminin. Breakdown of the nuclear envelope before segregation of the sister chromatids in mitosis is partly due to the phosphorylation of laminin by a CDK, which causes laminin filaments to depolymerize.

A second kinase target is the ATP-driven actin-myosin contractile machinery that pinches a dividing cell into two equal parts during cytokinesis. After the division, CDK phosphorylates a small regulatory subunit of myosin, causing dissociation of myosin from actin filaments and inactivating the contractile machinery. Subsequent dephosphorylation allows reassembly of the contractile apparatus for the next round of cytokinesis.

A third and very important CDK substrate is the **retinoblastoma protein, pRb**; when DNA damage is detected, this protein participates in a mechanism that arrests cell division in G1 (Fig. 12-46). Named for the retinal tumor cell line in which it was discovered, pRb functions in most, perhaps all, cell types to regulate cell division in response to a variety of stimuli. Unphosphorylated pRb binds the transcription factor E2F; while bound to pRb, E2F cannot promote transcription of a group of genes necessary for DNA synthesis (the genes for DNA polymerase α , ribonucleotide reductase, and other proteins; Chapter 25). In this state, the cell cycle cannot proceed from the G1 to the S phase, the step that commits a cell to mitosis and cell division. The pRb-E2F blocking mechanism is relieved when pRb is phosphorylated by cyclin E-CDK2, which occurs in response to a signal for cell division to proceed.

When the protein kinases ATM and ATR detect damage to DNA, such as a single-strand break, they activate p53 to serve as a transcription factor that stimulates the synthesis of the protein p21 (Fig. 12-46). This protein inhibits the protein kinase activity of cyclin E-CDK2. In the presence of p21, pRb remains unphosphorylated and bound to E2F, blocking the activity of this transcription factor, and the cell cycle is arrested in G1. This gives the cell time to repair its DNA before entering the S

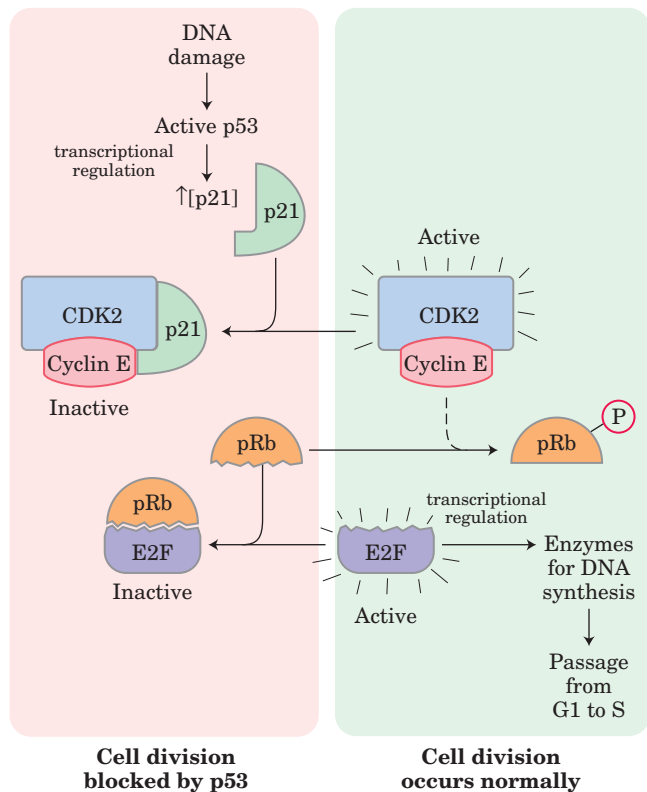


FIGURE 12-46 Regulation of passage from G1 to S by phosphorylation of pRb. When the retinoblastoma protein, pRb, is phosphorylated, it cannot bind and inactivate E2F, a transcription factor that promotes synthesis of enzymes essential to DNA synthesis. If the regulatory protein p53 is activated by ATM and ATR, protein kinases that detect damaged DNA, it stimulates the synthesis of p21, which can bind to and inhibit cyclin E-CDK2 and thus prevent phosphorylation of pRb. Unphosphorylated pRb binds and inactivates E2F, blocking passage from G1 to S until the DNA has been repaired.

phase, thereby avoiding the potentially disastrous transfer of a defective genome to one or both daughter cells.

SUMMARY 12.9 Regulation of the Cell Cycle by Protein Kinases

- Progression through the cell cycle is regulated by the cyclin-dependent protein kinases (CDKs), which act at specific points in the cycle, phosphorylating key proteins and modulating their activities. The catalytic subunit of CDKs is inactive unless associated with the regulatory cyclin subunit.
- The activity of a cyclin-CDK complex changes during the cell cycle through differential synthesis of CDKs, specific degradation of cyclin, phosphorylation and dephosphorylation of critical residues in CDKs, and binding of inhibitory proteins to specific cyclin-CDKs.

12.10 Oncogenes, Tumor Suppressor Genes, and Programmed Cell Death

Tumors and cancer are the result of uncontrolled cell division. Normally, cell division is regulated by a family of extracellular growth factors, proteins that cause resting cells to divide and, in some cases, differentiate. Defects in the synthesis, regulation, or recognition of growth factors can lead to cancer.

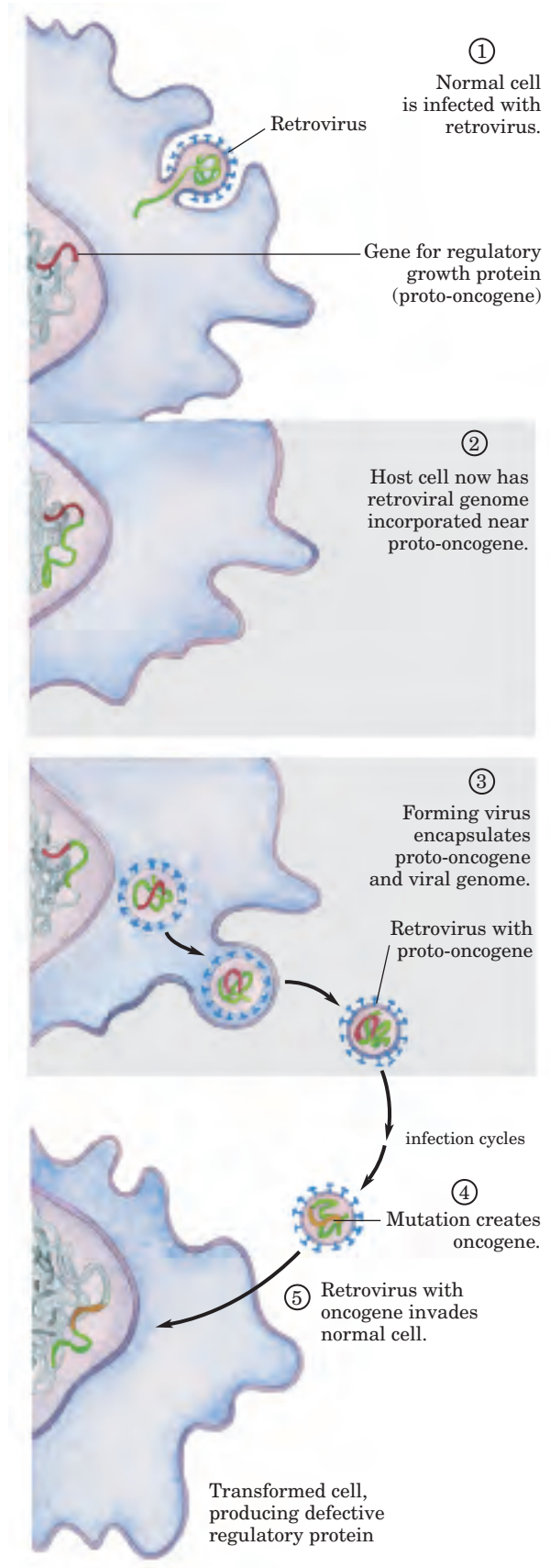
Oncogenes Are Mutant Forms of the Genes for Proteins That Regulate the Cell Cycle

Oncogenes were originally discovered in tumor-causing viruses, then later found to be closely similar to or derived from genes in the animal host cells, proto-oncogenes, which encode growth-regulating proteins. During viral infections, the DNA sequence of a proto-oncogene is sometimes copied by the virus and incorporated into its genome (Fig. 12-47). At some point during the viral infection cycle, the gene can become defective by truncation or mutation. When this viral oncogene is expressed in its host cell during a subsequent infection, the abnormal protein product interferes with normal regulation of cell growth, sometimes resulting in a tumor.

Proto-oncogenes can become oncogenes without a viral intermediary. Chromosomal rearrangements, chemical agents, and radiation are among the factors that can cause oncogenic mutations. The mutations that produce oncogenes are genetically dominant; if either of a pair of chromosomes contains a defective gene, that gene product sends the signal “divide” and a tumor will result. The oncogenic defect can be in any of the proteins involved in communicating the “divide” signal. We know of oncogenes that encode secreted proteins, growth factors, transmembrane proteins (receptors), cytoplasmic proteins (G proteins and protein kinases), and the nuclear transcription factors that control the expression of genes essential for cell division (Jun, Fos).

FIGURE 12-47 Conversion of a regulatory gene to a viral oncogene.

① A normal cell is infected by a retrovirus (Chapter 26), which ② inserts its own genome into the chromosome of the host cell, near the gene for a regulatory protein (the proto-oncogene). ③ Viral particles released from the infected cell sometimes “capture” a host gene, in this case a proto-oncogene. ④ During several cycles of infection, a mutation occurs in the viral proto-oncogene, converting it to an oncogene. ⑤ When the virus subsequently infects a cell, it introduces the oncogene into the cell’s DNA. Transcription of the oncogene leads to the production of a defective regulatory protein that continuously gives the signal for cell division, overriding normal regulatory mechanisms. Host cells infected with oncogene-carrying viruses undergo unregulated cell division—they form tumors. Proto-oncogenes can also undergo mutation to oncogenes without the intervention of a retrovirus, as described in the text.



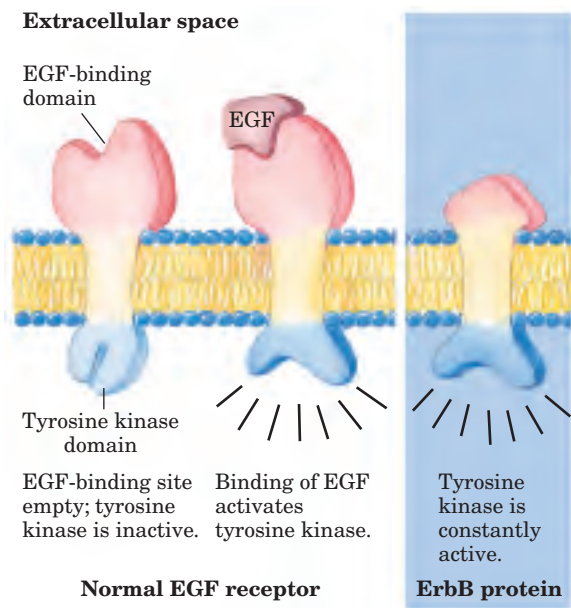


FIGURE 12-48 Oncogene-encoded defective EGF receptor. The product of the *erbB* oncogene (the ErbB protein) is a truncated version of the normal receptor for epidermal growth factor (EGF). Its intracellular domain has the structure normally induced by EGF binding, but the protein lacks the extracellular binding site for EGF. Unregulated by EGF, ErbB continuously signals cell division.

Some oncogenes encode surface receptors with defective or missing signal-binding sites such that their intrinsic Tyr kinase activity is unregulated. For example, the protein ErbB is essentially identical to the normal receptor for epidermal growth factor, except that ErbB lacks the amino-terminal domain that normally binds EGF (Fig. 12-48) and as a result sends the “divide” signal whether EGF is present or not. Mutations in *erbB2*, the gene for a receptor Tyr kinase related to ErbB, are commonly associated with cancers of the glandular epithelium in breast, stomach, and ovary. (For an explanation of the use of abbreviations in naming genes and their products, see Chapter 25.)

Mutant forms of the G protein Ras are common in tumor cells. The *ras* oncogene encodes a protein with normal GTP binding but no GTPase activity. The mutant Ras protein is therefore always in its activated (GTP-bound) form, regardless of the signals arriving through normal receptors. The result can be unregulated growth. Mutations in *ras* are associated with 30% to 50% of lung and colon carcinomas and more than 90% of pancreatic carcinomas.

Defects in Tumor Suppressor Genes Remove Normal Restraints on Cell Division



Tumor suppressor genes encode proteins that normally restrain cell division. Mutation in one or more of these genes can lead to tumor formation.

Unregulated growth due to defective tumor suppressor genes, unlike that due to oncogenes, is genetically recessive; tumors form only if *both* chromosomes of a pair contain a defective gene. In a person who inherits one correct copy and one defective copy, every cell has one defective copy of the gene. If any one of those 10^{12} somatic cells undergoes mutation in the one good copy, a tumor may grow from that doubly mutant cell. Mutations in both copies of the genes for pRb, p53, or p21 yield cells in which the normal restraint on cell division is lost and a tumor forms.

Retinoblastoma is a cancer of the retina that occurs in children who have two defective *Rb* alleles. Very young children who develop retinoblastoma commonly have multiple tumors in both eyes. Each tumor is derived from a single retinal cell that has undergone a mutation in its one good copy of the *Rb* gene. (A fetus with two mutant alleles in every cell is nonviable.) Retinoblastoma patients also have a high incidence of cancers of the lung, prostate, and breast.

A far less likely event is that a person born with two good copies of a gene will have two independent mutations in the *same* gene in the *same* cell, but this does occur. Some individuals develop retinoblastomas later in childhood, usually with only one tumor in only one eye. These individuals were presumably born with two good copies of *Rb* in every cell, but both *Rb* genes in a single retinal cell have undergone mutation, leading to a tumor.

Mutations in the gene for p53 also cause tumors; in more than 90% of human cutaneous squamous cell carcinomas (skin cancers) and about 50% of all other human cancers, *p53* is defective. Those very rare individuals who *inherit* one defective copy of *p53* commonly have the Li-Fraumeni cancer syndrome, in which multiple cancers (of the breast, brain, bone, blood, lung, and skin) occur at high frequency and at an early age. The explanation for multiple tumors in this case is the same as that for *Rb* mutations: an individual born with one defective copy of *p53* in every somatic cell is likely to suffer a second *p53* mutation in more than one cell in his or her lifetime.

Mutations in oncogenes and tumor suppressor genes do not have an all-or-none effect. In some cancers, perhaps in all, the progression from a normal cell to a malignant tumor requires an accumulation of mutations (sometimes over several decades), none of which, alone, is responsible for the end effect. For example, the development of colorectal cancer has several recognizable stages, each associated with a mutation (Fig. 12-49). If a normal epithelial cell in the colon undergoes mutation of both copies of the tumor suppressor gene *APC* (adenomatous polyposis coli), it begins to divide faster than normal and produces a clone of itself, a benign polyp (early adenoma). For reasons not yet known, the *APC* mutation results in chromosomal instability; whole regions of a chromosome are lost or re-

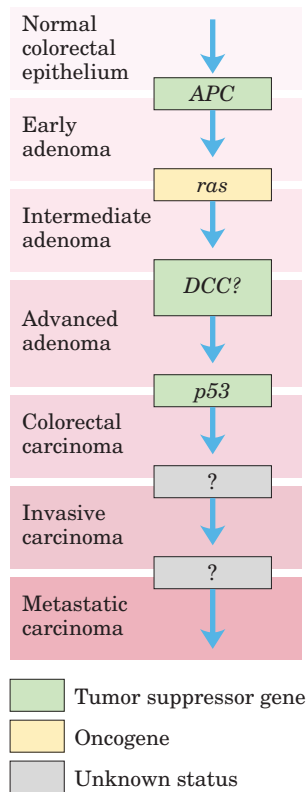


FIGURE 12-49 From normal epithelial cell to colorectal cancer. In the colon, mutations in both copies of the tumor suppressor gene *APC* lead to benign clusters of epithelial cells that multiply too rapidly (early adenoma). If a cell already defective in *APC* suffers a second mutation in the proto-oncogene *ras*, the doubly mutant cell gives rise to an intermediate adenoma, forming a benign polyp of the colon. When one of these cells undergoes further mutations in the tumor suppressor genes *DCC* (probably) and *p53*, increasingly aggressive tumors form. Finally, mutations in genes not yet characterized lead to a malignant tumor and finally to a metastatic tumor that can spread to other tissues. Most malignant tumors probably result from a series of mutations such as this.

arranged during cell division. This instability can lead to another mutation, commonly in *ras*, that converts the clone into an intermediate adenoma. A third mutation (probably in the tumor suppressor gene *DCC*) leads to a late adenoma. Only when both copies of *p53* become defective does this cell mass become a carcinoma, a malignant, life-threatening cancer. The full sequence therefore requires at least seven genetic “hits”: two on each of three tumor suppressor genes (*APC*, *DCC*, and *p53*) and one on the protooncogene *ras*. There are probably several other routes to colorectal cancer as well, but the principle that full malignancy results only from multiple mutations is likely to hold. When a polyp is detected in the early adenoma stage and the cells containing the first mutations are removed surgically, late adenomas and carcinomas will not develop; hence the importance of early detection. ■

Apoptosis Is Programmed Cell Suicide

Many cells can precisely control the time of their own death by the process of **programmed cell death**, or **apoptosis** (app'-a-toe'-sis; from the Greek for “dropping off,” as in leaves dropping in the fall). In the development of an embryo, for example, some cells must die. Carving fingers from stubby limb buds requires the precisely timed death of cells between developing finger bones. During development of the nematode *Caenorhabditis elegans* from a fertilized egg, exactly 131 cells (of a total of 1,090 somatic cells in the embryo) must undergo programmed death in order to construct the adult body.

Apoptosis also has roles in processes other than development. When an antibody-producing cell begins to make antibodies against an antigen normally present in the body, that cell undergoes programmed death in the thymus gland—an essential mechanism for eliminating anti-self antibodies. The monthly sloughing of cells of the uterine wall (menstruation) is another case of apoptosis mediating normal cell death. Sometimes cell suicide is not programmed but occurs in response to biological circumstances that threaten the rest of the organism. For example, a virus-infected cell that dies before completion of the infection cycle prevents spread of the virus to nearby cells. Severe stresses such as heat, hyperosmolarity, UV light, and gamma irradiation also trigger cell suicide; presumably the organism is better off with aberrant cells dead.

The regulatory mechanisms that trigger apoptosis involve some of the same proteins that regulate the cell cycle. The signal for suicide often comes from outside, through a surface receptor. Tumor necrosis factor (TNF), produced by cells of the immune system, interacts with cells through specific TNF receptors. These receptors have TNF-binding sites on the outer face of the plasma membrane and a “death domain” of about 80 amino acid residues that passes the self-destruct signal through the membrane to cytosolic proteins such as TRADD (TNF receptor-associated death domain) (Fig. 12-50). Another receptor, Fas, has a similar death domain that allows it to interact with the cytosolic protein FADD (Fas-associated death domain), which activates a cytosolic protease called caspase 8. This enzyme belongs to a family of proteases that participate in apoptosis; all are synthesized as inactive proenzymes, all have a critical Cys residue at the active site, and all hydrolyze their target proteins on the carboxyl-terminal side of specific Asp residues (hence the name caspase).

When caspase 8, an “initiator” caspase, is activated by an apoptotic signal carried through FADD, it further self-activates by cleaving its own proenzyme form. Mitochondria are one target of active caspase 8. The protease causes the release of certain proteins contained between the inner and outer mitochondrial membranes:

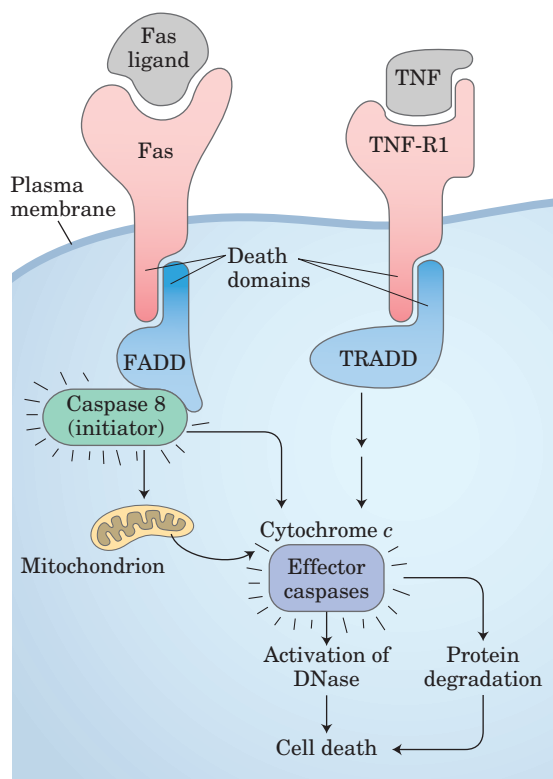


FIGURE 12-50 Initial events of apoptosis. Receptors in the plasma membrane (Fas, TNF-R1) receive signals from outside the cell (the Fas ligand or tumor necrosis factor (TNF), respectively). Activated receptors foster interaction between the “death domain” (an 80 amino acid sequence) in Fas or TNF-R1 and a similar death domain in the cytosolic proteins FADD or TRADD. FADD activates a cytosolic protease, caspase 8, that proteolytically activates other cellular proteases. TRADD also activates proteases. The resulting proteolysis is a primary factor in cell death.

cytochrome *c* (Chapter 19) and several “effector” caspases. Cytochrome *c* binds to the proenzyme form of the effector enzyme caspase 9 and stimulates its proteolytic activation. The activated caspase 9 in turn catalyzes wholesale destruction of cellular proteins—a major cause of apoptotic cell death. One specific target of caspase action is a caspase-activated deoxyribonuclease.

In apoptosis, the monomeric products of protein and DNA degradation (amino acids and nucleotides) are released in a controlled process that allows them to be taken up and reused by neighboring cells. Apoptosis thus allows the organism to eliminate a cell without wasting its components.

SUMMARY 12.10 Oncogenes, Tumor Suppressor Genes, and Programmed Cell Death

- Oncogenes encode defective signaling proteins. By continually giving the signal for cell division, they lead to tumor formation. Oncogenes are genetically dominant and may encode defective growth factors, receptors, G proteins, protein kinases, or nuclear regulators of transcription.
- Tumor suppressor genes encode regulatory proteins that normally inhibit cell division; mutations in these genes are genetically recessive but can lead to tumor formation.
- Cancer is generally the result of an accumulation of mutations in oncogenes and tumor suppressor genes.
- Apoptosis can be triggered by extracellular signals such as TNF through plasma membrane receptors.

Key Terms

Terms in bold are defined in the glossary.

signal transduction 421

enzyme cascade 422

desensitization 422

ligand-gated receptor channel 426

voltage-gated ion channel 427

second messenger 428

autophosphorylation 429

SH2 domain 429

G proteins 429

MAPK cascade 430

receptor Tyr kinase 432

serpentine receptors 435

G protein-coupled receptors

(GPCR) 435

7 transmembrane segment (7tm)

receptors 435

stimulatory G protein (G_s) 436

β -adrenergic receptor kinase

(β ARK) 441

β -arrestin (β arr; arrestin 2) 441

G protein-coupled receptor kinases

(GRKs) 441

scaffold proteins 441

inhibitory G protein (G_i) 441

calmodulin (CaM) 444

Ca^{2+} /calmodulin-dependent protein kinases (CaM kinases I–IV) 444

two-component signaling

systems 452

receptor His kinase 452

response regulator 452

receptorlike kinase (RLK) 455

hormone response element (HRE) 465

tamoxifen 465

RU486 466

cyclin 467

cyclin-dependent protein kinase (CDK) 467

ubiquitin 469

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Problems



1. Therapeutic Effects of Albuterol The respiratory symptoms of asthma result from constriction of the bronchi and bronchioles of the lungs due to contraction of the smooth muscle of their walls. This constriction can be reversed by raising the [cAMP] in the smooth muscle. Explain the therapeutic effects of albuterol, a β -adrenergic agonist taken (by inhalation) for asthma. Would you expect this drug to have any side effects? How might one design a better drug that did not have these effects?

2. Amplification of Hormonal Signals Describe all the sources of amplification in the insulin receptor system.

3. Termination of Hormonal Signals Signals carried by hormones must eventually be terminated. Describe several different mechanisms for signal termination.

4. Specificity of a Signal for a Single Cell Type Discuss the validity of the following proposition. A signaling molecule (hormone, growth factor, or neurotransmitter) elicits identical responses in different types of target cells if they contain identical receptors.

5. Resting Membrane Potential A variety of unusual invertebrates, including giant clams, mussels, and polychaete worms, live on the fringes of hydrothermal vents on the ocean bottom, where the temperature is 60 °C.

(a) The adductor muscle of a deep-sea giant clam has a resting membrane potential of -95 mV. Given the intracellular and extracellular ionic compositions shown below, would you have predicted this membrane potential? Why or why not?

Ion	Concentration (mM)	
	Intracellular	Extracellular
Na ⁺	50	440
K ⁺	400	20
Cl [−]	21	560
Ca ²⁺	0.4	10

(b) Assume that the adductor muscle membrane is permeable to only one of the ions listed above. Which ion could determine the V_m ?

6. Membrane Potentials in Frog Eggs Fertilization of a frog oocyte by a sperm cell triggers ionic changes similar to those observed in neurons (during movement of the action potential) and initiates the events that result in cell division and development of the embryo. Oocytes can be stimulated to divide without fertilization by suspending them in 80 mM KCl (normal pond water contains 9 mM KCl).

(a) Calculate how much the change in extracellular [KCl] changes the resting membrane potential of the oocyte. (Hint: Assume the oocyte contains 120 mM K^+ and is permeable *only* to K^+ .) Assume a temperature of 20 °C.

(b) When the experiment is repeated in Ca^{2+} -free water, elevated [KCl] has no effect. What does this suggest about the mechanism of the KCl effect?

7. Excitation Triggered by Hyperpolarization In most neurons, membrane *depolarization* leads to the opening of voltage-dependent ion channels, generation of an action potential, and ultimately an influx of Ca^{2+} , which causes release of neurotransmitter at the axon terminus. Devise a cellular strategy by which *hyperpolarization* in rod cells could produce excitation of the visual pathway and passage of visual signals to the brain. (Hint: The neuronal signaling pathway in higher organisms consists of a *series* of neurons that relay information to the brain (see Fig. 12–31). The signal released by one neuron can be either excitatory or inhibitory to the following, postsynaptic neuron.)

8. Hormone Experiments in Cell-Free Systems In the 1950s, Earl W. Sutherland, Jr., and his colleagues carried out pioneering experiments to elucidate the mechanism of action of epinephrine and glucagon. Given what you have learned in this chapter about hormone action, interpret each of the experiments described below. Identify substance X and indicate the significance of the results.

(a) Addition of epinephrine to a homogenate of normal liver resulted in an increase in the activity of glycogen phosphorylase. However, if the homogenate was first centrifuged at a high speed and epinephrine or glucagon was added to the clear supernatant fraction that contains phosphorylase, no increase in the phosphorylase activity occurred.

(b) When the particulate fraction from the centrifugation in (a) was treated with epinephrine, substance X was produced. The substance was isolated and purified. Unlike epinephrine, substance X activated glycogen phosphorylase when added to the clear supernatant fraction of the centrifuged homogenate.

(c) Substance X was heat-stable; that is, heat treatment did not affect its capacity to activate phosphorylase. (Hint: Would this be the case if substance X were a protein?) Substance X was nearly identical to a compound obtained when pure ATP was treated with barium hydroxide. (Fig. 8–6 will be helpful.)



9. Effect of Cholera Toxin on Adenylyl Cyclase

The gram-negative bacterium *Vibrio cholerae* produces a protein, cholera toxin (M_r 90,000), that is responsible for the characteristic symptoms of cholera: extensive loss of body water and Na^+ through continuous, debilitating diarrhea. If body fluids and Na^+ are not replaced, severe dehydration results; untreated, the disease is often fatal. When

the cholera toxin gains access to the human intestinal tract it binds tightly to specific sites in the plasma membrane of the epithelial cells lining the small intestine, causing adenylyl cyclase to undergo prolonged activation (hours or days).

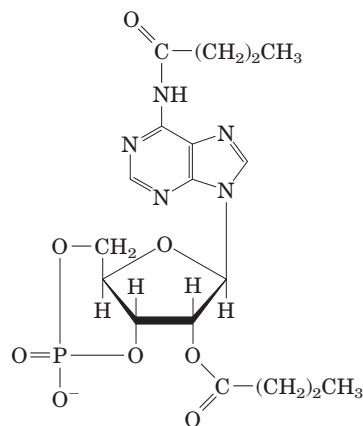
(a) What is the effect of cholera toxin on [cAMP] in the intestinal cells?

(b) Based on the information above, suggest how cAMP normally functions in intestinal epithelial cells.

(c) Suggest a possible treatment for cholera.

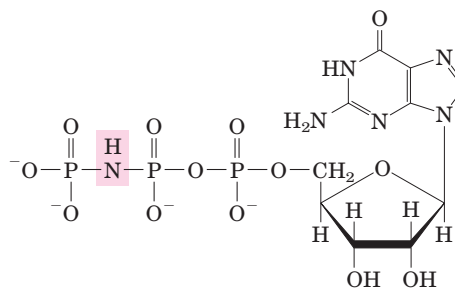
10. Effect of Dibutyryl cAMP versus cAMP on Intact Cells

The physiological effects of epinephrine should in principle be mimicked by addition of cAMP to the target cells. In practice, addition of cAMP to intact target cells elicits only a minimal physiological response. Why? When the structurally related derivative dibutyryl cAMP (shown below) is added to intact cells, the expected physiological response is readily apparent. Explain the basis for the difference in cellular response to these two substances. Dibutyryl cAMP is widely used in studies of cAMP function.



Dibutyryl cAMP
($N^6,O^{2'}$ -Dibutyryl adenosine 3',5'-cyclic monophosphate)

11. Nonhydrolyzable GTP Analogs Many enzymes can hydrolyze GTP between the β and γ phosphates. The GTP analog β,γ -imidoguanosine 5'-triphosphate Gpp(NH)p, shown below, cannot be hydrolyzed between the β and γ phosphates. Predict the effect of microinjection of Gpp(NH)p into a myocyte on the cell's response to β -adrenergic stimulation.



Gpp(NH)p
(β,γ -imidoguanosine 5'-triphosphate)

12. G Protein Differences Compare the G proteins G_s , which acts in transducing the signal from β -adrenergic receptors, and Ras. What properties do they share? How do they differ? What is the functional difference between G_s and G_i ?

13. EGTA Injection EGTA (ethylene glycol-bis(β -aminoethyl ether)- N,N,N',N' -tetraacetic acid) is a chelating agent with high affinity and specificity for Ca^{2+} . By microinjecting a cell with an appropriate Ca^{2+} -EDTA solution, an experimenter can prevent cytosolic $[\text{Ca}^{2+}]$ from rising above 10^{-7} M. How would EGTA microinjection affect a cell's response to vasopressin (see Table 12-5)? To glucagon?



14. Visual Desensitization Oguchi's disease is an inherited form of night blindness. Affected individuals are slow to recover vision after a flash of bright light against a dark background, such as the headlights of a car on the freeway. Suggest what the molecular defect(s) might be in Oguchi's disease. Explain in molecular terms how this defect accounts for the night blindness.

15. Mutations in PKA Explain how mutations in the R or C subunit of cAMP-dependent protein kinase (PKA) might lead to (a) a constantly active PKA or (b) a constantly inactive PKA.

16. Mechanisms for Regulating Protein Kinases Identify eight general types of protein kinases found in eukaryotic cells, and explain what factor is *directly* responsible for activating each type.

17. Mutations in Tumor Suppressor Genes and Oncogenes Explain why mutations in tumor suppressor genes are recessive (both copies of the gene must be defective for the regulation of cell division to be defective) whereas mutations in oncogenes are dominant.



18. Retinoblastoma in Children Explain why some children with retinoblastoma develop multiple tumors of the retina in both eyes, whereas others have a single tumor in only one eye.

19. Mutations in *ras* How does a mutation in the *ras* gene that leads to formation of a Ras protein with no GTPase activity affect a cell's response to insulin?



BIOENERGETICS AND METABOLISM

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- 23 Integration and Hormonal Regulation of Mammalian Metabolism 891

Metabolism is a highly coordinated cellular activity in which many multienzyme systems (metabolic pathways) cooperate to (1) obtain chemical energy by capturing solar energy or degrading energy-rich nutrients from the environment; (2) convert nutrient molecules into the cell's own characteristic molecules, including precursors of macromolecules; (3) polymerize monomeric precursors into macromolecules: proteins, nucleic acids, and polysaccharides; and (4) synthesize and degrade biomolecules required for specialized cellular functions, such as membrane lipids, intracellular messengers, and pigments.

Although metabolism embraces hundreds of different enzyme-catalyzed reactions, our major concern in Part II is the central metabolic pathways, which are few in number and remarkably similar in all forms of life. Living organisms can be divided into two large groups according to the chemical form in which they obtain carbon from the environment. **Autotrophs** (such as photosynthetic bacteria and vascular plants) can use carbon dioxide from the atmosphere as their sole source of carbon, from which they construct all their carbon-containing biomolecules (see Fig. 1–5). Some autotrophic organisms, such as cyanobacteria, can also use atmospheric nitrogen to generate all their nitrogenous components. **Heterotrophs** cannot use atmospheric carbon dioxide and must obtain carbon from their environment in the form of relatively complex organic molecules such as glucose. Multicellular animals and most microorganisms are heterotrophic. Autotrophic cells and organisms are relatively self-sufficient, whereas heterotrophic cells and organisms, with their requirements for carbon in more complex forms, must subsist on the products of other organisms.

Many autotrophic organisms are photosynthetic and obtain their energy from sunlight, whereas heterotrophic organisms obtain their energy from the degradation of organic nutrients produced by autotrophs. In our biosphere, autotrophs and heterotrophs live together in a vast, interdependent cycle in which autotrophic organisms use atmospheric carbon dioxide to build their organic biomolecules, some of them generating oxygen from water in the process. Heterotrophs in turn use the organic products of autotrophs as nutrients and return carbon dioxide to the atmosphere. Some of the oxidation reactions that produce carbon dioxide also consume oxygen, converting it to water. Thus carbon, oxygen, and water are constantly cycled between the heterotrophic and autotrophic worlds, with

solar energy as the driving force for this global process (Fig. 1).

All living organisms also require a source of nitrogen, which is necessary for the synthesis of amino acids, nucleotides, and other compounds. Plants can generally use either ammonia or nitrate as their sole source of nitrogen, but vertebrates must obtain nitrogen in the form of amino acids or other organic compounds. Only a few organisms—the cyanobacteria and many species of soil bacteria that live symbiotically on the roots of some plants—are capable of converting (“fixing”) atmospheric nitrogen (N_2) into ammonia. Other bacteria (the nitrifying bacteria) oxidize ammonia to nitrites and nitrates; yet others convert nitrate to N_2 . Thus, in addition to the global carbon and oxygen cycle, a nitrogen cycle operates in the biosphere, turning over huge amounts of nitrogen (Fig. 2). The cycling of carbon, oxygen, and nitrogen, which ultimately involves all species, depends on a proper balance between the activities of the producers (autotrophs) and consumers (heterotrophs) in our biosphere.

These cycles of matter are driven by an enormous flow of energy into and through the biosphere, beginning with the capture of solar energy by photosynthetic organisms and use of this energy to generate energy-rich carbohydrates and other organic nutrients; these nutrients are then used as energy sources by heterotrophic organisms. In metabolic processes, and in all energy transformations, there is a loss of useful energy (free energy) and an inevitable increase in the amount of unusable energy (heat and entropy). In contrast to the cycling of matter, therefore, energy flows one way

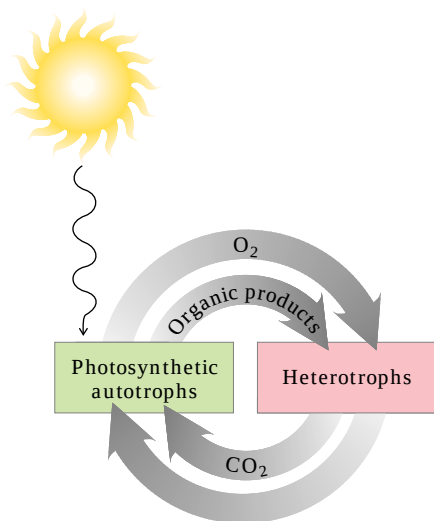


FIGURE 1 Cycling of carbon dioxide and oxygen between the autotrophic (photosynthetic) and heterotrophic domains in the biosphere. The flow of mass through this cycle is enormous; about 4×10^{11} metric tons of carbon are turned over in the biosphere annually.

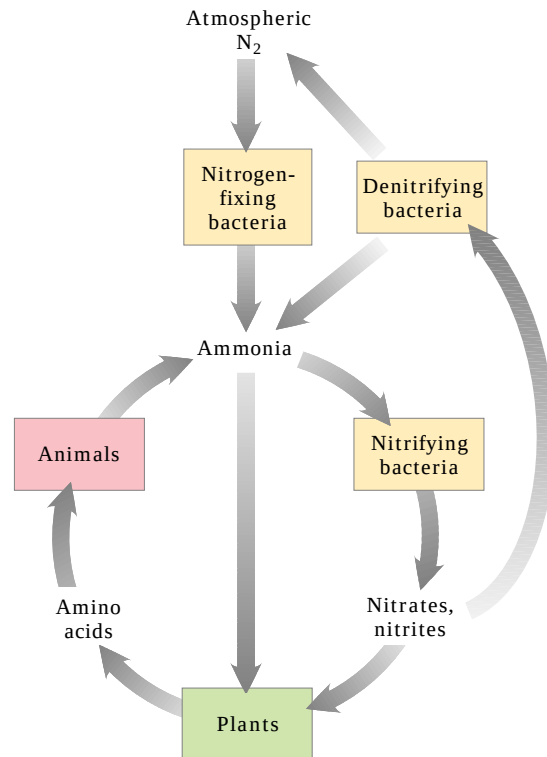


FIGURE 2 Cycling of nitrogen in the biosphere. Gaseous nitrogen (N_2) makes up 80% of the earth's atmosphere.

through the biosphere; organisms cannot regenerate useful energy from energy dissipated as heat and entropy. Carbon, oxygen, and nitrogen recycle continuously, but energy is constantly transformed into unusable forms such as heat.

Metabolism, the sum of all the chemical transformations taking place in a cell or organism, occurs through a series of enzyme-catalyzed reactions that constitute **metabolic pathways**. Each of the consecutive steps in a metabolic pathway brings about a specific, small chemical change, usually the removal, transfer, or addition of a particular atom or functional group. The precursor is converted into a product through a series of metabolic intermediates called **metabolites**. The term **intermediary metabolism** is often applied to the combined activities of all the metabolic pathways that interconvert precursors, metabolites, and products of low molecular weight (generally, $M_r < 1,000$).

Catabolism is the degradative phase of metabolism in which organic nutrient molecules (carbohydrates, fats, and proteins) are converted into smaller, simpler end products (such as lactic acid, CO_2 , NH_3). Catabolic pathways release energy, some of which is conserved in the formation of ATP and reduced electron carriers (NADH, NADPH, and $FADH_2$); the rest is lost as heat. In **anabolism**, also called biosynthesis, small, simple precursors are built up into larger and more complex

molecules, including lipids, polysaccharides, proteins, and nucleic acids. Anabolic reactions require an input of energy, generally in the form of the phosphoryl group transfer potential of ATP and the reducing power of NADH, NADPH, and FADH₂ (Fig. 3).

Some metabolic pathways are linear, and some are branched, yielding multiple useful end products from a single precursor or converting several starting materials into a single product. In general, catabolic pathways are convergent and anabolic pathways divergent (Fig. 4). Some pathways are cyclic: one starting component of the pathway is regenerated in a series of reactions that converts another starting component into a product. We shall see examples of each type of pathway in the following chapters.

Most cells have the enzymes to carry out both the degradation and the synthesis of the important categories of biomolecules—fatty acids, for example. The

simultaneous synthesis and degradation of fatty acids would be wasteful, however, and this is prevented by reciprocally regulating the anabolic and catabolic reaction sequences: when one sequence is active, the other is suppressed. Such regulation could not occur if anabolic and catabolic pathways were catalyzed by exactly the same set of enzymes, operating in one direction for anabolism, the opposite direction for catabolism: inhibition of an enzyme involved in catabolism would also inhibit the reaction sequence in the anabolic direction. Catabolic and anabolic pathways that connect the same two end points (glucose → → pyruvate and pyruvate → → glucose, for example) may employ many of the same enzymes, but invariably at least one of the steps is catalyzed by different enzymes in the catabolic and anabolic directions, and these enzymes are the sites of separate regulation. Moreover, for both anabolic and catabolic pathways to be essentially irreversible, the reactions unique to each direction must include at least one that is thermodynamically very favorable—in other words, a reaction for which the reverse reaction is very unfavorable. As a further contribution to the separate regulation of catabolic and anabolic reaction sequences, paired catabolic and anabolic pathways commonly take place in different cellular compartments: for example, fatty acid catabolism in mitochondria, fatty acid synthesis in the cytosol. The concentrations of intermediates, enzymes, and regulators can be maintained at different levels in these different compartments. Because metabolic pathways are subject to kinetic control by substrate concentration, separate pools of anabolic and catabolic intermediates also contribute to the control of metabolic rates. Devices that separate anabolic and catabolic processes will be of particular interest in our discussions of metabolism.

Metabolic pathways are regulated at several levels, from within the cell and from outside. The most immediate regulation is by the availability of substrate; when the intracellular concentration of an enzyme's substrate is near or below K_m (as is commonly the case), the rate of the reaction depends strongly upon substrate concentration (see Fig. 6–11). A second type of rapid control from within is allosteric regulation (p. 225) by a metabolic intermediate or coenzyme—an amino acid or ATP, for example—that signals the cell's internal metabolic state. When the cell contains an amount of, say, aspartate sufficient for its immediate needs, or when the cellular level of ATP indicates that further fuel consumption is unnecessary at the moment, these signals allosterically inhibit the activity of one or more enzymes in the relevant pathway. In multicellular organisms the metabolic activities of different tissues are regulated and integrated by growth factors and hormones that act from outside the cell. In some cases this regulation occurs virtually instantaneously (sometimes in less than a millisecond) through changes in the levels of intracellular

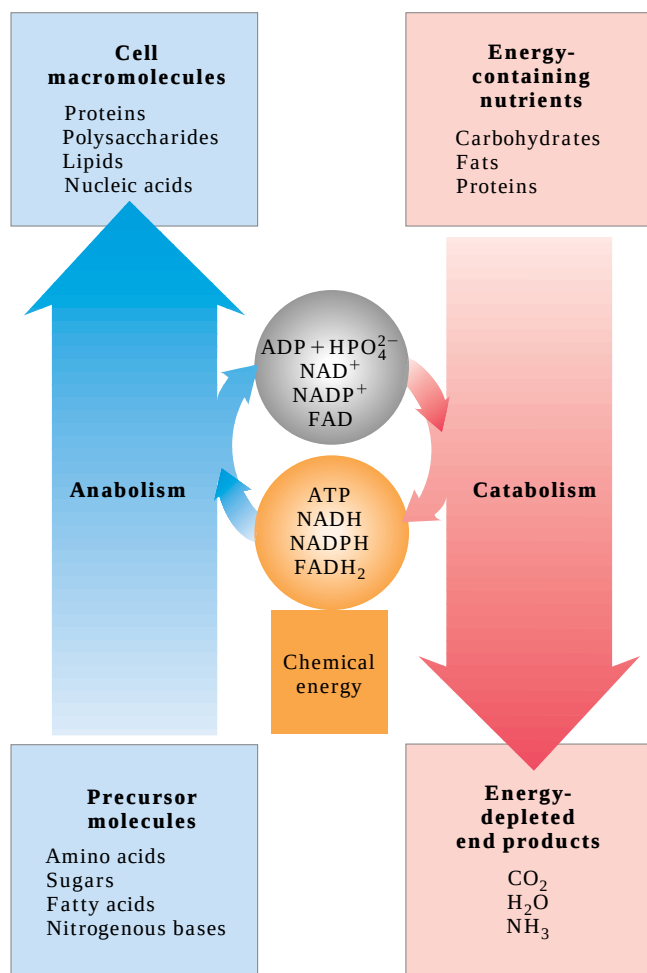


FIGURE 3 Energy relationships between catabolic and anabolic pathways. Catabolic pathways deliver chemical energy in the form of ATP, NADH, NADPH, and FADH₂. These energy carriers are used in anabolic pathways to convert small precursor molecules into cell macromolecules.

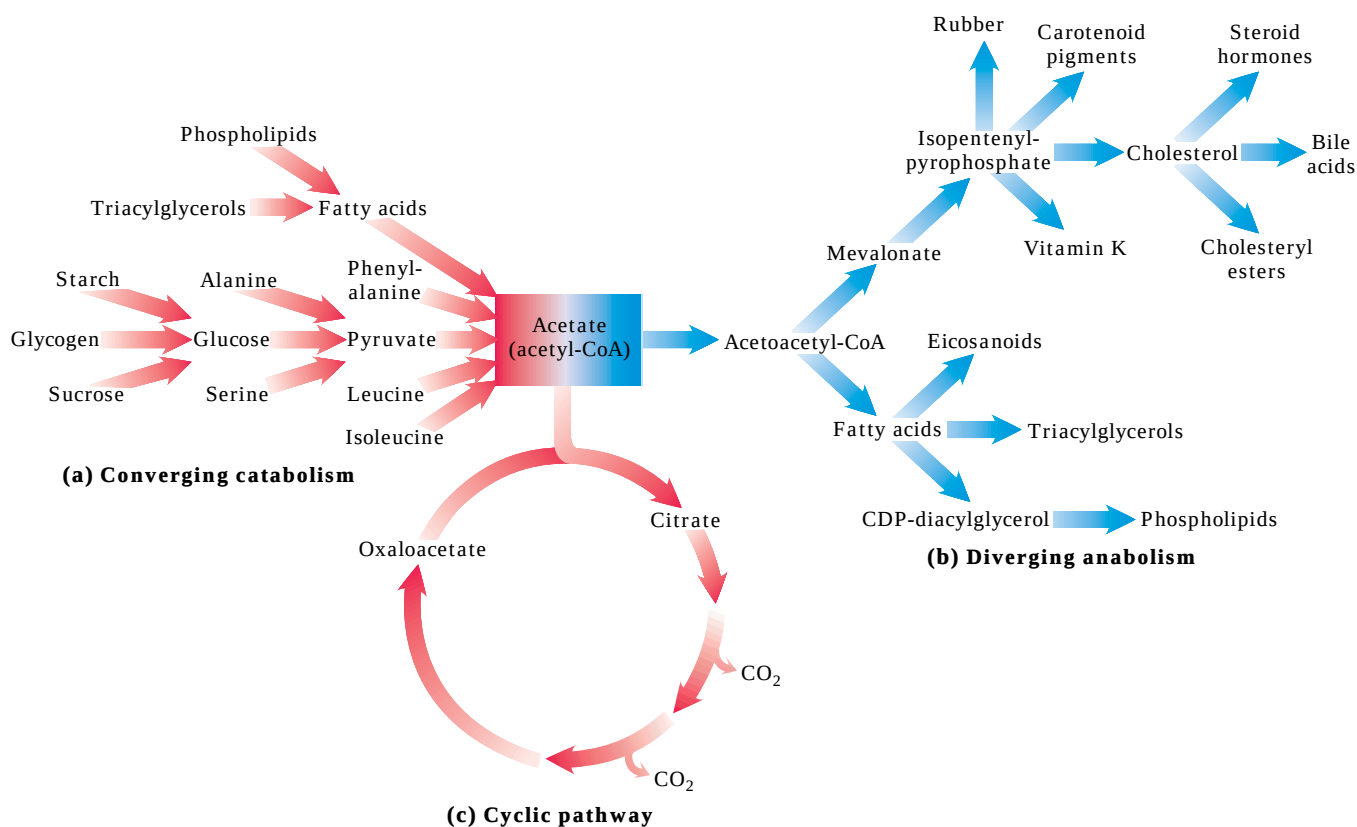


FIGURE 4 Three types of nonlinear metabolic pathways. **(a)** Converging, catabolic; **(b)** diverging, anabolic; and **(c)** cyclic, in which one of the starting materials (oxaloacetate in this case) is regenerated and reenters the pathway. Acetate, a key metabolic intermediate, is

the breakdown product of a variety of fuels **(a)**, serves as the precursor for an array of products **(b)**, and is consumed in the catabolic pathway known as the citric acid cycle **(c)**.

messengers that modify the activity of existing enzyme molecules by allosteric mechanisms or by covalent modification such as phosphorylation. In other cases, the extracellular signal changes the cellular concentration of an enzyme by altering the rate of its synthesis or degradation, so the effect is seen only after minutes or hours.

The number of metabolic transformations taking place in a typical cell can seem overwhelming to a beginning student. Most cells have the capacity to carry out thousands of specific, enzyme-catalyzed reactions; for example, transformation of a simple nutrient such as glucose into amino acids, nucleotides, or lipids; extraction of energy from fuels by oxidation; or polymerization of monomeric subunits into macromolecules. Fortunately for the student of biochemistry, there are patterns within this multitude of reactions; you do not need to learn all these reactions to comprehend the molecular logic of biochemistry. Most of the reactions in living cells fall into one of five general categories: (1) oxidation-reductions; (2) reactions that make or break carbon-carbon bonds; (3) internal rearrangements, isomerizations, and eliminations; (4) group transfers; and (5) free radical reactions. Reactions within each general category usually proceed by a limited set of mechanisms and often employ characteristic cofactors.

Before reviewing the five main reaction classes of biochemistry, let's consider two basic chemical principles. First, a covalent bond consists of a shared pair of electrons, and the bond can be broken in two general ways (Fig. 5). In homolytic cleavage, each atom leaves the bond as a radical, carrying one of the two electrons (now unpaired) that held the bonded atoms together. In the more common, heterolytic cleavage, one atom retains both bonding electrons. The species generated when C—C and C—H bonds are cleaved are illustrated in Figure 5. Carbanions, carbocations, and hydride ions are highly unstable; this instability shapes the chemistry of these ions, as described further below.

The second chemical principle of interest here is that many biochemical reactions involve interactions between nucleophiles (functional groups rich in electrons and capable of donating them) and electrophiles (electron-deficient functional groups that seek electrons). Nucleophiles combine with, and give up electrons to, electrophiles. Common nucleophiles and electrophiles are listed in Figure 6–21. Note that a carbon atom can act as either a nucleophile or an electrophile, depending on which bonds and functional groups surround it.

We now consider the five main reaction classes you will encounter in upcoming chapters.

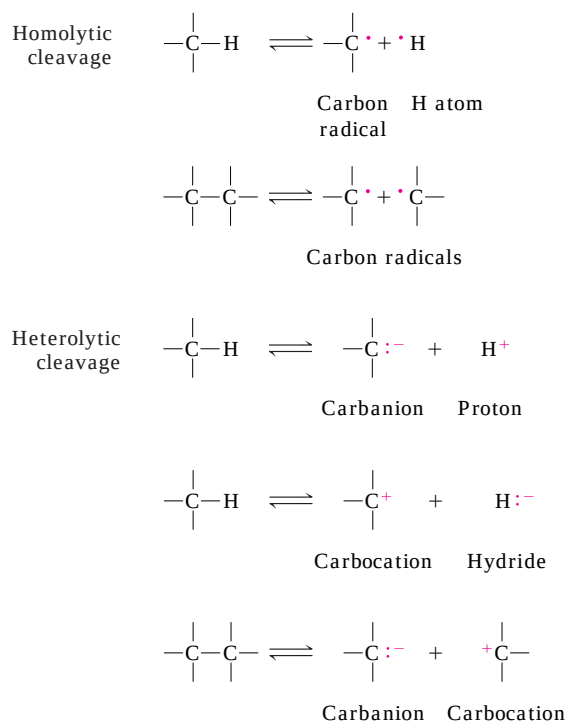


FIGURE 5 Two mechanisms for cleavage of a C—C or C—H bond.

In homolytic cleavages, each atom keeps one of the bonding electrons, resulting in the formation of carbon radicals (carbons having unpaired electrons) or uncharged hydrogen atoms. In heterolytic cleavages, one of the atoms retains both bonding electrons. This can result in the formation of carbanions, carbocations, protons, or hydride ions.

1. Oxidation-reduction reactions Carbon atoms encountered in biochemistry can exist in five oxidation states, depending on the elements with which carbon shares electrons (Fig. 6). In many biological oxidations, a compound loses two electrons and two hydrogen ions (that is, two hydrogen atoms); these reactions are commonly called dehydrogenations and the enzymes that catalyze them are called dehydrogenases (Fig. 7). In some, but not all, biological oxidations, a carbon atom becomes covalently bonded to an oxygen atom. The enzymes that

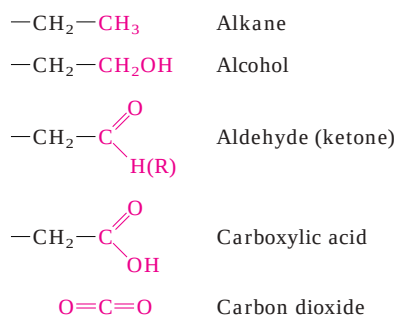


FIGURE 6 The oxidation states of carbon in biomolecules. Each compound is formed by oxidation of the red carbon in the compound listed above it. Carbon dioxide is the most highly oxidized form of carbon found in living systems.

catalyze these oxidations are generally called oxidases or, if the oxygen atom is derived directly from molecular oxygen (O_2), oxygenases.

Every oxidation must be accompanied by a reduction, in which an electron acceptor acquires the electrons removed by oxidation. Oxidation reactions generally release energy (think of camp fires: the compounds in wood are oxidized by oxygen molecules in the air). Most living cells obtain the energy needed for cellular work by oxidizing metabolic fuels such as carbohydrates or fat; photosynthetic organisms can also trap and use the energy of sunlight. The catabolic (energy-yielding) pathways described in Chapters 14 through 19 are oxidative reaction sequences that result in the transfer of electrons from fuel molecules, through a series of electron carriers, to oxygen. The high affinity of O_2 for electrons makes the overall electron-transfer process highly exergonic, providing the energy that drives ATP synthesis—the central goal of catabolism.

2. Reactions that make or break carbon–carbon bonds Heterolytic cleavage of a C—C bond yields a carbanion and a carbocation (Fig. 5). Conversely, the formation of a C—C bond involves the combination of a nucleophilic carbanion and an electrophilic carbocation. Groups with electronegative atoms play key roles in these reactions. Carbonyl groups are particularly important in the chemical transformations of metabolic pathways. As noted above, the carbon of a carbonyl group has a partial positive charge due to the electron-withdrawing nature of the adjacent bonded oxygen, and thus is an electrophilic carbon. The presence of a carbonyl group can also facilitate the formation of a carbanion on an adjoining carbon, because the carbonyl group can delocalize electrons through resonance (Fig. 8a, b). The importance of a carbonyl group is evident in three major classes of reactions in which C—C bonds are formed or broken (Fig 8c): aldol condensations (such as the aldolase reaction; see Fig. 14–5), Claisen condensations (as in the citrate synthase reaction; see Fig. 16–9), and

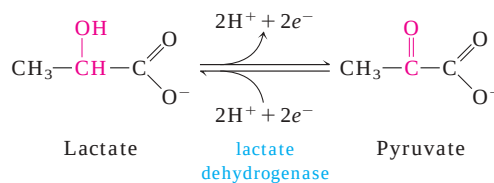


FIGURE 7 An oxidation-reduction reaction. Shown here is the oxidation of lactate to pyruvate. In this dehydrogenation, two electrons and two hydrogen ions (the equivalent of two hydrogen atoms) are removed from C-2 of lactate, an alcohol, to form pyruvate, a ketone. In cells the reaction is catalyzed by lactate dehydrogenase and the electrons are transferred to a cofactor called nicotinamide adenine dinucleotide. This reaction is fully reversible; pyruvate can be reduced by electrons from the cofactor. In Chapter 13 we discuss the factors that determine the direction of a reaction.

decarboxylations (as in the acetoacetate decarboxylase reaction; see Fig. 17–18). Entire metabolic pathways are organized around the introduction of a carbonyl group in a particular location so that a nearby carbon–carbon bond can be formed or cleaved. In some reactions, this role is played by an imine group or a specialized cofactor such as pyridoxal phosphate, rather than by a carbonyl group.

3. Internal rearrangements, isomerizations, and eliminations

Another common type of cellular reaction is an intramolecular rearrangement, in which redistribution of

electrons results in isomerization, transposition of double bonds, or cis-trans rearrangements of double bonds. An example of isomerization is the formation of fructose 6-phosphate from glucose 6-phosphate during sugar metabolism (Fig 9a; this reaction is discussed in detail in Chapter 14). Carbon-1 is reduced (from aldehyde to alcohol) and C-2 is oxidized (from alcohol to ketone). Figure 9b shows the details of the electron movements that result in isomerization.

A simple transposition of a C=C bond occurs during metabolism of the common fatty acid oleic acid (see Fig. 17–9), and you will encounter some spectacular examples of double-bond repositioning in the synthesis of cholesterol (see Fig. 21–35).

Elimination of water introduces a C=C bond between two carbons that previously were saturated (as in the enolase reaction; see Fig. 6–23). Similar reactions can result in the elimination of alcohols and amines.

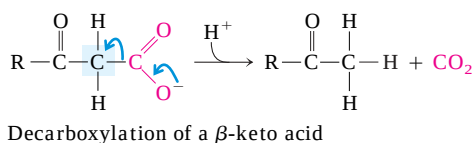
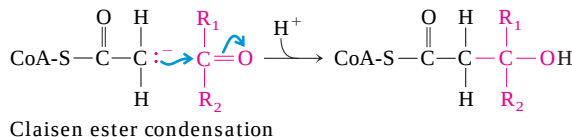
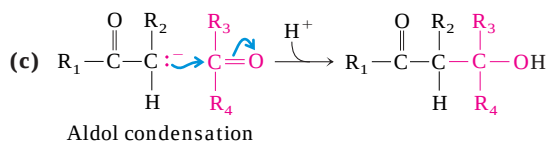
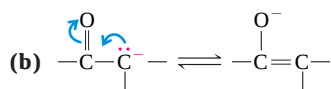
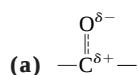
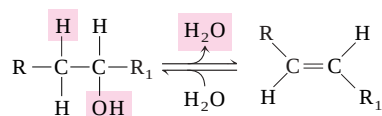
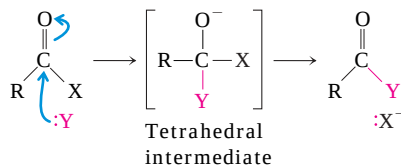


FIGURE 8 Carbon–carbon bond formation reactions. (a) The carbon atom of a carbonyl group is an electrophile by virtue of the electron-withdrawing capacity of the electronegative oxygen atom, which results in a resonance hybrid structure in which the carbon has a partial positive charge. (b) Within a molecule, delocalization of electrons into a carbonyl group facilitates the transient formation of a carbanion on an adjacent carbon. (c) Some of the major reactions involved in the formation and breakage of C—C bonds in biological systems. For both the aldol condensation and the Claisen condensation, a carbanion serves as nucleophile and the carbon of a carbonyl group serves as electrophile. The carbanion is stabilized in each case by another carbonyl at the carbon adjoining the carbanion carbon. In the decarboxylation reaction, a carbanion is formed on the carbon shaded blue as the CO₂ leaves. The reaction would not occur at an appreciable rate but for the stabilizing effect of the carbonyl adjacent to the carbanion carbon. Wherever a carbanion is shown, a stabilizing resonance with the adjacent carbonyl, as shown in (a), is assumed. The formation of the carbanion is highly disfavored unless the stabilizing carbonyl group, or a group of similar function such as an imine, is present.



4. Group transfer reactions The transfer of acyl, glycosyl, and phosphoryl groups from one nucleophile to another is common in living cells. Acyl group transfer generally involves the addition of a nucleophile to the carbonyl carbon of an acyl group to form a tetrahedral intermediate.



The chymotrypsin reaction is one example of acyl group transfer (see Fig. 6–21). Glycosyl group transfers involve nucleophilic substitution at C-1 of a sugar ring, which is the central atom of an acetal. In principle, the substitution could proceed by an S_N1 or S_N2 path, as described for the enzyme lysozyme (see Fig. 6–25).

Phosphoryl group transfers play a special role in metabolic pathways. A general theme in metabolism is the attachment of a good leaving group to a metabolic intermediate to “activate” the intermediate for subsequent reaction. Among the better leaving groups in nucleophilic substitution reactions are inorganic orthophosphate (the ionized form of H₃PO₄ at neutral pH, a mixture of H₂PO₄[−] and HPO₄^{2−}, commonly abbreviated P_i) and inorganic pyrophosphate (P₂O₇^{4−}, abbreviated PP_i); esters and anhydrides of phosphoric acid are effectively activated for reaction. Nucleophilic substitution is made more favorable by the attachment of a phosphoryl group to an otherwise poor leaving group such as —OH. Nucleophilic substitutions in which the

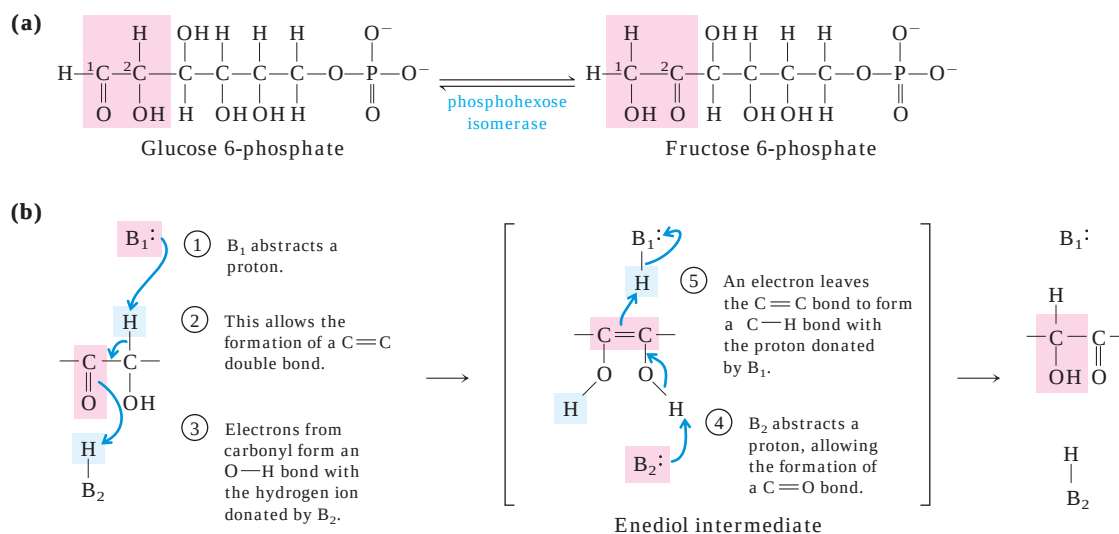


FIGURE 9 Isomerization and elimination reactions. (a) The conversion of glucose 6-phosphate to fructose 6-phosphate, a reaction of sugar metabolism catalyzed by phosphohexose isomerase. (b) This reaction proceeds through an enediol intermediate. The curved blue arrows represent the movement of bonding electrons from nucleophile (pink) to electrophile (blue). B_1 and B_2 are basic groups on the enzyme; they are capable of donating and accepting hydrogen ions (protons) as the reaction progresses.

phosphoryl group ($-\text{PO}_3^{2-}$) serves as a leaving group occur in hundreds of metabolic reactions.

Phosphorus can form five covalent bonds. The conventional representation of P_i (Fig. 10a), with three $P-O$ bonds and one $P=O$ bond, is not an accurate picture. In P_i , four equivalent phosphorus-oxygen bonds share some double-bond character, and the anion has a tetrahedral structure (Fig. 10b). As oxygen is more electronegative than phosphorus, the sharing of electrons is unequal: the central phosphorus bears a partial positive

charge and can therefore act as an electrophile. In a very large number of metabolic reactions, a phosphoryl group ($-\text{PO}_3^{2-}$) is transferred from ATP to an alcohol (forming a phosphate ester) (Fig. 10c) or to a carboxylic acid (forming a mixed anhydride). When a nucleophile attacks the electrophilic phosphorus atom in ATP, a relatively stable pentacoordinate structure is formed as a reaction intermediate (Fig. 10d). With departure of the leaving group (ADP), the transfer of a phosphoryl group is complete. The large family of enzymes that catalyze

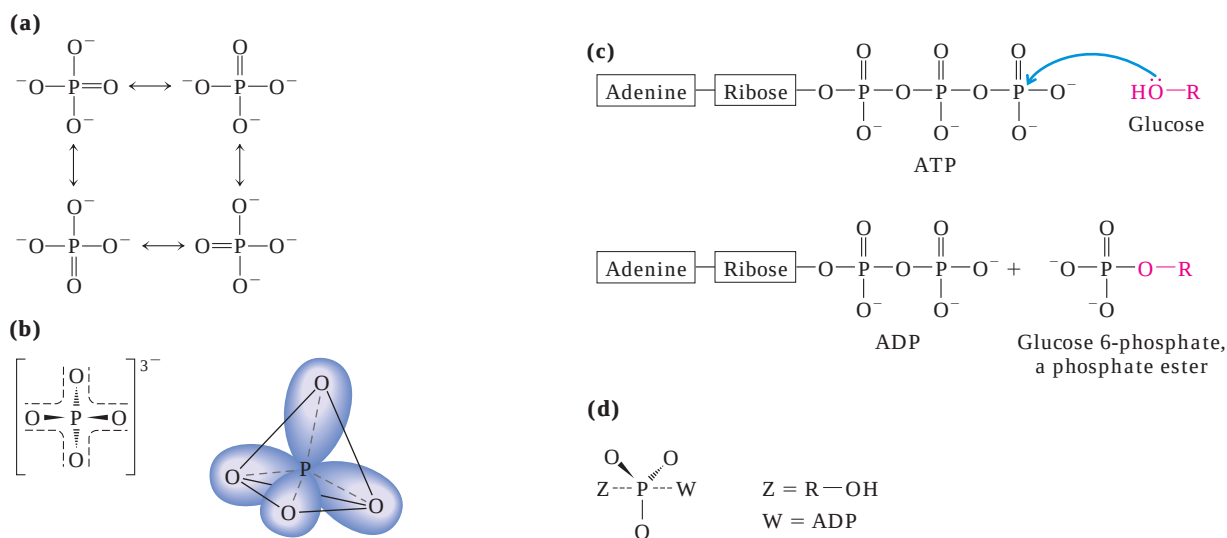


FIGURE 10 Alternative ways of showing the structure of inorganic orthophosphate. (a) In one (inadequate) representation, three oxygens are single-bonded to phosphorus, and the fourth is double-bonded, allowing the four different resonance structures shown. (b) The four resonance structures can be represented more accurately by showing

all four phosphorus-oxygen bonds with some double-bond character; the hybrid orbitals so represented are arranged in a tetrahedron with P at its center. (c) When a nucleophile Z (in this case, the $-\text{OH}$ on C-6 of glucose) attacks ATP, it displaces ADP (W). In this S_N2 reaction, a pentacoordinate intermediate (d) forms transiently.

phosphoryl group transfers with ATP as donor are called kinases (Greek *kinein*, “to move”). Hexokinase, for example, “moves” a phosphoryl group from ATP to glucose.

Phosphoryl groups are not the only activators of this type. Thioalcohols (thiols), in which the oxygen atom of an alcohol is replaced with a sulfur atom, are also good leaving groups. Thiols activate carboxylic acids by forming thioesters (thiol esters) with them. We will discuss a number of cases, including the reactions catalyzed by the fatty acyl transferases in lipid synthesis (see Fig. 21–2), in which nucleophilic substitution at the carbonyl carbon of a thioester results in transfer of the acyl group to another moiety.

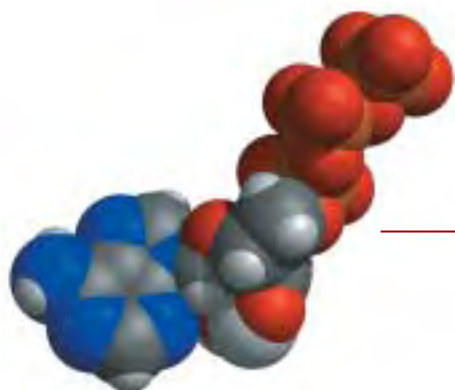
5. Free radical reactions Once thought to be rare, the homolytic cleavage of covalent bonds to generate free radicals has now been found in a range of biochemical processes. Some examples are the reactions of methylmalonyl-CoA mutase (see Box 17–2), ribonucleotide reductase (see Fig. 22–41), and DNA photolyase (see Fig. 25–25).

We begin Part II with a discussion of the basic energetic principles that govern all metabolism (Chapter 13). We then consider the major catabolic pathways by which cells obtain energy from the oxidation of various fuels (Chapters 14 through 19). Chapter 19 is the pivotal point of our discussion of metabolism; it concerns

chemiosmotic energy coupling, a universal mechanism in which a transmembrane electrochemical potential, produced either by substrate oxidation or by light absorption, drives the synthesis of ATP.

Chapters 20 through 22 describe the major anabolic pathways by which cells use the energy in ATP to produce carbohydrates, lipids, amino acids, and nucleotides from simpler precursors. In Chapter 23 we step back from our detailed look at the metabolic pathways—as they occur in all organisms, from *Escherichia coli* to humans—and consider how they are regulated and integrated in mammals by hormonal mechanisms.

As we undertake our study of intermediary metabolism, a final word. Keep in mind that the myriad reactions described in these pages take place in, and play crucial roles in, living organisms. As you encounter each reaction and each pathway ask, What does this chemical transformation do for the organism? How does this pathway interconnect with the other pathways operating simultaneously in the same cell to produce the energy and products required for cell maintenance and growth? How do the multilayered regulatory mechanisms cooperate to balance metabolic and energy inputs and outputs, achieving the dynamic steady state of life? Studied with this perspective, metabolism provides fascinating and revealing insights into life, with countless applications in medicine, agriculture, and biotechnology.



chapter 13

PRINCIPLES OF BIOENERGETICS

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- 13.2 Phosphoryl Group Transfers and ATP 496
- 13.3 Biological Oxidation-Reduction Reactions 507

The total energy of the universe is constant; the total entropy is continually increasing.

—Rudolf Clausius, *The Mechanical Theory of Heat with Its Applications to the Steam-Engine and to the Physical Properties of Bodies*, 1865 (trans. 1867)

The isomorphism of entropy and information establishes a link between the two forms of power: the power to do and the power to direct what is done.

—François Jacob, *La logique du vivant: une histoire de l'hérédité* (The Logic of Life: A History of Heredity), 1970

Living cells and organisms must perform work to stay alive, to grow, and to reproduce. The ability to harness energy and to channel it into biological work is a fundamental property of all living organisms; it must have been acquired very early in cellular evolution. Modern organisms carry out a remarkable variety of energy transductions, conversions of one form of energy to another. They use the chemical energy in fuels to bring about the synthesis of complex, highly ordered macromolecules from simple precursors. They also convert the chemical energy of fuels into concentration gradients and electrical gradients, into motion and heat, and, in a few organisms such as fireflies and some deep-sea fish, into light. Photosynthetic organisms transduce light energy into all these other forms of energy.

The chemical mechanisms that underlie biological energy transductions have fascinated and challenged biologists for centuries. Antoine Lavoisier, before he lost his head in the French Revolution, recognized that animals somehow transform chemical fuels (foods) into

heat and that this process of respiration is essential to life. He observed that

... in general, respiration is nothing but a slow combustion of carbon and hydrogen, which is entirely similar to that which occurs in a lighted lamp or candle, and that, from this point of view, animals that respire are true combustible bodies that burn and consume themselves ...



Antoine Lavoisier,
1743–1794

One may say that this analogy between combustion and respiration has not escaped the notice of the poets, or rather the philosophers of antiquity, and which they had expounded and interpreted. This fire stolen from heaven, this torch of Prometheus, does not only represent an ingenious and poetic idea, it is a faithful picture of the operations of nature, at least for animals that breathe; one may therefore say, with the ancients, that the torch of life lights itself at the moment the infant breathes for the first time, and it does not extinguish itself except at death.*

In this century, biochemical studies have revealed much of the chemistry underlying that “torch of life.” Biological energy transductions obey the same physical laws that govern all other natural processes. It is therefore essential for a student of biochemistry to understand these laws and how they apply to the flow of energy in the biosphere. In this chapter we first review the laws of thermodynamics and the quantitative relationships among free energy, enthalpy, and entropy. We then describe the special role of ATP in biological

*From a memoir by Armand Seguin and Antoine Lavoisier, dated 1789, quoted in Lavoisier, A. (1862) *Oeuvres de Lavoisier*, Imprimerie Impériale, Paris.

energy exchanges. Finally, we consider the importance of oxidation-reduction reactions in living cells, the energetics of electron-transfer reactions, and the electron carriers commonly employed as cofactors of the enzymes that catalyze these reactions.

13.1 Bioenergetics and Thermodynamics

Bioenergetics is the quantitative study of the energy transductions that occur in living cells and of the nature and function of the chemical processes underlying these transductions. Although many of the principles of thermodynamics have been introduced in earlier chapters and may be familiar to you, a review of the quantitative aspects of these principles is useful here.

Biological Energy Transformations Obey the Laws of Thermodynamics

Many quantitative observations made by physicists and chemists on the interconversion of different forms of energy led, in the nineteenth century, to the formulation of two fundamental laws of thermodynamics. The first law is the principle of the conservation of energy: *for any physical or chemical change, the total amount of energy in the universe remains constant; energy may change form or it may be transported from one region to another, but it cannot be created or destroyed.* The second law of thermodynamics, which can be stated in several forms, says that the universe always tends toward increasing disorder: *in all natural processes, the entropy of the universe increases.*

Living organisms consist of collections of molecules much more highly organized than the surrounding materials from which they are constructed, and organisms maintain and produce order, seemingly oblivious to the second law of thermodynamics. But living organisms do

not violate the second law; they operate strictly within it. To discuss the application of the second law to biological systems, we must first define those systems and their surroundings.

The reacting system is the collection of matter that is undergoing a particular chemical or physical process; it may be an organism, a cell, or two reacting compounds. The reacting system and its surroundings together constitute the universe. In the laboratory, some chemical or physical processes can be carried out in isolated or closed systems, in which no material or energy is exchanged with the surroundings. Living cells and organisms, however, are open systems, exchanging both material and energy with their surroundings; living systems are never at equilibrium with their surroundings, and the constant transactions between system and surroundings explain how organisms can create order within themselves while operating within the second law of thermodynamics.

In Chapter 1 (p. 23) we defined three thermodynamic quantities that describe the energy changes occurring in a chemical reaction:

Gibbs free energy, G , expresses the amount of energy capable of doing work during a reaction at constant temperature and pressure. When a reaction proceeds with the release of free energy (that is, when the system changes so as to possess less free energy), the free-energy change, ΔG , has a negative value and the reaction is said to be exergonic. In endergonic reactions, the system gains free energy and ΔG is positive.

Enthalpy, H , is the heat content of the reacting system. It reflects the number and kinds of chemical bonds in the reactants and products. When a chemical reaction releases heat, it is said to be exothermic; the heat content of the products is less than that of the reactants and ΔH has, by convention, a negative value. Reacting systems that take up heat from their surroundings are endothermic and have positive values of ΔH .

Entropy, S , is a quantitative expression for the randomness or disorder in a system (see Box 1–3). When the products of a reaction are less complex and more disordered than the reactants, the reaction is said to proceed with a gain in entropy.

The units of ΔG and ΔH are joules/mole or calories/mole (recall that 1 cal = 4.184 J); units of entropy are joules/mole · Kelvin (J/mol · K) (Table 13–1).

Under the conditions existing in biological systems (including constant temperature and pressure), changes in free energy, enthalpy, and entropy are related to each other quantitatively by the equation

$$\Delta G = \Delta H - T \Delta S \quad (13-1)$$

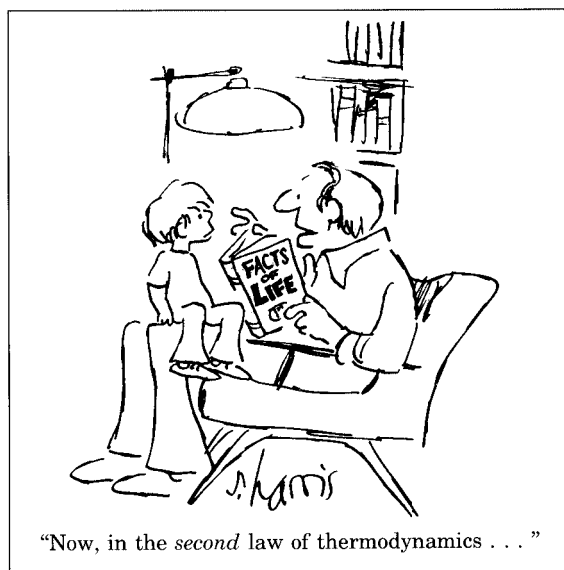


TABLE 13–1 Some Physical Constants and Units Used in Thermodynamics

Boltzmann constant, k	$= 1.381 \times 10^{-23} \text{ J/K}$
Avogadro's number, N	$= 6.022 \times 10^{23} \text{ mol}^{-1}$
Faraday constant, $\mathcal{F}$	$= 96,480 \text{ J/V} \cdot \text{mol}$
Gas constant, R	$= 8.315 \text{ J/mol} \cdot \text{K}$ $(= 1.987 \text{ cal/mol} \cdot \text{K})$
Units of ΔG and ΔH	are J/mol (or cal/mol)
Units of ΔS	are J/mol $\cdot$ K (or cal/mol $\cdot$ K)
	$1 \text{ cal} = 4.184 \text{ J}$
Units of absolute temperature, T , are Kelvin, K	
	$25^\circ \text{C} = 298 \text{ K}$
At 25°C , RT	$= 2.479 \text{ kJ/mol}$ $(= 0.592 \text{ kcal/mol})$

in which ΔG is the change in Gibbs free energy of the reacting system, ΔH is the change in enthalpy of the system, T is the absolute temperature, and ΔS is the change in entropy of the system. By convention, ΔS has a positive sign when entropy increases and ΔH , as noted above, has a negative sign when heat is released by the system to its surroundings. Either of these conditions, which are typical of favorable processes, tend to make ΔG negative. In fact, ΔG of a spontaneously reacting system is always negative.

The second law of thermodynamics states that the entropy of the universe increases during all chemical and physical processes, but it does not require that the entropy increase take place in the reacting system itself. The order produced within cells as they grow and divide is more than compensated for by the disorder they create in their surroundings in the course of growth and division (see Box 1–3, case 2). In short, living organisms preserve their internal order by taking from the surroundings free energy in the form of nutrients or sunlight, and returning to their surroundings an equal amount of energy as heat and entropy.

Cells Require Sources of Free Energy

Cells are isothermal systems—they function at essentially constant temperature (they also function at constant pressure). Heat flow is not a source of energy for cells, because heat can do work only as it passes to a zone or object at a lower temperature. The energy that cells can and must use is free energy, described by the Gibbs free-energy function G , which allows prediction of the direction of chemical reactions, their exact equilibrium position, and the amount of work they can in theory perform at constant temperature and pressure. Heterotrophic cells acquire free energy from nutrient molecules, and photosynthetic cells acquire it from absorbed solar radiation. Both kinds of cells transform this

free energy into ATP and other energy-rich compounds capable of providing energy for biological work at constant temperature.

The Standard Free-Energy Change Is Directly Related to the Equilibrium Constant

The composition of a reacting system (a mixture of chemical reactants and products) tends to continue changing until equilibrium is reached. At the equilibrium concentration of reactants and products, the rates of the forward and reverse reactions are exactly equal and no further net change occurs in the system. The concentrations of reactants and products at equilibrium define the equilibrium constant, K_{eq} (p. 26). In the general reaction $aA + bB \rightleftharpoons cC + dD$, where a , b , c , and d are the number of molecules of A, B, C, and D participating, the equilibrium constant is given by

$$K_{\text{eq}} = \frac{[C]^c[D]^d}{[A]^a[B]^b} \quad (13-2)$$

where $[A]$, $[B]$, $[C]$, and $[D]$ are the molar concentrations of the reaction components at the point of equilibrium.

When a reacting system is not at equilibrium, the tendency to move toward equilibrium represents a driving force, the magnitude of which can be expressed as the free-energy change for the reaction, ΔG . Under standard conditions ($298 \text{ K} = 25^\circ \text{C}$), when reactants and products are initially present at 1 M concentrations or, for gases, at partial pressures of 101.3 kilopascals (kPa), or 1 atm, the force driving the system toward equilibrium is defined as the standard free-energy change, ΔG° . By this definition, the standard state for reactions that involve hydrogen ions is $[\text{H}^+] = 1 \text{ M}$, or pH 0. Most biochemical reactions, however, occur in well-buffered aqueous solutions near pH 7; both the pH and the concentration of water (55.5 M) are essentially constant. For convenience of calculations, biochemists therefore define a different standard state, in which the concentration of H^+ is 10^{-7} M (pH 7) and that of water is 55.5 M; for reactions that involve Mg^{2+} (including most in which ATP is a reactant), its concentration in solution is commonly taken to be constant at 1 mM. Physical constants based on this biochemical standard state are called **standard transformed constants** and are written with a prime (such as $\Delta G'^\circ$ and K'_{eq}) to distinguish them from the untransformed constants used by chemists and physicists. (Notice that most other textbooks use the symbol ΔG° rather than $\Delta G'^\circ$. Our use of $\Delta G'^\circ$, recommended by an international committee of chemists and biochemists, is intended to emphasize that the transformed free energy G' is the criterion for equilibrium.) By convention, when H_2O , H^+ , and/or Mg^{2+} are reactants or products, their concentrations are not included in equations such as Equation 13–2 but are instead incorporated into the constants K'_{eq} and $\Delta G'^\circ$.

Just as K'_{eq} is a physical constant characteristic for each reaction, so too is $\Delta G'^{\circ}$ a constant. As we noted in Chapter 6, there is a simple relationship between K'_{eq} and $\Delta G'^{\circ}$:

$$\Delta G'^{\circ} = -RT \ln K'_{\text{eq}}$$

The standard free-energy change of a chemical reaction is simply an alternative mathematical way of expressing its equilibrium constant. Table 13–2 shows the relationship between $\Delta G'^{\circ}$ and K'_{eq} . If the equilibrium constant for a given chemical reaction is 1.0, the standard free-energy change of that reaction is 0.0 (the natural logarithm of 1.0 is zero). If K'_{eq} of a reaction is greater than 1.0, its $\Delta G'^{\circ}$ is negative. If K'_{eq} is less than 1.0, $\Delta G'^{\circ}$ is positive. Because the relationship between $\Delta G'^{\circ}$ and K'_{eq} is exponential, relatively small changes in $\Delta G'^{\circ}$ correspond to large changes in K'_{eq} .

It may be helpful to think of the standard free-energy change in another way. $\Delta G'^{\circ}$ is the difference between the free-energy content of the products and the free-energy content of the reactants, under standard conditions. When $\Delta G'^{\circ}$ is negative, the products contain less free energy than the reactants and the reaction will proceed spontaneously under standard conditions; all chemical reactions tend to go in the direction that results in a decrease in the free energy of the system. A positive value of $\Delta G'^{\circ}$ means that the products of the reaction contain more free energy than the reactants, and this reaction will tend to go in the reverse direction if we start with 1.0 M concentrations of all components (standard conditions). Table 13–3 summarizes these points.

TABLE 13–2 Relationship between the Equilibrium Constants and Standard Free-Energy Changes of Chemical Reactions

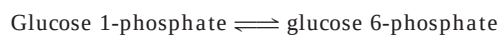
K'_{eq}	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)*
10^3	–17.1	–4.1
10^2	–11.4	–2.7
10^1	–5.7	–1.4
1	0.0	0.0
10^{-1}	5.7	1.4
10^{-2}	11.4	2.7
10^{-3}	17.1	4.1
10^{-4}	22.8	5.5
10^{-5}	28.5	6.8
10^{-6}	34.2	8.2

*Although joules and kilojoules are the standard units of energy and are used throughout this text, biochemists sometimes express $\Delta G'^{\circ}$ values in kilocalories per mole. We have therefore included values in both kilojoules and kilocalories in this table and in Tables 13–4 and 13–6. To convert kilojoules to kilocalories, divide the number of kilojoules by 4.184.

TABLE 13–3 Relationships among K'_{eq} , $\Delta G'^{\circ}$, and the Direction of Chemical Reactions under Standard Conditions

When K'_{eq} is . . .	$\Delta G'^{\circ}$ is . . .	Starting with all components at 1 M, the reaction . . .
>1.0	negative	proceeds forward
1.0	zero	is at equilibrium
<1.0	positive	proceeds in reverse

As an example, let's make a simple calculation of the standard free-energy change of the reaction catalyzed by the enzyme phosphoglucomutase:



Chemical analysis shows that whether we start with, say, 20 mM glucose 1-phosphate (but no glucose 6-phosphate) or with 20 mM glucose 6-phosphate (but no glucose 1-phosphate), the final equilibrium mixture at 25 °C and pH 7.0 will be the same: 1 mM glucose 1-phosphate and 19 mM glucose 6-phosphate. (Remember that enzymes do not affect the point of equilibrium of a reaction; they merely hasten its attainment.) From these data we can calculate the equilibrium constant:

$$K'_{\text{eq}} = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose 1-phosphate}]} = \frac{19 \text{ mM}}{1 \text{ mM}} = 19$$

From this value of K'_{eq} we can calculate the standard free-energy change:

$$\begin{aligned} \Delta G'^{\circ} &= -RT \ln K'_{\text{eq}} \\ &= -(8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K})(\ln 19) \\ &= -7.3 \text{ kJ/mol} \end{aligned}$$

Because the standard free-energy change is negative, when the reaction starts with 1.0 M glucose 1-phosphate and 1.0 M glucose 6-phosphate, the conversion of glucose 1-phosphate to glucose 6-phosphate proceeds with a loss (release) of free energy. For the reverse reaction (the conversion of glucose 6-phosphate to glucose 1-phosphate), $\Delta G'^{\circ}$ has the same magnitude but the opposite sign.

Table 13–4 gives the standard free-energy changes for some representative chemical reactions. Note that hydrolysis of simple esters, amides, peptides, and glycosides, as well as rearrangements and eliminations, proceed with relatively small standard free-energy changes, whereas hydrolysis of acid anhydrides is accompanied by relatively large decreases in standard free energy. The complete oxidation of organic compounds such as glucose or palmitate to CO_2 and H_2O , which in cells requires many steps, results in very large decreases in standard free energy. However, standard free-energy

TABLE 13–4 Standard Free-Energy Changes of Some Chemical Reactions at pH 7.0 and 25 °C (298 K)

Reaction type	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Hydrolysis reactions		
Acid anhydrides		
Acetic anhydride + H ₂ O $\longrightarrow$ 2 acetate	−91.1	−21.8
ATP + H ₂ O $\longrightarrow$ ADP + P _i	−30.5	−7.3
ATP + H ₂ O $\longrightarrow$ AMP + PP _i	−45.6	−10.9
PP _i + H ₂ O $\longrightarrow$ 2P _i	−19.2	−4.6
UDP-glucose + H ₂ O $\longrightarrow$ UMP + glucose 1-phosphate	−43.0	−10.3
Esters		
Ethyl acetate + H ₂ O $\longrightarrow$ ethanol + acetate	−19.6	−4.7
Glucose 6-phosphate + H ₂ O $\longrightarrow$ glucose + P _i	−13.8	−3.3
Amides and peptides		
Glutamine + H ₂ O $\longrightarrow$ glutamate + NH ₄ ⁺	−14.2	−3.4
Glycylglycine + H ₂ O $\longrightarrow$ 2 glycine	−9.2	−2.2
Glycosides		
Maltose + H ₂ O $\longrightarrow$ 2 glucose	−15.5	−3.7
Lactose + H ₂ O $\longrightarrow$ glucose + galactose	−15.9	−3.8
Rearrangements		
Glucose 1-phosphate $\longrightarrow$ glucose 6-phosphate	−7.3	−1.7
Fructose 6-phosphate $\longrightarrow$ glucose 6-phosphate	−1.7	−0.4
Elimination of water		
Malate $\longrightarrow$ fumarate + H ₂ O	3.1	0.8
Oxidations with molecular oxygen		
Glucose + 6O ₂ $\longrightarrow$ 6CO ₂ + 6H ₂ O	−2,840	−686
Palmitate + 23O ₂ $\longrightarrow$ 16CO ₂ + 16H ₂ O	−9,770	−2,338

changes such as those in Table 13–4 indicate how much free energy is available from a reaction under *standard conditions*. To describe the energy released under the conditions existing in cells, an expression for the *actual* free-energy change is essential.

Actual Free-Energy Changes Depend on Reactant and Product Concentrations

We must be careful to distinguish between two different quantities: the free-energy change, ΔG , and the standard free-energy change, $\Delta G'^{\circ}$. Each chemical reaction has a characteristic standard free-energy change, which may be positive, negative, or zero, depending on the equilibrium constant of the reaction. The standard free-

energy change tells us in which direction and how far a given reaction must go to reach equilibrium *when the initial concentration of each component is 1.0 M*, the pH is 7.0, the temperature is 25 °C, and the pressure is 101.3 kPa. Thus $\Delta G'^{\circ}$ is a constant: it has a characteristic, unchanging value for a given reaction. But the *actual* free-energy change, ΔG , is a function of reactant and product concentrations and of the temperature prevailing during the reaction, which will not necessarily match the standard conditions as defined above. Moreover, the ΔG of any reaction proceeding spontaneously toward its equilibrium is always negative, becomes less negative as the reaction proceeds, and is zero at the point of equilibrium, indicating that no more work can be done by the reaction.

ΔG and $\Delta G'^{\circ}$ for any reaction $A + B \rightleftharpoons C + D$ are related by the equation

$$\Delta G = \Delta G'^{\circ} + RT \ln \frac{[C][D]}{[A][B]} \quad (13-3)$$

in which the terms in red are those *actually prevailing* in the system under observation. The concentration terms in this equation express the effects commonly called mass action, and the term $[C][D]/[A][B]$ is called the **mass-action ratio, Q** . As an example, let us suppose that the reaction $A + B \rightleftharpoons C + D$ is taking place at the standard conditions of temperature (25 °C) and pressure (101.3 kPa) but that the concentrations of A, B, C, and D are *not* equal and none of the components is present at the standard concentration of 1.0 M. To determine the actual free-energy change, ΔG , under these nonstandard conditions of concentration as the reaction proceeds from left to right, we simply enter the *actual* concentrations of A, B, C, and D in Equation 13-3; the values of R , T , and $\Delta G'^{\circ}$ are the standard values. ΔG is negative and approaches zero as the reaction proceeds because the actual concentrations of A and B decrease and the concentrations of C and D increase. Notice that when a reaction is at equilibrium—when there is no force driving the reaction in either direction and ΔG is zero—Equation 13-3 reduces to

$$0 = \Delta G = \Delta G'^{\circ} + RT \ln \frac{[C]_{\text{eq}}[D]_{\text{eq}}}{[A]_{\text{eq}}[B]_{\text{eq}}}$$

or

$$\Delta G'^{\circ} = -RT \ln K'_{\text{eq}}$$

which is the equation relating the standard free-energy change and equilibrium constant given earlier.

The criterion for spontaneity of a reaction is the value of ΔG , not $\Delta G'^{\circ}$. A reaction with a positive $\Delta G'^{\circ}$ can go in the forward direction *if ΔG is negative*. This is possible if the term $RT \ln ([\text{products}]/[\text{reactants}])$ in Equation 13-3 is negative and has a larger absolute value than $\Delta G'^{\circ}$. For example, the immediate removal of the products of a reaction can keep the ratio $[\text{products}]/[\text{reactants}]$ well below 1, such that the term $RT \ln ([\text{products}]/[\text{reactants}])$ has a large, negative value.

$\Delta G'^{\circ}$ and ΔG are expressions of the *maximum* amount of free energy that a given reaction can *theoretically* deliver—an amount of energy that could be realized only if a perfectly efficient device were available to trap or harness it. Given that no such device is possible (some free energy is always lost to entropy during any process), the amount of work done by the reaction at constant temperature and pressure is always less than the theoretical amount.

Another important point is that some thermodynamically favorable reactions (that is, reactions for which $\Delta G'^{\circ}$ is large and negative) do not occur at meas-

urable rates. For example, combustion of firewood to CO_2 and H_2O is very favorable thermodynamically, but firewood remains stable for years because the activation energy (see Figs 6-2 and 6-3) for the combustion reaction is higher than the energy available at room temperature. If the necessary activation energy is provided (with a lighted match, for example), combustion will begin, converting the wood to the more stable products CO_2 and H_2O and releasing energy as heat and light. The heat released by this exothermic reaction provides the activation energy for combustion of neighboring regions of the firewood; the process is self-perpetuating.

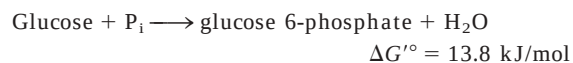
In living cells, reactions that would be extremely slow *if uncatalyzed* are caused to proceed, not by supplying additional heat but by lowering the activation energy with an enzyme. An enzyme provides an alternative reaction pathway with a lower activation energy than the uncatalyzed reaction, so that at room temperature a large fraction of the substrate molecules have enough thermal energy to overcome the activation barrier, and the reaction rate increases dramatically. *The free-energy change for a reaction is independent of the pathway by which the reaction occurs*; it depends only on the nature and concentration of the initial reactants and the final products. Enzymes cannot, therefore, change equilibrium constants; but they can and do increase the *rate* at which a reaction proceeds in the direction dictated by thermodynamics.

Standard Free-Energy Changes Are Additive

In the case of two sequential chemical reactions, $A \rightleftharpoons B$ and $B \rightleftharpoons C$, each reaction has its own equilibrium constant and each has its characteristic standard free-energy change, $\Delta G_1'^{\circ}$ and $\Delta G_2'^{\circ}$. As the two reactions are sequential, B cancels out to give the overall reaction $A \rightleftharpoons C$, which has its own equilibrium constant and thus its own standard free-energy change, $\Delta G'_{\text{total}}$. *The $\Delta G'^{\circ}$ values of sequential chemical reactions are additive*. For the overall reaction $A \rightleftharpoons C$, $\Delta G'_{\text{total}}$ is the sum of the individual standard free-energy changes, $\Delta G_1'^{\circ}$ and $\Delta G_2'^{\circ}$, of the two reactions: $\Delta G'_{\text{total}} = \Delta G_1'^{\circ} + \Delta G_2'^{\circ}$.

(1)	$A \longrightarrow B$	$\Delta G_1'^{\circ}$
(2)	$B \longrightarrow C$	$\Delta G_2'^{\circ}$
<i>Sum:</i>	$A \longrightarrow C$	$\Delta G_1'^{\circ} + \Delta G_2'^{\circ}$

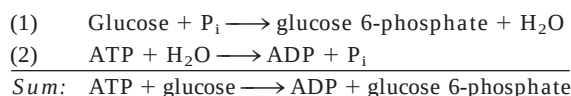
This principle of bioenergetics explains how a thermodynamically unfavorable (endergonic) reaction can be driven in the forward direction by coupling it to a highly exergonic reaction through a common intermediate. For example, the synthesis of glucose 6-phosphate is the first step in the utilization of glucose by many organisms:



The positive value of $\Delta G'^{\circ}$ predicts that under standard conditions the reaction will tend not to proceed spontaneously in the direction written. Another cellular reaction, the hydrolysis of ATP to ADP and P_i , is very exergonic:



These two reactions share the common intermediates P_i and H_2O and may be expressed as sequential reactions:



The overall standard free-energy change is obtained by adding the $\Delta G'^{\circ}$ values for individual reactions:

$$\Delta G'^{\circ} = 13.8 \text{ kJ/mol} + (-30.5 \text{ kJ/mol}) = -16.7 \text{ kJ/mol}$$

The overall reaction is exergonic. In this case, energy stored in ATP is used to drive the synthesis of glucose 6-phosphate, even though its formation from glucose and inorganic phosphate (P_i) is endergonic. The *pathway* of glucose 6-phosphate formation by phosphoryl transfer from ATP is different from reactions (1) and (2) above, but the net result is the same as the sum of the two reactions. In thermodynamic calculations, all that matters is the state of the system at the beginning of the process and its state at the end; the route between the initial and final states is immaterial.

We have said that $\Delta G'^{\circ}$ is a way of expressing the equilibrium constant for a reaction. For reaction (1) above,

$$K'_{\text{eq}_1} = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose}][P_i]} = 3.9 \times 10^{-3} \text{ M}^{-1}$$

Notice that H_2O is not included in this expression, as its concentration (55.5 M) is assumed to remain unchanged by the reaction. The equilibrium constant for the hydrolysis of ATP is

$$K'_{\text{eq}_2} = \frac{[\text{ADP}][P_i]}{[\text{ATP}]} = 2.0 \times 10^5 \text{ M}$$

The equilibrium constant for the two coupled reactions is

$$\begin{aligned} K'_{\text{eq}_3} &= \frac{[\text{glucose 6-phosphate}][\text{ADP}][P_i]}{[\text{glucose}][P_i][\text{ATP}]} \\ &= (K'_{\text{eq}_1})(K'_{\text{eq}_2}) = (3.9 \times 10^{-3} \text{ M}^{-1})(2.0 \times 10^5 \text{ M}) \\ &= 7.8 \times 10^2 \end{aligned}$$

This calculation illustrates an important point about equilibrium constants: although the $\Delta G'^{\circ}$ values for two reactions that sum to a third are *additive*, the K'_{eq} for a reaction that is the sum of two reactions is the *product* of their individual K'_{eq} values. Equilibrium constants are *multiplicative*. By coupling ATP hydrolysis to glu-

cose 6-phosphate synthesis, the K'_{eq} for formation of glucose 6-phosphate has been raised by a factor of about 2×10^5 .

This common-intermediate strategy is employed by all living cells in the synthesis of metabolic intermediates and cellular components. Obviously, the strategy works only if compounds such as ATP are continuously available. In the following chapters we consider several of the most important cellular pathways for producing ATP.

SUMMARY 13.1 Bioenergetics and Thermodynamics

- Living cells constantly perform work. They require energy for maintaining their highly organized structures, synthesizing cellular components, generating electric currents, and many other processes.
- Bioenergetics is the quantitative study of energy relationships and energy conversions in biological systems. Biological energy transformations obey the laws of thermodynamics.
- All chemical reactions are influenced by two forces: the tendency to achieve the most stable bonding state (for which enthalpy, H , is a useful expression) and the tendency to achieve the highest degree of randomness, expressed as entropy, S . The net driving force in a reaction is ΔG , the free-energy change, which represents the net effect of these two factors: $\Delta G = \Delta H - T \Delta S$.
- The standard transformed free-energy change, $\Delta G'^{\circ}$, is a physical constant that is characteristic for a given reaction and can be calculated from the equilibrium constant for the reaction: $\Delta G'^{\circ} = -RT \ln K'_{\text{eq}}$.
- The actual free-energy change, ΔG , is a variable that depends on $\Delta G'^{\circ}$ and on the concentrations of reactants and products: $\Delta G = \Delta G'^{\circ} + RT \ln ([\text{products}]/[\text{reactants}])$.
- When ΔG is large and negative, the reaction tends to go in the forward direction; when ΔG is large and positive, the reaction tends to go in the reverse direction; and when $\Delta G = 0$, the system is at equilibrium.
- The free-energy change for a reaction is independent of the pathway by which the reaction occurs. Free-energy changes are additive; the net chemical reaction that results from successive reactions sharing a common intermediate has an overall free-energy change that is the sum of the ΔG values for the individual reactions.

13.2 Phosphoryl Group Transfers and ATP

Having developed some fundamental principles of energy changes in chemical systems, we can now examine the energy cycle in cells and the special role of ATP as the energy currency that links catabolism and anabolism (see Fig. 1–28). Heterotrophic cells obtain free energy in a chemical form by the catabolism of nutrient molecules, and they use that energy to make ATP from ADP and P_i . ATP then donates some of its chemical energy to endergonic processes such as the synthesis of metabolic intermediates and macromolecules from smaller precursors, the transport of substances across membranes against concentration gradients, and mechanical motion. This donation of energy from ATP generally involves the covalent participation of ATP in the reaction that is to be driven, with the eventual result that ATP is converted to ADP and P_i or, in some reactions, to AMP and 2 P_i . We discuss here the chemical basis for the large free-energy changes that accompany hydrolysis of ATP and other high-energy phosphate compounds, and we show that most cases of energy donation by ATP involve group transfer, not simple hydrolysis of ATP. To illustrate the range of energy transductions in which ATP provides the energy, we consider the synthesis of information-rich macromolecules, the transport of solutes across membranes, and motion produced by muscle contraction.

The Free-Energy Change for ATP Hydrolysis Is Large and Negative

Figure 13–1 summarizes the chemical basis for the relatively large, negative, standard free energy of hydrolysis of ATP. The hydrolytic cleavage of the terminal phosphoric acid anhydride (phosphoanhydride) bond in ATP separates one of the three negatively charged phosphates and thus relieves some of the electrostatic repulsion in ATP; the P_i (HPO_4^{2-}) released is stabilized by the formation of several resonance forms not possible in ATP; and ADP^{2-} , the other direct product of hydrolysis, immediately ionizes, releasing H^+ into a medium of very low $[H^+]$ ($\sim 10^{-7}$ M). Because the concentrations of the direct products of ATP hydrolysis are, in the cell, far below the concentrations at equilibrium (Table 13–5), mass action favors the hydrolysis reaction in the cell.

Although the hydrolysis of ATP is highly exergonic ($\Delta G'^{\circ} = -30.5$ kJ/mol), the molecule is kinetically stable at pH 7 because the activation energy for ATP hydrolysis is relatively high. Rapid cleavage of the phosphoanhydride bonds occurs only when catalyzed by an enzyme.

The free-energy change for ATP hydrolysis is -30.5 kJ/mol under standard conditions, but the *actual* free energy of hydrolysis (ΔG) of ATP in living cells is very different: the cellular concentrations of ATP, ADP,

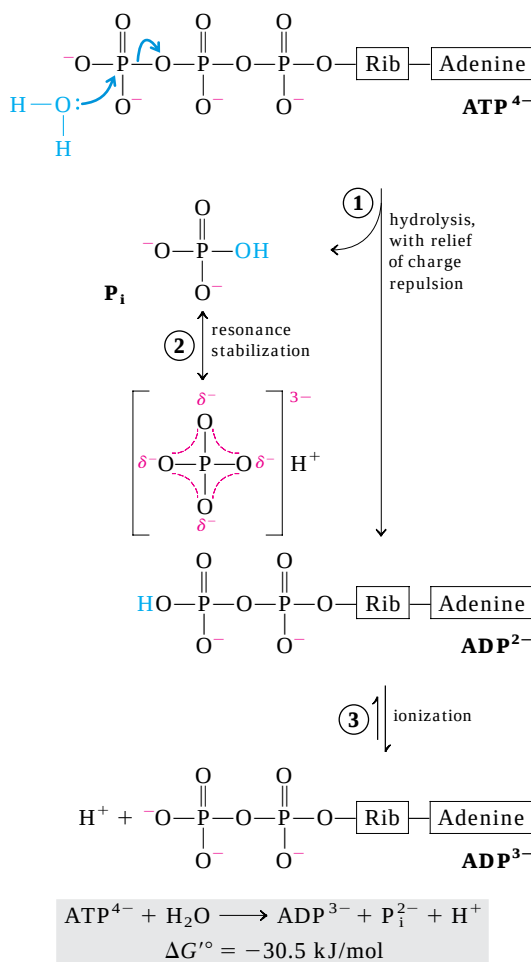


FIGURE 13–1 Chemical basis for the large free-energy change associated with ATP hydrolysis. ① The charge separation that results from hydrolysis relieves electrostatic repulsion among the four negative charges on ATP. ② The product inorganic phosphate (P_i) is stabilized by formation of a resonance hybrid, in which each of the four phosphorus–oxygen bonds has the same degree of double-bond character and the hydrogen ion is not permanently associated with any one of the oxygens. (Some degree of resonance stabilization also occurs in phosphates involved in ester or anhydride linkages, but fewer resonance forms are possible than for P_i .) ③ The product ADP^{2-} immediately ionizes, releasing a proton into a medium of very low $[H^+]$ (pH 7). A fourth factor (not shown) that favors ATP hydrolysis is the greater degree of solvation (hydration) of the products P_i and ADP relative to ATP, which further stabilizes the products relative to the reactants.

and P_i are not identical and are much lower than the 1.0 M of standard conditions (Table 13–5). Furthermore, Mg^{2+} in the cytosol binds to ATP and ADP (Fig. 13–2), and for most enzymatic reactions that involve ATP as phosphoryl group donor, the true substrate is $MgATP^{2-}$. The relevant $\Delta G'^{\circ}$ is therefore that for $MgATP^{2-}$ hydrolysis. Box 13–1 shows how ΔG for ATP hydrolysis in the intact erythrocyte can be calculated from the data in Table 13–5. In intact cells, ΔG for ATP hydrolysis, usually designated ΔG_p , is much more negative than

TABLE 13–5 Adenine Nucleotide, Inorganic Phosphate, and Phosphocreatine Concentrations in Some Cells

	Concentration (mM)*				
	ATP	ADP [†]	AMP	P _i	PCr
Rat hepatocyte	3.38	1.32	0.29	4.8	0
Rat myocyte	8.05	0.93	0.04	8.05	28
Rat neuron	2.59	0.73	0.06	2.72	4.7
Human erythrocyte	2.25	0.25	0.02	1.65	0
<i>E. coli</i> cell	7.90	1.04	0.82	7.9	0

*For erythrocytes the concentrations are those of the cytosol (human erythrocytes lack a nucleus and mitochondria). In the other types of cells the data are for the entire cell contents, although the cytosol and the mitochondria have very different concentrations of ADP. PCr is phosphocreatine, discussed on p. 489.

[†]This value reflects total concentration; the true value for free ADP may be much lower (see Box 13–1).

$\Delta G'^{\circ}$, ranging from -50 to -65 kJ/mol. ΔG_p is often called the **phosphorylation potential**. In the following discussions we use the standard free-energy change for ATP hydrolysis, because this allows comparison, on the same basis, with the energetics of other cellular reactions. Remember, however, that in living cells ΔG is the relevant quantity—for ATP hydrolysis and all other reactions—and may be quite different from $\Delta G'^{\circ}$.

Other Phosphorylated Compounds and Thioesters Also Have Large Free Energies of Hydrolysis

Phosphoenolpyruvate (Fig. 13–3) contains a phosphate ester bond that undergoes hydrolysis to yield the enol form of pyruvate, and this direct product can immediately tautomerize to the more stable keto form of pyruvate. Because the reactant (phosphoenolpyruvate) has only one form (enol) and the product (pyruvate) has two possible forms, the product is stabilized relative to the reactant. This is the greatest contributing factor to the high standard free energy of hydrolysis of phosphoenolpyruvate: $\Delta G'^{\circ} = -61.9$ kJ/mol.

Another three-carbon compound, 1,3-bisphosphoglycerate (Fig. 13–4), contains an anhydride bond between the carboxyl group at C-1 and phosphoric acid. Hydrolysis of this acyl phosphate is accompanied by a large, negative, standard free-energy change ($\Delta G'^{\circ} =$

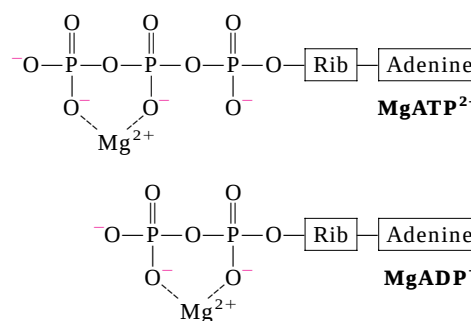


FIGURE 13–2 Mg²⁺ and ATP. Formation of Mg²⁺ complexes partially shields the negative charges and influences the conformation of the phosphate groups in nucleotides such as ATP and ADP.

-49.3 kJ/mol), which can, again, be explained in terms of the structure of reactant and products. When H₂O is added across the anhydride bond of 1,3-bisphosphoglycerate, one of the direct products, 3-phosphoglyceric acid, can immediately lose a proton to give the carboxylate ion, 3-phosphoglycerate, which has two equally probable resonance forms (Fig. 13–4). Removal of the direct product (3-phosphoglyceric acid) and formation of the resonance-stabilized ion favor the forward reaction.

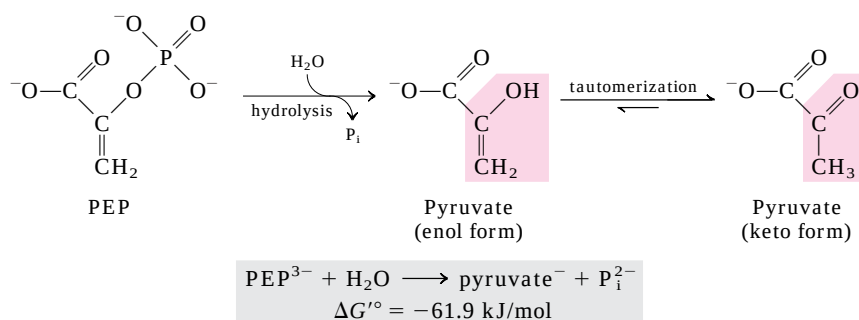


FIGURE 13–3 Hydrolysis of phosphoenolpyruvate (PEP). Catalyzed by pyruvate kinase, this reaction is followed by spontaneous tautomerization of the product, pyruvate. Tautomerization is not possible in PEP, and thus the products of hydrolysis are stabilized relative to the reactants. Resonance stabilization of P_i also occurs, as shown in Figure 13–1.

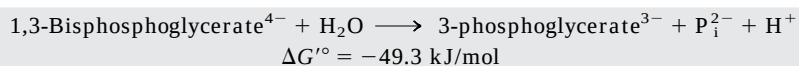
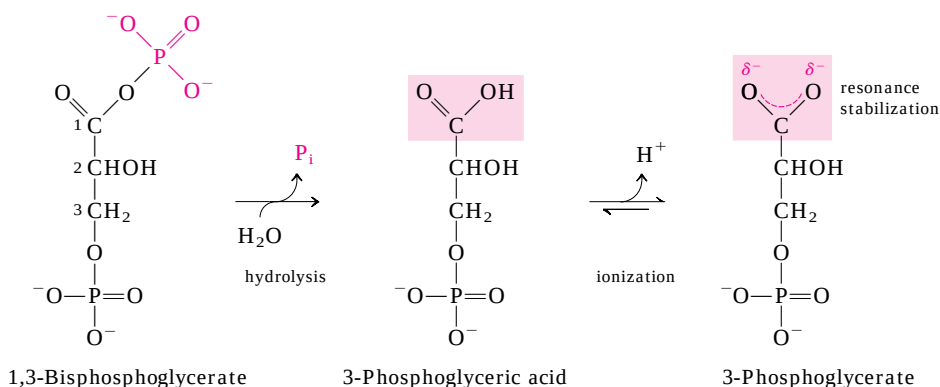


FIGURE 13-4 Hydrolysis of 1,3-bisphosphoglycerate. The direct product of hydrolysis is 3-phosphoglyceric acid, with an undissociated carboxylic acid group, but dissociation occurs immediately. This ionization and the resonance structures it makes possible stabilize the product relative to the reactants. Resonance stabilization of P_i further contributes to the negative free-energy change.

BOX 13-1 WORKING IN BIOCHEMISTRY

The Free Energy of Hydrolysis of ATP within Cells: The Real Cost of Doing Metabolic Business

The standard free energy of hydrolysis of ATP is -30.5 kJ/mol . In the cell, however, the concentrations of ATP, ADP, and P_i are not only unequal but much lower than the standard 1 M concentrations (see Table 13-5). Moreover, the cellular pH may differ somewhat from the standard pH of 7.0. Thus the *actual* free energy of hydrolysis of ATP under intracellular conditions (ΔG_p) differs from the standard free-energy change, $\Delta G'^{\circ}$. We can easily calculate ΔG_p .

In human erythrocytes, for example, the concentrations of ATP, ADP, and P_i are 2.25, 0.25, and 1.65 mM, respectively. Let us assume for simplicity that the pH is 7.0 and the temperature is 25°C , the standard pH and temperature. The actual free energy of hydrolysis of ATP in the erythrocyte under these conditions is given by the relationship

$$\Delta G_p = \Delta G'^{\circ} + RT \ln \frac{[\text{ADP}][\text{P}_i]}{[\text{ATP}]}$$

Substituting the appropriate values we obtain

$$\begin{aligned} \Delta G_p &= -30.5 \text{ kJ/mol} + \\ &\left[(8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K}) \ln \frac{(0.25 \times 10^{-3})(1.65 \times 10^{-3})}{2.25 \times 10^{-3}} \right] \\ &= -30.5 \text{ kJ/mol} + (2.48 \text{ kJ/mol}) \ln 1.8 \times 10^{-4} \\ &= -30.5 \text{ kJ/mol} - 21 \text{ kJ/mol} \\ &= -52 \text{ kJ/mol} \end{aligned}$$

Thus ΔG_p , the actual free-energy change for ATP hydrolysis in the intact erythrocyte (-52 kJ/mol), is

much larger than the standard free-energy change (-30.5 kJ/mol). By the same token, the free energy required to *synthesize* ATP from ADP and P_i under the conditions prevailing in the erythrocyte would be 52 kJ/mol .

Because the concentrations of ATP, ADP, and P_i differ from one cell type to another (see Table 13-5), ΔG_p for ATP hydrolysis likewise differs among cells. Moreover, in any given cell, ΔG_p can vary from time to time, depending on the metabolic conditions in the cell and how they influence the concentrations of ATP, ADP, P_i , and H^+ (pH). We can calculate the actual free-energy change for any given metabolic reaction as it occurs in the cell, providing we know the concentrations of all the reactants and products of the reaction and know about the other factors (such as pH, temperature, and concentration of Mg^{2+}) that may affect the $\Delta G'^{\circ}$ and thus the calculated free-energy change, ΔG_p .

To further complicate the issue, the total concentrations of ATP, ADP, P_i , and H^+ may be substantially higher than the *free* concentrations, which are the thermodynamically relevant values. The difference is due to tight binding of ATP, ADP, and P_i to cellular proteins. For example, the concentration of free ADP in resting muscle has been variously estimated at between 1 and $37 \mu\text{M}$. Using the value $25 \mu\text{M}$ in the calculation outlined above, we get a ΔG_p of -58 kJ/mol .

Calculation of the exact value of ΔG_p is perhaps less instructive than the generalization we can make about actual free-energy changes: *in vivo*, the energy released by ATP hydrolysis is greater than the standard free-energy change, $\Delta G'^{\circ}$.

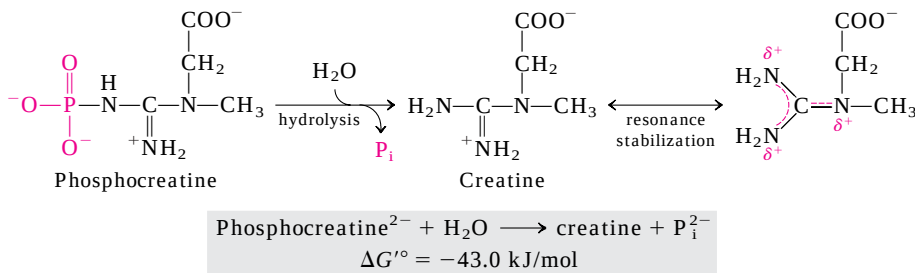


FIGURE 13-5 Hydrolysis of phosphocreatine. Breakage of the P—N bond in phosphocreatine produces creatine, which is stabilized by formation of a resonance hybrid. The other product, P_i , is also resonance stabilized.

In phosphocreatine (Fig. 13-5), the P—N bond can be hydrolyzed to generate free creatine and P_i . The release of P_i and the resonance stabilization of creatine favor the forward reaction. The standard free-energy change of phosphocreatine hydrolysis is again large, -43.0 kJ/mol .

In all these phosphate-releasing reactions, the several resonance forms available to P_i (Fig. 13-1) stabilize this product relative to the reactant, contributing to an already negative free-energy change. Table 13-6 lists the standard free energies of hydrolysis for a number of phosphorylated compounds.

Thioesters, in which a sulfur atom replaces the usual oxygen in the ester bond, also have large, negative, standard free energies of hydrolysis. Acetyl-coenzyme A, or acetyl-CoA (Fig. 13-6), is one of many thioesters important in metabolism. The acyl group in

these compounds is activated for transacylation, condensation, or oxidation-reduction reactions. Thioesters undergo much less resonance stabilization than do oxygen esters; consequently, the difference in free energy between the reactant and its hydrolysis products, which *are* resonance-stabilized, is greater for thioesters than for comparable oxygen esters (Fig. 13-7). In both cases, hydrolysis of the ester generates a carboxylic acid, which can ionize and assume several resonance forms. Together, these factors result in the large, negative $\Delta G'^{\circ}$ (-31 kJ/mol) for acetyl-CoA hydrolysis.

To summarize, for hydrolysis reactions with large, negative, standard free-energy changes, the products are more stable than the reactants for one or more of the following reasons: (1) the bond strain in reactants due to electrostatic repulsion is relieved by charge separation, as for ATP; (2) the products are stabilized by

TABLE 13-6 Standard Free Energies of Hydrolysis of Some Phosphorylated Compounds and Acetyl-CoA (a Thioester)

	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Phosphoenolpyruvate	-61.9	-14.8
1,3-bisphosphoglycerate ($\rightarrow$ 3-phosphoglycerate + P_i)	-49.3	-11.8
Phosphocreatine	-43.0	-10.3
ADP ($\rightarrow$ AMP + P_i)	-32.8	-7.8
ATP ($\rightarrow$ ADP + P_i)	-30.5	-7.3
ATP ($\rightarrow$ AMP + PP_i)	-45.6	-10.9
AMP ($\rightarrow$ adenosine + P_i)	-14.2	-3.4
PP_i ($\rightarrow$ 2 P_i)	-19.2	-4.0
Glucose 1-phosphate	-20.9	-5.0
Fructose 6-phosphate	-15.9	-3.8
Glucose 6-phosphate	-13.8	-3.3
Glycerol 1-phosphate	-9.2	-2.2
Acetyl-CoA	-31.4	-7.5

Source: Data mostly from Jencks, W.P. (1976) in *Handbook of Biochemistry and Molecular Biology*, 3rd edn (Fasman, G.D., ed.), Physical and Chemical Data, Vol. 1, pp. 296-304, CRC Press, Boca Raton, FL. The value for the free energy of hydrolysis of PP_i is from Frey, P.A. & Arabshahi, A. (1995) Standard free-energy change for the hydrolysis of the α - β -phosphoanhydride bridge in ATP. *Biochemistry* **34**, 11,307-11,310.

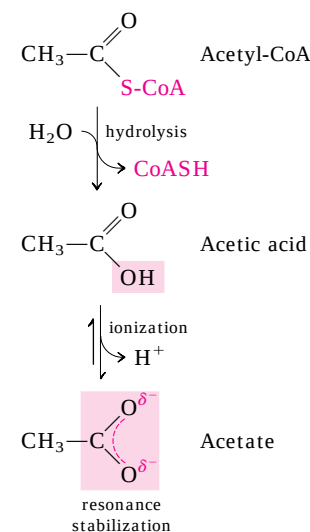


FIGURE 13-6 Hydrolysis of acetyl-coenzyme A. Acetyl-CoA is a thioester with a large, negative, standard free energy of hydrolysis. Thioesters contain a sulfur atom in the position occupied by an oxygen atom in oxygen esters. The complete structure of coenzyme A (CoA, or CoASH) is shown in Figure 8-41.

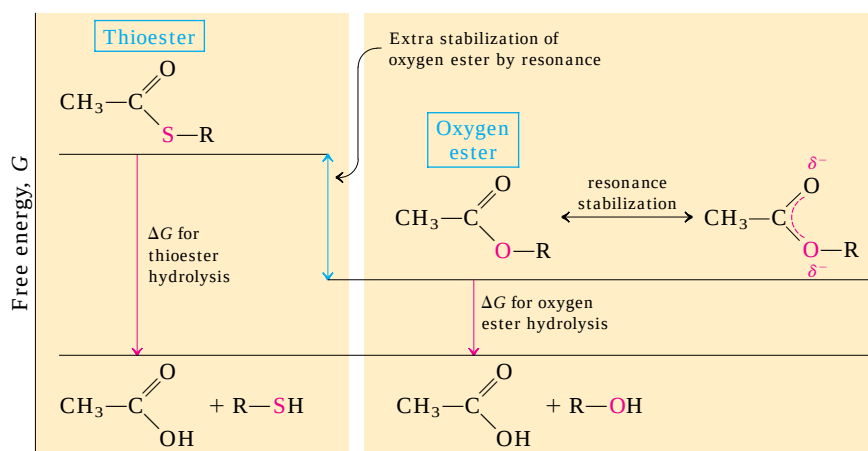


FIGURE 13-7 Free energy of hydrolysis for thioesters and oxygen esters. The products of both types of hydrolysis reaction have about the same free-energy content (G), but the thioester has a higher free-energy content than the oxygen ester. Orbital overlap between the O and C atoms allows resonance stabilization in oxygen esters; orbital overlap between S and C atoms is poorer and provides little resonance stabilization.

ionization, as for ATP, acyl phosphates, and thioesters; (3) the products are stabilized by isomerization (tautomerization), as for phosphoenolpyruvate; and/or (4) the products are stabilized by resonance, as for creatine released from phosphocreatine, carboxylate ion released from acyl phosphates and thioesters, and phosphate (P_i) released from anhydride or ester linkages.

ATP Provides Energy by Group Transfers, Not by Simple Hydrolysis

Throughout this book you will encounter reactions or processes for which ATP supplies energy, and the contribution of ATP to these reactions is commonly indicated as in Figure 13-8a, with a single arrow showing the conversion of ATP to ADP and P_i (or, in some cases, of ATP to AMP and pyrophosphate, PP_i). When written this way, these reactions of ATP appear to be simple hydrolysis reactions in which water displaces P_i (or PP_i), and one is tempted to say that an ATP-dependent reaction is “driven by the hydrolysis of ATP.” This is *not* the case. ATP hydrolysis per se usually accomplishes nothing but the liberation of heat, which cannot drive a chemical process in an isothermal system. A single reaction arrow such as that in Figure 13-8a almost invariably represents a two-step process (Fig. 13-8b) in which part of the ATP molecule, a phosphoryl or pyrophosphoryl group or the adenylate moiety (AMP), is first transferred to a substrate molecule or to an amino acid residue in an enzyme, becoming covalently attached to the substrate or the enzyme and raising its free-energy content. Then, in a second step, the phosphate-containing moiety transferred in the first step is displaced, generating P_i , PP_i , or AMP. Thus ATP participates *covalently* in the enzyme-catalyzed reaction to which it contributes free energy.

Some processes *do* involve direct hydrolysis of ATP (or GTP), however. For example, noncovalent binding of ATP (or of GTP), followed by its hydrolysis to ADP (or GDP) and P_i , can provide the energy to cycle some proteins between two conformations, producing me-

chanical motion. This occurs in muscle contraction and in the movement of enzymes along DNA or of ribosomes along messenger RNA. The energy-dependent reactions catalyzed by helicases, RecA protein, and some topoisomerases (Chapter 25) also involve direct hydrolysis of phosphoanhydride bonds. GTP-binding proteins that act in signaling pathways directly hydrolyze GTP to drive conformational changes that terminate signals

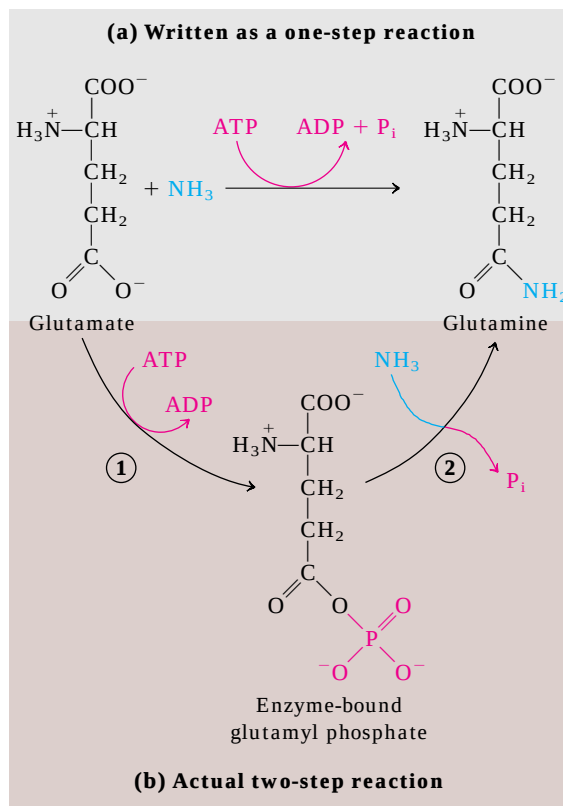


FIGURE 13-8 ATP hydrolysis in two steps. **(a)** The contribution of ATP to a reaction is often shown as a single step, but is almost always a two-step process. **(b)** Shown here is the reaction catalyzed by ATP-dependent glutamine synthetase. ① A phosphoryl group is transferred from ATP to glutamate, then ② the phosphoryl group is displaced by NH_3 and released as P_i .

triggered by hormones or by other extracellular factors (Chapter 12).

The phosphate compounds found in living organisms can be divided somewhat arbitrarily into two groups, based on their standard free energies of hydrolysis (Fig. 13–9). “High-energy” compounds have a $\Delta G'^{\circ}$ of hydrolysis more negative than -25 kJ/mol; “low-energy” compounds have a less negative $\Delta G'^{\circ}$. Based on this criterion, ATP, with a $\Delta G'^{\circ}$ of hydrolysis of -30.5 kJ/mol (-7.3 kcal/mol), is a high-energy compound; glucose 6-phosphate, with a $\Delta G'^{\circ}$ of hydrolysis of -13.8 kJ/mol (-3.3 kcal/mol), is a low-energy compound.

The term “high-energy phosphate bond,” long used by biochemists to describe the P—O bond broken in hydrolysis reactions, is incorrect and misleading as it wrongly suggests that the bond itself contains the energy. In fact, the breaking of all chemical bonds requires an *input* of energy. The free energy released by hydrolysis of phosphate compounds does not come from the specific bond that is broken; it results from the products of the reaction having a lower free-energy content than the reactants. For simplicity, we will sometimes use the term “high-energy phosphate compound” when referring to ATP or other phosphate compounds with a large, negative, standard free energy of hydrolysis.

As is evident from the additivity of free-energy changes of sequential reactions, any phosphorylated compound can be synthesized by coupling the synthesis to the breakdown of another phosphorylated compound with a more negative free energy of hydrolysis. For example, because cleavage of P_i from phosphoenolpyruvate (PEP) releases more energy than is needed to drive the condensation of P_i with ADP, the

direct donation of a phosphoryl group from PEP to ADP is thermodynamically feasible:

		$\Delta G'^{\circ}$ (kJ/mol)
(1)	$\text{PEP} + \text{H}_2\text{O} \longrightarrow \text{pyruvate} + P_i$	-61.9
(2)	$\text{ADP} + P_i \longrightarrow \text{ATP} + \text{H}_2\text{O}$	$+30.5$
Sum:	$\text{PEP} + \text{ADP} \longrightarrow \text{pyruvate} + \text{ATP}$	-31.4

Notice that while the overall reaction above is represented as the algebraic sum of the first two reactions, the overall reaction is actually a third, distinct reaction that does not involve P_i ; PEP donates a *phosphoryl* group *directly* to ADP. We can describe phosphorylated compounds as having a high or low phosphoryl group transfer potential, on the basis of their standard free energies of hydrolysis (as listed in Table 13–6). The phosphoryl group transfer potential of phosphoenolpyruvate is very high, that of ATP is high, and that of glucose 6-phosphate is low (Fig. 13–9).

Much of catabolism is directed toward the synthesis of high-energy phosphate compounds, but their formation is not an end in itself; they are the means of activating a very wide variety of compounds for further chemical transformation. The transfer of a phosphoryl group to a compound effectively puts free energy into that compound, so that it has more free energy to give up during subsequent metabolic transformations. We described above how the synthesis of glucose 6-phosphate is accomplished by phosphoryl group transfer from ATP. In the next chapter we see how this phosphorylation of glucose activates, or “primes,” the glucose for catabolic reactions that occur in nearly every living cell. Because of its intermediate position on the scale of group transfer potential, ATP can carry energy from high-energy

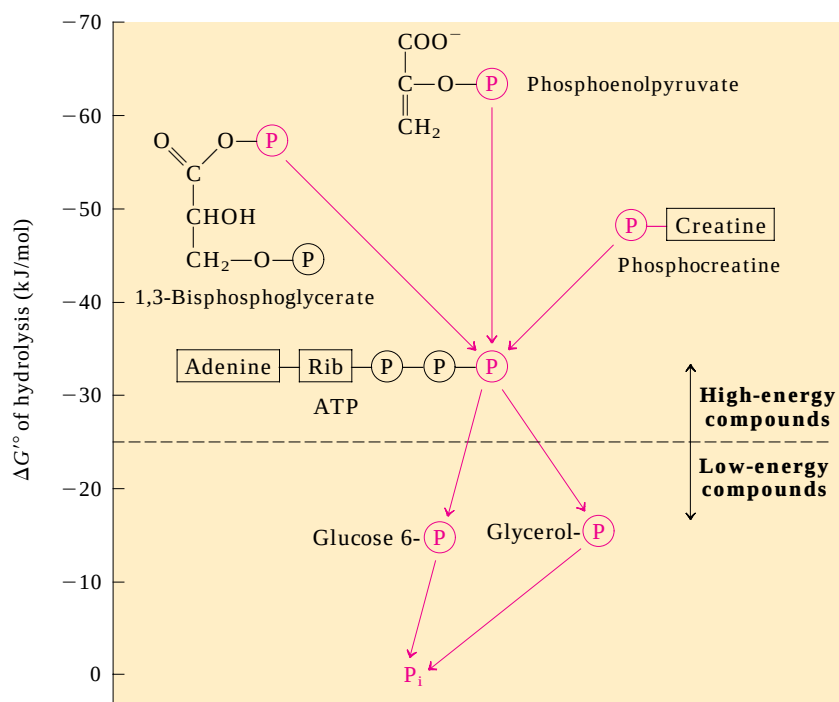


FIGURE 13–9 Ranking of biological phosphate compounds by standard free energies of hydrolysis. This shows the flow of phosphoryl groups, represented by P , from high-energy phosphoryl donors via ATP to acceptor molecules (such as glucose and glycerol) to form their low-energy phosphate derivatives. This flow of phosphoryl groups, catalyzed by enzymes called kinases, proceeds with an overall loss of free energy under intracellular conditions. Hydrolysis of low-energy phosphate compounds releases P_i , which has an even lower phosphoryl group transfer potential (as defined in the text).

phosphate compounds produced by catabolism to compounds such as glucose, converting them into more reactive species. ATP thus serves as the universal energy currency in all living cells.

One more chemical feature of ATP is crucial to its role in metabolism: although in aqueous solution ATP is thermodynamically unstable and is therefore a good phosphoryl group donor, it is *kinetically* stable. Because of the huge activation energies (200 to 400 kJ/mol) required for uncatalyzed cleavage of its phosphoanhydride bonds, ATP does not spontaneously donate phosphoryl groups to water or to the hundreds of other potential acceptors in the cell. Only when specific enzymes are present to lower the energy of activation does phosphoryl group transfer from ATP proceed. The cell is therefore able to regulate the disposition of the energy carried by ATP by regulating the various enzymes that act on it.

ATP Donates Phosphoryl, Pyrophosphoryl, and Adenylyl Groups

The reactions of ATP are generally S_N2 nucleophilic displacements (p. II.8), in which the nucleophile may be, for example, the oxygen of an alcohol or carboxylate, or a nitrogen of creatine or of the side chain of arginine or histidine. Each of the three phosphates of ATP is susceptible to nucleophilic attack (Fig. 13–10), and each position of attack yields a different type of product.

Nucleophilic attack by an alcohol on the γ phosphate (Fig. 13–10a) displaces ADP and produces a new phosphate ester. Studies with ^{18}O -labeled reactants have shown that the bridge oxygen in the new compound is derived from the alcohol, not from ATP; the group transferred from ATP is a phosphoryl ($-\text{PO}_3^{2-}$), not a phosphate ($-\text{OPO}_3^{2-}$). Phosphoryl group transfer from ATP to glutamate (Fig. 13–8) or to glucose

(p. 218) involves attack at the γ position of the ATP molecule.

Attack at the β phosphate of ATP displaces AMP and transfers a pyrophosphoryl (not pyrophosphate) group to the attacking nucleophile (Fig. 13–10b). For example, the formation of 5'-phosphoribosyl-1-pyrophosphate (p. XXX), a key intermediate in nucleotide synthesis, results from attack of an $-\text{OH}$ of the ribose on the β phosphate.

Nucleophilic attack at the α position of ATP displaces PP_i and transfers adenylyl (5'-AMP) as an adenylyl group (Fig. 13–10c); the reaction is an **adenylylation** (a-den'-i-li-lä'-shun, probably the most ungainly word in the biochemical language). Notice that hydrolysis of the α - β phosphoanhydride bond releases considerably more energy (~ 46 kJ/mol) than hydrolysis of the β - γ bond (~ 31 kJ/mol) (Table 13–6). Furthermore, the PP_i formed as a byproduct of the adenylylation is hydrolyzed to two P_i by the ubiquitous enzyme **inorganic pyrophosphatase**, releasing 19 kJ/mol and thereby providing a further energy “push” for the adenylylation reaction. In effect, both phosphoanhydride bonds of ATP are split in the overall reaction. Adenylylation reactions are therefore thermodynamically very favorable. When the energy of ATP is used to drive a particularly unfavorable metabolic reaction, adenylylation is often the mechanism of energy coupling. Fatty acid activation is a good example of this energy-coupling strategy.

The first step in the activation of a fatty acid—either for energy-yielding oxidation or for use in the synthesis of more complex lipids—is the formation of its thiol ester (see Fig. 17–5). The direct condensation of a fatty acid with coenzyme A is endergonic, but the formation of fatty acyl-CoA is made exergonic by stepwise removal of two phosphoryl groups from ATP. First, adenylyl (AMP) is transferred from ATP to the carboxyl group of the fatty acid, forming a mixed anhydride

Three positions on ATP for attack by the nucleophile R^{18}O

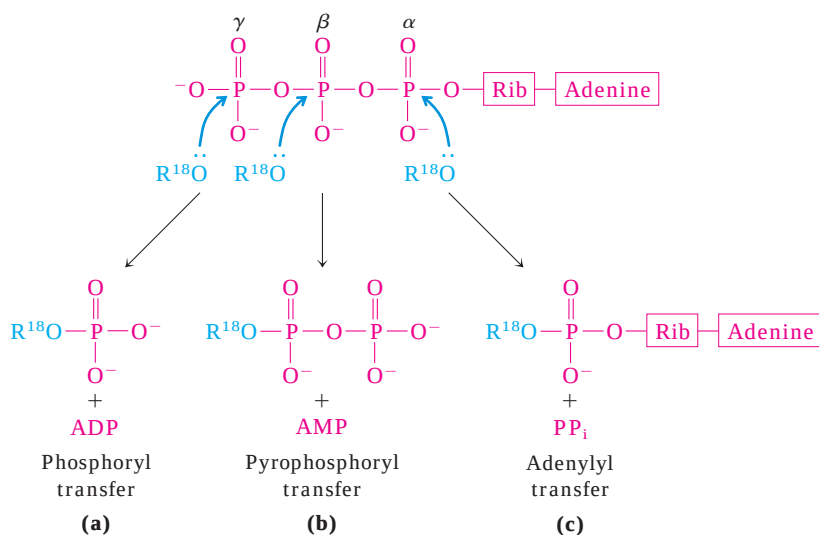


FIGURE 13–10 Nucleophilic displacement reactions of ATP. Any of the three P atoms (α , β , or γ) may serve as the electrophilic target for nucleophilic attack—in this case, by the labeled nucleophile R^{18}O . The nucleophile may be an alcohol (ROH), a carboxyl group (RCOO^-), or a phosphoanhydride (a nucleoside mono- or diphosphate, for example). **(a)** When the oxygen of the nucleophile attacks the γ position, the bridge oxygen of the product is labeled, indicating that the group transferred from ATP is a phosphoryl ($-\text{PO}_3^{2-}$), not a phosphate ($-\text{OPO}_3^{2-}$). **(b)** Attack on the β position displaces AMP and leads to the transfer of a pyrophosphoryl (not pyrophosphate) group to the nucleophile. **(c)** Attack on the α position displaces PP_i and transfers the adenylyl group to the nucleophile.

(fatty acyl adenylate) and liberating PP_i . The thiol group of coenzyme A then displaces the adenylate group and forms a thioester with the fatty acid. The sum of these two reactions is energetically equivalent to the exergonic hydrolysis of ATP to AMP and PP_i ($\Delta G'^{\circ} = -45.6$ kJ/mol) and the endergonic formation of fatty acyl-CoA ($\Delta G'^{\circ} = 31.4$ kJ/mol). The formation of fatty acyl-CoA is made energetically favorable by hydrolysis of the PP_i by inorganic pyrophosphatase. Thus, in the activation of a fatty acid, both phosphoanhydride bonds of ATP are broken. The resulting $\Delta G'^{\circ}$ is the sum of the $\Delta G'^{\circ}$ values

for the breakage of these bonds, or -45.6 kJ/mol + (-19.2) kJ/mol:

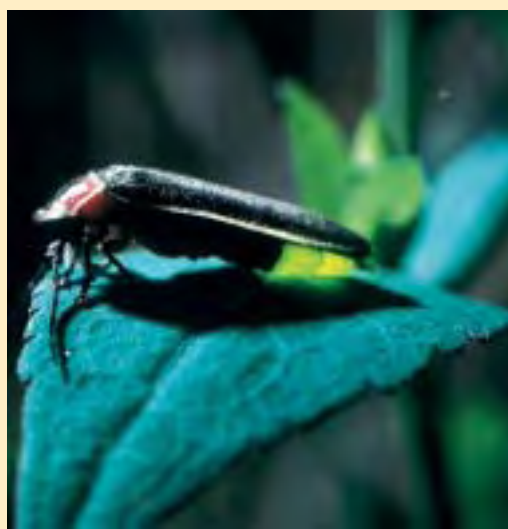


The activation of amino acids before their polymerization into proteins (see Fig. 27–14) is accomplished by an analogous set of reactions in which a transfer RNA molecule takes the place of coenzyme A. An interesting use of the cleavage of ATP to AMP and PP_i occurs in the firefly, which uses ATP as an energy source to produce light flashes (Box 13–2).

BOX 13–2 THE WORLD OF BIOCHEMISTRY

Firefly Flashes: Glowing Reports of ATP

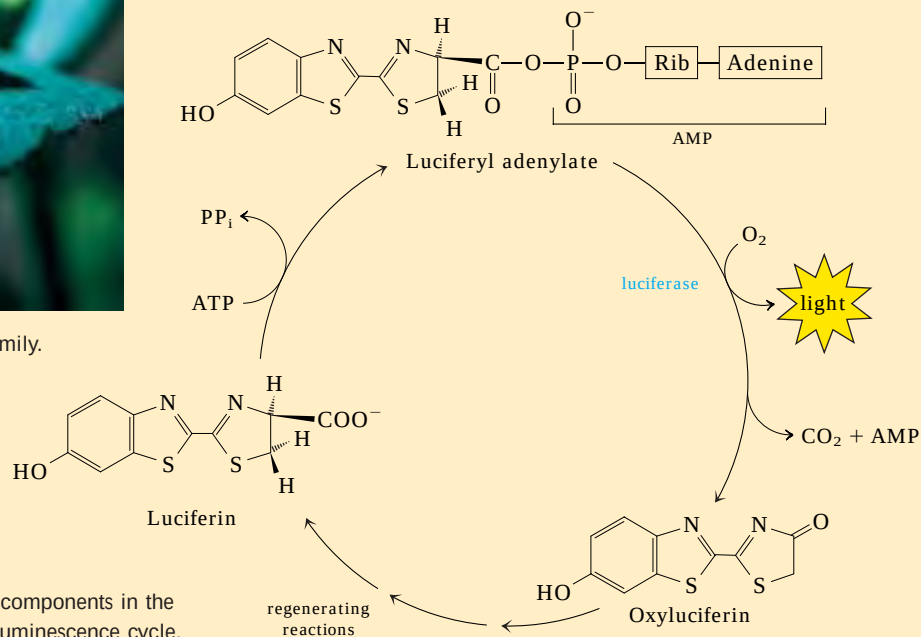
Bioluminescence requires considerable amounts of energy. In the firefly, ATP is used in a set of reactions that converts chemical energy into light energy. In the 1950s, from many thousands of fireflies collected by children in and around Baltimore, William McElroy and his colleagues at The Johns Hopkins University isolated the principal biochemical components: luciferin, a complex carboxylic acid, and luciferase, an enzyme. The generation of a light flash requires activation of luciferin by an enzymatic reaction involving



The firefly, a beetle of the Lampyridae family.

pyrophosphate cleavage of ATP to form luciferyl adenylate. In the presence of molecular oxygen and luciferase, the luciferin undergoes a multistep oxidative decarboxylation to oxyluciferin. This process is accompanied by emission of light. The color of the light flash differs with the firefly species and appears to be determined by differences in the structure of the luciferase. Luciferin is regenerated from oxyluciferin in a subsequent series of reactions.

In the laboratory, pure firefly luciferin and luciferase are used to measure minute quantities of ATP by the intensity of the light flash produced. As little as a few picomoles (10^{-12} mol) of ATP can be measured in this way. An enlightening extension of the studies in luciferase was the cloning of the luciferase gene into tobacco plants. When watered with a solution containing luciferin, the plants glowed in the dark (see Fig. 9–29).



Important components in the firefly bioluminescence cycle.

Assembly of Informational Macromolecules Requires Energy

When simple precursors are assembled into high molecular weight polymers with defined sequences (DNA, RNA, proteins), as described in detail in Part III, energy is required both for the condensation of monomeric units and for the creation of *ordered* sequences. The precursors for DNA and RNA synthesis are nucleoside triphosphates, and polymerization is accompanied by cleavage of the phosphoanhydride linkage between the α and β phosphates, with the release of PP_i (Fig. 13–11). The moieties transferred to the growing polymer in these reactions are adenylate (AMP), guanylate (GMP), cytidylate (CMP), or uridylylate (UMP) for RNA synthesis, and their deoxy analogs (with TMP in place of UMP) for DNA synthesis. As noted above, the activation of amino acids for protein synthesis involves the donation of adenylate groups from ATP, and we shall see in Chapter 27 that several steps of protein synthesis on the ribosome are also accompanied by GTP hydrolysis. In all these cases, the exergonic breakdown of a nucleoside triphosphate is coupled to the endergonic process of synthesizing a polymer of a specific sequence.

ATP Energizes Active Transport and Muscle Contraction

ATP can supply the energy for transporting an ion or a molecule across a membrane into another aqueous compartment where its concentration is higher (see Fig. 11–36). Transport processes are major consumers of energy; in human kidney and brain, for example, as much as two-thirds of the energy consumed at rest is used to pump Na^+ and K^+ across plasma membranes via the Na^+K^+ ATPase. The transport of Na^+ and K^+ is driven by cyclic phosphorylation and dephosphorylation of the transporter protein, with ATP as the phosphoryl group donor (see Fig. 11–37). Na^+ -dependent phosphorylation of the Na^+K^+ ATPase forces a change in the protein's conformation, and K^+ -dependent dephosphorylation favors return to the original conformation. Each cycle in the transport process results in the conversion of ATP to ADP and P_i , and it is the free-energy change of ATP hydrolysis that drives the cyclic changes in protein conformation that result in the electrogenic pumping of Na^+ and K^+ . Note that in this case ATP interacts covalently by phosphoryl group transfer to the enzyme, not the substrate.

In the contractile system of skeletal muscle cells, myosin and actin are specialized to transduce the chemical energy of ATP into motion (see Fig. 5–33). ATP binds tightly but noncovalently to one conformation of myosin, holding the protein in that conformation. When myosin catalyzes the hydrolysis of its bound ATP, the ADP and P_i dissociate from the protein, allowing it to

relax into a second conformation until another molecule of ATP binds. The binding and subsequent hydrolysis of ATP (by myosin ATPase) provide the energy that forces cyclic changes in the conformation of the myosin head. The change in conformation of many individual myosin

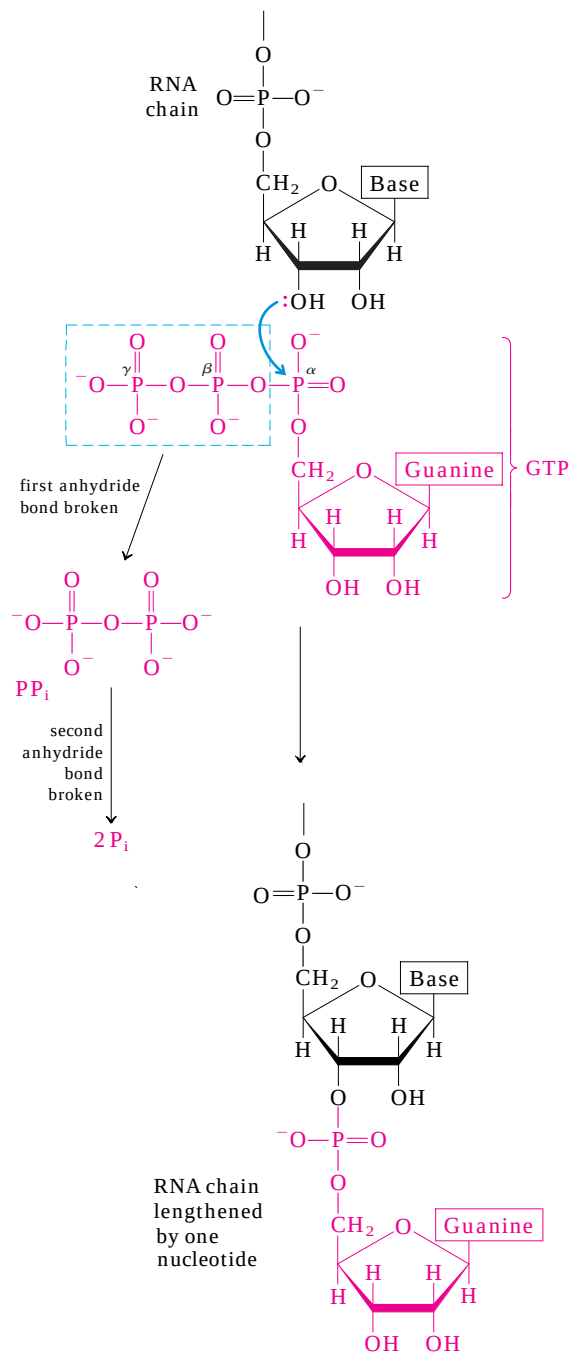


FIGURE 13–11 Nucleoside triphosphates in RNA synthesis. With each nucleoside monophosphate added to the growing chain, one PP_i is released and hydrolyzed to two P_i . The hydrolysis of two phosphoanhydride bonds for each nucleotide added provides the energy for forming the bonds in the RNA polymer and for assembling a specific sequence of nucleotides.

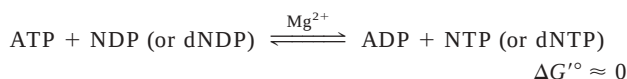
molecules results in the sliding of myosin fibrils along actin filaments (see Fig. 5–32), which translates into macroscopic contraction of the muscle fiber.

As we noted earlier, this production of mechanical motion at the expense of ATP is one of the few cases in which ATP hydrolysis per se, rather than group transfer from ATP, is the source of the chemical energy in a coupled process.

Transphosphorylations between Nucleotides Occur in All Cell Types

Although we have focused on ATP as the cell's energy currency and donor of phosphoryl groups, all other nucleoside triphosphates (GTP, UTP, and CTP) and all the deoxynucleoside triphosphates (dATP, dGTP, dTTP, and dCTP) are energetically equivalent to ATP. The free-energy changes associated with hydrolysis of their phosphoanhydride linkages are very nearly identical with those shown in Table 13–6 for ATP. In preparation for their various biological roles, these other nucleotides are generated and maintained as the nucleoside triphosphate (NTP) forms by phosphoryl group transfer to the corresponding nucleoside diphosphates (NDPs) and monophosphates (NMPs).

ATP is the primary high-energy phosphate compound produced by catabolism, in the processes of glycolysis, oxidative phosphorylation, and, in photosynthetic cells, photophosphorylation. Several enzymes then carry phosphoryl groups from ATP to the other nucleotides. **Nucleoside diphosphate kinase**, found in all cells, catalyzes the reaction



Although this reaction is fully reversible, the relatively high [ATP]/[ADP] ratio in cells normally drives the reaction to the right, with the net formation of NTPs and dNTPs. The enzyme actually catalyzes a two-step phosphoryl transfer, which is a classic case of a double-displacement (Ping-Pong) mechanism (Fig. 13–12; see also Fig. 6–13b). First, phosphoryl group transfer from ATP to an active-site His residue produces a phosphoenzyme

intermediate; then the phosphoryl group is transferred from the P-His residue to an NDP acceptor. Because the enzyme is nonspecific for the base in the NDP and works equally well on dNDPs and NDPs, it can synthesize all NTPs and dNTPs, given the corresponding NDPs and a supply of ATP.

Phosphoryl group transfers from ATP result in an accumulation of ADP; for example, when muscle is contracting vigorously, ADP accumulates and interferes with ATP-dependent contraction. During periods of intense demand for ATP, the cell lowers the ADP concentration, and at the same time acquires ATP, by the action of **adenylate kinase**:



This reaction is fully reversible, so after the intense demand for ATP ends, the enzyme can recycle AMP by converting it to ADP, which can then be phosphorylated to ATP in mitochondria. A similar enzyme, guanylate kinase, converts GMP to GDP at the expense of ATP. By pathways such as these, energy conserved in the catabolic production of ATP is used to supply the cell with all required NTPs and dNTPs.

Phosphocreatine (Fig. 13–5), also called creatine phosphate, serves as a ready source of phosphoryl groups for the quick synthesis of ATP from ADP. The phosphocreatine (PCr) concentration in skeletal muscle is approximately 30 mM, nearly ten times the concentration of ATP, and in other tissues such as smooth muscle, brain, and kidney [PCr] is 5 to 10 mM. The enzyme **creatine kinase** catalyzes the reversible reaction



When a sudden demand for energy depletes ATP, the PCr reservoir is used to replenish ATP at a rate considerably faster than ATP can be synthesized by catabolic pathways. When the demand for energy slackens, ATP produced by catabolism is used to replenish the PCr reservoir by reversal of the creatine kinase reaction. Organisms in the lower phyla employ other PCr-like molecules (collectively called **phosphagens**) as phosphoryl reservoirs.

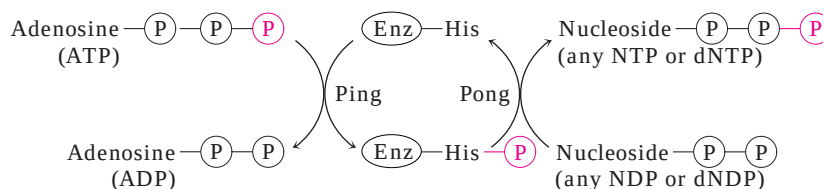
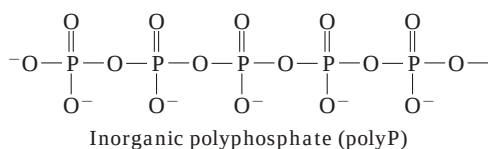


FIGURE 13–12 Ping-Pong mechanism of nucleoside diphosphate kinase. The enzyme binds its first substrate (ATP in our example), and a phosphoryl group is transferred to the side chain of a His residue. ADP departs, and another nucleoside (or deoxynucleoside) diphos-

phate replaces it, and this is converted to the corresponding triphosphate by transfer of the phosphoryl group from the phosphohistidine residue.

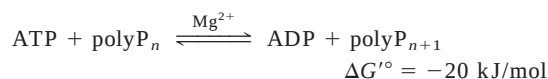
Inorganic Polyphosphate Is a Potential Phosphoryl Group Donor

Inorganic polyphosphate (polyP) is a linear polymer composed of many tens or hundreds of P_i residues linked through phosphoanhydride bonds. This polymer, present in all organisms, may accumulate to high levels in some cells. In yeast, for example, the amount of polyP that accumulates in the vacuoles would represent, if distributed uniformly throughout the cell, a concentration of 200 mM! (Compare this with the concentrations of other phosphoryl donors listed in Table 13–5.)

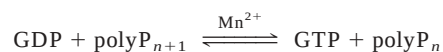


One potential role for polyP is to serve as a phosphagen, a reservoir of phosphoryl groups that can be used to generate ATP, as creatine phosphate is used in muscle. PolyP has about the same phosphoryl group transfer potential as PP_i . The shortest polyphosphate, PP_i ($n = 2$), can serve as the energy source for active transport of H^+ in plant vacuoles. For at least one form of the enzyme phosphofructokinase in plants, PP_i is the phosphoryl group donor, a role played by ATP in animals and microbes (p. XXX). The finding of high concentrations of polyP in volcanic condensates and steam vents suggests that it could have served as an energy source in prebiotic and early cellular evolution.

In prokaryotes, the enzyme **polyphosphate kinase-1** (PPK-1) catalyzes the reversible reaction



by a mechanism involving an enzyme-bound phosphohistidine intermediate (recall the mechanism of nucleoside diphosphate kinase, described above). A second enzyme, **polyphosphate kinase-2** (PPK-2), catalyzes the reversible synthesis of GTP (or ATP) from polyphosphate and GDP (or ADP):



PPK-2 is believed to act primarily in the direction of GTP and ATP synthesis, and PPK-1 in the direction of polyphosphate synthesis. PPK-1 and PPK-2 are present in a wide variety of prokaryotes, including many pathogenic bacteria.

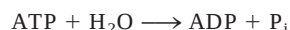
In prokaryotes, elevated levels of polyP have been shown to promote expression of a number of genes involved in adaptation of the organism to conditions of starvation or other threats to survival. In *Escherichia coli*, for example, polyP accumulates when cells are starved for amino acids or P_i , and this accumulation con-

fers a survival advantage. Deletion of the genes for polyphosphate kinases diminishes the ability of certain pathogenic bacteria to invade animal tissues. The enzymes may therefore prove to be vulnerable targets in the development of new antimicrobial drugs.

No gene in yeast encodes a PPK-like protein, but four genes—unrelated to bacterial PPK genes—are necessary for the synthesis of polyphosphate. The mechanism for polyphosphate synthesis in eukaryotes seems to be quite different from that in prokaryotes.

Biochemical and Chemical Equations Are Not Identical

Biochemists write metabolic equations in a simplified way, and this is particularly evident for reactions involving ATP. Phosphorylated compounds can exist in several ionization states and, as we have noted, the different species can bind Mg^{2+} . For example, at pH 7 and 2 mM Mg^{2+} , ATP exists in the forms ATP^{4-} , $HATP^{3-}$, H_2ATP^{2-} , $MgHATP^-$, and Mg_2ATP . In thinking about the biological role of ATP, however, we are not always interested in all this detail, and so we consider ATP as an entity made up of a sum of species, and we write its hydrolysis as the biochemical equation



where ATP, ADP, and P_i are sums of species. The corresponding apparent equilibrium constant, $K'_{eq} = [\text{ADP}][P_i]/[\text{ATP}]$, depends on the pH and the concentration of free Mg^{2+} . Note that H^+ and Mg^{2+} do not appear in the biochemical equation because they are held constant. Thus a biochemical equation does not balance H, Mg, or charge, although it does balance all other elements involved in the reaction (C, N, O, and P in the equation above).

We can write a chemical equation that *does* balance for all elements and for charge. For example, when ATP is hydrolyzed at a pH above 8.5 in the absence of Mg^{2+} , the chemical reaction is represented by



The corresponding equilibrium constant, $K'_{eq} = [\text{ADP}^{3-}][\text{HPO}_4^{2-}][H^+]/[\text{ATP}^{4-}]$, depends only on temperature, pressure, and ionic strength.

Both ways of writing a metabolic reaction have value in biochemistry. Chemical equations are needed when we want to account for all atoms and charges in a reaction, as when we are considering the mechanism of a chemical reaction. Biochemical equations are used to determine in which direction a reaction will proceed spontaneously, given a specified pH and $[Mg^{2+}]$, or to calculate the equilibrium constant of such a reaction.

Throughout this book we use biochemical equations, unless the focus is on chemical mechanism, and we use values of $\Delta G'^{\circ}$ and K'_{eq} as determined at pH 7 and 1 mM Mg^{2+} .

SUMMARY 13.2 Phosphoryl Group Transfers and ATP

- ATP is the chemical link between catabolism and anabolism. It is the energy currency of the living cell. The exergonic conversion of ATP to ADP and P_i , or to AMP and PP_i , is coupled to many endergonic reactions and processes.
- Direct hydrolysis of ATP is the source of energy in the conformational changes that produce muscle contraction but, in general, it is not ATP hydrolysis but the transfer of a phosphoryl, pyrophosphoryl, or adenylyl group from ATP to a substrate or enzyme molecule that couples the energy of ATP breakdown to endergonic transformations of substrates.
- Through these group transfer reactions, ATP provides the energy for anabolic reactions, including the synthesis of informational molecules, and for the transport of molecules and ions across membranes against concentration gradients and electrical potential gradients.
- Cells contain other metabolites with large, negative, free energies of hydrolysis, including phosphoenolpyruvate, 1,3-bisphosphoglycerate, and phosphocreatine. These high-energy compounds, like ATP, have a high phosphoryl group transfer potential; they are good donors of the phosphoryl group. Thioesters also have high free energies of hydrolysis.
- Inorganic polyphosphate, present in all cells, may serve as a reservoir of phosphoryl groups with high group transfer potential.

13.3 Biological Oxidation-Reduction Reactions

The transfer of phosphoryl groups is a central feature of metabolism. Equally important is another kind of transfer, electron transfer in oxidation-reduction reactions. These reactions involve the loss of electrons by one chemical species, which is thereby oxidized, and the gain of electrons by another, which is reduced. The flow of electrons in oxidation-reduction reactions is responsible, directly or indirectly, for all work done by living organisms. In nonphotosynthetic organisms, the sources of electrons are reduced compounds (foods); in photosynthetic organisms, the initial electron donor is a chemical species excited by the absorption of light. The path of electron flow in metabolism is complex. Electrons move from various metabolic intermediates to specialized electron carriers in enzyme-catalyzed reactions.

The carriers in turn donate electrons to acceptors with higher electron affinities, with the release of energy. Cells contain a variety of molecular energy transducers, which convert the energy of electron flow into useful work.

We begin our discussion with a description of the general types of metabolic reactions in which electrons are transferred. After considering the theoretical and experimental basis for measuring the energy changes in oxidation reactions in terms of electromotive force, we discuss the relationship between this force, expressed in volts, and the free-energy change, expressed in joules. We conclude by describing the structures and oxidation-reduction chemistry of the most common of the specialized electron carriers, which you will encounter repeatedly in later chapters.

The Flow of Electrons Can Do Biological Work

Every time we use a motor, an electric light or heater, or a spark to ignite gasoline in a car engine, we use the flow of electrons to accomplish work. In the circuit that powers a motor, the source of electrons can be a battery containing two chemical species that differ in affinity for electrons. Electrical wires provide a pathway for electron flow from the chemical species at one pole of the battery, through the motor, to the chemical species at the other pole of the battery. Because the two chemical species differ in their affinity for electrons, electrons flow spontaneously through the circuit, driven by a force proportional to the difference in electron affinity, the **electromotive force (emf)**. The electromotive force (typically a few volts) can accomplish work if an appropriate energy transducer—in this case a motor—is placed in the circuit. The motor can be coupled to a variety of mechanical devices to accomplish useful work.

Living cells have an analogous biological “circuit,” with a relatively reduced compound such as glucose as the source of electrons. As glucose is enzymatically oxidized, the released electrons flow spontaneously through a series of electron-carrier intermediates to another chemical species, such as O_2 . This electron flow is exergonic, because O_2 has a higher affinity for electrons than do the electron-carrier intermediates. The resulting electromotive force provides energy to a variety of molecular energy transducers (enzymes and other proteins) that do biological work. In the mitochondrion, for example, membrane-bound enzymes couple electron flow to the production of a transmembrane pH difference, accomplishing osmotic and electrical work. The proton gradient thus formed has potential energy, sometimes called the proton-motive force by analogy with electromotive force. Another enzyme, ATP synthase in the inner mitochondrial membrane, uses the proton-motive force to do chemical work: synthesis of ATP from ADP and P_i as protons flow spontaneously across the membrane. Similarly, membrane-localized enzymes in

E. coli convert electromotive force to proton-motive force, which is then used to power flagellar motion.

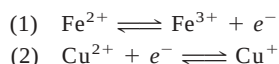
The principles of electrochemistry that govern energy changes in the macroscopic circuit with a motor and battery apply with equal validity to the molecular processes accompanying electron flow in living cells. We turn now to a discussion of those principles.

Oxidation-Reductions Can Be Described as Half-Reactions

Although oxidation and reduction must occur together, it is convenient when describing electron transfers to consider the two halves of an oxidation-reduction reaction separately. For example, the oxidation of ferrous ion by cupric ion,

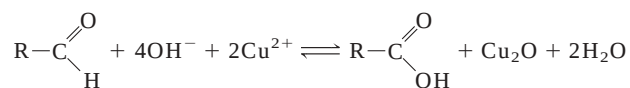


can be described in terms of two half-reactions:

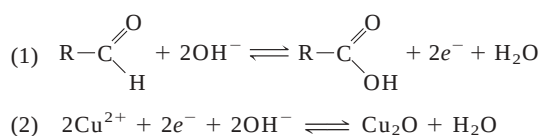


The electron-donating molecule in an oxidation-reduction reaction is called the reducing agent or reductant; the electron-accepting molecule is the oxidizing agent or oxidant. A given agent, such as an iron cation existing in the ferrous (Fe^{2+}) or ferric (Fe^{3+}) state, functions as a conjugate reductant-oxidant pair (redox pair), just as an acid and corresponding base function as a conjugate acid-base pair. Recall from Chapter 2 that in acid-base reactions we can write a general equation: proton donor $\rightleftharpoons \text{H}^+$ + proton acceptor. In redox reactions we can write a similar general equation: electron donor $\rightleftharpoons e^-$ + electron acceptor. In the reversible half-reaction (1) above, Fe^{2+} is the electron donor and Fe^{3+} is the electron acceptor; together, Fe^{2+} and Fe^{3+} constitute a **conjugate redox pair**.

The electron transfers in the oxidation-reduction reactions of organic compounds are not fundamentally different from those of inorganic species. In Chapter 7 we considered the oxidation of a reducing sugar (an aldehyde or ketone) by cupric ion (see Fig. 7–10a):



This overall reaction can be expressed as two half-reactions:



Because two electrons are removed from the aldehyde carbon, the second half-reaction (the one-electron reduction of cupric to cuprous ion) must be doubled to balance the overall equation.

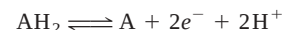
Biological Oxidations Often Involve Dehydrogenation

The carbon in living cells exists in a range of oxidation states (Fig. 13–13). When a carbon atom shares an electron pair with another atom (typically H, C, S, N, or O), the sharing is unequal in favor of the more electronegative atom. The order of increasing electronegativity is $\text{H} < \text{C} < \text{S} < \text{N} < \text{O}$. In oversimplified but useful terms, the more electronegative atom “owns” the bonding electrons it shares with another atom. For example, in methane (CH_4), carbon is more electronegative than the four hydrogens bonded to it, and the C atom therefore “owns” all eight bonding electrons (Fig. 13–13). In ethane, the electrons in the C—C bond are shared equally, so each C atom owns only seven of its eight bonding electrons. In ethanol, C-1 is less electronegative than the oxygen to which it is bonded, and the O atom therefore “owns” both electrons of the C—O bond, leaving C-1 with only five bonding electrons. With each formal loss of electrons, the carbon atom has undergone oxidation—even when no oxygen is involved, as in the conversion of an alkane ($-\text{CH}_2-\text{CH}_2-$) to an alkene ($-\text{CH}=\text{CH}-$). In this case, oxidation (loss of electrons) is coincident with the loss of hydrogen. In biological systems, oxidation is often synonymous with **dehydrogenation**, and many enzymes that catalyze oxidation reactions are **dehydrogenases**. Notice that the more reduced compounds in Figure 13–13 (top) are richer in hydrogen than in oxygen, whereas the more oxidized compounds (bottom) have more oxygen and less hydrogen.

Not all biological oxidation-reduction reactions involve carbon. For example, in the conversion of molecular nitrogen to ammonia, $6\text{H}^+ + 6e^- + \text{N}_2 \rightarrow 2\text{NH}_3$, the nitrogen atoms are reduced.

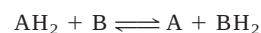
Electrons are transferred from one molecule (electron donor) to another (electron acceptor) in one of four different ways:

1. Directly as *electrons*. For example, the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox pair can transfer an electron to the $\text{Cu}^+/\text{Cu}^{2+}$ redox pair:
2. As *hydrogen atoms*. Recall that a hydrogen atom consists of a proton (H^+) and a single electron (e^-). In this case we can write the general equation



where AH_2 is the hydrogen/electron donor.

(Do not mistake the above reaction for an acid dissociation; the H^+ arises from the removal of a hydrogen atom, $\text{H}^+ + e^-$.) AH_2 and A together constitute a conjugate redox pair (A/AH_2), which can reduce another compound B (or redox pair, B/BH_2) by transfer of hydrogen atoms:



3. As a *hydride ion* (:H^-), which has two electrons. This occurs in the case of NAD-linked dehydrogenases, described below.
4. Through direct *combination with oxygen*. In this case, oxygen combines with an organic reductant and is covalently incorporated in the product, as in the oxidation of a hydrocarbon to an alcohol:

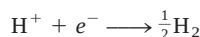


The hydrocarbon is the electron donor and the oxygen atom is the electron acceptor.

All four types of electron transfer occur in cells. The neutral term **reducing equivalent** is commonly used to designate a single electron equivalent participating in an oxidation-reduction reaction, no matter whether this equivalent is an electron per se, a hydrogen atom, or a hydride ion, or whether the electron transfer takes place in a reaction with oxygen to yield an oxygenated product. Because biological fuel molecules are usually enzymatically dehydrogenated to lose *two* reducing equivalents at a time, and because each oxygen atom can accept two reducing equivalents, biochemists by convention regard the unit of biological oxidations as two reducing equivalents passing from substrate to oxygen.

Reduction Potentials Measure Affinity for Electrons

When two conjugate redox pairs are together in solution, electron transfer from the electron donor of one pair to the electron acceptor of the other may proceed spontaneously. The tendency for such a reaction depends on the relative affinity of the electron acceptor of each redox pair for electrons. The **standard reduction potential, E°** , a measure (in volts) of this affinity, can be determined in an experiment such as that described in Figure 13–14. Electrochemists have chosen as a standard of reference the half-reaction



The electrode at which this half-reaction occurs (called a half-cell) is arbitrarily assigned a standard reduction

potential of 0.00 V. When this hydrogen electrode is connected through an external circuit to another half-cell in which an oxidized species and its corresponding reduced species are present at standard concentrations (each solute at 1 M, each gas at 101.3 kPa), electrons tend to flow through the external circuit from the half-cell of

Methane	$\begin{array}{c} \text{H} \\ \vdots \\ \text{H} : \text{C} : \text{H} \\ \vdots \\ \text{H} \end{array}$	8
Ethane (alkane)	$\begin{array}{c} \text{H} \quad \text{H} \\ \vdots \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{H} \\ \vdots \quad \vdots \\ \text{H} \quad \text{H} \end{array}$	7
Ethene (alkene)	$\begin{array}{c} \text{H} \quad \quad \text{H} \\ \vdots \quad \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{H} \\ \vdots \quad \quad \vdots \\ \text{H} \quad \quad \text{H} \end{array}$	6
Ethanol (alcohol)	$\begin{array}{c} \text{H} \quad \text{H} \\ \vdots \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{O} : \text{H} \\ \vdots \quad \vdots \\ \text{H} \quad \text{H} \end{array}$	5
Acetylene (alkyne)	$\text{H} : \text{C} : \text{C} : \text{H}$	5
Formaldehyde	$\begin{array}{c} \text{H} \\ \vdots \\ \text{H} : \text{C} : \text{O} : \\ \vdots \\ \text{H} \end{array}$	4
Acetaldehyde (aldehyde)	$\begin{array}{c} \text{H} \quad \quad \text{H} \\ \vdots \quad \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{O} : \\ \vdots \quad \quad \vdots \\ \text{H} \quad \quad \text{H} \end{array}$	3
Acetone (ketone)	$\begin{array}{c} \text{H} \quad \text{O} \quad \text{H} \\ \vdots \quad \vdots \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{C} : \text{H} \\ \vdots \quad \vdots \quad \vdots \\ \text{H} \quad \text{H} \end{array}$	2
Formic acid (carboxylic acid)	$\begin{array}{c} \text{O} \\ \vdots \\ \text{H} : \text{C} : \text{O} : \text{H} \\ \vdots \\ \text{H} \end{array}$	2
Carbon monoxide	$\text{:C} : \text{O} :$	2
Acetic acid (carboxylic acid)	$\begin{array}{c} \text{H} \quad \text{O} \\ \vdots \quad \vdots \\ \text{H} : \text{C} : \text{C} : \text{O} : \text{H} \\ \vdots \quad \vdots \\ \text{H} \quad \text{H} \end{array}$	1
Carbon dioxide	$\text{:O} : \text{C} : \text{O} :$	0

FIGURE 13–13 Oxidation states of carbon in the biosphere. The oxidation states are illustrated with some representative compounds. Focus on the red carbon atom and its bonding electrons. When this carbon is bonded to the less electronegative H atom, both bonding electrons (red) are assigned to the carbon. When carbon is bonded to another carbon, bonding electrons are shared equally, so one of the two electrons is assigned to the red carbon. When the red carbon is bonded to the more electronegative O atom, the bonding electrons are assigned to the oxygen. The number to the right of each compound is the number of electrons “owned” by the red carbon, a rough expression of the oxidation state of that carbon. When the red carbon undergoes oxidation (loses electrons), the number gets smaller. Thus the oxidation state increases from top to bottom of the list.

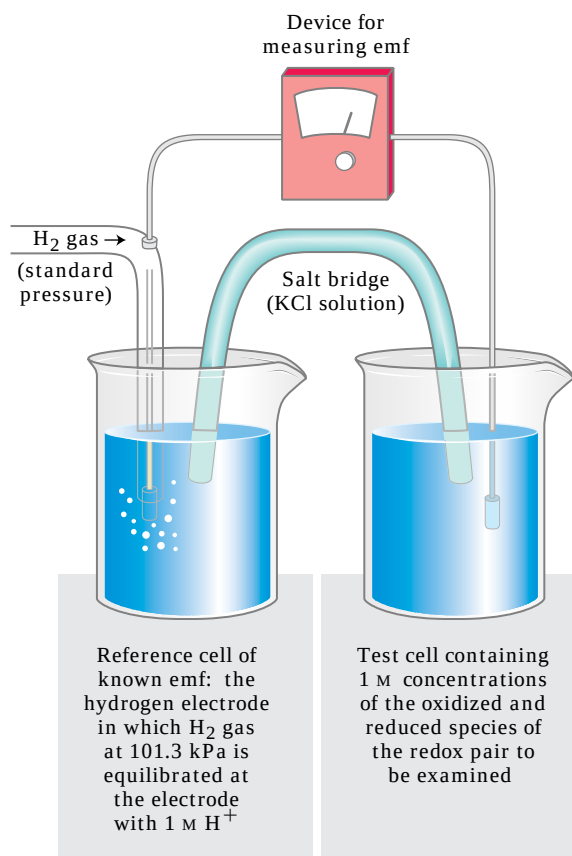


FIGURE 13-14 Measurement of the standard reduction potential (E'°) of a redox pair. Electrons flow from the test electrode to the reference electrode, or vice versa. The ultimate reference half-cell is the hydrogen electrode, as shown here, at pH 0. The electromotive force (emf) of this electrode is designated 0.00 V. At pH 7 in the test cell, E'° for the hydrogen electrode is -0.414 V. The direction of electron flow depends on the relative electron “pressure” or potential of the two cells. A salt bridge containing a saturated KCl solution provides a path for counter-ion movement between the test cell and the reference cell. From the observed emf and the known emf of the reference cell, the experimenter can find the emf of the test cell containing the redox pair. The cell that gains electrons has, by convention, the more positive reduction potential.

lower standard reduction potential to the half-cell of higher standard reduction potential. By convention, the half-cell with the stronger tendency to acquire electrons is assigned a positive value of E° .

The reduction potential of a half-cell depends not only on the chemical species present but also on their activities, approximated by their concentrations. About a century ago, Walther Nernst derived an equation that relates standard reduction potential (E°) to the reduction potential (E) at any concentration of oxidized and reduced species in the cell:

$$E = E^{\circ} + \frac{RT}{n\mathcal{F}} \ln \frac{[\text{electron acceptor}]}{[\text{electron donor}]} \quad (13-4)$$

where R and T have their usual meanings, n is the number of electrons transferred per molecule, and $\mathcal{F}$ is the Faraday constant (Table 13-1). At 298 K (25 °C), this expression reduces to

$$E = E^{\circ} + \frac{0.026 \text{ V}}{n} \ln \frac{[\text{electron acceptor}]}{[\text{electron donor}]} \quad (13-5)$$

Many half-reactions of interest to biochemists involve protons. As in the definition of $\Delta G'^{\circ}$, biochemists define the standard state for oxidation-reduction reactions as pH 7 and express reduction potential as E'° , the standard reduction potential at pH 7. The standard reduction potentials given in Table 13-7 and used throughout this book are values for E'° and are therefore valid only for systems at neutral pH. Each value represents the potential difference when the conjugate redox pair, at 1 M concentrations and pH 7, is connected with the standard (pH 0) hydrogen electrode. Notice in Table 13-7 that when the conjugate pair $2\text{H}^{+}/\text{H}_2$ at pH 7 is connected with the standard hydrogen electrode (pH 0), electrons tend to flow from the pH 7 cell to the standard (pH 0) cell; the measured E'° for the $2\text{H}^{+}/\text{H}_2$ pair is -0.414 V.

Standard Reduction Potentials Can Be Used to Calculate the Free-Energy Change

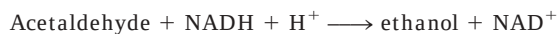
The usefulness of reduction potentials stems from the fact that when E values have been determined for any two half-cells, relative to the standard hydrogen electrode, their reduction potentials relative to each other are also known. We can then predict the direction in which electrons will tend to flow when the two half-cells are connected through an external circuit or when components of both half-cells are present in the same solution. Electrons tend to flow to the half-cell with the more positive E , and the strength of that tendency is proportional to the difference in reduction potentials, ΔE .

The energy made available by this spontaneous electron flow (the free-energy change for the oxidation-reduction reaction) is proportional to ΔE :

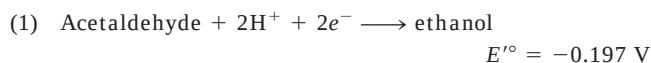
$$\Delta G = -n\mathcal{F} \Delta E \quad \text{or} \quad \Delta G'^{\circ} = -n\mathcal{F} \Delta E'^{\circ} \quad (13-6)$$

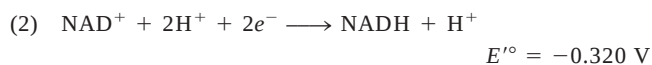
Here n represents the number of electrons transferred in the reaction. With this equation we can calculate the free-energy change for any oxidation-reduction reaction from the values of E'° in a table of reduction potentials (Table 13-7) and the concentrations of the species participating in the reaction.

Consider the reaction in which acetaldehyde is reduced by the biological electron carrier NADH:



The relevant half-reactions and their E'° values are:





By convention, $\Delta E'^{\circ}$ is expressed as E'° of the electron acceptor minus E'° of the electron donor. Because acetaldehyde is accepting electrons from NADH in our example, $\Delta E'^{\circ} = -0.197 \text{ V} - (-0.320 \text{ V}) = 0.123 \text{ V}$, and n is 2. Therefore,

$$\Delta G'^{\circ} = -nF \Delta E'^{\circ} = -2(96.5 \text{ kJ/V} \cdot \text{mol})(0.123 \text{ V}) \\ = -23.7 \text{ kJ/mol}$$

This is the free-energy change for the oxidation-reduction reaction at pH 7, when acetaldehyde, ethanol, NAD^+ , and NADH are all present at 1.00 M concentrations. If, instead, acetaldehyde and NADH were present at 1.00 M but ethanol and NAD^+ were present at 0.100 M, the value for ΔG would be calculated as follows. First,

the values of E for both reductants are determined (Eqn 13-4):

$$E_{\text{acetaldehyde}} = E^{\circ} + \frac{RT}{nF} \ln \frac{[\text{acetaldehyde}]}{[\text{ethanol}]} \\ = -0.197 \text{ V} + \frac{0.026 \text{ V}}{2} \ln \frac{1.00}{0.100} = -0.167 \text{ V} \\ E_{\text{NADH}} = E^{\circ} + \frac{RT}{nF} \ln \frac{[\text{NAD}^+]}{[\text{NADH}]} \\ = -0.320 \text{ V} + \frac{0.026 \text{ V}}{2} \ln \frac{1.00}{0.100} = -0.350 \text{ V}$$

Then ΔE is used to calculate ΔG (Eqn 13-5):

$$\Delta E = -0.167 \text{ V} - (-0.350 \text{ V}) = 0.183 \text{ V} \\ \Delta G = -nF \Delta E \\ = -2(96.5 \text{ kJ/V} \cdot \text{mol})(0.183 \text{ V}) \\ = -35.3 \text{ kJ/mol}$$

TABLE 13-7 Standard Reduction Potentials of Some Biologically Important Half-Reactions, at pH 7.0 and 25°C (298 K)

Half-reaction	E'° (V)
$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{O}$	0.816
$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	0.771
$\text{NO}_3^- + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{NO}_2^- + \text{H}_2\text{O}$	0.421
Cytochrome <i>f</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>f</i> (Fe^{2+})	0.365
$\text{Fe}(\text{CN})_6^{3-}$ (ferricyanide) + $\text{e}^- \longrightarrow \text{Fe}(\text{CN})_6^{4-}$	0.36
Cytochrome a_3 (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{O}_2$	0.295
Cytochrome <i>a</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>a</i> (Fe^{2+})	0.29
Cytochrome <i>c</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>c</i> (Fe^{2+})	0.254
Cytochrome c_1 (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome <i>b</i> (Fe^{3+}) + $\text{e}^- \longrightarrow$ cytochrome <i>b</i> (Fe^{2+})	0.077
Ubiquinone + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ ubiquinol + H_2	0.045
$\text{Fumarate}^{2-} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{succinate}^{2-}$	0.031
$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$ (at standard conditions, pH 0)	0.000
Crotonyl-CoA + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ butyryl-CoA	-0.015
$\text{Oxaloacetate}^{2-} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{malate}^{2-}$	-0.166
$\text{Pyruvate}^- + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{lactate}^-$	-0.185
Acetaldehyde + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ ethanol	-0.197
$\text{FAD} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{FADH}_2$	-0.219*
Glutathione + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ 2 reduced glutathione	-0.23
$\text{S} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{S}$	-0.243
Lipoic acid + $2\text{H}^+ + 2\text{e}^- \longrightarrow$ dihydrolipoic acid	-0.29
$\text{NAD}^+ + \text{H}^+ + 2\text{e}^- \longrightarrow \text{NADH}$	-0.320
$\text{NADP}^+ + \text{H}^+ + 2\text{e}^- \longrightarrow \text{NADPH}$	-0.324
Acetoacetate + $2\text{H}^+ + 2\text{e}^- \longrightarrow \beta$ -hydroxybutyrate	-0.346
α -Ketoglutarate + $\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow$ isocitrate	-0.38
$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$ (at pH 7)	-0.414
Ferredoxin (Fe^{3+}) + $\text{e}^- \longrightarrow$ ferredoxin (Fe^{2+})	-0.432

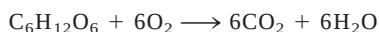
Source: Data mostly from Loach, P.A. (1976) In *Handbook of Biochemistry and Molecular Biology*, 3rd edn (Fasman, G.D., ed.), Physical and Chemical Data, Vol. I, pp. 122-130, CRC Press, Boca Raton, FL.

* This is the value for free FAD; FAD bound to a specific flavoprotein (for example succinate dehydrogenase) has a different E'° that depends on its protein environments.

It is thus possible to calculate the free-energy change for any biological redox reaction at any concentrations of the redox pairs.

Cellular Oxidation of Glucose to Carbon Dioxide Requires Specialized Electron Carriers

The principles of oxidation-reduction energetics described above apply to the many metabolic reactions that involve electron transfers. For example, in many organisms, the oxidation of glucose supplies energy for the production of ATP. The complete oxidation of glucose:



has a $\Delta G'^{\circ}$ of $-2,840$ kJ/mol. This is a much larger release of free energy than is required for ATP synthesis (50 to 60 kJ/mol; see Box 13–1). Cells convert glucose to CO_2 not in a single, high-energy-releasing reaction, but rather in a series of controlled reactions, some of which are oxidations. The free energy released in these oxidation steps is of the same order of magnitude as that required for ATP synthesis from ADP, with some energy to spare. Electrons removed in these oxidation steps are transferred to coenzymes specialized for carrying electrons, such as NAD^+ and FAD (described below).

A Few Types of Coenzymes and Proteins Serve as Universal Electron Carriers

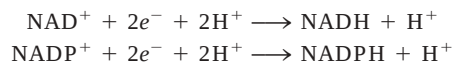
The multitude of enzymes that catalyze cellular oxidations channel electrons from their hundreds of different substrates into just a few types of universal electron carriers. The reduction of these carriers in catabolic processes results in the conservation of free energy released by substrate oxidation. NAD^+ , NADP^+ , FMN, and FAD are water-soluble coenzymes that undergo reversible oxidation and reduction in many of the electron-transfer reactions of metabolism. The nucleotides NAD^+ and NADP^+ move readily from one enzyme to another; the flavin nucleotides FMN and FAD are usually very tightly bound to the enzymes, called flavoproteins, for which they serve as prosthetic groups. Lipid-soluble quinones such as ubiquinone and plastoquinone act as electron carriers and proton donors in the nonaqueous environment of membranes. Iron-sulfur proteins and cytochromes, which have tightly bound prosthetic groups that undergo reversible oxidation and reduction, also serve as electron carriers in many oxidation-reduction reactions. Some of these proteins are water-soluble, but others are peripheral or integral membrane proteins (see Fig. 11–6).

We conclude this chapter by describing some chemical features of nucleotide coenzymes and some of the enzymes (dehydrogenases and flavoproteins) that use them. The oxidation-reduction chemistry of quinones, iron-sulfur proteins, and cytochromes is discussed in Chapter 19.

NADH and NADPH Act with Dehydrogenases as Soluble Electron Carriers

Nicotinamide adenine dinucleotide (NAD^+ in its oxidized form) and its close analog nicotinamide adenine dinucleotide phosphate (NADP^+) are composed of two nucleotides joined through their phosphate groups by a phosphoanhydride bond (Fig. 13–15a). Because the nicotinamide ring resembles pyridine, these compounds are sometimes called **pyridine nucleotides**. The vitamin niacin is the source of the nicotinamide moiety in nicotinamide nucleotides.

Both coenzymes undergo reversible reduction of the nicotinamide ring (Fig. 13–15). As a substrate molecule undergoes oxidation (dehydrogenation), giving up two hydrogen atoms, the oxidized form of the nucleotide (NAD^+ or NADP^+) accepts a hydride ion ($:\text{H}^-$, the equivalent of a proton and two electrons) and is transformed into the reduced form (NADH or NADPH). The second proton removed from the substrate is released to the aqueous solvent. The half-reaction for each type of nucleotide is therefore



Reduction of NAD^+ or NADP^+ converts the benzenoid ring of the nicotinamide moiety (with a fixed positive charge on the ring nitrogen) to the quinonoid form (with no charge on the nitrogen). Note that the reduced nucleotides absorb light at 340 nm; the oxidized forms do not (Fig. 13–15b). The plus sign in the abbreviations NAD^+ and NADP^+ does *not* indicate the net charge on these molecules (they are both negatively charged); rather, it indicates that the nicotinamide ring is in its oxidized form, with a positive charge on the nitrogen atom. In the abbreviations NADH and NADPH , the “H” denotes the added hydride ion. To refer to these nucleotides without specifying their oxidation state, we use NAD and NADP.

The total concentration of $\text{NAD}^+ + \text{NADH}$ in most tissues is about 10^{-5} M; that of $\text{NADP}^+ + \text{NADPH}$ is about 10^{-6} M. In many cells and tissues, the ratio of NAD^+ (oxidized) to NADH (reduced) is high, favoring hydride transfer from a substrate to NAD^+ to form NADH . By contrast, NADPH (reduced) is generally present in greater amounts than its oxidized form, NADP^+ , favoring hydride transfer *from* NADPH to a substrate. This reflects the specialized metabolic roles of the two coenzymes: NAD^+ generally functions in oxidations—usually as part of a catabolic reaction; and NADPH is the usual coenzyme in reductions—nearly always as part of an anabolic reaction. A few enzymes can use either coenzyme, but most show a strong preference for one over the other. The processes in which these two cofactors function are also segregated in specific organelles of eukaryotic cells: oxidations of fuels such as pyruvate, fatty acids, and α -keto acids derived from

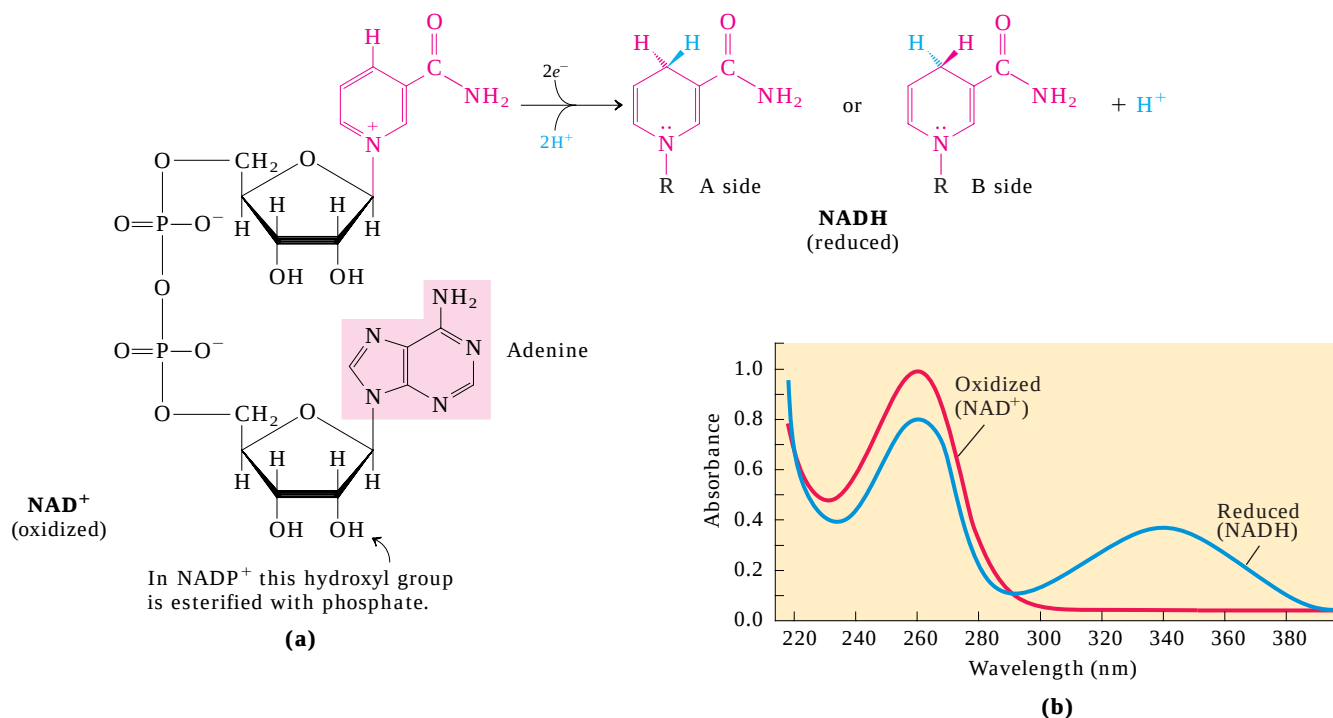
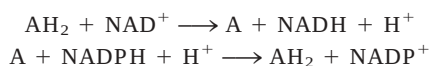


FIGURE 13-15 NAD and NADP. (a) Nicotinamide adenine dinucleotide, NAD^+ , and its phosphorylated analog NADP^+ undergo reduction to NADH and NADPH , accepting a hydride ion (two electrons and one proton) from an oxidizable substrate. The hydride ion is added to either the front (the A side) or the back (the B side) of the planar nicotinamide ring (see Table 13-8). (b) The UV absorption spec-

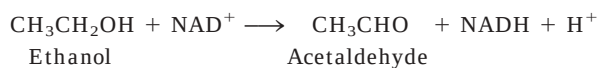
tra of NAD^+ and NADH . Reduction of the nicotinamide ring produces a new, broad absorption band with a maximum at 340 nm. The production of NADH during an enzyme-catalyzed reaction can be conveniently followed by observing the appearance of the absorbance at 340 nm (the molar extinction coefficient $\epsilon_{340} = 6,200 \text{ M}^{-1}\text{cm}^{-1}$).

amino acids occur in the mitochondrial matrix, whereas reductive biosynthesis processes such as fatty acid synthesis take place in the cytosol. This functional and spatial specialization allows a cell to maintain two distinct pools of electron carriers, with two distinct functions.

More than 200 enzymes are known to catalyze reactions in which NAD^+ (or NADP^+) accepts a hydride ion from a reduced substrate, or NADPH (or NADH) donates a hydride ion to an oxidized substrate. The general reactions are



where AH_2 is the reduced substrate and A the oxidized substrate. The general name for an enzyme of this type is **oxidoreductase**; they are also commonly called **dehydrogenases**. For example, alcohol dehydrogenase catalyzes the first step in the catabolism of ethanol, in which ethanol is oxidized to acetaldehyde:



Notice that one of the carbon atoms in ethanol has lost a hydrogen; the compound has been oxidized from an alcohol to an aldehyde (refer again to Fig. 13-13 for the oxidation states of carbon).

When NAD^+ or NADP^+ is reduced, the hydride ion could in principle be transferred to either side of the nicotinamide ring: the front (A side) or the back (B side), as represented in Figure 13-15a. Studies with isotopically labeled substrates have shown that a given enzyme catalyzes either an A-type or a B-type transfer, but not both. For example, yeast alcohol dehydrogenase and lactate dehydrogenase of vertebrate heart transfer a hydride ion to (or remove a hydride ion from) the A side of the nicotinamide ring; they are classed as type A dehydrogenases to distinguish them from another group of enzymes that transfer a hydride ion to (or remove a hydride ion from) the B side of the nicotinamide ring (Table 13-8). The specificity for one side or another can be very striking; lactate dehydrogenase, for example, prefers the A side over the B side by a factor of 5×10^7 !

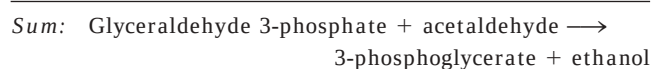
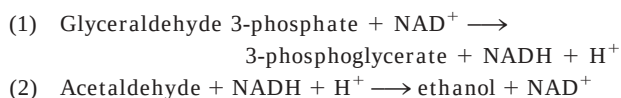
Most dehydrogenases that use NAD or NADP bind the cofactor in a conserved protein domain called the Rossmann fold (named for Michael Rossmann, who deduced the structure of lactate dehydrogenase and first described this structural motif). The Rossmann fold typically consists of a six-stranded parallel β sheet and four associated α helices (Fig. 13-16).

The association between a dehydrogenase and NAD or NADP is relatively loose; the coenzyme readily diffuses from one enzyme to another, acting as a water-soluble

TABLE 13–8 Stereospecificity of Dehydrogenases That Employ NAD⁺ or NADP⁺ as Coenzymes

Enzyme	Coenzyme	Stereochemical specificity for nicotinamide ring (A or B)	Text page(s)
Isocitrate dehydrogenase	NAD ⁺	A	XXX–XXX
α-Ketoglutarate dehydrogenase	NAD ⁺	B	XXX
Glucose 6-phosphate dehydrogenase	NADP ⁺	B	XXX
Malate dehydrogenase	NAD ⁺	A	XXX
Glutamate dehydrogenase	NAD ⁺ or NADP ⁺	B	XXX
Glyceraldehyde 3-phosphate dehydrogenase	NAD ⁺	B	XXX
Lactate dehydrogenase	NAD ⁺	A	XXX
Alcohol dehydrogenase	NAD ⁺	A	XXX

carrier of electrons from one metabolite to another. For example, in the production of alcohol during fermentation of glucose by yeast cells, a hydride ion is removed from glyceraldehyde 3-phosphate by one enzyme (glyceraldehyde 3-phosphate dehydrogenase, a type B enzyme) and transferred to NAD⁺. The NADH produced then leaves the enzyme surface and diffuses to another enzyme (alcohol dehydrogenase, a type A enzyme), which transfers a hydride ion to acetaldehyde, producing ethanol:



Notice that in the overall reaction there is no net production or consumption of NAD⁺ or NADH; the coenzymes function catalytically and are recycled repeatedly without a net change in the concentration of NAD⁺ + NADH.

Dietary Deficiency of Niacin, the Vitamin Form of NAD and NADP, Causes Pellagra



The pyridine-like rings of NAD and NADP are derived from the vitamin **niacin** (nicotinic acid; Fig. 13–17), which is synthesized from tryptophan. Humans generally cannot synthesize niacin in sufficient quantities, and this is especially so for those with diets low in tryptophan (maize, for example, has a low tryptophan content). Niacin deficiency, which affects all the NAD(P)-dependent dehydrogenases, causes the serious human disease pellagra (Italian for “rough skin”) and a related disease in dogs, blacktongue. These diseases are characterized by the “three Ds”: dermatitis, diarrhea, and dementia, followed in many cases by death. A century ago, pellagra was a common human disease; in the southern United States, where maize was a dietary staple, about 100,000 people were afflicted and about 10,000 died between 1912 and 1916. In 1920 Joseph Goldberger showed pellagra to be caused by a dietary insufficiency, and in 1937 Frank Strong, D. Wayne Wolley, and Conrad Elvehjem identified niacin as the curative agent for blacktongue. Supplementation of the human diet with this inexpensive compound led to the eradication of pellagra in the populations of the developed world—with one significant exception. Pellagra is still found among alcoholics, whose intestinal absorption of niacin is much

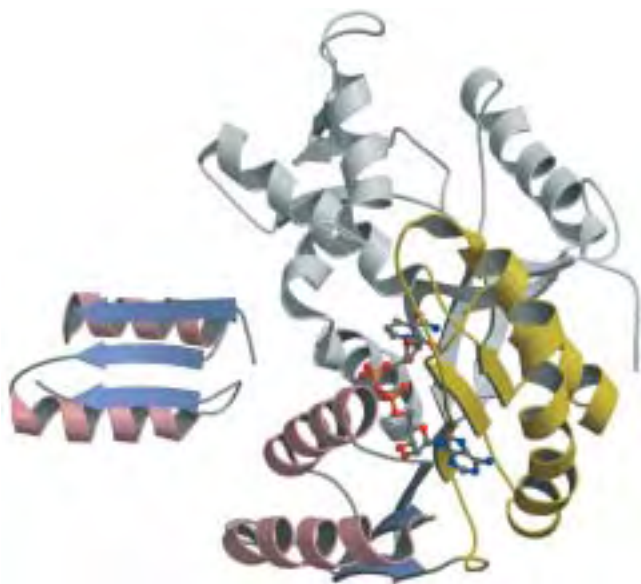


FIGURE 13–16 The nucleotide binding domain of the enzyme **lactate dehydrogenase**. (a) The Rossmann fold is a structural motif found in the NAD-binding site of many dehydrogenases. It consists of a six-stranded parallel β sheet and four α helices; inspection reveals the arrangement to be a pair of structurally similar β - α - β - α motifs. (b) The dinucleotide NAD binds in an extended conformation through hydrogen bonds and salt bridges (derived from PDB ID 3LDH).

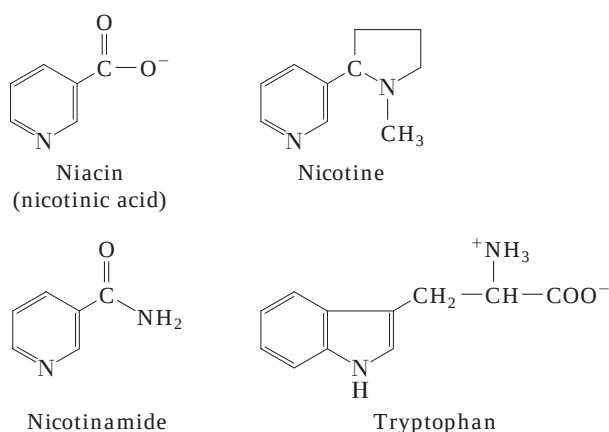


FIGURE 13-17 Structures of niacin (nicotinic acid) and its derivative nicotinamide. The biosynthetic precursor of these compounds is tryptophan. In the laboratory, nicotinic acid was first produced by oxidation of the natural product nicotine—thus the name. Both nicotinic acid and nicotinamide cure pellagra, but nicotine (from cigarettes or elsewhere) has no curative activity.

reduced, and whose caloric needs are often met with distilled spirits that are virtually devoid of vitamins, including niacin. In a few places, including the Deccan Plateau in India, pellagra still occurs, especially among the poor. ■

Flavin Nucleotides Are Tightly Bound in Flavoproteins

Flavoproteins (Table 13-9) are enzymes that catalyze oxidation-reduction reactions using either flavin mononucleotide (FMN) or flavin adenine dinucleotide (FAD) as coenzyme (Fig. 13-18). These coenzymes, the **flavin nucleotides**, are derived from the vitamin riboflavin. The fused ring structure of flavin nucleotides (the isoalloxazine ring) undergoes reversible reduction, accepting either one or two electrons in the form of one or two hydrogen atoms (each atom an electron plus a proton) from a reduced substrate. The fully reduced forms are abbreviated FADH₂ and FMNH₂. When a fully oxidized flavin nucleotide accepts only one electron (one hydrogen atom), the semiquinone form of the isoal-

TABLE 13-9 Some Enzymes (Flavoproteins) That Employ Flavin Nucleotide Coenzymes

Enzyme	Flavin nucleotide	Text page(s)
Acyl-CoA dehydrogenase	FAD	XXX
Dihydrolipoyl dehydrogenase	FAD	XXX
Succinate dehydrogenase	FAD	XXX
Glycerol 3-phosphate dehydrogenase	FAD	XXX
Thioredoxin reductase	FAD	XXX-XXX
NADH dehydrogenase (Complex I)	FMN	XXX
Glycolate oxidase	FMN	XXX

loxazine ring is produced, abbreviated FADH[•] and FMNH[•]. Because flavoproteins can participate in either one- or two-electron transfers, this class of proteins is involved in a greater diversity of reactions than the NAD (P)-linked dehydrogenases.

Like the nicotinamide coenzymes (Fig. 13-15), the flavin nucleotides undergo a shift in a major absorption band on reduction. Flavoproteins that are fully reduced (two electrons accepted) generally have an absorption maximum near 360 nm. When partially reduced (one electron), they acquire another absorption maximum at about 450 nm; when fully oxidized, the flavin has maxima at 370 and 440 nm. The intermediate radical form, reduced by one electron, has absorption maxima at 380, 480, 580, and 625 nm. These changes can be used to assay reactions involving a flavoprotein.

The flavin nucleotide in most flavoproteins is bound rather tightly to the protein, and in some enzymes, such as succinate dehydrogenase, it is bound covalently. Such tightly bound coenzymes are properly called prosthetic groups. They do not transfer electrons by diffusing from one enzyme to another; rather, they provide a means by which the flavoprotein can temporarily hold electrons while it catalyzes electron transfer from a reduced substrate to an electron acceptor. One important feature of the flavoproteins is the variability in the standard reduction potential (E'°) of the bound flavin nucleotide. Tight association between the enzyme and prosthetic group confers on the flavin ring a reduction potential typical of that particular flavoprotein, sometimes quite different from the reduction potential of the free flavin nucleotide. FAD bound to succinate dehydrogenase, for example, has an E'° close to 0.0 V, compared with -0.219 V for free FAD; E'° for other flavoproteins ranges from -0.40 V to $+0.06$ V.



Frank Strong,
1908–1993



D. Wayne Woolley,
1914–1966



Conrad Elvehjem,
1901–1962

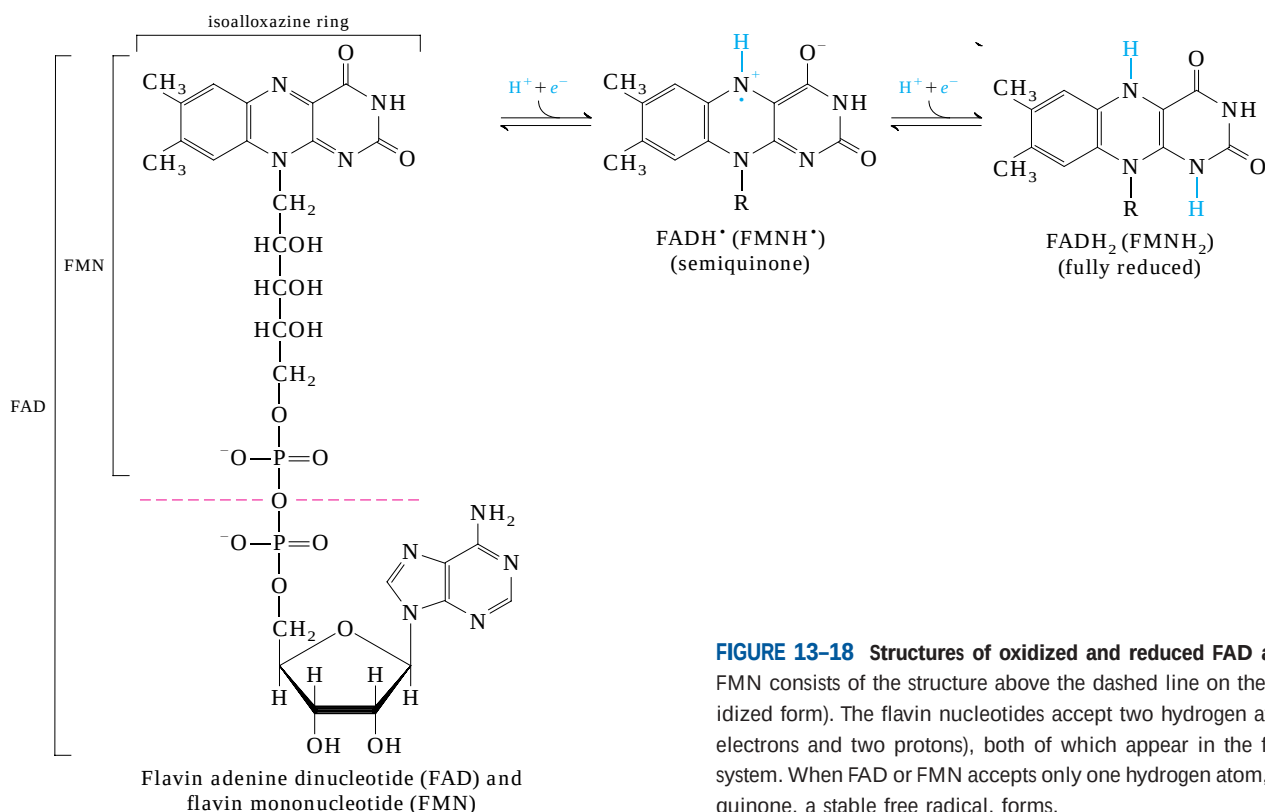


FIGURE 13-18 Structures of oxidized and reduced FAD and FMN. FMN consists of the structure above the dashed line on the FAD (oxidized form). The flavin nucleotides accept two hydrogen atoms (two electrons and two protons), both of which appear in the flavin ring system. When FAD or FMN accepts only one hydrogen atom, the semiquinone, a stable free radical, forms.

Flavoproteins are often very complex; some have, in addition to a flavin nucleotide, tightly bound inorganic ions (iron or molybdenum, for example) capable of participating in electron transfers.

Certain flavoproteins act in a quite different role as light receptors. **Cryptochromes** are a family of flavoproteins, widely distributed in the eukaryotic phyla, that mediate the effects of blue light on plant development and the effects of light on mammalian circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period). The cryptochromes are homologs of another family of flavoproteins, the photolyases. Found in both prokaryotes and eukaryotes, **photolyases** use the energy of absorbed light to repair chemical defects in DNA.

We examine the function of flavoproteins as electron carriers in Chapter 19, when we consider their roles in oxidative phosphorylation (in mitochondria) and photophosphorylation (in chloroplasts), and we describe the photolyase reactions in Chapter 25.

SUMMARY 13.3 Biological Oxidation-Reduction Reactions

- In many organisms, a central energy-conserving process is the stepwise oxidation of glucose to CO₂, in which some of the energy of oxidation is conserved in ATP as electrons are passed to O₂.

- Biological oxidation-reduction reactions can be described in terms of two half-reactions, each with a characteristic standard reduction potential, E'° .
- When two electrochemical half-cells, each containing the components of a half-reaction, are connected, electrons tend to flow to the half-cell with the higher reduction potential. The strength of this tendency is proportional to the difference between the two reduction potentials (ΔE) and is a function of the concentrations of oxidized and reduced species.
- The standard free-energy change for an oxidation-reduction reaction is directly proportional to the difference in standard reduction potentials of the two half-cells: $\Delta G'^{\circ} = -n\mathcal{F}\Delta E'^{\circ}$.
- Many biological oxidation reactions are dehydrogenations in which one or two hydrogen atoms ($H^+ + e^-$) are transferred from a substrate to a hydrogen acceptor. Oxidation-reduction reactions in living cells involve specialized electron carriers.
- NAD and NADP are the freely diffusible coenzymes of many dehydrogenases. Both NAD⁺ and NADP⁺ accept two electrons and

one proton. NAD and NADP are bound to dehydrogenases in a widely conserved structural motif called the Rossmann fold.

- FAD and FMN, the flavin nucleotides, serve as tightly bound prosthetic groups of

flavoproteins. They can accept either one or two electrons. Flavoproteins also serve as light receptors in cryptochromes and photolyases.

Key Terms

Terms in bold are defined in the glossary.

autotroph XXX	thioester XXX	dehydrogenation XXX
heterotroph XXX	adenylation XXX	dehydrogenases XXX
metabolism XXX	inorganic pyrophosphatase XXX	reducing equivalent XXX
metabolic pathways XXX	nucleoside diphosphate	standard reduction potential
metabolite XXX	kinase XXX	(E'°) XXX
intermediary metabolism XXX	adenylate kinase XXX	pyridine nucleotide XXX
catabolism XXX	creatine kinase XXX	oxidoreductase XXX
anabolism XXX	phosphagens XXX	flavoprotein XXX
standard transformed constants XXX	polyphosphate kinase-1, -2 XXX	flavin nucleotides XXX
phosphorylation potential	electromotive force (emf) XXX	cryptochrome XXX
(ΔG_p) XXX	conjugate redox pair XXX	photolyase XXX

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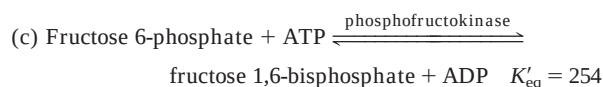
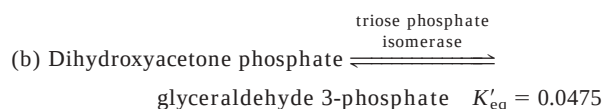
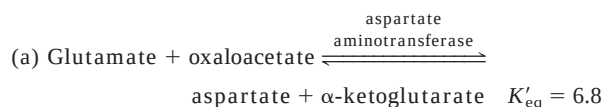
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Problems

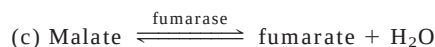
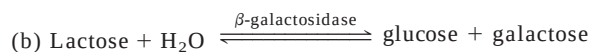
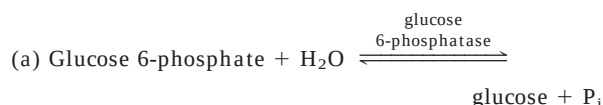
1. Entropy Changes during Egg Development Consider a system consisting of an egg in an incubator. The white and yolk of the egg contain proteins, carbohydrates, and lipids. If fertilized, the egg is transformed from a single cell to a complex organism. Discuss this irreversible process in terms of the entropy changes in the system, surroundings, and universe. Be sure that you first clearly define the system and surroundings.

2. Calculation of $\Delta G'^{\circ}$ from an Equilibrium Constant Calculate the standard free-energy changes of the following metabolically important enzyme-catalyzed reactions at 25 °C and pH 7.0, using the equilibrium constants given.

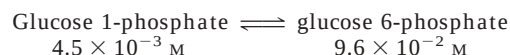


3. Calculation of the Equilibrium Constant from $\Delta G'^{\circ}$ Calculate the equilibrium constants K'_{eq} for each of the fol-

lowing reactions at pH 7.0 and 25 °C, using the $\Delta G'^{\circ}$ values in Table 13–4.



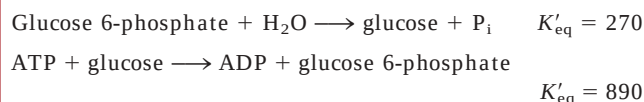
4. Experimental Determination of K'_{eq} and $\Delta G'^{\circ}$ If a 0.1 M solution of glucose 1-phosphate is incubated with a catalytic amount of phosphoglucomutase, the glucose 1-phosphate is transformed to glucose 6-phosphate. At equilibrium, the concentrations of the reaction components are



Calculate K'_{eq} and $\Delta G'^{\circ}$ for this reaction at 25 °C.

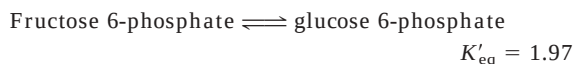
5. Experimental Determination of $\Delta G'^{\circ}$ for ATP Hydrolysis A direct measurement of the standard free-energy change associated with the hydrolysis of ATP is technically demanding because the minute amount of ATP remaining at equilibrium is difficult to measure accurately. The value of $\Delta G'^{\circ}$ can be calculated indirectly, however, from the equilib-

rium constants of two other enzymatic reactions having less favorable equilibrium constants:



Using this information, calculate the standard free energy of hydrolysis of ATP at 25 °C.

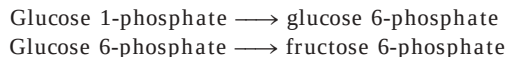
6. Difference between $\Delta G'^{\circ}$ and ΔG Consider the following interconversion, which occurs in glycolysis (Chapter 14):



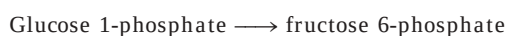
- What is $\Delta G'^{\circ}$ for the reaction (at 25 °C)?
- If the concentration of fructose 6-phosphate is adjusted to 1.5 M and that of glucose 6-phosphate is adjusted to 0.50 M, what is ΔG ?
- Why are $\Delta G'^{\circ}$ and ΔG different?

7. Dependence of ΔG on pH The free energy released by the hydrolysis of ATP under standard conditions at pH 7.0 is -30.5 kJ/mol. If ATP is hydrolyzed under standard conditions but at pH 5.0, is more or less free energy released? Explain.

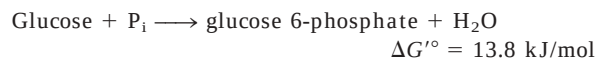
8. The $\Delta G'^{\circ}$ for Coupled Reactions Glucose 1-phosphate is converted into fructose 6-phosphate in two successive reactions:



Using the $\Delta G'^{\circ}$ values in Table 13–4, calculate the equilibrium constant, K'_{eq} , for the sum of the two reactions at 25 °C:



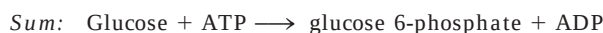
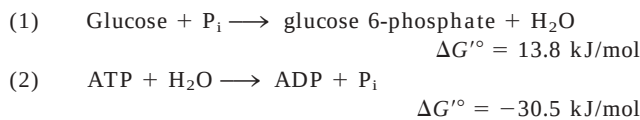
9. Strategy for Overcoming an Unfavorable Reaction: ATP-Dependent Chemical Coupling The phosphorylation of glucose to glucose 6-phosphate is the initial step in the catabolism of glucose. The direct phosphorylation of glucose by P_i is described by the equation



(a) Calculate the equilibrium constant for the above reaction. In the rat hepatocyte the physiological concentrations of glucose and P_i are maintained at approximately 4.8 mM. What is the equilibrium concentration of glucose 6-phosphate obtained by the direct phosphorylation of glucose by P_i ? Does this reaction represent a reasonable metabolic step for the catabolism of glucose? Explain.

(b) In principle, at least, one way to increase the concentration of glucose 6-phosphate is to drive the equilibrium reaction to the right by increasing the intracellular concentrations of glucose and P_i . Assuming a fixed concentration of P_i at 4.8 mM, how high would the intracellular concentration of glucose have to be to give an equilibrium concentration of glucose 6-phosphate of 250 μM (the normal physiological concentration)? Would this route be physiologically reasonable, given that the maximum solubility of glucose is less than 1 M?

(c) The phosphorylation of glucose in the cell is coupled to the hydrolysis of ATP; that is, part of the free energy of ATP hydrolysis is used to phosphorylate glucose:



Calculate K'_{eq} for the overall reaction. For the ATP-dependent phosphorylation of glucose, what concentration of glucose is needed to achieve a 250 μM intracellular concentration of glucose 6-phosphate when the concentrations of ATP and ADP are 3.38 mM and 1.32 mM, respectively? Does this coupling process provide a feasible route, at least in principle, for the phosphorylation of glucose in the cell? Explain.

(d) Although coupling ATP hydrolysis to glucose phosphorylation makes thermodynamic sense, we have not yet specified how this coupling is to take place. Given that coupling requires a common intermediate, one conceivable route is to use ATP hydrolysis to raise the intracellular concentration of P_i and thus drive the unfavorable phosphorylation of glucose by P_i . Is this a reasonable route? (Think about the solubility products of metabolic intermediates.)

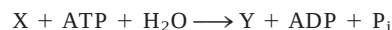
(e) The ATP-coupled phosphorylation of glucose is catalyzed in hepatocytes by the enzyme glucokinase. This enzyme binds ATP and glucose to form a glucose-ATP-enzyme complex, and the phosphoryl group is transferred directly from ATP to glucose. Explain the advantages of this route.

10. Calculations of $\Delta G'^{\circ}$ for ATP-Coupled Reactions From data in Table 13–6 calculate the $\Delta G'^{\circ}$ value for the reactions



11. Coupling ATP Cleavage to an Unfavorable Reaction To explore the consequences of coupling ATP hydrolysis under physiological conditions to a thermodynamically unfavorable biochemical reaction, consider the hypothetical transformation $\text{X} \rightarrow \text{Y}$, for which $\Delta G'^{\circ} = 20$ kJ/mol.

- What is the ratio $[\text{Y}]/[\text{X}]$ at equilibrium?
- Suppose X and Y participate in a sequence of reactions during which ATP is hydrolyzed to ADP and P_i . The overall reaction is



Calculate $[\text{Y}]/[\text{X}]$ for this reaction at equilibrium. Assume that the equilibrium concentrations of ATP, ADP, and P_i are 1 M.

(c) We know that $[\text{ATP}]$, $[\text{ADP}]$, and $[\text{P}_i]$ are *not* 1 M under physiological conditions. Calculate $[\text{Y}]/[\text{X}]$ for the ATP-coupled reaction when the values of $[\text{ATP}]$, $[\text{ADP}]$, and $[\text{P}_i]$ are those found in rat myocytes (Table 13–5).

12. Calculations of ΔG at Physiological Concentrations Calculate the physiological ΔG (not $\Delta G'^{\circ}$) for the reaction



at 25 °C, as it occurs in the cytosol of neurons, with phosphocreatine at 4.7 mM, creatine at 1.0 mM, ADP at 0.73 mM, and ATP at 2.6 mM.

13. Free Energy Required for ATP Synthesis under Physiological Conditions In the cytosol of rat hepatocytes, the mass-action ratio, Q , is

$$\frac{[\text{ATP}]}{[\text{ADP}][\text{P}_i]} = 5.33 \times 10^2 \text{ M}^{-1}$$

Calculate the free energy required to synthesize ATP in a rat hepatocyte.

14. Daily ATP Utilization by Human Adults

(a) A total of 30.5 kJ/mol of free energy is needed to synthesize ATP from ADP and P_i when the reactants and products are at 1 M concentrations (standard state). Because the actual physiological concentrations of ATP, ADP, and P_i are not 1 M, the free energy required to synthesize ATP under physiological conditions is different from $\Delta G'^{\circ}$. Calculate the free energy required to synthesize ATP in the human hepatocyte when the physiological concentrations of ATP, ADP, and P_i are 3.5, 1.50, and 5.0 mM, respectively.

(b) A 68 kg (150 lb) adult requires a caloric intake of 2,000 kcal (8,360 kJ) of food per day (24 h). The food is metabolized and the free energy is used to synthesize ATP, which then provides energy for the body's daily chemical and mechanical work. Assuming that the efficiency of converting food energy into ATP is 50%, calculate the weight of ATP used by a human adult in 24 h. What percentage of the body weight does this represent?

(c) Although adults synthesize large amounts of ATP daily, their body weight, structure, and composition do not change significantly during this period. Explain this apparent contradiction.

15. Rates of Turnover of γ and β Phosphates of ATP

If a small amount of ATP labeled with radioactive phosphorus in the terminal position, $[\gamma\text{-}^{32}\text{P}]\text{ATP}$, is added to a yeast extract, about half of the ^{32}P activity is found in P_i within a few minutes, but the concentration of ATP remains unchanged. Explain. If the same experiment is carried out using ATP labeled with ^{32}P in the central position, $[\beta\text{-}^{32}\text{P}]\text{ATP}$, the ^{32}P does not appear in P_i within such a short time. Why?

16. Cleavage of ATP to AMP and PP_i during Metabolism The synthesis of the activated form of acetate (acetyl-CoA) is carried out in an ATP-dependent process:



(a) The $\Delta G'^{\circ}$ for the hydrolysis of acetyl-CoA to acetate and CoA is -32.2 kJ/mol and that for hydrolysis of ATP to AMP and PP_i is -30.5 kJ/mol. Calculate $\Delta G'^{\circ}$ for the ATP-dependent synthesis of acetyl-CoA.

(b) Almost all cells contain the enzyme inorganic pyrophosphatase, which catalyzes the hydrolysis of PP_i to P_i . What effect does the presence of this enzyme have on the synthesis of acetyl-CoA? Explain.

17. Energy for H^+ Pumping The parietal cells of the stomach lining contain membrane "pumps" that transport hydrogen ions from the cytosol of these cells (pH 7.0) into the stomach, contributing to the acidity of gastric juice (pH 1.0). Calculate the free energy required to transport 1 mol of hydrogen ions through these pumps. (Hint: See Chapter 11.) Assume a temperature of 25°C .

18. Standard Reduction Potentials The standard reduction potential, E'° , of any redox pair is defined for the half-cell reaction:



The E'° values for the NAD^+/NADH and pyruvate/lactate conjugate redox pairs are -0.32 V and -0.19 V, respectively.

(a) Which conjugate pair has the greater tendency to lose electrons? Explain.

(b) Which is the stronger oxidizing agent? Explain.

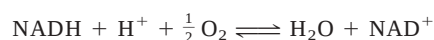
(c) Beginning with 1 M concentrations of each reactant and product at pH 7, in which direction will the following reaction proceed?



(d) What is the standard free-energy change ($\Delta G'^{\circ}$) at 25°C for the conversion of pyruvate to lactate?

(e) What is the equilibrium constant (K'_{eq}) for this reaction?

19. Energy Span of the Respiratory Chain Electron transfer in the mitochondrial respiratory chain may be represented by the net reaction equation



(a) Calculate the value of $\Delta E'^{\circ}$ for the net reaction of mitochondrial electron transfer. Use E'° values from Table 13-7.

(b) Calculate $\Delta G'^{\circ}$ for this reaction.

(c) How many ATP molecules can *theoretically* be generated by this reaction if the free energy of ATP synthesis under cellular conditions is 52 kJ/mol?

20. Dependence of Electromotive Force on Concentrations Calculate the electromotive force (in volts) registered by an electrode immersed in a solution containing the following mixtures of NAD^+ and NADH at pH 7.0 and 25°C , with reference to a half-cell of E'° 0.00 V.

(a) 1.0 mM NAD^+ and 10 mM NADH

(b) 1.0 mM NAD^+ and 1.0 mM NADH

(c) 10 mM NAD^+ and 1.0 mM NADH

21. Electron Affinity of Compounds List the following substances in order of increasing tendency to accept electrons: (a) α -ketoglutarate + CO_2 (yielding isocitrate); (b) oxaloacetate; (c) O_2 ; (d) NADP^+ .

22. Direction of Oxidation-Reduction Reactions Which of the following reactions would you expect to proceed in the direction shown, under standard conditions, assuming that the appropriate enzymes are present to catalyze them?

(a) Malate + $\text{NAD}^+ \longrightarrow$ oxaloacetate + NADH + H^+

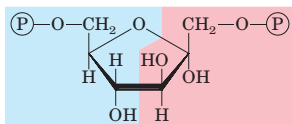
(b) Acetoacetate + NADH + $\text{H}^+ \longrightarrow$
 β -hydroxybutyrate + NAD^+

(c) Pyruvate + NADH + $\text{H}^+ \longrightarrow$ lactate + NAD^+

(d) Pyruvate + β -hydroxybutyrate $\longrightarrow$
lactate + acetoacetate

(e) Malate + pyruvate $\longrightarrow$ oxaloacetate + lactate

(f) Acetaldehyde + succinate $\longrightarrow$ ethanol + fumarate



chapter 14

GLYCOLYSIS, GLUCONEOGENESIS, AND THE PENTOSE PHOSPHATE PATHWAY

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- 14.2 Feeder Pathways for Glycolysis 534**
- 14.3 Fates of Pyruvate under Anaerobic Conditions: Fermentation 538**
- 14.4 Gluconeogenesis 543**
- 14.5 Pentose Phosphate Pathway of Glucose Oxidation 549**

The problem of alcoholic fermentation, of the origin and nature of that mysterious and apparently spontaneous change, which converted the insipid juice of the grape into stimulating wine, seems to have exerted a fascination over the minds of natural philosophers from the very earliest times.

—Arthur Harden, *Alcoholic Fermentation*, 1923

Glucose occupies a central position in the metabolism of plants, animals, and many microorganisms. It is relatively rich in potential energy, and thus a good fuel; the complete oxidation of glucose to carbon dioxide and water proceeds with a standard free-energy change of $-2,840$ kJ/mol. By storing glucose as a high molecular weight polymer such as starch or glycogen, a cell can stockpile large quantities of hexose units while maintaining a relatively low cytosolic osmolarity. When energy demands increase, glucose can be released from these intracellular storage polymers and used to produce ATP either aerobically or anaerobically.

Glucose is not only an excellent fuel, it is also a remarkably versatile precursor, capable of supplying a huge array of metabolic intermediates for biosynthetic reactions. A bacterium such as *Escherichia coli* can obtain from glucose the carbon skeletons for every amino acid, nucleotide, coenzyme, fatty acid, or other metabolic intermediate it needs for growth. A comprehensive study of the metabolic fates of glucose would encompass hundreds or thousands of transformations. In animals and vascular plants, glucose has three major fates: it may be stored (as a polysaccharide or as sucrose); oxidized to a three-carbon compound (pyruvate) via glycolysis to provide ATP and metabolic intermediates; or oxidized via the pentose phosphate (phosphogluconate) pathway to yield ribose 5-phosphate for nucleic acid synthesis and NADPH for reductive biosynthetic processes (Fig. 14–1).

Organisms that do not have access to glucose from other sources must make it. Photosynthetic organisms make glucose by first reducing atmospheric CO_2 to trioses, then converting the trioses to glucose. Non-photosynthetic cells make glucose from simpler three- and four-carbon precursors by the process of gluconeogenesis, effectively reversing glycolysis in a pathway that uses many of the glycolytic enzymes.

In this chapter we describe the individual reactions of glycolysis, gluconeogenesis, and the pentose phosphate pathway and the functional significance of each pathway. We also describe the various fates of the pyruvate produced by glycolysis; they include the fermentations that are used by many organisms in anaerobic niches to produce ATP and that are exploited industrially as sources of ethanol, lactic acid, and other

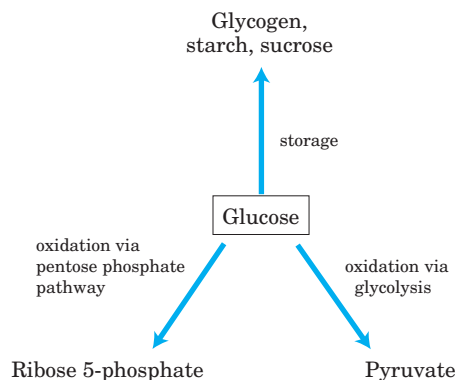


FIGURE 14-1 Major pathways of glucose utilization. Although not the only possible fates for glucose, these three pathways are the most significant in terms of the amount of glucose that flows through them in most cells.

commercially useful products. And we look at the pathways that feed various sugars from mono-, di-, and polysaccharides into the glycolytic pathway. The discussion of glucose metabolism continues in Chapter 15, where we describe the opposing anabolic and catabolic pathways that connect glucose and glycogen, and use the processes of carbohydrate synthesis and degradation as examples of the many mechanisms by which organisms regulate metabolic pathways.

14.1 Glycolysis

In **glycolysis** (from the Greek *glykys*, meaning “sweet,” and *lysis*, meaning “splitting”), a molecule of glucose is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon compound pyruvate. During the sequential reactions of glycolysis, some of the free energy released from glucose is conserved in the form of ATP and NADH. Glycolysis was the first metabolic pathway to be elucidated and is probably the best understood. From Eduard Buchner’s discovery in 1897 of fermentation in broken extracts of yeast cells until the elucidation of the whole pathway in yeast (by Otto Warburg and Hans von Euler-Chelpin)

and in muscle (by Gustav Embden and Otto Meyerhof) in the 1930s, the reactions of glycolysis in extracts of yeast and muscle were a major focus of biochemical research. The philosophical shift that accompanied these discoveries was announced by Jacques Loeb in 1906:

Through the discovery of Buchner, Biology was relieved of another fragment of mysticism. The splitting up of sugar into CO_2 and alcohol is no more the effect of a “vital principle” than the splitting up of cane sugar by invertase. The history of this problem is instructive, as it warns us against considering problems as beyond our reach because they have not yet found their solution.

The development of methods of enzyme purification, the discovery and recognition of the importance of coenzymes such as NAD, and the discovery of the pivotal metabolic role of ATP and other phosphorylated compounds all came out of studies of glycolysis. The glycolytic enzymes of many species have long since been purified and thoroughly studied.

Glycolysis is an almost universal central pathway of glucose catabolism, the pathway with the largest flux of carbon in most cells. The glycolytic breakdown of glucose is the sole source of metabolic energy in some mammalian tissues and cell types (erythrocytes, renal medulla, brain, and sperm, for example). Some plant tissues that are modified to store starch (such as potato tubers) and some aquatic plants (watercress, for example) derive most of their energy from glycolysis; many anaerobic microorganisms are entirely dependent on glycolysis.

Fermentation is a general term for the *anaerobic* degradation of glucose or other organic nutrients to obtain energy, conserved as ATP. Because living organisms first arose in an atmosphere without oxygen, anaerobic breakdown of glucose is probably the most ancient biological mechanism for obtaining energy from organic fuel molecules. In the course of evolution, the chemistry of this reaction sequence has been completely conserved; the glycolytic enzymes of vertebrates are closely

similar, in amino acid sequence and three-dimensional structure, to their homologs in yeast and spinach. Glycolysis differs among species only in the details of its regulation and in the subsequent metabolic fate of the pyruvate formed. The thermodynamic principles and the types of regulatory mechanisms that govern glycolysis are common to all pathways of cell metabolism. A study of glycolysis can therefore serve as a model for many aspects of the pathways discussed throughout this book.



Hans von Euler-Chelpin,
1873–1964



Gustav Embden,
1874–1933



Otto Meyerhof,
1884–1951

Before examining each step of the pathway in some detail, we take a look at glycolysis as a whole.

An Overview: Glycolysis Has Two Phases

The breakdown of the six-carbon glucose into two molecules of the three-carbon pyruvate occurs in ten steps, the first five of which constitute the *preparatory phase* (Fig. 14-2a). In these reactions, glucose is first phosphorylated at the hydroxyl group on C-6 (step ①). The D-glucose 6-phosphate thus formed is converted to D-fructose 6-phosphate (step ②), which is again phosphorylated, this time at C-1, to yield D-fructose 1,6-bisphosphate (step ③). For both phosphorylations, ATP is the phosphoryl group donor. As all sugar derivatives in glycolysis are the D isomers, we will usually omit the D designation except when emphasizing stereochemistry.

Fructose 1,6-bisphosphate is split to yield two three-carbon molecules, dihydroxyacetone phosphate and glyceraldehyde 3-phosphate (step ④); this is the “lysis” step that gives the pathway its name. The dihydroxyacetone phosphate is isomerized to a second molecule of glyceraldehyde 3-phosphate (step ⑤), ending the first phase of glycolysis. From a chemical perspective, the isomerization in step ② is critical for setting up the phosphorylation and C—C bond cleavage reactions in steps ③ and ④, as detailed later. Note that two molecules of ATP are invested before the cleavage of glucose into two three-carbon pieces; later there will be a good return on this investment. To summarize: in the preparatory phase of glycolysis the energy of ATP is invested, raising the free-energy content of the intermediates, and the carbon chains of all the metabolized hexoses are converted into a common product, glyceraldehyde 3-phosphate.

The energy gain comes in the *payoff phase* of glycolysis (Fig. 14-2b). Each molecule of glyceraldehyde 3-phosphate is oxidized and phosphorylated by inorganic phosphate (*not* by ATP) to form 1,3-bisphosphoglycerate (step ⑥). Energy is then released as the two molecules of 1,3-bisphosphoglycerate are converted to two molecules of pyruvate (steps ⑦ through ⑩). Much of this energy is conserved by the coupled phosphorylation of four molecules of ADP to ATP. The net yield is two molecules of ATP per molecule of glucose used, because two molecules of ATP were invested in the preparatory phase. Energy is also conserved in the payoff phase in the formation of two molecules of NADH per molecule of glucose.

In the sequential reactions of glycolysis, three types of chemical transformations are particularly noteworthy: (1) degradation of the carbon skeleton of glucose to yield pyruvate, (2) phosphorylation of ADP to ATP by high-energy phosphate compounds formed during glycolysis, and (3) transfer of a hydride ion to NAD^+ , forming NADH.

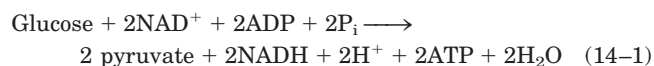
Fates of Pyruvate With the exception of some interesting variations in the bacterial realm, the pyruvate formed by glycolysis is further metabolized via one of three catabolic routes. In aerobic organisms or tissues, under aerobic conditions, glycolysis is only the first stage in the complete degradation of glucose (Fig. 14-3). Pyruvate is oxidized, with loss of its carboxyl group as CO_2 , to yield the acetyl group of acetyl-coenzyme A; the acetyl group is then oxidized completely to CO_2 by the citric acid cycle (Chapter 16). The electrons from these oxidations are passed to O_2 through a chain of carriers in the mitochondrion, to form H_2O . The energy from the electron-transfer reactions drives the synthesis of ATP in the mitochondrion (Chapter 19).

The second route for pyruvate is its reduction to lactate via **lactic acid fermentation**. When vigorously contracting skeletal muscle must function under low-oxygen conditions (**hypoxia**), NADH cannot be reoxidized to NAD^+ , but NAD^+ is required as an electron acceptor for the further oxidation of pyruvate. Under these conditions pyruvate is reduced to lactate, accepting electrons from NADH and thereby regenerating the NAD^+ necessary for glycolysis to continue. Certain tissues and cell types (retina and erythrocytes, for example) convert glucose to lactate even under aerobic conditions, and lactate is also the product of glycolysis under anaerobic conditions in some microorganisms (Fig. 14-3).

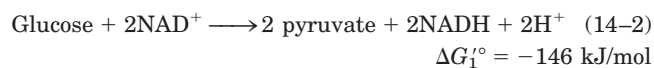
The third major route of pyruvate catabolism leads to ethanol. In some plant tissues and in certain invertebrates, protists, and microorganisms such as brewer's yeast, pyruvate is converted under hypoxic or anaerobic conditions into ethanol and CO_2 , a process called **ethanol (alcohol) fermentation** (Fig. 14-3).

The oxidation of pyruvate is an important catabolic process, but pyruvate has anabolic fates as well. It can, for example, provide the carbon skeleton for the synthesis of the amino acid alanine. We return to these anabolic reactions of pyruvate in later chapters.

ATP Formation Coupled to Glycolysis During glycolysis some of the energy of the glucose molecule is conserved in ATP, while much remains in the product, pyruvate. The overall equation for glycolysis is



For each molecule of glucose degraded to pyruvate, two molecules of ATP are generated from ADP and P_i . We can now resolve the equation of glycolysis into two processes—the conversion of glucose to pyruvate, which is exergonic:



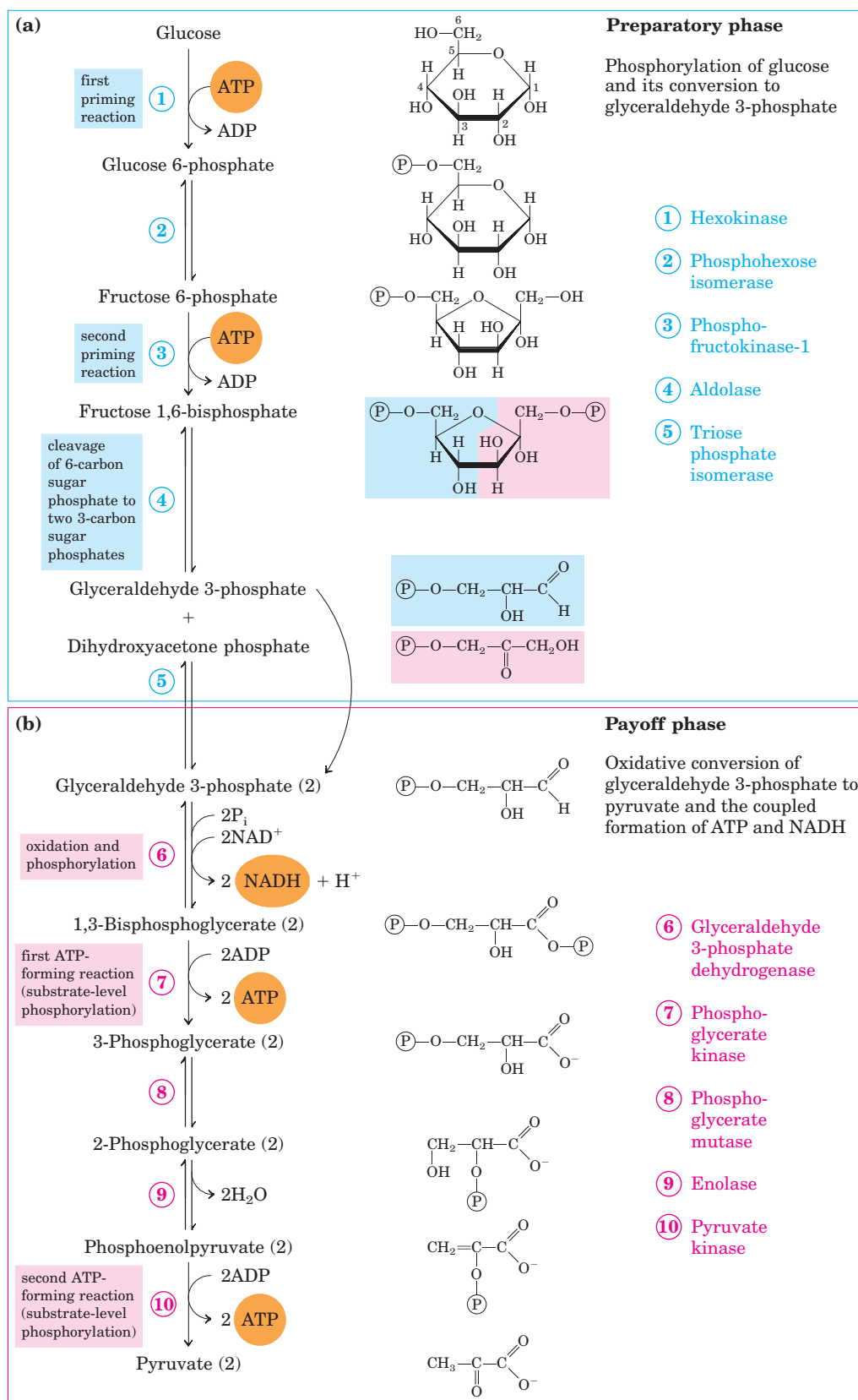
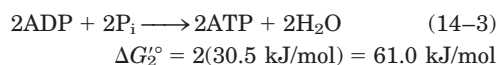


FIGURE 14-2 The two phases of glycolysis. For each molecule of glucose that passes through the preparatory phase (a), two molecules of glyceraldehyde 3-phosphate are formed; both pass through the payoff phase (b). Pyruvate is the end product of the second phase of glycolysis. For each glucose molecule, two ATP are consumed in the preparatory phase and four ATP are produced in the payoff phase, giving a

net yield of two ATP per molecule of glucose converted to pyruvate. The numbered reaction steps are catalyzed by the enzymes listed on the right, and also correspond to the numbered headings in the text discussion. Keep in mind that each phosphoryl group, represented here as P , has two negative charges ($-\text{PO}_3^{2-}$).

and the formation of ATP from ADP and P_i , which is endergonic:



The sum of Equations 14-2 and 14-3 gives the overall standard free-energy change of glycolysis, ΔG_s° :

$$\Delta G_s^{\circ} = \Delta G_1^{\circ} + \Delta G_2^{\circ} = -146 \text{ kJ/mol} + 61.0 \text{ kJ/mol}$$

$$= -85 \text{ kJ/mol}$$

Under standard conditions and in the cell, glycolysis is an essentially irreversible process, driven to completion by a large net decrease in free energy. At the actual intracellular concentrations of ATP, ADP, and P_i (see Box 13-1) and of glucose and pyruvate, the energy released in glycolysis (with pyruvate as the end product) is recovered as ATP with an efficiency of more than 60%.

Energy Remaining in Pyruvate Glycolysis releases only a small fraction of the total available energy of the glucose molecule; the two molecules of pyruvate formed by glycolysis still contain most of the chemical potential energy of glucose, energy that can be extracted by oxidative reactions in the citric acid cycle (Chapter 16) and oxidative phosphorylation (Chapter 19).

Importance of Phosphorylated Intermediates Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated (Fig. 14-2). The phosphoryl groups appear to have three functions.

1. Because the plasma membrane generally lacks transporters for phosphorylated sugars, the phosphorylated glycolytic intermediates cannot leave the cell. After the initial phosphorylation, no further energy is necessary to retain phosphorylated intermediates in the cell, despite the large difference in their intracellular and extracellular concentrations.
2. Phosphoryl groups are essential components in the enzymatic conservation of metabolic energy. Energy released in the breakage of phosphoanhydride bonds (such as those in ATP) is partially conserved in the formation of phosphate esters such as glucose 6-phosphate. High-energy phosphate compounds formed in glycolysis (1,3-bisphoglycerate and phosphoenolpyruvate) donate phosphoryl groups to ADP to form ATP.
3. Binding energy resulting from the binding of phosphate groups to the active sites of enzymes lowers the activation energy and increases the specificity of the enzymatic reactions (Chapter 6). The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with Mg^{2+} , and the substrate binding sites of many glycolytic enzymes are specific for these Mg^{2+} complexes. Most glycolytic enzymes require Mg^{2+} for activity.

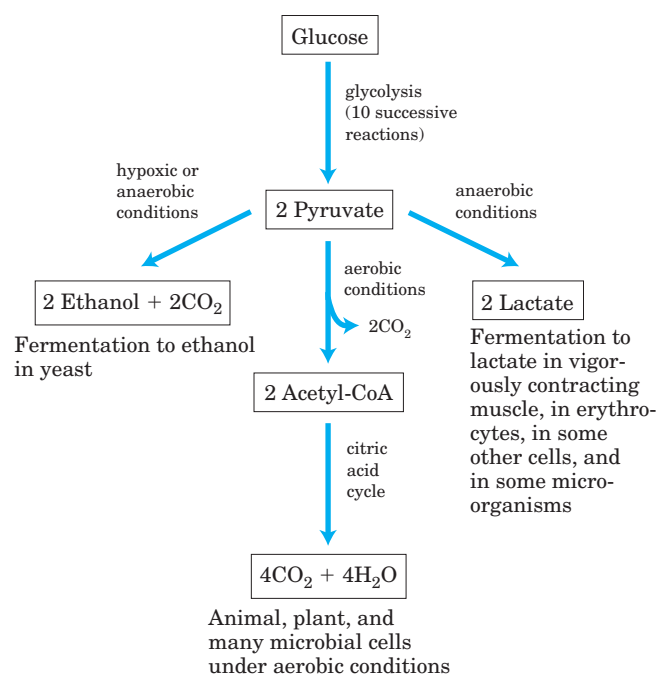


FIGURE 14-3 Three possible catabolic fates of the pyruvate formed in glycolysis. Pyruvate also serves as a precursor in many anabolic reactions, not shown here.

The Preparatory Phase of Glycolysis Requires ATP

In the preparatory phase of glycolysis, two molecules of ATP are invested and the hexose chain is cleaved into two triose phosphates. The realization that *phosphorylated* hexoses were intermediates in glycolysis came slowly and serendipitously. In 1906, Arthur Harden and William Young tested their hypothesis that inhibitors of proteolytic enzymes would stabilize the glucose-fermenting enzymes in yeast extract. They added blood serum (known to contain inhibitors of proteolytic enzymes) to yeast extracts and observed the predicted stimulation of glucose metabolism. However, in a control experiment intended to show that boiling the serum destroyed the stimulatory activity, they discovered that boiled serum was just as effective at stimulating glycolysis. Careful examination and testing of the contents of



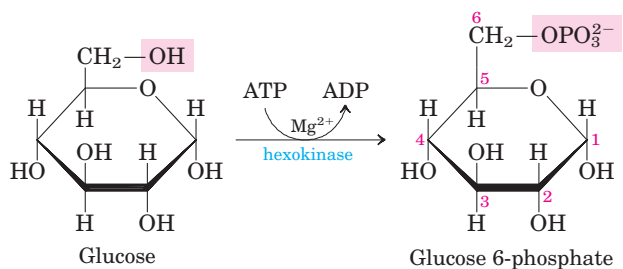
Arthur Harden,
1865–1940



William Young,
1878–1942

the boiled serum revealed that inorganic phosphate was responsible for the stimulation. Harden and Young soon discovered that glucose added to their yeast extract was converted to a hexose biphosphate (the “Harden-Young ester,” eventually identified as fructose 1,6-bisphosphate). This was the beginning of a long series of investigations on the role of organic esters of phosphate in biochemistry, which has led to our current understanding of the central role of phosphoryl group transfer in biology.

① Phosphorylation of Glucose In the first step of glycolysis, glucose is activated for subsequent reactions by its phosphorylation at C-6 to yield **glucose 6-phosphate**, with ATP as the phosphoryl donor:



$$\Delta G'^{\circ} = -16.7 \text{ kJ/mol}$$

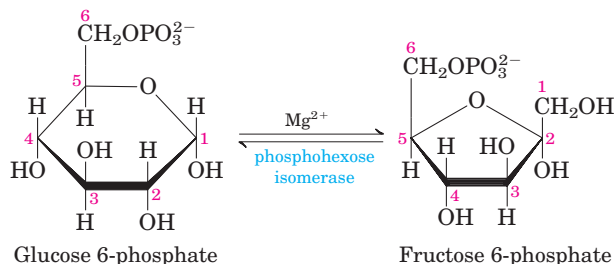
This reaction, which is irreversible under intracellular conditions, is catalyzed by **hexokinase**. Recall that kinases are enzymes that catalyze the transfer of the terminal phosphoryl group from ATP to an acceptor nucleophile (see Fig. 13–10). Kinases are a subclass of transferases (see Table 6–3). The acceptor in the case of hexokinase is a hexose, normally D-glucose, although hexokinase also catalyzes the phosphorylation of other common hexoses, such as D-fructose and D-mannose.

Hexokinase, like many other kinases, requires Mg^{2+} for its activity, because the true substrate of the enzyme is not ATP^{4-} but the MgATP^{2-} complex (see Fig. 13–2). Mg^{2+} shields the negative charges of the phosphoryl groups in ATP, making the terminal phosphorus atom an easier target for nucleophilic attack by an —OH of glucose. Hexokinase undergoes a profound change in shape, an induced fit, when it binds glucose; two domains of the protein move about 8 Å closer to each other when ATP binds (see Fig. 6–22). This movement brings bound ATP closer to a molecule of glucose also bound to the enzyme and blocks the access of water (from the solvent), which might otherwise enter the active site and attack (hydrolyze) the phosphoanhydride bonds of ATP. Like the other nine enzymes of glycolysis, hexokinase is a soluble, cytosolic protein.

Hexokinase is present in all cells of all organisms. Hepatocytes also contain a form of hexokinase called hexokinase IV or glucokinase, which differs from other forms of hexokinase in kinetic and regulatory properties (see Box 15–2). Two enzymes that catalyze the

same reaction but are encoded in different genes are called **isozymes**.

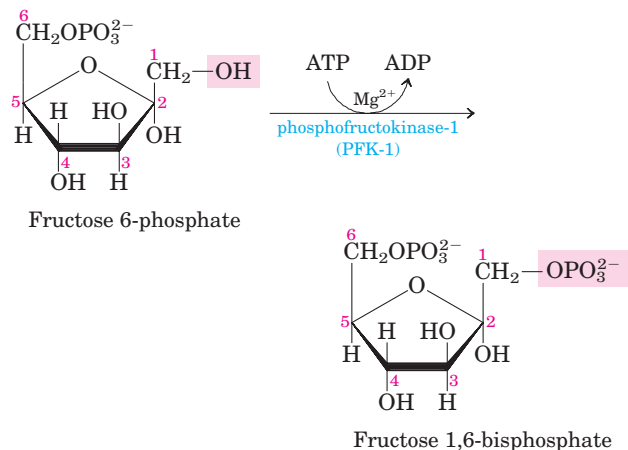
② Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate The enzyme **phosphohexose isomerase (phosphoglucose isomerase)** catalyzes the reversible isomerization of glucose 6-phosphate, an aldose, to **fructose 6-phosphate**, a ketose:



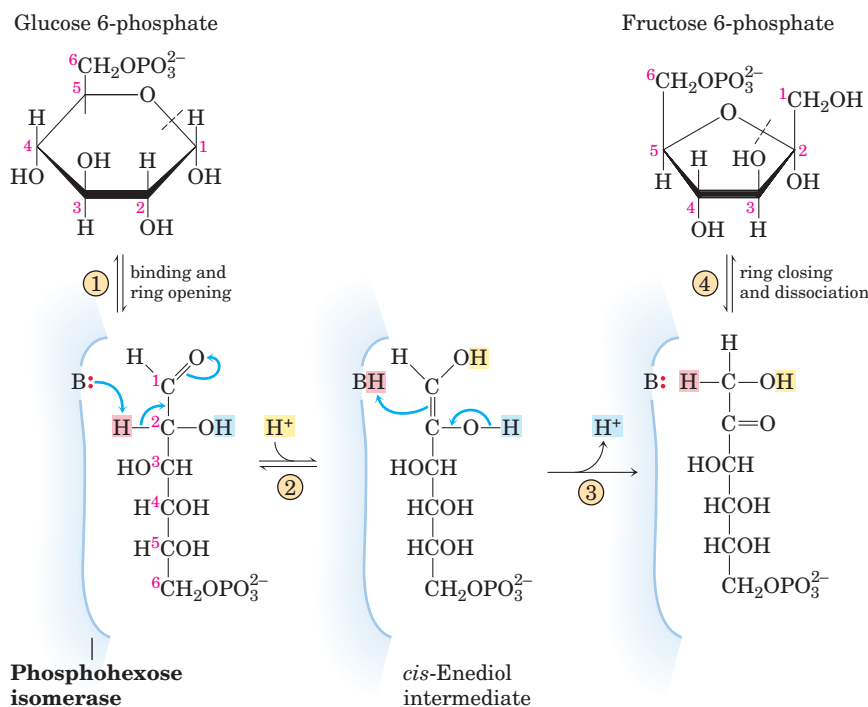
$$\Delta G'^{\circ} = 1.7 \text{ kJ/mol}$$

The mechanism for this reaction is shown in Figure 14–4. The reaction proceeds readily in either direction, as might be expected from the relatively small change in standard free energy. This isomerization has a critical role in the overall chemistry of the glycolytic pathway, as the rearrangement of the carbonyl and hydroxyl groups at C-1 and C-2 is a necessary prelude to the next two steps. The phosphorylation that occurs in the next reaction (step ③) requires that the group at C-1 first be converted from a carbonyl to an alcohol, and in the subsequent reaction (step ④) cleavage of the bond between C-3 and C-4 requires a carbonyl group at C-2 (p. 485).

③ Phosphorylation of Fructose 6-Phosphate to Fructose 1,6-Bisphosphate In the second of the two priming reactions of glycolysis, **phosphofructokinase-1 (PFK-1)** catalyzes the transfer of a phosphoryl group from ATP to fructose 6-phosphate to yield **fructose 1,6-bisphosphate**:



$$\Delta G'^{\circ} = -14.2 \text{ kJ/mol}$$

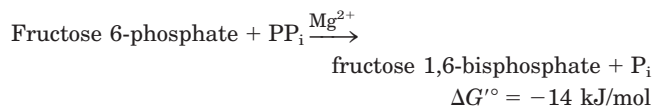


MECHANISM FIGURE 14-4 The phosphohexose isomerase reaction. The ring opening and closing reactions (steps ① and ④) are catalyzed by an active-site His residue, by mechanisms omitted here for simplicity. The movement of the proton between C-2 and C-1 (steps ② and ③) is base-catalyzed by an active-site Glu residue (shown as B:). The proton (pink) initially at C-2 is made more easily abstractable by electron withdrawal by the adjacent carbonyl and the nearby hydroxyl group. After its transfer from C-2 to the active-site Glu residue, the proton is freely exchanged with the surrounding solution; that is, the proton abstracted from C-2 in step ② is not necessarily the same one that is added to C-1 in step ③. (The additional exchange of protons (yellow and blue) between the hydroxyl groups and solvent is shown for completeness. The hydroxyl groups are weak acids and can exchange protons with the surrounding water whether the isomerization reaction is underway or not.)

Phosphohexose Isomerase Mechanism

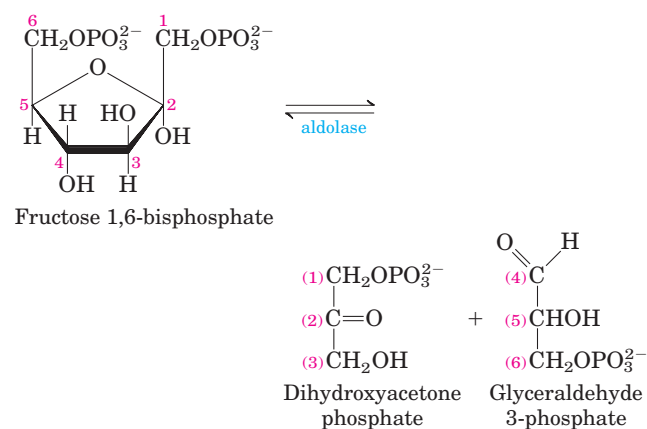
This enzyme is called PFK-1 to distinguish it from a second enzyme (PFK-2) that catalyzes the formation of fructose 2,6-bisphosphate from fructose 6-phosphate in a separate pathway. The PFK-1 reaction is essentially irreversible under cellular conditions, and it is the first “committed” step in the glycolytic pathway; glucose 6-phosphate and fructose 6-phosphate have other possible fates, but fructose 1,6-bisphosphate is targeted for glycolysis.

Some bacteria and protists and perhaps all plants have a phosphofructokinase that uses pyrophosphate (PP_i), not ATP, as the phosphoryl group donor in the synthesis of fructose 1,6-bisphosphate:



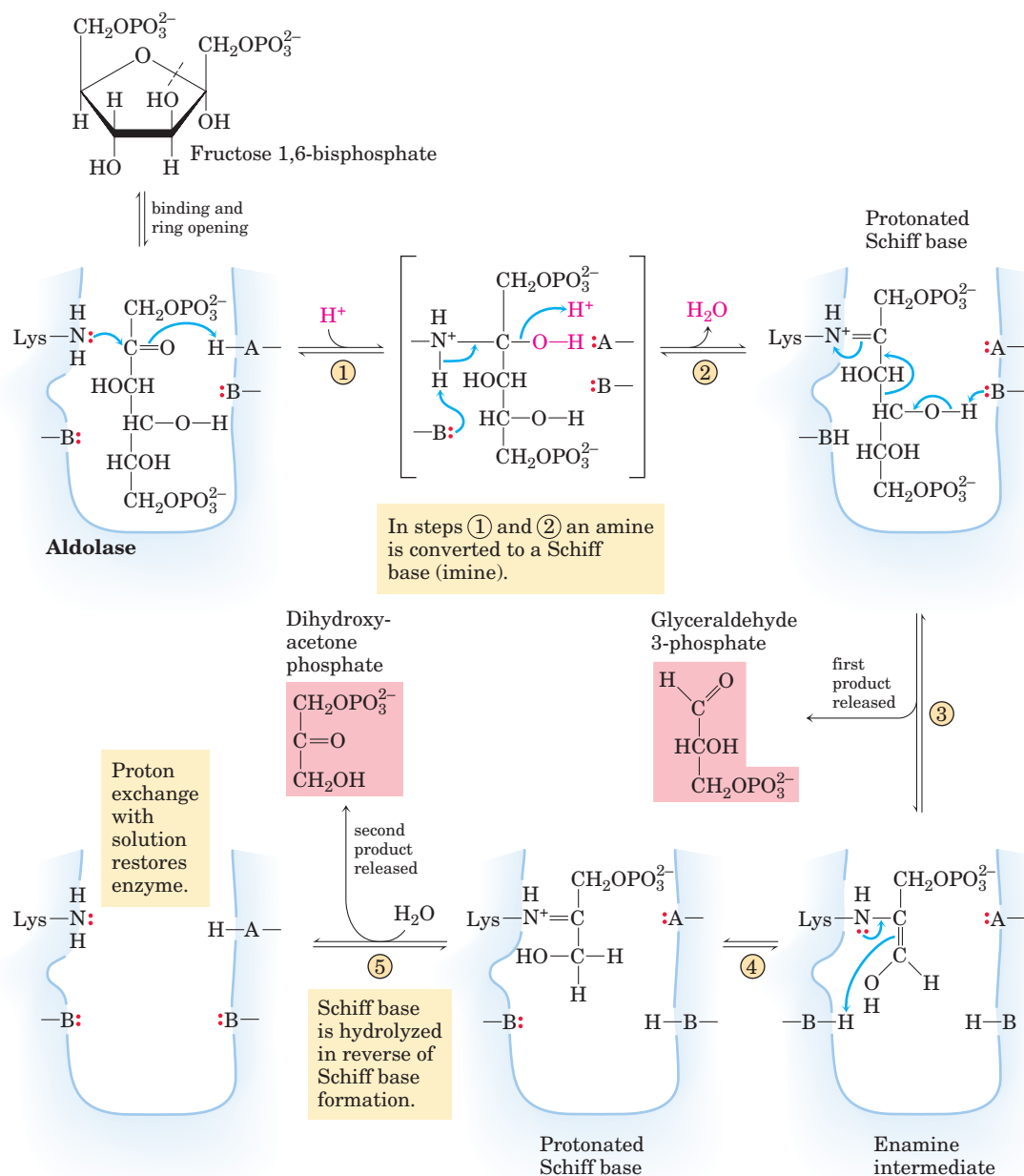
Phosphofructokinase-1 is a regulatory enzyme (Chapter 6), one of the most complex known. It is the major point of regulation in glycolysis. The activity of PFK-1 is increased whenever the cell's ATP supply is depleted or when the ATP breakdown products, ADP and AMP (particularly the latter), are in excess. The enzyme is inhibited whenever the cell has ample ATP and is well supplied by other fuels such as fatty acids. In some organisms, fructose 2,6-bisphosphate (not to be confused with the PFK-1 reaction product, fructose 1,6-bisphosphate) is a potent allosteric activator of PFK-1. The regulation of this step in glycolysis is discussed in greater detail in Chapter 15.

④ **Cleavage of Fructose 1,6-Bisphosphate** The enzyme **fructose 1,6-bisphosphate aldolase**, often called simply **aldolase**, catalyzes a reversible aldol condensation (p. 485). Fructose 1,6-bisphosphate is cleaved to yield two different triose phosphates, **glyceraldehyde 3-phosphate**, an aldose, and **dihydroxyacetone phosphate**, a ketose:



$$\Delta G'^{\circ} = 23.8 \text{ kJ/mol}$$

There are two classes of aldolases. Class I aldolases, found in animals and plants, use the mechanism shown in Figure 14-5. Class II enzymes, in fungi and bacteria, do not form the Schiff base intermediate. Instead, a zinc ion at the active site is coordinated with the carbonyl oxygen at C-2; the Zn^{2+} polarizes the carbonyl group



MECHANISM FIGURE 14-5 The class I aldolase reaction. The reaction shown here is the reverse of an aldol condensation. Note that cleavage between C-3 and C-4 depends on the presence of the carbonyl group at C-2. ① and ② The carbonyl reacts with an active-site Lys residue to form an imine, which stabilizes the carbanion generated by the bond cleavage—an imine delocalizes electrons even better than

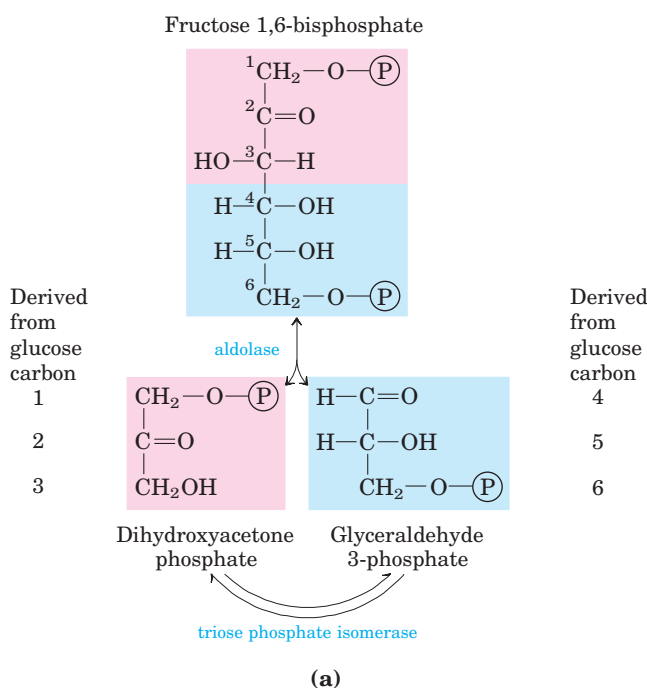
does a carbonyl. ③ Bond cleavage releases glyceraldehyde 3-phosphate as the first product. ④ The resulting enamine covalently linked to the enzyme is isomerized to a protonated Schiff base, and ⑤ hydrolysis of the Schiff base generates dihydroxyacetone phosphate as the second product. A and B represent amino acid residues that serve as general acid (A) or base (B).

and stabilizes the enolate intermediate created in the C—C bond cleavage step.

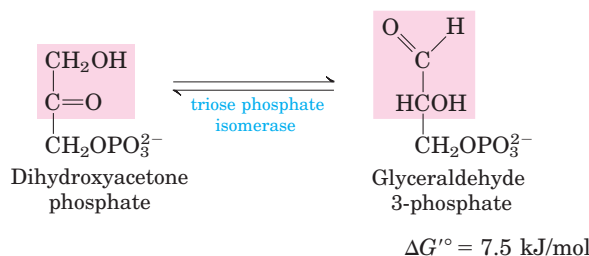
Although the aldolase reaction has a strongly positive standard free-energy change in the direction of fructose 1,6-bisphosphate cleavage, at the lower concentrations of reactants present in cells, the actual free-energy change is small and the aldolase reaction is readily reversible. We shall see later that aldolase acts in the re-

verse direction during the process of gluconeogenesis (see Fig. 14-16).

⑤ Interconversion of the Triose Phosphates Only one of the two triose phosphates formed by aldolase, glyceraldehyde 3-phosphate, can be directly degraded in the subsequent steps of glycolysis. The other product, dihydroxyacetone phosphate, is rapidly and reversibly



converted to glyceraldehyde 3-phosphate by the fifth enzyme of the sequence, **triose phosphate isomerase**:



The reaction mechanism is similar to the reaction promoted by phosphohexose isomerase in step ② of glycolysis (Fig. 14-4). After the triose phosphate isomerase reaction, C-1, C-2, and C-3 of the starting glucose are chemically indistinguishable from C-6, C-5, and C-4, respectively (Fig. 14-6), setting up the efficient metabolism of the entire six-carbon glucose molecule.

This reaction completes the preparatory phase of glycolysis. The hexose molecule has been phosphorylated at C-1 and C-6 and then cleaved to form two molecules of glyceraldehyde 3-phosphate.

The Payoff Phase of Glycolysis Yields ATP and NADH

The payoff phase of glycolysis (Fig. 14-2b) includes the energy-conserving phosphorylation steps in which some of the free energy of the glucose molecule is conserved in the form of ATP. Remember that one molecule of glucose yields two molecules of glyceraldehyde 3-phosphate; both halves of the glucose molecule follow the

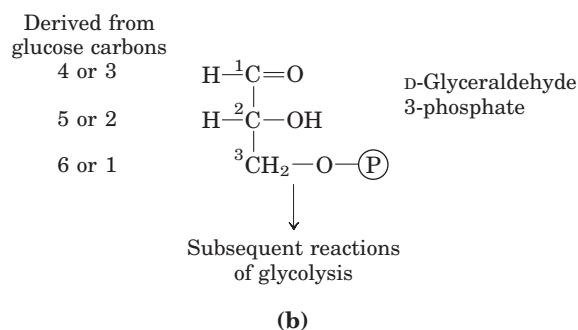
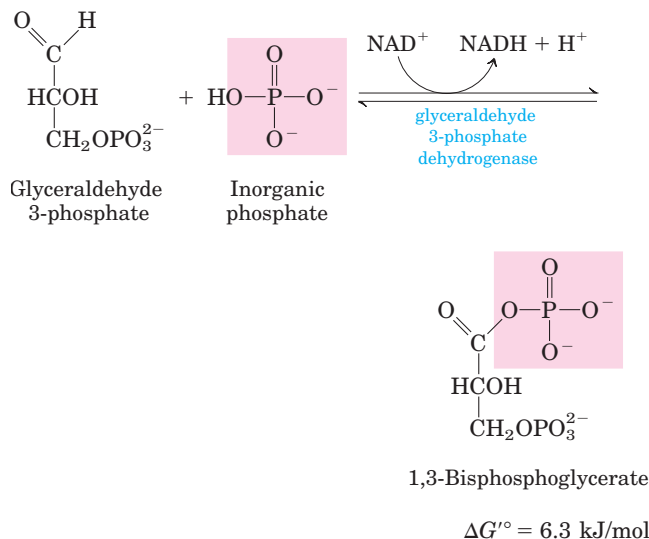


FIGURE 14-6 Fate of the glucose carbons in the formation of glyceraldehyde 3-phosphate. (a) The origin of the carbons in the two three-carbon products of the aldolase and triose phosphate isomerase reactions. The end product of the two reactions is glyceraldehyde 3-phosphate (two molecules). **(b)** Each carbon of glyceraldehyde 3-phosphate is derived from either of two specific carbons of glucose. Note that the numbering of the carbon atoms of glyceraldehyde 3-phosphate differs from that of the glucose from which it is derived. In glyceraldehyde 3-phosphate, the most complex functional group (the carbonyl) is specified as C-1. This numbering change is important for interpreting experiments with glucose in which a single carbon is labeled with a radioisotope. (See Problems 3 and 5 at the end of this chapter.)

same pathway in the second phase of glycolysis. The conversion of two molecules of glyceraldehyde 3-phosphate to two molecules of pyruvate is accompanied by the formation of four molecules of ATP from ADP. However, the net yield of ATP per molecule of glucose degraded is only two, because two ATP were invested in the preparatory phase of glycolysis to phosphorylate the two ends of the hexose molecule.

⑥ Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

The first step in the payoff phase is the oxidation of glyceraldehyde 3-phosphate to **1,3-bisphosphoglycerate**, catalyzed by **glyceraldehyde 3-phosphate dehydrogenase**:



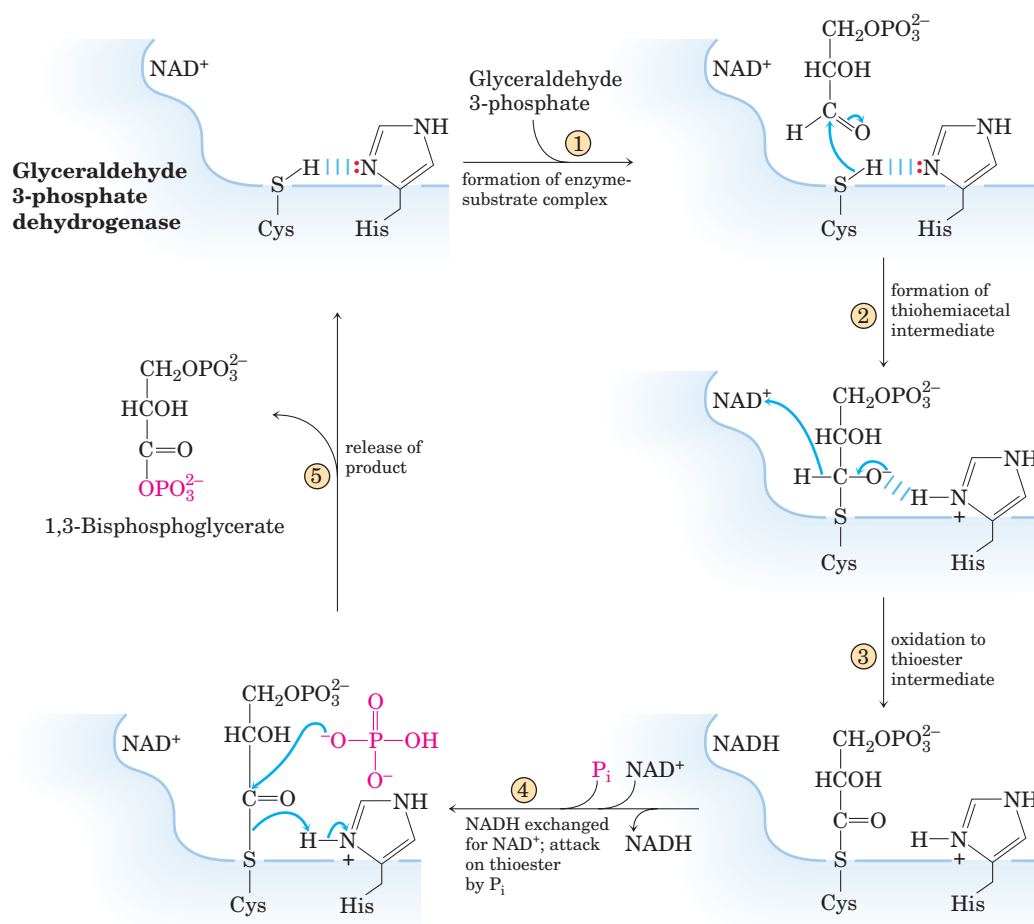
This is the first of the two energy-conserving reactions of glycolysis that eventually lead to the formation of ATP. The aldehyde group of glyceraldehyde 3-phosphate is oxidized, not to a free carboxyl group but to a carboxylic acid anhydride with phosphoric acid. This type of anhydride, called an **acyl phosphate**, has a very high standard free energy of hydrolysis ($\Delta G'^{\circ} = -49.3$ kJ/mol; see Fig. 13–4, Table 13–6). Much of the free energy of oxidation of the aldehyde group of glyceraldehyde 3-phosphate is conserved by formation of the acyl phosphate group at C-1 of 1,3-bisphosphoglycerate.

The acceptor of hydrogen in the glyceraldehyde 3-phosphate dehydrogenase reaction is NAD^+ (see Fig. 13–15), bound to a Rossmann fold as shown in Figure 13–16. The reduction of NAD^+ proceeds by the enzymatic transfer of a hydride ion (:H^-) from the aldehyde group of glyceraldehyde 3-phosphate to the nicoti-

namide ring of NAD^+ , yielding the reduced coenzyme NADH. The other hydrogen atom of the substrate molecule is released to the solution as H^+ .

Glyceraldehyde 3-phosphate is covalently bound to the dehydrogenase during the reaction (Fig. 14–7). The aldehyde group of glyceraldehyde 3-phosphate reacts with the —SH group of an essential Cys residue in the active site, in a reaction analogous to the formation of a hemiacetal (see Fig. 7–5), in this case producing a *thio*-hemiacetal. Reaction of the essential Cys residue with a heavy metal such as Hg^{2+} irreversibly inhibits the enzyme.

Because cells maintain only limited amounts of NAD^+ , glycolysis would soon come to a halt if the NADH formed in this step of glycolysis were not continuously reoxidized. The reactions in which NAD^+ is regenerated anaerobically are described in detail in Section 14.3, in our discussion of the alternative fates of pyruvate.

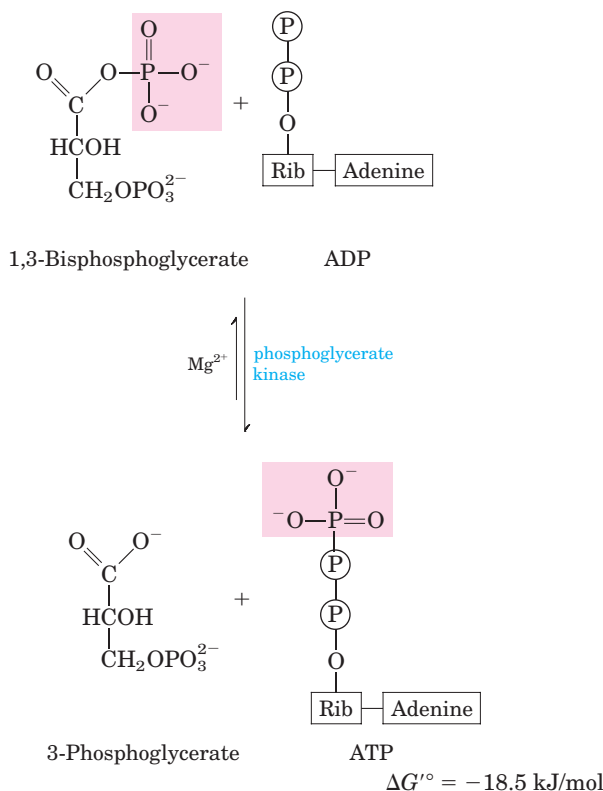


MECHANISM FIGURE 14–7 The glyceraldehyde 3-phosphate dehydrogenase reaction. After ① formation of the enzyme-substrate complex, ② a covalent thiohemiacetal linkage forms between the substrate and the —SH group of a Cys residue—facilitated by acid-base catalysis with a neighboring base catalyst, probably a His residue. ③ This enzyme-substrate intermediate is oxidized by NAD^+ bound to the active site, forming a covalent acyl-enzyme intermediate, a

thioester. ④ The newly formed NADH leaves the active site and is replaced by another NAD^+ molecule. The bond between the acyl group and the thiol group of the enzyme has a very high standard free energy of hydrolysis. ⑤ This bond undergoes phosphorolysis (attack by P_i), releasing the acyl phosphate product, 1,3-bisphosphoglycerate. Formation of this product conserves much of the free energy liberated during oxidation of the aldehyde group of glyceraldehyde 3-phosphate.

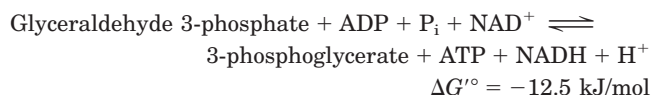
⑦ Phosphoryl Transfer from 1,3-Bisphosphoglycerate to ADP

The enzyme **phosphoglycerate kinase** transfers the high-energy phosphoryl group from the carboxyl group of 1,3-bisphosphoglycerate to ADP, forming ATP and **3-phosphoglycerate**:



Notice that phosphoglycerate kinase is named for the reverse reaction. Like all enzymes, it catalyzes the reaction in both directions. This enzyme acts in the direction suggested by its name during gluconeogenesis (see Fig. 14–16) and during photosynthetic CO_2 assimilation (see Fig. 20–4).

Steps ⑥ and ⑦ of glycolysis together constitute an energy-coupling process in which 1,3-bisphosphoglycerate is the common intermediate; it is formed in the first reaction (which would be endergonic in isolation), and its acyl phosphate group is transferred to ADP in the second reaction (which is strongly exergonic). The sum of these two reactions is



Thus the overall reaction is exergonic.

Recall from Chapter 13 that the actual free-energy change, ΔG , is determined by the standard free-energy change, $\Delta G'^{\circ}$, and the mass-action ratio, Q , which is the ratio [products]/[reactants] (see Eqn 13–3). For step ⑥

$$\Delta G = \Delta G'^{\circ} + RT \ln Q$$

$$= \Delta G'^{\circ} + RT \ln \frac{[\text{1,3-bisphosphoglycerate}][\text{NADH}]}{[\text{glyceraldehyde 3-phosphate}][\text{P}_i][\text{NAD}^+]}$$

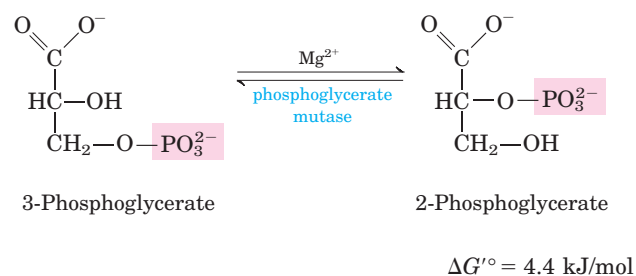
Notice that $[\text{H}^+]$ is not included in Q . In biochemical calculations, $[\text{H}^+]$ is assumed to be a constant (10^{-7} M), and this constant is included in the definition of $\Delta G'^{\circ}$ (p. 491).

When the mass-action ratio is less than 1.0, its natural logarithm has a negative sign. Step ⑦, by consuming the product of step ⑥ (1,3-bisphosphoglycerate), keeps [1,3-bisphosphoglycerate] relatively low in the steady state and thereby keeps Q for the overall energy-coupling process small. When Q is small, the contribution of $\ln Q$ can make ΔG strongly negative. This is simply another way of showing how the two reactions, steps ⑥ and ⑦, are coupled through a common intermediate.

The outcome of these coupled reactions, both reversible under cellular conditions, is that the energy released on oxidation of an aldehyde to a carboxylate group is conserved by the coupled formation of ATP from ADP and P_i . The formation of ATP by phosphoryl group transfer from a substrate such as 1,3-bisphosphoglycerate is referred to as a **substrate-level phosphorylation**, to distinguish this mechanism from **respiration-linked phosphorylation**. Substrate-level phosphorylations involve soluble enzymes and chemical intermediates (1,3-bisphosphoglycerate in this case). Respiration-linked phosphorylations, on the other hand, involve membrane-bound enzymes and transmembrane gradients of protons (Chapter 19).

⑧ Conversion of 3-Phosphoglycerate to 2-Phosphoglycerate

The enzyme **phosphoglycerate mutase** catalyzes a reversible shift of the phosphoryl group between C-2 and C-3 of glycerate; Mg^{2+} is essential for this reaction:



The reaction occurs in two steps (Fig. 14–8). A phosphoryl group initially attached to a His residue of the mutase is transferred to the hydroxyl group at C-2 of 3-phosphoglycerate, forming 2,3-bisphosphoglycerate (2,3-BPG). The phosphoryl group at C-3 of 2,3-BPG is then transferred to the same His residue, producing 2-phosphoglycerate and regenerating the phosphorylated enzyme. Phosphoglycerate mutase is initially phosphorylated by phosphoryl transfer from 2,3-BPG, which is required in small quantities to initiate the catalytic cycle and is continuously regenerated by that cycle. Although in most cells 2,3-BPG is present in only trace amounts, it is a major component ($\sim 5 \text{ mM}$) of erythrocytes, where it regulates the affinity of hemoglobin for

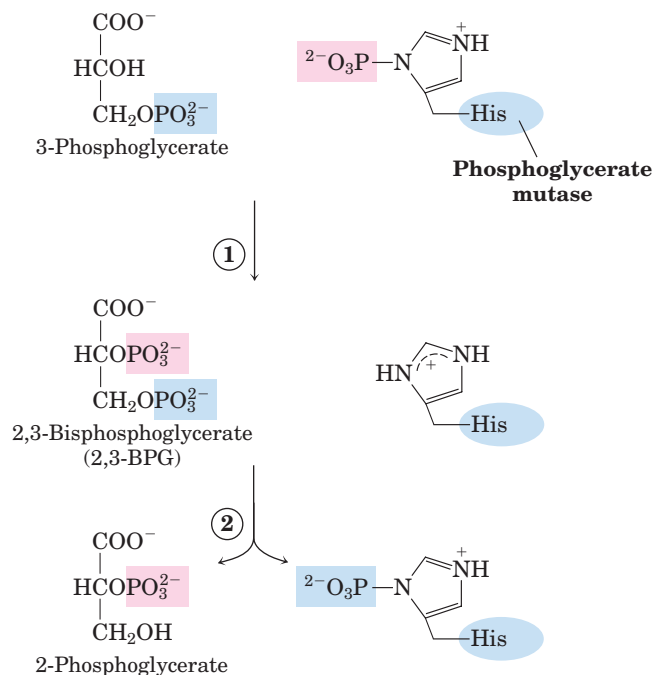
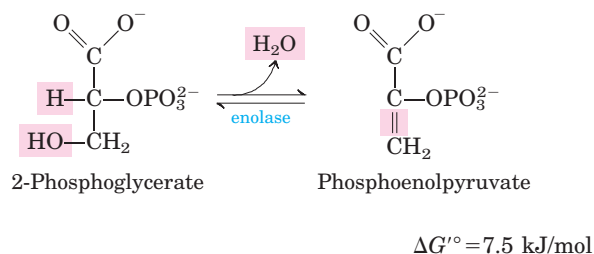


FIGURE 14-8 The phosphoglycerate mutase reaction. The enzyme is initially phosphorylated on a His residue. ① The phosphoenzyme transfers its phosphoryl group to 3-phosphoglycerate, forming 2,3-BPG. ② The phosphoryl group at C-3 of 2,3-BPG is transferred to the same His residue on the enzyme, producing 2-phosphoglycerate and regenerating the phosphoenzyme.

oxygen (see Fig. 5-17; note that in the context of hemoglobin regulation, 2,3-bisphosphoglycerate is usually abbreviated as simply BPG).

⑨ Dehydration of 2-Phosphoglycerate to Phosphoenolpyruvate

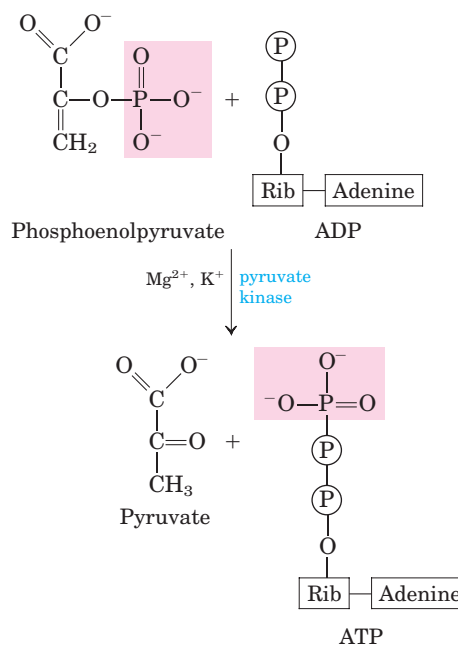
In the second glycolytic reaction that generates a compound with high phosphoryl group transfer potential, **enolase** promotes reversible removal of a molecule of water from 2-phosphoglycerate to yield **phosphoenolpyruvate (PEP)**:



The mechanism of the enolase reaction is presented in Figure 6-23. Despite the relatively small standard free-energy change of this reaction, there is a very large difference in the standard free energy of hydrolysis of

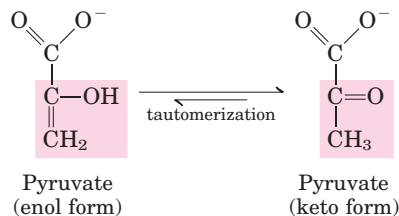
the phosphoryl groups of the reactant and product: -17.6 kJ/mol for 2-phosphoglycerate (a low-energy phosphate ester) and -61.9 kJ/mol for phosphoenolpyruvate (a compound with a very high standard free energy of hydrolysis) (see Fig. 13-3, Table 13-6). Although 2-phosphoglycerate and phosphoenolpyruvate contain nearly the same *total* amount of energy, the loss of the water molecule from 2-phosphoglycerate causes a redistribution of energy within the molecule, greatly increasing the standard free energy of hydrolysis of the phosphoryl group.

⑩ Transfer of the Phosphoryl Group from Phosphoenolpyruvate to ADP The last step in glycolysis is the transfer of the phosphoryl group from phosphoenolpyruvate to ADP, catalyzed by **pyruvate kinase**, which requires K^+ and either Mg^{2+} or Mn^{2+} :



$$\Delta G'^{\circ} = -31.4 \text{ kJ/mol}$$

In this substrate-level phosphorylation, the product **pyruvate** first appears in its enol form, then tautomerizes rapidly and nonenzymatically to its keto form, which predominates at pH 7:

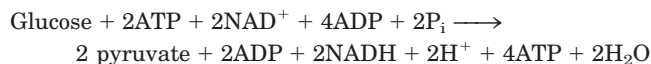


The overall reaction has a large, negative standard free-energy change, due in large part to the spontaneous conversion of the enol form of pyruvate to the keto form (see Fig. 13-3). The $\Delta G'^{\circ}$ of phosphoenolpyruvate

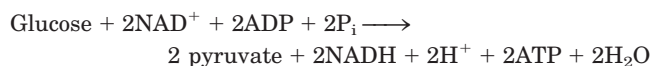
hydrolysis is -61.9 kJ/mol; about half of this energy is conserved in the formation of the phosphoanhydride bond of ATP ($\Delta G'^{\circ} = -30.5$ kJ/mol), and the rest (-31.4 kJ/mol) constitutes a large driving force pushing the reaction toward ATP synthesis. The pyruvate kinase reaction is essentially irreversible under intracellular conditions and is an important site of regulation, as described in Chapter 15.

The Overall Balance Sheet Shows a Net Gain of ATP

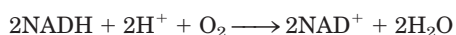
We can now construct a balance sheet for glycolysis to account for (1) the fate of the carbon skeleton of glucose, (2) the input of P_i and ADP and the output of ATP, and (3) the pathway of electrons in the oxidation-reduction reactions. The left-hand side of the following equation shows all the inputs of ATP, NAD^+ , ADP, and P_i (consult Fig. 14-2), and the right-hand side shows all the outputs (keep in mind that each molecule of glucose yields two molecules of pyruvate):



Canceling out common terms on both sides of the equation gives the overall equation for glycolysis under aerobic conditions:



The two molecules of NADH formed by glycolysis in the cytosol are, under aerobic conditions, reoxidized to NAD^+ by transfer of their electrons to the electron-transfer chain, which in eukaryotic cells is located in the mitochondria. The electron-transfer chain passes these electrons to their ultimate destination, O_2 :



Electron transfer from NADH to O_2 in mitochondria provides the energy for synthesis of ATP by respiration-linked phosphorylation (Chapter 19).

In the overall glycolytic process, one molecule of glucose is converted to two molecules of pyruvate (the pathway of carbon). Two molecules of ADP and two of P_i are converted to two molecules of ATP (the pathway of phosphoryl groups). Four electrons, as two hydride ions, are transferred from two molecules of glyceraldehyde 3-phosphate to two of NAD^+ (the pathway of electrons).

Glycolysis Is under Tight Regulation

During his studies on the fermentation of glucose by yeast, Louis Pasteur discovered that both the rate and the total amount of glucose consumption were many times greater under anaerobic than aerobic conditions. Later studies of muscle showed the same large differ-

ence in the rates of anaerobic and aerobic glycolysis. The biochemical basis of this “Pasteur effect” is now clear. The ATP yield from glycolysis under anaerobic conditions (2 ATP per molecule of glucose) is much smaller than that from the complete oxidation of glucose to CO_2 under aerobic conditions (30 or 32 ATP per glucose; see Table 19-5). About 15 times as much glucose must therefore be consumed anaerobically as aerobically to yield the same amount of ATP.

The flux of glucose through the glycolytic pathway is regulated to maintain nearly constant ATP levels (as well as adequate supplies of glycolytic intermediates that serve biosynthetic roles). The required adjustment in the rate of glycolysis is achieved by a complex interplay among ATP consumption, NADH regeneration, and allosteric regulation of several glycolytic enzymes—including hexokinase, PFK-1, and pyruvate kinase—and by second-to-second fluctuations in the concentration of key metabolites that reflect the cellular balance between ATP production and consumption. On a slightly longer time scale, glycolysis is regulated by the hormones glucagon, epinephrine, and insulin, and by changes in the expression of the genes for several glycolytic enzymes. We return to a more detailed discussion of the regulation of glycolysis in Chapter 15.

Cancerous Tissue Has Deranged Glucose Catabolism



Glucose uptake and glycolysis proceed about ten times faster in most solid tumors than in non-cancerous tissues. Tumor cells commonly experience hypoxia (limited oxygen supply), because they initially lack an extensive capillary network to supply the tumor with oxygen. As a result, cancer cells more than 100 to 200 μm from the nearest capillaries depend on anaerobic glycolysis for much of their ATP production. They take up more glucose than normal cells, converting it to pyruvate and then to lactate as they recycle NADH. The high glycolytic rate may also result in part from smaller numbers of mitochondria in tumor cells; less ATP made by respiration-linked phosphorylation in mitochondria means more ATP is needed from glycolysis. In addition, some tumor cells overproduce several glycolytic enzymes, including an isozyme of hexokinase that associates with the cytosolic face of the mitochondrial inner membrane and is insensitive to feedback inhibition by glucose 6-phosphate. This enzyme may monopolize the ATP produced in mitochondria, using it to convert glucose to glucose 6-phosphate and committing the cell to continued glycolysis. The hypoxia-inducible transcription factor (HIF-1) is a protein that acts at the level of mRNA synthesis to stimulate the synthesis of at least eight of the glycolytic enzymes. This gives the tumor cell the capacity to survive anaerobic conditions until the supply of blood vessels has caught up with tumor growth.

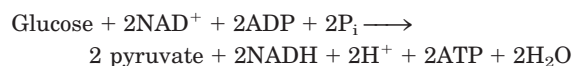
The German biochemist Otto Warburg was the first to show, as early as 1928, that tumors have a higher rate of glucose metabolism than other tissues. With his associates, Warburg purified and crystallized seven of the enzymes of glycolysis. In these studies he developed and used an experimental tool that revolutionized biochemical studies of oxidative metabolism: the Warburg manometer, which measured directly the consumption of oxygen by monitoring changes in gas volume, and therefore allowed quantitative measurement of any enzyme with oxidase activity.

Warburg, considered by many the preeminent biochemist of the first half of the twentieth century, made seminal contributions to many other areas of biochemistry, including respiration, photosynthesis, and the enzymology of intermediary metabolism. Trained in carbohydrate chemistry in the laboratory of the great Emil Fischer (who won the Nobel Prize in Chemistry in 1902), Warburg himself won the Nobel Prize in Physiology or Medicine in 1931. A number of Warburg's students and colleagues also were awarded Nobel Prizes:

Otto Meyerhof in 1922, Hans Krebs and Fritz Lipmann in 1953, and Hugo Theorell in 1955. Meyerhof's laboratory provided training for Lipmann, and for several other Nobel Prize winners: Severo Ochoa (1959), Andre Lwoff (1965), and George Wald (1967). ■

SUMMARY 14.1 Glycolysis

- Glycolysis is a near-universal pathway by which a glucose molecule is oxidized to two molecules of pyruvate, with energy conserved as ATP and NADH.
- All ten glycolytic enzymes are in the cytosol, and all ten intermediates are phosphorylated compounds of three or six carbons.
- In the preparatory phase of glycolysis, ATP is invested to convert glucose to fructose 1,6-bisphosphate. The bond between C-3 and C-4 is then broken to yield two molecules of triose phosphate.
- In the payoff phase, each of the two molecules of glyceraldehyde 3-phosphate derived from glucose undergoes oxidation at C-1; the energy of this oxidation reaction is conserved in the formation of one NADH and two ATP per triose phosphate oxidized. The net equation for the overall process is



- Glycolysis is tightly regulated in coordination with other energy-yielding pathways to assure a steady supply of ATP. Hexokinase, PFK-1, and pyruvate kinase are all subject to allosteric regulation that controls the flow of carbon through the pathway and maintains constant levels of metabolic intermediates.

14.2 Feeder Pathways for Glycolysis

Many carbohydrates besides glucose meet their catabolic fate in glycolysis, after being transformed into one of the glycolytic intermediates. The most significant are the storage polysaccharides glycogen and starch; the disaccharides maltose, lactose, trehalose, and sucrose; and the monosaccharides fructose, mannose, and galactose (Fig. 14–9).

Glycogen and Starch Are Degraded by Phosphorolysis

Glycogen in animal tissues and in microorganisms (and starch in plants) can be mobilized for use within the same cell by a phosphorolytic reaction catalyzed by **glycogen phosphorylase** (**starch phosphorylase** in plants). These enzymes catalyze an attack by P_i on the ($\alpha 1 \rightarrow 4$) glycosidic linkage that joins the last two glucose residues at a nonreducing end, generating glucose 1-phosphate and a polymer one glucose unit shorter (Fig. 14–10). *Phosphorolysis* preserves some of the energy of the glycosidic bond in the phosphate ester glucose 1-phosphate. Glycogen phosphorylase (or starch phosphorylase) acts repetitively until it approaches an ($\alpha 1 \rightarrow 6$) branch point (see Fig. 7–15), where its action stops. A **debranching enzyme** removes the branches. The mechanisms and control of glycogen degradation are described in detail in Chapter 15.

Glucose 1-phosphate produced by glycogen phosphorylase is converted to glucose 6-phosphate by **phosphoglucomutase**, which catalyzes the reversible reaction



The glucose 6-phosphate thus formed can enter glycolysis or another pathway such as the pentose phosphate pathway, described in Section 14.5. Phosphoglucomutase employs essentially the same mechanism as phosphoglycerate mutase (p. 531). The general name **mutase** is given to enzymes that catalyze the transfer of a functional group from one position to another in the same molecule. Mutases are a subclass of **isomerases**, enzymes that interconvert stereoisomers or structural or positional isomers (see Table 6–3).

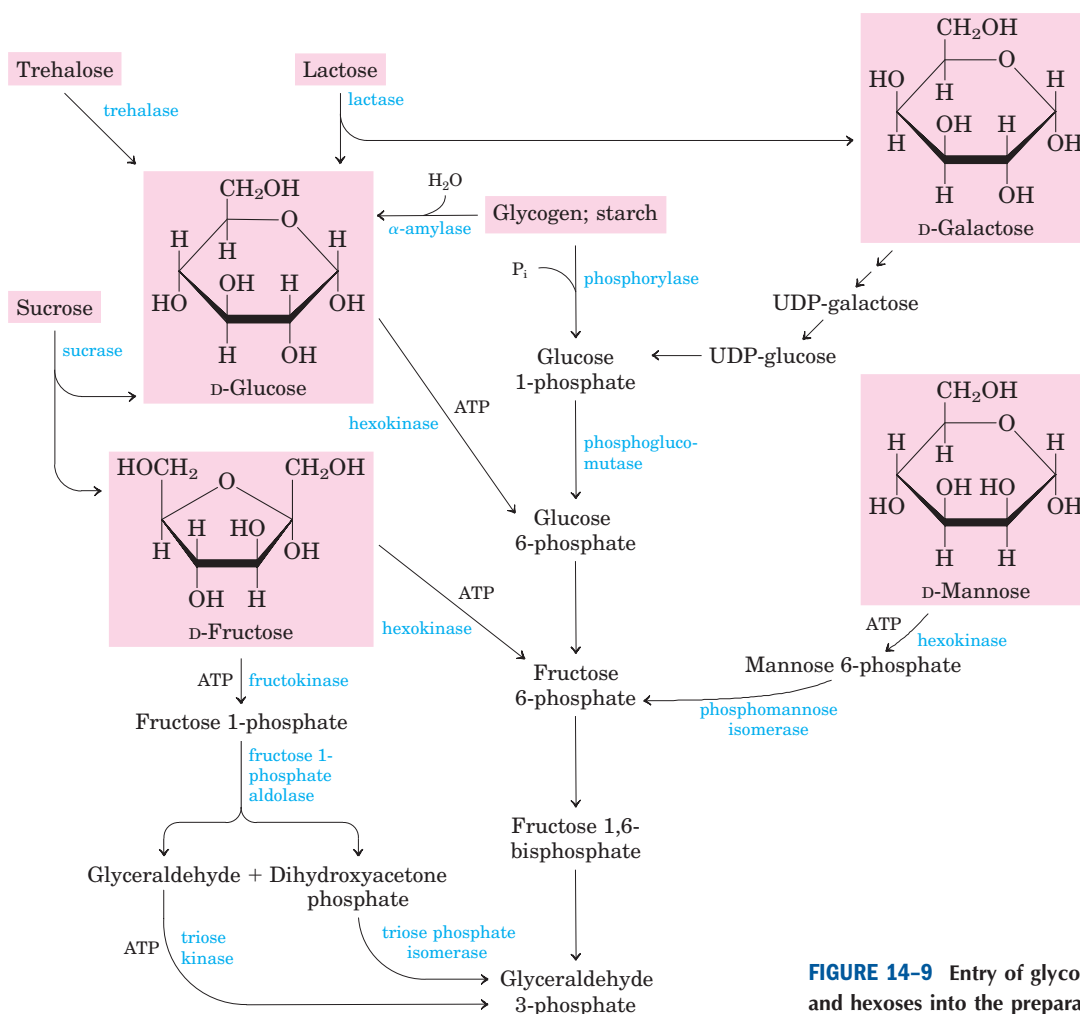
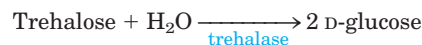
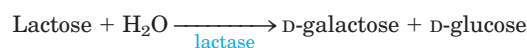


FIGURE 14-9 Entry of glycogen, starch, disaccharides, and hexoses into the preparatory stage of glycolysis.

Dietary Polysaccharides and Disaccharides Undergo Hydrolysis to Monosaccharides

For most humans, starch is the major source of carbohydrates in the diet. Digestion begins in the mouth, where salivary **α -amylase** (Fig. 14-9) hydrolyzes the internal glycosidic linkages of starch, producing short polysaccharide fragments or oligosaccharides. (Note that in this *hydrolysis* reaction, water, not P_i , is the attacking species.) In the stomach, salivary α -amylase is inactivated by the low pH, but a second form of α -amylase, secreted by the pancreas into the small intestine, continues the breakdown process. Pancreatic α -amylase yields mainly maltose and maltotriose (the di- and trisaccharides of $\alpha(1\rightarrow4)$ glucose) and oligosaccharides called limit dextrins, fragments of amylopectin containing $\alpha(1\rightarrow6)$ branch points. Maltose and dextrins are degraded by enzymes of the intestinal brush border (the fingerlike microvilli of intestinal epithelial cells, which greatly increase the area of the intestinal surface). Dietary glycogen has essentially the same structure as starch, and its digestion proceeds by the same pathway.

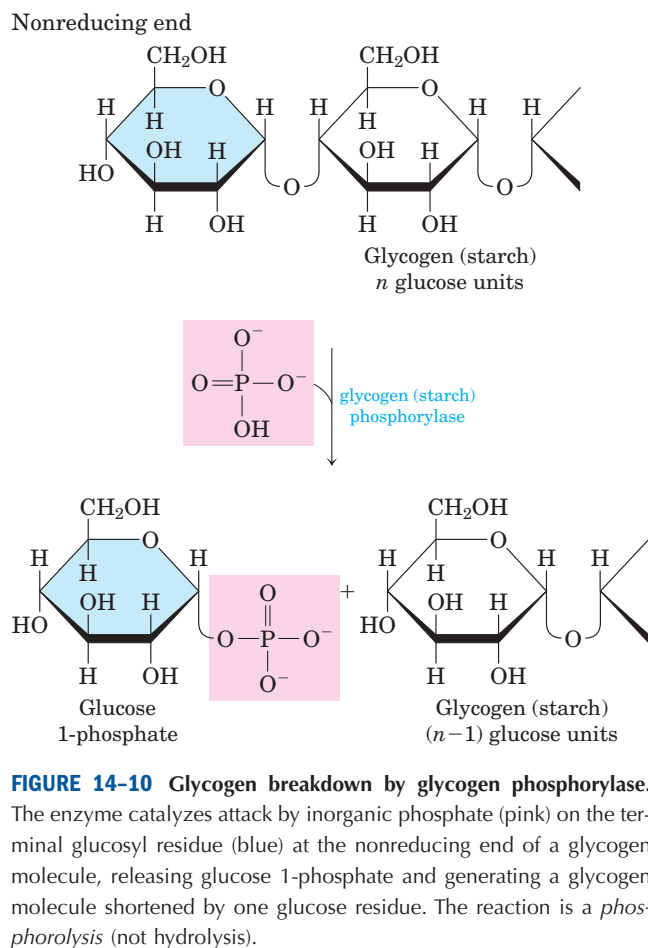
Disaccharides must be hydrolyzed to monosaccharides before entering cells. Intestinal disaccharides and dextrins are hydrolyzed by enzymes attached to the outer surface of the intestinal epithelial cells:



The monosaccharides so formed are actively transported into the epithelial cells (see Fig. 11-44), then passed into the blood to be carried to various tissues, where they are phosphorylated and funneled into the glycolytic sequence.



Lactose intolerance, common among adults of most human populations except those originating



in Northern Europe and some parts of Africa, is due to the disappearance after childhood of most or all of the lactase activity of the intestinal cells. Lactose cannot be completely digested and absorbed in the small intestine and passes into the large intestine, where bacteria convert it to toxic products that cause abdominal cramps and diarrhea. The problem is further complicated because undigested lactose and its metabolites increase the osmolarity of the intestinal contents, favoring the retention of water in the intestine. In most parts of the world where lactose intolerance is prevalent, milk is not used as a food by adults, although milk products predigested with lactase are commercially available in some countries. In certain human disorders, several or all of the intestinal disaccharidases are missing. In these cases, the digestive disturbances triggered by dietary disaccharides can sometimes be minimized by a controlled diet. ■

Other Monosaccharides Enter the Glycolytic Pathway at Several Points

In most organisms, hexoses other than glucose can undergo glycolysis after conversion to a phosphorylated

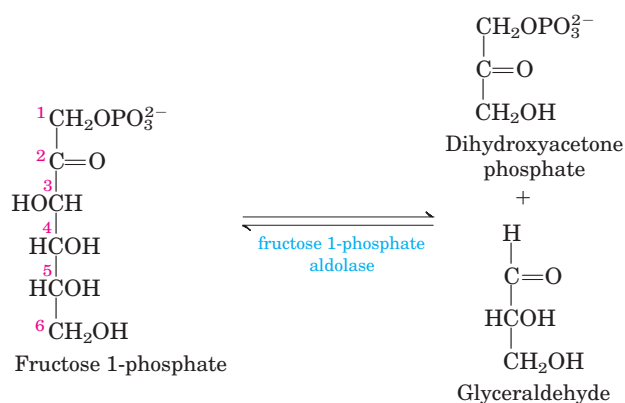
derivative. D-Fructose, present in free form in many fruits and formed by hydrolysis of sucrose in the small intestine of vertebrates, is phosphorylated by hexokinase:



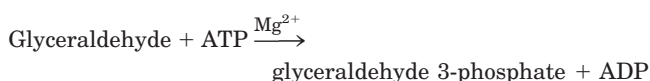
This is a major pathway of fructose entry into glycolysis in the muscles and kidney. In the liver, however, fructose enters by a different pathway. The liver enzyme **fructokinase** catalyzes the phosphorylation of fructose at C-1 rather than C-6:



The fructose 1-phosphate is then cleaved to glyceraldehyde and dihydroxyacetone phosphate by **fructose 1-phosphate aldolase**:



Dihydroxyacetone phosphate is converted to glyceraldehyde 3-phosphate by the glycolytic enzyme triose phosphate isomerase. Glyceraldehyde is phosphorylated by ATP and **triose kinase** to glyceraldehyde 3-phosphate:



Thus both products of fructose 1-phosphate hydrolysis enter the glycolytic pathway as glyceraldehyde 3-phosphate.

D-Galactose, a product of hydrolysis of the disaccharide lactose (milk sugar), passes in the blood from the intestine to the liver, where it is first phosphorylated at C-1, at the expense of ATP, by the enzyme **galactokinase**:



The galactose 1-phosphate is then converted to its epimer at C-4, glucose 1-phosphate, by a set of reactions in which **uridine diphosphate (UDP)** functions as a coenzyme-like carrier of hexose groups (Fig. 14-11). The epimerization involves first the oxidation of the C-4 —OH group to a ketone, then reduction of the ketone to an —OH, with inversion of the configuration at C-4. NAD is the cofactor for both the oxidation and the reduction.

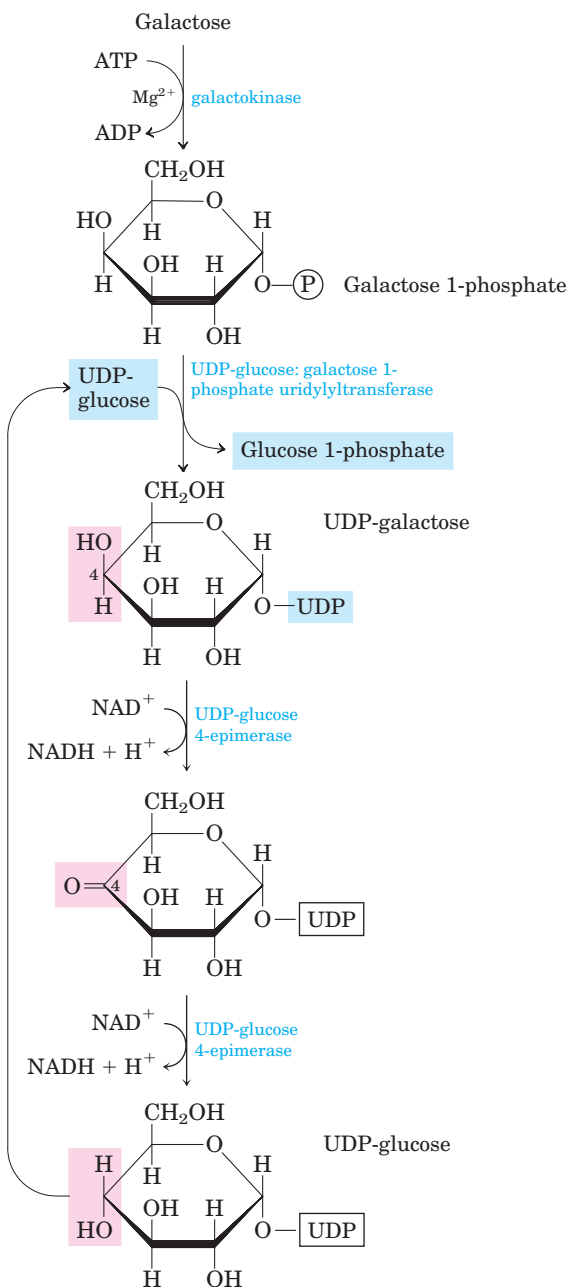
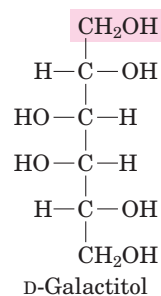


FIGURE 14-11 Conversion of galactose to glucose 1-phosphate. The conversion proceeds through a sugar-nucleotide derivative, UDP-galactose, which is formed when galactose 1-phosphate displaces glucose 1-phosphate from UDP-glucose. UDP-galactose is then converted by UDP-glucose 4-epimerase to UDP-glucose, in a reaction that involves oxidation of C-4 (pink) by NAD^+ , then reduction of C-4 by $NADH$; the result is inversion of the configuration at C-4. The UDP-glucose is recycled through another round of the same reaction. The net effect of this cycle is the conversion of galactose 1-phosphate to glucose 1-phosphate; there is no net production or consumption of UDP-galactose or UDP-glucose.

Defects in any of the three enzymes in this pathway cause **galactosemia** in humans. In galactokinase-deficiency galactosemia, high galactose concentrations are found in blood and urine. Infants develop cataracts, caused by deposition of the galactose metabolite galactitol in the lens.



The symptoms in this disorder are relatively mild, and strict limitation of galactose in the diet greatly diminishes their severity.

Transferase-deficiency galactosemia is more serious; it is characterized by poor growth in children, speech abnormality, mental deficiency, and liver damage that may be fatal, even when galactose is withheld from the diet. Epimerase-deficiency galactosemia leads to similar symptoms, but is less severe when dietary galactose is carefully controlled. ■

D-Mannose, released in the digestion of various polysaccharides and glycoproteins of foods, can be phosphorylated at C-6 by hexokinase:



Mannose 6-phosphate is isomerized by **phosphomannose isomerase** to yield fructose 6-phosphate, an intermediate of glycolysis.

SUMMARY 14.2 Feeder Pathways for Glycolysis

- Glycogen and starch, polymeric storage forms of glucose, enter glycolysis in a two-step process. Phosphorolytic cleavage of a glucose residue from an end of the polymer, forming glucose 1-phosphate, is catalyzed by glycogen phosphorylase or starch phosphorylase. Phosphoglucomutase then converts the glucose 1-phosphate to glucose 6-phosphate, which can enter glycolysis.
- Ingested polysaccharides and disaccharides are converted to monosaccharides by intestinal hydrolytic enzymes, and the monosaccharides then enter intestinal cells and are transported to the liver or other tissues.
- A variety of D-hexoses, including fructose, galactose, and mannose, can be funneled into glycolysis. Each is phosphorylated and converted to either glucose 6-phosphate or fructose 6-phosphate.
- Conversion of galactose 1-phosphate to glucose 1-phosphate involves two nucleotide derivatives: UDP-galactose and UDP-glucose. Genetic defects in any of the three enzymes that catalyze conversion of galactose to glucose 1-phosphate result in galactosemias of varying severity.

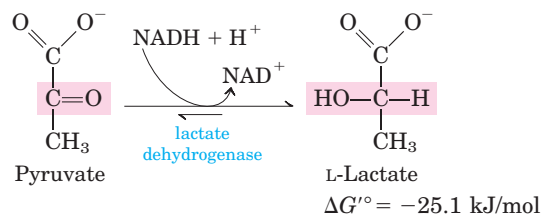
14.3 Fates of Pyruvate under Anaerobic Conditions: Fermentation

Pyruvate occupies an important junction in carbohydrate catabolism (Fig. 14–3). Under aerobic conditions pyruvate is oxidized to acetate, which enters the citric acid cycle and is oxidized to CO_2 and H_2O , and NADH formed by the dehydrogenation of glyceraldehyde 3-phosphate is ultimately reoxidized to NAD^+ by passage of its electrons to O_2 in mitochondrial respiration. However, under hypoxic conditions, as in very active skeletal muscle, in submerged plant tissues, or in lactic acid bacteria, NADH generated by glycolysis cannot be reoxidized by O_2 . Failure to regenerate NAD^+ would leave the cell with no electron acceptor for the oxidation of glyceraldehyde 3-phosphate, and the energy-yielding reactions of glycolysis would stop. NAD^+ must therefore be regenerated in some other way.

The earliest cells lived in an atmosphere almost devoid of oxygen and had to develop strategies for deriving energy from fuel molecules under anaerobic conditions. Most modern organisms have retained the ability to constantly regenerate NAD^+ during anaerobic glycolysis by transferring electrons from NADH to form a reduced end product such as lactate or ethanol.

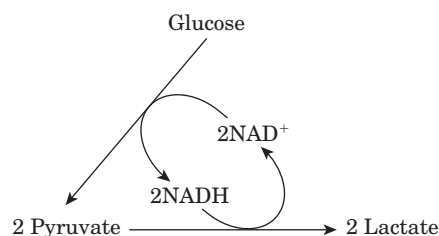
Pyruvate Is the Terminal Electron Acceptor in Lactic Acid Fermentation

When animal tissues cannot be supplied with sufficient oxygen to support aerobic oxidation of the pyruvate and NADH produced in glycolysis, NAD^+ is regenerated from NADH by the reduction of pyruvate to **lactate**. As mentioned earlier, some tissues and cell types (such as erythrocytes, which have no mitochondria and thus cannot oxidize pyruvate to CO_2) produce lactate from glucose even under aerobic conditions. The reduction of pyruvate is catalyzed by **lactate dehydrogenase**, which forms the L isomer of lactate at pH 7:



The overall equilibrium of this reaction strongly favors lactate formation, as shown by the large negative standard free-energy change.

In glycolysis, dehydrogenation of the two molecules of glyceraldehyde 3-phosphate derived from each molecule of glucose converts two molecules of NAD^+ to two of NADH . Because the reduction of two molecules of pyruvate to two of lactate regenerates two molecules of NAD^+ , there is no net change in NAD^+ or NADH :

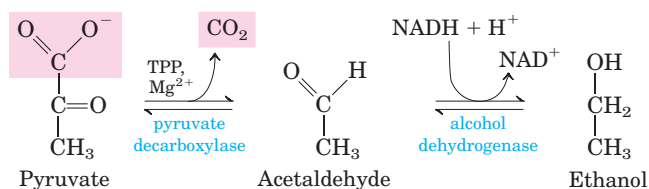


The lactate formed by active skeletal muscles (or by erythrocytes) can be recycled; it is carried in the blood to the liver, where it is converted to glucose during the recovery from strenuous muscular activity. When lactate is produced in large quantities during vigorous muscle contraction (during a sprint, for example), the acidification that results from ionization of lactic acid in muscle and blood limits the period of vigorous activity. The best-conditioned athletes can sprint at top speed for no more than a minute (Box 14–1).

Although conversion of glucose to lactate includes two oxidation-reduction steps, there is no net change in the oxidation state of carbon; in glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and lactic acid ($\text{C}_3\text{H}_6\text{O}_3$), the H:C ratio is the same. Nevertheless, some of the energy of the glucose molecule has been extracted by its conversion to lactate—enough to give a net yield of two molecules of ATP for every glucose molecule consumed. **Fermentation** is the general term for such processes, which extract energy (as ATP) but do not consume oxygen or change the concentrations of NAD^+ or NADH . Fermentations are carried out by a wide range of organisms, many of which occupy anaerobic niches, and they yield a variety of end products, some of which find commercial uses.

Ethanol Is the Reduced Product in Ethanol Fermentation

Yeast and other microorganisms ferment glucose to ethanol and CO_2 , rather than to lactate. Glucose is converted to pyruvate by glycolysis, and the pyruvate is converted to ethanol and CO_2 in a two-step process:



In the first step, pyruvate is decarboxylated in an irreversible reaction catalyzed by **pyruvate decarboxylase**. This reaction is a simple decarboxylation and does not involve the net oxidation of pyruvate. Pyruvate decarboxylase requires Mg^{2+} and has a tightly bound coenzyme, thiamine pyrophosphate, discussed below. In the second step, acetaldehyde is reduced to ethanol through the action of **alcohol dehydrogenase**, with

BOX 14-1 THE WORLD OF BIOCHEMISTRY**Athletes, Alligators, and Coelacanths: Glycolysis at Limiting Concentrations of Oxygen**

Most vertebrates are essentially aerobic organisms; they convert glucose to pyruvate by glycolysis, then use molecular oxygen to oxidize the pyruvate completely to CO_2 and H_2O . Anaerobic catabolism of glucose to lactate occurs during short bursts of extreme muscular activity, for example in a 100 m sprint, during which oxygen cannot be carried to the muscles fast enough to oxidize pyruvate. Instead, the muscles use their stored glucose (glycogen) as fuel to generate ATP by fermentation, with lactate as the end product. In a sprint, lactate in the blood builds up to high concentrations. It is slowly converted back to glucose by gluconeogenesis in the liver in the subsequent rest or recovery period, during which oxygen is consumed at a gradually diminishing rate until the breathing rate returns to normal. The excess oxygen consumed in the recovery period represents a repayment of the oxygen debt. This is the amount of oxygen required to supply ATP for gluconeogenesis during recovery respiration, in order to regenerate the glycogen “borrowed” from liver and muscle to carry out intense muscular activity in the sprint. The cycle of reactions that includes glucose conversion to lactate in muscle and lactate conversion to glucose in liver is called the Cori cycle, for Carl and Gerty Cori, whose studies in the 1930s and 1940s clarified the pathway and its role (see Box 15-1).

The circulatory systems of most small vertebrates can carry oxygen to their muscles fast enough to avoid having to use muscle glycogen anaerobically. For example, migrating birds often fly great distances at high speeds without rest and without incurring an oxygen debt. Many running animals of moderate size also maintain an essentially aerobic metabolism in their skeletal muscle. However, the circulatory systems of larger animals, including humans, cannot completely sustain aerobic metabolism in skeletal muscles over long periods of intense muscular activity. These animals generally are slow-moving under normal circumstances and engage in intense muscular activity only in the gravest emergencies, because such bursts of activity require long recovery periods to repay the oxygen debt.

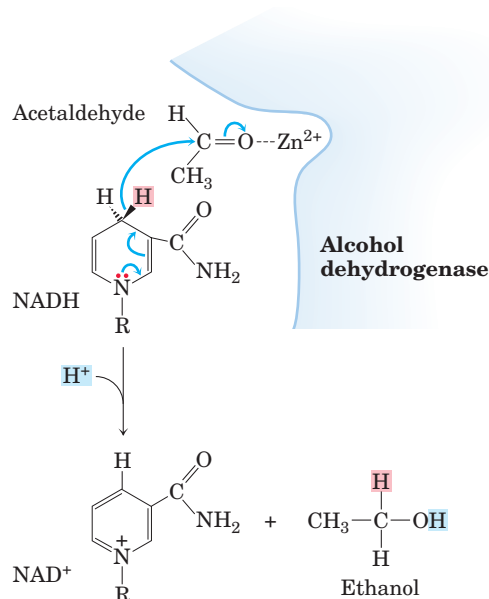
Alligators and crocodiles, for example, are normally sluggish animals. Yet when provoked they are capable of lightning-fast charges and dangerous lashings of their powerful tails. Such intense bursts of activity are short and must be followed by long periods of recovery. The fast emergency movements require

lactic acid fermentation to generate ATP in skeletal muscles. The stores of muscle glycogen are rapidly expended in intense muscular activity, and lactate reaches very high concentrations in muscles and extracellular fluid. Whereas a trained athlete can recover from a 100 m sprint in 30 min or less, an alligator may require many hours of rest and extra oxygen consumption to clear the excess lactate from its blood and regenerate muscle glycogen after a burst of activity.

Other large animals, such as the elephant and rhinoceros, have similar metabolic characteristics, as do diving mammals such as whales and seals. Dinosaurs and other huge, now-extinct animals probably had to depend on lactic acid fermentation to supply energy for muscular activity, followed by very long recovery periods during which they were vulnerable to attack by smaller predators better able to use oxygen and thus better adapted to continuous, sustained muscular activity.

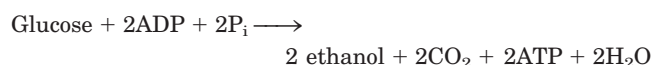
Deep-sea explorations have revealed many species of marine life at great ocean depths, where the oxygen concentration is near zero. For example, the primitive coelacanth, a large fish recovered from depths of 4,000 m or more off the coast of South Africa, has an essentially anaerobic metabolism in virtually all its tissues. It converts carbohydrates to lactate and other products, most of which must be excreted. Some marine vertebrates ferment glucose to ethanol and CO_2 in order to generate ATP.





MECHANISM FIGURE 14-12 The alcohol dehydrogenase reaction. A Zn^{2+} at the active site polarizes the carbonyl oxygen of acetaldehyde, allowing transfer of a hydride ion (red) from the reduced cofactor NADH. The reduced intermediate acquires a proton from the medium (blue) to form ethanol. [Alcohol Dehydrogenase Mechanism](#)

the reducing power furnished by NADH derived from the dehydrogenation of glyceraldehyde 3-phosphate. This reaction is a well-studied case of hydride transfer from NADH (Fig. 14-12). Ethanol and CO_2 are thus the end products of ethanol fermentation, and the overall equation is



As in lactic acid fermentation, there is no net change in the ratio of hydrogen to carbon atoms when glucose (H:C ratio = $12/6 = 2$) is fermented to two ethanol and

two CO_2 (combined H:C ratio = $12/6 = 2$). In all fermentations, the H:C ratio of the reactants and products remains the same.

Pyruvate decarboxylase is present in brewer's and baker's yeast and in all other organisms that ferment glucose to ethanol, including some plants. The CO_2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characteristic carbonation of champagne. The ancient art of brewing beer involves a number of enzymatic processes in addition to the reactions of ethanol fermentation (Box 14-2). In baking, CO_2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sugar causes dough to rise. The enzyme is absent in vertebrate tissues and in other organisms that carry out lactic acid fermentation.

Alcohol dehydrogenase is present in many organisms that metabolize ethanol, including humans. In human liver it catalyzes the oxidation of ethanol, either ingested or produced by intestinal microorganisms, with the concomitant reduction of NAD^+ to NADH.

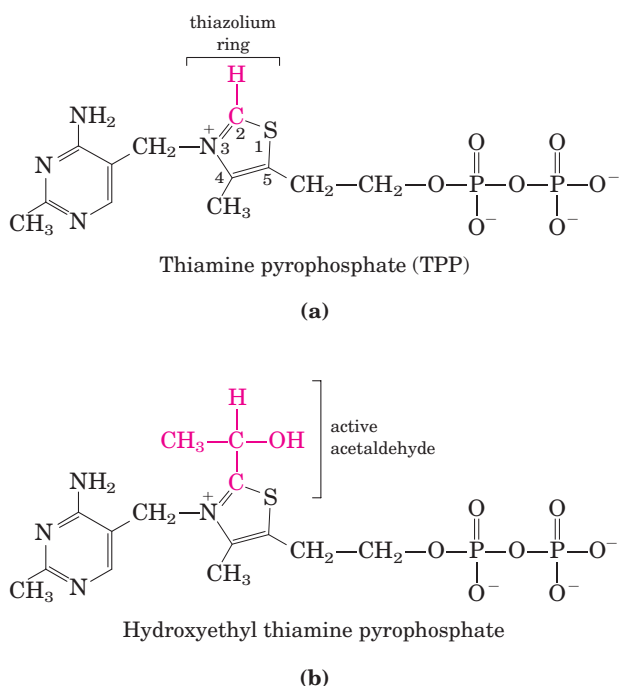
Thiamine Pyrophosphate Carries “Active Acetaldehyde” Groups

The pyruvate decarboxylase reaction provides our first encounter with **thiamine pyrophosphate (TPP)** (Fig. 14-13), a coenzyme derived from vitamin B_1 . Lack of vitamin B_1 in the human diet leads to the condition known as beriberi, characterized by an accumulation of body fluids (swelling), pain, paralysis, and ultimately death.

Thiamine pyrophosphate plays an important role in the cleavage of bonds adjacent to a carbonyl group, such as the decarboxylation of α -keto acids, and in chemical rearrangements in which an activated acetaldehyde group is transferred from one carbon atom to another (Table 14-1). The functional part of TPP, the thiazolium ring, has a relatively acidic proton at C-2. Loss of this

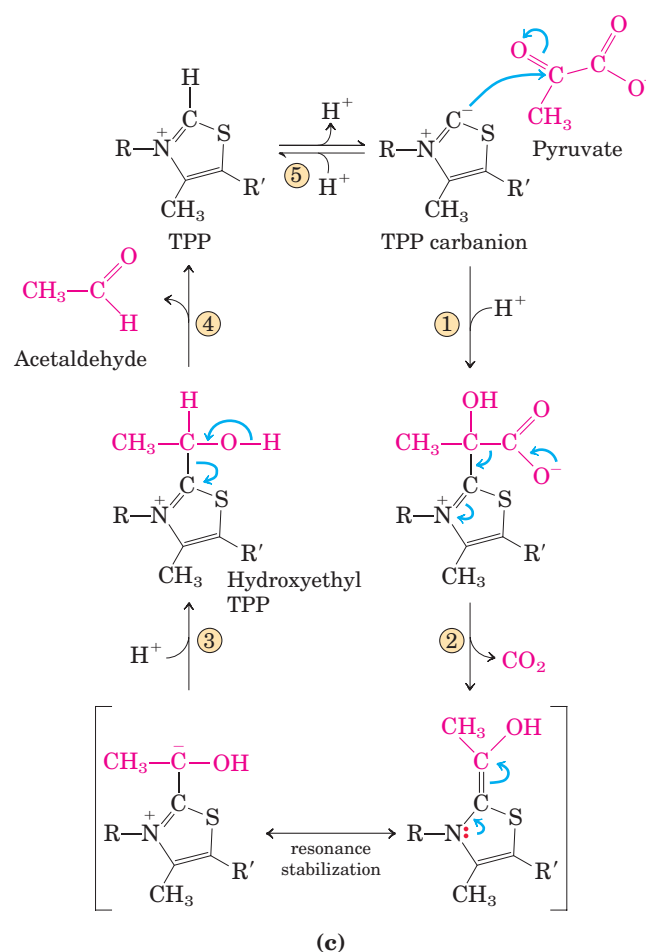
TABLE 14-1 Some TPP-Dependent Reactions

Enzyme	Pathway(s)	Bond cleaved	Bond formed
Pyruvate decarboxylase	Ethanol fermentation	$\text{R}^1-\text{C}(=\text{O})-\text{C}(=\text{O})\text{O}^-$	$\text{R}^1-\text{C}(=\text{O})\text{H}$
Pyruvate dehydrogenase α -Ketoglutarate dehydrogenase	Synthesis of acetyl-CoA Citric acid cycle	$\text{R}^2-\text{C}(=\text{O})-\text{C}(=\text{O})\text{O}^-$	$\text{R}^2-\text{C}(=\text{O})\text{S-CoA}$
Transketolase	Carbon-assimilation reactions Pentose phosphate pathway	$\text{R}^3-\text{C}(=\text{O})-\text{C}(\text{OH})(\text{H})-\text{R}^4$	$\text{R}^3-\text{C}(=\text{O})-\text{C}(\text{OH})(\text{H})-\text{R}^5$



MECHANISM FIGURE 14-13 Thiamine pyrophosphate (TPP) and its role in pyruvate decarboxylation. (a) TPP is the coenzyme form of vitamin B₁ (thiamine). The reactive carbon atom in the thiazolium ring of TPP is shown in red. In the reaction catalyzed by pyruvate decarboxylase, two of the three carbons of pyruvate are carried transiently on TPP in the form of a hydroxyethyl, or “active acetaldehyde,” group (b), which is subsequently released as acetaldehyde. (c) After cleavage of a carbon–carbon bond, one product often has a free electron pair, or carbanion, which because of its strong tendency to form a new bond is generally unstable. The thiazolium ring of TPP stabilizes carbanion intermediates by providing an electrophilic (electron-deficient) structure into which the carbanion electrons can be delocalized by resonance. Structures with this property, often called “electron sinks,” play a role in many biochemical reactions. This principle is illustrated here for the reaction catalyzed by pyruvate decarboxylase. ① The TPP carbanion acts as a nucleophile, attacking the carbonyl group of pyruvate. ② Decarboxylation produces a carbanion that is stabilized by the thiazolium ring. ③ Protonation to form hydroxyethyl TPP is followed by ④ release of acetaldehyde. ⑤ A proton dissociates to regenerate the carbanion.

Thiamine Pyrophosphate Mechanism



proton produces a carbanion that is the active species in TPP-dependent reactions (Fig. 14-13). The carbanion readily adds to carbonyl groups, and the thiazolium ring is thereby positioned to act as an “electron sink” that greatly facilitates reactions such as the decarboxylation catalyzed by pyruvate decarboxylase.

Fermentations Yield a Variety of Common Foods and Industrial Chemicals

Our progenitors learned millennia ago to use fermentation in the production and preservation of foods. Certain microorganisms present in raw food products fer-

ment the carbohydrates and yield metabolic products that give the foods their characteristic forms, textures, and tastes. Yogurt, already known in Biblical times, is produced when the bacterium *Lactobacillus bulgaricus* ferments the carbohydrate in milk, producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt. Another bacterium, *Propionibacterium freudenreichii*, ferments milk to produce propionic acid and CO₂; the propionic acid precipitates milk proteins, and bubbles of CO₂ cause the holes characteristic of Swiss cheese. Many other food products are the result of fermentations: pickles, sauerkraut, sausage, soy sauce, and a variety of national favorites, such as kimchi (Korea), tempoyak (Indonesia), kefir (Russia), dahi (India), and pozol (Mexico). The drop in pH associated with fermentation also helps to preserve foods, because most of the microorganisms that cause food spoilage cannot grow at low pH. In agriculture, plant byproducts such as corn stalks are preserved for use as animal feed by packing them into a large container (a silo) with limited access to air; microbial fermentation produces acids that lower the pH. The silage that results from this fermentation

BOX 14-2 THE WORLD OF BIOCHEMISTRY

Brewing Beer

Brewers prepare beer by ethanol fermentation of the carbohydrates in cereal grains (seeds) such as barley, carried out by yeast glycolytic enzymes. The carbohydrates, largely polysaccharides, must first be degraded to disaccharides and monosaccharides. In a process called malting, the barley seeds are allowed to germinate until they form the hydrolytic enzymes required to break down their polysaccharides, at which point germination is stopped by controlled heating. The product is malt, which contains enzymes that catalyze the hydrolysis of the β linkages of cellulose and other cell wall polysaccharides of the barley husks, and enzymes such as α -amylase and maltase.

The brewer next prepares the wort, the nutrient medium required for fermentation by yeast cells. The malt is mixed with water and then mashed or crushed. This allows the enzymes formed in the malting process to act on the cereal polysaccharides to form maltose, glucose, and other simple sugars, which are soluble in the aqueous medium. The remaining cell matter is then separated, and the liquid wort is boiled with hops to give flavor. The wort is cooled and then aerated.

Now the yeast cells are added. In the aerobic wort the yeast grows and reproduces very rapidly, using energy obtained from available sugars. No ethanol forms during this stage, because the yeast, amply supplied with oxygen, oxidizes the pyruvate formed by glycolysis to CO_2 and H_2O via the citric acid cycle. When all the dissolved oxygen in the vat of wort has been consumed, the yeast cells switch to anaerobic metabolism, and from this point they ferment the sugars into ethanol and CO_2 . The fermentation process is controlled in part by the concentration of the ethanol formed, by the pH, and by the amount of remaining sugar. After fermentation has been stopped, the cells are removed and the “raw” beer is ready for final processing.

In the final steps of brewing, the amount of foam or head on the beer, which results from dissolved proteins, is adjusted. Normally this is controlled by proteolytic enzymes that arise in the malting process. If these enzymes act on the proteins too long, the beer will have very little head and will be flat; if they do not act long enough, the beer will not be clear when it is cold. Sometimes proteolytic enzymes from other sources are added to control the head.

process can be kept as animal feed for long periods without spoilage.

In 1910 Chaim Weizmann (later to become the first president of Israel) discovered that the bacterium *Clostridium acetobutyricum* ferments starch to butanol and acetone. This discovery opened the field of industrial fermentations, in which some readily available material rich in carbohydrate (corn starch or molasses, for example) is supplied to a pure culture of a specific microorganism, which ferments it into a product of greater value. The methanol used to make “gasohol” is produced by microbial fermentation, as are formic, acetic, propionic, butyric, and succinic acids, and glycerol, ethanol, isopropanol, butanol, and butanediol. These fermentations are generally carried out in huge closed vats in which temperature and access to air are adjusted to favor the multiplication of the desired microorganism and to exclude contaminating organisms (Fig. 14-14). The beauty of industrial fermentations is that complicated, multistep chemical transformations are carried out in high yields and with few side products by chemical factories that reproduce themselves—microbial cells. For some industrial fermentations, technology has been developed to immobilize the cells in an inert support, to pass the starting material continuously through the bed of immobilized cells, and to collect the desired product in the effluent—an engineer’s dream!



FIGURE 14-14 Industrial-scale fermentation. Microorganisms are cultured in a sterilizable vessel containing thousands of liters of growth medium—an inexpensive source of both carbon and energy—under carefully controlled conditions, including low oxygen concentration and constant temperature. After centrifugal separation of the cells from the growth medium, the valuable products of the fermentation are recovered from the cells or from the supernatant fluid.

SUMMARY 14.3 Fates of Pyruvate under Anaerobic Conditions: Fermentation

- The NADH formed in glycolysis must be recycled to regenerate NAD^+ , which is required as an electron acceptor in the first step of the payoff phase. Under aerobic conditions, electrons pass from NADH to O_2 in mitochondrial respiration.
- Under anaerobic or hypoxic conditions, many organisms regenerate NAD^+ by transferring electrons from NADH to pyruvate, forming lactate. Other organisms, such as yeast, regenerate NAD^+ by reducing pyruvate to ethanol and CO_2 . In these anaerobic processes (fermentations), there is no *net* oxidation or reduction of the carbons of glucose.
- A variety of microorganisms can ferment sugar in fresh foods, resulting in changes in pH, taste, and texture, and preserving food from spoilage. Fermentations are used in industry to produce a wide variety of commercially valuable organic compounds from inexpensive starting materials.

14.4 Gluconeogenesis

The central role of glucose in metabolism arose early in evolution, and this sugar remains the nearly universal fuel and building block in modern organisms, from microbes to humans. In mammals, some tissues depend almost completely on glucose for their metabolic energy. For the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the blood is the sole or major fuel source. The brain alone requires about 120 g of glucose each day—more than half of all the glucose stored as glycogen in muscle and liver. However, the supply of glucose from these stores is not always sufficient; between meals and during longer fasts, or after vigorous exercise, glycogen is depleted. For these times, organisms need a method for synthesizing glucose from noncarbohydrate precursors. This is accomplished by a pathway called **gluconeogenesis** (“formation of new sugar”), which converts pyruvate and related three- and four-carbon compounds to glucose.

Gluconeogenesis occurs in all animals, plants, fungi, and microorganisms. The reactions are essentially the same in all tissues and all species. The important precursors of glucose in animals are three-carbon compounds such as lactate, pyruvate, and glycerol, as well as certain amino acids (Fig. 14–15). In mammals, gluconeogenesis takes place mainly in the liver, and to a lesser extent in renal cortex. The glucose produced passes into the blood to supply other tissues. After vigorous exercise, lactate produced by anaerobic glycoly-

sis in skeletal muscle returns to the liver and is converted to glucose, which moves back to muscle and is converted to glycogen—a circuit called the Cori cycle (Box 14–1; see also Fig. 23–18). In plant seedlings, stored fats and proteins are converted, via paths that include gluconeogenesis, to the disaccharide sucrose for transport throughout the developing plant. Glucose and its derivatives are precursors for the synthesis of plant cell walls, nucleotides and coenzymes, and a variety of other essential metabolites. In many microorganisms, gluconeogenesis starts from simple organic compounds of two or three carbons, such as acetate, lactate, and propionate, in their growth medium.

Although the reactions of gluconeogenesis are the same in all organisms, the metabolic context and the regulation of the pathway differ from one species to another and from tissue to tissue. In this section we focus on gluconeogenesis as it occurs in the mammalian liver. In Chapter 20 we show how photosynthetic organisms use this pathway to convert the primary products of photosynthesis into glucose, to be stored as sucrose or starch.

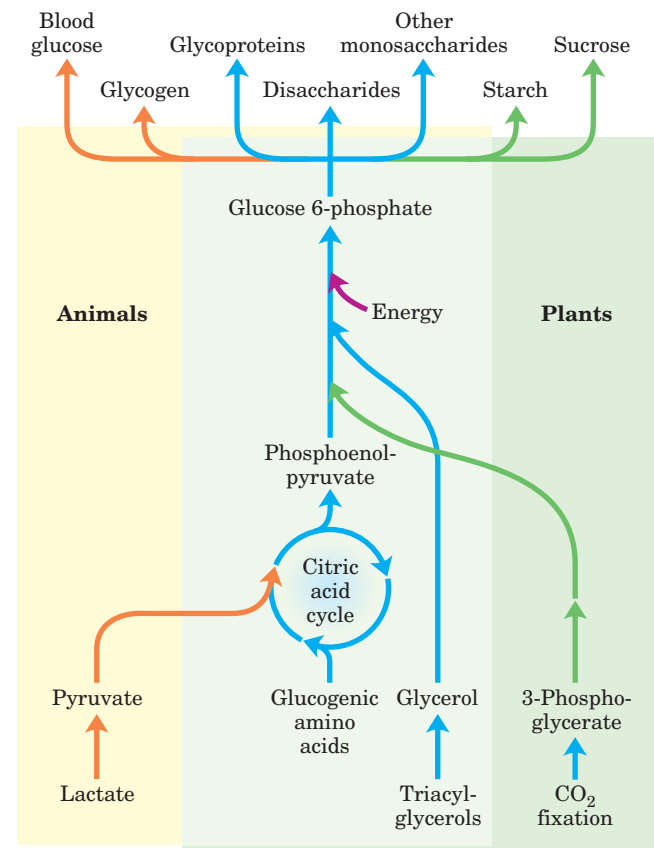
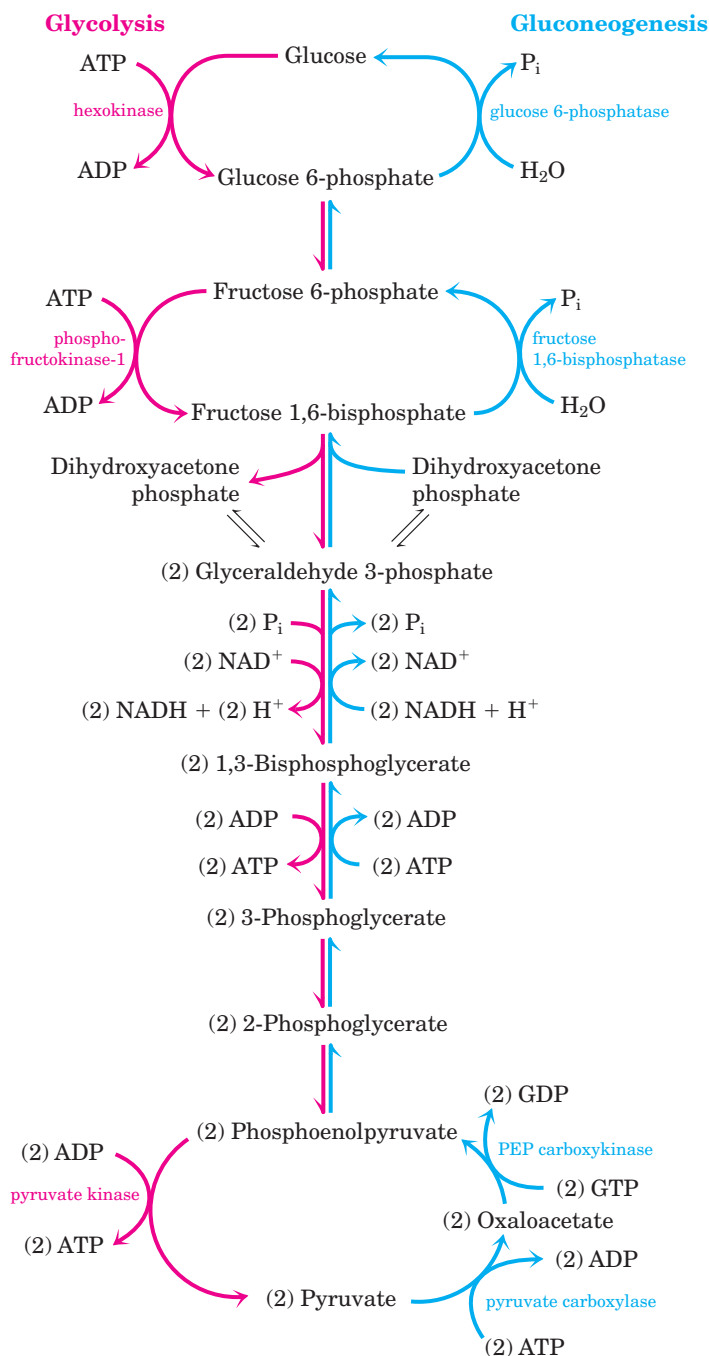


FIGURE 14–15 Carbohydrate synthesis from simple precursors. The pathway from phosphoenolpyruvate to glucose 6-phosphate is common to the biosynthetic conversion of many different precursors of carbohydrates in animals and plants. Plants and photosynthetic bacteria are uniquely able to convert CO_2 to carbohydrates.



Gluconeogenesis and glycolysis are not identical pathways running in opposite directions, although they do share several steps (Fig. 14-16); seven of the ten enzymatic reactions of gluconeogenesis are the reverse of glycolytic reactions. However, three reactions of glycolysis are essentially irreversible *in vivo* and cannot be used in gluconeogenesis: the conversion of glucose to glucose 6-phosphate by hexokinase, the phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate by phosphofructokinase-1, and the conversion of phosphoenolpyruvate to pyruvate by pyruvate kinase (Fig. 14-16). In cells, these three reactions are charac-

FIGURE 14-16 Opposing pathways of glycolysis and gluconeogenesis in rat liver. The reactions of glycolysis are shown on the left side in blue; the opposing pathway of gluconeogenesis is shown on the right in red. The major sites of regulation of gluconeogenesis shown here are discussed later in this chapter, and in detail in Chapter 15. Figure 14-19 illustrates an alternative route for oxaloacetate produced in mitochondria.

terized by a large negative free-energy change, ΔG , whereas other glycolytic reactions have a ΔG near 0 (Table 14-2). In gluconeogenesis, the three irreversible steps are bypassed by a separate set of enzymes, catalyzing reactions that are sufficiently exergonic to be effectively irreversible in the direction of glucose synthesis. Thus, both glycolysis and gluconeogenesis are irreversible processes in cells. In animals, both pathways occur largely in the cytosol, necessitating their reciprocal and coordinated regulation. Separate regulation of the two pathways is brought about through controls exerted on the enzymatic steps unique to each.

We begin by considering the three bypass reactions of gluconeogenesis. (Keep in mind that “bypass” refers throughout to the bypass of irreversible glycolytic reactions.)

Conversion of Pyruvate to Phosphoenolpyruvate Requires Two Exergonic Reactions

The first of the bypass reactions in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP). This reaction cannot occur by reversal of the pyruvate kinase reaction of glycolysis (p. 532), which has a large, negative standard free-energy change and is irreversible under the conditions prevailing in intact cells (Table 14-2, step 10). Instead, the phosphorylation of pyruvate is achieved by a roundabout sequence of reactions that in eukaryotes requires enzymes in both the cytosol and mitochondria. As we shall see, the pathway shown in Figure 14-16 and described in detail here is one of two routes from pyruvate to PEP; it is the predominant path when pyruvate or alanine is the glucogenic precursor. A second pathway, described later, predominates when lactate is the glucogenic precursor.

Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine within mitochondria by transamination, in which the α -amino group is removed from alanine (leaving pyruvate) and added to an α -keto carboxylic acid (transamination reactions are discussed in detail in Chapter 18). Then **pyruvate carboxylase**, a mitochondrial enzyme that requires the coenzyme **biotin**, converts the pyruvate to oxaloacetate (Fig. 14-17):

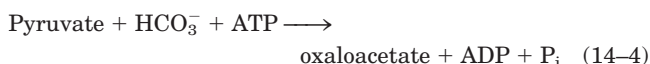


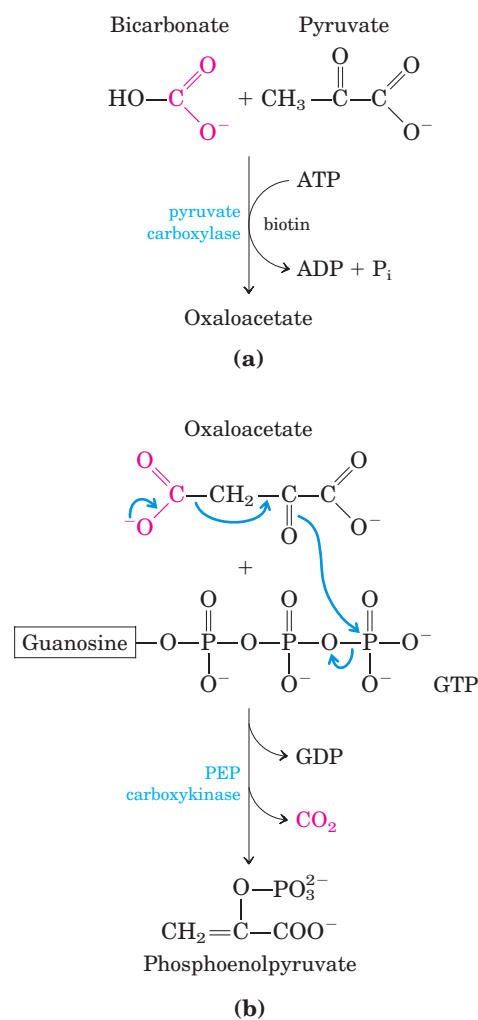
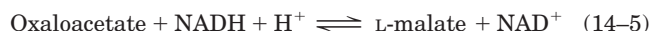
TABLE 14-2 Free-Energy Changes of Glycolytic Reactions in Erythrocytes

Glycolytic reaction step	$\Delta G'^{\circ}$ (kJ/mol)	ΔG (kJ/mol)
① Glucose + ATP $\longrightarrow$ glucose 6-phosphate + ADP	-16.7	-33.4
② Glucose 6-phosphate $\rightleftharpoons$ fructose 6-phosphate	1.7	0 to 25
③ Fructose 6-phosphate + ATP $\longrightarrow$ fructose 1,6-bisphosphate + ADP	-14.2	-22.2
④ Fructose 1,6-bisphosphate $\rightleftharpoons$ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	23.8	0 to -6
⑤ Dihydroxyacetone phosphate $\rightleftharpoons$ glyceraldehyde 3-phosphate	7.5	0 to 4
⑥ Glyceraldehyde 3-phosphate + P_i + NAD^+ $\rightleftharpoons$ 1,3-bisphosphoglycerate + $NADH$ + H^+	6.3	-2 to 2
⑦ 1,3-Bisphosphoglycerate + ADP $\rightleftharpoons$ 3-phosphoglycerate + ATP	-18.8	0 to 2
⑧ 3-Phosphoglycerate $\rightleftharpoons$ 2-phosphoglycerate	4.4	0 to 0.8
⑨ 2-Phosphoglycerate $\rightleftharpoons$ phosphoenolpyruvate + H_2O	7.5	0 to 3.3
⑩ Phosphoenolpyruvate + ADP $\longrightarrow$ pyruvate + ATP	-31.4	-16.7

Note: $\Delta G'^{\circ}$ is the standard free-energy change, as defined in Chapter 13 (p. 491). ΔG is the free-energy change calculated from the actual concentrations of glycolytic intermediates present under physiological conditions in erythrocytes, at pH 7. The glycolytic reactions bypassed in gluconeogenesis are shown in red. Biochemical equations are not necessarily balanced for H or charge (p. 506).

The reaction involves biotin as a carrier of activated HCO_3^- (Fig. 14-18). The reaction mechanism is shown in Figure 16-16. Pyruvate carboxylase is the first regulatory enzyme in the gluconeogenic pathway, requiring acetyl-CoA as a positive effector. (Acetyl-CoA is produced by fatty acid oxidation (Chapter 17), and its accumulation signals the availability of fatty acids as fuel.) As we shall see in Chapter 16 (see Fig. 16-15), the pyruvate carboxylase reaction can replenish intermediates in another central metabolic pathway, the citric acid cycle.

Because the mitochondrial membrane has no transporter for oxaloacetate, before export to the cytosol the oxaloacetate formed from pyruvate must be reduced to malate by mitochondrial **malate dehydrogenase**, at the expense of $NADH$:

**FIGURE 14-17** Synthesis of phosphoenolpyruvate from pyruvate.

(a) In mitochondria, pyruvate is converted to oxaloacetate in a biotin-requiring reaction catalyzed by pyruvate carboxylase. **(b)** In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP carboxykinase. The CO_2 incorporated in the pyruvate carboxylase reaction is lost here as CO_2 . The decarboxylation leads to a rearrangement of electrons that facilitates attack of the carbonyl oxygen of the pyruvate moiety on the γ phosphate of GTP.

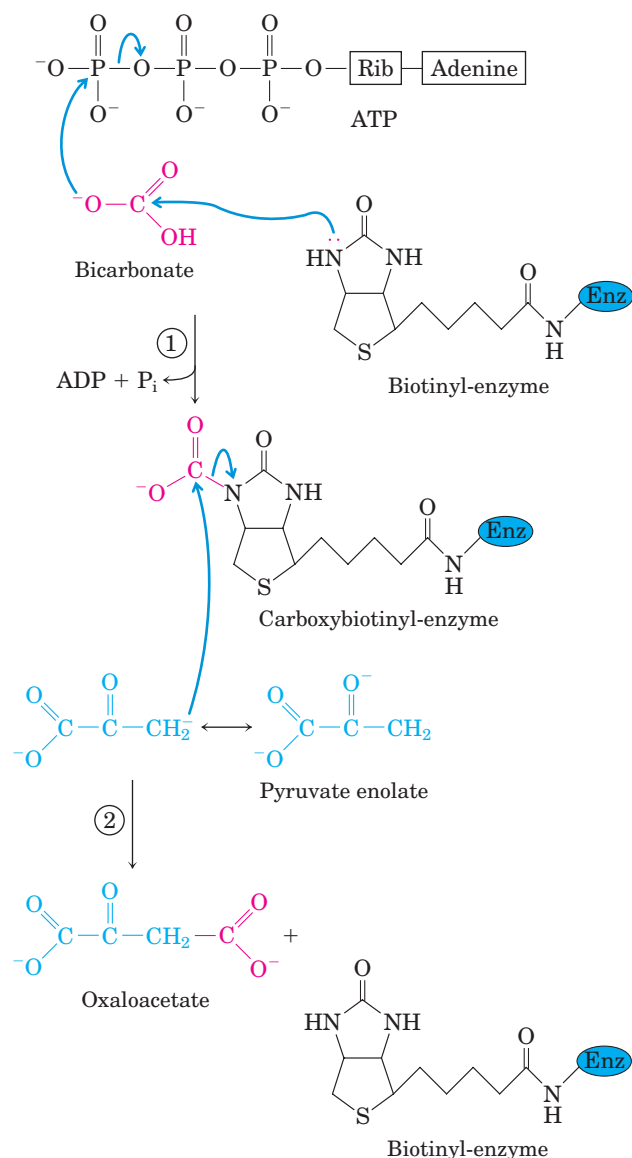


FIGURE 14-18 Role of biotin in the pyruvate carboxylase reaction.

The cofactor biotin is covalently attached to the enzyme through an amide linkage to the ϵ -amino group of a Lys residue, forming a biotinyl-enzyme. The reaction occurs in two phases, which occur at two different sites in the enzyme. At catalytic site 1, bicarbonate ion is converted to CO_2 at the expense of ATP. Then CO_2 reacts with biotin, forming carboxybiotinyl-enzyme. The long arm composed of biotin and the side chain of the Lys to which it is attached then carry the CO_2 of carboxybiotinyl-enzyme to catalytic site 2 on the enzyme surface, where CO_2 is released and reacts with the pyruvate, forming oxaloacetate and regenerating the biotinyl-enzyme. The general role of flexible arms in carrying reaction intermediates between enzyme active sites is described in Figure 16-17, and the mechanistic details of the pyruvate carboxylase reaction are shown in Figure 16-16. Similar mechanisms occur in other biotin-dependent carboxylation reactions, such as those catalyzed by propionyl-CoA carboxylase (see Fig. 17-11) and acetyl-CoA carboxylase (see Fig. 21-1).

The standard free-energy change for this reaction is quite high, but under physiological conditions (including a very low concentration of oxaloacetate) $\Delta G \approx 0$ and the reaction is readily reversible. Mitochondrial malate dehydrogenase functions in both gluconeogenesis and the citric acid cycle, but the overall flow of metabolites in the two processes is in opposite directions.

Malate leaves the mitochondrion through a specific transporter in the inner mitochondrial membrane (see Fig. 19-27), and in the cytosol it is reoxidized to oxaloacetate, with the production of cytosolic NADH:

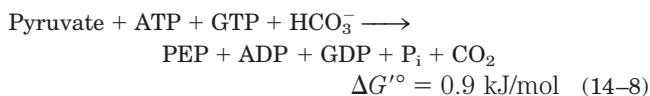


The oxaloacetate is then converted to PEP by **phosphoenolpyruvate carboxykinase** (Fig. 14-17). This Mg^{2+} -dependent reaction requires GTP as the phosphoryl group donor :



The reaction is reversible under intracellular conditions; the formation of one high-energy phosphate compound (PEP) is balanced by the hydrolysis of another (GTP).

The overall equation for this set of bypass reactions, the sum of Equations 14-4 through 14-7, is



Two high-energy phosphate equivalents (one from ATP and one from GTP), each yielding about 50 kJ/mol under cellular conditions, must be expended to phosphorylate one molecule of pyruvate to PEP. In contrast, when PEP is converted to pyruvate during glycolysis, only one ATP is generated from ADP. Although the standard free-energy change ($\Delta G'^{\circ}$) of the two-step path from pyruvate to PEP is 0.9 kJ/mol, the actual free-energy change (ΔG), calculated from measured cellular concentrations of intermediates, is very strongly negative (-25 kJ/mol); this results from the ready consumption of PEP in other reactions such that its concentration remains relatively low. The reaction is thus effectively irreversible in the cell.

Note that the CO_2 added to pyruvate in the pyruvate carboxylase step is the same molecule that is lost in the PEP carboxykinase reaction (Fig. 14-17). This carboxylation-decarboxylation sequence represents a way of “activating” pyruvate, in that the decarboxylation of oxaloacetate facilitates PEP formation. In Chapter 21 we shall see how a similar carboxylation-decarboxylation sequence is used to activate acetyl-CoA for fatty acid biosynthesis (see Fig. 21-1).

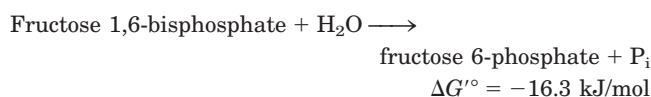
There is a logic to the route of these reactions through the mitochondrion. The $[\text{NADH}]/[\text{NAD}^+]$ ratio in the cytosol is 8×10^{-4} , about 10^5 times lower than in mitochondria. Because cytosolic NADH is consumed in gluconeogenesis (in the conversion of 1,3-bisphos-

phoglycerate to glyceraldehyde 3-phosphate; Fig. 14–16), glucose biosynthesis cannot proceed unless NADH is available. The transport of malate from the mitochondrion to the cytosol and its reconversion there to oxaloacetate effectively moves reducing equivalents to the cytosol, where they are scarce. This path from pyruvate to PEP therefore provides an important balance between NADH produced and consumed in the cytosol during gluconeogenesis.

A second pyruvate → PEP bypass predominates when lactate is the glucogenic precursor (Fig. 14–19). This pathway makes use of lactate produced by glycolysis in erythrocytes or anaerobic muscle, for example, and it is particularly important in large vertebrates after vigorous exercise (Box 14–1). The conversion of lactate to pyruvate in the cytosol of hepatocytes yields NADH, and the export of reducing equivalents (as malate) from mitochondria is therefore unnecessary. After the pyruvate produced by the lactate dehydrogenase reaction is transported into the mitochondrion, it is converted to oxaloacetate by pyruvate carboxylase, as described above. This oxaloacetate, however, is converted directly to PEP by a mitochondrial isozyme of PEP carboxykinase, and the PEP is transported out of the mitochondrion to continue on the gluconeogenic path. The mitochondrial and cytosolic isozymes of PEP carboxykinase are encoded by separate genes in the nuclear chromosomes, providing another example of two distinct enzymes catalyzing the same reaction but having different cellular locations or metabolic roles (recall the isozymes of hexokinase).

Conversion of Fructose 1,6-Bisphosphate to Fructose 6-Phosphate Is the Second Bypass

The second glycolytic reaction that cannot participate in gluconeogenesis is the phosphorylation of fructose 6-phosphate by PFK-1 (Table 14–2, step ③). Because this reaction is highly exergonic and therefore irreversible in intact cells, the generation of fructose 6-phosphate from fructose 1,6-bisphosphate (Fig. 14–16) is catalyzed by a different enzyme, Mg^{2+} -dependent **fructose 1,6-bisphosphatase (FBPase-1)**, which promotes the essentially irreversible *hydrolysis* of the C-1 phosphate (*not* phosphoryl group transfer to ADP):



Conversion of Glucose 6-Phosphate to Glucose Is the Third Bypass

The third bypass is the final reaction of gluconeogenesis, the dephosphorylation of glucose 6-phosphate to yield glucose (Fig. 14–16). Reversal of the hexokinase reaction (p. 526) would require phosphoryl group trans-

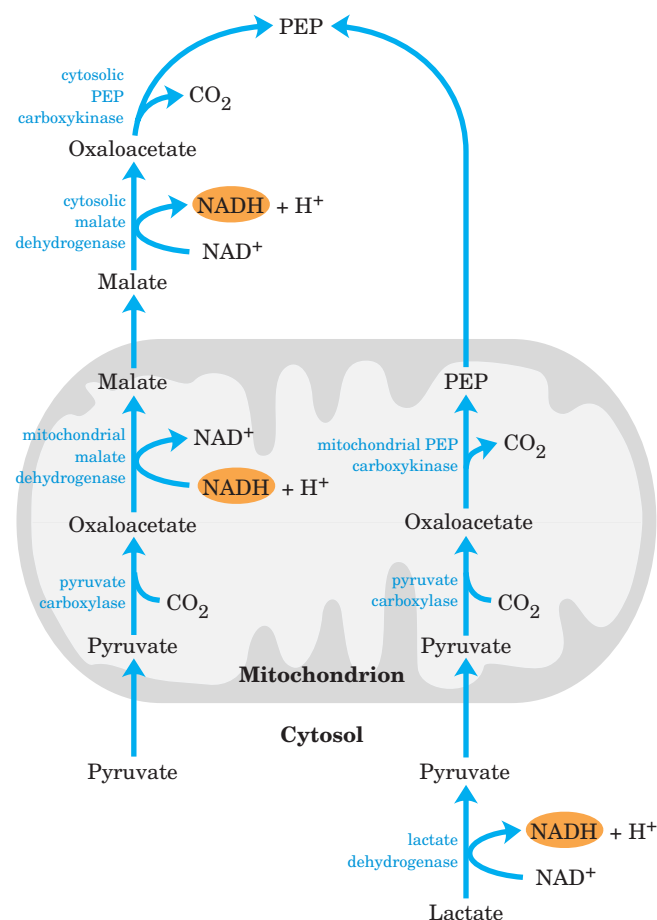
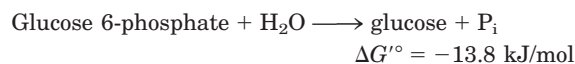


FIGURE 14–19 Alternative paths from pyruvate to phosphoenolpyruvate. The path that predominates depends on the glucogenic precursor (lactate or pyruvate). The path on the right predominates when lactate is the precursor, because cytosolic NADH is generated in the lactate dehydrogenase reaction and does not have to be shuttled out of the mitochondrion (see text). The relative importance of the two pathways depends on the availability of lactate and the cytosolic requirements for NADH by gluconeogenesis.

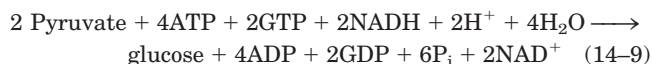
fer from glucose 6-phosphate to ADP, forming ATP, an energetically unfavorable reaction (Table 14–2, step ①). The reaction catalyzed by **glucose 6-phosphatase** does not require synthesis of ATP; it is a simple hydrolysis of a phosphate ester:



This Mg^{2+} -activated enzyme is found on the luminal side of the endoplasmic reticulum of hepatocytes and renal cells (see Fig. 15–6). Muscle and brain tissue do not contain this enzyme and so cannot carry out gluconeogenesis. Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to brain and muscle through the bloodstream.

Gluconeogenesis Is Energetically Expensive, but Essential

The sum of the biosynthetic reactions leading from pyruvate to free blood glucose (Table 14–3) is



For each molecule of glucose formed from pyruvate, six high-energy phosphate groups are required, four from ATP and two from GTP. In addition, two molecules of NADH are required for the reduction of two molecules of 1,3-bisphosphoglycerate. Clearly, Equation 14–9 is not simply the reverse of the equation for conversion of glucose to pyruvate by glycolysis, which requires only two molecules of ATP:



The synthesis of glucose from pyruvate is a relatively expensive process. Much of this high energy cost is necessary to ensure the irreversibility of gluconeogenesis. Under intracellular conditions, the overall free-energy change of glycolysis is at least -63 kJ/mol . Under the same conditions the overall ΔG of gluconeogenesis is -16 kJ/mol . Thus both glycolysis and gluconeogenesis are essentially irreversible processes in cells.

Citric Acid Cycle Intermediates and Many Amino Acids Are Glucogenic

The biosynthetic pathway to glucose described above allows the net synthesis of glucose not only from pyruvate but also from the four-, five-, and six-carbon intermediates of the citric acid cycle (Chapter 16). Citrate,

isocitrate, α -ketoglutarate, succinyl-CoA, succinate, fumarate, and malate—all are citric acid cycle intermediates that can undergo oxidation to oxaloacetate (see Fig. 16–7). Some or all of the carbon atoms of most amino acids derived from proteins are ultimately catabolized to pyruvate or to intermediates of the citric acid cycle. Such amino acids can therefore undergo net conversion to glucose and are said to be **glucogenic** (Table 14–4). Alanine and glutamine, the principal molecules that transport amino groups from extrahepatic tissues to the liver (see Fig. 18–9), are particularly important glucogenic amino acids in mammals. After removal of their amino groups in liver mitochondria, the carbon skeletons remaining (pyruvate and α -ketoglutarate, respectively) are readily funneled into gluconeogenesis.

In contrast, no net conversion of fatty acids to glucose occurs in mammals. As we shall see in Chapter 17, the catabolism of most fatty acids yields only acetyl-CoA. Mammals cannot use acetyl-CoA as a precursor of glucose, because the pyruvate dehydrogenase reaction is irreversible and cells have no other pathway to convert acetyl-CoA to pyruvate. Plants, yeast, and many bacteria do have a pathway (the glyoxylate cycle; see Fig. 16–20) for converting acetyl-CoA to oxaloacetate, so these organisms can use fatty acids as the starting material for gluconeogenesis. This is especially important during the germination of seedlings, before photosynthesis can serve as a source of glucose.

Glycolysis and Gluconeogenesis Are Regulated Reciprocally

If glycolysis (the conversion of glucose to pyruvate) and gluconeogenesis (the conversion of pyruvate to glucose) were allowed to proceed simultaneously at high rates,

TABLE 14–3 Sequential Reactions in Gluconeogenesis Starting from Pyruvate

Pyruvate + HCO_3^- + ATP $\longrightarrow$ oxaloacetate + ADP + P_i	×2
Oxaloacetate + GTP $\rightleftharpoons$ phosphoenolpyruvate + CO_2 + GDP	×2
Phosphoenolpyruvate + H_2O $\rightleftharpoons$ 2-phosphoglycerate	×2
2-Phosphoglycerate $\rightleftharpoons$ 3-phosphoglycerate	×2
3-Phosphoglycerate + ATP $\rightleftharpoons$ 1,3-bisphosphoglycerate + ADP	×2
1,3-Bisphosphoglycerate + NADH + H^+ $\rightleftharpoons$ glyceraldehyde 3-phosphate + NAD^+ + P_i	×2
Glyceraldehyde 3-phosphate $\rightleftharpoons$ dihydroxyacetone phosphate	
Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate $\rightleftharpoons$ fructose 1,6-bisphosphate	
Fructose 1,6-bisphosphate $\longrightarrow$ fructose 6-phosphate + P_i	
Fructose 6-phosphate $\rightleftharpoons$ glucose 6-phosphate	
Glucose 6-phosphate + H_2O $\longrightarrow$ glucose + P_i	
Sum: 2 Pyruvate + 4ATP + 2GTP + 2NADH + 2H ⁺ + 4H ₂ O $\longrightarrow$ glucose + 4ADP + 2GDP + 6P _i + 2NAD ⁺	

Note: The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the right indicate that the reaction is to be counted twice, because two three-carbon precursors are required to make a molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceraldehyde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge (p. 506).

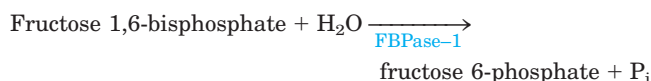
TABLE 14–4 Glucogenic Amino Acids, Grouped by Site of Entry

Pyruvate	Succinyl-CoA
Alanine	Isoleucine*
Cysteine	Methionine
Glycine	Threonine
Serine	Valine
Threonine	Fumarate
Tryptophan*	Phenylalanine*
α-Ketoglutarate	Tyrosine*
Arginine	Oxaloacetate
Glutamate	Asparagine
Glutamine	Aspartate
Histidine	
Proline	

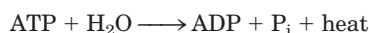
Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis.

*These amino acids are also ketogenic (see Fig. 18–21).

the result would be the consumption of ATP and the production of heat. For example, PFK-1 and FBPase-1 catalyze opposing reactions:



The sum of these two reactions is



These two enzymatic reactions, and a number of others in the two pathways, are regulated allosterically and by covalent modification (phosphorylation). In Chapter 15 we take up the mechanisms of this regulation in detail. For now, suffice it to say that the pathways are regulated so that when the flux of glucose through glycolysis goes up, the flux of pyruvate toward glucose goes down, and vice versa.

SUMMARY 14.4 Gluconeogenesis

- Gluconeogenesis is a ubiquitous multistep process in which pyruvate or a related three-carbon compound (lactate, alanine) is converted to glucose. Seven of the steps in gluconeogenesis are catalyzed by the same enzymes used in glycolysis; these are the reversible reactions.
- Three irreversible steps in the glycolytic pathway are bypassed by reactions catalyzed by gluconeogenic enzymes: (1) conversion of

pyruvate to PEP via oxaloacetate, catalyzed by pyruvate carboxylase and PEP carboxykinase; (2) dephosphorylation of fructose 1,6-bisphosphate by FBPase-1; and (3) dephosphorylation of glucose 6-phosphate by glucose 6-phosphatase.

- Formation of one molecule of glucose from pyruvate requires 4 ATP, 2 GTP, and 2 NADH; it is expensive.
- In mammals, gluconeogenesis in the liver and kidney provides glucose for use by the brain, muscles, and erythrocytes.
- Pyruvate carboxylase is stimulated by acetyl-CoA, increasing the rate of gluconeogenesis when the cell already has adequate supplies of other substrates (fatty acids) for energy production.
- Animals cannot convert acetyl-CoA derived from fatty acids into glucose; plants and microorganisms can.
- Glycolysis and gluconeogenesis are reciprocally regulated to prevent wasteful operation of both pathways at the same time.

14.5 Pentose Phosphate Pathway of Glucose Oxidation



In most animal tissues, the major catabolic fate of glucose 6-phosphate is glycolytic breakdown to pyruvate, much of which is then oxidized via the citric acid cycle, ultimately leading to the formation of ATP. Glucose 6-phosphate does have other catabolic fates, however, which lead to specialized products needed by the cell. Of particular importance in some tissues is the oxidation of glucose 6-phosphate to pentose phosphates by the **pentose phosphate pathway** (also called the **phosphogluconate pathway** or the **hexose monophosphate pathway**; Fig. 14–20). In this oxidative pathway, NADP^+ is the electron acceptor, yielding NADPH. Rapidly dividing cells, such as those of bone marrow, skin, and intestinal mucosa, use the pentoses to make RNA, DNA, and such coenzymes as ATP, NADH, FADH_2 , and coenzyme A.

In other tissues, the essential product of the pentose phosphate pathway is not the pentoses but the electron donor NADPH, needed for reductive biosynthesis or to counter the damaging effects of oxygen radicals. Tissues that carry out extensive fatty acid synthesis (liver, adipose, lactating mammary gland) or very active synthesis of cholesterol and steroid hormones (liver, adrenal gland, gonads) require the NADPH provided by the pathway. Erythrocytes and the cells of the lens and cornea are directly exposed to oxygen and thus to the damaging free radicals generated by oxygen.

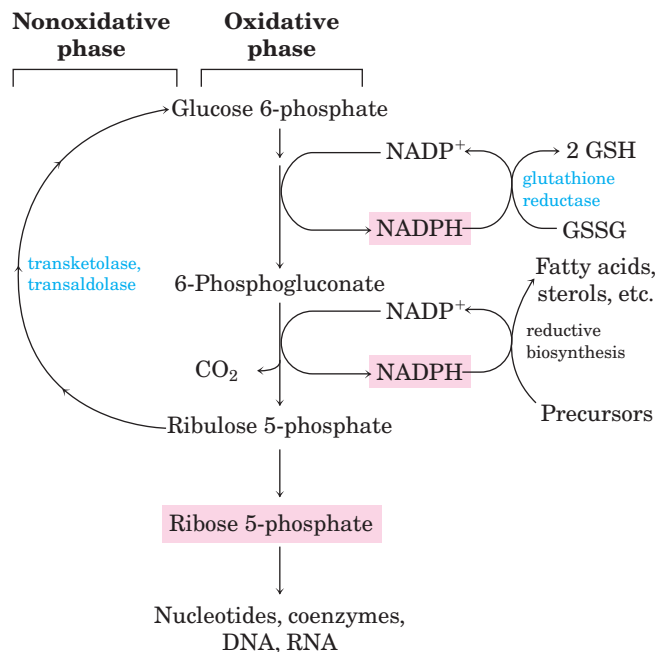


FIGURE 14-20 General scheme of the pentose phosphate pathway. NADPH formed in the oxidative phase is used to reduce glutathione, GSSG (see Box 14-3) and to support reductive biosynthesis. The other product of the oxidative phase is ribose 5-phosphate, which serves as precursor for nucleotides, coenzymes, and nucleic acids. In cells that are not using ribose 5-phosphate for biosynthesis, the nonoxidative phase recycles six molecules of the pentose into five molecules of the hexose glucose 6-phosphate, allowing continued production of NADPH and converting glucose 6-phosphate (in six cycles) to CO_2 .

By maintaining a reducing atmosphere (a high ratio of NADPH to NADP^+ and a high ratio of reduced to oxidized glutathione), they can prevent or undo oxidative damage to proteins, lipids, and other sensitive molecules. In erythrocytes, the NADPH produced by the pentose phosphate pathway is so important in preventing oxidative damage that a genetic defect in glucose 6-phosphate dehydrogenase, the first enzyme of the pathway, can have serious medical consequences (Box 14-3). ■

The Oxidative Phase Produces Pentose Phosphates and NADPH

The first reaction of the pentose phosphate pathway (Fig. 14-21) is the oxidation of glucose 6-phosphate by **glucose 6-phosphate dehydrogenase (G6PD)** to form 6-phosphoglucono- δ -lactone, an intramolecular ester. NADP^+ is the electron acceptor, and the overall equilibrium lies far in the direction of NADPH formation. The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific **lactonase**, then 6-phosphogluconate undergoes oxidation and decarboxylation by **6-phosphogluconate dehydrogenase** to form the ketopentose ribulose 5-phosphate. This reaction generates

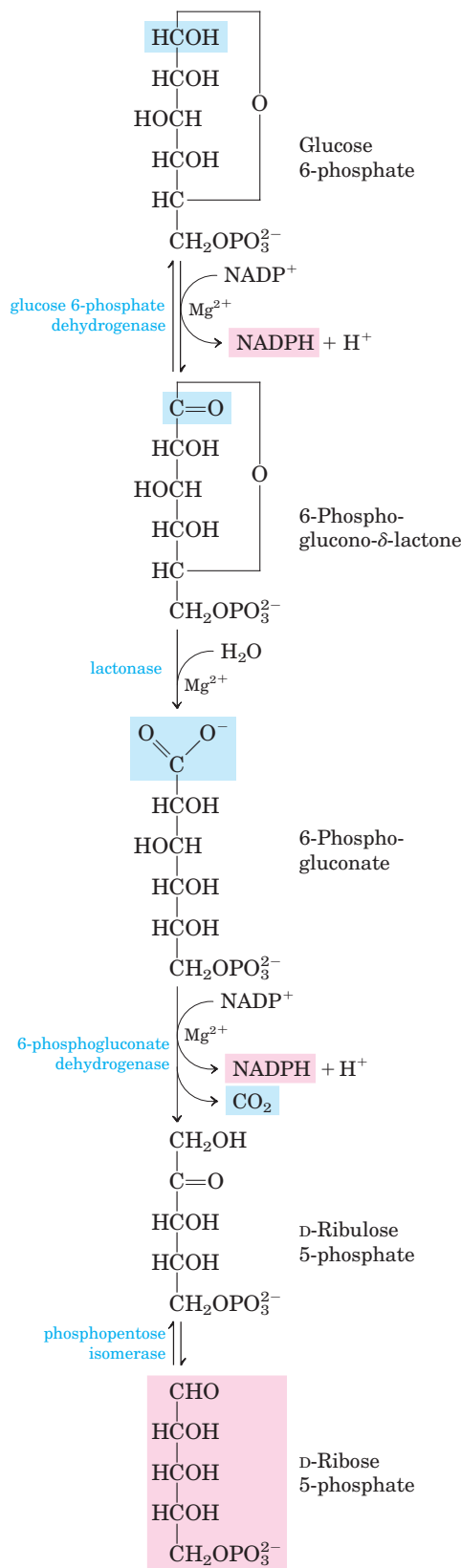


FIGURE 14-21 Oxidative reactions of the pentose phosphate pathway. The end products are ribose 5-phosphate, CO_2 , and NADPH.



BOX 14-3 BIOCHEMISTRY IN MEDICINE

Why Pythagoras Wouldn't Eat Falafel: Glucose 6-Phosphate Dehydrogenase Deficiency

Fava beans, an ingredient of falafel, have been an important food source in the Mediterranean and Middle East since antiquity. The Greek philosopher and mathematician Pythagoras prohibited his followers from dining on fava beans, perhaps because they make many people sick with a condition called favism, which can be fatal. In favism, erythrocytes begin to lyse 24 to 48 hours after ingestion of the beans, releasing free hemoglobin into the blood. Jaundice and sometimes kidney failure can result. Similar symptoms can occur with ingestion of the antimalarial drug primaquine or of sulfa antibiotics or following exposure to certain herbicides. These symptoms have a genetic basis: glucose 6-phosphate dehydrogenase (G6PD) deficiency, which affects about 400 million people. Most G6PD-deficient individuals are asymptomatic; only the combination of G6PD deficiency and certain environmental factors produces the clinical manifestations.

G6PD catalyzes the first step in the pentose phosphate pathway (see Fig. 14-21), which produces NADPH. This reductant, essential in many biosynthetic pathways, also protects cells from oxidative damage by hydrogen peroxide (H_2O_2) and superoxide free radicals, highly reactive oxidants generated as metabolic byproducts and through the actions of drugs such as primaquine and natural products such as divicine—the toxic ingredient of fava beans. During normal detoxification, H_2O_2 is converted to H_2O by reduced glutathione and glutathione peroxidase, and the oxidized glutathione is converted back to the reduced form by glutathione reductase and NADPH (Fig. 1). H_2O_2 is also broken down to H_2O and O_2 by catalase, which also requires NADPH. In G6PD-deficient individuals, the NADPH production is diminished and detoxification of H_2O_2 is inhibited. Cellular damage results: lipid peroxidation leading to breakdown of erythrocyte membranes and oxidation of proteins and DNA.

The geographic distribution of G6PD deficiency is instructive. Frequencies as high as 25% occur in tropical Africa, parts of the Middle East, and Southeast Asia, areas where malaria is most prevalent. In addition to such epidemiological observations, in vitro studies show that growth of one malaria parasite, *Plasmodium falciparum*, is inhibited in G6PD-deficient erythrocytes. The parasite is very sensitive to oxidative damage and is killed by a level of oxidative stress that is tolerable to a G6PD-deficient human host. Because the advantage of resistance to malaria balances the disadvantage of lowered resistance to oxidative damage, natural selection sustains the G6PD-deficient genotype in human populations where malaria is prevalent. Only under overwhelming oxidative stress, caused by drugs, herbicides, or divicine, does G6PD deficiency cause serious medical problems.

An antimalarial drug such as primaquine is believed to act by causing oxidative stress to the parasite. It is ironic that antimalarial drugs can cause illness through the same biochemical mechanism that provides resistance to malaria. Divicine also acts as an antimalarial drug, and ingestion of fava beans may protect against malaria. By refusing to eat falafel, many Pythagoreans with normal G6PD activity may have unwittingly increased their risk of malaria!

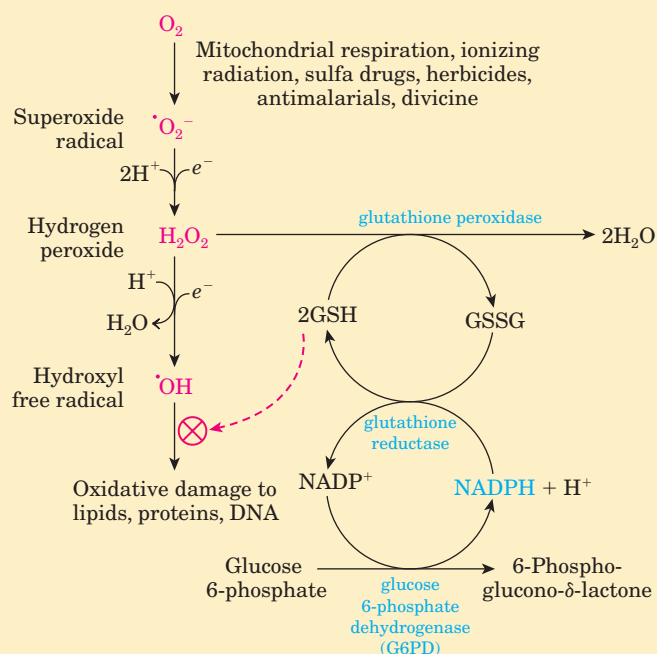
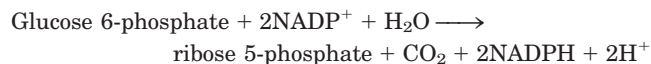


FIGURE 1 Role of NADPH and glutathione in protecting cells against highly reactive oxygen derivatives. Reduced glutathione (GSH) protects the cell by destroying hydrogen peroxide and hydroxyl free radicals. Regeneration of GSH from its oxidized form (GSSG) requires the NADPH produced in the glucose 6-phosphate dehydrogenase reaction.

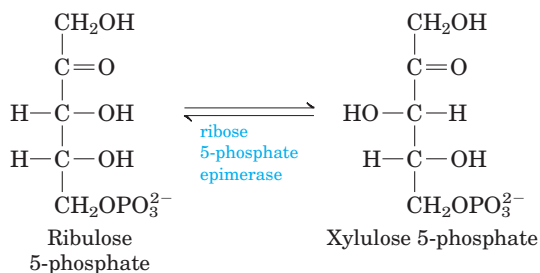
a second molecule of NADPH. **Phosphopentose isomerase** converts ribulose 5-phosphate to its aldose isomer, ribose 5-phosphate. In some tissues, the pentose phosphate pathway ends at this point, and its overall equation is



The net result is the production of NADPH, a reductant for biosynthetic reactions, and ribose 5-phosphate, a precursor for nucleotide synthesis.

The Nonoxidative Phase Recycles Pentose Phosphates to Glucose 6-Phosphate

In tissues that require primarily NADPH, the pentose phosphates produced in the oxidative phase of the pathway are recycled into glucose 6-phosphate. In this non-oxidative phase, ribulose 5-phosphate is first epimerized to xylulose 5-phosphate:



Then, in a series of rearrangements of the carbon skeletons (Fig. 14-22), six five-carbon sugar phosphates are

converted to five six-carbon sugar phosphates, completing the cycle and allowing continued oxidation of glucose 6-phosphate with production of NADPH. Continued recycling leads ultimately to the conversion of glucose 6-phosphate to six CO_2 . Two enzymes unique to the pentose phosphate pathway act in these interconversions of sugars: transketolase and transaldolase.

Transketolase catalyzes the transfer of a two-carbon fragment from a ketose donor to an aldose acceptor (Fig. 14-23a). In its first appearance in the pentose phosphate pathway, transketolase transfers C-1 and C-2 of xylulose 5-phosphate to ribose 5-phosphate, forming the seven-carbon product sedoheptulose 7-phosphate (Fig. 14-23b). The remaining three-carbon fragment from xylulose is glyceraldehyde 3-phosphate.

Next, **transaldolase** catalyzes a reaction similar to the aldolase reaction of glycolysis: a three-carbon fragment is removed from sedoheptulose 7-phosphate and condensed with glyceraldehyde 3-phosphate, forming fructose 6-phosphate and the tetrose erythrose 4-phosphate (Fig. 14-24). Now transketolase acts again, forming fructose 6-phosphate and glyceraldehyde 3-phosphate from erythrose 4-phosphate and xylulose 5-phosphate (Fig. 14-25). Two molecules of glyceraldehyde 3-phosphate formed by two iterations of these reactions can be converted to a molecule of fructose 1,6-bisphosphate as in gluconeogenesis (Fig. 14-16), and finally FBPase-1 and phosphohexose isomerase convert fructose 1,6-bisphosphate to glucose 6-phosphate. The cycle is complete: six pentose phosphates have been converted to five hexose phosphates (Fig. 14-22b).

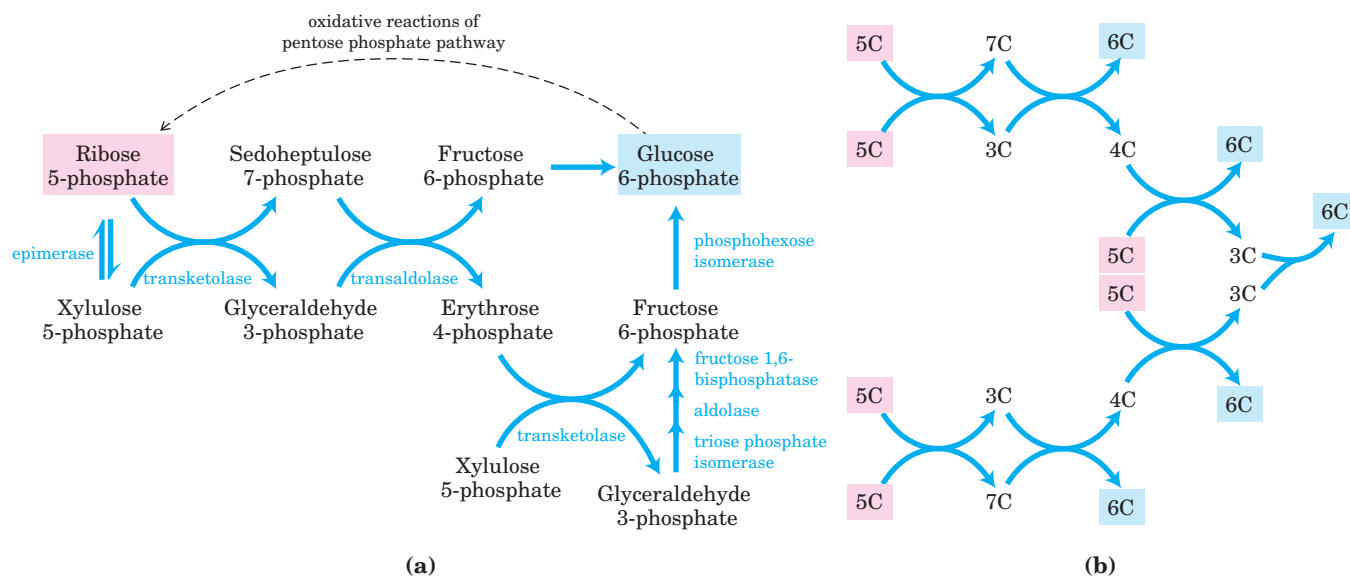
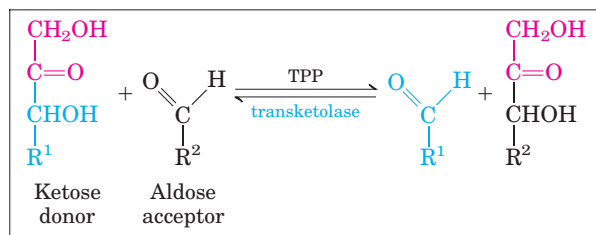
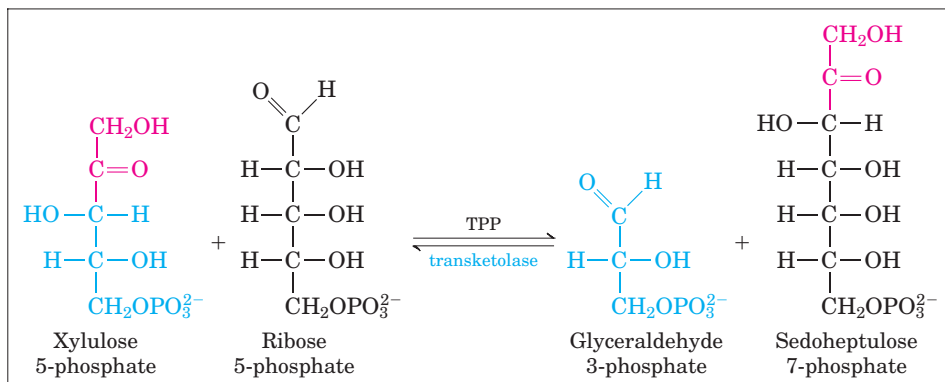


FIGURE 14-22 Nonoxidative reactions of the pentose phosphate pathway. (a) These reactions convert pentose phosphates to hexose phosphates, allowing the oxidative reactions (see Fig. 14-21) to continue. The enzymes transketolase and transaldolase are specific to this pathway; the other enzymes also serve in the glycolytic or gluconeogenic pathways. (b) A schematic diagram showing the pathway

from six pentoses (5C) to five hexoses (6C). Note that this involves two sets of the interconversions shown in (a). Every reaction shown here is reversible; unidirectional arrows are used only to make clear the direction of the reactions during continuous oxidation of glucose 6-phosphate. In the light-independent reactions of photosynthesis, the direction of these reactions is reversed (see Fig. 20-10).



(a)



(b)

FIGURE 14-23 The first reaction catalyzed by transketolase. (a) The general reaction catalyzed by transketolase is the transfer of a two-carbon group, carried temporarily on enzyme-bound TPP, from a ketose donor to an aldose acceptor. (b) Conversion of two pentose phosphates to a triose phosphate and a seven-carbon sugar phosphate, sedoheptulose 7-phosphate.

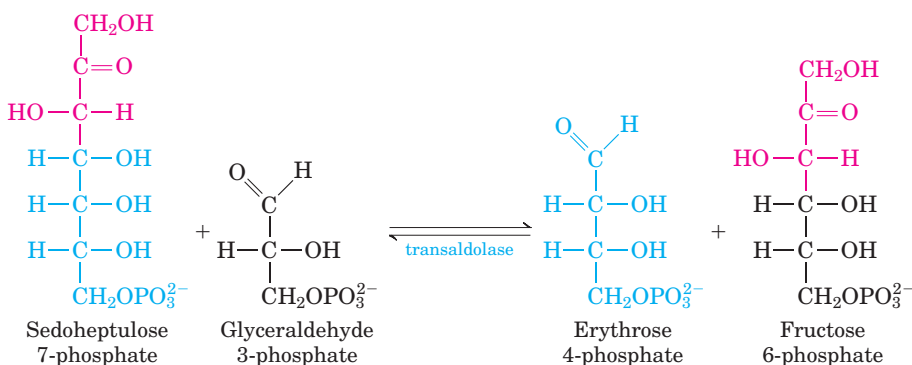


FIGURE 14-24 The reaction catalyzed by transaldolase.

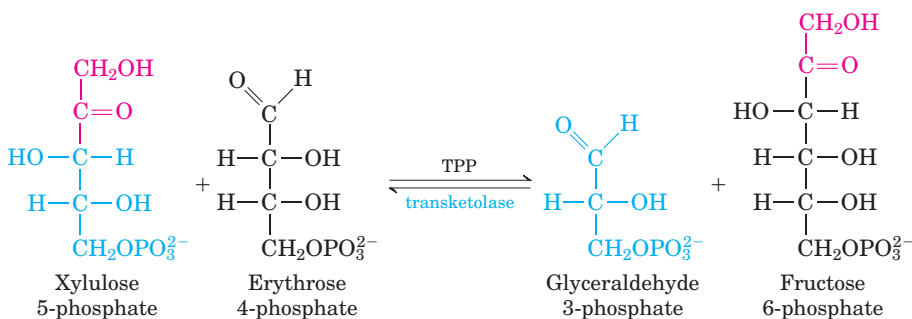
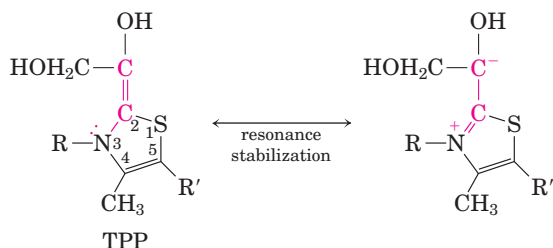
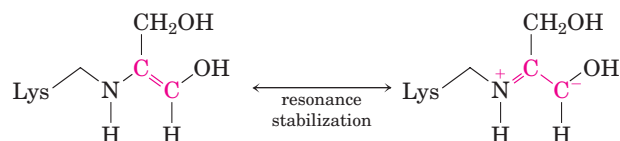


FIGURE 14-25 The second reaction catalyzed by transketolase.

Transketolase requires the cofactor thiamine pyrophosphate (TPP), which stabilizes a two-carbon carbanion in this reaction (Fig. 14-26a), just as it does in the pyruvate decarboxylase reaction (Fig. 14-13). Transaldolase uses a Lys side chain to form a Schiff base with the carbonyl group of its substrate, a ketose,

thereby stabilizing a carbanion (Fig. 14-26b) that is central to the reaction mechanism.

The process described in Figure 14-21 is known as the **oxidative pentose phosphate pathway**. The first two steps are oxidations with large, negative standard free-energy changes and are essentially irreversible in

(a) Transketolase**(b) Transaldolase**

Protonated Schiff base

FIGURE 14-26 Carbanion intermediates stabilized by covalent interactions with transketolase and transaldolase. **(a)** The ring of TPP stabilizes the two-carbon carbanion carried by transketolase; see Fig. 14-13 for the chemistry of TPP action. **(b)** In the transaldolase reaction, the protonated Schiff base formed between the ϵ -amino group of a Lys side chain and the substrate stabilizes a three-carbon carbanion.

the cell. The reactions of the nonoxidative part of the pentose phosphate pathway (Fig. 14-22) are readily reversible and thus also provide a means of converting hexose phosphates to pentose phosphates. As we shall see in Chapter 20, a process that converts hexose phosphates to pentose phosphates is crucial to the photosynthetic assimilation of CO_2 by plants. That pathway, the **reductive pentose phosphate pathway**, is essentially the reversal of the reactions shown in Figure 14-22 and employs many of the same enzymes.

All the enzymes in the pentose phosphate pathway are located in the cytosol, like those of glycolysis and most of those of gluconeogenesis. In fact, these three pathways are connected through several shared intermediates and enzymes. The glyceraldehyde 3-phosphate formed by the action of transketolase is readily converted to dihydroxyacetone phosphate by the glycolytic enzyme triose phosphate isomerase, and these two trioses can be joined by the aldolase as in gluconeogenesis, forming fructose 1,6-bisphosphate. Alternatively, the triose phosphates can be oxidized to pyruvate by the glycolytic reactions. The fate of the trioses is determined by the cell's relative needs for pentose phosphates, NADPH, and ATP.

Wernicke-Korsakoff Syndrome Is Exacerbated by a Defect in Transketolase



In humans with Wernicke-Korsakoff syndrome, a mutation in the gene for transketolase results in

an enzyme having an affinity for its coenzyme TPP that is one-tenth that of the normal enzyme. Although moderate deficiencies in the vitamin thiamine have little effect on individuals with an unmutated transketolase gene, in those with the altered gene, thiamine deficiency drops the level of TPP below that needed to saturate the enzyme. The lowering of transketolase activity slows the whole pentose phosphate pathway, and the result is the Wernicke-Korsakoff syndrome: severe memory loss, mental confusion, and partial paralysis. The syndrome is more common among alcoholics than in the general population; chronic alcohol consumption interferes with the intestinal absorption of some vitamins, including thiamine. ■

Glucose 6-Phosphate Is Partitioned between Glycolysis and the Pentose Phosphate Pathway

Whether glucose 6-phosphate enters glycolysis or the pentose phosphate pathway depends on the current needs of the cell and on the concentration of NADP^+ in the cytosol. Without this electron acceptor, the first reaction of the pentose phosphate pathway (catalyzed by G6PD) cannot proceed. When a cell is rapidly converting NADPH to NADP^+ in biosynthetic reductions, the level of NADP^+ rises, allosterically stimulating G6PD and thereby increasing the flux of glucose 6-phosphate through the pentose phosphate pathway (Fig. 14-27). When the demand for NADPH slows, the level of NADP^+ drops, the pentose phosphate pathway slows, and glucose 6-phosphate is instead used to fuel glycolysis.

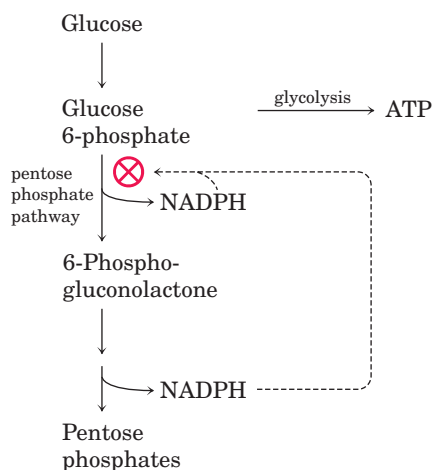


FIGURE 14-27 Role of NADPH in regulating the partitioning of glucose 6-phosphate between glycolysis and the pentose phosphate pathway. When NADPH is forming faster than it is being used for biosynthesis and glutathione reduction (see Fig. 14-20), $[\text{NADPH}]$ rises and inhibits the first enzyme in the pentose phosphate pathway. As a result, more glucose 6-phosphate is available for glycolysis.

SUMMARY 14.5 Pentose Phosphate Pathway of Glucose Oxidation

- The *oxidative* pentose phosphate pathway (phosphogluconate pathway, or hexose monophosphate pathway) brings about oxidation and decarboxylation at C-1 of glucose 6-phosphate, reducing NADP^+ to NADPH and producing pentose phosphates.
- NADPH provides reducing power for biosynthetic reactions, and ribose 5-phosphate is a precursor for nucleotide and nucleic acid synthesis. Rapidly growing tissues and tissues carrying out active biosynthesis of fatty acids, cholesterol, or steroid hormones send more glucose 6-phosphate through the pentose phosphate pathway than do tissues with less demand for pentose phosphates and reducing power.
- The first phase of the pentose phosphate pathway consists of two oxidations that convert glucose 6-phosphate to ribulose 5-phosphate and reduce NADP^+ to NADPH. The second phase comprises nonoxidative steps that convert pentose phosphates to glucose 6-phosphate, which begins the cycle again.
- In the second phase, transaldolase (with TPP as cofactor) and transketolase catalyze the interconversion of three-, four-, five-, six-, and seven-carbon sugars, with the reversible conversion of six pentose phosphates to five hexose phosphates. In the carbon-assimilating reactions of photosynthesis, the same enzymes catalyze the reverse process, called the *reductive* pentose phosphate pathway: conversion of five hexose phosphates to six pentose phosphates.
- A genetic defect in transketolase that lowers its affinity for TPP exacerbates the Wernicke-Korsakoff syndrome.
- Entry of glucose 6-phosphate either into glycolysis or into the pentose phosphate pathway is largely determined by the relative concentrations of NADP^+ and NADPH.

Key Terms

Terms in bold are defined in the glossary.

glycolysis 522	acyl phosphate 530	mutases 534	biotin 544
fermentation 522	substrate-level phosphorylation 531	isomerases 534	pentose phosphate pathway 549
lactic acid fermentation	respiration-linked phosphorylation 531	lactose intolerance	phosphogluconate pathway 549
hypoxia 523	phosphoenolpyruvate (PEP) 532	galactosemia 537	hexose monophosphate pathway 549
ethanol (alcohol) fermentation 523		thiamine pyrophosphate (TPP) 540	
isozymes 526		gluconeogenesis 543	

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Problems

1. Equation for the Preparatory Phase of Glycolysis

Write balanced biochemical equations for all the reactions in the catabolism of glucose to two molecules of glyceraldehyde 3-phosphate (the preparatory phase of glycolysis), including the standard free-energy change for each reaction. Then write the overall or net equation for the preparatory phase of glycolysis, with the net standard free-energy change.

2. The Payoff Phase of Glycolysis in Skeletal Muscle

In working skeletal muscle under anaerobic conditions, glyceraldehyde 3-phosphate is converted to pyruvate (the payoff phase of glycolysis), and the pyruvate is reduced to lactate. Write balanced biochemical equations for all the reactions in this process, with the standard free-energy change for each reaction. Then write the overall or net equation for the payoff phase of glycolysis (with lactate as the end product), including the net standard free-energy change.

3. Pathway of Atoms in Fermentation A “pulse-chase” experiment using ^{14}C -labeled carbon sources is carried out on a yeast extract maintained under strictly anaerobic conditions to produce ethanol. The experiment consists of incubating a small amount of ^{14}C -labeled substrate (the pulse) with the yeast extract just long enough for each intermediate in the fermentation pathway to become labeled. The label is then “chased” through the pathway by the addition of excess unlabeled glucose. The chase effectively prevents any further entry of labeled glucose into the pathway.

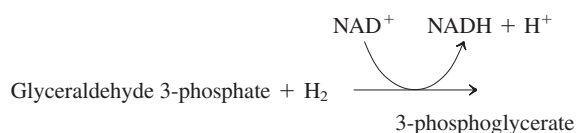
(a) If $[1\text{-}^{14}\text{C}]\text{glucose}$ (glucose labeled at C-1 with ^{14}C) is used as a substrate, what is the location of ^{14}C in the product ethanol? Explain.

(b) Where would ^{14}C have to be located in the starting glucose to ensure that all the ^{14}C activity is liberated as $^{14}\text{CO}_2$ during fermentation to ethanol? Explain.

4. Fermentation to Produce Soy Sauce Soy sauce is prepared by fermenting a salted mixture of soybeans and wheat with several microorganisms, including yeast, over a period of 8 to 12 months. The resulting sauce (after solids are removed) is rich in lactate and ethanol. How are these two compounds produced? To prevent the soy sauce from having a strong vinegar taste (vinegar is dilute acetic acid), oxygen must be kept out of the fermentation tank. Why?

5. Equivalence of Triose Phosphates ^{14}C -Labeled glyceraldehyde 3-phosphate was added to a yeast extract. After a short time, fructose 1,6-bisphosphate labeled with ^{14}C at C-3 and C-4 was isolated. What was the location of the ^{14}C label in the starting glyceraldehyde 3-phosphate? Where did the second ^{14}C label in fructose 1,6-bisphosphate come from? Explain.

6. Glycolysis Shortcut Suppose you discovered a mutant yeast whose glycolytic pathway was shorter because of the presence of a new enzyme catalyzing the reaction:



Would shortening the glycolytic pathway in this way benefit the cell? Explain.

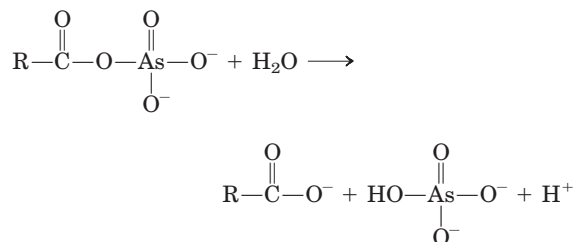
7. Role of Lactate Dehydrogenase During strenuous activity, the demand for ATP in muscle tissue is vastly increased. In rabbit leg muscle or turkey flight muscle, the ATP is produced almost exclusively by lactic acid fermentation. ATP is formed in the payoff phase of glycolysis by two reactions, promoted by phosphoglycerate kinase and pyruvate kinase. Suppose skeletal muscle were devoid of lactate dehydrogenase. Could it carry out strenuous physical activity; that is, could it generate ATP at a high rate by glycolysis? Explain.

8. Efficiency of ATP Production in Muscle The transformation of glucose to lactate in myocytes releases only about 7% of the free energy released when glucose is completely oxidized to CO_2 and H_2O . Does this mean that anaerobic glycolysis in muscle is a wasteful use of glucose? Explain.

9. Free-Energy Change for Triose Phosphate Oxidation

The oxidation of glyceraldehyde 3-phosphate to 1,3-bisphosphoglycerate, catalyzed by glyceraldehyde 3-phosphate dehydrogenase, proceeds with an unfavorable equilibrium constant ($K'_{\text{eq}} = 0.08$; $\Delta G'^{\circ} = 6.3 \text{ kJ/mol}$), yet the flow through this point in the glycolytic pathway proceeds smoothly. How does the cell overcome the unfavorable equilibrium?

10. Arsenate Poisoning Arsenate is structurally and chemically similar to inorganic phosphate (P_i), and many enzymes that require phosphate will also use arsenate. Organic compounds of arsenate are less stable than analogous phosphate compounds, however. For example, acyl *arsenates* decompose rapidly by hydrolysis:



On the other hand, acyl *phosphates*, such as 1,3-bisphosphoglycerate, are more stable and undergo further enzyme-catalyzed transformation in cells.

(a) Predict the effect on the net reaction catalyzed by glyceraldehyde 3-phosphate dehydrogenase if phosphate were replaced by arsenate.

(b) What would be the consequence to an organism if arsenate were substituted for phosphate? Arsenate is very toxic to most organisms. Explain why.

11. Requirement for Phosphate in Ethanol Fermentation

In 1906 Harden and Young, in a series of classic studies on the fermentation of glucose to ethanol and CO_2 by extracts of brewer's yeast, made the following observations. (1) Inorganic phosphate was essential to fermentation; when the supply of phosphate was exhausted, fermentation ceased before all the glucose was used. (2) During fermentation under these conditions, ethanol, CO_2 , and a hexose bisphosphate

accumulated. (3) When arsenate was substituted for phosphate, no hexose biphosphate accumulated, but the fermentation proceeded until all the glucose was converted to ethanol and CO_2 .

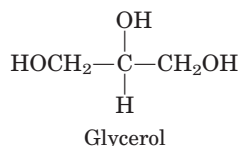
(a) Why did fermentation cease when the supply of phosphate was exhausted?

(b) Why did ethanol and CO_2 accumulate? Was the conversion of pyruvate to ethanol and CO_2 essential? Why? Identify the hexose biphosphate that accumulated. Why did it accumulate?

(c) Why did the substitution of arsenate for phosphate prevent the accumulation of the hexose biphosphate yet allow fermentation to ethanol and CO_2 to go to completion? (See Problem 10.)

12. Role of the Vitamin Niacin Adults engaged in strenuous physical activity require an intake of about 160 g of carbohydrate daily but only about 20 mg of niacin for optimal nutrition. Given the role of niacin in glycolysis, how do you explain the observation?

13. Metabolism of Glycerol Glycerol obtained from the breakdown of fat is metabolized by conversion to dihydroxyacetone phosphate, a glycolytic intermediate, in two enzyme-catalyzed reactions. Propose a reaction sequence for glycerol metabolism. On which known enzyme-catalyzed reactions is your proposal based? Write the net equation for the conversion of glycerol to pyruvate according to your scheme.

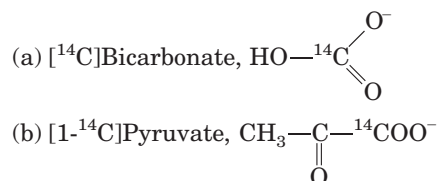


14. Severity of Clinical Symptoms Due to Enzyme Deficiency

The clinical symptoms of two forms of galactosemia—deficiency of galactokinase or of UDP-glucose:galactose 1-phosphate uridylyltransferase—show radically different severity. Although both types produce gastric discomfort after milk ingestion, deficiency of the transferase also leads to liver, kidney, spleen, and brain dysfunction and eventual death. What products accumulate in the blood and tissues with each type of enzyme deficiency? Estimate the relative toxicities of these products from the above information.

15. Muscle Wasting in Starvation One consequence of starvation is a reduction in muscle mass. What happens to the muscle proteins?

16. Pathway of Atoms in Gluconeogenesis A liver extract capable of carrying out all the normal metabolic reactions of the liver is briefly incubated in separate experiments with the following ^{14}C -labeled precursors:



Trace the pathway of each precursor through gluconeogenesis. Indicate the location of ^{14}C in all intermediates and in the product, glucose.

17. Pathway of CO_2 in Gluconeogenesis In the first bypass step of gluconeogenesis, the conversion of pyruvate to phosphoenolpyruvate, pyruvate is carboxylated by pyruvate carboxylase to oxaloacetate, which is subsequently decarboxylated by PEP carboxykinase to yield phosphoenolpyruvate. The observation that the addition of CO_2 is directly followed by the loss of CO_2 suggests that ^{14}C of $^{14}\text{CO}_2$ would not be incorporated into PEP, glucose, or any intermediates in gluconeogenesis. However, when a rat liver preparation synthesizes glucose in the presence of $^{14}\text{CO}_2$, ^{14}C slowly appears in PEP and eventually at C-3 and C-4 of glucose. How does the ^{14}C label get into PEP and glucose? (Hint: During gluconeogenesis in the presence of $^{14}\text{CO}_2$, several of the four-carbon citric acid cycle intermediates also become labeled.)

18. Energy Cost of a Cycle of Glycolysis and Gluconeogenesis What is the cost (in ATP equivalents) of transforming glucose to pyruvate via glycolysis and back again to glucose via gluconeogenesis?

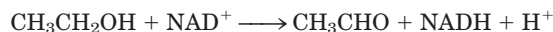
19. Glucogenic Substrates A common procedure for determining the effectiveness of compounds as precursors of glucose in mammals is to starve the animal until the liver glycogen stores are depleted and then administer the compound in question. A substrate that leads to a *net* increase in liver glycogen is termed glucogenic, because it must first be converted to glucose 6-phosphate. Show by means of known enzymatic reactions which of the following substances are glucogenic:

- (a) Succinate, $^- \text{OOC}-\text{CH}_2-\text{CH}_2-\text{COO}^-$
- (b) Glycerol, $\begin{array}{c} \text{OH} \quad \text{OH} \quad \text{OH} \\ | \quad | \quad | \\ \text{CH}_2-\text{C}-\text{CH}_2 \\ | \\ \text{H} \end{array}$
- (c) Acetyl-CoA, $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{S-CoA} \end{array}$
- (d) Pyruvate, $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{COO}^- \end{array}$
- (e) Butyrate, $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{COO}^-$



20. Ethanol Affects Blood Glucose Levels

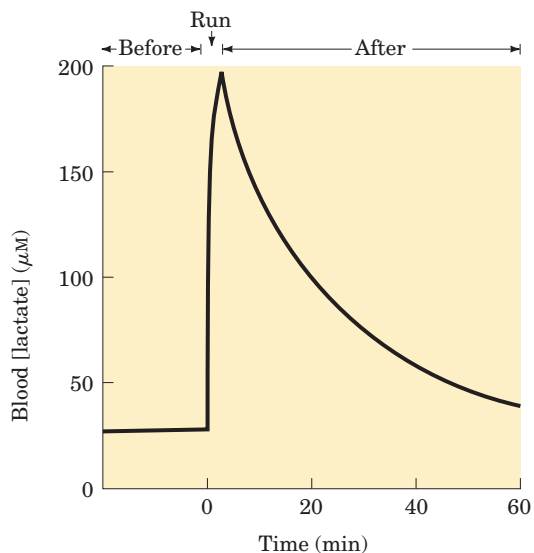
The consumption of alcohol (ethanol), especially after periods of strenuous activity or after not eating for several hours, results in a deficiency of glucose in the blood, a condition known as hypoglycemia. The first step in the metabolism of ethanol by the liver is oxidation to acetaldehyde, catalyzed by liver alcohol dehydrogenase:



Explain how this reaction inhibits the transformation of lactate to pyruvate. Why does this lead to hypoglycemia?

21. Blood Lactate Levels during Vigorous Exercise

The concentrations of lactate in blood plasma before, during, and after a 400 m sprint are shown in the graph.

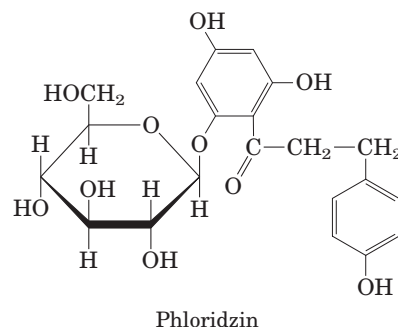


- What causes the rapid rise in lactate concentration?
- What causes the decline in lactate concentration after completion of the sprint? Why does the decline occur more slowly than the increase?
- Why is the concentration of lactate not zero during the resting state?

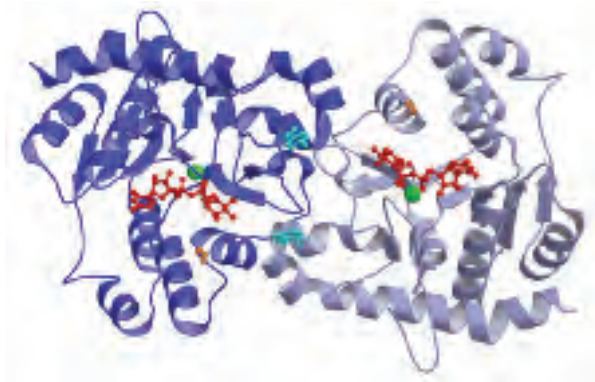
22. Relationship between Fructose 1,6-Bisphosphatase and Blood Lactate Levels A congenital defect in the liver enzyme fructose 1,6-bisphosphatase results in abnormally high levels of lactate in the blood plasma. Explain.

23. Effect of Phloridzin on Carbohydrate Metabolism

Phloridzin, a toxic glycoside from the bark of the pear tree, blocks the normal reabsorption of glucose from the kidney tubule, thus causing blood glucose to be almost completely excreted in the urine. In an experiment, rats fed phloridzin and sodium succinate excreted about 0.5 mol of glucose (made by gluconeogenesis) for every 1 mol of sodium succinate ingested. How is the succinate transformed to glucose? Explain the stoichiometry.



24. Excess O_2 Uptake during Gluconeogenesis Lactate absorbed by the liver is converted to glucose, with the input of 6 mol of ATP for every mole of glucose produced. The extent of this process in a rat liver preparation can be monitored by administering $[^{14}C]$ lactate and measuring the amount of $[^{14}C]$ glucose produced. Because the stoichiometry of O_2 consumption and ATP production is known (about 5 ATP per O_2), we can predict the extra O_2 consumption above the normal rate when a given amount of lactate is administered. However, when the extra O_2 used in the synthesis of glucose from lactate is actually measured, it is always higher than predicted by known stoichiometric relationships. Suggest a possible explanation for this observation.



chapter 15

PRINCIPLES OF METABOLIC REGULATION: GLUCOSE AND GLYCOGEN

- 15.1 The Metabolism of Glycogen in Animals 562**
- 15.2 Regulation of Metabolic Pathways 571**
- 15.3 Coordinated Regulation of Glycolysis and Gluconeogenesis 575**
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- 15.5 Analysis of Metabolic Control 591**

Formation of liver glycogen from lactic acid is thus seen to establish an important connection between the metabolism of the muscle and that of the liver. Muscle glycogen becomes available as blood sugar through the intervention of the liver, and blood sugar in turn is converted into muscle glycogen. There exists therefore a complete cycle of the glucose molecule in the body . . . Epinephrine was found to accelerate this cycle in the direction of muscle glycogen to liver glycogen . . . Insulin, on the other hand, was found to accelerate the cycle in the direction of blood glucose to muscle glycogen.

—C. F. Cori and G. T. Cori, *article in Journal of Biological Chemistry*, 1929

Metabolic regulation, a central theme in biochemistry, is one of the most remarkable features of a living cell. Of the thousands of enzyme-catalyzed reactions that can take place in a cell, there is probably not one that escapes some form of regulation. Although it

is convenient (and perhaps essential) in writing a textbook to divide metabolic processes into “pathways” that play discrete roles in the cell’s economy, no such separation exists inside the cell. Rather, each of the pathways we discuss in this book is inextricably intertwined with all the other cellular pathways in a multidimensional network of reactions (Fig. 15–1). For example, in Chapter 14 we discussed three possible fates for glucose 6-phosphate in a hepatocyte: passage into glycolysis for the production of ATP, passage into the pentose phosphate pathway for the production of NADPH and pentose phosphates, or hydrolysis to glucose and phosphate to replenish blood glucose. In fact, glucose 6-phosphate has a number of other possible fates; it may, for example, be used to synthesize other sugars, such as glucosamine, galactose, galactosamine, fucose, and neuraminic acid, for use in protein glycosylation, or it may be partially degraded to provide acetyl-CoA for fatty acid and sterol synthesis. In the extreme case, the bacterium *Escherichia coli* can use glucose to produce the carbon skeleton of every one of its molecules. When a cell “decides” to use glucose 6-phosphate for one purpose, that decision affects all the other pathways for which glucose 6-phosphate is a precursor or intermediate; any change in the allocation of glucose 6-phosphate to one pathway affects, directly or indirectly, the metabolite flow through all the others.

Such changes in allocation are common in the life of a cell. Louis Pasteur was the first to describe the large (greater than tenfold) increase in glucose consumption by a yeast culture when it was shifted from aerobic to anaerobic conditions. This phenomenon, called the

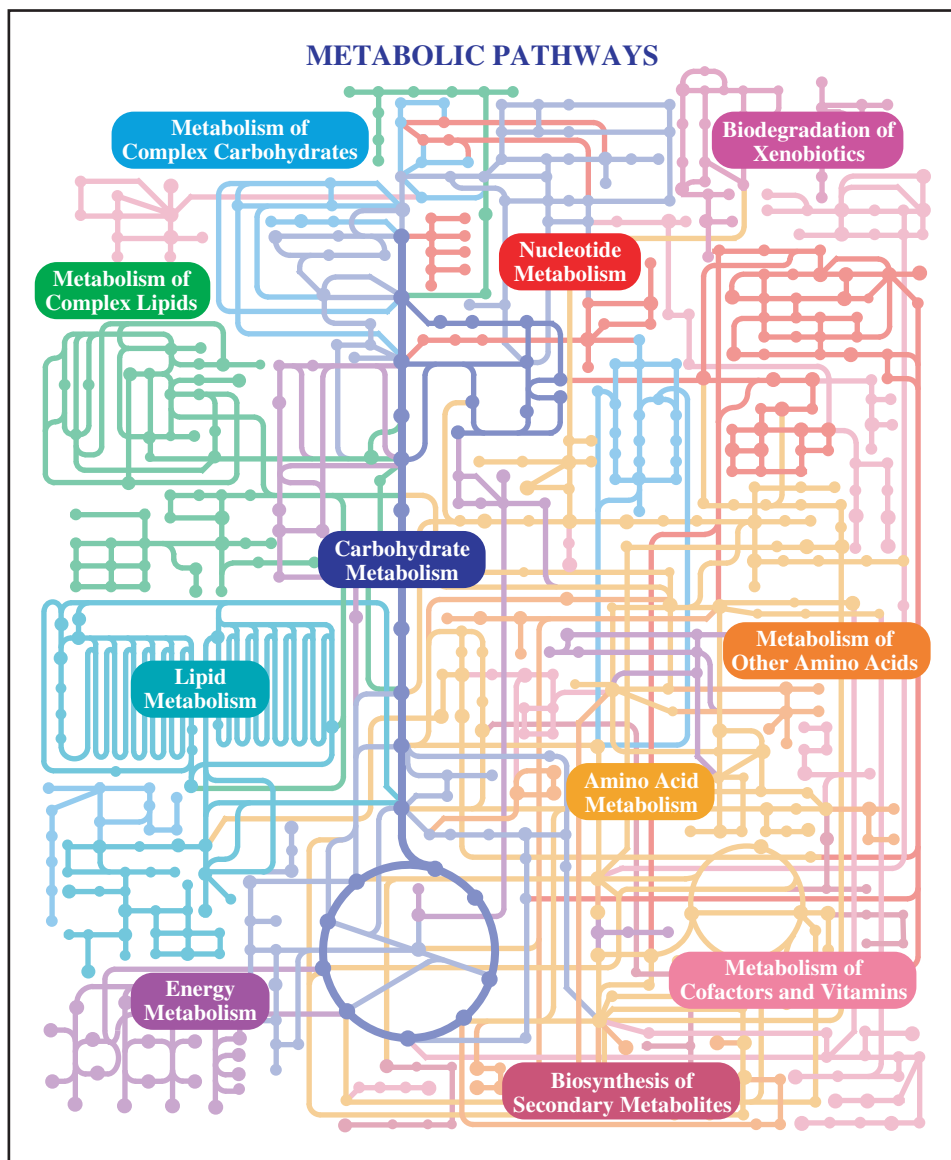


FIGURE 15-1 Metabolism as a three-dimensional meshwork. A typical eukaryotic cell has the capacity to make about 30,000 different proteins, which catalyze thousands of different reactions involving many hundreds of metabolites, most shared by more than one “pathway.” This overview image of metabolic pathways is from the online KEGG (Kyoto Encyclopedia of Genes and Genomes) PATHWAY database (www.genome.ad.jp/kegg/pathway/map/map01100.html). Each area can be further expanded for increasingly detailed information, to the level of specific enzymes and intermediates.

Pasteur effect, occurs without a significant change in the concentration of ATP or any of the hundreds of metabolic intermediates and products derived from glucose. A similar change takes place in cells of skeletal muscle when a sprinter leaves the starting blocks. The ability of a cell to carry out all these interlocking metabolic processes simultaneously—obtaining every product in the amount needed and at the right time, in the face of major perturbations from outside, and without generating leftovers—is an *astounding* accomplishment.

In this chapter we look at mechanisms of metabolic regulation, using the pathways in which glucose is an intermediate to illustrate some general principles. First we consider the pathways by which glycogen is synthesized and broken down, a very well-studied case of meta-

bolic regulation. Then we look at the general roles of regulation in achieving metabolic homeostasis. Focusing on the pathways that connect pyruvate with glycogen in both directions, we next consider the specific regulatory properties of the participating enzymes and the ways in which the cell accomplishes coordinated regulation of catabolic and anabolic pathways. Finally, we discuss metabolic control analysis, a system for treating complex metabolic interactions quantitatively, and consider some surprising results of its application.

In selecting carbohydrate metabolism to illustrate the principles of metabolic regulation, we have artificially separated the metabolism of fats and carbohydrates. In fact, these two activities are very tightly integrated, as we shall see in Chapter 23.

15.1 The Metabolism of Glycogen in Animals

In a wide range of organisms, excess glucose is converted to polymeric forms for storage—glycogen in vertebrates and many microorganisms, starch in plants. In vertebrates, glycogen is found primarily in the liver and skeletal muscle; it may represent up to 10% of the weight of liver and 1% to 2% of the weight of muscle. If this much glucose were dissolved in the cytosol of a hepatocyte, its concentration would be about 0.4 M, enough to dominate the osmotic properties of the cell. When stored as a long polymer (glycogen), however, the same mass of glucose has a concentration of only 0.01 μM . Glycogen is stored in large cytosolic granules. The elementary particle of glycogen, the β -particle, about 21 nm in diameter, consists of up to 55,000 glucose residues with about 2,000 nonreducing ends. Twenty to 40 of these particles cluster together to form α -rosettes, easily seen with the microscope in tissue samples from well-fed animals (Fig. 15–2) but essentially absent after a 24-hour fast.

The glycogen in muscle is there to provide a quick source of energy for either aerobic or anaerobic metabolism. Muscle glycogen can be exhausted in less than an hour during vigorous activity. Liver glycogen serves as a reservoir of glucose for other tissues when dietary glucose is not available (between meals or during a fast); this is especially important for the neurons of the brain, which cannot use fatty acids as fuel. Liver glycogen can be depleted in 12 to 24 hours. In humans, the total amount of energy stored as glycogen is far less than the

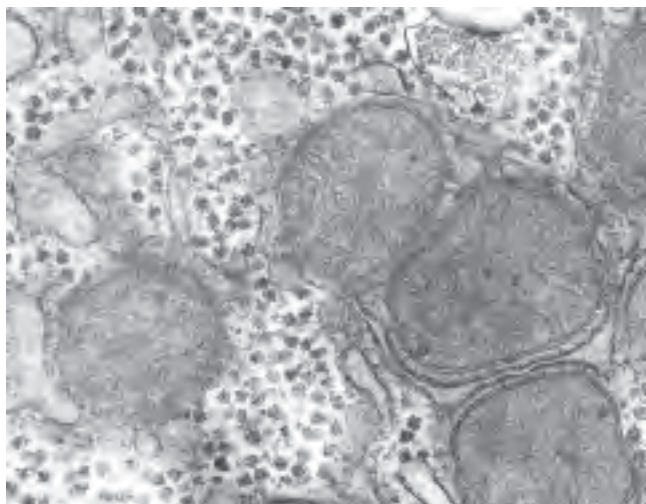


FIGURE 15–2 Glycogen granules in a hepatocyte. Glycogen is a storage form of carbohydrate in cells, especially hepatocytes, as illustrated here. Glycogen appears as electron-dense particles, often in aggregates or rosettes. In hepatocytes the glycogen is closely associated with tubules of the smooth endoplasmic reticulum. Many mitochondria are also present.

amount stored as fat (triacylglycerol) (see Table 23–5), but fats cannot be converted to glucose in mammals and cannot be catabolized anaerobically.

Glycogen granules are complex aggregates of glycogen and the enzymes that synthesize it and degrade it, as well as the machinery for regulating these enzymes. The general mechanisms for storing and mobilizing glycogen are the same in muscle and liver, but the enzymes differ in subtle yet important ways that reflect the different roles of glycogen in the two tissues. Glycogen is also obtained in the diet and broken down in the gut, and this involves a separate set of hydrolytic enzymes that convert glycogen (and starch) to free glucose.

The transformations of glucose discussed in this chapter and in Chapter 14 are central to the metabolism of most organisms, microbial, animal, or plant. We begin with a discussion of the *catabolic* pathways from glycogen to glucose 6-phosphate (**glycogenolysis**) and from glucose 6-phosphate to pyruvate (**glycolysis**), then turn to the *anabolic* pathways from pyruvate to glucose (**gluconeogenesis**) and from glucose to glycogen (**glycogenesis**).

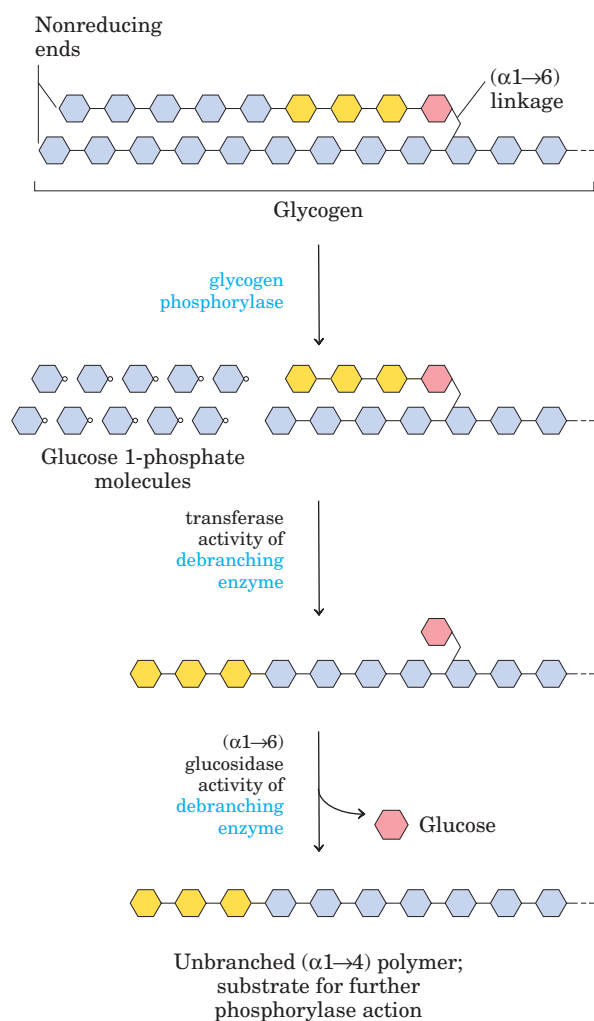
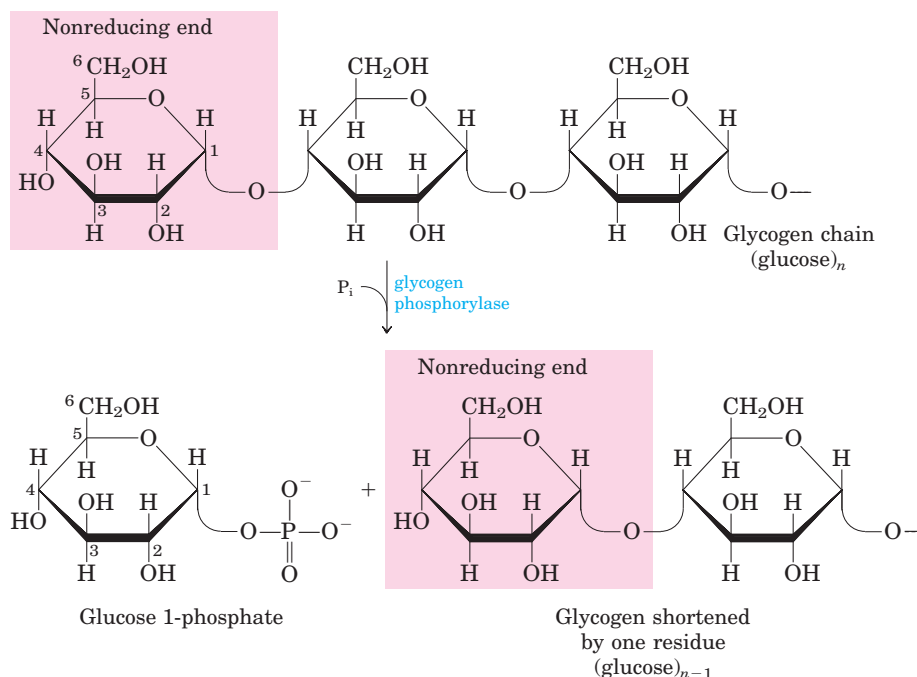
Glycogen Breakdown Is Catalyzed by Glycogen Phosphorylase

In skeletal muscle and liver, the glucose units of the outer branches of glycogen enter the glycolytic pathway through the action of three enzymes: glycogen phosphorylase, glycogen debranching enzyme, and phosphoglucomutase. Glycogen phosphorylase catalyzes the reaction in which an ($\alpha 1 \rightarrow 4$) glycosidic linkage between two glucose residues at a nonreducing end of glycogen undergoes attack by inorganic phosphate (P_i), removing the terminal glucose residue as α -D-glucose 1-phosphate (Fig. 15–3). This *phosphorolysis* reaction is different from the *hydrolysis* of glycosidic bonds by amylase during intestinal degradation of dietary glycogen and starch. In phosphorolysis, some of the energy of the glycosidic bond is preserved in the formation of the phosphate ester, glucose 1-phosphate.

Pyridoxal phosphate is an essential cofactor in the glycogen phosphorylase reaction; its phosphate group acts as a general acid catalyst, promoting attack by P_i on the glycosidic bond. (This is an unusual role for this cofactor; its more typical role is as a cofactor in amino acid metabolism; see Fig. 18–6.)

Glycogen phosphorylase acts repetitively on the nonreducing ends of glycogen branches until it reaches a point four glucose residues away from an ($\alpha 1 \rightarrow 6$) branch point (see Fig. 7–15), where its action stops. Further degradation by glycogen phosphorylase can occur only after the **debranching enzyme**, formally known as **oligo ($\alpha 1 \rightarrow 6$) to ($\alpha 1 \rightarrow 4$) glucantransferase**, catalyzes two successive reactions that transfer

FIGURE 15-3 Removal of a terminal glucose residue from the nonreducing end of a glycogen chain by glycogen phosphorylase. This process is repetitive; the enzyme removes successive glucose residues until it reaches the fourth glucose unit from a branch point (see Fig. 15-4).



branches (Fig. 15-4). Once these branches are transferred and the glucosyl residue at C-6 is hydrolyzed, glycogen phosphorylase activity can continue.

Glucose 1-Phosphate Can Enter Glycolysis or, in Liver, Replenish Blood Glucose

Glucose 1-phosphate, the end product of the glycogen phosphorylase reaction, is converted to glucose 6-phosphate by **phosphoglucomutase**, which catalyzes the reversible reaction



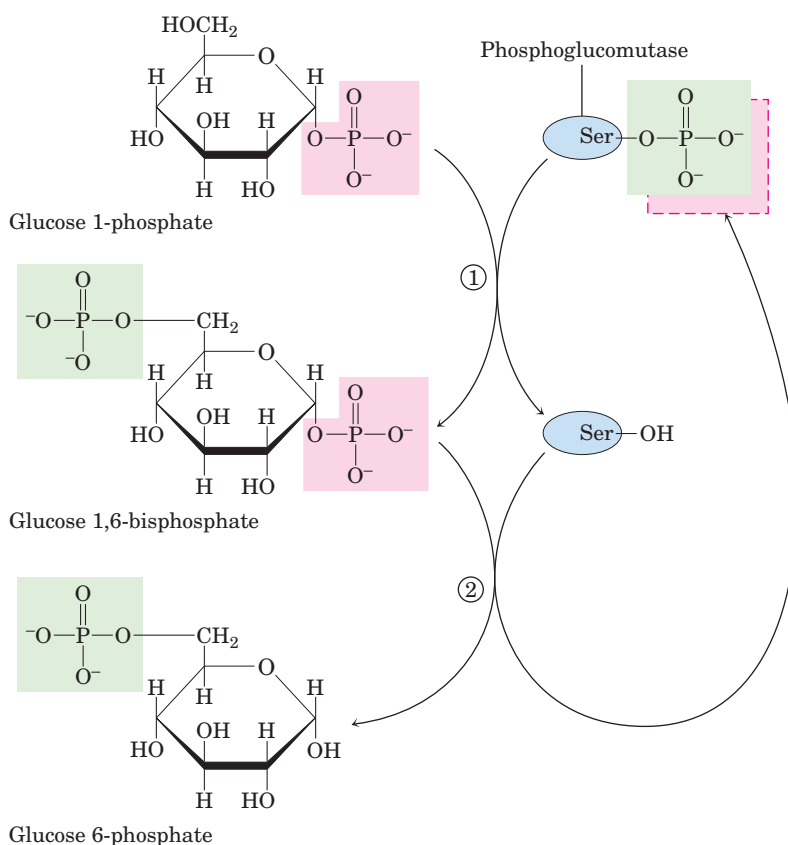
Initially phosphorylated at a Ser residue, the enzyme donates a phosphoryl group to C-6 of the substrate, then accepts a phosphoryl group from C-1 (Fig. 15-5).

The glucose 6-phosphate formed from glycogen in skeletal muscle can enter glycolysis and serve as an energy source to support muscle contraction. In liver,

FIGURE 15-4 Glycogen breakdown near an (α1→6) branch point.

Following sequential removal of terminal glucose residues by glycogen phosphorylase (see Fig. 15-3), glucose residues near a branch are removed in a two-step process that requires a bifunctional “debranching enzyme.” First, the transferase activity of the enzyme shifts a block of three glucose residues from the branch to a nearby nonreducing end, to which they are reattached in (α1→4) linkage. The single glucose residue remaining at the branch point, in (α1→6) linkage, is then released as free glucose by the enzyme’s (α1→6) glucosidase activity. The glucose residues are shown in shorthand form, which omits the —H, —OH, and —CH₂OH groups from the pyranose rings.

FIGURE 15-5 Reaction catalyzed by phosphoglucomutase. The reaction begins with the enzyme phosphorylated on a Ser residue. In step ①, the enzyme donates its phosphoryl group (green) to glucose 1-phosphate, producing glucose 1,6-bisphosphate. In step ②, the phosphoryl group at C-1 of glucose 1,6-bisphosphate (red) is transferred back to the enzyme, re-forming the phosphoenzyme and producing glucose 6-phosphate.



glycogen breakdown serves a different purpose: to release glucose into the blood when the blood glucose level drops, as it does between meals. This requires an enzyme, glucose 6-phosphatase, that is present in liver and kidney but not in other tissues. The enzyme is an integral membrane protein of the endoplasmic reticulum, predicted to contain nine transmembrane helices,

with its active site on the luminal side of the ER. Glucose 6-phosphate formed in the cytosol is transported into the ER lumen by a specific transporter (T1) (Fig. 15-6) and hydrolyzed at the luminal surface by the glucose 6-phosphatase. The resulting P_i and glucose are thought to be carried back into the cytosol by two different transporters (T2 and T3), and the glucose leaves

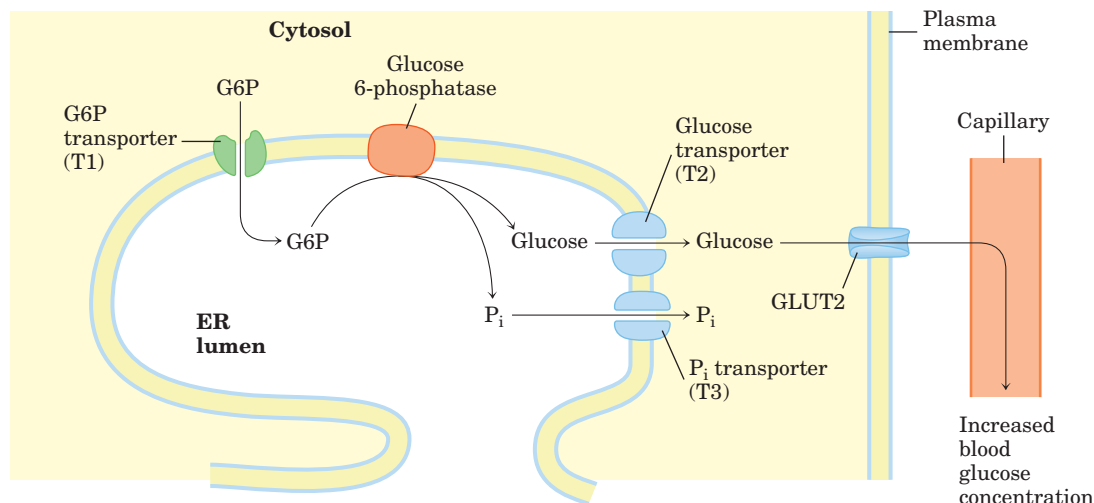


FIGURE 15-6 Hydrolysis of glucose 6-phosphate by glucose 6-phosphatase of the ER. The catalytic site of glucose 6-phosphatase faces the lumen of the ER. A glucose 6-phosphate (G6P) transporter (T1) carries the substrate from the cytosol to the lumen, and the prod-

ucts glucose and P_i pass to the cytosol on specific transporters (T2 and T3). Glucose leaves the cell via the GLUT2 transporter in the plasma membrane.

the hepatocyte via yet another transporter in the plasma membrane (GLUT2). Notice that by having the active site of glucose 6-phosphatase inside the ER lumen, the cell separates this reaction from the process of glycolysis, which takes place in the cytosol and would be aborted by the action of glucose 6-phosphatase. Genetic defects in either glucose 6-phosphatase or T1 lead to serious derangement of glycogen metabolism, resulting in type Ia glycogen storage disease (Box 15-1).

Because muscle and adipose tissue lack glucose 6-phosphatase, they cannot convert the glucose 6-phosphate formed by glycogen breakdown to glucose, and these tissues therefore do not contribute glucose to the blood.

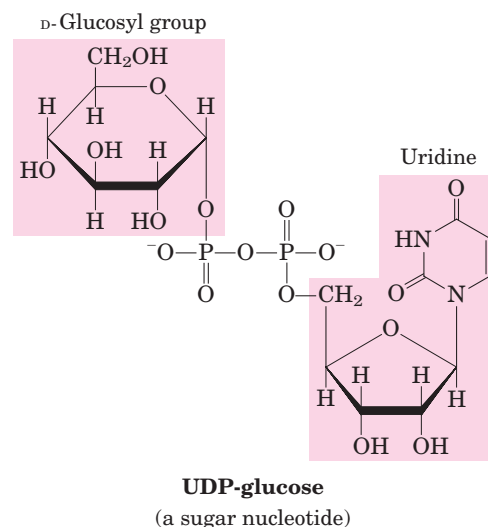
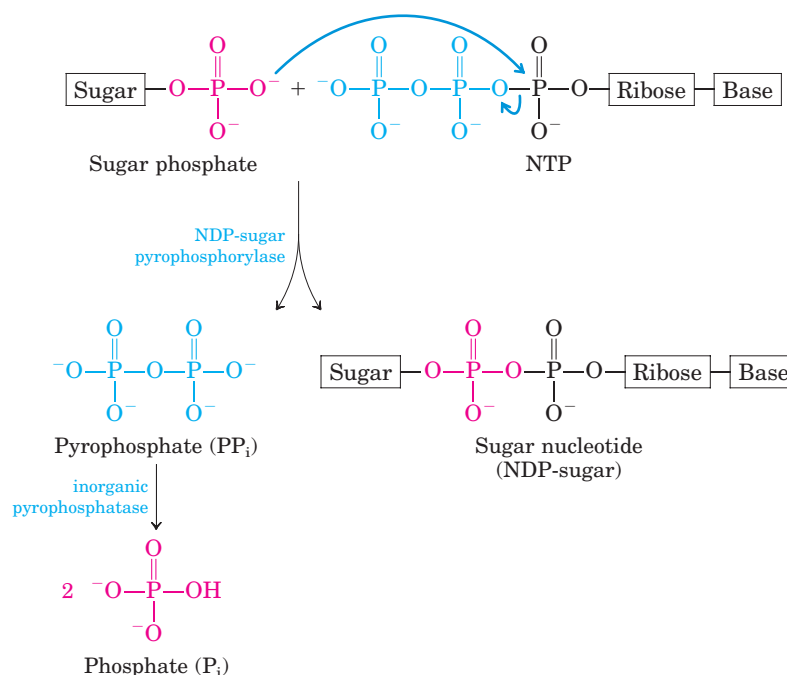
The Sugar Nucleotide UDP-Glucose Donates Glucose for Glycogen Synthesis

Many of the reactions in which hexoses are transformed or polymerized involve **sugar nucleotides**, compounds in which the anomeric carbon of a sugar is activated by attachment to a nucleotide through a phosphate ester linkage. Sugar nucleotides are the substrates for polymerization of monosaccharides into disaccharides, glycogen, starch, cellulose, and more complex extracellular polysaccharides. They are also key intermediates in the production of the aminohexoses and deoxyhexoses found in some of these polysaccharides, and in the synthesis of vitamin C (L-ascorbic acid). The role of sugar nucleotides in the biosynthesis of glycogen and many other carbohydrate derivatives was first discovered by the Argentine biochemist Luis Leloir.



Luis Leloir, 1906–1987

FIGURE 15-7 Formation of a sugar nucleotide. A condensation reaction occurs between a nucleoside triphosphate (NTP) and a sugar phosphate. The negatively charged oxygen on the sugar phosphate serves as a nucleophile, attacking the α phosphate of the nucleoside triphosphate and displacing pyrophosphate. The reaction is pulled in the forward direction by the hydrolysis of PP_i by inorganic pyrophosphatase.



The suitability of sugar nucleotides for biosynthetic reactions stems from several properties:

1. Their formation is metabolically irreversible, contributing to the irreversibility of the synthetic pathways in which they are intermediates. The condensation of a nucleoside triphosphate with a hexose 1-phosphate to form a sugar nucleotide has a small positive free-energy change, but the reaction releases PP_i , which is rapidly hydrolyzed by inorganic pyrophosphatase in a reaction that is strongly exergonic ($\Delta G'^{\circ} = -19.2$ kJ/mol; Fig. 15-7). This keeps the cellular concentration of PP_i low, ensuring that the actual free-energy change in

BOX 15-1 WORKING IN BIOCHEMISTRY

Carl and Gerty Cori: Pioneers in Glycogen Metabolism and Disease

Much of what is written in present-day biochemistry textbooks about the metabolism of glycogen was discovered between about 1925 and 1950 by the remarkable husband and wife team of Carl F. Cori and Gerty T. Cori. Both trained in medicine in Europe at the end of World War I (she completed premedical studies and medical school in one year!). They left Europe together in 1922 to establish research laboratories in the United States, first for nine years in Buffalo, New York, at what is now the Roswell Park Memorial Institute, then from 1931 until the end of their lives at Washington University in St. Louis.



The Coris in Gerty Cori's laboratory, around 1947.

In their early physiological studies of the origin and fate of glycogen in animal muscle, the Coris demonstrated the conversion of glycogen to lactate in tissues, movement of lactate in the blood to the liver, and, in the liver, reconversion of lactate to glycogen—a pathway that came to be known as the Cori cycle (see Fig. 23-18). Pursuing these observations at the biochemical level, they showed that glycogen was mobilized in a phosphorolysis reaction catalyzed by the enzyme they discovered, glycogen phosphorylase. They identified the product of this reaction (the “Cori ester”) as glucose 1-phosphate and showed that it could be reincorporated into glycogen in the reverse reaction. Although this did not prove to be the reaction by which glycogen is synthesized in cells, it was the first *in vitro* demonstration of the synthesis of a macromolecule from simple monomeric subunits, and it inspired others to search for polymerizing enzymes. Arthur Kornberg, discoverer of the first DNA polymerase, has said of his experience in the Coris' lab, “Glycogen phosphorylase, not base pairing, was what led me to DNA polymerase.”



Gerty Cori became interested in human genetic diseases in which too much glycogen is stored in the liver. She was able to identify the biochemical defect in several of these diseases and to show that these diseases could be diagnosed by assays of the enzymes of glycogen metabolism in small samples of tissue obtained by biopsy. Table 1 summarizes what we now know about 13 genetic diseases of this sort. ■

Carl and Gerty Cori shared the Nobel Prize in Physiology or Medicine in 1947 with Bernardo Houssay of Argentina, who was cited for his studies of hormonal regulation of carbohydrate metabolism. The Cori laboratories in St. Louis became an international center of biochemical research in the 1940s and 1950s, and at least six scientists who trained with the Coris became Nobel laureates: Arthur Kornberg (for DNA synthesis, 1959), Severo Ochoa (for RNA synthesis, 1959), Luis Leloir (for the role of sugar nucleotides in

the cell is favorable. In effect, rapid removal of the product, driven by the large, negative free-energy change of PP_i hydrolysis, pulls the synthetic reaction forward, a common strategy in biological polymerization reactions.

2. Although the chemical transformations of sugar nucleotides do not involve the atoms of the nucleotide itself, the nucleotide moiety has many

groups that can undergo noncovalent interactions with enzymes; the additional free energy of binding can contribute significantly to catalytic activity (Chapter 6; see also p. 301).

3. Like phosphate, the nucleotidyl group (UMP or AMP, for example) is an excellent leaving group, facilitating nucleophilic attack by activating the sugar carbon to which it is attached.

polysaccharide synthesis, 1970), Earl Sutherland (for the discovery of cAMP in the regulation of carbohydrate metabolism, 1971), Christian de Duve (for sub-

cellular fractionation, 1974), and Edwin Krebs (for the discovery of phosphorylase kinase, 1991).

TABLE 1 Glycogen Storage Diseases of Humans

Type (name)	Enzyme affected	Primary organ affected	Symptoms
Type 0	Glycogen synthase	Liver	Low blood glucose, high ketone bodies, early death
Type Ia (von Gierke's)	Glucose 6-phosphatase	Liver	Enlarged liver, kidney failure
Type Ib	Microsomal glucose 6-phosphate translocase	Liver	As in Ia; also high susceptibility to bacterial infections
Type Ic	Microsomal P _i transporter	Liver	As in Ia
Type II (Pompe's)	Lysosomal glucosidase	Skeletal and cardiac muscle	Infantile form: death by age 2; juvenile form: muscle defects (myopathy); adult form: as in muscular dystrophy
Type IIIa (Cori's or Forbes's)	Debranching enzyme	Liver, skeletal and cardiac muscle	Enlarged liver in infants; myopathy
Type IIIb	Liver debranching enzyme (muscle enzyme normal)	Liver	Enlarged liver in infants
Type IV (Andersen's)	Branching enzyme	Liver, skeletal muscle	Enlarged liver and spleen, myoglobin in urine
Type V (McArdle's)	Muscle phosphorylase	Skeletal muscle	Exercise-induced cramps and pain; myoglobin in urine
Type VI (Hers's)	Liver phosphorylase	Liver	Enlarged liver
Type VII (Tarui's)	Muscle PFK-1	Muscle, erythrocytes	As in V; also hemolytic anemia
Type VIb, VIII, or IX	Phosphorylase kinase	Liver, leukocytes, muscle	Enlarged liver
Type XI (Fanconi-Bickel)	Glucose transporter (GLUT2)	Liver	Failure to thrive, enlarged liver, rickets, kidney dysfunction

- By “tagging” some hexoses with nucleotidyl groups, cells can set them aside in a pool for one purpose (glycogen synthesis, for example), separate from hexose phosphates destined for another purpose (such as glycolysis).

Glycogen synthesis takes place in virtually all animal tissues but is especially prominent in the liver and skeletal muscles. The starting point for synthesis of glycogen

is **glucose 6-phosphate**. As we saw in Chapter 14, this can be derived from free glucose in a reaction catalyzed by the isozymes **hexokinase I** and **hexokinase II** in muscle and **hexokinase IV** (glucokinase) in liver:



However, some ingested glucose takes a more roundabout path to glycogen. It is first taken up by erythrocytes and converted to lactate glycolytically; the lactate is then

taken up by the liver and converted to glucose 6-phosphate by gluconeogenesis.

To initiate glycogen synthesis, the glucose 6-phosphate is converted to **glucose 1-phosphate** in the phosphoglucomutase reaction:



The product of this reaction is converted to UDP-glucose by the action of **UDP-glucose pyrophosphorylase**, in a key step of glycogen biosynthesis:



Notice that this enzyme is named for the reverse reaction; in the cell, the reaction proceeds in the direction of UDP-glucose formation, because pyrophosphate is rapidly hydrolyzed by inorganic pyrophosphatase (Fig. 15-7).

UDP-glucose is the immediate donor of glucose residues in the reaction catalyzed by **glycogen synthase**, which promotes the transfer of the glucose residue from UDP-glucose to a nonreducing end of a branched glyco-

gen molecule (Fig. 15-8). The overall equilibrium of the path from glucose 6-phosphate to lengthened glycogen greatly favors synthesis of glycogen.

Glycogen synthase cannot make the (α 1 $\rightarrow$ 6) bonds found at the branch points of glycogen; these are formed by the glycogen-branching enzyme, also called **amylo (1 $\rightarrow$ 4) to (1 $\rightarrow$ 6) transglycosylase** or glycosyl-(4 $\rightarrow$ 6)-transferase. The glycogen-branching enzyme catalyzes transfer of a terminal fragment of 6 or 7 glucose residues from the nonreducing end of a glycogen branch having at least 11 residues to the C-6 hydroxyl group of a glucose residue at a more interior position of the same or another glycogen chain, thus creating a new branch (Fig. 15-9). Further glucose residues may be added to the new branch by glycogen synthase. The biological effect of branching is to make the glycogen molecule more soluble and to increase the number of nonreducing ends. This increases the number of sites accessible to glycogen phosphorylase and glycogen synthase, both of which act only at nonreducing ends.

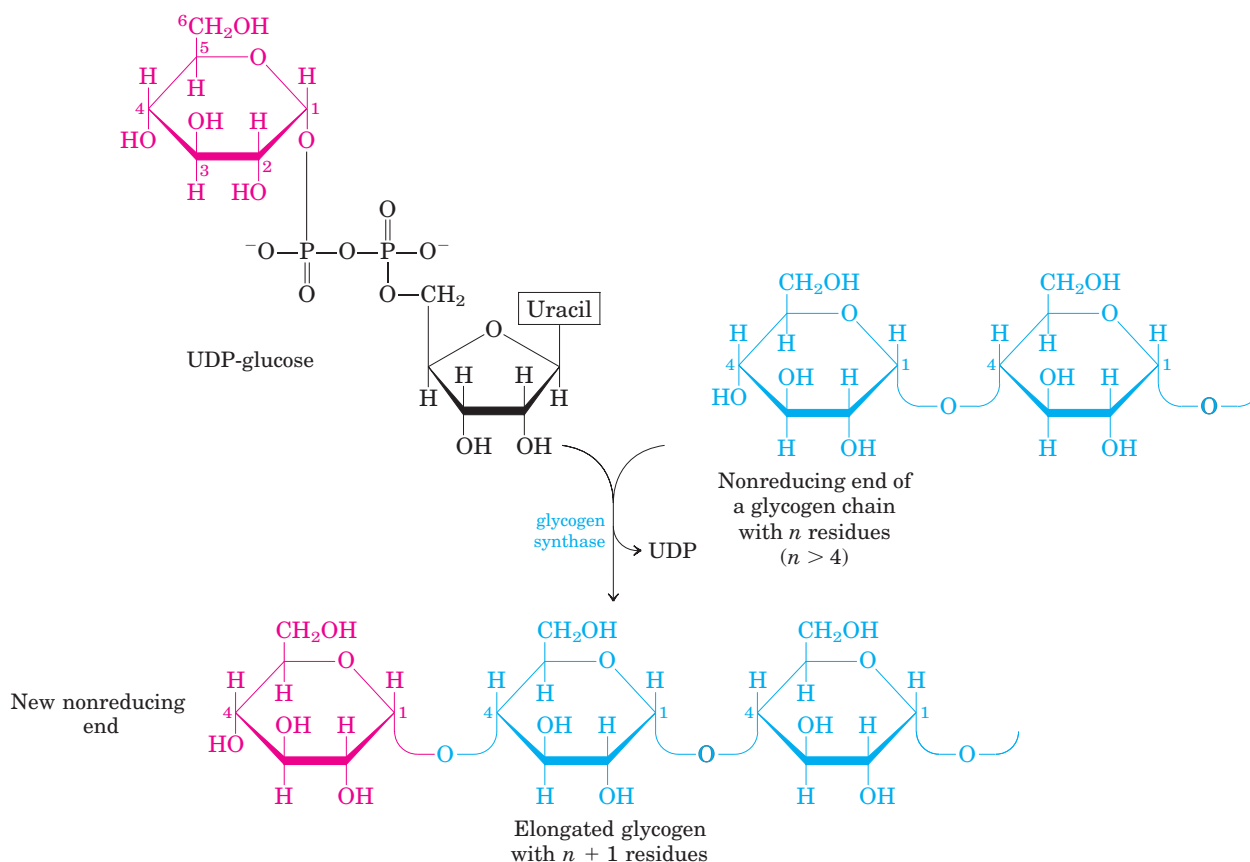


FIGURE 15-8 Glycogen synthesis. A glycogen chain is elongated by glycogen synthase. The enzyme transfers the glucose residue of UDP-glucose to the nonreducing end of a glycogen branch (see Fig. 7-15) to make a new (α 1 $\rightarrow$ 4) linkage.

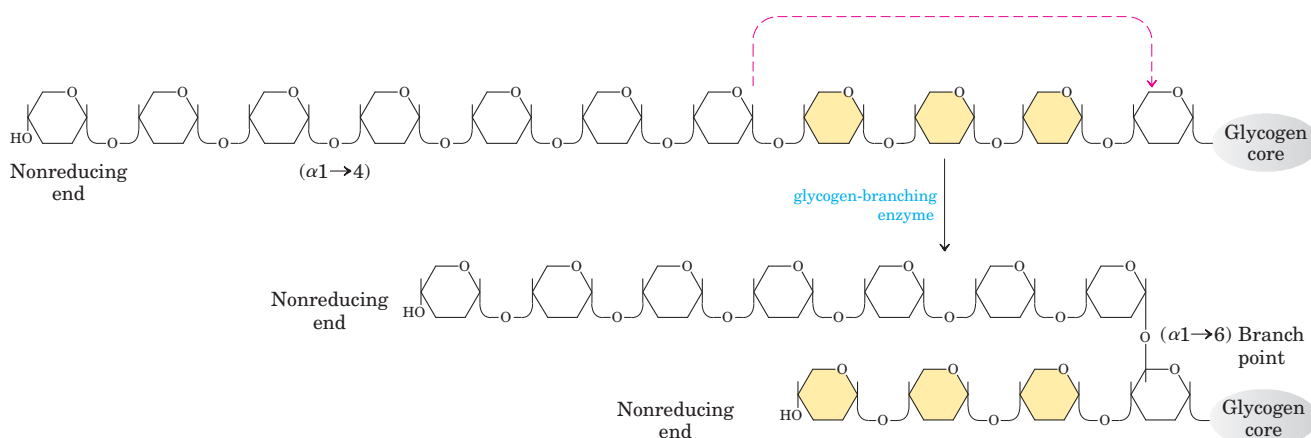


FIGURE 15-9 Branch synthesis in glycogen. The glycogen-branching enzyme (also called amylo (1 $\rightarrow$ 4) to (1 $\rightarrow$ 6) transglycosylase or glycosyl-(4 $\rightarrow$ 6)-transferase) forms a new branch point during glycogen synthesis.

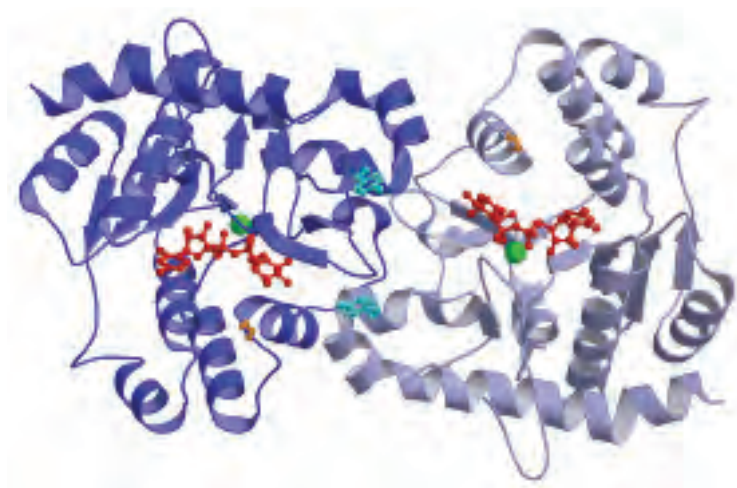
Glycogenin Primes the Initial Sugar Residues in Glycogen

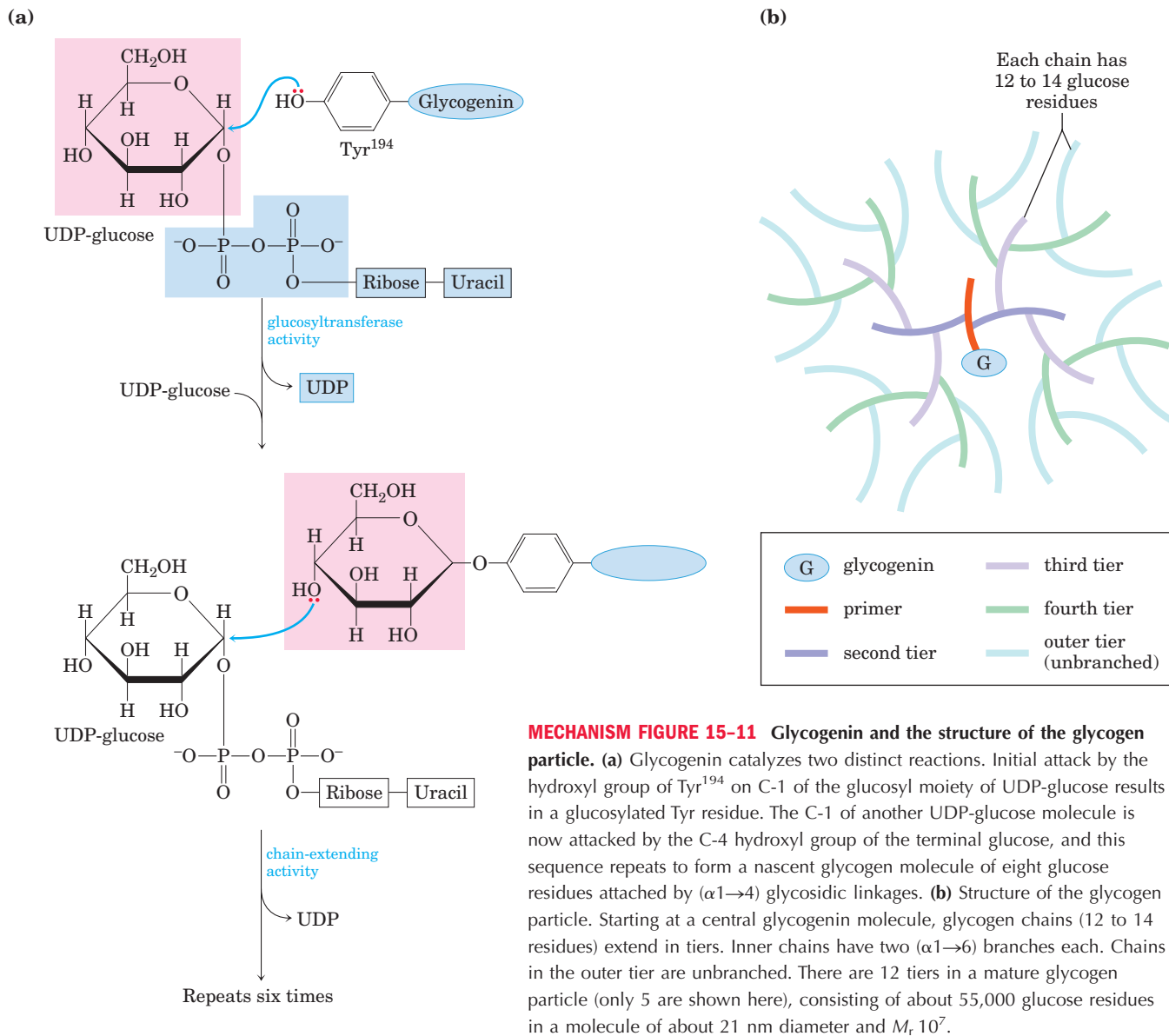
Glycogen synthase cannot initiate a new glycogen chain *de novo*. It requires a primer, usually a preformed (α 1 $\rightarrow$ 4) polyglucose chain or branch having at least eight glucose residues. How is a *new* glycogen molecule initiated? The intriguing protein **glycogenin** (Fig. 15-10) is both the primer on which new chains are assembled and the enzyme that catalyzes their assembly. The first step in the synthesis of a new glycogen mole-

cule is the transfer of a glucose residue from UDP-glucose to the hydroxyl group of Tyr¹⁹⁴ of glycogenin, catalyzed by the protein's intrinsic glucosyltransferase activity (Fig. 15-11a). The nascent chain is extended by the sequential addition of seven more glucose residues, each derived from UDP-glucose; the reactions are catalyzed by the chain-extending activity of glycogenin. At this point, glycogen synthase takes over, further extending the glycogen chain. Glycogenin remains buried within the particle, covalently attached to the single reducing end of the glycogen molecule (Fig. 15-11b).

FIGURE 15-10 Glycogenin structure. (PDB 1D 1772)

Muscle glycogenin (M_r 37,000) forms dimers in solution. Humans have a second isoform in liver, glycogenin-2. The substrate, UDP-glucose (shown as a red ball-and-stick structure), is bound to a Rossman fold near the amino terminus and is some distance from the Tyr¹⁹⁴ residues (turquoise)—15 Å from that in the same monomer, 12 Å from that in the dimeric partner. Each UDP-glucose is bound through its phosphates to a Mn²⁺ ion (green) that is essential to catalysis. Mn²⁺ is believed to function as an electron-pair acceptor (Lewis acid) to stabilize the leaving group, UDP. The glycosidic bond in the product has the same configuration about the C-1 of glucose as the substrate UDP-glucose, suggesting that the transfer of glucose from UDP to Tyr¹⁹⁴ occurs in two steps. The first step is probably a nucleophilic attack by Asp¹⁶² (orange), forming a temporary intermediate with inverted configuration. A second nucleophilic attack by Tyr¹⁹⁴ then restores the starting configuration.





SUMMARY 15.1 The Metabolism of Glycogen in Animals

- Glycogen is stored in muscle and liver as large particles. Contained within the particles are the enzymes that metabolize glycogen, as well as regulatory enzymes.
- Glycogen phosphorylase catalyzes phosphorolytic cleavage at the nonreducing ends of glycogen chains, producing glucose 1-phosphate. The debranching enzyme transfers branches onto main chains and releases the residue at the (α 1 $\rightarrow$ 6) branch as free glucose.
- Phosphoglucomutase interconverts glucose 1-phosphate and glucose 6-phosphate. Glucose

6-phosphate can enter glycolysis or, in liver, can be converted to free glucose by glucose 6-phosphatase in the endoplasmic reticulum, then released to replenish blood glucose.

- The sugar nucleotide UDP-glucose donates glucose residues to the nonreducing end of glycogen in the reaction catalyzed by glycogen synthase. A separate branching enzyme produces the (α 1 $\rightarrow$ 6) linkages at branch points.
- New glycogen particles begin with the auto-catalytic formation of a glycosidic bond between the glucose of UDP-glucose and a Tyr residue in the protein glycogenin, followed by addition of several glucose residues to form a primer that can be acted upon by glycogen synthase.

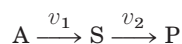
15.2 Regulation of Metabolic Pathways

The pathways of glycogen metabolism provide, in the catabolic direction, the energy essential to oppose the forces of entropy and, in the anabolic direction, biosynthetic precursors and a storage form of metabolic energy. These reactions are so important to survival that very complex regulatory mechanisms have evolved to ensure that metabolites move through each pathway in the correct direction and at the correct rate to exactly match the cell's or the organism's current circumstances, and that appropriate adjustments are made in the rate of metabolite flow through the whole pathway if external circumstances change.

Circumstances do change, sometimes dramatically. The demand for ATP production in muscle may increase 100-fold in a few seconds in response to exercise. The availability of oxygen may decrease due to hypoxia (diminished delivery of oxygen to tissues) or ischemia (diminished flow of blood to tissues). The relative proportions of carbohydrate, fat, and protein in the diet vary from meal to meal, and the supply of fuels obtained in the diet is intermittent, requiring metabolic adjustments between meals and during starvation. Wound healing requires huge amounts of energy and biosynthetic precursors.

Living Cells Maintain a Dynamic Steady State

Fuels such as glucose enter a cell, and waste products such as CO_2 leave, but the mass and the gross composition of a typical cell do not change appreciably over time; cells and organisms exist in a dynamic steady state, but not at equilibrium with their surroundings. At the molecular level, this means that for each metabolic reaction in a pathway, the substrate is provided by the preceding reaction at the same rate at which it is converted to product. Thus, although the rate of metabolite flow, or **flux**, through this step of the pathway may be high, the concentration of substrate, S , remains constant. For the reaction



when $v_1 = v_2$, $[S]$ is constant.

When the steady state is disturbed by some change in external circumstances or energy supply, the temporarily altered fluxes through individual metabolic pathways trigger regulatory mechanisms intrinsic to each pathway. The net effect of all these adjustments is to return the organism to a new steady state—to achieve **homeostasis**.

Regulatory Mechanisms Evolved under Strong Selective Pressures

In the course of evolution, organisms have acquired a remarkable collection of regulatory mechanisms for

maintaining homeostasis at the molecular, cellular, and organismal level. The importance of metabolic regulation to an organism is reflected in the relative proportion of genes that encode regulatory machinery—in humans, about 4,000 genes (~12% of all genes) encode regulatory proteins, including a variety of receptors, regulators of gene expression, and about 500 different protein kinases! These regulatory mechanisms act over different time scales (from seconds to days) and have different sensitivities to external changes. In many cases, the mechanisms overlap: one enzyme is subject to regulation by several different mechanisms.

After the protection of its DNA from damage, perhaps nothing is more important to a cell than maintaining a constant supply and concentration of ATP. Many ATP-using enzymes have K_m values between 0.1 and 1 mM, and the ATP concentration in a typical cell is about 5 mM. If $[ATP]$ were to drop significantly, the rates of hundreds of reactions that involve ATP would decrease, and the cell would probably not survive. Furthermore, because ATP is converted to ADP or AMP when “spent” to accomplish cellular work, the $[ATP]/[ADP]$ ratio profoundly affects all reactions that employ these cofactors. The same is true for other important cofactors, such as $NADH/NAD^+$ and $NADPH/NADP^+$. For example, consider the reaction catalyzed by hexokinase:



$$K'_{eq} = \frac{[ADP]_{eq}[\text{glucose 6-phosphate}]_{eq}}{[ATP]_{eq}[\text{glucose}]_{eq}} = 2 \times 10^3$$

Note that this expression is true *only* when reactants and products are at their *equilibrium* concentrations, where $\Delta G' = 0$. At any other set of concentrations, $\Delta G'$ is not zero. Recall (from Chapter 13) that the ratio of products to substrates (the mass action ratio, Q) determines the magnitude and sign of $\Delta G'$ and therefore the amount of free energy released during the reaction:

$$\Delta G' = \Delta G'^{\circ} + RT \ln \frac{[ADP][\text{glucose 6-phosphate}]}{[ATP][\text{glucose}]}$$

Because an alteration of this mass action ratio profoundly influences every reaction that involves ATP, organisms have been under strong evolutionary pressure to develop regulatory mechanisms that respond to the $[ATP]/[ADP]$ ratio. Similar arguments show the importance of maintaining appropriate $[NADH]/[NAD^+]$ and $[NADPH]/[NADP^+]$ ratios.

AMP concentration is a much more sensitive indicator of a cell's energetic state than is ATP. Normally cells have a far higher concentration of ATP (5 to 10 mM) than of AMP (<0.1 mM). When some process (say, muscle contraction) consumes ATP, AMP is produced in two steps. First, hydrolysis of ATP produces ADP, then the reaction catalyzed by **adenylate kinase** produces AMP:

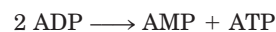


TABLE 15–1 Relative Changes in [ATP] and [AMP] When ATP Is Consumed

Adenine nucleotide	Concentration before ATP depletion (mM)	Concentration after ATP depletion (mM)	Relative change
ATP	5.0	4.5	10%
ADP	1.0	1.0	0
AMP	0.1	0.6	600%

If [ATP] drops by 10%, producing ADP and AMP in the same amounts, the relative change in [AMP] is much greater (Table 15–1). It is not surprising, therefore, that many regulatory processes hinge on changes in [AMP]. One important mediator of regulation by AMP is **AMP-dependent protein kinase (AMPK)**, which responds to an increase in [AMP] by phosphorylating key proteins, thereby regulating their activities. The rise in [AMP] may be caused by a reduced nutrient supply or increased exercise. The action of AMPK (not to be confused with the *cyclic* AMP-dependent protein kinase; see Section 15.4) increases glucose transport and activates glycolysis and fatty acid oxidation, while suppressing energy-requiring processes such as the synthesis of fatty acids, cholesterol, and protein. We discuss this enzyme further, and the detailed mechanisms by which it effects these changes, in Chapter 23.

In addition to the cofactors ATP, NADH, and NADPH, hundreds of metabolic intermediates also must be present at appropriate concentrations in the cell. For example, the glycolytic intermediates dihydroxyacetone phosphate and 3-phosphoglycerate are precursors of triacylglycerols and serine, respectively. When these products are needed, the rate of glycolysis must be adjusted to provide them without reducing the glycolytic production of ATP.

Priorities at the *organismal* level have also driven the evolution of regulatory mechanisms. In mammals, the brain has virtually no stored source of energy, depending instead on a constant supply of glucose from the blood. If glucose in the blood drops from its normal concentration of 4 to 5 mM to half that level, mental confusion results, and a fivefold reduction in blood glucose can lead to coma and death. To buffer against changes in blood glucose concentration, release of the hormones insulin and glucagon, elicited by high or low blood glucose, respectively, triggers metabolic changes that tend to return the blood glucose concentration to normal.

Other selective pressures must also have operated throughout evolution, selecting for regulatory mechanisms that

1. maximize the efficiency of fuel utilization by preventing the simultaneous operation of

pathways in opposite directions (such as glycolysis and gluconeogenesis);

2. partition metabolites appropriately between alternative pathways (such as glycolysis and the pentose phosphate pathway);
3. draw on the fuel best suited for the immediate needs of the organism (glucose, fatty acids, glycogen, or amino acids); and
4. shut down biosynthetic pathways when their products accumulate.

The importance of effective metabolic regulation is clear from the consequences of failed regulation: in many cases, serious disease (as described in Box 15–1, for example).

Regulatory Enzymes Respond to Changes in Metabolite Concentration

Flux through a biochemical pathway depends on the activities of the enzymes that catalyze each reaction in that pathway. For some steps, the reaction is close to equilibrium within the cell (Fig. 15–12). The net flow of metabolites through these steps is the small difference between the rates of the forward and reverse reactions, rates that are very similar when the reaction is near equilibrium. Small changes in substrate or product concentration can produce large changes in the net rate, and can even change the direction of the net flow. We can identify these near-equilibrium reactions in a cell by comparing the **mass action ratio, Q** , with the equilibrium constant for the reaction, K'_{eq} . Recall that for the reaction $A + B \rightarrow C + D$, $Q = [C][D]/[A][B]$. When Q and K'_{eq} are within a few orders of magnitude, the reaction is near equilibrium. This is the case for six of the ten reactions in the glycolytic pathway (Table 15–2).

Other reactions are far from equilibrium in the cell. For example, K'_{eq} for the phosphofructokinase-1 (PFK-1) reaction in glycolysis is about 1,000, but Q ($[\text{fructose 1,6-bisphosphate}][\text{ADP}]/[\text{fructose 6-phosphate}][\text{ATP}]$) in a typical cell in the steady state is about 0.1 (Table 15–2). It is *because* the reaction is so far from equilibrium that the process is exergonic under cellular con-

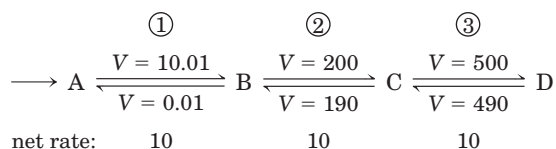


FIGURE 15-12 Near-equilibrium and nonequilibrium steps in a metabolic pathway. Steps 2 and 3 of this pathway are near equilibrium in the cell; their forward rates are only slightly greater than their reverse rates, so the net forward rates (10) are relatively low and the free-energy change $\Delta G'$ for each step is close to zero. An increase in the intracellular concentration of metabolite C or D can reverse the direction of these steps. Step 1 is maintained in the cell far from equilibrium; its forward rate greatly exceeds its reverse rate. The net rate of step 1 (10) is much larger than the reverse rate (0.09) and is identical to the net rates of steps 2 and 3 when the pathway is operating in the steady state. Step 1 has a large, negative $\Delta G'$.

ditions and tends to go in the forward direction. The reaction is held far from equilibrium because, under prevailing cellular conditions of substrate, product, and effector concentrations, the rate of conversion of fructose 6-phosphate to fructose 1,6-bisphosphate is limited by the activity of PFK-1, which is itself limited by the number of PFK-1 molecules present and by the actions of effectors. Thus the net forward rate of the enzyme-catalyzed reaction is equal to the net flow of glycolytic intermediates through other steps in the pathway, and the reverse flow through PFK-1 remains near zero.

The cell *cannot* allow reactions with large equilibrium constants to reach equilibrium. If [fructose 6-phosphate], [ATP], and [ADP] in the cell were held at their usual level (low millimolar) and the PFK-1 reaction were allowed to reach equilibrium by an increase in [fructose 1,6-bisphosphate], the concentration of fructose 1,6-bisphosphate would rise into the molar range, wreaking osmotic havoc on the cell. Consider another case: if the reaction $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ were allowed to approach equilibrium in the cell, the actual free-energy change ($\Delta G'$) for that reaction (see Box 13-1) would approach zero, and ATP would lose the high phosphoryl group transfer potential that makes it valuable to the cell as an energy source. It is therefore essential that enzymes catalyzing ATP breakdown and other highly exergonic reactions in a cell be sensitive to regulation, so that when metabolic changes are forced by external circumstances, the flow through these enzymes will be adjusted to ensure that [ATP] remains far above its equilibrium level. When such metabolic changes occur, enzymatic activities in all interconnected pathways adjust to keep these critical steps away from equilibrium. Thus, not surprisingly, many enzymes that catalyze highly exergonic reactions (such as PFK-1) are subject to a variety of subtle regulatory mechanisms. The multiplicity of these adjustments is so great that we cannot predict by examining the properties of any one enzyme in a pathway whether that enzyme has a strong influence on net

TABLE 15-2 Equilibrium Constants, Mass Action Coefficients, and Free-Energy Changes for Enzymes of Carbohydrate Metabolism

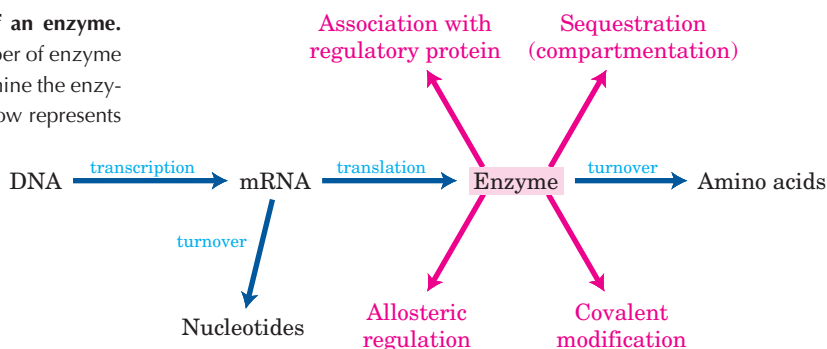
Enzyme	K'_{eq}	Mass action ratio, Q		Reaction near equilibrium in vivo?*	$\Delta G'^0$ (kJ/mol)	$\Delta G'$ (kJ/mol) in heart
		Liver	Heart			
Hexokinase	2×10^3	2×10^{-2}	8×10^{-2}	No	-17	-27
PFK-1	1.0×10^3	9×10^{-2}	3×10^{-2}	No	-14.2	-23
Aldolase	1.0×10^{-4}	1.2×10^{-6}	9×10^{-6}	Yes	+23.8	-6.0
Triose phosphate isomerase	4×10^{-2}	—	2.4×10^{-1}	Yes	+7.5	+3.8
Glyceraldehyde 3-phosphate dehydrogenase + phosphoglycerate kinase	2×10^3	6×10^2	9.0	Yes	-13	+3.5
Phosphoglycerate mutase	1×10^{-1}	1×10^{-1}	1.2×10^{-1}	Yes	+4.4	+0.6
Enolase	3	2.9	1.4	Yes	-3.2	-0.5
Pyruvate kinase	2×10^4	7×10^{-1}	40	No	-31.0	-17
Phosphoglucose isomerase	4×10^{-1}	3.1×10^{-1}	2.4×10^{-1}	Yes	+2.2	-1.4
Pyruvate carboxylase + PEP carboxykinase	7	1×10^{-3}	—	No	-5.0	-22.8
Glucose 6-phosphatase	8.5×10^2	1.2×10^2	—	Yes	-17.3	-5.0

*For simplicity, any reaction for which the absolute value of the calculated $\Delta G'$ is less than 6 is considered near equilibrium.

Source: K'_{eq} and Q from Newsholme, E.A. & Start, C. (1973) *Regulation in Metabolism*, Wiley Press, New York, pp. 97, 263 $\Delta G'$ and $\Delta G'^0$ were calculated from these data.

FIGURE 15-13 Factors that determine the activity of an enzyme.

Blue arrows represent processes that determine the number of enzyme molecules in the cell; red arrows show factors that determine the enzymatic activity of an existing enzyme molecule. Each arrow represents a point at which regulation can occur.



flow through the entire pathway. This complex problem can be approached by metabolic control analysis, as described in Section 15.5.

Enzyme Activity Can Be Altered in Several Ways

The activity of an enzyme can be modulated by changes in the number of enzyme molecules in the cell or by changes in the catalytic activity of each enzyme molecule already present, for example through allosteric regulation or covalent alteration (Fig. 15-13). The number of enzyme molecules in the cell is a function of the rates of synthesis and degradation, both of which, for many enzymes, are tightly controlled. The rate of synthesis of a protein can be adjusted by the production or alteration (in response to some outside signal) of a transcription factor, a protein that binds to a regulatory region of DNA adjacent to the gene in question and increases the likelihood of its transcription into mRNA (Chapter 28). The stability of mRNAs—their resistance to degradation by a ribonuclease—varies, so the amount of a given mRNA in the cell is a function of its rates of synthesis and degradation (Chapter 26). Finally, the rate at which an mRNA is translated on the ribosome depends on several factors, described in detail in Chapter 27.

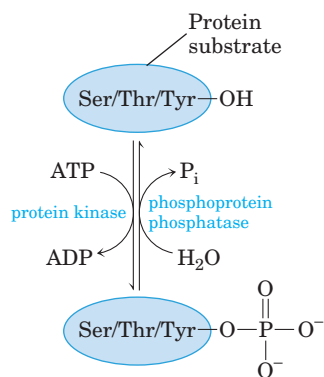
The rate of protein degradation also differs from enzyme to enzyme and depends on the conditions in the cell; protein half-lives vary from a few minutes to many days. Some proteins are tagged for degradation in proteasomes (discussed in Chapter 28) by the covalent attachment of ubiquitin (recall the case of cyclin; see Fig. 12-44). Some proteins are synthesized as inactive forms, or proenzymes, that become active only when a proteolytic event removes an inhibitory sequence in the proenzyme.

As a result of these several mechanisms of regulating enzyme level, cells can change their complement of enzymes in response to changes in metabolic circumstances. In vertebrates, liver is the most adaptable tissue; a change in diet from high carbohydrate to high lipid, for example, affects the transcription into mRNA of hundreds of genes and thus the levels of hundreds of proteins. These global changes in gene expression can be quantified by the use of DNA microarrays (see Fig. 9-22) or two-dimensional gel electrophoresis (see Fig.

3-22) to display the protein complement of a tissue. Both techniques provide great insights into metabolic regulation.

Changes in the number of molecules of an enzyme are generally relatively slow, occurring over seconds to hours. Covalent modifications of existing proteins are faster, taking seconds to minutes. Various types of covalent modifications are known, such as adenylation, methylation, or attachment of lipids (p. 228). By far the most common type is phosphorylation and dephosphorylation (Fig. 15-14); up to half of a eukaryotic cell's proteins are targets of phosphorylation under some circumstances. Phosphorylation may alter the electrostatic features of the active site, cause movement of an inhibitory region of the protein out of the active site, alter the protein's interaction with other proteins, or force conformational changes that translate into changes in $V_{\max}$ or K_m . For covalent modification to be useful in regulation, the cell must be able to restore the altered enzyme to its original condition. A family of phosphoprotein phosphatases, at least some of which are themselves under regulation, catalyze dephosphorylation of proteins that have been phosphorylated by protein kinases.

Alteration of the number of enzyme molecules and covalent modifications are generally triggered by some signal from outside the cell—a hormone or growth factor, for example—and result in a change of metabolite

**FIGURE 15-14** Protein phosphorylation and dephosphorylation. Protein kinases transfer a phosphoryl group from ATP to a Ser, Thr, or Tyr residue in a protein. Protein phosphatases remove the phosphoryl group as P_i.

flux through one or more pathways. In contrast, the very rapid (milliseconds) allosteric changes in enzyme activity are generally triggered locally, by changes in the level of a metabolite within the cell. The allosteric effector may be a substrate of the affected pathway (glucose for glycolysis, for example), a product of a pathway (ATP from glycolysis), or a key metabolite or cofactor (such as NADH) that indicates the cell's metabolic state. A single enzyme is commonly regulated in several ways—for example, by modulation of its synthesis, by covalent alteration, and by allosteric effectors.

Yet another way to alter the effective activity of an enzyme is to change the accessibility of its substrate. The hexokinase of muscle cannot act on glucose until the sugar enters the myocyte from the blood, and the rate at which it enters depends on the activity of glucose transporters in the plasma membrane. Within cells, membrane-bounded compartments segregate certain enzymes and enzyme systems, and the transport of substrate into these compartments may be the limiting factor in enzyme action.

In the discussion that follows, it is useful to think of changes in enzymatic activity as serving two distinct though complementary roles. We use the term **metabolic regulation** to refer to processes that serve to maintain homeostasis at the molecular level—to hold some cellular parameter (concentration of a metabolite, for example) at a steady level over time, even as the flow of metabolites through the pathway changes. The term **metabolic control** refers to a process that leads to a change in the output of a metabolic pathway over time, in response to some outside signal or change in circumstances. The distinction, although useful, is not always easy to make.

SUMMARY 15.2 Regulation of Metabolic Pathways

- In a metabolically active cell in a steady state, intermediates are formed and consumed at equal rates. When a perturbation alters the rate of formation or consumption of a metabolite, compensating changes in enzyme activities return the system to the steady state.
- Regulatory mechanisms maintain nearly constant levels of key metabolites such as ATP and NADH in cells and glucose in the blood, while matching the use or storage of glycogen to the organism's changing needs.
- In multistep processes such as glycolysis, certain reactions are essentially at equilibrium in the steady state; the rates of these substrate-limited reactions rise and fall with substrate concentration. Other reactions are far from equilibrium; their rates are too slow to produce instant equilibration of substrate and product. These enzyme-limited reactions are

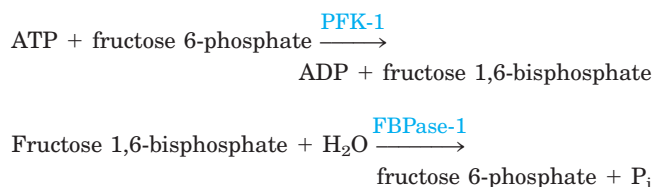
often highly exergonic and therefore metabolically irreversible, and the enzymes that catalyze them are commonly the points at which flux through the pathway is regulated.

- The activity of an enzyme can be regulated by changing the rate of its synthesis or degradation, by allosteric or covalent alteration of existing enzyme molecules, or by separating the enzyme from its substrate in subcellular compartments.
- Fast metabolic adjustments (on the time scale of seconds or less) at the intracellular level are generally allosteric. The effects of hormones and growth factors are generally slower (seconds to hours) and are commonly achieved by covalent modification or changes in enzyme synthesis.

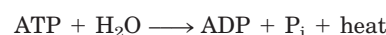
15.3 Coordinated Regulation of Glycolysis and Gluconeogenesis

In mammals, gluconeogenesis occurs primarily in the liver, where its role is to provide glucose for export to other tissues when glycogen stores are exhausted. Gluconeogenesis employs most of the enzymes that act in glycolysis, but it is not simply the reversal of glycolysis. Seven of the glycolytic reactions are freely reversible, and the enzymes that catalyze these reactions also function in gluconeogenesis (Fig. 15–15). Three reactions of glycolysis are so exergonic as to be essentially irreversible: those catalyzed by hexokinase, PFK-1, and pyruvate kinase. Notice in Table 15–2 that all three reactions have a large, negative $\Delta G'$. Gluconeogenesis uses detours around each of these irreversible steps; for example, the conversion of fructose 1,6-bisphosphate to fructose 6-phosphate is catalyzed by fructose 1,6-bisphosphatase (FBPase-1; Fig. 15–15). Note that each of these bypass reactions also has a large, negative $\Delta G'$.

At each of the three points where glycolytic reactions are bypassed by alternative, gluconeogenic reactions, simultaneous operation of both pathways would consume ATP without accomplishing any chemical or biological work. For example, PFK-1 and FBPase-1 catalyze opposing reactions:



The sum of these two reactions is



that is, hydrolysis of ATP without any useful metabolic work being done. Clearly, if these two reactions were

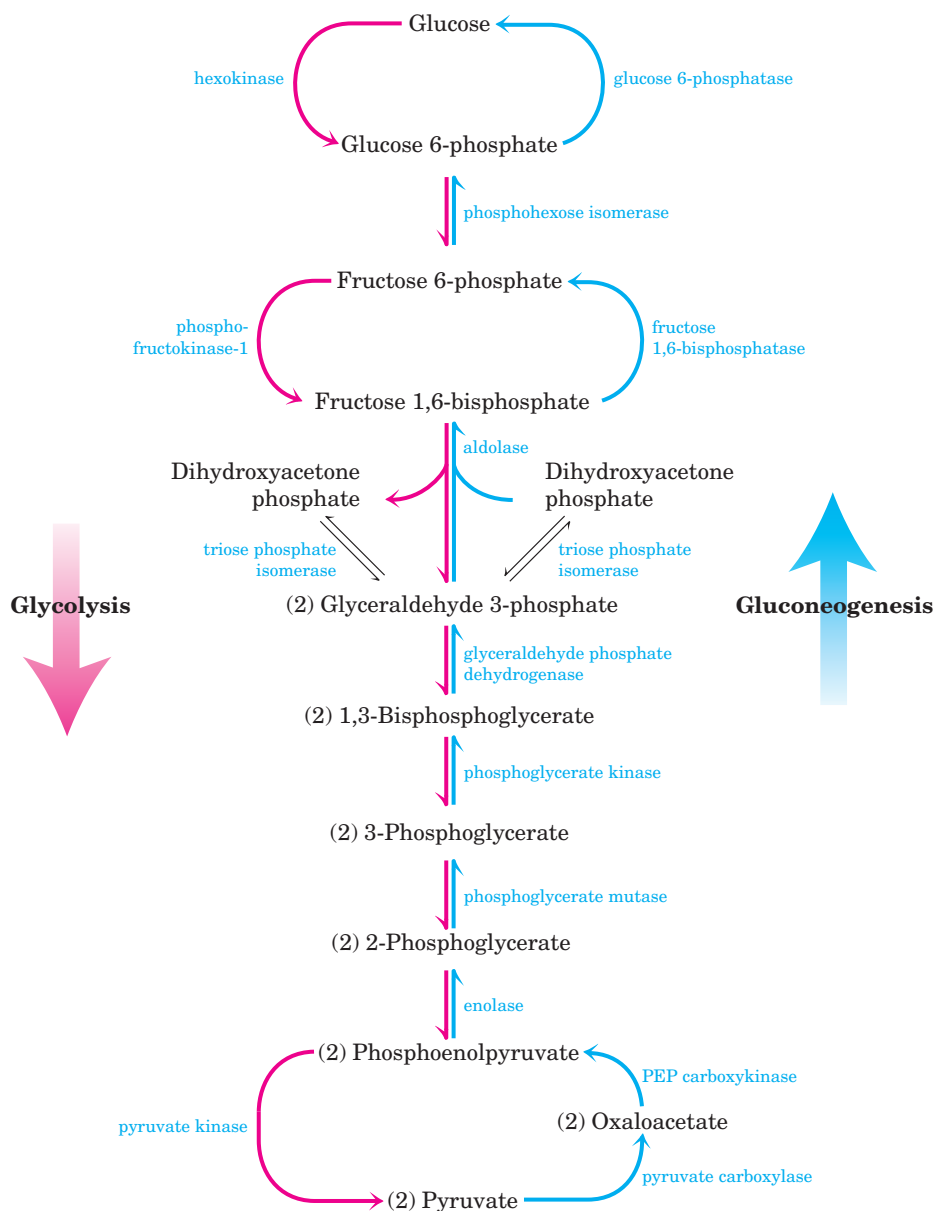


FIGURE 15-15 Glycolysis and gluconeogenesis. Opposing pathways of glycolysis (pink) and gluconeogenesis (blue) in rat liver. Three steps are catalyzed by different enzymes in gluconeogenesis

(the “bypass reactions”) and glycolysis; seven steps are catalyzed by the same enzymes in the two pathways. Cofactors have been omitted for simplicity.

allowed to proceed simultaneously at a high rate in the same cell, a large amount of chemical energy would be dissipated as heat. This uneconomical process has been called a **futile cycle**. However, as we shall see later, such cycles may provide advantages for controlling pathways, and the term **substrate cycle** is a better description. Similar substrate cycles also occur with the other two sets of bypass reactions of gluconeogenesis (Fig. 15-15).

We begin our examination of the coordinated regulation of glycolysis and gluconeogenesis by considering the regulatory patterns seen at the three main control

points of glycolysis. We then look at the regulation of the enzymes of gluconeogenesis, leading to a consideration of how the regulation of both pathways is tightly, reciprocally coordinated.

Hexokinase Isozymes of Muscle and Liver Are Affected Differently by Their Product, Glucose 6-Phosphate

Hexokinase, which catalyzes the entry of free glucose into the glycolytic pathway, is a regulatory enzyme. There are four isozymes (designated I to IV), encoded

by four different genes. Isozymes are different proteins that catalyze the same reaction (Box 15–2). The predominant hexokinase isozyme of myocytes (hexokinase II) has a high affinity for glucose—it is half-saturated at about 0.1 mM. Because glucose entering myocytes from the blood (where the glucose concentration is 4 to 5 mM) produces an intracellular glucose concentration high enough to saturate hexokinase II, the enzyme normally acts at or near its maximal rate. Muscle hexokinases I and II are allosterically inhibited by their product, glucose 6-phosphate, so whenever the cellular concentra-

tion of glucose 6-phosphate rises above its normal level, these isozymes are temporarily and reversibly inhibited, bringing the rate of glucose 6-phosphate formation into balance with the rate of its utilization and reestablishing the steady state.

The different hexokinase isozymes of liver and muscle reflect the different roles of these organs in carbohydrate metabolism: muscle consumes glucose, using it for energy production, whereas liver maintains blood glucose homeostasis by removing or producing glucose, depending on the prevailing glucose concentration. The

BOX 15–2 WORKING IN BIOCHEMISTRY

Isozymes: Different Proteins That Catalyze the Same Reaction

The four forms of hexokinase found in mammalian tissues are but one example of a common biological situation: the same reaction catalyzed by two or more different molecular forms of an enzyme. These multiple forms, called isozymes or isoenzymes, may occur in the same species, in the same tissue, or even in the same cell. The different forms of the enzyme generally differ in kinetic or regulatory properties, in the cofactor they use (NADH or NADPH for dehydrogenase isozymes, for example), or in their subcellular distribution (soluble or membrane-bound). Isozymes may have similar, but not identical, amino acid sequences, and in many cases they clearly share a common evolutionary origin.

One of the first enzymes found to have isozymes was lactate dehydrogenase (LDH) (p. 538), which, in vertebrate tissues, exists as at least five different isozymes separable by electrophoresis. All LDH isozymes contain four polypeptide chains (each of M_r 33,500), each type containing a different ratio of two kinds of polypeptides. The M (for muscle) chain and the H (for heart) chain are encoded by two different genes.

In skeletal muscle the predominant isozyme contains four M chains, and in heart the predominant isozyme contains four H chains. Other tissues have some combination of the five possible types of LDH isozymes:

Type	Composition	Location
LDH ₁	HHHH	Heart and erythrocyte
LDH ₂	HHHM	Heart and erythrocyte
LDH ₃	HHMM	Brain and kidney
LDH ₄	HMMM	Skeletal muscle and liver
LDH ₅	MMMM	Skeletal muscle and liver



These differences in the isozyme content of tissues can be used to assess the timing and extent of heart damage due to myocardial infarction (heart attack). Damage to heart tissue results in the

release of heart LDH into the blood. Shortly after a heart attack, the blood level of total LDH increases, and there is more LDH₂ than LDH₁. After 12 hours the amounts of LDH₁ and LDH₂ are very similar, and after 24 hours there is more LDH₁ than LDH₂. This switch in the LDH₁/LDH₂ ratio, combined with increased concentrations in the blood of another heart enzyme, creatine kinase, is very strong evidence of a recent myocardial infarction. ■

The different LDH isozymes have significantly different values of V_{max} and K_M , particularly for pyruvate. The properties of LDH₄ favor rapid reduction of very low concentrations of pyruvate to lactate in skeletal muscle, whereas those of isozyme LDH₁ favor rapid oxidation of lactate to pyruvate in the heart.

In general, the distribution of different isozymes of a given enzyme reflects at least four factors:

1. *Different metabolic patterns in different organs.* For glycogen phosphorylase, the isozymes in skeletal muscle and liver have different regulatory properties, reflecting the different roles of glycogen breakdown in these two tissues.
2. *Different locations and metabolic roles for isozymes in the same cell.* The isocitrate dehydrogenase isozymes of the cytosol and the mitochondrion are an example (Chapter 16).
3. *Different stages of development in embryonic or fetal tissues and in adult tissues.* For example, the fetal liver has a characteristic isozyme distribution of LDH, which changes as the organ develops into its adult form. Some enzymes of glucose catabolism in malignant (cancer) cells occur as their fetal, not adult, isozymes.
4. *Different responses of isozymes to allosteric modulators.* This difference is useful in fine-tuning metabolic rates. Hexokinase IV (glucokinase) of liver and the hexokinase isozymes of other tissues differ in their sensitivity to inhibition by glucose 6-phosphate.

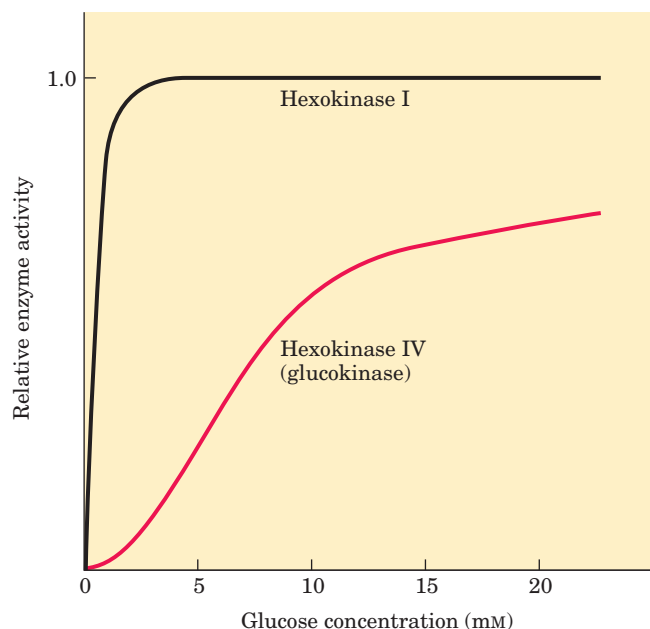


FIGURE 15-16 Comparison of the kinetic properties of hexokinase IV (glucokinase) and hexokinase I. Note the sigmoidicity for hexokinase IV and the much lower K_m for hexokinase I. When blood glucose rises above 5 mM, hexokinase IV activity increases, but hexokinase I is already operating near V_{max} at 5 mM glucose and cannot respond to an increase in glucose concentration. Hexokinase I, II, and III have similar kinetic properties.

predominant hexokinase isozyme of liver is hexokinase IV (glucokinase), which differs in three important respects from hexokinases I–III of muscle. First, the glucose concentration at which hexokinase IV is half-saturated (about 10 mM) is higher than the usual concentration of glucose in the blood. Because an efficient glucose transporter in hepatocytes (**GLUT2**; see Fig. 11–31) rapidly equilibrates the glucose concentrations in cytosol and blood, the high K_m of hexokinase IV allows its direct regulation by the level of blood glucose (Fig. 15–16). When the blood glucose concentration is high, as it is after a meal rich in carbohydrates, excess glucose is transported into hepatocytes, where hexokinase IV

converts it to glucose 6-phosphate. Because hexokinase IV is not saturated at 10 mM glucose, its activity continues to increase as the glucose concentration rises to 10 mM or more.

Second, hexokinase IV is subject to inhibition by the reversible binding of a regulatory protein specific to liver (Fig. 15–17). The binding is much tighter in the presence of the allosteric effector fructose 6-phosphate. Glucose competes with fructose 6-phosphate for binding and causes dissociation of the regulatory protein from hexokinase IV, relieving the inhibition. Immediately after a carbohydrate-rich meal, when blood glucose is high, glucose enters the hepatocyte via GLUT2 and activates hexokinase IV by this mechanism. During a fast, when blood glucose drops below 5 mM, fructose 6-phosphate triggers the inhibition of hexokinase IV by the regulatory protein, so the liver does not compete with other organs for the scarce glucose. The mechanism of inhibition by the regulatory protein is interesting: the protein anchors hexokinase IV inside the nucleus, where it is segregated from the other enzymes of glycolysis in the cytosol (Fig. 15–17). When the glucose concentration in the cell rises, it equilibrates with glucose in the nucleus by transport through the nuclear pores. Glucose causes dissociation of the regulatory protein, and hexokinase IV enters the cytosol and begins to phosphorylate glucose.

Third, hexokinase IV is not inhibited by glucose 6-phosphate, and it can therefore continue to operate when the accumulation of glucose 6-phosphate completely inhibits hexokinases I–III.

Phosphofructokinase-1 Is under Complex Allosteric Regulation

As we have noted, glucose 6-phosphate can flow either into glycolysis or through any of several other pathways, including glycogen synthesis and the pentose phosphate pathway. The metabolically irreversible reaction catalyzed by PFK-1 is the step that commits glucose to glycolysis. In addition to its substrate-binding sites, this

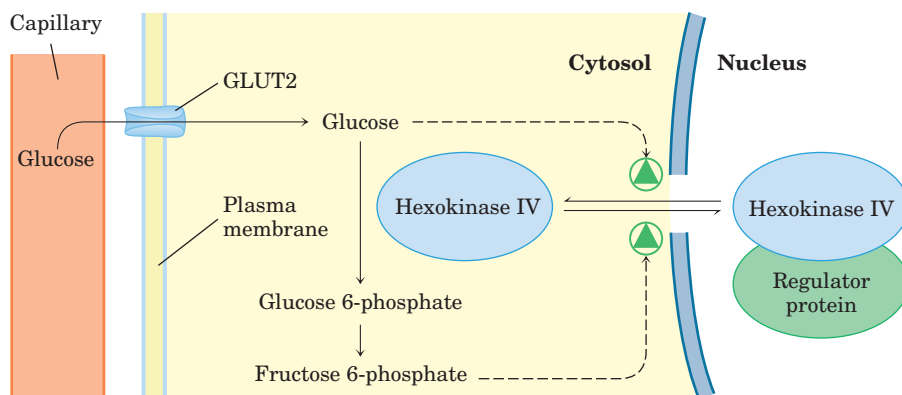
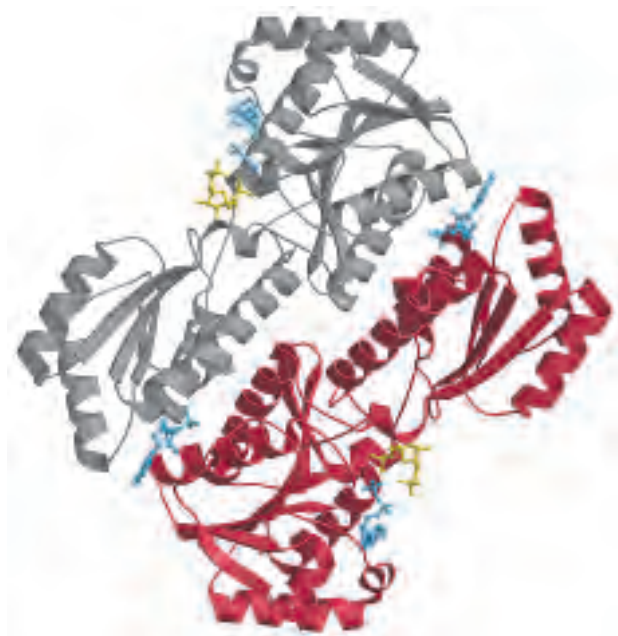


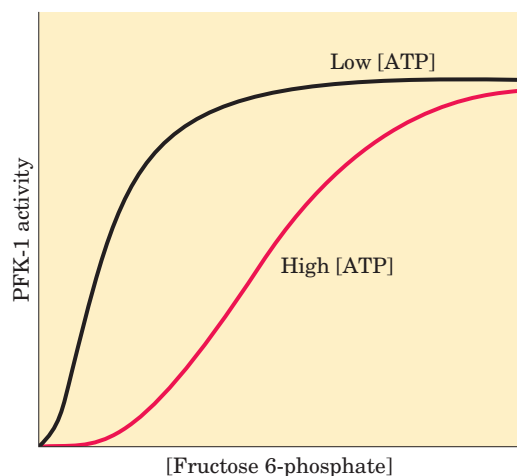
FIGURE 15-17 Regulation of hexokinase IV (glucokinase) by sequestration in the nucleus. The protein inhibitor of hexokinase IV is a nuclear binding protein that draws hexokinase IV into the nucleus when the fructose 6-phosphate concentration in liver is high and releases it to the cytosol when the glucose concentration is high.

complex enzyme has several regulatory sites at which allosteric activators or inhibitors bind.

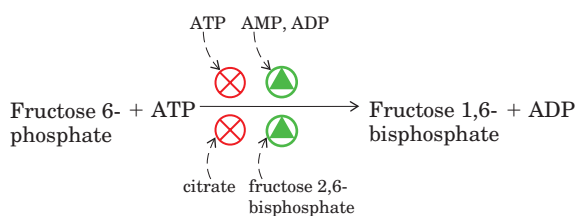
ATP is not only a substrate for PFK-1 but also an end product of the glycolytic pathway. When high cellular [ATP] signals that ATP is being produced faster than it is being consumed, ATP inhibits PFK-1 by bind-



(a)



(b)



(c)

ing to an allosteric site and lowering the affinity of the enzyme for fructose 6-phosphate (Fig. 15–18). ADP and AMP, which increase in concentration as consumption of ATP outpaces production, act allosterically to relieve this inhibition by ATP. These effects combine to produce higher enzyme activity when ADP or AMP accumulates and lower activity when ATP accumulates.

Citrate (the ionized form of citric acid), a key intermediate in the aerobic oxidation of pyruvate, fatty acids, and amino acids, also serves as an allosteric regulator of PFK-1; high citrate concentration increases the inhibitory effect of ATP, further reducing the flow of glucose through glycolysis. In this case, as in several others encountered later, citrate serves as an intracellular signal that the cell is meeting its current needs for energy-yielding metabolism by the oxidation of fats and proteins.

The most significant allosteric regulator of PFK-1 is fructose 2,6-bisphosphate, which strongly activates the enzyme. We return to this role of fructose 2,6-bisphosphate later.

Pyruvate Kinase Is Allosterically Inhibited by ATP

At least three isozymes of pyruvate kinase are found in vertebrates, differing in their tissue distribution and their response to modulators. High concentrations of ATP, acetyl-CoA, and long-chain fatty acids (signs of abundant energy supply) allosterically inhibit all isozymes of pyruvate kinase (Fig. 15–19). The liver isozyme (L form), but not the muscle isozyme (M form), is subject to further regulation by phosphorylation. When low blood glucose causes glucagon release, cAMP-dependent protein kinase phosphorylates the L isozyme of pyruvate kinase, inactivating it. This slows the use of glucose as a fuel in liver, sparing it for export to the brain and other organs. In muscle, the effect of increased [cAMP] is quite different. In response to epinephrine, cAMP activates glycogen breakdown and glycolysis and provides the fuel needed for the fight-or-flight response.

FIGURE 15–18 Phosphofructokinase-1 (PFK-1) and its regulation.

(a) Ribbon diagram of *E. coli* phosphofructokinase-1, showing two of its four identical subunits (PDB ID 1PFK). Each subunit has its own catalytic site, where ADP (blue) and fructose 1,6-bisphosphate (yellow) are almost in contact, and its own binding sites for the allosteric regulator ADP (blue), located at the interface between subunits. (b) Allosteric regulation of muscle PFK-1 by ATP, shown by a substrate-activity curve. At low concentrations of ATP, the $K_{0.5}$ for fructose 6-phosphate is relatively low, enabling the enzyme to function at a high rate at relatively low concentrations of fructose 6-phosphate. (Recall from Chapter 6 that $K_{0.5}$ or K_m is equivalent to the substrate concentration at which half-maximal enzyme activity occurs.) When the concentration of ATP is high, $K_{0.5}$ for fructose 6-phosphate is greatly increased, as indicated by the sigmoid relationship between substrate concentration and enzyme activity. (c) Summary of the regulators affecting PFK-1 activity.

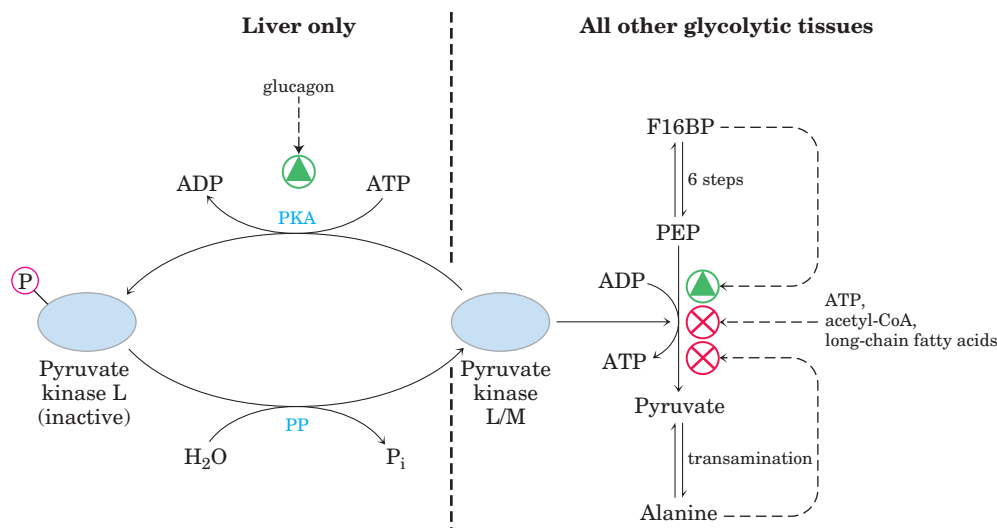


FIGURE 15-19 Regulation of pyruvate kinase. The enzyme is allosterically inhibited by ATP, acetyl-CoA, and long-chain fatty acids (all signs of an abundant energy supply), and the accumulation of fructose 1,6-bisphosphate triggers its activation. Accumulation of alanine, which can be synthesized from pyruvate in one step, allosterically inhibits pyruvate kinase, slowing the production of pyruvate by glycolysis. The liver isozyme (L form) is also regulated hormonally; glucagon

activates cAMP-dependent protein kinase (PKA; see Fig. 15-25), which phosphorylates the pyruvate kinase L isozyme, inactivating it. When the glucagon level drops, a protein phosphatase (PP) dephosphorylates pyruvate kinase, activating it. This mechanism prevents the liver from consuming glucose by glycolysis when the blood glucose concentration is low; instead, liver exports glucose. The muscle isozyme (M form) is not affected by this phosphorylation mechanism.

Gluconeogenesis Is Regulated at Several Steps

In the pathway leading from pyruvate to glucose, the first control point determines the fate of pyruvate in the mitochondrion. Pyruvate can be converted either to acetyl-CoA (by the pyruvate dehydrogenase complex; Chapter 16) to fuel the citric acid cycle, or to oxaloacetate (by pyruvate carboxylase) to start the process of gluconeogenesis (Fig. 15-20). When fatty acids are readily available as fuels, their breakdown in liver mitochondria yields acetyl-CoA, a signal that further oxidation of glucose for fuel is not necessary. Acetyl-CoA is a positive allosteric modulator of pyruvate carboxylase and a negative modulator of pyruvate dehydrogenase, through stimulation of a protein kinase that inactivates the dehydrogenase. When the cell's energetic needs are being met, oxidative phosphorylation slows, NADH rises relative to NAD⁺ and inhibits the citric acid cycle, and acetyl-CoA accumulates. The increased concentration of acetyl-CoA inhibits the pyruvate dehydrogenase complex, slowing the formation of acetyl-CoA from pyruvate, and stimulates gluconeogenesis by activating pyruvate carboxylase, allowing excess pyruvate to be converted to glucose.

The second control point in gluconeogenesis is the reaction catalyzed by FBPase-1 (Fig. 15-21), which is strongly inhibited by AMP. The corresponding glycolytic enzyme, PFK-1, is stimulated by AMP and ADP but inhibited by citrate and ATP. Thus these opposing steps

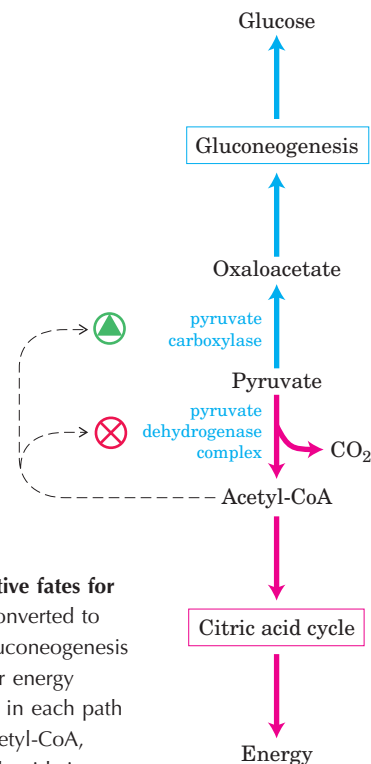


FIGURE 15-20 Two alternative fates for pyruvate. Pyruvate can be converted to glucose and glycogen via gluconeogenesis or oxidized to acetyl-CoA for energy production. The first enzyme in each path is regulated allosterically; acetyl-CoA, produced either by fatty acid oxidation or by the pyruvate dehydrogenase complex, stimulates pyruvate carboxylase and inhibits pyruvate dehydrogenase.

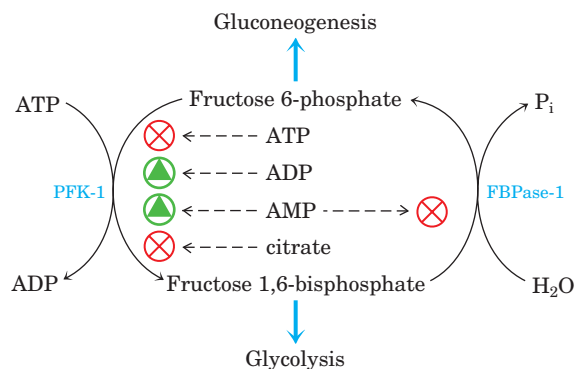


FIGURE 15-21 Regulation of fructose 1,6-bisphosphatase-1 (FBPase-1) and phosphofructokinase-1 (PFK-1). The important role of fructose 2,6-bisphosphate in the regulation of this substrate cycle is detailed in subsequent figures.

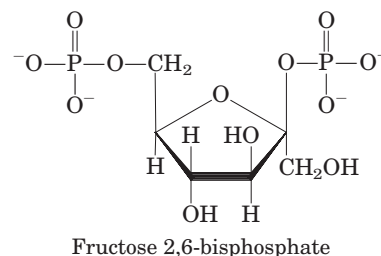
in the two pathways are regulated in a coordinated and reciprocal manner. In general, when sufficient concentrations of acetyl-CoA or citrate (the product of acetyl-CoA condensation with oxaloacetate) are present, or when a high proportion of the cell's adenylate is in the form of ATP, gluconeogenesis is favored. AMP promotes glycogen degradation and glycolysis by activating glycogen phosphorylase (via activation of phosphorylase kinase) and stimulating the activity of PFK-1.

All the regulatory actions discussed here are triggered by changes inside the cell and are mediated by very rapid, instantly reversible, allosteric mechanisms. Another set of regulatory processes is triggered from outside the cell by the hormones insulin and glucagon, which signal too much or too little glucose in the blood, respectively, or by epinephrine, which signals the impending need for fuel for a fight-or-flight response. These hormonal signals bring about covalent modification (phosphorylation or dephosphorylation) of target proteins inside the cell; this takes place on a somewhat longer time scale than the internally driven allosteric mechanisms—seconds or minutes, rather than milliseconds.

Fructose 2,6-Bisphosphate Is a Potent Regulator of Glycolysis and Gluconeogenesis

The special role of liver in maintaining a constant blood glucose level requires additional regulatory mechanisms to coordinate glucose production and consumption. When the blood glucose level decreases, the hormone **glucagon** signals the liver to produce and release more glucose and to stop consuming it for its own needs. One source of glucose is glycogen stored in the liver; another source is gluconeogenesis.

The hormonal regulation of glycolysis and gluconeogenesis is mediated by **fructose 2,6-bisphosphate**, an allosteric effector for the enzymes PFK-1 and FBPase-1 (Fig. 15-22):



When fructose 2,6-bisphosphate binds to its allosteric site on PFK-1, it increases that enzyme's affinity for its substrate, fructose 6-phosphate, and reduces its affinity for the allosteric inhibitors ATP and citrate. At the physiological concentrations of its substrates ATP and fructose 6-phosphate and of its other positive and negative effectors (ATP, AMP, citrate), PFK-1 is virtually inactive in the absence of fructose 2,6-bisphosphate. Fructose 2,6-bisphosphate *activates* PFK-1 and stimulates glycolysis in liver and, at the same time, *inhibits* FBPase-1, thereby slowing gluconeogenesis.

Although structurally related to fructose 1,6-bisphosphate, fructose 2,6-bisphosphate is not an intermediate in gluconeogenesis or glycolysis; it is a *regulator* whose cellular level reflects the level of glucagon in the blood, which rises when blood glucose falls. The cellular concentration of fructose 2,6-bisphosphate is set by the relative rates of its formation and breakdown (Fig. 15-23a). It is formed by phosphorylation of fructose 6-phosphate, catalyzed by **phosphofructokinase-2 (PFK-2)**, and is broken down by **fructose 2,6-bisphosphatase (FBPase-2)**. (Note that these enzymes are distinct from PFK-1 and FBPase-1, which catalyze the formation and breakdown, respectively, of fructose 1,6-bisphosphate.) PFK-2 and FBPase-2 are two distinct enzymatic activities of a single, bifunctional protein. The balance of these two activities in the liver, which determines the cellular level of fructose 2,6-bisphosphate, is regulated by glucagon and insulin (Fig. 15-23b). As we saw in Chapter 12 (p. 441), glucagon stimulates the adenylyl cyclase of liver to synthesize 3',5'-cyclic AMP (cAMP) from ATP. Then cAMP activates cAMP-dependent protein kinase, which transfers a phosphoryl group from ATP to the bifunctional protein PFK-2/FBPase-2. Phosphorylation of this protein enhances its FBPase-2 activity and inhibits its PFK-2 activity. Glucagon thereby lowers the cellular level of fructose 2,6-bisphosphate, inhibiting glycolysis and stimulating gluconeogenesis. The resulting production of more glucose enables the liver to replenish blood glucose in response to glucagon. Insulin has the opposite effect, stimulating the activity of a phosphoprotein phosphatase that catalyzes removal of the phosphoryl group from the bifunctional protein PFK-2/FBPase-2, activating its PFK-2 activity, increasing the level of fructose 2,6-bisphosphate, stimulating glycolysis, and inhibiting gluconeogenesis.

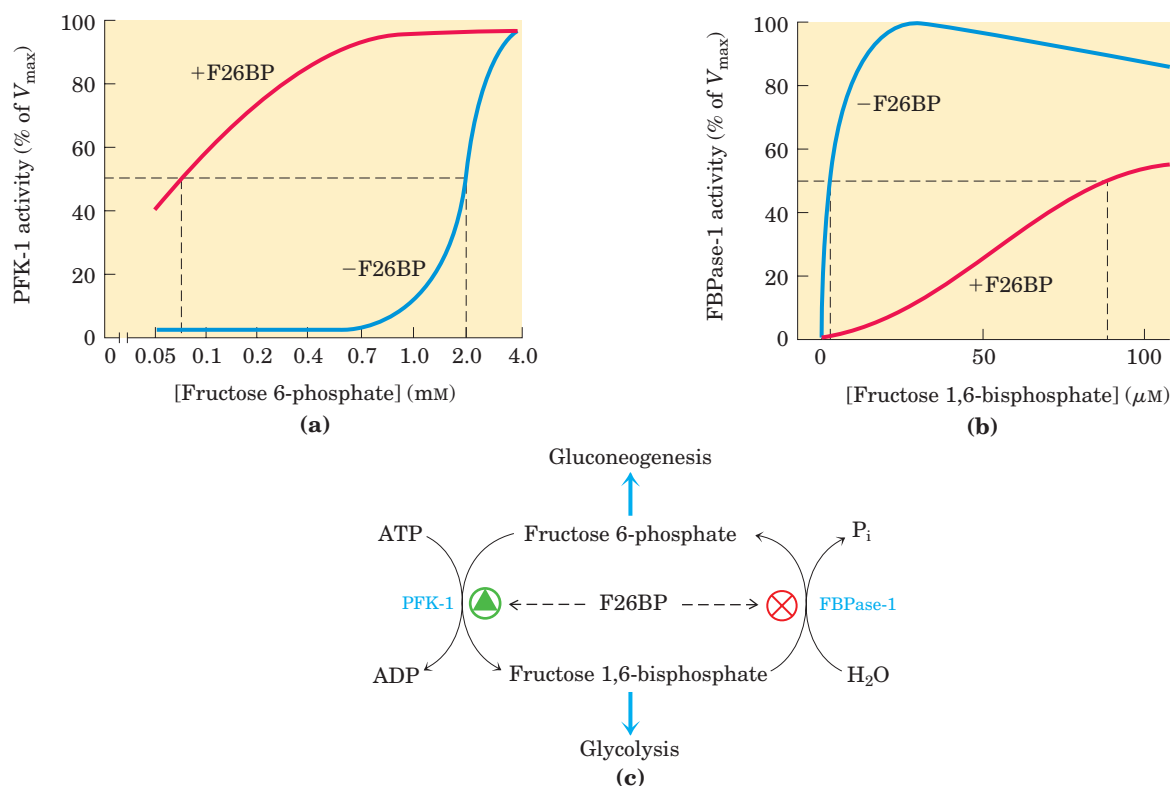


FIGURE 15-22 Role of fructose 2,6-bisphosphate in regulation of glycolysis and gluconeogenesis. Fructose 2,6-bisphosphate (F26BP) has opposite effects on the enzymatic activities of phosphofructokinase-1 (PFK-1, a glycolytic enzyme) and fructose 1,6-bisphosphatase (FBPase-1, a gluconeogenic enzyme). **(a)** PFK-1 activity in the absence of F26BP (blue curve) is half-maximal when the concentration of fructose 6-phosphate is 2 mM (that is, $K_{0.5} = 2$ mM). When 0.13 μ M F26BP is present (red curve), the $K_{0.5}$ for fructose 6-phosphate is only

0.08 mM. Thus F26BP activates PFK-1 by increasing its apparent affinity (Fig. 15-18) for fructose 6-phosphate. **(b)** FBPase-1 activity is inhibited by as little as 1 μ M F26BP and is strongly inhibited by 25 μ M. In the absence of this inhibitor (blue curve) the $K_{0.5}$ for fructose 1,6-bisphosphate is 5 μ M, but in the presence of 25 μ M F26BP (red curve) the $K_{0.5}$ is > 70 μ M. Fructose 2,6-bisphosphate also makes FBPase-1 more sensitive to inhibition by another allosteric regulator, AMP. **(c)** Summary of regulation by F26BP.

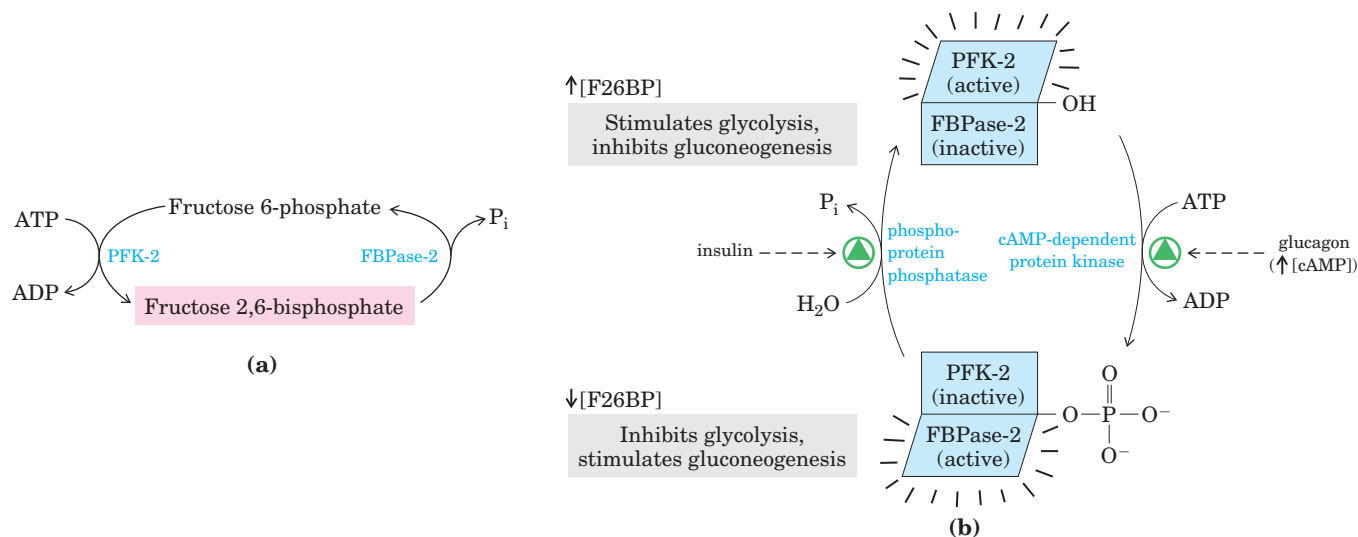


FIGURE 15-23 Regulation of fructose 2,6-bisphosphate level. **(a)** The cellular concentration of the regulator fructose 2,6-bisphosphate (F26BP) is determined by the rates of its synthesis by phosphofructokinase-2 (PFK-2) and breakdown by fructose 2,6-bisphosphatase

(FBPase-2). **(b)** Both enzymes are part of the same polypeptide chain, and both are regulated, in a reciprocal fashion, by insulin and glucagon. Here and elsewhere, arrows are used to indicate increasing ($\uparrow$) and decreasing ($\downarrow$) levels of metabolites.

Are Substrate Cycles Futile?

We noted above that substrate cycles (sometimes called futile cycles) occur at several points in the pathways that interconnect glycogen and pyruvate. For reactions such as those catalyzed by PFK-1 and FBPase-1 (Fig. 15–22c) to take place at the same time, each must be exergonic under the conditions prevailing in the cell. The PFK-1 reaction is exergonic because it involves a phosphoryl group transfer from ATP, and the FBPase-1 reaction is exergonic because it entails hydrolysis of a phosphate ester. Because the cycle involves two different enzymes, not simply one working in both directions, each activity can be regulated separately: fructose 2,6-bisphosphate activates PFK-1, favoring glycolysis, and inhibits FBPase-1, inhibiting gluconeogenesis. The two-enzyme cycle thus provides a means of controlling the *direction* of net metabolite flow. The apparent energetic disadvantage of the “futile” cycle is evidently outweighed by the advantage of allowing this type of control of pathway direction.

Xylulose 5-Phosphate Is a Key Regulator of Carbohydrate and Fat Metabolism

Another recently discovered regulatory mechanism also acts by controlling the level of fructose 2,6-bisphosphate. In the mammalian liver, xylulose 5-phosphate (see Fig. 14–23), a product of the hexose monophosphate pathway, mediates the increase in glycolysis that follows ingestion of a high-carbohydrate meal. The xylulose 5-phosphate concentration rises as glucose entering the liver is converted to glucose 6-phosphate and enters both the glycolytic and hexose monophosphate pathways. Xylulose 5-phosphate activates a phosphoprotein phosphatase, PP2A, that dephosphorylates the bifunctional PFK-2/FBPase-2 enzyme. Dephosphorylation activates PFK-2 and inhibits FBPase-2, and the resulting rise in [fructose 2,6-bisphosphate] stimulates glycolysis and inhibits gluconeogenesis. The increased glycolysis boosts the production of acetyl-CoA, while the increased flow of hexose through the hexose monophosphate pathway generates NADPH. Acetyl-CoA and NADPH are the starting materials for fatty acid synthesis, which has long been known to increase dramatically in response to intake of a high-carbohydrate meal. Xylulose 5-phosphate also increases the synthesis of all the enzymes required for fatty acid synthesis; we shall return to this effect in our discussion of the integration of carbohydrate and lipid metabolism (Chapter 23).

SUMMARY 15.3 Coordinated Regulation of Glycolysis and Gluconeogenesis

- Three glycolytic enzymes are subject to allosteric regulation: hexokinase IV, phosphofructokinase-1 (PFK-1), and pyruvate kinase.

- Hexokinase IV (glucokinase) is sequestered in the nucleus of the hepatocyte, but is released when the cytosolic glucose concentration rises.
- PFK-1 is allosterically inhibited by ATP and citrate. In most mammalian tissues, including liver, PFK-1 is allosterically activated by fructose 2,6-bisphosphate.
- Pyruvate kinase is allosterically inhibited by ATP, and the liver isozyme is inhibited by cAMP-dependent phosphorylation.
- Gluconeogenesis is regulated at the level of pyruvate carboxylase (which is activated by acetyl-CoA) and FBPase-1 (which is inhibited by fructose 2,6-bisphosphate and AMP).
- To limit futile cycling between glycolysis and gluconeogenesis, the two pathways are under reciprocal allosteric control, mainly achieved by the opposite effects of fructose 2,6-bisphosphate on PFK-1 and FBPase-1.
- Glucagon or epinephrine decreases [fructose 2,6-bisphosphate]. The hormones do this by raising [cAMP] and bringing about phosphorylation of the bifunctional enzyme that makes and breaks down fructose 2,6-bisphosphate. Phosphorylation inactivates PFK-2 and activates FBPase-2, leading to breakdown of fructose 2,6-bisphosphate. Insulin increases [fructose 2,6-bisphosphate] by activating a phosphoprotein phosphatase that dephosphorylates (activates) PFK-2.

15.4 Coordinated Regulation of Glycogen Synthesis and Breakdown

As we have seen, the mobilization of stored glycogen is brought about by glycogen phosphorylase, which degrades glycogen to glucose 1-phosphate (Fig. 15–3). Glycogen phosphorylase provides an especially instructive case of enzyme regulation. It was one of the first known examples of an allosterically regulated enzyme and the first enzyme shown to be controlled by reversible phosphorylation. It was also one of the first allosteric enzymes for which the detailed three-dimensional structures of the active and inactive forms were revealed by x-ray crystallographic studies. Glycogen phosphorylase also illustrates how isozymes play their tissue-specific roles.

Glycogen Phosphorylase Is Regulated Allosterically and Hormonally

In the late 1930s, Carl and Gerty Cori (Box 15–1) discovered that the glycogen phosphorylase of skeletal muscle exists in two interconvertible forms: **glycogen phosphorylase a**, which is catalytically active, and

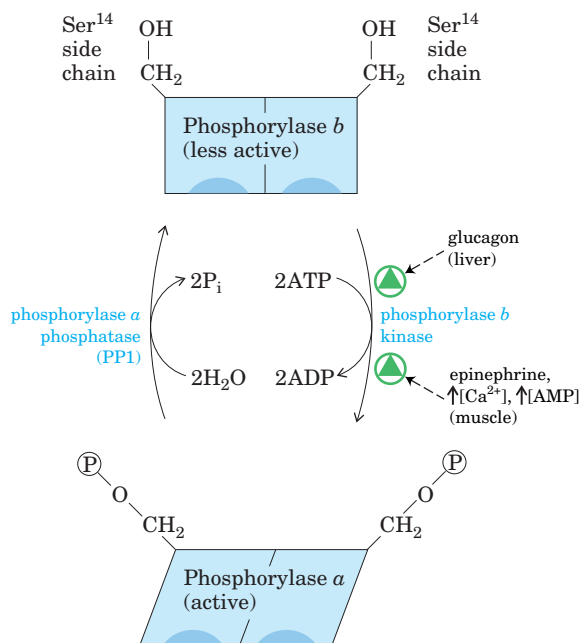


FIGURE 15-24 Regulation of muscle glycogen phosphorylase by covalent modification. In the more active form of the enzyme, phosphorylase *a*, Ser¹⁴ residues, one on each subunit, are phosphorylated. Phosphorylase *a* is converted to the less active form, phosphorylase *b*, by enzymatic loss of these phosphoryl groups, catalyzed by phosphorylase *a* phosphatase (PP1). Phosphorylase *b* can be reconverted (reactivated) to phosphorylase *a* by the action of phosphorylase *b* kinase.

glycogen phosphorylase *b*, which is less active (Fig. 15-24). Subsequent studies by Earl Sutherland showed that phosphorylase *b* predominates in resting muscle, but during vigorous muscular activity the hormone epinephrine triggers phosphorylation of a specific Ser residue in phosphorylase *b*, converting it to its more active form, phosphorylase *a*. (Note that glycogen phosphorylase is often referred to simply as phosphorylase—so honored because it was the first phosphorylase to be discovered; the shortened name has persisted in common usage and in the literature.)



Earl W. Sutherland, Jr.,
1915–1974

The enzyme (phosphorylase *b* kinase) responsible for activating phosphorylase by transferring a phosphoryl group to its Ser residue is itself activated by epinephrine or glucagon through a series of steps shown in Figure 15-25. Sutherland discovered the second messenger cAMP, which increases in concentration in response to stimulation by epinephrine (in muscle) or glucagon (in liver). Elevated [cAMP] ini-

tiates an **enzyme cascade**, in which a catalyst activates a catalyst, which activates a catalyst. Such cascades allow for large amplification of the initial signal (see pink boxes in Fig. 15-25). The rise in [cAMP] activates cAMP-dependent protein kinase, also called protein kinase A (PKA). PKA then phosphorylates and activates **phosphorylase *b* kinase**, which catalyzes the phosphorylation of Ser residues in each of the two identical subunits of glycogen phosphorylase, activating it and thus stimulating glycogen breakdown. In muscle, this provides fuel for glycolysis to sustain muscle contraction for the fight-or-flight response signaled by epinephrine. In liver, glycogen breakdown counters the low blood glucose signaled by glucagon, releasing glucose. These different roles are reflected in subtle differences in the regulatory mechanisms in muscle and liver. The glycogen phosphorylases of liver and muscle are isozymes, encoded by different genes and differing in their regulatory properties.

In muscle, superimposed on the regulation of phosphorylase by covalent modification are two allosteric control mechanisms (Fig. 15-25). Ca^{2+} , the signal for muscle contraction, binds to and activates phosphorylase *b* kinase, promoting conversion of phosphorylase *b* to the active *a* form. Ca^{2+} binds to phosphorylase *b* kinase through its δ subunit, which is calmodulin (see Fig. 12-21). AMP, which accumulates in vigorously contracting muscle as a result of ATP breakdown, binds to and activates phosphorylase, speeding the release of glucose 1-phosphate from glycogen. When ATP levels are adequate, ATP blocks the allosteric site to which AMP binds, inactivating phosphorylase.

When the muscle returns to rest, a second enzyme, **phosphorylase *a* phosphatase**, also called **phosphoprotein phosphatase 1 (PP1)**, removes the phosphoryl groups from phosphorylase *a*, converting it to the less active form, phosphorylase *b*.

Like the enzyme of muscle, the glycogen phosphorylase of liver is regulated hormonally (by phosphorylation/dephosphorylation) and allosterically. The dephosphorylated form is essentially inactive. When the blood glucose level is too low, glucagon (acting by the same cascade mechanism shown in Fig. 15-25) activates phosphorylase *b* kinase, which in turn converts phosphorylase *b* to its active *a* form, initiating the release of glucose into the blood. When blood glucose levels return to normal, glucose enters hepatocytes and binds to an inhibitory allosteric site on phosphorylase *a*. This binding also produces a conformational change that exposes the phosphorylated Ser residues to PP1, which catalyzes their dephosphorylation and inactivates the phosphorylase (Fig. 15-26). The allosteric site for glucose allows liver glycogen phosphorylase to act as its own glucose sensor and to respond appropriately to changes in blood glucose.

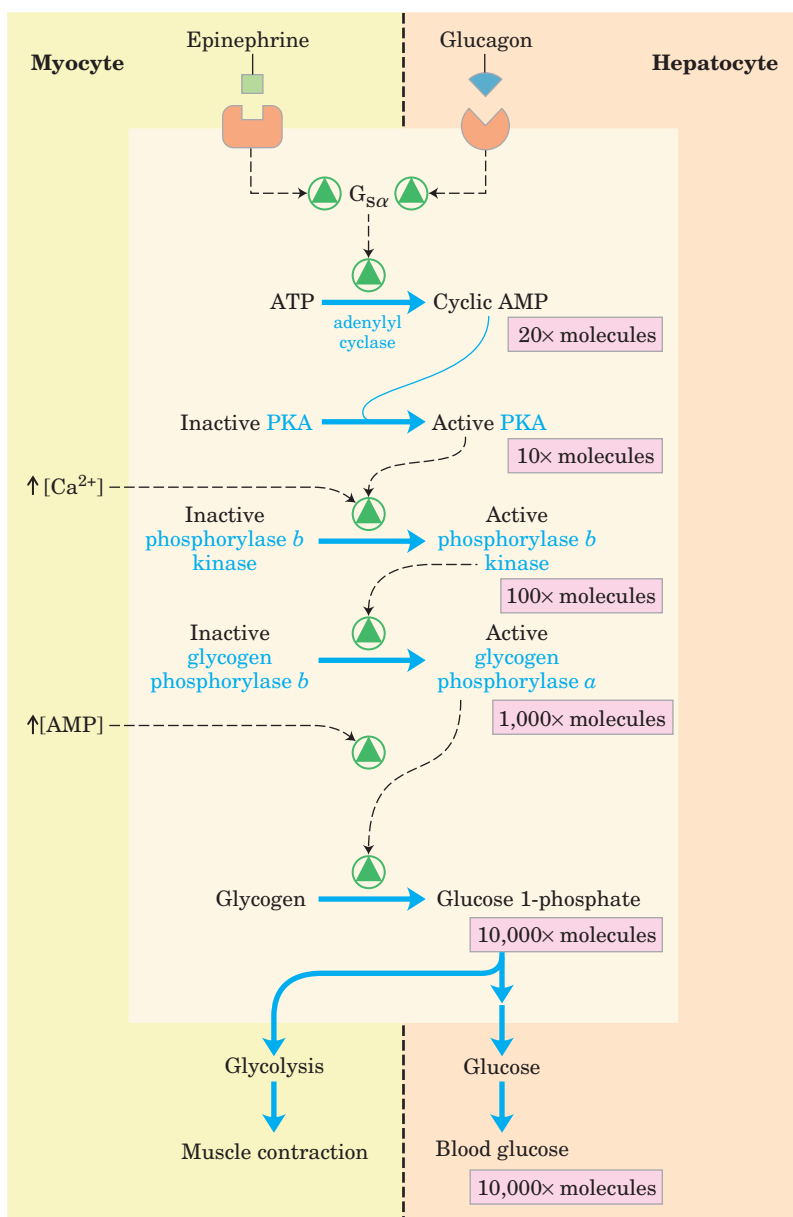


FIGURE 15-25 Cascade mechanism of epinephrine and glucagon action. By binding to specific surface receptors, either epinephrine acting on a myocyte (left) or glucagon acting on a hepatocyte (right) activates a GTP-binding protein G_{sα} (see Fig. 12-12). Active G_{sα} triggers a rise in [cAMP], activating PKA. This sets off a cascade of phosphorylations; PKA activates phosphorylase b kinase, which then activates glycogen phosphorylase. Such cascades effect a large amplification of the initial signal; the figures in pink boxes are probably low estimates of the actual increase in number of molecules at each stage of the cascade. The resulting breakdown of glycogen provides glucose, which in the myocyte can supply ATP (via glycolysis) for muscle contraction and in the hepatocyte is released into the blood to counter the low blood glucose.

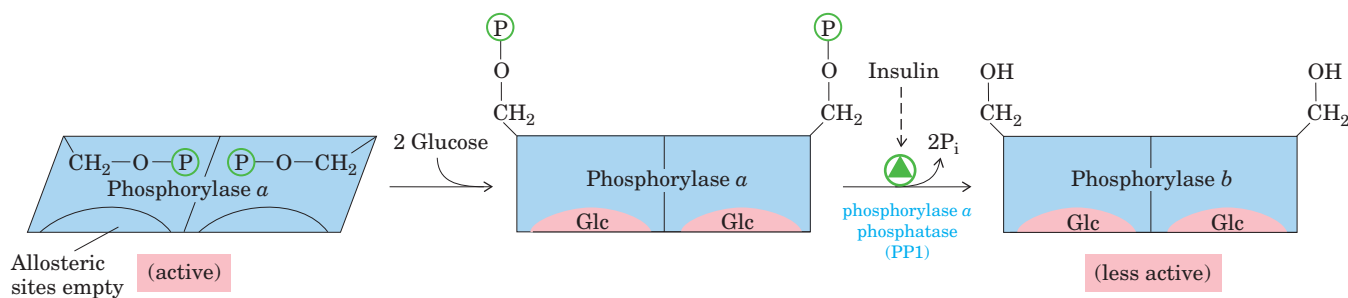
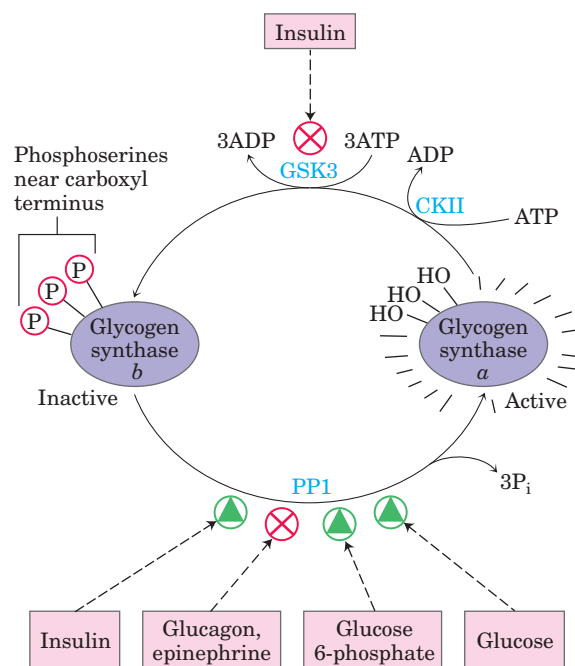


FIGURE 15-26 Glycogen phosphorylase of liver as a glucose sensor. Glucose binding to an allosteric site of the phosphorylase a isozyme of liver induces a conformational change that exposes its phosphorylated Ser residues to the action of phosphorylase a phosphatase 1 (PP1).

This phosphatase converts phosphorylase a to phosphorylase b, sharply reducing the activity of phosphorylase and slowing glycogen breakdown in response to high blood glucose. Insulin also acts indirectly to stimulate PP1 and slow glycogen breakdown.

FIGURE 15-27 Effects of GSK3 on glycogen synthase activity.

Glycogen synthase *a*, the active form, has three Ser residues near its carboxyl terminus, which are phosphorylated by glycogen synthase kinase 3 (GSK3). This converts glycogen synthase to the inactive (*b*) form (GS*b*). GSK3 action requires prior phosphorylation (priming) by casein kinase (CKII). Insulin triggers activation of glycogen synthase *b* by blocking the activity of GSK3 (see the pathway for this action in Fig. 12-8) and activating a phosphoprotein phosphatase (PP1 in muscle, another phosphatase in liver). In muscle, epinephrine activates PKA, which phosphorylates the glycogen-targeting protein G α (see Fig. 15-30) on a site that causes dissociation of PP1 from glycogen. Glucose 6-phosphate favors dephosphorylation of glycogen synthase by binding to it and promoting a conformation that is a good substrate for PP1. Glucose also promotes dephosphorylation; the binding of glucose to glycogen phosphorylase *a* forces a conformational change that favors dephosphorylation to glycogen phosphorylase *b*, thus relieving its inhibition of PP1 (see Fig. 15-29).



Glycogen Synthase Is Also Regulated by Phosphorylation and Dephosphorylation

Like glycogen phosphorylase, glycogen synthase can exist in phosphorylated and dephosphorylated forms (Fig. 15-27). Its active form, **glycogen synthase *a***, is unphosphorylated. Phosphorylation of the hydroxyl side chains of several Ser residues of both subunits converts glycogen synthase *a* to **glycogen synthase *b***, which is inactive unless its allosteric activator, glucose 6-phosphate, is present. Glycogen synthase is remarkable for its ability to be phosphorylated on various residues by at least 11 different protein kinases. The most important regulatory kinase is **glycogen synthase kinase 3 (GSK3)**, which adds phosphoryl groups to three Ser residues near the carboxyl terminus of glycogen synthase, strongly inactivating it. The action of GSK3 is hierarchical; it cannot phosphorylate glycogen synthase until another protein kinase, **casein kinase II (CKII)**, has first phosphorylated the glycogen synthase on a nearby residue, an event called **priming** (Fig. 15-28a).

In liver, conversion of glycogen synthase *b* to the active form is promoted by PP1, which is bound to the glycogen particle. PP1 removes the phosphoryl groups from the three Ser residues phosphorylated by GSK3. Glucose 6-phosphate binds to an allosteric site on glycogen synthase *b*, making the enzyme a better substrate for dephosphorylation by PP1 and causing its activation. By analogy with glycogen phosphorylase, which acts as a glucose sensor, glycogen synthase can be regarded as

a glucose 6-phosphate sensor. In muscle, a different phosphatase may have the role played by PP1 in liver, activating glycogen synthase by dephosphorylating it.

Glycogen Synthase Kinase 3 Mediates the Actions of Insulin

As we saw in Chapter 12, one way in which insulin triggers intracellular changes is by activating a protein kinase (protein kinase B, or PKB) that in turn phosphorylates and inactivates GSK3 (Fig. 15-29; see also Fig. 12-8). Phosphorylation of a Ser residue near the amino terminus of GSK3 converts that region of the protein to a pseudosubstrate, which folds into the site at which the priming phosphorylated Ser residue normally binds (Fig. 15-28b). This prevents GSK3 from binding the priming site of a real substrate, thereby inactivating the enzyme and tipping the balance in favor of dephosphorylation of glycogen synthase by PP1. Glycogen phosphorylase can also affect the phosphorylation of glycogen synthase: active glycogen phosphorylase directly inhibits PP1, preventing it from activating glycogen synthase (Fig. 15-27).

Although first discovered in its role in glycogen metabolism (hence the name glycogen synthase kinase), GSK3 clearly has a much broader role than the regulation of glycogen synthase. It mediates signaling by insulin and other growth factors and nutrients, and it acts in the specification of cell fates during embryonic development. Among its targets are cytoskeletal proteins

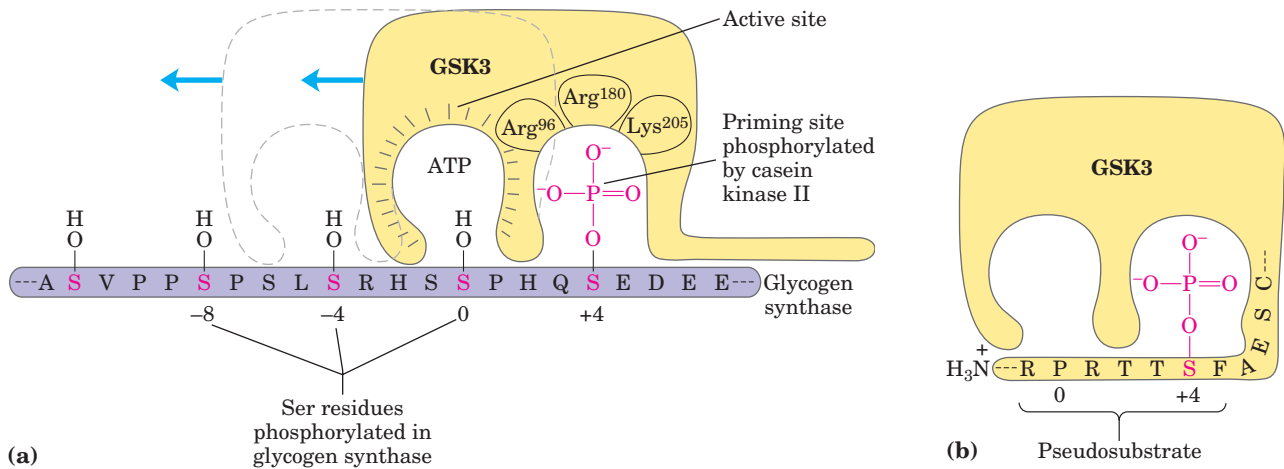


FIGURE 15-28 Priming of GSK3 phosphorylation of glycogen synthase. (a) Glycogen synthase kinase 3 first associates with its substrate (glycogen synthase) by interaction between three positively charged residues (Arg⁹⁶, Arg¹⁸⁰, Lys²⁰⁵) and a phosphoserine residue at position +4 in the substrate. (For orientation, the Ser or Thr residue to be phosphorylated in the substrate is assigned the index 0. Residues on the amino-terminal side of this residue are numbered -1, -2, and so forth; residues on the carboxyl-terminal side are numbered +1, +2, and so forth.) This association aligns the active site of the enzyme with a Ser residue at position 0, which it phosphorylates. This creates a new

priming site, and the enzyme moves down the protein to phosphorylate the Ser residue at position -4, and then the Ser at -8. (b) GSK3 has a Ser residue near its amino terminus that can be phosphorylated by PKA or PKB (see Fig. 15-29). This produces a “pseudosubstrate” region in GSK3 that folds into the priming site and makes the active site inaccessible to another protein substrate, inhibiting GSK3 until the priming phosphoryl group of its pseudosubstrate region is removed by PP1. Other proteins that are substrates for GSK3 also have a priming site at position +4, which must be phosphorylated by another protein kinase before GSK3 can act on them.

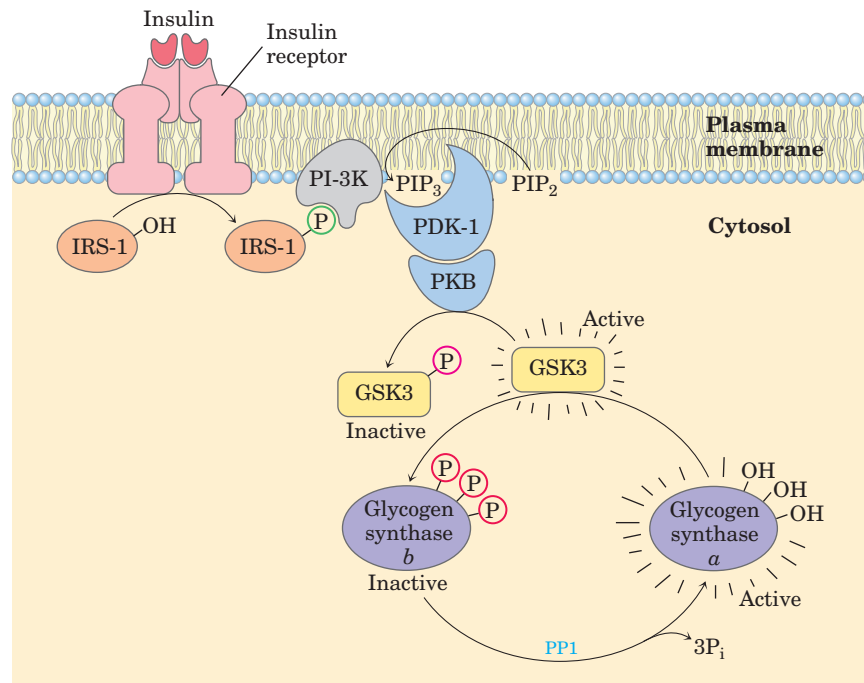


FIGURE 15-29 The path from insulin to GSK3 and glycogen synthase. Insulin binding to its receptor activates a tyrosine protein kinase in the receptor, which phosphorylates insulin receptor substrate-1 (IRS-1). The phosphotyrosine in this protein is then bound by phosphatidylinositol 3-kinase (PI-3K), which converts phosphatidylinositol 4,5-bisphosphate (PIP₂) in the membrane to phosphatidylinositol 3,4,5-trisphosphate (PIP₃). A protein kinase (PDK-1) that is activated

when bound to PIP₃ activates a second protein kinase (PKB), which phosphorylates glycogen synthase kinase 3 (GSK3) in its pseudosubstrate region, inactivating it by the mechanisms shown in Figure 15-28b. The inactivation of GSK3 allows phosphoprotein phosphatase 1 (PP1) to dephosphorylate glycogen synthase, converting it to its active form. In this way, insulin stimulates glycogen synthesis. (See Fig. 12-8 for more details on insulin action.)

and proteins essential for mRNA and protein synthesis. These targets, like glycogen synthase, must first undergo a priming phosphorylation by another protein kinase before they can be phosphorylated by GSK3.

Phosphoprotein Phosphatase 1 Is Central to Glycogen Metabolism

A single enzyme, PP1, can remove phosphoryl groups from all three of the enzymes phosphorylated in response to glucagon (liver) and epinephrine (liver and muscle): phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. Insulin stimulates glycogen synthesis by activating PP1 and by inactivating GSK3.

PP1 does not exist free in the cytosol, but is tightly bound to its target proteins by one of a family of **glycogen-targeting proteins** that bind glycogen and each of the three enzymes, glycogen phosphorylase, phosphorylase kinase, and glycogen synthase (Fig. 15–30). PP1 is itself subject to covalent and allosteric regulation; it is inactivated when phosphorylated by PKA and is allosterically activated by glucose 6-phosphate.

Transport into Cells Can Limit Glucose Utilization

The passive uptake of glucose by muscle and adipose tissue is catalyzed by the GLUT4 transporter described in Box 11–2. In the absence of insulin, most GLUT4 molecules are sequestered in membrane vesicles within the cell, but when blood glucose rises, release of insulin triggers GLUT4 movement to the plasma membrane. Glucose transport into hepatocytes involves a different, high-capacity transporter, GLUT2, which is always present in the plasma membrane. It catalyzes facilitated diffusion of glucose in both directions, at a rate high enough to ensure virtually instantaneous equilibration of glucose concentration in the blood and in the hepa-

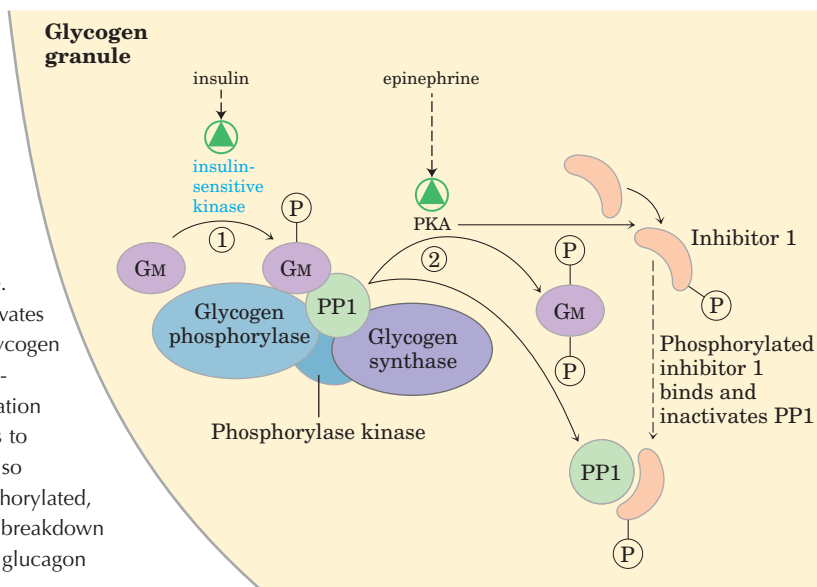
tocyte cytosol. In its role as a glucose sensor, the glycogen phosphorylase of hepatocytes is essentially measuring the glucose level in blood.

Allosteric and Hormonal Signals Coordinate Carbohydrate Metabolism

Having looked at the mechanisms that regulate individual enzymes, we can now consider the overall shifts in carbohydrate metabolism that occur in the well-fed state, during fasting, and in the fight-or-flight response—signaled by insulin, glucagon, and epinephrine, respectively. We need to contrast two cases in which regulation serves different ends: (1) the role of hepatocytes in supplying glucose to the blood, and (2) the selfish use of carbohydrate fuels by nonhepatic tissues, typified by skeletal muscle (the myocyte), to support their own activities.

After ingestion of a carbohydrate-rich meal, the elevation of blood glucose triggers insulin release (Fig. 15–31, top). In a hepatocyte, insulin has two immediate effects: it inactivates GSK3, acting through the cascade shown in Figure 15–29, and activates a protein phosphatase, perhaps PP1. These two actions fully activate glycogen synthase. PP1 also inactivates glycogen phosphorylase α and phosphorylase kinase by dephosphorylating both, effectively stopping glycogen breakdown. Glucose enters the hepatocyte through the high-capacity transporter GLUT2, always present in the plasma membrane, and the elevated intracellular glucose leads to dissociation of hexokinase IV (glucokinase) from its nuclear regulator protein. Hexokinase IV enters the cytosol and phosphorylates glucose, stimulating glycolysis and supplying the precursor for glycogen synthesis. Under these conditions, hepatocytes use the excess glucose in the blood to synthesize glycogen, up to the limit of about 10% of the total weight of the liver.

FIGURE 15–30 Glycogen-targeting protein GM. The glycogen-targeting protein GM is one of a family of proteins that bind other proteins (including PP1) to glycogen particles. GM can be phosphorylated in two different positions in response to insulin or epinephrine. ① Insulin-stimulated phosphorylation of GM site 1 activates PP1, which dephosphorylates phosphorylase kinase, glycogen phosphorylase, and glycogen synthase. ② Epinephrine-stimulated phosphorylation of GM site 2 causes dissociation of PP1 from the glycogen particle, preventing its access to glycogen phosphorylase and glycogen synthase. PKA also phosphorylates a protein (inhibitor 1) that, when phosphorylated, inhibits PP1. By these means, insulin inhibits glycogen breakdown and stimulates glycogen synthesis, and epinephrine (or glucagon in the liver) has the opposite effects.



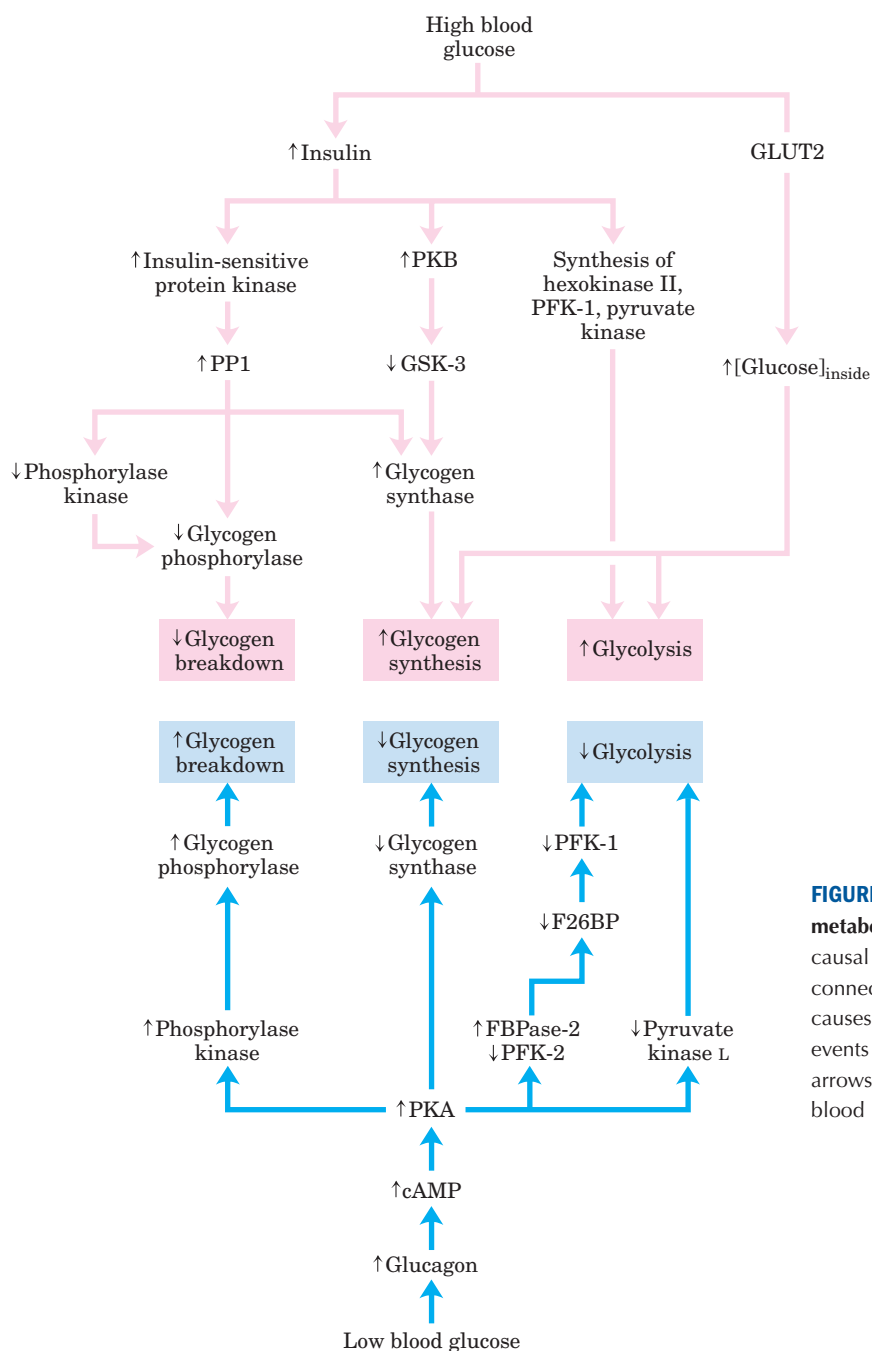


FIGURE 15-31 Regulation of carbohydrate metabolism in the hepatocyte. Arrows indicate causal relationships between the changes they connect. $\downarrow A \rightarrow \uparrow B$ means that a decrease in A causes an increase in B. Pink arrows connect events that result from high blood glucose; blue arrows connect events that result from low blood glucose.

Between meals, or during an extended fast, the drop in blood glucose triggers the release of glucagon, which, acting through the cascade shown in Figure 15-25, activates PKA. PKA mediates all the effects of glucagon (Fig. 15-31, bottom). It phosphorylates phosphorylase kinase, activating it and leading to the activation of glycogen phosphorylase. It phosphorylates glycogen synthase, inactivating it and blocking glycogen synthesis. It phosphorylates PFK-2/FBPase-2, leading to a drop in the concentration of the regulator fructose 2,6-bisphosphate, which has the effect of inactivating the glycolytic enzyme PFK-1 and activating the gluconeogenic enzyme FBPase-1. And it phosphorylates and inactivates the glycolytic enzyme pyruvate kinase. Under these conditions, the

liver produces glucose 6-phosphate by glycogen breakdown and by gluconeogenesis, and it stops using glucose to fuel glycolysis or make glycogen, maximizing the amount of glucose it can release to the blood. This release of glucose is possible only in liver, because other tissues lack glucose 6-phosphatase (Fig. 15-6).

The physiology of skeletal muscle differs from that of liver in three ways important to our discussion of metabolic regulation (Fig. 15-32): (1) muscle uses its stored glycogen only for its own needs; (2) as it goes from rest to vigorous contraction, muscle undergoes very large changes in its demand for ATP, which is supported by glycolysis; (3) muscle lacks the enzymatic machinery for gluconeogenesis. The regulation of carbohydrate

TABLE 15–3 Some of the Genes Regulated by Insulin

<i>Change in gene expression</i>	<i>Pathway</i>
Increased expression	
Hexokinase II	Glycolysis
Hexokinase IV	Glycolysis
Phosphofructokinase-1 (PFK-1)	Glycolysis
Pyruvate kinase	Glycolysis
PFK-2/FBPase-2	Regulation of glycolysis/gluconeogenesis
Glucose 6-phosphate dehydrogenase	Pentose phosphate pathway (NADPH)
6-Phosphogluconate dehydrogenase	Pentose phosphate pathway (NADPH)
Pyruvate dehydrogenase	Fatty acid synthesis
Acetyl-CoA carboxylase	Fatty acid synthesis
Malic enzyme	Fatty acid synthesis (NADPH)
ATP-citrate lyase	Fatty acid synthesis (provides acetyl-CoA)
Fatty acid synthase complex	Fatty acid synthesis
Stearoyl-CoA dehydrogenase	Fatty acid desaturation
Acyl-CoA-glycerol transferases	Triacylglycerol synthesis
Decreased expression	
PEP carboxykinase	Gluconeogenesis
Glucose 6-phosphatase (catalytic subunit)	Glucose release to blood

metabolism in muscle reflects these differences from liver. First, myocytes lack receptors for glucagon. Second, the muscle isozyme of pyruvate kinase is not phosphorylated by PKA, so glycolysis is not turned off when [cAMP] is high. In fact, cAMP *increases* the rate of glycolysis in muscle, probably by activating glycogen phosphorylase. When epinephrine is released into the blood in a fight-or-flight situation, PKA is activated by the rise in [cAMP], and phosphorylates and activates glycogen phosphorylase kinase. The resulting phosphorylation and activation of glycogen phosphorylase results in faster glycogen breakdown. Epinephrine is not released under low-stress conditions, but with each neuronal stimulation of muscle contraction, cytosolic $[Ca^{2+}]$ rises briefly and activates phosphorylase kinase through its calmodulin subunit.

Elevated insulin triggers increased glycogen synthesis in myocytes by activating PP1 and inactivating GSK3. Unlike hepatocytes, myocytes have a reserve of GLUT4 sequestered in intracellular vesicles. Insulin triggers their movement to the plasma membrane, where they allow increased glucose uptake. In response to insulin, therefore, myocytes help to lower blood glucose by increasing their rates of glucose uptake, glycogen synthesis, and glycolysis.

Insulin Changes the Expression of Many Genes Involved in Carbohydrate and Fat Metabolism

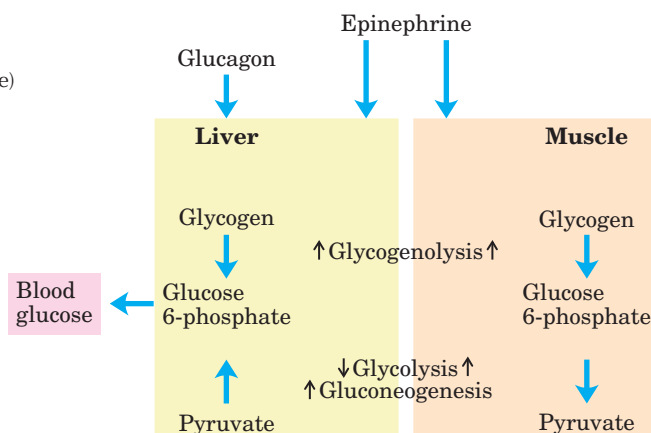
In addition to its effects on the activity of existing enzymes, insulin also regulates the expression of as many as 150 genes, including some related to fuel metabolism

(Fig. 15–31; Table 15–3). Insulin stimulates the transcription of the genes that encode hexokinases II and IV, PFK-1, pyruvate kinase, and the bifunctional enzyme PFK-2/FBPase-2 (all involved in glycolysis and its regulation), several enzymes involved in fatty acid synthesis, and two enzymes that generate the reductant for fatty acid synthesis (NADPH) via the pentose phosphate pathway (glucose 6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase). Insulin also slows the expression of the genes for two enzymes of gluconeogenesis (PEP carboxykinase and glucose 6-phosphatase). These effects take place on a longer time scale (minutes to hours) than those mediated by covalent alteration of enzymes, but the impact on metabolism can be very significant. When the diet provides an excess of glucose, the resulting rise in insulin increases the synthesis of glucose-metabolizing proteins, and glucose becomes the fuel of choice (via glycolysis) for liver, adipose tissue, and muscle. In liver and adipose tissue, glucose is converted to glycogen and triacylglycerols for temporary storage.

Carbohydrate and Lipid Metabolism Are Integrated by Hormonal and Allosteric Mechanisms

As complex as the regulation of carbohydrate metabolism is, it is far from the whole story of fuel metabolism. The metabolism of fats and fatty acids is very closely tied to that of carbohydrates. Hormonal signals such as insulin and changes in diet or exercise are equally important in regulating fat metabolism and integrating it with that of carbohydrates. We shall return to this overall metabolic integration in mammals in Chapter 23,

FIGURE 15-32 Difference in the regulation of carbohydrate metabolism in liver and muscle. In liver, either glucagon (indicating low blood glucose) or epinephrine (signaling the need to fight or flee) has the effect of maximizing the output of glucose into the bloodstream. In muscle, epinephrine increases glycogen breakdown and glycolysis, which together provide fuel to produce the ATP needed for muscle contraction.



after first considering the metabolic pathways for fats and amino acids (Chapters 17 and 18). The message we wish to convey here is that metabolic pathways are overlaid with complex regulatory controls that are exquisitely sensitive to changes in metabolic circumstances. These mechanisms act to adjust the flow of metabolites through various metabolic pathways as needed by the cell and organism, and do so without causing major changes in the concentrations of intermediates shared with other pathways.

SUMMARY 15.4 Coordinated Regulation of Glycogen Synthesis and Breakdown

- Glycogen phosphorylase is activated in response to glucagon or epinephrine, which raise [cAMP] and activate PKA. PKA phosphorylates and activates phosphorylase kinase, which converts glycogen phosphorylase *b* to its active *a* form. Phosphoprotein phosphatase 1 (PP1) reverses the phosphorylation of glycogen phosphorylase *a*, inactivating it. Glucose binds to the liver isozyme of glycogen phosphorylase *a*, favoring its dephosphorylation and inactivation.
- Glycogen synthase *a* is inactivated by phosphorylation catalyzed by GSK3. Insulin blocks GSK3. PP1, which is activated by insulin, reverses the inhibition by dephosphorylating glycogen synthase *b*.
- Insulin increases glucose uptake into myocytes and adipocytes by triggering movement of the glucose transporter GLUT4 to the plasma membrane.
- Insulin stimulates the synthesis of hexokinases II and IV, PFK-1, pyruvate kinase, and several enzymes involved in lipid synthesis. Insulin stimulates glycogen synthesis in muscle and liver.

- In liver, glucagon stimulates glycogen breakdown and gluconeogenesis while blocking glycolysis, thereby sparing glucose for export to the brain and other tissues.
- In muscle, epinephrine stimulates glycogen breakdown and glycolysis, providing ATP to support contraction.

15.5 Analysis of Metabolic Control

For every complex problem there is a simple solution. And it is always wrong.

—H. L. Mencken, *A Mencken Chrestomathy*, 1949

Beginning with Eduard Buchner's discovery (c. 1900) that an extract of broken yeast cells could convert glucose to ethanol and CO₂, a major thrust of biochemical research was to deduce the steps by which this transformation occurred and to purify and characterize the enzymes that catalyzed each step. By the middle of the twentieth century, all ten enzymes of the glycolytic pathway had been purified and characterized. In the next 50 years much was learned about the regulation of these enzymes by intracellular and extracellular signals, through the kinds of allosteric and covalent mechanisms we have described in this chapter. The conventional wisdom was that in a linear pathway such as glycolysis, catalysis by one enzyme must be the slowest and must therefore determine the rate of metabolite flow, or flux, through the whole pathway. For glycolysis, PFK-1 was considered the rate-limiting enzyme, because it was known to be closely regulated by fructose 2,6-bisphosphate and other allosteric effectors.



Eduard Buchner,
1860–1917

With the advent of genetic engineering technology, it became possible to test this “single rate-determining step” hypothesis by increasing the concentration of the enzyme that catalyzes the “rate-limiting step” in a pathway and determining whether flux through the pathway increases proportionally. More often than not, it does not do so: the simple solution (a single rate-determining step) is wrong. It has now become clear that in most pathways the control of flux is distributed among several enzymes, and the extent to which each contributes to the control varies with metabolic circumstances—the supply of the starting material (say, glucose), the supply of oxygen, the need for other products derived from intermediates of the pathway (say, glucose 6-phosphate for the pentose phosphate pathway in cells synthesizing large amounts of nucleotides), the effects of metabolites with regulatory roles, and the hormonal status of the organism (the levels of insulin and glucagon), among other factors.

Why are we interested in what limits the flux through a pathway? To understand the action of hormones or drugs, or the pathology that results from a failure of metabolic regulation, we must know where control is exercised. If researchers wish to develop a drug that stimulates or inhibits a pathway, the logical target is the enzyme that has the greatest impact on the flux through that pathway. And the bioengineering of a microorganism to overproduce a product of commercial value (p. 315) requires a knowledge of what limits the flux of metabolites toward that product.

The Contribution of Each Enzyme to Flux through a Pathway Is Experimentally Measurable

There are several ways to determine experimentally how a change in the activity of one enzyme in a pathway affects metabolite flux through that pathway. Consider the experimental results shown in Figure 15–33. When a sample of rat liver was homogenized to release all soluble enzymes, the extract carried out the glycolytic conversion of glucose to fructose 1,6-bisphosphate at a measurable rate. (This experiment, for simplicity, focused on just the first part of the glycolytic pathway.) When increasing amounts of purified hexokinase IV were added to the extract, the rate of glycolysis progressively increased. The addition of purified PFK-1 to the extract also increased the rate of glycolysis, but not as dramatically as did hexokinase. Purified phosphohexose isomerase was without effect. These results suggest that hexokinase and PFK-1 both contribute to setting the flux through the pathway (hexokinase more than PFK-1), and that phosphohexose isomerase does not.

Similar experiments can be done on intact cells or organisms, using specific inhibitors or activators to change the activity of one enzyme while observing the effect on flux through the pathway. The amount of an

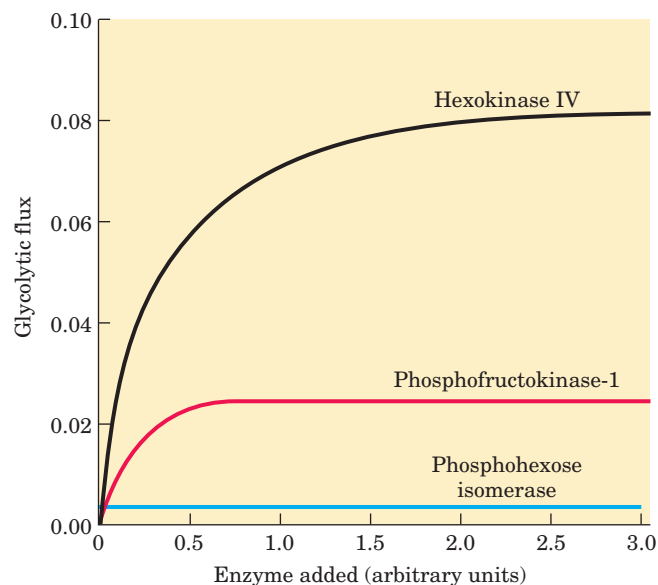


FIGURE 15–33 Dependence of glycolytic flux in a rat liver homogenate on added enzymes. Purified enzymes in the amounts shown on the x axis were added to an extract of liver carrying out glycolysis in vitro. The increase in flux through the pathway is shown on the y axis.

enzyme can also be altered genetically; bioengineering can produce a cell that makes extra copies of the enzyme under investigation or has a version of the enzyme that is less active than the normal enzyme. Increasing the concentration of an enzyme genetically sometimes has significant effects on flux; sometimes it has no effect.

Three critical parameters, which together describe the responsiveness of a pathway to changes in metabolic circumstances, lie at the center of **metabolic control analysis**. We turn now to a qualitative description of these parameters and their meaning in the context of a living cell. In Box 15–3 we will provide a more rigorous quantitative discussion.

The Control Coefficient Quantifies the Effect of a Change in Enzyme Activity on Metabolite Flux through a Pathway

Quantitative data obtained as described in Figure 15–33 can be used to calculate a **flux control coefficient**, C , for each enzyme in a pathway. This coefficient expresses the relative contribution of each enzyme to setting the rate at which metabolites flow through the pathway—that is, the **flux**, J . C can have any value from 0.0 (for an enzyme with no impact on the flux) to 1.0 (for an enzyme that wholly determines the flux). An enzyme can also have a *negative* flux control coefficient. In a branched pathway, an enzyme in one branch, by drawing intermediates away from the other branch, can have a negative impact on the flux through that other branch (Fig. 15–34). C is not a constant, and it is not

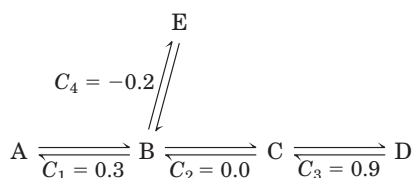


FIGURE 15-34 Flux control coefficient, C , in a branched metabolic pathway. In this simple pathway, the intermediate B has two alternative fates. To the extent that reaction $\text{B} \rightarrow \text{E}$ draws B away from the pathway $\text{A} \rightarrow \text{D}$, it controls that pathway, which will result in a *negative* flux control coefficient for the enzyme that catalyzes step $\text{B} \rightarrow \text{E}$. Note that the sum of all four coefficients equals 1.0, as it must.

intrinsic to a single enzyme; it is a function of the whole system of enzymes, and its value depends on the concentrations of substrates and effectors.

When real data from the experiment on glycolysis in a rat liver extract (Fig. 15-33) were subjected to this kind of analysis, investigators found flux control coefficients (for enzymes at the concentrations found in the extract) of 0.79 for hexokinase, 0.21 for PFK-1, and 0.0 for phosphohexose isomerase. It is not just fortuitous that these values add up to 1.0; we can show that for any complete pathway, the sum of the flux control coefficients must equal unity.

The Elasticity Coefficient Is Related to an Enzyme's Responsiveness to Changes in Metabolite or Regulator Concentrations

A second parameter, the **elasticity coefficient**, ϵ , expresses quantitatively the responsiveness of a single enzyme to changes in the concentration of a metabolite or regulator; it is a function of the enzyme's intrinsic kinetic properties. For example, an enzyme with typical Michaelis-Menten kinetics shows a hyperbolic response to increasing substrate concentration (Fig. 15-35). At low concentrations of substrate (say, $0.1 K_m$) each increment in substrate concentration results in a comparable increase in enzymatic activity, yielding an ϵ near 1.0. At relatively high substrate concentrations (say, $10 K_m$), increasing the substrate concentration has little effect on the reaction rate, because the enzyme is already saturated with substrate. The elasticity in this case approaches zero. For allosteric enzymes that show positive cooperativity, ϵ may exceed 1.0, but it cannot exceed the Hill coefficient. Recall that the Hill coefficient is a measure of the degree of cooperativity, typically between 1.0 and 4.0 (p. 167).

The Response Coefficient Expresses the Effect of an Outside Controller on Flux through a Pathway

We can also derive a quantitative expression for the relative impact of an outside factor (such as a hormone or growth factor), which is neither a metabolite nor an en-

zyme in the pathway, on the flux through the pathway. The experiment would measure the flux through the pathway (glycolysis, in this case) at various levels of the parameter P (the insulin concentration, for example) to obtain the **response coefficient**, R , which expresses the change in pathway flux when P ([insulin]) changes.

The three coefficients C , ϵ , and R are related in a simple way: the responsiveness (R) of a pathway to an outside factor that affects a certain enzyme is a function of (1) how sensitive the pathway is to changes in the activity of that enzyme (the control coefficient, C) and (2) how sensitive that specific enzyme is to changes in the outside factor (the elasticity, ϵ):

$$R = C \cdot \epsilon$$

Each enzyme in the pathway can be examined in this way, and the effects of any of several outside factors on flux through the pathway can be separately determined. Thus, in principle, we can predict how the flux of substrate through a series of enzymatic steps will change when there is a change in one or more controlling factors external to the pathway. Box 15-3 shows how these qualitative concepts are treated quantitatively.

Metabolic Control Analysis Has Been Applied to Carbohydrate Metabolism, with Surprising Results

Metabolic control analysis provides a framework within which we can think quantitatively about regulation, interpret the significance of the regulatory properties of each enzyme in a pathway, identify the steps that most affect the flux through the pathway, and distinguish between *regulatory* mechanisms that act to maintain metabolite concentrations and *control* mechanisms that actually alter the flux through the pathway. Analysis of the glycolytic pathway in yeast, for example, has

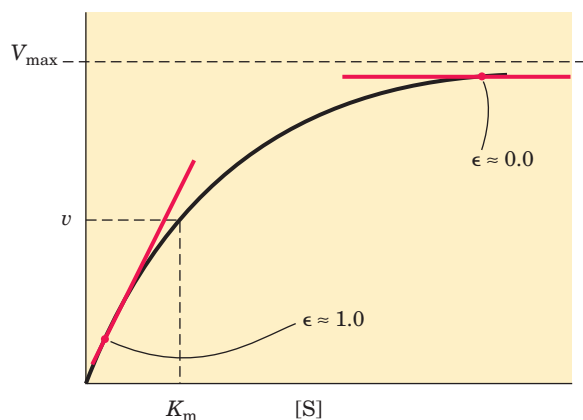


FIGURE 15-35 Elasticity coefficient, ϵ , of an enzyme with typical Michaelis-Menten kinetics. At substrate concentrations far below the K_m , each increase in $[S]$ produces a correspondingly large increase in the reaction velocity, v . For this region of the curve, the enzyme has an elasticity, ϵ , of about 1.0. At $[S] \gg K_m$, increasing $[S]$ has little effect on v ; ϵ here is close to 0.0.

BOX 15-3 WORKING IN BIOCHEMISTRY

Metabolic Control Analysis: Quantitative Aspects

The factors that influence the flow of intermediates (flux) through a pathway may be determined quantitatively by experiment and expressed in terms useful for predicting the change in flux when some factor involved in the pathway changes. Consider the simple reaction sequence in Figure 1, in which a substrate X (say, glucose) is converted in several steps to a product Z (perhaps pyruvate, formed glycolytically). A later enzyme in the pathway is a dehydrogenase (ydh) that acts on substrate Y. Because the action of a dehydrogenase is easily measured (see Fig. 13–15), we can use the flux (J) through this step (J_{ydh}) to measure the flux through the whole path. We manipulate experimentally the level of an early enzyme in the pathway (xase, which acts on the substrate X) and measure the flux through the path (J_{ydh}) for several levels of the enzyme xase.

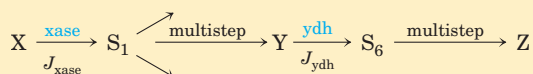


FIGURE 1 Flux through a hypothetical multienzyme pathway.

The relationship between the flux through the pathway from X to Z in the intact cell and the concentration of each enzyme in the path should be hyperbolic, with virtually no flux at infinitely low enzyme and near-maximum flux at very high enzyme activity. In a plot of J_{ydh} against the concentration of xase, E_{xase} , the change of flux with a small change of enzyme is $\partial J_{ydh} / \partial E_{xase}$, which is simply the slope of the tangent to the curve at any concentration of enzyme, E_{xase} , and which tends toward zero at saturating E_{xase} . At low E_{xase} , the slope is steep; the flux increases with each incremental increase in enzyme activity. At very high E_{xase} , the slope is much smaller; the system is less responsive to added xase, because it is already present in excess over the other enzymes in the pathway.

To show quantitatively the dependence of flux through the pathway, ∂J_{ydh} , on ∂E_{xase} , we could use

the ratio $\partial J_{ydh} / \partial E_{xase}$. However, its usefulness is limited because its value depends on the units used to express flux and enzyme activity. By expressing the *fractional* changes in flux and enzyme activity, $\partial J_{ydh} / J_{ydh}$, and $\partial E_{xase} / E_{xase}$, we obtain a unitless expression for the flux control coefficient, $C_{xase}^{J_{ydh}}$:

$$C_{xase}^{J_{ydh}} \approx \frac{\partial J_{ydh}}{J_{ydh}} \bigg/ \frac{\partial E_{xase}}{E_{xase}} \quad (1)$$

This can be rearranged to

$$C_{xase}^{J_{ydh}} \approx \frac{\partial J_{ydh}}{\partial E_{xase}} \cdot \frac{E_{xase}}{J_{ydh}}$$

which is mathematically identical to

$$C_{xase}^{J_{ydh}} = \frac{\partial \ln J_{ydh}}{\partial \ln E_{xase}}$$

This equation suggests a simple graphical means for determining the flux control coefficient: $C_{xase}^{J_{ydh}}$ is the slope of the tangent to the plot of $\ln J_{ydh}$ versus $\ln E_{xase}$, which can be obtained by replotting the experimental data in Figure 2a to obtain Figure 2b. Notice that $C_{xase}^{J_{ydh}}$ is not a constant; it depends on the starting E_{xase} from which the change in enzyme level takes place. For the cases shown in Figure 2, $C_{xase}^{J_{ydh}}$ is about 1.0 at the lowest E_{xase} , but only about 0.2 at high E_{xase} . A value near 1.0 for $C_{xase}^{J_{ydh}}$ means that the enzyme's concentration wholly determines the flux through the pathway; a value near 0.0 means that the enzyme's concentration does not limit the flux through the path. Unless the flux control coefficient is greater than about 0.5, changes in the activity of the enzyme will not have a strong effect on the flux.

The **elasticity**, ϵ , of an enzyme is a measure of how that enzyme's catalytic activity changes when the concentration of a metabolite—substrate, product, or effector—changes. It is obtained from an experimental plot of the rate of the reaction catalyzed by the enzyme versus the concentration of the metabolite, at metabolite concentrations that prevail in the cell. By arguments analogous to those used to derive C , we can show ϵ to be the slope of the tangent to a plot of

revealed an unexpectedly low flux control coefficient for PFK-1, which, because of its known elaborate allosteric regulation, has been viewed as the main point of flux control—the “rate-determining step”—in glycolysis. Experimentally raising the level of PFK-1 fivefold led to a change in flux through glycolysis of less than 10%,

suggesting that the real role of PFK-1 regulation is not to control flux through glycolysis but to mediate metabolite homeostasis—to prevent large changes in metabolite concentrations when the flux through glycolysis increases in response to elevated blood glucose or insulin. Recall that the study of glycolysis in a liver

$\ln V$ versus $\ln$ [substrate, or product, or effector]:

$$\begin{aligned}\varepsilon_S^{\text{xase}} &= \frac{\partial V_{\text{xase}}}{\partial S} \cdot \frac{S}{V_{\text{xase}}} \\ &= \frac{\partial \ln |V_{\text{xase}}|}{\partial \ln S}\end{aligned}$$

For an enzyme with typical Michaelis-Menten kinetics, the value of ε ranges from about 1 at substrate concentrations far below K_m to near 0 as V_{max} is approached. Allosteric enzymes can have elasticities greater than 1.0, but not larger than their Hill coefficients (p. 167).

Finally, the effect of controllers outside the pathway itself (that is, not metabolites) can be measured and expressed as the **response coefficient, R** . The change in flux through the pathway is measured for changes in the concentration of the controlling parameter P , and R is defined in a form analogous to that

of Equation 1, yielding the expression

$$R_P^{J_{\text{ydh}}} = \frac{\partial J_{\text{ydh}}}{\partial P} \cdot \frac{P}{J_{\text{ydh}}}$$

Using the same logic and graphical methods as described above for determining C , we can obtain R as the slope of the tangent to the plot of $\ln J$ versus $\ln P$.

The three coefficients we have described are related in this simple way:

$$R_P^{J_{\text{ydh}}} = C_{\text{xase}}^{J_{\text{ydh}}} \cdot \varepsilon_P^{\text{xase}}$$

Thus the responsiveness of each enzyme in a pathway to a change in an outside controlling factor is a simple function of two things: the control coefficient, a variable that expresses the extent to which that enzyme influences the flux under a given set of conditions, and the elasticity, an intrinsic property of the enzyme that reflects its sensitivity to substrate and effector concentrations.

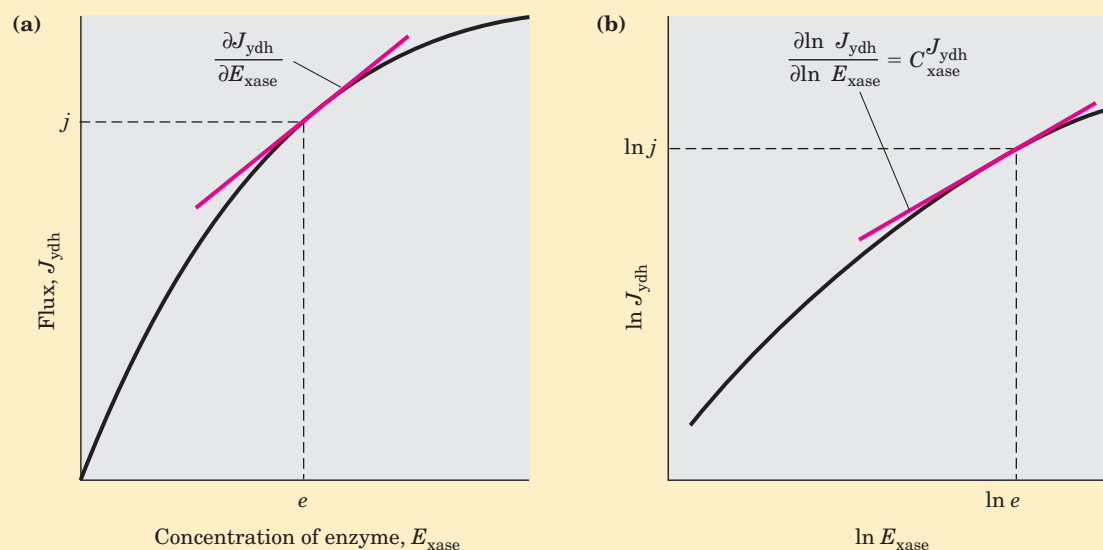


FIGURE 2 The flux control coefficient. (a) Typical variation of the pathway flux, J_{ydh} , measured at the step catalyzed by the enzyme ydh, as a function of the amount of the enzyme xase, E_{xase} , which catalyzes an earlier step in the pathway. The flux control coefficient at (e, j) is the slope of the product of the tangent to the curve, $\partial J_{\text{ydh}}/\partial E_{\text{xase}}$, and the ratio (scaling factor), e/j . (b) On a double-logarithmic plot of the same curve, the flux control coefficient is the slope of the tangent to the curve.

extract (Fig. 15–33) also yielded a flux control coefficient that contradicted the conventional wisdom; it showed that hexokinase, not PFK-1, is most influential in setting the flux through glycolysis. We must note here that a liver extract is far from equivalent to a hepatocyte; the ideal way to study flux control is by manipu-

lating one enzyme at a time in the living cell. This is already feasible in some cases.

Investigators have used nuclear magnetic resonance (NMR) as a noninvasive means to determine the concentration of glycogen and metabolites in the five-step pathway from glucose in the blood to glycogen in myocytes

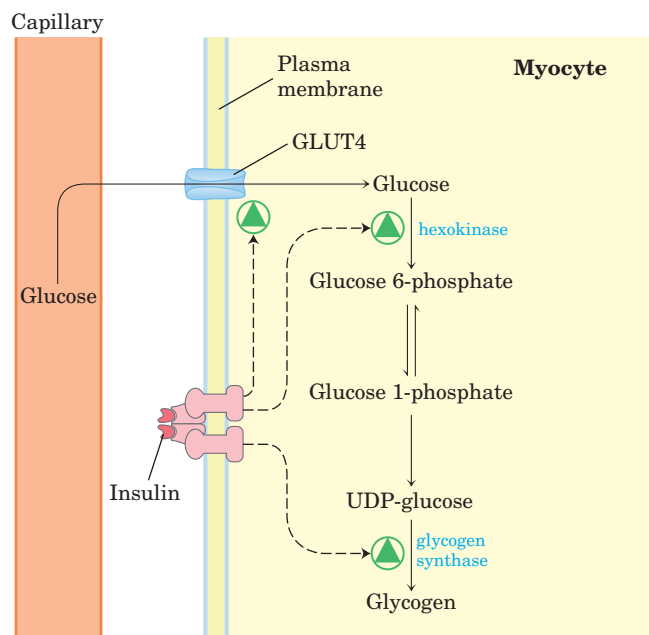


FIGURE 15-36 Control of glycogen synthesis from blood glucose in myocytes. Insulin affects three of the five steps in this pathway, but it is the effects on transport and hexokinase activity, not the change in glycogen synthase activity, that increase the flux toward glycogen.

(Fig. 15-36) in rat and human muscle. They found that the flux control coefficient for glycogen synthase was smaller than that for the steps catalyzed by the glucose transporter and hexokinase. This finding, too, contradicts the conventional wisdom that glycogen synthase is the locus of flux control and suggests that the importance of the phosphorylation/dephosphorylation of glycogen synthase is related instead to the maintenance of metabolite homeostasis—that is, *regulation*, not *control*. Two metabolites in this pathway, glucose and glucose 6-phosphate, are key intermediates in other pathways, including glycolysis, the pentose phosphate pathway, and the synthesis of glucosamine. Metabolic control analysis suggests that when the blood glucose level rises, insulin acts in muscle to (1) increase glucose transport into cells by bringing GLUT4 to the plasma membrane, (2) induce the synthesis of hexokinase, and (3) activate glycogen synthase by covalent alteration (Fig. 15-29). The first two effects of insulin increase glucose flux through the pathway (control), and the third serves to adapt the activity of glycogen synthase so that metabolite levels (glucose 6-phosphate, for example) will not change dramatically with the increased flux (regulation).

Metabolic Control Analysis Suggests a General Method for Increasing Flux through a Pathway

How could an investigator engineer a cell to increase the flux through one pathway without altering the concen-

trations of other metabolites or the fluxes through other pathways? More than two decades ago Henrik Kacser predicted, on the basis of metabolic control analysis, that this could be accomplished by increasing the concentrations of every enzyme in a pathway. The prediction has been confirmed in several experimental tests, and it also fits with the way cells normally control fluxes through a pathway. For example, when rats are fed a high-protein diet, they dispose of excess amino groups by converting them to urea in the urea cycle (Chapter 18). After such a dietary shift, the urea output increases fourfold, and the amount of all eight enzymes in the urea cycle increases two- to threefold. Similarly, when increased fatty acid oxidation is triggered by activation of the enzyme peroxisome proliferator-activated receptor γ (PPAR γ ; see Fig. 21-22), synthesis of the *whole set* of oxidative enzymes is increased. With the growing use of DNA microarrays to study the expression of whole sets of genes in response to various perturbations, we should soon learn whether this is the general mechanism by which cells make long-term adjustments in the fluxes through specific pathways.

SUMMARY 15.5 Analysis of Metabolic Control

- Metabolic control analysis shows that control of the rate of metabolite flux through a pathway is distributed among several of the enzymes in that path.
- The flux control coefficient, C , is an experimentally determined measure of the effect of an enzyme's concentration on flux through a multienzyme pathway. It is characteristic of the whole system, not intrinsic to the enzyme.
- The elasticity coefficient, ε , of an enzyme is an experimentally determined measure of how responsive the enzyme is to changes in the concentration of a metabolite or regulator molecule.
- The response coefficient, R , is the expression for the experimentally determined change in flux through a pathway in response to a regulatory hormone or second messenger. It is a function of C and ε : $R = C \cdot \varepsilon$.
- Some regulated enzymes control the flux through a pathway, while others rebalance the level of metabolites in response to the change in flux. This latter, rebalancing activity is *regulation*; the former activity is *control*.
- Metabolic control analysis predicts that flux toward a desired product is most effectively increased by raising the concentration of all enzymes in the pathway.

Key Terms

Terms in bold are defined in the glossary.

glycogenolysis	562	metabolic control	575	glycogen synthase <i>a</i>	586
glycolysis	562	futile cycle	576	glycogen synthase <i>b</i>	586
gluconeogenesis	562	substrate cycle	576	glycogen synthase kinase 3	
glycogenesis	562	GLUT	578	(GSK3)	586
debranching enzyme	562	glucagon	581	priming	586
sugar nucleotides	565	fructose 2,6-bisphosphate	581	glycogen-targeting proteins	588
glycogenin	569	glycogen phosphorylase <i>a</i>	583	flux control coefficient, <i>C</i>	592
homeostasis	571	glycogen phosphorylase <i>b</i>	584	flux, <i>J</i>	592
adenylate kinase	571	enzyme cascade	584	elasticity coefficient, ε	593
mass action ratio, <i>Q</i>	572	phosphoprotein phosphatase 1		response coefficient, <i>R</i>	593
metabolic regulation	575	(PP1)	584		

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Problems

1. Measurement of Intracellular Metabolite Concentrations Measuring the concentrations of metabolic intermediates in a living cell presents great experimental difficulties—usually a cell must be destroyed before metabolite concentrations can be measured. Yet enzymes catalyze metabolic interconversions very rapidly, so a common problem associated with these types of measurements is that the findings reflect not the physiological concentrations of metabolites but the equilibrium concentrations. A reliable experimental technique requires all enzyme-catalyzed reactions to be instantaneously stopped in the intact tissue so that the metabolic intermediates do not undergo change. This objective is accomplished by rapidly compressing the tissue between large aluminum plates cooled with liquid nitrogen (-190°C), a process called **freeze-clamping**. After freezing, which stops enzyme action instantly, the tissue is powdered and the enzymes are inactivated by precipitation with perchloric acid. The precipitate is removed by centrifugation, and

the clear supernatant extract is analyzed for metabolites. To calculate intracellular concentrations, the intracellular volume is determined from the total water content of the tissue and a measurement of the extracellular volume.

The intracellular concentrations of the substrates and products of the phosphofructokinase-1 reaction in isolated rat heart tissue are given in the table below.

Metabolite	Concentration (μM) [*]
Fructose 6-phosphate	87.0
Fructose 1,6-bisphosphate	22.0
ATP	11,400
ADP	1,320

Source: From Williamson, J.R. (1965) Glycolytic control mechanisms I: inhibition of glycolysis by acetate and pyruvate in the isolated, perfused rat heart. *J. Biol. Chem.* **240**, 2308–2321.

^{*}Calculated as $\mu\text{mol/mL}$ of intracellular water.

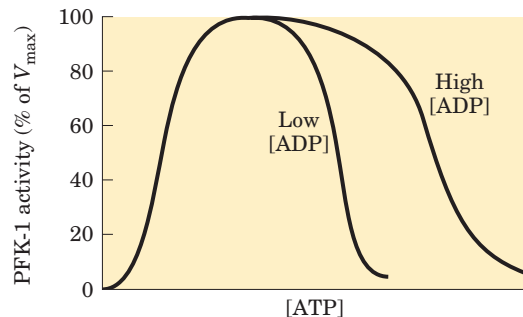
(a) Calculate Q , $[\text{fructose 1,6-bisphosphate}][\text{ADP}] / [\text{fructose 6-phosphate}][\text{ATP}]$, for the PFK-1 reaction under physiological conditions.

(b) Given a $\Delta G'^{\circ}$ for the PFK-1 reaction of -14.2 kJ/mol , calculate the equilibrium constant for this reaction.

(c) Compare the values of Q and K'_{eq} . Is the physiological reaction near or far from equilibrium? Explain. What does this experiment suggest about the role of PFK-1 as a regulatory enzyme?

2. Effect of O_2 Supply on Glycolytic Rates The regulated steps of glycolysis in intact cells can be identified by studying the catabolism of glucose in whole tissues or organs. For example, the glucose consumption by heart muscle can be measured by artificially circulating blood through an isolated intact heart and measuring the concentration of glucose before and after the blood passes through the heart. If the circulating blood is deoxygenated, heart muscle consumes glucose at a steady rate. When oxygen is added to the blood, the rate of glucose consumption drops dramatically, then is maintained at the new, lower rate. Why?

3. Regulation of PFK-1 The effect of ATP on the allosteric enzyme PFK-1 is shown below. For a given concentration of fructose 6-phosphate, the PFK-1 activity increases with increasing concentrations of ATP, but a point is reached beyond which increasing the concentration of ATP inhibits the enzyme.



(a) Explain how ATP can be both a substrate and an inhibitor of PFK-1. How is the enzyme regulated by ATP?

(b) In what ways is glycolysis regulated by ATP levels?

(c) The inhibition of PFK-1 by ATP is diminished when the ADP concentration is high, as shown in the illustration. How can this observation be explained?

4. Are All Metabolic Reactions at Equilibrium?

(a) Phosphoenolpyruvate (PEP) is one of the two phosphoryl group donors in the synthesis of ATP during glycolysis. In human erythrocytes, the steady-state concentration of ATP is 2.24 mM , that of ADP is 0.25 mM , and that of pyruvate is 0.051 mM . Calculate the concentration of PEP at 25°C , assuming that the pyruvate kinase reaction (see Fig. 13–3) is at equilibrium in the cell.

(b) The physiological concentration of PEP in human erythrocytes is 0.023 mM . Compare this with the value obtained in (a). Explain the significance of this difference.

5. Cellular Glucose Concentration The concentration of glucose in human blood plasma is maintained at about 5 mM . The concentration of free glucose inside a myocyte is much lower. Why is the concentration so low in the cell? What

happens to glucose after entry into the cell? Glucose is administered intravenously as a food source in certain clinical situations. Given that the transformation of glucose to glucose 6-phosphate consumes ATP, why not administer intravenous glucose 6-phosphate instead?

6. Enzyme Activity and Physiological Function The V_{max} of the enzyme glycogen phosphorylase from skeletal muscle is much greater than the V_{max} of the same enzyme from liver tissue.

(a) What is the physiological function of glycogen phosphorylase in skeletal muscle? In liver tissue?

(b) Why does the V_{max} of the muscle enzyme need to be greater than that of the liver enzyme?

7. Glycogen Phosphorylase Equilibrium Glycogen phosphorylase catalyzes the removal of glucose from glycogen. The $\Delta G'^{\circ}$ for this reaction is 3.1 kJ/mol . (a) Calculate the ratio of $[\text{P}_i]$ to $[\text{glucose 1-phosphate}]$ when the reaction is at equilibrium. (Hint: The removal of glucose units from glycogen does not change the glycogen concentration.) (b) The measured ratio $[\text{P}_i]/[\text{glucose 1-phosphate}]$ in myocytes under physiological conditions is more than 100:1. What does this indicate about the direction of metabolite flow through the glycogen phosphorylase reaction in muscle? (c) Why are the equilibrium and physiological ratios different? What is the possible significance of this difference?

8. Regulation of Glycogen Phosphorylase In muscle tissue, the rate of conversion of glycogen to glucose 6-phosphate is determined by the ratio of phosphorylase *a* (active) to phosphorylase *b* (less active). Determine what happens to the rate of glycogen breakdown if a muscle preparation containing glycogen phosphorylase is treated with (a) phosphorylase kinase and ATP; (b) PP1; (c) epinephrine.

9. Glycogen Breakdown in Rabbit Muscle The intracellular use of glucose and glycogen is tightly regulated at four points. In order to compare the regulation of glycolysis when oxygen is plentiful and when it is depleted, consider the utilization of glucose and glycogen by rabbit leg muscle in two physiological settings: a resting rabbit, with low ATP demands, and a rabbit that sights its mortal enemy, the coyote, and dashes into its burrow. For each setting, determine the relative levels (high, intermediate, or low) of AMP, ATP, citrate, and acetyl-CoA and how these levels affect the flow of metabolites through glycolysis by regulating specific enzymes. In periods of stress, rabbit leg muscle produces much of its ATP by anaerobic glycolysis (lactate fermentation) and very little by oxidation of acetyl-CoA derived from fat breakdown.

10. Glycogen Breakdown in Migrating Birds Unlike the rabbit with its short dash, migratory birds require energy for extended periods of time. For example, ducks generally fly several thousand miles during their annual migration. The flight muscles of migratory birds have a high oxidative capacity and obtain the necessary ATP through the oxidation of acetyl-CoA (obtained from fats) via the citric acid cycle. Compare the regulation of muscle glycolysis during short-term intense activity, as in the fleeing rabbit, and during extended activity, as in the migrating duck. Why must the regulation in these two settings be different?



11. Enzyme Defects in Carbohydrate Metabolism

Summaries of four clinical case studies follow. For each case determine which enzyme is defective and designate the appropriate treatment, from the lists provided at the end of the problem. Justify your choices. Answer the questions contained in each case study. (You may need to refer to information in Chapter 14.)

Case A The patient develops vomiting and diarrhea shortly after milk ingestion. A lactose tolerance test is administered. (The patient ingests a standard amount of lactose, and the glucose and galactose concentrations of blood plasma are measured at intervals. In normal individuals the levels increase to a maximum in about 1 hour, then decline.) The patient's blood glucose and galactose concentrations do not increase during the test. Why do blood glucose and galactose increase and then decrease during the test in normal individuals? Why do they fail to rise in the patient?

Case B The patient develops vomiting and diarrhea after ingestion of milk. His blood is found to have a low concentration of glucose but a much higher than normal concentration of reducing sugars. The urine test for galactose is positive. Why is the concentration of reducing sugar in the blood high? Why does galactose appear in the urine?

Case C The patient complains of painful muscle cramps when performing strenuous physical exercise but has no

other symptoms. A muscle biopsy indicates a muscle glycogen concentration much higher than normal. Why does glycogen accumulate?

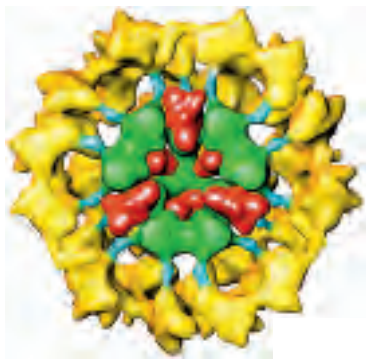
Case D The patient is lethargic, her liver is enlarged, and a biopsy of the liver shows large amounts of excess glycogen. She also has a lower than normal blood glucose level. What is the reason for the low blood glucose in this patient?

Defective Enzyme

- (a) Muscle PFK-1
- (b) Phosphomannose isomerase
- (c) Galactose 1-phosphate uridylyltransferase
- (d) Liver glycogen phosphorylase
- (e) Triose kinase
- (f) Lactase in intestinal mucosa
- (g) Maltase in intestinal mucosa
- (h) Muscle-debranching enzyme

Treatment

- 1. Jogging 5 km each day
- 2. Fat-free diet
- 3. Low-lactose diet
- 4. Avoiding strenuous exercise
- 5. Large doses of niacin (the precursor of NAD^+)
- 6. Frequent regular feedings



chapter 16

THE CITRIC ACID CYCLE

- 16.1 Production of Acetyl-CoA (Activated Acetate) 602
- 16.2 Reactions of the Citric Acid Cycle 606
- 16.3 Regulation of the Citric Acid Cycle 621
- 16.4 The Glyoxylate Cycle 623

If citrate is added the rate of respiration is often increased . . . the extra oxygen uptake is by far greater than can be accounted for by the complete oxidation of citrate . . . Since citric acid reacts catalytically in the tissue it is probable that it is removed by a primary reaction but regenerated by a subsequent reaction.

—H. A. Krebs and W. A. Johnson, *article in Enzymologia*, 1937

As we saw in Chapter 14, some cells obtain energy (ATP) by fermentation, breaking down glucose in the absence of oxygen. For most eukaryotic cells and many bacteria, which live under aerobic conditions and oxidize their organic fuels to carbon dioxide and water, glycolysis is but the first stage in the complete oxidation of glucose. Rather than being reduced to lactate, ethanol, or some other fermentation product, the pyruvate produced by glycolysis is further oxidized to H_2O and CO_2 . This aerobic phase of catabolism is called **respiration**. In the broader physiological or macroscopic sense, respiration refers to a multicellular organism's uptake of O_2 and release of CO_2 . Biochemists and cell biologists, however, use the term in a narrower sense to refer to the molecular processes by which *cells* consume O_2 and produce CO_2 —processes more precisely termed **cellular respiration**.

Cellular respiration occurs in three major stages (Fig. 16–1). In the first, organic fuel molecules—glucose, fatty acids, and some amino acids—are oxidized to yield two-carbon fragments in the form of the acetyl group of acetyl-coenzyme A (acetyl-CoA). In the sec-

ond stage, the acetyl groups are fed into the citric acid cycle, which enzymatically oxidizes them to CO_2 ; the energy released is conserved in the reduced electron carriers NADH and FADH_2 . In the third stage of respiration, these reduced coenzymes are themselves oxidized, giving up protons (H^+) and electrons. The electrons are transferred to O_2 —the final electron acceptor—via a chain of electron-carrying molecules known as the respiratory chain. In the course of electron transfer, the large amount of energy released is conserved in the form of ATP, by a process called oxidative phosphorylation (Chapter 19). Respiration is more complex than glycolysis and is believed to have evolved much later, after the appearance of cyanobacteria. The metabolic activities of cyanobacteria account for the rise of oxygen levels in the earth's atmosphere, a dramatic turning point in evolutionary history.

We consider first the conversion of pyruvate to acetyl groups, then the entry of those groups into the **citric acid cycle**, also called the **tricarboxylic acid (TCA) cycle** or the **Krebs cycle** (after its discoverer, Hans Krebs). We next examine the cycle reactions and the enzymes that catalyze them. Because intermediates of the citric acid cycle are also siphoned off as biosynthetic precursors, we go on to consider some ways in which these intermediates are replenished. The citric acid cycle is a hub in metabolism, with degradative pathways leading in and anabolic pathways leading out, and it is closely regulated in coordination with other pathways. The chapter ends with a description of the glyoxylate pathway, a metabolic sequence in some organisms that employs several of the same enzymes and reactions used in the citric acid cycle, bringing about the net synthesis of glucose from stored triacylglycerols.



Hans Krebs, 1900–1981

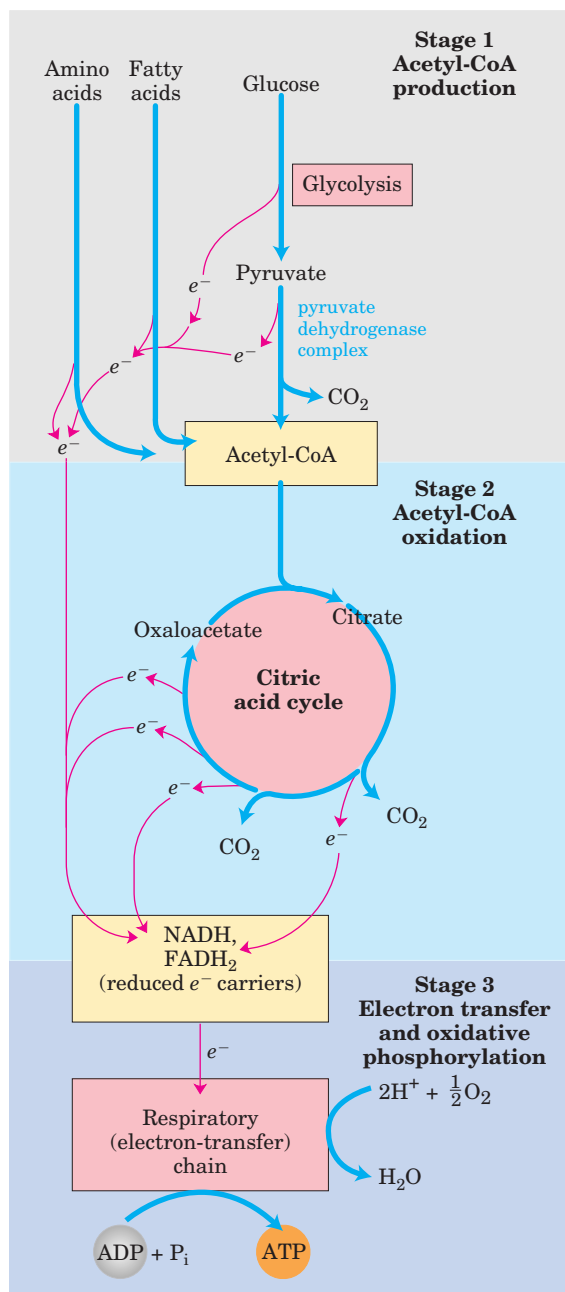


FIGURE 16-1 Catabolism of proteins, fats, and carbohydrates in the three stages of cellular respiration. Stage 1: oxidation of fatty acids, glucose, and some amino acids yields acetyl-CoA. Stage 2: oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted. Stage 3: electrons carried by NADH and FADH₂ are funneled into a chain of mitochondrial (or, in bacteria, plasma membrane-bound) electron carriers—the respiratory chain—ultimately reducing O₂ to H₂O. This electron flow drives the production of ATP.

and other sugars by glycolysis, is oxidized to acetyl-CoA and CO₂ by the **pyruvate dehydrogenase (PDH) complex**, a cluster of enzymes—multiple copies of each of three enzymes—located in the mitochondria of eukaryotic cells and in the cytosol of prokaryotes.

A careful examination of this enzyme complex is rewarding in several respects. The PDH complex is a classic, much-studied example of a multienzyme complex in which a series of chemical intermediates remain bound to the enzyme molecules as a substrate is transformed into the final product. Five cofactors, four derived from vitamins, participate in the reaction mechanism. The regulation of this enzyme complex also illustrates how a combination of covalent modification and allosteric regulation results in precisely regulated flux through a metabolic step. Finally, the PDH complex is the prototype for two other important enzyme complexes: α -ketoglutarate dehydrogenase, of the citric acid cycle, and the branched-chain α -keto acid dehydrogenase, of the oxidative pathways of several amino acids (see Fig. 18–28). The remarkable similarity in the protein structure, cofactor requirements, and reaction mechanisms of these three complexes doubtless reflects a common evolutionary origin.

Pyruvate Is Oxidized to Acetyl-CoA and CO₂

The overall reaction catalyzed by the pyruvate dehydrogenase complex is an **oxidative decarboxylation**, an irreversible oxidation process in which the carboxyl group is removed from pyruvate as a molecule of CO₂

16.1 Production of Acetyl-CoA (Activated Acetate)

In aerobic organisms, glucose and other sugars, fatty acids, and most amino acids are ultimately oxidized to CO₂ and H₂O via the citric acid cycle and the respiratory chain. Before entering the citric acid cycle, the carbon skeletons of sugars and fatty acids are degraded to the acetyl group of acetyl-CoA, the form in which the cycle accepts most of its fuel input. Many amino acid carbons also enter the cycle this way, although several amino acids are degraded to other cycle intermediates. Here we focus on how pyruvate, derived from glucose

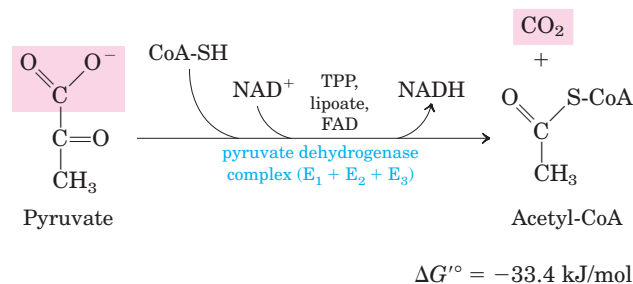


FIGURE 16-2 Overall reaction catalyzed by the pyruvate dehydrogenase complex. The five coenzymes participating in this reaction, and the three enzymes that make up the enzyme complex, are discussed in the text.

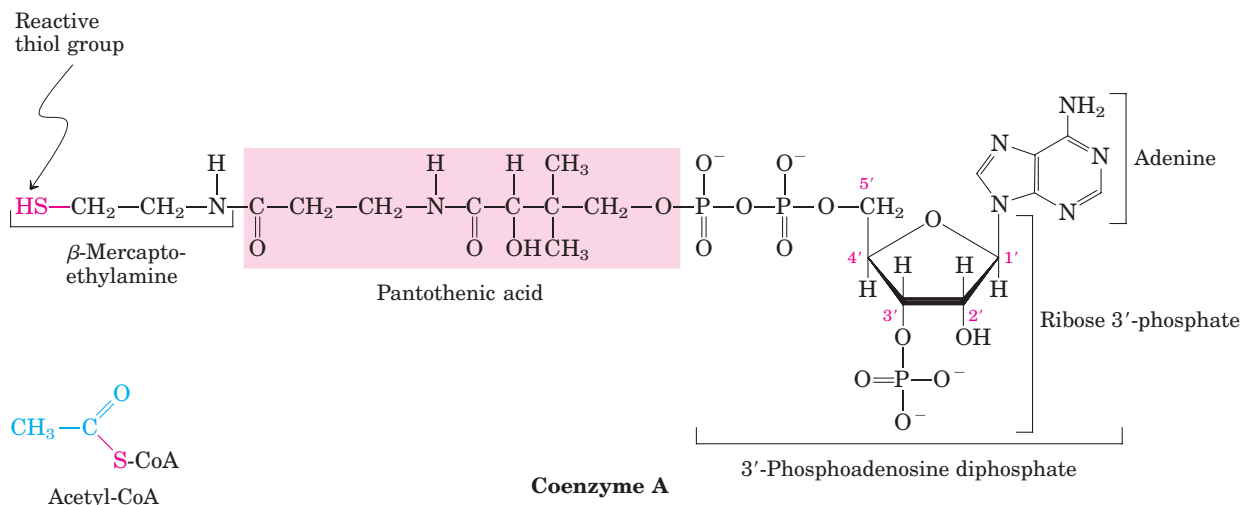


FIGURE 16-3 Coenzyme A (CoA). A hydroxyl group of pantothenic acid is joined to a modified ADP moiety by a phosphate ester bond, and its carboxyl group is attached to β -mercaptoethylamine in amide linkage. The hydroxyl group at the 3' position of the ADP moiety has a phosphoryl group not present in free ADP. The —SH group of the mercaptoethylamine moiety forms a thioester with acetate in acetyl-coenzyme A (acetyl-CoA) (lower left).

and the two remaining carbons become the acetyl group of acetyl-CoA (Fig. 16-2). The NADH formed in this reaction gives up a hydride ion (:H^-) to the respiratory chain (Fig. 16-1), which carries the two electrons to oxygen or, in anaerobic microorganisms, to an alternative electron acceptor such as nitrate or sulfate. The transfer of electrons from NADH to oxygen ultimately generates 2.5 molecules of ATP per pair of electrons. The irreversibility of the PDH complex reaction has been demonstrated by isotopic labeling experiments: the complex cannot reattach radioactively labeled CO_2 to acetyl-CoA to yield carboxyl-labeled pyruvate.

The Pyruvate Dehydrogenase Complex Requires Five Coenzymes

The combined dehydrogenation and decarboxylation of pyruvate to the acetyl group of acetyl-CoA (Fig. 16-2) requires the sequential action of three different enzymes and five different coenzymes or prosthetic groups—thiamine pyrophosphate (TPP), flavin adenine dinucleotide (FAD), coenzyme A (CoA, sometimes denoted CoA-SH, to emphasize the role of the —SH group), nicotinamide adenine dinucleotide (NAD), and lipoate. Four different vitamins required in human nutrition are vital components of this system: thiamine (in TPP), riboflavin (in FAD), niacin (in NAD), and pantothenate (in CoA). We have already described the roles of FAD and NAD as electron carriers (Chapter 13), and we have encountered TPP as the coenzyme of pyruvate decarboxylase (see Fig. 14-13).

Coenzyme A (Fig. 16-3) has a reactive thiol (—SH) group that is critical to the role of CoA as an acyl car-

rier in a number of metabolic reactions. Acyl groups are covalently linked to the thiol group, forming **thioesters**. Because of their relatively high standard free energies of hydrolysis (see Figs 13-6, 13-7), thioesters have a high acyl group transfer potential and can donate their acyl groups to a variety of acceptor molecules. The acyl group attached to coenzyme A may thus be thought of as “activated” for group transfer.

The fifth cofactor of the PDH complex, **lipoate** (Fig. 16-4), has two thiol groups that can undergo reversible oxidation to a disulfide bond (—S—S—), similar to that between two Cys residues in a protein. Because of its capacity to undergo oxidation-reduction reactions, lipoate can serve both as an electron hydrogen carrier and as an acyl carrier, as we shall see.

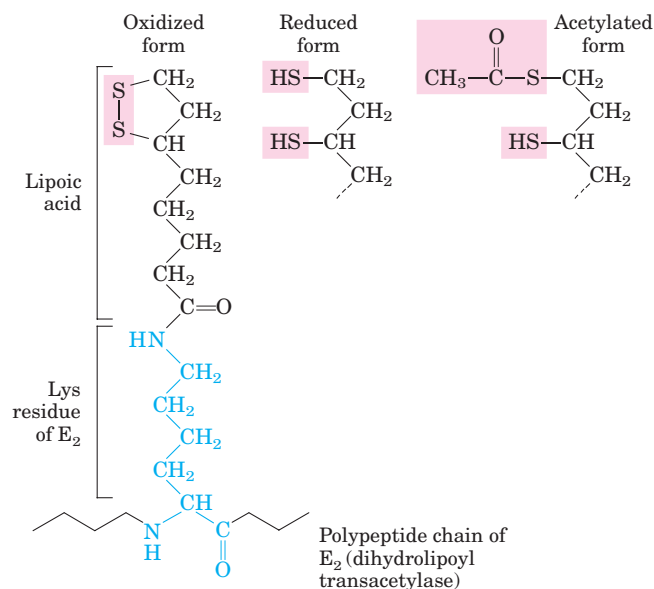


FIGURE 16-4 Lipoic acid (lipoate) in amide linkage with a Lys residue. The lipoyllysyl moiety is the prosthetic group of dihydrolipoyl transacetylase (E_2 of the PDH complex). The lipoyl group occurs in oxidized (disulfide) and reduced (dithiol) forms and acts as a carrier of both hydrogen and an acetyl (or other acyl) group.

The Pyruvate Dehydrogenase Complex Consists of Three Distinct Enzymes

The PDH complex contains three enzymes—**pyruvate dehydrogenase** (E_1), **dihydrolipoyl transacetylase** (E_2), and **dihydrolipoyl dehydrogenase** (E_3)—each present in multiple copies. The number of copies of each enzyme and therefore the size of the complex varies among species. The PDH complex isolated from mammals is about 50 nm in diameter—more than five times the size of an entire ribosome and big enough to be visualized with the electron microscope (Fig. 16–5a). In the bovine enzyme, 60 identical copies of E_2 form a pentagonal dodecahedron (the core) with a diameter of about 25 nm (Fig. 16–5b). (The core of the *Escherichia coli* enzyme contains 24 copies of E_2 .) E_2 is the point of

connection for the prosthetic group lipoate, attached through an amide bond to the ϵ -amino group of a Lys residue (Fig. 16–4). E_2 has three functionally distinct domains (Fig. 16–5c): the amino-terminal *lipoyl domain*, containing the lipoyl-Lys residue(s); the central E_1 - and E_3 -*binding domain*; and the inner-core *acyltransferase domain*, which contains the acyltransferase active site. The yeast PDH complex has a single lipoyl domain with a lipoate attached, but the mammalian complex has two, and *E. coli* has three (Fig. 16–5c). The domains of E_2 are separated by linkers, sequences of 20 to 30 amino acid residues, rich in Ala and Pro and interspersed with charged residues; these linkers tend to assume their extended forms, holding the three domains apart.

The active site of E_1 has bound TPP, and that of E_3 has bound FAD. Also part of the complex are two reg-

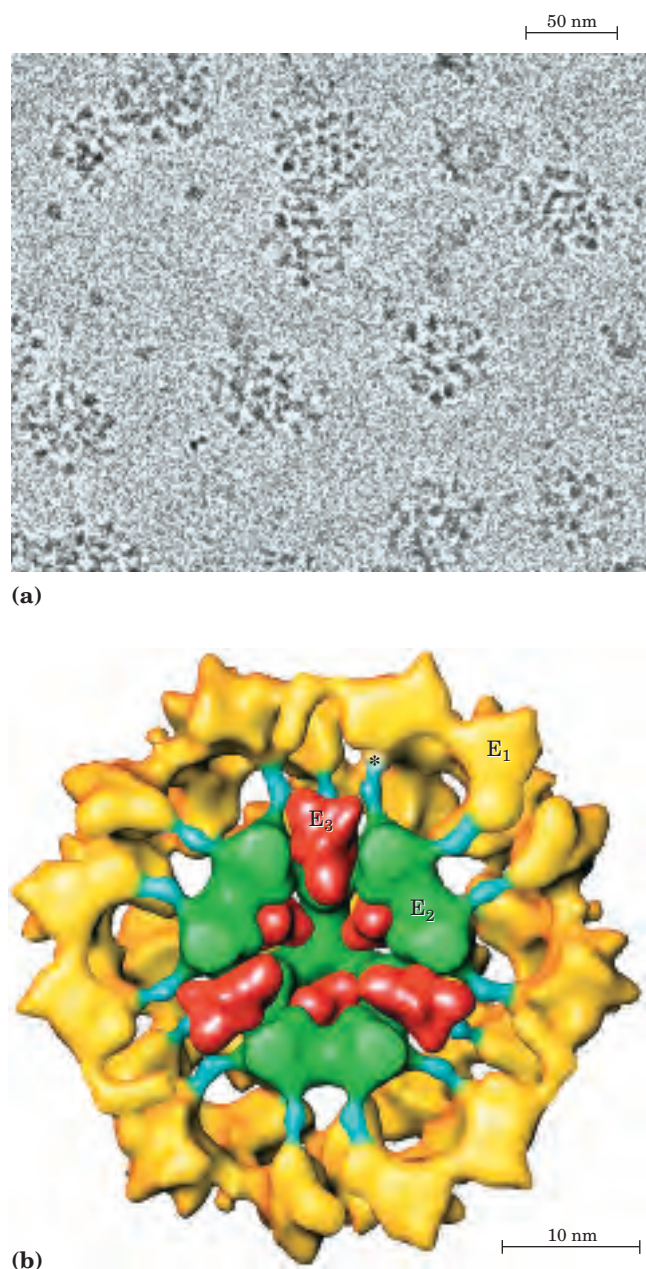
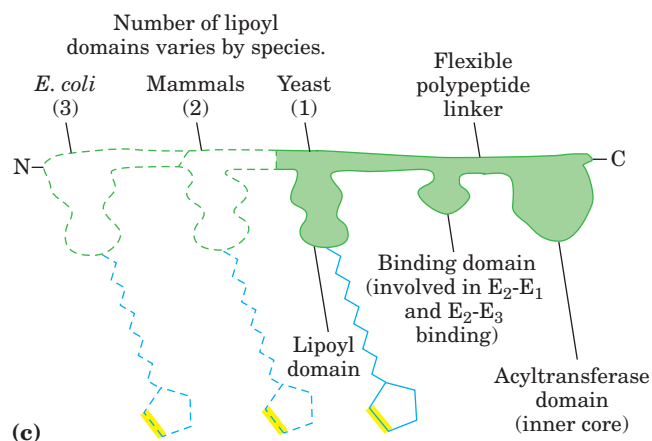


FIGURE 16–5 Structure of the pyruvate dehydrogenase complex (a) Cryoelectron micrograph of PDH complexes isolated from bovine kidney. In cryoelectron microscopy, biological samples are viewed at extremely low temperatures; this avoids potential artifacts introduced by the usual process of dehydrating, fixing, and staining. (b) Three-dimensional image of PDH complex, showing the subunit structure: E_1 , pyruvate dehydrogenase; E_2 , dihydrolipoyl transacetylase; and E_3 , dihydrolipoyl dehydrogenase. This image is reconstructed by analysis of a large number of images such as those in (a), combined with crystallographic studies of individual subunits. The core (green) consists of 60 molecules of E_2 , arranged in 20 trimers to form a pentagonal dodecahedron. The lipoyl domain of E_2 (blue) reaches outward to touch the active sites of E_1 molecules (yellow) arranged on the E_2 core. A number of E_3 subunits (red) are also bound to the core, where the swinging arm on E_2 can reach their active sites. An asterisk marks the site where a lipoyl group is attached to the lipoyl domain of E_2 . To make the structure clearer, about half of the complex has been cut away from the front. This model was prepared by Z. H. Zhou et al. (2001); in another model, proposed by J. L. S. Milne et al. (2002), the E_3 subunits are located more toward the periphery (see Further Reading). (c) E_2 consists of three types of domains linked by short polypeptide linkers: a catalytic acyltransferase domain; a binding domain, involved in the binding of E_2 to E_1 and E_3 ; and one or more (depending on the species) lipoyl domains.



ulatory proteins, a protein kinase and a phosphoprotein phosphatase, discussed below. This basic E_1 - E_2 - E_3 structure has been conserved during evolution and used in a number of similar metabolic reactions, including the oxidation of α -ketoglutarate in the citric acid cycle (described below) and the oxidation of α -keto acids derived from the breakdown of the branched-chain amino acids valine, isoleucine, and leucine (see Fig. 18–28). Within a given species, E_3 of PDH is identical to E_3 of the other two enzyme complexes. The attachment of lipoate to the end of a Lys side chain in E_2 produces a long, flexible arm that can move from the active site of E_1 to the active sites of E_2 and E_3 , a distance of perhaps 5 nm or more.

In Substrate Channeling, Intermediates Never Leave the Enzyme Surface

Figure 16–6 shows schematically how the pyruvate dehydrogenase complex carries out the five consecutive reactions in the decarboxylation and dehydrogenation of pyruvate. Step ① is essentially identical to the reaction catalyzed by pyruvate decarboxylase (see Fig. 14–13c); C-1 of pyruvate is released as CO_2 , and C-2, which in pyruvate has the oxidation state of an aldehyde, is attached to TPP as a hydroxyethyl group. This first step is the slowest and therefore limits the rate of the overall reaction. It is also the point at which the PDH complex exercises its substrate specificity. In step ② the hydroxyethyl group is oxidized to the level of a car-

boxylic acid (acetate). The two electrons removed in this reaction reduce the $-\text{S}-\text{S}-$ of a lipoyl group on E_2 to two thiol ($-\text{SH}$) groups. The acetyl moiety produced in this oxidation-reduction reaction is first esterified to one of the lipoyl $-\text{SH}$ groups, then transesterified to CoA to form acetyl-CoA (step ③). Thus the energy of oxidation drives the formation of a high-energy thioester of acetate. The remaining reactions catalyzed by the PDH complex (by E_3 , in steps ④ and ⑤) are electron transfers necessary to regenerate the oxidized (disulfide) form of the lipoyl group of E_2 to prepare the enzyme complex for another round of oxidation. The electrons removed from the hydroxyethyl group derived from pyruvate pass through FAD to NAD^+ .

Central to the mechanism of the PDH complex are the swinging lipoyllysyl arms of E_2 , which accept from E_1 the two electrons and the acetyl group derived from pyruvate, passing them to E_3 . All these enzymes and coenzymes are clustered, allowing the intermediates to react quickly without diffusing away from the surface of the enzyme complex. The five-reaction sequence shown in Figure 16–6 is thus an example of **substrate channeling**. The intermediates of the multistep sequence never leave the complex, and the local concentration of the substrate of E_2 is kept very high. Channeling also prevents theft of the activated acetyl group by other enzymes that use this group as substrate. As we shall see, a similar tethering mechanism for the channeling of substrate between active

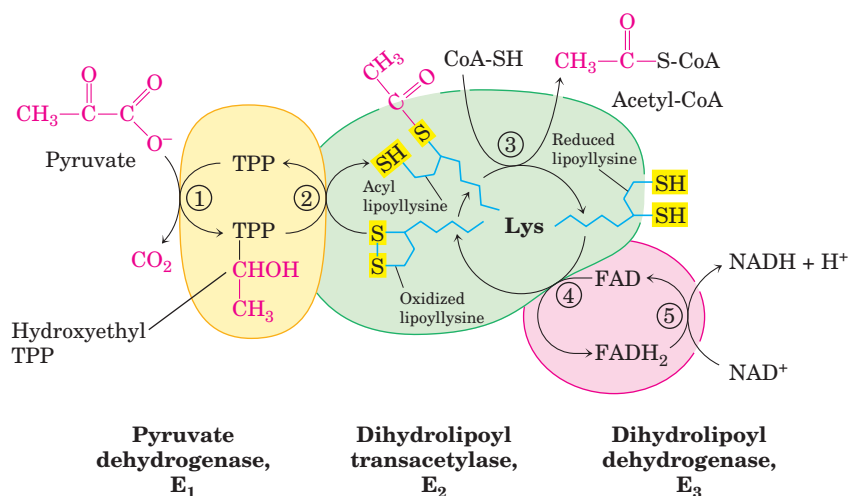


FIGURE 16–6 Oxidative decarboxylation of pyruvate to acetyl-CoA by the PDH complex. The fate of pyruvate is traced in red. In step ① pyruvate reacts with the bound thiamine pyrophosphate (TPP) of pyruvate dehydrogenase (E_1), undergoing decarboxylation to the hydroxyethyl derivative (see Fig. 14–13). Pyruvate dehydrogenase also carries out step ②, the transfer of two electrons and the acetyl group from TPP to the oxidized form of the lipoyllysyl group of the core enzyme, dihydrolipoyl transacetylase (E_2), to form the acetyl thioester of the reduced lipoyl group. Step ③ is a transesterification in which the

$-\text{SH}$ group of CoA replaces the $-\text{SH}$ group of E_2 to yield acetyl-CoA and the fully reduced (dithiol) form of the lipoyl group. In step ④ dihydrolipoyl dehydrogenase (E_3) promotes transfer of two hydrogen atoms from the reduced lipoyl groups of E_2 to the FAD prosthetic group of E_3 , restoring the oxidized form of the lipoyllysyl group of E_2 . In step ⑤ the reduced FADH_2 of E_3 transfers a hydride ion to NAD^+ , forming NADH . The enzyme complex is now ready for another catalytic cycle. (Subunit colors correspond to those in Fig. 16–5b.)

sites is used in some other enzymes, with lipoate, biotin, or a CoA-like moiety serving as cofactors.



As one might predict, mutations in the genes for the subunits of the PDH complex, or a dietary thiamine deficiency, can have severe consequences. Thiamine-deficient animals are unable to oxidize pyruvate normally. This is of particular importance to the brain, which usually obtains all its energy from the aerobic oxidation of glucose in a pathway that necessarily includes the oxidation of pyruvate. Beriberi, a disease that results from thiamine deficiency, is characterized by loss of neural function. This disease occurs primarily in populations that rely on a diet consisting mainly of white (polished) rice, which lacks the hulls in which most of the thiamine of rice is found. People who habitually consume large amounts of alcohol can also develop thiamine deficiency, because much of their dietary intake consists of the vitamin-free “empty calories” of distilled spirits. An elevated level of pyruvate in the blood is often an indicator of defects in pyruvate oxidation due to one of these causes. ■

SUMMARY 16.1 Production of Acetyl-CoA (Activated Acetate)

- Pyruvate, the product of glycolysis, is converted to acetyl-CoA, the starting material for the citric acid cycle, by the pyruvate dehydrogenase complex.
- The PDH complex is composed of multiple copies of three enzymes: pyruvate dehydrogenase, E_1 (with its bound cofactor TPP); dihydrolipoyl transacetylase, E_2 (with its covalently bound lipoyl group); and dihydrolipoyl dehydrogenase, E_3 (with its cofactors FAD and NAD).
- E_1 catalyzes first the decarboxylation of pyruvate, producing hydroxyethyl-TPP, and then the oxidation of the hydroxyethyl group to an acetyl group. The electrons from this oxidation reduce the disulfide of lipoate bound to E_2 , and the acetyl group is transferred into thioester linkage with one —SH group of reduced lipoate.
- E_2 catalyzes the transfer of the acetyl group to coenzyme A, forming acetyl-CoA.
- E_3 catalyzes the regeneration of the disulfide (oxidized) form of lipoate; electrons pass first to FAD, then to NAD^+ .
- The long lipoyllysine arm swings from the active site of E_1 to E_2 to E_3 , tethering the intermediates to the enzyme complex to allow substrate channeling.

- The organization of the PDH complex is very similar to that of the enzyme complexes that catalyze the oxidation of α -ketoglutarate and the branched-chain α -keto acids.

16.2 Reactions of the Citric Acid Cycle

We are now ready to trace the process by which acetyl-CoA undergoes oxidation. This chemical transformation is carried out by the citric acid cycle, the first *cyclic* pathway we have encountered (Fig. 16–7). To begin a turn of the cycle, acetyl-CoA donates its acetyl group to the four-carbon compound oxaloacetate to form the six-carbon citrate. Citrate is then transformed into isocitrate, also a six-carbon molecule, which is dehydrogenated with loss of CO_2 to yield the five-carbon compound α -ketoglutarate (also called oxoglutarate). α -Ketoglutarate undergoes loss of a second molecule of CO_2 and ultimately yields the four-carbon compound succinate. Succinate is then enzymatically converted in three steps into the four-carbon oxaloacetate—which is then ready to react with another molecule of acetyl-CoA. In each turn of the cycle, one acetyl group (two carbons) enters as acetyl-CoA and two molecules of CO_2 leave; one molecule of oxaloacetate is used to form citrate and one molecule of oxaloacetate is regenerated. No net removal of oxaloacetate occurs; one molecule of oxaloacetate can theoretically bring about oxidation of an infinite number of acetyl groups, and, in fact, oxaloacetate is present in cells in very low concentrations. Four of the eight steps in this process are oxidations, in which the energy of oxidation is very efficiently conserved in the form of the reduced coenzymes NADH and $FADH_2$.

As noted earlier, although the citric acid cycle is central to energy-yielding metabolism its role is not limited to energy conservation. Four- and five-carbon intermediates of the cycle serve as precursors for a wide variety of products. To replace intermediates removed for this purpose, cells employ anaplerotic (replenishing) reactions, which are described below.

Eugene Kennedy and Albert Lehninger showed in 1948 that, in eukaryotes, the entire set of reactions of the citric acid cycle takes place in mitochondria. Isolated mitochondria were found to contain not only all the enzymes and coenzymes required for the citric acid cycle, but also all the enzymes and proteins necessary for the last stage of respiration—electron transfer and ATP synthesis by oxidative phosphorylation. As we shall see in later chapters, mitochondria also contain the enzymes for the oxidation of fatty acids and some amino acids to acetyl-CoA, and the oxidative degradation of other amino acids to α -ketoglutarate, succinyl-CoA, or oxaloacetate. Thus, in nonphotosynthetic eukaryotes, the mitochondrion is the site of most energy-yielding

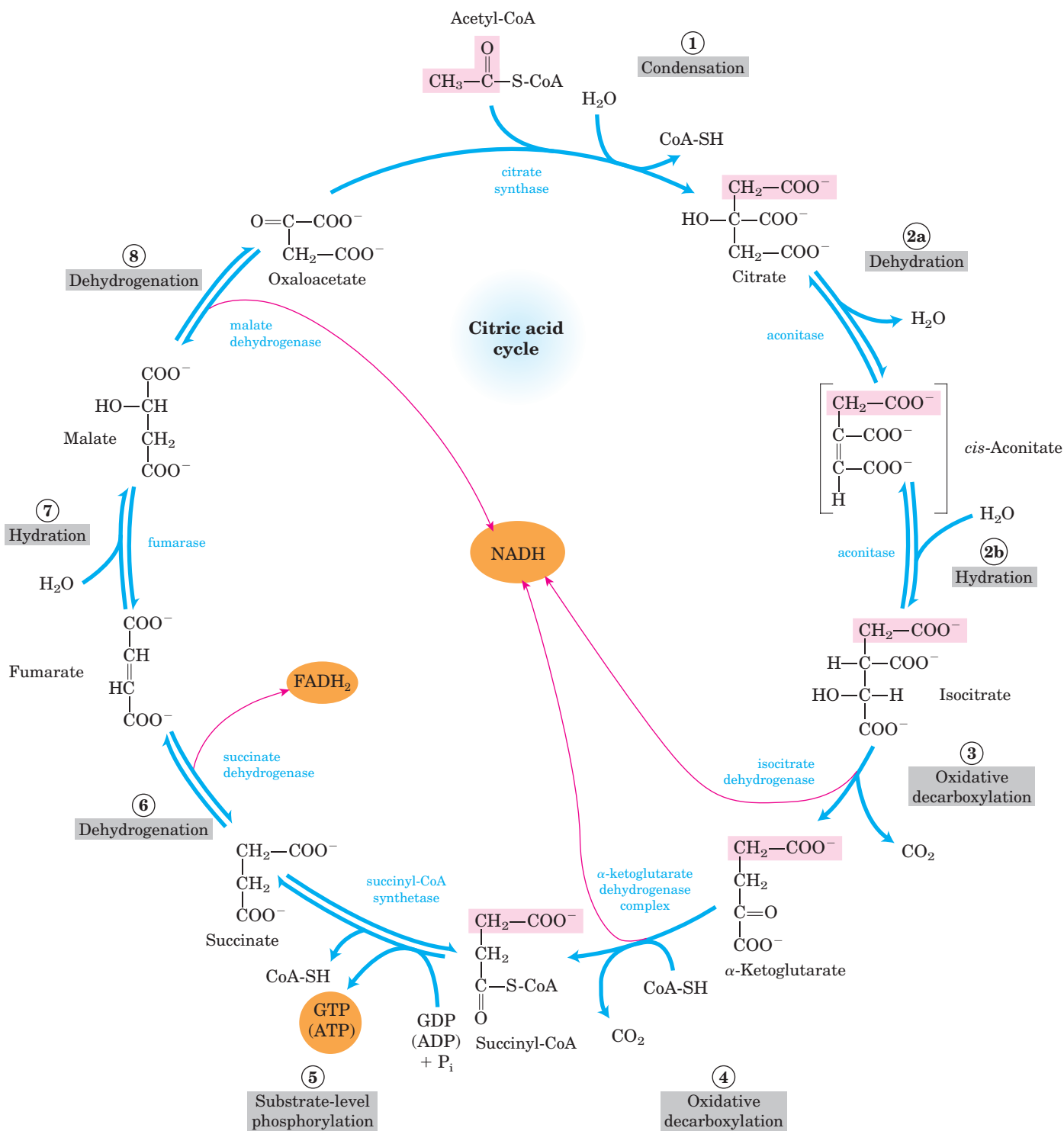


FIGURE 16-7 Reactions of the citric acid cycle. The carbon atoms shaded in pink are those derived from the acetate of acetyl-CoA in the first turn of the cycle; these are *not* the carbons released as CO₂ in the first turn. Note that in succinate and fumarate, the two-carbon group derived from acetate can no longer be specifically denoted; because succinate and fumarate are symmetric molecules, C-1 and C-2 are indistinguishable from C-4 and C-3. The number beside each

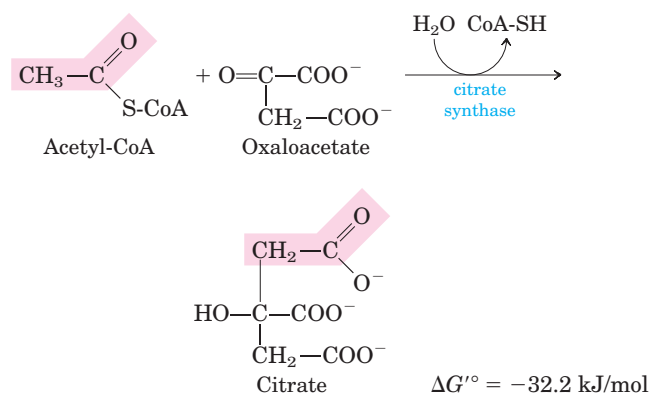
reaction step corresponds to a numbered heading on pages 608–612. The red arrows show where energy is conserved by electron transfer to FAD or NAD⁺, forming FADH₂ or NADH + H⁺. Steps ①, ③, and ④ are essentially irreversible in the cell; all other steps are reversible. The product of step ⑤ may be either ATP or GTP, depending on which succinyl-CoA synthetase isozyme is the catalyst.

oxidative reactions and of the coupled synthesis of ATP. In photosynthetic eukaryotes, mitochondria are the major site of ATP production in the dark, but in daylight chloroplasts produce most of the organism's ATP. In most prokaryotes, the enzymes of the citric acid cycle are in the cytosol, and the plasma membrane plays a role analogous to that of the inner mitochondrial membrane in ATP synthesis (Chapter 19).

The Citric Acid Cycle Has Eight Steps

In examining the eight successive reaction steps of the citric acid cycle, we place special emphasis on the chemical transformations taking place as citrate formed from acetyl-CoA and oxaloacetate is oxidized to yield CO_2 and the energy of this oxidation is conserved in the reduced coenzymes NADH and FADH_2 .

① Formation of Citrate The first reaction of the cycle is the condensation of acetyl-CoA with **oxaloacetate** to form **citrate**, catalyzed by **citrate synthase**:



In this reaction the methyl carbon of the acetyl group is joined to the carbonyl group (C-2) of oxaloacetate. Citroyl-CoA is a transient intermediate formed on the active site of the enzyme (see Fig. 16–9). It rapidly undergoes hydrolysis to free CoA and citrate, which are released from the active site. The hydrolysis of this high-energy thioester intermediate makes the forward reaction highly exergonic. The large, negative standard free-energy change of the citrate synthase reaction is essential to the operation of the cycle because, as noted earlier, the concentration of oxaloacetate is normally very low. The CoA liberated in this reaction is recycled to participate in the oxidative decarboxylation of another molecule of pyruvate by the PDH complex.

Citrate synthase from mitochondria has been crystallized and visualized by x-ray diffraction in the presence and absence of its substrates and inhibitors (Fig. 16–8). Each subunit of the homodimeric enzyme is a single polypeptide with two domains, one large and rigid, the other smaller and more flexible, with the active site between them. Oxaloacetate, the first substrate to bind to the enzyme, induces a large conformational

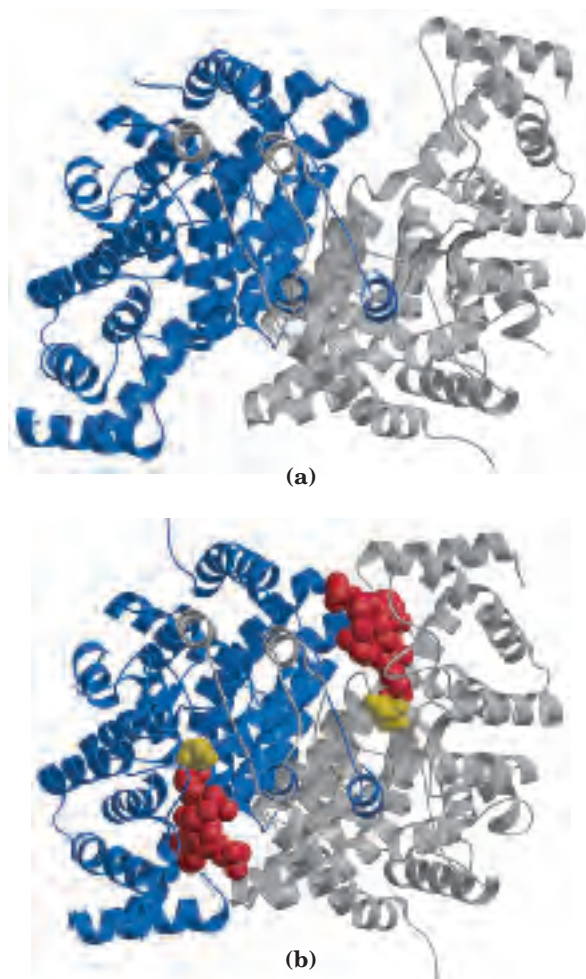
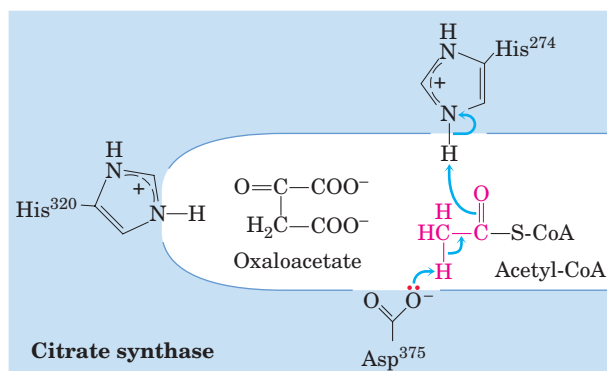


FIGURE 16–8 Structure of citrate synthase. The flexible domain of each subunit undergoes a large conformational change on binding oxaloacetate creating a binding site for acetyl-CoA. **(a)** open form of the enzyme alone (PDB ID 5CSC); **(b)** closed form with bound oxaloacetate (yellow) and a stable analog of acetyl-CoA (carboxymethyl-CoA; red) (derived from PDB ID 5CTS).

change in the flexible domain, creating a binding site for the second substrate, acetyl-CoA. When citroyl-CoA has formed in the enzyme active site, another conformational change brings about thioester hydrolysis, releasing CoA-SH. This induced fit of the enzyme first to its substrate and then to its reaction intermediate decreases the likelihood of premature and unproductive cleavage of the thioester bond of acetyl-CoA. Kinetic studies of the enzyme are consistent with this ordered bisubstrate mechanism (see Fig. 6–13). The reaction catalyzed by citrate synthase is essentially a Claisen condensation (p. 485), involving a thioester (acetyl-CoA) and a ketone (oxaloacetate) (Fig. 16–9).

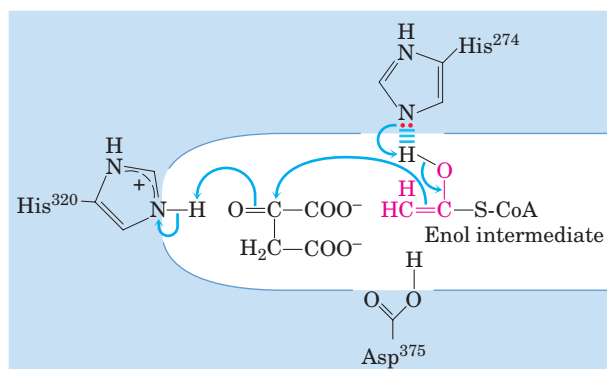
② Formation of Isocitrate via cis-Aconitate The enzyme **aconitase** (more formally, **aconitate hydratase**) catalyzes the reversible transformation of citrate to **isocitrate**, through the intermediary formation of the tricarboxylic acid **cis-aconitate**, which normally does



The thioester linkage in acetyl-CoA activates the methyl hydrogens, and Asp³⁷⁵ abstracts a proton from the methyl group, forming an enolate intermediate.

①

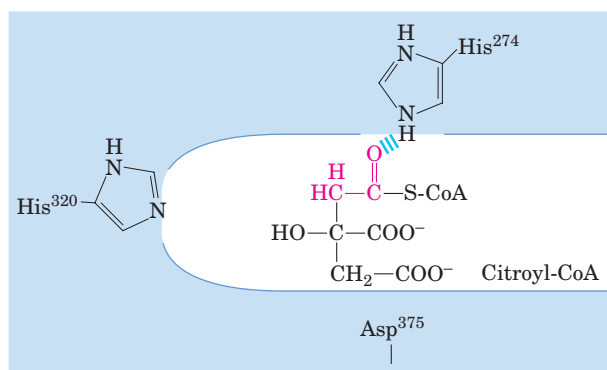
The intermediate is stabilized by hydrogen bonding to and/or protonation by His²⁷⁴ (full protonation is shown).



The enol(ate) rearranges to attack the carbonyl carbon of oxaloacetate, with His²⁷⁴ positioned to abstract the proton it had previously donated. His³²⁰ acts as a general acid.

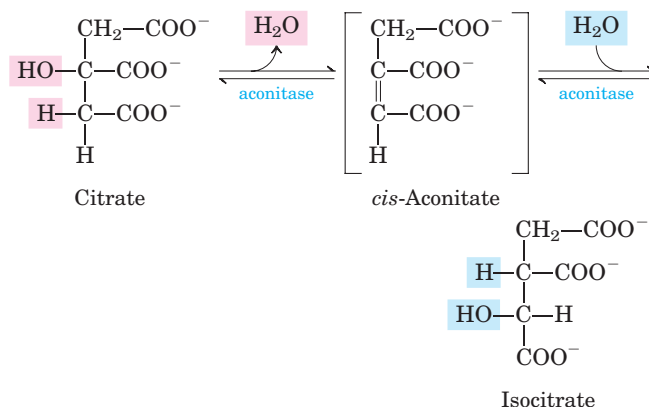
②

The resulting condensation generates citryl-CoA.



The thioester is subsequently hydrolyzed, regenerating CoA-SH and producing citrate.

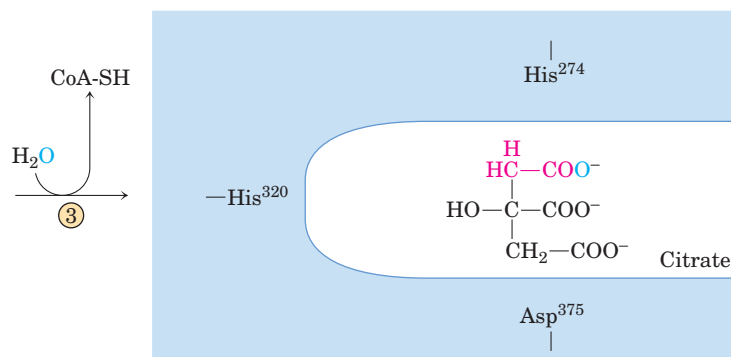
not dissociate from the active site. Aconitase can promote the reversible addition of H₂O to the double bond of enzyme-bound *cis*-aconitate in two different ways, one leading to citrate and the other to isocitrate:



$$\Delta G'^{\circ} = 13.3 \text{ kJ/mol}$$

Although the equilibrium mixture at pH 7.4 and 25 °C contains less than 10% isocitrate, in the cell the reaction is pulled to the right because isocitrate is rapidly consumed in the next step of the cycle, lowering its steady-state concentration. Aconitase contains an **iron-sulfur center** (Fig. 16–10), which acts both in the binding of the substrate at the active site and in the catalytic addition or removal of H₂O.

MECHANISM FIGURE 16-9 Citrate synthase. In the mammalian citrate synthase reaction, oxaloacetate binds first, in a strictly ordered reaction sequence. This binding triggers a conformation change that opens up the binding site for acetyl-CoA. Oxaloacetate is specifically oriented in the active site of citrate synthase by interaction of its two carboxylates with two positively charged Arg residues (not shown here). The details of the mechanism are described in the figure.  **Citrate Synthase Mechanism**



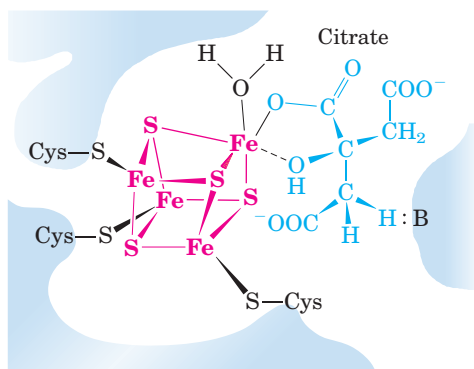
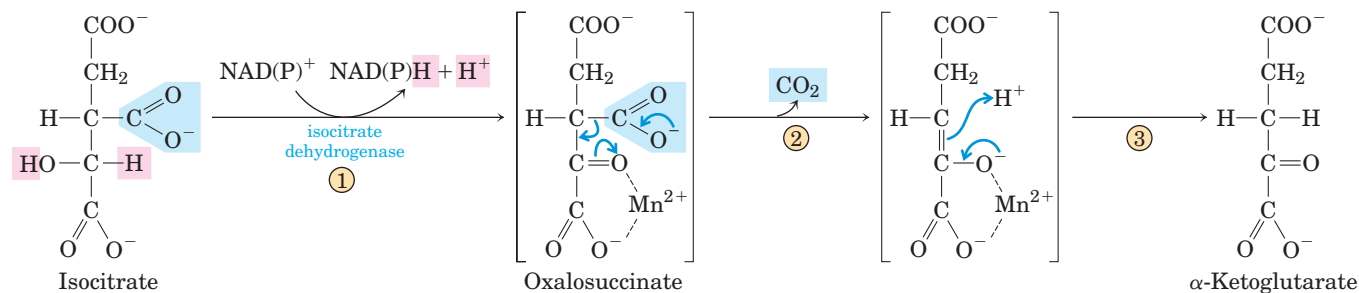


FIGURE 16-10 Iron-sulfur center in aconitase. The iron-sulfur center is in red, the citrate molecule in blue. Three Cys residues of the enzyme bind three iron atoms; the fourth iron is bound to one of the carboxyl groups of citrate and also interacts noncovalently with a hydroxyl group of citrate (dashed bond). A basic residue (:B) on the enzyme helps to position the citrate in the active site. The iron-sulfur center acts in both substrate binding and catalysis. The general properties of iron-sulfur proteins are discussed in Chapter 19 (see Fig. 19-5).

③ Oxidation of Isocitrate to α -Ketoglutarate and CO_2 In the next step, **isocitrate dehydrogenase** catalyzes oxidative decarboxylation of isocitrate to form **α -ketoglutarate** (Fig. 16-11). Mn^{2+} in the active site interacts with the carbonyl group of the intermediate oxalosuccinate, which is formed transiently but does not leave the binding site until decarboxylation converts it to α -ketoglutarate. Mn^{2+} also stabilizes the enol formed transiently by decarboxylation.

There are two different forms of isocitrate dehydrogenase in all cells, one requiring NAD^+ as electron acceptor and the other requiring NADP^+ . The overall reactions are otherwise identical. In eukaryotic cells, the NAD-dependent enzyme occurs in the mitochondrial matrix and serves in the citric acid cycle. The main function of the NADP-dependent enzyme, found in both the

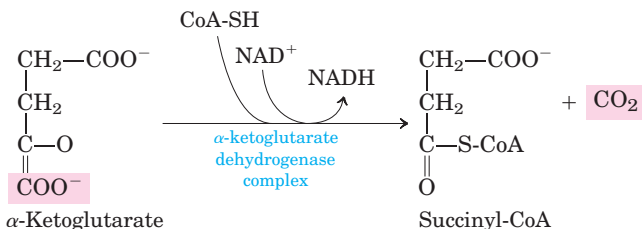


MECHANISM FIGURE 16-11 Isocitrate dehydrogenase. In this reaction, the substrate, isocitrate, loses one carbon by oxidative decarboxylation. In step ①, isocitrate binds to the enzyme and is oxidized by hydride transfer to NAD^+ or NADP^+ , depending on the isocitrate dehydrogenase isozyme. (See Fig. 14-12 for more information on hydride transfer reactions involving NAD^+ and NADP^+ .) The resulting

mitochondrial matrix and the cytosol, may be the generation of NADPH, which is essential for reductive anabolic reactions.

④ Oxidation of α -Ketoglutarate to Succinyl-CoA and CO_2

The next step is another oxidative decarboxylation, in which α -ketoglutarate is converted to **succinyl-CoA** and CO_2 by the action of the **α -ketoglutarate dehydrogenase complex**; NAD^+ serves as electron acceptor and CoA as the carrier of the succinyl group. The energy of oxidation of α -ketoglutarate is conserved in the formation of the thioester bond of succinyl-CoA:

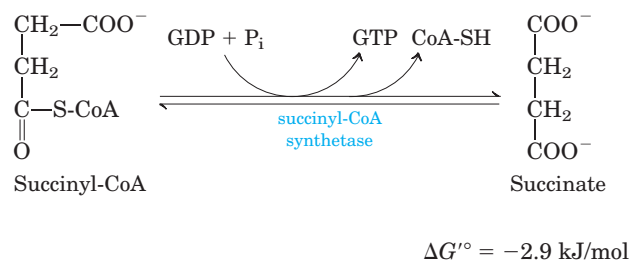


$$\Delta G'^{\circ} = -33.5 \text{ kJ/mol}$$

This reaction is virtually identical to the pyruvate dehydrogenase reaction discussed above, and the α -ketoglutarate dehydrogenase complex closely resembles the PDH complex in both structure and function. It includes three enzymes, homologous to E_1 , E_2 , and E_3 of the PDH complex, as well as enzyme-bound TPP, bound lipoate, FAD, NAD, and coenzyme A. Both complexes are certainly derived from a common evolutionary ancestor. Although the E_1 components of the two complexes are structurally similar, their amino acid sequences differ and, of course, they have different binding specificities: E_1 of the PDH complex binds pyruvate, and E_1 of the α -ketoglutarate dehydrogenase complex binds α -ketoglutarate. The E_2 components of the two complexes are also very similar, both having covalently bound lipoyl moieties. The subunits of E_3 are identical in the two enzyme complexes.

carbonyl group sets up the molecule for decarboxylation in step ②. Interaction of the carbonyl oxygen with a bound Mn^{2+} ion increases the electron-withdrawing capacity of the carbonyl group and facilitates the decarboxylation step. The reaction is completed in step ③ by rearrangement of the enol intermediate to generate α -ketoglutarate.

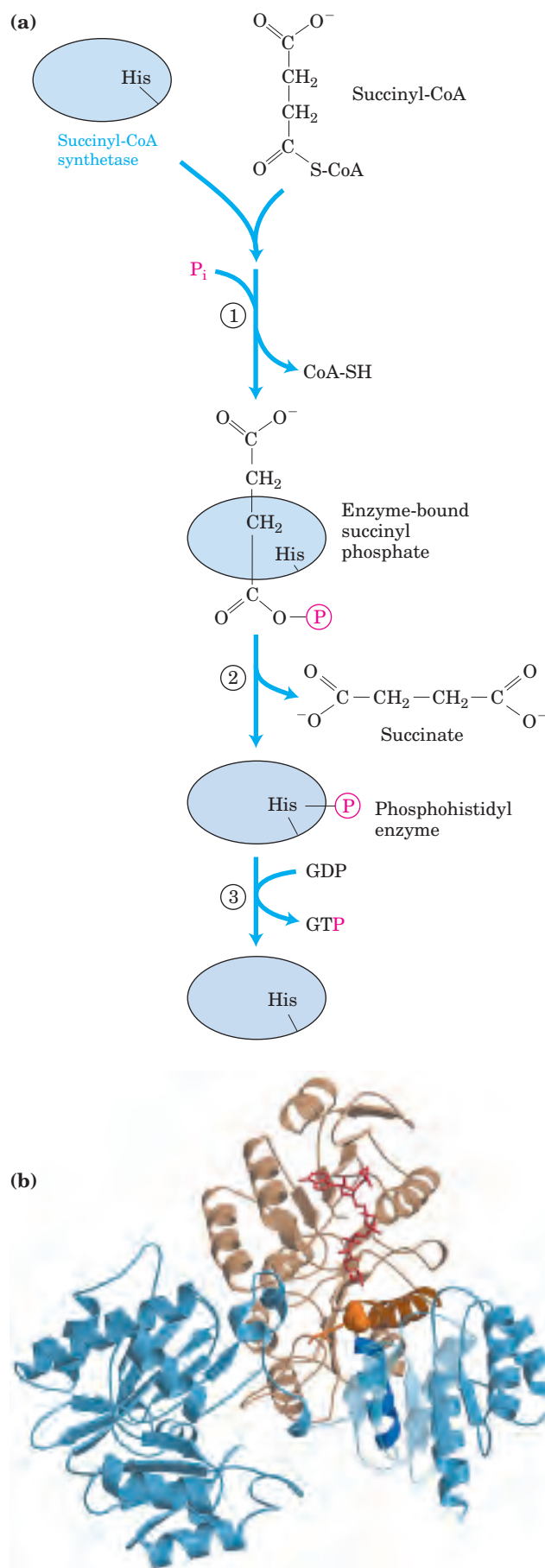
⑤ Conversion of Succinyl-CoA to Succinate Succinyl-CoA, like acetyl-CoA, has a thioester bond with a strongly negative standard free energy of hydrolysis ($\Delta G'^{\circ} \approx -36$ kJ/mol). In the next step of the citric acid cycle, energy released in the breakage of this bond is used to drive the synthesis of a phosphoanhydride bond in either GTP or ATP, with a net $\Delta G'^{\circ}$ of only -2.9 kJ/mol. **Succinate** is formed in the process:



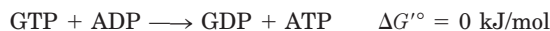
The enzyme that catalyzes this reversible reaction is called **succinyl-CoA synthetase** or **succinic thiokinase**; both names indicate the participation of a nucleoside triphosphate in the reaction (Box 16-1).

This energy-conserving reaction involves an intermediate step in which the enzyme molecule itself becomes phosphorylated at a His residue in the active site (Fig. 16-12a). This phosphoryl group, which has a high group transfer potential, is transferred to ADP (or GDP) to form ATP (or GTP). Animal cells have two isozymes of succinyl-CoA synthetase, one specific for ADP and the other for GDP. The enzyme has two subunits, α (M_r 32,000), which has the P-His residue (His^{246}) and the binding site for CoA, and β (M_r 42,000), which confers specificity for either ADP or GDP. The active site is at the interface between subunits. The crystal structure of succinyl-CoA synthetase reveals two “power helices” (one from each subunit), oriented so that their electric dipoles situate partial positive charges close to the negatively charged P-His (Fig. 16-12b), stabilizing the phosphoenzyme intermediate. (Recall the similar role of helix dipoles in stabilizing K^+ ions in the K^+ channel (see Fig. 11-48).)

FIGURE 16-12 The succinyl-CoA synthetase reaction. (a) In step ① a phosphoryl group replaces the CoA of succinyl-CoA bound to the enzyme, forming a high-energy acyl phosphate. In step ② the succinyl phosphate donates its phosphoryl group to a His residue on the enzyme, forming a high-energy phosphohistidyl enzyme. In step ③ the phosphoryl group is transferred from the His residue to the terminal phosphate of GDP (or ADP), forming GTP (or ATP). (b) Succinyl-CoA synthetase of *E. coli* (derived from PDB ID 1SCU). The bacterial and mammalian enzymes have similar amino acid sequences and presumably have very similar three-dimensional structures. The active site includes part of both the α (blue) and β (brown) subunits. The power helices (bright blue, dark brown) situate the partial positive charges of the helix dipole near the phosphate group (orange) on His^{246} of the α chain, stabilizing the phosphohistidyl enzyme. Coenzyme A is shown here as a red stick structure. (To improve the visibility of the power helices, some nearby secondary structures have been made transparent.)

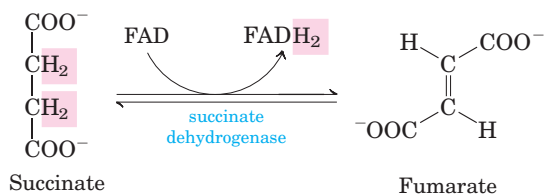


The formation of ATP (or GTP) at the expense of the energy released by the oxidative decarboxylation of α -ketoglutarate is a substrate-level phosphorylation, like the synthesis of ATP in the glycolytic reactions catalyzed by glyceraldehyde 3-phosphate dehydrogenase and pyruvate kinase (see Fig. 14–2). The GTP formed by succinyl-CoA synthetase can donate its terminal phosphoryl group to ADP to form ATP, in a reversible reaction catalyzed by **nucleoside diphosphate kinase** (p. 505):



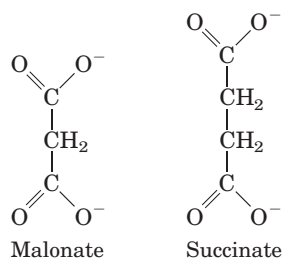
Thus the net result of the activity of either isozyme of succinyl-CoA synthetase is the conservation of energy as ATP. There is no change in free energy for the nucleoside diphosphate kinase reaction; ATP and GTP are energetically equivalent.

⑥ Oxidation of Succinate to Fumarate The succinate formed from succinyl-CoA is oxidized to **fumarate** by the flavoprotein **succinate dehydrogenase**:



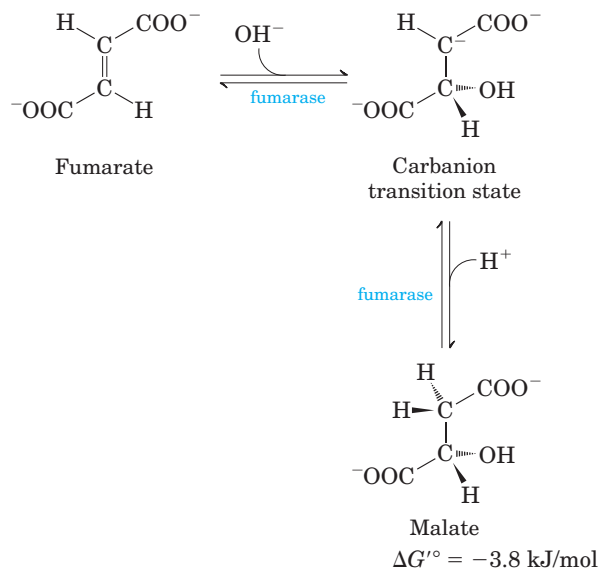
$$\Delta G'^{\circ} = 0 \text{ kJ/mol}$$

In eukaryotes, succinate dehydrogenase is tightly bound to the inner mitochondrial membrane; in prokaryotes, to the plasma membrane. The enzyme contains three different iron-sulfur clusters and one molecule of covalently bound FAD (see Fig. 19–xx). Electrons pass from succinate through the FAD and iron-sulfur centers before entering the chain of electron carriers in the mitochondrial inner membrane (or the plasma membrane in bacteria). Electron flow from succinate through these carriers to the final electron acceptor, O_2 , is coupled to the synthesis of about 1.5 ATP molecules per pair of electrons (respiration-linked phosphorylation). Malonate, an analog of succinate not normally present in cells, is a strong competitive inhibitor of succinate dehydrogenase and its addition to mitochondria blocks the activity of the citric acid cycle.

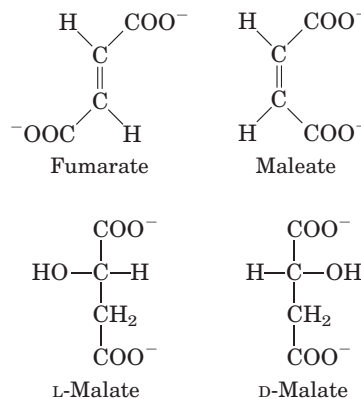


⑦ Hydration of Fumarate to Malate The reversible hydration of fumarate to **L-malate** is catalyzed by **fumarase**

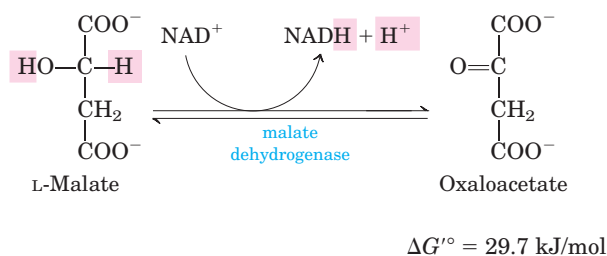
(formally, **fumarate hydratase**). The transition state in this reaction is a carbanion:



This enzyme is highly stereospecific; it catalyzes hydration of the trans double bond of fumarate but not the cis double bond of maleate (the cis isomer of fumarate). In the reverse direction (from L-malate to fumarate), fumarase is equally stereospecific: D-malate is not a substrate.



⑧ Oxidation of Malate to Oxaloacetate In the last reaction of the citric acid cycle, NAD-linked **L-malate dehydrogenase** catalyzes the oxidation of L-malate to oxaloacetate:



The equilibrium of this reaction lies far to the left under standard thermodynamic conditions, but in intact cells

BOX 16-1 WORKING IN BIOCHEMISTRY

Synthases and Synthetases; Ligases and Lyases; Kinases, Phosphatases, and Phosphorylases: Yes, the Names Are Confusing!

Citrate synthase is one of many enzymes that catalyze condensation reactions, yielding a product more chemically complex than its precursors. **Synthases** catalyze condensation reactions in which no nucleoside triphosphate (ATP, GTP, and so forth) is required as an energy source. **Synthetases** catalyze condensations that *do* use ATP or another nucleoside triphosphate as a source of energy for the synthetic reaction. Succinyl-CoA synthetase is such an enzyme. **Ligases** (from the Latin *ligare*, “to tie together”) are enzymes that catalyze condensation reactions in which two atoms are joined using ATP or another energy source. (Thus synthetases are ligases.) DNA ligase, for example, closes breaks in DNA molecules, using energy supplied by either ATP or NAD⁺; it is widely used in joining DNA pieces for genetic engineering. Ligases are not to be confused with **lyases**, enzymes that catalyze cleavages (or, in the reverse direction, additions) in which electronic rearrangements occur. The PDH complex, which oxidatively cleaves CO₂ from pyruvate, is a member of the large class of lyases.

The name **kinase** is applied to enzymes that transfer a phosphoryl group from a nucleoside triphosphate such as ATP to an acceptor molecule—a sugar (as in hexokinase and glucokinase), a protein (as in glycogen phosphorylase kinase), another nucleotide (as in nucleoside diphosphate kinase), or a metabolic intermediate such as oxaloacetate (as in PEP carboxykinase). The reaction catalyzed by a kinase is a *phosphorylation*. On the other hand, *phosphorolysis* is a displacement reaction in which phosphate is the attacking species and becomes covalently attached at the point of bond breakage. Such reactions are catalyzed by **phosphorylases**. Glycogen phosphorylase, for example, catalyzes the phosphorolysis of glycogen, producing glucose 1-phosphate. *Dephos-*

phorylation, the removal of a phosphoryl group from a phosphate ester, is catalyzed by **phosphatases**, with water as the attacking species. Fructose bisphosphatase-1 converts fructose 1,6-bisphosphate to fructose 6-phosphate in gluconeogenesis, and phosphorylase *a* phosphatase removes phosphoryl groups from phosphoserine in phosphorylated glycogen phosphorylase. Whew!

Unfortunately, these descriptions of enzyme types overlap, and many enzymes are commonly called by two or more names. Succinyl-CoA synthetase, for example, is also called succinate thiokinase; the enzyme is both a synthetase in the citric acid cycle and a kinase when acting in the direction of succinyl-CoA synthesis. This raises another source of confusion in the naming of enzymes. An enzyme may have been discovered by the use of an assay in which, say, A is converted to B. The enzyme is then named for that reaction. Later work may show, however, that in the cell, the enzyme functions primarily in converting B to A. Commonly, the first name continues to be used, although the metabolic role of the enzyme would be better described by naming it for the reverse reaction. The glycolytic enzyme pyruvate kinase illustrates this situation (p. 532). To a beginner in biochemistry, this duplication in nomenclature can be bewildering. International committees have made heroic efforts to systematize the nomenclature of enzymes (see Table 6-3 for a brief summary of the system), but some systematic names have proved too long and cumbersome and are not frequently used in biochemical conversation.

We have tried throughout this book to use the enzyme name most commonly used by working biochemists and to point out cases in which an enzyme has more than one widely used name. For current information on enzyme nomenclature, refer to the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (www.chem.qmw.ac.uk/iubmb/nomenclature/).

oxaloacetate is continually removed by the highly exergonic citrate synthase reaction (step ① of Fig. 16-7). This keeps the concentration of oxaloacetate in the cell extremely low ($<10^{-6}$ M), pulling the malate dehydrogenase reaction toward the formation of oxaloacetate.

Although the individual reactions of the citric acid cycle were initially worked out *in vitro*, using minced muscle tissue, the pathway and its regulation have also been studied extensively *in vivo*. By using radioactively la-

beled precursors such as [¹⁴C]pyruvate and [¹⁴C]acetate, researchers have traced the fate of individual carbon atoms through the citric acid cycle. Some of the earliest experiments with isotopes produced an unexpected result, however, which aroused considerable controversy about the pathway and mechanism of the citric acid cycle. In fact, these experiments at first seemed to show that citrate was not the first tricarboxylic acid to be formed. Box 16-2 gives some details of this episode in the history of citric acid cycle research. Metabolic flux

through the cycle can now be monitored in living tissue by using ^{13}C -labeled precursors and whole-tissue NMR spectroscopy. Because the NMR signal is unique to the compound containing the ^{13}C , biochemists can trace the movement of precursor carbons into each cycle intermediate and into compounds derived from the intermediates. This technique has great promise for studies of regulation of the citric acid cycle and its interconnections with other metabolic pathways such as glycolysis.

The Energy of Oxidations in the Cycle Is Efficiently Conserved

We have now covered one complete turn of the citric acid cycle (Fig. 16–13). A two-carbon acetyl group entered the cycle by combining with oxaloacetate. Two carbon atoms emerged from the cycle as CO_2 from the oxidation of isocitrate and α -ketoglutarate. The energy released by these oxidations was conserved in the reduction of three NAD^+ and one FAD and the produc-

tion of one ATP or GTP. At the end of the cycle a molecule of oxaloacetate was regenerated. Note that the two carbon atoms appearing as CO_2 are not the same two carbons that entered in the form of the acetyl group; additional turns around the cycle are required to release these carbons as CO_2 (Fig. 16–7).

Although the citric acid cycle directly generates only one ATP per turn (in the conversion of succinyl-CoA to succinate), the four oxidation steps in the cycle provide a large flow of electrons into the respiratory chain via NADH and FADH_2 and thus lead to formation of a large number of ATP molecules during oxidative phosphorylation.

We saw in Chapter 14 that the energy yield from the production of two molecules of pyruvate from one molecule of glucose in glycolysis is 2 ATP and 2 NADH . In oxidative phosphorylation (Chapter 19), passage of two electrons from NADH to O_2 drives the formation of about 2.5 ATP, and passage of two electrons from FADH_2 to O_2 yields about 1.5 ATP. This stoichiometry allows us to calculate the overall yield of ATP from the complete

BOX 16-2 WORKING IN BIOCHEMISTRY

Citrate: A Symmetrical Molecule That Reacts Asymmetrically

When compounds enriched in the heavy-carbon isotope ^{13}C and the radioactive carbon isotopes ^{11}C and ^{14}C became available about 60 years ago, they were soon put to use in tracing the pathway of carbon atoms through the citric acid cycle. One such experiment initiated the controversy over the role of citrate. Acetate labeled in the carboxyl group (designated $[1-^{14}\text{C}]$ acetate) was incubated aerobically with an animal tissue preparation. Acetate is enzymatically converted to acetyl-CoA in animal tissues, and the pathway of the

labeled carboxyl carbon of the acetyl group in the cycle reactions could thus be traced. α -Ketoglutarate was isolated from the tissue after incubation, then degraded by known chemical reactions to establish the position(s) of the isotopic carbon.

Condensation of unlabeled oxaloacetate with carboxyl-labeled acetate would be expected to produce citrate labeled in one of the two primary carboxyl groups. Citrate is a symmetric molecule, its two terminal carboxyl groups being chemically indistinguishable. Therefore, half the labeled citrate molecules were expected to yield α -ketoglutarate labeled in

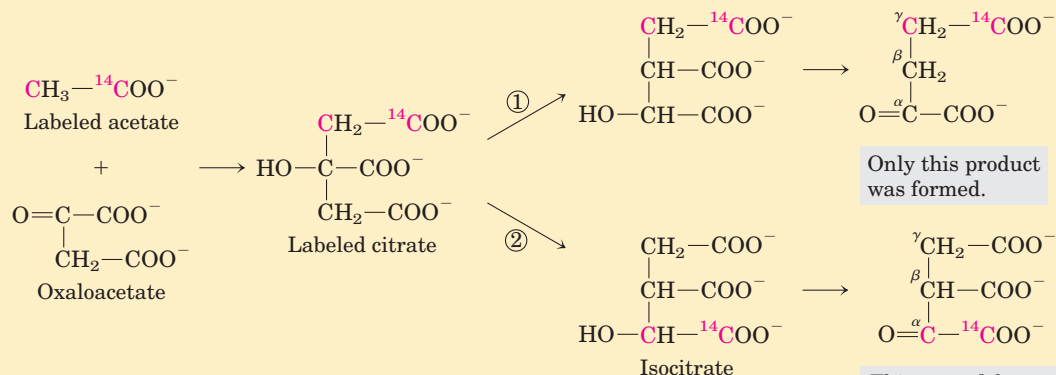


FIGURE 1 Incorporation of the isotopic carbon (^{14}C) of the labeled acetyl group into α -ketoglutarate by the citric acid cycle. The carbon atoms of the entering acetyl group are shown in red.

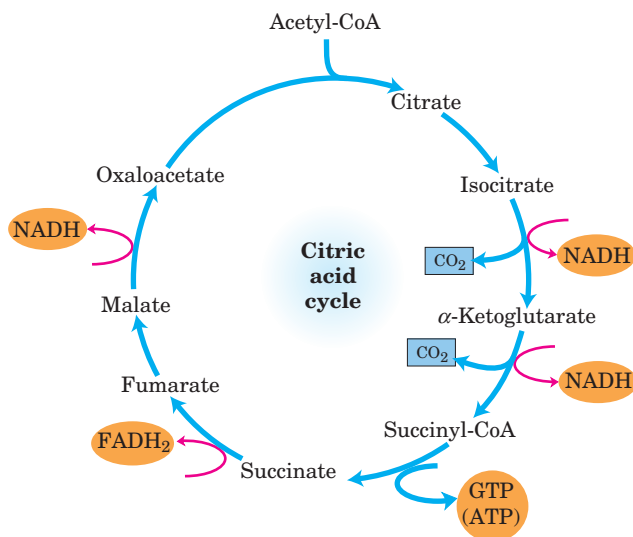


FIGURE 16-13 Products of one turn of the citric acid cycle. At each turn of the cycle, three NADH, one FADH₂, one GTP (or ATP), and two CO₂ are released in oxidative decarboxylation reactions. Here and in several following figures, all cycle reactions are shown as proceeding in one direction only, but keep in mind that most of the reactions are reversible (see Fig. 16-7).

oxidation of glucose. When both pyruvate molecules are oxidized to 6 CO₂ via the pyruvate dehydrogenase complex and the citric acid cycle, and the electrons are transferred to O₂ via oxidative phosphorylation, as many as 32 ATP are obtained per glucose (Table 16-1). In round numbers, this represents the conservation of $32 \times 30.5 \text{ kJ/mol} = 976 \text{ kJ/mol}$, or 34% of the theoretical maximum of about 2,840 kJ/mol available from the complete oxidation of glucose. These calculations employ the standard free-energy changes; when corrected for the actual free energy required to form ATP within cells (see Box 13-1), the calculated efficiency of the process is closer to 65%.

Why Is the Oxidation of Acetate So Complicated?

The eight-step cyclic process for oxidation of simple two-carbon acetyl groups to CO₂ may seem unnecessarily cumbersome and not in keeping with the biological principle of maximum economy. The role of the citric acid cycle is not confined to the oxidation of acetate, however.

the α -carboxyl group and the other half to yield α -keto-glutarate labeled in the γ -carboxyl group; that is, the α -ketoglutarate isolated was expected to be a mixture of the two types of labeled molecules (Fig. 1, pathways ① and ②). Contrary to this expectation, the labeled α -ketoglutarate isolated from the tissue suspension contained ¹⁴C only in the γ -carboxyl group (Fig. 1, pathway ①). The investigators concluded that citrate (or any other symmetric molecule) could not be an intermediate in the pathway from acetate to α -ketoglutarate. Rather, an asymmetric tricarboxylic acid, presumably *cis*-aconitate or isocitrate, must be the first product formed from condensation of acetate and oxaloacetate.

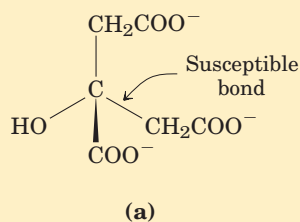
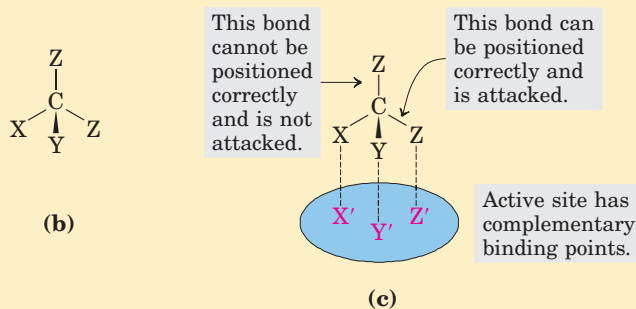


FIGURE 2 The prochiral nature of citrate. (a) Structure of citrate; (b) schematic representation of citrate: X = —OH; Y = —COO[−]; Z = —CH₂COO[−]. (c) Correct complementary fit of citrate to the binding site of aconitase. There is only one way in which the three specified groups of citrate can fit on the three points of the binding site. Thus only one of the two —CH₂COO[−] groups is bound by aconitase.

In 1948, however, Alexander Ogston pointed out that although citrate has no chiral center (see Fig. 1-19), it has the *potential* to react asymmetrically if an enzyme with which it interacts has an active site that is asymmetric. He suggested that the active site of aconitase may have three points to which the citrate must be bound and that the citrate must undergo a specific three-point attachment to these binding points. As seen in Figure 2, the binding of citrate to three such points could happen in only one way, and this would account for the formation of only one type of labeled α -ketoglutarate. Organic molecules such as citrate that have no chiral center but are potentially capable of reacting asymmetrically with an asymmetric active site are now called **prochiral** molecules.



This pathway is the hub of intermediary metabolism. Four- and five-carbon end products of many catabolic processes feed into the cycle to serve as fuels. Oxaloacetate and α -ketoglutarate, for example, are produced from aspartate and glutamate, respectively, when proteins are degraded. Under some metabolic circumstances, intermediates are drawn out of the cycle to be used as precursors in a variety of biosynthetic pathways.

The citric acid cycle, like all other metabolic pathways, is the product of evolution, and much of this evolution occurred before the advent of aerobic organisms. It does not necessarily represent the *shortest* pathway from acetate to CO_2 , but it is the pathway that has, over time, conferred the greatest selective advantage. Early anaerobes most probably used some of the reactions of the citric acid cycle in linear biosynthetic processes. In fact, some modern anaerobic microorganisms use an incomplete citric acid cycle as a source of, not energy, but biosynthetic precursors (Fig. 16–14). These organisms use the first three reactions of the cycle to make α -ketoglutarate but, lacking α -ketoglutarate dehydrogenase, they cannot carry out the complete set of citric acid cycle reactions. They do have the four enzymes that catalyze the reversible conversion of oxaloacetate to succinyl-CoA and can produce malate, fumarate, succinate, and succinyl-CoA from oxaloacetate in a reversal of the “normal” (oxidative) direction of flow through the cycle. This pathway is a fermentation, with the NADH produced by isocitrate oxidation recycled to NAD^+ by reduction of oxaloacetate to succinate.

With the evolution of cyanobacteria that produced O_2 from water, the earth’s atmosphere became aerobic and organisms were under selective pressure to develop aerobic metabolism, which, as we have seen, is much more efficient than anaerobic fermentation.

Citric Acid Cycle Components Are Important Biosynthetic Intermediates

In aerobic organisms, the citric acid cycle is an **amphibolic pathway**, one that serves in both catabolic and anabolic processes. Besides its role in the oxidative catabolism of carbohydrates, fatty acids, and amino acids, the cycle provides precursors for many biosynthetic pathways (Fig. 16–15), through reactions that served the same purpose in anaerobic ancestors. α -Ketoglutarate and oxaloacetate can, for example, serve as precursors of the amino acids aspartate and glutamate by simple transamination (Chapter 22). Through aspartate and glutamate, the carbons of oxaloacetate and α -ketoglutarate are then used to build other amino acids, as well as purine and pyrimidine nucleotides. Oxaloacetate is converted to glucose in gluconeogenesis (see Fig. 15–15). Succinyl-CoA is a central intermediate in the synthesis of the porphyrin ring of heme groups, which serve as oxygen carriers (in hemoglobin and myoglobin) and electron carriers (in cytochromes) (see Fig. 22–23). And the citrate produced in some organisms is used commercially for a variety of purposes (Box 16–3).

Anaplerotic Reactions Replenish Citric Acid Cycle Intermediates

As intermediates of the citric acid cycle are removed to serve as biosynthetic precursors, they are replenished by **anaplerotic reactions** (Fig. 16–15; Table 16–2). Under normal circumstances, the reactions by which cycle intermediates are siphoned off into other pathways and those by which they are replenished are in dynamic balance, so that the concentrations of the citric acid cycle intermediates remain almost constant.

TABLE 16-1 Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

Reaction	Number of ATP or reduced coenzyme directly formed	Number of ATP ultimately formed*
Glucose $\longrightarrow$ glucose 6-phosphate	–1 ATP	–1
Fructose 6-phosphate $\longrightarrow$ fructose 1,6-bisphosphate	–1 ATP	–1
2 Glyceraldehyde 3-phosphate $\longrightarrow$ 2 1,3-bisphosphoglycerate	2 NADH	3 or 5 [†]
2 1,3-Bisphosphoglycerate $\longrightarrow$ 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate $\longrightarrow$ 2 pyruvate	2 ATP	2
2 Pyruvate $\longrightarrow$ 2 acetyl-CoA	2 NADH	5
2 Isocitrate $\longrightarrow$ 2 α -ketoglutarate	2 NADH	5
2 α -Ketoglutarate $\longrightarrow$ 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA $\longrightarrow$ 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate $\longrightarrow$ 2 fumarate	2 FADH_2	3
2 Malate $\longrightarrow$ 2 oxaloacetate	2 NADH	5
Total		30–32

* This is calculated as 2.5 ATP per NADH and 1.5 ATP per FADH_2 . A negative value indicates consumption.

[†] This number is either 3 or 5, depending on the mechanism used to shuttle NADH equivalents from the cytosol to the mitochondrial matrix; see Figures 19–27 and 19–28.

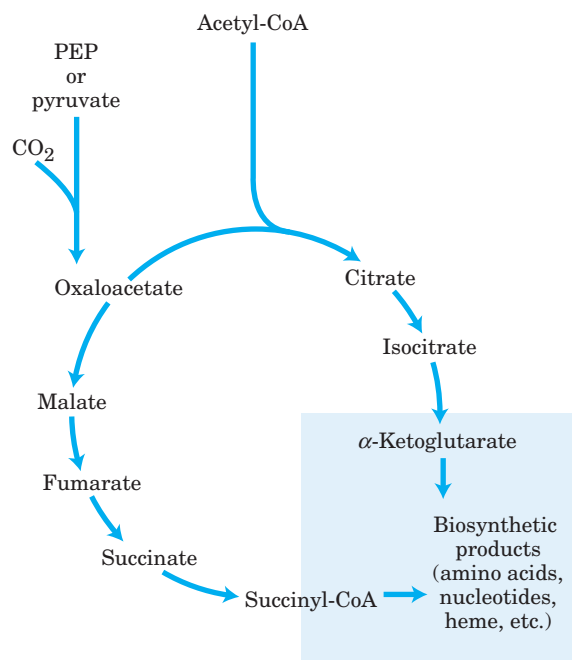


FIGURE 16-14 Biosynthetic precursors produced by an incomplete citric acid cycle in anaerobic bacteria. These anaerobes lack α -ketoglutarate dehydrogenase and therefore cannot carry out the complete citric acid cycle. α -Ketoglutarate and succinyl-CoA serve as precursors in a variety of biosynthetic pathways. (See Fig. 16-13 for the “normal” direction of these reactions in the citric acid cycle.)

Table 16-2 shows the most common anaplerotic reactions, all of which, in various tissues and organisms, convert either pyruvate or phosphoenolpyruvate to oxaloacetate or malate. The most important anaplerotic reaction in mammalian liver and kidney is the reversible carboxylation of pyruvate by CO₂ to form oxaloacetate, catalyzed by **pyruvate carboxylase**. When the citric acid cycle is deficient in oxaloacetate or any other intermediates, pyruvate is carboxylated to produce more oxaloacetate. The enzymatic addition of a carboxyl group to pyruvate requires energy, which is supplied by ATP—the free energy required to attach a carboxyl group to pyruvate is about equal to the free energy available from ATP.

Pyruvate carboxylase is a regulatory enzyme and is virtually inactive in the absence of acetyl-CoA, its positive allosteric modulator. Whenever acetyl-CoA, the fuel for the citric acid cycle, is present in excess, it stimulates the pyruvate carboxylase reaction to produce more oxaloacetate, enabling the cycle to use more acetyl-CoA in the citrate synthase reaction.

The other anaplerotic reactions shown in Table 16-2 are also regulated to keep the level of intermediates high enough to support the activity of the citric acid cycle. Phosphoenolpyruvate (PEP) carboxylase, for example, is activated by the glycolytic intermediate fructose 1,6-bisphosphate, which accumulates when the citric acid cycle operates too slowly to process the pyruvate generated by glycolysis.

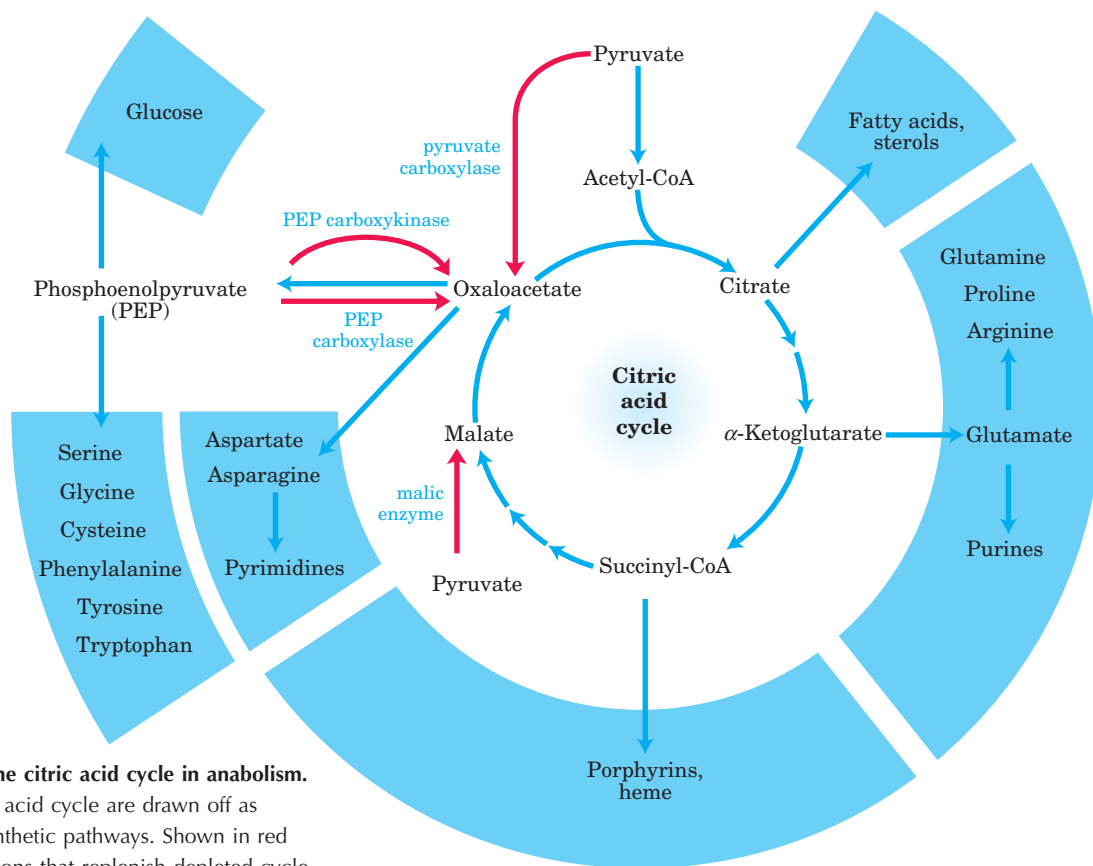


FIGURE 16-15 Role of the citric acid cycle in anabolism. Intermediates of the citric acid cycle are drawn off as precursors in many biosynthetic pathways. Shown in red are four anaplerotic reactions that replenish depleted cycle intermediates (see Table 16-2).

BOX 16-3 THE WORLD OF BIOCHEMISTRY

Citrate Synthase, Soda Pop, and the World Food Supply

Citrate has a number of important industrial applications. A quick examination of the ingredients in most soft drinks reveals the common use of citric acid to provide a tart or fruity flavor. Citric acid is also used as a plasticizer and foam inhibitor in the manufacture of certain resins, as a mordant to brighten colors, and as an antioxidant to preserve the flavors of foods. Citric acid is produced industrially by growing the fungus *Aspergillus niger* in the presence of an inexpensive sugar source, usually beet molasses. Culture conditions are designed to inhibit the reactions of the citric acid cycle such that citrate accumulates.

On a grander scale, citric acid may one day play a spectacular role in the alleviation of world hunger. With its three negatively charged carboxyl groups, citrate is a good chelator of metal ions, and some plants exploit this property by releasing citrate into the soil, where it binds metal ions and prevents their absorption by the plant. Of particular importance is the aluminum ion (Al^{3+}), which is toxic to many plants and causes decreased crop yields on 30% to 40% of the world's arable land. Aluminum is the most abundant metal in the earth's crust, yet it occurs mostly in chemical compounds, such as $\text{Al}(\text{OH})_3$, that are biologically inert. However, when soil pH is less than 5, Al^{3+} be-

comes soluble and thus can be absorbed by plant roots. Acidic soil and Al^{3+} toxicity are most prevalent in the tropics, where maize yields can be depressed by as much as 80%. In Mexico, Al^{3+} toxicity limits papaya production to 20,000 hectares, instead of the 3 million hectares that could theoretically be cultivated. One solution would be to raise soil pH with lime, but this is economically and environmentally unsound. An alternative would be to breed Al^{3+} -resistant plants. Naturally resistant plants do exist, and these provide the means for a third solution: transferring resistance to crop plants by genetic engineering.

A group of researchers in Mexico has genetically engineered tobacco and papaya plants to express elevated levels of bacterial citrate synthase. These plants secrete five to six times their normal amount of Al^{3+} -chelating citric acid and can grow in soils with Al^{3+} levels ten times those at which control plants can grow. This degree of resistance would allow Mexico to grow papaya on the 3 million hectares of land currently rendered unsuitable by Al^{3+} .

Given projected levels of population growth, world food production must more than triple in the next 50 years to adequately feed 9.6 billion people. A long-term solution may turn on increasing crop productivity on the arable land affected by aluminum toxicity, and citric acid may play an important role in achieving this goal.

Biotin in Pyruvate Carboxylase Carries CO_2 Groups

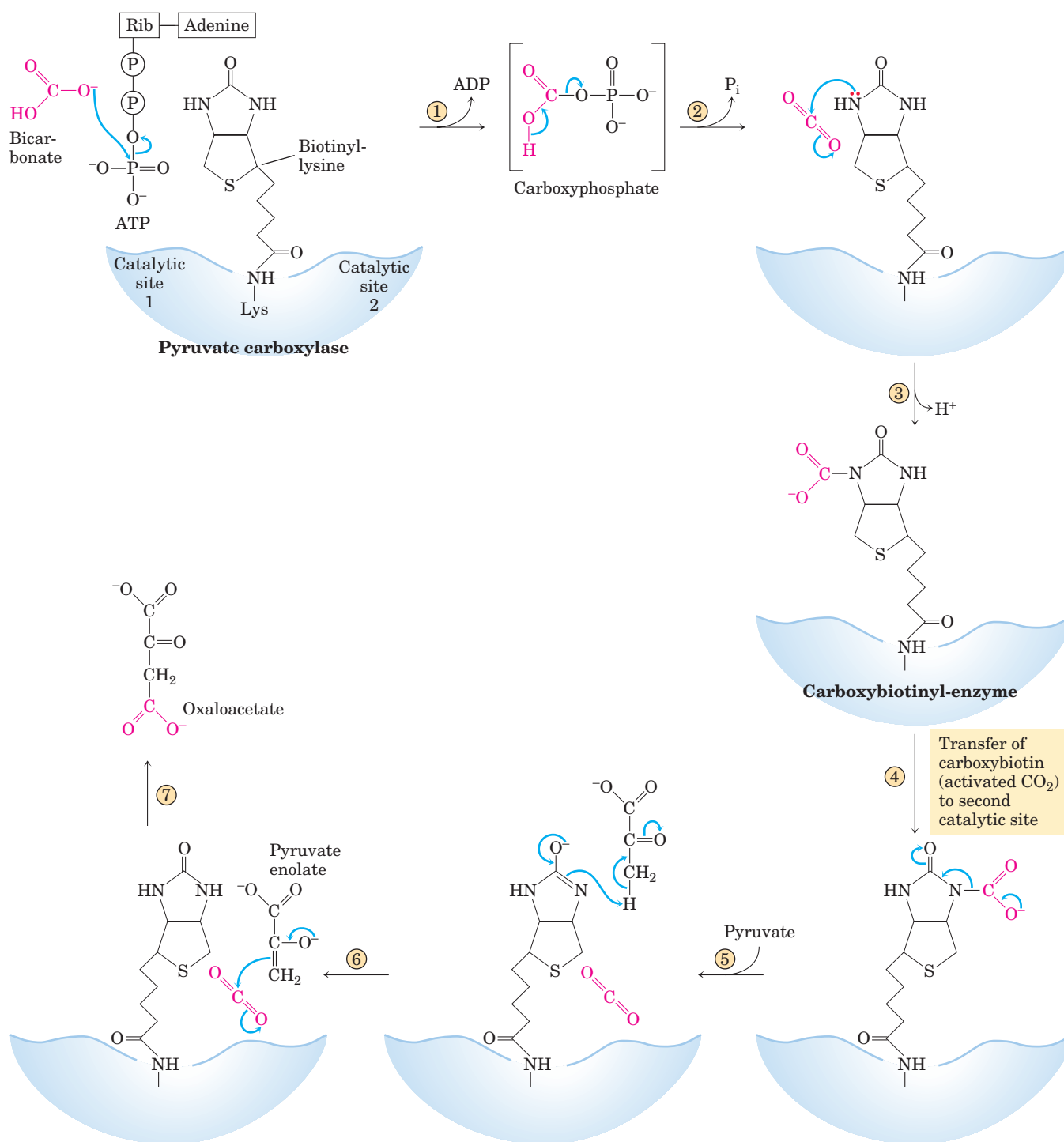
The pyruvate carboxylase reaction requires the vitamin **biotin** (Fig. 16-16), which is the prosthetic group of the enzyme. Biotin plays a key role in many carboxylation reactions. It is a specialized carrier of one-carbon groups in their most oxidized form: CO_2 . (The transfer of one-carbon groups in more reduced forms is mediated by other cofactors, notably tetrahydrofolate and *S*-adenosylmethionine, as described in Chapter 18.)

Carboxyl groups are activated in a reaction that splits ATP and joins CO_2 to enzyme-bound biotin. This "activated" CO_2 is then passed to an acceptor (pyruvate in this case) in a carboxylation reaction.

Pyruvate carboxylase has four identical subunits, each containing a molecule of biotin covalently attached through an amide linkage to the ϵ -amino group of a specific Lys residue in the enzyme active site. Carboxylation of pyruvate proceeds in two steps (Fig. 16-16): first, a carboxyl group derived from HCO_3^- is attached to biotin,

TABLE 16-2 Anaplerotic Reactions

Reaction	Tissue(s)/organism(s)
Pyruvate + HCO_3^- + ATP $\xrightleftharpoons{\text{pyruvate carboxylase}}$ oxaloacetate + ADP + P_i	Liver, kidney
Phosphoenolpyruvate + CO_2 + GDP $\xrightleftharpoons{\text{PEP carboxykinase}}$ oxaloacetate + GTP	Heart, skeletal muscle
Phosphoenolpyruvate + HCO_3^- $\xrightleftharpoons{\text{PEP carboxylase}}$ oxaloacetate + P_i	Higher plants, yeast, bacteria
Pyruvate + HCO_3^- + NAD(P)H $\xrightleftharpoons{\text{malic enzyme}}$ malate + NAD(P) $^+$	Widely distributed in eukaryotes and prokaryotes



MECHANISM FIGURE 16-16 The role of biotin in the reaction catalyzed by pyruvate carboxylase. Biotin is attached to the enzyme through an amide bond with the ϵ -amino group of a Lys residue, forming biotinyl-enzyme. Biotin-mediated carboxylation reactions occur in two phases, generally catalyzed by separate active sites on the enzyme as exemplified by the pyruvate carboxylase reaction. In the first phase (steps ① to ③), bicarbonate is converted to the more activated CO₂, and then used to carboxylate biotin. The bicarbonate is first activated by reaction with ATP to form carboxyphosphate (step ①), which breaks down to carbon dioxide (step ②). In effect, the

bicarbonate is dehydrated by its reaction with ATP, and the CO₂ can react with biotin to form carboxybiotin (step ③). The biotin acts as a carrier to transport the CO₂ from one active site to another on the same enzyme (step ④). In the second phase of the reaction (steps ⑤ to ⑦), catalyzed in a second active site, the CO₂ reacts with pyruvate to form oxaloacetate. The CO₂ is released in the second active site (step ⑤). Pyruvate is converted to its enolate form in step ⑥, transferring a proton to biotin. The enolate then attacks the CO₂ to generate oxaloacetate in the final step of the reaction (step ⑦).

then the carboxyl group is transferred to pyruvate to form oxaloacetate. These two steps occur at separate active sites; the long flexible arm of biotin transfers activated carboxyl groups from the first active site to the second, functioning much like the long lipoyllysine arm of E_2 in the PDH complex (Fig. 16–6) and the long arm of the CoA-like moiety in the acyl carrier protein involved in fatty acid synthesis (see Fig. 21–5); these are compared in Figure 16–17. Lipoate, biotin, and pantothenate all enter cells on the same transporter; all become covalently attached to proteins by similar reactions; and all provide a flexible tether that allows bound reaction intermediates to move from one active site to another in an enzyme complex, without dissociating from it—all, that is, participate in substrate channeling.

Biotin is a vitamin required in the human diet; it is abundant in many foods and is synthesized by intestinal bacteria. Biotin deficiency is rare, but can sometimes be caused by a diet rich in raw eggs. Egg whites contain a large amount of the protein **avidin** (M_r 70,000), which binds very tightly to biotin and prevents its absorption in the intestine. The avidin of egg whites may be a defense mechanism for the potential chick embryo, inhibiting the growth of bacteria. When eggs are cooked, avidin is denatured (and thereby inactivated) along with all other egg white proteins. Purified avidin is a useful reagent in biochemistry and cell biology. A protein that contains covalently bound biotin (derived experimentally or produced *in vivo*) can be recovered by affinity

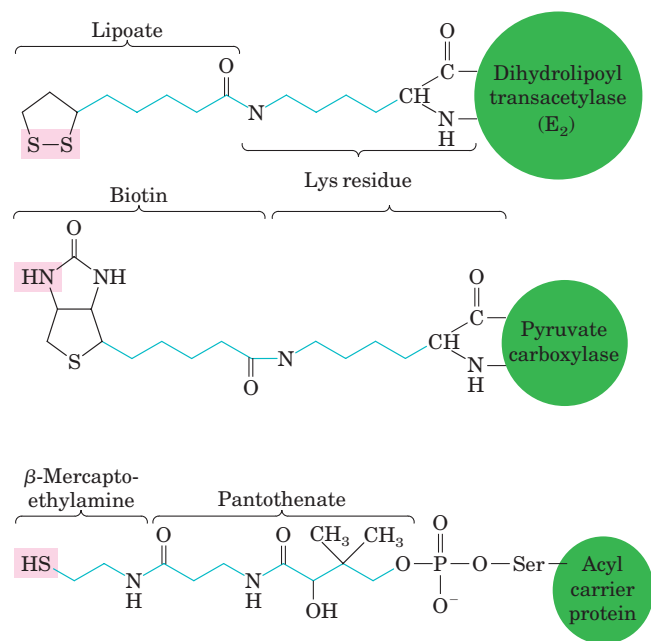


FIGURE 16–17 Biological tethers. The cofactors lipoate, biotin, and the combination of β -mercaptoethylamine and pantothenate form long, flexible arms in the enzymes to which they are covalently bound, acting as tethers that move intermediates from one active site to the next. The group shaded pink is in each case the point of attachment of the activated intermediate to the tether.

chromatography (see Fig. 3–18c) based on biotin's strong affinity for avidin. The protein is then eluted from the column with an excess of free biotin. The very high affinity of biotin for avidin is also used in the laboratory in the form of a molecular glue that can hold two structures together (see Fig. 19–25).

SUMMARY 16.2 Reactions of the Citric Acid Cycle

- The citric acid cycle (Krebs cycle, TCA cycle) is a nearly universal central catabolic pathway in which compounds derived from the breakdown of carbohydrates, fats, and proteins are oxidized to CO_2 , with most of the energy of oxidation temporarily held in the electron carriers FADH_2 and NADH . During aerobic metabolism, these electrons are transferred to O_2 and the energy of electron flow is trapped as ATP.
- Acetyl-CoA enters the citric acid cycle (in the mitochondria of eukaryotes, the cytosol of prokaryotes) as citrate synthase catalyzes its condensation with oxaloacetate to form citrate.
- In seven sequential reactions, including two decarboxylations, the citric acid cycle converts citrate to oxaloacetate and releases two CO_2 . The pathway is cyclic in that the intermediates of the cycle are not used up; for each oxaloacetate consumed in the path, one is produced.
- For each acetyl-CoA oxidized by the citric acid cycle, the energy gain consists of three molecules of NADH , one FADH_2 , and one nucleoside triphosphate (either ATP or GTP).
- Besides acetyl-CoA, any compound that gives rise to a four- or five-carbon intermediate of the citric acid cycle—for example, the breakdown products of many amino acids—can be oxidized by the cycle.
- The citric acid cycle is amphibolic, serving in both catabolism and anabolism; cycle intermediates can be drawn off and used as the starting material for a variety of biosynthetic products.
- When intermediates are shunted from the citric acid cycle to other pathways, they are replenished by several anaplerotic reactions, which produce four-carbon intermediates by carboxylation of three-carbon compounds; these reactions are catalyzed by pyruvate carboxylase, PEP carboxykinase, PEP carboxylase, and malic enzyme. Enzymes that catalyze carboxylations commonly employ biotin to activate CO_2 and to carry it to acceptors such as pyruvate or phosphoenolpyruvate.

16.3 Regulation of the Citric Acid Cycle

As we have seen in Chapter 15, the regulation of key enzymes in metabolic pathways, by allosteric effectors and by covalent modification, ensures the production of intermediates at the rates required to keep the cell in a stable steady state while avoiding wasteful overproduction. The flow of carbon atoms from pyruvate into and through the citric acid cycle is under tight regulation at two levels: the conversion of pyruvate to acetyl-CoA, the starting material for the cycle (the pyruvate dehydrogenase complex reaction), and the entry of acetyl-CoA into the cycle (the citrate synthase reaction). Acetyl-CoA is also produced by pathways other than the PDH complex reaction—most cells produce acetyl-CoA from the oxidation of fatty acids and certain amino acids—and the availability of intermediates from these other pathways is important in the regulation of pyruvate oxidation and of the citric acid cycle. The cycle is also regulated at the isocitrate dehydrogenase and α -ketoglutarate dehydrogenase reactions.

Production of Acetyl-CoA by the Pyruvate Dehydrogenase Complex Is Regulated by Allosteric and Covalent Mechanisms

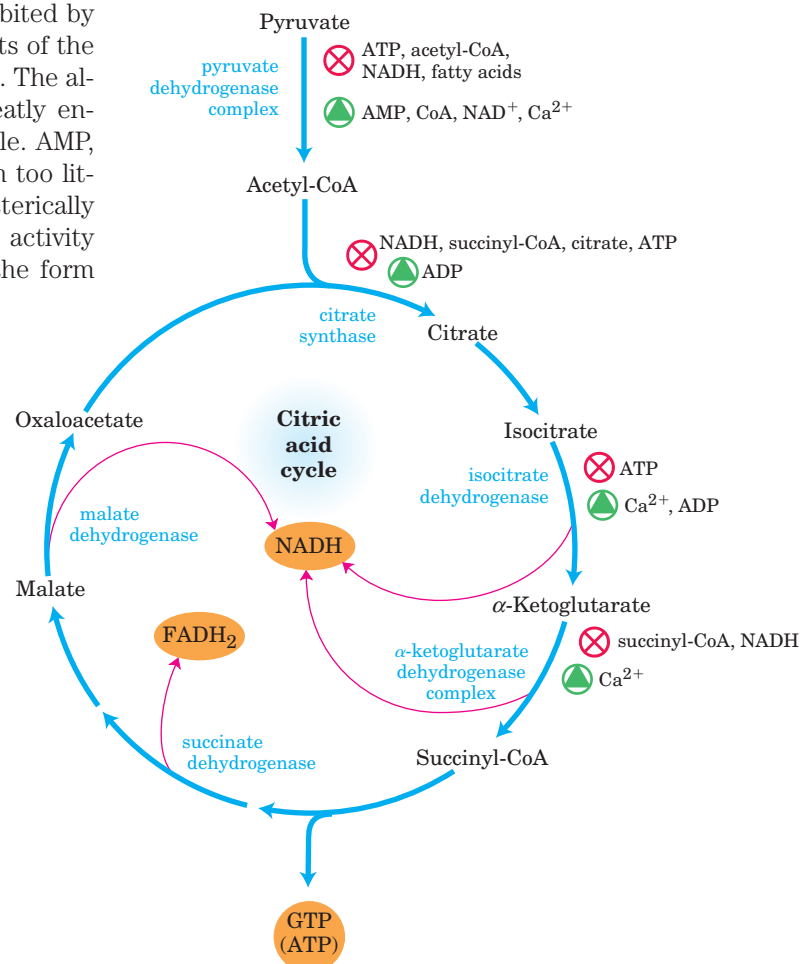
The PDH complex of mammals is strongly inhibited by ATP and by acetyl-CoA and NADH, the products of the reaction catalyzed by the complex (Fig. 16–18). The allosteric inhibition of pyruvate oxidation is greatly enhanced when long-chain fatty acids are available. AMP, CoA, and NAD^+ , all of which accumulate when too little acetate flows into the citric acid cycle, allosterically activate the PDH complex. Thus, this enzyme activity is turned off when ample fuel is available in the form

of fatty acids and acetyl-CoA and when the cell's $[\text{ATP}]/[\text{ADP}]$ and $[\text{NADH}]/[\text{NAD}^+]$ ratios are high, and it is turned on again when energy demands are high and the cell requires greater flux of acetyl-CoA into the citric acid cycle.

In mammals, these allosteric regulatory mechanisms are complemented by a second level of regulation: covalent protein modification. The PDH complex is inhibited by reversible phosphorylation of a specific Ser residue on one of the two subunits of E_1 . As noted earlier, in addition to the enzymes E_1 , E_2 , and E_3 , the mammalian PDH complex contains two regulatory proteins whose sole purpose is to regulate the activity of the complex. A specific protein kinase phosphorylates and thereby inactivates E_1 , and a specific phosphoprotein phosphatase removes the phosphoryl group by hydrolysis and thereby activates E_1 . The kinase is allosterically activated by ATP: when $[\text{ATP}]$ is high (reflecting a sufficient supply of energy), the PDH complex is inactivated by phosphorylation of E_1 . When $[\text{ATP}]$ declines, kinase activity decreases and phosphatase action removes the phosphoryl groups from E_1 , activating the complex.

The PDH complex of plants, located in the mitochondrial matrix and in plastids, is inhibited by its products, NADH and acetyl-CoA. The plant mitochondrial

FIGURE 16–18 Regulation of metabolite flow from the PDH complex through the citric acid cycle. The PDH complex is allosterically inhibited when $[\text{ATP}]/[\text{ADP}]$, $[\text{NADH}]/[\text{NAD}^+]$, and $[\text{acetyl-CoA}]/[\text{CoA}]$ ratios are high, indicating an energy-sufficient metabolic state. When these ratios decrease, allosteric activation of pyruvate oxidation results. The rate of flow through the citric acid cycle can be limited by the availability of the citrate synthase substrates, oxaloacetate and acetyl-CoA, or of NAD^+ , which is depleted by its conversion to NADH, slowing the three NAD-dependent oxidation steps. Feedback inhibition by succinyl-CoA, citrate, and ATP also slows the cycle by inhibiting early steps. In muscle tissue, Ca^{2+} signals contraction and, as shown here, stimulates energy-yielding metabolism to replace the ATP consumed by contraction.



enzyme is also regulated by reversible phosphorylation; pyruvate inhibits the kinase, thus activating the PDH complex, and NH_4^+ stimulates the kinase, causing inactivation of the complex. The PDH complex of *E. coli* is under allosteric regulation similar to that of the mammalian enzyme, but it does not seem to be regulated by phosphorylation.

The Citric Acid Cycle Is Regulated at Its Three Exergonic Steps

The flow of metabolites through the citric acid cycle is under stringent regulation. Three factors govern the rate of flux through the cycle: substrate availability, inhibition by accumulating products, and allosteric feedback inhibition of the enzymes that catalyze early steps in the cycle.

Each of the three strongly exergonic steps in the cycle—those catalyzed by citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase (Fig. 16–18)—can become the rate-limiting step under some circumstances. The availability of the substrates for citrate synthase (acetyl-CoA and oxaloacetate) varies with the metabolic state of the cell and sometimes limits the rate of citrate formation. NADH, a product of isocitrate and α -ketoglutarate oxidation, accumulates under some conditions, and at high $[\text{NADH}]/[\text{NAD}^+]$ both dehydrogenase reactions are severely inhibited by mass action. Similarly, in the cell, the malate dehydrogenase reaction is essentially at equilibrium (that is, it is substrate-limited, and when $[\text{NADH}]/[\text{NAD}^+]$ is high the concentration of oxaloacetate is low, slowing the first step in the cycle. Product accumulation inhibits all three limiting steps of the cycle: succinyl-CoA inhibits α -ketoglutarate dehydrogenase (and also citrate synthase); citrate blocks citrate synthase; and the end product, ATP, inhibits both citrate synthase and isocitrate dehydrogenase. The inhibition of citrate synthase by ATP is relieved by ADP, an allosteric activator of this enzyme. In vertebrate muscle, Ca^{2+} , the signal for contraction and for a concomitant increase in demand for ATP, activates both isocitrate dehydrogenase and α -ketoglutarate dehydrogenase, as well as the PDH complex. In short, the concentrations of substrates and intermediates in the citric acid cycle set the flux through this pathway at a rate that provides optimal concentrations of ATP and NADH.

Under normal conditions, the rates of glycolysis and of the citric acid cycle are integrated so that only as much glucose is metabolized to pyruvate as is needed to supply the citric acid cycle with its fuel, the acetyl groups of acetyl-CoA. Pyruvate, lactate, and acetyl-CoA are normally maintained at steady-state concentrations. The rate of glycolysis is matched to the rate of the citric acid cycle not only through its inhibition by high levels of ATP and NADH, which are common to both the

glycolytic and respiratory stages of glucose oxidation, but also by the concentration of citrate. Citrate, the product of the first step of the citric acid cycle, is an important allosteric inhibitor of phosphofructokinase-1 in the glycolytic pathway (see Fig. 15–18).

Substrate Channeling through Multienzyme Complexes May Occur in the Citric Acid Cycle

Although the enzymes of the citric acid cycle are usually described as soluble components of the mitochondrial matrix (except for succinate dehydrogenase, which is membrane-bound), growing evidence suggests that within the mitochondrion these enzymes exist as multi-enzyme complexes. The classic approach of enzymology—purification of individual proteins from extracts of broken cells—was applied with great success to the citric acid cycle enzymes. However, the first casualty of cell breakage is higher-level organization within the cell—the noncovalent, reversible interaction of one protein with another, or of an enzyme with some structural component such as a membrane, microtubule, or microfilament. When cells are broken open, their contents, including enzymes, are diluted 100- or 1,000-fold (Fig. 16–19).

Several types of evidence suggest that, in cells, multi-enzyme complexes ensure efficient passage of the product of one enzyme reaction to the next enzyme in the pathway. Such complexes are called **metabolons**. Certain enzymes of the citric acid cycle have been isolated together as supramolecular aggregates, or have been found associated with the inner mitochondrial membrane, or have been shown to diffuse in the mitochondrial matrix more slowly than expected for the individual protein in solution. There is strong evidence for substrate channeling through multienzyme complexes in

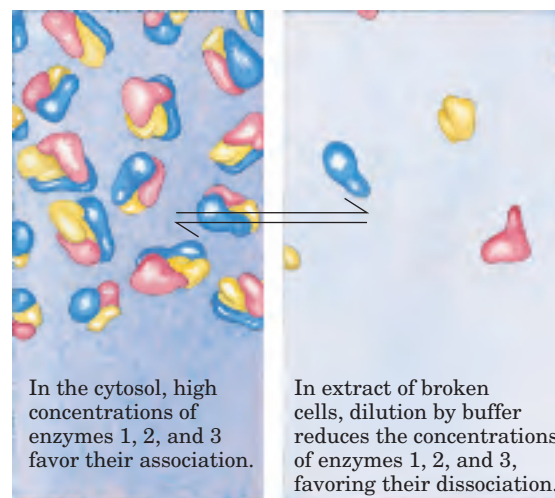


FIGURE 16-19 Dilution of a solution containing a noncovalent protein complex—such as one consisting of three enzymes—favors dissociation of the complex into its constituents.

other metabolic pathways, and many enzymes thought of as “soluble” probably function in the cell as highly organized complexes that channel intermediates. We will encounter other examples of channeling when we discuss the biosynthesis of amino acids and nucleotides in Chapter 22.

SUMMARY 16.3 Regulation of the Citric Acid Cycle

- The overall rate of the citric acid cycle is controlled by the rate of conversion of pyruvate to acetyl-CoA and by the flux through citrate synthase, isocitrate dehydrogenase, and α -ketoglutarate dehydrogenase. These fluxes are largely determined by the concentrations of substrates and products: the end products ATP and NADH are inhibitory, and the substrates NAD⁺ and ADP are stimulatory.
- The production of acetyl-CoA for the citric acid cycle by the PDH complex is inhibited allosterically by metabolites that signal a sufficiency of metabolic energy (ATP, acetyl-CoA, NADH, and fatty acids) and stimulated by metabolites that indicate a reduced energy supply (AMP, NAD⁺, CoA).

16.4 The Glyoxylate Cycle

Vertebrates cannot convert fatty acids, or the acetate derived from them, to carbohydrates. Conversion of phosphoenolpyruvate to pyruvate (p. 532) and of pyruvate to acetyl-CoA (Fig. 16–2) are so exergonic as to be essentially irreversible. If a cell cannot convert acetate into phosphoenolpyruvate, acetate cannot serve as the starting material for the gluconeogenic pathway, which leads from phosphoenolpyruvate to glucose (see Fig. 15–15). Without this capacity, then, a cell or organism is unable to convert fuels or metabolites that are degraded to acetate (fatty acids and certain amino acids) into carbohydrates.

As noted in the discussion of anaplerotic reactions (Table 16–2), phosphoenolpyruvate can be synthesized from oxaloacetate in the reversible reaction catalyzed by PEP carboxykinase:

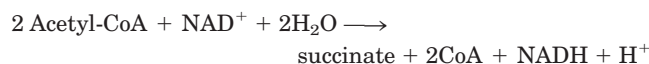


Because the carbon atoms of acetate molecules that enter the citric acid cycle appear eight steps later in oxaloacetate, it might seem that this pathway could generate oxaloacetate from acetate and thus generate phosphoenolpyruvate for gluconeogenesis. However, as an examination of the stoichiometry of the citric acid cycle shows, there is no *net* conversion of acetate to ox-

aloacetate; in vertebrates, for every two carbons that enter the cycle as acetyl-CoA, two leave as CO₂. In many organisms other than vertebrates, the glyoxylate cycle serves as a mechanism for converting acetate to carbohydrate.

The Glyoxylate Cycle Produces Four-Carbon Compounds from Acetate

In plants, certain invertebrates, and some microorganisms (including *E. coli* and yeast) acetate can serve both as an energy-rich fuel and as a source of phosphoenolpyruvate for carbohydrate synthesis. In these organisms, enzymes of the **glyoxylate cycle** catalyze the net conversion of acetate to succinate or other four-carbon intermediates of the citric acid cycle:



In the glyoxylate cycle, acetyl-CoA condenses with oxaloacetate to form citrate, and citrate is converted to isocitrate, exactly as in the citric acid cycle. The next step, however, is not the breakdown of isocitrate by isocitrate dehydrogenase but the cleavage of isocitrate by **isocitrate lyase**, forming succinate and **glyoxylate**. The glyoxylate then condenses with a second molecule of acetyl-CoA to yield malate, in a reaction catalyzed by **malate synthase**. The malate is subsequently oxidized to oxaloacetate, which can condense with another molecule of acetyl-CoA to start another turn of the cycle (Fig. 16–20). Each turn of the glyoxylate cycle consumes two molecules of acetyl-CoA and produces one molecule of succinate, which is then available for biosynthetic purposes. The succinate may be converted through fumarate and malate into oxaloacetate, which can then be converted to phosphoenolpyruvate by PEP carboxykinase, and thus to glucose by gluconeogenesis. Vertebrates do not have the enzymes specific to the glyoxylate cycle (isocitrate lyase and malate synthase) and therefore cannot bring about the net synthesis of glucose from lipids.

In plants, the enzymes of the glyoxylate cycle are sequestered in membrane-bounded organelles called glyoxysomes, which are specialized peroxisomes (Fig. 16–21). Those enzymes common to the citric acid and glyoxylate cycles have two isozymes, one specific to mitochondria, the other to glyoxysomes. Glyoxysomes are not present in all plant tissues at all times. They develop in lipid-rich seeds during germination, before the developing plant acquires the ability to make glucose by photosynthesis. In addition to glyoxylate cycle enzymes, glyoxysomes contain all the enzymes needed for the degradation of the fatty acids stored in seed oils (see Fig. 17–13). Acetyl-CoA formed from lipid breakdown is converted to succinate via the glyoxylate cycle, and the succinate is exported to mitochondria, where citric

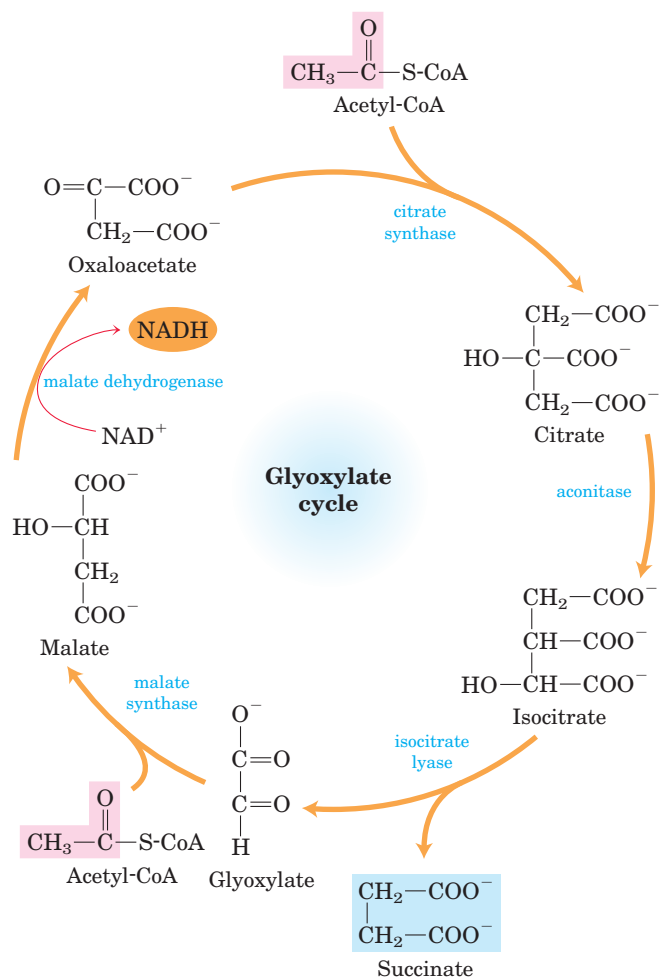


FIGURE 16-20 Glyoxylate cycle. The citrate synthase, aconitase, and malate dehydrogenase of the glyoxylate cycle are isozymes of the citric acid cycle enzymes; isocitrate lyase and malate synthase are unique to the glyoxylate cycle. Notice that two acetyl groups (pink) enter the cycle and four carbons leave as succinate (blue). The glyoxylate cycle was elucidated by Hans Kornberg and Neil Madsen in the laboratory of Hans Krebs.

acid cycle enzymes transform it to malate. A cytosolic isozyme of malate dehydrogenase oxidizes malate to oxaloacetate, a precursor for gluconeogenesis. Germinating seeds can therefore convert the carbon of stored lipids into glucose.

The Citric Acid and Glyoxylate Cycles Are Coordinately Regulated

In germinating seeds, the enzymatic transformations of dicarboxylic and tricarboxylic acids occur in three intracellular compartments: mitochondria, glyoxysomes, and the cytosol. There is a continuous interchange of metabolites among these compartments (Fig. 16-22).

The carbon skeleton of oxaloacetate from the citric acid cycle (in the mitochondrion) is carried to the glyoxysome in the form of aspartate. Aspartate is converted

to oxaloacetate, which condenses with acetyl-CoA derived from fatty acid breakdown. The citrate thus formed is converted to isocitrate by aconitase, then split into glyoxylate and succinate by isocitrate lyase. The succinate returns to the mitochondrion, where it reenters the citric acid cycle and is transformed into malate, which enters the cytosol and is oxidized (by cytosolic malate dehydrogenase) to oxaloacetate. Oxaloacetate is converted via gluconeogenesis into hexoses and sucrose, which can be transported to the growing roots and shoot. Four distinct pathways participate in these conversions: fatty acid breakdown to acetyl-CoA (in glyoxysomes), the glyoxylate cycle (in glyoxysomes), the citric acid cycle (in mitochondria), and gluconeogenesis (in the cytosol).

The sharing of common intermediates requires that these pathways be coordinately regulated. Isocitrate is a crucial intermediate, at the branch point between the glyoxylate and citric acid cycles (Fig. 16-23). Isocitrate dehydrogenase is regulated by covalent modification: a specific protein kinase phosphorylates and thereby inactivates the dehydrogenase. This inactivation shunts isocitrate to the glyoxylate cycle, where it begins the synthetic route toward glucose. A phosphoprotein phosphatase removes the phosphoryl group from isocitrate dehydrogenase, reactivating the enzyme and sending more isocitrate through the energy-yielding citric acid cycle. The regulatory protein kinase and phosphoprotein phosphatase are separate enzymatic activities of a single polypeptide.

Some bacteria, including *E. coli*, have the full complement of enzymes for the glyoxylate and citric acid cycles in the cytosol and can therefore grow on acetate as their sole source of carbon and energy. The phosphoprotein phosphatase that activates isocitrate dehydrogenase is stimulated by intermediates of the citric acid cycle and glycolysis and by indicators of reduced cellular energy supply (Fig. 16-23). The same metabolites *inhibit* the protein kinase activity of the bifunctional polypeptide. Thus, the accumulation of intermediates of

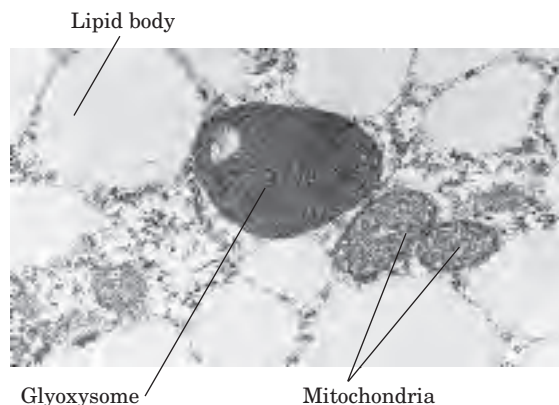


FIGURE 16-21 Electron micrograph of a germinating cucumber seed, showing a glyoxysome, mitochondria, and surrounding lipid bodies.

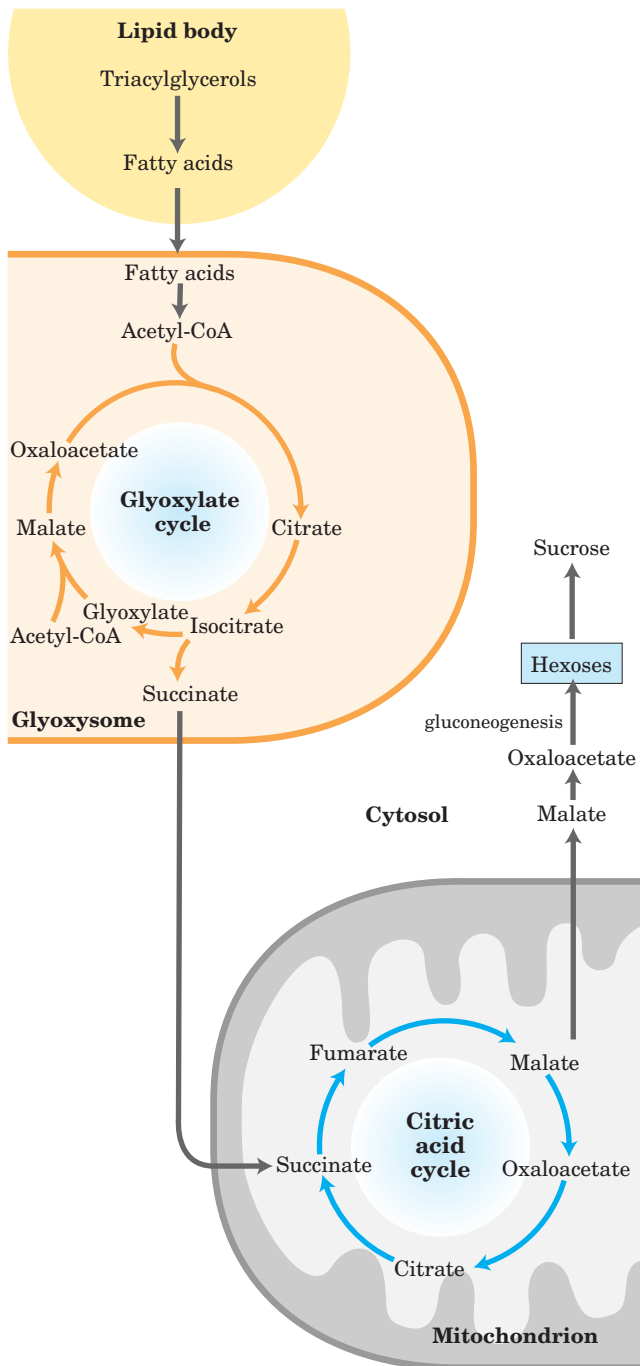


FIGURE 16-22 Relationship between the glyoxylate and citric acid cycles. The reactions of the glyoxylate cycle (in glyoxysomes) proceed simultaneously with, and mesh with, those of the citric acid cycle (in mitochondria), as intermediates pass between these compartments. The conversion of succinate to oxaloacetate is catalyzed by citric acid cycle enzymes. The oxidation of fatty acids to acetyl-CoA is described in Chapter 17; the synthesis of hexoses from oxaloacetate is described in Chapter 20.

the central energy-yielding pathways—indicating energy depletion—results in the activation of isocitrate dehydrogenase. When the concentration of these regulators falls, signaling a sufficient flux through the energy-yielding citric acid cycle, isocitrate dehydrogenase is inactivated by the protein kinase.

The same intermediates of glycolysis and the citric acid cycle that activate isocitrate dehydrogenase are allosteric inhibitors of isocitrate lyase. When energy-yielding metabolism is sufficiently fast to keep the concentrations of glycolytic and citric acid cycle intermediates low, isocitrate dehydrogenase is inactivated, the inhibition of isocitrate lyase is relieved, and isocitrate flows into the glyoxylate pathway, to be used in the biosynthesis of carbohydrates, amino acids, and other cellular components.

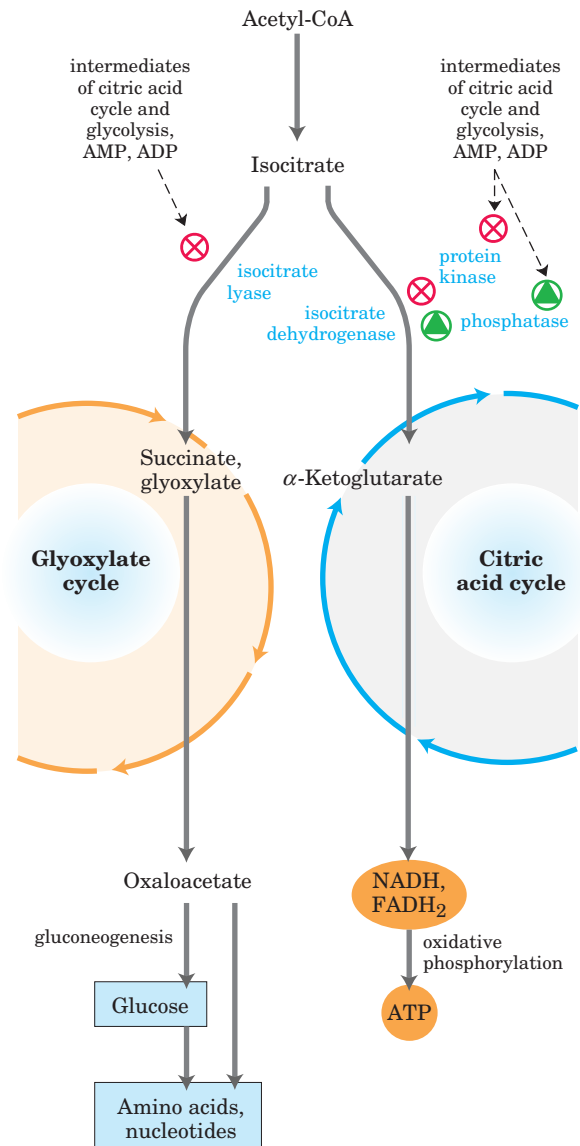


FIGURE 16-23 Coordinated regulation of glyoxylate and citric acid cycles. Regulation of isocitrate dehydrogenase activity determines the partitioning of isocitrate between the glyoxylate and citric acid cycles. When the enzyme is inactivated by phosphorylation (by a specific protein kinase), isocitrate is directed into biosynthetic reactions via the glyoxylate cycle. When the enzyme is activated by dephosphorylation (by a specific phosphatase), isocitrate enters the citric acid cycle and ATP is produced.

SUMMARY 16.4 The Glyoxylate Cycle

- The glyoxylate cycle is active in the germinating seeds of some plants and in certain microorganisms that can live on acetate as the sole carbon source. In plants, the pathway takes place in glyoxysomes in seedlings. It involves several citric acid cycle enzymes and two additional enzymes: isocitrate lyase and malate synthase.
- In the glyoxylate cycle, the bypassing of the two decarboxylation steps of the citric acid

cycle makes possible the *net* formation of succinate, oxaloacetate, and other cycle intermediates from acetyl-CoA. Oxaloacetate thus formed can be used to synthesize glucose via gluconeogenesis.

- The partitioning of isocitrate between the citric acid cycle and the glyoxylate cycle is controlled at the level of isocitrate dehydrogenase, which is regulated by reversible phosphorylation.
- Vertebrates lack the glyoxylate cycle and cannot synthesize glucose from acetate or the fatty acids that give rise to acetyl-CoA.

Key Terms

Terms in bold are defined in the glossary.

respiration 601	substrate channeling 605	phosphorylases 613
cellular respiration 601	iron-sulfur center 609	phosphatases 613
citric acid cycle 601	α -ketoglutarate dehydrogenase complex 610	prochiral molecule 615
tricarboxylic acid (TCA) cycle 601	nucleoside diphosphate kinase 612	amphibolic pathway 616
Krebs cycle 601	synthases 613	anaplerotic reaction 616
pyruvate dehydrogenase (PDH) complex 602	synthetases 613	biotin 618
oxidative decarboxylation 602	ligases 613	avidin 620
thioester 603	lyases 613	metabolon 622
lipoate 603	kinases 613	glyoxylate cycle 623

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Problems

1. Balance Sheet for the Citric Acid Cycle The citric acid cycle has eight enzymes: citrate synthase, aconitase, isocitrate dehydrogenase, α -ketoglutarate dehydrogenase, succinyl-CoA synthetase, succinate dehydrogenase, fumarase, and malate dehydrogenase.

(a) Write a balanced equation for the reaction catalyzed by each enzyme.

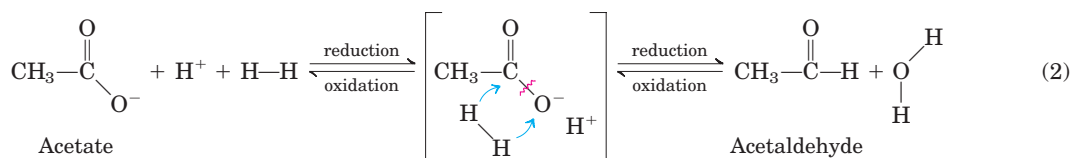
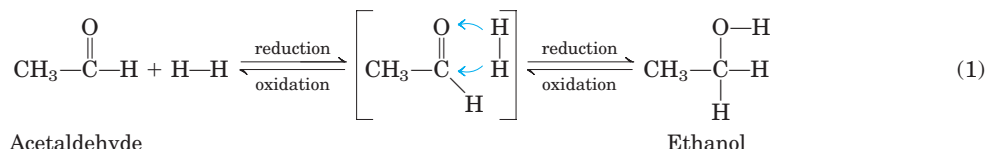
(b) Name the cofactor(s) required by each enzyme reaction.

(c) For each enzyme determine which of the following describes the type of reaction(s) catalyzed: condensation (carbon-carbon bond formation); dehydration (loss of water); hydration (addition of water); decarboxylation (loss of CO₂); oxidation-reduction; substrate-level phosphorylation; isomerization.

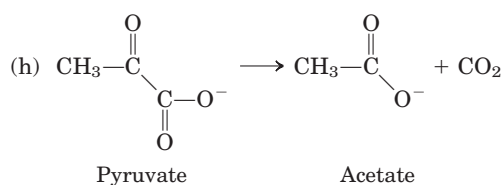
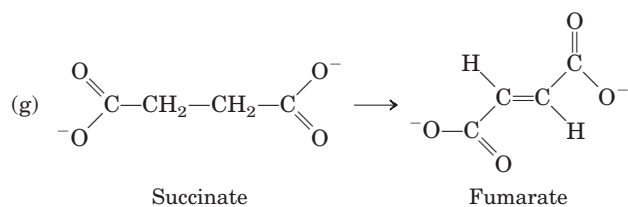
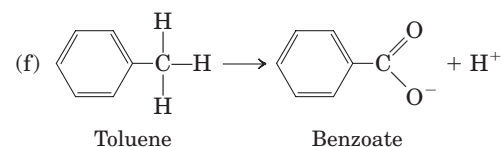
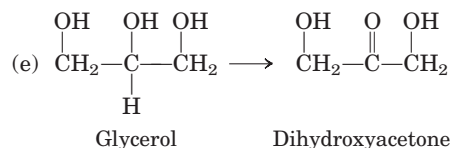
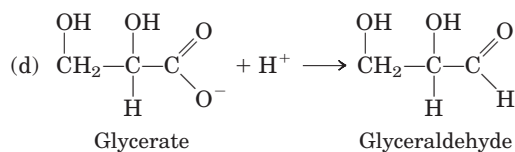
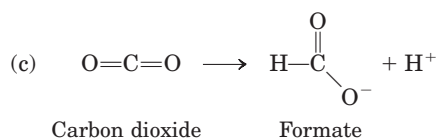
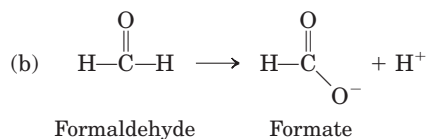
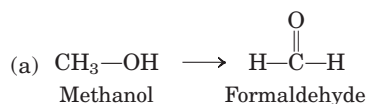
(d) Write a balanced net equation for the catabolism of acetyl-CoA to CO₂.

2. Recognizing Oxidation and Reduction Reactions

One biochemical strategy of many living organisms is the step-wise oxidation of organic compounds to CO₂ and H₂O and the conservation of a major part of the energy thus produced in the form of ATP. It is important to be able to recognize oxidation-reduction processes in metabolism. Reduction of an organic molecule results from the hydrogenation of a double bond (Eqn 1, below) or of a single bond with accompanying cleavage (Eqn 2). Conversely, oxidation results from dehydrogenation. In biochemical redox reactions, the coenzymes NAD and FAD dehydrogenate/hydrogenate organic molecules in the presence of the proper enzymes.



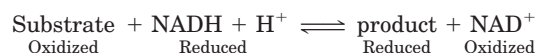
For each of the metabolic transformations in (a) through (h), determine whether oxidation or reduction has occurred. Balance each transformation by inserting H—H and, where necessary, H₂O.



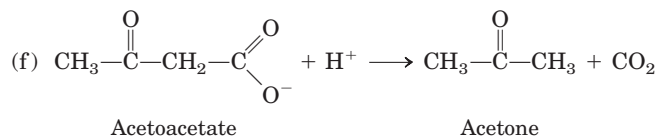
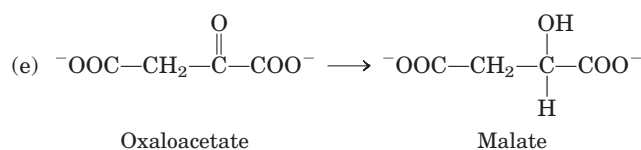
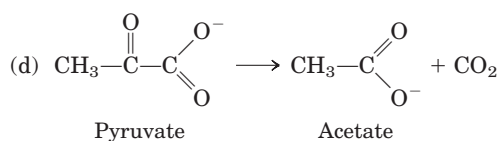
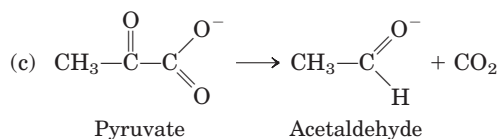
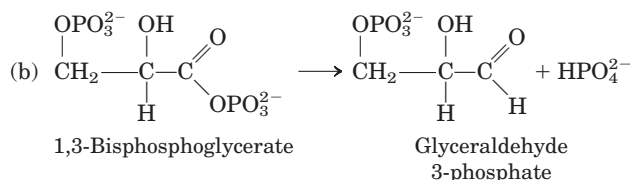
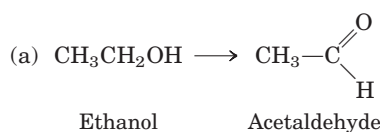
3. Relationship between Energy Release and the Oxidation State of Carbon A eukaryotic cell can use glucose (C₆H₁₂O₆) and hexanoic acid (C₆H₁₄O₂) as fuels for cellular respiration. On the basis of their structural formulas, which substance releases more energy per gram on complete combustion to CO₂ and H₂O?

4. Nicotinamide Coenzymes as Reversible Redox Carriers The nicotinamide coenzymes (see Fig. 13-15) can undergo reversible oxidation-reduction reactions with specific

substrates in the presence of the appropriate dehydrogenase. In these reactions, NADH + H⁺ serves as the hydrogen source, as described in Problem 2. Whenever the coenzyme is oxidized, a substrate must be simultaneously reduced:

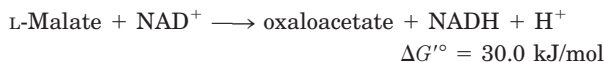


For each of the reactions in (a) through (f), determine whether the substrate has been oxidized or reduced or is unchanged in oxidation state (see Problem 2). If a redox change has occurred, balance the reaction with the necessary amount of NAD⁺, NADH, H⁺, and H₂O. The objective is to recognize when a redox coenzyme is necessary in a metabolic reaction.



5. Stimulation of Oxygen Consumption by Oxaloacetate and Malate In the early 1930s, Albert Szent Györgyi reported the interesting observation that the addition of small amounts of oxaloacetate or malate to suspensions of minced pigeon-breast muscle stimulated the oxygen consumption of the preparation. Surprisingly, the amount of oxygen consumed was about seven times more than the amount necessary for complete oxidation (to CO₂ and H₂O) of the added oxaloacetate or malate. Why did the addition of oxaloacetate or malate stimulate oxygen consumption? Why was the amount of oxygen consumed so much greater than the amount necessary to completely oxidize the added oxaloacetate or malate?

6. Formation of Oxaloacetate in a Mitochondrion In the last reaction of the citric acid cycle, malate is dehydrogenated to regenerate the oxaloacetate necessary for the entry of acetyl-CoA into the cycle:



(a) Calculate the equilibrium constant for this reaction at 25 °C.

(b) Because $\Delta G'^{\circ}$ assumes a standard pH of 7, the equilibrium constant calculated in (a) corresponds to

$$K'_{\text{eq}} = \frac{[\text{oxaloacetate}][\text{NADH}]}{[\text{L-malate}][\text{NAD}^+]}$$

The measured concentration of L-malate in rat liver mitochondria is about 0.20 mM when $[\text{NAD}^+]/[\text{NADH}]$ is 10. Calculate the concentration of oxaloacetate at pH 7 in these mitochondria.

(c) To appreciate the magnitude of the mitochondrial oxaloacetate concentration, calculate the number of oxaloacetate molecules in a single rat liver mitochondrion. Assume the mitochondrion is a sphere of diameter 2.0 μm .

7. Energy Yield from the Citric Acid Cycle The reaction catalyzed by succinyl-CoA synthetase produces the high-energy compound GTP. How is the free energy contained in GTP incorporated into the cellular ATP pool?

8. Respiration Studies in Isolated Mitochondria Cellular respiration can be studied in isolated mitochondria by measuring oxygen consumption under different conditions. If 0.01 M sodium malonate is added to actively respiring mitochondria that are using pyruvate as fuel source, respiration soon stops and a metabolic intermediate accumulates.

- What is the structure of this intermediate?
- Explain why it accumulates.
- Explain why oxygen consumption stops.
- Aside from removal of the malonate, how can this inhibition of respiration be overcome? Explain.

9. Labeling Studies in Isolated Mitochondria The metabolic pathways of organic compounds have often been delineated by using a radioactively labeled substrate and following the fate of the label.

(a) How can you determine whether glucose added to a suspension of isolated mitochondria is metabolized to CO_2 and H_2O ?

(b) Suppose you add a brief pulse of $[3\text{-}^{14}\text{C}]$ pyruvate (labeled in the methyl position) to the mitochondria. After one turn of the citric acid cycle, what is the location of the ^{14}C in the oxaloacetate? Explain by tracing the ^{14}C label through the pathway. How many turns of the cycle are required to release all the $[3\text{-}^{14}\text{C}]$ pyruvate as CO_2 ?

10. $[1\text{-}^{14}\text{C}]$ Glucose Catabolism An actively respiring bacterial culture is briefly incubated with $[1\text{-}^{14}\text{C}]$ glucose, and the glycolytic and citric acid cycle intermediates are isolated. Where is the ^{14}C in each of the intermediates listed below? Consider only the initial incorporation of ^{14}C , in the first pass of labeled glucose through the pathways.

- Fructose 1,6-bisphosphate
- Glyceraldehyde 3-phosphate

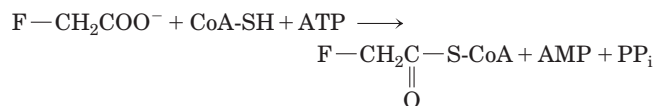
- Phosphoenolpyruvate
- Acetyl-CoA
- Citrate
- α -Ketoglutarate
- Oxaloacetate



11. Role of the Vitamin Thiamine People with beriberi, a disease caused by thiamine deficiency, have elevated levels of blood pyruvate and α -ketoglutarate, especially after consuming a meal rich in glucose. How are these effects related to a deficiency of thiamine?

12. Synthesis of Oxaloacetate by the Citric Acid Cycle Oxaloacetate is formed in the last step of the citric acid cycle by the NAD^+ -dependent oxidation of L-malate. Can a net synthesis of oxaloacetate from acetyl-CoA occur using only the enzymes and cofactors of the citric acid cycle, without depleting the intermediates of the cycle? Explain. How is oxaloacetate that is lost from the cycle (to biosynthetic reactions) replenished?

13. Mode of Action of the Rodenticide Fluoroacetate Fluoroacetate, prepared commercially for rodent control, is also produced by a South African plant. After entering a cell, fluoroacetate is converted to fluoroacetyl-CoA in a reaction catalyzed by the enzyme acetate thiokinase:



The toxic effect of fluoroacetate was studied in an experiment using intact isolated rat heart. After the heart was perfused with 0.22 mM fluoroacetate, the measured rate of glucose uptake and glycolysis decreased, and glucose 6-phosphate and fructose 6-phosphate accumulated. Examination of the citric acid cycle intermediates revealed that their concentrations were below normal, except for citrate, with a concentration 10 times higher than normal.

(a) Where did the block in the citric acid cycle occur? What caused citrate to accumulate and the other cycle intermediates to be depleted?

(b) Fluoroacetyl-CoA is enzymatically transformed in the citric acid cycle. What is the structure of the end product of fluoroacetate metabolism? Why does it block the citric acid cycle? How might the inhibition be overcome?

(c) In the heart perfusion experiments, why did glucose uptake and glycolysis decrease? Why did hexose monophosphates accumulate?

(d) Why is fluoroacetate poisoning fatal?

14. Synthesis of L-Malate in Wine Making The tartness of some wines is due to high concentrations of L-malate. Write a sequence of reactions showing how yeast cells synthesize L-malate from glucose under anaerobic conditions in the presence of dissolved CO_2 (HCO_3^-). Note that the overall reaction for this fermentation cannot involve the consumption of nicotinamide coenzymes or citric acid cycle intermediates.

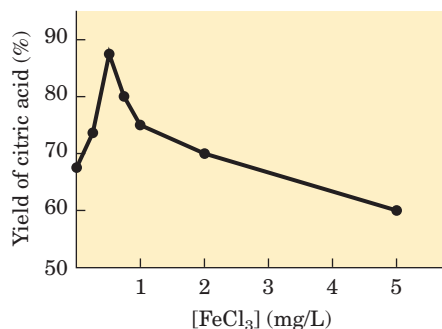
15. Net Synthesis of α -Ketoglutarate α -Ketoglutarate plays a central role in the biosynthesis of several amino acids. Write a sequence of enzymatic reactions that could result in

the net synthesis of α -ketoglutarate from pyruvate. Your proposed sequence must not involve the net consumption of other citric acid cycle intermediates. Write an equation for the overall reaction and identify the source of each reactant.

16. Regulation of the Pyruvate Dehydrogenase Complex In animal tissues, the rate of conversion of pyruvate to acetyl-CoA is regulated by the ratio of active, phosphorylated to inactive, unphosphorylated PDH complex. Determine what happens to the rate of this reaction when a preparation of rabbit muscle mitochondria containing the PDH complex is treated with (a) pyruvate dehydrogenase kinase, ATP, and NADH; (b) pyruvate dehydrogenase phosphatase and Ca^{2+} ; (c) malonate.

17. Commercial Synthesis of Citric Acid Citric acid is used as a flavoring agent in soft drinks, fruit juices, and many other foods. Worldwide, the market for citric acid is valued at hundreds of millions of dollars per year. Commercial production uses the mold *Aspergillus niger*, which metabolizes sucrose under carefully controlled conditions.

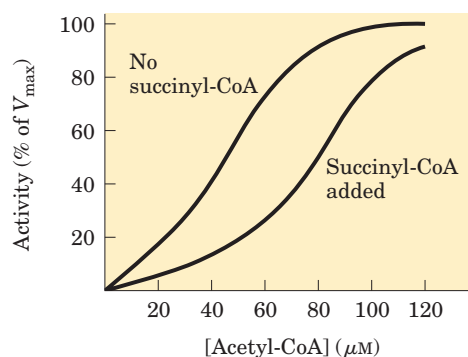
(a) The yield of citric acid is strongly dependent on the concentration of FeCl_3 in the culture medium, as indicated in the graph. Why does the yield decrease when the concentration of Fe^{3+} is above or below the optimal value of 0.5 mg/L?



(b) Write the sequence of reactions by which *A. niger* synthesizes citric acid from sucrose. Write an equation for the overall reaction.

(c) Does the commercial process require the culture medium to be aerated—that is, is this a fermentation or an aerobic process? Explain.

18. Regulation of Citrate Synthase In the presence of saturating amounts of oxaloacetate, the activity of citrate synthase from pig heart tissue shows a sigmoid dependence on the concentration of acetyl-CoA, as shown in the graph. When succinyl-CoA is added, the curve shifts to the right and the sigmoid dependence is more pronounced.



On the basis of these observations, suggest how succinyl-CoA regulates the activity of citrate synthase. (Hint: See Fig. 6–29.) Why is succinyl-CoA an appropriate signal for regulation of the citric acid cycle? How does the regulation of citrate synthase control the rate of cellular respiration in pig heart tissue?

19. Regulation of Pyruvate Carboxylase The carboxylation of pyruvate by pyruvate carboxylase occurs at a very low rate unless acetyl-CoA, a positive allosteric modulator, is present. If you have just eaten a meal rich in fatty acids (triacylglycerols) but low in carbohydrates (glucose), how does this regulatory property shut down the oxidation of glucose to CO_2 and H_2O but increase the oxidation of acetyl-CoA derived from fatty acids?

20. Relationship between Respiration and the Citric Acid Cycle Although oxygen does not participate directly in the citric acid cycle, the cycle operates only when O_2 is present. Why?

21. Thermodynamics of Citrate Synthase Reaction in Cells Citrate is formed by the condensation of acetyl-CoA with oxaloacetate, catalyzed by citrate synthase:



In rat heart mitochondria at pH 7.0 and 25 °C, the concentrations of reactants and products are: oxaloacetate, 1 μM ; acetyl-CoA, 1 μM ; citrate, 220 μM ; and CoA, 65 μM . The standard free-energy change for the citrate synthase reaction is -32.2 kJ/mol . What is the direction of metabolite flow through the citrate synthase reaction in rat heart cells? Explain.

22. Reactions of the Pyruvate Dehydrogenase Complex Two of the steps in the oxidative decarboxylation of pyruvate (steps ④ and ⑤ in Fig. 16–6) do not involve any of the three carbons of pyruvate yet are essential to the operation of the PDH complex. Explain.



FATTY ACID CATABOLISM

17.1 Digestion, Mobilization, and Transport of Fats 632

17.2 Oxidation of Fatty Acids 637

17.3 Ketone Bodies 650

Jack Sprat could eat no fat,
His wife could eat no lean,
And so between them both you see,
They licked the platter clean.

—John Clarke, *Paroemiologia Anglo-Latina*
(Proverbs English and Latin), 1639

The oxidation of long-chain fatty acids to acetyl-CoA is a central energy-yielding pathway in many organisms and tissues. In mammalian heart and liver, for example, it provides as much as 80% of the energetic needs under all physiological circumstances. The electrons removed from fatty acids during oxidation pass through the respiratory chain, driving ATP synthesis; the acetyl-CoA produced from the fatty acids may be completely oxidized to CO₂ in the citric acid cycle, resulting in further energy conservation. In some species and in some tissues, the acetyl-CoA has alternative fates. In liver, acetyl-CoA may be converted to ketone bodies—water-soluble fuels exported to the brain and other tissues when glucose is not available. In higher plants, acetyl-CoA serves primarily as a biosynthetic precursor, only secondarily as fuel. Although the biological role of fatty acid oxidation differs from organism to organism, the mechanism is essentially the same. The repetitive four-step process, called **β oxidation**, by which fatty acids are converted into acetyl-CoA is the main topic of this chapter.

In Chapter 10 we described the properties of triacylglycerols (also called triglycerides or neutral fats)

that make them especially suitable as storage fuels. The long alkyl chains of their constituent fatty acids are essentially hydrocarbons, highly reduced structures with an energy of complete oxidation (~38 kJ/g) more than twice that for the same weight of carbohydrate or protein. This advantage is compounded by the extreme insolubility of lipids in water; cellular triacylglycerols aggregate in lipid droplets, which do not raise the osmolarity of the cytosol, and they are unsolvated. (In storage polysaccharides, by contrast, water of solvation can account for two-thirds of the overall weight of the stored molecules.) And because of their relative chemical inertness, triacylglycerols can be stored in large quantity in cells without the risk of undesired chemical reactions with other cellular constituents.

The properties that make triacylglycerols good storage compounds, however, present problems in their role as fuels. Because they are insoluble in water, ingested triacylglycerols must be emulsified before they can be digested by water-soluble enzymes in the intestine, and triacylglycerols absorbed in the intestine or mobilized from storage tissues must be carried in the blood bound to proteins that counteract their insolubility. To overcome the relative stability of the C—C bonds in a fatty acid, the carboxyl group at C-1 is activated by attachment to coenzyme A, which allows stepwise oxidation of the fatty acyl group at the C-3, or β , position—hence the name β oxidation.

We begin this chapter with a brief discussion of the sources of fatty acids and the routes by which they travel to the site of their oxidation, with special emphasis on the process in vertebrates. We then describe the chemical steps of fatty acid oxidation in mitochondria. The complete oxidation of fatty acids to CO₂ and H₂O takes place in three stages: the oxidation of long-chain fatty acids to two-carbon fragments, in the form of acetyl-CoA (β oxidation); the oxidation of acetyl-CoA to CO₂ in the citric acid cycle (Chapter 16); and the transfer of

electrons from reduced electron carriers to the mitochondrial respiratory chain (Chapter 19). In this chapter we focus on the first of these stages. We begin our discussion of β oxidation with the simple case in which a fully saturated fatty acid with an even number of carbon atoms is degraded to acetyl-CoA. We then look briefly at the extra transformations necessary for the degradation of unsaturated fatty acids and fatty acids with an odd number of carbons. Finally, we discuss variations on the β -oxidation theme in specialized organelles—peroxisomes and glyoxysomes—and two less common pathways of fatty acid catabolism, ω and α oxidation. The chapter concludes with a description of an alternative fate for the acetyl-CoA formed by β oxidation in vertebrates: the production of ketone bodies in the liver.

17.1 Digestion, Mobilization, and Transport of Fats

Cells can obtain fatty acid fuels from three sources: fats consumed in the diet, fats stored in cells as lipid droplets, and fats synthesized in one organ for export to another. Some species use all three sources under various circumstances, others use one or two. Vertebrates, for example, obtain fats in the diet, mobilize fats stored in specialized tissue (adipose tissue, consisting of cells called adipocytes), and, in the liver, convert excess dietary carbohydrates to fats for export to other tissues. On average, 40% or more of the daily energy requirement of humans in highly industrialized countries is supplied by dietary triacylglycerols (although most nutritional guidelines recommend no more than 30% of daily caloric intake from fats). Triacylglycerols provide more than half the energy requirements of some organs, particularly the liver, heart, and resting skeletal muscle. Stored triacylglycerols are virtually the sole source of energy in hibernating animals and migrating birds. Protists obtain fats by consuming organisms lower in the food chain, and some also store fats as cytosolic lipid droplets. Vascular plants mobilize fats stored in seeds during germination, but do not otherwise depend on fats for energy.

Dietary Fats Are Absorbed in the Small Intestine

In vertebrates, before ingested triacylglycerols can be absorbed through the intestinal wall they must be converted from insoluble macroscopic fat particles to finely dispersed microscopic micelles. This solubilization is carried out by bile salts, such as taurocholic acid (p. 355), which are synthesized from cholesterol in the liver, stored in the gallbladder, and released into the small intestine after ingestion of a fatty meal. Bile salts are amphipathic compounds that act as biological detergents, converting dietary fats into mixed micelles of bile salts and triacylglycerols (Fig. 17–1, step ①).

Micelle formation enormously increases the fraction of lipid molecules accessible to the action of water-soluble lipases in the intestine, and lipase action converts triacylglycerols to monoacylglycerols (monoglycerides) and diacylglycerols (diglycerides), free fatty acids, and glycerol (step ②). These products of lipase action diffuse into the epithelial cells lining the intestinal surface (the intestinal mucosa) (step ③), where they are reconverted to triacylglycerols and packaged with dietary cholesterol and specific proteins into lipoprotein aggregates called **chylomicrons** (Fig. 17–2; see also Fig. 17–1, step ④).

Apolipoproteins are lipid-binding proteins in the blood, responsible for the transport of triacylglycerols, phospholipids, cholesterol, and cholesteryl esters between organs. Apolipoproteins (“apo” means “detached” or “separate,” designating the protein in its lipid-free form) combine with lipids to form several classes of lipoprotein particles, spherical aggregates with hydrophobic lipids at the core and hydrophilic protein side chains and lipid head groups at the surface. Various combinations of lipid and protein produce particles of different densities, ranging from chylomicrons and very-low-density lipoproteins (VLDL) to very-high-density lipoproteins (VHDL), which can be separated by ultracentrifugation. The structures of these lipoprotein particles and their roles in lipid transport are detailed in Chapter 21.

The protein moieties of lipoproteins are recognized by receptors on cell surfaces. In lipid uptake from the intestine, chylomicrons, which contain apolipoprotein C-II (apoC-II), move from the intestinal mucosa into the lymphatic system, and then enter the blood, which carries them to muscle and adipose tissue (Fig. 17–1, step ⑤). In the capillaries of these tissues, the extracellular enzyme **lipoprotein lipase**, activated by apoC-II, hydrolyzes triacylglycerols to fatty acids and glycerol (step ⑥), which are taken up by cells in the target tissues (step ⑦). In muscle, the fatty acids are oxidized for energy; in adipose tissue, they are reesterified for storage as triacylglycerols (step ⑧).

The remnants of chylomicrons, depleted of most of their triacylglycerols but still containing cholesterol and apolipoproteins, travel in the blood to the liver, where they are taken up by endocytosis, mediated by receptors for their apolipoproteins. Triacylglycerols that enter the liver by this route may be oxidized to provide energy or to provide precursors for the synthesis of ketone bodies, as described in Section 17.3. When the diet contains more fatty acids than are needed immediately for fuel or as precursors, the liver converts them to triacylglycerols, which are packaged with specific apolipoproteins into VLDLs. The VLDLs are transported in the blood to adipose tissues, where the triacylglycerols are removed and stored in lipid droplets within adipocytes.

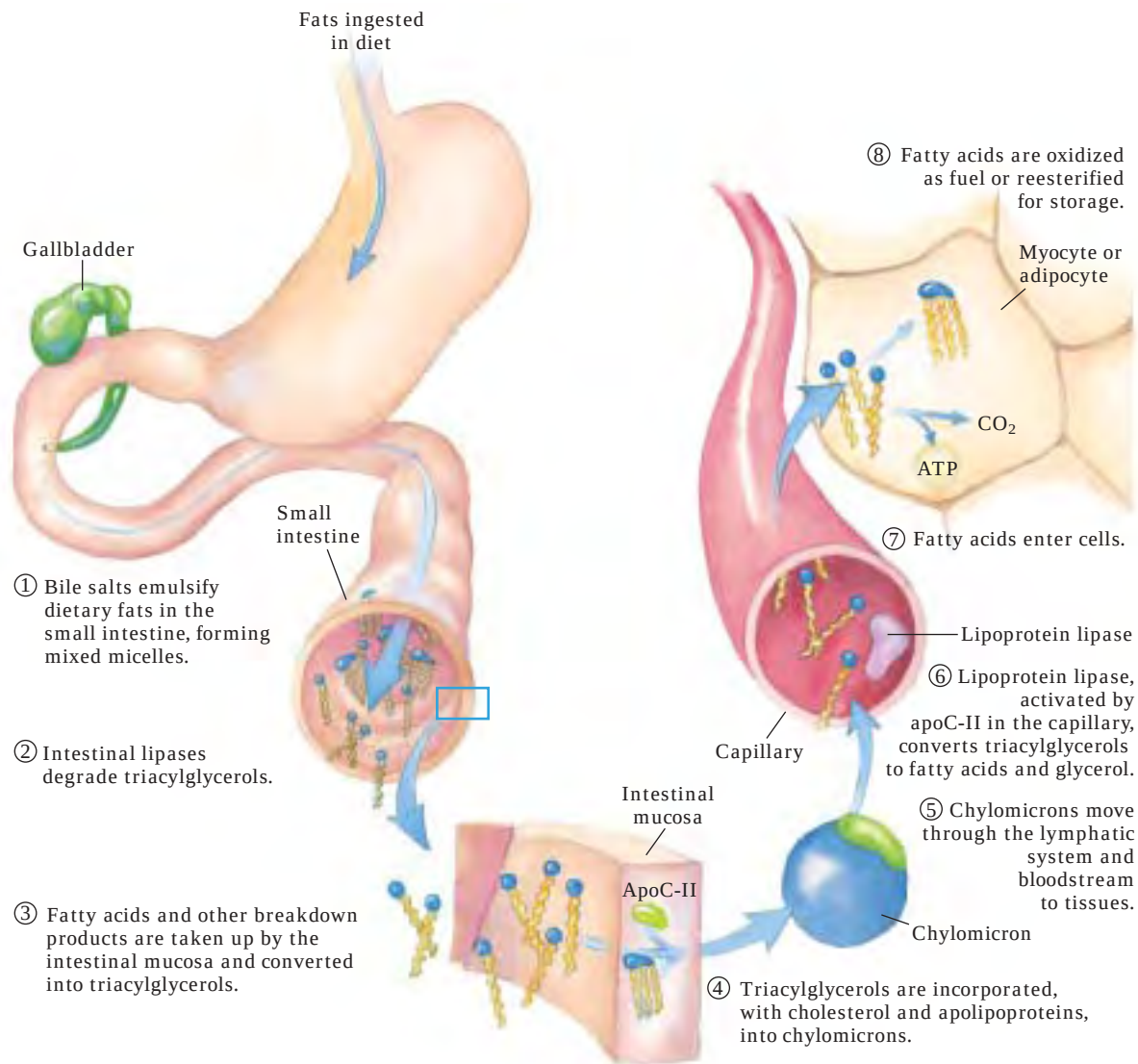


FIGURE 17-1 Processing of dietary lipids in vertebrates. Digestion and absorption of dietary lipids occur in the small intestine, and the fatty acids released from triacylglycerols are packaged and delivered to muscle and adipose tissues. The eight steps are discussed in the text.

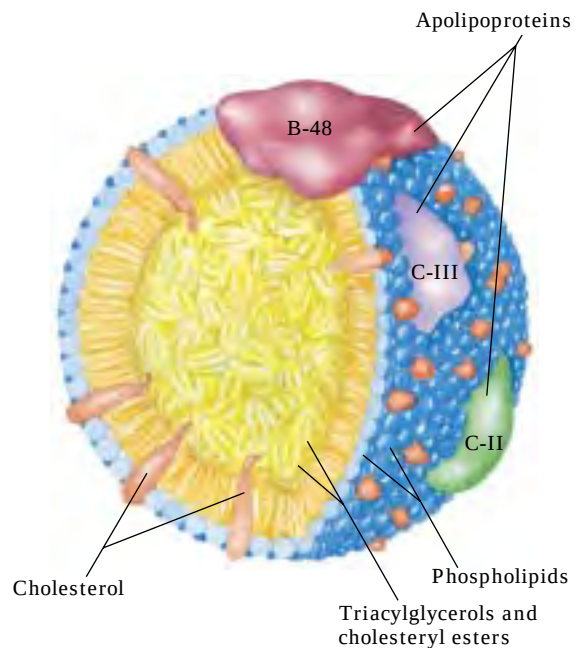


FIGURE 17-2 Molecular structure of a chylomicron. The surface is a layer of phospholipids, with head groups facing the aqueous phase. Triacylglycerols sequestered in the interior (yellow) make up more than 80% of the mass. Several apolipoproteins that protrude from the surface (B-48, C-III, C-II) act as signals in the uptake and metabolism of chylomicron contents. The diameter of chylomicrons ranges from about 100 to 500 nm.

Hormones Trigger Mobilization of Stored Triacylglycerols

Neutral lipids are stored in adipocytes (and in steroid-synthesizing cells of the adrenal cortex, ovary, and testes) in the form of lipid droplets, with a core of sterol esters and triacylglycerols surrounded by a monolayer of phospholipids. The surface of these droplets is coated with **perilipins**, a family of proteins that restrict access to lipid droplets, preventing untimely lipid mobilization. When hormones signal the need for metabolic energy, triacylglycerols stored in adipose tissue are mobilized (brought out of storage) and transported to tissues (skeletal muscle, heart, and renal cortex) in which fatty

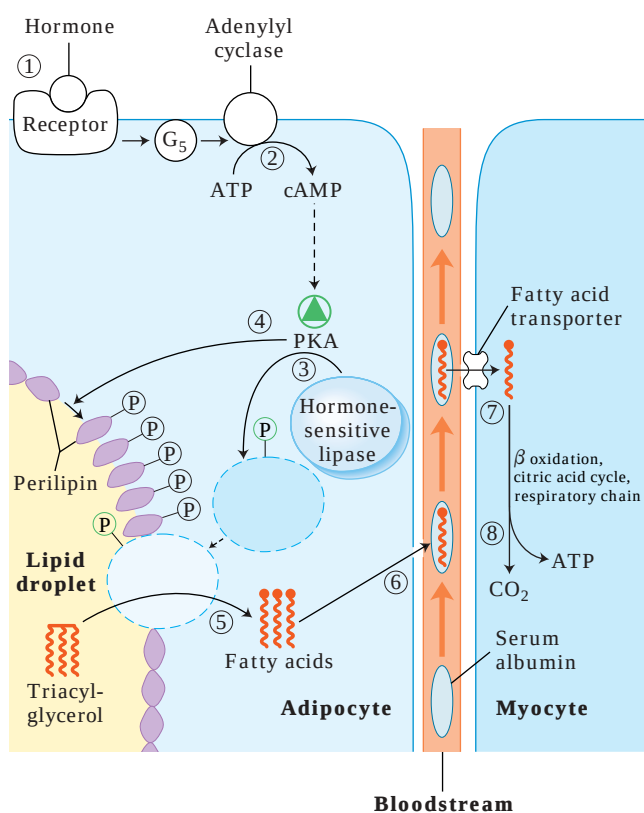


FIGURE 17-3 Mobilization of triacylglycerols stored in adipose tissue. When low levels of glucose in the blood trigger the release of glucagon, ① the hormone binds its receptor in the adipocyte membrane and thus ② stimulates adenyl cyclase, via a G protein, to produce cAMP. This activates PKA, which phosphorylates ③ the hormone-sensitive lipase and ④ perilipin molecules on the surface of the lipid droplet. Phosphorylation of perilipin permits hormone-sensitive lipase access to the surface of the lipid droplet, where ⑤ it hydrolyzes triacylglycerols to free fatty acids. ⑥ Fatty acids leave the adipocyte, bind serum albumin in the blood, and are carried in the blood; they are released from the albumin and ⑦ enter a myocyte via a specific fatty acid transporter. ⑧ In the myocyte, fatty acids are oxidized to CO₂, and the energy of oxidation is conserved in ATP, which fuels muscle contraction and other energy requiring metabolism in the myocyte.

acids can be oxidized for energy production. The hormones epinephrine and glucagon, secreted in response to low blood glucose levels, activate the enzyme adenyl cyclase in the adipocyte plasma membrane (Fig. 17-3), which produces the intracellular second messenger cyclic AMP (cAMP; see Fig. 12-13). Cyclic AMP-dependent protein kinase (PKA) phosphorylates perilipin A, and the phosphorylated perilipin causes **hormone-sensitive lipase** in the cytosol to move to the lipid droplet surface, where it can begin hydrolyzing triacylglycerols to free fatty acids and glycerol. PKA also phosphorylates hormone-sensitive lipase, doubling or tripling its activity, but the more than 50-fold increase in fat mobilization triggered by epinephrine is due primarily to perilipin phosphorylation. Cells with defective perilipin genes have almost no response to increases in cAMP concentration; their hormone-sensitive lipase does not associate with lipid droplets.

As hormone-sensitive lipase hydrolyzes triacylglycerol in adipocytes, the fatty acids thus released (**free fatty acids, FFA**) pass from the adipocyte into the blood, where they bind to the blood protein **serum albumin**. This protein (M_r 66,000), which makes up about half of the total serum protein, noncovalently binds as many as 10 fatty acids per protein monomer. Bound to this soluble protein, the otherwise insoluble fatty acids are carried to tissues such as skeletal muscle, heart, and renal cortex. In these target tissues, fatty acids dissociate from albumin and are moved by plasma membrane transporters into cells to serve as fuel.

About 95% of the biologically available energy of triacylglycerols resides in their three long-chain fatty acids; only 5% is contributed by the glycerol moiety. The glycerol released by lipase action is phosphorylated by **glycerol kinase** (Fig. 17-4), and the resulting glycerol 3-phosphate is oxidized to dihydroxyacetone phosphate. The glycolytic enzyme triose phosphate isomerase converts this compound to glyceraldehyde 3-phosphate, which is oxidized via glycolysis.

Fatty Acids Are Activated and Transported into Mitochondria

The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix, as demonstrated in 1948 by Eugene P. Kennedy and Albert Lehninger. The fatty acids with chain lengths of 12 or fewer carbons enter mitochondria without the help of membrane transporters. Those with 14 or more carbons, which constitute the majority of the FFA obtained in the diet or released from adipose tissue, cannot pass directly through the mitochondrial membranes—they must first undergo the three enzymatic reactions of the **carnitine shuttle**. The first reaction is catalyzed by a family of isozymes (different isozymes specific for fatty acids having short, intermediate, or long carbon chains) present

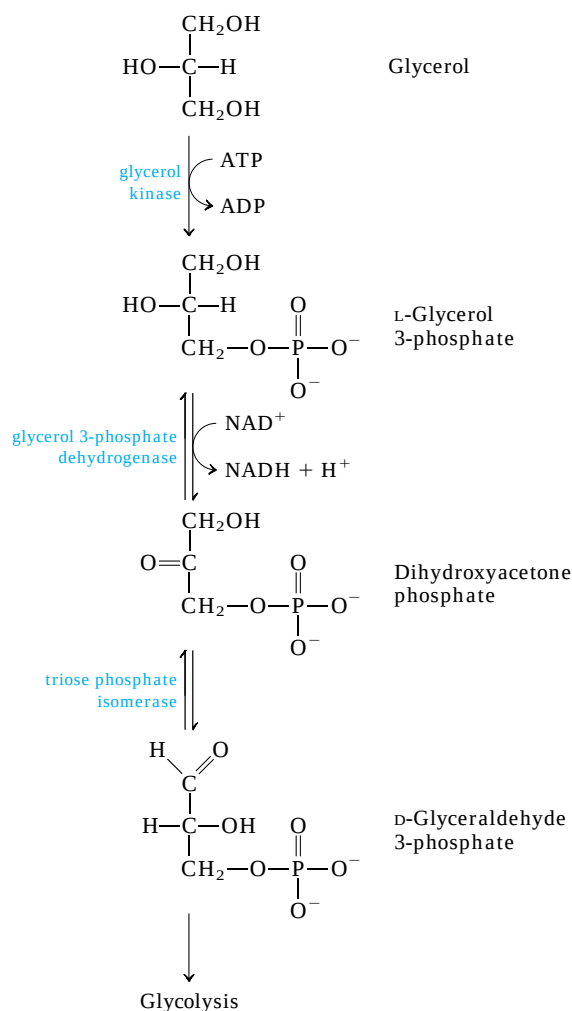
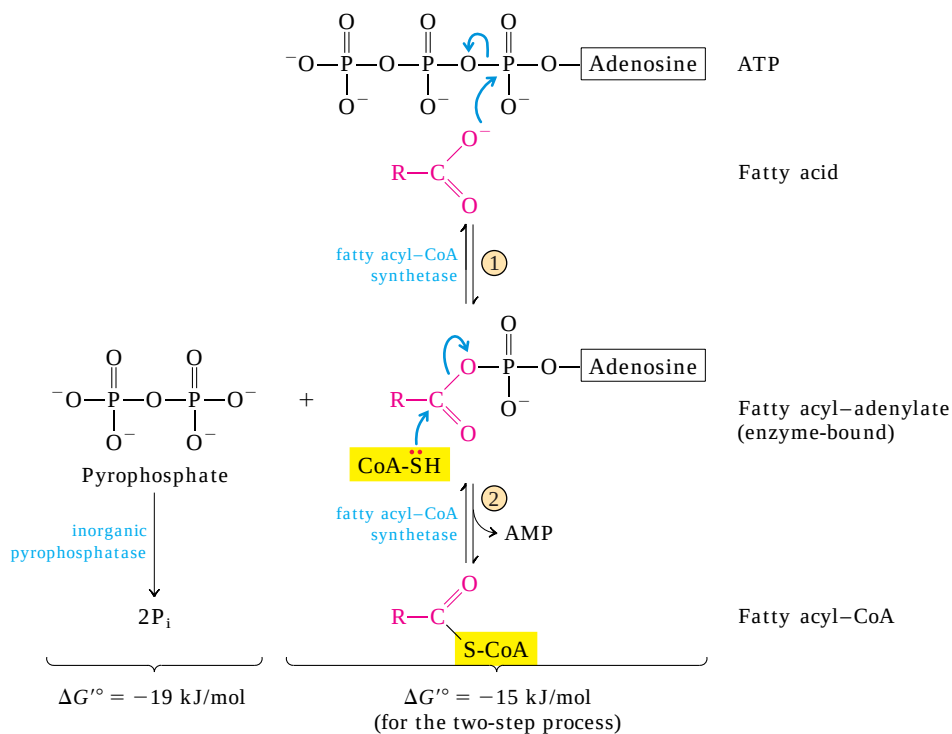


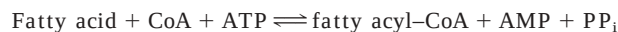
FIGURE 17-4 Entry of glycerol into the glycolytic pathway.

MECHANISM FIGURE 17-5

Conversion of a fatty acid to a fatty acyl-CoA. The conversion is catalyzed by fatty acyl-CoA synthetase and inorganic pyrophosphatase. Fatty acid activation by formation of the fatty acyl-CoA derivative occurs in two steps. In step ①, the carboxylate ion displaces the outer two (β and γ) phosphates of ATP to form a fatty acyl-adenylate, the mixed anhydride of a carboxylic acid and a phosphoric acid. The other product is PP_i, an excellent leaving group that is immediately hydrolyzed to two P_i, pulling the reaction in the forward direction. In step ②, the thiol group of coenzyme A carries out nucleophilic attack on the enzyme-bound mixed anhydride, displacing AMP and forming the thioester fatty acyl-CoA. The overall reaction is highly exergonic.

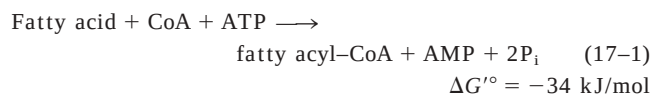


in the outer mitochondrial membrane, the **acyl-CoA synthetases**, which promote the general reaction



Thus, acyl-CoA synthetases catalyze the formation of a thioester linkage between the fatty acid carboxyl group and the thiol group of coenzyme A to yield a **fatty acyl-CoA**, coupled to the cleavage of ATP to AMP and PP_i. (Recall the description of this reaction in Chapter 13, to illustrate how the free energy released by cleavage of phosphoanhydride bonds in ATP could be coupled to the formation of a high-energy compound; p. XXX.) The reaction occurs in two steps and involves a fatty acyl-adenylate intermediate (Fig. 17-5).

Fatty acyl-CoAs, like acetyl-CoA, are high-energy compounds; their hydrolysis to FFA and CoA has a large, negative standard free-energy change ($\Delta G'^{\circ} \approx -31 \text{ kJ/mol}$). The formation of a fatty acyl-CoA is made more favorable by the hydrolysis of *two* high-energy bonds in ATP; the pyrophosphate formed in the activation reaction is immediately hydrolyzed by inorganic pyrophosphatase (left side of Fig. 17-5), which pulls the preceding activation reaction in the direction of fatty acyl-CoA formation. The overall reaction is



Fatty acyl-CoA esters formed at the cytosolic side of the outer mitochondrial membrane can be transported into the mitochondrion and oxidized to produce ATP, or they can be used in the cytosol to synthesize

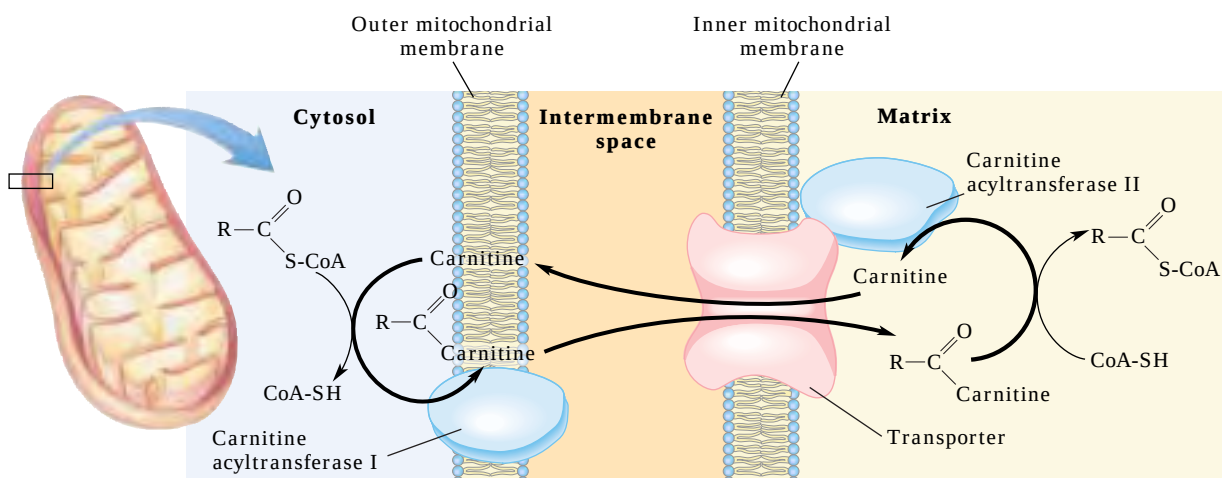
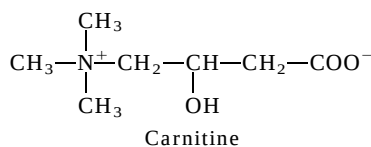


FIGURE 17-6 Fatty acid entry into mitochondria via the acyl-carnitine/carnitine transporter. After fatty acyl-carnitine is formed at the outer membrane or in the intermembrane space, it moves into the matrix by facilitated diffusion through the transporter in the inner membrane. In the matrix, the acyl group is transferred to mitochondrial coenzyme

A, freeing carnitine to return to the intermembrane space through the same transporter. Acyltransferase I is inhibited by malonyl-CoA, the first intermediate in fatty acid synthesis (see Fig. 21-1). This inhibition prevents the simultaneous synthesis and degradation of fatty acids.

membrane lipids. Fatty acids destined for mitochondrial oxidation are transiently attached to the hydroxyl group of **carnitine** to form fatty acyl-carnitine—the second reaction of the shuttle. This transesterification is catalyzed by **carnitine acyltransferase I** (M_r 88,000), in the outer membrane. Either the acyl-CoA passes through the outer membrane and is converted to the carnitine ester in the intermembrane space (Fig. 17-6), or the carnitine ester is formed on the cytosolic face of the outer membrane, then moved across the outer membrane to the intermembrane space—the current evidence does not reveal which. In either case, passage into the intermembrane space (the space between the outer and inner membranes) occurs through large pores (formed by the protein porin) in the outer membrane. The fatty acyl-carnitine ester then enters the matrix by facilitated diffusion through the **acyl-carnitine/carnitine transporter** of the inner mitochondrial membrane (Fig. 17-6).



In the third and final step of the carnitine shuttle, the fatty acyl group is enzymatically transferred from carnitine to intramitochondrial coenzyme A by **carnitine acyltransferase II**. This isozyme, located on the inner face of the inner mitochondrial membrane, regenerates fatty acyl-CoA and releases it, along with free carnitine, into the matrix (Fig. 17-6). Carnitine reenters the intermembrane space via the acyl-carnitine/carnitine transporter.

This three-step process for transferring fatty acids into the mitochondrion—esterification to CoA, transesterification to carnitine followed by transport, and transesterification back to CoA—links two separate pools of coenzyme A and of fatty acyl-CoA, one in the cytosol, the other in mitochondria. These pools have different functions. Coenzyme A in the mitochondrial matrix is largely used in oxidative degradation of pyruvate, fatty acids, and some amino acids, whereas cytosolic coenzyme A is used in the biosynthesis of fatty acids (see Fig. 21-10). Fatty acyl-CoA in the cytosolic pool can be used for membrane lipid synthesis or can be moved into the mitochondrial matrix for oxidation and ATP production. Conversion to the carnitine ester commits the fatty acyl moiety to the oxidative fate.

The carnitine-mediated entry process is the rate-limiting step for oxidation of fatty acids in mitochondria and, as discussed later, is a regulation point. Once inside the mitochondrion, the fatty acyl-CoA is acted upon by a set of enzymes in the matrix.

SUMMARY 17.1 Digestion, Mobilization, and Transport of Fats

- The fatty acids of triacylglycerols furnish a large fraction of the oxidative energy in animals. Dietary triacylglycerols are emulsified in the small intestine by bile salts, hydrolyzed by intestinal lipases, absorbed by intestinal epithelial cells, reconverted into triacylglycerols, then formed into chylomicrons by combination with specific apolipoproteins.

- Chylomicrons deliver triacylglycerols to tissues, where lipoprotein lipase releases free fatty acids for entry into cells. Triacylglycerols stored in adipose tissue are mobilized by a hormone-sensitive triacylglycerol lipase. The released fatty acids bind to serum albumin and are carried in the blood to the heart, skeletal muscle, and other tissues that use fatty acids for fuel.
- Once inside cells, fatty acids are activated at the outer mitochondrial membrane by conversion to fatty acyl-CoA thioesters. Fatty acyl-CoA to be oxidized enters mitochondria in three steps, via the carnitine shuttle.

17.2 Oxidation of Fatty Acids

As noted earlier, mitochondrial oxidation of fatty acids takes place in three stages (Fig. 17-7). In the first stage— β oxidation—fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-CoA, starting from the carboxyl end of the fatty acyl chain. For example, the 16-carbon palmitic acid (palmitate at pH 7) undergoes seven passes through the oxidative sequence, in each pass losing two carbons as acetyl-CoA. At the end of seven cycles the last two carbons of palmitate (originally C-15 and C-16) remain as acetyl-CoA. The overall result is the conversion of the 16-carbon chain of palmitate to eight two-carbon acetyl groups of acetyl-CoA molecules. Formation of each acetyl-CoA requires removal of four hydrogen atoms (two pairs of electrons and four H^+) from the fatty acyl moiety by dehydrogenases.

In the second stage of fatty acid oxidation, the acetyl groups of acetyl-CoA are oxidized to CO_2 in the citric acid cycle, which also takes place in the mitochondrial matrix. Acetyl-CoA derived from fatty acids thus enters a final common pathway of oxidation with the acetyl-CoA derived from glucose via glycolysis and pyruvate oxidation (see Fig. 16-1). The first two stages of fatty acid oxidation produce the reduced electron carriers NADH and $FADH_2$, which in the third stage donate electrons to the mitochondrial respiratory chain, through which the electrons pass to oxygen with the concomitant phosphorylation of ADP to ATP (Fig. 17-7). The energy released by fatty acid oxidation is thus conserved as ATP.

We now take a closer look at the first stage of fatty acid oxidation, beginning with the simple case of a saturated fatty acyl chain with an even number of carbons, then turning to the slightly more complicated cases of unsaturated and odd-number chains. We also consider the regulation of fatty acid oxidation, the β -oxidative processes as they occur in organelles other than mitochondria, and, finally, two less-general modes of fatty acid catabolism, α oxidation and ω oxidation.

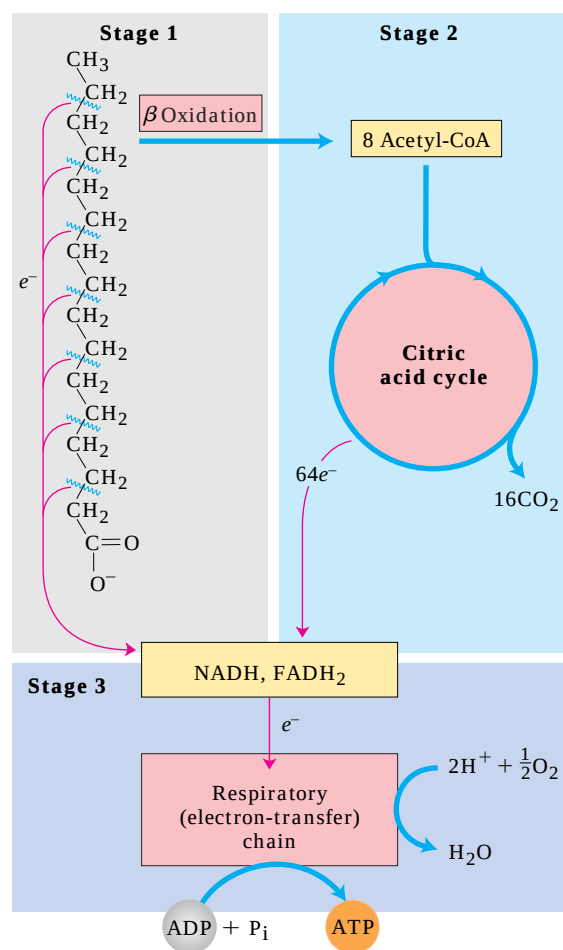


FIGURE 17-7 Stages of fatty acid oxidation. Stage 1: A long-chain fatty acid is oxidized to yield acetyl residues in the form of acetyl-CoA. This process is called β oxidation. Stage 2: The acetyl groups are oxidized to CO_2 via the citric acid cycle. Stage 3: Electrons derived from the oxidations of stages 1 and 2 pass to O_2 via the mitochondrial respiratory chain, providing the energy for ATP synthesis by oxidative phosphorylation.

The β Oxidation of Saturated Fatty Acids Has Four Basic Steps

Four enzyme-catalyzed reactions make up the first stage of fatty acid oxidation (Fig. 17-8a). First, dehydrogenation of fatty acyl-CoA produces a double bond between the α and β carbon atoms (C-2 and C-3), yielding a ***trans*- Δ^2 -enoyl-CoA** (the symbol Δ^2 designates the position of the double bond; you may want to review fatty acid nomenclature, p. 343.) Note that the new double bond has the *trans* configuration, whereas the double bonds in naturally occurring unsaturated fatty acids are normally in the *cis* configuration. We consider the significance of this difference later.

This first step is catalyzed by three isozymes of **acyl-CoA dehydrogenase**, each specific for a range of fatty-acyl chain lengths: very-long-chain acyl-CoA dehydrogenase (VLCAD), acting on fatty acids of 12 to 18

carbons; medium-chain (MCAD), acting on fatty acids of 4 to 14 carbons; and short-chain (SCAD), acting on fatty acids of 4 to 8 carbons. All three isozymes are flavo-proteins with FAD (see Fig. 13–18) as a prosthetic group. The electrons removed from the fatty acyl-CoA are transferred to FAD, and the reduced form of the dehydrogenase immediately donates its electrons to an electron carrier of the mitochondrial respiratory chain, the **electron-transferring flavoprotein (ETF)** (see Fig. 19–8). The oxidation catalyzed by an acyl-CoA dehydrogenase is analogous to succinate dehydrogenation

in the citric acid cycle (p. XXX); in both reactions the enzyme is bound to the inner membrane, a double bond is introduced into a carboxylic acid between the α and β carbons, FAD is the electron acceptor, and electrons from the reaction ultimately enter the respiratory chain and pass to O_2 , with the concomitant synthesis of about 1.5 ATP molecules per electron pair.

In the second step of the β -oxidation cycle (Fig. 17–8a), water is added to the double bond of the *trans*- Δ^2 -enoyl-CoA to form the L stereoisomer of **β -hydroxyacyl-CoA (3-hydroxyacyl-CoA)**. This reaction, catalyzed by **enoyl-CoA hydratase**, is formally analogous to the fumarase reaction in the citric acid cycle, in which H_2O adds across an α - β double bond (p. XXX).

In the third step, L- β -hydroxyacyl-CoA is dehydrogenated to form **β -ketoacyl-CoA**, by the action of **β -hydroxyacyl-CoA dehydrogenase**; NAD^+ is the electron acceptor. This enzyme is absolutely specific for the L stereoisomer of hydroxyacyl-CoA. The NADH formed in the reaction donates its electrons to **NADH dehydrogenase**, an electron carrier of the respiratory chain, and ATP is formed from ADP as the electrons pass to O_2 . The reaction catalyzed by β -hydroxyacyl-CoA dehydrogenase is closely analogous to the malate dehydrogenase reaction of the citric acid cycle (p. XXX).

The fourth and last step of the β -oxidation cycle is catalyzed by **acyl-CoA acetyltransferase**, more commonly called **thiolase**, which promotes reaction of β -ketoacyl-CoA with a molecule of free coenzyme A to split off the carboxyl-terminal two-carbon fragment of the original fatty acid as acetyl-CoA. The other product is the coenzyme A thioester of the fatty acid, now shortened by two carbon atoms (Fig. 17–8a). This reaction is called **thiolysis**, by analogy with the process of hydrolysis, because the β -ketoacyl-CoA is cleaved by reaction with the thiol group of coenzyme A.

The last three steps of this four-step sequence are catalyzed by either of two sets of enzymes, with the enzymes employed depending on the length of the fatty acyl chain. For fatty acyl chains of 12 or more carbons, the reactions are catalyzed by a multienzyme complex associated with the inner mitochondrial membrane, the **trifunctional protein (TFP)**. TFP is a heterooctamer of $\alpha_4\beta_4$ subunits. Each α subunit contains two activities, the enoyl-CoA hydratase and the β -hydroxyacyl-CoA dehydrogenase; the β subunits contain the thiolase activity. This tight association of three enzymes may allow efficient substrate channeling from one active site to the

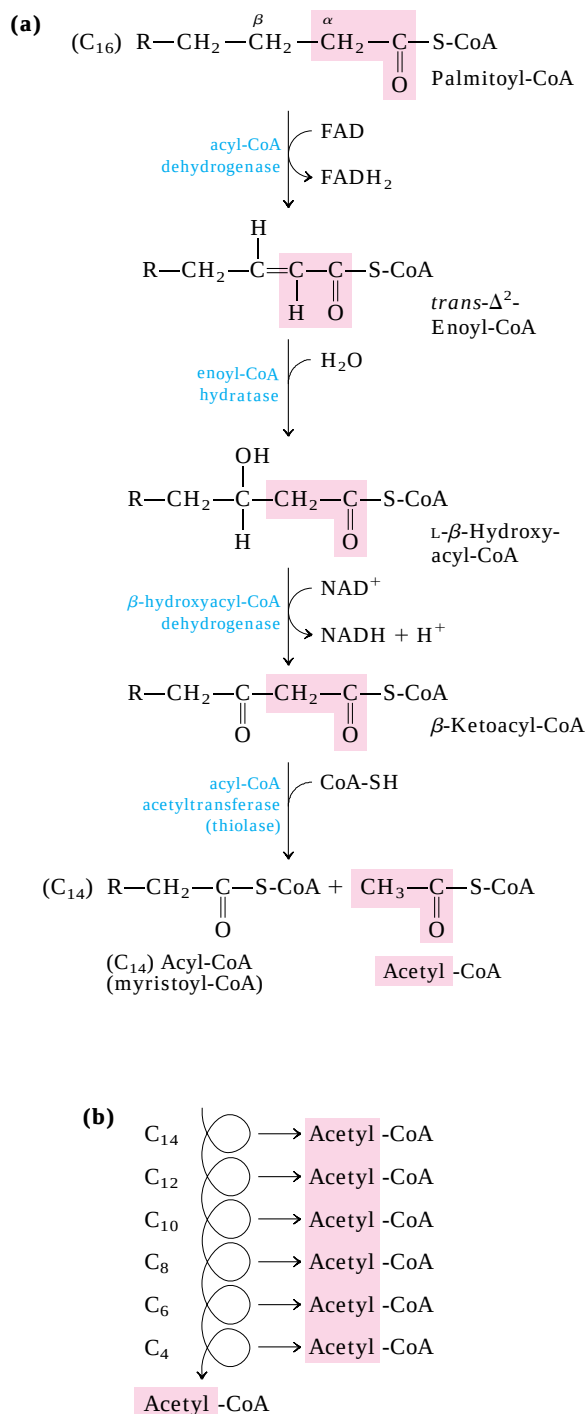


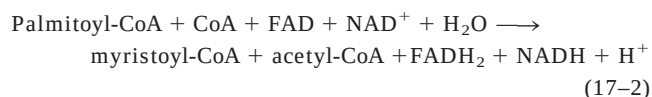
FIGURE 17–8 The β -oxidation pathway. (a) In each pass through this four-step sequence, one acetyl residue (shaded in pink) is removed in the form of acetyl-CoA from the carboxyl end of the fatty acyl chain—in this example palmitate (C_{16}), which enters as palmitoyl-CoA. (b) Six more passes through the pathway yield seven more molecules of acetyl-CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain. Eight molecules of acetyl-CoA are formed in all.

next, without diffusion of the intermediates away from the enzyme surface. When TFP has shortened the fatty acyl chain to 12 or fewer carbons, further oxidations are catalyzed by a set of four soluble enzymes in the matrix.

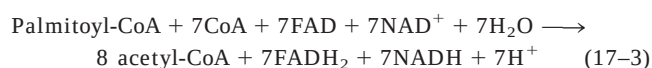
As noted earlier, the single bond between methylene ($-\text{CH}_2-$) groups in fatty acids is relatively stable. The β -oxidation sequence is an elegant mechanism for destabilizing and breaking these bonds. The first three reactions of β oxidation create a much less stable C—C bond, in which the α carbon (C-2) is bonded to two carbonyl carbons (the β -ketoacyl-CoA intermediate). The ketone function on the β carbon (C-3) makes it a good target for nucleophilic attack by the $-\text{SH}$ of coenzyme A, catalyzed by thiolase. The acidity of the α hydrogen and the resonance stabilization of the carbanion generated by the departure of this hydrogen make the terminal $-\text{CH}_2-\text{CO}-\text{S-CoA}$ a good leaving group, facilitating breakage of the α - β bond.

The Four β -Oxidation Steps Are Repeated to Yield Acetyl-CoA and ATP

In one pass through the β -oxidation sequence, one molecule of acetyl-CoA, two pairs of electrons, and four protons (H^+) are removed from the long-chain fatty acyl-CoA, shortening it by two carbon atoms. The equation for one pass, beginning with the coenzyme A ester of our example, palmitate, is



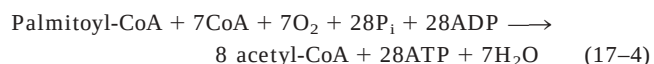
Following removal of one acetyl-CoA unit from palmitoyl-CoA, the coenzyme A thioester of the shortened fatty acid (now the 14-carbon myristate) remains. The myristoyl-CoA can now go through another set of four β -oxidation reactions, exactly analogous to the first, to yield a second molecule of acetyl-CoA and lauroyl-CoA, the coenzyme A thioester of the 12-carbon laurate. Altogether, seven passes through the β -oxidation sequence are required to oxidize one molecule of palmitoyl-CoA to eight molecules of acetyl-CoA (Fig. 17-8b). The overall equation is



Each molecule of FADH_2 formed during oxidation of the fatty acid donates a pair of electrons to ETF of the respiratory chain, and about 1.5 molecules of ATP are generated during the ensuing transfer of each electron pair to O_2 . Similarly, each molecule of NADH formed delivers a pair of electrons to the mitochondrial NADH dehydrogenase, and the subsequent transfer of each pair of electrons to O_2 results in formation of about 2.5 molecules of ATP. Thus four molecules of ATP are formed for each two-carbon unit removed in one pass through the sequence. Note that water is also produced in this

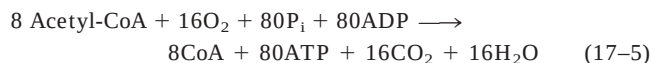
process. Transfer of electrons from NADH or FADH_2 to O_2 yields one H_2O per electron pair. Reduction of O_2 by NADH also consumes one H^+ per NADH molecule: $\text{NADH} + \text{H}^+ + \frac{1}{2}\text{O}_2 \longrightarrow \text{NAD}^+ + \text{H}_2\text{O}$. In hibernating animals, fatty acid oxidation provides metabolic energy, heat, and water—all essential for survival of an animal that neither eats nor drinks for long periods (Box 17-1). Camels obtain water to supplement the meager supply available in their natural environment by oxidation of fats stored in their hump.

The overall equation for the oxidation of palmitoyl-CoA to eight molecules of acetyl-CoA, including the electron transfers and oxidative phosphorylations, is



Acetyl-CoA Can Be Further Oxidized in the Citric Acid Cycle

The acetyl-CoA produced from the oxidation of fatty acids can be oxidized to CO_2 and H_2O by the citric acid cycle. The following equation represents the balance sheet for the second stage in the oxidation of palmitoyl-CoA, together with the coupled phosphorylations of the third stage:



Combining Equations 17-4 and 17-5, we obtain the overall equation for the complete oxidation of palmitoyl-CoA to carbon dioxide and water:

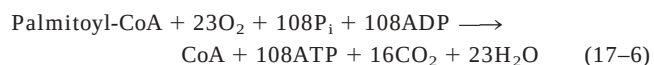


Table 17-1 summarizes the yields of NADH, FADH_2 , and ATP in the successive steps of palmitoyl-CoA oxidation. Note that because the activation of palmitate to palmitoyl-CoA breaks both phosphoanhydride bonds in ATP (Fig. 17-5), the energetic cost of activating a fatty acid is equivalent to two ATP, and the net gain per molecule of palmitate is 106 ATP. The standard free-energy change for the oxidation of palmitate to CO_2 and H_2O is about 9,800 kJ/mol. Under standard conditions, the energy recovered as the phosphate bond energy of ATP is $106 \times 30.5 \text{ kJ/mol} = 3,230 \text{ kJ/mol}$, about 33% of the theoretical maximum. However, when the free-energy changes are calculated from actual concentrations of reactants and products under intracellular conditions (see Box 13-1), the free-energy recovery is more than 60%; the energy conservation is remarkably efficient.

Oxidation of Unsaturated Fatty Acids Requires Two Additional Reactions

The fatty acid oxidation sequence just described is typical when the incoming fatty acid is saturated (that is, has only single bonds in its carbon chain). However,

BOX 17-1 THE WORLD OF BIOCHEMISTRY

Fat Bears Carry Out β Oxidation in Their Sleep

Many animals depend on fat stores for energy during hibernation, during migratory periods, and in other situations involving radical metabolic adjustments. One of the most pronounced adjustments of fat metabolism occurs in hibernating grizzly bears. These animals remain in a continuous state of dormancy for periods as long as seven months. Unlike most hibernating species, the bear maintains a body temperature of between 32 and 35 °C, close to the normal (nonhibernating) level.



A grizzly bear prepares its hibernation nest, near the McNeil River in Canada.

Although expending about 25,000 kJ/day (6,000 kcal/day), the bear does not eat, drink, urinate, or defecate for months at a time.

Experimental studies have shown that hibernating grizzly bears use body fat as their sole fuel. Fat oxidation yields sufficient energy for maintenance of body temperature, active synthesis of amino acids and proteins, and other energy-requiring activities, such as membrane transport. Fat oxidation also releases large amounts of water, as described in the text, which replenishes water lost in breathing. The glycerol released by degradation of triacylglycerols is converted into blood glucose by gluconeogenesis. Urea formed during breakdown of amino acids is reabsorbed in the kidneys and recycled, the amino groups reused to make new amino acids for maintaining body proteins.

Bears store an enormous amount of body fat in preparation for their long sleep. An adult grizzly consumes about 38,000 kJ/day during the late spring and summer, but as winter approaches it feeds 20 hours a day, consuming up to 84,000 kJ daily. This change in feeding is a response to a seasonal change in hormone secretion. Large amounts of triacylglycerols are formed from the huge intake of carbohydrates during the fattening-up period. Other hibernating species, including the tiny dormouse, also accumulate large amounts of body fat.

most of the fatty acids in the triacylglycerols and phospholipids of animals and plants are unsaturated, having one or more double bonds. These bonds are in the cis configuration and cannot be acted upon by enoyl-CoA hydratase, the enzyme catalyzing the addition of H₂O to

the trans double bond of the Δ^2 -enoyl-CoA generated during β oxidation. Two auxiliary enzymes are needed for β oxidation of the common unsaturated fatty acids: an isomerase and a reductase. We illustrate these auxiliary reactions with two examples.

TABLE 17-1 Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO₂ and H₂O

Enzyme catalyzing the oxidation step	Number of NADH or FADH ₂ formed	Number of ATP ultimately formed*
Acyl-CoA dehydrogenase	7 FADH ₂	10.5
β -Hydroxyacyl-CoA dehydrogenase	7 NADH	17.5
Isocitrate dehydrogenase	8 NADH	20
α -Ketoglutarate dehydrogenase	8 NADH	20
Succinyl-CoA synthetase		8 [†]
Succinate dehydrogenase	8 FADH ₂	12
Malate dehydrogenase	8 NADH	20
Total		108

*These calculations assume that mitochondrial oxidative phosphorylation produces 1.5 ATP per FADH₂ oxidized and 2.5 ATP per NADH oxidized.

[†]GTP produced directly in this step yields ATP in the reaction catalyzed by nucleoside diphosphate kinase (p. XXX).

Oleate is an abundant 18-carbon monounsaturated fatty acid with a *cis* double bond between C-9 and C-10 (denoted Δ^9). In the first step of oxidation, oleate is converted to oleoyl-CoA and, like the saturated fatty acids, enters the mitochondrial matrix via the carnitine shuttle (Fig. 17–6). Oleoyl-CoA then undergoes three passes through the fatty acid oxidation cycle to yield three molecules of acetyl-CoA and the coenzyme A ester of a Δ^3 , 12-carbon unsaturated fatty acid, *cis*- Δ^3 -dodecenoyl-CoA (Fig. 17–9). This product cannot serve as a substrate for enoyl-CoA hydratase, which acts only on *trans* double bonds. The auxiliary enzyme **Δ^3, Δ^2 -enoyl-CoA isomerase** isomerizes the *cis*- Δ^3 -enoyl-CoA to the *trans*- Δ^2 -enoyl-CoA, which is converted by enoyl-CoA hydratase into the corresponding L- β -hydroxyacyl-CoA (*trans*- Δ^2 -dodecenoyl-CoA). This intermediate is now acted upon by the remaining enzymes of β oxidation to yield acetyl-CoA and the coenzyme A ester of a 10-carbon saturated fatty acid, decanoyl-CoA. The latter undergoes four more passes through the pathway to yield five more molecules of acetyl-CoA. Altogether, nine acetyl-CoAs are produced from one molecule of the 18-carbon oleate.

The other auxiliary enzyme (a reductase) is required for oxidation of polyunsaturated fatty acids—for

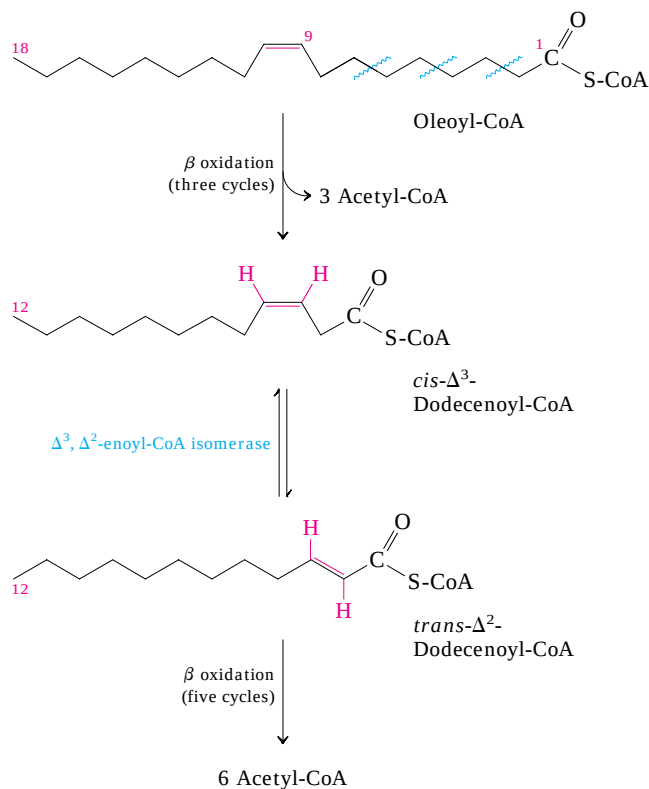


FIGURE 17–9 Oxidation of a monounsaturated fatty acid. Oleic acid, as oleoyl-CoA (Δ^9), is the example used here. Oxidation requires an additional enzyme, enoyl-CoA isomerase, to reposition the double bond, converting the *cis* isomer to a *trans* isomer, a normal intermediate in β oxidation.

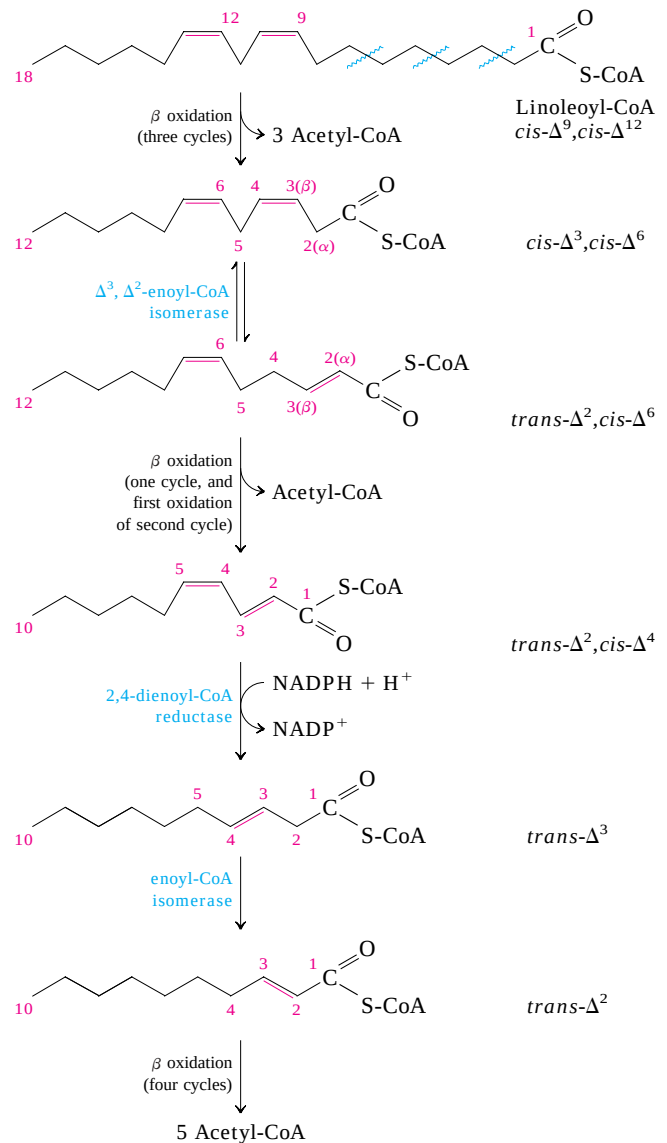


FIGURE 17–10 Oxidation of a polyunsaturated fatty acid. The example here is linoleic acid, as linoleoyl-CoA (Δ^9, Δ^{12}). Oxidation requires a second auxiliary enzyme in addition to enoyl-CoA isomerase: NADPH-dependent 2,4-dienoyl-CoA reductase. The combined action of these two enzymes converts a *trans*- Δ^2, Δ^4 -dienoyl-CoA intermediate to the *trans*- Δ^2 -enoyl-CoA substrate necessary for β oxidation.

example, the 18-carbon linoleate, which has a *cis*- Δ^9, Δ^{12} configuration (Fig. 17–10). Linoleoyl-CoA undergoes three passes through the β -oxidation sequence to yield three molecules of acetyl-CoA and the coenzyme A ester of a 12-carbon unsaturated fatty acid with a *cis*- Δ^3, Δ^6 configuration. This intermediate cannot be used by the enzymes of the β -oxidation pathway; its double bonds are in the wrong position and have the wrong configuration (*cis*, not *trans*). However, the combined action of enoyl-CoA isomerase and **2,4-dienoyl-CoA reductase**, as shown in Figure 17–10, allows reentry of this intermediate into the β -oxidation pathway

and its degradation to six acetyl-CoAs. The overall result is conversion of linoleate to nine molecules of acetyl-CoA.

Complete Oxidation of Odd-Number Fatty Acids Requires Three Extra Reactions

Although most naturally occurring lipids contain fatty acids with an even number of carbon atoms, fatty acids with an odd number of carbons are common in the lipids of many plants and some marine organisms. Cattle and other ruminant animals form large amounts of the three-carbon **propionate** ($\text{CH}_3\text{—CH}_2\text{—COO}^-$) during fermentation of carbohydrates in the rumen. The propionate is absorbed into the blood and oxidized by the liver and other tissues. And small quantities of propionate are added as a mold inhibitor to some breads and cereals, thus entering the human diet.

Long-chain odd-number fatty acids are oxidized in the same pathway as the even-number acids, beginning at the carboxyl end of the chain. However, the substrate for the last pass through the β -oxidation sequence is a fatty acyl-CoA with a five-carbon fatty acid. When this is oxidized and cleaved, the products are acetyl-CoA and **propionyl-CoA**. The acetyl-CoA can be oxidized in the citric acid cycle, of course, but propionyl-CoA enters a different pathway involving three enzymes.

Propionyl-CoA is first carboxylated to form the D stereoisomer of **methylmalonyl-CoA** (Fig. 17–11) by **propionyl-CoA carboxylase**, which contains the cofactor biotin. In this enzymatic reaction, as in the pyruvate carboxylase reaction (see Fig. 16–16), CO_2 (or its hydrated ion, HCO_3^-) is activated by attachment to biotin before its transfer to the substrate, in this case the propionate moiety. Formation of the carboxybiotin intermediate requires energy, which is provided by the cleavage of ATP to ADP and P_i . The D-methylmalonyl-CoA thus formed is enzymatically epimerized to its L stereoisomer by **methylmalonyl-CoA epimerase** (Fig. 17–11). The L-methylmalonyl-CoA then undergoes an intramolecular rearrangement to form succinyl-CoA, which can enter the citric acid cycle. This rearrangement is catalyzed by **methylmalonyl-CoA mutase**, which requires as its coenzyme **5'-deoxyadenosylcobalamin**, or **coenzyme B₁₂**, which is derived from vitamin B₁₂ (cobalamin). Box 17–2 describes the role of coenzyme B₁₂ in this remarkable exchange reaction.

Fatty Acid Oxidation Is Tightly Regulated

Oxidation of fatty acids consumes a precious fuel, and it is regulated so as to occur only when the need for energy requires it. In the liver, fatty acyl-CoA formed in the cytosol has two major pathways open to it: (1) β oxidation by enzymes in mitochondria or (2) conversion into triacylglycerols and phospholipids by enzymes in the cytosol. The pathway taken depends on the rate of

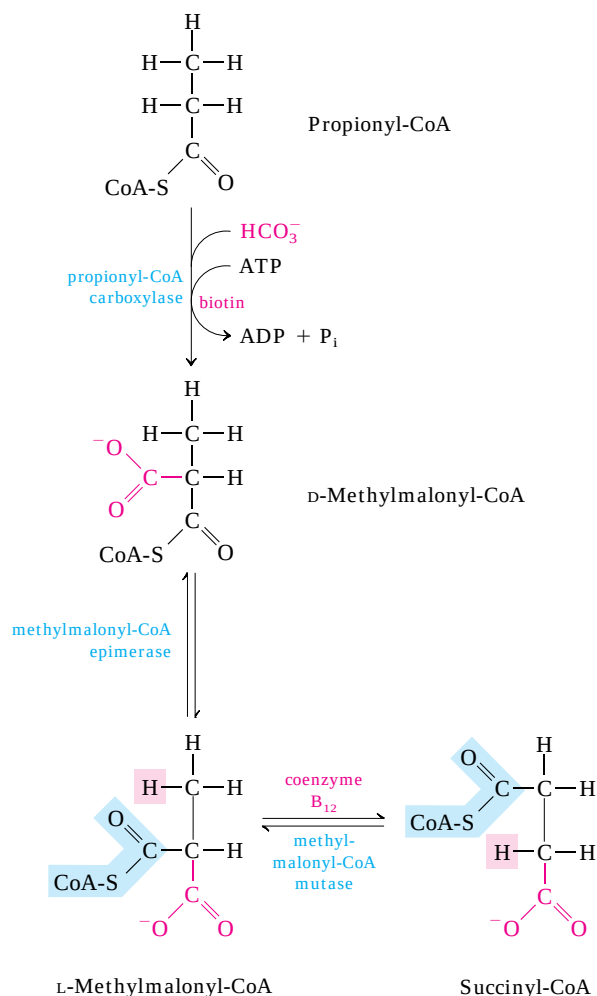
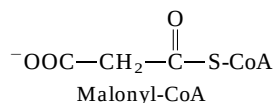


FIGURE 17–11 Oxidation of propionyl-CoA produced by β oxidation of odd-number fatty acids. The sequence involves the carboxylation of propionyl-CoA to D-methylmalonyl-CoA and conversion of the latter to succinyl-CoA. This conversion requires epimerization of D- to L-methylmalonyl-CoA, followed by a remarkable reaction in which substituents on adjacent carbon atoms exchange positions (see Box 17–2).

transfer of long-chain fatty acyl-CoA into mitochondria. The three-step process (carnitine shuttle) by which fatty acyl groups are carried from cytosolic fatty acyl-CoA into the mitochondrial matrix (Fig. 17–6) is rate-limiting for fatty acid oxidation and is an important point of regulation. Once fatty acyl groups have entered the mitochondrion, they are committed to oxidation to acetyl-CoA.

Malonyl-CoA, the first intermediate in the cytosolic biosynthesis of long-chain fatty acids from acetyl-CoA (see Fig. 21–1), increases in concentration whenever the animal is well supplied with carbohydrate; excess glucose that cannot be oxidized or stored as glycogen is converted in the cytosol into fatty acids for storage as triacylglycerol. The inhibition of carnitine acyltransferase I by malonyl-CoA ensures that the oxidation of

fatty acids is inhibited whenever the liver is amply supplied with glucose as fuel and is actively making triacylglycerols from excess glucose.



Two of the enzymes of β oxidation are also regulated by metabolites that signal energy sufficiency. When the $[\text{NADH}]/[\text{NAD}^+]$ ratio is high, β -hydroxyacyl-CoA dehydrogenase is inhibited; in addition, high concentrations of acetyl-CoA inhibit thiolase (Fig. 17-12).

Genetic Defects in Fatty Acyl-CoA Dehydrogenases Cause Serious Disease



Stored triacylglycerols are typically the chief source of energy for muscle contraction, and an inability to oxidize fatty acids from triacylglycerols has serious consequences for health. The most common genetic defect in fatty acid catabolism in U.S. and northern European populations is due to a mutation in the gene encoding the **medium-chain acyl-CoA dehydrogenase (MCAD)**. Among northern Europeans, the

frequency of carriers (individuals with this recessive mutation on one of the two homologous chromosomes) is about 1 in 40, and about 1 individual in 10,000 has the disease—that is, has two copies of the mutant MCAD allele and is unable to oxidize fatty acids of 6 to 12 carbons. The disease is characterized by recurring episodes of a syndrome that includes fat accumulation in the liver, high blood levels of octanoic acid, low blood glucose (hypoglycemia), sleepiness, vomiting, and coma. The pattern of organic acids in the urine helps in the diagnosis of this disease: the urine commonly contains high levels of 6-carbon to 10-carbon dicarboxylic acids (produced by ω oxidation) and low levels of urinary ketone bodies (we discuss ω oxidation below and ketone bodies in Section 17.3). Although individuals may have no symptoms between episodes, the episodes are very serious; mortality from this disease is 25% to 60% in early childhood. If the genetic defect is detected shortly after birth, the infant can be started on a low-fat, high-carbohydrate diet. With early detection and careful management of the diet—including avoiding long intervals between meals, to prevent the body from turning to its fat reserves for energy—the prognosis for these individuals is good.

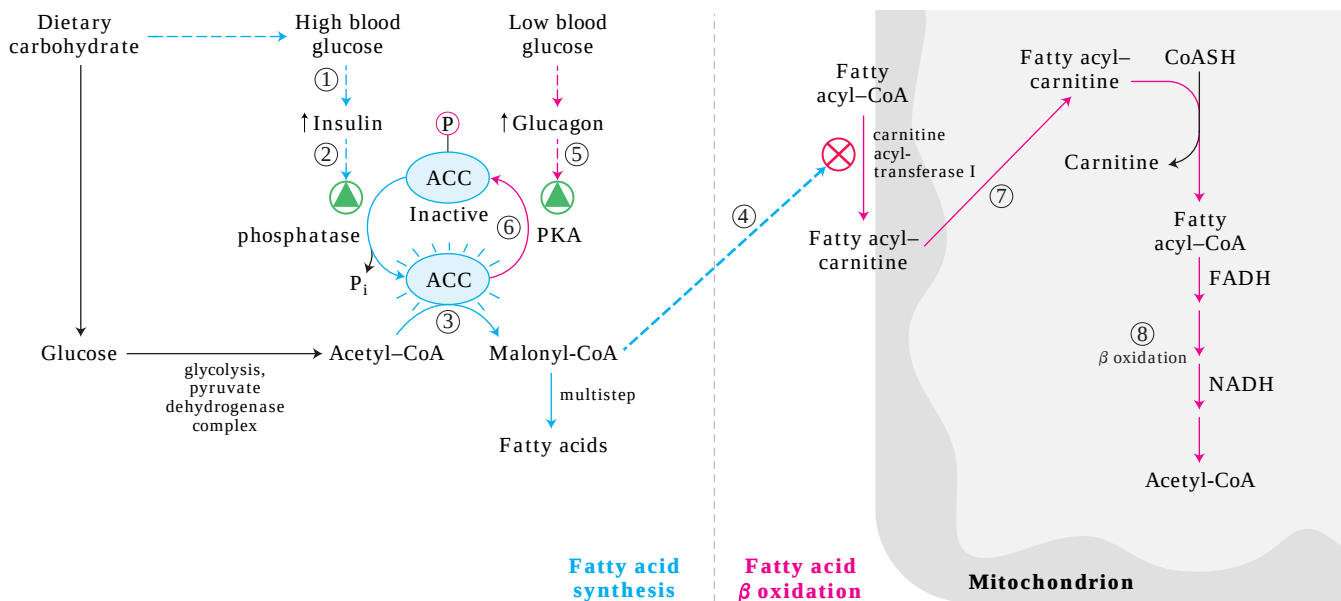


FIGURE 17-12 Coordinated regulation of fatty acid synthesis and breakdown. When the diet provides a ready source of carbohydrate as fuel, β oxidation of fatty acids is unnecessary and is therefore down-regulated. Two enzymes are key to the coordination of fatty acid metabolism: acetyl-CoA carboxylase (ACC), the first enzyme in the synthesis of fatty acids (see Fig. 21-1), and carnitine acyl transferase I, which limits the transport of fatty acids into the mitochondrial matrix for β oxidation (see Fig. 17-6). Ingestion of a high-carbohydrate meal raises the blood glucose level and thus (1) triggers the release of insulin. (2) Insulin-dependent protein phosphatase dephosphorylates ACC, activating it. (3) ACC catalyzes the formation of malonyl-CoA

(the first intermediate of fatty acid synthesis), and (4) malonyl-CoA inhibits carnitine acyltransferase I, thereby preventing fatty acid entry into the mitochondrial matrix.

When blood glucose levels drop between meals, (5) glucagon release activates cAMP-dependent protein kinase (PKA), which (6) phosphorylates and inactivates ACC. The concentration of malonyl-CoA falls, the inhibition of fatty acid entry into mitochondria is relieved, and (7) fatty acids enter the mitochondrial matrix and (8) become the major fuel. Because glucagon also triggers the mobilization of fatty acids in adipose tissue, a supply of fatty acids begins arriving in the blood.

Coenzyme B₁₂: A Radical Solution to a Perplexing Problem

In the methylmalonyl-CoA mutase reaction (see Fig. 17-11), the group —CO—S-CoA at C-2 of the original propionate exchanges position with a hydrogen atom at C-3 of the original propionate (Fig. 1a). Coenzyme B₁₂ is the cofactor for this reaction, as it is for almost all enzymes that catalyze reactions of this general type (Fig. 1b). These coenzyme B₁₂-dependent processes are among the very few enzymatic reactions in biology in which there is an exchange of an alkyl or substituted alkyl group (X) with a hydrogen atom on an adjacent carbon, *with no mixing of the transferred hydrogen atom with the hydrogen of the solvent, H₂O*. How can the hydrogen atom move between two carbons without mixing with the enormous excess of hydrogen atoms in the solvent?

Coenzyme B₁₂ is the cofactor form of vitamin B₁₂, which is unique among all the vitamins in that it contains not only a complex organic molecule but an essential trace element, cobalt. The complex **corrin ring system** of vitamin B₁₂ (colored blue in Fig. 2), to which cobalt (as Co³⁺) is coordinated, is chemically related to the porphyrin ring system of heme and heme proteins (see Fig. 5-1). A fifth coordination position of cobalt is filled by dimethylbenzimidazole ribonucleotide (shaded yellow), bound covalently by its 3'-phosphate group to a side chain of the corrin ring, through aminoisopropanol. The formation of this complex cofactor occurs in one of only two known reactions in which triphosphate is cleaved from ATP (Fig. 3); the other reaction is the formation of S-adenosylmethionine from ATP and methionine (see Fig. 18-18).

Vitamin B₁₂ as usually isolated is called **cyanocobalamin**, because it contains a cyano group (picked up during purification) attached to cobalt in the sixth coordination position. In **5'-deoxyadenosylcobalamin**, the cofactor for methylmalonyl-CoA mutase, the cyano group is replaced by the **5'-deoxyadenosyl group** (red in Fig. 2), covalently bound through C-5' to the cobalt. The three-dimensional structure of the cofactor was determined by Dorothy Crowfoot Hodgkin in 1956, using x-ray crystallography.

The key to understanding how coenzyme B₁₂ catalyzes hydrogen exchange lies in the properties of the covalent bond between cobalt and C-5' of the deoxyadeno-

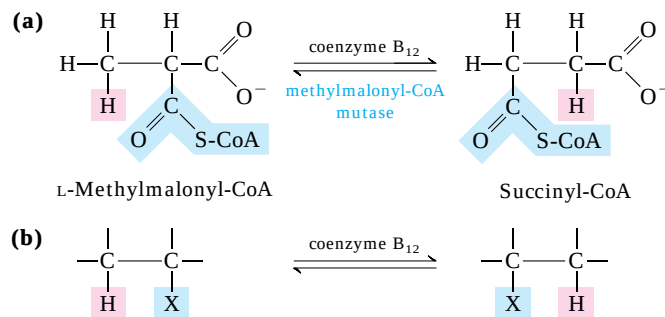


FIGURE 1

syl group (Fig. 2). This is a relatively weak bond; its bond dissociation energy is about 110 kJ/mol, compared with 348 kJ/mol for a typical C—C bond or 414 kJ/mol for a C—H bond. Merely illuminating the compound with visible light is enough to break this Co—C bond. (This extreme photolability probably accounts for the absence of vitamin B₁₂ in plants.) Dissociation produces a 5'-deoxyadenosyl radical and the Co²⁺

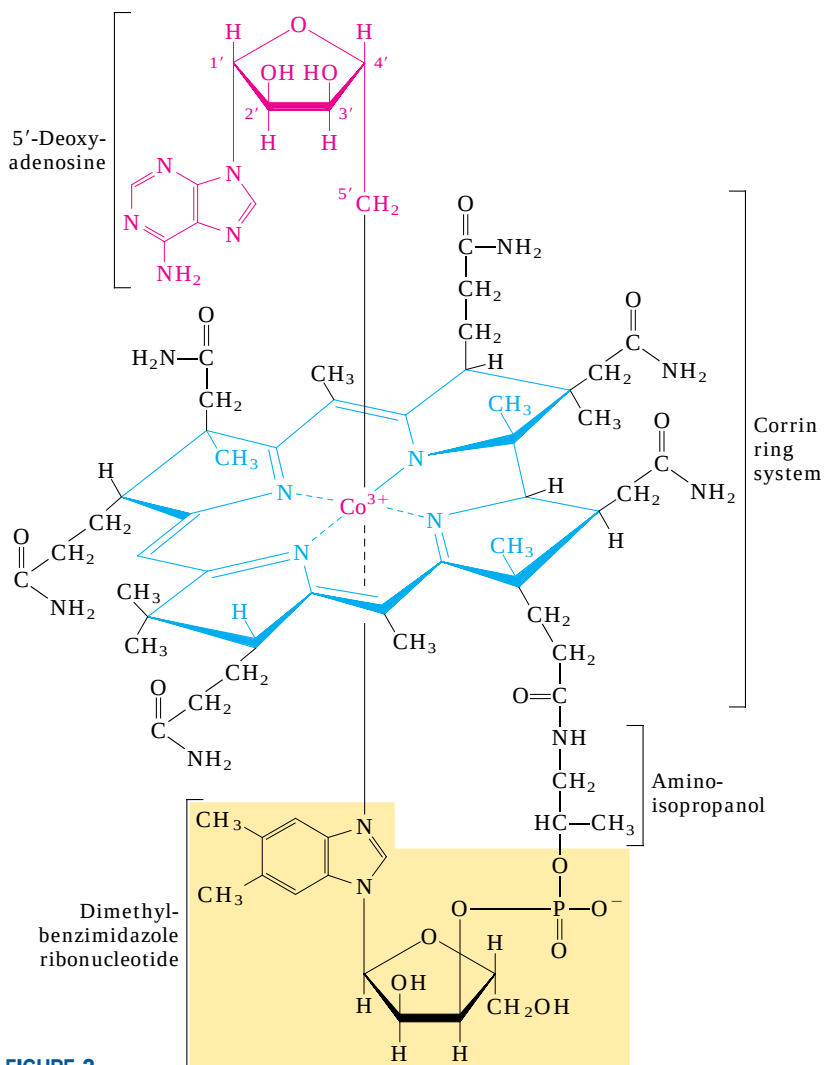


FIGURE 2



Dorothy Crowfoot Hodgkin,
1910–1994

form of the vitamin. The chemical function of 5'-deoxyadenosylcobalamin is to generate free radicals in this way, thus initiating a series of transformations such as that illustrated in Figure 4—a postulated mechanism for the reaction catalyzed by methylmalonyl-CoA mutase and a number of other coenzyme B₁₂-dependent transformations.

① The enzyme first breaks the Co—C bond in the cofactor, leaving the coenzyme in its Co²⁺ form and producing the 5'-deoxyadenosyl free radical. ② This radical now abstracts a hydrogen atom from the substrate, converting the substrate to a radical and producing 5'-deoxyadenosine. ③ Rearrangement of the substrate radical yields another radical, in which the migrating group X (—CO—S-CoA for methylmalonyl-CoA mutase) has moved to the adjacent carbon to form a radical that has the carbon skeleton of the eventual product (a four-carbon straight chain). The hydrogen atom initially abstracted from the substrate is now part of the —CH₃ group of 5'-deoxyadenosine. ④ One of the hydrogens from this same —CH₃ group (it can be the same one originally abstracted) is now returned to

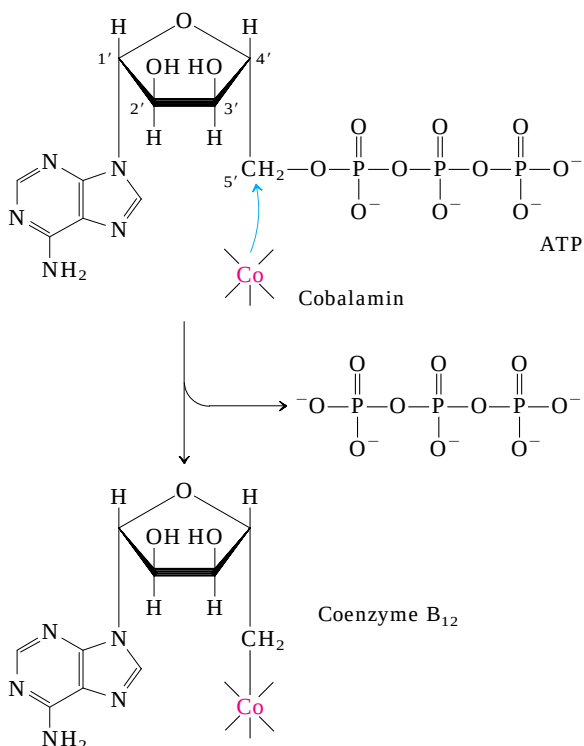
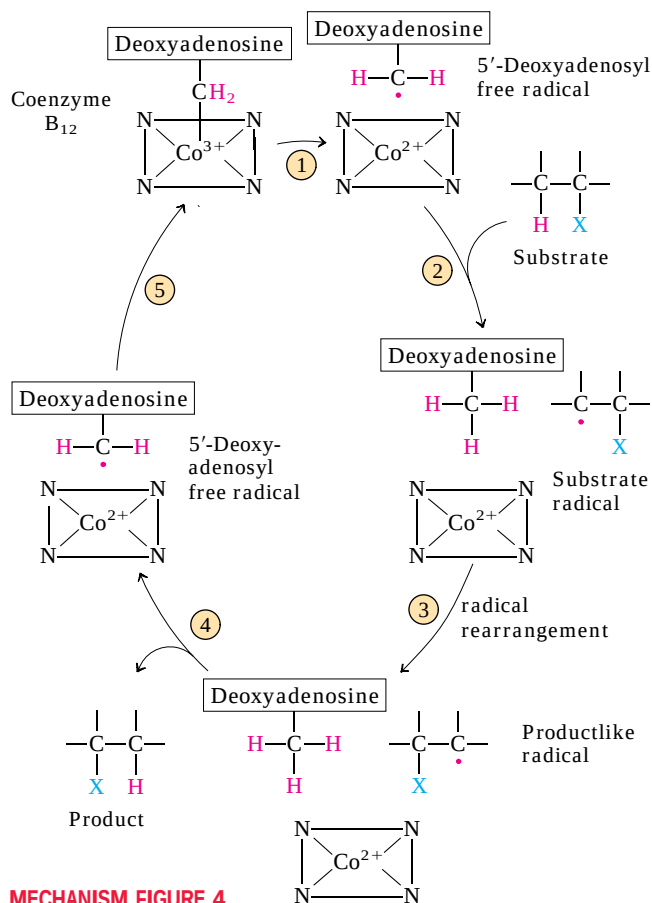


FIGURE 3

the productlike radical, generating the product and regenerating the deoxyadenosyl free radical. ⑤ The bond re-forms between cobalt and the —CH₂ group of the deoxyadenosyl radical, destroying the free radical and regenerating the cofactor in its Co³⁺ form, ready to undergo another catalytic cycle. In this postulated mechanism, the migrating hydrogen atom never exists as a free species and is thus never free to exchange with the hydrogen of surrounding water molecules.



Vitamin B₁₂ deficiency results in serious disease. This vitamin is not made by plants or animals and can be synthesized only by a few species of microorganisms. It is required by healthy people in only minute amounts, about 3 μg/day. The severe disease **pernicious anemia** results from failure to absorb vitamin B₁₂ efficiently from the intestine, where it is synthesized by intestinal bacteria or obtained from digestion of meat. Individuals with this disease do not produce sufficient amounts of **intrinsic factor**, a glycoprotein essential to vitamin B₁₂ absorption. The pathology in pernicious anemia includes reduced production of erythrocytes, reduced levels of hemoglobin, and severe, progressive impairment of the central nervous system. Administration of large doses of vitamin B₁₂ alleviates these symptoms in at least some cases. ■



MECHANISM FIGURE 4

More than 20 other human genetic defects in fatty acid transport or oxidation have been documented, most much less common than the defect in MCAD. One of the most severe disorders results from loss of the long-chain β -hydroxyacyl-CoA dehydrogenase activity of the tri-functional protein, TFP. Other disorders include defects in the α or β subunits that affect all three activities of TFP and cause serious heart disease and abnormal skeletal muscle. ■

Peroxisomes Also Carry Out β Oxidation

The mitochondrial matrix is the major site of fatty acid oxidation in animal cells, but in certain cells other compartments also contain enzymes capable of oxidizing fatty acids to acetyl-CoA, by a pathway similar to, but not identical with, that in mitochondria. In plant cells, the major site of β oxidation is not mitochondria but peroxisomes.

In **peroxisomes**, membrane-enclosed organelles of animal and plant cells, the intermediates for β oxidation of fatty acids are coenzyme A derivatives, and the process consists of four steps, as in mitochondrial β oxidation (Fig. 17-13): (1) dehydrogenation, (2) addition of water to the resulting double bond, (3) oxidation of the β -hydroxyacyl-CoA to a ketone, and (4) thiolytic cleavage by coenzyme A. (The identical reactions also occur in glyoxysomes, as discussed below.)

One difference between the peroxisomal and mitochondrial pathways is in the chemistry of the first step. In peroxisomes, the flavoprotein acyl-CoA oxidase that introduces the double bond passes electrons directly to O_2 , producing H_2O_2 (Fig. 17-13). This strong and potentially damaging oxidant is immediately cleaved to H_2O and O_2 by **catalase**. Recall that in mitochondria, the electrons removed in the first oxidation step pass through the respiratory chain to O_2 to produce H_2O , and this process is accompanied by ATP synthesis. In peroxisomes, the energy released in the first oxidative step of fatty acid breakdown is not conserved as ATP, but is dissipated as heat.



A second important difference between mitochondrial and peroxisomal β oxidation in mammals is in the specificity for fatty acyl-CoAs; the peroxisomal system is much more active on very-long-chain fatty acids such as hexacosanoic acid (26:0) and on branched-chain fatty acids such as phytanic acid and pristanic acid (see Fig. 17-17). These less-common fatty acids are obtained in the diet from dairy products, the fat of ruminant animals, meat, and fish. Their catabolism in the peroxisome involves several auxiliary enzymes unique to this organelle. The inability to oxidize these compounds is responsible for several serious human diseases. Individuals with **Zellweger syndrome** are unable to make peroxisomes and therefore lack all the metabolism unique to that organelle. In **X-linked adrenoleukodystrophy (XALD)**, peroxisomes fail to

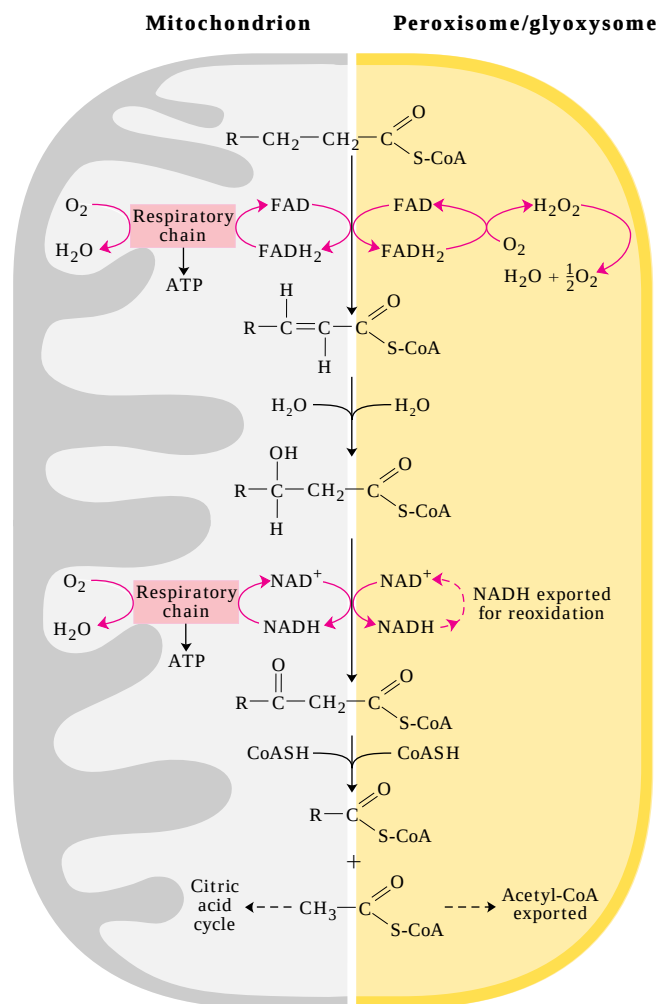


FIGURE 17-13 Comparison of β oxidation in mitochondria and in peroxisomes and glyoxysomes. The peroxisomal/glyoxysomal system differs from the mitochondrial system in two respects: (1) in the first oxidative step electrons pass directly to O_2 , generating H_2O_2 , and (2) the NADH formed in the second oxidative step cannot be reoxidized in the peroxisome or glyoxysome, so reducing equivalents are exported to the cytosol, eventually entering mitochondria. The acetyl-CoA produced by peroxisomes and glyoxysomes is also exported; the acetate from glyoxysomes (organelles found only in germinating seeds) serves as a biosynthetic precursor (see Fig. 17-14). Acetyl-CoA produced in mitochondria is further oxidized in the citric acid cycle.

oxidize very-long-chain fatty acids, apparently for lack of a functional transporter for these fatty acids in the peroxisomal membrane. Both defects lead to accumulation in the blood of very-long-chain fatty acids, especially 26:0. XALD affects young boys before the age of 10 years, causing loss of vision, behavioral disturbances, and death within a few years. ■

In mammals, high concentrations of fats in the diet result in increased synthesis of the enzymes of peroxisomal β oxidation in the liver. Liver peroxisomes do not contain the enzymes of the citric acid cycle and cannot catalyze the oxidation of acetyl-CoA to CO_2 . Instead,

long-chain or branched fatty acids are catabolized to shorter-chain products, such as hexanoyl-CoA, which are exported to mitochondria and completely oxidized.

Plant Peroxisomes and Glyoxysomes Use Acetyl-CoA from β Oxidation as a Biosynthetic Precursor

In plants, fatty acid oxidation does not occur primarily in mitochondria (as noted earlier) but in the peroxisomes of leaf tissue and in the glyoxysomes of germinating seeds. Plant peroxisomes and glyoxysomes are similar in structure and function; glyoxysomes, which occur only in germinating seeds, may be considered specialized peroxisomes. The biological role of β oxidation in these organelles is to use stored lipids primarily to provide but biosynthetic precursors, not energy.

During seed germination, stored triacylglycerols are converted into glucose, sucrose, and a wide variety of essential metabolites (Fig. 17–14). Fatty acids released from the triacylglycerols are first activated to their coenzyme A derivatives and oxidized in glyoxysomes by the same four-step process that takes place in peroxisomes (Fig. 17–13). The acetyl-CoA produced is converted via the glyoxylate cycle (see Fig. 16–20) to four-carbon precursors for gluconeogenesis (see Fig. 14–18). Glyoxysomes, like peroxisomes, contain high concentrations of catalase, which converts the H_2O_2 produced by β oxidation to H_2O and O_2 .

The β -Oxidation Enzymes of Different Organelles Have Diverged during Evolution

Although the β -oxidation reactions in mitochondria are essentially the same as those in peroxisomes and glyoxysomes, the enzymes (isozymes) differ significantly between the two types of organelles. The differences apparently reflect an evolutionary divergence that occurred very early, with the separation of gram-positive and gram-negative bacteria (see Fig. 1–6).

In mitochondria, the four β -oxidation enzymes that act on short-chain fatty acyl-CoAs are separate, soluble proteins (as noted earlier), similar in structure to the analogous enzymes of gram-positive bacteria (Fig. 17–15a). The gram-negative bacteria have four activities in three soluble subunits (Fig. 17–15b), and the eukaryotic enzyme system that acts on long-chain fatty acids—the trifunctional protein, TFP—has three enzyme activities in two subunits that are membrane-associated (Fig. 17–15c). The β -oxidation enzymes of plant peroxisomes and glyoxysomes, however, form a complex of proteins, one of which contains four enzymatic activities in a single polypeptide chain (Fig. 17–15d). The first enzyme, acyl-CoA oxidase, is a single polypeptide chain; the **multifunctional protein (MFP)** contains the second and third enzyme activities (enoyl-CoA hydratase and hydroxyacyl-CoA dehydrogenase) as well as two auxiliary activities needed for the oxidation of unsaturated

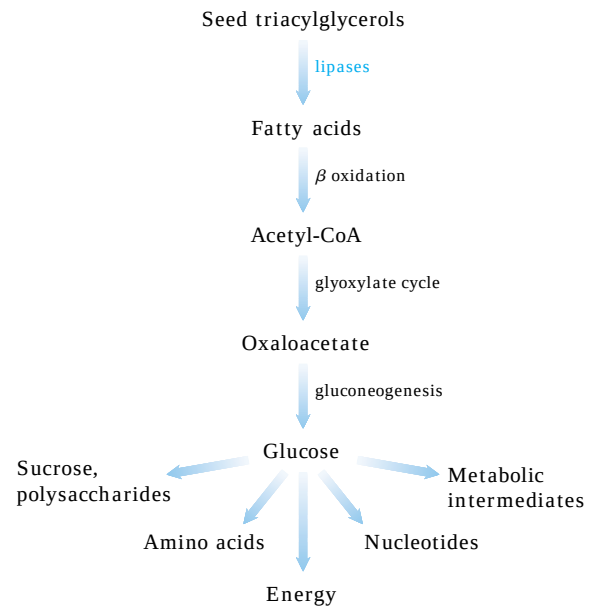


FIGURE 17–14 Triacylglycerols as glucose source in seeds. β Oxidation is one stage in a pathway that converts stored triacylglycerols to glucose in germinating seeds. For more detail, see Figure 16–22.

fatty acids (Δ^3 -hydroxyacyl-CoA epimerase and Δ^3, Δ^2 -enoyl-CoA isomerase); the fourth enzyme, thiolase, is a separate, soluble polypeptide.

It is interesting that the enzymes that catalyze essentially the reversal of β oxidation in the synthesis of fatty acids are also organized differently in prokaryotes and eukaryotes; in bacteria, the seven enzymes needed for fatty acid synthesis are separate polypeptides, but in mammals, all seven activities are part of a single, huge polypeptide chain (see Fig. 21–7). One advantage to the cell in having several enzymes of the same pathway encoded in a single polypeptide chain is that this solves the problem of regulating the synthesis of enzymes that must interact functionally; regulation of the expression of *one* gene ensures production of the same number of active sites for all enzymes in the path. When each enzyme activity is on a separate polypeptide, some mechanism is required to coordinate the synthesis of all the gene products. The *disadvantage* of having several activities on the same polypeptide is that the longer the polypeptide chain, the greater is the probability of a mistake in its synthesis: a single incorrect amino acid in the chain may make all the enzyme activities in that chain useless. Comparison of the gene structures for these proteins in many species may shed light on the reasons for the selection of one or the other strategy in evolution.

The ω Oxidation of Fatty Acids Occurs in the Endoplasmic Reticulum

Although mitochondrial β oxidation, in which enzymes act at the carboxyl end of a fatty acid, is by far the most

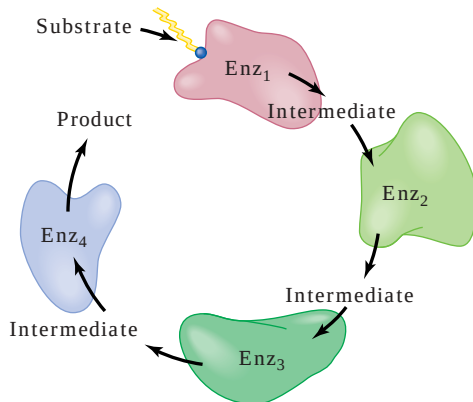
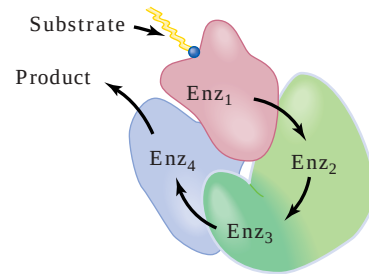
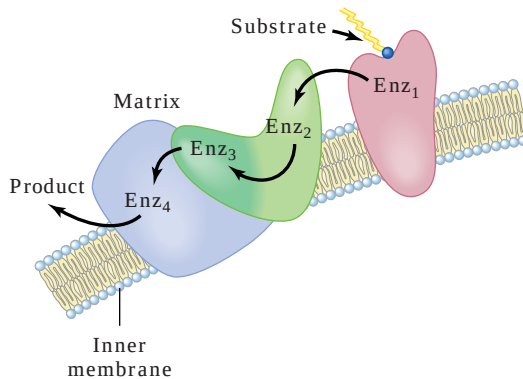
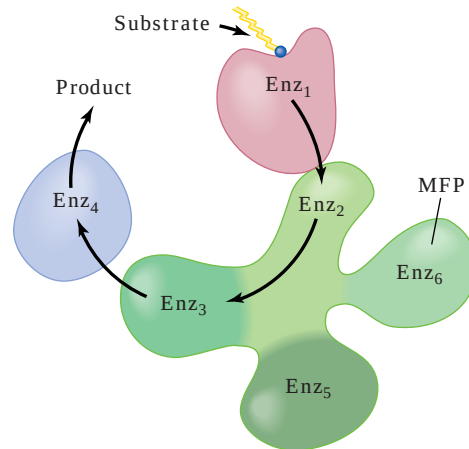
(a) Gram-positive bacteria and mitochondrial short-chain-specific system**(b)** Gram-negative bacteria**(c)** Mitochondrial very-long-chain-specific system**(d)** Peroxisomal and glyoxysomal system of plants

FIGURE 17-15 The enzymes of β oxidation. Shown here are the different subunit structures of the enzymes of β oxidation in gram-positive and gram-negative bacteria, mitochondria, and plant peroxisomes and glyoxysomes. Enz₁ is acyl-CoA dehydrogenase; Enz₂, enoyl-CoA hydratase; Enz₃, L- β -hydroxyacyl-CoA dehydrogenase; Enz₄, thiolase; Enz₅, D-3-hydroxyacyl-CoA epimerase, and Enz₆, Δ^3, Δ^2 -enoyl-CoA isomerase. **(a)** The four enzymes of β oxidation in gram-positive bacteria are separate, soluble entities, as are those of the short-chain-specific system of mitochondria. **(b)** In gram-negative bacteria, the four enzyme

activities reside in three polypeptides; enzymes 2 and 3 are parts of a single polypeptide chain. **(c)** The very-long-chain-specific system of mitochondria is also composed of three polypeptides, one of which includes the activities of enzymes 2 and 3; in this case, the system is bound to the inner mitochondrial membrane. **(d)** In the peroxisomal and glyoxysomal β -oxidation systems of plants, enzymes 1 and 4 are separate polypeptides, but enzymes 2 and 3, as well as two auxiliary enzymes, are part of a single polypeptide chain, the multifunctional protein, MFP.

important catabolic fate for fatty acids in animal cells, there is another pathway in some species, including vertebrates, that involves oxidation of the ω (omega) carbon—the carbon most distant from the carboxyl group. The enzymes unique to ω oxidation are located (in vertebrates) in the endoplasmic reticulum of liver and kidney, and the preferred substrates are fatty acids of 10 or 12 carbon atoms. In mammals ω oxidation is normally a minor pathway for fatty acid degradation, but when β oxidation is defective (because of mutation or a carnitine deficiency, for example) it becomes more important.

The first step introduces a hydroxyl group onto the ω carbon (Fig. 17-16). The oxygen for this group comes from molecular oxygen (O_2) in a complex reaction that involves cytochrome P450 and the electron donor NADPH. Reactions of this type are catalyzed by **mixed-function oxidases**, described in Box 21-1. Two more enzymes now act on the ω carbon: **alcohol dehydrogenase** oxidizes the hydroxyl group to an aldehyde, and **aldehyde dehydrogenase** oxidizes the aldehyde group to a carboxylic acid, producing a fatty acid with a carboxyl group at each end. At this point, either end can be attached to coenzyme A, and the molecule can en-

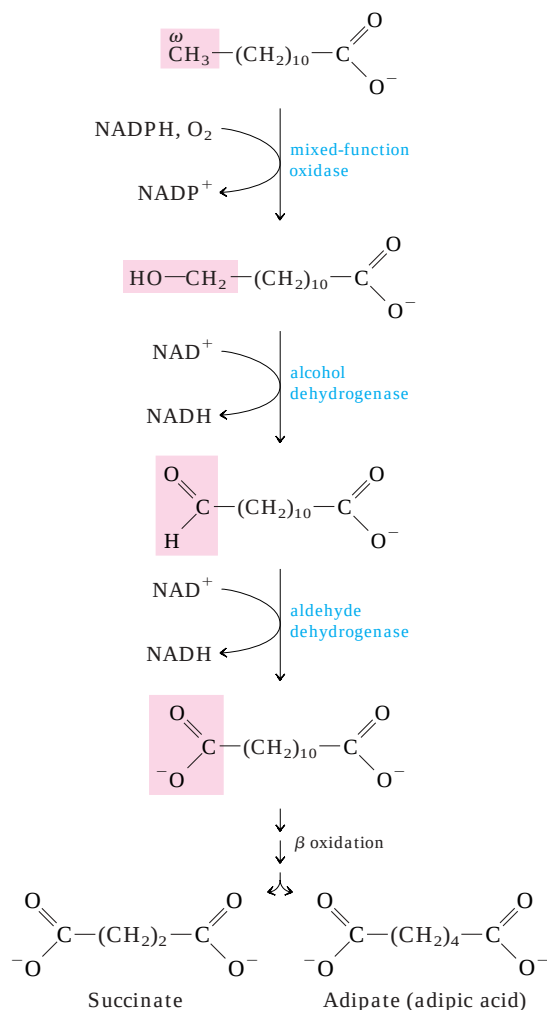


FIGURE 17-16 The ω oxidation of fatty acids in the endoplasmic reticulum. This alternative to β oxidation begins with oxidation of the carbon most distant from the α carbon—the ω (omega) carbon. The substrate is usually a medium-chain fatty acid; shown here is lauric acid (laurate). This pathway is generally not the major route for oxidative catabolism of fatty acids.

ter the mitochondrion and undergo β oxidation by the normal route. In each pass through the β -oxidation pathway, the “double-ended” fatty acid yields dicarboxylic acids such as succinic acid, which can enter the citric acid cycle, and adipic acid (Fig. 17-16).

Phytanic Acid Undergoes α Oxidation in Peroxisomes



The presence of a methyl group on the β carbon of a fatty acid makes β oxidation impossible, and these branched fatty acids are catabolized in peroxisomes of animal cells by **α oxidation**. In the oxidation of phytanic acid, for example (Fig. 17-17), phytanoyl-CoA is hydroxylated on its α carbon, in a reaction that involves molecular oxygen; decarboxylated to form an aldehyde one carbon shorter; and then oxidized to the

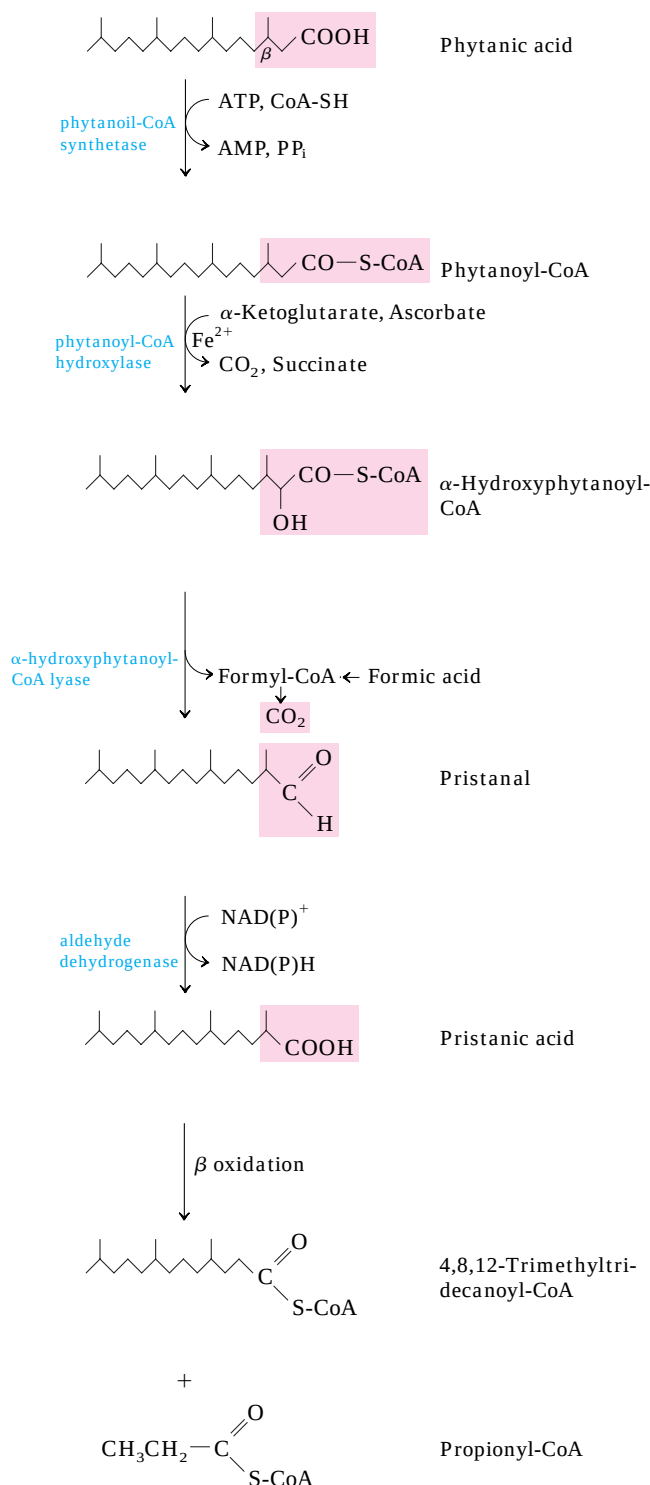


FIGURE 17-17 The α oxidation of a branched-chain fatty acid (phytanic acid) in peroxisomes. Phytanic acid has a methyl-substituted β carbon and therefore cannot undergo β oxidation. The combined action of the enzymes shown here removes the carboxyl carbon of phytanic acid, to produce pristanic acid, in which the β carbon is unsubstituted, allowing oxidation. Notice that β oxidation of pristanic acid releases propionyl-CoA, not acetyl-CoA. This is further catabolized as in Figure 17-11. (The details of the reaction that produces pristanal remain controversial.)

corresponding carboxylic acid, which now has no substituent on the β carbon and can be oxidized further by β oxidation. **Refsum's disease**, resulting from a genetic defect in phytanoyl-CoA hydroxylase, leads to very high blood levels of phytanic acid and severe neurological problems including blindness and deafness. ■

SUMMARY 17.2 Oxidation of Fatty Acids

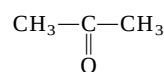
- In the first stage of β oxidation, four reactions remove each acetyl-CoA unit from the carboxyl end of a saturated fatty acyl-CoA:
 - (1) dehydrogenation of the α and β carbons (C-2 and C-3) by FAD-linked acyl-CoA dehydrogenases, (2) hydration of the resulting *trans*- Δ^2 double bond by enoyl-CoA hydratase, (3) dehydrogenation of the resulting L- β -hydroxyacyl-CoA by NAD-linked β -hydroxyacyl-CoA dehydrogenase, and (4) CoA-requiring cleavage of the resulting β -ketoacyl-CoA by thiolase, to form acetyl-CoA and a fatty acyl-CoA shortened by two carbons. The shortened fatty acyl-CoA then reenters the sequence.
- In the second stage of fatty acid oxidation, the acetyl-CoA is oxidized to CO_2 in the citric acid cycle. A large fraction of the theoretical yield of free energy from fatty acid oxidation is recovered as ATP by oxidative phosphorylation, the final stage of the oxidative pathway.
- Malonyl-CoA, an early intermediate of fatty acid synthesis, inhibits carnitine acyltransferase I, preventing fatty acid entry into mitochondria. This blocks fatty acid breakdown while synthesis is occurring.
- Genetic defects in the medium-chain acyl-CoA dehydrogenase result in serious human disease, as do mutations in other components of the β -oxidation system.
- Oxidation of unsaturated fatty acids requires two additional enzymes: enoyl-CoA isomerase and 2,4-dienoyl-CoA reductase. Odd-number fatty acids are oxidized by the β -oxidation pathway to yield acetyl-CoA and a molecule of propionyl-CoA. This is carboxylated to methylmalonyl-CoA, which is isomerized to succinyl-CoA in a reaction catalyzed by methylmalonyl-CoA mutase, an enzyme requiring coenzyme B_{12} .
- Peroxisomes of plants and animals, and glyoxysomes of plants, carry out β oxidation in four steps similar to those of the mitochondrial pathway in animals. The first oxidation step, however, transfers electrons directly to O_2 , generating H_2O_2 . Peroxisomes of animal tissues

specialize in the oxidation of very-long-chain fatty acids and branched fatty acids. In glyoxysomes, in germinating seeds, β oxidation is one step in the conversion of stored lipids into a variety of intermediates and products.

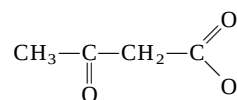
- The reactions of ω oxidation, occurring in the endoplasmic reticulum, produce dicarboxylic fatty acyl intermediates, which can undergo β oxidation at either end to yield short dicarboxylic acids such as succinate.

17.3 Ketone Bodies

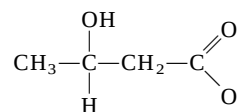
In humans and most other mammals, acetyl-CoA formed in the liver during oxidation of fatty acids can either enter the citric acid cycle (stage 2 of Fig. 17-7) or undergo conversion to the “ketone bodies,” **acetone**, **acetoacetate**, and **D- β -hydroxybutyrate**, for export to other tissues. (The term “bodies” is a historical artifact; the term is occasionally applied to insoluble particles, but these compounds are quite soluble in blood and urine.) Acetone, produced in smaller quantities than the other ketone bodies, is exhaled. Acetoacetate and D- β -hydroxybutyrate are transported by the blood to tissues other than the liver (extrahepatic tissues), where they are converted to acetyl-CoA and oxidized in the citric acid cycle, providing much of the energy required by tissues such as skeletal and heart muscle and the renal cortex. The brain, which preferentially uses glucose as fuel, can adapt to the use of acetoacetate or D- β -hydroxybutyrate under starvation conditions, when glucose is unavailable. The production and export of ketone bodies from the liver to extrahepatic tissues allow continued oxidation of fatty acids in the liver when acetyl-CoA is not being oxidized in the citric acid cycle.



Acetone



Acetoacetate



D- β -Hydroxybutyrate

Ketone Bodies, Formed in the Liver, Are Exported to Other Organs as Fuel

The first step in the formation of acetoacetate, occurring in the liver (Fig. 17-18), is the enzymatic condensation of two molecules of acetyl-CoA, catalyzed by thiolase;

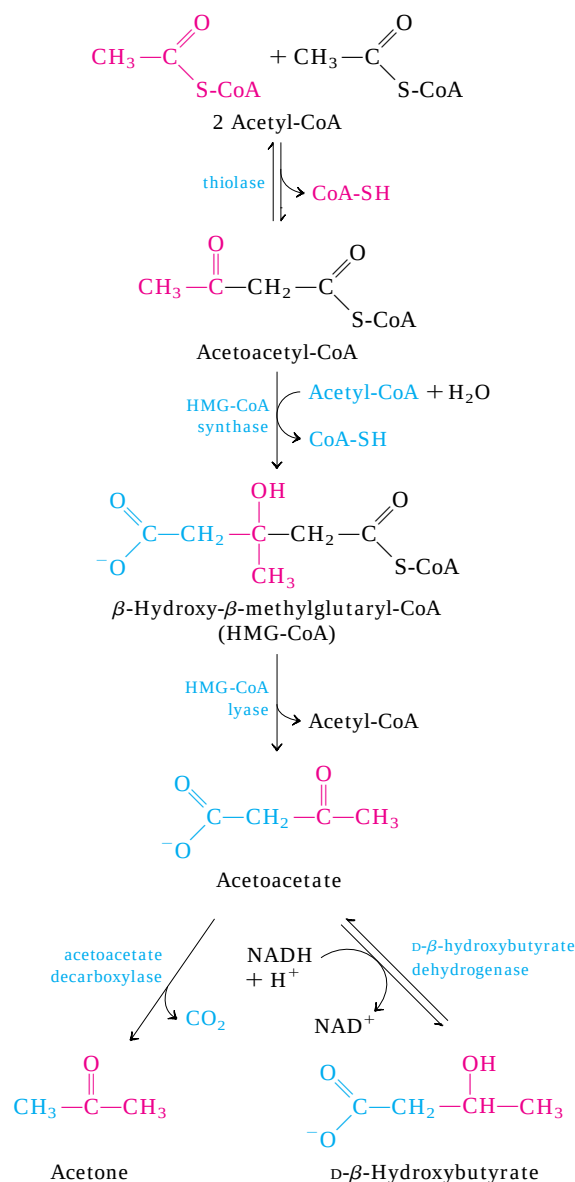


FIGURE 17-18 Formation of ketone bodies from acetyl-CoA.

Healthy, well-nourished individuals produce ketone bodies at a relatively low rate. When acetyl-CoA accumulates (as in starvation or untreated diabetes, for example), thiolase catalyzes the condensation of two acetyl-CoA molecules to acetoacetyl-CoA, the parent compound of the three ketone bodies. The reactions of ketone body formation occur in the matrix of liver mitochondria. The six-carbon compound β -hydroxy- β -methylglutaryl-CoA (HMG-CoA) is also an intermediate of sterol biosynthesis, but the enzyme that forms HMG-CoA in that pathway is cytosolic. HMG-CoA lyase is present only in the mitochondrial matrix.

this is simply the reversal of the last step of β oxidation. The acetoacetyl-CoA then condenses with acetyl-CoA to form **β -hydroxy- β -methylglutaryl-CoA (HMG-CoA)**, which is cleaved to free acetoacetate and acetyl-CoA. The acetoacetate is reversibly reduced by D- β -hydroxybutyrate dehydrogenase, a mitochondrial enzyme, to D- β -hydroxybutyrate. This enzyme is specific for the

D stereoisomer; it does not act on L- β -hydroxyacyl-CoAs and is not to be confused with L- β -hydroxyacyl-CoA dehydrogenase of the β -oxidation pathway.



In healthy people, acetone is formed in very small amounts from acetoacetate, which is easily decarboxylated, either spontaneously or by the action of **acetoacetate decarboxylase** (Fig. 17-18). Because individuals with untreated diabetes produce large quantities of acetoacetate, their blood contains significant amounts of acetone, which is toxic. Acetone is volatile and imparts a characteristic odor to the breath, which is sometimes useful in diagnosing diabetes.

In extrahepatic tissues, D- β -hydroxybutyrate is oxidized to acetoacetate by D- β -hydroxybutyrate dehydrogenase (Fig. 17-19). The acetoacetate is activated to its coenzyme A ester by transfer of CoA from succinyl-CoA, an intermediate of the citric acid cycle (see Fig. 16-7), in a reaction catalyzed by **β -ketoacyl-CoA transferase**. The acetoacetyl-CoA is then cleaved by thiolase to yield two acetyl-CoAs, which enter the citric acid cycle. Thus the ketone bodies are used as fuels.

The production and export of ketone bodies by the liver allow continued oxidation of fatty acids with only minimal oxidation of acetyl-CoA. When intermediates of the citric acid cycle are being siphoned off for glucose

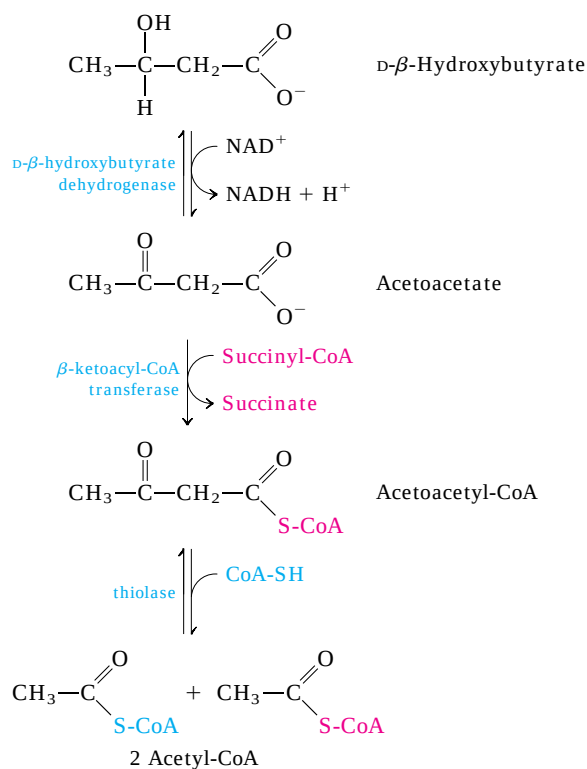


FIGURE 17-19 D- β -Hydroxybutyrate as a fuel. D- β -Hydroxybutyrate, synthesized in the liver, passes into the blood and thus to other tissues, where it is converted in three steps to acetyl-CoA. It is first oxidized to acetoacetate, which is activated with coenzyme A donated from succinyl-CoA, then split by thiolase. The acetyl-CoA thus formed is used for energy production.

synthesis by gluconeogenesis, for example, oxidation of cycle intermediates slows—and so does acetyl-CoA oxidation. Moreover, the liver contains only a limited amount of coenzyme A, and when most of it is tied up in acetyl-CoA, β oxidation slows for want of the free coenzyme. The production and export of ketone bodies free coenzyme A, allowing continued fatty acid oxidation.

Ketone Bodies Are Overproduced in Diabetes and during Starvation

 Starvation and untreated diabetes mellitus lead to overproduction of ketone bodies, with several associated medical problems. During starvation, gluconeogenesis depletes citric acid cycle intermediates, diverting acetyl-CoA to ketone body production (Fig. 17–20). In untreated diabetes, when the insulin level is insufficient, extrahepatic tissues cannot take up glucose efficiently from the blood, either for fuel or for conversion to fat. Under these conditions, levels of malonyl-CoA (the starting material for fatty acid synthesis) fall, inhibition of carnitine acyltransferase I is relieved, and fatty acids enter mitochondria to be degraded to acetyl-CoA—which cannot pass through the citric acid cycle because cycle intermediates have been drawn off for use as substrates in gluconeogenesis. The resulting accumulation of acetyl-CoA accelerates the formation of ketone bodies beyond the capacity of extrahepatic tissues to oxidize them. The increased blood levels of acetoacetate and D- β -hydroxybutyrate lower the blood pH, causing the condition known as **acidosis**. Extreme acidosis can lead to coma and in some cases death. Ketone bodies in the blood and urine of untreated diabetics can reach extraordinary levels—a blood concentration of 90 mg/100 mL (compared with a normal level of <3 mg/100 mL) and urinary excretion of 5,000 mg/24 hr (compared with a normal rate of $\leq$ 125 mg/24 hr). This condition is called **ketosis**.

Individuals on very low-calorie diets, using the fats stored in adipose tissue as their major energy source, also have increased levels of ketone bodies in their blood and urine. These levels must be monitored to avoid the dangers of acidosis and ketosis (ketoacidosis). ■

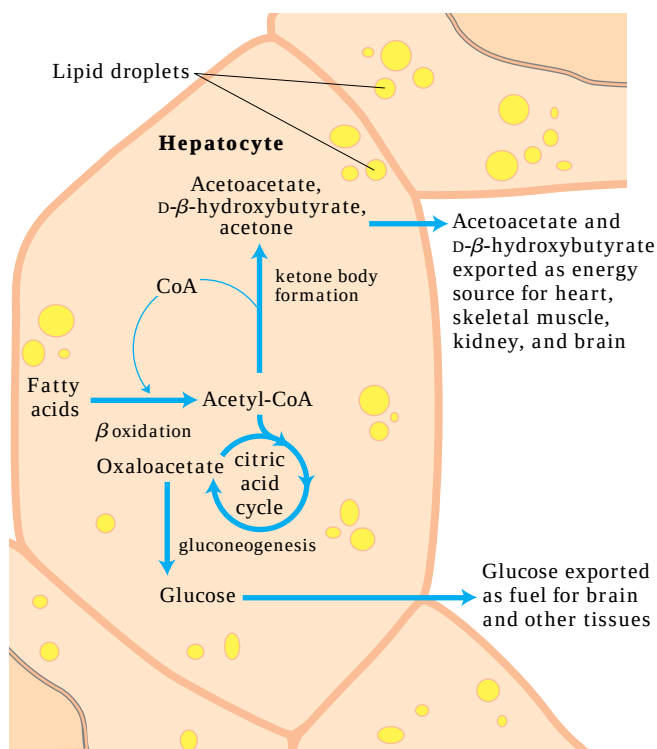


FIGURE 17–20 Ketone body formation and export from the liver.

Conditions that promote gluconeogenesis (untreated diabetes, severely reduced food intake) slow the citric acid cycle (by drawing off oxaloacetate) and enhance the conversion of acetyl-CoA to acetoacetate. The released coenzyme A allows continued β oxidation of fatty acids.

SUMMARY 17.3 Ketone Bodies

- The ketone bodies—acetone, acetoacetate, and D- β -hydroxybutyrate—are formed in the liver. The latter two compounds serve as fuel molecules in extrahepatic tissues, through oxidation to acetyl-CoA and entry into the citric acid cycle.
- Overproduction of ketone bodies in uncontrolled diabetes or severely reduced calorie intake can lead to acidosis or ketosis.

Key Terms

Terms in bold are defined in the glossary.

β oxidation XXX	carnitine shuttle XXX
chylomicron XXX	carnitine acyltransferase I XXX
apolipoprotein XXX	acyl-carnitine/carnitine transporter XXX
lipoprotein XXX	carnitine acyltransferase II XXX
perilipins XXX	trifunctional protein (TFP) XXX
hormone-sensitive lipase XXX	
free fatty acids XXX	
serum albumin XXX	

methylmalonyl-CoA mutase XXX
coenzyme B₁₂ XXX
pernicious anemia XXX
intrinsic factor XXX
malonyl-CoA XXX
medium-chain acyl-CoA dehydrogenase (MCAD) XXX

multifunctional protein (MFP) XXX
ω oxidation XXX
mixed-function oxidases XXX
α oxidation XXX
acidosis XXX
ketosis XXX

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Problems

1. Energy in Triacylglycerols On a per-carbon basis, where does the largest amount of biologically available energy in triacylglycerols reside: in the fatty acid portions or the glycerol portion? Indicate how knowledge of the chemical structure of triacylglycerols provides the answer.

2. Fuel Reserves in Adipose Tissue Triacylglycerols, with their hydrocarbon-like fatty acids, have the highest energy content of the major nutrients.

(a) If 15% of the body mass of a 70.0 kg adult consists of triacylglycerols, what is the total available fuel reserve, in

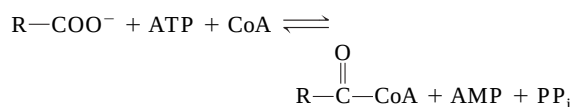
both kilojoules and kilocalories, in the form of triacylglycerols? Recall that $1.00 \text{ kcal} = 4.18 \text{ kJ}$.

(b) If the basal energy requirement is approximately $8,400 \text{ kJ/day}$ ($2,000 \text{ kcal/day}$), how long could this person survive if the oxidation of fatty acids stored as triacylglycerols were the only source of energy?

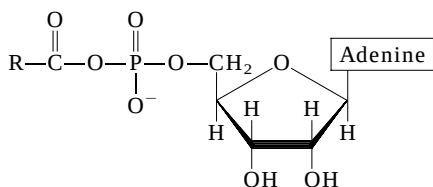
(c) What would be the weight loss in pounds per day under such starvation conditions ($1 \text{ lb} = 0.454 \text{ kg}$)?

3. Common Reaction Steps in the Fatty Acid Oxidation Cycle and Citric Acid Cycle Cells often use the same enzyme reaction pattern for analogous metabolic conversions. For example, the steps in the oxidation of pyruvate to acetyl-CoA and of α -ketoglutarate to succinyl-CoA, although catalyzed by different enzymes, are very similar. The first stage of fatty acid oxidation follows a reaction sequence closely resembling a sequence in the citric acid cycle. Use equations to show the analogous reaction sequences in the two pathways.

4. Chemistry of the Acyl-CoA Synthetase Reaction Fatty acids are converted to their coenzyme A esters in a reversible reaction catalyzed by acyl-CoA synthetase:



(a) The enzyme-bound intermediate in this reaction has been identified as the mixed anhydride of the fatty acid and adenosine monophosphate (AMP), acyl-AMP:



Write two equations corresponding to the two steps of the reaction catalyzed by acyl-CoA synthetase.

(b) The acyl-CoA synthetase reaction is readily reversible, with an equilibrium constant near 1. How can this reaction be made to favor formation of fatty acyl-CoA?

5. Oxidation of Tritiated Palmitate Palmitate uniformly labeled with tritium (^3H) to a specific activity of 2.48×10^8 counts per minute (cpm) per micromole of palmitate is added to a mitochondrial preparation that oxidizes it to acetyl-CoA. The acetyl-CoA is isolated and hydrolyzed to acetate. The specific activity of the isolated acetate is 1.00×10^7 cpm/ μmol . Is this result consistent with the β -oxidation pathway? Explain. What is the final fate of the removed tritium?

6. Compartmentation in β Oxidation Free palmitate is activated to its coenzyme A derivative (palmitoyl-CoA) in the cytosol before it can be oxidized in the mitochondrion. If palmitate and [^{14}C]coenzyme A are added to a liver homogenate, palmitoyl-CoA isolated from the cytosolic fraction is radioactive, but that isolated from the mitochondrial fraction is not. Explain.

7. Comparative Biochemistry: Energy-Generating Pathways in Birds One indication of the relative impor-

tance of various ATP-producing pathways is the V_{max} of certain enzymes of these pathways. The values of V_{max} of several enzymes from the pectoral muscles (chest muscles used for flying) of pigeon and pheasant are listed below.

Enzyme	V_{max} ($\mu\text{mol substrate/min/g tissue}$)	
	Pigeon	Pheasant
Hexokinase	3.0	2.3
Glycogen phosphorylase	18.0	120.0
Phosphofructokinase-1	24.0	143.0
Citrate synthase	100.0	15.0
Triacylglycerol lipase	0.07	0.01

(a) Discuss the relative importance of glycogen metabolism and fat metabolism in generating ATP in the pectoral muscles of these birds.

(b) Compare oxygen consumption in the two birds.

(c) Judging from the data in the table, which bird is the long-distance flyer? Justify your answer.

(d) Why were these particular enzymes selected for comparison? Would the activities of triose phosphate isomerase and malate dehydrogenase be equally good bases for comparison? Explain.



8. Effect of Carnitine Deficiency

An individual developed a condition characterized by progressive muscular weakness and aching muscle cramps. The symptoms were aggravated by fasting, exercise, and a high-fat diet. The homogenate of a skeletal muscle specimen from the patient oxidized added oleate more slowly than did control homogenates, consisting of muscle specimens from healthy individuals. When carnitine was added to the patient's muscle homogenate, the rate of oleate oxidation equaled that in the control homogenates. The patient was diagnosed as having a carnitine deficiency.

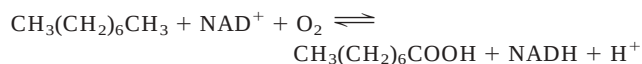
(a) Why did added carnitine increase the rate of oleate oxidation in the patient's muscle homogenate?

(b) Why were the patient's symptoms aggravated by fasting, exercise, and a high-fat diet?

(c) Suggest two possible reasons for the deficiency of muscle carnitine in this individual.

9. Fatty Acids as a Source of Water Contrary to legend, camels do not store water in their humps, which actually consist of large fat deposits. How can these fat deposits serve as a source of water? Calculate the amount of water (in liters) that a camel can produce from 1.0 kg of fat. Assume for simplicity that the fat consists entirely of tripalmitoylglycerol.

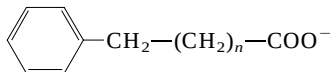
10. Petroleum as a Microbial Food Source Some microorganisms of the genera *Nocardia* and *Pseudomonas* can grow in an environment where hydrocarbons are the only food source. These bacteria oxidize straight-chain aliphatic hydrocarbons, such as octane, to their corresponding carboxylic acids:



How could these bacteria be used to clean up oil spills? What

would be some of the limiting factors to the efficiency of this process?

11. Metabolism of a Straight-Chain Phenylated Fatty Acid A crystalline metabolite was isolated from the urine of a rabbit that had been fed a straight-chain fatty acid containing a terminal phenyl group:



A 302 mg sample of the metabolite in aqueous solution was completely neutralized by 22.2 mL of 0.100 M NaOH.

(a) What is the probable molecular weight and structure of the metabolite?

(b) Did the straight-chain fatty acid contain an even or an odd number of methylene ($\text{—CH}_2\text{—}$) groups (i.e., is n even or odd)? Explain.



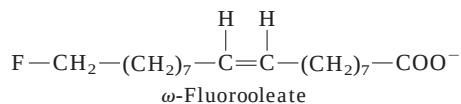
12. Fatty Acid Oxidation in Uncontrolled Diabetes When the acetyl-CoA produced during β oxidation in the liver exceeds the capacity of the citric acid cycle, the excess acetyl-CoA forms ketone bodies—acetone, acetoacetate, and $\text{D-}\beta$ -hydroxybutyrate. This occurs in severe, uncontrolled diabetes: because the tissues cannot use glucose, they oxidize large amounts of fatty acids instead. Although acetyl-CoA is not toxic, the mitochondrion must divert the acetyl-CoA to ketone bodies. What problem would arise if acetyl-CoA were not converted to ketone bodies? How does the diversion to ketone bodies solve the problem?

13. Consequences of a High-Fat Diet with No Carbohydrates Suppose you had to subsist on a diet of whale blubber and seal blubber, with little or no carbohydrate.

(a) What would be the effect of carbohydrate deprivation on the utilization of fats for energy?

(b) If your diet were totally devoid of carbohydrate, would it be better to consume odd- or even-numbered fatty acids? Explain.

14. Metabolic Consequences of Ingesting ω -Fluorooleate The shrub *Dichapetalum toxicarium*, native to Sierra Leone, produces ω -fluorooleate, which is highly toxic to warm-blooded animals.



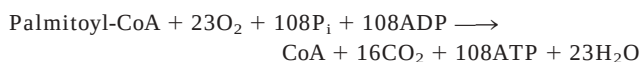
This substance has been used as an arrow poison, and powdered fruit from the plant is sometimes used as a rat poison (hence the plant's common name, ratsbane). Why is this substance so toxic? (Hint: Review Chapter 16, Problem 13.)

15. Role of FAD as Electron Acceptor Acyl-CoA dehydrogenase uses enzyme-bound FAD as a prosthetic group to dehydrogenate the α and β carbons of fatty acyl-CoA. What is the advantage of using FAD as an electron acceptor rather than NAD^+ ? Explain in terms of the standard reduction potentials for the Enz-FAD/FADH_2 ($E'^\circ = -0.219 \text{ V}$) and NAD^+/NADH ($E'^\circ = -0.320 \text{ V}$) half-reactions.

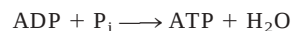
16. β Oxidation of Arachidic Acid How many turns of the fatty acid oxidation cycle are required for complete oxidation of arachidic acid (see Table 10–1) to acetyl-CoA?

17. Fate of Labeled Propionate If $[3\text{-}^{14}\text{C}]$ propionate (^{14}C in the methyl group) is added to a liver homogenate, ^{14}C -labeled oxaloacetate is rapidly produced. Draw a flow chart for the pathway by which propionate is transformed to oxaloacetate, and indicate the location of the ^{14}C in oxaloacetate.

18. Sources of H_2O Produced in β Oxidation The complete oxidation of palmitoyl-CoA to carbon dioxide and water is represented by the overall equation



Water is also produced in the reaction



but is not included as a product in the overall equation. Why?

19. Biological Importance of Cobalt In cattle, deer, sheep, and other ruminant animals, large amounts of propionate are produced in the rumen through the bacterial fermentation of ingested plant matter. Propionate is the principal source of glucose for these animals, via the route $\text{propionate} \rightarrow \text{oxaloacetate} \rightarrow \text{glucose}$. In some areas of the world, notably Australia, ruminant animals sometimes show symptoms of anemia with concomitant loss of appetite and retarded growth, resulting from an inability to transform propionate to oxaloacetate. This condition is due to a cobalt deficiency caused by very low cobalt levels in the soil and thus in plant matter. Explain.

20. Fat Loss during Hibernation Bears expend about $25 \times 10^6 \text{ J/day}$ during periods of hibernation, which may last as long as seven months. The energy required to sustain life is obtained from fatty acid oxidation. How much weight loss (in kilograms) has occurred after seven months? How might ketosis be minimized during hibernation? (Assume the oxidation of fat yields 38 kJ/g .)



chapter 18

AMINO ACID OXIDATION AND THE PRODUCTION OF UREA

- 18.1 Metabolic Fates of Amino Groups 657**
- 18.2 Nitrogen Excretion and the Urea Cycle 665**
- 18.3 Pathways of Amino Acid Degradation 671**

I chose the study of the synthesis of urea in the liver because it appeared to be a relatively simple problem.

—Hans Krebs, *article in Perspectives in Biology and Medicine*, 1970

We now turn our attention to the amino acids, the final class of biomolecules that, through their oxidative degradation, make a significant contribution to the generation of metabolic energy. The fraction of metabolic energy obtained from amino acids, whether they are derived from dietary protein or from tissue protein, varies greatly with the type of organism and with metabolic conditions. Carnivores can obtain (immediately following a meal) up to 90% of their energy requirements from amino acid oxidation, whereas herbivores may fill only a small fraction of their energy needs by this route. Most microorganisms can scavenge amino acids from their environment and use them as fuel when required by metabolic conditions. Plants, however, rarely if ever oxidize amino acids to provide energy; the carbohydrate produced from CO_2 and H_2O in photosynthesis is generally their sole energy source. Amino acid concentrations in plant tissues are carefully regulated to just meet the requirements for biosynthesis of proteins, nucleic acids, and other molecules needed to support growth. Amino acid catabolism does occur in plants, but its purpose is to produce metabolites for other biosynthetic pathways.

In animals, amino acids undergo oxidative degradation in three different metabolic circumstances:

1. During the normal synthesis and degradation of cellular proteins (protein turnover; Chapter 27), some amino acids that are released from protein breakdown and are not needed for new protein synthesis undergo oxidative degradation.
2. When a diet is rich in protein and the ingested amino acids exceed the body's needs for protein synthesis, the surplus is catabolized; amino acids cannot be stored.
3. During starvation or in uncontrolled diabetes mellitus, when carbohydrates are either unavailable or not properly utilized, cellular proteins are used as fuel.

Under all these metabolic conditions, amino acids lose their amino groups to form α -keto acids, the “carbon skeletons” of amino acids. The α -keto acids undergo oxidation to CO_2 and H_2O or, often more importantly, provide three- and four-carbon units that can be converted by gluconeogenesis into glucose, the fuel for brain, skeletal muscle, and other tissues.

The pathways of amino acid catabolism are quite similar in most organisms. The focus of this chapter is on the pathways in vertebrates, because these have received the most research attention. As in carbohydrate and fatty acid catabolism, the processes of amino acid degradation converge on the central catabolic pathways, with the carbon skeletons of most amino acids finding their way to the citric acid cycle. In some cases the reaction pathways of amino acid breakdown closely parallel steps in the catabolism of fatty acids (Chapter 17).

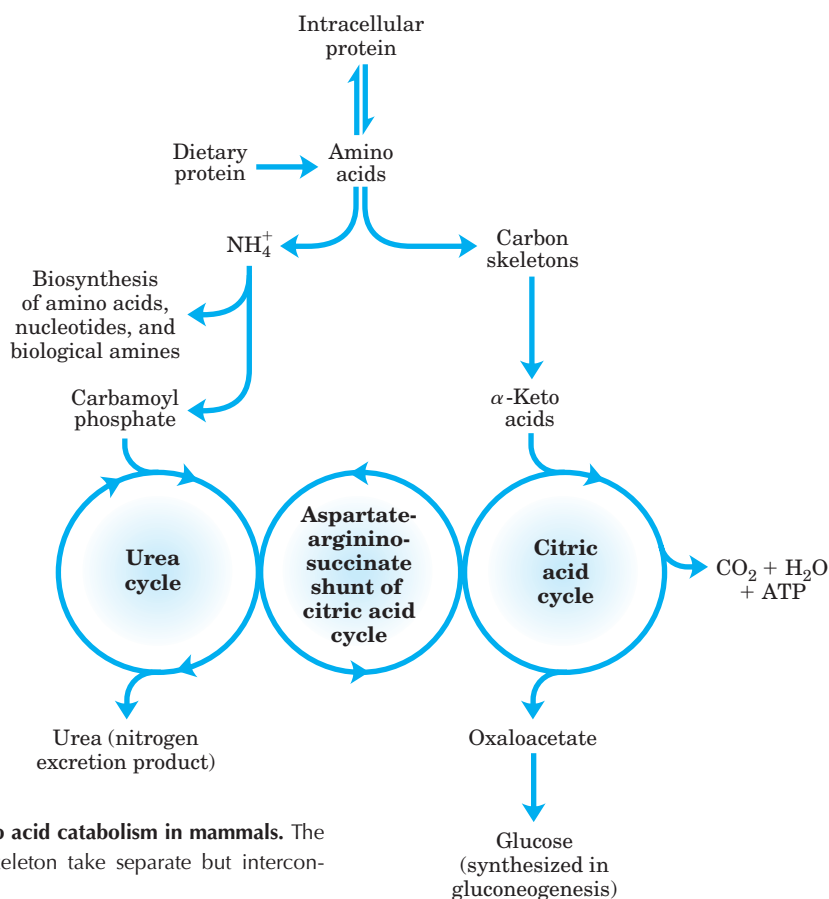


FIGURE 18-1 Overview of amino acid catabolism in mammals. The amino groups and the carbon skeleton take separate but interconnected pathways.

One important feature distinguishes amino acid degradation from other catabolic processes described to this point: every amino acid contains an amino group, and the pathways for amino acid degradation therefore include a key step in which the α -amino group is separated from the carbon skeleton and shunted into the pathways of amino group metabolism (Fig. 18-1). We deal first with amino group metabolism and nitrogen excretion, then with the fate of the carbon skeletons derived from the amino acids; along the way we see how the pathways are interconnected.

18.1 Metabolic Fates of Amino Groups

Nitrogen, N_2 , is abundant in the atmosphere but is too inert for use in most biochemical processes. Because only a few microorganisms can convert N_2 to biologically useful forms such as NH_3 (Chapter 22), amino groups are carefully husbanded in biological systems.

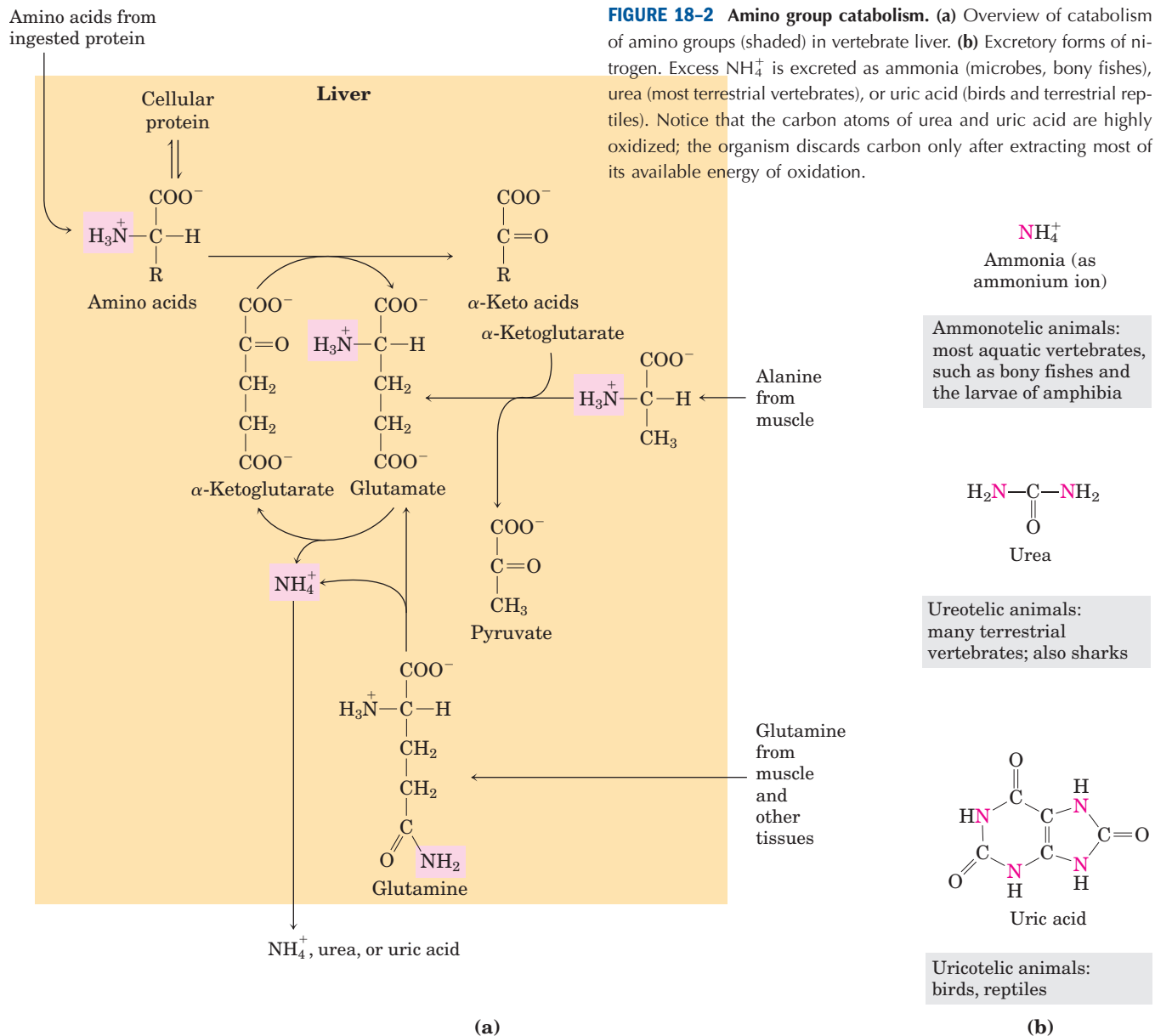
Figure 18-2a provides an overview of the catabolic pathways of ammonia and amino groups in vertebrates. Amino acids derived from dietary protein are the source of most amino groups. Most amino acids are metabolized in the liver. Some of the ammonia generated in this

process is recycled and used in a variety of biosynthetic pathways; the excess is either excreted directly or converted to urea or uric acid for excretion, depending on the organism (Fig. 18-2b). Excess ammonia generated in other (extrahepatic) tissues travels to the liver (in the form of amino groups, as described below) for conversion to the excretory form.

Glutamate and glutamine play especially critical roles in nitrogen metabolism, acting as a kind of general collection point for amino groups. In the cytosol of hepatocytes, amino groups from most amino acids are transferred to α -ketoglutarate to form glutamate, which enters mitochondria and gives up its amino group to form NH_4^+ . Excess ammonia generated in most other tissues is converted to the amide nitrogen of glutamine, which passes to the liver, then into liver mitochondria. Glutamine or glutamate or both are present in higher concentrations than other amino acids in most tissues.

In skeletal muscle, excess amino groups are generally transferred to pyruvate to form alanine, another important molecule in the transport of amino groups to the liver.

We begin with a discussion of the breakdown of dietary proteins, then give a general description of the metabolic fates of amino groups.



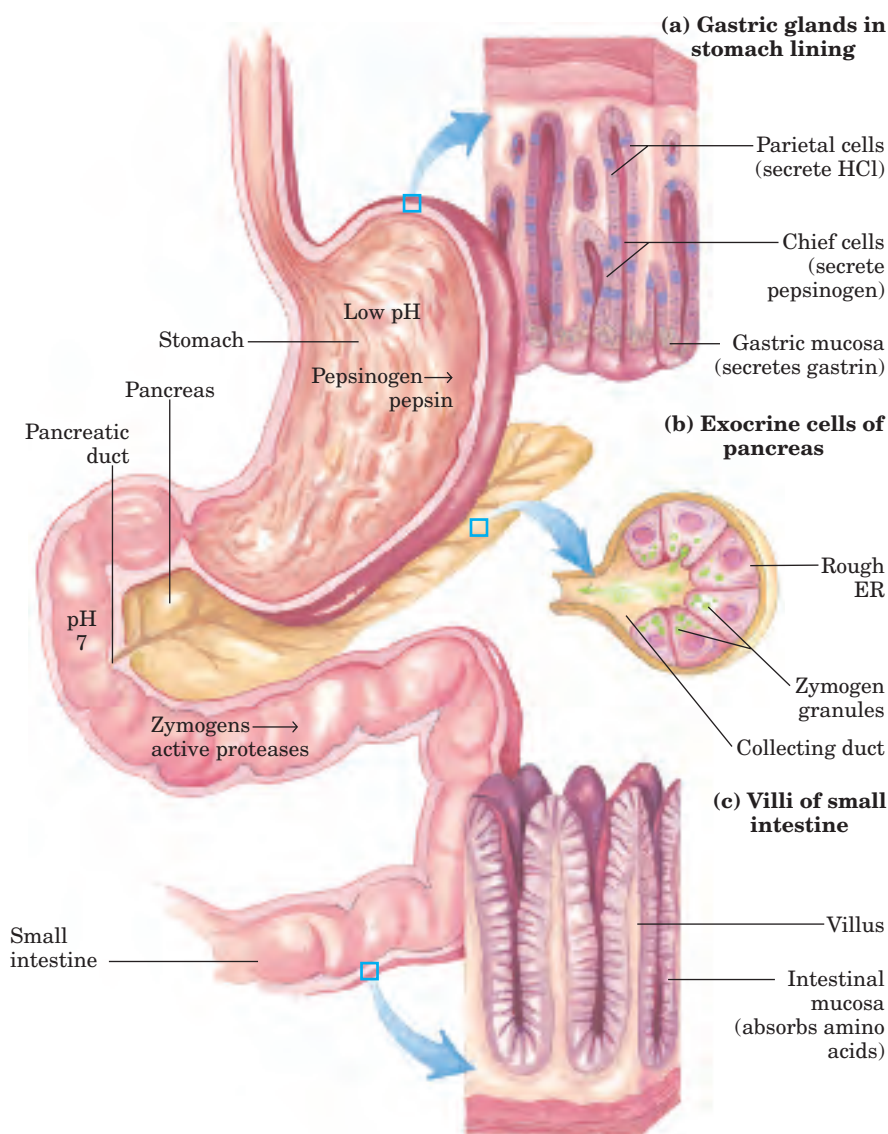
Dietary Protein Is Enzymatically Degraded to Amino Acids

In humans, the degradation of ingested proteins to their constituent amino acids occurs in the gastrointestinal tract. Entry of dietary protein into the stomach stimulates the gastric mucosa to secrete the hormone **gastrin**, which in turn stimulates the secretion of hydrochloric acid by the parietal cells and pepsinogen by the chief cells of the gastric glands (Fig. 18-3a). The acidic gastric juice (pH 1.0 to 2.5) is both an antiseptic, killing most bacteria and other foreign cells, and a denaturing agent, unfolding globular proteins and rendering their internal peptide bonds more accessible to enzymatic hydrolysis. **Pepsinogen** (M_r 40,554), an inactive precursor, or zymogen (p. 231), is converted to active pepsin

(M_r 34,614) by the enzymatic action of pepsin itself. In the stomach, pepsin hydrolyzes ingested proteins at peptide bonds on the amino-terminal side of the aromatic amino acid residues Phe, Trp, and Tyr (see Table 3-7), cleaving long polypeptide chains into a mixture of smaller peptides.

As the acidic stomach contents pass into the small intestine, the low pH triggers secretion of the hormone **secretin** into the blood. Secretin stimulates the pancreas to secrete bicarbonate into the small intestine to neutralize the gastric HCl, abruptly increasing the pH to about 7. (All pancreatic secretions pass into the small intestine through the pancreatic duct.) The digestion of proteins now continues in the small intestine. Arrival of amino acids in the upper part of the intestine (duodenum) causes release into the blood of the hormone

FIGURE 18-3 Part of the human digestive (gastrointestinal) tract. (a) The parietal cells and chief cells of the gastric glands secrete their products in response to the hormone gastrin. Pepsin begins the process of protein degradation in the stomach. (b) The cytoplasm of exocrine cells is completely filled with rough endoplasmic reticulum, the site of synthesis of the zymogens of many digestive enzymes. The zymogens are concentrated in membrane-enclosed transport particles called zymogen granules. When an exocrine cell is stimulated, its plasma membrane fuses with the zymogen granule membrane and zymogens are released into the lumen of the collecting duct by exocytosis. The collecting ducts ultimately lead to the pancreatic duct and thence to the small intestine. (c) Amino acids are absorbed through the epithelial cell layer (intestinal mucosa) of the villi and enter the capillaries. Recall that the products of lipid hydrolysis in the small intestine enter the lymphatic system after their absorption by the intestinal mucosa (see Fig. 17-1).



cholecystikinin, which stimulates secretion of several pancreatic enzymes with activity optima at pH 7 to 8. **Trypsinogen**, **chymotrypsinogen**, and **procarboxypeptidases A and B**, the zymogens of **trypsin**, **chymotrypsin**, and **carboxypeptidases A and B**, are synthesized and secreted by the exocrine cells of the pancreas (Fig. 18-3b). Trypsinogen is converted to its active form, trypsin, by **enteropeptidase**, a proteolytic enzyme secreted by intestinal cells. Free trypsin then catalyzes the conversion of additional trypsinogen to trypsin (see Fig. 6-33). Trypsin also activates chymotrypsinogen, the procarboxypeptidases, and proelastase.

Why this elaborate mechanism for getting active digestive enzymes into the gastrointestinal tract? Synthesis of the enzymes as inactive precursors protects the exocrine cells from destructive proteolytic attack. The pancreas further protects itself against self-digestion by making a specific inhibitor, a protein called **pancreatic trypsin inhibitor** (p. 231), that effectively prevents

premature production of active proteolytic enzymes within the pancreatic cells.

Trypsin and chymotrypsin further hydrolyze the peptides that were produced by pepsin in the stomach. This stage of protein digestion is accomplished very efficiently, because pepsin, trypsin, and chymotrypsin have different amino acid specificities (see Table 3-7). Degradation of the short peptides in the small intestine is then completed by other intestinal peptidases. These include carboxypeptidases A and B (both of which are zinc-containing enzymes), which remove successive carboxyl-terminal residues from peptides, and an **aminopeptidase** that hydrolyzes successive amino-terminal residues from short peptides. The resulting mixture of free amino acids is transported into the epithelial cells lining the small intestine (Fig. 18-3c), through which the amino acids enter the blood capillaries in the villi and travel to the liver. In humans, most globular proteins from animal

sources are almost completely hydrolyzed to amino acids in the gastrointestinal tract, but some fibrous proteins, such as keratin, are only partly digested. In addition, the protein content of some plant foods is protected against breakdown by indigestible cellulose husks.

Acute pancreatitis is a disease caused by obstruction of the normal pathway by which pancreatic secretions enter the intestine. The zymogens of the proteolytic enzymes are converted to their catalytically active forms prematurely, *inside* the pancreatic cells, and attack the pancreatic tissue itself. This causes excruciating pain and damage to the organ that can prove fatal. ■

Pyridoxal Phosphate Participates in the Transfer of α -Amino Groups to α -Ketoglutarate

The first step in the catabolism of most L-amino acids, once they have reached the liver, is removal of the α -amino groups, promoted by enzymes called **aminotransferases** or **transaminases**. In these **transamination** reactions, the α -amino group is transferred to the α -carbon atom of α -ketoglutarate, leaving behind the corresponding α -keto acid analog of the amino acid (Fig. 18–4). There is no net deamination (loss of amino groups) in these reactions, because the α -ketoglutarate becomes aminated as the α -amino acid is deaminated. The effect of transamination reactions is to collect the amino groups from many different amino acids in the form of L-glutamate. The glutamate then functions as an amino group donor for biosynthetic pathways or for

excretion pathways that lead to the elimination of nitrogenous waste products.

Cells contain different types of aminotransferases. Many are specific for α -ketoglutarate as the amino group acceptor but differ in their specificity for the L-amino acid. The enzymes are named for the amino group donor (alanine aminotransferase, aspartate aminotransferase, for example). The reactions catalyzed by aminotransferases are freely reversible, having an equilibrium constant of about 1.0 ($\Delta G'^{\circ} \approx 0$ kJ/mol).

All aminotransferases have the same prosthetic group and the same reaction mechanism. The prosthetic group is **pyridoxal phosphate (PLP)**, the coenzyme form of pyridoxine, or vitamin B₆. We encountered pyridoxal phosphate in Chapter 15, as a coenzyme in the glycogen phosphorylase reaction, but its role in that reaction is not representative of its usual coenzyme function. Its primary role in cells is in the metabolism of molecules with amino groups.

Pyridoxal phosphate functions as an intermediate carrier of amino groups at the active site of aminotransferases. It undergoes reversible transformations between its aldehyde form, pyridoxal phosphate, which can accept an amino group, and its aminated form, pyridoxamine phosphate, which can donate its amino group to an α -keto acid (Fig. 18–5a). Pyridoxal phosphate is generally covalently bound to the enzyme's active site through an aldimine (Schiff base) linkage to the ϵ -amino group of a Lys residue (Fig. 18–5b, d).

Pyridoxal phosphate participates in a variety of reactions at the α , β , and γ carbons (C-2 to C-4) of amino acids. Reactions at the α carbon (Fig. 18–6) include racemizations (interconverting L- and D-amino acids) and decarboxylations, as well as transaminations. Pyridoxal phosphate plays the same chemical role in each of these reactions. A bond to the α carbon of the substrate is broken, removing either a proton or a carboxyl group. The electron pair left behind on the α carbon would form a highly unstable carbanion, but pyridoxal phosphate provides resonance stabilization of this intermediate (Fig. 18–6 inset). The highly conjugated structure of PLP (an electron sink) permits delocalization of the negative charge.

Aminotransferases (Fig. 18–5) are classic examples of enzymes catalyzing bimolecular Ping-Pong reactions (see Fig. 6–13b), in which the first substrate reacts and the product must leave the active site before the second substrate can bind. Thus the incoming amino acid binds to the active site, donates its amino group to pyridoxal phosphate, and departs in the form of an α -keto acid. The incoming α -keto acid then binds, accepts the amino group from pyridoxamine phosphate, and departs in the form of an amino acid. As described in Box 18–1 on page 664, measurement of the alanine aminotransferase and aspartate aminotransferase levels in blood serum is important in some medical diagnoses.

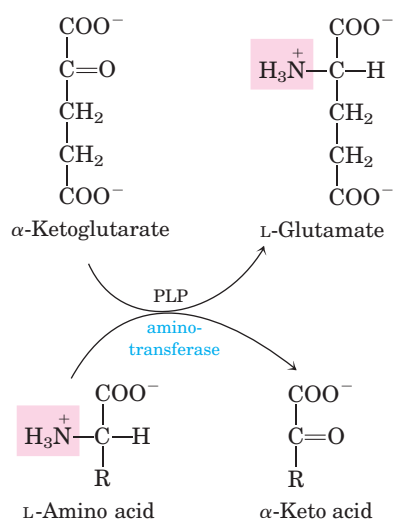


FIGURE 18–4 Enzyme-catalyzed transaminations. In many aminotransferase reactions, α -ketoglutarate is the amino group acceptor. All aminotransferases have pyridoxal phosphate (PLP) as cofactor. Although the reaction is shown here in the direction of transfer of the amino group to α -ketoglutarate, it is readily reversible.

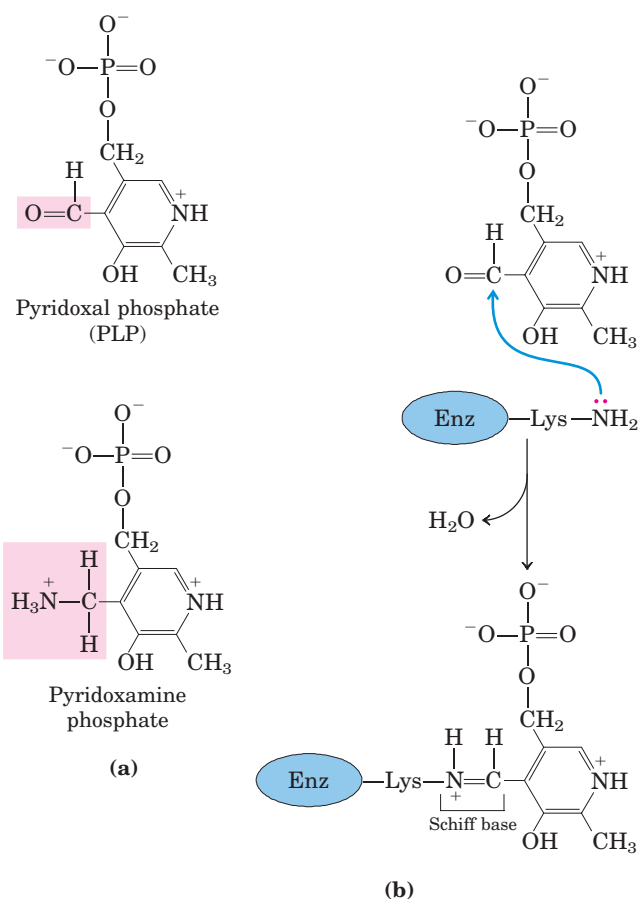
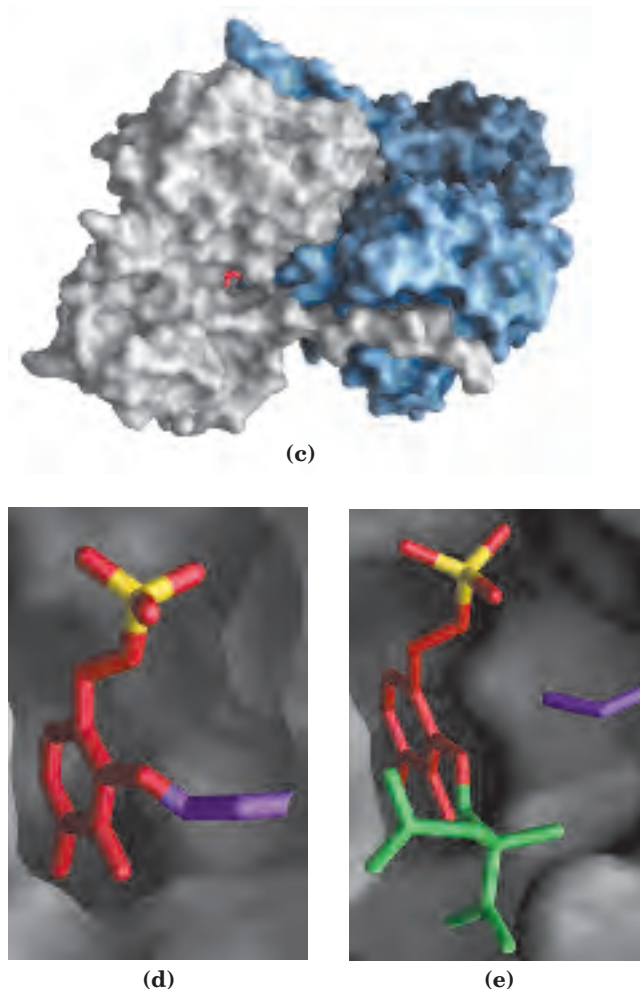


FIGURE 18-5 Pyridoxal phosphate, the prosthetic group of amino-transferases. **(a)** Pyridoxal phosphate (PLP) and its aminated form, pyridoxamine phosphate, are the tightly bound coenzymes of amino-transferases. The functional groups are shaded. **(b)** Pyridoxal phosphate is bound to the enzyme through noncovalent interactions and a Schiff-base linkage to a Lys residue at the active site. The steps in the formation of a Schiff base from a primary amine and a carbonyl group

Glutamate Releases Its Amino Group as Ammonia in the Liver

As we have seen, the amino groups from many of the α -amino acids are collected in the liver in the form of the amino group of L-glutamate molecules. These amino groups must next be removed from glutamate to prepare them for excretion. In hepatocytes, glutamate is transported from the cytosol into mitochondria, where it undergoes **oxidative deamination** catalyzed by **L-glutamate dehydrogenase** (M_r 330,000). In mammals, this enzyme is present in the mitochondrial matrix. It is the only enzyme that can use either NAD⁺ or NADP⁺ as the acceptor of reducing equivalents (Fig. 18-7).

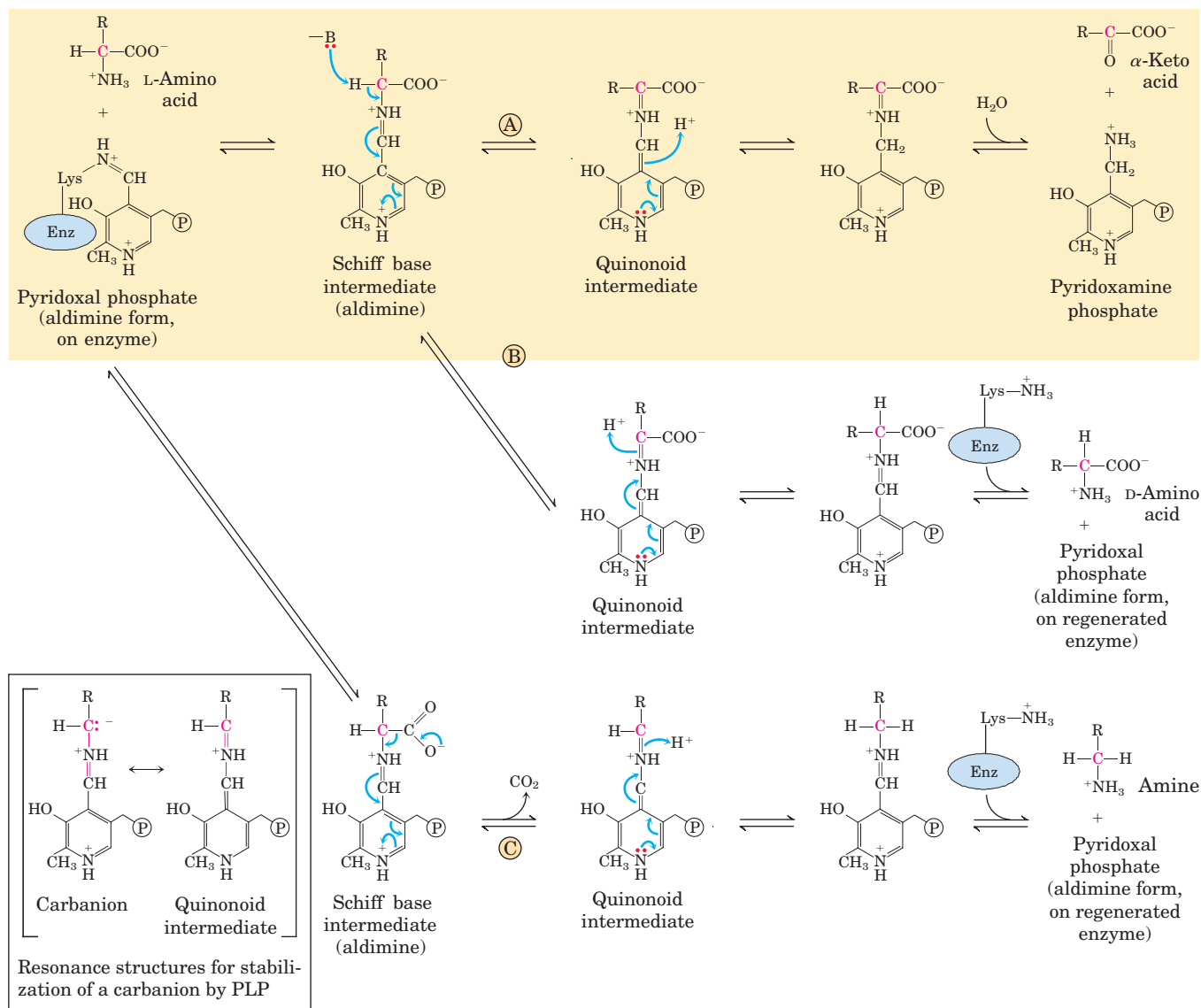
The combined action of an aminotransferase and glutamate dehydrogenase is referred to as **transdeamination**. A few amino acids bypass the transdeamina-



are detailed in Figure 14-5. **(c)** PLP (red) bound to one of the two active sites of the dimeric enzyme aspartate aminotransferase, a typical aminotransferase; **(d)** close-up view of the active site, with PLP (red, with yellow phosphorus) in aldimine linkage with the side chain of Lys²⁵⁸ (purple); **(e)** another close-up view of the active site, with PLP linked to the substrate analog 2-methylaspartate (green) via a Schiff base (PDB ID 1AJS).

tion pathway and undergo direct oxidative deamination. The fate of the NH₄⁺ produced by any of these deamination processes is discussed in detail in Section 18.2. The α -ketoglutarate formed from glutamate deamination can be used in the citric acid cycle and for glucose synthesis.

Glutamate dehydrogenase operates at an important intersection of carbon and nitrogen metabolism. An allosteric enzyme with six identical subunits, its activity is influenced by a complicated array of allosteric modulators. The best-studied of these are the positive modulator ADP and the negative modulator GTP. The metabolic rationale for this regulatory pattern has not been elucidated in detail. Mutations that alter the allosteric binding site for GTP or otherwise cause permanent activation of glutamate dehydrogenase lead to a human genetic disorder called hyperinsulinism-hyperammonemia



MECHANISM FIGURE 18-6 Some amino acid transformations at the α carbon that are facilitated by pyridoxal phosphate. Pyridoxal phosphate is generally bonded to the enzyme through a Schiff base (see Fig. 18-5b, d). Reactions begin (top left) with formation of a new Schiff base (aldimine) between the α -amino group of the amino acid and PLP, which substitutes for the enzyme-PLP linkage. Three alternative fates for this Schiff base are shown: **(A)** transamination, **(B)** racemization, and **(C)** decarboxylation. The Schiff base formed between PLP and the amino acid is in conjugation with the pyridine ring, an electron sink that permits delocalization of an electron pair to avoid formation

of an unstable carbanion on the α carbon (inset). A quinonoid intermediate is involved in all three types of reactions. The transamination route **(A)** is especially important in the pathways described in this chapter. The pathway highlighted here (shown left to right) represents only part of the overall reaction catalyzed by aminotransferases. To complete the process, a second α -keto acid replaces the one that is released, and this is converted to an amino acid in a reversal of the reaction steps (right to left). Pyridoxal phosphate is also involved in certain reactions at the β and γ carbons of some amino acids (not shown).  **Pyridoxal Phosphate Reaction Mechanisms**

syndrome, characterized by elevated levels of ammonia in the bloodstream and hypoglycemia.

Glutamine Transports Ammonia in the Bloodstream

Ammonia is quite toxic to animal tissues (we examine some possible reasons for this toxicity later), and the levels present in blood are regulated. In many tissues, including the brain, some processes such as nucleotide

degradation generate free ammonia. In most animals much of the free ammonia is converted to a nontoxic compound before export from the extrahepatic tissues into the blood and transport to the liver or kidneys. For this transport function, glutamate, critical to *intracellular* amino group metabolism, is supplanted by L-glutamine. The free ammonia produced in tissues is combined with glutamate to yield glutamine by the action of **glutamine synthetase**. This reaction requires

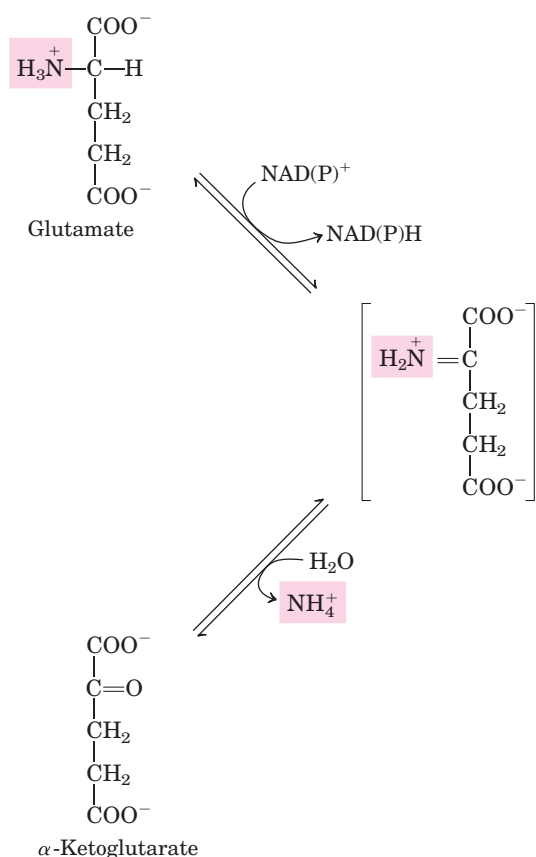


FIGURE 18-7 Reaction catalyzed by glutamate dehydrogenase. The glutamate dehydrogenase of mammalian liver has the unusual capacity to use either NAD^+ or NADP^+ as cofactor. The glutamate dehydrogenases of plants and microorganisms are generally specific for one or the other. The mammalian enzyme is allosterically regulated by GTP and ADP.

ATP and occurs in two steps (Fig. 18-8). First, glutamate and ATP react to form ADP and a γ -glutamyl phosphate intermediate, which then reacts with ammonia to produce glutamine and inorganic phosphate. Glutamine is a nontoxic transport form of ammonia; it is normally present in blood in much higher concentrations than other amino acids. Glutamine also serves as a source of amino groups in a variety of biosynthetic reactions. Glutamine synthetase is found in all organisms, always playing a central metabolic role. In microorganisms, the enzyme serves as an essential portal for the entry of fixed nitrogen into biological systems. (The roles of glutamine and glutamine synthetase in metabolism are further discussed in Chapter 22.)

In most terrestrial animals, glutamine in excess of that required for biosynthesis is transported in the blood to the intestine, liver, and kidneys for processing. In these tissues, the amide nitrogen is released as ammonium ion in the mitochondria, where the enzyme **glutaminase** converts glutamine to glutamate and NH_4^+ (Fig. 18-8). The NH_4^+ from intestine and kidney is transported in the

blood to the liver. In the liver, the ammonia from all sources is disposed of by urea synthesis. Some of the glutamate produced in the glutaminase reaction may be further processed in the liver by glutamate dehydrogenase, releasing more ammonia and producing carbon skeletons for metabolic fuel. However, most glutamate enters the transamination reactions required for amino acid biosynthesis and other processes (Chapter 22).

 In metabolic acidosis (p. 652) there is an increase in glutamine processing by the kidneys. Not all the excess NH_4^+ thus produced is released into the bloodstream or converted to urea; some is excreted directly into the urine. In the kidney, the NH_4^+ forms salts with metabolic acids, facilitating their removal in the urine. Bicarbonate produced by the decarboxylation of α -ketoglutarate in the citric acid cycle can also serve as a buffer in blood plasma. Taken together, these effects of glutamine metabolism in the kidney tend to counteract acidosis. ■

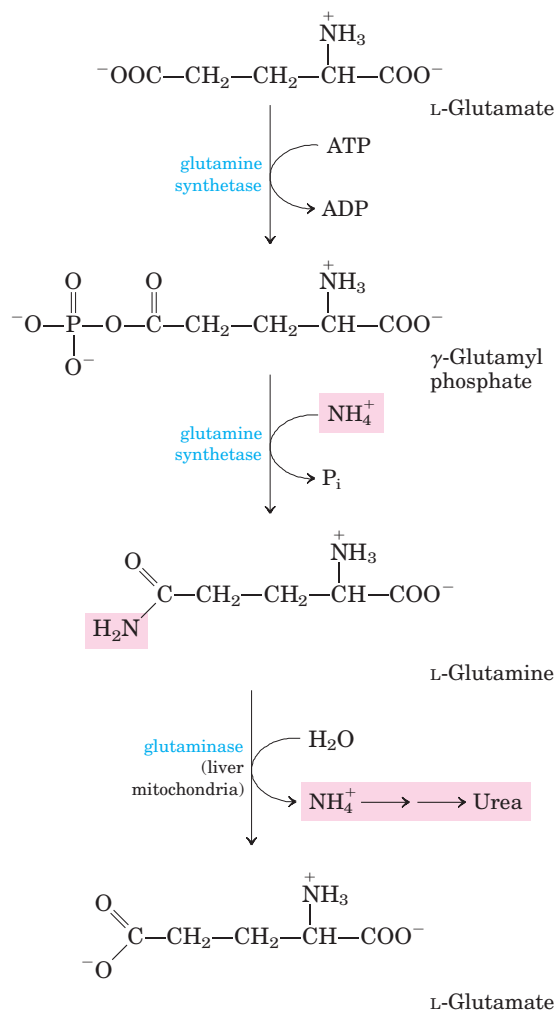


FIGURE 18-8 Ammonia transport in the form of glutamine. Excess ammonia in tissues is added to glutamate to form glutamine, a process catalyzed by glutamine synthetase. After transport in the bloodstream, the glutamine enters the liver and NH_4^+ is liberated in mitochondria by the enzyme glutaminase.



BOX 18-1 BIOCHEMISTRY IN MEDICINE

Assays for Tissue Damage

Analyses of certain enzyme activities in blood serum give valuable diagnostic information for a number of disease conditions.

Alanine aminotransferase (ALT; also called glutamate-pyruvate transaminase, GPT) and aspartate aminotransferase (AST; also called glutamate-oxaloacetate transaminase, GOT) are important in the diagnosis of heart and liver damage caused by heart attack, drug toxicity, or infection. After a heart attack, a variety of enzymes, including these aminotransferases, leak from the injured heart cells into the bloodstream. Measurements of the blood serum concentrations of the two aminotransferases by the SGPT and SGOT tests (S for serum)—and of another enzyme, **creatinine kinase**, by the SCK test—can pro-

vide information about the severity of the damage. Creatine kinase is the first heart enzyme to appear in the blood after a heart attack; it also disappears quickly from the blood. GOT is the next to appear, and GPT follows later. Lactate dehydrogenase also leaks from injured or anaerobic heart muscle.

The SGOT and SGPT tests are also important in occupational medicine, to determine whether people exposed to carbon tetrachloride, chloroform, or other industrial solvents have suffered liver damage. Liver degeneration caused by these solvents is accompanied by leakage of various enzymes from injured hepatocytes into the blood. Aminotransferases are most useful in the monitoring of people exposed to these chemicals, because these enzyme activities are high in liver and can be detected in very small amounts.

Alanine Transports Ammonia from Skeletal Muscles to the Liver

Alanine also plays a special role in transporting amino groups to the liver in a nontoxic form, via a pathway called the **glucose-alanine cycle** (Fig. 18-9). In muscle and certain other tissues that degrade amino acids for fuel, amino groups are collected in the form of glutamate by transamination (Fig. 18-2a). Glutamate can be converted to glutamine for transport to the liver, as described above, or it can transfer its α -amino group to pyruvate, a readily available product of muscle glycolysis, by the action of **alanine aminotransferase** (Fig. 18-9). The alanine so formed passes into the blood and travels to the liver. In the cytosol of hepatocytes, alanine aminotransferase transfers the amino group from alanine to α -ketoglutarate, forming pyruvate and glutamate. Glutamate can then enter mitochondria, where the glutamate dehydrogenase reaction releases NH_4^+ (Fig. 18-7), or can undergo transamination with oxaloacetate to form aspartate, another nitrogen donor in urea synthesis, as we shall see.

The use of alanine to transport ammonia from skeletal muscles to the liver is another example of the intrinsic economy of living organisms. Vigorously contracting skeletal muscles operate anaerobically, producing pyruvate and lactate from glycolysis as well as

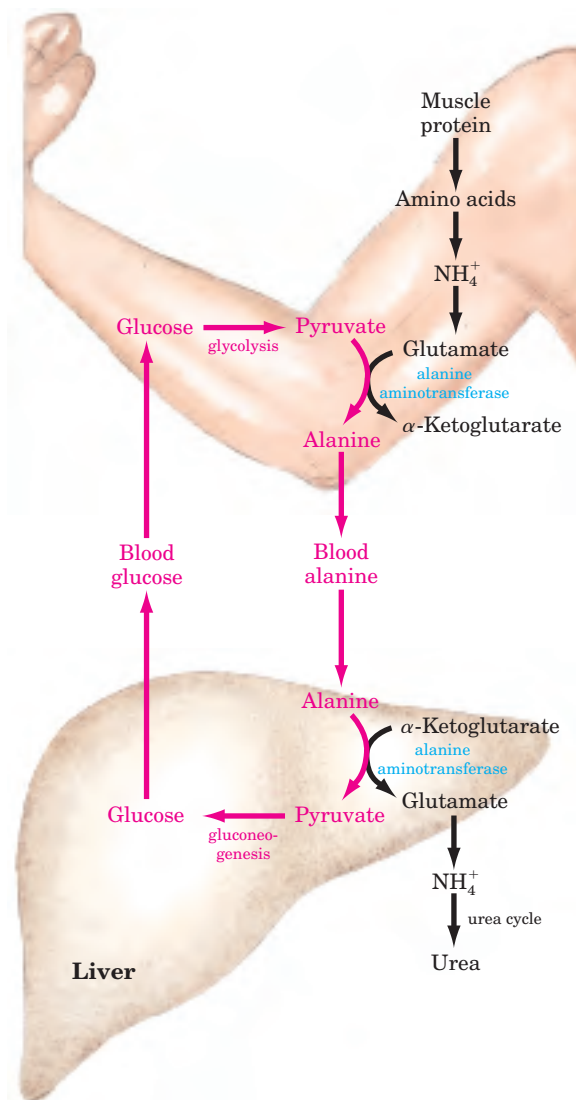


FIGURE 18-9 Glucose-alanine cycle. Alanine serves as a carrier of ammonia and of the carbon skeleton of pyruvate from skeletal muscle to liver. The ammonia is excreted and the pyruvate is used to produce glucose, which is returned to the muscle.

ammonia from protein breakdown. These products must find their way to the liver, where pyruvate and lactate are incorporated into glucose, which is returned to the muscles, and ammonia is converted to urea for excretion. The glucose-alanine cycle, in concert with the Cori cycle (see Box 14–1 and Fig. 23–18), accomplishes this transaction. The energetic burden of gluconeogenesis is thus imposed on the liver rather than the muscle, and all available ATP in muscle is devoted to muscle contraction.

Ammonia Is Toxic to Animals

 The catabolic production of ammonia poses a serious biochemical problem, because ammonia is very toxic. The molecular basis for this toxicity is not entirely understood. The terminal stages of ammonia intoxication in humans are characterized by onset of a comatose state accompanied by cerebral edema (an increase in the brain's water content) and increased cranial pressure, so research and speculation on ammonia toxicity have focused on this tissue. Speculation centers on a potential depletion of ATP in brain cells.

Ridding the cytosol of excess ammonia requires reductive amination of α -ketoglutarate to glutamate by glutamate dehydrogenase (the reverse of the reaction described earlier; Fig. 18–7) and conversion of glutamate to glutamine by glutamine synthetase. Both enzymes are present at high levels in the brain, although the glutamine synthetase reaction is almost certainly the more important pathway for removal of ammonia. High levels of NH_4^+ lead to increased levels of glutamine, which acts as an osmotically active solute (osmolyte) in brain astrocytes, star-shaped cells of the nervous system that provide nutrients, support, and insulation for neurons. This triggers an uptake of water into the astrocytes to maintain osmotic balance, leading to swelling and the symptoms noted above.

Depletion of glutamate in the glutamine synthetase reaction may have additional effects on the brain. Glutamate and its derivative γ -aminobutyrate (GABA; see Fig. 22–29) are important neurotransmitters; the sensitivity of the brain to ammonia may reflect a depletion of neurotransmitters as well as changes in cellular osmotic balance. ■

As we close this discussion of amino group metabolism, note that we have described several processes that deposit excess ammonia in the mitochondria of hepatocytes (Fig. 18–2). We now look at the fate of that ammonia.

SUMMARY 18.1 Metabolic Fates of Amino Groups

- Humans derive a small fraction of their oxidative energy from the catabolism of amino acids. Amino acids are derived from the normal

breakdown (recycling) of cellular proteins, degradation of ingested proteins, and breakdown of body proteins in lieu of other fuel sources during starvation or in uncontrolled diabetes mellitus.

- Proteases degrade ingested proteins in the stomach and small intestine. Most proteases are initially synthesized as inactive zymogens.
- An early step in the catabolism of amino acids is the separation of the amino group from the carbon skeleton. In most cases, the amino group is transferred to α -ketoglutarate to form glutamate. This transamination reaction requires the coenzyme pyridoxal phosphate.
- Glutamate is transported to liver mitochondria, where glutamate dehydrogenase liberates the amino group as ammonium ion (NH_4^+). Ammonia formed in other tissues is transported to the liver as the amide nitrogen of glutamine or, in transport from skeletal muscle, as the amino group of alanine.
- The pyruvate produced by deamination of alanine in the liver is converted to glucose, which is transported back to muscle as part of the glucose-alanine cycle.

18.2 Nitrogen Excretion and the Urea Cycle

If not reused for the synthesis of new amino acids or other nitrogenous products, amino groups are channeled into a single excretory end product (Fig. 18–10). Most aquatic species, such as the bony fishes, are **ammonotelic**, excreting amino nitrogen as ammonia. The toxic ammonia is simply diluted in the surrounding water. Terrestrial animals require pathways for nitrogen excretion that minimize toxicity and water loss. Most terrestrial animals are **ureotelic**, excreting amino nitrogen in the form of urea; birds and reptiles are **uricotelic**, excreting amino nitrogen as uric acid. (The pathway of uric acid synthesis is described in Fig. 22–45.) Plants recycle virtually all amino groups, and nitrogen excretion occurs only under very unusual circumstances.

In ureotelic organisms, the ammonia deposited in the mitochondria of hepatocytes is converted to urea in the **urea cycle**. This pathway was discovered in 1932 by Hans Krebs (who later also discovered the citric acid cycle) and a medical student associate, Kurt Henseleit. Urea production occurs almost exclusively in the liver and is the fate of most of the ammonia channeled there. The urea passes into the bloodstream and thus to the kidneys and is excreted into the urine. The production of urea now becomes the focus of our discussion.

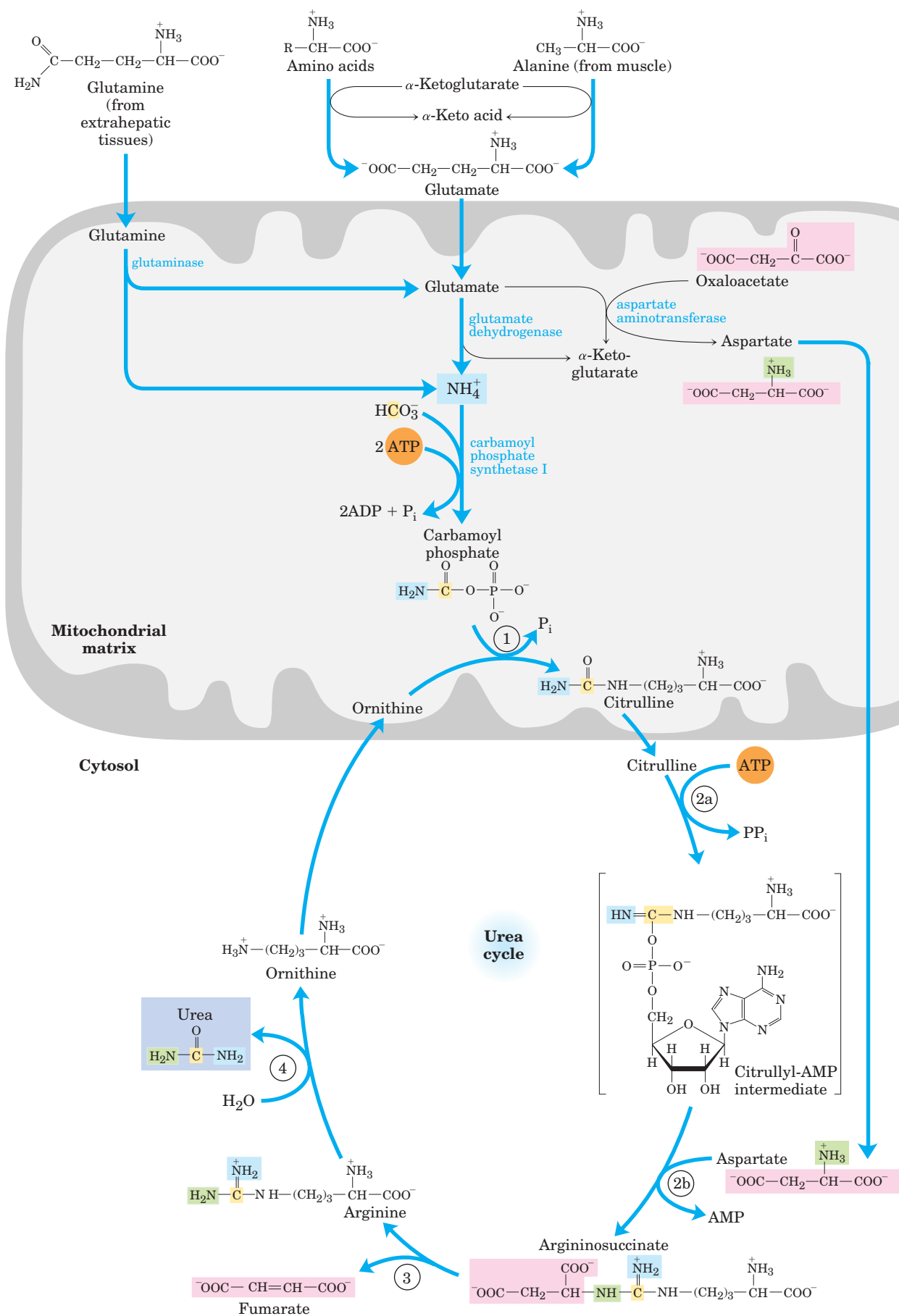


FIGURE 18-10 (facing page) **Urea cycle and reactions that feed amino groups into the cycle.** The enzymes catalyzing these reactions (named in the text) are distributed between the mitochondrial matrix and the cytosol. One amino group enters the urea cycle as carbamoyl phosphate, formed in the matrix; the other enters as aspartate, formed in the matrix by transamination of oxaloacetate and glutamate, catalyzed by aspartate aminotransferase. The urea cycle consists of four steps. ① Formation of citrulline from ornithine and carbamoyl phosphate (entry of the first amino group); the citrulline passes into the cytosol. ② Formation of argininosuccinate through a citrullinyl-AMP intermediate (entry of the second amino group). ③ Formation of arginine from argininosuccinate; this reaction releases fumarate, which enters the citric acid cycle. ④ Formation of urea; this reaction also regenerates, ornithine. The pathways by which NH_4^+ arrives in the mitochondrial matrix of hepatocytes were discussed in Section 18.1.

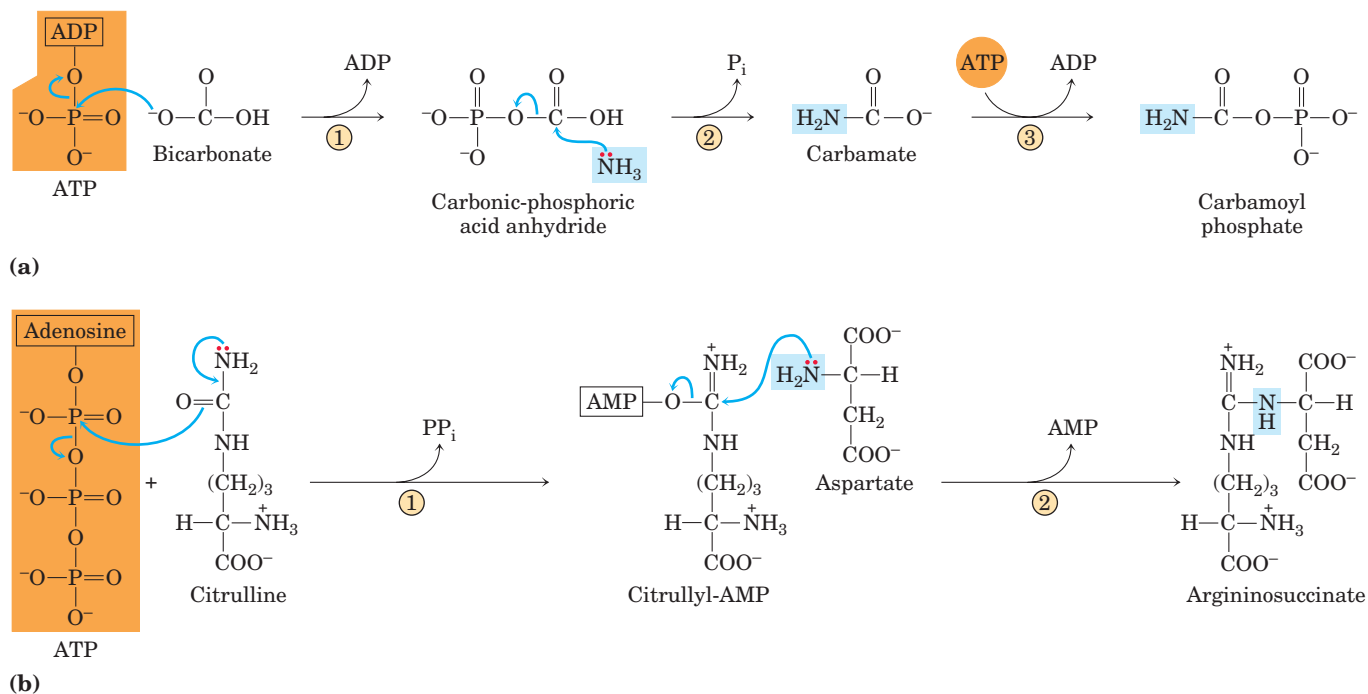
Urea Is Produced from Ammonia in Five Enzymatic Steps

The urea cycle begins inside liver mitochondria, but three of the subsequent steps take place in the cytosol; the cycle thus spans two cellular compartments (Fig. 18-10). The first amino group to enter the urea cycle is derived from ammonia in the mitochondrial matrix— NH_4^+ arising by the pathways described above.

The liver also receives some ammonia via the portal vein from the intestine, from the bacterial oxidation of amino acids. Whatever its source, the NH_4^+ generated in liver mitochondria is immediately used, together with CO_2 (as HCO_3^-) produced by mitochondrial respiration, to form carbamoyl phosphate in the matrix (Fig. 18-11a; see also Fig. 18-10). This ATP-dependent reaction is catalyzed by **carbamoyl phosphate synthetase I**, a regulatory enzyme (see below). The mitochondrial form of the enzyme is distinct from the cytosolic (II) form, which has a separate function in pyrimidine biosynthesis (Chapter 22).

The carbamoyl phosphate, which functions as an activated carbamoyl group donor, now enters the urea cycle. The cycle has four enzymatic steps. First, carbamoyl phosphate donates its carbamoyl group to ornithine to form citrulline, with the release of P_i (Fig. 18-10, step ①). Ornithine plays a role resembling that of oxaloacetate in the citric acid cycle, accepting material at each turn of the cycle. The reaction is catalyzed by **ornithine transcarbamoylase**, and the citrulline passes from the mitochondrion to the cytosol.

The second amino group now enters from aspartate (generated in mitochondria by transamination and transported into the cytosol) by a condensation reaction between the amino group of aspartate and the ureido



MECHANISM FIGURE 18-11 **Nitrogen-acquiring reactions in the synthesis of urea.** The urea nitrogens are acquired in two reactions, each requiring ATP. **(a)** In the reaction catalyzed by carbamoyl phosphate synthetase I, the first nitrogen enters from ammonia. The terminal phosphate groups of two molecules of ATP are used to form one molecule of carbamoyl phosphate. In other words, this reaction has two activa-

tion steps (① and ③). **Carbamoyl Phosphate Synthetase I Mechanism** **(b)** In the reaction catalyzed by argininosuccinate synthetase, the second nitrogen enters from aspartate. The ureido oxygen of citrulline is activated by the addition of AMP in step ①; this sets up the addition of aspartate in step ②, with AMP (including the ureido oxygen) as the leaving group. **Argininosuccinate Synthetase Mechanism**

(carbonyl) group of citrulline, forming argininosuccinate (step ② in Fig. 18–10). This cytosolic reaction, catalyzed by **argininosuccinate synthetase**, requires ATP and proceeds through a citrullyl-AMP intermediate (Fig. 18–11b). The argininosuccinate is then cleaved by **argininosuccinase** (step ③ in Fig. 18–10) to form free arginine and fumarate, the latter entering mitochondria to join the pool of citric acid cycle intermediates. This is the only reversible step in the urea cycle. In the last reaction of the urea cycle (step ④), the cytosolic enzyme **arginase** cleaves arginine to yield **urea** and ornithine. Ornithine is transported into the mitochondrion to initiate another round of the urea cycle.

As we noted in Chapter 16, the enzymes of many metabolic pathways are clustered (p. 605), with the product of one enzyme reaction being channeled directly to the next enzyme in the pathway. In the urea cycle, the mitochondrial and cytosolic enzymes appear to be clustered in this way. The citrulline transported out of the mitochondrion is not diluted into the general pool of metabolites in the cytosol but is passed directly to the active site of argininosuccinate synthetase. This channeling between enzymes continues for argininosuccinate, arginine, and ornithine. Only urea is released into the general cytosolic pool of metabolites.

The Citric Acid and Urea Cycles Can Be Linked

Because the fumarate produced in the argininosuccinase reaction is also an intermediate of the citric acid cycle, the cycles are, in principle, interconnected—in a process dubbed the “Krebs bicycle” (Fig. 18–12). However, each cycle can operate independently and communication between them depends on the transport of key intermediates between the mitochondrion and cytosol. Several enzymes of the citric acid cycle, including fumarase (fumarate hydratase) and malate dehydrogenase (p. 612), are also present as isozymes in the cytosol. The fumarate generated in cytosolic arginine synthesis can therefore be converted to malate in the cytosol, and these intermediates can be further metabolized in the cytosol or transported into mitochondria for use in the citric acid cycle. Aspartate formed in mitochondria by transamination between oxaloacetate and glutamate can be transported to the cytosol, where it serves as nitrogen donor in the urea cycle reaction catalyzed by argininosuccinate synthetase. These reactions, making up the **aspartate-argininosuccinate shunt**, provide metabolic links between the separate pathways by which the amino groups and carbon skeletons of amino acids are processed.

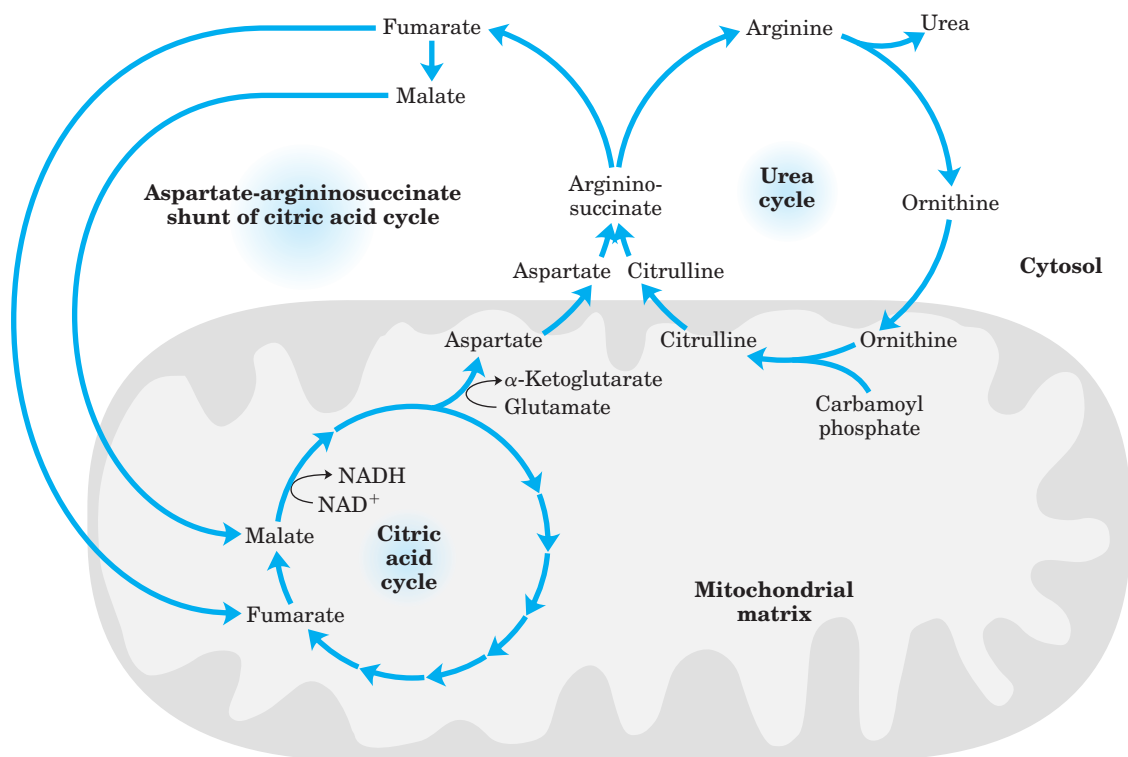


FIGURE 18–12 Links between the urea cycle and citric acid cycle. The interconnected cycles have been called the “Krebs bicycle.” The pathways linking the citric acid and urea cycles are called the aspartate-argininosuccinate shunt; these effectively link the fates of the amino groups and the carbon skeletons of amino acids. The interconnections are even more elaborate than the arrows suggest. For

example, some citric acid cycle enzymes, such as fumarase and malate dehydrogenase, have both cytosolic and mitochondrial isozymes. Fumarate produced in the cytosol—whether by the urea cycle, purine biosynthesis, or other processes—can be converted to cytosolic malate, which is used in the cytosol or transported into mitochondria (via the malate-aspartate shuttle; see Fig. 19–27) to enter the citric acid cycle.

The Activity of the Urea Cycle Is Regulated at Two Levels

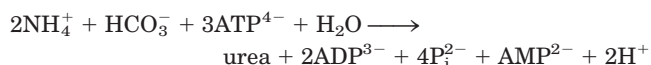
The flux of nitrogen through the urea cycle in an individual animal varies with diet. When the dietary intake is primarily protein, the carbon skeletons of amino acids are used for fuel, producing much urea from the excess amino groups. During prolonged starvation, when breakdown of muscle protein begins to supply much of the organism's metabolic energy, urea production also increases substantially.

These changes in demand for urea cycle activity are met over the long term by regulation of the rates of synthesis of the four urea cycle enzymes and carbamoyl phosphate synthetase I in the liver. All five enzymes are synthesized at higher rates in starving animals and in animals on very-high-protein diets than in well-fed animals eating primarily carbohydrates and fats. Animals on protein-free diets produce lower levels of urea cycle enzymes.

On a shorter time scale, allosteric regulation of at least one key enzyme adjusts the flux through the urea cycle. The first enzyme in the pathway, carbamoyl phosphate synthetase I, is allosterically activated by ***N*-acetylglutamate**, which is synthesized from acetyl-CoA and glutamate by ***N*-acetylglutamate synthase** (Fig. 18–13). In plants and microorganisms this enzyme catalyzes the first step in the *de novo* synthesis of arginine from glutamate (see Fig. 22–10), but in mammals *N*-acetylglutamate synthase activity in the liver has a purely regulatory function (mammals lack the other enzymes needed to convert glutamate to arginine). The steady-state levels of *N*-acetylglutamate are determined by the concentrations of glutamate and acetyl-CoA (the substrates for *N*-acetylglutamate synthase) and arginine (an activator of *N*-acetylglutamate synthase, and thus an activator of the urea cycle).

Pathway Interconnections Reduce the Energetic Cost of Urea Synthesis

If we consider the urea cycle in isolation, we see that the synthesis of one molecule of urea requires four high-energy phosphate groups (Fig. 18–10). Two ATP molecules are required to make carbamoyl phosphate, and one ATP to make argininosuccinate—the latter ATP undergoing a pyrophosphate cleavage to AMP and PP_i, which is hydrolyzed to two P_i. The overall equation of the urea cycle is



However, the urea cycle also causes a net conversion of oxaloacetate to fumarate (via aspartate), and the regeneration of oxaloacetate (Fig. 18–12) produces NADH in the malate dehydrogenase reaction. Each NADH molecule can generate up to 2.5 ATP during mitochondrial

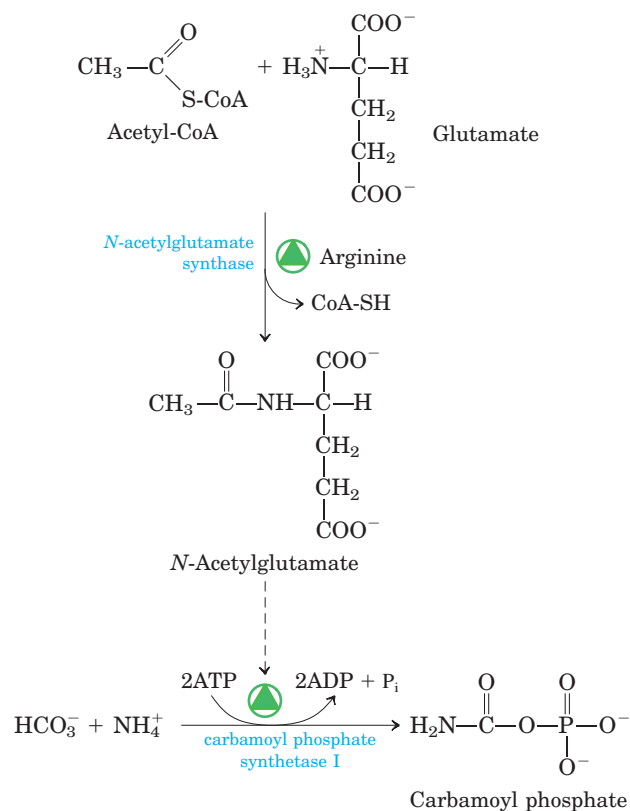


FIGURE 18–13 Synthesis of *N*-acetylglutamate and its activation of carbamoyl phosphate synthetase I.

respiration (Chapter 19), greatly reducing the overall energetic cost of urea synthesis.

Genetic Defects in the Urea Cycle Can Be Life-Threatening



People with genetic defects in any enzyme involved in urea formation cannot tolerate protein-rich diets. Amino acids ingested in excess of the minimum daily requirements for protein synthesis are deaminated in the liver, producing free ammonia that cannot be converted to urea and exported into the bloodstream, and, as we have seen, ammonia is highly toxic. The absence of a urea cycle enzyme can result in hyperammonemia or in the build-up of one or more urea cycle intermediates, depending on the enzyme that is missing. Given that most urea cycle steps are irreversible, the absent enzyme activity can often be identified by determining which cycle intermediate is present in especially elevated concentration in the blood and/or urine. Although the breakdown of amino acids can have serious health consequences in individuals with urea cycle deficiencies, a protein-free diet is not a treatment option. Humans are incapable of synthesizing half of the 20 common amino acids, and these **essential amino acids** (Table 18–1) must be provided in the diet.

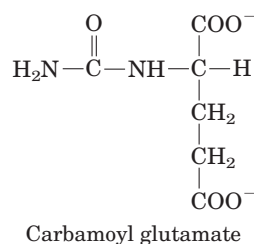
TABLE 18-1 Nonessential and Essential Amino Acids for Humans and the Albino Rat

Nonessential	Conditionally essential*	Essential
Alanine	Arginine	Histidine
Asparagine	Cysteine	Isoleucine
Aspartate	Glutamine	Leucine
Glutamate	Glycine	Lysine
Serine	Proline	Methionine
	Tyrosine	Phenylalanine
		Threonine
		Tryptophan
		Valine

*Required to some degree in young, growing animals, and/or sometimes during illness.

A variety of treatments are available for individuals with urea cycle defects. Careful administration of the aromatic acids benzoate or phenylbutyrate in the diet can help lower the level of ammonia in the blood. Benzoate is converted to benzoyl-CoA, which combines with glycine to form hippurate (Fig. 18-14, left). The glycine used up in this reaction must be regenerated, and ammonia is thus taken up in the glycine synthase reaction. Phenylbutyrate is converted to phenylacetate by β oxidation. The phenylacetate is then converted to phenylacetyl-CoA, which combines with glutamine to form phenylacetylglutamine (Fig. 18-14, right). The resulting removal of glutamine triggers its further synthesis by glutamine synthetase (see Eqn 22-1) in a reaction that takes up ammonia. Both hippurate and phenylacetylglutamine are nontoxic compounds that are excreted in the urine. The pathways shown in Figure 18-14 make only minor contributions to normal metabolism, but they become prominent when aromatic acids are ingested.

Other therapies are more specific to a particular enzyme deficiency. Deficiency of *N*-acetylglutamate synthase results in the absence of the normal activator of carbamoyl phosphate synthetase I (Fig. 18-13). This condition can be treated by administering carbamoyl glutamate, an analog of *N*-acetylglutamate that is effective in activating carbamoyl phosphate synthetase I.



Supplementing the diet with arginine is useful in treating deficiencies of ornithine transcarbamoylase, argininosuccinate synthetase, and argininosuccinase. Many

of these treatments must be accompanied by strict dietary control and supplements of essential amino acids. In the rare cases of arginase deficiency, arginine, the substrate of the defective enzyme, must be excluded from the diet. ■

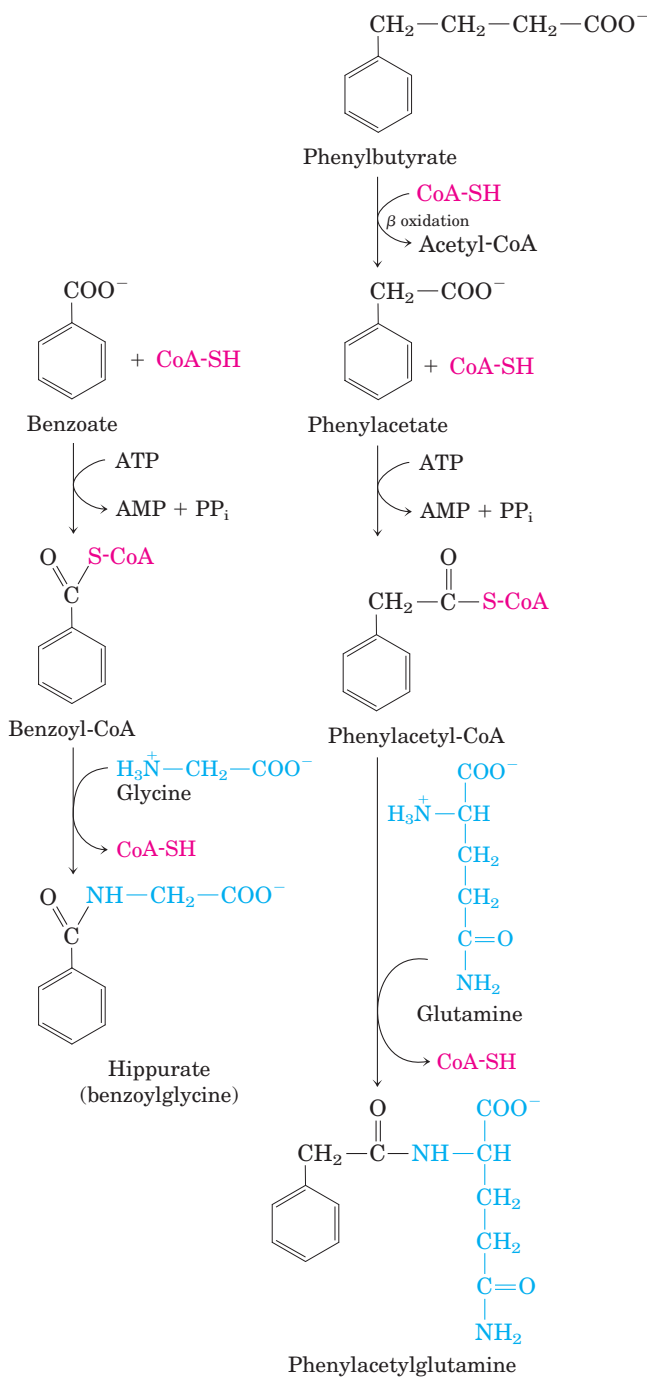


FIGURE 18-14 Treatment for deficiencies in urea cycle enzymes. The aromatic acids benzoate and phenylbutyrate, administered in the diet, are metabolized and combine with glycine and glutamine, respectively. The products are excreted in the urine. Subsequent synthesis of glycine and glutamine to replenish the pool of these intermediates removes ammonia from the bloodstream.

SUMMARY 18.2 Nitrogen Excretion and the Urea Cycle

- Ammonia is highly toxic to animal tissues. In the urea cycle, ornithine combines with ammonia, in the form of carbamoyl phosphate, to form citrulline. A second amino group is transferred to citrulline from aspartate to form arginine—the immediate precursor of urea. Arginase catalyzes hydrolysis of arginine to urea and ornithine; thus ornithine is regenerated in each turn of the cycle.
- The urea cycle results in a net conversion of oxaloacetate to fumarate, both of which are intermediates in the citric acid cycle. The two cycles are thus interconnected.
- The activity of the urea cycle is regulated at the level of enzyme synthesis and by allosteric regulation of the enzyme that catalyzes the formation of carbamoyl phosphate.

18.3 Pathways of Amino Acid Degradation

The pathways of amino acid catabolism, taken together, normally account for only 10% to 15% of the human body's energy production; these pathways are not nearly as active as glycolysis and fatty acid oxidation. Flux through these catabolic routes also varies greatly, depending on the balance between requirements for bio-

synthetic processes and the availability of a particular amino acid. The 20 catabolic pathways converge to form only six major products, all of which enter the citric acid cycle (Fig. 18–15). From here the carbon skeletons are diverted to gluconeogenesis or ketogenesis or are completely oxidized to CO_2 and H_2O .

All or part of the carbon skeletons of seven amino acids are ultimately broken down to acetyl-CoA. Five amino acids are converted to α -ketoglutarate, four to succinyl-CoA, two to fumarate, and two to oxaloacetate. Parts or all of six amino acids are converted to pyruvate, which can be converted to either acetyl-CoA or oxaloacetate. We later summarize the individual pathways for the 20 amino acids in flow diagrams, each leading to a specific point of entry into the citric acid cycle. In these diagrams the carbon atoms that enter the citric acid cycle are shown in color. Note that some amino acids appear more than once, reflecting different fates for different parts of their carbon skeletons. Rather than examining every step of every pathway in amino acid catabolism, we single out for special discussion some enzymatic reactions that are particularly noteworthy for their mechanisms or their medical significance.

Some Amino Acids Are Converted to Glucose, Others to Ketone Bodies

The seven amino acids that are degraded entirely or in part to acetoacetyl-CoA and/or acetyl-CoA—phenylalanine, tyrosine, isoleucine, leucine, tryptophan, threonine, and lysine—can yield ketone bodies in the liver,

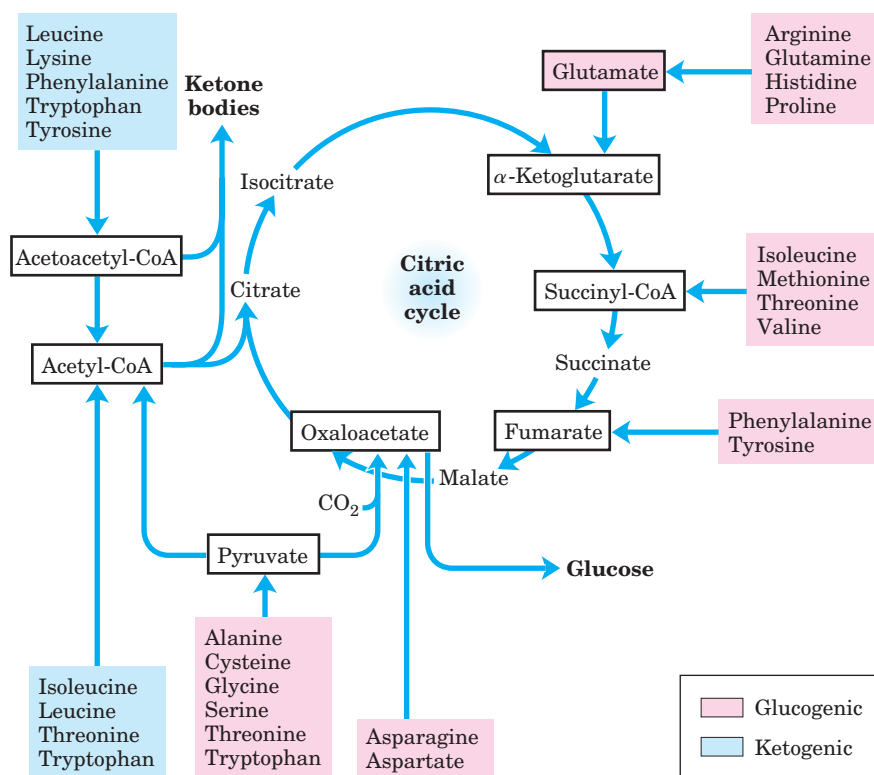


FIGURE 18-15 Summary of amino acid catabolism. Amino acids are grouped according to their major degradative end product. Some amino acids are listed more than once because different parts of their carbon skeletons are degraded to different end products. The figure shows the most important catabolic pathways in vertebrates, but there are minor variations among vertebrate species. Threonine, for instance, is degraded via at least two different pathways (see Figs 18–19, 18–27), and the importance of a given pathway can vary with the organism and its metabolic conditions. The glucogenic and ketogenic amino acids are also delineated in the figure, by color shading. Notice that five of the amino acids are both glucogenic and ketogenic. The amino acids degraded to pyruvate are also potentially ketogenic. Only two amino acids, leucine and lysine, are exclusively ketogenic.

where acetoacetyl-CoA is converted to acetoacetate and then to acetone and β -hydroxybutyrate (see Fig. 17–18). These are the **ketogenic** amino acids (Fig. 18–15). Their ability to form ketone bodies is particularly evident in uncontrolled diabetes mellitus, in which the liver produces large amounts of ketone bodies from both fatty acids and the ketogenic amino acids.

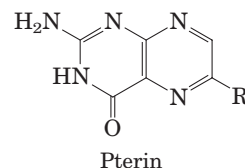
The amino acids that are degraded to pyruvate, α -ketoglutarate, succinyl-CoA, fumarate, and/or oxaloacetate can be converted to glucose and glycogen by pathways described in Chapters 14 and 15. They are the **glucogenic** amino acids. The division between ketogenic and glucogenic amino acids is not sharp; five amino acids—tryptophan, phenylalanine, tyrosine, threonine, and isoleucine—are both ketogenic and glucogenic. Catabolism of amino acids is particularly critical to the survival of animals with high-protein diets or during starvation. Leucine is an exclusively ketogenic amino acid that is very common in proteins. Its degradation makes a substantial contribution to ketosis under starvation conditions.

Several Enzyme Cofactors Play Important Roles in Amino Acid Catabolism

A variety of interesting chemical rearrangements occur in the catabolic pathways of amino acids. It is useful to begin our study of these pathways by noting the classes of reactions that recur and introducing their enzyme cofactors. We have already considered one important class: transamination reactions requiring pyridoxal phosphate. Another common type of reaction in amino acid catabolism is one-carbon transfers, which usually involve one of three cofactors: biotin, tetrahydrofolate, or *S*-adenosylmethionine (Fig. 18–16). These cofactors transfer one-carbon groups in different oxidation states: biotin transfers carbon in its most oxidized state, CO_2

(see Fig. 14–18); tetrahydrofolate transfers one-carbon groups in intermediate oxidation states and sometimes as methyl groups; and *S*-adenosylmethionine transfers methyl groups, the most reduced state of carbon. The latter two cofactors are especially important in amino acid and nucleotide metabolism.

Tetrahydrofolate (H_4 folate), synthesized in bacteria, consists of substituted pterin (6-methylpterin),



p-aminobenzoate, and glutamate moieties (Fig. 18–16). The oxidized form, folate, is a vitamin for mammals; it is converted in two steps to tetrahydrofolate by the enzyme dihydrofolate reductase. The one-carbon group undergoing transfer, in any of three oxidation states, is bonded to N-5 or N-10 or both. The most reduced form of the cofactor carries a methyl group, a more oxidized form carries a methylene group, and the most oxidized forms carry a methenyl, formyl, or formimino group (Fig. 18–17). Most forms of tetrahydrofolate are interconvertible and serve as donors of one-carbon units in a variety of metabolic reactions. The primary source of one-carbon units for tetrahydrofolate is the carbon removed in the conversion of serine to glycine, producing $\text{N}^5, \text{N}^{10}$ -methylenetetrahydrofolate.

Although tetrahydrofolate can carry a methyl group at N-5, the transfer potential of this methyl group is insufficient for most biosynthetic reactions. ***S*-Adenosylmethionine (adoMet)** is the preferred cofactor for biological methyl group transfers. It is synthesized from ATP and methionine by the action of **methionine**

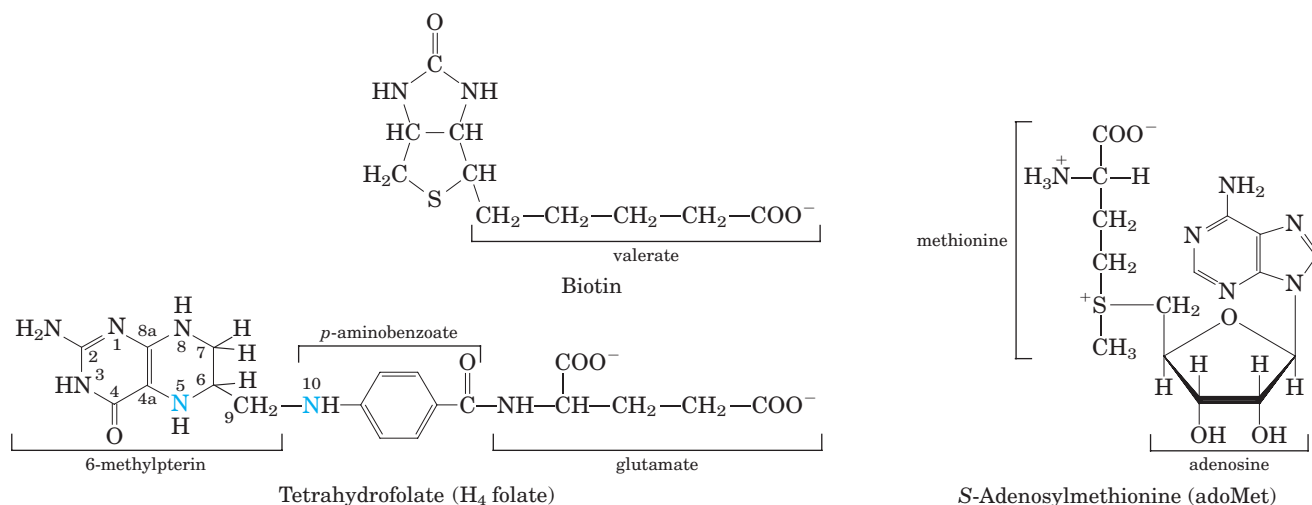
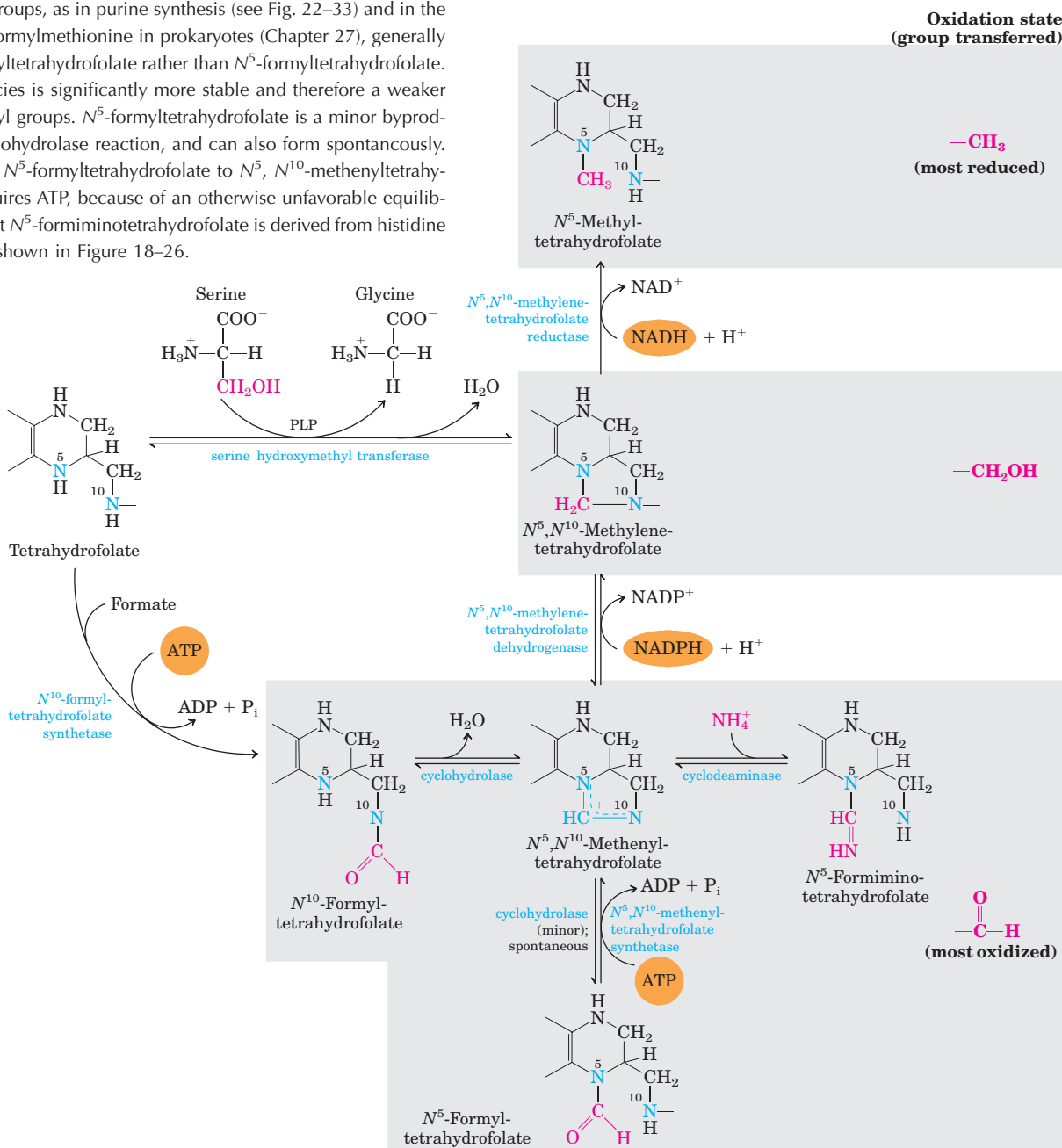


FIGURE 18–16 Some enzyme cofactors important in one-carbon transfer reactions. The nitrogen atoms to which one-carbon groups are attached in tetrahydrofolate are shown in blue.

FIGURE 18-17 Conversions of one-carbon units on tetrahydrofolate.

The different molecular species are grouped according to oxidation state, with the most reduced at the top and most oxidized at the bottom. All species within a single shaded box are at the same oxidation state. The conversion of N^5, N^{10} -methylenetetrahydrofolate to N^5 -methyltetrahydrofolate is effectively irreversible. The enzymatic transfer of formyl groups, as in purine synthesis (see Fig. 22–33) and in the formation of formylmethionine in prokaryotes (Chapter 27), generally uses N^{10} -formyltetrahydrofolate rather than N^5 -formyltetrahydrofolate. The latter species is significantly more stable and therefore a weaker donor of formyl groups. N^5 -formyltetrahydrofolate is a minor byproduct of the cyclohydrolase reaction, and can also form spontaneously. Conversion of N^5 -formyltetrahydrofolate to N^5, N^{10} -methenyltetrahydrofolate, requires ATP, because of an otherwise unfavorable equilibrium. Note that N^5 -formiminotetrahydrofolate is derived from histidine in a pathway shown in Figure 18–26.



adenosyl transferase (Fig. 18–18, step ①). This reaction is unusual in that the nucleophilic sulfur atom of methionine attacks the 5' carbon of the ribose moiety of ATP rather than one of the phosphorus atoms. Triphosphate is released and is cleaved to P_i and PP_i on the enzyme, and the PP_i is cleaved by inorganic pyrophosphatase; thus three bonds, including two bonds of

high-energy phosphate groups, are broken in this reaction. The only other known reaction in which triphosphate is displaced from ATP occurs in the synthesis of coenzyme B_{12} (see Box 17–2, Fig. 3).

S-Adenosylmethionine is a potent alkylating agent by virtue of its destabilizing sulfonium ion. The methyl group is subject to attack by nucleophiles and is about

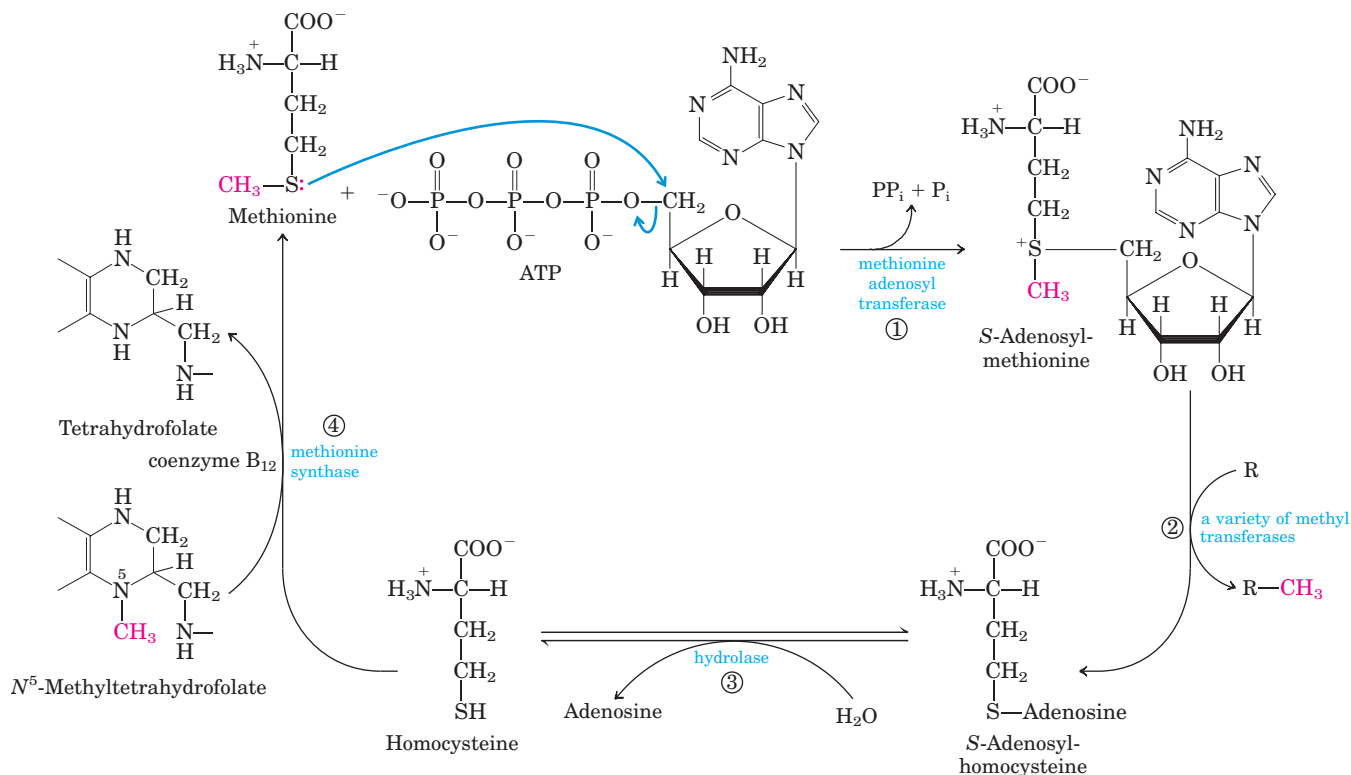


FIGURE 18-18 Synthesis of methionine and S-adenosylmethionine in an activated-methyl cycle. The steps are described in the text. In the methionine synthase reaction (step ④), the methyl group is transferred to cobalamin to form methylcobalamin, which in turn is

methyl donor in the formation of methionine. S-Adenosylmethionine, which has a positively charged sulfur (and is thus a sulfonium ion), is a powerful methylating agent in a number of biosynthetic reactions. The methyl group acceptor (step ②) is designated R.

1,000 times more reactive than the methyl group of N⁵-methyltetrahydrofolate.

Transfer of the methyl group from S-adenosylmethionine to an acceptor yields **S-adenosylhomocysteine** (Fig. 18-18, step ②), which is subsequently broken down to homocysteine and adenosine (step ③). Methionine is regenerated by transfer of a methyl group to homocysteine in a reaction catalyzed by methionine synthase (step ④), and methionine is reconverted to S-adenosylmethionine to complete an activated-methyl cycle.



One form of methionine synthase common in bacteria uses N⁵-methyltetrahydrofolate as a methyl donor. Another form of the enzyme present in some bacteria and mammals uses N⁵-methyltetrahydrofolate, but the methyl group is first transferred to cobalamin, derived from coenzyme B₁₂, to form methylcobalamin as the methyl donor in methionine formation. This reaction and the rearrangement of L-methylmalonyl-CoA to succinyl-CoA (see Box 17-2, Fig. 1a) are the only known coenzyme B₁₂-dependent reactions in mammals. In cases of vitamin B₁₂ deficiency, some symptoms can be alleviated by administering not only vitamin B₁₂ but folate. As noted above, the methyl group of methylcobalamin is derived from N⁵-methyltetrahydrofolate. Because the reaction converting the N⁵,N¹⁰-methylene form to the N⁵-methyl form of tetrahydrofo-

late is irreversible (Fig. 18-17), if coenzyme B₁₂ is not available for the synthesis of methylcobalamin, then no acceptor is available for the methyl group of N⁵-methyltetrahydrofolate and metabolic folates become trapped in the N⁵-methyl form. This sequestering of folates in one form may be the cause of some symptoms of the vitamin B₁₂ deficiency disease pernicious anemia. However, we do not know whether this is the only effect of insufficient vitamin B₁₂. ■

Tetrahydrobiopterin, another cofactor of amino acid catabolism, is similar to the pterin moiety of tetrahydrofolate, but it is not involved in one-carbon transfers; instead it participates in oxidation reactions. We consider its mode of action when we discuss phenylalanine degradation (see Fig. 18-24).

Six Amino Acids Are Degraded to Pyruvate

The carbon skeletons of six amino acids are converted in whole or in part to pyruvate. The pyruvate can then be converted to either acetyl-CoA (a ketone body precursor) or oxaloacetate (a precursor for gluconeogenesis). Thus amino acids catabolized to pyruvate are both ketogenic and glucogenic. The six are alanine, tryptophan, cysteine, serine, glycine, and threonine (Fig. 18-19). **Alanine** yields pyruvate directly on transamination with

α -ketoglutarate, and the side chain of **tryptophan** is cleaved to yield alanine and thus pyruvate. **Cysteine** is converted to pyruvate in two steps; one removes the sulfur atom, the other is a transamination. **Serine** is converted to pyruvate by serine dehydratase. Both the β -hydroxyl and the α -amino groups of serine are removed in this single pyridoxal phosphate–dependent reaction (Fig. 18–20a).

Glycine is degraded via three pathways, only one of which leads to pyruvate. Glycine is converted to serine by enzymatic addition of a hydroxymethyl group (Figs 18–19 and 18–20b). This reaction, catalyzed by **serine hydroxymethyl transferase**, requires the coenzymes tetrahydrofolate and pyridoxal phosphate. The serine is converted to pyruvate as described above. In the second pathway, which predominates in animals, glycine undergoes oxidative cleavage to CO_2 , NH_4^+ , and a methylene group ($-\text{CH}_2-$) (Fig. 18–19). This readily reversible reaction, catalyzed by **glycine cleavage enzyme** (also called glycine synthase), also requires tetrahydrofolate, which accepts the methylene group. In

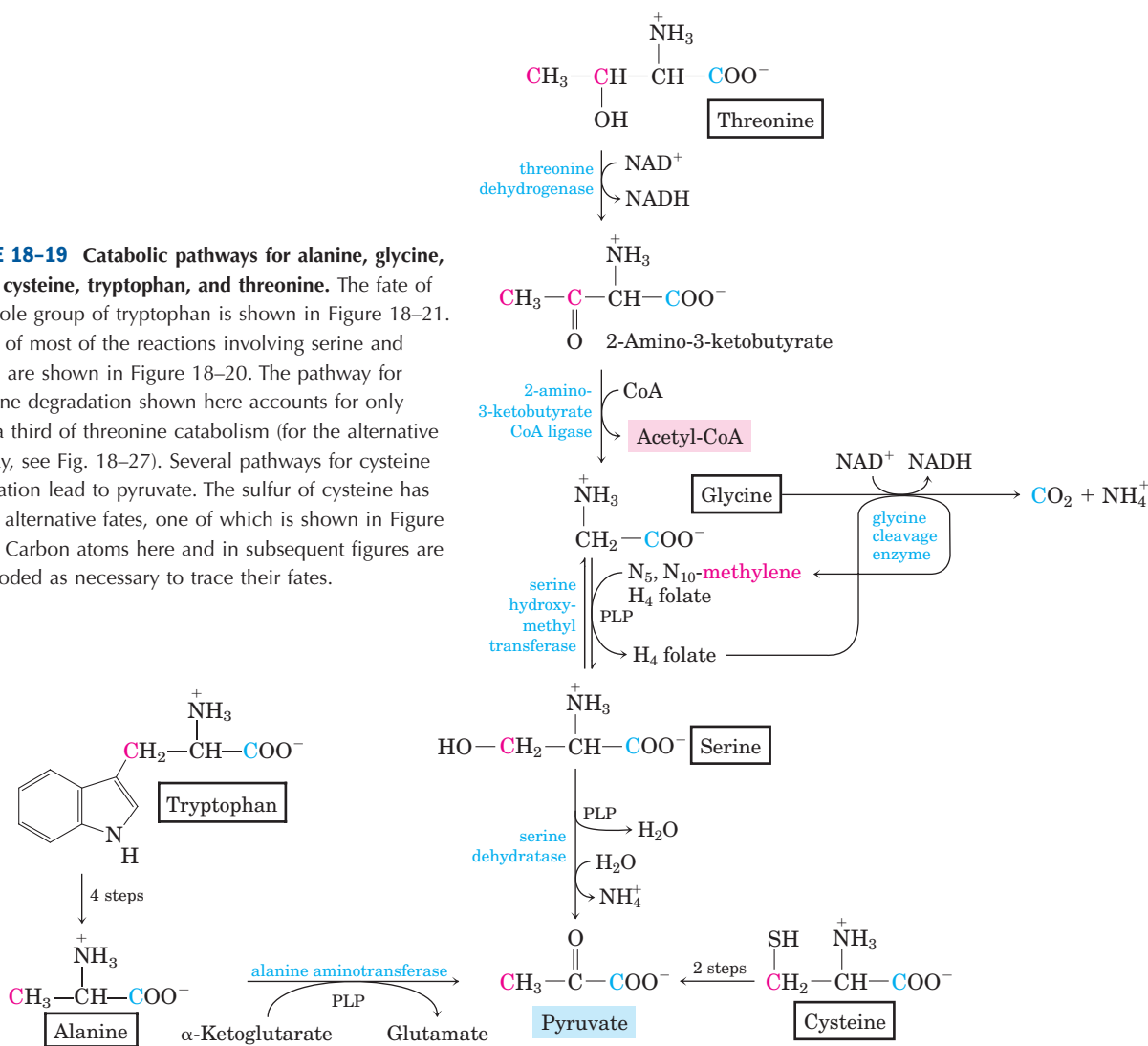
this oxidative cleavage pathway the two carbon atoms of glycine do not enter the citric acid cycle. One carbon is lost as CO_2 and the other becomes the methylene group of N^5, N^{10} -methylenetetrahydrofolate (Fig. 18–17), a one-carbon group donor in certain biosynthetic pathways.

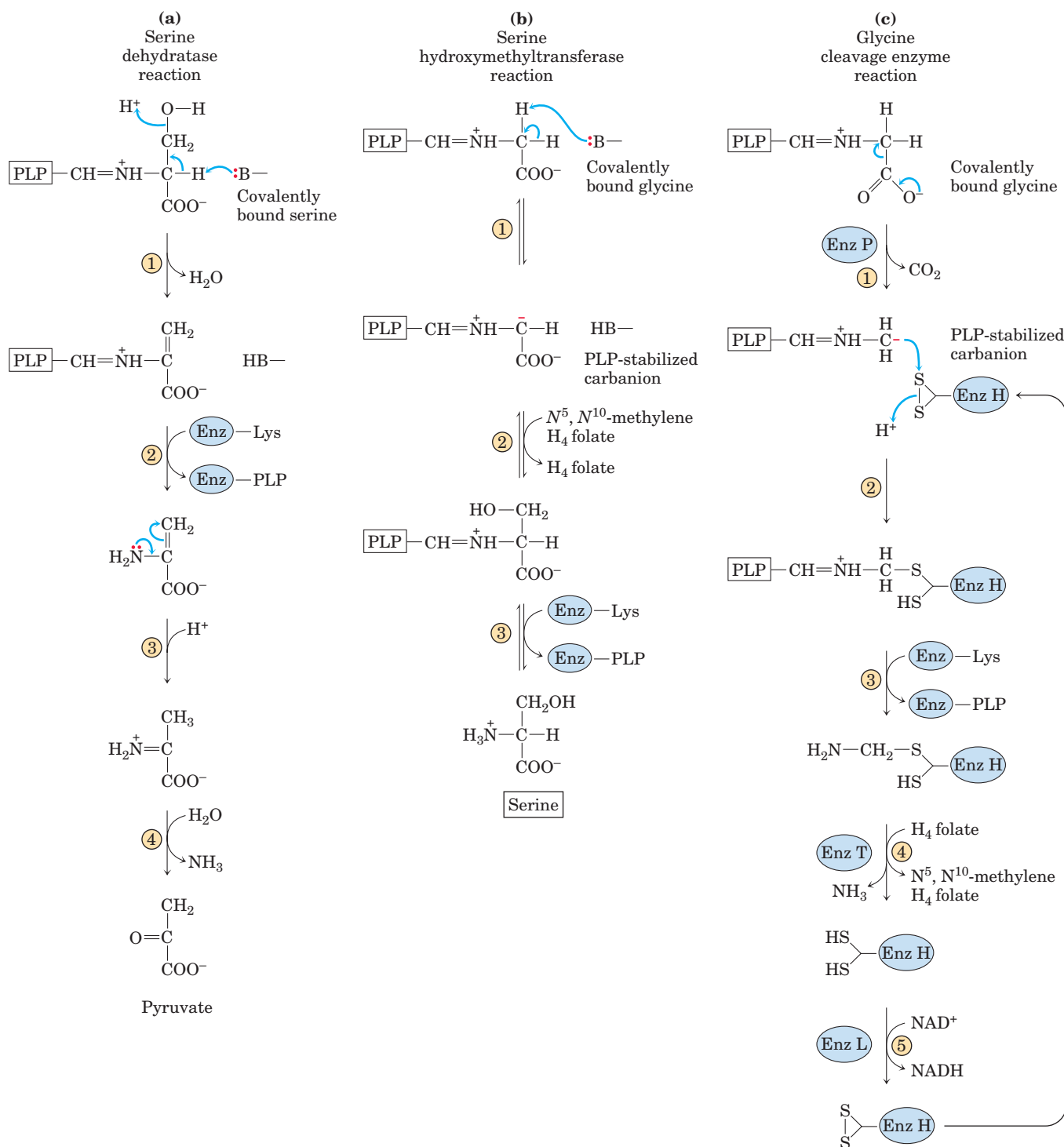


This second pathway for glycine degradation appears to be critical in mammals. Humans with serious defects in glycine cleavage enzyme activity suffer from a condition known as nonketotic hyperglycinemia. The condition is characterized by elevated serum levels of glycine, leading to severe mental deficiencies and death in very early childhood. At high levels, glycine is an inhibitory neurotransmitter, perhaps explaining the neurological effects of the disease. Many genetic defects of amino acid metabolism have been identified in humans (Table 18–2). We will encounter several more in this chapter. ■

In the third and final pathway of glycine degradation, the achiral glycine molecule is a substrate for the enzyme D-amino acid oxidase. The glycine is converted to glyoxylate, an alternative substrate for hepatic lactate

FIGURE 18–19 Catabolic pathways for alanine, glycine, serine, cysteine, tryptophan, and threonine. The fate of the indole group of tryptophan is shown in Figure 18–21. Details of most of the reactions involving serine and glycine are shown in Figure 18–20. The pathway for threonine degradation shown here accounts for only about a third of threonine catabolism (for the alternative pathway, see Fig. 18–27). Several pathways for cysteine degradation lead to pyruvate. The sulfur of cysteine has several alternative fates, one of which is shown in Figure 22–15. Carbon atoms here and in subsequent figures are color-coded as necessary to trace their fates.





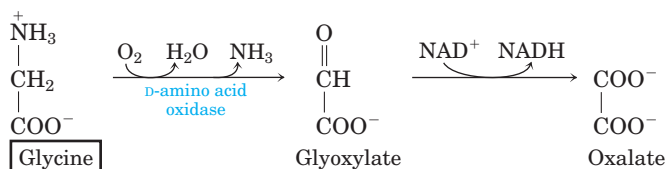
MECHANISM FIGURE 18-20 Interplay of the pyridoxal phosphate and tetrahydrofolate cofactors in serine and glycine metabolism. The first step in each of these reactions (not shown) involves the formation of a covalent imine linkage between enzyme-bound PLP and the substrate amino acid—serine in (a), glycine in (b) and (c). (a) The serine dehydratase reaction entails a PLP-catalyzed elimination of water across the bond between the α and β carbons (step 1), leading eventually to the production of pyruvate (steps 2 through 4). (b) In the serine hydroxymethyltransferase reaction, a PLP-stabilized carbanion on the α carbon of glycine (product of step 1) is a key intermediate

in the transfer of the methylene group (as $-\text{CH}_2-\text{OH}$) from N^5, N^{10} -methylenetetrahydrofolate to form serine. This reaction is reversible. (c) The glycine cleavage enzyme is a multienzyme complex, with components P, H, T, and L. The overall reaction, which is reversible, converts glycine to CO_2 and NH_4^+ , with the second glycine carbon taken up by tetrahydrofolate to form N^5, N^{10} -methylenetetrahydrofolate. Pyridoxal phosphate activates the α carbon of amino acids at critical stages in all these reactions, and tetrahydrofolate carries one-carbon units in two of them (see Figs 18-6, 18-17).

TABLE 18-2 Some Human Genetic Disorders Affecting Amino Acid Catabolism

Medical condition	Approximate incidence (per 100,000 births)	Defective process	Defective enzyme	Symptoms and effects
Albinism	<3	Melanin synthesis from tyrosine	Tyrosine 3-monooxygenase (tyrosinase)	Lack of pigmentation: white hair, pink skin
Alkaptonuria	<0.4	Tyrosine degradation	Homogentisate 1,2-dioxygenase	Dark pigment in urine; late-developing arthritis
Argininemia	<0.5	Urea synthesis	Arginase	Mental retardation
Argininosuccinic acidemia	<1.5	Urea synthesis	Argininosuccinase	Vomiting; convulsions
Carbamoyl phosphate synthetase I deficiency	<0.5	Urea synthesis	Carbamoyl phosphate synthetase I	Lethargy; convulsions; early death
Homocystinuria	<0.5	Methionine degradation	Cystathionine β -synthase	Faulty bone development; mental retardation
Maple syrup urine disease (branched-chain ketoaciduria)	<0.4	Isoleucine, leucine, and valine degradation	Branched-chain α -keto acid dehydrogenase complex	Vomiting; convulsions; mental retardation; early death
Methylmalonic acidemia	<0.5	Conversion of propionyl-CoA to succinyl-CoA	Methylmalonyl-CoA mutase	Vomiting; convulsions; mental retardation; early death
Phenylketonuria	<8	Conversion of phenylalanine to tyrosine	Phenylalanine hydroxylase	Neonatal vomiting; mental retardation

dehydrogenase (p. 538). Glyoxylate is oxidized in an NAD^+ -dependent reaction to oxalate:



The primary function of D-amino acid oxidase, present at high levels in the kidney, is thought to be the detoxification of ingested D-amino acids derived from bacterial cell walls and from cooked foodstuffs (heat causes some spontaneous racemization of the L-amino acids in proteins). Oxalate, whether obtained in foods or produced enzymatically in the kidneys, has medical significance. Crystals of calcium oxalate account for up to 75% of all kidney stones. ■

There are two significant pathways for **threonine** degradation. One pathway leads to pyruvate via glycine (Fig. 18–19). The conversion to glycine occurs in two steps, with threonine first converted to 2-amino-3-

ketobutyrate by the action of threonine dehydrogenase. This is a relatively minor pathway in humans, accounting for 10% to 30% of threonine catabolism, but is more important in some other mammals. The major pathway in humans leads to succinyl-CoA and is described later.

In the laboratory, serine hydroxymethyltransferase will catalyze the conversion of threonine to glycine and acetaldehyde in one step, but this is not a significant pathway for threonine degradation in mammals.

Seven Amino Acids Are Degraded to Acetyl-CoA

Portions of the carbon skeletons of seven amino acids—**tryptophan, lysine, phenylalanine, tyrosine, leucine, isoleucine, and threonine**—yield acetyl-CoA and/or acetoacetyl-CoA, the latter being converted to acetyl-CoA (Fig. 18–21). Some of the final steps in the degradative pathways for leucine, lysine, and tryptophan resemble steps in the oxidation of fatty acids. Threonine (not shown in Fig. 18–21) yields some acetyl-CoA via the minor pathway illustrated in Figure 18–19.

The degradative pathways of two of these seven amino acids deserve special mention. Tryptophan breakdown is the most complex of all the pathways of amino

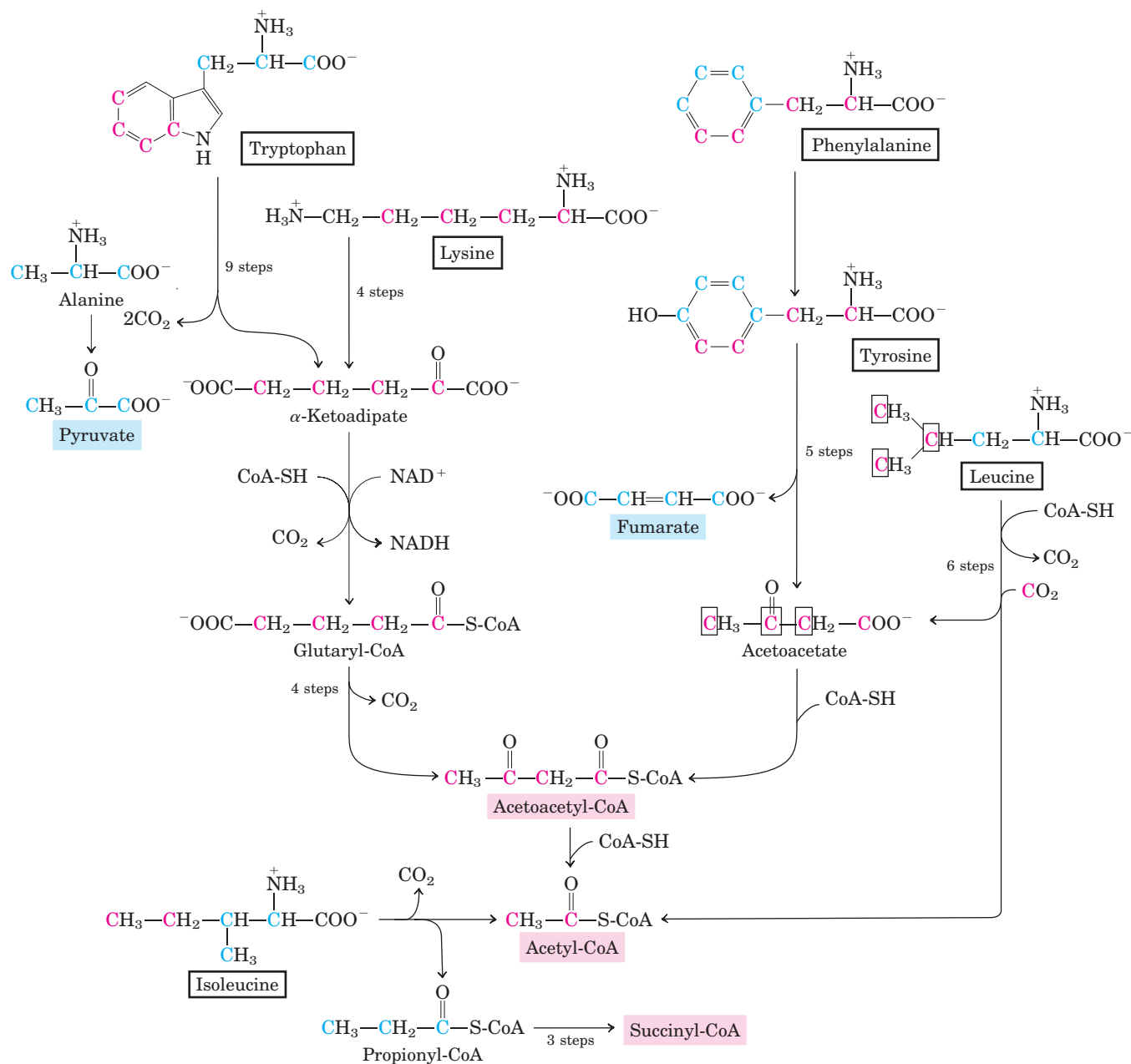


FIGURE 18-21 Catabolic pathways for tryptophan, lysine, phenylalanine, tyrosine, leucine, and isoleucine. These amino acids donate some of their carbons (red) to acetyl-CoA. Tryptophan, phenylalanine, tyrosine, and isoleucine also contribute carbons (blue) to pyruvate or

citric acid cycle intermediates. The phenylalanine pathway is described in more detail in Figure 18-23. The fate of nitrogen atoms is not traced in this scheme; in most cases they are transferred to α -ketoglutarate to form glutamate.

acid catabolism in animal tissues; portions of tryptophan (four of its carbons) yield acetyl-CoA via acetoacetyl-CoA. Some of the intermediates in tryptophan catabolism are precursors for the synthesis of other biomolecules (Fig. 18-22), including nicotinate, a precursor of NAD and NADP in animals; serotonin, a neurotransmitter in vertebrates; and indoleacetate, a growth factor in plants. Some of these biosynthetic pathways are described in more detail in Chapter 22 (see Figs 22-28, 22-29).

The breakdown of phenylalanine is noteworthy because genetic defects in the enzymes of this pathway lead to several inheritable human diseases (Fig. 18-23), as discussed below. Phenylalanine and its oxidation

product tyrosine (both with nine carbons) are degraded into two fragments, both of which can enter the citric acid cycle: four of the nine carbon atoms yield free acetoacetate, which is converted to acetoacetyl-CoA and thus acetyl-CoA, and a second four-carbon fragment is recovered as fumarate. Eight of the nine carbons of these two amino acids thus enter the citric acid cycle; the remaining carbon is lost as CO₂. Phenylalanine, after its hydroxylation to tyrosine, is also the precursor of dopamine, a neurotransmitter, and of norepinephrine and epinephrine, hormones secreted by the adrenal medulla (see Fig. 22-29). Melanin, the black pigment of skin and hair, is also derived from tyrosine.

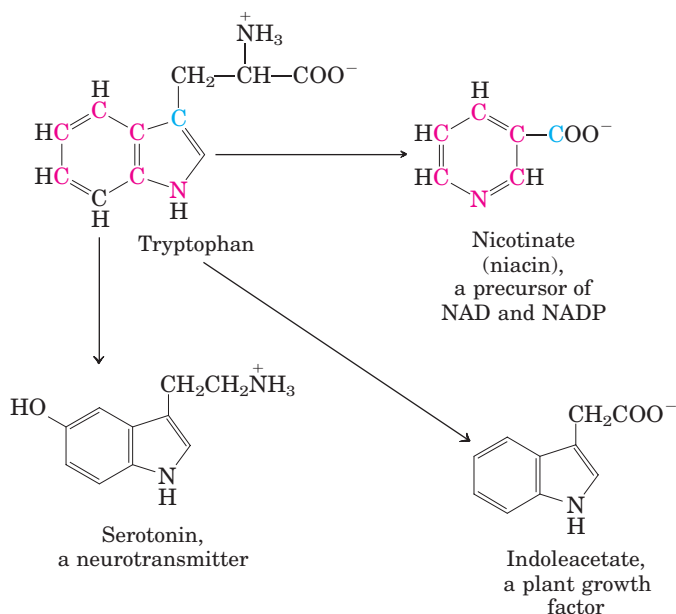


FIGURE 18-22 Tryptophan as precursor. The aromatic rings of tryptophan give rise to nicotinate, indoleacetate, and serotonin. Colored atoms trace the source of the ring atoms in nicotinate.

Phenylalanine Catabolism Is Genetically Defective in Some People



Given that many amino acids are either neurotransmitters or precursors or antagonists of neurotransmitters, genetic defects of amino acid metabolism can cause defective neural development and mental retardation. In most such diseases specific intermediates accumulate. For example, a genetic defect in **phenylalanine hydroxylase**, the first enzyme in the catabolic pathway for phenylalanine (Fig. 18-23), is responsible for the disease **phenylketonuria (PKU)**, the most common cause of elevated levels of phenylalanine (hyperphenylalaninemia).

Phenylalanine hydroxylase (also called phenylalanine-4-monooxygenase) is one of a general class of enzymes called **mixed-function oxidases** (see Box 21-1), all of which catalyze simultaneous hydroxylation of a substrate by an oxygen atom of O_2 and reduction of the other oxygen atom to H_2O . Phenylalanine hydroxylase

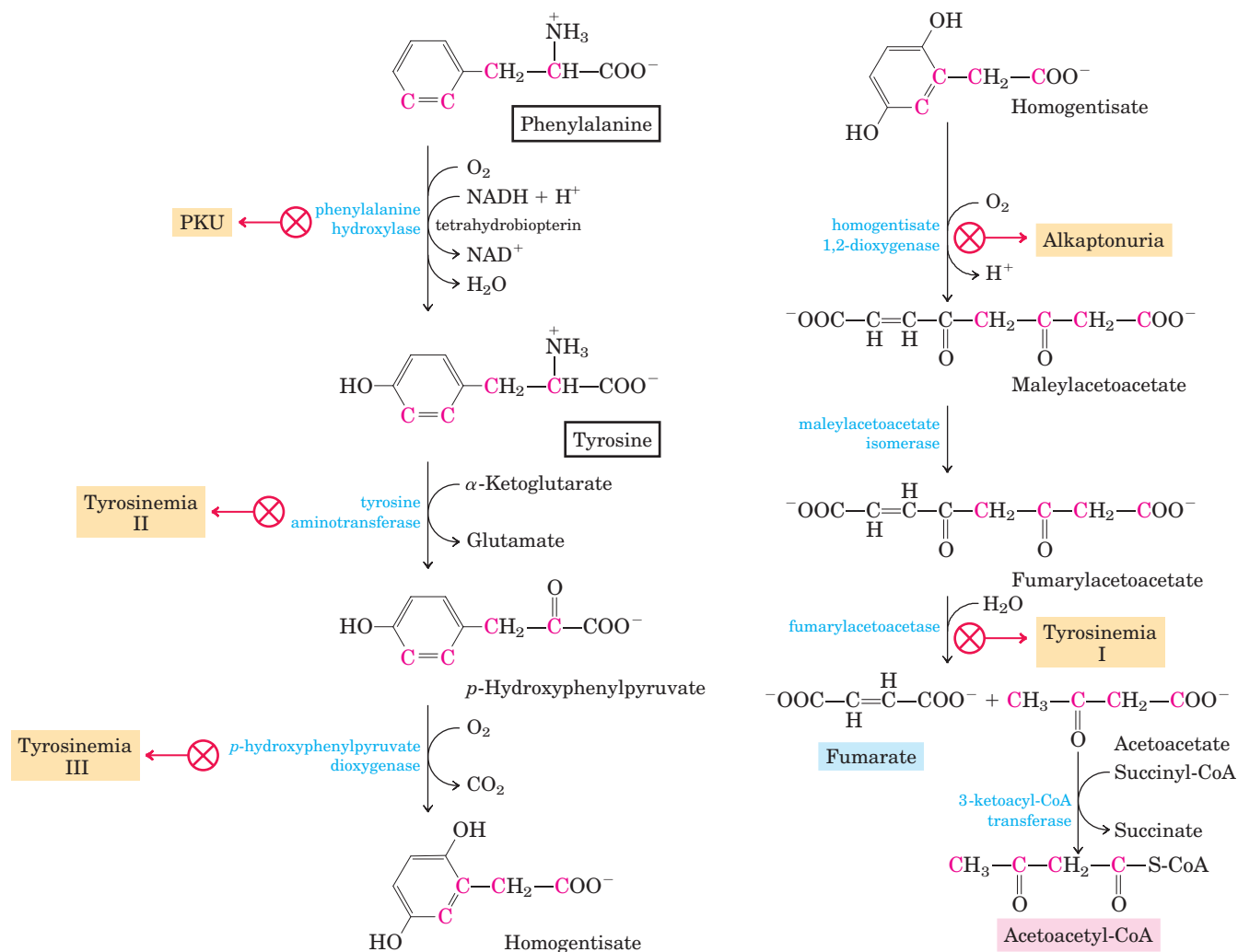


FIGURE 18-23 Catabolic pathways for phenylalanine and tyrosine. In humans these amino acids are normally con-

verted to acetoacetyl-CoA and fumarate. Genetic defects in many of these enzymes cause inheritable human diseases (shaded yellow).

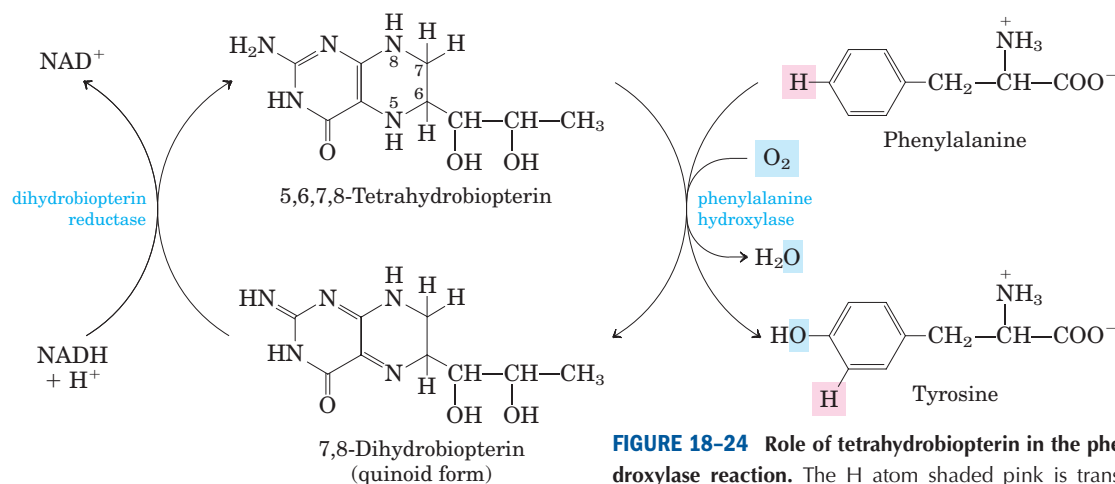


FIGURE 18-24 Role of tetrahydrobiopterin in the phenylalanine hydroxylase reaction. The H atom shaded pink is transferred directly from C-4 to C-3 in the reaction. This feature, discovered at the NIH, is called the NIH Shift.

requires the cofactor tetrahydrobiopterin, which carries electrons from NADH to O_2 and becomes oxidized to dihydrobiopterin in the process (Fig. 18-24). It is subsequently reduced by the enzyme **dihydrobiopterin reductase** in a reaction that requires NADH.

In individuals with PKU, a secondary, normally little-used pathway of phenylalanine metabolism comes into play. In this pathway phenylalanine undergoes transamination with pyruvate to yield **phenylpyruvate** (Fig. 18-25). Phenylalanine and phenylpyruvate accumulate in the blood and tissues and are excreted in the urine—hence the name “phenylketonuria.” Much of the phenylpyruvate, rather than being excreted as such, is either decarboxylated to phenylacetate or reduced to phenyllactate. Phenylacetate imparts a characteristic odor to the urine, which nurses have traditionally used to detect PKU in infants. The accumulation of phenylalanine or its metabolites in early life impairs normal development of the brain, causing severe mental retardation. This may be caused by excess phenylalanine competing with other amino acids for transport across the blood-brain barrier, resulting in a deficit of required metabolites.

Phenylketonuria was among the first inheritable metabolic defects discovered in humans. When this condition is recognized early in infancy, mental retardation can largely be prevented by rigid dietary control. The diet must supply only enough phenylalanine and tyrosine to meet the needs for protein synthesis. Consumption of protein-rich foods must be curtailed. Natural proteins, such as casein of milk, must first be hydrolyzed and much of the phenylalanine removed to provide an appropriate diet, at least through childhood. Because the artificial sweetener aspartame is a dipeptide of aspartate and the methyl ester of phenylalanine (see Fig. 1-23b), foods sweetened with aspartame bear warnings addressed to individuals on phenylalanine-controlled diets.

Phenylketonuria can also be caused by a defect in the enzyme that catalyzes the regeneration of tetrahydrobiopterin (Fig. 18-24). The treatment in this case is

more complex than restricting the intake of phenylalanine and tyrosine. Tetrahydrobiopterin is also required for the formation of L-3,4-dihydroxyphenylalanine (L-dopa) and 5-hydroxytryptophan—precursors of the neurotransmitters norepinephrine and serotonin, respectively—and in phenylketonuria of this type, these precursors must be supplied in the diet. Supplementing the diet with tetrahydrobiopterin itself is ineffective because it is unstable and does not cross the blood-brain barrier.

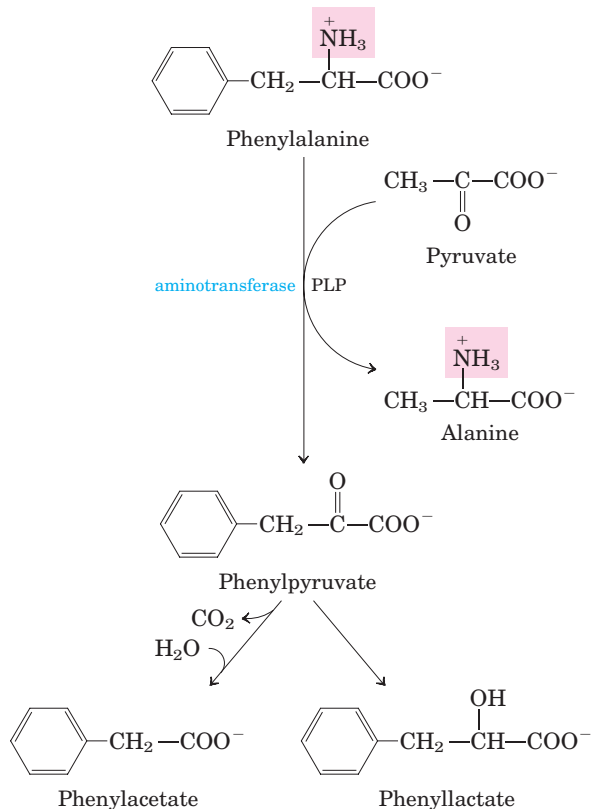


FIGURE 18-25 Alternative pathways for catabolism of phenylalanine in phenylketonuria. In PKU, phenylpyruvate accumulates in the tissues, blood, and urine. The urine may also contain phenylacetate and phenyllactate.

Screening newborns for genetic diseases can be highly cost-effective, especially in the case of PKU. The tests (no longer relying on urine odor) are relatively inexpensive, and the detection and early treatment of PKU in infants (eight to ten cases per 100,000 newborns) saves millions of dollars in later health care costs each year. More importantly, the emotional trauma avoided by early detection with these simple tests is inestimable.

Another inborn disease of phenylalanine catabolism is **alkaptonuria**, in which the defective enzyme is **homogentisate dioxygenase** (Fig. 18–23). Less serious than PKU, this condition produces few ill effects, although large amounts of homogentisate are excreted and its oxidation turns the urine black. Individuals with alkaptonuria are also prone to develop a form of arthri-

tis. Alkaptonuria is of considerable historical interest. Archibald Garrod discovered in the early 1900s that this condition is inherited, and he traced the cause to the absence of a single enzyme. Garrod was the first to make a connection between an inheritable trait and an enzyme, a great advance on the path that ultimately led to our current understanding of genes and the information pathways described in Part III. ■

Five Amino Acids Are Converted to α -Ketoglutarate

The carbon skeletons of five amino acids (proline, glutamate, glutamine, arginine, and histidine) enter the citric acid cycle as α -ketoglutarate (Fig. 18–26). **Proline**, **glutamate**, and **glutamine** have five-carbon skeletons. The cyclic structure of proline is opened by oxidation

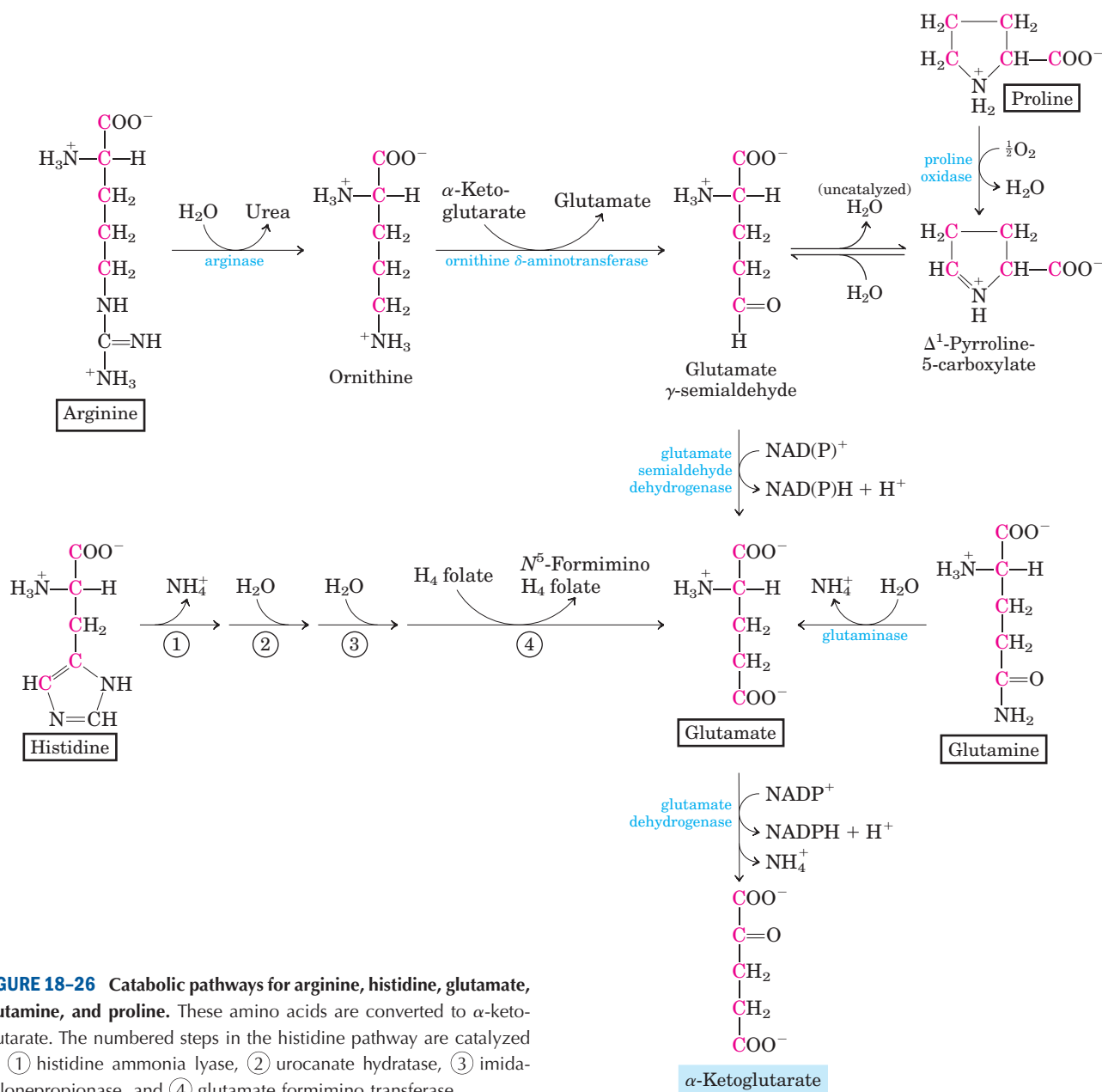


FIGURE 18–26 Catabolic pathways for arginine, histidine, glutamate, glutamine, and proline. These amino acids are converted to α -ketoglutarate. The numbered steps in the histidine pathway are catalyzed by ① histidine ammonia lyase, ② urocanate hydratase, ③ imidazolepropionase, and ④ glutamate formimino transferase.

of the carbon most distant from the carboxyl group to create a Schiff base, then hydrolysis of the Schiff base to a linear semialdehyde, glutamate γ -semialdehyde. This intermediate is further oxidized at the same carbon to produce glutamate. The action of glutaminase, or any of several enzyme reactions in which glutamine donates its amide nitrogen to an acceptor, converts glutamine to glutamate. Transamination or deamination of glutamate produces α -ketoglutarate.

Arginine and **histidine** contain five adjacent carbons and a sixth carbon attached through a nitrogen atom. The catabolic conversion of these amino acids to glutamate is therefore slightly more complex than the path from proline or glutamine (Fig. 18–26). Arginine is

converted to the five-carbon skeleton of ornithine in the urea cycle (Fig. 18–10), and the ornithine is transaminated to glutamate γ -semialdehyde. Conversion of histidine to the five-carbon glutamate occurs in a multistep pathway; the extra carbon is removed in a step that uses tetrahydrofolate as cofactor.

Four Amino Acids Are Converted to Succinyl-CoA

The carbon skeletons of methionine, isoleucine, threonine, and valine are degraded by pathways that yield succinyl-CoA (Fig. 18–27), an intermediate of the citric acid cycle. **Methionine** donates its methyl group to one of several possible acceptors through *S*-adenosylmethionine,

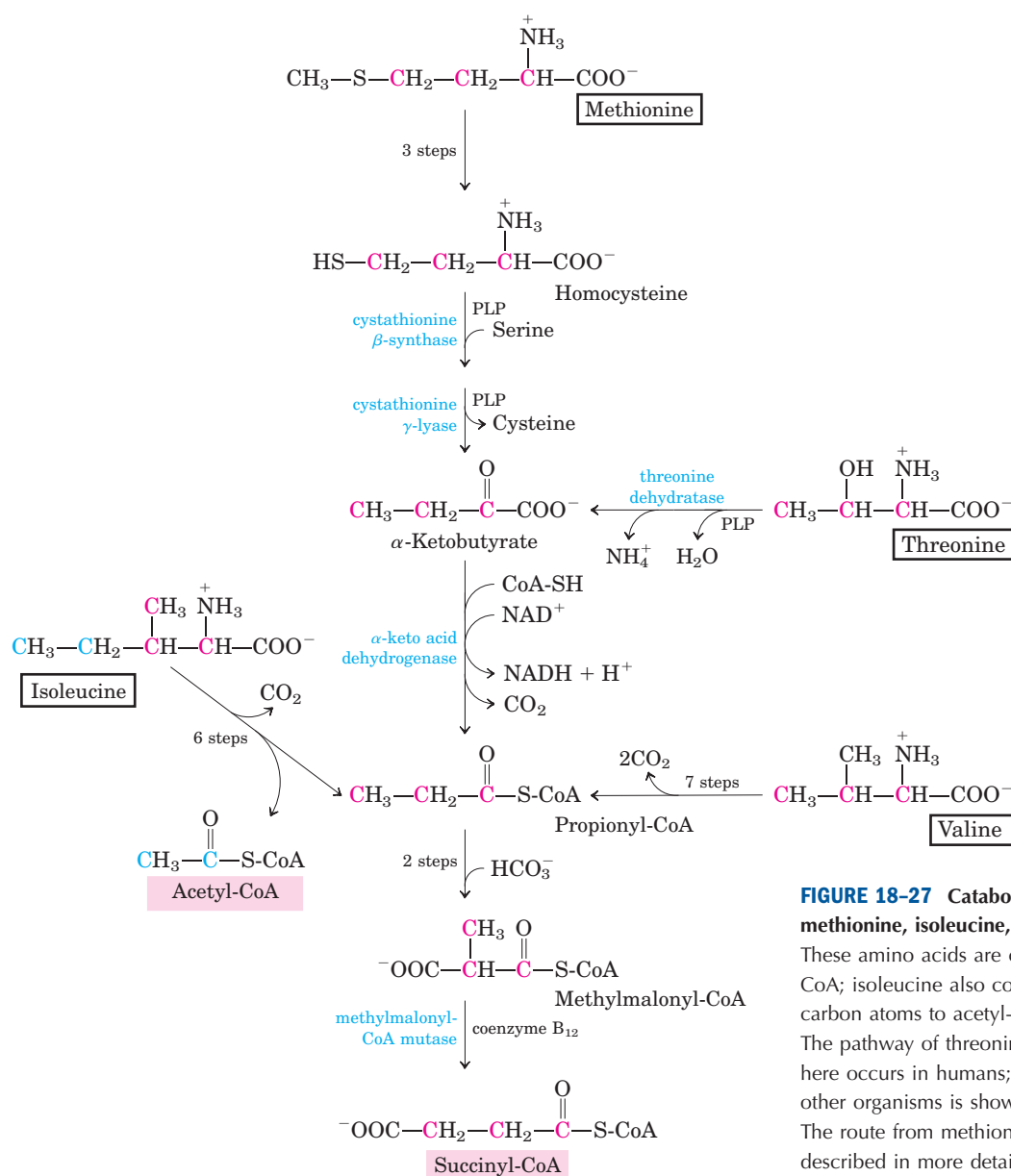


FIGURE 18–27 Catabolic pathways for methionine, isoleucine, threonine, and valine. These amino acids are converted to succinyl-CoA; isoleucine also contributes two of its carbon atoms to acetyl-CoA (see Fig. 18–21). The pathway of threonine degradation shown here occurs in humans; a pathway found in other organisms is shown in Figure 18–19. The route from methionine to homocysteine is described in more detail in Figure 18–18; the conversion of homocysteine to α -ketobutyrate in Figure 22–14; the conversion of propionyl-CoA to succinyl-CoA in Figure 17–11.

and three of its four remaining carbon atoms are converted to the propionate of propionyl-CoA, a precursor of succinyl-CoA. **Isoleucine** undergoes transamination, followed by oxidative decarboxylation of the resulting α -keto acid. The remaining five-carbon skeleton is further oxidized to acetyl-CoA and propionyl-CoA. **Valine** undergoes transamination and decarboxylation, then a series of oxidation reactions that convert the remaining four carbons to propionyl-CoA. Some parts of the valine and isoleucine degradative pathways closely parallel steps in fatty acid degradation (see Fig. 17–8a). In human tissues, **threonine** is also converted in two steps to propionyl-CoA. This is the primary pathway for threonine degradation in humans (see Fig. 18–19 for the alternative pathway). The mechanism of the first step is analogous to that catalyzed by serine dehydratase, and the serine and threonine dehydratases may actually be the same enzyme.

The propionyl-CoA derived from these three amino acids is converted to succinyl-CoA by a pathway described in Chapter 17: carboxylation to methylmalonyl-CoA, epimerization of the methylmalonyl-CoA, and conversion to succinyl-CoA by the coenzyme B₁₂-dependent methylmalonyl-CoA mutase (see Fig. 17–11). In the rare genetic disease known as methylmalonic acidemia, methylmalonyl-CoA mutase is lacking—with serious metabolic consequences (Table 18–2; Box 18–2).

Branched-Chain Amino Acids Are Not Degraded in the Liver

Although much of the catabolism of amino acids takes place in the liver, the three amino acids with branched side chains (leucine, isoleucine, and valine) are oxidized as fuels primarily in muscle, adipose, kidney, and brain tissue. These extrahepatic tissues contain an aminotransferase, absent in liver, that acts on all three branched-chain amino acids to produce the corresponding α -keto acids (Fig. 18–28). The **branched-chain α -keto acid dehydrogenase complex** then catalyzes oxidative decarboxylation of all three α -keto acids, in each case releasing the carboxyl group as CO₂ and producing the acyl-CoA derivative. This reaction is formally analogous to two other oxidative decarboxylations encountered in Chapter 16: oxidation of pyruvate to acetyl-CoA by the pyruvate dehydrogenase complex (see Fig. 16–6) and oxidation of α -ketoglutarate to succinyl-CoA by the α -ketoglutarate dehydrogenase complex (p. 610). In fact, all three enzyme complexes are similar in structure and share essentially the same reaction mechanism. Five cofactors (thiamine pyrophosphate, FAD, NAD, lipoate, and coenzyme A) participate, and the three proteins in each complex catalyze homologous reactions. This is clearly a case in which enzymatic machinery that evolved to catalyze

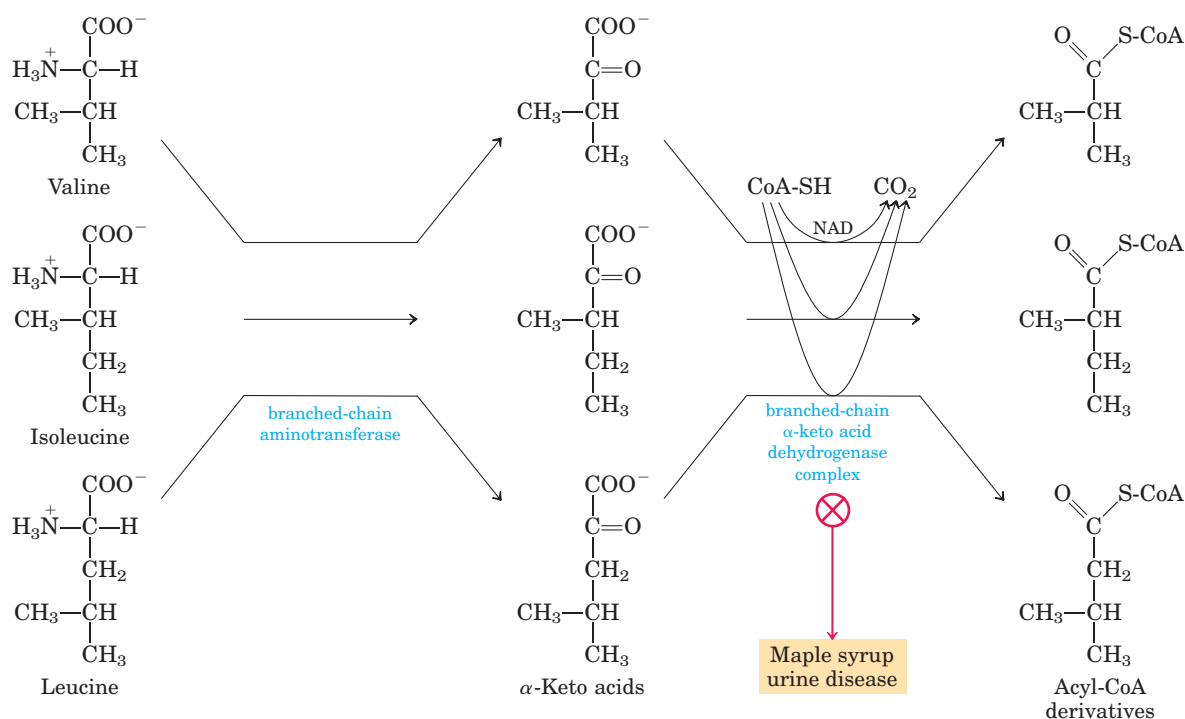


FIGURE 18–28 Catabolic pathways for the three branched-chain amino acids: valine, isoleucine, and leucine. The three pathways, which occur in extrahepatic tissues, share the first two enzymes, as shown here. The branched-chain α -keto acid dehydrogenase complex

is analogous to the pyruvate and α -ketoglutarate dehydrogenase complexes and requires the same five cofactors (some not shown here). This enzyme is defective in people with maple syrup urine disease.



BOX 18-2 BIOCHEMISTRY IN MEDICINE

Scientific Sleuths Solve a Murder Mystery

Truth can sometimes be stranger than fiction—or at least as strange as a made-for-TV movie. Take, for example, the case of Patricia Stallings. Convicted of the murder of her infant son, she was sentenced to life in prison—but was later found innocent, thanks to the medical sleuthing of three persistent researchers.

The story began in the summer of 1989 when Stallings brought her three-month-old son, Ryan, to the emergency room of Cardinal Glennon Children's Hospital in St. Louis. The child had labored breathing, uncontrollable vomiting, and gastric distress. According to the attending physician, a toxicologist, the child's symptoms indicated that he had been poisoned with ethylene glycol, an ingredient of antifreeze, a conclusion apparently confirmed by analysis at a commercial lab.

After he recovered, the child was placed in a foster home, and Stallings and her husband, David, were allowed to see him in supervised visits. But when the infant became ill, and subsequently died, after a visit in which Stallings had been briefly left alone with him, she was charged with first-degree murder and held without bail. At the time, the evidence seemed compelling as both the commercial lab and the hospital lab found large amounts of ethylene glycol in the boy's blood and traces of it in a bottle of milk Stallings had fed her son during the visit.

But without knowing it, Stallings had performed a brilliant experiment. While in custody, she learned she was pregnant; she subsequently gave birth to another son, David Stallings Jr., in February 1990. He was placed immediately in a foster home, but within two weeks he started having symptoms similar to Ryan's. David was eventually diagnosed with a rare metabolic disorder called methylmalonic acidemia (MMA). A recessive genetic disorder of amino acid metabolism, MMA affects about 1 in 48,000 newborns and presents symptoms almost identical with those caused by ethylene glycol poisoning.

Stallings couldn't possibly have poisoned her second son, but the Missouri state prosecutor's office was not impressed by the new developments and pressed forward with her trial anyway. The court wouldn't allow the MMA diagnosis of the second child to be introduced as evidence, and in January 1991 Patricia Stallings was convicted of assault with a deadly weapon and sentenced to life in prison.

Fortunately for Stallings, however, William Sly, chairman of the Department of Biochemistry and Mol-

ecular Biology at St. Louis University, and James Shoemaker, head of a metabolic screening lab at the university, got interested in her case when they heard about it from a television broadcast. Shoemaker performed his own analysis of Ryan's blood and didn't detect ethylene glycol. He and Sly then contacted Piero Rinaldo, a metabolic disease expert at Yale University School of Medicine whose lab is equipped to diagnose MMA from blood samples.

When Rinaldo analyzed Ryan's blood serum, he found high concentrations of methylmalonic acid, a breakdown product of the branched-chain amino acids isoleucine and valine, which accumulates in MMA patients because the enzyme that should convert it to the next product in the metabolic pathway is defective. And particularly telling, he says, the child's blood and urine contained massive amounts of ketones, another metabolic consequence of the disease. Like Shoemaker, he did not find any ethylene glycol in a sample of the baby's bodily fluids. The bottle couldn't be tested, since it had mysteriously disappeared. Rinaldo's analyses convinced him that Ryan had died from MMA, but how to account for the results from two labs, indicating that the boy had ethylene glycol in his blood? Could they both be wrong?

When Rinaldo obtained the lab reports, what he saw was, he says, "scary." One lab said that Ryan Stallings' blood contained ethylene glycol, even though the blood sample analysis did not match the lab's own profile for a known sample containing ethylene glycol. "This was not just a matter of questionable interpretation. The quality of their analysis was unacceptable," Rinaldo says. And the second laboratory? According to Rinaldo, that lab detected an abnormal component in Ryan's blood and just "assumed it was ethylene glycol." Samples from the bottle had produced nothing unusual, says Rinaldo, yet the lab claimed evidence of ethylene glycol in that, too.

Rinaldo presented his findings to the case's prosecutor, George McElroy, who called a press conference the very next day. "I no longer believe the laboratory data," he told reporters. Having concluded that Ryan Stallings had died of MMA after all, McElroy dismissed all charges against Patricia Stallings on September 20, 1991.

By Michelle Hoffman (1991). *Science* **253**, 931. Copyright 1991 by the American Association for the Advancement of Science.

one reaction was “borrowed” by gene duplication and further evolved to catalyze similar reactions in other pathways.

Experiments with rats have shown that the branched-chain α -keto acid dehydrogenase complex is regulated by covalent modification in response to the content of branched-chain amino acids in the diet. With little or no excess dietary intake of branched-chain amino acids, the enzyme complex is phosphorylated and thereby inactivated by a protein kinase. Addition of excess branched-chain amino acids to the diet results in dephosphorylation and consequent activation of the enzyme. Recall that the pyruvate dehydrogenase complex is subject to similar regulation by phosphorylation and dephosphorylation (p. 621).

 There is a relatively rare genetic disease in which the three branched-chain α -keto acids (as well as their precursor amino acids, especially leucine) accumulate in the blood and “spill over” into the urine. This condition, called **maple syrup urine disease** because of the characteristic odor imparted to the urine by the α -keto acids, results from a defective branched-chain α -keto acid dehydrogenase complex. Untreated, the disease results in abnormal development of the brain, mental retardation, and death in early infancy. Treatment entails rigid control of the diet, limiting the intake of valine, isoleucine, and leucine to the minimum required to permit normal growth. ■

Asparagine and Aspartate Are Degraded to Oxaloacetate

The carbon skeletons of **asparagine** and **aspartate** ultimately enter the citric acid cycle as oxaloacetate. The enzyme **asparaginase** catalyzes the hydrolysis of asparagine to aspartate, which undergoes transamination with α -ketoglutarate to yield glutamate and oxaloacetate (Fig. 18–29).

We have now seen how the 20 common amino acids, after losing their nitrogen atoms, are degraded by dehydrogenation, decarboxylation, and other reactions to yield portions of their carbon backbones in the form of six central metabolites that can enter the citric acid cycle. Those portions degraded to acetyl-CoA are completely oxidized to carbon dioxide and water, with generation of ATP by oxidative phosphorylation.

As was the case for carbohydrates and lipids, the degradation of amino acids results ultimately in the generation of reducing equivalents (NADH and FADH₂) through the action of the citric acid cycle. Our survey of catabolic processes concludes in the next chapter with a discussion of respiration, in which these reducing equivalents fuel the ultimate oxidative and energy-generating process in aerobic organisms.

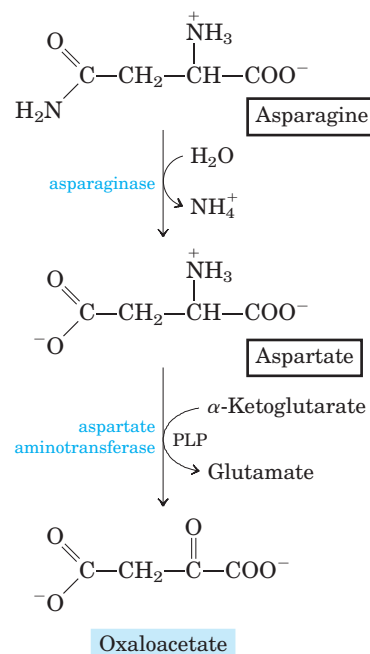


FIGURE 18–29 Catabolic pathway for asparagine and aspartate. Both amino acids are converted to oxaloacetate.

SUMMARY 18.3 Pathways of Amino Acid Degradation

- After removal of their amino groups, the carbon skeletons of amino acids undergo oxidation to compounds that can enter the citric acid cycle for oxidation to CO₂ and H₂O. The reactions of these pathways require a number of cofactors, including tetrahydrofolate and *S*-adenosylmethionine in one-carbon transfer reactions and tetrahydrobiopterin in the oxidation of phenylalanine by phenylalanine hydroxylase.
- Depending on their degradative end product, some amino acids can be converted to ketone bodies, some to glucose, and some to both. Thus amino acid degradation is integrated into intermediary metabolism and can be critical to survival under conditions in which amino acids are a significant source of metabolic energy.
- The carbon skeletons of amino acids enter the citric acid cycle through five intermediates: acetyl-CoA, α -ketoglutarate, succinyl-CoA, fumarate, and oxaloacetate. Some are also degraded to pyruvate, which can be converted to either acetyl-CoA or oxaloacetate.

- The amino acids producing pyruvate are alanine, cysteine, glycine, serine, threonine, and tryptophan. Leucine, lysine, phenylalanine, and tryptophan yield acetyl-CoA via acetoacetyl-CoA. Isoleucine, leucine, threonine, and tryptophan also form acetyl-CoA directly.
- Arginine, glutamate, glutamine, histidine, and proline produce α -ketoglutarate; isoleucine, methionine, threonine, and valine produce succinyl-CoA; four carbon atoms of phenylalanine and tyrosine give rise to fumarate; and asparagine and aspartate produce oxaloacetate.
- The branched-chain amino acids (isoleucine, leucine, and valine), unlike the other amino acids, are degraded only in extrahepatic tissues.
- A number of serious human diseases can be traced to genetic defects in the enzymes of amino acid catabolism.

Key Terms

Terms in bold are defined in the glossary.

aminotransferases 660	glucose-alanine cycle 664	tetrahydrofolate 672
transaminases 660	ammonotelic 665	S-adenosylmethionine (adoMet) 672
transamination 660	ureotelic 665	tetrahydrobiopterin 674
pyridoxal phosphate (PLP) 660	uricotelic 665	phenylketonuria (PKU) 679
oxidative deamination 661	urea cycle 665	mixed-function oxidases 679
L-glutamate dehydrogenase 661	urea 668	alkaptonuria 681
glutamine synthetase 662	essential amino acids 669	maple syrup urine disease 685
glutaminase 663	ketogenic 672	
creatine kinase 664	glucogenic 672	

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Problems

1. Products of Amino Acid Transamination Name and draw the structure of the α -keto acid resulting when each of the following amino acids undergoes transamination with α -ketoglutarate: (a) aspartate, (b) glutamate, (c) alanine, (d) phenylalanine.

2. Measurement of Alanine Aminotransferase Activity The activity (reaction rate) of alanine aminotransferase is usually measured by including an excess of pure lactate dehydrogenase and NADH in the reaction system. The rate of alanine disappearance is equal to the rate of NADH disappearance measured spectrophotometrically. Explain how this assay works.

3. Distribution of Amino Nitrogen If your diet is rich in alanine but deficient in aspartate, will you show signs of aspartate deficiency? Explain.



4. A Genetic Defect in Amino Acid Metabolism: A Case History A two-year-old child was taken to the hospital. His mother said that he vomited frequently, especially after feedings. The child's weight and physical development were below normal. His hair, although dark, contained patches of white. A urine sample treated with ferric chloride (FeCl_3) gave a green color characteristic of the presence of phenylpyruvate. Quantitative analysis of urine samples gave the results shown in the table.

Substance	Concentration (mM)	
	Patient's urine	Normal urine
Phenylalanine	7.0	0.01
Phenylpyruvate	4.8	0
Phenyllactate	10.3	0

(a) Suggest which enzyme might be deficient in this child. Propose a treatment.

(b) Why does phenylalanine appear in the urine in large amounts?

(c) What is the source of phenylpyruvate and phenyllactate? Why does this pathway (normally not functional)

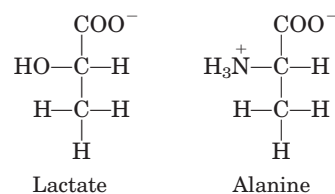
come into play when the concentration of phenylalanine rises?

(d) Why does the boy's hair contain patches of white?

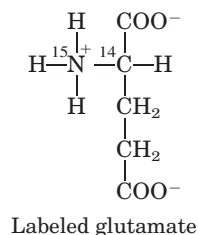


5. Role of Cobalamin in Amino Acid Catabolism Pernicious anemia is caused by impaired absorption of vitamin B_{12} . What is the effect of this impairment on the catabolism of amino acids? Are all amino acids equally affected? (Hint: See Box 17–2.)

6. Lactate versus Alanine as Metabolic Fuel: The Cost of Nitrogen Removal The three carbons in lactate and alanine have identical oxidation states, and animals can use either carbon source as a metabolic fuel. Compare the net ATP yield (moles of ATP per mole of substrate) for the complete oxidation (to CO_2 and H_2O) of lactate versus alanine when the cost of nitrogen excretion as urea is included.

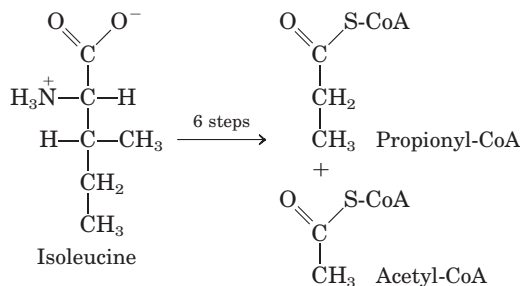


7. Pathway of Carbon and Nitrogen in Glutamate Metabolism When $[2\text{-}^{14}\text{C}, ^{15}\text{N}]$ glutamate undergoes oxidative degradation in the liver of a rat, in which atoms of the following metabolites will each isotope be found: (a) urea, (b) succinate, (c) arginine, (d) citrulline, (e) ornithine, (f) aspartate?

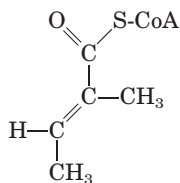
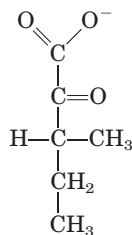
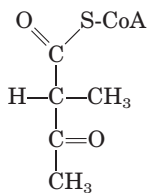
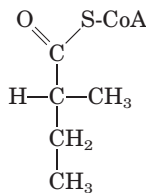
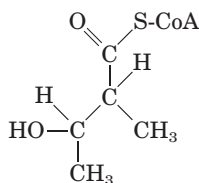


8. Chemical Strategy of Isoleucine Catabolism

Isoleucine is degraded in six steps to propionyl-CoA and acetyl-CoA:



(a) The chemical process of isoleucine degradation includes strategies analogous to those used in the citric acid cycle and the β oxidation of fatty acids. The intermediates of isoleucine degradation (I to V) shown below are not in the proper order. Use your knowledge and understanding of the citric acid cycle and β -oxidation pathway to arrange the intermediates in the proper metabolic sequence for isoleucine degradation.

**I****II****III****IV****V**

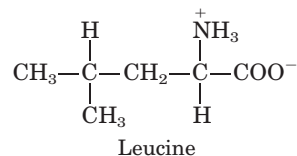
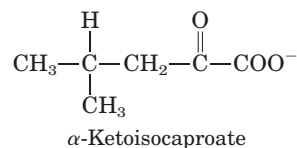
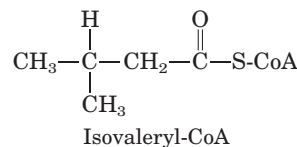
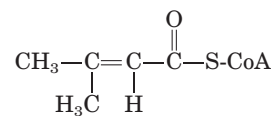
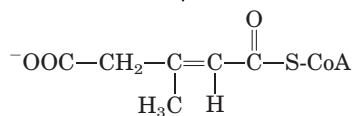
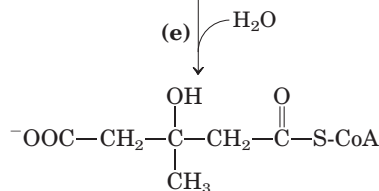
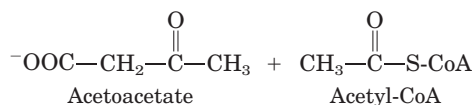
(b) For each step you propose, describe the chemical process, provide an analogous example from the citric acid cycle or β -oxidation pathway (where possible), and indicate any necessary cofactors.

9. Role of Pyridoxal Phosphate in Glycine Metabolism

The enzyme serine hydroxymethyltransferase requires pyridoxal phosphate as cofactor. Propose a mechanism for the reaction catalyzed by this enzyme, in the direction of serine degradation (glycine production). (Hint: See Figs 18–19 and 18–20b.)

10. Parallel Pathways for Amino Acid and Fatty Acid Degradation

The carbon skeleton of leucine is degraded by a series of reactions closely analogous to those of the citric acid cycle and β oxidation. For each reaction, (a) through (f), indicate its type, provide an analogous example from the citric acid cycle or β -oxidation pathway (where possible), and note any necessary cofactors.

**(a)****(b)** CoA-SH $\rightarrow \text{CO}_2$ **(c)****(d)** HCO_3^- **(e)** H_2O **(f)**

11. Ammonia Toxicity Resulting from an Arginine-Deficient Diet In a study conducted some years ago, cats were fasted overnight then given a single meal complete in all amino acids except arginine. Within 2 hours, blood ammonia levels increased from a normal level of $18 \mu\text{g/L}$ to $140 \mu\text{g/L}$, and the cats showed the clinical symptoms of ammonia toxicity. A control group fed a complete amino acid diet or an amino acid diet in which arginine was replaced by ornithine showed no unusual clinical symptoms.

- What was the role of fasting in the experiment?
- What caused the ammonia levels to rise in the experimental group? Why did the absence of arginine lead to ammonia toxicity? Is arginine an essential amino acid in cats? Why or why not?
- Why can ornithine be substituted for arginine?

12. Oxidation of Glutamate Write a series of balanced equations, and an overall equation for the net reaction, describing the oxidation of 2 mol of glutamate to 2 mol of α -ketoglutarate and 1 mol of urea.

13. Transamination and the Urea Cycle Aspartate aminotransferase has the highest activity of all the mammalian liver aminotransferases. Why?

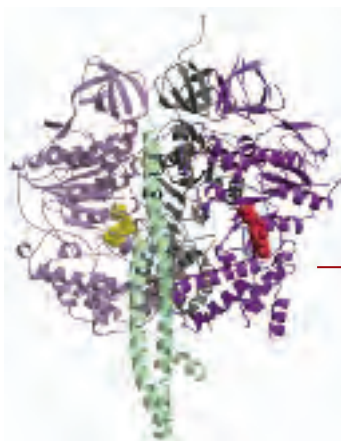


14. The Case against the Liquid Protein Diet A weight-reducing diet heavily promoted some years ago required the daily intake of “liquid protein” (soup of hydrolyzed gelatin), water, and an assortment of vitamins. All other food and drink were to be avoided. People on this diet typically lost 10 to 14 lb in the first week.

(a) Opponents argued that the weight loss was almost entirely due to water loss and would be regained very soon after a normal diet was resumed. What is the biochemical basis for this argument?

(b) A number of people on this diet died. What are some of the dangers inherent in the diet, and how can they lead to death?

15. Alanine and Glutamine in the Blood Normal human blood plasma contains all the amino acids required for the synthesis of body proteins, but not in equal concentrations. Alanine and glutamine are present in much higher concentrations than any other amino acids. Suggest why.



chapter 19

OXIDATIVE PHOSPHORYLATION AND PHOTOPHOSPHORYLATION

OXIDATIVE PHOSPHORYLATION

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If an idea presents itself to us, we must not reject it simply because it does not agree with the logical deductions of a reigning theory.

—Claude Bernard, *An Introduction to the Study of Experimental Medicine*, 1813

The aspect of the present position of consensus that I find most remarkable and admirable, is the altruism and generosity with which former opponents of the chemiosmotic hypothesis have not only come to accept it, but have actively promoted it to the status of a theory.

—Peter Mitchell, *Nobel Address*, 1978

Oxidative phosphorylation is the culmination of energy-yielding metabolism in aerobic organisms. All oxidative steps in the degradation of carbohydrates, fats, and amino acids converge at this final stage of cellular respiration, in which the energy of oxidation drives the synthesis of ATP. Photophosphorylation is the means by which photosynthetic organisms capture the energy of sunlight—the ultimate source of energy in the biosphere—and harness it to make ATP. Together, oxidative phosphorylation and photophosphorylation account for most of the ATP synthesized by most organisms most of the time.

In eukaryotes, oxidative phosphorylation occurs in mitochondria, photophosphorylation in chloroplasts. Oxidative phosphorylation involves the *reduction* of O_2 to H_2O with electrons donated by NADH and $FADH_2$; it occurs equally well in light or darkness. Photophosphorylation involves the *oxidation* of H_2O to O_2 , with $NADP^+$ as ultimate electron acceptor; it is absolutely dependent on the energy of light. Despite their differences, these two highly efficient energy-converting processes have fundamentally similar mechanisms.

Our current understanding of ATP synthesis in mitochondria and chloroplasts is based on the hypothesis, introduced by Peter Mitchell in 1961, that transmembrane differences in proton concentration are the reservoir for the energy extracted from biological oxidation reactions. This **chemiosmotic theory** has been accepted as one of the great unifying principles of twentieth century biology. It provides insight into the processes of oxidative phosphorylation and photophosphorylation, and into such apparently disparate energy transductions as active transport across membranes and the motion of bacterial flagella.

Oxidative phosphorylation and photophosphorylation are mechanistically similar in three respects. (1) Both

processes involve the flow of electrons through a chain of membrane-bound carriers. (2) The free energy made available by this “downhill” (exergonic) electron flow is coupled to the “uphill” transport of protons across a proton-impermeable membrane, conserving the free energy of fuel oxidation as a transmembrane electrochemical potential (p. 391). (3) The transmembrane flow of protons down their concentration gradient through specific protein channels provides the free energy for synthesis of ATP, catalyzed by a membrane protein complex (ATP synthase) that couples proton flow to phosphorylation of ADP.

We begin this chapter with oxidative phosphorylation. We first describe the components of the electron-transfer chain, their organization into large functional complexes in the inner mitochondrial membrane, the path of electron flow through them, and the proton movements that accompany this flow. We then consider the remarkable enzyme complex that, by “rotational catalysis,” captures the energy of proton flow in ATP, and the regulatory mechanisms that coordinate oxidative phosphorylation with the many catabolic pathways by which fuels are oxidized. With this understanding of mitochondrial oxidative phosphorylation, we turn to photophosphorylation, looking first at the absorption of light by photosynthetic pigments, then at the light-driven flow of electrons from H_2O to NADP^+ and the molecular basis for coupling electron and proton flow. We also consider the similarities of structure and mechanism between the ATP synthases of chloroplasts and mitochondria, and the evolutionary basis for this conservation of mechanism.

OXIDATIVE PHOSPHORYLATION

19.1 Electron-Transfer Reactions in Mitochondria

The discovery in 1948 by Eugene Kennedy and Albert Lehninger that mitochondria are the site of oxidative phosphorylation in eukaryotes marked the beginning of the modern phase of studies in biological energy transductions. Mitochondria, like gram-negative bacteria, have two membranes (Fig. 19–1). The outer mitochondrial membrane is readily permeable to small molecules ($M_r < 5,000$) and ions, which move freely through transmembrane channels formed by a family of integral membrane proteins called porins. The inner membrane is impermeable to most small



Albert L. Lehninger,
1917–1986

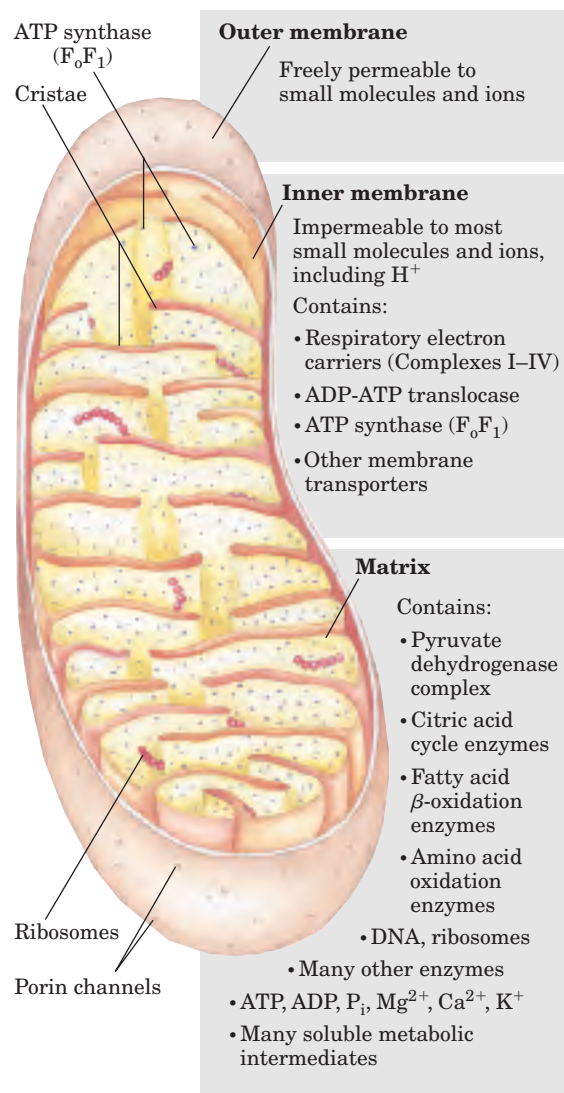


FIGURE 19–1 Biochemical anatomy of a mitochondrion. The convolutions (cristae) of the inner membrane provide a very large surface area. The inner membrane of a single liver mitochondrion may have more than 10,000 sets of electron-transfer systems (respiratory chains) and ATP synthase molecules, distributed over the membrane surface. Heart mitochondria, which have more profuse cristae and thus a much larger area of inner membrane, contain more than three times as many sets of electron-transfer systems as liver mitochondria. The mitochondrial pool of coenzymes and intermediates is functionally separate from the cytosolic pool. The mitochondria of invertebrates, plants, and microbial eukaryotes are similar to those shown here, but with much variation in size, shape, and degree of convolution of the inner membrane.

molecules and ions, including protons (H^+); the only species that cross this membrane do so through specific transporters. The inner membrane bears the components of the respiratory chain and the ATP synthase.

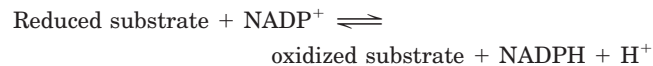
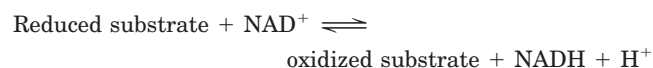
The mitochondrial matrix, enclosed by the inner membrane, contains the pyruvate dehydrogenase complex and the enzymes of the citric acid cycle, the fatty

acid β -oxidation pathway, and the pathways of amino acid oxidation—all the pathways of fuel oxidation except glycolysis, which takes place in the cytosol. The selectively permeable inner membrane segregates the intermediates and enzymes of cytosolic metabolic pathways from those of metabolic processes occurring in the matrix. However, specific transporters carry pyruvate, fatty acids, and amino acids or their α -keto derivatives into the matrix for access to the machinery of the citric acid cycle. ADP and P_i are specifically transported into the matrix as newly synthesized ATP is transported out.

Electrons Are Funneled to Universal Electron Acceptors

Oxidative phosphorylation begins with the entry of electrons into the respiratory chain. Most of these electrons arise from the action of dehydrogenases that collect electrons from catabolic pathways and funnel them into universal electron acceptors—nicotinamide nucleotides (NAD^+ or $NADP^+$) or flavin nucleotides (FMN or FAD).

Nicotinamide nucleotide-linked dehydrogenases catalyze reversible reactions of the following general types:



Most dehydrogenases that act in catabolism are specific for NAD^+ as electron acceptor (Table 19–1). Some are

in the cytosol, others are in mitochondria, and still others have mitochondrial and cytosolic isozymes.

NAD-linked dehydrogenases remove two hydrogen atoms from their substrates. One of these is transferred as a hydride ion ($:H^-$) to NAD^+ ; the other is released as H^+ in the medium (see Fig. 13–15). NADH and NADPH are water-soluble electron carriers that associate *reversibly* with dehydrogenases. NADH carries electrons from catabolic reactions to their point of entry into the respiratory chain, the NADH dehydrogenase complex described below. NADPH generally supplies electrons to anabolic reactions. Cells maintain separate pools of NADPH and NADH, with different redox potentials. This is accomplished by holding the ratios of [reduced form]/[oxidized form] relatively high for NADPH and relatively low for NADH. Neither NADH nor NADPH can cross the inner mitochondrial membrane, but the electrons they carry can be shuttled across indirectly, as we shall see.

Flavoproteins contain a very tightly, sometimes covalently, bound flavin nucleotide, either FMN or FAD (see Fig. 13–18). The oxidized flavin nucleotide can accept either one electron (yielding the semiquinone form) or two (yielding $FADH_2$ or $FMNH_2$). Electron transfer occurs because the flavoprotein has a higher reduction potential than the compound oxidized. The standard reduction potential of a flavin nucleotide, unlike that of NAD or NADP, depends on the protein with which it is associated. Local interactions with functional groups in the protein distort the electron orbitals in the flavin ring, changing the relative stabilities of oxidized and reduced forms. The relevant standard reduction

TABLE 19–1 Some Important Reactions Catalyzed by NAD(P)H-Linked Dehydrogenases

Reaction*	Location†
NAD-linked	
α -Ketoglutarate + CoA + NAD^+ $\rightleftharpoons$ succinyl-CoA + CO_2 + NADH + H^+	M
L-Malate + NAD^+ $\rightleftharpoons$ oxaloacetate + NADH + H^+	M and C
Pyruvate + CoA + NAD^+ $\rightleftharpoons$ acetyl-CoA + CO_2 + NADH + H^+	M
Glyceraldehyde 3-phosphate + P_i + NAD^+ $\rightleftharpoons$ 1,3-bisphosphoglycerate + NADH + H^+	C
Lactate + NAD^+ $\rightleftharpoons$ pyruvate + NADH + H^+	C
β -Hydroxyacyl-CoA + NAD^+ $\rightleftharpoons$ β -ketoacyl-CoA + NADH + H^+	M
NADP-linked	
Glucose 6-phosphate + $NADP^+$ $\rightleftharpoons$ 6-phosphogluconate + NADPH + H^+	C
NAD- or NADP-linked	
L-Glutamate + H_2O + $NAD(P)^+$ $\rightleftharpoons$ α -ketoglutarate + NH_4^+ + NAD(P)H	M
Isocitrate + $NAD(P)^+$ $\rightleftharpoons$ α -ketoglutarate + CO_2 + NAD(P)H + H^+	M and C

*These reactions and their enzymes are discussed in Chapters 14 through 18.

†M designates mitochondria; C, cytosol.

potential is therefore that of the particular flavoprotein, not that of isolated FAD or FMN. The flavin nucleotide should be considered part of the flavoprotein's active site rather than a reactant or product in the electron-transfer reaction. Because flavoproteins can participate in either one- or two-electron transfers, they can serve as intermediates between reactions in which two electrons are donated (as in dehydrogenations) and those in which only one electron is accepted (as in the reduction of a quinone to a hydroquinone, described below).

Electrons Pass through a Series of Membrane-Bound Carriers

The mitochondrial respiratory chain consists of a series of sequentially acting electron carriers, most of which are integral proteins with prosthetic groups capable of accepting and donating either one or two electrons. Three types of electron transfers occur in oxidative phosphorylation: (1) direct transfer of electrons, as in the reduction of Fe^{3+} to Fe^{2+} ; (2) transfer as a hydrogen atom ($\text{H}^+ + e^-$); and (3) transfer as a hydride ion ($:\text{H}^-$), which bears two electrons. The term **reducing equivalent** is used to designate a single electron equivalent transferred in an oxidation-reduction reaction.

In addition to NAD and flavoproteins, three other types of electron-carrying molecules function in the respiratory chain: a hydrophobic quinone (ubiquinone) and two different types of iron-containing proteins (cytochromes and iron-sulfur proteins). **Ubiquinone** (also called **coenzyme Q**, or simply **Q**) is a lipid-soluble benzoquinone with a long isoprenoid side chain (Fig. 19-2). The closely related compounds plastoquinone (of plant chloroplasts) and menaquinone (of bacteria) play roles analogous to that of ubiquinone, carrying electrons in membrane-associated electron-transfer chains. Ubiquinone can accept one electron to become the semiquinone radical ($^{\bullet}\text{QH}$) or two electrons to form ubiquinol (QH_2) (Fig. 19-2) and, like flavoprotein carriers, it can act at the junction between a two-electron donor and a one-electron acceptor. Because ubiquinone is both small and hydrophobic, it is freely diffusible within the lipid bilayer of the inner mitochondrial membrane and can shuttle reducing equivalents between other, less mobile electron carriers in the membrane. And because it carries both electrons and protons, it plays a central role in coupling electron flow to proton movement.

The **cytochromes** are proteins with characteristic strong absorption of visible light, due to their iron-containing heme prosthetic groups (Fig. 19-3). Mitochondria contain three classes of cytochromes, designated *a*, *b*, and *c*, which are distinguished by differences in their light-absorption spectra. Each type of cytochrome in its reduced (Fe^{2+}) state has three absorption bands in the visible range (Fig. 19-4). The longest-wavelength band is near 600 nm in type *a* cytochromes,

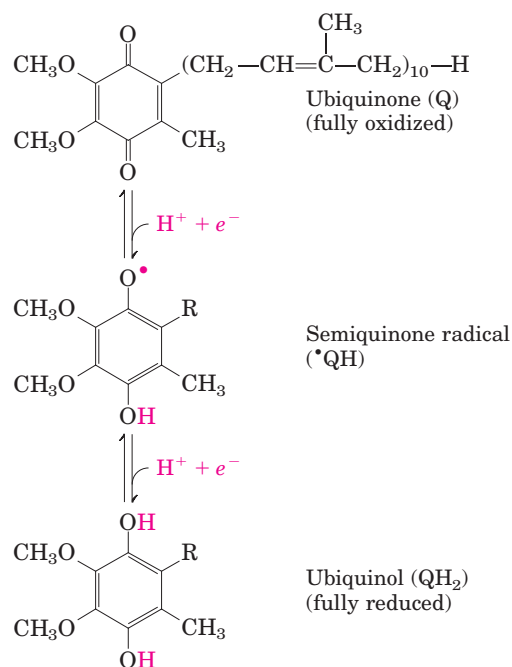


FIGURE 19-2 Ubiquinone (Q, or coenzyme Q). Complete reduction of ubiquinone requires two electrons and two protons, and occurs in two steps through the semiquinone radical intermediate.

near 560 nm in type *b*, and near 550 nm in type *c*. To distinguish among closely related cytochromes of one type, the exact absorption maximum is sometimes used in the names, as in cytochrome *b*₅₆₂.

The heme cofactors of *a* and *b* cytochromes are tightly, but not covalently, bound to their associated proteins; the hemes of *c*-type cytochromes are covalently attached through Cys residues (Fig. 19-3). As with the flavoproteins, the standard reduction potential of the heme iron atom of a cytochrome depends on its interaction with protein side chains and is therefore different for each cytochrome. The cytochromes of type *a* and *b* and some of type *c* are integral proteins of the inner mitochondrial membrane. One striking exception is the cytochrome *c* of mitochondria, a soluble protein that associates through electrostatic interactions with the outer surface of the inner membrane. We encountered cytochrome *c* in earlier discussions of protein structure (see Fig. 4-18).

In **iron-sulfur proteins**, first discovered by Helmut Beinert, the iron is present not in heme but in association with inorganic sulfur atoms or with the sulfur atoms of Cys residues in the protein, or both. These iron-sulfur (Fe-S)



Helmut Beinert

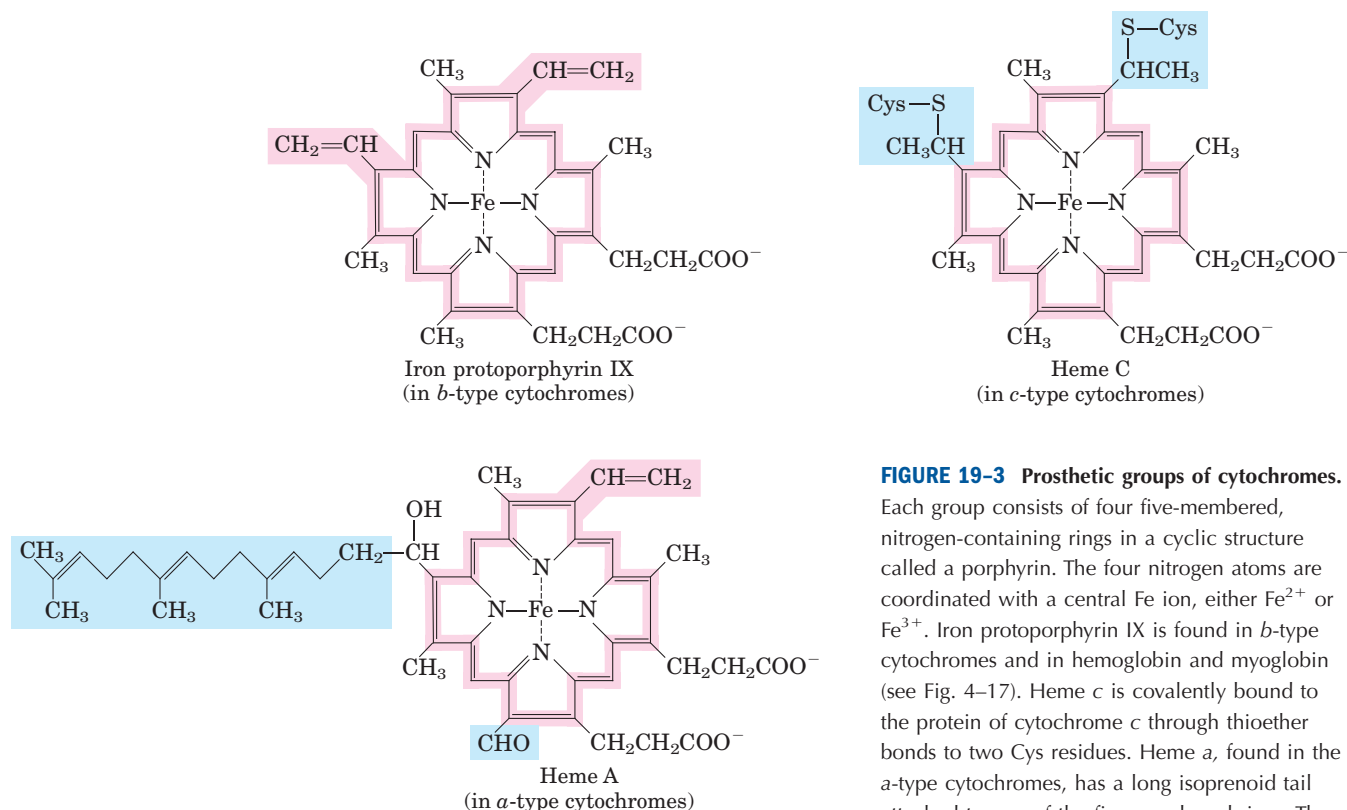


FIGURE 19-3 Prosthetic groups of cytochromes.

Each group consists of four five-membered, nitrogen-containing rings in a cyclic structure called a porphyrin. The four nitrogen atoms are coordinated with a central Fe ion, either Fe^{2+} or Fe^{3+} . Iron protoporphyrin IX is found in *b*-type cytochromes and in hemoglobin and myoglobin (see Fig. 4-17). Heme *c* is covalently bound to the protein of cytochrome *c* through thioether bonds to two Cys residues. Heme *a*, found in the *a*-type cytochromes, has a long isoprenoid tail attached to one of the five-membered rings. The conjugated double-bond system (shaded pink) of the porphyrin ring accounts for the absorption of visible light by these hemes.

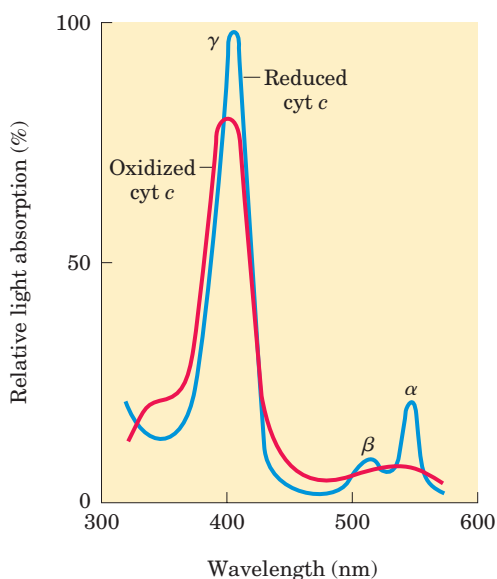
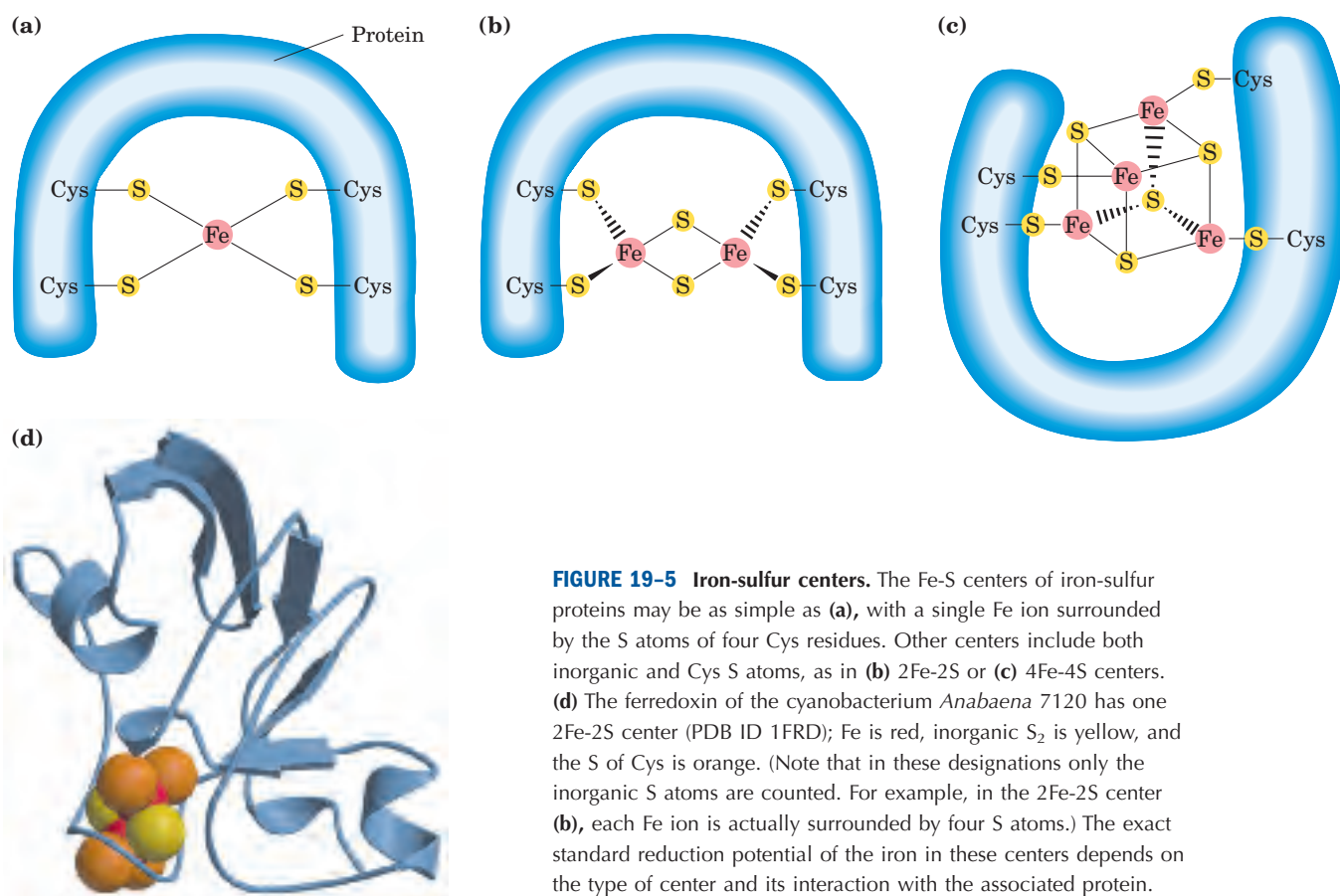


FIGURE 19-4 Absorption spectra of cytochrome *c* (cyt *c*) in its oxidized (red) and reduced (blue) forms. Also labeled are the characteristic α , β , and γ bands of the reduced form.

centers range from simple structures with a single Fe atom coordinated to four Cys —SH groups to more complex Fe-S centers with two or four Fe atoms (Fig. 19-5). **Rieske iron-sulfur proteins** (named after their discoverer, John S. Rieske) are a variation on this theme, in which one Fe atom is coordinated to two His residues rather than two Cys residues. All iron-sulfur proteins participate in one-electron transfers in which one iron atom of the iron-sulfur cluster is oxidized or reduced. At least eight Fe-S proteins function in mitochondrial electron transfer. The reduction potential of Fe-S proteins varies from -0.65 V to $+0.45$ V, depending on the microenvironment of the iron within the protein.

In the overall reaction catalyzed by the mitochondrial respiratory chain, electrons move from NADH, succinate, or some other primary electron donor through flavoproteins, ubiquinone, iron-sulfur proteins, and cytochromes, and finally to O_2 . A look at the methods used to determine the sequence in which the carriers act is instructive, as the same general approaches have been used to study other electron-transfer chains, such as those of chloroplasts.



First, the standard reduction potentials of the individual electron carriers have been determined experimentally (Table 19–2). We would expect the carriers to function in order of increasing reduction potential, because electrons tend to flow spontaneously from carriers of lower E'° to carriers of higher E'° . The order of carriers deduced by this method is $\text{NADH} \rightarrow$

$\text{Q} \rightarrow \text{cytochrome } b \rightarrow \text{cytochrome } c_1 \rightarrow \text{cytochrome } c \rightarrow \text{cytochrome } a \rightarrow \text{cytochrome } a_3 \rightarrow \text{O}_2$. Note, however, that the order of standard reduction potentials is not necessarily the same as the order of *actual* reduction potentials under cellular conditions, which depend on the concentration of reduced and oxidized forms (p. 510). A second method for determining the sequence

TABLE 19-2 Standard Reduction Potentials of Respiratory Chain and Related Electron Carriers

Redox reaction (half-reaction)	E'° (V)
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	−0.414
$\text{NAD}^+ + \text{H}^+ + 2\text{e}^- \rightarrow \text{NADH}$	−0.320
$\text{NADP}^+ + \text{H}^+ + 2\text{e}^- \rightarrow \text{NADPH}$	−0.324
$\text{NADH dehydrogenase (FMN)} + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{NADH dehydrogenase (FMNH}_2\text{)}$	−0.30
$\text{Ubiquinone} + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{ubiquinol}$	0.045
$\text{Cytochrome } b (\text{Fe}^{3+}) + \text{e}^- \rightarrow \text{cytochrome } b (\text{Fe}^{2+})$	0.077
$\text{Cytochrome } c_1 (\text{Fe}^{3+}) + \text{e}^- \rightarrow \text{cytochrome } c_1 (\text{Fe}^{2+})$	0.22
$\text{Cytochrome } c (\text{Fe}^{3+}) + \text{e}^- \rightarrow \text{cytochrome } c (\text{Fe}^{2+})$	0.254
$\text{Cytochrome } a (\text{Fe}^{3+}) + \text{e}^- \rightarrow \text{cytochrome } a (\text{Fe}^{2+})$	0.29
$\text{Cytochrome } a_3 (\text{Fe}^{3+}) + \text{e}^- \rightarrow \text{cytochrome } a_3 (\text{Fe}^{2+})$	0.35
$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$	0.8166

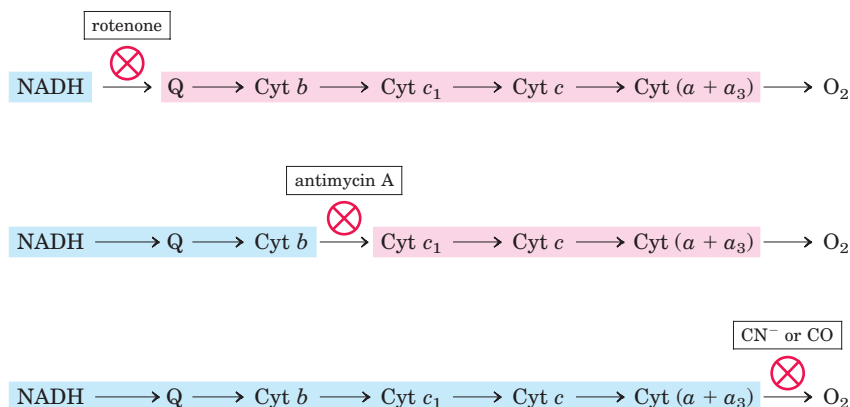


FIGURE 19-6 Method for determining the sequence of electron carriers. This method measures the effects of inhibitors of electron transfer on the oxidation state of each carrier. In the presence of an electron donor and O₂, each inhibitor causes a characteristic pattern of oxidized/reduced carriers: those before the block become reduced (blue), and those after the block become oxidized (pink).

of electron carriers involves reducing the entire chain of carriers experimentally by providing an electron source but no electron acceptor (no O₂). When O₂ is suddenly introduced into the system, the rate at which each electron carrier becomes oxidized (measured spectroscopically) reveals the order in which the carriers function. The carrier nearest O₂ (at the end of the chain) gives up its electrons first, the second carrier from the end is oxidized next, and so on. Such experiments have confirmed the sequence deduced from standard reduction potentials.

In a final confirmation, agents that inhibit the flow of electrons through the chain have been used in combination with measurements of the degree of oxidation of each carrier. In the presence of O₂ and an electron donor, carriers that function before the inhibited step become fully reduced, and those that function after this step are completely oxidized (Fig. 19-6). By using several inhibitors that block different steps in the chain, investigators have determined the entire sequence; it is the same as deduced in the first two approaches.

Electron Carriers Function in Multienzyme Complexes

The electron carriers of the respiratory chain are organized into membrane-embedded supramolecular

complexes that can be physically separated. Gentle treatment of the inner mitochondrial membrane with detergents allows the resolution of four unique electron-carrier complexes, each capable of catalyzing electron transfer through a portion of the chain (Table 19-3; Fig. 19-7). Complexes I and II catalyze electron transfer to ubiquinone from two different electron donors: NADH (Complex I) and succinate (Complex II). Complex III carries electrons from reduced ubiquinone to cytochrome *c*, and Complex IV completes the sequence by transferring electrons from cytochrome *c* to O₂.

We now look in more detail at the structure and function of each complex of the mitochondrial respiratory chain.

Complex I: NADH to Ubiquinone Figure 19-8 illustrates the relationship between Complexes I and II and ubiquinone. **Complex I**, also called **NADH:ubiquinone oxidoreductase** or **NADH dehydrogenase**, is a large enzyme composed of 42 different polypeptide chains, including an FMN-containing flavoprotein and at least six iron-sulfur centers. High-resolution electron microscopy shows Complex I to be L-shaped, with one arm of the L in the membrane and the other extending into the matrix. As shown in Figure 19-9, Complex I catalyzes two simultaneous and obligately coupled processes: (1) the

TABLE 19-3 The Protein Components of the Mitochondrial Electron-Transfer Chain

Enzyme complex/protein	Mass (kDa)	Number of subunits*	Prosthetic group(s)
I NADH dehydrogenase	850	43 (14)	FMN, Fe-S
II Succinate dehydrogenase	140	4	FAD, Fe-S
III Ubiquinone cytochrome <i>c</i> oxidoreductase	250	11	Hemes, Fe-S
Cytochrome <i>c</i> [†]	13	1	Heme
IV Cytochrome oxidase	160	13 (3-4)	Hemes; Cu _A , Cu _B

*Numbers of subunits in the bacterial equivalents in parentheses.

[†]Cytochrome *c* is not part of an enzyme complex; it moves between Complexes III and IV as a freely soluble protein.

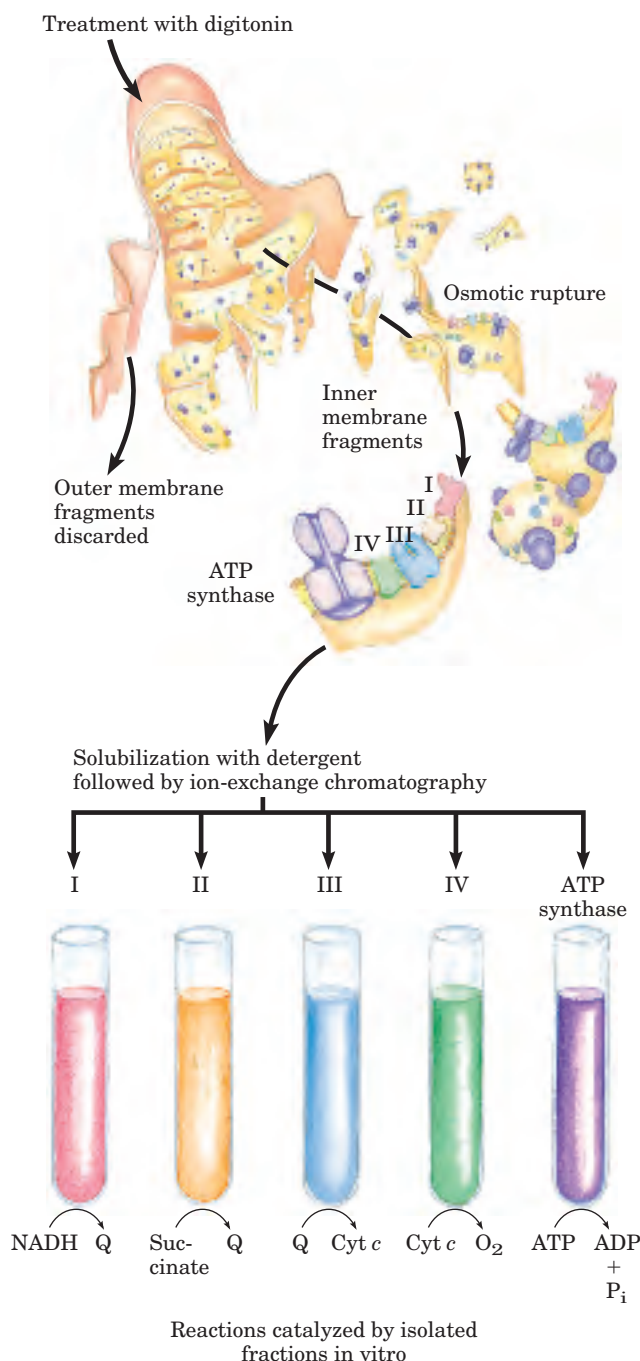


FIGURE 19-7 Separation of functional complexes of the respiratory chain. The outer mitochondrial membrane is first removed by treatment with the detergent digitonin. Fragments of inner membrane are then obtained by osmotic rupture of the mitochondria, and the fragments are gently dissolved in a second detergent. The resulting mixture of inner membrane proteins is resolved by ion-exchange chromatography into different complexes (I through IV) of the respiratory chain, each with its unique protein composition (see Table 19-3), and the enzyme ATP synthase (sometimes called Complex V). The isolated Complexes I through IV catalyze transfers between donors (NADH and succinate), intermediate carriers (Q and cytochrome *c*), and O₂, as shown. In vitro, isolated ATP synthase has only ATP-hydrolyzing (ATPase), not ATP-synthesizing, activity.

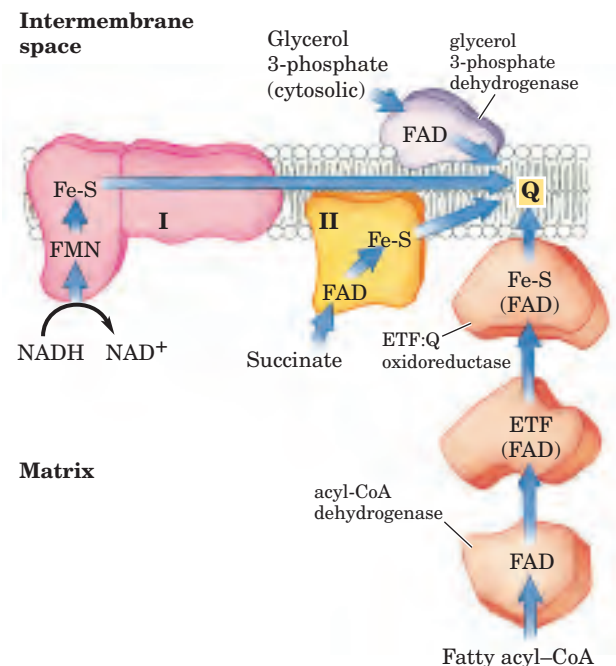
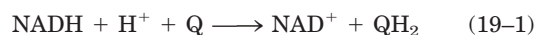


FIGURE 19-8 Path of electrons from NADH, succinate, fatty acyl-CoA, and glycerol 3-phosphate to ubiquinone. Electrons from NADH pass through a flavoprotein to a series of iron-sulfur proteins (in Complex I) and then to Q. Electrons from succinate pass through a flavoprotein and several Fe-S centers (in Complex II) on the way to Q. Glycerol 3-phosphate donates electrons to a flavoprotein (glycerol 3-phosphate dehydrogenase) on the outer face of the inner mitochondrial membrane, from which they pass to Q. Acyl-CoA dehydrogenase (the first enzyme of β oxidation) transfers electrons to electron-transferring flavoprotein (ETF), from which they pass to Q via ETF:ubiquinone oxidoreductase.

exergonic transfer to ubiquinone of a hydride ion from NADH and a proton from the matrix, expressed by



and (2) the endergonic transfer of four protons from the matrix to the intermembrane space. Complex I is therefore a proton pump driven by the energy of electron transfer, and the reaction it catalyzes is **vectorial**: it moves protons in a specific direction from one location (the matrix, which becomes negatively charged with the departure of protons) to another (the intermembrane space, which becomes positively charged). To emphasize the vectorial nature of the process, the overall reaction is often written with subscripts that indicate the location of the protons: *P* for the positive side of the inner membrane (the intermembrane space), *N* for the negative side (the matrix):



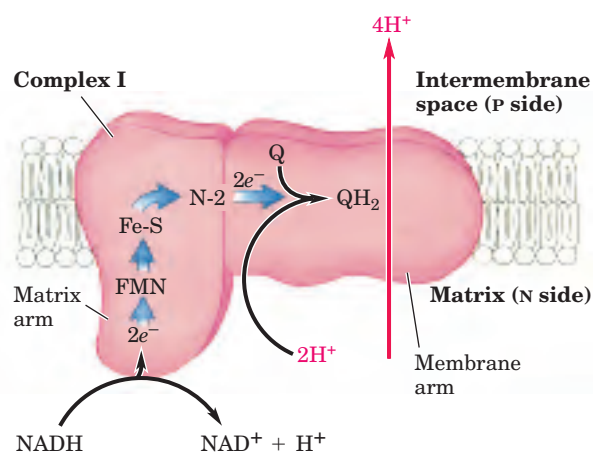


FIGURE 19-9 NADH:ubiquinone oxidoreductase (Complex I). Complex I catalyzes the transfer of a hydride ion from NADH to FMN, from which two electrons pass through a series of Fe-S centers to the iron-sulfur protein N-2 in the matrix arm of the complex. Electron transfer from N-2 to ubiquinone on the membrane arm forms QH₂, which diffuses into the lipid bilayer. This electron transfer also drives the expulsion from the matrix of four protons per pair of electrons. The detailed mechanism that couples electron and proton transfer in Complex I is not yet known, but probably involves a Q cycle similar to that in Complex III in which QH₂ participates twice per electron pair (see Fig. 19-12). Proton flux produces an electrochemical potential across the inner mitochondrial membrane (N side negative, P side positive), which conserves some of the energy released by the electron-transfer reactions. This electrochemical potential drives ATP synthesis.

Amytal (a barbiturate drug), rotenone (a plant product commonly used as an insecticide), and piericidin A (an antibiotic) inhibit electron flow from the Fe-S centers of Complex I to ubiquinone (Table 19-4) and therefore block the overall process of oxidative phosphorylation.

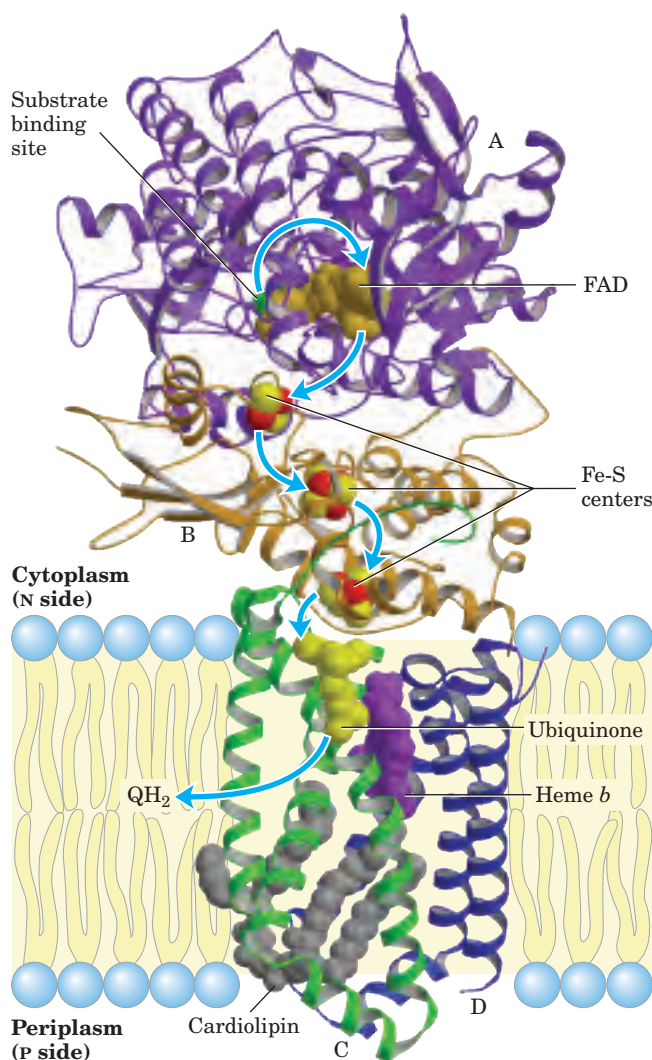
Ubiquinol (QH₂, the fully reduced form; Fig. 19-2) diffuses in the inner mitochondrial membrane from Complex I to Complex III, where it is oxidized to Q in a process that also involves the outward movement of H⁺.

Complex II: Succinate to Ubiquinone We encountered **Complex II** in Chapter 16 as **succinate dehydrogenase**, the only membrane-bound enzyme in the citric acid cycle (p. 612). Although smaller and simpler than Complex I, it contains five prosthetic groups of two types and four different protein subunits (Fig. 19-10). Subunits C and D are integral membrane proteins, each with three transmembrane helices. They contain a heme group, heme *b*, and a binding site for ubiquinone, the final electron acceptor in the reaction catalyzed by Complex II. Subunits A and B extend into the matrix (or the cytosol of a bacterium); they contain three 2Fe-2S centers, bound FAD, and a binding site for the substrate, succinate. The path of electron transfer from the succinate-binding site to FAD, then through the Fe-S centers to the Q-binding site, is more than 40 Å long, but none of the individual electron-transfer distances exceeds about 11 Å—a reasonable distance for rapid electron transfer (Fig. 19-10).

TABLE 19-4 Agents That Interfere with Oxidative Phosphorylation or Photophosphorylation

Type of interference	Compound*	Target/mode of action
Inhibition of electron transfer	Cyanide	Inhibit cytochrome oxidase
	Carbon monoxide	
	Antimycin A	Blocks electron transfer from cytochrome <i>b</i> to cytochrome <i>c</i> ₁
	Myxothiazol	
	Rotenone	Prevent electron transfer from Fe-S center to ubiquinone
	Amytal	
	Piericidin A	
Inhibition of ATP synthase	DCMU	Competes with Q _B for binding site in PSII
	Aurovertin	Inhibits F ₁
	Oligomycin	Inhibit F _o and CF _o
	Venturicidin	
Uncoupling of phosphorylation from electron transfer	DCCD	Blocks proton flow through F _o and CF _o
	FCCP	Hydrophobic proton carriers
	DNP	
	Valinomycin	K ⁺ ionophore
	Thermogenin	In brown fat, forms proton-conducting pores in inner mitochondrial membrane
Inhibition of ATP-ADP exchange	Atractyloside	Inhibits adenine nucleotide translocase

*DCMU is 3-(3,4-dichlorophenyl)-1,1-dimethylurea; DCCD, dicyclohexylcarbodiimide; FCCP, cyanide-*p*-trifluoromethoxyphenylhydrazine; DNP, 2,4-dinitrophenol.



The heme *b* of Complex II is apparently not in the direct path of electron transfer; it may serve instead to reduce the frequency with which electrons “leak” out of the system, moving from succinate to molecular oxygen to produce the **reactive oxygen species (ROS)** hydrogen peroxide (H_2O_2) and the **superoxide radical** (O_2^-) described in Section 19.5. Humans with point mutations in Complex II subunits near heme *b* or the quinone-binding site suffer from hereditary paraganglioma. This inherited condition is characterized by benign tumors of the head and neck, commonly in the carotid body, an organ that senses O_2 levels in the blood. These mutations result in greater production of ROS and perhaps greater tissue damage during succinate oxidation. ■

Other substrates for mitochondrial dehydrogenases pass electrons into the respiratory chain at the level of ubiquinone, but not through Complex II. The first step in the β oxidation of fatty acyl-CoA, catalyzed by the

FIGURE 19-10 Structure of Complex II (succinate dehydrogenase) of *E. coli* (PDB ID 1NEK). The enzyme has two transmembrane subunits, C (green) and D (blue); the cytoplasmic extensions contain subunits B (orange) and A (purple). Just behind the FAD in subunit A (gold) is the binding site for succinate (occupied in this crystal structure by the inhibitor oxaloacetate, green). Subunit B has three sets of Fe-S centers (yellow and red); ubiquinone (yellow) is bound to subunit C; and heme *b* (purple) is sandwiched between subunits C and D. A cardiolipin molecule is so tightly bound to subunit C that it shows up in the crystal structure (gray spacefilling). Electrons move (blue arrows) from succinate to FAD, then through the three Fe-S centers to ubiquinone. The heme *b* is not on the main path of electron transfer but protects against the formation of reactive oxygen species (ROS) by electrons that go astray.

flavoprotein **acyl-CoA dehydrogenase** (see Fig. 17-8), involves transfer of electrons from the substrate to the FAD of the dehydrogenase, then to electron-transferring flavoprotein (ETF), which in turn passes its electrons to **ETF:ubiquinone oxidoreductase** (Fig. 19-8). This enzyme transfers electrons into the respiratory chain by reducing ubiquinone. Glycerol 3-phosphate, formed either from glycerol released by triacylglycerol breakdown or by the reduction of dihydroxyacetone phosphate from glycolysis, is oxidized by **glycerol 3-phosphate dehydrogenase** (see Fig. 17-4). This enzyme is a flavoprotein located on the outer face of the inner mitochondrial membrane, and like succinate dehydrogenase and acyl-CoA dehydrogenase it channels electrons into the respiratory chain by reducing ubiquinone (Fig. 19-8). The important role of glycerol 3-phosphate dehydrogenase in shuttling reducing equivalents from cytosolic NADH into the mitochondrial matrix is described in Section 19.2 (see Fig. 19-28). The effect of each of these electron-transferring enzymes is to contribute to the pool of reduced ubiquinone. QH_2 from all these reactions is reoxidized by Complex III.

Complex III: Ubiquinone to Cytochrome *c* The next respiratory complex, **Complex III**, also called **cytochrome *bc*₁ complex** or **ubiquinone:cytochrome *c* oxidoreductase**, couples the transfer of electrons from ubiquinol (QH_2) to cytochrome *c* with the vectorial transport of protons from the matrix to the intermembrane space. The determination of the complete structure of this huge complex (Fig. 19-11) and of Complex IV (below) by x-ray crystallography, achieved between 1995 and 1998, were landmarks in the study of mitochondrial electron transfer, providing the structural framework to integrate the many biochemical observations on the functions of the respiratory complexes.

Based on the structure of Complex III and detailed biochemical studies of the redox reactions, a reasonable model has been proposed for the passage of electrons

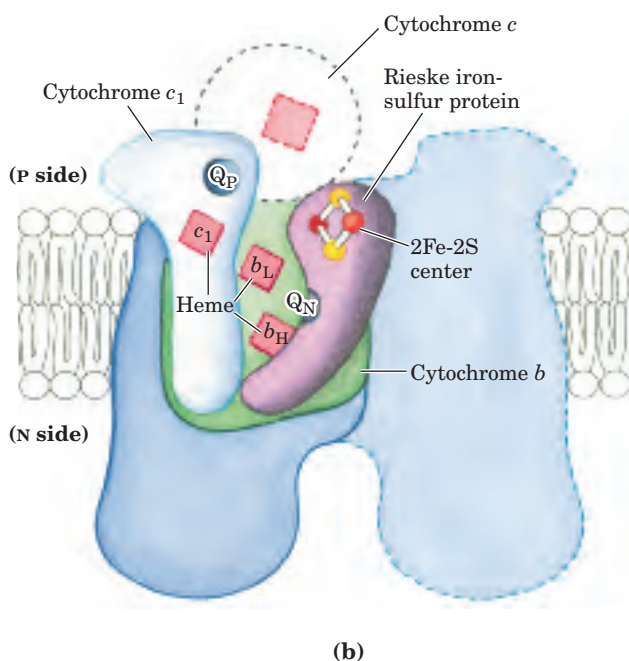
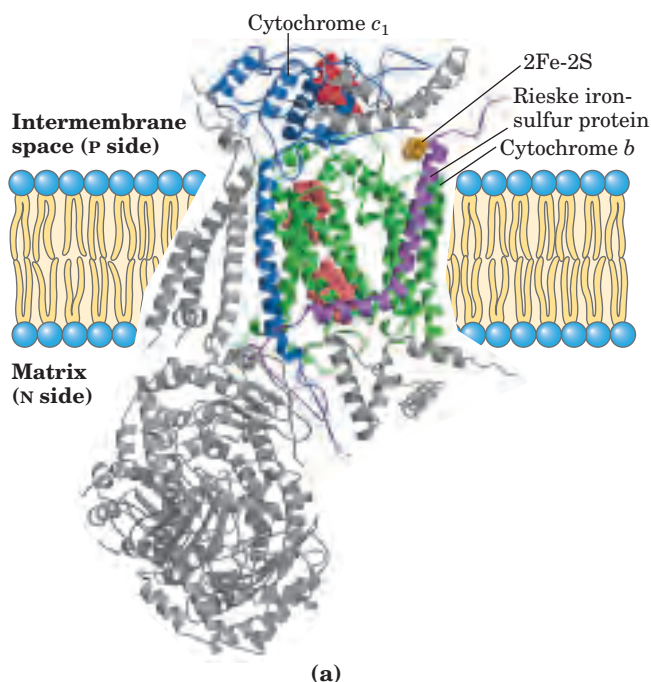
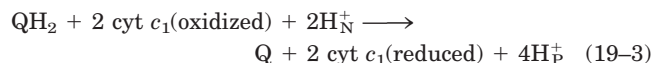


FIGURE 19-11 Cytochrome bc_1 complex (Complex III). The complex is a dimer of identical monomers, each with 11 different subunits. **(a)** Structure of a monomer. The functional core is three subunits: cytochrome b (green) with its two hemes (b_H and b_L , light red); the Rieske iron-sulfur protein (purple) with its 2Fe-2S centers (yellow); and cytochrome c_1 (blue) with its heme (red) (PDB ID 1BGY). **(b)** The dimeric functional unit. Cytochrome c_1 and the Rieske iron-sulfur protein project from the P surface and can interact with cytochrome c (not part of the functional complex) in the intermembrane space. The complex has two distinct binding sites for ubiquinone, Q_N and Q_P , which correspond to the sites of inhibition by two drugs that block oxidative phosphorylation. Antimycin A, which blocks electron flow from heme b_H to Q , binds at Q_N , close to heme b_H on the N (matrix) side of the membrane. Myxothiazol, which prevents electron flow from

QH_2 to the Rieske iron-sulfur protein, binds at Q_P , near the 2Fe-2S center and heme b_L on the P side. The dimeric structure is essential to the function of Complex III. The interface between monomers forms two pockets, each containing a Q_P site from one monomer and a Q_N site from the other. The ubiquinone intermediates move within these sheltered pockets.

Complex III crystallizes in two distinct conformations (not shown). In one, the Rieske Fe-S center is close to its electron acceptor, the heme of cytochrome c_1 , but relatively distant from cytochrome b and the QH_2 -binding site at which the Rieske Fe-S center receives electrons. In the other, the Fe-S center has moved away from cytochrome c_1 and toward cytochrome b . The Rieske protein is thought to oscillate between these two conformations as it is reduced, then oxidized.

and protons through the complex. The net equation for the redox reactions of this **Q cycle** (Fig. 19-12) is



The Q cycle accommodates the switch between the two-electron carrier ubiquinone and the one-electron carriers—cytochromes b_{562} , b_{566} , c_1 , and c —and explains the measured stoichiometry of four protons translocated per pair of electrons passing through the Complex III to cytochrome c . Although the path of electrons through this segment of the respiratory chain is complicated, the net effect of the transfer is simple: QH_2 is oxidized to Q and two molecules of cytochrome c are reduced.

Cytochrome c (see Fig. 4-18) is a soluble protein of the intermembrane space. After its single heme accepts an electron from Complex III, cytochrome c moves to Complex IV to donate the electron to a binuclear copper center.

Complex IV: Cytochrome c to O_2 In the final step of the respiratory chain, **Complex IV**, also called **cytochrome oxidase**, carries electrons from cytochrome c to molecular oxygen, reducing it to H_2O . Complex IV is a large enzyme (13 subunits; M_r 204,000) of the inner mitochondrial membrane. Bacteria contain a form that is much simpler, with only three or four subunits, but still capable of catalyzing both electron transfer and proton pumping. Comparison of the mitochondrial and bacterial complexes suggests that three subunits are critical to the function (Fig. 19-13).

Mitochondrial subunit II contains two Cu ions complexed with the —SH groups of two Cys residues in a binuclear center (Cu_A ; Fig. 19-13b) that resembles the 2Fe-2S centers of iron-sulfur proteins. Subunit I contains two heme groups, designated a and a_3 , and another copper ion (Cu_B). Heme a_3 and Cu_B form a second binuclear center that accepts electrons from heme a and transfers them to O_2 bound to heme a_3 .

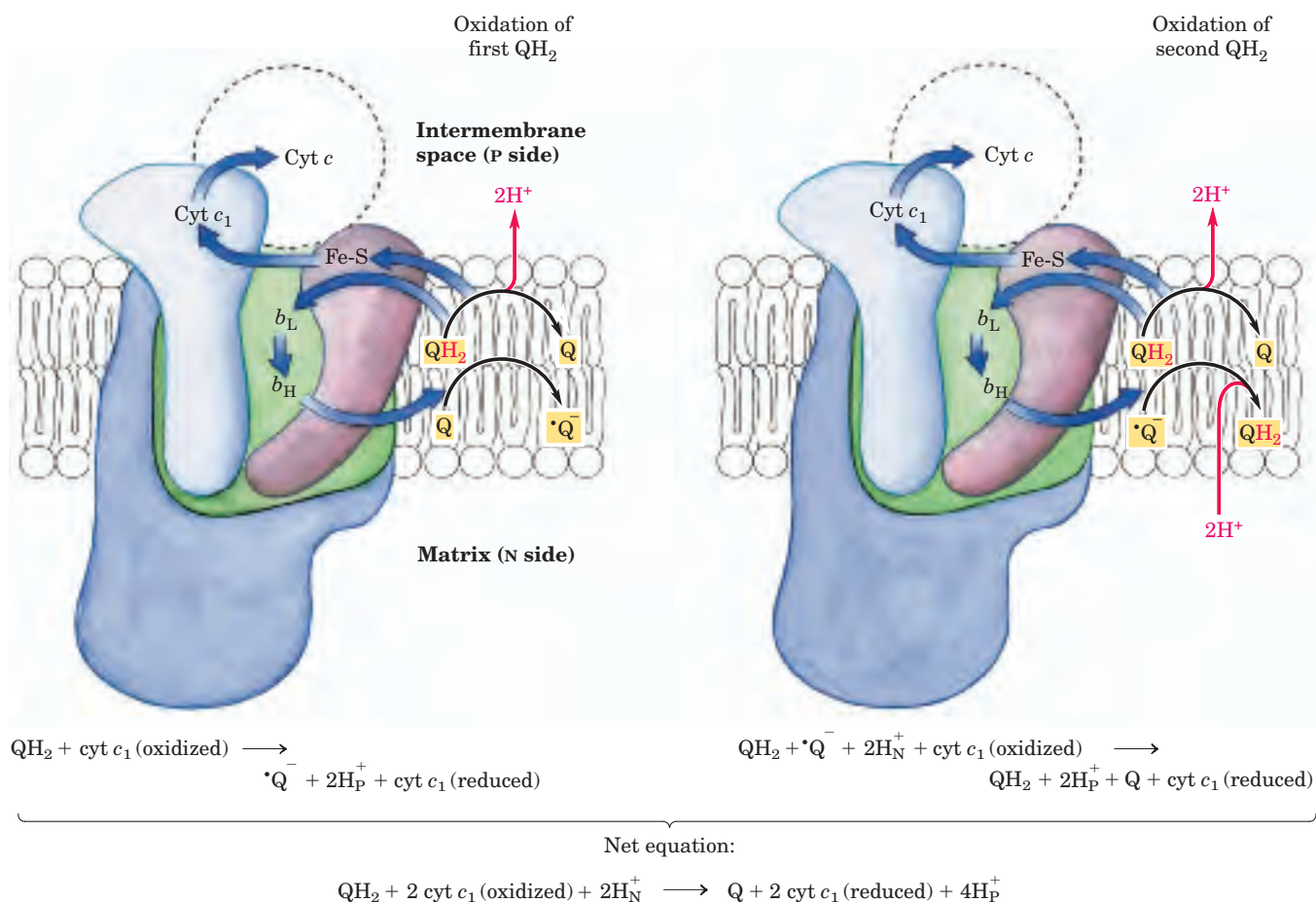
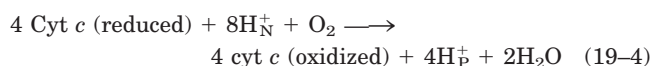


FIGURE 19-12 The Q cycle. The path of electrons through Complex III is shown by blue arrows. On the P side of the membrane, two molecules of QH₂ are oxidized to Q near the P side, releasing two protons per Q (four protons in all) into the intermembrane space. Each

QH₂ donates one electron (via the Rieske Fe-S center) to cytochrome c₁, and one electron (via cytochrome b) to a molecule of Q near the N side, reducing it in two steps to QH₂. This reduction also uses two protons per Q, which are taken up from the matrix.

Electron transfer through Complex IV is from cytochrome c to the Cu_A center, to heme a, to the heme a₃-Cu_B center, and finally to O₂ (Fig. 19-14). For every four electrons passing through this complex, the enzyme consumes four “substrate” H⁺ from the matrix (N side) in converting O₂ to 2H₂O. It also uses the energy of this redox reaction to pump one proton outward into the intermembrane space (P side) for each electron that passes through, adding to the electrochemical potential produced by redox-driven proton transport through Complexes I and III. The overall reaction catalyzed by Complex IV is



This four-electron reduction of O₂ involves redox centers that carry only one electron at a time, and it must occur without the release of incompletely reduced intermediates such as hydrogen peroxide or hydroxyl free radicals—very reactive species that would damage cellular components. The intermediates remain tightly

bound to the complex until completely converted to water.

The Energy of Electron Transfer Is Efficiently Conserved in a Proton Gradient

The transfer of two electrons from NADH through the respiratory chain to molecular oxygen can be written as



This net reaction is highly exergonic. For the redox pair NAD⁺/NADH, E'° is -0.320 V , and for the pair O₂/H₂O, E'° is 0.816 V . The $\Delta E'^\circ$ for this reaction is therefore 1.14 V , and the standard free-energy change (see Eqn 13-6, p. 510) is

$$\begin{aligned} \Delta G'^\circ &= -n \mathcal{F} \Delta E'^\circ \\ &= -2(96.5 \text{ kJ/V} \cdot \text{mol})(1.14 \text{ V}) \\ &= -220 \text{ kJ/mol (of NADH)} \end{aligned} \quad (19-6)$$

This *standard* free-energy change is based on the assumption of equal concentrations (1 M) of NADH and

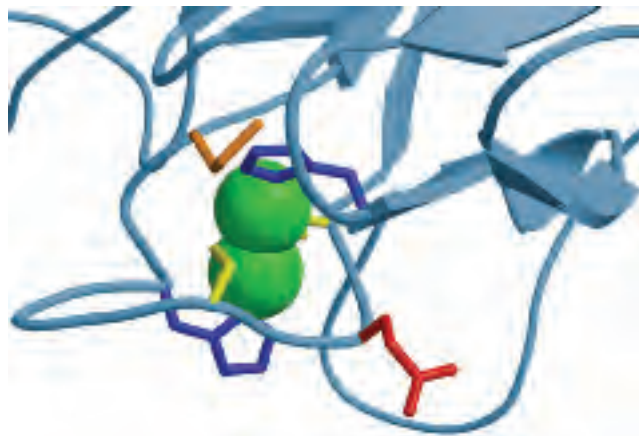
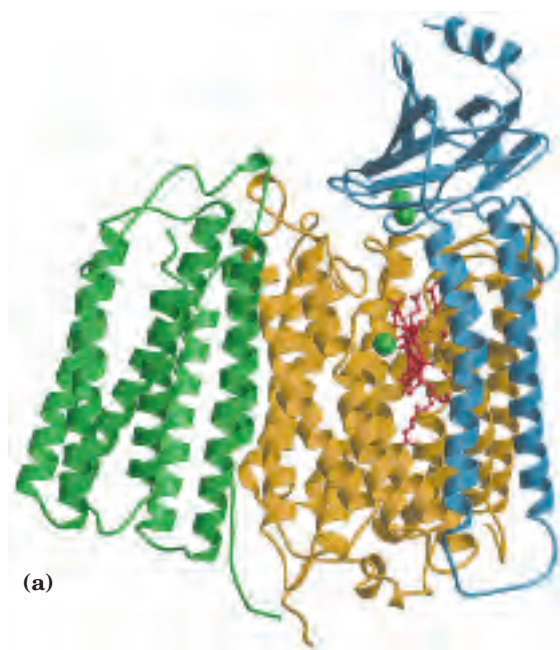
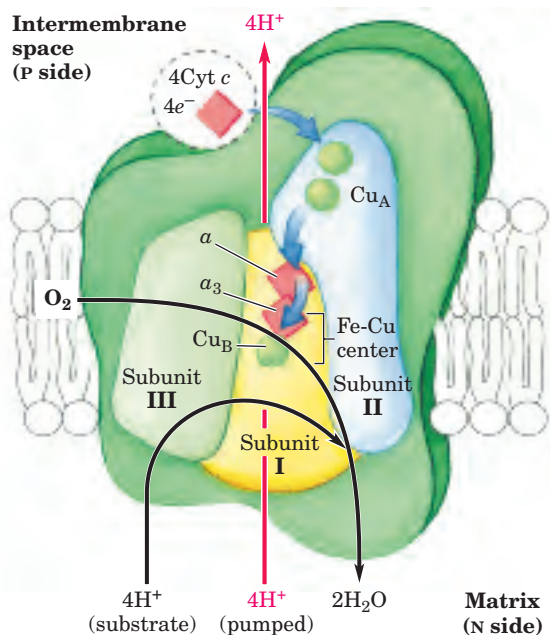


FIGURE 19-13 Critical subunits of cytochrome oxidase (Complex IV). The bovine complex is shown here (PDB ID 1OCC). **(a)** The core of Complex IV, with three subunits. Subunit I (yellow) has two heme groups, a and a_3 (red), and a copper ion, Cu_B (green sphere). Heme a_3 and Cu_B form a binuclear Fe-Cu center. Subunit II (blue) contains two Cu ions (green spheres) complexed with the —SH groups of two Cys residues in a binuclear center, Cu_A , that resembles the 2Fe-2S centers of iron-sulfur proteins. This binuclear center and the cytochrome

c -binding site are located in a domain of subunit II that protrudes from the P side of the inner membrane (into the intermembrane space). Subunit III (green) seems to be essential for Complex IV function, but its role is not well understood. **(b)** The binuclear center of Cu_A . The Cu ions (green spheres) share electrons equally. When the center is reduced they have the formal charges $Cu^{1+}Cu^{1+}$; when oxidized, $Cu^{1.5+}Cu^{1.5+}$. Ligands around the Cu ions include two His (dark blue), two Cys (yellow), an Asp (red), and Met (orange) residues.



NAD^+ . In actively respiring mitochondria, the actions of many dehydrogenases keep the actual $[NADH]/[NAD^+]$ ratio well above unity, and the real free-energy change for the reaction shown in Equation 19-5 is therefore substantially greater (more negative) than -220 kJ/mol. A similar calculation for the oxidation of succinate shows that electron transfer from succinate (E'° for fumarate/succinate = 0.031 V) to O_2 has a smaller, but still negative, standard free-energy change of about -150 kJ/mol.

FIGURE 19-14 Path of electrons through Complex IV. The three proteins critical to electron flow are subunits I, II, and III. The larger green structure includes the other ten proteins in the complex. Electron transfer through Complex IV begins when two molecules of reduced cytochrome c (top) each donate an electron to the binuclear center Cu_A . From here electrons pass through heme a to the Fe-Cu center (cytochrome a_3 and Cu_B). Oxygen now binds to heme a_3 and is reduced to its peroxy derivative (O_2^{2-}) by two electrons from the Fe-Cu center. Delivery of two more electrons from cytochrome c (making four electrons in all) converts the O_2^{2-} to two molecules of water, with consumption of four “substrate” protons from the matrix. At the same time, four more protons are pumped from the matrix by an as yet unknown mechanism.

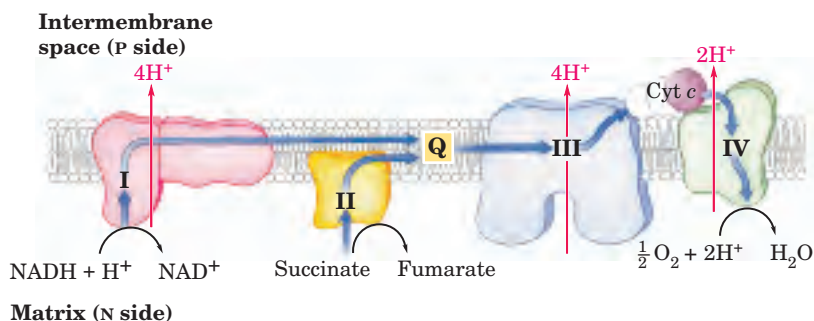


FIGURE 19-15 Summary of the flow of electrons and protons through the four complexes of the respiratory chain. Electrons reach Q through Complexes I and II. QH₂ serves as a mobile carrier of electrons and protons. It passes electrons to Complex III, which passes them to another mobile connecting link, cytochrome c. Complex IV

then transfers electrons from reduced cytochrome c to O₂. Electron flow through Complexes I, III, and IV is accompanied by proton flow from the matrix to the intermembrane space. Recall that electrons from β oxidation of fatty acids can also enter the respiratory chain through Q (see Fig. 19-8).

Much of this energy is used to pump protons out of the matrix. For each pair of electrons transferred to O₂, four protons are pumped out by Complex I, four by Complex III, and two by Complex IV (Fig. 19-15). The *vectorial* equation for the process is therefore



The electrochemical energy inherent in this difference in proton concentration and separation of charge represents a temporary conservation of much of the energy of electron transfer. The energy stored in such a gradient, termed the **proton-motive force**, has two components: (1) the *chemical potential energy* due to the difference in concentration of a chemical species (H⁺) in the two regions separated by the membrane, and (2) the *electrical potential energy* that results from the separation of charge when a proton moves across the membrane without a counterion (Fig. 19-16).

As we showed in Chapter 11, the free-energy change for the creation of an electrochemical gradient by an ion pump is

$$\Delta G = RT \ln \left(\frac{C_2}{C_1} \right) + Z\mathcal{F}\Delta\psi \quad (19-8)$$

where C_2 and C_1 are the concentrations of an ion in two regions, and $C_2 > C_1$; Z is the absolute value of its electrical charge (1 for a proton), and $\Delta\psi$ is the transmembrane difference in electrical potential, measured in volts.

For protons at 25 °C,

$$\begin{aligned} \ln \left(\frac{C_2}{C_1} \right) &= 2.3(\log [\text{H}^+]_{\text{P}} - \log [\text{H}^+]_{\text{N}}) \\ &= 2.3(\text{pH}_{\text{N}} - \text{pH}_{\text{P}}) = 2.3 \Delta\text{pH} \end{aligned}$$

and Equation 19-8 reduces to

$$\begin{aligned} \Delta G &= 2.3RT \Delta\text{pH} + \mathcal{F}\Delta\psi \\ &= (5.70 \text{ kJ/mol})\Delta\text{pH} + (96.5 \text{ kJ/V} \cdot \text{mol})\Delta\psi \end{aligned} \quad (19-9)$$

In actively respiring mitochondria, the measured $\Delta\psi$ is 0.15 to 0.20 V and the pH of the matrix is about 0.75

units more alkaline than that of the intermembrane space, so the calculated free-energy change for pumping protons outward is about 20 kJ/mol (of H⁺), most of which is contributed by the electrical portion of the electrochemical potential. Because the transfer of two electrons from NADH to O₂ is accompanied by the outward pumping of 10 H⁺ (Eqn 19-7), roughly 200 kJ of the 220 kJ released by oxidation of a mole of NADH is conserved in the proton gradient.

When protons flow spontaneously *down* their electrochemical gradient, energy is made available to do work. In mitochondria, chloroplasts, and aerobic bacteria, the electrochemical energy in the proton gradient drives the synthesis of ATP from ADP and P_i. We return to the energetics and stoichiometry of ATP synthesis driven by the electrochemical potential of the proton gradient in Section 19.2.

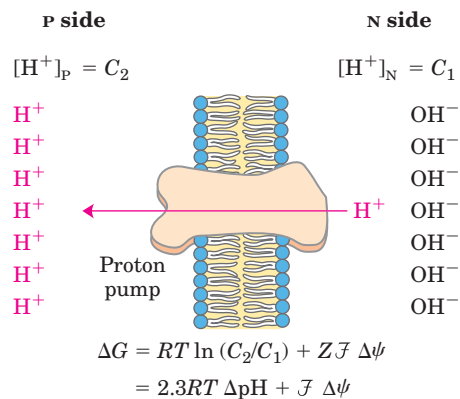
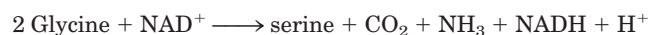


FIGURE 19-16 Proton-motive force. The inner mitochondrial membrane separates two compartments of different [H⁺], resulting in differences in chemical concentration (ΔpH) and charge distribution ($\Delta\psi$) across the membrane. The net effect is the proton-motive force (ΔG), which can be calculated as shown here. This is explained more fully in the text.

Plant Mitochondria Have Alternative Mechanisms for Oxidizing NADH

Plant mitochondria supply the cell with ATP during periods of low illumination or darkness by mechanisms entirely analogous to those used by nonphotosynthetic organisms. In the light, the principal source of mitochondrial NADH is a reaction in which glycine, produced by a process known as photorespiration, is converted to serine (see Fig. 20–21):



For reasons discussed in Chapter 20, plants must carry out this reaction even when they do not need NADH for ATP production. To regenerate NAD^+ from unneeded NADH, plant mitochondria transfer electrons from NADH directly to ubiquinone and from ubiquinone directly to O_2 , bypassing Complexes III and IV and their proton pumps. In this process the energy in NADH is dissipated as heat, which can sometimes be of value to the plant (Box 19–1). Unlike cytochrome oxidase (Complex IV), the alternative QH_2 oxidase is not inhibited by cyanide. Cyanide-resistant NADH oxidation is therefore the hallmark of this unique plant electron-transfer pathway.

SUMMARY 19.1 Electron-Transfer Reactions in Mitochondria

- Chemiosmotic theory provides the intellectual framework for understanding many biological energy transductions, including oxidative phosphorylation and photophosphorylation. The mechanism of energy coupling is similar in both cases: the energy of electron flow is conserved by the concomitant pumping of protons across the membrane, producing an electrochemical gradient, the proton-motive force.
- In mitochondria, hydride ions removed from substrates by NAD-linked dehydrogenases donate electrons to the respiratory (electron-transfer) chain, which transfers the electrons to molecular O_2 , reducing it to H_2O .
- Shuttle systems convey reducing equivalents from cytosolic NADH to mitochondrial NADH. Reducing equivalents from all NAD-linked dehydrogenations are transferred to mitochondrial NADH dehydrogenase (Complex I).
- Reducing equivalents are then passed through a series of Fe-S centers to ubiquinone, which transfers the electrons to cytochrome *b*, the first carrier in Complex III. In this complex, electrons take two separate paths through two *b*-type cytochromes and cytochrome c_1 to an Fe-S center. The Fe-S center passes electrons, one at a time, through cytochrome *c* and into

Complex IV, cytochrome oxidase. This copper-containing enzyme, which also contains cytochromes *a* and a_3 , accumulates electrons, then passes them to O_2 , reducing it to H_2O .

- Some electrons enter this chain of carriers through alternative paths. Succinate is oxidized by succinate dehydrogenase (Complex II), which contains a flavoprotein that passes electrons through several Fe-S centers to ubiquinone. Electrons derived from the oxidation of fatty acids pass to ubiquinone via the electron-transferring flavoprotein.
- Plants also have an alternative, cyanide-resistant NADH oxidation pathway.

19.2 ATP Synthesis

How is a concentration gradient of protons transformed into ATP? We have seen that electron transfer releases, and the proton-motive force conserves, more than enough free energy (about 200 kJ) per “mole” of electron pairs to drive the formation of a mole of ATP, which requires about 50 kJ (see Box 13–1). Mitochondrial oxidative phosphorylation therefore poses no thermodynamic problem. But what is the chemical mechanism that couples proton flux with phosphorylation?

The **chemiosmotic model**, proposed by Peter Mitchell, is the paradigm for this mechanism. According to the model (Fig. 19–17), the electrochemical energy inherent in the difference in proton concentration and separation of charge across the inner mitochondrial membrane—the proton-motive force—drives the synthesis of ATP as protons flow passively back into the matrix through a proton pore associated with **ATP synthase**. To emphasize this crucial role of the proton-motive force, the equation for ATP synthesis is sometimes written



Mitchell used “chemiosmotic” to describe enzymatic reactions that involve, simultaneously, a chemical reaction and a transport process. The operational definition of “coupling” is shown in Figure 19–18. When isolated mitochondria are suspended in a buffer containing ADP, P_i , and an oxidizable substrate such as succinate, three easily measured processes occur: (1) the substrate is oxidized (succinate yields fumarate), (2) O_2 is consumed, and (3) ATP is synthesized. Oxygen consumption and ATP synthesis depend on the presence of an oxidizable substrate (succinate in this case) as well as ADP and P_i .



Peter Mitchell,
1920–1992

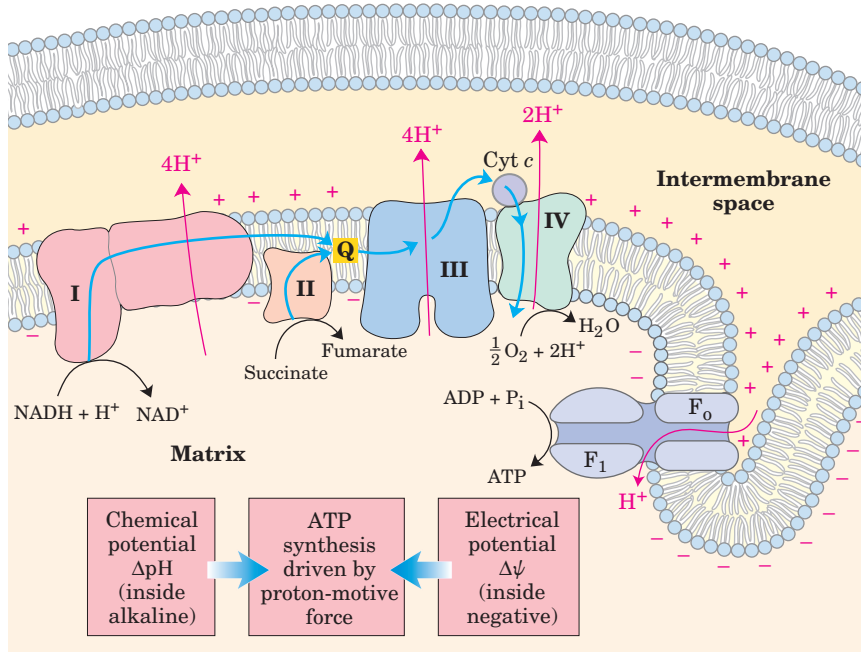


FIGURE 19-17 Chemiosmotic model. In this simple representation of the chemiosmotic theory applied to mitochondria, electrons from NADH and other oxidizable substrates pass through a chain of carriers arranged asymmetrically in the inner membrane. Electron flow is accompanied by proton transfer across the membrane, producing both a chemical gradient (ΔpH) and an electrical gradient ($\Delta\psi$). The inner mitochondrial membrane is impermeable to protons; protons can reenter the matrix only through proton-specific channels (F_0). The proton-motive force that drives protons back into the matrix provides the energy for ATP synthesis, catalyzed by the F_1 complex associated with F_0 .

Because the energy of substrate oxidation drives ATP synthesis in mitochondria, we would expect inhibitors of the passage of electrons to O_2 (such as cyanide, carbon monoxide, and antimycin A) to block ATP synthesis (Fig. 19-18a). More surprising is the finding that the converse is also true: inhibition of ATP synthesis blocks electron transfer in intact mitochondria. This obligatory coupling can be demonstrated in isolated mitochondria by providing O_2 and oxidizable substrates, but not ADP (Fig. 19-18b). Under these conditions, no ATP synthesis can occur and electron transfer to O_2 does not proceed. Coupling of oxidation and phosphorylation can also be demonstrated using oligomycin or venturicidin, toxic antibiotics that bind to the ATP synthase in mitochondria. These compounds are potent in-

hibitors of both ATP synthesis *and* the transfer of electrons through the chain of carriers to O_2 (Fig. 19-18b). Because oligomycin is known to interact not directly with the electron carriers but with ATP synthase, it follows that electron transfer and ATP synthesis are obligately coupled; neither reaction occurs without the other.

Chemiosmotic theory readily explains the dependence of electron transfer on ATP synthesis in mitochondria. When the flow of protons into the matrix through the proton channel of ATP synthase is blocked (with oligomycin, for example), no path exists for the return of protons to the matrix, and the continued extrusion of protons driven by the activity of the respiratory chain generates a large proton gradient. The proton-motive force builds up until the cost (free energy) of pumping

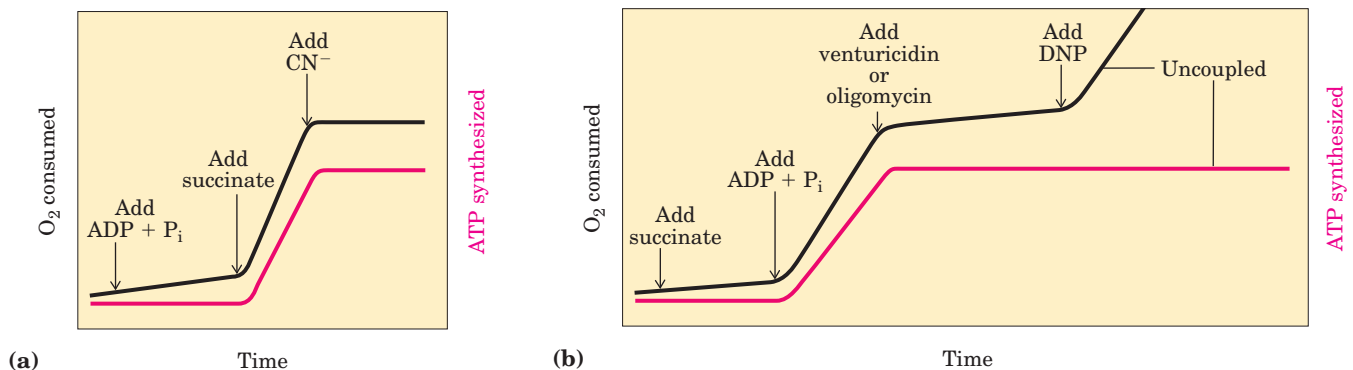


FIGURE 19-18 Coupling of electron transfer and ATP synthesis in mitochondria. In experiments to demonstrate coupling, mitochondria are suspended in a buffered medium and an O_2 electrode monitors O_2 consumption. At intervals, samples are removed and assayed for the presence of ATP. **(a)** Addition of ADP and P_i alone results in little or no increase in either respiration (O_2 consumption; black) or ATP synthesis (red). When succinate is added, respiration begins immediately and

ATP is synthesized. Addition of cyanide (CN^-), which blocks electron transfer between cytochrome oxidase and O_2 , inhibits both respiration and ATP synthesis. **(b)** Mitochondria provided with succinate respire and synthesize ATP only when ADP and P_i are added. Subsequent addition of venturicidin or oligomycin, inhibitors of ATP synthase, blocks both ATP synthesis and respiration. Dinitrophenol (DNP) is an uncoupler, allowing respiration to continue without ATP synthesis.

BOX 19-1 THE WORLD OF BIOCHEMISTRY

Hot, Stinking Plants and Alternative Respiratory Pathways

Many flowering plants attract insect pollinators by releasing odorant molecules that mimic an insect's natural food sources or potential egg-laying sites. Plants pollinated by flies or beetles that normally feed on or lay their eggs in dung or carrion sometimes use foul-smelling compounds to attract these insects.

One family of stinking plants is the Araceae, which includes philodendrons, arum lilies, and skunk cabbages. These plants have tiny flowers densely packed on an erect structure, the spadix, surrounded by a modified leaf, the spathe. The spadix releases odors of rotting flesh or dung. Before pollination the spadix also heats up, in some species to as much as 20 to 40 °C above the ambient temperature. Heat production (thermogenesis) helps evaporate odorant molecules for better dispersal, and because rotting flesh and dung are usually warm from the hyperactive metabolism of scavenging microbes, the heat itself might also attract insects. In the case of the eastern skunk cabbage (Fig. 1), which flowers in late winter or early spring when snow still covers the ground, thermogenesis allows the spadix to grow up through the snow.

How does a skunk cabbage heat its spadix? The mitochondria of plants, fungi, and unicellular eukaryotes have electron-transfer systems that are essentially the same as those in animals, but they also have an alternative respiratory pathway. A cyanide-resistant QH_2 oxidase transfers electrons from the ubiquinone pool directly to oxygen, bypassing the two proton-translocating steps of Complexes III and IV (Fig. 2). Energy that might have been conserved as



FIGURE 1 Eastern skunk cabbage.

ATP is instead released as heat. Plant mitochondria also have an alternative NADH dehydrogenase, insensitive to the Complex I inhibitor rotenone (see Table 19-4), that transfers electrons from NADH in the matrix directly to ubiquinone, bypassing Complex I and its associated proton pumping. And plant mitochondria have yet another NADH dehydrogenase, on the external face of the inner membrane, that transfers electrons from NADPH or NADH in the intermembrane space to ubiquinone, again bypassing Complex I. Thus when electrons enter the alternative respiratory pathway through the rotenone-insensitive NADH dehydrogenase, the external NADH dehydrogenase, or succinate dehydrogenase (Complex II), and pass to O_2 via the cyanide-resistant alternative oxidase, energy is not conserved as ATP but is released as heat. A skunk cabbage can use the heat to melt snow, produce a foul stench, or attract beetles or flies.

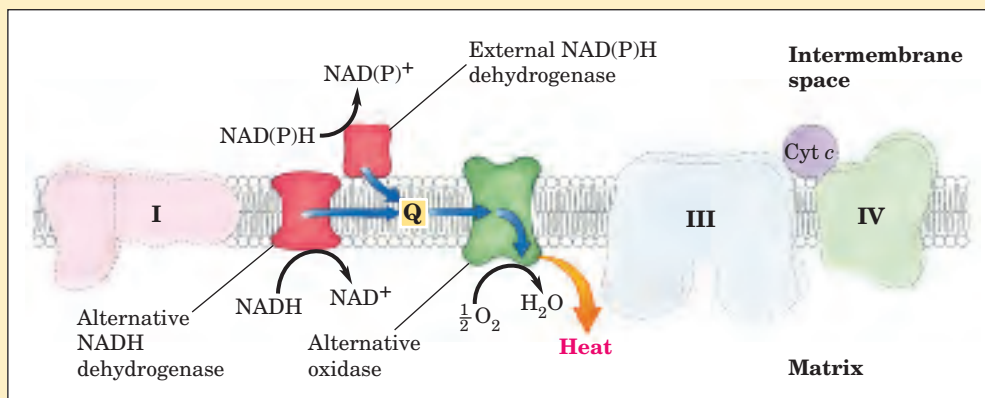


FIGURE 2 Electron carriers of the inner membrane of plant mitochondria. Electrons can flow through Complexes I, III, and IV, as in animal mitochondria, or through plant-specific alternative carriers by the paths shown with blue arrows.

protons out of the matrix against this gradient equals or exceeds the energy released by the transfer of electrons from NADH to O_2 . At this point electron flow must stop; the free energy for the overall process of electron flow coupled to proton pumping becomes zero, and the system is at equilibrium.

Certain conditions and reagents, however, can uncouple oxidation from phosphorylation. When intact mitochondria are disrupted by treatment with detergent or by physical shear, the resulting membrane fragments can still catalyze electron transfer from succinate or NADH to O_2 , but no ATP synthesis is coupled to this respiration. Certain chemical compounds cause uncoupling without disrupting mitochondrial structure. Chemical uncouplers include 2,4-dinitrophenol (DNP) and carbonylcyanide-*p*-trifluoromethoxyphenylhydrazone (FCCP) (Table 19-4; Fig. 19-19), weak acids with hydrophobic properties that permit them to diffuse readily across mitochondrial membranes. After entering the matrix in the protonated form, they can release a proton, thus dissipating the proton gradient. Ionophores such as valinomycin (see Fig. 11-45) allow inorganic ions to pass easily through membranes. Ionophores uncouple electron transfer from oxidative phosphorylation by dissipating the electrical contribution to the electrochemical gradient across the mitochondrial membrane.

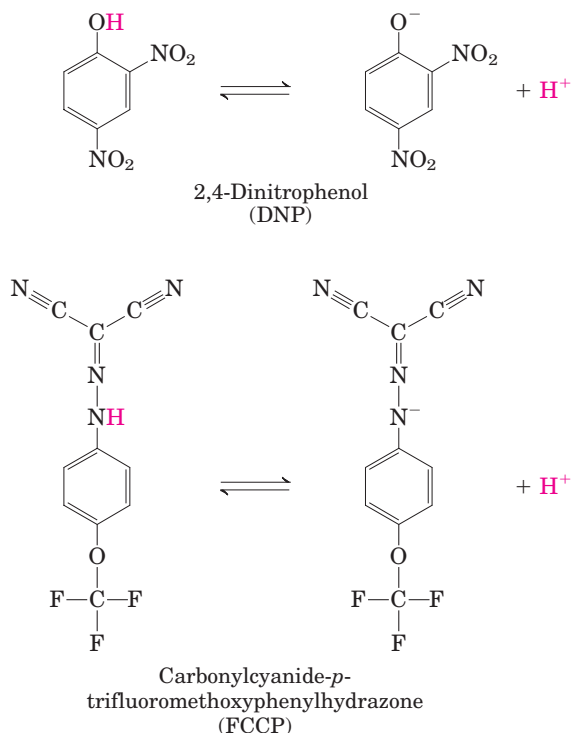


FIGURE 19-19 Two chemical uncouplers of oxidative phosphorylation. Both DNP and FCCP have a dissociable proton and are very hydrophobic. They carry protons across the inner mitochondrial membrane, dissipating the proton gradient. Both also uncouple phosphorylation (see Fig. 19-57).

A prediction of the chemiosmotic theory is that, because the role of electron transfer in mitochondrial ATP synthesis is simply to pump protons to create the electrochemical potential of the proton-motive force, an artificially created proton gradient should be able to replace electron transfer in driving ATP synthesis. This has been experimentally confirmed (Fig. 19-20). Mitochondria manipulated so as to impose a difference of proton concentration and a separation of charge across the inner membrane synthesize ATP *in the absence of an oxidizable substrate*; the proton-motive force alone suffices to drive ATP synthesis.

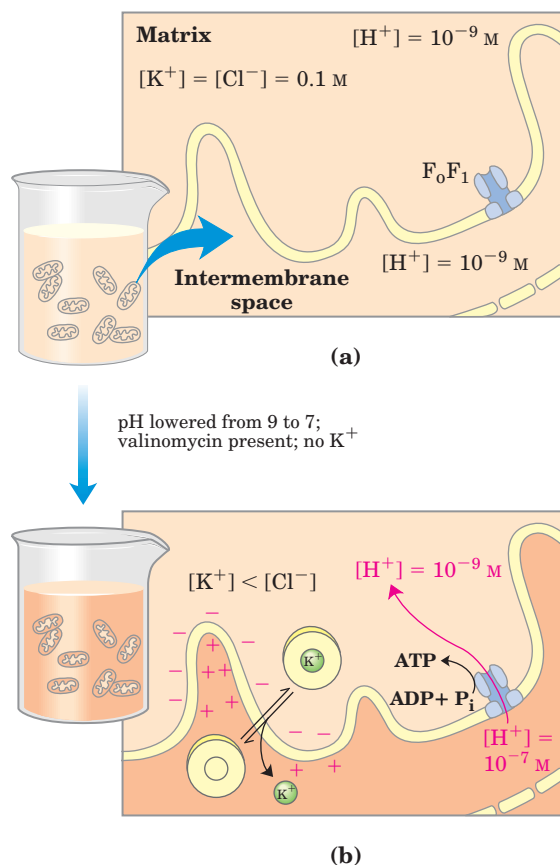


FIGURE 19-20 Evidence for the role of a proton gradient in ATP synthesis. An artificially imposed electrochemical gradient can drive ATP synthesis in the absence of an oxidizable substrate as electron donor. In this two-step experiment, (a) isolated mitochondria are first incubated in a pH 9 buffer containing 0.1 M KCl. Slow leakage of buffer and KCl into the mitochondria eventually brings the matrix into equilibrium with the surrounding medium. No oxidizable substrates are present. (b) Mitochondria are now separated from the pH 9 buffer and resuspended in pH 7 buffer containing valinomycin but no KCl. The change in buffer creates a difference of two pH units across the inner mitochondrial membrane. The outward flow of K^+ , carried (by valinomycin) down its concentration gradient without a counterion, creates a charge imbalance across the membrane (matrix negative). The sum of the chemical potential provided by the pH difference and the electrical potential provided by the separation of charges is a proton-motive force large enough to support ATP synthesis in the absence of an oxidizable substrate.

ATP Synthase Has Two Functional Domains, F_o and F_1

Mitochondrial **ATP synthase** is an F-type ATPase (see Fig. 11–39; Table 11–3) similar in structure and mechanism to the ATP synthases of chloroplasts and eubacteria. This large enzyme complex of the inner mitochondrial membrane catalyzes the formation of ATP



Efraim Racker,
1913–1991

from ADP and P_i , accompanied by the flow of protons from the P to the N side of the membrane (Eqn 19–10). ATP synthase, also called Complex V, has two distinct components: F_1 , a peripheral membrane protein, and F_o (*o* denoting oligomycin-sensitive), which is integral to the membrane. F_1 , the first factor recognized as essential for oxidative phosphorylation, was identified and purified by Efraim Racker and

his colleagues in the early 1960s.

In the laboratory, small membrane vesicles formed from inner mitochondrial membranes carry out ATP synthesis coupled to electron transfer. When F_1 is gently extracted, the “stripped” vesicles still contain intact respiratory chains and the F_o portion of ATP synthase. The vesicles can catalyze electron transfer from NADH to O_2

but cannot produce a proton gradient: F_o has a proton pore through which protons leak as fast as they are pumped by electron transfer, and without a proton gradient the F_1 -depleted vesicles cannot make ATP. Isolated F_1 catalyzes ATP hydrolysis (the reversal of synthesis) and was therefore originally called **F_1 ATPase**. When purified F_1 is added back to the depleted vesicles, it reassociates with F_o , plugging its proton pore and restoring the membrane’s capacity to couple electron transfer and ATP synthesis.

ATP Is Stabilized Relative to ADP on the Surface of F_1

Isotope exchange experiments with purified F_1 reveal a remarkable fact about the enzyme’s catalytic mechanism: on the enzyme surface, the reaction $ADP + P_i \rightleftharpoons ATP + H_2O$ is readily reversible—the free-energy change for ATP synthesis is close to zero! When ATP is hydrolyzed by F_1 in the presence of ^{18}O -labeled water, the P_i released contains an ^{18}O atom. Careful measurement of the ^{18}O content of P_i formed in vitro by F_1 -catalyzed hydrolysis of ATP reveals that the P_i has not one, but three or four ^{18}O atoms (Fig. 19–21). This indicates that the terminal pyrophosphate bond in ATP is cleaved and re-formed repeatedly before P_i leaves the enzyme surface. With P_i free to tumble in its binding site, each hydrolysis inserts ^{18}O randomly at one of the

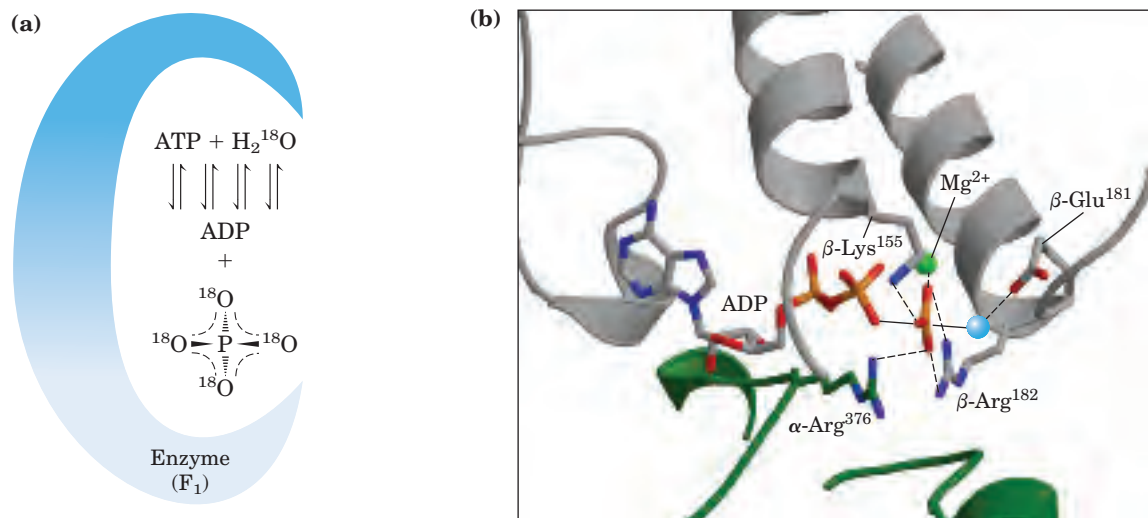


FIGURE 19–21 Catalytic mechanism of F_1 . (a) ^{18}O -exchange experiment. F_1 solubilized from mitochondrial membranes is incubated with ATP in the presence of ^{18}O -labeled water. At intervals, a sample of the solution is withdrawn and analyzed for the incorporation of ^{18}O into the P_i produced from ATP hydrolysis. In minutes, the P_i contains three or four ^{18}O atoms, indicating that both ATP hydrolysis and ATP synthesis have occurred several times during the incubation. (b) The likely transition state complex for ATP hydrolysis and synthesis in ATP

synthase (derived from PDB ID 1BMF). The α subunit is shown in green, β in gray. The positively charged residues β -Arg¹⁸² and α -Arg³⁷⁶ coordinate two oxygens of the pentavalent phosphate intermediate; β -Lys¹⁵⁵ interacts with a third oxygen, and the Mg^{2+} ion (green sphere) further stabilizes the intermediate. The blue sphere represents the leaving group (H_2O). These interactions result in the ready equilibration of ATP and ADP + P_i in the active site.

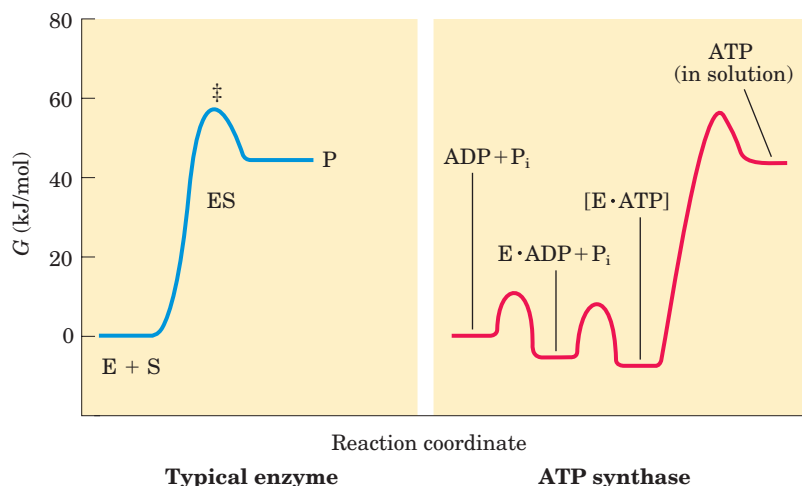


FIGURE 19-22 Reaction coordinate diagrams for ATP synthase and for a more typical enzyme. In a typical enzyme-catalyzed reaction (left), reaching the transition state ($\ddagger$) between substrate and product is the major energy barrier to overcome. In the reaction catalyzed by ATP synthase (right), release of ATP from the enzyme, not formation of ATP, is the major energy barrier. The free-energy change for the formation of ATP from ADP and P_i in aqueous solution is large and positive, but on the enzyme surface, the very tight binding of ATP provides sufficient binding energy to bring the free energy of the enzyme-bound ATP close to that of $ADP + P_i$, so the reaction is readily reversible. The equilibrium constant is near 1. The free energy required for the release of ATP is provided by the proton-motive force.

four positions in the molecule. This exchange reaction occurs in unenergized F_0F_1 complexes (with no proton gradient) and with isolated F_1 —the exchange does not require the input of energy.

Kinetic studies of the initial rates of ATP synthesis and hydrolysis confirm the conclusion that $\Delta G'^\circ$ for ATP synthesis on the enzyme is near zero. From the measured rates of hydrolysis ($k_1 = 10 \text{ s}^{-1}$) and synthesis ($k_{-1} = 24 \text{ s}^{-1}$), the calculated equilibrium constant for the reaction



is

$$K'_{\text{eq}} = \frac{k_{-1}}{k_1} = \frac{24 \text{ s}^{-1}}{10 \text{ s}^{-1}} = 2.4$$

From this K'_{eq} , the calculated apparent $\Delta G'^\circ$ is close to zero. This is much different from the K'_{eq} of about 10^5 ($\Delta G'^\circ = -30.5 \text{ kJ/mol}$) for the hydrolysis of ATP free in solution (not on the enzyme surface).

What accounts for the huge difference? ATP synthase stabilizes ATP relative to $ADP + P_i$ by binding ATP more tightly, releasing enough energy to counterbalance the cost of making ATP. Careful measurements of the binding constants show that F_0F_1 binds ATP with very high affinity ($K_d \leq 10^{-12} \text{ M}$) and ADP with much lower affinity ($K_d \approx 10^{-5} \text{ M}$). The difference in K_d corresponds to a difference of about 40 kJ/mol in binding energy, and this binding energy drives the equilibrium toward formation of the product ATP.

The Proton Gradient Drives the Release of ATP from the Enzyme Surface

Although ATP synthase equilibrates ATP with $ADP + P_i$, in the absence of a proton gradient the newly synthesized ATP does not leave the surface of the enzyme.

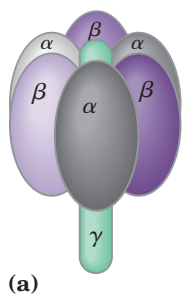
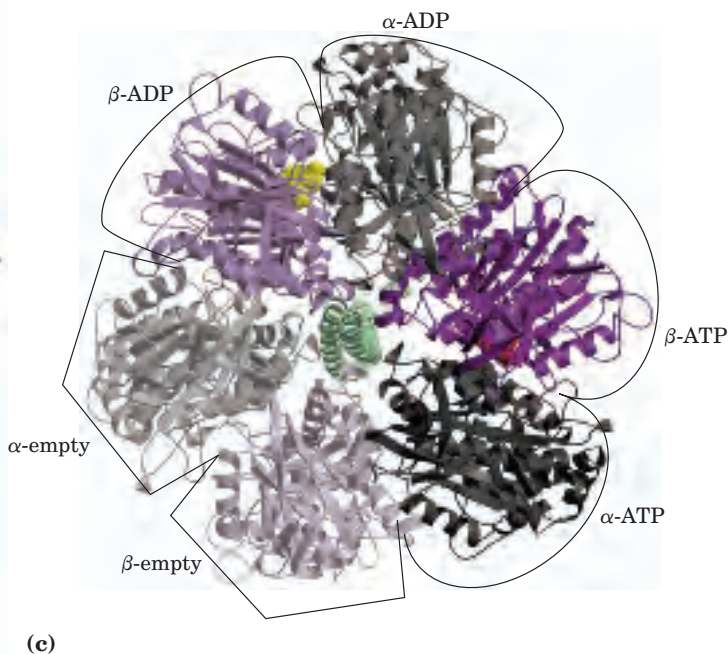
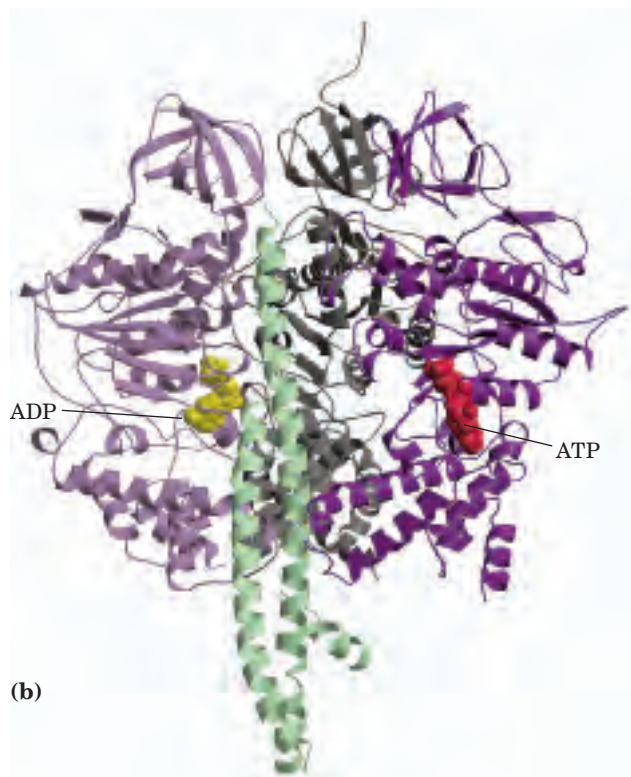
It is the proton gradient that causes the enzyme to release the ATP formed on its surface. The reaction coordinate diagram of the process (Fig. 19-22) illustrates the difference between the mechanism of ATP synthase and that of many other enzymes that catalyze endergonic reactions.

For the continued synthesis of ATP, the enzyme must cycle between a form that binds ATP very tightly and a form that releases ATP. Chemical and crystallographic studies of the ATP synthase have revealed the structural basis for this alternation in function.

Each β Subunit of ATP Synthase Can Assume Three Different Conformations

Mitochondrial F_1 has nine subunits of five different types, with the composition $\alpha_3\beta_3\gamma\delta\epsilon$. Each of the three β subunits has one catalytic site for ATP synthesis. The crystallographic determination of the F_1 structure by John E. Walker and colleagues revealed structural details very helpful in explaining the catalytic mechanism of the enzyme. The knoblike portion of F_1 is a flattened sphere, 8 nm high and 10 nm across, consisting of alternating α and β subunits arranged like the sections of an orange (Fig. 19-23a–c). The polypeptides that make up the stalk in the F_1 crystal structure are asymmetrically arranged, with one domain of the single γ subunit making up a central shaft that passes through F_1 , and another domain of γ associated primarily with one of the three β subunits, designated β -empty (Fig. 19-23c). Although the amino acid sequences of the three β subunits are identical, *their conformations differ*, in part because of the association of the γ subunit with just one of the three. The structures of the δ and ϵ subunits are not revealed in these crystallographic studies.

The conformational differences among β subunits extend to differences in their ATP/ADP-binding sites.



John E. Walker

When researchers crystallized the protein in the presence of ADP and App(NH)p, a close structural analog of ATP that cannot be hydrolyzed by the ATPase activity of F_1 , the binding site of one of the three β subunits was filled with App(NH)p, the second was filled with

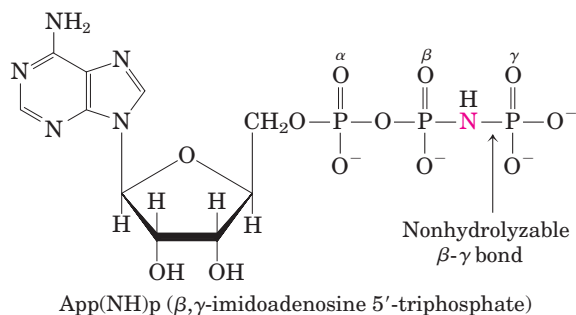
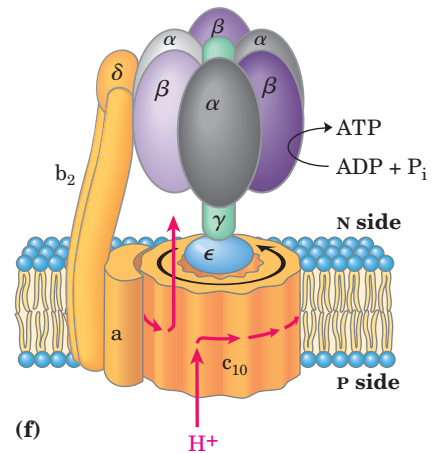
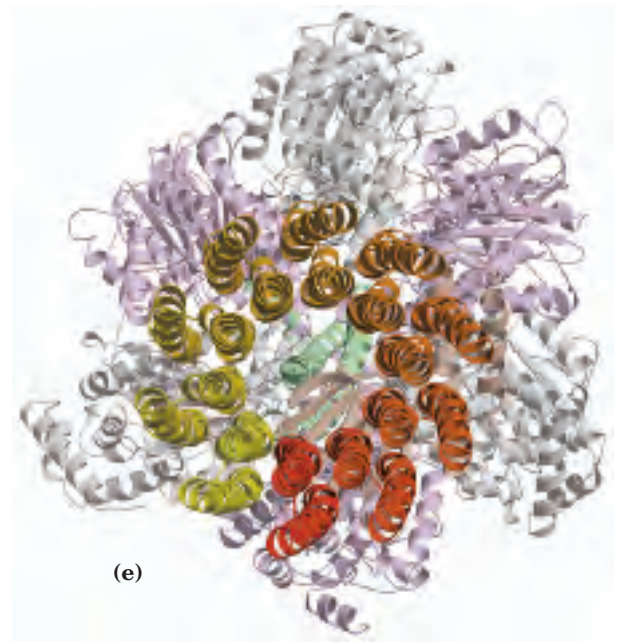
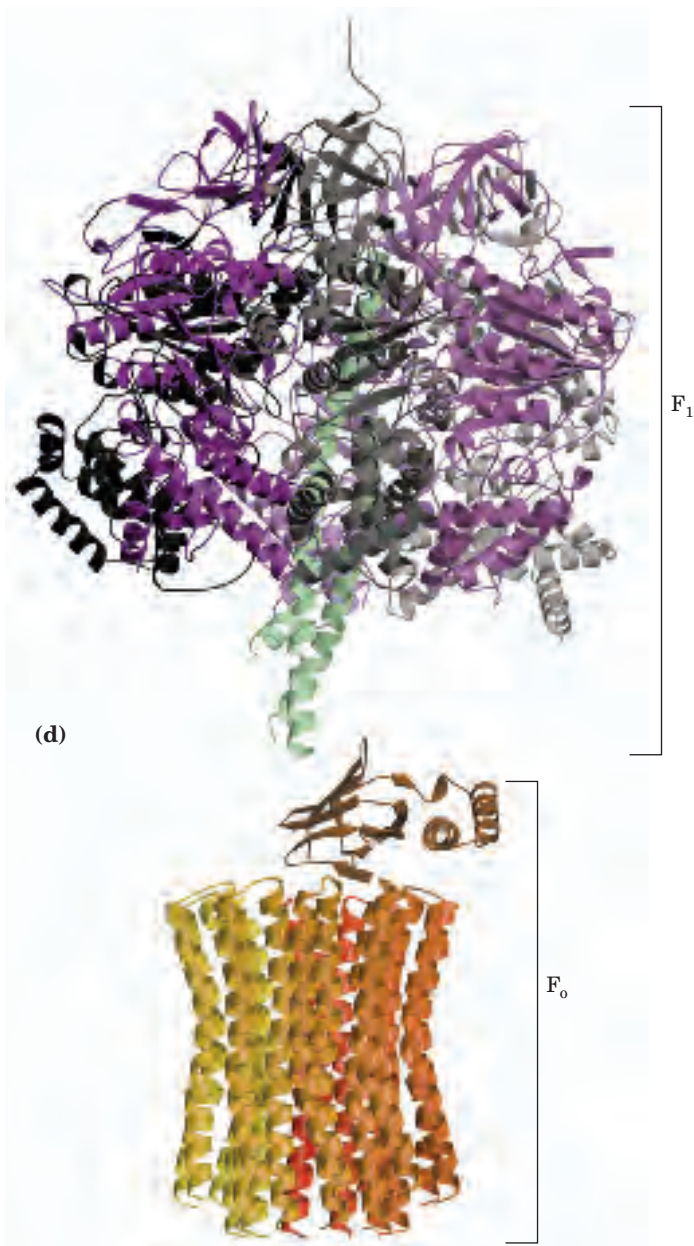


FIGURE 19-23 Mitochondrial ATP synthase complex. (a) Structure of the F_1 complex, deduced from crystallographic and biochemical studies. In F_1 , three α and three β subunits are arranged like the segments of an orange, with alternating α (shades of gray) and β (shades of purple) subunits around a central shaft, the γ subunit (green). (b) Crystal structure of bovine F_1 (PDB ID 1BMF), viewed from the side. Two α subunits and one β subunit have been omitted to reveal the central shaft (γ subunit) and the binding sites for ATP (red) and ADP (yellow) on the β subunits. The δ and ϵ subunits are not shown here. (c) F_1 viewed from above (that is, from the N side of the membrane), showing the three β and three α subunits and the central shaft (γ subunit, green). Each β subunit, near its interface with the neighboring α subunit, has a nucleotide-binding site critical to the catalytic activity. The single γ subunit associates primarily with one of the three $\alpha\beta$ pairs, forcing each of the three β subunits into slightly different conformations, with different nucleotide-binding sites. In the crystalline enzyme, one subunit (β -ADP) has ADP (yellow) in its binding site, the next (β -ATP) has ATP (red), and the third (β -empty) has no bound nucleotide. (d) Side view of the F_0F_1 structure. This is a composite, in which the crystallographic coordinates of bovine mitochondrial F_1 (shades of purple and gray) have been combined with those of yeast mitochondrial F_0 (shades of yellow and orange) (PDB ID 1QO1). Subunits a, b, δ , and ϵ were not part of the crystal structure shown here. (e) The F_0F_1 structure, viewed end-on in the direction P side to N side. The major structures visible in this cross section are the two trans-membrane helices of each of ten c subunits arranged in concentric circles. (f) Diagram of the F_0F_1 complex, deduced from biochemical and crystallographic studies. The two b subunits of F_0 associate firmly with the α and β subunits of F_1 , holding them fixed relative to the membrane. In F_0 , the membrane-embedded cylinder of c subunits is attached to the shaft made up of F_1 subunits γ and ϵ . As protons flow through the membrane from the P side to the N side through F_0 , the cylinder and shaft rotate, and the β subunits of F_1 change conformation as the γ subunit associates with each in turn.



ADP, and the third was empty. The corresponding β subunit conformations were designated β -ATP, β -ADP, and β -empty (Fig. 19-23c). This difference in nucleotide binding among the three subunits is critical to the mechanism of the complex.

The F_0 complex making up the proton pore is composed of three subunits, a, b, and c, in the proportion ab_2c_{10-12} . Subunit c is a small (M_r 8,000), very hydrophobic polypeptide, consisting almost entirely of two transmembrane helices, with a small loop extending from the matrix side of the membrane. The crystal structure of the yeast F_0F_1 , solved in 1999, shows the arrangement of the c subunits. The yeast complex has ten c subunits, each with two transmembrane helices roughly perpendicular to the plane of the membrane and arranged in two concentric circles (Fig. 19-23d, e). The

inner circle is made up of the amino-terminal helices of each c subunit; the outer circle, about 55 Å in diameter, is made up of the carboxyl-terminal helices. The ϵ and γ subunits of F_1 form a leg-and-foot that projects from the bottom (membrane) side of F_1 and stands firmly on the ring of c subunits. The schematic drawing in Figure 19-23f combines the structural information from studies of bovine F_1 and yeast F_0F_1 .

Rotational Catalysis Is Key to the Binding-Change Mechanism for ATP Synthesis

On the basis of detailed kinetic and binding studies of the reactions catalyzed by F_0F_1 , Paul Boyer proposed a **rotational catalysis** mechanism in which the three active sites of F_1 take turns catalyzing ATP synthesis



Paul Boyer

(Fig. 19–24). A given β subunit starts in the β -ADP conformation, which binds ADP and P_i from the surrounding medium. The subunit now changes conformation, assuming the β -ATP form that tightly binds and stabilizes ATP, bringing about the ready equilibration of $ADP + P_i$ with ATP on the enzyme surface.

Finally, the subunit changes to the β -empty conformation, which has very low affinity for ATP, and the newly synthesized ATP leaves the enzyme surface. Another round of catalysis begins when

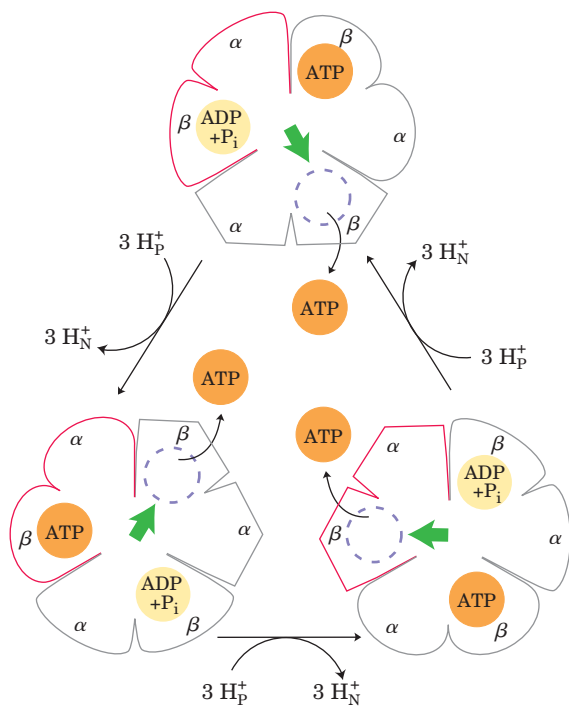


FIGURE 19–24 Binding-change model for ATP synthase. The F_1 complex has three nonequivalent adenine nucleotide-binding sites, one for each pair of α and β subunits. At any given moment, one of these sites is in the β -ATP conformation (which binds ATP tightly), a second is in the β -ADP (loose-binding) conformation, and a third is in the β -empty (very-loose-binding) conformation. The proton-motive force causes rotation of the central shaft—the γ subunit, shown as a green arrowhead—which comes into contact with each $\alpha\beta$ subunit pair in succession. This produces a cooperative conformational change in which the β -ATP site is converted to the β -empty conformation, and ATP dissociates; the β -ADP site is converted to the β -ATP conformation, which promotes condensation of bound $ADP + P_i$ to form ATP; and the β -empty site becomes a β -ADP site, which loosely binds $ADP + P_i$ entering from the solvent. This model, based on experimental findings, requires that at least two of the three catalytic sites alternate in activity; ATP cannot be released from one site unless and until ADP and P_i are bound at the other.

this subunit again assumes the β -ADP form and binds ADP and P_i .

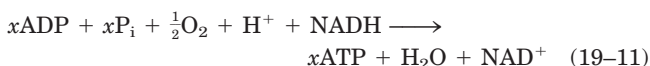
The conformational changes central to this mechanism are driven by the passage of protons through the F_o portion of ATP synthase. The streaming of protons through the F_o “pore” causes the cylinder of c subunits and the attached γ subunit to rotate about the long axis of γ , which is perpendicular to the plane of the membrane. The γ subunit passes through the center of the $\alpha_3\beta_3$ spheroid, which is held stationary relative to the membrane surface by the b_2 and δ subunits (Fig. 19–23f). With each rotation of 120° , γ comes into contact with a different β subunit, and the contact forces that β subunit into the β -empty conformation.

The three β subunits interact in such a way that when one assumes the β -empty conformation, its neighbor to one side *must* assume the β -ADP form, and the other neighbor the β -ATP form. Thus one complete rotation of the γ subunit causes each β subunit to cycle through all three of its possible conformations, and for each rotation, three ATP are synthesized and released from the enzyme surface.

One strong prediction of this binding-change model is that the γ subunit should rotate in one direction when F_oF_1 is synthesizing ATP and in the opposite direction when the enzyme is hydrolyzing ATP. This prediction was confirmed in elegant experiments in the laboratories of Masasuke Yoshida and Kazuhiko Kinosita, Jr. The rotation of γ in a single F_1 molecule was observed microscopically by attaching a long, thin, fluorescent actin polymer to γ and watching it move relative to $\alpha_3\beta_3$ immobilized on a microscope slide, as ATP was hydrolyzed. When the entire F_oF_1 complex (not just F_1) was used in a similar experiment, the entire ring of c subunits rotated with γ (Fig. 19–25). The “shaft” rotated in the predicted direction through 360° . The rotation was not smooth, but occurred in three discrete steps of 120° . As calculated from the known rate of ATP hydrolysis by one F_1 molecule and from the frictional drag on the long actin polymer, the efficiency of this mechanism in converting chemical energy into motion is close to 100%. It is, in Boyer’s words, “a splendid molecular machine!”

Chemiosmotic Coupling Allows Nonintegral Stoichiometries of O_2 Consumption and ATP Synthesis

Before the general acceptance of the chemiosmotic model for oxidative phosphorylation, the assumption was that the overall reaction equation would take the following form:



with the value of x —sometimes called the **P/O ratio** or the **P/2e[−] ratio**—always an integer. When intact mito-

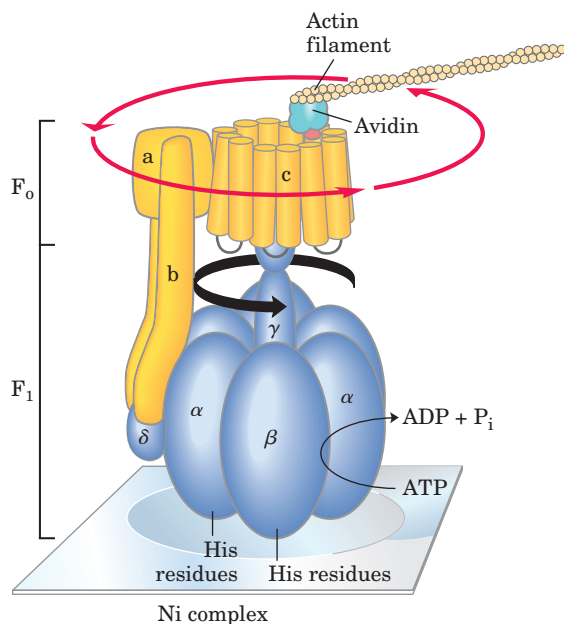
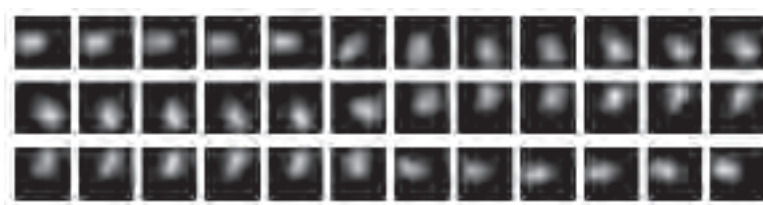


FIGURE 19-25 Rotation of F_0 and γ experimentally demonstrated.

F_1 genetically engineered to contain a run of His residues adheres tightly to a microscope slide coated with a Ni complex; biotin is covalently attached to a c subunit of F_0 . The protein avidin, which binds biotin very tightly, is covalently attached to long filaments of actin labeled with a fluorescent probe. Biotin-avidin binding now attaches the actin filaments to the c subunit. When ATP is provided as substrate for the ATPase activity of F_1 , the labeled filament is seen to rotate continuously in one direction, proving that the F_0 cylinder of c subunits rotates. In another experiment, a fluorescent actin filament was attached directly to the γ subunit. The series of fluorescence micrographs shows the position of the actin filament at intervals of 133 ms. Note that as the filament rotates, it makes a discrete jump about every eleventh frame. Presumably the cylinder and shaft move as one unit.



chondria are suspended in solution with an oxidizable substrate such as succinate or NADH and are provided with O_2 , ATP synthesis is readily measurable, as is the decrease in O_2 . Measurement of P/O, however, is complicated by the fact that intact mitochondria consume ATP in many reactions taking place in the matrix, and they consume O_2 for purposes other than oxidative phosphorylation. Most experiments have yielded P/O (ATP to $\frac{1}{2}O_2$) ratios of between 2 and 3 when NADH was the electron donor, and between 1 and 2 when succinate was the donor. Given the assumption that P/O should have an integral value, most experimenters agreed that the P/O ratios must be 3 for NADH and 2 for succinate, and for years those values have appeared in research papers and textbooks.

With introduction of the chemiosmotic paradigm for coupling ATP synthesis to electron transfer, there was no theoretical requirement for P/O to be integral. The relevant questions about stoichiometry became, how many protons are pumped outward by electron transfer from one NADH to O_2 , and how many protons must flow inward through the F_0F_1 complex to drive the synthesis of one ATP? The measurement of proton fluxes is technically complicated; the investigator must take into account the buffering capacity of mitochondria, non-productive leakage of protons across the inner membrane, and use of the proton gradient for functions other than ATP synthesis, such as driving the transport of substrates across the inner mitochondrial membrane (described below). The consensus values for number of protons pumped out per pair of electrons are 10 for NADH

and 6 for succinate. The most widely accepted experimental value for number of protons required to drive the synthesis of an ATP molecule is 4, of which 1 is used in transporting P_i , ATP, and ADP across the mitochondrial membrane (see below). If 10 protons are pumped out per NADH and 4 must flow in to produce 1 ATP, the proton-based P/O ratio is 2.5 for NADH as the electron donor and 1.5 ($6/4$) for succinate. We use the P/O values of 2.5 and 1.5 throughout this book, but the values 3.0 and 2.0 are still common in the biochemical literature. The final word on proton stoichiometry will probably not be written until we know the full details of the F_0F_1 reaction mechanism.

The Proton-Motive Force Energizes Active Transport

Although the primary role of the proton gradient in mitochondria is to furnish energy for the synthesis of ATP, the proton-motive force also drives several transport processes essential to oxidative phosphorylation. The inner mitochondrial membrane is generally impermeable to charged species, but two specific systems transport ADP and P_i into the matrix and ATP out to the cytosol (Fig. 19-26).

The **adenine nucleotide translocase**, integral to the inner membrane, binds ADP^{3-} in the intermembrane space and transports it into the matrix in exchange for an ATP^{4-} molecule simultaneously transported outward (see Fig. 13-1 for the ionic forms of ATP and ADP). Because this antiporter moves four negative charges out for every three moved in, its activity is favored by the

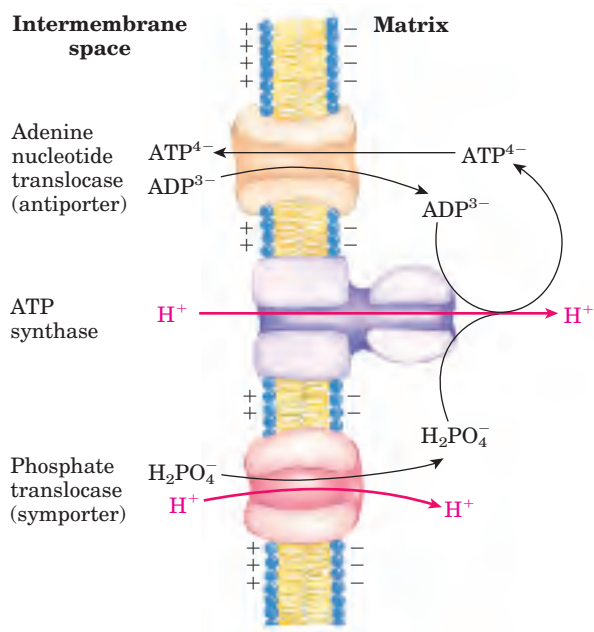


FIGURE 19-26 Adenine nucleotide and phosphate translocases.

Transport systems of the inner mitochondrial membrane carry ADP and P_i into the matrix and newly synthesized ATP into the cytosol. The adenine nucleotide translocase is an antiporter; the same protein moves ADP into the matrix and ATP out. The effect of replacing ATP⁴⁻ with ADP³⁻ is the net efflux of one negative charge, which is favored by the charge difference across the inner membrane (outside positive). At pH 7, P_i is present as both HPO₄²⁻ and H₂PO₄⁻; the phosphate translocase is specific for H₂PO₄⁻. There is no net flow of charge during symport of H₂PO₄⁻ and H⁺, but the relatively low proton concentration in the matrix favors the inward movement of H⁺. Thus the proton-motive force is responsible both for providing the energy for ATP synthesis and for transporting substrates (ADP and P_i) in and product (ATP) out of the mitochondrial matrix. All three of these transport systems can be isolated as a single membrane-bound complex (ATP synthasome).

transmembrane electrochemical gradient, which gives the matrix a net negative charge; the proton-motive force drives ATP-ADP exchange. Adenine nucleotide translocase is specifically inhibited by atractyloside, a toxic glycoside formed by a species of thistle. If the transport of ADP into and ATP out of mitochondria is inhibited, cytosolic ATP cannot be regenerated from ADP, explaining the toxicity of atractyloside.

A second membrane transport system essential to oxidative phosphorylation is the **phosphate translocase**, which promotes symport of one H₂PO₄⁻ and one H⁺ into the matrix. This transport process, too, is favored by the transmembrane proton gradient (Fig. 19-26). Notice that the process requires movement of one proton from the P to the N side of the inner membrane, consuming some of the energy of electron transfer. A complex of the ATP synthase and both translocases, the **ATP synthasome**, can be isolated from

mitochondria by gentle dissection with detergents, suggesting that the functions of these three proteins are very tightly integrated.

Shuttle Systems Indirectly Convey Cytosolic NADH into Mitochondria for Oxidation

The NADH dehydrogenase of the inner mitochondrial membrane of animal cells can accept electrons only from NADH in the matrix. Given that the inner membrane is not permeable to NADH, how can the NADH generated by glycolysis in the cytosol be reoxidized to NAD⁺ by O₂ via the respiratory chain? Special shuttle systems carry reducing equivalents from cytosolic NADH into mitochondria by an indirect route. The most active NADH shuttle, which functions in liver, kidney, and heart mitochondria, is the **malate-aspartate shuttle** (Fig. 19-27). The reducing equivalents of cytosolic NADH are first transferred to cytosolic oxaloacetate to yield malate, catalyzed by cytosolic malate dehydrogenase. The malate thus formed passes through the inner membrane via the malate- α -ketoglutarate transporter. Within the matrix the reducing equivalents are passed to NAD⁺ by the action of matrix malate dehydrogenase, forming NADH; this NADH can pass electrons directly to the respiratory chain. About 2.5 molecules of ATP are generated as this pair of electrons passes to O₂. Cytosolic oxaloacetate must be regenerated by transamination reactions and the activity of membrane transporters to start another cycle of the shuttle.

Skeletal muscle and brain use a different NADH shuttle, the **glycerol 3-phosphate shuttle** (Fig. 19-28). It differs from the malate-aspartate shuttle in that it delivers the reducing equivalents from NADH to ubiquinone and thus into Complex III, not Complex I (Fig. 19-8), providing only enough energy to synthesize 1.5 ATP molecules per pair of electrons.

The mitochondria of plants have an *externally* oriented NADH dehydrogenase that can transfer electrons directly from cytosolic NADH into the respiratory chain at the level of ubiquinone. Because this pathway bypasses the NADH dehydrogenase of Complex I and the associated proton movement, the yield of ATP from cytosolic NADH is less than that from NADH generated in the matrix (Box 19-1).

SUMMARY 19.2 ATP Synthesis

- The flow of electrons through Complexes I, III, and IV results in pumping of protons across the inner mitochondrial membrane, making the matrix alkaline relative to the intermembrane space. This proton gradient provides the energy (in the form of the proton-motive force) for ATP synthesis from ADP and P_i by ATP synthase (F₀F₁ complex) in the inner membrane.

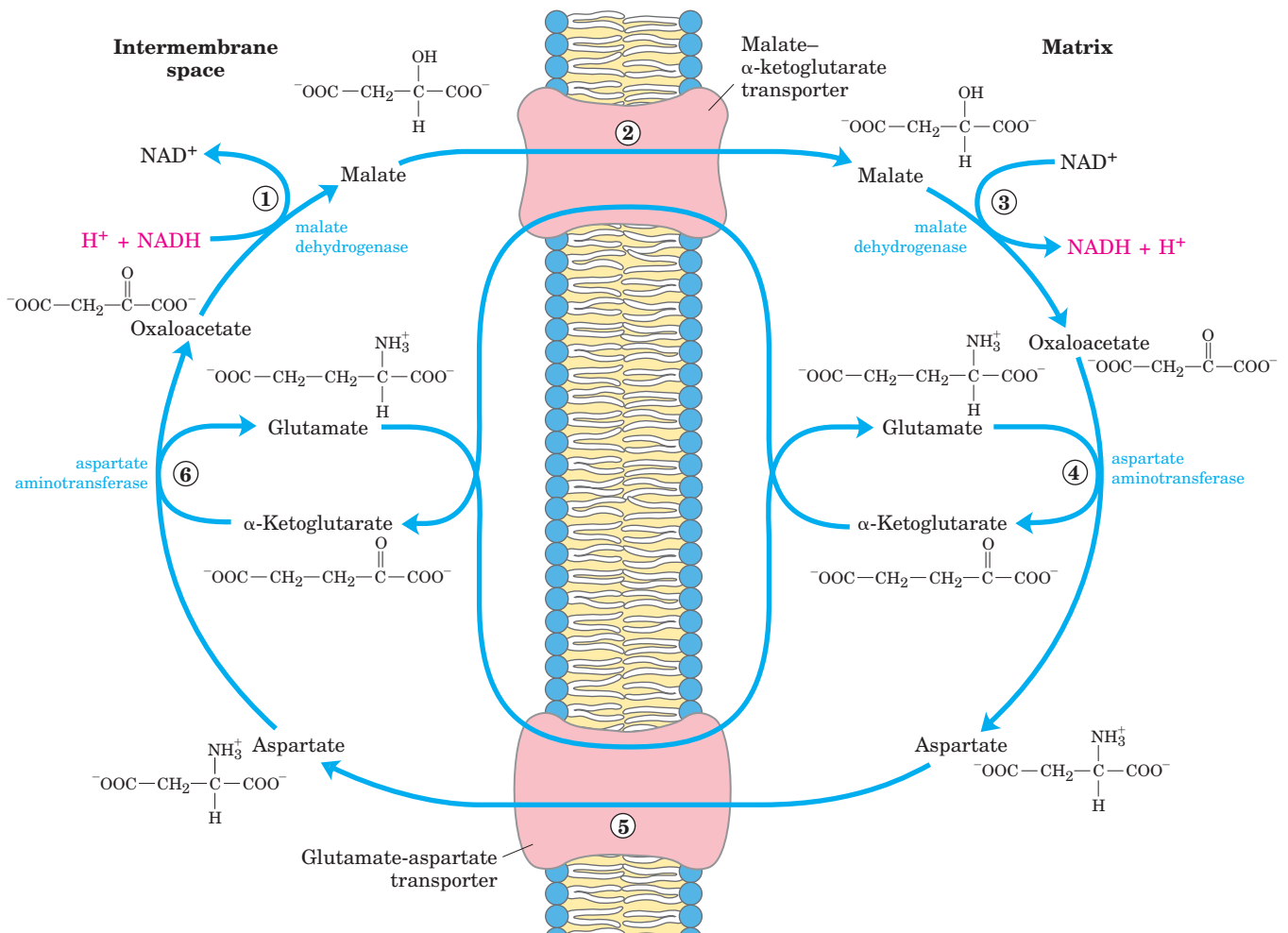
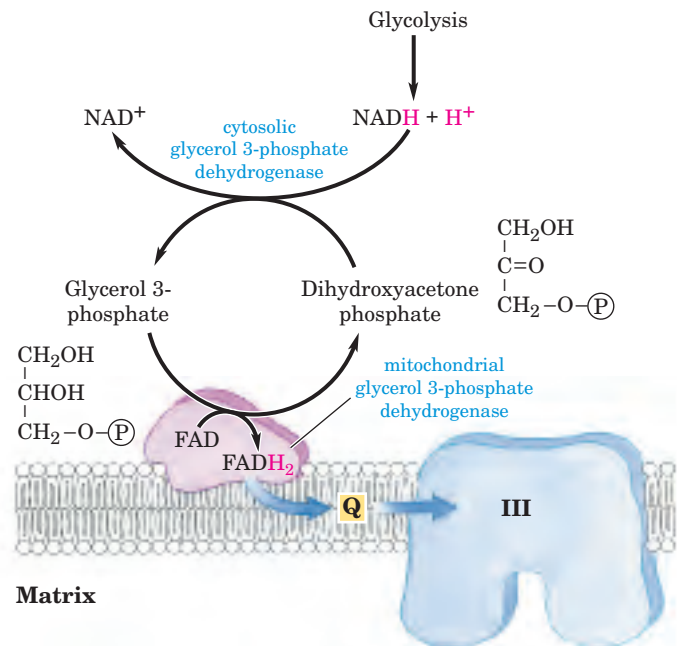


FIGURE 19-27 Malate-aspartate shuttle. This shuttle for transporting reducing equivalents from cytosolic NADH into the mitochondrial matrix is used in liver, kidney, and heart. ① NADH in the cytosol (intermembrane space) passes two reducing equivalents to oxaloacetate, producing malate. ② Malate crosses the inner membrane via the malate- α -ketoglutarate transporter. ③ In the matrix, malate passes

two reducing equivalents to NAD^+ , and the resulting NADH is oxidized by the respiratory chain. The oxaloacetate formed from malate cannot pass directly into the cytosol. ④ It is first transaminated to aspartate, which ⑤ can leave via the glutamate-aspartate transporter. ⑥ Oxaloacetate is regenerated in the cytosol, completing the cycle.

FIGURE 19-28 Glycerol 3-phosphate shuttle. This alternative means of moving reducing equivalents from the cytosol to the mitochondrial matrix operates in skeletal muscle and the brain. In the cytosol, dihydroxyacetone phosphate accepts two reducing equivalents from NADH in a reaction catalyzed by cytosolic glycerol 3-phosphate dehydrogenase. An isozyme of glycerol 3-phosphate dehydrogenase bound to the outer face of the inner membrane then transfers two reducing equivalents from glycerol 3-phosphate in the intermembrane space to ubiquinone. Note that this shuttle does not involve membrane transport systems.



- ATP synthase carries out “rotational catalysis,” in which the flow of protons through F_o causes each of three nucleotide-binding sites in F_1 to cycle from (ADP + P_i)-bound to ATP-bound to empty conformations.
- ATP formation on the enzyme requires little energy; the role of the proton-motive force is to push ATP from its binding site on the synthase.
- The ratio of ATP synthesized per $\frac{1}{2}O_2$ reduced to H_2O (the P/O ratio) is about 2.5 when electrons enter the respiratory chain at Complex I, and 1.5 when electrons enter at CoQ.
- Energy conserved in a proton gradient can drive solute transport uphill across a membrane.
- The inner mitochondrial membrane is impermeable to NADH and NAD^+ , but NADH equivalents are moved from the cytosol to the matrix by either of two shuttles. NADH equivalents moved in by the malate-aspartate shuttle enter the respiratory chain at Complex I and yield a P/O ratio of 2.5; those moved in by the glycerol 3-phosphate shuttle enter at CoQ and give a P/O ratio of 1.5.

19.3 Regulation of Oxidative Phosphorylation

Oxidative phosphorylation produces most of the ATP made in aerobic cells. Complete oxidation of a molecule of glucose to CO_2 yields 30 or 32 ATP (Table 19–5). By comparison, glycolysis under anaerobic conditions (lactate fermentation) yields only 2 ATP per glucose. Clearly, the evolution of oxidative phosphorylation provided a tremendous increase in the energy efficiency of catabolism. Complete oxidation to CO_2 of the coenzyme A derivative of palmitate (16:0), which also occurs in the mitochondrial matrix, yields 108 ATP per palmitoyl-

CoA (see Table 17–1). A similar calculation can be made for the ATP yield from oxidation of each of the amino acids (Chapter 18). Aerobic oxidative pathways that result in electron transfer to O_2 accompanied by oxidative phosphorylation therefore account for the vast majority of the ATP produced in catabolism, so the regulation of ATP production by oxidative phosphorylation to match the cell’s fluctuating needs for ATP is absolutely essential.

Oxidative Phosphorylation Is Regulated by Cellular Energy Needs

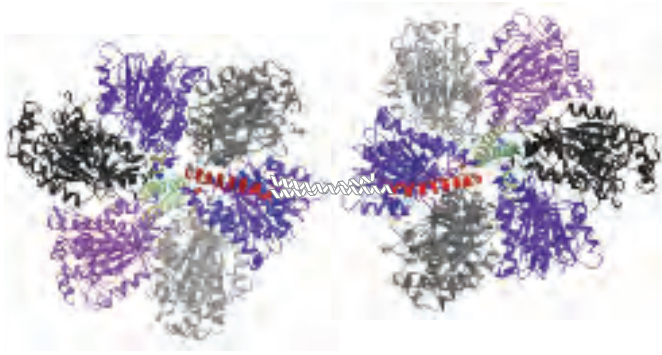
The rate of respiration (O_2 consumption) in mitochondria is tightly regulated; it is generally limited by the availability of ADP as a substrate for phosphorylation. Dependence of the rate of O_2 consumption on the availability of the P_i acceptor ADP (Fig. 19–18b), the **acceptor control** of respiration, can be remarkable. In some animal tissues, the **acceptor control ratio**, the ratio of the maximal rate of ADP-induced O_2 consumption to the basal rate in the absence of ADP, is at least ten.

The intracellular concentration of ADP is one measure of the energy status of cells. Another, related measure is the **mass-action ratio** of the ATP-ADP system, $[ATP]/([ADP][P_i])$. Normally this ratio is very high, so the ATP-ADP system is almost fully phosphorylated. When the rate of some energy-requiring process (protein synthesis, for example) increases, the rate of breakdown of ATP to ADP and P_i increases, lowering the mass-action ratio. With more ADP available for oxidative phosphorylation, the rate of respiration increases, causing regeneration of ATP. This continues until the mass-action ratio returns to its normal high level, at which point respiration slows again. The rate of oxidation of cellular fuels is regulated with such sensitivity and precision that the $[ATP]/([ADP][P_i])$ ratio fluctuates only slightly in most tissues, even during extreme variations in energy demand. In short, ATP is formed only as fast as it is used in energy-requiring cellular activities.

TABLE 19–5 ATP Yield from Complete Oxidation of Glucose

Process	Direct product	Final ATP
Glycolysis	2 NADH (cytosolic) 2 ATP	3 or 5* 2
Pyruvate oxidation (two per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl-CoA oxidation in citric acid cycle (two per glucose)	6 NADH (mitochondrial matrix) 2 $FADH_2$ 2 ATP or 2 GTP	15 3 2
Total yield per glucose		30 or 32

*The number depends on which shuttle system transfers reducing equivalents into the mitochondrion.



An Inhibitory Protein Prevents ATP Hydrolysis during Ischemia

We have already encountered ATP synthase as an ATP-driven proton pump (see Fig. 11–39; Table 11–3), catalyzing the reverse of ATP synthesis. When a cell is ischemic (deprived of oxygen), as in a heart attack or stroke, electron transfer to oxygen ceases, and so does the pumping of protons. The proton-motive force soon collapses. Under these conditions, the ATP synthase could operate in reverse, hydrolyzing ATP to pump protons outward and causing a disastrous drop in ATP levels. This is prevented by a small (84 amino acids) protein inhibitor, IF₁, which simultaneously binds to two ATP synthase molecules, inhibiting their ATPase activity (Fig. 19–29). IF₁ is inhibitory only in its dimeric form, which is favored at pH lower than 6.5. In a cell starved for oxygen, the main source of ATP becomes glycolysis, and the pyruvic or lactic acid thus formed lowers the pH in the cytosol and the mitochondrial matrix. This favors IF₁ dimerization, leading to inhibition of the ATPase activity of ATP synthase, thereby preventing wasteful hydrolysis of ATP. When aerobic metabolism resumes, production of pyruvic acid slows, the pH of the cytosol rises, the IF₁ dimer is destabilized, and the inhibition of ATP synthase is lifted.

Uncoupled Mitochondria in Brown Fat Produce Heat

There is a remarkable and instructive exception to the general rule that respiration slows when the ATP supply is adequate. Most newborn mammals, including humans, have a type of adipose tissue called **brown fat** in which fuel oxidation serves not to produce ATP but to generate heat to keep the newborn warm. This specialized adipose tissue is brown because of the presence of large numbers of mitochondria and thus large amounts of

FIGURE 19–29 Structure of bovine F₁-ATPase in a complex with its regulatory protein IF₁. (Derived from PDB ID 1OHH) Two F₁ molecules are viewed here as in Figure 19–23c. The inhibitor IF₁ (red) binds to the $\alpha\beta$ interface of the subunits in the diphosphate (ADP) conformation (α ADP and β ADP), freezing the two F₁ complexes and thereby blocking ATP hydrolysis (and synthesis). (Parts of IF₁ that failed to resolve in crystals of F₁ are shown in white outline as they occur in crystals of isolated IF₁.) This complex is stable only at the low cytosolic pH characteristic of cells that are producing ATP by glycolysis; when aerobic metabolism resumes, the cytosolic pH rises, the inhibitor is destabilized, and ATP synthase becomes active.

cytochromes, whose heme groups are strong absorbers of visible light.

The mitochondria of brown fat are like those of other mammalian cells in all respects, except that they have a unique protein in their inner membrane. **Thermogenin**, also called the **uncoupling protein** (Table 19–4), provides a path for protons to return to the matrix without passing through the F₀F₁ complex (Fig. 19–30).

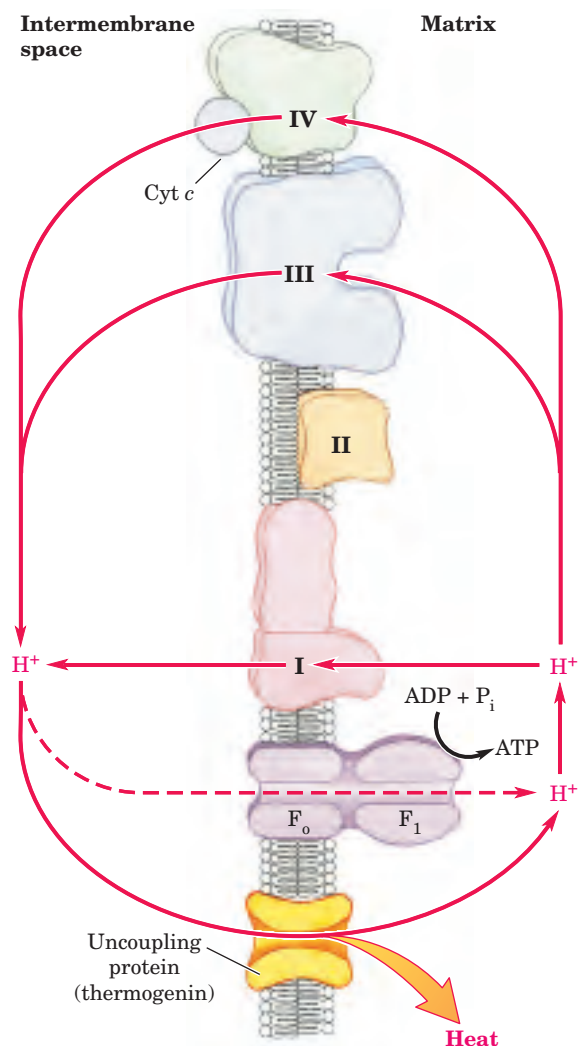


FIGURE 19–30 Heat generation by uncoupled mitochondria. The uncoupling protein (thermogenin) of brown fat mitochondria, by providing an alternative route for protons to reenter the mitochondrial matrix, causes the energy conserved by proton pumping to be dissipated as heat.

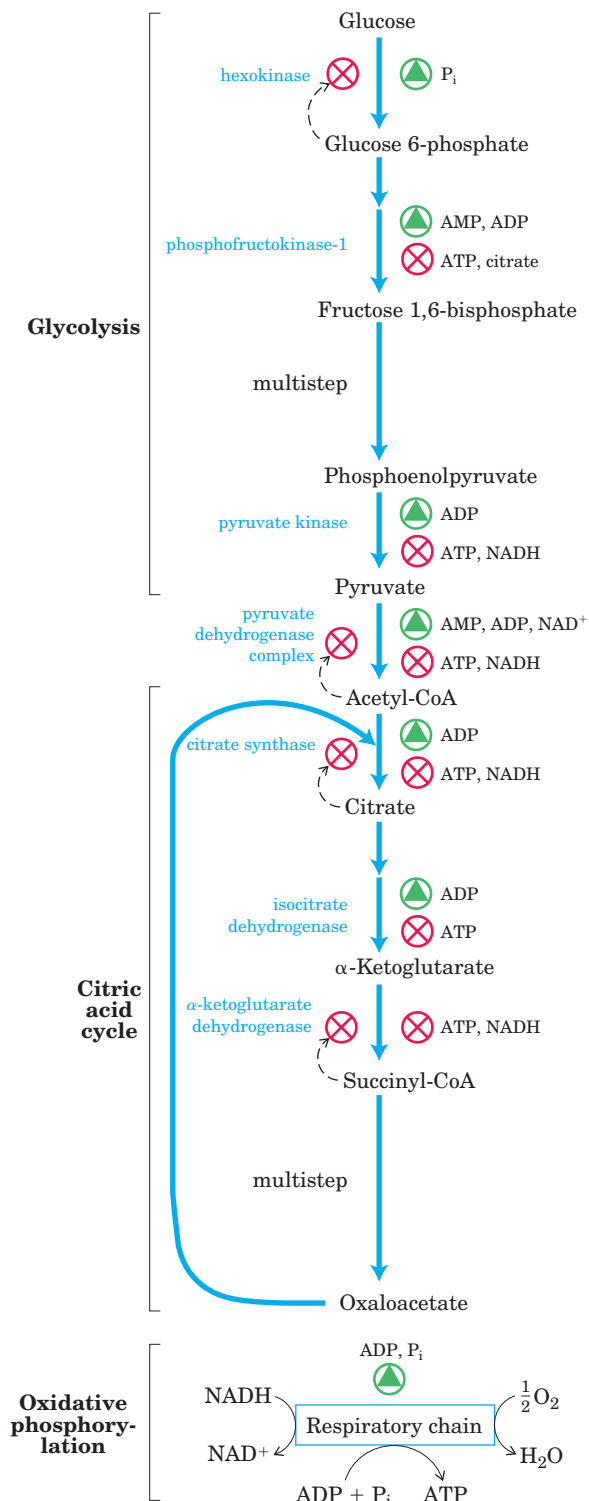


FIGURE 19-31 Regulation of the ATP-producing pathways. This diagram shows the interlocking regulation of glycolysis, pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation by the relative concentrations of ATP, ADP, and AMP, and by NADH. High [ATP] (or low [ADP] and [AMP]) produces low rates of glycolysis, pyruvate oxidation, acetate oxidation via the citric acid cycle, and oxidative phosphorylation. All four pathways are accelerated when the use of ATP and the formation of ADP, AMP, and P_i increase. The interlocking of glycolysis and the citric acid cycle by citrate, which inhibits glycolysis, supplements the action of the adenine nucleotide system. In addition, increased levels of NADH and acetyl-CoA also inhibit the oxidation of pyruvate to acetyl-CoA, and a high $[NADH]/[NAD^+]$ ratio inhibits the dehydrogenase reactions of the citric acid cycle (see Fig. 16-18).

ATP-Producing Pathways Are Coordinately Regulated

The major catabolic pathways have interlocking and concerted regulatory mechanisms that allow them to function together in an economical and self-regulating manner to produce ATP and biosynthetic precursors. The relative concentrations of ATP and ADP control not only the rates of electron transfer and oxidative phosphorylation but also the rates of the citric acid cycle, pyruvate oxidation, and glycolysis (Fig. 19-31). Whenever ATP consumption increases, the rate of electron transfer and oxidative phosphorylation increases. Simultaneously, the rate of pyruvate oxidation via the citric acid cycle increases, increasing the flow of electrons into the respiratory chain. These events can in turn evoke an increase in the rate of glycolysis, increasing the rate of pyruvate formation. When conversion of ADP to ATP lowers the ADP concentration, acceptor control slows electron transfer and thus oxidative phosphorylation. Glycolysis and the citric acid cycle are also slowed, because ATP is an allosteric inhibitor of the glycolytic enzyme phosphofructokinase-1 (see Fig. 15-18) and of pyruvate dehydrogenase (see Fig. 16-18).

Phosphofructokinase-1 is also inhibited by citrate, the first intermediate of the citric acid cycle. When the cycle is “idling,” citrate accumulates within mitochondria, then spills into the cytosol. When the concentrations of both ATP and citrate rise, they produce a concerted allosteric inhibition of phosphofructokinase-1 that is greater than the sum of their individual effects, slowing glycolysis.

SUMMARY 19.3 Regulation of Oxidative Phosphorylation

- Oxidative phosphorylation is regulated by cellular energy demands. The intracellular $[ADP]$ and the mass-action ratio $[ATP]/([ADP][P_i])$ are measures of a cell's energy status.

As a result of this short-circuiting of protons, the energy of oxidation is not conserved by ATP formation but is dissipated as heat, which contributes to maintaining the body temperature of the newborn. Hibernating animals also depend on uncoupled mitochondria of brown fat to generate heat during their long dormancy (see Box 17-1).

- In ischemic (oxygen-deprived) cells, a protein inhibitor blocks ATP hydrolysis by the ATP synthase operating in reverse, preventing a drastic drop in [ATP].
- In brown fat, which is specialized for the production of metabolic heat, electron transfer is uncoupled from ATP synthesis and the energy of fatty acid oxidation is dissipated as heat.
- ATP and ADP concentrations set the rate of electron transfer through the respiratory chain via a series of interlocking controls on respiration, glycolysis, and the citric acid cycle.

19.4 Mitochondrial Genes: Their Origin and the Effects of Mutations

Mitochondria contain their own genome, a circular, double-stranded DNA molecule. Each of the hundreds or thousands of mitochondria in a typical cell has about five copies of this genome. The human mitochondrial chromosome (Fig. 19–32) contains 37 genes (16,569 bp), including 13 that encode subunits of proteins of the respiratory chain (Table 19–6); the remaining genes code for rRNA and tRNA molecules essential to the protein-synthesizing machinery of mitochondria. About 900 different mitochondrial proteins are encoded by nuclear genes, synthesized on cytoplasmic ribosomes, then imported and assembled within the mitochondria (Chapter 27).

Mutations in Mitochondrial Genes Cause Human Disease

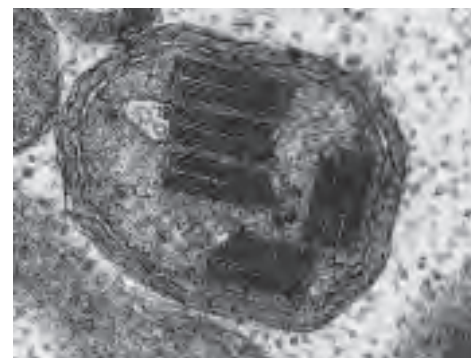
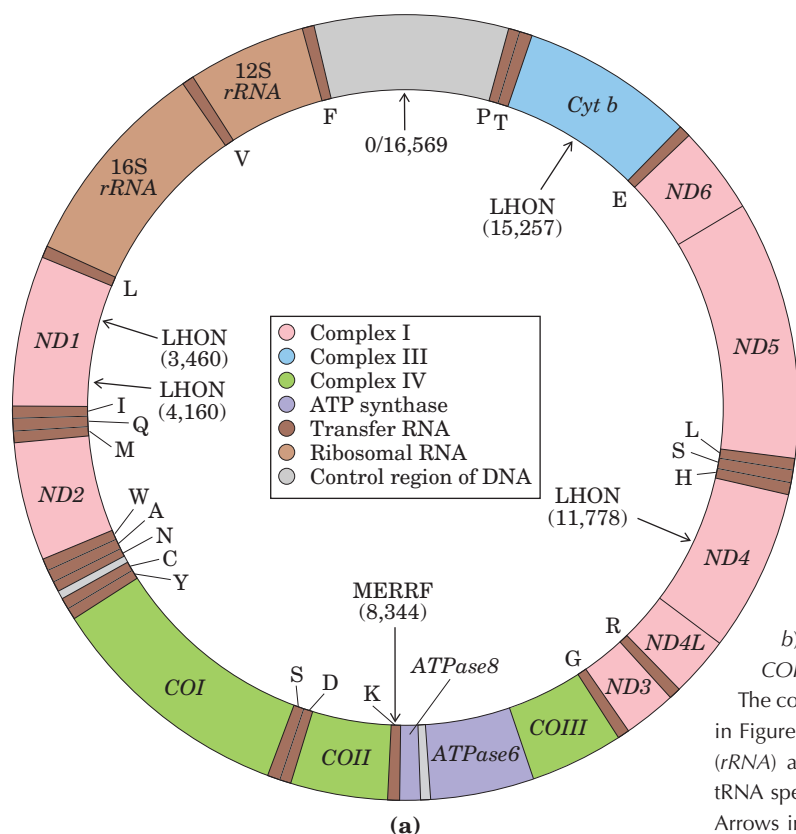
 A growing number of human diseases can be attributed to mutations in mitochondrial genes. Many of these diseases, those known as the **mitochondrial encephalomyopathies**, affect primarily the brain and skeletal muscle (both heavily dependent on an abundant supply of ATP). These diseases are invariably inherited from the mother, because a developing em-

bryo derives all its mitochondria from the mother's egg. The rare disease **Leber's hereditary optic neuropathy (LHON)** affects the central nervous system, including the optic nerves, causing bilateral loss of vision in early adulthood. A single base change in the mitochondrial gene *ND4* (Fig. 19–32a) changes an Arg residue to a His residue in a polypeptide of Complex I, and the result is mitochondria partially defective in electron transfer from NADH to ubiquinone. Although these mitochondria can produce some ATP by electron transfer from succinate, they apparently cannot supply sufficient ATP to support the very active metabolism of neurons. One result is damage to the optic nerve, leading to blindness. A single base change in the mitochondrial gene for cytochrome *b*, a component of Complex III, also produces LHON, demonstrating that the pathology results from a general reduction of mitochondrial function, not specifically from a defect in electron transfer through Complex I.

Myoclonic epilepsy and ragged-red fiber disease (MERRF) is caused by a mutation in the mitochondrial gene that encodes a transfer RNA specific for lysine (lysyl-tRNA). This disease, characterized by uncontrollable muscular jerking, apparently results from defective production of several of the proteins whose synthesis involves mitochondrial tRNAs. Skeletal muscle fibers of individuals with MERRF have abnormally shaped mitochondria that sometimes contain paracrystalline structures (Fig. 19–32b). Mutations in the mitochondrial lysyl-tRNA gene are also one of the causes of adult-onset (type II) diabetes mellitus. Other mutations in mitochondrial genes are believed to be responsible for the progressive muscular weakness that characterizes mitochondrial myopathy and for enlargement and deterioration of the heart muscle in hypertrophic cardiomyopathy. According to one hypothesis on the progressive changes that accompany aging, the accumulation of mutations in mitochondrial DNA during a lifetime of exposure to DNA-damaging agents such as O_2^- (see below) results in mitochondria that cannot supply sufficient ATP for normal cellular function. Mitochondrial disease can also result from mutations in any of the 900 nuclear genes that encode mitochondrial proteins. ■

TABLE 19–6 Respiratory Proteins Encoded by Mitochondrial Genes in Humans

Complex	Number of subunits	Number of subunits encoded by mitochondrial DNA
I NADH dehydrogenase	>43	7
II Succinate dehydrogenase	4	0
III Ubiquinone:cytochrome <i>c</i> oxidoreductase	11	1
IV Cytochrome oxidase	13	3
V ATP synthase	8	2

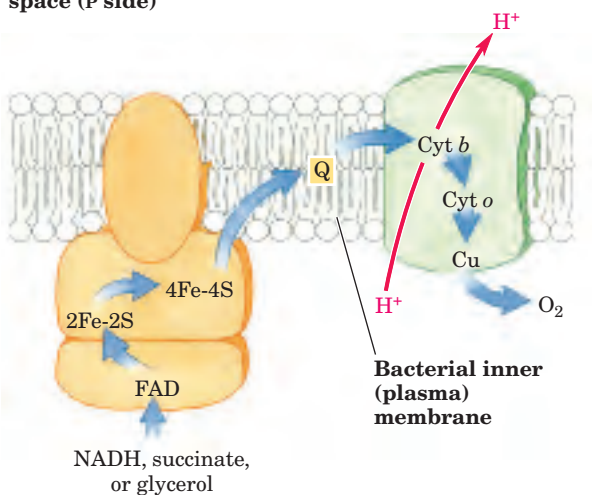


(b)

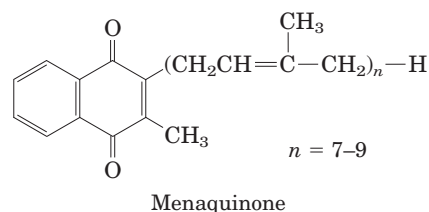
FIGURE 19-32 Mitochondrial genes and mutations.

(a) Map of human mitochondrial DNA, showing the genes that encode proteins of Complex I, the NADH dehydrogenase (ND1 to ND6); the cytochrome *b* of Complex III (Cyt *b*); the subunits of cytochrome oxidase (Complex IV) (COI to COIII); and two subunits of ATP synthase (ATPase6 and ATPase8). The colors of the genes correspond to those of the complexes shown in Figure 19-7. Also included here are the genes for ribosomal RNAs (*rRNA*) and for a number of mitochondrion-specific transfer RNAs; tRNA specificity is indicated by the one-letter codes for amino acids. Arrows indicate the positions of mutations that cause Leber's hereditary optic neuropathy (LHON) and myoclonic epilepsy and ragged-red fiber disease (MERRF). Numbers in parentheses indicate the position of the altered nucleotides (nucleotide 1 is at the top of the circle and numbering proceeds counterclockwise). (b) Electron micrograph of an abnormal mitochondrion from the muscle of an individual with MERRF, showing the paracrystalline protein inclusions sometimes present in the mutant mitochondria.

Periplasmic space (P side)



(a)



(b)

FIGURE 19-33 Bacterial respiratory chain. (a) Shown here are the respiratory carriers of the inner membrane of *E. coli*. Eubacteria contain a minimal form of Complex I, containing all the prosthetic groups normally associated with the mitochondrial complex but only 14 polypeptides. This plasma membrane complex transfers electrons from NADH to ubiquinone or to (b) menaquinone, the bacterial equivalent of ubiquinone, while pumping protons outward and creating an electrochemical potential that drives ATP synthesis.

Mitochondria Evolved from Endosymbiotic Bacteria

The existence of mitochondrial DNA, ribosomes, and tRNAs supports the hypothesis of the endosymbiotic origin of mitochondria (see Fig. 1–36), which holds that the first organisms capable of aerobic metabolism, including respiration-linked ATP production, were prokaryotes. Primitive eukaryotes that lived anaerobically (by fermentation) acquired the ability to carry out oxidative phosphorylation when they established a symbiotic relationship with bacteria living in their cytosol. After much evolution and the movement of many bacterial genes into the nucleus of the “host” eukaryote, the endosymbiotic bacteria eventually became mitochondria.

This hypothesis presumes that early free-living prokaryotes had the enzymatic machinery for oxidative phosphorylation and predicts that their modern prokaryotic descendants must have respiratory chains closely similar to those of modern eukaryotes. They do. Aerobic bacteria carry out NAD-linked electron transfer from substrates to O_2 , coupled to the phosphorylation of cytosolic ADP. The dehydrogenases are located in the bacterial cytosol and the respiratory chain in the plasma membrane. The electron carriers are similar to some mitochondrial electron carriers (Fig. 19–33). They translocate protons outward across the plasma membrane as electrons are transferred to O_2 . Bacteria such as *Escherichia coli* have F_oF_1 complexes in their plasma membranes; the F_1 portion protrudes into the cytosol and catalyzes ATP synthesis from ADP and P_i as protons flow back into the cell through the proton channel of F_o .

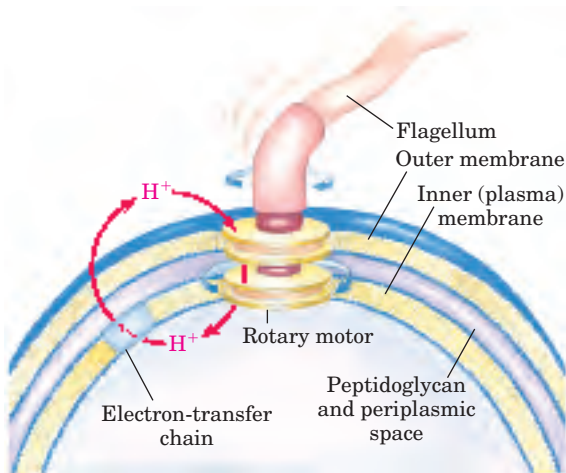


FIGURE 19–34 Rotation of bacterial flagella by proton-motive force.

The shaft and rings at the base of the flagellum make up a rotary motor that has been called a “proton turbine.” Protons ejected by electron transfer flow back into the cell through the turbine, causing rotation of the shaft of the flagellum. This motion differs fundamentally from the motion of muscle and of eukaryotic flagella and cilia, for which ATP hydrolysis is the energy source.

The respiration-linked extrusion of protons across the bacterial plasma membrane also provides the driving force for other processes. Certain bacterial transport systems bring about uptake of extracellular nutrients (lactose, for example) against a concentration gradient, in symport with protons (see Fig. 11–42). And the rotary motion of bacterial flagella is provided by “proton turbines,” molecular rotary motors driven not by ATP but directly by the transmembrane electrochemical potential generated by respiration-linked proton pumping (Fig. 19–34). It appears likely that the chemiosmotic mechanism evolved early, before the emergence of eukaryotes.

SUMMARY 19.4 Mitochondrial Genes: Their Origin and the Effects of Mutations

- A small proportion of human mitochondrial proteins (13 proteins) are encoded in the mitochondrial genome and synthesized within mitochondria. About 900 mitochondrial proteins are encoded in nuclear genes and imported into mitochondria after their synthesis.
- Mutations in the genes that encode components of the respiratory chain, whether in the mitochondrial genes or in the nuclear genes that encode mitochondrial proteins, cause a variety of human diseases, which often affect muscle and brain most severely.
- Mitochondria most likely arose from aerobic prokaryotes that entered into an endosymbiotic relationship with ancestral eukaryotes.

19.5 The Role of Mitochondria in Apoptosis and Oxidative Stress

Besides their central role in ATP synthesis, mitochondria also participate in processes associated with cellular damage and death. **Apoptosis** is a controlled process by which cells die for the good of the organism, while the organism conserves the molecular components (amino acids, nucleotides, and so forth) of the dead cells. Apoptosis may be triggered by an external signal, acting at a receptor in the plasma membrane, or by internal events such as a viral infection. When a cell receives a signal for apoptosis, one consequence is an increase in the permeability of the outer mitochondrial membrane, allowing escape of the cytochrome *c* normally confined in the intermembrane space (see Fig. 12–50). The released cytochrome *c* activates one of the proteolytic enzymes (caspase 9) responsible for protein degradation during apoptosis. This is a dramatic case of

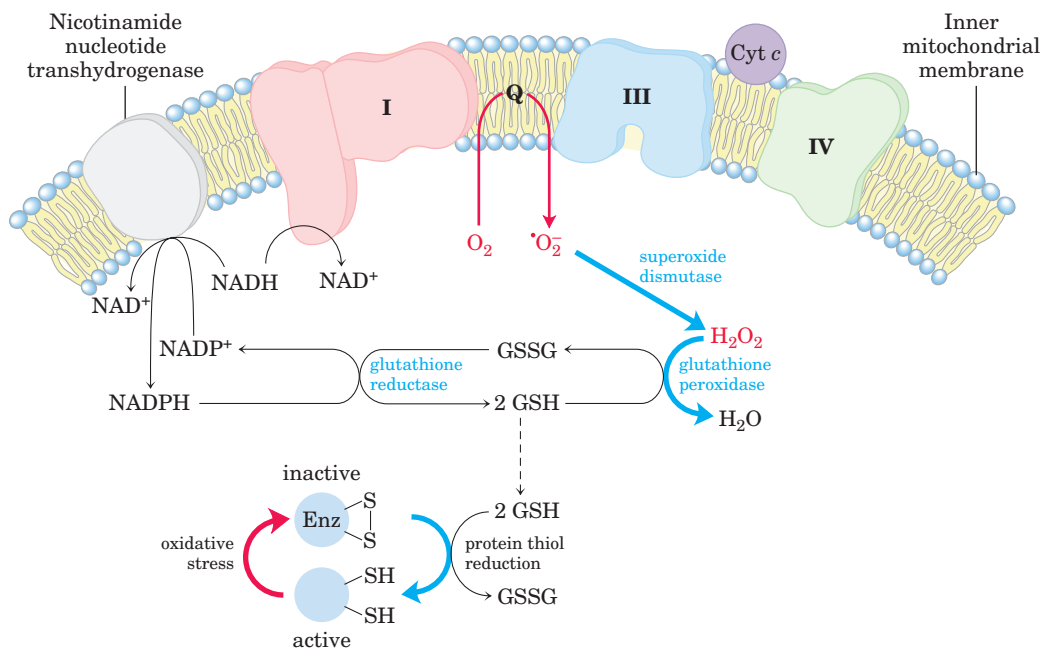
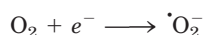


FIGURE 19-35 Mitochondrial production and disposal of superoxide. Superoxide radical, O_2^- , is formed in side reactions at Complexes I and III, as the partially reduced ubiquinol radical (Q^-) donates an electron to O_2 . The reactions shown in blue defend the cell against the damaging effects of superoxide. Reduced glutathione

(GSH; see Fig. 22-27) donates electrons for the reduction of hydrogen peroxide (H_2O_2) and of oxidized Cys residues ($-\text{S}-\text{S}-$) in proteins, and GSH is regenerated from the oxidized form (GSSG) by reduction with NADPH.

one protein (cytochrome *c*) playing two very different roles in the cell.

Mitochondria are also involved in the cell's response to oxidative stress. As we have seen, several steps in the path of oxygen reduction in mitochondria have the potential to produce highly reactive free radicals that can damage cells. The passage of electrons from QH_2 to cytochrome b_L through Complex III, and passage of electrons from Complex I to QH_2 , involve the radical Q^- as an intermediate. The Q^- can, with a low probability, pass an electron to O_2 in the reaction



The superoxide free radical thus generated, O_2^- , is very reactive and can damage enzymes, membrane lipids, and nucleic acids. Antimycin A, an inhibitor of Complex III, may act by occupying the Q_N site (Fig. 19-11), thus blocking the Q cycle and prolonging the binding of Q^- to the Q_p site; this would increase the likelihood of superoxide radical formation and cellular damage. From 0.1% to as much as 4% of the O_2 used by actively respiring mitochondria forms O_2^- —more than enough to have lethal effects on a cell unless the free radical is quickly disposed of.

To prevent oxidative damage by O_2^- , cells have several forms of the enzyme **superoxide dismutase**, which catalyzes the reaction



The hydrogen peroxide (H_2O_2) generated by this reaction is rendered harmless by the action of **glutathione peroxidase** (Fig. 19-35). This enzyme is remarkable for the presence of a selenocysteine residue (see Fig. 3-8a), in which an atom of selenium replaces the sulfur atom normally present in the thiol of the side chain. The selenol group ($-\text{SeH}$) is more acidic than the thiol ($-\text{SH}$); its pKa is about 5, so at neutral pH, the selenocysteine side chain is essentially fully ionized ($-\text{CH}_2\text{Se}^-$). Glutathione reductase recycles oxidized glutathione to its reduced form, using electrons from the NADPH formed by nicotinamide nucleotide transhydrogenase or by the pentose phosphate pathway (see Fig. 14-20). Reduced glutathione also serves in keeping protein sulfhydryl groups in their reduced state, preventing some of the deleterious effects of oxidative stress (Fig. 19-35).

SUMMARY 19.5 The Role of Mitochondria in Apoptosis and Oxidative Stress

- Mitochondrial cytochrome *c*, released into the cytosol, participates in activation of one of the proteases (caspase 9) involved in apoptosis.
- Reactive oxygen species produced in mitochondria are inactivated by a set of protective enzymes, including superoxide dismutase and glutathione peroxidase.

PHOTOSYNTHESIS: HARVESTING LIGHT ENERGY

We now turn to another reaction sequence in which the flow of electrons is coupled to the synthesis of ATP: light-driven phosphorylation. The capture of solar energy by photosynthetic organisms and its conversion to the chemical energy of reduced organic compounds is the ultimate source of nearly all biological energy. Photosynthetic and heterotrophic organisms live in a balanced steady state in the biosphere (Fig. 19–36). Photosynthetic organisms trap solar energy and form ATP and NADPH, which they use as energy sources to make carbohydrates and other organic compounds from CO_2 and H_2O ; simultaneously, they release O_2 into the atmosphere. Aerobic heterotrophs (humans, for example, as well as plants during dark periods) use the O_2 so formed to degrade the energy-rich organic products of photosynthesis to CO_2 and H_2O , generating ATP. The CO_2 returns to the atmosphere, to be used again by photosynthetic organisms. Solar energy thus provides the driving force for the continuous cycling of CO_2 and O_2 through the biosphere and provides the reduced substrates—fuels, such as glucose—on which nonphotosynthetic organisms depend.

Photosynthesis occurs in a variety of bacteria and in unicellular eukaryotes (algae) as well as in vascular plants. Although the process in these organisms differs in detail, the underlying mechanisms are remarkably similar, and much of our understanding of photosynthesis in vascular plants is derived from studies of simpler organisms. The overall equation for photosynthesis in vascular plants describes an oxidation-reduction reaction in which H_2O donates electrons (as hydrogen) for the reduction of CO_2 to carbohydrate (CH_2O):

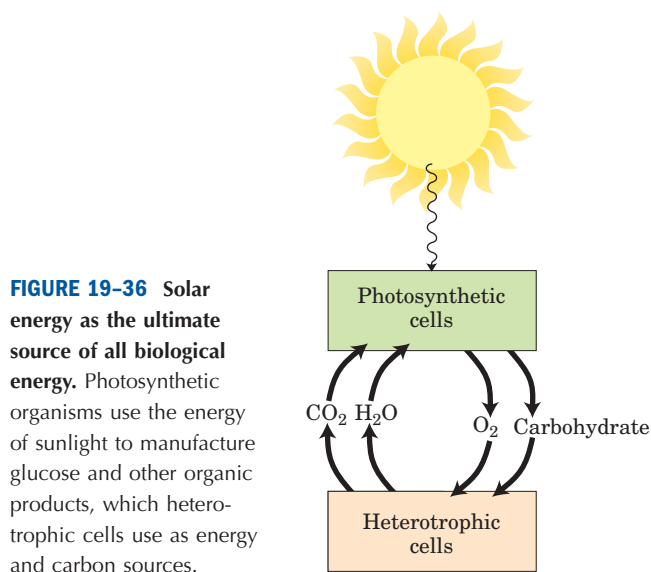
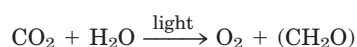


FIGURE 19–36 Solar energy as the ultimate source of all biological energy. Photosynthetic organisms use the energy of sunlight to manufacture glucose and other organic products, which heterotrophic cells use as energy and carbon sources.

19.6 General Features of Photophosphorylation

Unlike NADH (the major electron donor in oxidative phosphorylation), H_2O is a poor donor of electrons; its standard reduction potential is 0.816 V, compared with -0.320 V for NADH. Photophosphorylation differs from oxidative phosphorylation in requiring the input of energy in the form of light to *create* a good electron donor and a good electron acceptor. In photophosphorylation, electrons flow through a series of membrane-bound carriers including cytochromes, quinones, and iron-sulfur proteins, while protons are pumped across a membrane to create an electrochemical potential. Electron transfer and proton pumping are catalyzed by membrane complexes homologous in structure and function to Complex III of mitochondria. The electrochemical potential they produce is the driving force for ATP synthesis from ADP and P_i , catalyzed by a membrane-bound ATP synthase complex closely similar to that of oxidative phosphorylation.

Photosynthesis in plants encompasses two processes: the **light-dependent reactions**, or **light reactions**, which occur only when plants are illuminated, and the **carbon-assimilation reactions** (or **carbon-fixation reactions**), sometimes misleadingly called the dark reactions, which are driven by products of the light reactions (Fig. 19–37). In the light reactions, chlorophyll and other pigments of photosynthetic cells absorb light energy and conserve it as ATP and NADPH; simultaneously, O_2 is evolved. In the carbon-assimilation reactions, ATP and NADPH are used to reduce CO_2 to form triose phosphates, starch, and sucrose, and other products derived from them. In this chapter we are concerned only with the light-dependent reactions that lead to the synthesis of ATP and NADPH. The reduction of CO_2 is described in Chapter 20.

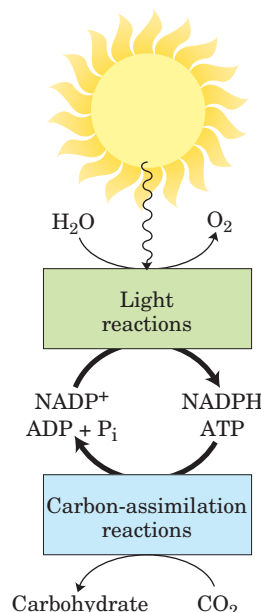


FIGURE 19–37 The light reactions of photosynthesis generate energy-rich NADPH and ATP at the expense of solar energy. These products are used in the carbon-assimilation reactions, which occur in light or darkness, to reduce CO_2 to form trioses and more complex compounds (such as glucose) derived from trioses.

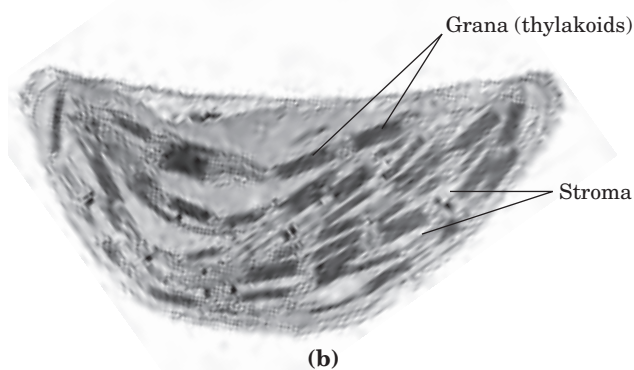
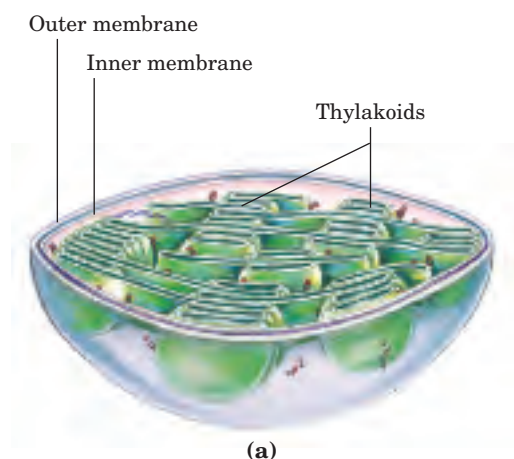


FIGURE 19-38 Chloroplast. (a) Schematic diagram. (b) Electron micrograph at high magnification showing grana, stacks of thylakoid membranes.

Photosynthesis in Plants Takes Place in Chloroplasts

In photosynthetic eukaryotic cells, both the light-dependent and the carbon-assimilation reactions take place in the chloroplasts (Fig. 19-38), membrane-bounded intracellular organelles that are variable in shape and generally a few micrometers in diameter. Like mitochondria, they are surrounded by two membranes, an outer membrane that is permeable to small molecules and ions, and an inner membrane that encloses the internal compartment. This compartment contains many flattened, membrane-surrounded vesicles or sacs, the **thylakoids**, usually arranged in stacks called **grana** (Fig. 19-38b). Embedded in the thylakoid membranes (commonly called **lamellae**) are the photosynthetic pigments and the enzyme complexes that carry out the light reactions and ATP synthesis. The **stroma** (the aqueous phase enclosed by the inner membrane) contains most of the enzymes required for the carbon-assimilation reactions.

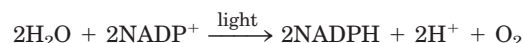
Light Drives Electron Flow in Chloroplasts

In 1937 Robert Hill found that when leaf extracts containing chloroplasts were illuminated, they (1) evolved O_2 and (2) reduced a nonbiological electron acceptor added to the medium, according to the **Hill reaction**:

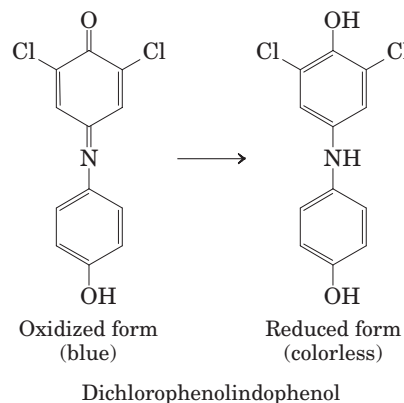


where A is the artificial electron acceptor, or **Hill reagent**. One Hill reagent, the dye 2,6-dichlorophenol-indophenol, is blue when oxidized (A) and colorless when reduced (AH_2), making the reaction easy to follow. When a leaf extract supplemented with the dye was illuminated, the blue dye became colorless and O_2 was evolved. In the dark, neither O_2 evolution nor dye reduction took place. This was the first evidence that absorbed light energy causes electrons to flow from H_2O to an electron acceptor. Moreover, Hill found that CO_2

was neither required nor reduced to a stable form under these conditions; O_2 evolution could be dissociated from CO_2 reduction. Several years later Severo Ochoa showed that $NADP^+$ is the biological electron acceptor in chloroplasts, according to the equation



To understand this photochemical process, we must first consider the more general topic of the effects of light absorption on molecular structure.



SUMMARY 19.6 General Features of Photophosphorylation

- The light reactions of photosynthesis are those directly dependent on the absorption of light; the resulting photochemistry takes electrons from H_2O and drives them through a series of membrane-bound carriers, producing NADPH and ATP.
- The carbon-assimilation reactions of photosynthesis reduce CO_2 with electrons from NADPH and energy from ATP.

19.7 Light Absorption

Visible light is electromagnetic radiation of wavelengths 400 to 700 nm, a small part of the electromagnetic spectrum (Fig. 19–39), ranging from violet to red. The energy of a single **photon** (a quantum of light) is greater at the violet end of the spectrum than at the red end; shorter wavelength (and higher frequency) corresponds to higher energy. The energy, E , in a “mole” of photons (1 einstein, or 6×10^{23} photons) of visible light is 170 to 300 kJ, as given by the Planck equation:

$$E = h\nu$$

where h is Planck’s constant (6.626×10^{-34} J·s) and ν is the wavelength. These amounts of energy are almost an order of magnitude greater than the 30 to 50 kJ required to synthesize a mole of ATP from ADP and P_i .

When a photon is absorbed, an electron in the absorbing molecule (chromophore) is lifted to a higher energy level. This is an all-or-nothing event; to be absorbed, the photon must contain a quantity of energy (a **quantum**) that exactly matches the energy of the electronic transition. A molecule that has absorbed a photon is in an **excited state**, which is generally unstable. An electron lifted into a higher-energy orbital usually returns rapidly to its normal lower-energy orbital; the excited molecule decays to the stable **ground state**, giving up the absorbed quantum as light or heat or using it to do chemical work. Light emission accompanying decay of excited molecules (called **fluorescence**) is always at a longer wavelength (lower energy) than that of the absorbed light (see Box 12–2). An alternative mode of decay important in photosynthesis involves direct transfer of excitation energy from an excited molecule to a neighboring molecule. Just as the photon is a quantum of light energy, so the **exciton** is a quantum of energy passed from an excited molecule to another molecule in a process called **exciton transfer**.

Chlorophylls Absorb Light Energy for Photosynthesis

The most important light-absorbing pigments in the thylakoid membranes are the **chlorophylls**, green pigments with polycyclic, planar structures resembling the protoporphyrin of hemoglobin (see Fig. 5–1), except that Mg^{2+} , not Fe^{2+} , occupies the central position (Fig. 19–40). The four inward-oriented nitrogen atoms of chlorophyll are coordinated with the Mg^{2+} . All chlorophylls have a long **phytol** side chain, esterified to a carboxyl-group substituent in ring IV, and chlorophylls also have a fifth five-membered ring not present in heme.

The heterocyclic five-ring system that surrounds the Mg^{2+} has an extended polyene structure, with alternating single and double bonds. Such polyenes characteristically show strong absorption in the visible region of the spectrum (Fig. 19–41); the chlorophylls have unusually high molar extinction coefficients (see Box 3–1) and are therefore particularly well-suited for absorbing visible light during photosynthesis.

Chloroplasts always contain both chlorophyll *a* and chlorophyll *b* (Fig. 19–40a). Although both are green, their absorption spectra are sufficiently different (Fig. 19–41) that they complement each other’s range of light absorption in the visible region. Most plants contain about twice as much chlorophyll *a* as chlorophyll *b*. The pigments in algae and photosynthetic bacteria include chlorophylls that differ only slightly from the plant pigments.

Chlorophyll is always associated with specific binding proteins, forming **light-harvesting complexes (LHCs)** in which chlorophyll molecules are fixed in relation to each other, to other protein complexes, and to the membrane. The detailed structure of one light-harvesting complex is known from x-ray crystallography (Fig. 19–42). It contains seven molecules of chlorophyll *a*, five of chlorophyll *b*, and two of the accessory pigment lutein (see below).

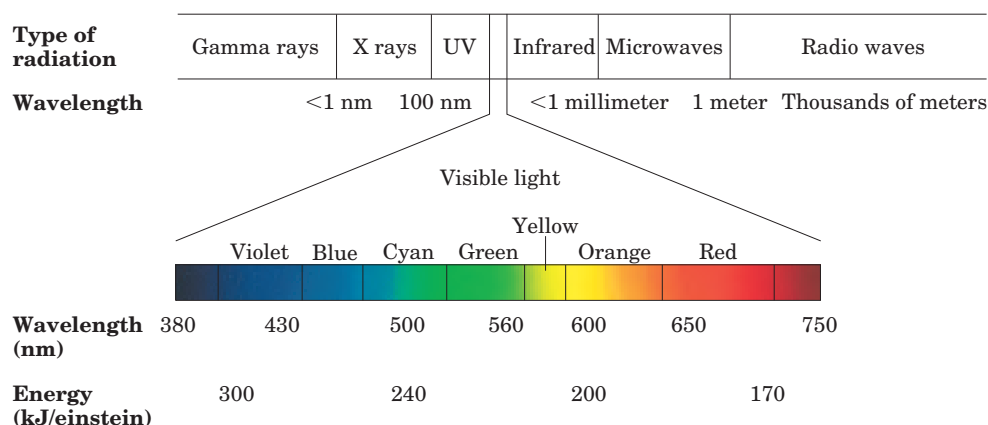


FIGURE 19–39 Electromagnetic radiation. The spectrum of electromagnetic radiation, and the energy of photons in the visible range of the spectrum. One einstein is 6×10^{23} photons.

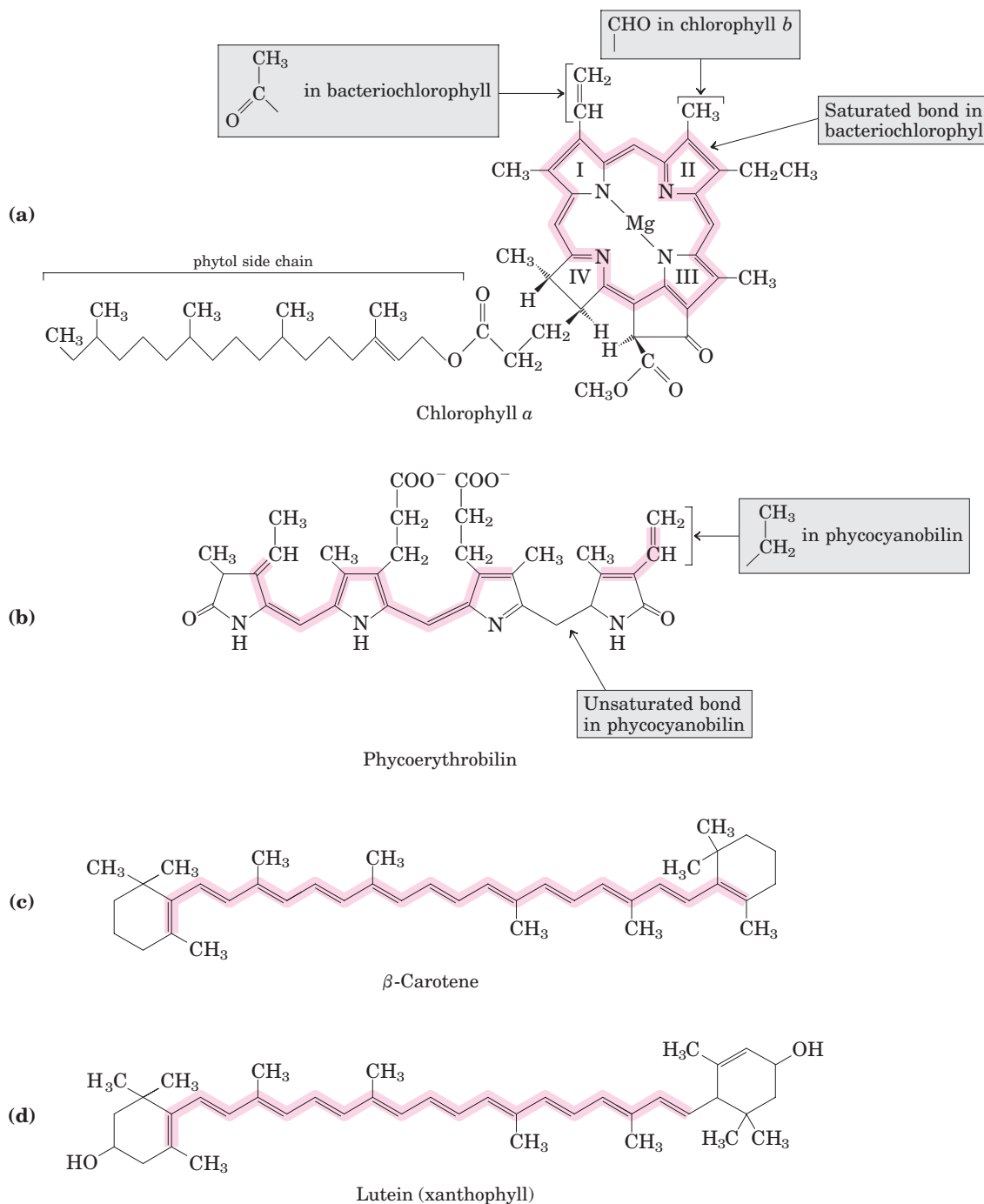


FIGURE 19-40 Primary and secondary photopigments. (a) Chlorophylls *a* and *b* and bacteriochlorophyll are the primary gatherers of light energy. (b) Phycoerythrobilin and phycocyanobilin (phycobilins) are the antenna pigments in cyanobacteria and red algae. (c) β-

Carotene (a carotenoid) and (d) lutein (a xanthophyll) are accessory pigments in plants. The areas shaded pink are the conjugated systems (alternating single and double bonds) that largely account for the absorption of visible light.

Cyanobacteria and red algae employ **phycobilins** such as phycoerythrobilin and phycocyanobilin (Fig. 19-40b) as their light-harvesting pigments. These open-chain tetrapyrroles have the extended polyene system found in chlorophylls, but not their cyclic structure or

central Mg²⁺. Phycobilins are covalently linked to specific binding proteins, forming **phycobiliproteins**, which associate in highly ordered complexes called phycobilisomes (Fig. 19-43) that constitute the primary light-harvesting structures in these microorganisms.

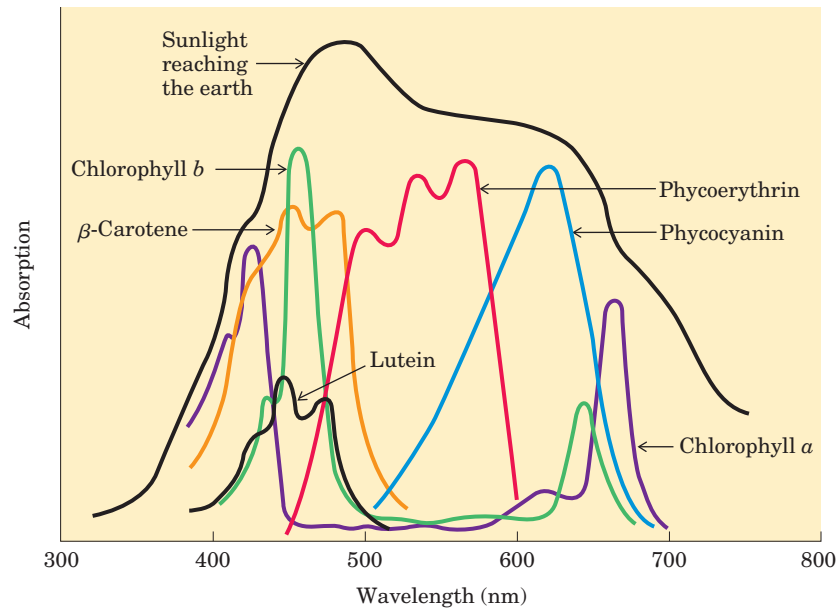


FIGURE 19-41 Absorption of visible light by photopigments. Plants are green because their pigments absorb light from the red and blue regions of the spectrum, leaving primarily green light to be reflected or transmitted. Compare the absorption spectra of the pigments with the spectrum of sunlight reaching the earth's surface; the combination of chlorophylls (*a* and *b*) and accessory pigments enables plants to harvest most of the energy available in sunlight.

The relative amounts of chlorophylls and accessory pigments are characteristic of a particular plant species. Variation in the proportions of these pigments is responsible for the range of colors of photosynthetic organisms, from the deep blue-green of spruce needles, to the greener green of maple leaves, to the red, brown, or purple color of some species of multicellular algae and the leaves of some foliage plants favored by gardeners.

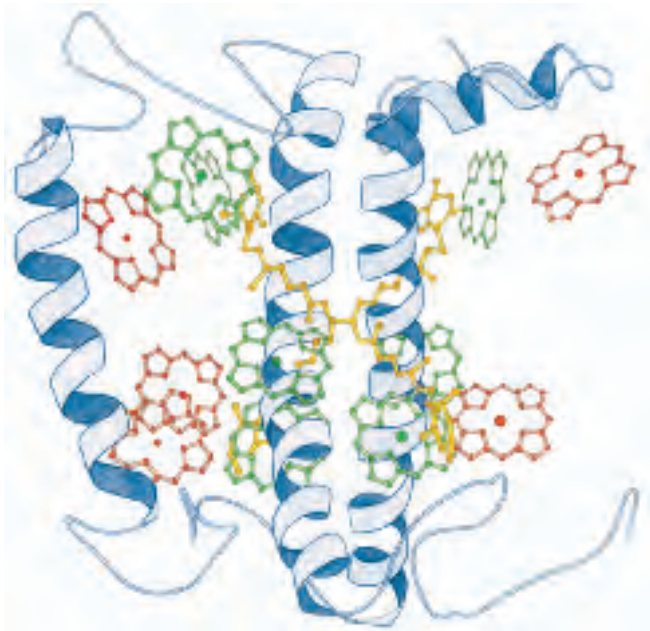


FIGURE 19-42 A light-harvesting complex, LHCII. The functional unit is an LHC trimer, with 36 chlorophyll and 6 lutein molecules. Shown here is a monomer, viewed in the plane of the membrane, with its three transmembrane α -helical segments, seven chlorophyll *a* molecules (green), five chlorophyll *b* molecules (red), and two molecules of the accessory pigment lutein (yellow), which form an internal cross-brace.

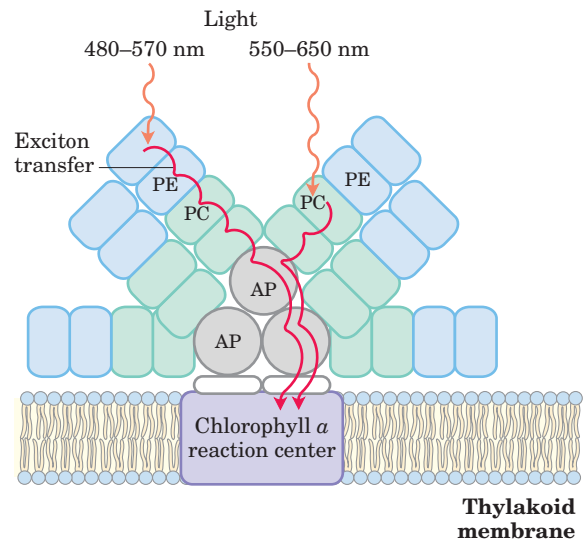
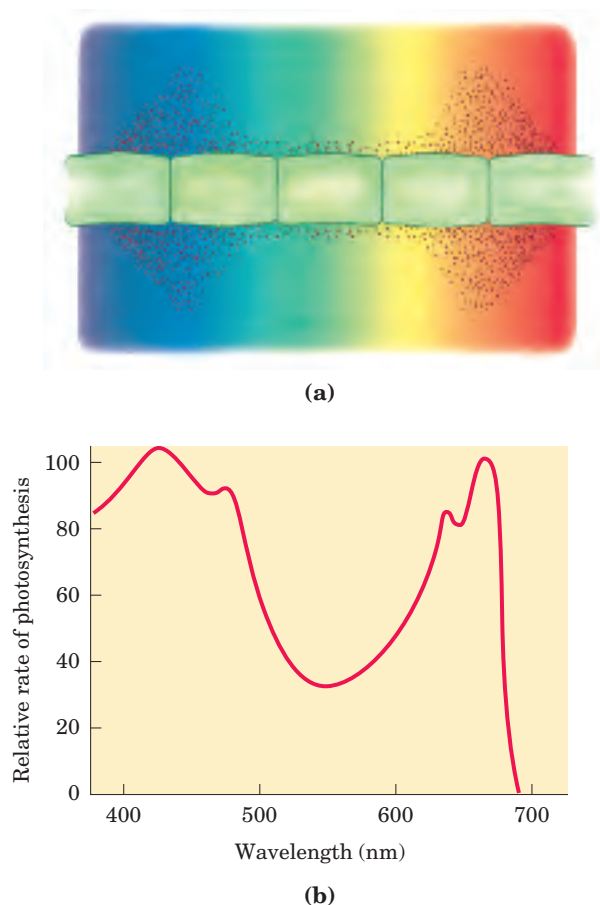


FIGURE 19-43 A phycobilisome. In these highly structured assemblies found in cyanobacteria and red algae, phycobilin pigments bound to specific proteins form complexes called phycoerythrin (PE), phycocyanin (PC), and allophycocyanin (AP). The energy of photons absorbed by PE or PC is conveyed through AP (a phycocyanobilin-binding protein) to chlorophyll *a* of the reaction center by exciton transfer, a process discussed in the text.

Accessory Pigments Extend the Range of Light Absorption

In addition to chlorophylls, thylakoid membranes contain secondary light-absorbing pigments, or **accessory pigments**, called carotenoids. **Carotenoids** may be yellow, red, or purple. The most important are **β -carotene**, which is a red-orange isoprenoid, and the yellow carotenoid **lutein** (Fig. 19–40c, d). The carotenoid pigments absorb light at wavelengths not absorbed by the chlorophylls (Fig. 19–41) and thus are supplementary light receptors.

Experimental determination of the effectiveness of light of different colors in promoting photosynthesis yields an **action spectrum** (Fig. 19–44), often useful in identifying the pigment primarily responsible for a biological effect of light. By capturing light in a region of the spectrum not used by other organisms, a photosynthetic organism can claim a unique ecological niche. For example, the phycobilins in red algae and cyanobacteria absorb light in the range 520 to 630 nm (Fig. 19–41), allowing them to occupy niches where light of lower or higher wavelength has been filtered out by the pigments of other organisms living in the water above them, or by the water itself.



Chlorophyll Funnels the Absorbed Energy to Reaction Centers by Exciton Transfer

The light-absorbing pigments of thylakoid or bacterial membranes are arranged in functional arrays called **photosystems**. In spinach chloroplasts, for example, each photosystem contains about 200 chlorophyll and 50 carotenoid molecules. All the pigment molecules in a photosystem can absorb photons, but only a few chlorophyll molecules associated with the **photochemical reaction center** are specialized to transduce light into chemical energy. The other pigment molecules in a photosystem are called **light-harvesting** or **antenna molecules**. They absorb light energy and transmit it rapidly and efficiently to the reaction center (Fig. 19–45).

The chlorophyll molecules in light-harvesting complexes have light-absorption properties that are subtly different from those of free chlorophyll. When isolated chlorophyll molecules *in vitro* are excited by light, the absorbed energy is quickly released as fluorescence and heat, but when chlorophyll in intact leaves is excited by visible light (Fig. 19–46, step ①), very little fluorescence is observed. Instead, the excited antenna chlorophyll transfers energy directly to a neighboring chlorophyll molecule, which becomes excited as the first molecule returns to its ground state (step ②). This transfer of energy, exciton transfer, extends to a third, fourth, or subsequent neighbor, until one of a special pair of chlorophyll *a* molecules at the photochemical reaction center is excited (step ③). In this excited chlorophyll molecule, an electron is promoted to a higher-energy orbital. This electron then passes to a nearby electron acceptor that is part of the electron-transfer chain, leaving the reaction-center chlorophyll with a

FIGURE 19–44 Two ways to determine the action spectrum for photosynthesis. (a) Results of a classic experiment performed by T. W. Engelmann in 1882 to determine the wavelength of light that is most effective in supporting photosynthesis. Engelmann placed cells of a filamentous photosynthetic alga on a microscope slide and illuminated them with light from a prism, so that one part of the filament received mainly blue light, another part yellow, another red. To determine which algal cells carried out photosynthesis most actively, Engelmann also placed on the microscope slide bacteria known to migrate toward regions of high O_2 concentration. After a period of illumination, the distribution of bacteria showed highest O_2 levels (produced by photosynthesis) in the regions illuminated with violet and red light.

(b) Results of a similar experiment that used modern techniques (an oxygen electrode) for the measurement of O_2 production. An action spectrum (as shown here) describes the relative rate of photosynthesis for illumination with a constant number of photons of different wavelengths. An action spectrum is useful because, by comparison with absorption spectra (such as those in Fig. 19–41), it suggests which pigments can channel energy into photosynthesis.

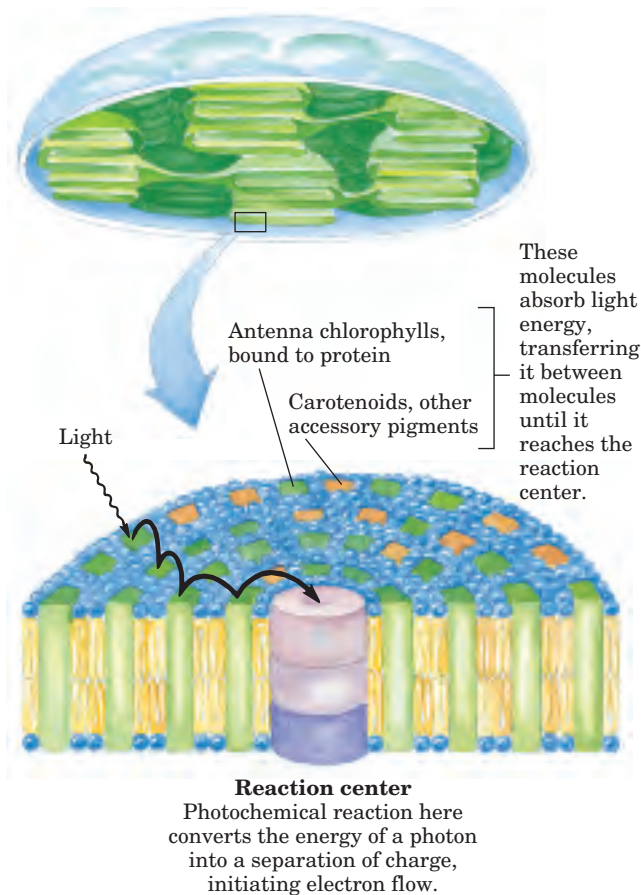
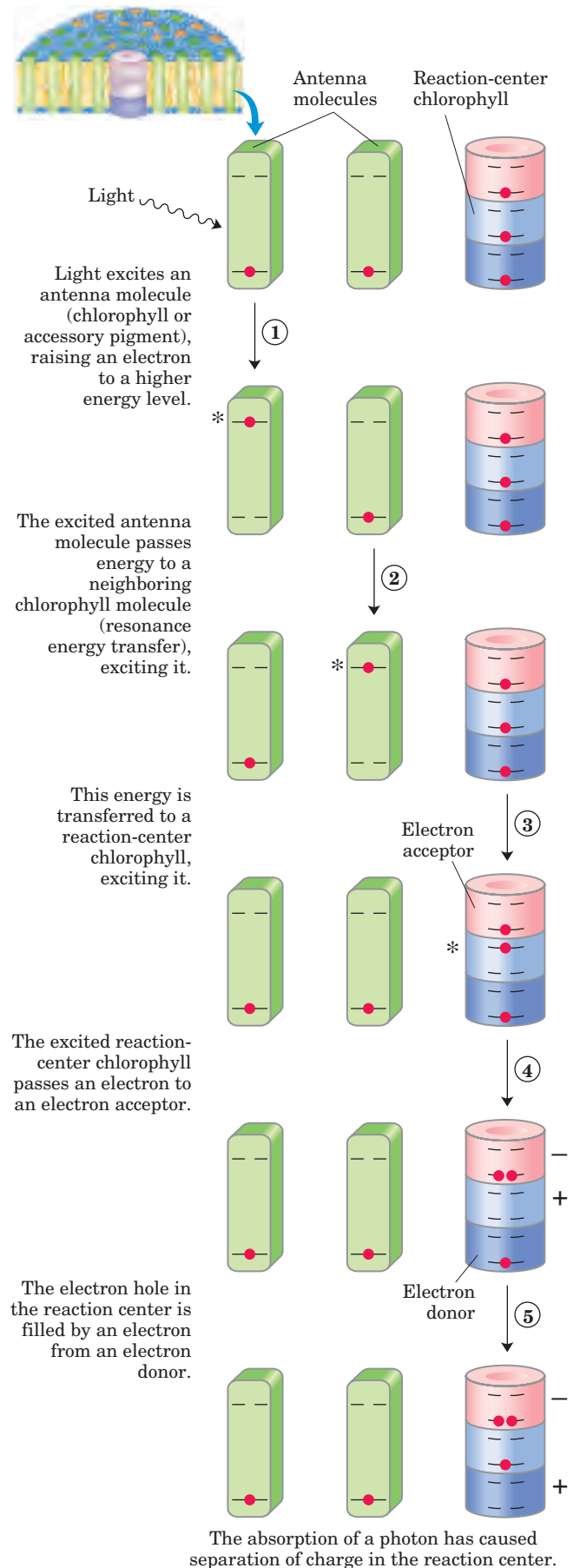


FIGURE 19-45 Organization of photosystems in the thylakoid membrane. Photosystems are tightly packed in the thylakoid membrane, with several hundred antenna chlorophylls and accessory pigments surrounding a photoreaction center. Absorption of a photon by any of the antenna chlorophylls leads to excitation of the reaction center by exciton transfer (black arrow). Also embedded in the thylakoid membrane are the cytochrome b_6f complex and ATP synthase (see Fig. 19-52).

missing electron (an “electron hole,” denoted by + in Fig. 19-46) (step ④). The electron acceptor acquires a negative charge in this transaction. The electron lost by the reaction-center chlorophyll is replaced by an electron from a neighboring electron-donor molecule (step ⑤), which thereby becomes positively charged. In this way, *excitation by light causes electric charge separation and initiates an oxidation-reduction chain*.

FIGURE 19-46 Exciton and electron transfer. This generalized scheme shows conversion of the energy of an absorbed photon into separation of charges at the reaction center. The steps are further described in the text. Note that step ① may repeat between successive antenna molecules until the exciton reaches a reaction-center chlorophyll. The asterisk (*) represents the excited state of an antenna molecule.



SUMMARY 19.7 Light Absorption

- Photophosphorylation in the chloroplasts of green plants and in cyanobacteria involves electron flow through a series of membrane-bound carriers.
- In the light reactions of plants, absorption of a photon excites chlorophyll molecules and other (accessory) pigments, which funnel the energy into reaction centers in the thylakoid membranes. In the reaction centers, photo-excitation results in a charge separation that produces a strong electron donor (reducing agent) and a strong electron acceptor.

19.8 The Central Photochemical Event: Light-Driven Electron Flow

Light-driven electron transfer in plant chloroplasts during photosynthesis is accomplished by multienzyme systems in the thylakoid membrane. Our current picture of photosynthetic mechanisms is a composite, drawn from studies of plant chloroplasts and a variety of bacteria and algae. Determination of the molecular structures of bacterial photosynthetic complexes (by x-ray crystallography) has given us a much improved understanding of the molecular events in photosynthesis in general.

Bacteria Have One of Two Types of Single Photochemical Reaction Center

One major insight from studies of photosynthetic bacteria came in 1952 when Louis Duysens found that illumination of the photosynthetic membranes of the purple bacterium *Rhodospirillum rubrum* with a pulse of light of a specific wavelength (870 nm) caused a temporary decrease in the absorption of light at that wavelength; a pigment was “bleached” by 870 nm light. Later studies by Bessel Kok and Horst Witt showed similar bleaching of plant chloroplast pigments by light of 680 and 700 nm. Furthermore, addition of the (nonbiological) electron acceptor $[\text{Fe}(\text{CN})_6]^{3-}$ (ferricyanide) caused bleaching at these wavelengths *without illumination*. These findings indicated that bleaching of the pigments was due to the loss of an electron from a photochemical reaction center. The pigments were named for the wavelength of maximum bleaching: P870, P680, and P700.

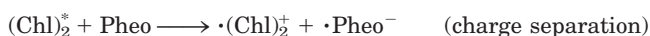
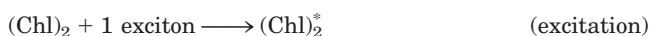
Photosynthetic bacteria have relatively simple phototransduction machinery, with one of two general types of reaction center. One type (found in purple bacteria) passes electrons through **pheophytin** (chlorophyll lacking the central Mg^{2+} ion) to a quinone. The other (in green sulfur bacteria) passes electrons through a quinone to an iron-sulfur center. Cyanobacteria and plants have two photosystems (PSI, PSII), one of each type, acting in tandem. Biochemical and biophysical

studies have revealed many of the molecular details of reaction centers of bacteria, which therefore serve as prototypes for the more complex phototransduction systems of plants.

The Pheophytin-Quinone Reaction Center (Type II Reaction Center) The photosynthetic machinery in purple bacteria consists of three basic modules (Fig. 19–47a): a single reaction center (P870), a cytochrome bc_1 electron-transfer complex similar to Complex III of the mitochondrial electron-transfer chain, and an ATP synthase, also similar to that of mitochondria. Illumination drives electrons through pheophytin and a quinone to the cytochrome bc_1 complex; after passing through the complex, electrons flow through cytochrome c_2 back to the reaction center, restoring its preillumination state. This light-driven cyclic flow of electrons provides the energy for proton pumping by the cytochrome bc_1 complex. Powered by the resulting proton gradient, ATP synthase produces ATP, exactly as in mitochondria.

The three-dimensional structures of the reaction centers of purple bacteria (*Rhodopseudomonas viridis* and *Rhodobacter sphaeroides*), deduced from x-ray crystallography, shed light on how phototransduction takes place in a pheophytin-quinone reaction center. The *R. viridis* reaction center (Fig. 19–48a) is a large protein complex containing four polypeptide subunits and 13 cofactors: two pairs of bacterial chlorophylls, a pair of pheophytins, two quinones, a nonheme iron, and four hemes in the associated *c*-type cytochrome.

The extremely rapid sequence of electron transfers shown in Figure 19–48b has been deduced from physical studies of the bacterial pheophytin-quinone centers, using brief flashes of light to trigger phototransduction and a variety of spectroscopic techniques to follow the flow of electrons through several carriers. A pair of bacteriochlorophylls—the “special pair,” designated $(\text{Chl})_2$ —is the site of the initial photochemistry in the bacterial reaction center. Energy from a photon absorbed by one of the many antenna chlorophyll molecules surrounding the reaction center reaches $(\text{Chl})_2$ by exciton transfer. When these two chlorophyll molecules—so close that their bonding orbitals overlap—absorb an exciton, the redox potential of $(\text{Chl})_2$ is shifted, by an amount equivalent to the energy of the photon, converting the special pair to a very strong electron donor. The $(\text{Chl})_2$ donates an electron that passes through a neighboring chlorophyll monomer to pheophytin (Pheo). This produces two radicals, one positively charged (the special pair of chlorophylls) and one negatively charged (the pheophytin):



The pheophytin radical now passes its electron to a tightly bound molecule of quinone (Q_A), converting it to a semiquinone radical, which immediately donates its

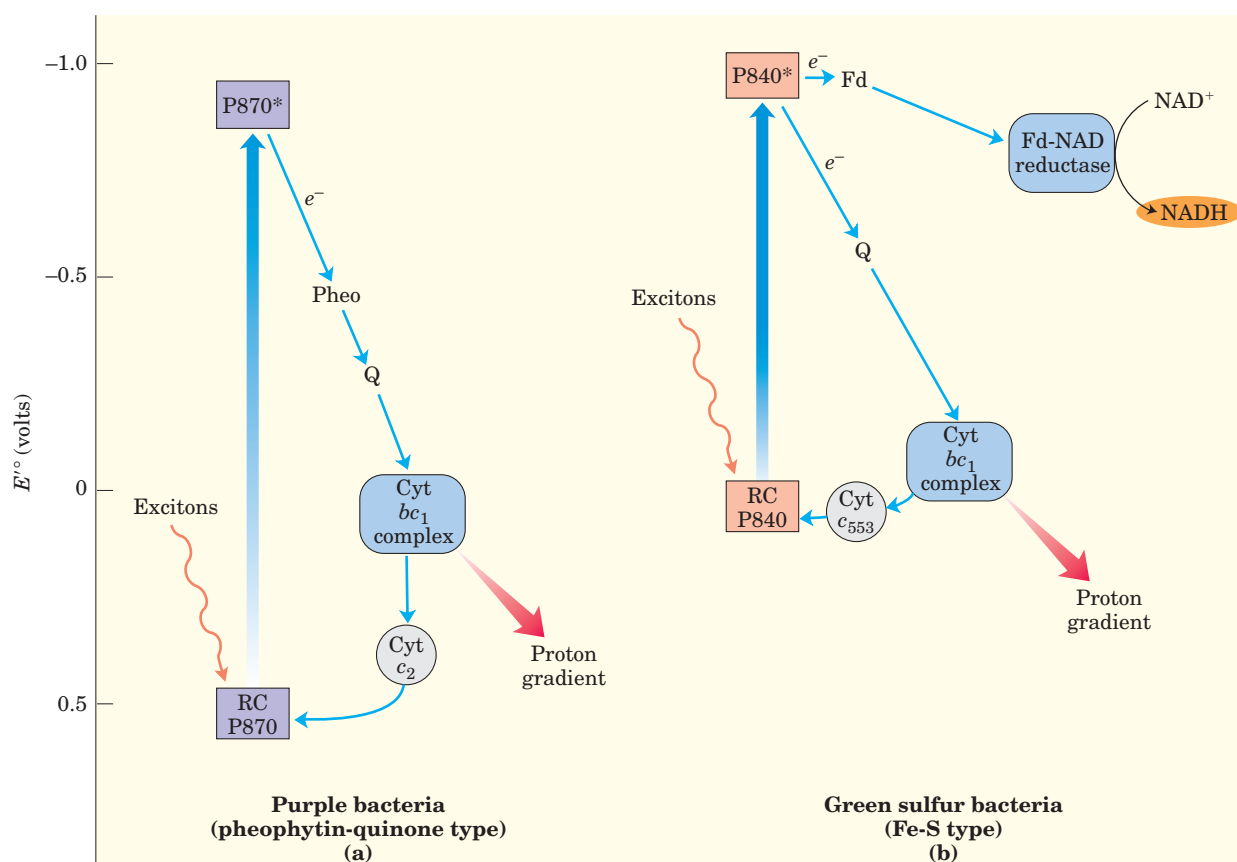
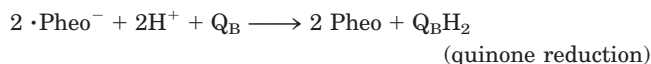


FIGURE 19-47 Functional modules of photosynthetic machinery in purple bacteria and green sulfur bacteria. (a) In purple bacteria, light energy drives electrons from the reaction center P870 through pheophytin (Pheo), a quinone (Q), and the cytochrome bc_1 complex, then through cytochrome c_2 back to the reaction center. Electron flow through the cytochrome bc_1 complex causes proton pumping, creating an elec-

trochemical potential that powers ATP synthesis. **(b)** Green sulfur bacteria have two routes for electrons driven by excitation of P840: a cyclic route passes through a quinone to the cytochrome bc_1 complex and back to the reaction center via cytochrome c , and a noncyclic route from the reaction center through the iron-sulfur protein ferredoxin (Fd), then to NAD^+ in a reaction catalyzed by ferredoxin:NAD reductase.

extra electron to a second, loosely bound quinone (Q_B). Two such electron transfers convert Q_B to its fully reduced form, Q_{BH_2} , which is free to diffuse in the membrane bilayer, away from the reaction center:



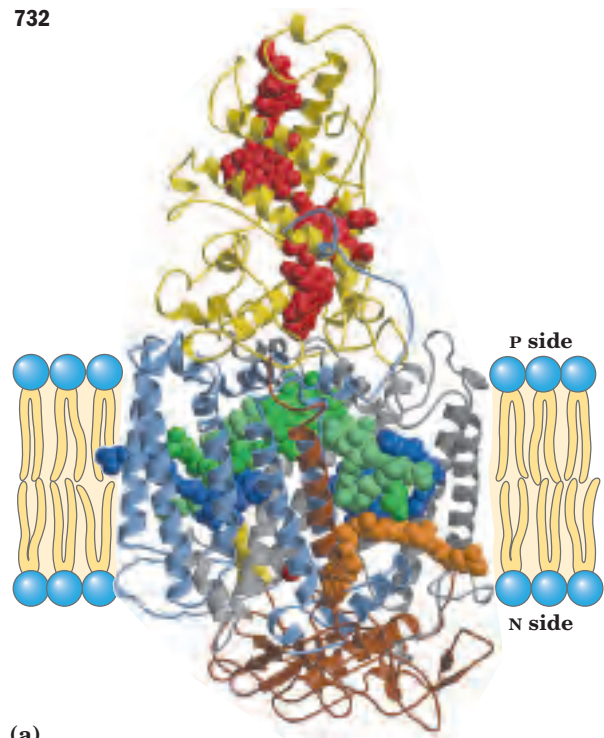
The hydroquinone (Q_{BH_2}), carrying in its chemical bonds some of the energy of the photons that originally excited P870, enters the pool of reduced quinone (QH_2) dissolved in the membrane and moves through the lipid phase of the bilayer to the cytochrome bc_1 complex.

Like the homologous Complex III in mitochondria, the cytochrome bc_1 complex of purple bacteria carries electrons from a quinol donor (QH_2) to an electron acceptor, using the energy of electron transfer to pump protons across the membrane, producing a proton-motive force. The path of electron flow through this complex is believed to be very similar to that through mitochondrial Complex III, involving a Q cycle (Fig. 19-12) in which protons are consumed on one side of the membrane and released on the other. The ultimate

electron acceptor in purple bacteria is the electron-depleted form of P870, $\cdot(\text{Chl})_2^+$ (Fig. 19-47a). Electrons move from the cytochrome bc_1 complex to P870 via a soluble c -type cytochrome, cytochrome c_2 . The electron-transfer process completes the cycle, returning the reaction center to its unbleached state, ready to absorb another exciton from antenna chlorophyll.

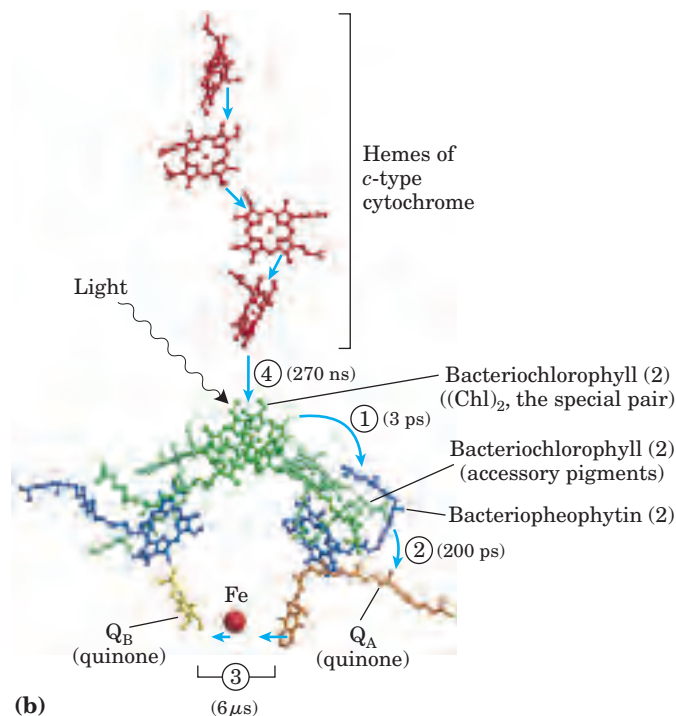
A remarkable feature of this system is that all the chemistry occurs in the *solid state*, with reacting species held close together in the right orientation for reaction. The result is a very fast and efficient series of reactions.

The Fe-S Reaction Center (Type I Reaction Center) Photosynthesis in green sulfur bacteria involves the same three modules as in purple bacteria, but the process differs in several respects and involves additional enzymatic reactions (Fig. 19-47b). Excitation causes an electron to move from the reaction center to the cytochrome bc_1 complex via a quinone carrier. Electron transfer through this complex powers proton transport and creates the proton-motive force used for ATP synthesis, just as in purple bacteria and in mitochondria.



(a)

FIGURE 19-48 Photoreaction center of the purple bacterium *Rhodospseudomonas viridis*. (PDB ID 1PRC) (a) The system has four components: three subunits, H, M, and L (brown, blue, and gray, respectively), with a total of 11 transmembrane helical segments, and a fourth protein, cytochrome c (yellow), associated with the complex at the membrane surface. Subunits L and M are paired transmembrane proteins that together form a cylindrical structure with roughly bilateral symmetry about its long axis. Shown as space-filling models (and in (b) as ball-and-stick structures) are the prosthetic groups that participate in the photochemical events. Bound to the L and M chains are two pairs of bacteriochlorophyll molecules (green); one of the pairs (the “special pair,” $(\text{Chl})_2$) is the site of the first photochemical changes after light absorption. Also incorporated in the system are a pair of pheophytin a (Pheo a) molecules (blue); two quinones, menaquinone (Q_A) and ubiquinone (Q_B) (orange and yellow), also arranged with bi-



(b)

lateral symmetry; and a single nonheme Fe (red) located approximately on the axis of symmetry between the quinones. Shown at the top of the figure are four heme groups (red) associated with the c-type cytochrome of the reaction center. The reaction center of another purple bacterium, *Rhodobacter sphaeroides*, is very similar, except that cytochrome c is not part of the crystalline complex.

(b) Sequence of events following excitation of the special pair of bacteriochlorophylls (all components colored as in (a)), with the time scale of the electron transfers in parentheses. ① The excited special pair passes an electron to pheophytin, ② from which the electron moves rapidly to the tightly bound menaquinone, Q_A . ③ This quinone passes electrons much more slowly to the diffusible ubiquinone, Q_B , through the nonheme Fe. Meanwhile, ④ the “electron hole” in the special pair is filled by an electron from a heme of cytochrome c.

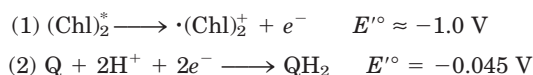
However, in contrast to the cyclic flow of electrons in purple bacteria, some electrons flow from the reaction center to an iron-sulfur protein, **ferredoxin**, which then passes electrons via ferredoxin:NAD reductase to NAD^+ , producing NADH. The electrons taken from the reaction center to reduce NAD^+ are replaced by the oxidation of H_2S to elemental S, then to SO_4^{2-} , in the reaction that defines the green sulfur bacteria. This oxidation of H_2S by bacteria is chemically analogous to the oxidation of H_2O by oxygenic plants.

Kinetic and Thermodynamic Factors Prevent the Dissipation of Energy by Internal Conversion

The complex construction of reaction centers is the product of evolutionary selection for efficiency in the photosynthetic process. The excited state $(\text{Chl})_2^*$ could in principle decay to its ground state by internal conversion, a very rapid process (10 picoseconds; $1 \text{ ps} = 10^{-12} \text{ s}$) in which the energy of the absorbed photon is

converted to heat (molecular motion). Reaction centers are constructed to prevent the inefficiency that would result from internal conversion. The proteins of the reaction center hold the bacteriochlorophylls, bacterio-pheophytins, and quinones in a fixed orientation relative to each other, allowing the photochemical reactions to take place in a virtually solid state. This accounts for the high efficiency and rapidity of the reactions; nothing is left to chance collision or random diffusion. Exciton transfer from antenna chlorophyll to the special pair of the reaction center is accomplished in less than 100 ps with $>90\%$ efficiency. Within 3 ps of the excitation of P870, pheophytin has received an electron and become a negatively charged radical; less than 200 ps later, the electron has reached the quinone Q_B (Fig. 19-48b). The electron-transfer reactions not only are fast but are thermodynamically “downhill”; the excited special pair $(\text{Chl})_2^*$ is a very good electron donor ($E'^{\circ} -1 \text{ V}$), and each successive electron transfer is to an acceptor of substantially less negative E'° . The standard free-energy

change for the process is therefore negative and large; recall from Chapter 13 that $\Delta G'^{\circ} = -nF\Delta E'^{\circ}$; here, $\Delta E'^{\circ}$ is the difference between the standard reduction potentials of the two half-reactions



Thus

$$\Delta E'^{\circ} = -0.045 \text{ V} - (-1.0 \text{ V}) \approx 0.95 \text{ V}$$

and

$$\Delta G'^{\circ} = -2(96.5 \text{ kJ/V} \cdot \text{mol})(0.95 \text{ V}) = -180 \text{ kJ/mol}$$

The combination of fast kinetics and favorable thermodynamics makes the process virtually irreversible and highly efficient. The overall energy yield (the percentage of the photon's energy conserved in QH_2) is $>30\%$, with the remainder of the energy dissipated as heat.

In Plants, Two Reaction Centers Act in Tandem

The photosynthetic apparatus of modern cyanobacteria, algae, and vascular plants is more complex than the one-center bacterial systems, and it appears to have evolved through the combination of two simpler bacterial photocenters. The thylakoid membranes of chloroplasts

have two different kinds of photosystems, each with its own type of photochemical reaction center and set of antenna molecules. The two systems have distinct and complementary functions (Fig. 19–49). **Photosystem II (PSII)** is a pheophytin-quinone type of system (like the single photosystem of purple bacteria) containing roughly equal amounts of chlorophylls *a* and *b*. Excitation of its reaction center P680 drives electrons through the cytochrome *b₆f* complex with concomitant movement of protons across the thylakoid membrane. **Photosystem I (PSI)** is structurally and functionally related to the type I reaction center of green sulfur bacteria. It has a reaction center designated P700 and a high ratio of chlorophyll *a* to chlorophyll *b*. Excited P700 passes electrons to the Fe-S protein ferredoxin, then to NADP^+ , producing NADPH. The thylakoid membranes of a single spinach chloroplast have many hundreds of each kind of photosystem.

These two reaction centers in plants act in tandem to catalyze the light-driven movement of electrons from H_2O to NADP^+ (Fig. 19–49). Electrons are carried between the two photosystems by the soluble protein **plastocyanin**, a one-electron carrier functionally similar to cytochrome *c* of mitochondria. To replace the electrons that move from PSII through PSI to NADP^+ , cyanobacteria and plants oxidize H_2O (as green sulfur

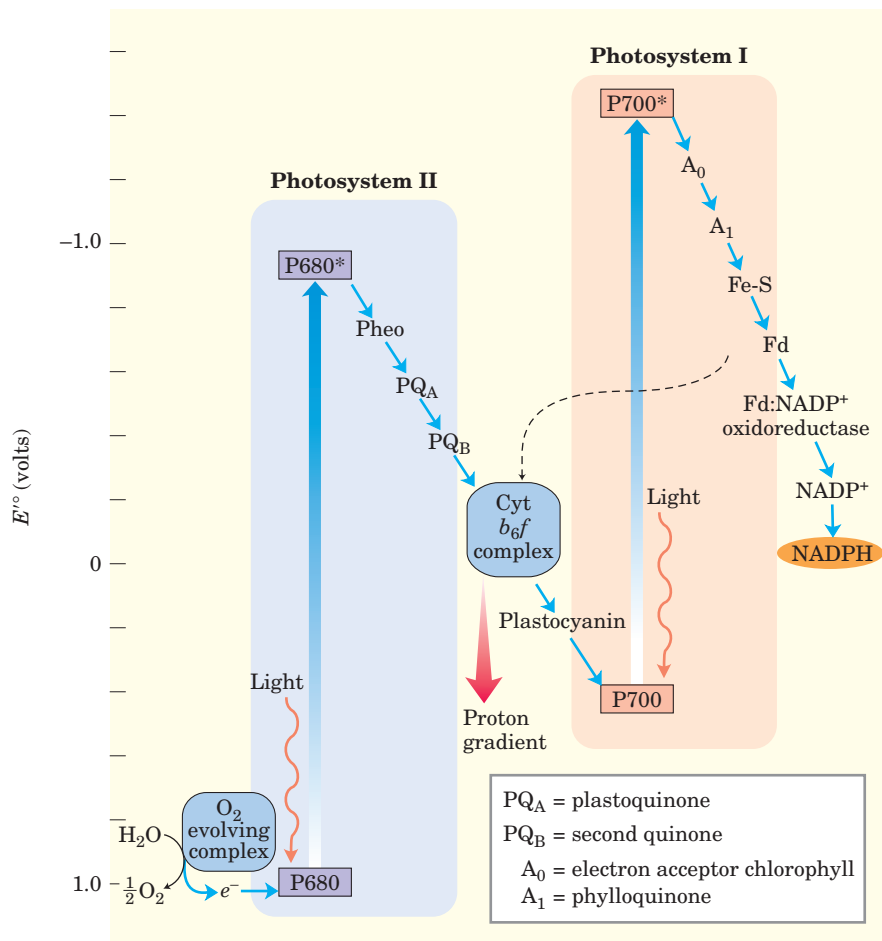


FIGURE 19–49 Integration of photosystems I and II in chloroplasts. This “Z scheme” shows the pathway of electron transfer from H_2O (lower left) to NADP^+ (far right) in noncyclic photosynthesis. The position on the vertical scale of each electron carrier reflects its standard reduction potential. To raise the energy of electrons derived from H_2O to the energy level required to reduce NADP^+ to NADPH, each electron must be “lifted” twice (heavy arrows) by photons absorbed in PSII and PSI. One photon is required per electron in each photosystem. After excitation, the high-energy electrons flow “downhill” through the carrier chains shown. Protons move across the thylakoid membrane during the water-splitting reaction and during electron transfer through the cytochrome *b₆f* complex, producing the proton gradient that is central to ATP formation. The dashed arrow is the path of cyclic electron transfer (discussed later in the text), which involves only PSI; electrons return via the cyclic pathway to PSI, instead of reducing NADP^+ to NADPH.

bacteria oxidize H_2S), producing O_2 (Fig. 19–49, bottom left). This process is called **oxygenic photosynthesis** to distinguish it from the anoxygenic photosynthesis of purple and green sulfur bacteria. All O_2 -evolving photosynthetic cells—those of plants, algae, and cyanobacteria—contain both PSI and PSII; organisms with only one photosystem do not evolve O_2 . The diagram in Figure 19–49, often called the **Z scheme** because of its overall form, outlines the pathway of electron flow between the two photosystems and the energy relationships in the light reactions. The Z scheme thus describes the complete route by which electrons flow from H_2O to NADP^+ , according to the equation



For every two photons absorbed (one by each photosystem), one electron is transferred from H_2O to NADP^+ . To form one molecule of O_2 , which requires transfer of four electrons from two H_2O to two NADP^+ , a total of eight photons must be absorbed, four by each photosystem.

The mechanistic details of the photochemical reactions in PSII and PSI are essentially similar to those of the two bacterial photosystems, with several important additions. In PSII, two very similar proteins, D1 and D2, form an almost symmetrical dimer, to which all the electron-carrying cofactors are bound (Fig. 19–50). Excitation of P680 in PSII produces P680^* , an excellent electron donor that, within picoseconds, transfers an electron to pheophytin, giving it a negative charge (Pheo^-). With the loss of its electron, P680^* is transformed into a radical cation, P680^+ . Pheo^- very rapidly passes its extra electron to a protein-bound **plastoquinone**, PQ_A (or Q_A), which in turn passes its electron to another, more loosely bound plastoquinone, PQ_B (or Q_B). When PQ_B has acquired two electrons in two such transfers from PQ_A and two protons from the solvent water, it is in its fully reduced quinol form, PQ_BH_2 . The overall reaction initiated by light in PSII is



Eventually, the electrons in PQ_BH_2 pass through the cytochrome b_6f complex (Fig. 19–49). The electron initially removed from P680 is replaced with an electron obtained from the oxidation of water, as described below. The binding site for plastoquinone is the point of action of many commercial herbicides that kill plants by blocking electron transfer through the cytochrome b_6f complex and preventing photosynthetic ATP production.

The photochemical events that follow excitation of PSI at the reaction center P700 are formally similar to those in PSII. The excited reaction center P700^* loses an electron to an acceptor, A_0 (believed to be a special form of chlorophyll, functionally homologous to the pheophytin of PSII), creating A_0^- and P700^+ (Fig. 19–49, right side); again, excitation results in charge separation

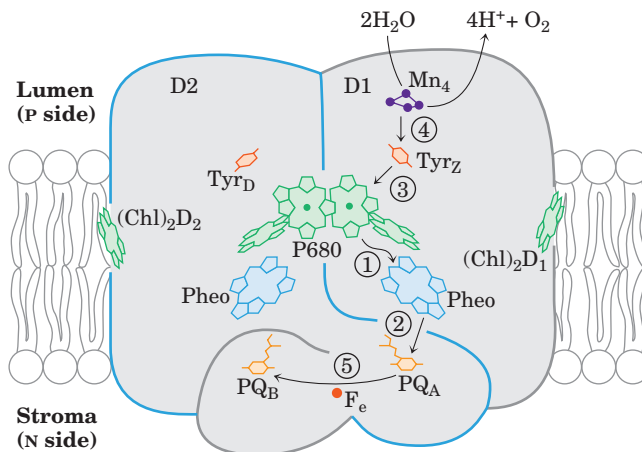


FIGURE 19–50 Photosystem II of the cyanobacterium *Synechococcus elongates*. The monomeric form of the complex shown here has two major transmembrane proteins, D1 and D2, each with its set of cofactors. Although the two subunits are nearly symmetric, electron flow occurs through only one of the two branches of cofactors, that on the right (on D1). The arrows show the path of electron flow from the Mn ion cluster (Mn_4 , purple) of the water-splitting enzyme to the quinone PQ_B (orange). The photochemical events occur in the sequence indicated by the step numbers. Notice the close similarity between the positions of cofactors here and the positions in the bacterial photoreaction center shown in Figure 19–48. The role of the Tyr residues is discussed later in the text.

at the photochemical reaction center. P700^+ is a strong oxidizing agent, which quickly acquires an electron from plastocyanin, a soluble Cu-containing electron-transfer protein. A_0^- is an exceptionally strong reducing agent that passes its electron through a chain of carriers that leads to NADP^+ . First, **phylloquinone** (A_1) accepts an electron and passes it to an iron-sulfur protein (through three Fe-S centers in PSI). From here, the electron moves to **ferredoxin** (Fd), another iron-sulfur protein loosely associated with the thylakoid membrane. Spinach ferredoxin (M_r 10,700) contains a 2Fe-2S center (Fig. 19–5) that undergoes one-electron oxidation and reduction reactions. The fourth electron carrier in the chain is the flavoprotein **ferredoxin:NADP⁺ oxidoreductase**, which transfers electrons from reduced ferredoxin (Fd_{red}) to NADP^+ :



This enzyme is homologous to the ferredoxin:NAD reductase of green sulfur bacteria (Fig. 19–47b).

Antenna Chlorophylls Are Tightly Integrated with Electron Carriers

The electron-carrying cofactors of PSI and the light-harvesting complexes are part of a supramolecular complex (Fig. 19–51a), the structure of which has been solved crystallographically. The protein consists of three identical complexes, each composed of 11 different proteins (Fig. 19–51b). In this remarkable structure the many antenna chlorophyll and carotenoid molecules are

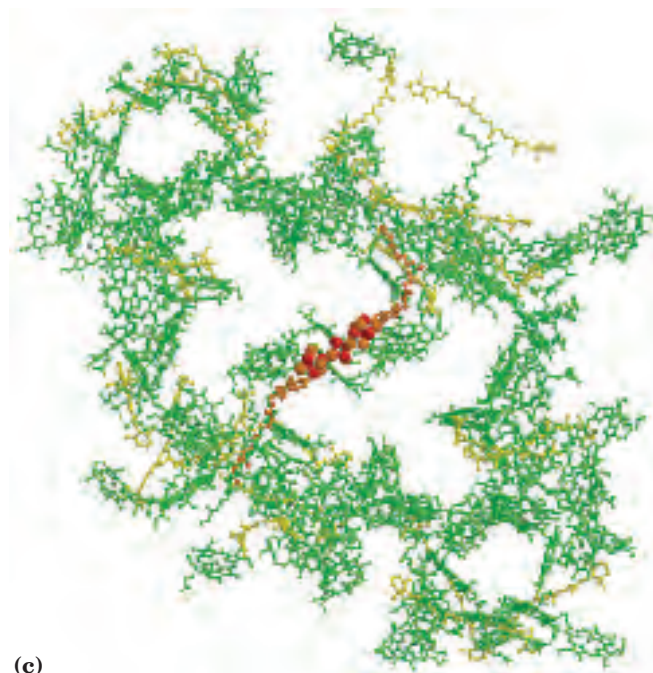
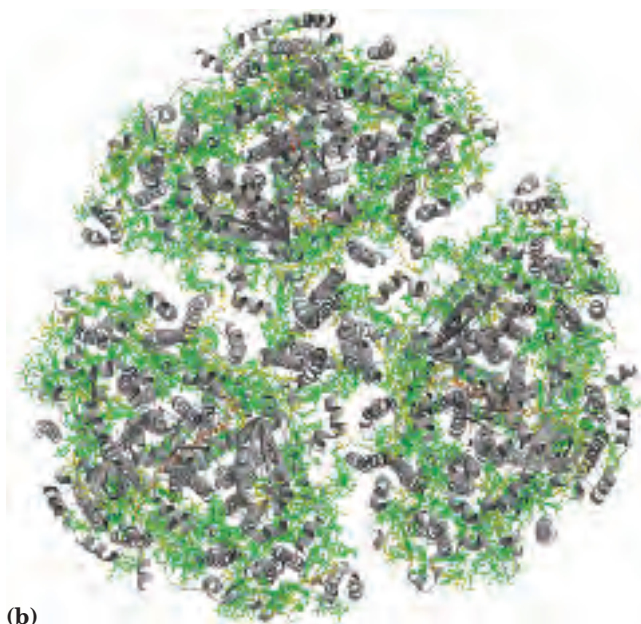
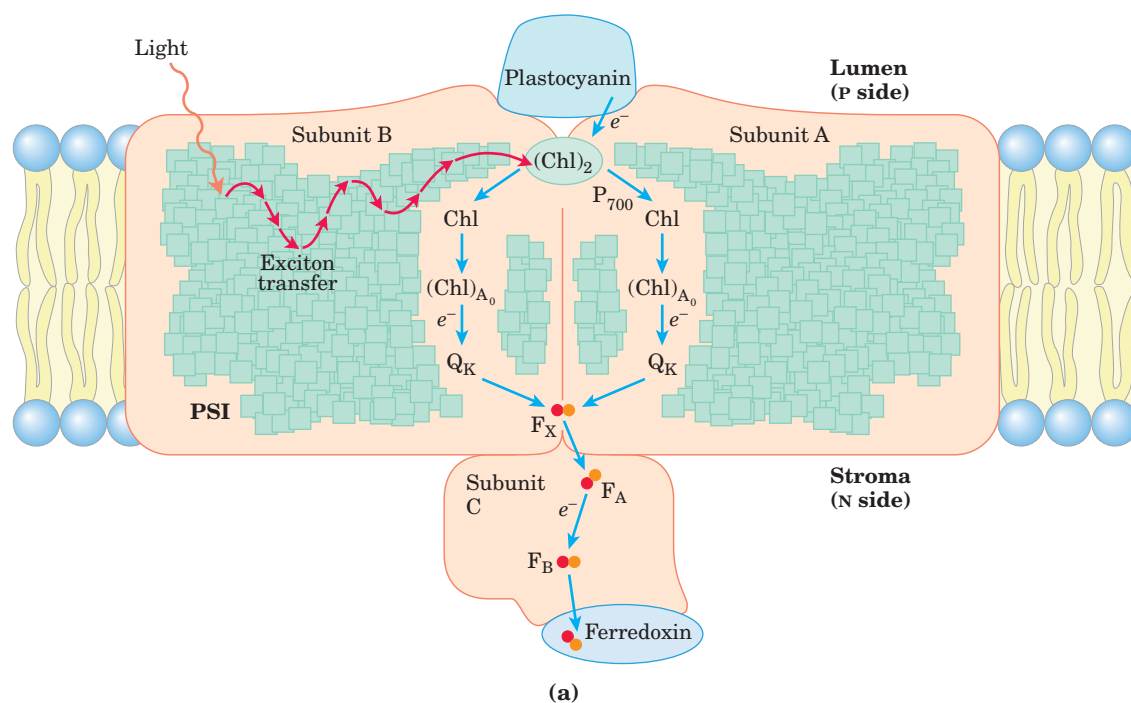


FIGURE 19-51 The supramolecular complex of PSI and its associated antenna chlorophylls. (a) Schematic drawing of the essential proteins and cofactors in a single unit of PSI. A large number of antenna chlorophylls surround the reaction center and convey to it (red arrows) the energy of photons they have absorbed. The result is excitation of the pair of chlorophyll molecules that constitute P700. Excitation of P700 greatly decreases its reduction potential, and it passes an electron through two nearby chlorophylls to phyloquinone (Q_K ; also called A_1). Reduced phyloquinone is reoxidized as it passes two electrons, one at a time, to an Fe-S protein (F_X) near the N side of the membrane. From F_X , electrons move through two more Fe-S centers (F_A and F_B), then to the Fe-S protein ferredoxin in the stroma. Ferredoxin

then donates electrons to NADP^+ (not shown), reducing it to NADPH, one of the forms in which the energy of photons is trapped in chloroplasts. (b) The trimeric structure (derived from PDB ID 1JBO), viewed from the thylakoid lumen perpendicular to the membrane, showing all protein subunits (gray) and cofactors (described in (c)). (c) A monomer of PSI with all the proteins omitted, revealing the antenna and reaction center chlorophylls (green with dark green Mg^{2+} ions in the center), carotenoids (yellow), and Fe-S centers of the reaction center (space-filling red and orange structures). The proteins in the complex hold the components rigidly in orientations that maximize efficient exciton transfers between excited antenna molecules and the reaction center.

precisely arrayed around the reaction center (Fig. 19–51c). The reaction center's electron-carrying cofactors are therefore tightly integrated with antenna chlorophylls. This arrangement allows very rapid and efficient exciton transfer from antenna chlorophylls to the reaction center. In contrast to the single path of electrons in PSII, the electron flow initiated by absorption of a photon is believed to occur through both branches of carriers in PSI.

Spatial Separation of Photosystems I and II Prevents Exciton Larceny

The energy required to excite PSI (P700) is less than that needed to excite PSII (P680) (shorter wavelength corresponds to higher energy). If PSI and PSII were physically contiguous, excitons originating in the antenna system of PSII would migrate to the reaction center of PSI, leaving PSII chronically underexcited and interfering with the operation of the two-center system. This “exciton larceny” is prevented by separation of PSI and PSII in the thylakoid membrane (Fig. 19–52). PSII is located almost exclusively in the tightly appressed membrane stacks of thylakoid grana (granal lamellae); its associated light-harvesting complex (LHCII) mediates the tight association of adjacent membranes in the grana. PSI and the ATP synthase complex are located almost exclusively in the thylakoid membranes that are not appressed (the stromal lamellae), where both have access to the contents of the stroma, including ADP and NADP^+ . The cytochrome b_6f complex is present throughout the thylakoid membrane.

The association of LHCII with PSII is regulated by light intensity and wavelength. In bright sunlight (with a large component of blue light), PSII absorbs more light

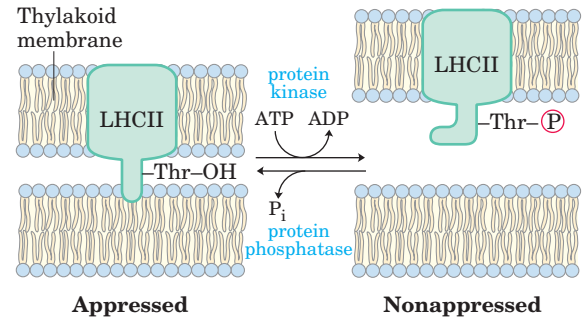
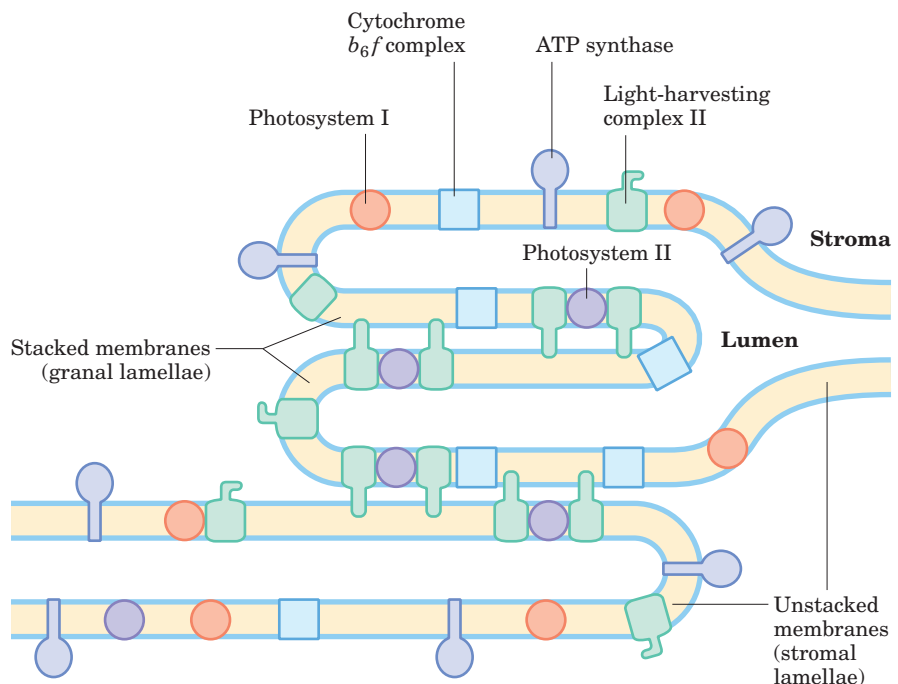


FIGURE 19–53 Equalization of electron flow in PSI and PSII by modulation of granal stacking. A hydrophobic domain of LHCII in one thylakoid lamella inserts into the neighboring lamella and closely appresses the two membranes. Accumulation of plastoquinol (not shown) stimulates a protein kinase that phosphorylates a Thr residue in the hydrophobic domain of LHCII, which reduces its affinity for the neighboring thylakoid membrane and converts appressed regions (granal lamellae) to nonappressed regions (stromal lamellae). A specific protein phosphatase reverses this regulatory phosphorylation when the $[\text{PQ}]/[\text{PQH}_2]$ ratio increases.

than PSI and produces reduced plastoquinone (plastoquinol, PQH_2) faster than PSI can oxidize it. The resulting accumulation of PQH_2 activates a protein kinase that phosphorylates a Thr residue on LHCII (Fig. 19–53). Phosphorylation weakens the interaction of LHCII with PSII, and some LHCII dissociates and moves to the stromal lamellae; here it captures photons for PSI, speeding the oxidation of PQH_2 and reversing the imbalance between electron flow in PSI and PSII. In less intense light (in the shade, with more red light), PSI oxidizes PQH_2 faster than PSII can make it, and the resulting increase in PQ concentration triggers dephosphorylation of LHCII, reversing the effect of phosphorylation.

FIGURE 19–52 Localization of PSI and PSII in thylakoid membranes. Light-harvesting complex LHCII and ATP synthase are located in regions of the thylakoid membrane that are appressed (granal lamellae, in which several membranes are in contact) and in regions that are not appressed (stromal lamellae) and have ready access to ADP and NADP^+ in the stroma. Photosystem II is present almost exclusively in the appressed regions, and photosystem I almost exclusively in nonappressed regions exposed to the stroma. LHCII is the “adhesive” that holds appressed lamellae together (see Fig. 19–53).



The Cytochrome b_6f Complex Links Photosystems II and I

Electrons temporarily stored in plastoquinol as a result of the excitation of P680 in PSII are carried to P700 of PSI via the cytochrome b_6f complex and the soluble protein plastocyanin (Fig. 19–49, center). Like Complex III

of mitochondria, the cytochrome b_6f complex (Fig. 19–54) contains a b -type cytochrome with two heme groups (designated b_H and b_L), a Rieske iron-sulfur protein (M_r 20,000), and cytochrome f (for the Latin *frons*, “leaf”). Electrons flow through the cytochrome b_6f complex from PQ_BH_2 to cytochrome f , then to plastocyanin, and finally to P700, thereby reducing it.

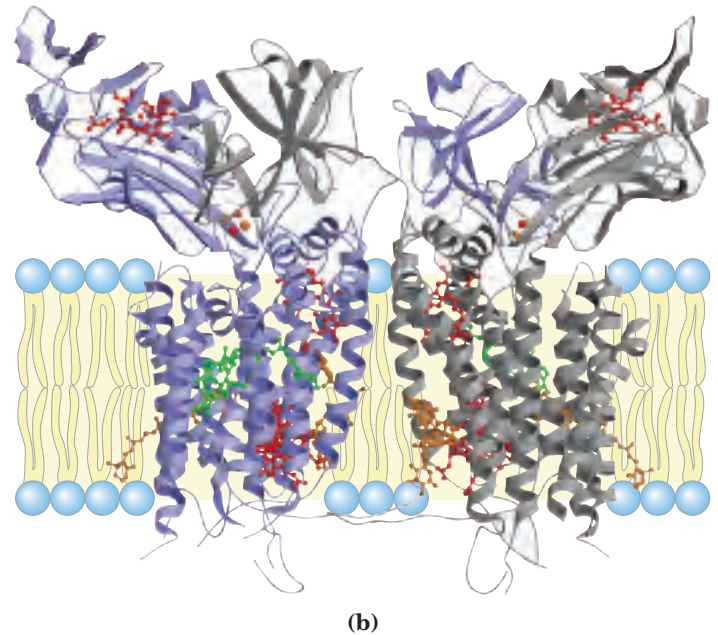
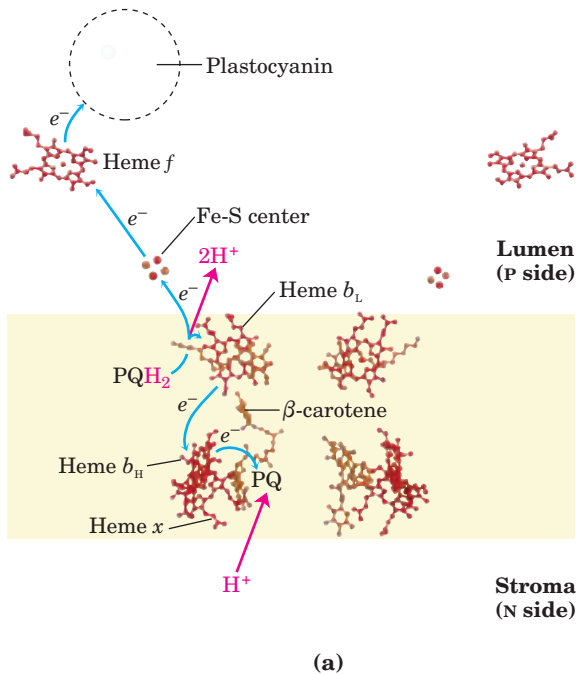
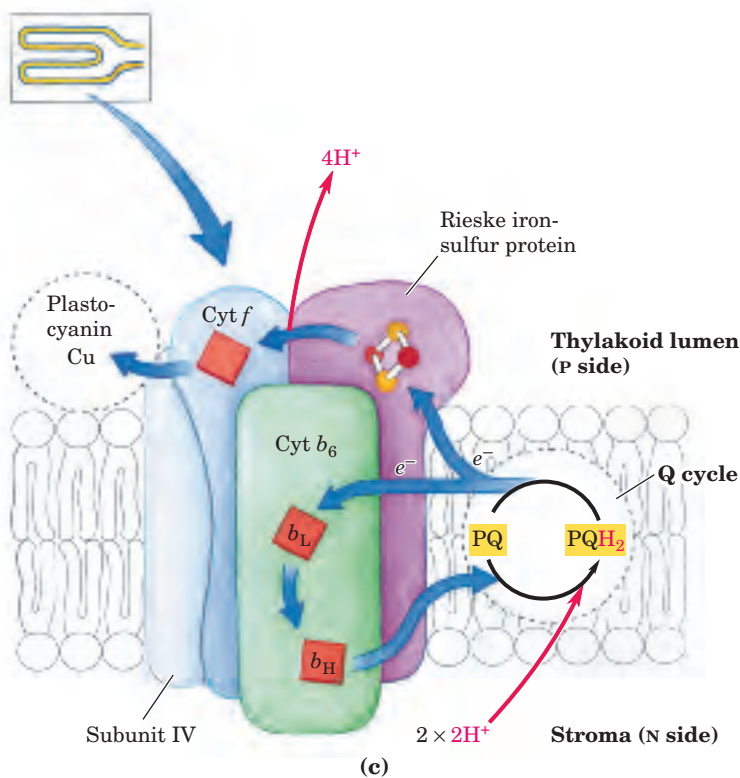


FIGURE 19–54 Electron and proton flow through the cytochrome b_6f complex. (a) The crystal structure of the complex (PDB ID 1UM3) reveals the positions of the cofactors involved in electron transfers. In addition to the hemes of cytochrome b (heme b_H and b_L ; also called heme b_N and b_P , respectively, because of their proximity to the N and P sides of the bilayer) and that of cytochrome f (heme f), there is a fourth (heme x) near heme b_H , and there is a β -carotene of unknown function. Two sites bind plastoquinone: the PQH_2 site near the P side of the bilayer, and the PQ site near the N side. The Fe-S center of the Rieske protein lies just outside the bilayer on the P side, and the heme f site is on a protein domain that extends well into the thylakoid lumen. (b) The complex is a homodimer arranged to create a cavern connecting the PQH_2 and PQ sites (compare with the structure of mitochondrial Complex III in Fig. 19–12). This cavern allows plastoquinone movement between the sites of its oxidation and reduction.

(c) Plastoquinol (PQH_2) formed in PSII is oxidized by the cytochrome b_6f complex in a series of steps like those of the Q cycle in the cytochrome bc_1 complex (Complex III) of mitochondria (see Fig. 19–11). One electron from PQH_2 passes to the Fe-S center of the Rieske protein (purple), the other to heme b_L of cytochrome b_6 (green). The net effect is passage of electrons from PQH_2 to the soluble protein plastocyanin, which carries them to PSI.



Like Complex III of mitochondria, cytochrome b_6f conveys electrons from a reduced quinone—a mobile, lipid-soluble carrier of two electrons (Q in mitochondria, PQ_B in chloroplasts)—to a water-soluble protein that carries one electron (cytochrome c in mitochondria, plastocyanin in chloroplasts). As in mitochondria, the function of this complex involves a Q cycle (Fig. 19–12) in which electrons pass, one at a time, from PQ_BH_2 to cytochrome b_6 . This cycle results in the pumping of protons across the membrane; in chloroplasts, the direction of proton movement is from the stromal compartment to the thylakoid lumen, up to four protons moving for each pair of electrons. The result is production of a proton gradient across the thylakoid membrane as electrons pass from PSII to PSI. Because the volume of the flattened thylakoid lumen is small, the influx of a small number of protons has a relatively large effect on luminal pH. The measured difference in pH between the stroma (pH 8) and the thylakoid lumen (pH 5) represents a 1,000-fold difference in proton concentration—a powerful driving force for ATP synthesis.

Cyanobacteria Use the Cytochrome b_6f Complex and Cytochrome c_6 in Both Oxidative Phosphorylation and Photophosphorylation

Cyanobacteria can synthesize ATP by oxidative phosphorylation or by photophosphorylation, although they have neither mitochondria nor chloroplasts. The enzymatic machinery for both processes is in a highly convoluted plasma membrane (see Fig. 1–6). Two protein components function in both processes (Fig. 19–55). The proton-pumping cytochrome b_6f complex carries electrons from plastoquinone to cytochrome c_6 in photosynthesis, and also carries electrons from ubiquinone to cytochrome c_6 in oxidative phosphorylation—the role played by cytochrome bc_1 in mitochondria. Cytochrome c_6 , homologous to mitochondrial cytochrome c , carries electrons from Complex III to Complex IV in cyanobacteria; it can also carry electrons from the cytochrome b_6f complex to PSI—a role performed in plants by plastocyanin. We therefore see the functional homology between the cyanobacterial cytochrome b_6f complex and the mitochondrial cytochrome bc_1 complex, and between cyanobacterial cytochrome c_6 and plant plastocyanin.

Water Is Split by the Oxygen-Evolving Complex

The ultimate source of the electrons passed to NADPH in plant (oxygenic) photosynthesis is water. Having given up an electron to pheophytin, $P680^+$ (of PSII) must acquire an electron to return to its ground state in preparation for capture of another photon. In principle, the required electron might come from any number of organic or inorganic compounds. Photosynthetic bacteria use a variety of electron donors for this purpose—acetate, succinate, malate, or sulfide—depending on

what is available in a particular ecological niche. About 3 billion years ago, evolution of primitive photosynthetic bacteria (the progenitors of the modern cyanobacteria) produced a photosystem capable of taking electrons from a donor that is always available—water. Two water molecules are split, yielding four electrons, four protons, and molecular oxygen:



A single photon of visible light does not have enough energy to break the bonds in water; four photons are required in this photolytic cleavage reaction.

The four electrons abstracted from water do not pass directly to $P680^+$, which can accept only one electron at a time. Instead, a remarkable molecular device, the **oxygen-evolving complex** (also called the **water-splitting complex**), passes four electrons *one at a*

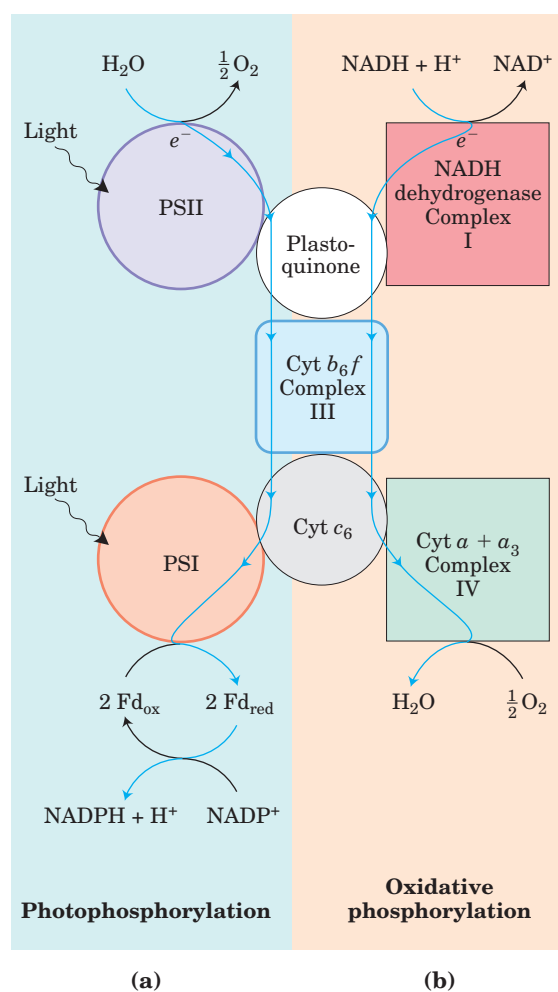


FIGURE 19–55 Dual roles of cytochrome b_6f and cytochrome c_6 in cyanobacteria. Cyanobacteria use cytochrome b_6f , cytochrome c_6 , and plastoquinone for both oxidative phosphorylation and photophosphorylation. (a) In photophosphorylation, electrons flow (top to bottom) from water to $NADP^+$. (b) In oxidative phosphorylation, electrons flow from $NADH$ to O_2 . Both processes are accompanied by proton movement across the membrane, accomplished by a Q cycle.

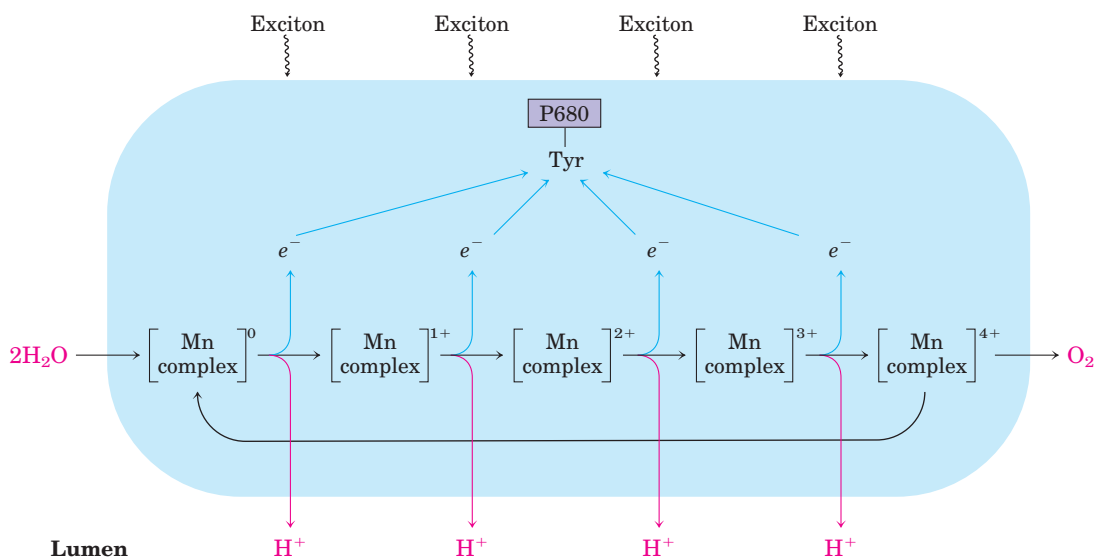


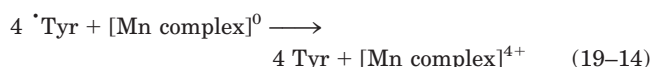
FIGURE 19-56 Water-splitting activity of the oxygen-evolving complex. Shown here is the process that produces a four-electron oxidizing agent—believed to be a multinuclear center with several Mn ions—in the water-splitting complex of PSII. The sequential absorption of four photons (excitons), each absorption causing the loss of one elec-

tron from the Mn center, produces an oxidizing agent that can remove four electrons from two molecules of water, producing O_2 . The electrons lost from the Mn center pass one at a time to an oxidized Tyr residue in a PSII protein, then to P680 $^+$.

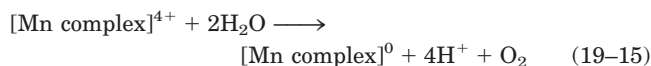
time to P680 $^+$ (Fig. 19-56). The immediate electron donor to P680 $^+$ is a Tyr residue (often designated Z or Tyr $_Z$) in protein subunit D1 of the PSII reaction center. The Tyr residue loses both a proton and an electron, generating the electrically neutral Tyr free radical, $^{\bullet}\text{Tyr}$:



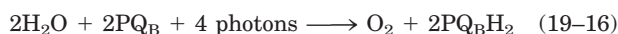
The Tyr radical regains its missing electron and proton by oxidizing a cluster of four manganese ions in the water-splitting complex. With each single-electron transfer, the Mn cluster becomes more oxidized; four single-electron transfers, each corresponding to the absorption of one photon, produce a charge of +4 on the Mn complex (Fig. 19-56):



In this state, the Mn complex can take four electrons from a pair of water molecules, releasing 4 H^+ and O_2 :



Because the four protons produced in this reaction are released into the thylakoid lumen, the oxygen-evolving complex acts as a proton pump, driven by electron transfer. The sum of Equations 19-12 through 19-15 is



The water-splitting activity associated with the PSII reaction center has proved exceptionally difficult to purify. A peripheral membrane protein (M_r 33,000) on the

luminal side of the thylakoid membrane is believed to stabilize the Mn complex. In the crystal structure (PDB ID 1FE1; see Fig. 19-50), four Mn ions are clustered with precise geometry near a Tyr residue on the D1 subunit, presumably the one involved in water oxidation. Manganese can exist in stable oxidation states from +2 to +7, so a cluster of four Mn ions can certainly donate or accept four electrons. The mechanism shown in Figure 19-56 is consistent with the experimental facts, but until the exact chemical structures of all the intermediates of the Mn cluster are known, the detailed mechanism remains elusive.

SUMMARY 19.8 The Central Photochemical Event: Light-Driven Electron Flow

- Bacteria have a single reaction center; in purple bacteria, it is of the pheophytin-quinone type, and in green sulfur bacteria, the Fe-S type.
- Structural studies of the reaction center of a purple bacterium have provided information about light-driven electron flow from an excited special pair of chlorophyll molecules, through pheophytin, to quinones. Electrons then pass from quinones through the cytochrome bc_1 complex, and back to the photoreaction center.
- An alternative path, in green sulfur bacteria, sends electrons from reduced quinones to NAD^+ .

- Cyanobacteria and plants have two different photoreaction centers, arranged in tandem.
- Plant photosystem I passes electrons from its excited reaction center, P700, through a series of carriers to ferredoxin, which then reduces NADP^+ to NADPH.
- The reaction center of plant photosystem II, P680, passes electrons to plastoquinone, and the electrons lost from P680 are replaced by electrons from H_2O (electron donors other than H_2O are used in other organisms).
- The light-driven splitting of H_2O is catalyzed by a Mn-containing protein complex; O_2 is produced. The reduced plastoquinone carries electrons to the cytochrome b_6f complex; from here they pass to plastocyanin, and then to P700 to replace those lost during its photoexcitation.
- Electron flow through the cytochrome b_6f complex drives protons across the plasma membrane, creating a proton-motive force that provides the energy for ATP synthesis by an ATP synthase.

19.9 ATP Synthesis by Photophosphorylation

The combined activities of the two plant photosystems move electrons from water to NADP^+ , conserving some of the energy of absorbed light as NADPH (Fig. 19–49). Simultaneously, protons are pumped across the thylakoid membrane and energy is conserved as an electrochemical potential. We turn now to the process by which this proton gradient drives the synthesis of ATP, the other energy-conserving product of the light-dependent reactions.



Daniel Arnon, 1910–1994

In 1954 Daniel Arnon and his colleagues discovered that ATP is generated from ADP and P_i during photosynthetic electron transfer in illuminated spinach chloroplasts. Support for these findings came from the work of Albert Frenkel, who detected light-dependent ATP production in pigment-containing membranous structures called **chromatophores**, derived from photosynthetic

bacteria. Investigators concluded that some of the light energy captured by the photosynthetic systems of these organisms is transformed into the phosphate bond en-

ergy of ATP. This process is called **photophosphorylation**, to distinguish it from oxidative phosphorylation in respiring mitochondria.

A Proton Gradient Couples Electron Flow and Phosphorylation

Several properties of photosynthetic electron transfer and photophosphorylation in chloroplasts indicate that a proton gradient plays the same role as in mitochondrial oxidative phosphorylation. (1) The reaction centers, electron carriers, and ATP-forming enzymes are located in a proton-impermeable membrane—the thylakoid membrane—which must be intact to support photophosphorylation. (2) Photophosphorylation can be uncoupled from electron flow by reagents that promote the passage of protons through the thylakoid membrane. (3) Photophosphorylation can be blocked by venturicidin and similar agents that inhibit the formation of ATP from ADP and P_i by the mitochondrial ATP synthase (Table 19–4). (4) ATP synthesis is catalyzed by F_0F_1 complexes, located on the outer surface of the thylakoid membranes, that are very similar in structure and function to the F_0F_1 complexes of mitochondria.

Electron-transferring molecules in the chain of carriers connecting PSII and PSI are oriented asymmetrically in the thylakoid membrane, so photoinduced electron flow results in the net movement of protons across the membrane, from the stromal side to the thylakoid lumen (Fig. 19–57). In 1966 André Jagendorf showed that a pH gradient across the thylakoid membrane (alkaline outside) could furnish the driving force to generate ATP. His early observations provided some of the most important experimental evidence in support of Mitchell's chemiosmotic hypothesis.



André Jagendorf

Jagendorf incubated chloroplasts, in the dark, in a pH 4 buffer; the buffer slowly penetrated into the inner compartment of the thylakoids, lowering their internal pH. He added ADP and P_i to the dark suspension of chloroplasts and then suddenly raised the pH of the outer medium to 8, momentarily creating a large pH gradient across the membrane. As protons moved out of the thylakoids into the medium, ATP was generated from ADP and P_i . Because the formation of ATP occurred in the dark (with no input of energy from light), this experiment showed that a pH gradient across the membrane is a high-energy state that, as in mitochondrial oxidative phosphorylation, can mediate the transduction of energy from electron transfer into the chemical energy of ATP.

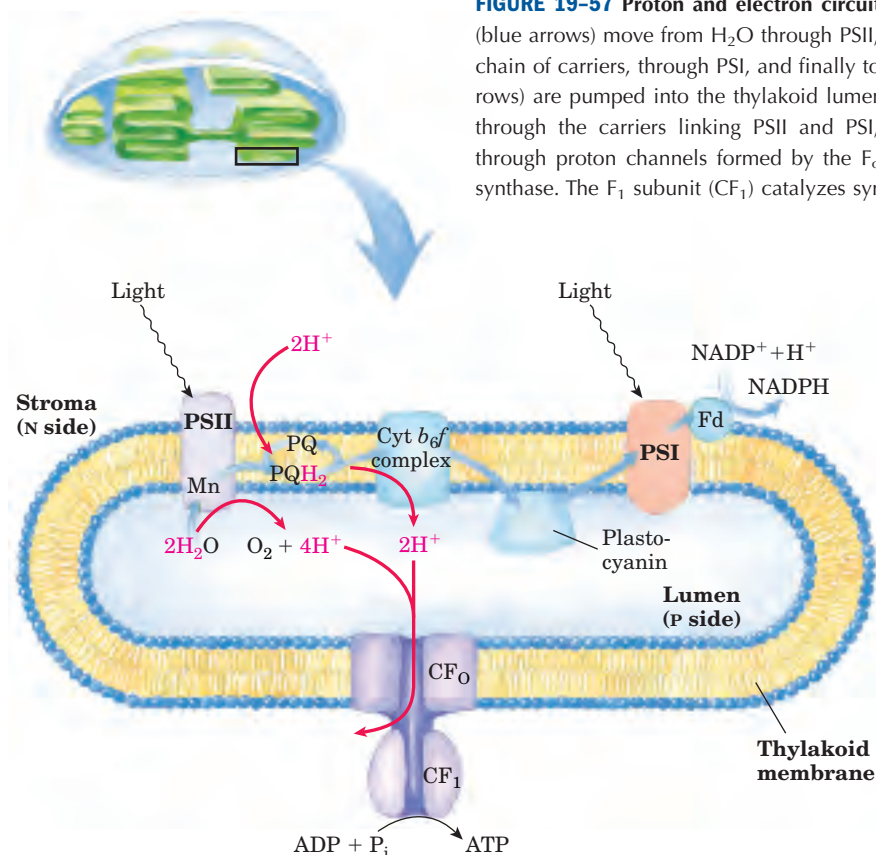


FIGURE 19-57 Proton and electron circuits in thylakoids. Electrons (blue arrows) move from H_2O through PSII, through the intermediate chain of carriers, through PSI, and finally to NADP^+ . Protons (red arrows) are pumped into the thylakoid lumen by the flow of electrons through the carriers linking PSII and PSI, and reenter the stroma through proton channels formed by the F_0 (designated CF_0) of ATP synthase. The F_1 subunit (CF_1) catalyzes synthesis of ATP.

The Approximate Stoichiometry of Photophosphorylation Has Been Established

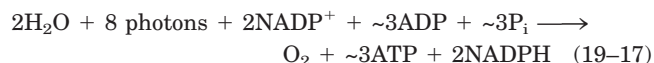
As electrons move from water to NADP^+ in plant chloroplasts, about 12 H^+ move from the stroma to the thylakoid lumen per four electrons passed (that is, per O_2 formed). Four of these protons are moved by the oxygen-evolving complex, and up to eight by the cytochrome b_6f complex. The measurable result is a 1,000-fold difference in proton concentration across the thylakoid membrane ($\Delta\text{pH} = 3$). Recall that the energy stored in a proton gradient (the electrochemical potential) has two components: a proton concentration difference (ΔpH) and an electrical potential ($\Delta\psi$) due to charge separation. In chloroplasts, ΔpH is the dominant component; counterion movement apparently dissipates most of the electrical potential. In illuminated chloroplasts, the energy stored in the proton gradient per mole of protons is

$$\Delta G = 2.3RT \Delta\text{pH} + ZF \Delta\psi = -17 \text{ kJ/mol}$$

so the movement of 12 mol of protons across the thylakoid membrane represents conservation of about 200 kJ of energy—enough energy to drive the synthesis of several moles of ATP ($\Delta G'^{\circ} = 30.5 \text{ kJ/mol}$). Experimental measurements yield values of about 3 ATP per O_2 produced.

At least eight photons must be absorbed to drive four electrons from H_2O to NADPH (one photon per electron at each reaction center). The energy in eight photons of visible light is more than enough for the synthesis of three molecules of ATP.

ATP synthesis is not the only energy-conserving reaction of photosynthesis in plants; the NADPH formed in the final electron transfer is (like its close analog NADH) also energetically rich. The overall equation for noncyclic photophosphorylation (a term explained below) is



Cyclic Electron Flow Produces ATP but Not NADPH or O_2

Using an alternative path of light-induced electron flow, plants can vary the ratio of NADPH to ATP formed in the light; this path is called **cyclic electron flow** to differentiate it from the normally unidirectional or **non-cyclic electron flow** from H_2O to NADP^+ , as discussed thus far. Cyclic electron flow (Fig. 19-49) involves only PSI. Electrons passing from P700 to ferredoxin do not continue to NADP^+ , but move back through the cytochrome b_6f complex to plastocyanin. The path of